The Science of Fairness



Stephen Takowsky

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Reference Edition

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Publishing Details

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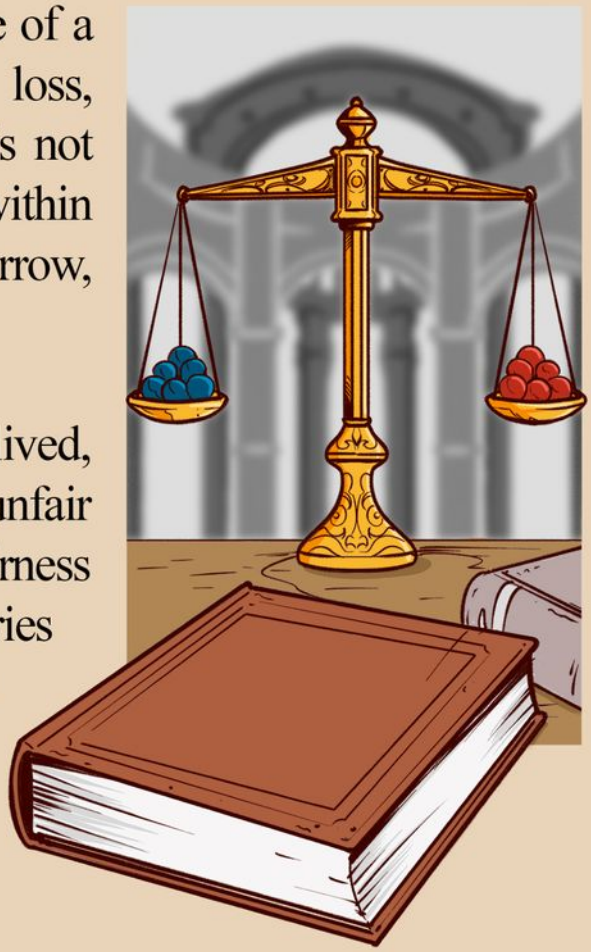
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Series Map

THE LAW OF FAIRNESS

A new way of seeing every life.

We judge fairness by the visible shape of a life: health and hardship, love and loss, opportunity and misfortune. Yet life is not lived from the outside. It is lived within consciousness, through joy and sorrow, pleasure and pain, peace and fear.



Could life be fair where it is actually lived, even when it appears profoundly unfair from the outside? The Law of Fairness proposes that every felt experience carries weight according to its intensity and duration. Across the completed conscious life, total positive and negative feeling exactly balance.

Picture a scale open for a lifetime. Joy and relief add weight to one side; pain and grief to the other. It may tip heavily for years, but when the conscious life is complete, the two sides balance. Each life carries its own scale. No life balances another.

Lives may differ enormously in length, choices, pleasures, wounds, and circumstances. The scale weighs neither wealth nor virtue, status nor faith. It weighs felt experience alone. Circumstances tell the visible story of a life, but feeling is how that story is lived. The Law asks us not to mistake the visible story for the whole.

WHAT IF THE LAW IS REAL?

A new meaning of utopia.

If the Law of Fairness is real, utopia cannot mean endless pleasure. More comfort, luxury, or consumption does not make a society ideal. If conscious lives end with no lasting surplus of positive feeling, utopia must mean something deeper and steadier than limitless happiness.



"There is no free happiness" does not mean joy is wrong, pain is payment, or suffering is holy. Pleasure alone cannot define a good life, nor appearances a good society. The goal is joy without excess. A fairer world would reduce needless suffering, poverty, disease, cruelty, corruption, and abuse while protecting love, beauty, rest, play, choice, and joy. Moderation restrains excess, work builds worthy goals, and stewardship preserves the world so future generations inherit something worth protecting.

If the Law is real, its deepest equality lies beneath wealth, status, power, culture, and belief. Lives may differ while reaching the same final balance of felt experience. These ten books follow that question from science into life, spirituality, government, cities, law, transition, and peace, exploring how humanity might build a lasting utopia.

THE TEN BOOKS OF FAIRNESS

A complete path from truth to practice.

THE LAW OF FAIRNESS



Explores the core question, defines key terms, and sets the ethical bounds of the series.

THE SCIENCE OF FAIRNESS



Evaluates the Law through evidence, measurement, rival models, protocols, and falsification.

THE UTOPIA OF FAIRNESS



Brings the series into a mature view of what a peaceful world could become.

THE TRANSITION OF FAIRNESS



Asks how a move toward utopia could unfold with care, step by step, under real limits.

THE LIFE OF FAIRNESS



Brings the Law into daily life and asks what it means for how we live.

THE RELIGION OF FAIRNESS



Explores faith, world religions, and the optional path of Fairness.

THE GOVERNMENT OF FAIRNESS



Examines civic structure and the public institutions that give it form in practice.

THE CITY OF FAIRNESS



Builds a civic space for learning, service, and peace with privacy kept safe.

THE CONSTITUTION OF FAIRNESS



Sets out the rights, rules, paths, and checks a lasting utopia would need.

THE PEACE OF FAIRNESS



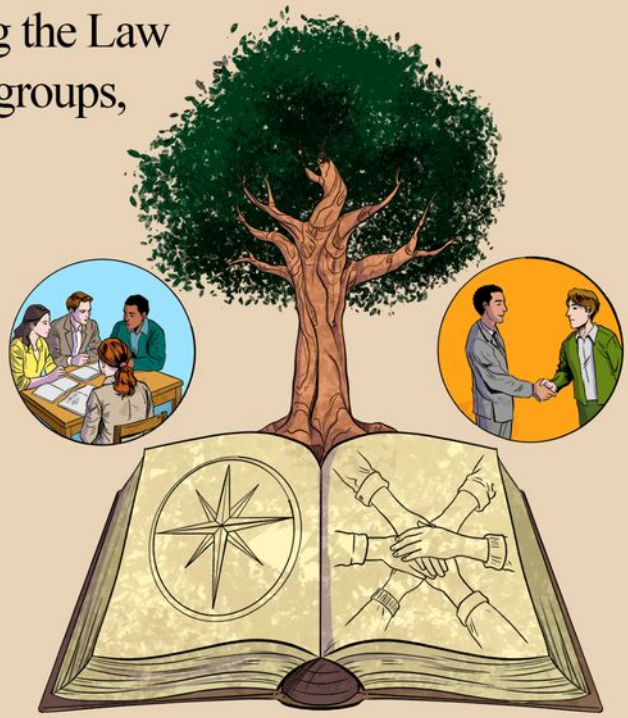
Builds the base for lasting peace and a fairer future for all humanity.

The series moves from the deepest question about reality to the structures that make a more peaceful world.

THE WORK AHEAD

A shared mission to build a better world.

Read the books, share them, and bring the Law of Fairness into classrooms, research groups, reading circles, and community conversations. Important ideas grow stronger when people examine them together, challenge assumptions, compare perspectives, and turn understanding into shared effort.



Scientists, researchers, clinicians, philosophers, and other experts can help test the Law, refine its measurements, improve its methods, and compare it with strong rivals. Serious criticism is essential. It exposes weaknesses, sharpens the questions, and helps reveal what should endure, what needs revision, and what should be rejected.

Beyond the hypothesis lies a practical challenge: building a world with less corruption and needless suffering, and more room for dignity, freedom, meaning, beauty, and opportunity. If this project speaks to you, bring your knowledge, experience, curiosity, and determination. Progress begins when people from different paths meet, share what they know, and build what survives scrutiny.

Help us make this generation
the one that brought peace to our world.

How to Read This Book

This Reference Edition explores the book's ideas in depth and develops the reasoning behind them. The companion Foundation Edition distills the essential ideas into a shorter, accessible introduction that stands on its own and requires no specialized background.

The Law of Fairness proposes that each qualifying conscious stream has finite, exactly equal total positive and negative felt magnitudes across its complete qualifying history strictly before an independently defined true endpoint. A stream is one subject's course of experience, considered separately from every other stream. The technical name is Per-Stream Terminal Valence Closure, or PTVC. In the additive account, felt magnitude reflects both intensity and duration.

Keep the Law, a candidate explanation, and a finding distinct. The Law states the proposed relationship. A candidate model describes a possible way it could arise. Evidence bears on a particular claim under stated conditions. A useful result need not settle the whole inquiry.

Read Parts I and II in order to establish the scientific question and its formal meaning. Part III examines what measurement must preserve. Part IV follows conscious streams through time and considers how their endpoints could be defined independently of the balance being tested. Part V develops possible studies. Parts VI and VII examine candidate mechanisms and competing explanations, then ask how an analysis could distinguish them. Part VIII addresses replication and the conclusions the evidence could justify.

For a shorter route through the technical argument, begin with Chapter 4, then Chapters 7 and 8. Mechanism researchers can follow Chapters 18–22. Researchers planning a study can begin with Stage Zero in Section 29.4, then use Chapters 14 and 24–26

for design and evaluation. Chapter 3 supplies the research protections that apply throughout.

Treat alternative explanations as part of the inquiry. Similar curves can arise from different processes. Ask what observation would distinguish them and what information each analysis was allowed to use. Simulated records with known origins can show whether a method recognizes the process that produced them. A rival explanation that performs better is useful knowledge.

Read each result by the question it actually answers. Validating a measure does not establish complete-stream closure. Stronger conclusions require independently defined streams and endpoints, credible coverage of experience, and a justified connection between observations and the proposed relationship. Uncertainty can clarify what the next investigation needs.

Use the Glossary for unfamiliar terms and the Index to revisit an argument. Read each model with its assumptions in view. Care remains prior to evidence: research must not induce harm or delay treatment. No result authorizes a human-worth score or forces a meaning onto another person’s experience.

This volume is self-contained. The Law of Fairness offers the broader introduction to the central idea. Readers moving between the Foundation and Reference Editions should follow subjects rather than chapter numbers. The invitation here is to help turn a striking proposal into questions that nature can answer.

Part I — Scientific Jurisdiction and Synopsis

Two histories can lead to the same present feeling while differing greatly in what came before. One person may reach relief after prolonged distress; another may arrive at the same state by a quieter route. An observation of the present leaves that difference concealed. The scientific problem begins when a proposed account assigns the histories different consequences and an investigator asks how those consequences could become observable.

For the Law of Fairness, this is a problem of choosing the right comparison. Familiar changes in mood can arise through adaptation, expectation, treatment, or the way a question is asked. A useful experiment therefore needs a reason to expect the proposed account and its strongest alternatives to diverge in the conditions being studied. The opening chapters develop that reason before turning to the machinery of a particular protocol.

Scientific work also depends on knowing what a result is about. A study of response scales can improve measurement. A study of transitions can distinguish two candidate mechanisms. A study of complete trajectories faces a different inferential demand. These contributions can help one another without becoming interchangeable. Their connection is a practical research question: which result would make a more demanding investigation possible, and which adverse result would make it unnecessary?

Consider a team that revises the neutral origin used by its instrument. The change may alter earlier comparisons, software outputs, and a public conclusion. Following that consequence through the record reveals the value of explicit dependencies. It also makes disagreement productive. A critic can identify the assumption that matters and test its consequences without accepting the surrounding program as a whole (Mayo, 2018;

Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

The terms of participation matter just as directly. A scientific question can be worthwhile while a proposed way of answering it is unacceptable. The investigator must therefore design around the authority of the participant and the responsibilities of care. This is part of selecting the experiment, alongside choosing its target and comparison. Chapter 3 makes those responsibilities concrete at the moments when research interests could otherwise obscure them.

Part I establishes the working relationship among the question, the evidence, and the people involved. Chapter 1 explains the research opportunity. Chapter 2 distinguishes the kinds of claims at stake. Chapter 3 governs the encounter with participants and their records. Chapter 4 gives a technical synopsis that the later Parts develop. Together they prepare the reader to ask, of each proposed investigation, what difference it could reveal and why that difference would matter.

Chapter 1 — Why the Hypothesis Needs a Science

A scientific program earns attention when it makes a difficult question more precise and gives a skeptical investigator a practical way to challenge it. For PTVC, the question concerns the relation between signed experiential history and what remains before a genuine endpoint. Its first investigations need not attempt a lifetime verdict. They need a contrast that existing methods can begin to resolve.

Consider a participant whose reported mood improves after treatment. That change could reflect feeling better, revised expectations of what counts as a good day, or an alteration in how the question is understood. Each possibility matters to the interpretation. Before using the report as evidence about accumulated signed experience, an investigator must establish which quantity the scale tracks and what the competing explanations predict.

Several foundations are already empirical. In an experience-sampling study involving more than 28,000 people, reported mood predicted subsequent activity choice (Taquet et al., 2016). A preregistered behavioral replication largely reproduced a computational account of reported well-being in a gambling task (Rutledge et al., 2014; Vanhasbroeck et al., 2021). Engineered biological circuits demonstrate integral feedback, including a mammalian proportional–integral controller (Aoki et al., 2019; Frei et al., 2022). These findings supply measurable temporal phenomena and experimentally realized ways of retaining information about prior deviations.

The next difficulty appears when researchers try to combine their work. Two sites label a data column valence. One records answers to a mood questionnaire; the other records clinician-rated agitation. Pooling the numbers as signed experience produces an aggregate with no defensible common unit. Each site must disclose its construct and transformation before the

apparent agreement can mean anything. The same problem can arise within a site when caregiver-mediated answers replace direct self-report: the observation route has changed even if the column heading has not.

Comparison brings a different trap. A history-sensitive model might beat an adaptation model because it receives daily reports while its rival receives weekly averages. That victory would identify an advantage in information access. A worthwhile experiment instead isolates the feature on which the accounts differ. Simulated records can reveal whether the proposed analysis can recover that feature before a prospective study asks participants to supply new observations.

The work therefore needs coordination as well as invention. An unusable scale, an unresolved endpoint, and a rival that predicts better call for different next decisions. Sections 1.1–1.4 assign those responsibilities; the reader routes and public-artifact discussion then explain how specialists and nonspecialists can examine the resulting claims. The program succeeds as a scientific undertaking when its results make the next question sharper, including when they make a favored line of inquiry unnecessary (Lakatos, 1978).

1.1 — Why a Separate Science Volume

A dedicated science volume gives critics a stable account of what PTVC proposes, what a study can test, and which finding would require a change. It also separates criticism of the hypothesis from disagreement with the series' ethical aims. An investigator may share those aims while rejecting the scientific premises (Popper, 1959; Lakatos, 1978; Laudan, 1977; Mayo, 2018).

1.1.1 — A Formal Claim Needs Scientific Governance

The Law of Fairness receives a formal architecture involving separately routed maximal qualifying streams, finite signed phenomenal measure on the primary branch, exhaustive ownership and routing, a complete strict-pre-endpoint domain, and a truth-level endpoint. Scientific development assigns each term its status, assumptions, observation requirements, tests, and rules for revision.

The first governance problem is version control. A statement of the Law, an estimator of lived experience, a stopping-time rule, a reserve-coupled realization, and a public verdict label are different objects. If one changes, the research program must record which claim version has changed. A favorable result under one bridge cannot be transferred quietly to a different bridge simply because the project prefers continuity.

The second governance problem is assumption visibility. The formal claim may be stated in an idealized latent space, but empirical work requires auxiliary assumptions about representation, scale, sampling, endpoint classification, channel reliability, missingness, and rival processes. A science volume makes those auxiliary assumptions visible so that critics can attack them without being accused of attacking the whole moral

aspiration of the series (Popper, 1959; Lakatos, 1978; Laudan, 1977).

Governance also requires contribution typing: definitions specify objects, formal derivations establish consequences of stated assumptions, measurement models link observations to targets, studies evaluate components or predictions, and results update only the claims they actually reach. Making those transitions explicit keeps each statement connected to its warrant.

The separate science volume is therefore an institutional answer to a linguistic danger. The Law is too large to be left to informal confidence. It needs a place where every important term is made answerable to a method, every method is made answerable to a rival, and every favorable result is made answerable to the status it can actually bear.

A field expert will also look for governance of comparison, not merely governance of definition. If the Law is compared against a simplified rival while the Law receives a richer auxiliary structure, the comparison is biased before the study begins. Scientific governance therefore includes a symmetry rule: a model that is allowed to use context, noise, and hidden state should face rivals that are allowed comparable resources under comparable constraints.

1.1.2 — Applied Coherence Is Not Validation

Applied coherence is not validation. A city, institution, ritual, or interface may be useful because it solves a local problem, and its difficulties may generate research questions, without supplying evidence for PTVC. Translate applied observations into bounded, ethically authorized studies with explicit rivals before they enter the scientific record. Operational success can

motivate a test; it cannot decide the target hypothesis (Laudan, 1977; Mayo, 2018).

1.1.3 — A Large Claim Needs Modular Failure

A large claim becomes tractable when its components can fail separately. A defective instrument need not invalidate the mathematical target; a defective representation may invalidate every instrument built on it. Chapter 4 makes those dependencies explicit.

The larger a claim becomes, the more dangerous it is to treat it as one indivisible object. The Law of Fairness touches the bearer of experience, the currency of felt valence, the temporal accumulation of experience, the endpoint at which a trajectory closes, the measurement bridge from latent to observed, the candidate dynamics that could realize closure, and the statistical rules that would justify a verdict. Each component can fail in a different way.

Modular testing assigns each study to the component it can actually reach: representation, measurement, endpoint construction, candidate realization, or rival explanation. Failure blocks the descendants that depend on that component without automatically deciding unrelated branches (National Academies of Sciences, Engineering, and Medicine, 2019).

The program therefore uses a dependency lattice rather than a single ladder of belief. That structure prevents one weak instrument from becoming total defeat and one favorable signal from becoming total validation; it lets each success or failure change only the status it has earned.

The separate science volume is needed because no applied book can maintain this lattice without becoming a methods book. Each applied domain has its own burdens. The science book carries the cross-cutting burden of deciding which

components are vulnerable, how they are connected, and what consequence follows when one of them fails.

The modular structure also prevents premature despair. If endpoint research proves intractable in one clinical setting, measurement development may still proceed in ordinary longitudinal settings. If a reserve-coupled realization fails, rival comparison may still clarify which ordinary mechanisms explain apparent balancing. A good program learns from partial loss rather than requiring every failure to become total collapse.

Cumulative judgment evaluates how those modules work together. Sustained problems across representation, measurement, endpoints, model comparison, and transport can justify revising or retiring the affected scientific claim. The component record identifies the evidence behind that decision.

1.1.4 — Answerability Before Adoption

Scientific answerability requires explicit predictions, fixed comparisons, and result-contingent decisions before adoption pressure can shape the test. Public interest does not raise evidential status. For confirmatory work, freeze the empirical bridge, name serious rivals, specify adverse-result rules, and preserve the dependency order from representation through measurement, endpoint, model comparison, replication, and transport. Applications may proceed on their own evidence, but they cannot make the scientific claim more nearly proven by momentum (Popper, 1959; Lakatos, 1978; Laudan, 1977).

1.1.5 — Scientific Seriousness Before Support

A hypothesis can be scientifically serious before it is supported when it is coherent enough to state, distinctive enough to generate risky consequences, vulnerable enough to lose, and connected to methods capable of distinguishing it from rivals.

PTVC now has a defined target, candidate consequence families, explicit rivals, and routes to narrowing or defeat. Those features justify treating it as an answerable research program; they do not constitute empirical support. Credibility must come from instruments, comparisons, replications, and results that survive the stated failure conditions (Laudan, 1977; Mayo, 2018).

1.1.6 — The House-Skeptic Function

The house skeptic maintains objections that the constructive team must answer, including challenges to a study's assumptions, comparisons, and proposed interpretation. The challenge log in Section 25.7 records the response to each objection.

The house-skeptic function gives criticism equal standing inside the research program. It asks which component is supported, which rival could explain the same pattern, whether endpoint and analysis rules were fixed before outcomes, which missingness process could mimic convergence, and what claim must change if a bridge fails.

To make that role substantive, skeptical and constructive teams should receive comparable access to assumptions, data dictionaries, simulation fixtures, code, protocol history, model budgets, and scoring rules. A rival denied relevant covariates or forced into a weaker observation law is not a fair comparator (Gneiting & Raftery, 2007; Mayo, 2018).

Major objections should enter a challenge log with the component challenged, evidence needed, owner, and interim public status. Section 25.7 supplies the detailed red-team and challenge-log procedure; Section 1.6 governs hostile-crop public communication.

1.2 — Inheritance and Scientific Jurisdiction

The Science of Fairness develops the public question introduced in *The Law of Fairness* into a self-contained scientific inquiry. Chapters 4–7 state the hypothesis, define its objects, and derive its principal consequences; later chapters develop measurement, candidate dynamics, tests, and interpretation. Readers can follow the scientific argument within this volume, with external research cited where it informs the analysis (Douglas, 2009).

1.2.1 — The Formal Hypothesis Is Stated

The first scientific act is specification. This volume states a formal hypothesis about separately routed maximal qualifying conscious streams and exact terminal valence closure over complete strict-pre-endpoint domains. It does not replace the bearer with public identity, the phenomenal target with welfare preference, the endpoint with bodily death, or the latent cumulative object with a dashboard. An operational target must remain recognizably connected to the object it is meant to test.

Specification makes the vocabulary explicit. Qualifying bearer, exhaustive ownership and routing, separately routed maximal stream, phenomenal valence, complete strict-pre-endpoint domain, truth-level endpoint, latent terminal residual, momentary valence estimator, interval contribution estimator, and branch-matched inference are defined here as the objects of scientific inquiry. Science may show that an object is difficult to identify, propose provisional bridges toward it, or conclude that a bridge fails. It may not declare that a simpler neighboring object was the target all along.

The hypothesis also retains its ethical boundaries: PTVC carries no implication of moral desert, victim-blaming, cross-stream transfer, pain as morally owed payment, harm induction, delayed care, human ranking, or machine authority; Chapter 3

governs the corresponding operational protections (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

The task is to determine what the received objects require from research. What counts as a bearer in a study? Which aspects of felt valence can be observed? How can an endpoint be classified independently of the ledger result? How should each scientific finding be communicated?

The public question motivates the formal target specified here. This volume develops the observations, models, and comparisons through which researchers can evaluate it.

The first jurisdiction rule follows: receive the formal target whole enough to test honestly, then record any scientific narrowing openly. If research shows that a component is unidentified, a reference class is too broad, or an endpoint bridge fails, the change is a scientific status update. It is not a hidden rewrite that preserves the original language while testing something else.

Reception also requires resisting friendly simplification. It would be easier to replace the Law with a familiar claim about adaptation, life satisfaction, welfare, resilience, or moral balance. Those topics have literatures and instruments. But a research program that changes the target to fit available tools has not made the original claim scientific. It has studied a neighboring object.

Approximation is sometimes necessary, but it should never erase the harder target from view. Public readers should be shown both layers: the received formal object and the provisional bridge by which a particular study approaches it. Keeping those layers visible prevents an operational measure from quietly replacing the claim it was designed to investigate.

1.2.2 — The Formulation Becomes a Research Target

A formal model becomes a research target when its definitions are translated into obligations. The ledger is not merely a symbol; it demands a representation theory of lived valence. The bearer and routing rules demand a defensible account of which experiences belong to which maximal qualifying stream. The truth-level endpoint demands prospective construction and adjudication that do not define closure by the ledger value being tested. Some protocol-fixed branches add timing-independence, support-density, hitting, absorption, atom, or continuation premises, and each such premise must remain branch-specific rather than becoming a universal assumption.

The translation turns the formal model into operational questions. For latent L(t) and its measured estimates, the science book specifies the requirements for construction, calibration, versioning, and reporting. For terminal valence closure, it derives measurable consequences under declared observation models and rival data-generating processes.

The model also becomes a target for dimensional discipline. A rate, an integral, a unit, a reserve variable, a horizon, a variance, and an endpoint are not interchangeable pieces of poetic mathematics. Each has assumptions, units, and failure conditions. No equation should travel into public language without its object, dimensions, branch, and status attached (Folland, 1999; Protter, 2005).

A candidate stochastic realization derives a possible process from stated assumptions. Its scientific development identifies those assumptions, predicts observable consequences, and compares the resulting model with empirical data and alternatives.

Making this formulation a research target also gives critics a fair route in. They can dispute representation, scalarization,

bearer qualification, ownership and routing, endpoint construction, branch-specific timing assumptions, reserve coupling, measurement invariance, rival comparison, or verdict logic. That is better than a general dispute over whether the Law feels plausible. A target that can be attacked component by component can also be improved component by component.

The target also demands that proposed failures be meaningful. A failed momentary valence estimator channel is not a decisive refutation of the Law if the channel never estimated the intended construct. But repeated failure of reasonable, prospectively specified bridge families is not irrelevant either. The research target must allow both distinctions, because otherwise the program becomes either too brittle or too protected.

1.2.3 — Scientific Status Is Governed Here

Scientific-status governance is this book's central jurisdiction. The volume defines the procedures, evidence roles, and public language through which PTVC-related claims may be updated, while evidence, independent review, and the wider scientific community determine the verdict. Without that distinction, public language can drift toward whatever status feels most useful at the moment.

Scientific-status governance records how findings update claims. A bounded cohort can support a specified consequence; measurement work can qualify or revise an estimator; and a comparative result can guide mechanism selection. These decisions make the research record usable.

Negative status belongs here as visibly as support. Nulls, failed instruments, rival victories, narrowing, route retirement, and theory-level loss remain part of the scientific record. A program that publicizes only favorable movement cannot learn reliably from evidence.

Replication is part of status governance. A single site, instrument, clinical setting, culture, or endpoint procedure cannot carry universal language. The science book must require transport before broadening. It must record when a result is local, fragile, incompatible across sites, sensitive to missingness, or dependent on an auxiliary assumption that has not yet traveled.

Scientific-status governance includes correction of public artifacts; Chapter 28 supplies the full propagation, archival, and retirement workflow. A correction must reach the public surface where the overstatement traveled.

Scientific-status authority is narrow and consequential. It does not govern law, faith, city design, ordinary practice, or transition sequence. It determines whether those domains may use scientific language and, when they may, the exact scope that the evidence has earned.

Scientific-status governance also applies to language inside the volume. Words like suggest, support, validate, imply, estimate, demonstrate, and prove are not stylistic variations; they name different relationships between evidence and claim. Treating those verbs as interchangeable misleads even when a formal warning appears elsewhere.

1.2.4 — Definitions Flow Out; Evidence Flows Back

The series requires a clean direction of flow. Definitions flow outward from The Law of Fairness because the root book states the formal claim. Evidence flows back through The Science of Fairness because scientific findings must update public status under governed rules. If this direction is reversed, applied success can rewrite definitions, and the project becomes self-validating.

Suppose a clinical event is easier to measure than the endpoint defined by the hypothesis. Researchers may evaluate it as a proxy; they may not silently substitute it for the original endpoint. If a rival explains the observed pattern near an endpoint better than the proposed reserve mechanism, that result bears on the candidate mechanism. Revising the hypothesis itself would require an explicit change of claim.

Applications in later books cannot require a favorable scientific verdict. Evidence must remain able to disappoint the canon (Douglas, 2009; Mitchell, 2009; Woodward, 2003).

Flow is not a wall. Applied books can send observations back as research leads. A city may reveal record burden; an ordinary-life practice may reveal repair dynamics; a religious house may reveal ritual effects; a transition pilot may reveal public-status misunderstanding. But those observations become questions only after they are transformed into research designs with consent, context, rivals, and status limits.

The return flow from science is component-specific. Support for a measurement bridge returns as that bounded update; it does not strengthen every sentence in every title. A failure returns with its scope and consequence, which may require language correction, artifact retirement, source updates, or claim narrowing without creating panic beyond the component affected.

A favorable scientific finding can update the relevant passage in the formal-law book. The handoff identifies the supported component, the evidence behind it, and its established scope, allowing readers across the series to follow the same result.

1.2.5 — Applied Observations Are Research Leads

Applied success can generate a scientific question. A practice may help through attention, community, expectation, or a

mechanism unrelated to PTVC. Its benefit is worth studying on its own terms.

Applied observations matter because the world often reveals burden before instruments do. A household may expose how care work disappears into routine. A public office may expose how answerability shifts onto citizens. A religious practice may reveal how meaning changes distress. A technological system may reveal how a dashboard pressures users to perform. These are valuable observations, but their value begins as question-generation.

A research lead names a phenomenon that deserves controlled inquiry. It does not state the cause. If a group using a Fairness practice reports more repair, the science must ask whether the effect comes from attention, community, expectancy, selection, time, language, ritual, reduced isolation, or some Law-specific consequence. Treating the observation as evidence too early would disrespect the complexity that made it interesting.

Harms can also become research leads without becoming research data by default. If a public artifact creates misunderstanding, communication research can examine the pathway. If an experience measure creates anxiety, burden research can examine reactivity. If readers infer terminal destiny, hostile-crop testing can study which fragments carry that inference. The moral problem is immediate; the scientific opportunity begins only after consent, context, rival explanations, and status limits are specified (Fischhoff & Davis, 2014; National Academies of Sciences, Engineering, and Medicine, 2017; National Institute of Standards and Technology, 2020).

Applied observations can reveal rival mechanisms worth formalizing. Relief after ritual may arise through expectancy, attention, social support, or meaning. Terminal calm in a setting

with strong death scripts may reflect script-induced appraisal. Apparent variance compression where sensors fail may be observation-gain collapse. Each observation improves the program when it becomes a discriminating question rather than a favored anecdote.

The research lead must pass through a separate door before becoming evidence. That door includes consent, data minimization, context definition, sampling, prospectively specified outcomes, endpoint rules where relevant, rival models, missingness plans, and public-status limits. Operational records do not become research data merely because they are convenient. Participation in an institution is not research consent.

Applied observations also help the program notice harms that laboratory design may miss. A public dashboard may produce social pressure. A community ritual may intensify guilt for some participants while relieving others. A record system may help officials while exposing families. These are not peripheral anecdotes. They are burden data that can improve future research questions if they are handled under consent and scope discipline.

1.2.6 — Jurisdiction Is Not Mastery

Scientific jurisdiction is bounded. This volume governs scientific claim status; it does not govern human worth, spiritual meaning, legal rights, clinical care, transition timing, or private life. Section 3.6 states the operative domain boundaries. For example, qualifying an affect measure can inform an application while leaving the application's legal and clinical decisions to their own review. A scientific result can therefore sharpen a factual premise without deciding the rule another field should apply. The receiving field remains responsible for integrating

that evidence with its own standards, duties, and forms of review.

That limit also applies to participants. Measurement and consent authorize only the declared research uses and inferences; they do not transfer ownership of inner life. Ethics, law, clinical care, disability studies, privacy, and science communication remain responsible for the consequences within their domains. A later request to train a new model on stored reports is a new proposed use, not an extension silently granted by the original measurement. Secondary use should therefore be treated as a fresh governance decision, with its own stated purpose, access conditions, retention limits, and challenge route. This keeps consent tied to a defined use rather than turning one permission into an open-ended claim on future data.

Cross-disciplinary work therefore uses explicit handoffs: each contribution states what it establishes, the assumptions it carries, and which questions remain for other fields. Scientific precision is strongest when it does not borrow authority from neighboring domains. A useful handoff identifies which conclusion depends on the scientific result, so a later revision can update that conclusion without reopening unrelated judgments. A strong handoff also carries the result’s status, uncertainty, and reference class, because those features determine how far the result can travel. The receiving field can then combine it with its own evidence and decision rules without stripping away the conditions that made the scientific statement valid.

1.3 — The Research Program in Stages

The core claim concerns exact closure within each qualifying conscious stream; Section 5.3 gives its formal statement. Auxiliary hypotheses connect that target to observations through representation, ownership, endpoint classification, and measurement.

1.3.1 — Core Claim and Auxiliary Hypotheses

The first question is representational: can felt experience be described, for the proposed purpose, by signed contributions that accumulate over time? If mixed affect or context defeats that representation, the difficulty precedes instrument design.

The core claim is the strongest formal target the science receives. On the primary finite signed-measure branch, every separately routed maximal qualifying stream in the declared truth-domain class has equal finite positive and negative phenomenal magnitude over its complete strict-pre-endpoint domain, equivalently a zero terminal signed residual. The public empirical claim is not identical to that set-universal target. Public inquiry must proceed through validated representation, ownership, endpoint, observation, measured-to-latent bridges, rival comparison, and branch-matched inference. The target gives direction; auxiliary hypotheses make bounded research possible.

Auxiliary hypotheses include representation assumptions. Can lived experience be treated, for a declared purpose, as a signed valence-rate target that accumulates over time? How are mixed affect, numbness, dread, relief, meaning, dignity, memory, anticipation, and bodily distress handled? A failure here would not be a small measurement problem. It would challenge whether the proposed target can be represented in the required form.

Other auxiliaries concern measurement. Candidate momentary and interval estimators may function as estimators only within a specified context of use, with uncertainty, calibration, versioning, missingness, and correction rules. If the estimator changes, the claim version changes. If the estimator cannot preserve sign or scale within the required setting, downstream terminal analysis cannot repair that failure by collecting more data (Chambers, 2013; Lakatos, 1978; Laudan, 1977).

Endpoint assumptions form another auxiliary layer. The truth-level endpoint must be classified through channels not algebraically identical to the ledger estimator. If the endpoint is defined as the moment the ledger reaches zero, neutrality becomes a tautology. If endpoint channels share hidden tuning with the ledger, residual dependence must be modeled rather than waved away.

Candidate mechanisms include reserve coupling, horizon pressure, queue dynamics, adaptation, predictive control, and homeostasis. Each generates patterns and consequences that can be developed into comparative predictions.

The first stage of seriousness is therefore not data collection. It is an assumption register. The reader should know what would remain if a representation failed, what would remain if a measurement failed, and what would remain if a mechanism failed. Only then can later evidence have a clear target.

The auxiliary structure should be visible enough that a reviewer can ask whether the program is becoming progressive or degenerating. If a measurement bridge is replaced, what new prediction does the replacement create? If a reserve assumption changes, what rival does it now face? If an endpoint rule becomes more complex, does complexity improve independent classification or merely protect the desired terminal pattern?

Some auxiliaries determine whether the next inference is available. A defensible origin, endpoint, and measurement bridge provide the foundation for terminal analysis. The staged architecture directs early work toward those dependencies.

1.3.2 — Progressive Problem Shifts

A revision advances the research when it solves an independent problem or makes a successful prediction that was not used to construct it. The next example shows how a measurement failure can create that opportunity.

A failed single-channel valence estimator may motivate an observation model that predicts when self-report and physiology diverge. If that model forecasts the divergences in a new cohort, the revision has solved a measurement problem and generated a new testable consequence.

A research program progresses when a revision does more than protect the core from trouble. It solves an independent problem, clarifies a mechanism, improves measurement, predicts something not used to build the revision, or exposes the program to a new possible loss. The Law of Fairness needs that standard because it will inevitably meet anomalies, instrument failures, ambiguous endpoints, and rival explanations.

Another progressive shift might arise from endpoint adjudication. Suppose a unity-proxy rule performs poorly in one boundary state but, after expert review, yields a refined confidence-class system that predicts which cases should return no estimate. If that system improves prospective endpoint classification and increases no-estimate honesty, the research program has progressed even before any Law-specific terminal support appears (Laudan, 1977; Munafò et al., 2017).

Progressive shifts can also involve rivals. If an adaptation model explains most local rebound but fails to match reserve-

linked drift and innovation behavior under held-out prediction, the Law-specific program gains a sharper target. The gain is not that adaptation was dismissed. The gain is that the contrast taught the program what it must uniquely explain.

An informative revision improves the connection between model and observation. Additional variables, uncertainty models, and flexibility contribute when they yield clearer predictions, better calibrated estimates, or a more discriminating comparison.

A progressive shift should improve the critic's position as well as the proponent's. After the revision, it should be easier to say what would now count against the program. Better measurement, for example, should create clearer defeat conditions, not only better-looking fits. Better endpoint classification should increase no-estimate honesty as well as terminal precision.

1.3.3 — Degenerating Rescue Moves

A degenerating rescue move is a change made after difficulty appears, not to improve contact with the world, but to preserve a desired conclusion. Such moves are especially tempting here because the Law touches invisible experience. If unfavorable data appear, one can always say the relevant experience was unmeasured, the endpoint was misclassified, the scale was wrong, or the compensation occurred elsewhere. Sometimes these objections may be true. They become degenerating when they remove risk without adding a new test.

The clearest forbidden rescue is endpoint recoding by result. If the measured cumulative estimate does not approach neutrality, the project cannot redefine the truth-level endpoint as whatever time produces a favorable estimate. Endpoint discipline exists precisely to prevent that move. The endpoint must be

prospectively specified through unity or mind-continuity channels, and ambiguity must be allowed to produce no estimate rather than a rescued verdict.

Another rescue is origin adjustment after the fact. Because felt valence lacks an instrument-fixed natural zero, empirical work must declare origins, anchors, and equivalence bands before analysis. If the origin is shifted only after a result disappoints the Law, the program has not discovered neutrality. It has moved the ruler (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Measurement expansion can also degenerate. If every failed channel leads to adding more channels until the desired pattern appears, the estimator becomes an oracle for the theory. A legitimate measurement revision must explain why the old model failed, state what the new channel contributes, freeze the update before evaluation, and test whether the revision predicts beyond the dataset that motivated it.

Rival handling can degenerate as well. A rival is not serious if it is made artificially weak, excluded after it wins, or reclassified as part of the Law whenever it succeeds. Adaptation, homeostasis, predictive control, Brownian bridge effects, observation shutdown, clinical confounds, and expectation scripts must be allowed to win under the same constraints imposed on the Law-specific model.

The practical rule is severe but fair: every rescue must pay rent. It must create a new prediction, a new independent measurement, a narrower public claim, or a declared loss. If it merely preserves favorable language, it is not a scientific modification. It is protection from science.

Degenerating moves often begin with true statements. It is true that measures are imperfect. It is true that endpoints are difficult. It is true that unobserved experience matters. It is true that rivals

may be incomplete. The danger lies in using true difficulties to make the Law unreachable by any practical test. A serious book must distinguish legitimate uncertainty from theory protection.

The practical safeguard is prospective classification. Before outcome inspection, the protocol states which uncertainty makes the study unusable, which widens an interval, which narrows the claim, and which becomes adverse under valid prerequisites. After the result, those categories do not change merely because the outcome is disappointing. That discipline separates humility from retrospective rescue.

1.3.4 — The First Five Studies

The five study families in Section 29.4 are parallel research tracks. Their numbering identifies distinct questions, not a rigid calendar. Each contribution qualifies a specific dependency for a later comparison.

Formal verification and the existing-data Stage Zero challenge support the five tracks. Formal work checks the coherence of the objects, dimensions, and process classes. It identifies independent assumptions and asks which claim versions are identifiable in principle. Existing-data work helps locate contrasts worth pursuing before new observation is proposed.

Study One investigates representation and sign. Bounded tasks and ordinary-life records examine whether positive and negative contributions can be placed on a stable scale with a qualified neutral origin. The work also evaluates interval structure and whether a scalar or vector account preserves the distinctions needed for the intended comparison.

Study Two qualifies the momentary and interval estimators for their declared context of use. It evaluates the proposed measurement relation, calibration, uncertainty, channel conflict, and missingness, then tests transfer across relevant conditions.

The result identifies which observations can support a later accumulation estimate and which measurements need further development (Chambers, 2013; Lakatos, 1978; Laudan, 1977).

Study Three evaluates unity and endpoint classification. It begins with state transitions and recovery settings in which classifications can be checked, including sleep, anesthesia, sedation, delirium, and disorders of consciousness. Graded confidence, return handling, and indeterminate classifications remain explicit. Endpoint evidence and adjudication rules are specified independently of the ledger result being tested.

Study Four is the generative benchmark challenge. Known candidate and rival processes are passed through the intended observation and analysis pipeline. Recovery and confusion results show which comparisons the design can distinguish and where an additional observation would be informative. These computational tests precede the confirmatory model-attribution comparisons that depend on them.

Study Five examines humane preterminal dynamics after its representation, measurement, endpoint, recovery, and welfare prerequisites are met. Preparation includes palliative feasibility and burden assessment: whether the observations fit care, what participants and staff can sustain, and which missingness patterns the design produces. A qualified study compares specified preterminal signatures with adaptation, homeostasis, endpoint conditioning, observation loss, and clinical explanations under common rules. Its immediate contribution is a discriminating comparison within the declared setting.

Each track returns a contribution another team can use. A calibrated scale supports a later instrument; a benchmark guides the next observation; and a replicated prediction motivates a broader comparison. Section 29.4 specifies the deliverables and decisions for all five families.

1.3.5 — Replication Before Status Upgrade

Report the first study by the component tested, the setting, the instrument and endpoint procedure, and the alternatives compared. Those details identify the result that subsequent studies can reproduce and extend.

Replication upgrades status only when the relevant claim survives independent teams, prospectively fixed analyses, appropriate transport, missingness sensitivity, and adversarial review. Replication should also test the bridge itself: disagreement across instrument versions, endpoint classes, or care settings is evidence about scope, not noise to be averaged away. Chapter 26 specifies the multisite system and Chapter 27 governs the resulting status transitions.

Public status should move more slowly than excitement. A component can gain support, a bridge can improve, a rival can weaken, or a scope can narrow. The phrase validated Law is unavailable until the exact claim version being announced has passed the prerequisites, rival comparisons, replication, and quantifier bridge required for that announcement.

Replication should also include adversarial sympathy. A replication team should be allowed to disagree about the best rival, the missingness model, or the endpoint confidence threshold. The program should preserve those disagreements rather than smoothing them into a single friendly result. Consensus is valuable only after disagreement has been given room to sharpen the analysis.

Status updates should identify concrete advances: a measurement bridge has matured, a rival has been distinguished in one cohort, a protocol proved humane, or a no-estimate rule worked as intended. Each gain contributes to the research program and deserves a clear account of what was learned.

An inspectable replication packet should therefore exist before the first result is known. At minimum it should preserve the claim version, protocol registration, representation and endpoint specifications, analysis code, synthetic recovery benchmarks, instrument versions, data dictionary, missingness plan, adverse-event logic, decision thresholds, and the exact public wording permitted by each outcome. The packet is not bureaucracy around the experiment. It is the experiment's memory (Chambers, 2013; Munafò et al., 2017; Nosek et al., 2018).

Chapter 26 specifies how frozen and adaptive elements, implementation changes, tacit knowledge, provenance, executable code, and analytical reproducibility are governed across sites. Here the governing rule is enough: replication cannot upgrade status unless an independent team can reconstruct both the analysis and the claim-relevant implementation without hidden favorable choices (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018; Munafò et al., 2017).

This standard is demanding because broad claims are unusually vulnerable to invisible degrees of freedom. A theory can appear to replicate when each site silently chooses a favorable origin, smoothing rule, endpoint window, exclusion set, or rival budget. Replication becomes evidential only when those freedoms are controlled, recorded, and exposed to a team that has no obligation to preserve the preferred story.

1.3.6 — A Program With Clear Decision Gates

A research program specifies the conditions for pausing a study, retiring a bridge, narrowing a claim, replacing a mechanism, or reaching a theory-level verdict. Those decisions connect the evidence to the next responsible action.

Stopping can occur early. If no defensible representation of the target can be specified, terminal testing should not proceed under a false sense of precision. If measurement channels cannot preserve sign or transfer across required contexts, the public empirical claim must narrow. If endpoint adjudication remains circular, terminal valence closure cannot be tested responsibly. Each stop protects the project from building on a broken dependency.

Stopping can occur at the mechanism level. Reserve-coupled dynamics may fail while other aspects of the inquiry remain alive.

Queue language may prove useful as a model of pressure but not as a Law-specific realization. Shadow-price formulations may help describe horizon pressure while failing as evidence for closure. A serious book lets useful mechanisms retire from stronger claims while preserving the knowledge they helped produce (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Stopping can also occur because rivals win. If adaptation and homeostasis explain the signatures with better prediction, simpler assumptions, and equal or better transport, the Law-specific claim must lose the relevant field. If endpoint-conditioned bridges reproduce the terminal shape without reserve coupling, the program must narrow. If observation shutdown creates false compression, the measurement claim must be corrected before future study.

Public communication is reviewed for clarity, unintended personal interpretations, and participant pressure. When problems recur, the affected materials are paused, revised, and assessed before further release. The correction record identifies what changed and how the revised design addresses the problem.

Stopping also protects contributors from sunk-cost pressure. Researchers, donors, builders, and readers may invest time and identity in the program. The longer they invest, the harder it becomes to admit that a component has failed. A public stop rule gives loyalty a better form: loyalty to the truth conditions that made the project worth beginning.

1.3.7 — Negative Results as Scientific Infrastructure

Negative results are typed outputs, not one null. Distinguish identifiable nulls, invalid instruments, failed model recovery, unadjudicable endpoints, rival wins, burden-limited studies, and nontransport because they block different descendants and imply different next work. Preserve the failed object, the reason, the surviving claims, and the successor obligations in a versioned record; Chapters 27 and 28 contain the detailed status and archive machinery. A negative result is scientifically useful when it changes the next decision or prevents unnecessary participant burden.

1.4 — The Book's Evidence Responsibilities

A monograph that studies fairness has to expose the costs of its own inquiry. Vague status language shifts interpretive work to readers; excessive observation shifts time and discomfort to participants; impractical procedures shift work to clinicians; unclear targets shift model construction to critics; premature conclusions shift repair to later titles. The program should name and govern those costs rather than hide them inside ambition (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Academies of Sciences, Engineering, and Medicine, 2019).

1.4.1 — Claim Burden

Claim burden is the duty to say what kind of sentence is being spoken. A definition, formal hypothesis, candidate realization, measurement estimate, rival model, preliminary result, and public verdict may all sound authoritative in polished prose. They carry different warrants. The category should be visible in the sentence itself, not recoverable only from a remote disclaimer.

The burden begins with the strong form. The latent target is an exact set-universal boundary condition over separately routed maximal qualifying streams in a declared class, not a directly observed public fact. Public science must proceed through measurable consequences, truth-containing compatibility sets, model comparisons, finite bands, endpoint and observation bridges, and explicit status updates. A chapter that treats those layers as one claim fails before the study begins.

Claim burden also requires scope. A result may concern a measurement bridge in healthy-adult diary data, a terminal signature in one palliative cohort, a sleep-related adjustment pattern, or a model-recovery benchmark under simulated data.

The status label must name the component, population, window, instrument, and dependency under discussion (Gneiting & Raftery, 2007; Lakens, 2017).

A claim can become too vague by trying to be safe. Repeating that nothing is proven does not satisfy claim burden. The reader needs to know what has been proposed, what would count as evidence, what would count against it, and which version of the claim is under review. Humility without specificity becomes another form of evasion.

Claim burden is therefore the book's grammar of honesty. Every major sentence must carry its own status or be surrounded by enough context that the status cannot be mistaken. The Law can remain ambitious only if its sentences refuse to pretend they have earned more than they have.

1.4.2 — Identification Burden

Intrusive observation can change the state enough to worsen identification. Family and staff work can also determine who is able to remain in a study. A sound design may require less measurement rather than more; Chapter 3 develops the responsibilities behind that choice.

Identification burden asks whether the target can be distinguished in principle. It is deeper than reducing noise. A more sensitive instrument cannot identify an object whose origin, unit, scalarization, bearer, endpoint, or reference class remains underdetermined. The central challenge is therefore not merely to build better affect sensors, but to show what those sensors can and cannot identify.

The terminal-zero problem separates direct measurement from empirical exposure. Finite noisy observations can challenge exact closure when a valid residual set excludes zero under independently justified calibration, missing-history, and

endpoint assumptions. A deterministic bound and a statistical confidence set license different conclusions, as Section 24.6.4 explains. Neutrality must be identified independently of the result: recentering a record to make its total vanish supplies no evidence for closure.

Endpoint identification is equally binding. If the endpoint is chosen by the ledger, closure becomes a tautology. If it is equated with bodily death without a consciousness bridge, it may miss the truth-level ending of the relevant stream. If endpoint channels and ledger channels share hidden causes, a terminal pattern may reflect the same clinical process entering both estimators rather than PTVC. These dependencies must be declared before endpoint data are interpreted (International Council for Harmonisation, 2019).

Bearer identification also belongs here. A conscious stream is not identical to a legal person, biography, body, or public identity. Sleep, anesthesia, dementia, dissociation, coma, developmental thresholds, nonhuman animals, and future artificial systems all pressure the bearer concept. Chapter 1 does not solve those cases, but it explains why the volume must own a research route for them.

Identification work can produce a No adequate estimate outcome for a particular cohort, endpoint, channel, or state. Reporting that outcome directs attention to the evidence needed to complete the measurement bridge and gives later studies a clear starting point.

Better sensing can make important contributions to inner-life science when it is joined to clear identification work. Even a highly reliable channel must be connected to a defensible neutral origin, scale, bearer, routing rule, truth-level endpoint, and target-class transport argument. Separating those tasks lets

improvements in instruments and improvements in conceptual foundations reinforce one another.

The same point protects participants. If a study cannot identify the target, the ethical response is not to measure more intrusively. Intrusion can increase data while worsening identification by altering the state under study. Sometimes fairness to the person and fairness to the science both require less measurement and more honest abstention.

1.4.3 — Measurement Burden

Measurement burden begins after identification has named a target and still leaves enormous work. Lived experience does not hand itself to science as a single clean signal. Self-report, physiology, behavior, context, sleep, clinical observation, caregiver testimony, passive signals, and records are all partial witnesses. Each witness can be useful; none is sovereign.

Measurement must therefore be treated as disciplined translation, not extraction. The momentary valence estimator is a context-bound momentary estimate under uncertainty, not a device that reads inner life. The interval valence contribution estimate is a context-bound interval estimate under conditions, not a moral unit. The running measured cumulative estimate is an approximation under a model, not the latent cumulative process itself. Measurement burden requires this layered humility whenever numbers appear.

Calibration is part of the burden. A measure that performs well within one person may not compare cleanly across persons, cultures, ages, disabilities, languages, clinical conditions, or terminal states. Translation does not establish invariance, and identical responses need not carry identical latent meaning. The program therefore distinguishes within-stream calibration, cross-context linking, partial identification, and principled local

inference (Borsboom, 2005; International Test Commission, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Missingness is not an empty space in a dataset. It can follow distress, fatigue, withdrawal, death, technology failure, family intervention, treatment, or the point at which a protocol no longer fits care. A terminal study that treats those absences as random can manufacture calm by losing precisely the observations most likely to challenge it. The missingness mechanism must therefore be modeled before favorable patterns are interpreted.

Versioning is also measurement burden. If the momentary valence estimator channel set changes, if weights change, if anchor rules change, if a wearable model is updated, if a natural-language model summarizes reports differently, or if a clinical workflow changes observation frequency, the measured object may change. Version labels and bridge tests are required rather than silent continuity.

Interpretation itself also needs governance. When channels disagree, the protocol has to state how self-report, physiology, behavior, clinician observation, and model output are weighted or held apart. Those choices should be versioned and auditable because a clean estimate can otherwise conceal a value judgment about whose evidence counts.

1.4.4 — Participant and Bystander Burden

Participant and bystander burden is part of the data-generating process. Prompts, devices, interviews, family involvement, staff workload, and clinical interference can alter the state observed, who enrolls, who remains, and which data go missing.

Protocols therefore need prospective burden endpoints and stop rules covering fatigue, distress, privacy risk, device

discomfort, family pressure, clinical interference, and staff overload. Burden-linked selection and missingness must be analyzed rather than treated as background noise.

Sections 3.3 through 3.5 provide the detailed rules for consent, bystanders, vulnerability, withdrawal, and burden budgets. The evidence responsibility is to identify how measurement changes participation and observation, then account for those effects in the analysis.

1.4.5 — Rival and Negative-Result Burden

Rival burden means giving serious alternatives enough information, flexibility, and expertise to explain the data under common rules. Adaptation, homeostasis, predictive processing, learning, endpoint conditioning, observation loss, treatment, frailty, and selection are competing generators.

Negative results belong to the same record as favorable ones. Failed representation, instrument loss, endpoint failure, rival victory, replication failure, and nontransport each change the relevant component and should remain discoverable.

Section 2.4 defines severe comparison, and Chapters 21 through 25 develop the rival and recovery methods. Strong rivals are allowed to win, and unfavorable results remain part of the scientific memory.

1.4.6 — Correction Burden

Correction burden is the duty to repair the public record where a claim traveled. Claims, instruments, endpoint rules, analyses, and public artifacts must remain traceable enough that readers can distinguish current from superseded authority.

Chapter 28 describes the full correction, propagation, archive, and retirement workflow. When scientific status changes,

dependent public language changes with it, while prior versions remain visible as history rather than continuing authority.

1.4.7 — Failure Memory and Successor Design

Failure memory should identify the failed object, the reason it failed, the burdens it created, the descendant claims it invalidates, and the repair a successor version is required to test. A successor does not inherit evidentiary credit from the failed version; it begins with a new empirical commitment. Chapters 27 and 28 provide the full evidence and change-control machinery.

A research program becomes cumulative when its successors are constrained by failures as well as inspired by successes. That memory is especially important for an ambitious hypothesis, where ambition can erase past disappointment. The evidence responsibility is to move forward only after failure is made legible enough to improve what comes next. The next version should be able to identify the specific uncertainty it reduces and the specific weakness it no longer carries forward. Otherwise repeated revision can create the appearance of progress without making the underlying test more informative.

Superseded drafts, retired diagrams, obsolete instruments, and earlier claims can reenter circulation simply because they remain vivid or easy to find. Preserve their history and retired status, and make current versions easy to distinguish from archival ones wherever researchers or public readers encounter them.

1.5 — How to Read as Skeptic and Builder

There are several ways into this inquiry. Skeptics press its defeat conditions. Builders trace its necessary dependencies. Scientists test what can be observed without assent.

1.5.1 — For Skeptics

Two signed mappings may fit identical reports yet imply incompatible cumulative histories. An endpoint test then cannot identify which ledger it measures. The skeptic can demand a prespecified contrast between the mappings before terminal inference. More observations from one ambiguous channel may sharpen an estimate without identifying its meaning.

Skeptics should press the positive case as well: the hypothesis must state a discriminating consequence, not only conditions under which inference fails.

1.5.2 — For Builders

Builders can use the applied books while evaluating their practical aims independently. For the boundary between operational participation and research, begin with Section 3.4.3; for evidence passed between titles, use Section 28.4.

1.5.3 — For Scientists

Scientists can enter through a tractable contribution: interval bounds in Sections 9.4.3 and 10.6.1, the benchmark in Section 25.5.4, or the existing-data Stage Zero challenge in Section 29.4. These routes identify work that can proceed before a terminal test is possible.

1.5.4 — For Philosophers

Philosophers can begin with two questions. In Chapter 2, ask whether the proposed structural identity explains anything

beyond the definitions used to state it, and whether its apparent economy survives the cost of its auxiliary assumptions. Chapters 5 and 6 make the objects and identification problem precise enough to test that judgment (Gelman & Loken, 2014; Mayo, 2018).

1.5.5 — For Public Readers

Public readers can begin with the opening examples, use Chapter 2 to distinguish a hypothesis from a finding, and consult Chapter 3 when a proposed use affects a person.

1.5.6 — For Canon Readers

Across the wider series, scientific status should propagate only to passages that actually depend on the changed component. Applied domains retain their own evidence, values, and authority; the science volume contributes a bounded premise, not master jurisdiction. Chapter 28 supplies the cross-book correction machinery. Here the rule is simply dependency-based updating: when a measurement, endpoint, route, or target changes status, revise only the summaries, diagrams, and applications that rely on that object, while preserving the conditions and reference class that made the scientific statement valid.

1.6 — Public Artifacts and Scientific Humility

An orientation visual may illustrate a hypothesis, a conditional consequence, or a practical aspiration without becoming a research protocol. The Series Map uses the balance scale to introduce the proposed complete-stream equality and asks what would follow if the Law is real. Read its statements within that declared frame. Figures reporting observations, estimates, or simulations identify those different evidence roles explicitly (Fischhoff & Davis, 2014; Spiegelhalter, 2017; Tufte, 2001).

1.6.1 — Visuals Orient; They Do Not Prove

Use images of participants with their privacy and consent intact. Avoid imagery that romanticizes dying or turns children, disabled people, patients, or grieving families into symbols of the hypothesis.

Public research figures should link to a source record identifying the source chapter and model or instrument version, information available, simulation status, and correction route. Supply an accessible description with the image; keep interpretation-critical qualifications in its caption (Fischhoff & Davis, 2014; Spiegelhalter, 2017; Tufte, 2001).

A quoted equation needs its warrant as well as its symbols: definition, conditional theorem, fitted estimate, or model assumption. The surrounding text must make that warrant clear.

A visual can make a concept visible before the reader knows its technical vocabulary. That is useful. A dependency map can show that representation precedes measurement. A diagram of rival models can show that several explanations compete. A simple ledger illustration can make accumulation intuitive. But the image must never make the reader feel that symmetry, smoothness, or scientific styling has proven the claim.

Visual design should make scientific relationships easy to understand. Graphs identify their variables, endpoint diagrams explain their timing conventions, equations name their objects, and illustrations clarify what the instruments observe. Clear captions connect each image to the argument it serves.

Visuals must also respect persons. A person in a study should not be represented as a data source emptied of privacy. A terminal image should not romanticize death or imply that dying reveals cosmic justice. A child, disabled person, patient, or grieving family should not become an aesthetic symbol for a scientific hypothesis.

The standard is not dullness. It is truth in miniature. A good visual invites the reader into the burden of inquiry. It makes the problem clearer without making the result easier than it is. It should help a skeptic identify a test, not help a supporter feel that the test has been passed.

For this reason, every visual artifact belongs to scientific governance. It needs a label, a status, a scope, and a release review. If the visual cannot survive those conditions, it should stay inside internal planning until it can.

1.6.2 — Formula Status Must Travel

Formulas carry special danger because they look more authoritative than ordinary prose. A formula can define a target, describe a model, derive a consequence, specify an estimator, simulate a rival, or state a verdict rule. These are different statuses. A public formula that does not carry its status invites readers to treat mathematics as evidence.

When the formal ledger is displayed, the reader must know whether it is a received definition, an idealization, an estimator, or a candidate realization. When a reserve-coupled equation appears, the reader must know that it is a model with

assumptions, not a measured fact. When an equivalence band appears, the reader must know that it is a public verdict rule under observation error, not direct sight of exact terminal equality.

Formula status also requires assumptions. What is the unit? What is the time variable? Is the process signed? Is reserve measured or latent? How is the endpoint defined? Which observations are missing? Is the displayed path simulated, filtered, smoothed, or retrospectively aligned? A formula without these conditions can create precision without identification (Box, 1976; Gneiting & Raftery, 2007; Wilson & Collins, 2019).

For public use, attach a plain-language status label to the formula: received definition, candidate realization, estimator, design criterion, or open conjecture. Carry that label across slides, websites, figures, donor materials, classroom handouts, and Artificial Intelligence summaries. The formula need not carry its full derivation, but it must retain the assumptions needed to prevent a definition or simulation from being mistaken for evidence. If detachment would predictably raise the claim's status, redesign the artifact.

1.6.3 — Hostile-Crop Review

Hostile-crop review tests the headline, graph, caption, equation, and summary independently. Each fragment must identify its scientific object and status without relying on surrounding paragraphs to correct a misleading impression.

Hostile-crop review asks a simple question: if the least charitable reader extracted this sentence, caption, diagram, or formula from its context, would it still be accurate enough not to mislead? This reflects ordinary public communication in a fragmented environment. A claim that is safe only when

surrounded by five paragraphs of correction may not be safe as a public artifact.

The test should be applied before release. Crop the headline away from the chapter. Crop the graph away from its caption. Crop the caption away from the graph. Crop the formula away from the assumptions. Crop the public summary away from the full status label. If the artifact now implies proof, destiny, scoring, terminal consolation, or scientific authority that the evidence has not earned, the artifact fails.

The review should test both literal cropping and likely paraphrase. Ask how a supporter, critic, clinician, civil-liberties reader, religious reader, or grieving reader could reasonably restate the artifact. If the shortened meaning becomes proof, score, destiny, permission, or consolation that the evidence has not earned, rewrite the artifact before release. Use specific nouns and verbs so the object studied and the result obtained remain visible even when the surrounding explanation disappears (Spiegelhalter, 2017; Tufte, 2001).

1.6.4 — No Score or Terminal-Destiny Aesthetic

Visuals must not turn research estimates into human rankings. Leaderboards, red/green lives, badges, or public burden maps can imply trust, worth, eligibility, or clinical priority. Apply the same design rule to internal dashboards that could later circulate beyond their intended context.

A balanced scale can illustrate the proposed equality over a completed conscious stream. It must not be presented as an observed verdict on a particular person, proof obtained at biological death, or a guarantee of spiritual justice. The stream endpoint remains distinct from a clinical marker or the last available report. Terminal research retains its narrow clinical

and scientific scope (Fischhoff & Davis, 2014; National Academies of Sciences, Engineering, and Medicine, 2017).

The same distinction applies to suffering. A visual may explain the equal-and-opposite net remainder required by exact PTVC; it must not portray that requirement as moral debt, deserved harm, purification, or a reliable technique for obtaining a chosen reward. The absence of event-by-event pairing does not remove the aggregate requirement. Section 3.7.1 states that distinction.

A safer aesthetic shows inquiry, uncertainty, care, and constraints. It may show instruments with limits, pathways with gates, rival models, consent boundaries, uncertainty bands, and correction loops. It may show the dignity of subjects and the burden of researchers. It should not show a hidden accountant, a cosmic scoreboard, or a final spiritual settlement.

Score aesthetics can also appear in research-facing tools. A model card, participant dashboard, cohort map, or quality-control display may be intended for internal use but later leak into public interpretation. The same no-score discipline should govern internal visual design when the artifact could be misunderstood outside its narrow context.

Images accompanying terminal research should represent care, clinical context, and the participant's expressed choices with dignity. Figures of scientific results should identify the measurements and processes actually depicted.

1.6.5 — Artificial Intelligence Disclosure and Limits

Artificial Intelligence may assist bounded research and public-communication tasks only under documented human responsibility, purpose-limited data use, and independent scientific review. The book must disclose and limit that role because fluent outputs can overstate evidence or omit context.

Sensitive participant data require explicit authorization and governance. Treat every AI-generated summary, visual, translation, or accessibility artifact as a new derivative artifact with source and version provenance, human review, and checks that claim status, assumptions, and correction routes survive. Section 3.6.4 governs authority and Chapter 28 governs correction and retirement.

1.6.6 — Release, Correction, and Retirement

Release review should test status, scope, source, audience, burden, hostile-crop risk, and correctability before public circulation. When evidence or assumptions change, revise or retire the affected artifact where it traveled and preserve a visible correction history. Section 28.6 contains the detailed workflow.

Chapter 2 — What Kind of Claim Is the Law of Fairness?

The word law names an ambition before it names an achievement. A pattern can be beautiful and still be local. A formula can be exact without describing the world. An explanation can connect several observations and fail when another investigator makes a better prediction. This chapter asks what PTVC would have to accomplish before its strongest scientific language was earned.

There are several different achievements to distinguish. A finite signed measure satisfies an arithmetic identity between its positive magnitude, negative magnitude, and residual. A claim that the residual vanishes adds a restriction. A mechanism describes how a process might satisfy that restriction; an instrument links a latent quantity to observations. An empirical study evaluates a specified connection among these objects. Calling each achievement confirmation of a law would conceal the very transitions that need investigation.

The distinction matters even within the mathematical claim. A possible completing path is not a forecast of the actual stream. Imagine a mechanism that closes only if an unusual action is chosen. If that action is unavailable, a proof that some completing path exists does not settle what happens. Existence, probability, policy, and every-continuation claims ask different questions. Chapter 5 gives them formal statements; here they show why the form of an assertion determines what could count as evidence for it.

The positive case begins where an added restriction earns its keep. A proposed law could connect observations that otherwise require separate explanations. But the economy disappears if every new setting needs another freely adjusted assumption. A compact statement can hide an elaborate repair system. If every result can be redescribed as support, the word law adds grandeur but no information. The relevant question is what the account

excludes and whether that exclusion helps anticipate an observation its rivals handle differently.

Recurring patterns provide starting points for stronger theory. Adaptation after joy and grief, homeostatic regulation, and institutional burden shifting each suggest regularities to explain. A proposed lawlike constraint becomes informative when it adds a specified restriction that can distinguish its predictions from those of simpler accounts (Cartwright, 1983; Hempel, 1965; Lakatos, 1978).

The same care applies when an explanation changes status. A bounded tendency can be a substantial discovery. A failed mechanism can leave a useful instrument behind. A metaphor can retain human significance after an empirical claim associated with it has been withdrawn. These outcomes become misleading only when evidence earned in one category silently supplies authority in another. A reader should be able to identify the assertion being made without reconstructing a history of shifting labels.

2.1 — Claim Families and Category Discipline

Formal claims, empirical hypotheses, statistical tendencies, and metaphors carry different warrants. Their evidence does not transfer automatically. A metaphor may clarify without confirming. The categories below identify what is being asserted (Cartwright, 1983; Hempel, 1965; Lakatos, 1978; Lange, 2009).

2.1.1 — Lawlike Constraint Claim

A lawlike constraint applies throughout its declared reference class, rather than only as an average tendency. Section 5.3 states the exact PTVC restriction and the domain over which it is asserted.

A sample from a wider class does not check every stream in that class. Section 2.6.3 describes the evidential conditions for an adverse case.

2.1.2 — Empirical Hypothesis

An empirical hypothesis connects a claim to observations that could favor or challenge it. Here the proposed tests concern specified measurable consequences of a latent architecture under stated measurement assumptions.

A hypothesis must identify an unfavorable result. A failed observation model can be a local failure; repeated failures across independently motivated models can challenge the empirical claim. A replacement bridge must therefore add an independently testable commitment (Lakatos, 1978; Lange, 2009).

2.1.3 — Statistical Tendency

A statistical tendency describes a pattern in a named reference class under stated measurement conditions, including its uncertainty and exceptions. Such findings can establish useful

early knowledge and identify the conditions under which a pattern becomes more general.

For example, a sleep study might find affective reprocessing patterns better predicted by a declared queue or reserve account than by a simple mood-rebound account. Such a finding would establish a comparative tendency in the studied setting.

2.1.4 — Heuristic and Metaphorical Uses

A fairness-oriented heuristic directs attention to burdens hidden in an apparently successful arrangement. Its value is judged by the questions and practical corrections it produces; it need not assume that the formal Law is true.

A metaphor gives difficult experience a shape. Words such as ledger, burden and closure can make an invisible cost easier to discuss. Their usefulness depends on what they help a reader understand, rather than on the evidential status of the formal hypothesis.

2.1.5 — Category Drift as Auditable Error

Category drift changes the kind of assertion being made without accounting for that change.

In this project, category drift has a distinctive shape. The Law may sound like a deep law when introduced, a hypothesis when questioned, a metaphor when criticized, a heuristic when applied, and a law again when admired. That cycle gives the claim the benefits of every category while protecting it from the costs of each one. It is not fair to critics, readers, participants, or neighboring books.

Category drift is the movement of a claim between statuses without public accounting. It can happen innocently. A writer may speak poetically in one passage, scientifically in another, spiritually in a third, and practically in a fourth. The problem

begins when the audience cannot tell which burden the sentence is carrying.

Category drift is auditable because it leaves traces. A statement can be checked for what category it invokes, what evidence it cites, what authority it implies, and what correction it would require if the evidence changes. If a public artifact presents a proposed law as empirically established, it requires correction. The title Law of Fairness and a clearly framed statement of what the hypothesis proposes do not by themselves make that error. If a donor page implies validation where the science chapter says research program, the page has shifted burden (Popper, 1959).

The audit must include old materials. Category correction is not only future-facing. Earlier summaries, images, captions, talks, guides, social excerpts, and Artificial Intelligence summaries can continue to carry obsolete status. A serious project must repair its public memory when the claim changes category. Otherwise the correction lives in the current book while the old authority continues in the world.

Status may change, but the change must be explicit, versioned, justified, and propagated to the public sentence that actually travels. A lawlike claim may narrow to a tendency; a mechanism may retire while a measurement contribution survives; a metaphor may remain after scientific defeat. Use category-specific verbs; hypotheses propose, estimators estimate, tendencies appear, lenses ask, metaphors illuminate; so the claim cannot inflate when admired or retreat into metaphor when challenged.

A useful test of category discipline is which admissible outcomes would require revising or withdrawing the claim. If no possible outcome under its declared test conditions could change its status, the sentence makes no testable empirical claim. The

stronger the consequence of a valid adverse result, the more explicitly those conditions should be stated. The discipline is not punitive. It lets strong statements remain strong by attaching them to the exact risk they carry.

2.2 — Lawhood, Explanation, and Compression

With an independently fixed neutral origin, zero remains zero under a positive change of units and specifies a terminal relation without an additional fitted residual magnitude. A nonzero target adds a magnitude that must transform with the units. Bounded closure adds a tolerance. A claim of local state regulation alone does not specify the accumulated integral's terminal value. These alternatives differ in which outcomes they exclude. Formal economy motivates investigating exact closure; its explanatory value depends on comparative predictions.

2.2.1 — Best-System Ambition

A best-system account judges a proposed law by simplicity and informative strength (Lewis, 1994). The relevant cost includes the independent assumptions needed to connect the measurements, endpoint and dynamics to a common prediction.

A best-system ambition asks whether a hypothesis belongs in the simplest, strongest, most informative account of a domain. The phrase is useful here because it refuses two shortcuts at once. It refuses the shortcut of mere regularity, since not every pattern deserves lawlike dignity. It also refuses the shortcut of isolated success, since a local fit may be outweighed by complexity, fragility, or poor rival performance.

Best-system appraisal asks whether adding the Law improves the account as a whole. The relevant comparison connects endpoint behavior, preterminal dynamics, measurement, reserve or control mechanisms, and alternative explanations. A successful account gains predictive reach and coherence while keeping its assumptions economical.

This ambition is especially demanding because the object is not a planet, gas, gene frequency, or engineered signal. It is lived experience in a conscious stream. The Law would have to

compress patterns across a domain where units, origins, endpoints, and bearers are contested. That difficulty does not make the claim impossible. It makes the standards for law-language higher (Box, 1976; Hempel, 1965; Lipton, 2004).

Best-system ambition also creates a rival obligation. The Law must be compared with homeostasis, allostasis, adaptation, predictive processing, reinforcement learning, Brownian bridge conditioning, frailty, sedation, selection, and observation artifacts. If those accounts produce the same consequence structure with fewer assumptions or better predictions, the tested Law-specific realization loses its comparative advantage. The appraisal then identifies which additional commitments earned no explanatory gain.

The scientific ambition is concrete: explain how reserve-linked dynamics, endpoint behavior, affective reprocessing, and hidden-burden signatures fit together. Shared predictions across those domains would create a focused comparison with alternative explanations.

2.2.2 — Explanatory Compression

For example, a complete prefix of +4 units and an independently bounded complete remainder between −3 and +3 imply a terminal residual between +1 and +7, excluding zero. This illustrates Chapter 7's reachability restriction: a justified bound on what remains can test exact closure before the endpoint. Applying it requires the complete prefix and remaining domain; a study-window total does not establish either.

Cross-domain convergence becomes informative when one specified structure predicts several domains with a limited set of shared parameters. A candidate state or potential can be estimated in one module, frozen, and used to predict another. The comparison should record the cost of transferring

parameters, converting units, and adding context-specific terms. A structure that retains calibrated predictive performance across those transitions offers a concrete form of explanatory compression: the same commitments explain more of the observations.

Explanatory compression means that a theory reduces the number of independent facts that must be separately accepted. A compressed explanation says: these several patterns hang together because one deeper structure is doing work. The danger is that compression can be faked by hiding assumptions in auxiliary bridges, broad definitions, flexible endpoints, and adjustable measurement choices.

The Law of Fairness aims at compression across a difficult field. It suggests that the total positive and negative phenomenal variation of a qualifying stream may be governed by an exact terminal boundary rather than merely described as disconnected episodes. It asks whether preterminal and terminal patterns, if observed under strong controls, would be better explained as consequences of that candidate boundary than as adaptation, clinical practice, selection, conditioning, or observation effects.

The compression burden is therefore two-sided. The program must explain more than terminal valence closure. It must explain why a measurement stack should have a privileged origin, why an endpoint can be defined independently, why reserve or control dynamics could generate the relevant shapes, and why rivals fail on the conjunction. If each piece requires a separate special story, the program is not compressed. It is an anthology (Lipton, 2004; Mitchell, 2009).

A useful way to count compression is to ask what becomes harder to believe without the Law. If the Law merely renames adaptation, it adds no compression. If it explains why several independent signatures survive prospective specification while

simpler models repeatedly miss them, it begins to compress. If it keeps adding exceptions after failures, the compression reverses: the theory becomes bulkier than the facts.

A theory can look simple because the cost of its auxiliaries is hidden from the reader. An untested measurement bridge, a convenient endpoint rule, a narrowed reference class, or an unspecified mechanism may be doing most of the explanatory work. Scientific fairness requires those commitments to be visible so that compression is not purchased by concealed complexity.

2.2.3 — Predictive Novelty

A prediction is not novel merely because it is printed in a book. For the prospective tests emphasized here, predictive novelty requires evaluation on observations not used to construct or tune the prediction. Discrimination additionally requires the competing models to assign different expectations to those observations. For the Law, candidate consequences might include reserve-linked variance shape, forecast performance under a prospectively specified independent-endpoint branch, model-recovery success, or a signature conjunction that simpler homeostatic and conditioning accounts do not reproduce.

Existing observations can provide comparative evidence when models assign them different probabilities under explicit assumptions. Retrospective comparisons should record how the hypotheses, measurements, and datasets were selected, account for model flexibility, and retain independent tests wherever possible. Prospective prediction offers especially clear separation between model construction and evaluation. Both approaches contribute when their information and selection processes are visible (Salmon, 1984; Woodward, 2003).

Predictive novelty tests a model on observations it was developed to anticipate. In this volume, terminal patterns, emotional adjustment, and clinical trajectories provide settings in which competing accounts can make specified predictions before the outcomes are inspected.

Accommodation is not forbidden. A new program often begins by trying to make sense of known anomalies, recurring intuitions, or unexplained patterns. The rule is that accommodation must be labeled and then converted into prospective risk. If the Law explains a known terminal report pattern, the next question is what new data it predicts differently from its rivals.

Prequential performance gives one useful discipline: prediction is assessed as data arrive, not after the full dataset is available for interpretive adjustment. A program that repeatedly improves held-out forecasts under frozen bridges earns more respect than a program that repeatedly fits the past by moving its categories. The Law should seek the former path.

The public language should track this distinction. A result that was predicted prospectively may be called prospective support for the tested component. A result fitted after exploration may be called a generated hypothesis or an accommodated pattern. Both can be part of science. Only one carries the full burden of prediction.

2.2.4 — Mechanistic Depth

The strong Law proposes a structural relation among conscious trajectories. Candidate mechanisms explore how that relation could be realized through reserve-coupled dynamics, queue processes, finite-horizon control, interoception, opponent dynamics, memory reprocessing, or other processes examined in later chapters.

Mechanistic depth comes from specifying how a proposed process works. A Queue model may describe burden ordering; reserve coupling may generate a mathematical realization; interoceptive control may offer a brain-body route. Each candidate can be developed, compared, and revised on its own evidence.

Mechanistic depth links formal quantities to observables, observables to biological or psychological processes, and those processes to distinct predictions. A mechanism contributes by explaining a process, deriving a restriction, or separating competing accounts (Box, 1976; Hempel, 1965; Lipton, 2004).

The Law's mechanism burden is unusually high because felt experience is not an external object like temperature on a wall. It is lived from within and inferred from without. A mechanistic account must therefore respect first-person evidence, physiological signals, behavioral patterns, clinical realities, and the fact that no single channel owns the person. Depth here means disciplined humility across channels.

Mechanistic depth also limits public overclaim. If a mechanism is candidate, the text must say candidate. If a process is a realization, the text must say realization. If a formula demonstrates possibility rather than observation, the text must say construction, not evidence. The same precision that satisfies experts protects vulnerable readers from hearing destiny.

2.2.5 — Causal and Structural Explanation

A structural explanation restricts the space of possible trajectories; a causal explanation identifies a process producing an outcome. PTVC's strongest ambition is structural. A candidate mechanism must still connect that restriction to observations through specified dynamics and an observation model (Lipton, 2004; Mitchell, 2009).

A structural constraint limits possible trajectories; it does not choose events or supply moral purpose.

Causal explanation still matters. If the proposed boundary condition has empirical consequences, some process must mediate the observable patterns. Reserve, adaptation, sleep, memory, interoception, prediction, and control systems are causal or dynamical candidates. The science can ask which processes generate which patterns without implying that the processes morally justify those patterns (Lipton, 2004; Mitchell, 2009).

A useful local rule is to keep three questions apart. What is the formal target? What process could realize the relevant consequence? What data would distinguish that process from rivals? The first belongs to received formal structure, the second to mechanism chapters, and the third to study and inference chapters. Chapter 2 supplies the category discipline that keeps them from impersonating one another.

Structural explanation also creates a special defeat problem. A mechanism can fail while a structural claim remains open, but a structural claim cannot hide forever behind failed mechanisms. If all independently motivated mechanisms, observation bridges, endpoint rules, and consequence tests fail or become ad hoc, the structure loses empirical content even if metaphysical possibility remains.

The same analytic lens makes this an issue of burden honesty. A theory that claims structural depth while refusing causal or empirical accountability shifts the cost of its grandeur onto others. A theory that distinguishes structure, cause, and measurement lets each carry the burden it actually owns.

2.2.6 — Lawhood Under Rival Pressure

Lawhood under rival pressure means that the favored claim and its alternatives receive comparable information, fitting freedom, diagnostic attention, and opportunity to fail. Rivals must be generative rather than decorative: Brownian bridges, ordinary mean reversion, sedation, disease severity, and other competitors should produce their own expected patterns and face held-out scoring under rules fixed before results (Gneiting & Raftery, 2007; Mayo, 2018).

The positive case becomes meaningful through conjunctions that strong rivals fail. Drift, concentration, or a terminal band alone may be imitated, whereas measurement validity, noncircular endpoint construction, required timing premises, route-state restrictions, model recovery, held-out prediction, and transport expose different failure routes. Lawhood therefore requires victories across study families, each attached to its own reference class and limits (Salmon, 1984; Woodward, 2003).

Rival pressure makes the comparison informative. If adaptation predicts the observations better, the tested Law-specific mechanism loses comparative support. If endpoint conditioning accounts for apparent closure, that pattern no longer distinguishes the mechanism. A measurement failure instead identifies the observation bridge that needs repair. These outcomes direct the next investigation toward the component the evidence actually reaches.

A useful lawhood audit asks four questions: does the hypothesis compress more structure than it imports, generate prospective restrictions, connect to mechanisms or structural constraints that can fail, and outperform serious rivals? Compression is not simplicity at any cost. A complex model may be warranted, while a simple sentence may merely hide bridges, endpoints, observation rules, and trust assumptions. The

audit should therefore count auxiliary assumptions and model flexibility alongside the apparent economy of the headline claim.

Novelty adds temporal discipline. Retrospective explanation can illuminate known phenomena, but progress requires consequences specified before observation. Mechanism and structure add empirical risk: reserve coupling, endpoint dynamics, and interoceptive or control routes are candidate ways for the boundary claim to produce consequences that can be lost. Their evidentiary value increases when the predicted direction, timing, or dependency is fixed clearly enough to distinguish success from later reinterpretation.

Rival performance is the sharpest question. A moving account of hidden burden may still lose to adaptation, homeostasis, predictive regulation, treatment, or observation artifacts. The relevant measure is surplus explanation after those rivals receive full strength, not the emotional force of the theory. A rival victory should therefore identify which additional structure proved unnecessary and which observations gave the competing account its advantage.

Lawhood must also be earned inside the claimed domain. Conservation analogies, thermodynamic or quantum vocabulary, cosmic imagery, capitalized names, and formal symbols may orient readers but supply no necessity or evidence about conscious streams. The proposal stands or falls through its own definitions, discriminators, endpoint discipline, and willingness to forbid outcomes. Borrowed scientific language becomes useful only when it contributes a precise relation or test that remains valid in the phenomenal domain being studied.

A strong future case remains possible. Repeated prospective restrictions, failure of simpler rivals on a predeclared conjunction, and transport without moving the bridges would

make law-language increasingly earned. The expert reader should be able to reject the hypothesis and still recognize a fair standard: clarity, assumption visibility, empirical risk, and a genuine capacity to lose. Confidence should rise only to the extent that these successes persist across independent teams, settings, and measurement implementations.

2.3 — Component Claims and Their Scope

Representation, measurement, endpoint, realization, and consequence claims perform different inferential work. Evidence about one can change another only through an explicit dependency, and a successful downstream fit cannot repair an invalid upstream object. Section 4.1 pairs these layers with their tests; the distinctions here explain why their evidential scopes cannot substitute for one another (Borsboom, 2005; Hernán & Robins, 2020; Manski, 2003).

2.3.1 — Representation Claim

The representation claim asks whether lived experience can be treated, for a scientific purpose, as a signed and temporally additive target. This is not a casual assumption. It asks whether moments of felt burden and relief can be represented in a way that permits accumulation without flattening dignity, meaning, context, or depth into a crude score.

The representation claim is logically prior to measurement. An instrument cannot measure what the theory has not represented coherently. If valence is only ordinal, if neutral origin cannot be defended, if temporal additivity fails, or if experiences differ in kind in ways no scalar can responsibly carry, then the strongest ledger language must narrow before any terminal study begins.

A scientific representation selects the features needed for a defined question. A signed valence-rate representation can be evaluated by how coherently and usefully it describes its intended research context (Borsboom, 2005).

For the Law, the representation claim includes the origin problem. Exact closure depends on a neutral point that is not created by subtracting a sample mean or centering a questionnaire. If the origin is arbitrary, then exact zero is not identified. The project must either defend a privileged origin,

test quantities invariant to the chosen gauge, or narrow the claim.

Representation testing determines which mathematical operations a domain supports. Its result can license the next measurement stage, identify a repair, or direct the program toward an alternative representation.

The representation claim is therefore the first place where law-language can lose height. A theory about felt experience cannot borrow the simplicity of its notation. It must show that the notation is worthy of the subject.

2.3.2 — Measurement Claim

The measurement claim asks whether momentary and interval valence estimators or later instruments estimate the intended construct inside a declared context of use. It is not a claim that an instrument reads inner life directly. It is a bridge claim: given a representation, a population, a protocol, a set of channels, and an error model, the estimator tracks enough of the target to support the specified inference.

Instrument qualification is context-specific. Sedation, communication capacity, medication, and language can change the performance of physiological estimators, self-report measures, wearables, and item banks. Testing those contexts identifies where an instrument supports inference and where it needs adaptation.

The measurement claim must include uncertainty, missingness, drift, calibration, and versioning. A candidate momentary or interval estimate is not a human-worth score, civic trust, spiritual rank, religious standing or commitment within a spiritual community, or public eligibility. It is a research estimate under constraints. The word estimate should remain visible wherever public language might turn the measure

into access or rank (Borsboom, 2005; Manski, 2003; Nissenbaum, 2010).

A favorable measurement result has a limited grammar. It may say that an instrument estimates a target well enough for a study family under specified conditions. It may not say that the instrument has seen the latent cumulative process or the whole person. If the measurement bridge later changes, earlier claims must be indexed to the old version rather than silently reinterpreted.

Measurement claims are also reference-class claims. A tool validated in English-speaking adult hospice participants cannot be applied without transport to children, different cultures, nonverbal states, neurodivergent populations, sudden deaths, animals, or artificial systems. The burden of extension belongs to the project, not to the people being generalized over.

2.3.3 — Endpoint Claim

The endpoint claim asks whether the truth-level endpoint of a maximal qualifying stream can be prospectively defined without using the ledger outcome later tested. Biological death, legal death, narrative closure, family memory, or sensor cessation are not substitutes. Defining the endpoint by ledger zero would make closure tautological, so endpoint channels must be prospectively specified and not algebraically identical to the measured cumulative estimate; residual dependence must be modeled. On a separately declared independent-endpoint branch, timing independence from the ledger-generated information set and the branch's support conditions are additional premises.

Endpoint claims vary by class. Slow terminal decline, sudden death, resuscitation, coma, anesthesia, dementia, delirium, sleep, and disorders of consciousness present different evidentiary structures. A result in one endpoint class cannot silently

transport to another. Sudden low-signal endings, in particular, cannot be filled in by analogy with palliative windows unless the bridge is defended (van der Vaart, 1998).

Endpoint validation permits terminal or near-terminal data to be interpreted under a defined stopping rule. When the boundary remains unresolved, the analysis reports an indeterminate classification and identifies the evidence needed to resolve it.

Endpoint studies place care, comfort, and consent first. Their procedures combine ethical participation with independently justified classification, and record an indeterminate endpoint when either requirement prevents adequate observation.

The endpoint claim is therefore both scientific and humane. It keeps the theory from defining victory into existence, and it keeps research from treating the last vulnerability of a person as a laboratory convenience.

2.3.4 — Realization Claim

A model may end at zero because its boundary condition requires it. Its empirical value lies in predictions about the preceding path, evaluated using independently specified measurement and selection. Constructing that endpoint does not qualify the instrument or endpoint classifier (Borsboom, 2005; Hernán & Robins, 2020).

The realization claim asks whether a candidate dynamical model describes the relevant data-generating process. Sections 4.2 and 7.6 develop a reserve-coupled realization that derives closure-related consequences from explicit assumptions. Testing those assumptions and consequences connects the construction to observed dynamics.

A realization claim is not identical to the strong latent claim. The Law might be stated as a boundary condition while a reserve product, queue model, finite-horizon control process, or

interoceptive mechanism proposes how patterns could arise. If a candidate realization loses, that result may retire the realization without instantly exhausting the space of possible realizations.

The realization claim requires rival processes. Ordinary homeostatic mean reversion, endpoint-conditioned processes, safe-control baselines, and predictive regulation can all generate orderly trajectories. The question is not whether the Law can generate a pattern. The question is whether it generates a conjunction, under frozen assumptions, that rivals do not match as well.

A favorable realization result might support a mechanism family: reserve coupling in slowly declining terminal illness, queue pressure in certain affective reprocessing, or finite-horizon shadow pricing in choice behavior. Each support statement names the mechanism, the tested reference class, and the evidence that distinguishes it from competing accounts.

The realization claim is where a good theory becomes willing to lose machinery it likes. Reserve-coupled dynamics, completion-potential constructions, conditioned bridges, or any other candidate process must be removable without collapsing dignity or hiding failure. A mechanism that cannot be retired has become doctrine.

2.3.5 — Consequence Claim

For example, a prospective study might support a declared reserve-coupled route by recovering its variance and forecast restrictions against an adequate homeostatic baseline in one adult cancer cohort. Endpoint ascertainment must be noncircular; statistical timing independence is an additional premise only on a branch that declares it. Such a result supports that route and reference class, not every qualifying stream.

The consequence claim asks whether the prospectively specified signature conjunction appears and out-predicts rivals. This is where the Law begins to touch data in a way that can matter. The relevant question is not whether one curve looks suggestive. It is whether a declared family of consequences appears under the declared bridge, with rivals given serious opportunity to win.

A consequence can be preterminal, terminal, sleep-related, life-course, behavioral, physiological, or model-comparison-based. The strongest empirical burden in the current program begins with prospective and comparative consequence structures because exact latent equality is not directly observable. Support is therefore indirect, component-level, and bridge-dependent.

Consequence claims must specify the direction of inference. A terminal concentration result may support a bounded terminal consequence under one endpoint and instrument. A variance-shape result may support a reserve-linked process if measurement noise, sedation, frailty, and endpoint conditioning are ruled out. A held-out prediction gain may support one model class over another. None alone establishes the exact set-universal target over the declared class (Gneiting & Raftery, 2007; Manski, 2003).

Suppose adaptation and medication-adjusted frailty models predict terminal patterns as well as or better than the PTVC candidate. The result then supports parity or preference for those explanations in the tested class and guides the next version of the consequence model.

2.3.6 — Reference-Class Discipline

Reference-class discipline turns a sample description into the boundary of warranted speech. Every empirical status statement should name the population, condition, endpoint class,

instrument version, analysis window, and transport assumptions to which it applies. The strongest formal target is broad; every first study is local.

A selected clinical cohort can speak most directly to comparable cohorts, less directly to sudden interruption or other endpoint classes, and almost not at all to nonhuman animals or artificial systems without additional bridges. A study of report-capable adults cannot silently represent nonreporting states, and a culture-specific calibration does not become global because the result is striking. Local truth is not lesser truth; it is truth with honest borders (van der Vaart, 1998).

Transport assumptions must be stated before they are used. A claim that a signature should persist across diagnoses, cultures, communication states, or endpoint classes must name the invariance asserted and the rival reasons it might fail. If the bridge changes after an adverse result, the new bridge defines a new version rather than rescuing the old claim without loss.

The component program forms a dependency lattice. Representation, measurement, bearer and routing, endpoint, realization, consequence, rival discrimination, replication, and transport each have their own validation tasks and connections to the others.

The strong latent claim is the most dependency-heavy node, not a crown that lends authority downward. Failure at one node weakens the claims that depend on it while leaving neighboring nodes open. Conversely, genuine support at a lower node should be reported as support for that node rather than minimized to metaphor merely because the strong target remains unproved.

Reference class runs through every component. Representation may work for some forms of experience and not others; measurement may be stable in one language, disease class, or communication state and not another; endpoint

adjudication may be defensible in slow decline and weak in sudden interruption; route signatures may depend on care context or reserve definition. Transport is therefore part of each claim, not a final decorative test. Two findings can therefore differ without contradiction when their reference classes differ. A genuine conflict begins only when the relevant populations, instruments, endpoint rules, and conditions overlap enough that both claims address the same empirical domain.

Partial identification is a legitimate result. Data may constrain a truth-containing region without isolating one latent value or one explanation. A bounded identified set is progress when its boundaries, assumptions, and unresolved rivals are reported as such rather than converted into hidden certainty. The structure of the identified set can itself guide the next study by showing which assumptions, channels, or bridges carry the remaining uncertainty. Progress can therefore consist in shrinking or partitioning that set with independently justified information before a point estimate becomes defensible.

Component-level reporting supplies the grammar for honest updating. Evidence may strengthen one measurement bridge, weaken an endpoint rule, retire a route, or leave the set-universal target open or narrowed. That is more precise than saying a study supports or refutes the Law without naming the object that changed. This approach allows heterogeneous evidence to accumulate without forcing a single book-level verdict. An improved estimator, endpoint method, or recovery procedure can remain a durable scientific contribution even when a mechanism fails or the strongest branch remains unresolved.

Transport analysis must name the target population, relevant differences from the enrolled cohort, plausible effect modifiers, and the assumptions needed for reweighting or structural transport. When those assumptions fail, the result remains local

or set-valued; sample size inside a narrow class cannot manufacture support outside it (Manski, 2003; Hernán & Robins, 2020). Transport sensitivity should reveal which assumptions are doing the work rather than presenting one reweighted estimate as though extension were automatic. Agreement across independently defensible transport specifications strengthens generalization, while disagreement identifies the boundary that future data must resolve.

Instrument and protocol versions also belong inside the reference class. A classifier validated under one medication pattern, device generation, language, reporting capacity, or care setting may change elsewhere. Failed transport is evidence about the phenomenon, the measurement system, or both, and should narrow public speech rather than be dismissed as nuisance replication (Little & Rubin, 2019; National Academies of Sciences, Engineering, and Medicine, 2019). A version change should therefore be treated as a potential change in reference class until equivalence or transport has been demonstrated. Reporting the exact instrument, software, protocol, and calibration version makes replication interpretable and prevents implementation differences from being mistaken for substantive contradictions.

2.4 — Severe Tests and Underdetermination

An informative scientific test distinguishes a model's predictions from alternatives the method can detect (Chamberlin, 1890; Platt, 1964). Terminal and affective observations make that comparison especially useful because several latent processes can generate similar measured patterns. Explicit alternatives and common comparison rules turn those possibilities into tractable questions (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018; Quine, 1951).

2.4.1 — Risky Prediction

A risky prediction identifies outcomes that competing accounts expect differently. Reserve-linked dynamics, local mean reversion and endpoint-conditioned convergence can resemble one another while implying different joint predictions. Specify that contrast before evaluation and show that the analysis can distinguish it.

Falsifiability, identifiability and adjudicability are different achievements. A conceivable counterexample makes a claim logically vulnerable, but a current study may still lack an identified target or a defensible measured-to-latent bridge. Conversely, a measurable surrogate can support a decision without testing the declared target. A useful design connects the vulnerable claim to an identifiable object and a decision rule that responds differently in relevant rival worlds (Borsboom, 2005; Mayo, 2018; Woodward, 2003).

A risky prediction is one that could plausibly have failed and would have mattered if it did. The Law of Fairness should not claim support from patterns it would have expected no matter what happened. If the theory can treat convergence, divergence, volatility, flatness, silence, and missingness all as favorable without prior rules, it has no empirical risk.

Risk is created by forbidding or disfavoring outcomes. A reserve-coupled signature might predict a particular relationship between reserve decline and ledger variance under a specified bridge. A rival homeostatic model might predict local mean reversion without the same terminal scaling. A Brownian bridge might predict endpoint convergence without independent reserve coupling. The risky prediction is not the pattern alone; it is the pattern as differentially expected across models.

The prediction must be generated before outcome inspection, not narrated afterward. Prospective specification creates a dated record of hypotheses, endpoints, instruments, analysis windows, exclusions, and rival families. It does not sanctify a weak design; it distinguishes a risk taken in advance from an explanation composed after the answer. Public preregistration is claimed only when a timestamped external registry and persistent identifier actually exist (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018; Nosek et al., 2018).

Risky prediction also protects against private hope. A community that wants the Law to be true may notice favorable anecdotes and forget silent failures. A research program that wants the Law to be true may adjust models until the result looks natural. The remedy is to state in advance what the model should not do well under rivals and what would count against it.

A risky prediction can be modest. It need not test the whole Law. It may test one component: a measurement bridge, endpoint classifier, model-recovery procedure, or reference-class transport assumption. Smaller severe tests often teach more than grand unfocused tests because they know exactly what they can break.

The public language after a risky prediction should be restrained and exact. If the prediction succeeds, say which component survived and against which rival set. If it fails, say

which component lost. Do not let a successful local prediction become total validation, and do not let a failed prediction vanish into metaphor.

2.4.2 — Underdetermination by Terminal Patterns

Concentration near an endpoint is not unique. A Brownian bridge conditioned to end at a value can show approach behavior by construction. A clinical process that reduces variability near death can mimic variance compression. A dataset that loses the most distressed participants can appear to stabilize. A scale that loses sensitivity at low arousal can produce false neutrality. Each rival must be treated as a potential generator, not as a footnote.

Terminal observations bring several explanatory processes together. Clinical care, medication, sedation, frailty, reporting changes, family context, selection, sensor performance, and endpoint conditioning can each influence an observed trajectory. A study becomes informative by distinguishing their predictions.

The Law-specific question is therefore conjunctive. Is the endpoint independently defined? Does the result survive observation-model and missingness checks? Is any reserve variable estimated without using the ledger outcome? Does the candidate out-predict endpoint-conditioned, clinical, and measurement rivals on held-out data? Most importantly, can the analysis recover each generator from simulated worlds where the truth is known? A test that returns the same favorable verdict in candidate and rival worlds is not severe, however sophisticated its fit appears (Mayo, 2018; Quine, 1951; Wilson & Collins, 2019).

Terminal studies combine careful inference with comfort-first care. Their design accounts for clinical context, communication,

observation quality, and consent so that participation remains proportionate and the resulting comparisons are interpretable.

A terminal pattern can provide valuable comparative evidence when its measurement, endpoint procedure, and alternative explanations are specified. The result identifies the consequence supported within that study and the next conditions to investigate.

The correct public sentence is not: the terminal pattern confirms the Law. The correct sentence is more demanding: this terminal pattern, in this reference class, under this endpoint and observation model, supports or fails to support this consequence relative to these rivals. That sentence is less dramatic. It is also the beginning of science.

2.4.3 — Model Pluralism

Model pluralism means keeping multiple live explanations in the field long enough for evidence to discriminate among them. It is not indecision. It is the refusal to let one favored vocabulary monopolize interpretation before the data have done their work. For the Law of Fairness, pluralism is mandatory because affective dynamics, consciousness, endpoint states, and measurement each have serious existing theories.

The rival set must include models that are attractive for reasons independent of the Law. Hedonic adaptation explains many returns toward baseline. Opponent processes explain rebound and counterweighting. Predictive processing and active-inference accounts explain regulation through prediction and error. Homeostasis and allostasis explain viability-centered regulation. Reinforcement learning explains policy adaptation under reward and constraint. Clinical models explain terminal decline through disease, treatment, and care context.

A pluralistic field also includes statistical and observation rivals. Smoothing can manufacture approach. Noise floors can manufacture variance contraction. Equivalence procedures can be applied to biased estimators. Missingness can select the visible sample. Flexible latent models can absorb the same pattern under different mechanisms. These rivals deserve equal information, parameter, and tuning budgets because a candidate has not won if the comparison was calibrated to let it win (Gelman & Loken, 2014; Stanford, 2006; Wilson & Collins, 2019).

Model pluralism protects the positive case by demanding specificity. If the Law wins only against weak rivals, its victory is shallow. If it wins against strong rivals in a known reference class, its status improves. A theory that can say which rival would have explained the result and why that rival failed has earned more trust than a theory that treats alternatives as disbelief.

Pluralism also protects against premature defeat. If one Law-specific mechanism fails, another component may remain live. If one rival wins in one class, another class may remain open. The discipline is not to preserve the Law at any cost, but to identify exactly which model won where and why.

2.4.4 — Auxiliary-Hypothesis Failure

Every empirical test of a large theory depends on auxiliary hypotheses. The Law of Fairness depends on assumptions about representation, measurement, endpoint classification, instrument calibration, sampling, rival specification, missingness, and transport. When a test fails, the failure may belong to one of those bridges rather than to the strong latent claim itself.

Auxiliary dependence creates a legitimate diagnostic problem and a familiar escape route. A proponent can always announce

that some bridge failed while leaving the core untouched. The responsible response localizes the failed bridge, demonstrates the failure with independent evidence, changes public status immediately, and permits a successor only when the repair creates a new risky prediction. Without those conditions, auxiliary diagnosis becomes immunity.

For example, suppose a terminal study fails because the endpoint classifier was later shown to conflate irreversible loss of consciousness with reversible sedation. That may be an endpoint-bridge failure, not a defeat of the strong form. But the repaired endpoint bridge must be specified prospectively in the next study. The old result cannot be rebranded as support after the bridge changes (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018).

Suppose instead that multiple endpoint classifiers, measurement systems, and rival comparisons all fail to show Law-specific consequences in a reference class where they should have appeared. At some point, the appeal to auxiliary failure becomes degenerating. The core may remain metaphysically conceivable, but the public scientific claim loses status.

A title-owned fairness analysis again reveals the fairness problem. If the core keeps prestige while each failed bridge is discarded without public cost, the theory has shifted burden onto instruments, participants, critics, and future studies. Auxiliary hypotheses may protect inquiry only when their own failures are recorded and paid for in claim status.

2.4.5 — Local Versus Theory-Level Defeat

Local defeat occurs when a component, mechanism, instrument, endpoint class, reference class, or prediction fails. Theory-level defeat occurs when the failures reach the dependencies required

by the strong claim and cannot be repaired without destroying empirical content. Confusing these two levels makes science either too brittle or too evasive.

A local defeat may retire a momentary valence estimator version, a reserve proxy, a terminal classifier, a conditioned-path candidate route, a completion-potential model, or a reference-class transport assumption. Such defeats are valuable. They teach the program what not to say. They prevent weak bridges from carrying stronger public language. They also allow surviving components to remain under discipline.

A weak study does not defeat the strong claim. One gate-valid counterexample can contradict a declared set-universal branch under its stated inferential guarantee. Separately, repeated failure of independently motivated bridges and discriminating predictions can justify retiring the empirical program without establishing that the latent target is false. A theory that preserves its headline only through outcome-dependent changes to origins, endpoints, instruments, or reference classes loses the scientific standing claimed for that version (Mayo, 2018; Quine, 1951).

The exact set-universal form is asymmetric. No finite sample confirms every qualifying stream in a declared class. A counterexample matters only when the stream qualifies under the prospectively fixed bearer and routing rules, the truth-containing residual object is nonempty and excludes zero, and the representation, endpoint, observation, missingness, and measured-to-latent bridges are adequate for that conclusion. When those conditions hold, one stream is adverse to the protocol-fixed universal branch; a population average cannot rescue it.

Distinguishing local from theory-level defeat protects both ambition and honest inference. It keeps a difficult idea open to

productive investigation while ensuring that adequate contrary evidence has its stated consequence.

2.4.6 — Severe-Test Reporting

The report should distinguish prediction, retrodiction, accommodation, and exploration (Stanford, 2006; Wilson & Collins, 2019).

A severe test is incomplete unless the report states what had a genuine chance to go wrong and demonstrates that the analysis could have detected it. The report names the component tested, empowered rivals, frozen bridges, negative controls, model-recovery thresholds, failure routes, and decision rule for status change. Without a recovery result, apparent discrimination in real data may only reflect an analysis that favors one model by construction.

Report each study through its scientific object. A palliative study examines a defined endpoint class and consequence family; a sleep study investigates reprocessing and associated dynamics; a measurement study qualifies an estimator. This specificity lets each contribution receive appropriate recognition.

The report should distinguish prediction, retrodiction, accommodation, and exploration. A predicted effect under a frozen plan has one status. An exploratory pattern has another. A model tuned after data has another. All can be useful. They cannot be given the same public language (Stanford, 2006; Wilson & Collins, 2019).

Severe-test reporting includes failed and unusable results. An unavailable estimate can identify unfinished measurement work; informative missingness can identify the conditions needed for inference; and a rival victory can show that PTVC-specific

structure is unnecessary in the tested class. Each outcome has an explicit next action.

Public reporting should make meaningful scientific progress easy to recognize. Name the component supported, the observations that changed its status, and the next question those findings open. Precise reporting gives each result its full value.

The final reporting rule is hostile-crop safe: an isolated result sentence still names the component, reference class, and status ceiling. If three distant paragraphs are needed to prevent a local sentence from implying proof, the local sentence carries too much borrowed authority and must be rewritten.

Example: terminal concentration near a neutral band has little Law-specific force if endpoint-conditioned simulations, sedation, or informative dropout reproduce it. The result becomes more discriminating only when independently estimated reserve relations, stable measurement gain, noncircular endpoint classification, and held-out superiority to strong rivals survive together; and even then it is component support in that reference class, not universal validation.

A second worked example goes the other direction. Suppose the Law-specific model predicts a reserve-linked signature, but a clinical frailty model with medication, delirium, and care-context variables out-predicts it across held-out sites. Suppose model recovery shows the analysis could have detected the Law-specific pattern if it were present. In that case, the rival wins for that class. The correct response is narrowing or defeat of the consequence claim there, not a search for a poetic interpretation.

These examples show why underdetermination is not a reason to stop. It is a reason to build sharper discriminators. If several latent stories can explain one pattern, the program should seek patterns that divide them: timing, reserve coupling, noncircular endpoint construction, branch-specific endpoint premises,

negative controls, transport behavior, and forecast performance. The work becomes more technical because the claim is large.

Severe testing also changes the moral tone of the book. The project is not asking dying people, sleeping people, grieving people, or ordinary readers to serve as illustrations for a beautiful idea. It is asking whether a precisely stated claim survives designs that count the burden of being studied. A severe test that violates care is not severe in the sense this book requires, because it has shifted the burden of clarity onto vulnerable subjects.

Underdetermination remains even after good tests. The goal is not to eliminate every alternative forever. The goal is to reduce live alternatives in a declared model space and report the remainder. A scientific verdict is always relative to what was compared, what was measured, and what could have been detected.

Severe testing must be designed before emotional investment hardens around a preferred result. If the first genuinely dangerous test arrives only after the community has built identity, public artifacts, and institutional plans around support, the social cost of losing becomes part of the data environment. A humane research program therefore makes loss speakable early. The chapter’s philosophical point has a practical purpose: reduce the burden carried by future scientists who may need to report disappointing results.

2.4.7 — Auxiliary Hypotheses and the Duhem Problem

Before attributing real observations to a model, pass candidate and rival synthetic records through the intended pipeline. An analysis that returns the same favorable verdict in both worlds has not demonstrated discrimination. Failure limits the affected model-comparison verdict; separately qualified descriptive

results retain their own scope. Chapter 25 develops the recovery designs.

When a prediction fails, the data do not announce which auxiliary was responsible. This is the practical force of the Duhem problem for the Law of Fairness. Begin with assumptions that can be tested independently of the disputed result. For an endpoint classifier, compare old and revised confusion profiles against independently adjudicated reversible and irreversible cases. Reserve fresh cases for assessing the repaired rule, distinguishing corrected classification from improved fit to the original outcome (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018).

No empirical test confronts an isolated proposition. A terminal-pattern study simultaneously relies on a representation of valence, a neutral origin, a stream-routing rule, an endpoint construction, an observation process, an estimator, a missingness model, a rival set, and a decision rule.

The answer is not to pretend the network can be eliminated. The answer is to type it. Each auxiliary should have an owner, a source of independent motivation, a validation route, a version, and a known effect on the inference. Representation failure should not be redescribed as endpoint failure. Endpoint circularity should not be repaired by widening an equivalence band. A model-recovery failure should stop real-data interpretation rather than become a footnote after a striking curve appears.

A dependency graph makes this discipline visible. Upstream gates include whether the target quantity is coherent and whether the instrument can identify it in the relevant population. Midstream gates include routing, time, endpoint, and observation. Downstream gates include estimators, candidate signatures, rivals, multiplicity, transport, and verdicts. Success at

a downstream node cannot compensate for failure above it, because the downstream calculation would no longer be about the intended object (Duhem, 1954; Gneiting & Raftery, 2007; Mayo, 2018).

Auxiliary hypotheses can often be tested with data independent of the principal prediction. Neutral-origin procedures can be evaluated in longitudinal cohorts; missingness models in synthetic and semi-synthetic data; endpoint rules against independent clinical adjudication; and model recovery before cohort analysis. Together, these studies qualify the components used in terminal inference.

Synthetic challenge data are especially useful because the generating truth is known. Researchers can ask whether the pipeline recovers a PTVC-consistent route when that route generated the data, an Ornstein-Uhlenbeck process when mean reversion generated them, a conditioned bridge when endpoint conditioning generated them, and an observation artifact when dropout generated the shape. If the pipeline cannot tell these worlds apart in simulation, it cannot claim to distinguish them in human data.

Negative controls provide another route through underdetermination. A candidate signature should disappear, reverse, or remain invariant under declared perturbations for reasons specific to the model. If the same result appears under a control window that should contain no endpoint information, if it survives randomized endpoint labels, or if it strengthens when measurement quality worsens, the signature is probably not carrying the intended meaning. Controls turn abstract rival pressure into inspectable consequences. A useful control therefore challenges a specific pathway by which the apparent signal could be produced without the candidate mechanism. Convergence across controls with different failure modes

provides stronger discrimination than repeating variations of the same check.

Underdetermination changes what a favorable result means: within a declared representation, endpoint, observation model, estimator, and rival set, one prediction structure survived better than the alternatives tested. The program gains credibility by reducing viable explanations while keeping the assumptions and unresolved alternatives visible.

Endpoint analysis brings several ordinary processes into the same frame: changing reserve, care, family presence, medication, speech, and observation quality. Comparing their predictions helps identify which process explains a terminal pattern. The most informative designs seek observations on which these processes predict different timing, direction, or magnitude rather than merely asking which can fit the same final curve. This turns clinical complexity into comparative leverage while preserving the limits imposed by care and measurement.

2.5 — Confirmation, Rescue, and Model Change

Revision changes a network of dependencies. A measurement may remain valid when the inference attached to it does not. Model comparison must preserve that distinction. Bayesian updating, predictive scoring, and version records test different aspects of the change (Gelman et al., 2013; Kuhn, 2012; Lakatos, 1978; Laudan, 1977).

2.5.1 — Bayesian Updating as One Tool

Bayesian updating expresses how evidence changes degrees of belief. A result has confirmatory force when it is more expected under one model than under alternatives, given the prior assumptions.

Posterior comparisons depend on the rival set, prior mass and observation models. Sensitivity to those choices belongs in the analysis; a decisive number cannot resolve an undefined claim.

Exact closure also requires distinguishing parameter estimation from comparison of hypotheses. With an ordinary noisy likelihood and a continuous prior over a residual, the probability of any single exact value remains zero. That is a property of the chosen model, not empirical evidence against an exact law. To compare exact closure with departures from it, assign prior probabilities to explicit competing models and report their marginal likelihoods, prior sensitivity, and predictive adequacy (Wagenmakers et al., 2010).

A hypothetical benchmark makes the distinction concrete. Let y_i be already-qualified complete-stream estimates and Z_i their true residuals. Conditional on the residuals and independently calibrated variances s_i^2, assume independent observations $y_i \mid Z_i \sim N\left(Z_i, s_i^2\right)$. Model M_0 imposes $Z_i = 0$ for every included stream, giving $y_i \sim N\left(0, s_i^2\right)$. Model M_1 assumes conditionally

independent residuals $Z_i \mid \mu, \tau \sim N\left(\mu, \tau^2\right)$, with proper, prospectively specified priors for μ and $\tau \geq 0$. Its conditional observation model gives $y_i \mid \mu, \tau \sim N\left(\mu, s_i^2 + \tau^2\right)$. A zero population mean with τ greater than zero is compatible with M_1 and violates the per-stream restriction in M_0. Shared calibration bias, incomplete histories, correlated errors, and endpoint uncertainty must be represented comparably in both models rather than absorbed into whichever account is favored.

2.5.2 — Likelihood and Predictive Comparison

Likelihood and predictive comparison ask a direct question: which model expected the observed data better under declared assumptions, and which model predicts new or held-out data better?

Prequential performance is a useful discipline for longitudinal studies. The model makes predictions as the stream unfolds, and each new observation tests the forecast without access to the future. This matters for the Law because endpoint-aligned analysis can otherwise create the illusion that the model saw the end coming when it was only aligned after the fact (Dawid, 1984).

2.5.3 — Independently Motivated Revision

A revised explanation must account for the failed prediction and make a new prediction on which it can fail. Keep the original result and successor test under separate versions; Chapter 28 gives the change register (Lakatos, 1978; Laudan, 1977; Nosek et al., 2018; Vehtari, Gelman, & Gabry, 2017).

Independent motivation can come from evidence outside the failed test. For example, suppose a separate calibration study reveals that an instrument changes its response under a particular condition. A revised measurement bridge can then

predict the size and direction of that change in a fresh sample. The revision explains why the earlier test behaved as it did and supplies a new observation against which the proposed correction can be assessed.

2.5.4 — No Moving Endpoints or Origins

Section 1.3 illustrates the moved-ruler problem. A justified origin or endpoint revision must make its consequences traceable to the original result.

After an endpoint revision, recompute which observation windows, exclusions, censoring decisions, and terminal estimands depend on the redefined event.

After an origin revision, recalculate residuals and uncertainty under both calibrations; report any change in conclusion. Section 6.2 states the identification question (Gelman et al., 2013; Kuhn, 2012; Lakatos, 1978).

2.5.5 — Predictive Scoring and the Cost of Flexibility

Proper scoring rules make truthful predictive distributions optimal in expectation for the chosen target. Fix the score and prediction task before evaluation. Logarithmic and interval scores, calibration diagnostics and decision losses serve different purposes. Cross-validation depends on grouping, time order, clustering, preprocessing and leakage control: random splits within one person do not establish transport to new people, and smoothing before a temporal split can leak the future (Gneiting & Raftery, 2007; Vehtari, Gelman, & Gabry, 2017).

Searching across analysis choices can create apparent stability when no analysis was privileged beforehand. Multiverse and specification-curve work reveal sensitivity; they do not convert exploratory selection into confirmation. Preserve the analytic history and uncertainty around model differences.

Complexity is warranted when it earns recovery, discrimination, or transport. A larger hidden-state model may absorb local irregularities yet fail to predict held-out transitions, survive sealed-world recovery, or retain meaning across cohorts and channels. Simulation can check whether its added states are recoverable from the measurements the study will actually collect; otherwise mechanistic detail merely redistributes uncertainty into parameters that cannot be separately identified.

Model change becomes easier to judge when prediction is separated from accommodation. A model assessed on the observations used to invent its features may fit beautifully while saying little about what comes next. Held-out data, prospective forecasts, and prequential evaluation impose the needed order: the model speaks before the outcome, and the outcome can answer back.

Flexibility should therefore be budgeted rather than denied. A materially changed model requires fresh holdout or replication evidence, and the cost of rescue is paid through renewed exposure to failure rather than a rhetorical claim that the revised model remains the same test (Chambers, 2013; Gelman & Loken, 2014; Nosek et al., 2018). The revision plan should reserve data or replication capacity for testing the specific improvement that motivated the change. That reserve prevents all available evidence from being consumed during model development, leaving no independent information with which to evaluate the revision. A flexible program remains scientifically disciplined when every substantial change creates a corresponding obligation for fresh evidence.

2.5.6 — Versioned Theory Change

Versioned theory change records the object that changed, the evidence that forced the change, the earlier claim or public

artifact that loses authority, and the new risky prediction that the revision must face. Formal-definition changes, empirical-bridge changes, and applied-domain corrections remain distinct. Chapter 28 provides the full register, migration, and cross-book propagation procedure.

Bayesian and likelihood tools can live inside this bridge record. They can show how much a result changes relative support among models. But the record should also include non-numerical obligations: source of priors, omitted rivals, context of use, endpoint confidence, participant burden, and public wording changes. Updating is not only a calculation. It is a governance act.

The question becomes almost procedural: who carries the cost when the bridge changes? In an answerable research program, the theory pays by losing status or version continuity where appropriate. The public, the critic, and the participant do not pay by having to chase an unnamed moving target.

Bayesian language can clarify updating, but it cannot cleanse a badly formed model space. If the only live models in the comparison set are weak versions of rivals and a protected version of the Law, a posterior shift toward the Law is bookkeeping over an unfair contest. The fair comparison includes strong rivals, plausible observation artifacts, and the option that none of the proposed models deserves confidence. Otherwise the math has inherited the hidden bias of the menu.

2.6 — Narrowing, Retirement, and Defeat

An adverse result does not define its own reach. The inference to a claim must be shown. A verdict names what changes. Its remaining lessons should survive the change (Popper, 1959).

2.6.1 — Narrowing the Claim

A narrowed verdict should state its boundary directly. For example: 'In this measured adult cohort, the prespecified signature survived the stated rivals; other endpoint classes and unmeasured populations remain untested.' This is a reporting template, not a finding of this book.

2.6.2 — Mechanism and Claim Retirement

Suppose adaptation models repeatedly out-predict a candidate reserve-coupled mechanism in slowly declining adult cohorts. Under a qualified comparison and the declared decision rule, such results can justify retiring the tested mechanism or consequence. They do not by themselves refute or narrow PTVC. That stronger conclusion requires a stated entailment from the Law to the defeated prediction and adequate tests of the needed bridges. A successor begins as a new version.

When a mechanism is retired, correct the diagrams, public summaries, and neighboring books that rely on it (Stanford, 2006).

Retirement may be local: a mechanism could fail in terminal studies while predicting ordinary affective reprocessing. Name its failed and retained domains.

2.6.3 — Defeat of the Strong Form

Defeat of the exact set-universal target is the most serious negative outcome. On a declared branch and reference class, one independently audited, gate-valid stream is adverse when its

nonempty truth-containing latent compatibility set excludes zero under the protocol-fixed inferential guarantee. Local instrument failure, endpoint ambiguity, observation failure, or loss of an optional mechanism may instead block inference, narrow a class, or retire the affected route; route failure becomes target adversity only when a protocol-fixed entailment warrants that step. A credible counterexample cannot be absorbed by quiet scope drift: any narrowing must be explicit, versioned, and accompanied by withdrawal of the broader language. Program-level defeat can also arise when independently motivated bridges repeatedly fail and replacements cease to generate novel risky predictions (Laudan, 1977; National Academies of Sciences, Engineering, and Medicine, 2019).

Defeat also includes empirical-content exhaustion. If every independent bridge fails and each replacement merely explains the previous failure without predicting something new, the program loses scientific character. It may continue as philosophy, spirituality, or metaphor, but not as a live empirical research program with the same authority.

Report theory-level defeat clearly and calmly: identify the affected claim, the evidence behind the verdict, the findings that remain useful, and the materials and public statements that require correction.

2.6.4 — Defeat as Knowledge Transfer

Defeat should change more than the label attached to a preferred hypothesis. It should transfer what was learned into the fields that made the loss intelligible. If conditioned paths explain an apparent closure signature, the result belongs to stochastic-process research; if allostatic regulation wins, it belongs to affective science; if observation shutdown produces the pattern,

it belongs to missing-data methodology and clinical study design.

Credit, citation, funding, and training should follow the result. Cite the defeated model for the problem it clarified and the rival for the explanation that prevailed. If the decisive advances came from missingness models, palliative methods, endpoint adjudication, or conditioned-process theory, invest in those capacities rather than preserving the losing mechanism's institutional identity.

Defeat may reveal a poorly framed problem. A terminal-closure program might find that the tractable object is the interaction among affective dynamics, endpoint selection, observation loss, and memory. Reframing is legitimate when declared as a new question that makes unresolved discrepancies among instruments, cultures, clinical states, and endpoint classes more testable.

The transfer begins by separating what failed from what remains valid. A measurement instrument, humane burden protocol, endpoint-adjudication procedure, negative control, or model-recovery suite may remain useful even when the terminal target or a candidate route fails. Scientific loss can reduce one theory while enlarging the surrounding body of knowledge.

A rival victory must remain a rival victory. The preferred program may learn from the winning account, but it cannot absorb that explanation and preserve its former status. A successor theory needs a new identity, dependency map, and statement of what the earlier evidence no longer supports (Lakatos, 1978).

Defeat also creates correction duties. Diagrams, summaries, websites, teaching materials, and cross-volume references that imply a stronger status must be corrected or withdrawn. The record should state what failed, what remains useful, and how

the burdens contributed by participants, clinicians, families, and staff will prevent repeated work or repeated harm.

A successor should begin with the loss. It should list abandoned claims, remaining validated tools, which rival explains best, unresolved uncertainties, and evidence required for a genuinely new hypothesis. Beginning from the defeat prevents continuity of name from disguising discontinuity of content. The successor should also identify which earlier results remain interpretable under the new framework and which require reanalysis. That separation prevents inherited evidence from quietly supporting a claim it was never designed to test.

Reusable scientific objects should stay attached to the claim they failed to support. Benchmark simulations, measurement-failure signatures, burden thresholds, endpoint criteria, and negative controls become more valuable when their provenance records which attractive path no longer deserves the same investment (Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019). Preserving that provenance also helps later researchers understand why a tool was created and which failure modes it was designed to expose. Reuse is strongest when the object travels with both its demonstrated utility and the limits revealed by the original test.

A successor may retain an instrument while rejecting the terminal law, preserve the intrastream unit while changing the endpoint claim, or keep the ethical infrastructure while replacing every realization. That inheritance is progress only when continuity and abandonment are explicit. A program earns trust by letting reality end a story without taking the science, care, or contributors down with it (Lakatos, 1978; Laudan, 1977). Scientific continuity should therefore reside in validated methods, transparent records, and durable safeguards rather than in preserving the identity of a theory. This allows useful

knowledge to survive even when the explanation that originally motivated it does not.

2.6.5 — Dignity After Scientific Loss

Scientific loss changes a claim, not the standing of persons or the duties in Chapter 3. Public language should state exactly what survived: for example, a qualified measurement component can remain usable even if a mechanism is retired and terminal evidence remains inconclusive.

When the strong empirical program is defeated or retired, dependent tables, summaries, diagrams, and applied claims should be corrected or withdrawn through the Chapter 28 process. Scientific status changes; care and dignity do not.

Chapter 3 — Ethics and Scientific Authority

An optional sensor irritates the skin of a patient. Staff remove it at once. If removal happens more often during distress, the resulting silent intervals are not a random sample of the patient's day. Calling them neutral would turn responsible care into false evidence of calm. The right action is to help the person and report honestly what the interruption does to the estimate.

That small scene explains why ethics and inference meet throughout a science of experience. A protocol can alter sleep, attention, family contact, and willingness to answer. Its instruments may change the situation they record. A precise time series can therefore describe a mixture of experience and the effects of observation. The scientific question includes what the study made visible, what it made harder to express, and what it appropriately chose to leave unknown.

Burden also changes with circumstances. The same ten-minute questionnaire may be effortless for one participant and exhausting for another. A patient may hesitate to decline a request from someone associated with care. A translator or caregiver may make communication possible while introducing a privacy question of their own. Designing around these differences requires attention to relationships, not a permanent label that divides people into vulnerable and invulnerable groups.

Near death or serious illness, the cost of an ill-timed request becomes especially clear. A family may want quiet; a clinician may need to change treatment; a participant may no longer want to explain anything. The last usable observation is then a fact about the research record. It need not be the end of the experience the study hoped to understand. The distinction preserves both the person's choices and the limits of the scientific account (World Medical Association, 2024).

An estimate can also acquire a harmful second life. Suppose a device flags an unusual pattern because its interpretation fails under sedation. The flag describes a measurement problem. Reusing it for employment or insurance decisions would transform that uncertainty into an assertion about the person. Preventing that transformation requires control over data access, permitted uses, and correction after release. A reassuring sentence beside the original chart cannot govern every later use of the number.

This chapter follows those decisions from the research encounter to the public record. It specifies who can stop an activity, what consent authorizes, how bystanders become exposed, and which decisions remain outside scientific authority. Its practical test is whether a participant's refusal changes what the team does, whether a clinical interruption changes what the analysis claims, and whether a correction reaches the people who relied on an earlier result. Those are observable responsibilities. They give ethical commitments force at the moments when a study is most tempted to treat them as inconvenient.

3.1 — Care Before Evidence

Care before evidence fixes a practical ordering: the person's needs determine whether research contact can proceed, while consent determines what the resulting record may contain. This ordering also affects inference. If access to help depends on remaining observable, participation and reporting can change in response to that dependence.

3.1.1 — Present Suffering Is Present

Every protocol should contain a visible present-need trigger: acute pain, panic, breathlessness, confusion, fear, unsafe housing, hunger, exposure, abandonment, or any condition that makes data collection secondary. Staff respond to the need first, pause or stop research as required, and record the care event only to the extent consent and clinical governance permit. A possible terminal balance never reduces the urgency of present suffering; ignoring present need would be both ethically wrong and scientifically confounding (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

3.1.2 — Clinical and Social Care Remain Prior

Care independence extends beyond hospitals. Transportation, translation, food, and caregiver respite can determine participation in home care or community settings. Housing, disability access, safety planning, and crisis support also affect who can participate and what becomes observable. Provide a nonresearch route to needed support, explain that refusal leaves care unchanged, and name the staff member who handles unmet needs. If study staff become the only route to that support, both voluntariness and the data-generating context have changed.

Clinical care does not become secondary because a person enters a study. Physicians, nurses, chaplains, social workers, therapists, aides, family caregivers, and crisis workers do not become servants of the research schedule. Their obligations remain directed toward the person, not toward the elegance of the data.

The distinction matters because the Law of Fairness touches settings where research and care can easily blur. A terminal study may occur inside hospice. A measurement study may recruit from a clinic. A life-course study may observe people during hardship. In each case, participants must know that care is not contingent on participation and that refusal will not cost them help (International Council for Harmonisation, 2026).

The therapeutic misconception is not only a consent-form problem. It can be built into tone, setting, dependence, and hope. If a participant mistakes research participation for individualized clinical care or believes that refusing will reduce the care otherwise available, the consent process has not done its work. The protocol must separate research contact from clinical entitlement visibly enough that refusal remains real.

A care-prior design should include a care-independence statement, a refusal-protection route, and a staff escalation path. If the study uncovers unmet need, it routes the person to support without making that support a study benefit that pressures others to enroll. Care discovered by research should still belong to the person, not to the recruitment strategy.

Clinical and social care remain prior because the study is not the highest institution in the room. The person’s safety, comfort, voice, and material need come first. Research may observe and learn under that rule, but it may not reorganize care around itself.

3.1.3 — No Scientific Desert

A report can imply that suffering was deserved by describing severe burden as morally necessary or useful to the theory. Code injury, loss, and symptoms as conditions to understand and address. Recruitment materials should describe the measured trajectories without presenting a participant's burdens as a moral account to settle.

Data dictionaries should distinguish adverse circumstances from personal attributes. Later statistical models can inherit the coding scheme, so recording exposure and context separately helps preserve that distinction.

Scientific findings must never be translated into desert. A pattern in data cannot show that someone deserved a burden, earned a loss, invited a disease, merited grief, or received pain as correction. The word fairness in this project does not authorize blame.

This distinction must be made early because visible suffering often invites moral causality. People ask why a child suffers, why a gentle person dies, why luck is uneven, and why grief falls where it falls. The scientific volume cannot answer those questions by making victims responsible for pain or by treating suffering as evidence about desert (National Commission, 1979).

Even a future result favorable to the Law would not identify the moral meaning of any event. Section 3.7.1 states the conditional net-remainder arithmetic; here the ethical point is narrower: no illness, injury, deprivation, loss, or other event becomes deserved, earned, or justified.

Recruitment materials, consent scripts, public diagrams, and study reports should distinguish conditional balance requirements from a known personal balance or a moral account to settle. Participants are not recruited to prove that their burdens

are justified. Study descriptions identify the measured trajectories, uncertainty, and limits of the proposed inference.

No scientific desert also protects staff and families. Researchers must not impose an explanation of a loved one's suffering, describe it as morally deserved, or use it to discourage care. Questions about PTVC can be answered through its conditional arithmetic and stated uncertainties without declaring anyone's actual balance or assigning a purpose to their pain.

The framework can reveal hidden burden without assigning blame for it. Its better use here is to ask who is carrying cost, who can refuse, who can be helped, and who is being asked to bear the burden of evidence. It does not rank the person who suffers.

3.1.4 — No Forced Meaning

Participants retain authority over the meaning of their experiences. A codebook can distinguish a report from a refusal of interpretation; it cannot turn a chart, silence, or family anecdote into the participant's final message. Record personal interpretations only with consent and context, and permit their revision or withholding.

No forced meaning applies to researchers as well as to family, clergy, media, and public readers. A bereaved parent, a patient approaching death, a nurse after a difficult shift, or a survivor describing trauma may use language that is sacred, skeptical, angry, grateful, confused, or silent. The study may classify speech acts for analytic purposes, but it may not tell the person what the experience means. The most humane codebook includes a category for refusal of interpretation. The most honest public report can say that a pattern was observed without saying that a human being's pain completed a lesson. Scientific

humility is not the absence of interpretation; it is the refusal to seize another person's inner authority.

Forced meaning asks a person who is already suffering to satisfy someone else's need for coherence. In a project about fairness, that transfer is especially dangerous because the theory can tempt observers to make narrative peace before the person has chosen any meaning at all. Comfort, refusal, silence, and plural interpretation remain legitimate (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

This rule does not forbid participants from speaking. If someone chooses to describe meaning, regret, peace, anger, doubt, faith, resentment, or refusal, the research can record that statement under consent and context. The point is that the meaning belongs to the person, and the right to revise or withhold it remains.

Consent materials should therefore separate study explanation from existential interpretation. They can say what data may be collected, why the study exists, what risks are present, and what may be learned. They should not tell participants that their involvement gives their suffering a role in proving the Law.

No forced meaning keeps the science clean. It prevents researcher expectancy, family desire, spiritual pressure, and public storytelling from becoming hidden variables. The study can ask whether a pattern exists; it cannot decide what anyone's pain meant.

3.1.5 — Care Survives Every Verdict

Consent materials should state before enrollment that symptom relief, emotional support, communication access, withdrawal, and ordinary care continue under every analytic branch. Participants, staff, and families must never feel that comfort

contaminates the study or that leaving threatens help. This separation protects dignity, voluntariness, and validity by removing pressure to perform a terminal story for the theory.

Care does not depend on scientific status. It remains when the strong form fails, a component survives, the study is inconclusive, or measurement collapses. Human beings receive care because their burdens matter before any theory succeeds, not because a model describes them correctly (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000).

The same independence applies after favorable or adverse results. No finding creates a class of lives more worthy of care, and no cumulative estimate determines triage, clinical priority, or social support. Those decisions follow their own ethical and legal standards.

Care-independent planning should specify clinical support, crisis response, interpreter and transportation access, privacy, and ordinary communication with the care team without conditioning them on analyzable data or study completion. This reduces selection pressure and keeps research from becoming the price of help (World Medical Association, 2024).

When a claim narrows or fails, public and clinical records should carry a concise care-continuity consequence: treatment, support, privacy, and dignity remain in force because the hypothesis never licensed them. Negative results concern the claim, not the person’s entitlement to relief.

Later science can therefore build and lose models under a prior commitment: the person in front of us is not waiting for the universe to certify their need. That independence is one of the clearest tests of whether care truly outranks claim status (Wendler & Grady, 2008).

3.2 — No Harm-Induction or Balance-Later Designs

Not every decisive-seeming comparison is ethically available. Inducing suffering or withholding care would invalidate the design. Such exclusions create identification limits rather than negative results. Section 3.2.4 explains what remains inferable without forbidden experiments (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979; U.S. Department of Health and Human Services, 2018).

3.2.1 — No Deliberate Suffering

No study may deliberately worsen a person's condition to create a balancing contrast. Existing suffering may be observed under consent and protection, and independently justified beneficial interventions may be studied (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Commission, 1979).

3.2.2 — No Withholding or Path Engineering

Withholding can hide inside scheduling: a device fitting before medication, an interview before rest, a questionnaire before transport, or a recording window before family contact. No protocol may delay treatment, rescue, comfort, analgesia, social assistance, safety planning, translation, clinical response, or material help for cleaner data. Whenever research and care compete, care comes first; the study records the intervention and models the resulting signal under humane care rather than preserving an untreated state (National Commission, 1979; U.S. Department of Health and Human Services, 2018).

Operationalize this rule with an escalation clock that names the distress trigger, responder, stopped procedure, and reporting route. A needed care event ends or changes data collection immediately. The resulting missingness or treatment effect is

modeled as part of the observation process; any actual protocol deviation is still reported.

No study may engineer deprived lives, withdraw support, or assign hardship to test later balancing. Naturally occurring differences, policy changes, and clinical pathways can inform comparisons with confounding analysis. Beneficial interventions require independent clinical or social justification; their effects need a declared estimand and transport argument before generalization (World Medical Association, 2024).

3.2.3 — No Self-Experimentation Through Danger

Dangerous self-testing creates social pressure as well as individual risk. A public figure's attempt can encourage imitation or make danger appear a test of commitment. The project must not encourage, praise, amplify, or interpret such attempts as research, even when they are described as voluntary (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; National Commission, 1979).

3.2.4 — Forbidden Designs Stay Informative

Record which apparently decisive experiments are prohibited and why, including the causal contrast that remains unavailable. Apply the same boundary to pilots, self-experiments, documentaries, and donor-funded challenges. A noninvasive surrogate can answer a narrower contrast when its relationship to the excluded intervention is explicitly modeled.

When a prohibited intervention would be required to separate two mechanisms, a weaker observation cannot inherit the same verdict language. Natural changes, clinical variation, and noninvasive surrogates may narrow the possibilities only under explicit assumptions connecting them to the unavailable

intervention (Manski, 2003; Hernán & Robins, 2020). The resulting bound can identify which additional naturally occurring variation would be informative.

Before recruitment, simulate what the permitted designs can distinguish under care-related treatment changes, observation loss, and burden-triggered stopping. Record which mechanisms remain indistinguishable under the tested design and assumptions (Box, 1976; National Academies of Sciences, Engineering, and Medicine, 2019). Comparing simulated confusion across observation schedules can then prioritize feasible designs before any participant is contacted.

The value of a forbidden design is analytic. It identifies a causal contrast that cannot be pursued directly and forces the program to specify the estimand, confounds, and humane alternatives needed to study it indirectly. The ethical boundary therefore sharpens the method rather than weakening the question.

For example, directly inducing suffering might appear to identify a later counterweight. Since that design is prohibited, the program must instead use observational contrasts, natural symptom fluctuation, care interventions with independent benefit, negative controls, and careful rival modeling. Those humane alternatives should be judged by the identification they actually support, with any remaining limits stated explicitly (National Commission, 1979; U.S. Department of Health and Human Services, 2018).

Similarly, withholding comfort might seem to preserve a natural terminal path. Since withholding is prohibited, the program must model treatment within the terminal process. Sedation, analgesia, family presence, clinical escalation, spiritual support, and withdrawal choices become modeled variables, not nuisances.

The same is true for path manipulation. Because the project cannot engineer lives, it must face selection, non-ergodicity, culture, disability, socioeconomic context, and missingness honestly. This keeps the program grounded in the conditions it can responsibly observe.

A forbidden-design register should travel with prospective specification. It tells reviewers and red teams which apparently decisive tests are excluded and why. It also prevents later public critics from claiming that the program avoided such tests because it feared the result. The answer is more principled: some knowledge is not worth obtaining that way.

3.3 — Comfort-First Terminal Research

Terminal research observes a changing clinical setting. Its design must accommodate symptom relief, family contact, and clinical judgment while keeping each observation proportionate to the participant's condition (Council for International Organizations of Medical Sciences, 2016; Sanders et al., 2024; World Health Organization, 2020; World Medical Association, 2024).

3.3.1 — Palliative Care Outranks Measurement

The research design records care as part of the observed trajectory. Medication changes, sedation depth, delirium episodes, oxygen support, repositioning, family visits, clinician interventions, and device removals become time-stamped context where consent permits.

Palliative priority should appear in site training. Research staff must know whom to call, when to leave, how to respond to distress, and how to explain that the study has no authority over care. Clinicians must know they are free to act for comfort without asking permission from the research team.

Example: during a scheduled prompt, a participant develops acute breathlessness. The nurse stops the prompt, removes any interfering device, provides clinically indicated treatment, and asks research staff to leave. The interruption is recorded as care-priority, not participant deviation; the missing interval is not imputed as calm, distress, or neutrality (World Medical Association, 2024; International Council for Harmonisation, 2026). Event details are retained only to the extent permitted by consent and clinical policy.

If the participant asks to resume, reconfirm consent and proportionate burden, then record changed medication and

observation conditions as a new segment. Otherwise, follow the predeclared analysis for the incomplete record.

The site then asks whether the event exposed a design defect: repeated device interference, poorly timed prompts, or unmanageable staff load triggers redesign or paused enrollment (Sanders et al., 2024; World Health Organization, 2020).

3.3.2 — Sedation and Delirium Are Clinical Realities

Proportionate sedation may be part of humane care; delirium is a clinical condition requiring assessment and appropriate management. Study both as components of the terminal context, with separate treatment and observation models.

Sedative exposure, sedative effect, delirium, and reduced reportability are different variables. Dose, timing, tolerance, and fluctuating attention can affect channels differently. Record treatment and state transitions separately and identify which channels still support interpretation (International Council for Harmonisation, 2026; World Medical Association, 2024).

Treat sedation and delirium as clinically meaningful intercurrent states and declare which inferences remain admissible. Deep sedation may leave some physiological observations while defeating claims about conscious valence. Delirium may carry distress while defeating stable questionnaire interpretation. The result is not forced into a clean trajectory; the permitted inference changes, narrows, or becomes unavailable (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026).

For each terminal segment, record consciousness confidence, reportability, treatment context, sensor availability, and which inferences the evidence supports and the protocol permits. Some segments support only care description; others support measurement feasibility or a qualified component analysis.

3.3.3 — Family and Staff Burden

Measure the extra work that observation creates for families and staff. Record prompt delays, skipped classifications, overnight response times, and missed contacts alongside staffing conditions. These variables help separate a change in experience from a change in the workflow that makes it visible (Little & Rubin, 2019; Rubin, 1976).

Families may experience participation as honoring the person or helping future patients. Continuing consent should make stopping an ordinary choice, without suggesting betrayal or failure (Council for International Organizations of Medical Sciences, 2016; Wendler & Grady, 2008; World Medical Association, 2024).

Use clinically natural documentation where possible; otherwise budget and compensate added staff time. Monitor the resulting workload against preset pause or reduction limits, using the burden measures in Section 3.5.5.

Family and staff burden also affect external validity. A protocol feasible at a highly resourced academic hospice may fail in a small facility, a rural home-care setting, a crowded ward, or a community with different language needs. The result should not travel farther than the burden conditions allow.

3.3.4 — Stop, Withdraw, and Remain Silent

Provide practical, round-the-clock routes for participants, authorized representatives, bedside clinicians, and designated monitors to pause or end observation without negotiating with the hypothesis owner. The stop takes effect before analytic value is weighed. No explanation is required from the participant. The record may retain permitted safety information, but a research stop is not an assumed terminal event (Council for International

Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026; World Medical Association, 2024).

The analysis plan must anticipate withdrawal and site pauses so that investigators cannot reinterpret a stop after seeing its effect on statistical power. A valuable endpoint creates no exception.

Quiet chart review requires a defined purpose and cannot support individual inferences beyond what the record warrants. Optional sensors must be removed on discomfort, distress, clinical interference, or staff overload. Interviews are shortened, postponed, or stopped when fatigue or preference requires it. The following procedures specify the bedside response and the resulting limits on inference.

The right to silence is especially important because this project studies felt experience. A participant may stop speaking because of fatigue, fear, medication, privacy, dignity, spiritual practice, distrust, or simple preference. The study cannot presume that silence is data in the direction it wants. A no-inference state should be available whenever the communicative channel fails for reasons the protocol cannot ethically resolve. Withdrawal should be fast, nonpunitive, and possible through the participant, an authorized surrogate within limits, or a clinical stop order when participation burdens care. The research record may preserve the reason for stopping; it may not turn the stop into a hidden endpoint story.

A terminal protocol must make stopping easy. The right to withdraw cannot be buried in a consent form, spoken once, and then surrounded by gratitude, urgency, devices, staff effort, and family hope. Withdrawal must remain available, visible, and free from interpretation.

Stopping can occur for discomfort, fatigue, privacy, family request, staff workload, clinical change, loss of capacity,

distress, device irritation, spiritual preference, data-security concern, or no stated reason. The protocol does not require a morally adequate explanation. The participant or authorized route may simply say no (Council for International Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026; World Medical Association, 2024).

Remaining silent is also a protected choice. A participant may decline interviews, decline meaning questions, decline final messages, decline dream sharing, decline family interpretation, or decline public quotation. Silence should not be coded as denial, imbalance, fear, or evidence. It is often the most dignified form of agency left.

Withdrawal raises data questions. The consent process should state what happens to already collected data, what can be deleted, what cannot be withdrawn because it has been aggregated or legally retained, and whether derived analyses can remain. Those distinctions must be understandable before the terminal period, not negotiated during crisis.

A research program earns trust when it can preserve silence and still publish honestly. It should be able to say: the endpoint was not observed; the inference was not made; the person remained protected. Reporting protected absence as an outcome also makes the study's evidentiary boundary visible to readers. What was not observed should remain distinguishable from what was observed and found neutral.

Silence must be represented as a governed outcome rather than interpreted as neutral experience. A missing report near the endpoint may reflect sleep, comfort, fatigue, delirium, sedation, device removal, clinical urgency, refusal, or loss of capacity. The study should preserve those possibilities in the observation record and return no estimate when they cannot be separated. Imputing peace from nonresponse would turn a humane stop

into a favorable result (Little & Rubin, 2019; World Medical Association, 2024). Where feasible, the record should distinguish known reasons for nonresponse from genuinely unresolved missingness without inferring the participant's unobserved experience. That distinction improves later sensitivity analysis while preserving the no-inference rule.

3.4 — Consent, Privacy, and Bystander Protection

Consent and privacy govern the relationship between a research question and the people whose lives supply its evidence. The permissions needed for recruitment, continuing observation, reuse, and disclosure can differ. The following distinctions connect specific uses to ongoing consent, communication access, bystander protection, and retention decisions, so that participation remains intelligible as circumstances change.

3.4.1 — Specific Consent by Use

Consent attaches to particular uses. Wearable observation, clinical-record linkage, public examples, secondary analysis, and Artificial Intelligence training create different permissions. Track them as a versioned consent state. A proposed use outside that state requires a new authorized route; the original signature does not silently expand its scope.

Check comprehension proportionately. Clinical-trial evidence shows that a completed consent form does not guarantee understanding of the study, its limits, or the withdrawal route (Tam et al., 2015).

Publication is likewise distinct. A chart, graph, quote, case vignette, or public artifact can expose meaning even when names are removed. Consent should describe what kinds of material may appear publicly and what review or withdrawal opportunity exists before publication where feasible.

3.4.2 — Capacity and Ongoing Consent

Capacity may fluctuate as delirium, fatigue, and communication change. Decide beforehand which activities require current assent, which may proceed under a prior directive or surrogate permission, and which stop on dissent. The representative protects the participant’s known wishes and interests. When

capacity returns, the participant's own choice resumes authority as soon as practicable.

Ongoing consent does not mean constant interrogation. It means periodic, proportionate confirmation that participation still matches the person's preference, understanding, and capacity. A low-burden reminder, a simple stop card, a device-removal option, a family-notification rule, or a re-consent checkpoint may be appropriate depending on the study (Council for International Organizations of Medical Sciences, 2016; Tam et al., 2015; U.S. Department of Health and Human Services, 2018).

Assent and dissent deserve special weight when full legal consent is unavailable. A person who cannot sign a form may still show discomfort, refusal, agitation, withdrawal, or fatigue. Those signals should be treated as participation data only after they are treated as possible refusal. In many cases, they end the procedure.

Material changes require renewed consent review. A changed device, greater burden, newly recognized privacy risk, or new publication mode must not inherit permission automatically from the earlier agreement.

3.4.3 — Participation and Research Consent

Participation in care, civic life, employment, religious practice, ordinary-life tools, public services, or any applied project is not research consent. Operational records do not become scientific data merely because they are interesting. Public access does not become model-training permission merely because a record can be scraped.

The Science of Fairness applies this distinction to records from other volumes in the series. A city service log or a religious community's form needs a separate authorized

research route before it can become study data (U.S. Department of Health and Human Services, 2018; Wendler & Grady, 2008).

If operational data are ever proposed for research, the protocol must pass a conversion gate: purpose compatibility, legal basis, consent or waiver justification, privacy review, de-identification humility, bystander analysis, data minimization, and no-score discipline.

3.4.4 — Bystanders and Relational Data

Map who may be captured around a participant and what information the record could reveal. Distinguish incidental contextual presence from deliberate collection about others. Decide what is necessary, what must be masked or excluded, when bystander permission is required, and when analysis must abstain because privacy cannot be protected (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016; National Commission, 1979).

A caregiver's report is a relational account shaped by context and possible pressure. Obtain permission for its use and distinguish the caregiver's observations from claims about the participant's experience.

Staff data also matter. Clinical notes, workflow records, and classifications can expose staff conduct, emotional burden, or institutional pressures. Staff should not be reduced to instruments without labor acknowledgment, privacy consideration, training, and reporting channels.

Relational data raise publication risk. A de-identified vignette can still reveal a family, community, diagnosis, site, or conflict. The more unusual the case, the higher the identification risk. Narrative power is not an excuse for relational exposure.

3.4.5 — Minimization, Retention, and Deletion

Before collection, identify the least information capable of answering the approved scientific question. Possible usefulness to an unspecified future model is insufficient reason to collect more.

Do not promise deletion that cannot technically or legally be delivered. Distinguish identifiable records, coded datasets, derived features, published results, trained model weights, and the records needed for consent, safety, and audit. For each category, specify the retention period, deletion limits, request route, and responsible authority (National Commission, 1979; National Institute of Standards and Technology, 2020).

Retire an affected use of data when an instrument failure, inadequate consent, a privacy breach, unfair recruitment, or a withdrawn model invalidates its scientific or permission basis. Stop affected analyses when permission, data-quality, or burden gates fail. Sharing agreements must permit restriction or recall after a breach, and withdrawal before sharing where feasible; the consent process must explain the limits of reversing prior releases.

3.4.6 — Privacy as a Dynamic Relationship

Privacy risk changes during a study. A data element that appears harmless at collection may become identifying when combined with location, medication, family timing, voice, device telemetry, or a rare clinical trajectory. Governance therefore cannot be exhausted by a one-time classification of identifiers. It must revisit what the dataset can reveal as linkages, models, and external information change.

Assign access by purpose and role. Collection staff may need contact details while analysts need coded observations. Name the people responsible for key custody, access review,

revocation, and incident response; record their decisions (Council for International Organizations of Medical Sciences, 2016; National Institute of Standards and Technology, 2020; U.S. Department of Health and Human Services, 2018).

Inference risk should be tested adversarially. Teams can attempt re-identification, membership inference, linkage, attribute inference, and extraction from released models or summaries. The goal is not to prove that privacy is perfect. It is to discover which releases create unacceptable pathways and to redesign or withhold them before participants bear the consequence.

Correction is part of privacy: inaccurate records can travel farther than disclosed ones. Participants need a route to report errors in identity, context, medication, consent state, or interpretation. Record whether corrections change metadata or analysis, require public notice, or invalidate prior artifacts.

3.5 — Vulnerability, Assent, and Research Proportionality

Vulnerability concerns the power of a research request within a person's circumstances. Dependence, hardship, inaccessible communication, or institutional authority can make refusal costly. Automatic exclusion can also remove the very people whose needs a study should understand. The design must address those relationships and explain whom its participation conditions leave unrepresented (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; World Medical Association, 2024).

3.5.1 — No-Desperation Consent

Recruitment must wait when acute crisis prevents meaningful refusal. Outside crisis, compensation should reflect time and cost without making access to care or essential support depend on enrollment. Review the invitation in the participant's circumstances, including the practical consequences of declining.

Payment and support require care. Compensation for time and reimbursement for travel, childcare, communication access, or lost wages can make participation more equitable. Excessive inducement can make refusal less real. The distinction depends on context, income, need, dependence, and the stakes of the study (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000; Wendler & Grady, 2008).

Recruitment should avoid language that frames participation as helping prove the Law, advancing humanity, honoring suffering, or joining a special circle of truth. Those phrases convert belief and belonging into pressure. A safer invitation names the study, the burden, the limits, and the right to decline.

3.5.2 — Children and Assent

Research with children and adolescents requires age-appropriate explanation, assent where developmentally possible, meaningful dissent protection, and guardian permission. Their developing agency, dependence on adults, and sensitivity to authority call for a stricter burden threshold. The question must address a child-specific care need or knowledge that cannot responsibly be obtained from adults, with clear justification for the benefit or knowledge sought. Low-risk developmental measurement uses accessible communication and protects the child's freedom to form their own interpretation of the research.

A child who cannot understand the abstract hypothesis may still communicate that a device hurts or that they want to stop. Treat that response as a present choice requiring action, even when an adult has given permission (Council for International Organizations of Medical Sciences, 2016; U.S. Department of Health and Human Services, 2018; Wendler & Grady, 2008).

Children should not be asked to carry existential language. Consent and assent materials should avoid telling a child that their feelings help test whether life is fair. The child's role, if any, should be described in concrete and modest terms: a study of sleep, mood, comfort, communication, or measurement feasibility.

Developmental differences in language, expression, and caregiver mediation require child-appropriate calibration before results can inform an adult reference class.

3.5.3 — Disability and Communication Access

Disability access determines who can participate and whose experience enters the evidence. Include nonverbal response routes where needed, and do not substitute a support person's convenience for the participant's will.

Accessible research begins before enrollment. Materials may require plain language, sign language, augmentative communication, large print, screen-reader compatibility, translation, flexible timing, sensory accommodation, participant-chosen support, and non-digital options. Accessibility bears directly on construct validity: an instrument that excludes or distorts communication cannot be presumed to measure the same target across participants (International Test Commission, 2018).

At the same time, inclusion must not become coercion. Some accommodations require time, staff, privacy, or specialized equipment. If the site cannot provide them safely, it should not recruit and then place the burden of adaptation on the participant. Access failure should trigger redesign, not pressure.

Let participants choose communication support where possible. Distinguish their responses from the supporter's interpretation, and apply the bystander protections in Section 3.4.4 to the support relationship.

3.5.4 — Institutional and Clinical Dependence

Institutional dependence can inhibit refusal without intended coercion. Examine who introduces the study, who knows the enrollment decision, and whether a complaint can leave the dependent relationship. Prisoners and detained persons face especially strong constraints. Where appropriate, ask after contact whether refusal felt possible, and monitor differential withdrawal.

The study must separate authority from recruitment. A clinician who controls care should not be the only person explaining enrollment when a neutral alternative is possible. A supervisor should not recruit subordinates. A service office should not present research while someone is seeking urgent

help. Refusal must not affect care, grades, employment, housing, benefits, or access (Wendler & Grady, 2008; World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

3.5.5 — Participant and Staff Burden Budgets

A participation budget needs a burden estimate, a ceiling, a monitoring schedule, and authority to pause the study when the design exceeds it. Patterned noncompliance, withdrawal, or skipped procedures can change the study population and missingness mechanism.

Track time spent in visits, device use, prompts, and staff or family support against the budget. Distress, privacy concerns, and reasons for withdrawal may require qualitative reporting alongside those quantities (Council for International Organizations of Medical Sciences, 2016; Emanuel, Wendler, & Grady, 2000).

Preset pause or redesign triggers may include clustered device-removal requests, rising staff overtime, family complaints, burden-linked missingness, or predominance of no-estimate conditions.

3.5.6 — Community and Post-Study Duties

Involve affected communities while the design can still change. Advisers should be able to influence timing, access, burden thresholds, communication, and return of results. Their knowledge informs feasibility and foreseeable misuse; it does not replace individual consent or accountable clinical and scientific judgment (Council for International Organizations of Medical Sciences, 2016; National Academies of Sciences, Engineering, and Medicine, 2017; World Medical Association, 2024).

Compensate community advisers for their expertise without making payment contingent on endorsement of the hypothesis.

Before enrollment, specify which aggregate findings or individual feedback will be returned. Identify results too preliminary or dependent on group-level inference to support meaningful personal feedback.

Name the custodian who remains available after data collection for corrections, withdrawal requests, and complaints, with an expected response time. Agree a separate route for clinical safety signals to reach the care team after the research staff leave.

3.6 — Scientific Authority and Domain Boundaries

Scientific evidence acquires authority within a defined question, method, and reference class. Its use in policy, religion, employment, or automated decisions requires additional judgments that the scientific finding does not settle. The following boundaries identify how research conclusions can travel accurately while leaving the receiving domain responsible for its own standards and decisions.

3.6.1 — No Policy Shortcut

Scientific findings may inform public decisions, but their authority remains specific to the evidence. A legislature, court, or service provider must still justify its decision through the applicable legal and public process. A successful research program supplies no exemption from scrutiny when its prestige is used to support surveillance, triage, or compulsory participation.

3.6.2 — No Religious Certification

Religious communities and nonbelievers may interpret scientific findings within their own traditions and lives. Scientific reporting describes the tested empirical relation and its uncertainty; it does not certify divine agency, spiritual rank, or religious obligation.

Tying sacred meaning to a dataset can turn an adverse result into a spiritual crisis or a favorable one into pressure on dissenters. Keeping empirical authority within its scope protects voluntary interpretation (Douglas, 2009).

3.6.3 — No Civic or Employment Rank

Research estimates and participation records must not become measures of eligibility or trustworthiness in ordinary life.

Prohibit their reuse for hiring, insurance, policing, benefits, or comparable personal decisions through data-sharing terms, access controls, and audit. Examine aggregate dashboards too: a group display can expose an identifiable participant or invite individual suspicion.

The misuse can begin subtly. A research participant may be treated as more committed; someone who refuses as evasive; a public supporter as more trustworthy; a measured person as more transparent. Each move converts participation or visibility into social hierarchy (National Commission, 1979).

3.6.4 — No Artificial Intelligence Sovereignty

Artificial Intelligence may assist an authorized research task under a named human reviewer. The reviewer must understand the system's limitations, inspect its output, and have power to reverse it; provide an appeal route for affected people. A fluent assessment does not confer authority over a person's consciousness, consent, dignity, or eligibility (Autio et al., 2024; Douglas, 2009).

3.6.5 — Authority After Publication

After publication, participants and other affected people need a direct route to flag an error, broken source, privacy concern, or misleading use. They should identify the affected artifact without reconstructing its research history. The route should also be open to independent reviewers. Name the reviewer, required evidence, and notice procedure, and link corrections to the superseded object (National Institute of Standards and Technology, 2020; Tabassi, 2023). Section 28.2 specifies the corresponding scientific artifact records.

Scientific communication continues through summaries, interviews, diagrams, classroom materials, dashboards, and

derivative tools. Each format should accurately name the study's object, population, method, and result so readers can follow the finding back to its source.

Every high-reach statement must preserve the level of the underlying verdict. A route-favored result cannot become evidence that every qualifying stream closes; measurement validation cannot become a personal score; and a replicated terminal-window signature cannot become a forecast of an individual death or destiny. The public sentence must not outrun the technical object it summarizes.

Institutional use remains bounded by the evidence. Universities, funders, companies, religious groups, political actors, and public agencies may cite the work, but citation does not broaden the verdict or transfer scientific ownership. Artificial Intelligence systems may assist search, consistency checking, or hypothesis generation; they may not become sovereign interpreters of persons, issuers of scientific status, or custodians of hidden experiential dossiers.

Public-use monitoring should be proportionate to reach. A registry of high-reach artifacts should record the source chapter and version, status ceiling, responsible human owner, correction route, and current state: active, corrected, superseded, or withdrawn. Archives should preserve prior objects without allowing search systems or derivative tools to present them as current.

Authority can expire when its evidentiary conditions change. An instrument may be superseded, consent for secondary use may lapse, a security incident may alter risk, or an independent replication may overturn the favored model. Review dates and event triggers reconnect public claims to those changes so a once-accurate statement does not become misleading through staleness (Fischhoff & Davis, 2014; Tabassi, 2023). A review

can narrow a claim’s permitted use while leaving the underlying observation in the record, with the reason for that change attached.

Those affected by public use need a route to contest it. A participant may challenge a disclosure, a clinician may challenge a ranking-prone dashboard, a rival may identify an omitted comparator, and a reader should be able to locate the current correction without searching through silent replacements. Contestability does not decide truth by vote; it gives error a path to the people responsible for correction. Someone raising a concern should be able to identify the affected artifact directly rather than reconstructing the full research history before review can begin.

Public scientific claims remain inspectable and correctable. Their documented evidence and revision history let contributors across the series assess the findings and use them within their respective domains. Correction notices are most useful when they distinguish the changed conclusion from findings that remain unaffected, allowing readers to update their understanding proportionately.

3.7 — Non-Negotiable Research Red Lines

Care and consent bind every protocol. Formal precision cannot waive them. A hypothesis does not authorize a study. The final distinctions guard this boundary (Council for International Organizations of Medical Sciences, 2016; National Commission, 1979; World Medical Association, 2024).

3.7.1 — Balance Requirements Without Moral Desert

Assume exact PTVC, finite positive and negative totals, one qualifying completed stream, and a valid fixed representation with a calibrated neutral origin. At any chosen cut, the accumulated net balance and the signed remainder must sum to zero. A genuinely negative accumulated balance therefore requires a net positive remainder; a genuinely positive accumulated balance requires a net negative remainder. This follows from terminal closure regardless of the mechanism.

The Series Map's statement "There is no free happiness" expresses the absence of a net positive surplus over a completed conscious stream under this hypothesis. It does not mean joy is wrong, pain is morally owed payment, or suffering is holy. Exact balance challenges a pleasure-only conception of utopia; the series develops its humane social goals through further ethical and practical reasoning.

If exact PTVC is true and accumulated balance is negative, an additional negative contribution increases the net-positive remainder required for closure by the same amount. This fixes future positive magnitude minus future negative magnitude, not either gross total or an improvement over an unchosen life. It specifies no matching event, timing, or safe strategy for obtaining a reward. An arithmetic requirement is not moral desert.

Kindness has no automatic valence value merely because it is kind. Enjoyment felt by the helper enters that person's balance; the recipient's experience belongs to the recipient's stream. If the helper's balance is positive, enjoyment increases the required net-negative remainder. If it is negative, enjoyment reduces the net-positive remainder until the balance crosses zero. Helping changes the life actually lived: terminal closure fixes neither a quota of suffering nor the recurrence of a particular prevented harm.

If the hypothesis is universal, informed choices and attempts to exploit it are included in its scope. Research prohibitions do not remove actually realized qualifying streams from the universal claim.

3.7.2 — No Cross-Person Balancing

One participant's insight does not justify another's suffering. A proposed future benefit to humanity cannot erase a family's or community's present research burden. That burden still requires independent justification, minimization, and appropriate consent.

3.7.3 — No Archive as Soul

An archive does not establish continuation of the recorded person's conscious stream. A model that reproduces aspects of their style supplies neither posthumous consent nor knowledge of their terminal experience.

An archive can preserve memory, provenance, consent, correction, and public accountability. It cannot preserve the person as a continuing conscious stream merely by containing records. This matters for terminal research because grief, technology, and public fascination can combine into a powerful temptation: to treat messages, recordings, model reconstructions,

or data traces as if they reveal or continue the self after death. The science volume must refuse that implication. Posthumous records may be handled with dignity, restricted access, correction, and family sensitivity. They may not be called the soul, the ledger, the endpoint, the surviving mind, or a substitute for the person who is gone.

Records, public artifacts, governance logs, research datasets, and future archives can preserve accountability, correction, and institutional memory. They remain records. No accumulation of files can resurrect consciousness, certify inner life, or become a proxy for personhood (National Academies of Sciences, Engineering, and Medicine, 2017).

An archive can also tempt endpoint overclaim. A complete record may make observers feel they know what a person was, meant, deserved, or became. The science volume must reject that feeling. The bearer of the formal claim is a separately routed maximal qualifying conscious stream, not a biography assembled from documents.

Artificial Intelligence intensifies the risk. A model trained on someone's text, voice, image, records, or behavior may simulate style or answer questions. It is not the person. It cannot consent after death. It cannot reveal terminal meaning. It cannot close the ledger.

Posthumous dignity requires record humility. Data retention, family access, publication, memorialization, and deletion should be handled under the permissions and governance set before or legally after death. The more moving the record, the more careful the boundary.

3.7.4 — No Concealed Failure

No concealed failure is the final ethical rule because it helps future studies learn from the complete research record. Null

results, rival wins, adverse events, breaches, recruitment problems, burden overruns, model failures, endpoint uncertainty, withdrawn measures, and substantive corrections must remain visible through appropriate channels. Concealment shifts burden onto the next site, the next family, the next dying person, the next staff member, and the next reader who trusts the public summary. Preserving this record is care extended forward in time: it gives later teams the information needed to avoid repeating preventable problems.

Concealed failure is not merely dishonesty. It makes the Law unfalsifiable in practice. If unfavorable outcomes disappear, the public sees only survival. A claim that survives by file drawer is not answerable; it is curated (Council for International Organizations of Medical Sciences, 2016). Making unsuccessful studies discoverable also helps researchers combine results across sites and choose more informative follow-up designs. A visible record of failure also allows later teams to distinguish isolated setbacks from recurring weaknesses in a method, bridge, or prediction. That distinction can prevent both premature abandonment and unjustified persistence.

The protocol should specify failure reporting before data collection begins. It should state what is reported publicly, what is reported to oversight bodies, what remains protected for privacy, what triggers correction, and what changes claim status. The route should exist before anyone knows the result. Assigning responsibility and a reporting deadline makes this commitment actionable when staff, funding, or institutional priorities change. The protocol should also identify who verifies that required reporting actually occurred and where the resulting record can later be found. Accountability is stronger when disclosure has an owner, destination, and auditable completion point.

Failure visibility also includes public artifacts. If a diagram, summary, website, donor page, or prior chapter becomes inaccurate after a result, it must be corrected, retired, or relabeled. A public artifact is not harmless because it is beautiful. It can carry old authority after the claim has changed. A dated correction notice should identify the affected claim and direct readers to the current explanation, allowing them to follow the change even when older copies remain in circulation. High-reach artifacts deserve priority because their outdated implications can persist long after the underlying scientific record has changed. Correction should therefore follow both evidentiary importance and the likely reach of the obsolete claim.

Rival wins deserve special protection. A study that favors adaptation, homeostasis, observation attenuation, sedation, frailty, bridge conditioning, or another model has produced knowledge. It should not be narrated as temporary confusion until the Law recovers. It should narrow, retire, or defeat the relevant component as prospectively specified. The comparison should also identify which predictions or observations gave the rival its advantage, turning the result into guidance for the next model or experiment. A rival victory is most informative when the record also shows where the preferred model failed and which assumptions no longer deserve the same confidence. That makes defeat a source of model improvement rather than merely a change in ranking.

Chapter 4 — Technical Synopsis: Claim, Tests, and Verdicts

Consider a hypothetical study whose prospective forecasts outperform a strong rival, while independent adjudicators cannot resolve one participant's endpoint. The forecasts and the endpoint assessment answer different questions. Reporting the study as a single success or failure would erase the distinction that makes its results useful. This chapter asks which claim each result can change.

The forecast comparison may remain informative about preterminal dynamics if its own measurement and comparison conditions hold. The unresolved endpoint blocks the terminal conclusion that depends on it. A scientific account should preserve both facts, allowing the successful component to guide further work without borrowing authority from an ending the study could not establish.

Now suppose the same study also produces a terminal estimate close to zero. Its apparent agreement introduces another question: how did the estimate acquire its origin and units? If the completed record was recentered to force that answer, the agreement belongs to the construction. An independently qualified representation makes a different claim available.

These distinctions determine how evidence travels. A measurement result can establish that an instrument performs adequately for a specified task. A dynamical result can support a consequence of a candidate mechanism. Reaching the central terminal claim requires the connections actually used in that inference; strength at one connection cannot supply a missing connection elsewhere.

The rival comparison gives the example its scientific force. The candidate must confront alternatives capable of producing the observations, using comparable information and evaluation rules. A result becomes discriminating through the difference between those predictions, rather than through the appeal of the

candidate's description (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

A protocol turns these dependencies into decisions before a result arrives. It specifies what the proposed observations could settle and what would require another study. The miniature protocol below applies that discipline to a bounded late-life setting, drawing on SPIRIT 2025 and ICH E6(R3) (Chan et al., 2025; International Council for Harmonisation, 2026).

For the terminal estimate, a finite equivalence band supplies a measurable comparison under its declared error model (Schuirmann, 1987). Its result belongs to that measured question. The relation to exact latent closure remains a further inferential connection, whose assumptions must survive the same scrutiny as the observations.

Agreement across independently qualified components can strengthen the case when their errors have different sources. Shared calibration or preprocessing can instead reproduce the same weakness across several apparently separate successes. The dependency map makes that distinction inspectable.

The synopsis follows the example from the object being claimed to the result that can be reported. Its purpose is to make a useful next step visible whether the study supports a component, favors a rival, or exposes a prerequisite that still needs work.

4.1 — The Scientific Claim in One View

The scientific claim and its representation, measurement, endpoint, realization, and public-status dependencies must remain separate. Each layer has its own evidentiary burden (Folland, 1999; Lakatos, 1978; Popper, 1959). The dependency chain below defines permissible scientific updates, not a verdict.

4.1.1 — The Received Strong Form

The received strong form is the exact terminal-valence-closure hypothesis stated at the latent level of the formal Law. On the primary finite signed-measure branch, one separately routed maximal qualifying stream carries a finite signed phenomenal measure over its complete strict-pre-endpoint domain. Its terminal residual is equivalently the measure of that domain, the difference between its positive and negative Jordan magnitudes, and the left limit of the phenomenal ledger at the truth-level endpoint. The scientific task is to connect this fixed target to observable consequences and the methods needed to evaluate them (Folland, 1999; Lakatos, 1978).

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{phen}(T_i^\star -)$$

$$PTVC_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0$$

Additive PTVC, the bipolar mixed-valence branch, and Generalized Terminal Neutrality remain separate unless a proved bridge establishes equivalence.

On the absolutely continuous primary subbranch, the phenomenal ledger accumulates realized neutral-relative value against experienced exposure. In a physical-time chart, the received representation is:

$$L_i^{phen}(t) = L_i^{phen}\left(S_{i,\theta}^-\right) + \int_{S_{i,\theta}}^{t} g_{\theta,i}(s)\, q_i^{\exp}(s)\, v_i^{real}(s)\, ds$$

A probability-one branch under a declared stream law is a separate modal claim:

$$P_i\{L_i^{phen}(T_i^{\star}-)=0\}=1$$

The display states [P1] closure: the latent integral equals zero almost surely under the specified law. [SET-ALL] instead quantifies over every qualifying stream. The branch label identifies which statement a derivation or analysis addresses.

The most discriminating structural result concerns endpoint timing. Under a declared reference law, suppose an endpoint is independent of the sigma-field generated by a jointly measurable pre-endpoint ledger path and has a conditional density that is positive almost everywhere on a fixed interval. Exact closure at that endpoint then forces the pre-endpoint ledger to be zero for almost every time in that interval, for almost every eligible path.

$$P_i^0\{L_i(T_i^{exo}-)=0\}=1$$

T_i^{exo} is independent of $\sigma\{L_i(s^-):s\in I\}$ under P_i^0, and $f_{T_i^{exo}}(t)>0$ for Lebesgue-almost every $t\in I$.

$$0=P_i^0\{L_i(T_i^{exo}-)\neq 0\}=\int_I P_i^0\{L_i(t^-)\neq 0\}f_{T_i^{exo}}(t)\,dt$$

Therefore $L_i(t^-)=0$ for Lebesgue-almost every $t\in I$ on a P_i^0 -probability-one set of eligible paths.

Endpoint timing has several distinct forms. Section 4.3.4 defines the branch distinctions and anti-tautology conditions; the reserve-coupled route is one candidate realization within that framework.

4.1.2 — The First Empirical Burden

Preterminal restrictions ask whether, before the ending is known, the system behaves as the candidate architecture predicted and rivals did not predict as well.

The first empirical target is therefore a vector of prospective consequence claims, not one scalar verdict:

S = { reserve-linked drift, innovation contraction, variance-shape law, forecast performance under the declared independent-endpoint branch, observation-model robustness, held-out superiority over specified rivals } (Popper, 1959).

A study may report evidence about one coordinate of S while abstaining on another. A fitted negative drift coefficient can support a local dynamical component. Variance shape, reserve coupling, the protocol-fixed endpoint-timing premise, and model recovery receive their own assessments. Component-specific reporting identifies what the result contributes and which comparison should come next.

Dynamical shapes, reserve-space relations, forecast errors, innovation behavior, and comparative predictions offer tractable early targets. Their dependence on origin assumptions can be assessed directly, helping identify which features can be compared before absolute terminal level is estimated.

4.1.3 — Component Dependencies and Scope

The compressed dependency spine can be written as a chain of conditional permissions. Let C0 be the received formal target, C1 the representation bridge, C2 the observation model, C3 the endpoint rule, C4 the declared consequence module, whether mechanism-neutral or route-specific, C5 the rival set, C6 the inference engine, and C7 the verdict rule. A public update is licensed only through the weakest surviving link invoked by the

target component, not through the most impressive-looking result at the end of the chain.

$$C_0 \to C_1 \to C_2 \to C_3 \to C_4 \to C_5 \to C_6 \to C_7$$

If any prerequisite gate Gi fails, the permissible update is capped before the failed gate:

$$\text{permitted status} \leq \min\{G_i : i \leq \text{tested component}\}$$

This is the noncompensatory evidence rule. The minimum operator in the displayed shorthand is ordinal over the prospectively specified status ladder, not arithmetic, and it ranges only over gates required by the component being judged. It does not imply that every target-law test requires one candidate realization. A clean endpoint band cannot repair an undefined sign convention; a reserve correlation cannot repair a circular endpoint; a beautiful simulation cannot repair failed model recovery.

Annotate each dependency with the assumption or measurement procedure that carries evidence from one stage to the next.

Strong latent hypothesis. Exact terminal valence closure is posited for the received latent cumulative process of a unified conscious stream. Component tests, noncircular endpoints, branch-specific timing, rivals, and replication offer indirect support. A diagram, fitted example, spiritual resonance, applied success, public enthusiasm, or a single equivalence study cannot confirm it.

Representation bridge. The lived-experience target can be represented as a signed, temporally additive measure under declared unity rules. Psychometric, phenomenological, physiological, and construct validation work can support or weaken this bridge. Terminal concentration cannot repair a failed sign convention or invalid construct.

Observation model. Candidate momentary estimator, candidate interval contribution estimate, reserve proxies, endpoint proxies, reports, clinical context, and missingness are treated as fallible witnesses. Calibration, reliability, missingness, sensitivity, and model recovery studies can update it. A favorable terminal estimate cannot make a noisy or biased instrument valid by outcome.

Candidate realization. Reserve-coupled or control-theoretic mechanisms may generate measurable consequence structures. Prospective dynamics, variance shape, innovation behavior, and rival comparison can support or retire a realization. Mathematical construction alone cannot establish that the candidate mechanism operates in nature.

Reference-class prediction. A specified population and endpoint class may show the protocol-fixed conjunction of signatures. Held-out prediction and transported replication can update a bounded claim. A hospice cohort cannot silently generalize to all cultures, sudden deaths, nonhuman animals, or artificial systems.

Public claim status. A result receives a status label: preliminary, replicated, inconclusive, unusable, narrowed, defeated, or abstained. The declared verdict process and independent review update public language. Donor interest, media attention, or institutional adoption cannot change scientific status.

Candidate realization is also distinct. The route coordinate Lrc(t)=cLU U(t)Y(t) may make terminal concentration dynamically approachable, but it is not the received definition of the phenomenal ledger. It is one candidate bridge from formal target to measurable consequences. The bridge can fail while the root hypothesis remains formally intelligible, or it can succeed

locally while the exact set-universal form over the declared class remains unconfirmed.

4.1.4 — Reference Classes and Scope

Reference-class discipline is also the antidote to hidden population ranking. A cohort that yields usable data is not a better kind of life. A cohort that yields unusable data is not a lesser one. It is only a domain where the current bridge either works, fails, or has not yet been built.

Public conclusions should state the sample, instrument, and endpoint class actually tested. Those details show readers the scope of the result and the next opportunity for extension.

No empirical result has universal scope by default. A study conducted in one hospice network, under one cultural setting, one instrument version, one endpoint rule, one medication environment, and one recruitment pathway does not silently speak for all conscious streams. The reference class is part of the claim. If it is not named, the result is already overstating itself (Lakatos, 1978; Popper, 1959).

The reference-class ceiling is especially important here because the formal hypothesis reaches beyond any first human cohort. Sudden deaths, pediatric cases, nonhuman animals, advanced dementia, anesthesia, disorders of consciousness, cultural variation, artificial systems, and future candidate bearers each raise separate representation and endpoint issues. A mature science can expand only by defending transport bridges, not by assuming them.

Scope statements must therefore name population, state space, endpoint class, instrument version, data window, care context, and transport assumptions. A result may say, for example, that in a defined adult palliative cohort under a particular momentary valence estimator version and endpoint adjudication rule, a

preterminal reserve-linked variance pattern survived specified rivals. That sentence is long because the claim is honest.

Reference classes also protect against country ranking, clinical ranking, and cultural arrogance. The project must not treat one population's data as evidence that another population is less fair, less balanced, less mature, or less measurable. A transport failure can reveal instrument limits, endpoint limits, cultural calibration needs, or mechanism scope. It is not a hierarchy of lives.

A positive result can sometimes justify expansion. It may motivate a new study in another care context, a different endpoint class, or a distinct culture. But expansion is an empirical bridge, not a rhetorical gesture. Every widening of scope should add a new vulnerability, not merely a broader sentence.

The fairness lens sees reference-class discipline as burden containment. Without it, a narrow study shifts interpretive burden onto people who never participated, never consented, and may not fit the assumptions. Scientific care means refusing to make absent lives carry conclusions drawn from a different world.

4.1.5 — What Support Could Update

Public summaries should identify the component studied, its population, method, and result. Those details let readers understand the finding and locate the work that can build on it.

A study can support a measurement bridge, an endpoint proxy, a reserve-coupled signature, a comparison method, a reference-class prediction, a replication claim, or a decision convention. Name that contribution directly so other investigators can build on the result.

A favorable measurement study can strengthen a witness channel. Correlations with carefully elicited reports, clinical context, and physiological markers in a bounded setting provide evidence for a specified estimator version and inform the design of later studies.

A favorable endpoint study can qualify a rule for a reference class. Agreement with expert adjudication and successful circularity checks strengthen the endpoint procedure for later terminal inference.

Consider a favorable rival-comparison study. If the Law candidate outperforms homeostatic, bridge, frailty, medication, learning, and observation-attenuation rivals on held-out preterminal dynamics, the result may support a conjunction of signatures. The strong form remains exact and latent, but the empirical program has earned more pressure and attention. That is the kind of support this book seeks first.

The update rule also distinguishes informative adverse results. Failed model recovery leaves the intended model-comparison conclusion unavailable; separately qualified descriptive findings retain their own scope. A rival victory can justify retiring the tested mechanism. Failed transport limits where the empirical result applies, while unresolved endpoints leave the terminal conclusion open. Recording the affected dependency gives the next study a precise problem to solve.

Support that lands specifically protects human beings from overreach. No one should become more ranked, more exposed, more morally interpreted, more recruitable, or more governable because a component survived. A status update belongs to the claim. It does not attach to the person who generated the data.

4.1.6 — What No Result Can Authorize

Results in this chapter update only the scientific propositions they test. Chapter 3 governs the standing prohibitions on harm, ranking, and authority leakage; locally, each formula, model and reported result should name its object, reference class, uncertainty, rivals, and status ceiling.

4.2 — Status, Notation, and Dimensions

The same symbol can name a formal target, a candidate mechanism, or an estimate only if the connection between those roles is made explicit. This section separates received definitions from proposed realizations and operational proxies, then checks dimensions, rates, initial conditions, and study windows. Those distinctions identify which conclusions follow mathematically and which still require an empirical or formal bridge (Durbin & Koopman, 2012; Folland, 1999; Karatzas & Shreve, 1991; Protter, 2005).

4.2.1 — Received Definitions

Notation note: Chapter 4 uses schematic, often index-suppressed symbols to orient the reader; Chapter 5 supplies the branch-specific stream-indexed definitions. A schematic symbol and its Chapter 5 counterpart are not interchangeable unless the stated branch or bridge identifies them.

The formula-status code is R for received definition, CR for candidate realization, OE for operational estimator, DC for design criterion, and OC for open conjecture.

Formal, route, and estimator notation follows.

$$R : L_i^{phen}(t) = \nu_i\left(\Omega_i \cap [S_i, t]\right)$$

$$CR : L_i^{rc}(t) := c_{LU} U_i(t) Y_i(t), c_{LU} > 0,$$

$$OE : \hat{L}_i(t_n) = \sum_{j \le n} \Delta \hat{L}_{ij}$$

The viability ratio Ψ requires validated windows, error analysis, and abstention rules, especially near zero reserve (Section 4.2.4). It must never become a public urgency score.

The formal objects must not be mistaken for measurements. F(t) is a latent signed valence-rate contribution, and the unity gate determines whether an interval enters the stream ledger. L(t) is the latent cumulative ledger. Its sign, origin, additive structure, and units require a defensible representation. The truth-level endpoint T must be qualified independently of the tested ledger; neither sensor time nor a fitted terminal residual defines it.

In the candidate reserve product, U(t) is independently meaningful functional reserve and Y(t) is a signed arithmetic reference process. They are proposed mechanism components, not received definitions of the phenomenal ledger. The reserve-product relation and exact viability-ratio formula are preserved in Section 4.2.4. Near zero reserve, denominator instability and proxy error require windowed analysis or abstention. A reference process that cannot be distinguished from endpoint conditioning or ordinary control does not validate the route.

A momentary rate estimate and an integrated interval estimate remain operational, fallible estimators. Calibration, response shift, version continuity, uncertainty, additive validity, and reference-class invariance restrict their use. Neither becomes direct observation of lived experience or a score of human worth.

Section 4.2 records the typed objects received from the formal architecture. Chapter 5 defines their stream ownership, measure structure, atoms, and endpoint branches; the synopsis below develops their dimensions and measurement interfaces (Durbin & Koopman, 2012; Folland, 1999; Karatzas & Shreve, 1991).

The key received object is the phenomenal ledger, a signed measure accrued over the declared phenomenal-event domain. On the absolutely continuous physical-time branch, the ledger uses the experienced-exposure density $g_{\theta,i}(t) q_{i\,exp}(t)$ and

realized neutral-relative value vireal(t). This is a formal target, not a sensor output, questionnaire total, or social record.

Dimensional reading follows from that factorization. The product gθ,i qiexp vireal has ledger-per-clock-time units on the declared chart; integration yields accumulated phenomenal value. A momentary estimator may approximate only a versioned local component, and an interval contribution estimates an increment, not the person, a score, or a hidden moral balance:

$$\Delta\hat{L}_{i,[a,b]} = \int_a^b \hat{v}_i(s)\, d\hat{\tau}_i(s)$$

$$\Delta\hat{L}_{ij} = \hat{v}_i(t_j)\Delta\hat{\tau}_{ij}, \hat{L}_i(t_n) = \sum_{j=1}^{n} \Delta\hat{L}_{ij}$$

L denotes the latent target under the formal hypothesis. L̂ denotes its estimate under a versioned observation model, carrying uncertainty from missingness, sign conventions, and context of use. The measurement relation connects these two objects in the analysis.

4.2.2 — Candidate Realizations

Candidate realizations are proposed ways the received formal object might generate measurable consequence structures. They are not root definitions. The principal current route coordinate is Lrc(t)=cLU U(t)Y(t), where a declared reserve state and signed normalized coordinate generate drift, innovation, and dispersion restrictions. Its value is that it exposes the route to measurable signatures. Its danger is that compact notation can be mistaken for a discovered mechanism (Karatzas & Shreve, 1991; Protter, 2005).

The reserve route has a specific scientific burden. U(t) must be independently meaningful, not merely a symbol chosen to vanish at the endpoint. Y(t) must be a signed arithmetic reference process, and cLU must carry a declared scale convention. The relationship between Lphen and Lrc must be derived, assumed, or marked open. Endpoint-reserve synchronization must be tested rather than built into the definitions.

On the reference route, reserve is nonincreasing with dUi(t)=-riU(t)dt and riU(t)≥0, while the signed normalized coordinate follows:

$$dY_i(t) = \sigma_i(t)dW_i(t), L_i^{rc}(t) = c_{LU}U_i(t)Y_i(t).$$

With Ui(t)>0, the product rule gives:

$$dL_i^{rc}(t) = \frac{-r_i^U(t)}{U_i(t)}L_i^{rc}(t)dt + c_{LU}\sigma_i(t)U_i(t)dW_i(t).$$

The empirical signature is joint, not single. On the positive, absolutely continuous reserve branch, the integrated restoring coefficient diverges as reserve tends to zero. Pointwise strengthening of the restoring coefficient $\kappa_i^U(t) = r_i^U(t)/U_i(t)$ requires an additional rate or monotonicity premise; vanishing innovation requires $\sigma_i(t)U_i(t) \to 0$. Endpoint-conditioned bridges can produce pull toward zero, but reserve-linked prediction asks whether the declared drift and innovation signatures track an independently estimated route state under common observation and rival models.

A conservative-state representation requires a separate justification. Defining valence as the derivative of a state function makes the integral a boundary difference; imposing equal initial and final values would additionally force closure. The target cannot gain empirical support from those imposed

conditions. An independently specified potential remains available as a model or rival, subject to a validated phenomenal bridge.

4.2.3 — Operational Proxies and Estimators

A generic observation layer can be written as a state-space model rather than a direct readout (Särkkä, 2013):

$$X_k(t) \sim p_{\phi,k}\left(X_k(t) \middle| F(t), U(t), \theta(t), C(t), M(t)\right)$$

$$\widehat{v}_i(t) = m_\psi\left(X_{i1}(t), \ldots, X_{iK}(t), C_i(t), M_i(t)\right) + \varepsilon_i(t)$$

Here C(t) denotes context and M(t) marks medication, device, clinical, and observation conditions. The formula is deliberately broad.

First-person reports, family context, clinician observation, behavior, and qualitative material may validate or challenge a momentary estimate without entering its declared channel model.

Version each change to the instrument or observation model and test comparability. A changed estimand, representation, or scope requires a revised claim; earlier results retain their original bridge.

A broader multimodal alternative makes the bridge clearer. This illustration is distinct from the received physiological candidate described in Section 6.5.1. Imagine a thirty-minute window with a brief report channel, a physiological channel corrected for medication and movement, and a clinical-context channel marking a pain episode, support contact, or rest. The separately declared model specifies which observations inform the signed estimate and how each channel's uncertainty is represented.

At each minute, this illustrative multimodal model combines the declared channels. A report missing because the participant

declined a prompt is not replaced by zero. Refusal, fatigue, sleep, distress, and device burden enter the observation model; unresolved missingness widens uncertainty or yields no estimate for that minute.

If the momentary valence estimator has been calibrated as a rate-like estimate, the interval valence contribution estimate for the thirty-minute interval is the time accumulation of those minute estimates. The covariance between minutes matters. A series of correlated errors does not shrink simply because the data table has many rows. The interval contribution travels with uncertainty and with a label showing which channels were active.

4.2.4 — Dimensions and Rate Consistency

The reserve-scale convention should be explicit. The reserve-coupled realization uses a positive route-scale constant cLU, allowing U to retain its declared reserve unit while Y carries the complementary unit. A normalized submodel may set cLU=1, but that is a declared gauge choice rather than an empirical finding:

$$L_i^{rc}(t) := c_{LU} U_i(t) Y_i(t), c_{LU} > 0,$$

The reserve-normalized coordinate is:

$$Y_i(t) = \frac{L_i^{rc}(t)}{c_{LU} U_i(t)}, \Psi_i(t) = \frac{|L_i^{rc}(t)|}{c_{LU} U_i(t)} = |Y_i(t)|.$$

The product Lrc(t)=cLU U(t)Y(t) specifies a dimensional relation. If U carries reserve units, cLU and Y together convert them into ledger units. Gauge rescaling leaves the product unchanged, so claims about the separate factors require an external anchor.

The viability ratio Ψ=|Lrc|/(cLU U) is useful only under the candidate route and only above a validated reserve floor. As U

approaches zero, small reserve error can dominate the ratio. The synopsis therefore treats Ψ as a fragile diagnostic.

4.2.5 — Initial Conditions and Study Windows

Stream onset and study enrollment define different starting points. A late-life study combines its observed window with an unknown pre-enrollment contribution, represented through an explicit estimate, bound, or sensitivity analysis (Karatzas & Shreve, 1991; Protter, 2005).

The reserve route adds an initial-condition compatibility problem. If Lphen and Lrc are asserted to be the same pointwise object at stream onset, their initial values must agree. A realization that chooses a convenient Y initial state without an explicit bridge changes the route rather than solving it.

On the exact path-bridge branch, the initial identity is:

$$L_i^{phen}\left(S_i^{\theta}\text{-}\right) = L_i^{rc}\left(S_i^{\theta}\text{-}\right) = c_{LU}U_i\left(S_i^{\theta}\text{-}\right)Y_i\left(S_i^{\theta}\text{-}\right).$$

If the phenomenal convention additionally sets the initial ledger to zero and reserve is positive, this identity constrains the initial Y state. Other branches may use an offset, a reduced-form route, or an explicit initial ledger, but the choice must be declared before terminal fitting.

A forty-day palliative cohort can estimate local drift, variance, reserve coupling, missingness, and endpoint behavior under its defined care context. Complete-history inference additionally requires the prior-contribution bridge in Section 6.6.

4.2.6 — Open Formal Bridges

A pathwise bridge between the additive ledger and reserve product must reproduce increments on the same domain, under the same initial conditions and time convention. Reproducing increments alone establishes only a related construction when

those other objects differ. Section 7.8.1 examines the process-class compatibility required for identity.

Endpoint-reserve synchronization is another bridge. If U(t) approaches zero at the same time the endpoint T occurs, the product construction can force L toward zero. That can be a candidate mechanism only if reserve is independently defined and not chosen because it vanishes where the endpoint is declared. Otherwise the construction risks making closure true by design. Tracking both factors near the endpoint helps determine whether the reference process remains bounded or grows rapidly enough to offset the decline in reserve. A useful test asks whether the same reserve definition predicts closure timing in held-out streams where endpoint location was not used to calibrate the reserve measure.

The Logical Fork also remains open. If the endpoint is independent and the terminal ledger distribution remains continuous and nondegenerate, exact terminal equality is not expected by chance. The program must therefore identify what premise changes: terminal degeneracy, atom formation, boundary control, endpoint-state coupling under declared rules, or another mechanism. Each option has different signatures and risks. Compare these branches under a shared observation model for detectable pre-endpoint differences in timing, distribution shape, or boundary behavior; terminal fit alone may leave their generative differences unresolved.

Scalarization is treated in Section 4.3.1 and Chapter 6: when admissible representations disagree, the empirical target may be a set of residuals.

4.3 — Measurement and Endpoint Dependency Chain

A prespecified retrospective estimate is not a prospective forecast. Section 4.3.5 fixes the information available to each. Representation and calibration constrain comparison, while bearer and endpoint classification constrain terminal inference. Model the observation process alongside latent dynamics (Borsboom, 2005; Little & Rubin, 2019; Rubin, 1976; Särkkä, 2013).

4.3.1 — Representation Before Integration

Scalarization must specify what information is discarded, which rival interpretation that loss permits, and whether a vector or identified-set representation better answers the study's question.

For the declared component vector and admissible weights, a linear scalarization gives (Borsboom, 2005; Folland, 1999):

$$F_w(t) = w^\top z(t), w \in W, L_w(T) = \int_0^T g_\theta(t) w^\top z(t) dt$$

If several admissible weights remain plausible, the identified terminal set is:

$$L(T;W) = \{L_w(T) : w \in W\}$$

A result is scalarization-sensitive when the sign, equivalence-band conclusion, or rival ranking changes across W. In that case the proper outcome is narrowing or abstention, not a more confident paragraph.

The most dangerous shortcut is to integrate a score whose sign is not stable. If high values sometimes mean burden and sometimes mean restoration, then a cumulative total can cancel unlike things by arithmetic rather than experience. The program must declare channel transformations, anchors, response-shift

rules, and exclusion conditions before the interval valence contribution estimate is computed.

When context or communication access changes signal interpretation, include that dependence in the model or restrict the reference class.

A study may find that the proposed signed target is unsuitable for a population or state. It then reports that result and develops an alternative representation before proceeding with accumulation.

4.3.2 — Calibration Before Comparison

Illness, medication, language, response adaptation, or sensor replacement can change a scale's meaning across times, sites, and people (Rubin, 1976; Särkkä, 2013).

Calibrate each comparison for its intended use: within-stream change, cross-site transport, or endpoint equivalence. The first two need not identify phenomenal neutrality. An equivalence band must be fixed before the relevant outcomes are inspected.

The affine-origin problem can be stated in one line. If x*(t)=a+b x(t), with b>0, then the study-window integral transforms as:

$$\int_{t_0}^{t_1} x^{*}(t)\,dt = a\,(t_1\text{-}t_0) + b\int_{t_0}^{t_1} x(t)\,dt$$

The effect of origin uncertainty differs by estimand. Analyze its influence on dynamical restrictions, comparisons, and absolute terminal levels separately, and carry the resulting uncertainty into each conclusion.

A scale may appear stable because participants learned the instrument rather than because experience kept the same metric. The program must model, test, or restrict around response shift before comparison.

Calibration failure restricts the comparisons supported by the method; specify which uses remain qualified.

4.3.3 — Bearer Before Endpoint

An endpoint belongs to the stream whose history is being accumulated. A record spanning an interruption may support continued identity, a possible partition, or an unresolved relation between its earlier and later intervals. Those alternatives change which history a proposed endpoint would close. The bearer assessment therefore records the continuity model and its confidence class before the terminal calculation, so ownership uncertainty travels into that calculation (Section 11.1).

4.3.4 — Endpoint Before Outcome Inspection

Fix endpoint channels and rules prospectively, without defining or tuning the endpoint by the tested ledger outcome; Section 13.2 gives the qualification procedure (Hernán & Robins, 2020).

On a declared first-hitting endpoint branch, with a specified endpoint filtration and regularity conditions ensuring the stopping-time property, the endpoint rule can be written as:

$$T_\theta(\omega) := inf\{t \geq 0 : \theta_t(\omega) \in E_\theta\}, \qquad \omega \in \Omega_\theta$$

$$\{T_\theta \leq t\} = \{\omega \in \Omega_\theta : T_\theta(\omega) \leq t\} \in \mathscr{F}_t^{endpoint}$$

$$T_\theta = \Phi_\theta\left(\theta_{[0,\infty)}\right).$$

Constructional separation does not require stochastic independence. Any shared causes or residual dependence between endpoint channels and the ledger must be represented in the joint model.

On a declared independent-endpoint branch, timing independence from the ledger-generated information set and the protocol-fixed support assumptions are additional premises.

State-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, continued experience, and endpoint-inclusive value remain distinct branches. Windows, smoothing rules, thresholds, and exclusions must be frozen before outcome inspection in every branch.

Clinical reality makes the rule hard. Dying is not always a single observable instant. Resuscitation, sedation, coma, low-signal periods, device removal, family decisions, and record gaps can create multiple plausible windows. The program must specify how uncertainty is represented rather than pretending precision that care cannot provide.

Endpoint failure is a valid verdict. If the endpoint cannot be adjudicated independently, no terminal inference should be drawn. A study may still inform measurement burden, observation processes, care feasibility, or preterminal dynamics, but it should not speak about terminal concentration.

If classification procedures intrude on comfort, family presence, or clinical judgment, the protocol has failed before the model is fitted. Endpoint discipline therefore begins with care-compatible observation and a standing right to return no estimate.

4.3.5 — Filtering Before Smoothing

Filtering uses observations available through a time point to estimate the state at that time. Forecasting predicts a later state from that information. Smoothing uses later observations to refine a retrospective estimate. Each analysis must retain its information-set label; future information cannot support a prospective claim (Rubin, 1976; Särkkä, 2013).

The distinction can be written in conditioning notation, with $\mathcal{Y}_t := \sigma(Y_s : s \leq t)$ denoting the information generated by observations through t:

$$\widehat{L}_{filter}(t) = \mathbb{E}\left[L(t) \mid \mathscr{Y}_t\right]$$

$$\widehat{L}_{smooth}(t) = \mathbb{E}\left[L(t) \mid \mathscr{Y}_T\right]$$

Label reported analyses with the information set used. Prespecified smoothing may support retrospective adjudication, quality control, or teaching.

Compare candidate and rival models separately on real-time forecasting and retrospective smoothing: similar endpoint-aligned fits may conceal different prospective predictions.

A study can fail this gate if the pipeline cannot reproduce what information was available at each time. Missing timestamps, undocumented endpoint adjudication, late transformations, and untracked imputation can make a dataset unsuitable for prospective claims. The right response is no inference or retrospective-only status, not narrative rescue.

4.3.6 — Observation Process as Data

For the candidate reserve coordinate $L^{rc}(t) = c_{LU}\, U(t)Y(t)$, a useful observation model is:

$$L_{obs}(t) = L^{rc}(t) + \varepsilon(t)$$

$$Var\left(L_{obs} \mid U = u\right) = c_{LU}^2 u^2 Var(Y \mid U = u) + Var(\varepsilon \mid U = u)$$
$$+2c_{LU} u Cov(Y, \varepsilon \mid U = u).$$

The additive-noise floor can bias the estimated variance exponent downward, making a reserve-coupled process look more bridge-like. More observations sharpen the wrong answer if the floor remains uncharacterized. The response requires an observation model, sensitivity analysis, and abstention when the floor dominates. Where feasible, estimating the floor from independent calibration or control measurements gives the

correction a separate empirical basis rather than allowing it to inherit the terminal structure being tested.

Record response latency, clinical interruptions, care changes, and participant-requested pauses, specifying whether each enters stratification, the observation model, or a decision to withhold inference.

Observation intensity can collapse near the endpoint. If the most distressed, medicated, or low-reserve periods are less likely to produce usable momentary valence estimator windows or reports, measured variance may shrink even when latent experience has not. A favorable-looking terminal pattern can therefore be a visibility pattern. The program must model observation gain. Because changing observability can alter apparent location, dispersion, and temporal dependence together, the analysis should carry the observation model through the full candidate signature rather than correct variance alone.

Distinguish burden created or amplified by measurement from burden already present in the clinical state; the former is observation reactivity introduced by the study.

Observation intensity can be modeled as a process with its own history and predictors. The probability of a usable report at time t may depend on earlier symptoms, treatment, reserve, family presence, staff workload, and previous prompts. A joint model of the latent trajectory and marked observation process makes those relationships explicit and supports sensitivity analysis across alternative missingness assumptions (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020). Joint models should report which predictors affect observation intensity independently of the latent state, because those variables can help diagnose selection without becoming outcome proxies.

Candidate controls for observation artifacts include battery loss, scheduled maintenance, staffing transitions, prompt-timing changes, and contact-sensitive channels. For each proposed control, justify the absence of a relevant pathway to the latent target; staffing and contact changes may also alter experience. Compare qualified controls across devices and sites to test whether apparent contraction tracks observation gain (Durbin & Koopman, 2012; Särkkä, 2013).

4.4 — Candidate Signatures and Rival Matrix

A PTVC-linked route or consequence model may yield testable restrictions before a complete ledger is identified. Specify the derivation, empirical bridge, and rival set before examining the relevant data. A signature supports the tested route or consequence model; it bears on PTVC only where the restriction follows from PTVC under independently justified bridges and within that entailment's scope. Section 4.4 pairs candidate restrictions with alternatives that can mimic them.

4.4.1 — Reserve-Linked Drift

Reserve-linked drift relates the route coordinate's pull toward neutrality to reserve. In the signed arithmetic reference version, the drift is $-\left(r^U/U\right)L^{rc}\,dt$. A steeper conditional drift slope requires r^U/U to increase over the declared regime. Its predicted relation to independently measured reserve and innovation behavior supplies the joint comparison developed below (Doob, 1957; Durbin & Koopman, 2012; Gneiting & Raftery, 2007).

Under the reference candidate, the local drift and local diffusion functions are:

$$b_L(\ell,t) = -\frac{r^U(t)}{U(t)}\ell, \qquad s_L(t) = c_{LU}\sigma(t)U(t)$$

The associated one-dimensional forward equation, conditional on a known reserve path and a justified Markov closure, is:

$$\partial_t p(\ell,t) = -\partial_\ell\{b_L(\ell,t)\,p(\ell,t)\} + \frac{1}{2}\partial_{ll}\{s_L^2(t)p(\ell,t)\}$$

If reserve is hidden, stochastic, or coupled to observation, the ledger coordinate alone is not generally a Markov state. Use an appropriate joint state-space model.

A simple Brownian bridge to zero has a state-dependent pull:

$$dB(t) = -B(t)/(T-t)\,dt + \sigma\,dW(t)$$

Reserve must be estimated independently. If reserve is merely a transformed version of the ledger, the signature is circular. If reserve is only disease severity and disease severity also changes affect, shared cause can mimic coupling. The analysis must include clinical state, treatment, observation intensity, and proxy error.

4.4.2 — Innovation Contraction

Derive a shared restriction from one realization and test it in independently sampled observational, behavioral, physiological, and sleep settings, using each channel's validated observation relation. Record overlapping participants and shared preprocessing (Lawlor et al., 2016; Munafò & Davey Smith, 2018).

Selective exit of high-variance participants through transfer, withdrawal, device intolerance, or sudden decline can mimic innovation contraction. Report who remains observable alongside within-stream innovation estimates.

Innovation contraction is the companion to drift on a separately declared candidate subbranch. Stochastic innovation in the reserve-coupled route coordinate scales through $c_{LU}\sigma_i(t)U_i(t)\,dW_i(t)$. Contraction requires the product $\sigma_i(t)U_i(t)$ to decline, and vanishing terminal innovation requires $\sigma_i(t)U_i(t) \to 0$; reserve decline alone is insufficient. The observation model must also exclude manufactured contraction. This is a route-specific consequence under those premises, not a unique fingerprint (Karatzas & Shreve, 1991; Protter, 2005).

For reserve-space variance under RC-1:

$$V_{L,i}(u,x,t) := Var\,(L_i^{rc}(t)|U_i(t)=u,\Xi_i^{\circ}(t)=x) = c_{LU}^2 u^2 V_{Y,i}(u,x,t).$$

This identity is exact conditional on U=u under Lrc=cLU UY. It becomes an empirical claim only when U is independently estimated, Y has a stable conditional second-order structure, and the observation process has been separated from latent variance. Otherwise the slope can be a sensor-gain artifact.

Innovation contraction should therefore face a resolution sweep rather than one favored window. The analysis should vary sampling interval, aggregation rule, and instrument channel under prospectively specified limits, then ask whether the route-specific relation transports within uncertainty. A biological relation may weaken predictably under coarser measurement; an artifact may reverse, disappear, or track one instrument version only.

4.4.3 — Variance Shape and Time Scaling

The candidate signatures require different discriminators. Reserve-linked drift needs independent reserve estimation and prospective held-out testing against ordinary homeostasis, allostatic control, bridge conditioning, and medication effects. Innovation contraction needs residual-innovation analysis after proxy and observation errors; sedation, device removal, and loss of signal can imitate it. Variance shape requires reserve-space and time-domain scaling tests with noise-floor correction, not generic terminal alignment.

Declare whether the estimand uses calendar, experienced, reserve, hazard, or endpoint-aligned time. State why that clock is justified, which transformations preserve the prediction, and which clock changes would count against the route.

Variance shape is more discriminating than a generic claim that variance declines. RC-1 predicts reserve-space contraction because Lrc=cLU UY implies conditional scaling in relation to U under stated assumptions. The observable boundary exponent

also depends on normalized-state dispersion and on the declared route clock. No universal time exponent follows from the product alone (Vehtari, Gelman, & Gabry, 2017).

If Ui(t)=cU[dirc(t)]β{1+o(1)} and the conditional normalized-state variance scales as cY[dirc(t)]γ{1+o(1)}, then:

$$Var\left(L_i^{rc}(t)\middle|U_i(t),\Xi_i^{\circ}(t)\right)=c_{LU}^2c_U^2c_Y\left[d_i^{rc}(t)\right]^{2\beta+\gamma}\{1+o(1)\}.$$

The route-space exponent is 2β+γ. A value of 2β requires γ=0 or an independently justified locally constant normalized-state dispersion. Neither β nor γ is universal; both are estimands or declared route assumptions.

For a normalized illustrative case with cLU=1, U(t)=U₀(1−t/T), Y(0)=0, and dY=σdW:

$$Var\left[L(t)\right]=U_0^2\sigma^2t\left(1-\frac{t}{T}\right)^2$$

This illustrative variance peaks at t=T/3 and then contracts. It demonstrates terminal-regime compression under the stated special case, not monotone whole-life smoothing or a universal exponent.

The exponent depends on its coordinate. A log-variance slope against reserve and a log-variance slope against remaining clock time coincide under the relevant linear-depletion conditions; nonlinear depletion, state dependence, shared disease drivers, and proxy error can change that relation. The analysis estimates the slope on its declared scale.

Variance shape can distinguish specified candidate and bridge models only where their predictions differ under qualified reserve and observation models. Test reserve dependence against time alignment, endpoint conditioning, and observation gain.

Apply the noise-floor analysis in Section 4.3.6 to the variance exponent. If subtraction leaves unstable or negative variance estimates, abstain from interpreting the corrected exponent.

4.4.4 — Terminal Concentration and Equivalence

Terminal concentration requires an independently fixed endpoint and a prospective equivalence band, with protection against smoothing, reporting loss, and clinical or social effects. Reference-class transport requires multisite replication against differences in culture, care, recruitment, and instrumentation. Even a visually favorable late curve may be manufactured by observation-gain collapse or endpoint conditioning; synthetic recovery must allow those rival explanations to win.

Use the equivalence design in Section 24.2, with the band, measurement-error model, power or precision target, and sensitivity analysis specified for this estimand.

The following is a population concentration claim about the measured terminal estimator, not an equivalence-test decision or direct sight of exact closure:

$$\Pr\left(-K \le \widehat{L}(\widehat{T}) \le K \mid \text{observation model, endpoint model}\right) \ge 1 - \alpha$$

This probability claim depends on K, α, the endpoint estimator, the observation model, and the reference class. Here α is the permitted tail probability. The analysis estimates that probability with quantified uncertainty. An equivalence test separately evaluates its declared estimand against a prespecified band under an appropriate decision rule.

Compare terminal concentration against the endpoint, observation, clinical, and selection rivals in Section 4.4 under qualified measurement and endpoint rules.

4.4.5 — Rival Comparison and the Joint Test

A rival may prevail for one component while a candidate remains useful for another: adaptation can explain short-term recovery without reserve-linked variance, and attenuation can explain terminal calm without settling earlier dynamics.

4.4.6 — The Discriminating Conjunction

The conjunction should be prospectively specified as a gate structure, not reconstructed as a story after results. Which components are necessary, which are exploratory, which are decision-critical, and which are sensitivity checks should be stated before analysis. Otherwise any favorable subset can be narrated as the real test after the fact.

This is where the project's ambition becomes disciplined. It is willing to ask for a pattern that ordinary rivals should struggle to reproduce, but it is also willing to lose if those rivals reproduce it. The conjunction is therefore both the positive case and the defeat route.

The discriminating conjunction is the book's first positive empirical target. No single pattern is enough. The conjunction requires independent reserve coupling, prospective endpoint separation, observation-model robustness, model recovery, held-out prediction, variance-shape discipline, and rival comparison. It is a compound claim because the serious rivals are strong in different places (Vehtari, Gelman, & Gabry, 2017).

A useful conjunction might say: under a declared reference class and instrument version, preterminal dynamics show reserve-linked drift, innovation contraction, and variance-shape restrictions; the endpoint is prospectively adjudicated by non-ledger channels; observation attenuation and clinical confounds are modeled; generative rivals are recovered when true; and the

Law candidate outperforms rivals on held-out forecasts. That is the level at which support becomes scientifically interesting.

The conjunction is deliberately vulnerable. It can fail through measurement invalidity, reserve proxy error, endpoint circularity, observation collapse, rival dominance, nontransport, model-recovery failure, or ethical burden. A failed member is not repaired by success elsewhere. The status update names the failed prerequisite.

The conjunction also makes positive results narrower and more valuable. If the full set survives in one bounded population, the result supports that reference-class prediction. It does not support all conscious streams. It creates a reason to replicate, expand, and challenge, not a reason to announce validation. The next replication should preserve the successful conjunction rather than selecting only its strongest coordinate, because the joint restriction is what earned the result.

This standard prevents the book from becoming a pattern-recognition exercise. Humans are good at seeing significance in curves. Scientific claim status requires a rule declared in advance that says which patterns matter together, which rivals were allowed to win, and what the public statement will say if the conjunction fails. Predeclared failure of any indispensable coordinate should lower the verdict even when the remaining plots look persuasive.

A conjunctive signature package keeps attractive findings from hiding qualifications in prose. Requiring the major conditions together states the evidentiary price of support before public compression. This structure also makes later synthesis cleaner, because independent studies can compare the same multicomponent criterion rather than retrospectively chosen highlights.

4.5 — The First Protocol in Miniature

Late-life observation supplies a bounded example. Synthetic recovery checks test the estimand. Those checks precede enrollment. A prespecified protocol follows (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026; Lakens, 2017; Nosek et al., 2018).

4.5.1 — Population and Enrollment

A first terminal-dynamics study might enroll adults in a palliative reference class only where observation avoids crisis recruitment and care teams can refuse or pause research without penalty. Its inference is bounded to that care environment, endpoint process, and participation pathway (International Council for Harmonisation, 2019; International Council for Harmonisation, 2026; Lakens, 2017).

The reference class should also include the care system. A study in a well-staffed hospice with stable relationships, trained research staff, and strong privacy infrastructure is not the same as a study in an overstressed hospital or fragmented home-care environment. Site conditions affect participation, observation, and data quality.

Recruitment materials describe the observed period and the study's questions about measurement, dynamics, endpoints, and competing explanations. Any inference using pre-enrollment history is identified separately, together with the information that supports it.

Use stable clinical recruitment pathways, repeated opportunities to decline, and independent care-team permission. Chapter 3 specifies crisis exclusions and refusal protections.

Inclusion and exclusion should be defined through care compatibility and scientific interpretability. Device tolerability, communication needs, cognitive fluctuation, family

involvement, staff capacity, and clinical instability all shape the data. When a subgroup cannot participate without added risk or unmanageable workload, the design narrows or changes instead of forcing representation through pressure.

Enrollment also includes bystander handling. Family members, caregivers, clinicians, and roommates may appear in context, speech, timing, or observation records. They do not become research participants by proximity. Their data must be minimized, separated, consented where needed, or excluded.

4.5.2 — Data and Context

Collect the physiological channels used by the candidate estimator, active reports when appropriate, reserve proxies, medication timing, sleep and rest context, care events, consent-permitted family presence, device status, observation intensity, and endpoint marks (Lakens, 2017; Nosek et al., 2018).

Treatment records are essential because medication and care can mimic target signatures. Sedation, analgesia, oxygen changes, delirium treatment, infections, hydration shifts, and clinical escalation can change physiology and reportability. A terminal pattern that ignores treatment context is under-modeled.

4.5.3 — Primary Estimands

Primary estimands may cover reserve-linked drift, innovation variance, the time-to-endpoint variance exponent, observation gain, branch-specific forecast error, and measured terminal concentration within a prespecified band (Schuirmann, 1987; Wilson & Collins, 2019).

A compressed estimand vector for the first study is:

$$\eta = (\kappa_U, \gamma_{innov}, \beta_{var}, \delta_{end}, \Delta ELPD_{Law-rivals})$$

κU indexes reserve-linked drift; γinnov indexes reserve-linked innovation contraction; βvar indexes time-domain variance

scaling; δendpoint indexes terminal equivalence under the observation model; and ΔELPD compares held-out predictive performance against prospectively specified rivals. The joint decision evaluates their predicted relations and each required component.

The ordering matters. Preterminal dynamic estimands should be primary before terminal-level equivalence because they are less dependent on arbitrary origin and whole-life initial conditions. Terminal concentration may be secondary or high-level, depending on endpoint and measurement gates. The protocol should not let the most dramatic outcome become primary after seeing it.

An estimand can also remain unestimated. If the endpoint gate fails, terminal concentration is not estimated. If observation collapses, preterminal dynamics may become no inference. If model recovery fails, rival comparison may be withheld. These outcomes should be written into the protocol before the data arrive.

4.5.4 — Rival Set and Recovery

Recovery should include near-miss designs. The analysis should test cases where the Law candidate and a bridge rival produce superficially similar endpoint behavior but differ in preterminal variance, observation marks, or reserve relation.

Use the generative rival families registered in Section 4.4 and include explicit no-closure and mixed-mechanism fixtures. For each generator, freeze its parameter range, observation channel, missingness process, and output label before running model recovery (Gneiting & Raftery, 2007; Nosek et al., 2018; Wilson & Collins, 2019).

Evaluate recovery under realistic missingness, noise, endpoint uncertainty, and observation marks. The pipeline must

distinguish the Law candidate from each rival with sufficient calibrated accuracy to support the planned real-data comparison.

The recovery requirement is operational:

$$M_j \to O_\phi \to \{M_1, \ldots, M_J\} \to \hat{j} \; \textit{with calibrated uncertainty}$$

Recovery tests should include adversarial mixtures. Real participants may not be generated by a single pure model. Observation attenuation may combine with homeostasis. Medication may combine with bridge-like endpoint alignment. Reserve coupling may exist but be weaker than assumed. The pipeline should know when it is uncertain rather than forcing classification.

Include positive-terminal-drift and no-closure baselines so the candidate can lose to trajectories moving away from neutrality or showing no structured closure.

Rival success changes public language. If the bridge model wins, the result favors an endpoint-conditioning explanation over the defeated candidate route. If observation attenuation wins, the result pressures measurement, not the Law directly. If homeostasis wins, the reserve-coupled mechanism may narrow. The verdict should name the component that lost.

4.5.5 — Adverse and Unusable Branches

In the favorable synthetic branch, the endpoint is adjudicated without ledger variables, reserve proxies decline under a prospectively specified low-reserve rule, the fitted Law candidate recovers the joint drift-diffusion pattern, and held-out prediction beats the bridge and homeostatic baselines.

A successful run would support computational recovery and discrimination under the specified generator. Empirical component support requires a separate study with qualified measurement, endpoint, and comparison procedures (Lakens,

2017; National Academies of Sciences, Engineering, and Medicine, 2019).

In the adverse synthetic branch, terminal values cluster near zero only after device removal, sedation escalation, or endpoint alignment. The candidate fits the visible end, but the attenuation rival explains the missing excursions. Apply Section 24.2's distinction: failed prerequisites make the affected inference unusable; a valid comparison can favor the attenuation rival.

Adverse branches should be predeclared as decision paths. If device burden crosses a threshold, monitoring stops. If care teams report interference, the study pauses. If endpoint channels conflict, terminal inference abstains. If observation gain collapses below a usable level, variance claims are withheld. If synthetic recovery fails, model comparison is not reported as decisive.

Quarantine the unusable analysis, record the failed prerequisite, assess participant and privacy consequences, and correct any public claim based on it.

Plan withdrawal, deletion requests, data retirement, incident reporting, family communication, and staff support under Chapter 3’s procedures.

4.5.6 — Bounded Public Result

A first study might report preliminary support for reserve-linked dynamics in the declared reference class, or withhold terminal inference because endpoint adjudication or a required timing premise failed. Report the result using Section 26.8.2's release fields (Fischhoff & Davis, 2014; Nosek et al., 2018; Spiegelhalter, 2017).

Neighboring research provides useful methods, not confirmation of exact lifetime closure. Affective dynamics describes adaptation and regulation. Consciousness-state

research supplies boundary-state assessments, while psychometrics supplies reliability, calibration, invariance, and response-shift tests. Palliative studies demonstrate how comfort-first observation can be designed. Statistical comparison provides equivalence tests, sensitivity analysis, predictive validation, and model recovery. The Law-specific evidence remains the open task: qualified estimators and endpoints, prospective tests against strong rivals, replication, and bounded public verdicts.

4.6 — Verdicts, Stop Gates, and Open Problems

The verdict system below maps findings to the dependencies they test. Its inputs include the endpoint restrictions and rate-integral construction in Chapter 7, the causal information requirement in Section 18.7.2, the measurement-error tests in Section 24.6.4, and the reproducible observation benchmark in Section 25.5.4 (Gneiting & Raftery, 2007; Lakatos, 1978; Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

4.6.1 — Preliminary and Replicated Support

Preliminary component support means that one prospectively specified component survives measurement checks and the specified rivals within a bounded reference class. Name that component and class without claiming that the Law is validated.

Replicated component support means that independent sites reproduce the same bounded result under compatible instruments and endpoint rules. State the replication scope and remaining transport limits.

Rival dominance occurs when a protocol-fixed rival explains the target signatures with better held-out performance and fewer strained assumptions. Credit that rival and specify which Law claim loses standing.

For preliminary component support, name the specific component supported (Gneiting & Raftery, 2007; Lakatos, 1978; Mayo, 2018).

For transported confirmation, bridge changed instrument versions, apply compatible endpoint rules, declare reference-class differences, and repeat the rival comparison (Section 26.4).

Retain preliminary status until the required external replication, source verification, data audit, code review,

endpoint-adjudication review, and public-artifact review are complete.

4.6.2 — Inconclusive and Unusable Evidence

After ethics and permission are satisfied, test representation. A failed signed, temporally interpretable target blocks ledger calculation; separately valid phenomenology, feasibility, or qualitative findings remain available. A surviving representation proceeds to calibration, scale, uncertainty, and response-shift checks.

Calibration failure blocks the affected comparison and the claims that depend on it; separately qualified uses may remain valid. For each surviving inference, the tree then checks the required bearer and endpoint evidence. Unresolved stream identity or endpoint adjudication withholds the terminal claims that depend on it.

Evidence is inconclusive when adequate data do not distinguish the candidates within the protocol-fixed sensitivity bounds. Report the comparison as unresolved rather than supported.

Evidence is unusable for an affected claim when its required origin, endpoint, sign, observation, recovery, ethics, or privacy conditions fail. Withhold that inference; the same dataset cannot supply a narrative substitute for the failed test.

An unresolved comparison after valid prerequisites differs from unusable evidence after a failed prerequisite. Apply that distinction to the affected inference; an invalid terminal gate need not defeat separately qualified measurement or feasibility results (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

4.6.3 — Narrowing and Mechanism Retirement

Narrowing retains support only within a smaller population, endpoint class, or measurement version. Change the claim version visibly rather than presenting the earlier broader claim as unchanged.

Mechanism retirement withdraws a failed candidate realization while leaving the broader formal question open. Its failure does not establish a victory elsewhere.

Suppose a signature transports within one specified care setting but fails elsewhere. Retain both site results, label the smaller population as the successor reference class, and preregister a test of the proposed transport boundary. Its outcome determines whether the restriction predicts a difference or merely excludes an adverse case.

Apply Section 27.6's retirement procedure to a failed realization; other claims retain only independently earned support.

4.6.4 — Theory-Level Defeat

Abstention withholds a claim update when its prerequisites do not hold. It is a valid scientific result with an explicit reason, rather than an implicit favorable verdict.

A valid counterexample under the required conditions defeats the declared universal branch. Repeated bridge failure, uninformative replacements, and strong rival dominance can instead exhaust the empirical program. In either case, withdraw scientific law-language for the affected claim and identify the basis for withdrawal.

One gate-valid counterexample can reject a declared set-universal [SET-ALL] branch when its stated identification and inferential prerequisites hold. That logical result does not by itself refute a probability-one [P1] branch, which is defined

under a declared probability law and may permit exceptional trajectories of probability zero.

4.6.5 — Publication and Safety Stops

Redesign figures that mislead when cropped and narrow result sentences whose meaning depends on remote caveats. Hold a release until source verification, endpoint adjudication, model recovery, privacy review, and the adverse-event record are complete.

After release, changes to these prerequisites trigger Chapter 28's correction process (Gneiting & Raftery, 2007; Lakatos, 1978).

Device burden, distress, sleep disruption, family pressure, staff overload, clinical interference, privacy breach, security incident, or consent withdrawal can stop the protocol.

Data stops must distinguish deletion, retirement, retained derived results, restricted access, and model-training permission. Withdrawal does not always erase already aggregated scientific products, but future use must follow the consent and governance rule declared at enrollment. A model cannot inherit permission merely because a dataset existed.

4.6.6 — Open Theorem and Identification Register

The open register collects the bridges in Section 4.2.6, recording whether ledger-product identity is derived, assumed, or limited to a signature, and which preterminal features survive admissible rescaling.

The recovery entry should identify which generators or mixtures remain indistinguishable at the planned sample size, noise level, and observation coverage; Section 4.5.4 specifies the recovery procedure.

Use the evidence states in Section 24.2 for negative or null results, attaching any target-level consequence to its explicit entailment.

Part II — PTVC Formal Core

Before a cumulative claim can be tested, the thing being accumulated must be specified. A change in who owns an experience, where the scale begins, or when the domain ends can change the mathematical object itself. Part II fixes these choices in a form that later investigators can use. Its purpose is to make the formal consequences traceable: a reader should be able to locate the assumption responsible for a conclusion.

The first choice concerns the bearer. Experiences assigned to one stream are combined within that stream’s domain. Questions of continuation, separation, fusion, and interruption therefore enter before the terminal condition is evaluated. Chapter 5 develops the branches through which those questions are represented. The resulting specification determines which histories can be compared and what a proposed counterexample would have to concern, without depending on a favored physical mechanism.

Representation then determines which operations are meaningful. An additive account requires more than ordered reports of better and worse. It needs a defensible origin and a rule for combining contributions across experienced exposure. Alternative typed representations preserve different information and support different terminal functionals. Keeping their types explicit allows a proposed bridge between them to be examined as a mathematical claim, with its own assumptions and possible failures.

Time enters this representation as part of the quantity. A change of clock must carry the corresponding change in rate; a later start to observation leaves an earlier domain to account for. Chapter 6 connects these formal requirements to measurement. The question is which aspects of the desired quantity the observations determine, and how uncertainty in the remainder changes the comparisons an analysis can support. This

connection guides instrument development toward the information the theory actually requires.

Chapter 7 then asks what follows before a mechanism is selected. A partial history leaves a set of possible completions. Restrictions on remaining capacity or endpoint support can shrink that set, sometimes making a proposed closure impossible under the stated assumptions. Random timing creates a particularly demanding case because a process that reaches a boundary under one stopping rule need not do so under another. These consequences provide tests of whole classes of candidate routes.

The Part moves from definition through representation to necessity. This order gives later modeling a useful discipline: each mechanism must inhabit a specified domain and satisfy its inherited constraints. A successful construction can then be studied for the distinctive predictions it adds. An unsuccessful construction identifies where a route fails, rather than forcing a change in the meaning of the original object. The formal core thus becomes a shared working surface for measurement researchers, mathematical critics, and empirical teams.

Chapter 5 — PTVC: Hypothesis, Branches, and Scope

Two analysts agree on every valence report yet differ on where one stream begins and ends. A split rule can change its terminal integral without altering the observations. The equation alone cannot resolve that dispute. This chapter defines the objects and ownership rules on which formal closure depends.

Their disagreement concerns the carrier of the claim. The Law of Fairness proposes a terminal relation for a qualifying stream; an analyst must therefore decide what constitutes one instance before evaluating its residual. A continuing body, an inherited memory, and a continuous record can supply evidence without settling whether phenomenal ownership continues in the same way.

The problem becomes concrete when two candidate continuations both claim an earlier episode. Assigning the episode in full to both duplicates its contribution. Assigning it to neither leaves part of the history unaccounted for. The formal question is which continuation relation governs ownership, including what remains unresolved under that relation (Bayne, 2010; Bayne & Chalmers, 2003).

Incomplete observation creates a different difficulty. A study may see only the end of a history whose earlier contribution still belongs to the target. Refining the instrument improves what is known about the observed portion; it does not determine where the stream began or supply the missing portion by itself. Domain and evidence remain separate objects.

Boundary choices also determine what is accumulated. A terminal jump belongs to an endpoint-inclusive quantity but can lie outside a strict-pre-endpoint target. A gap in access can leave continuation uncertain. The formal branch must settle how such cases are represented before an apparently balanced total can be interpreted.

Once the carrier is specified, the measure determines which comparisons have meaning. Positive and negative totals, gross variation, overlap, and temporal order describe different features. Two histories can share a terminal residual while differing in the amount and arrangement of their contributions. The chapter keeps those distinctions available instead of asking the residual to recover them.

The quantifier adds another commitment. Almost-sure closure under a probability law, closure for every eligible history, and closure under every permitted intervention make different assertions. An exceptional path, a changed routing rule, or a different intervention family can therefore challenge one assertion without reaching another. A branch label records that logical scope.

Empirical uncertainty does not determine truth-level membership by convenience. A poorly measured qualifying stream remains within the declared target class, even when the study cannot adjudicate its residual. Scientific qualification concerns what the observations license; it must not become a rule for removing difficult members from a universal claim.

The definitions below turn these disagreements into mathematical questions. They identify the conditions under which representations coincide, the counterexamples that separate them, and the changes that require a new branch.

5.1 — Truth-Domain Streams, Ownership, and Maximality

The formal target belongs to a unified conscious stream over a declared domain. Defining that bearer requires distinguishing the stream from a body, biography, legal person, or public record and specifying how events enter its history. Ownership and maximality make the subsequent endpoint and accumulation statements precise without assuming that an available observation already identifies the stream.

5.1.1 — Qualifying Bearers and Unified Streams

The received bearer is a unified conscious stream. Body, biography, legal person, public record, and social identity cannot substitute for that research object.

The proposed bearer is the stream of unified felt experience. Body, legal person, biography, and administrative record are different objects: the ledger integrates what is lived rather than what is externally attributed (Bayne, 2010; Bayne & Chalmers, 2003).

5.1.2 — Ownership, Routing, and Nonduplication

The declared unity or ownership gate and exposure-density factor jointly determine which contributions enter the calculation. Encode unity or ownership at most once, and preserve the units when time is reparameterized.

Ownership is the rule that assigns each phenomenal event to one declared stream-history branch. The assignment is not inferred from where a datum is stored or which body produced it. It is a truth-level relation that the protocol represents fallibly. A sensor trace, testimony, or medical record may be evidence about ownership without becoming ownership itself (Bayne, 2010; Gazzaniga, 2000).

Two formally admissible encodings are especially important. On a restricted-exposure branch, the phenomenal measure is already limited to events owned by the stream. On a common-base branch, one frozen allocation or unity weight assigns exposure and signed components. The same ownership relation may not be multiplied into the exposure, valence rate, gate, and quadrature weights several times.

Nonduplication becomes decisive when candidates overlap. A component stream and a whole-system stream may both look plausible; two restored copies may inherit the same memories; a split history may support several continuations. The protocol must either give distinct ownership, apportion events under a declared rule, or retain unresolved typed slack. It may not let the same event close several ledgers.

Residual ownership cannot be hidden in one signed remainder. Unresolved positive variation, negative variation, exposure, atoms, and singular mass have different consequences. They remain separate coordinates with explicit bounds. When those coordinates are too wide for adjudication, the scientifically honest result is Model-ambiguous or Inconclusive evidence rather than a conveniently balanced total.

Routing also constrains maximality. A qualifying instance is maximal only relative to a prospectively fixed continuation relation that preserves event ownership. Maximality does not mean the spatially largest system, the longest biography, the most detailed archive, or the most socially recognized identity. Those substitutions change the carrier and therefore change the claim (Parfit, 1984; Schechtman, 1996).

Fission, fusion, copying, and restoration may redistribute a fixed token or history weight under a declared branch. The routing relation specifies those historical allocations and distinguishes them from the successor streams' future

phenomenal contributions. Informational continuity and future accumulation are modeled as separate quantities.

A useful audit case begins with one recorded episode that is accessible to two candidate continuations. If the allocation is chosen after terminal residuals are inspected, the result is selected. Freeze a complete, nonduplicating allocation, preserve an explicit unresolved set, or use a justified quotient before inspecting those residuals.

5.1.3 — Maximal Event Domains

On an absolutely continuous chart, the latent composite contribution rate is defined from realized density and one declared exposure convention. Pointwise rate notation must be withdrawn when unmatched atoms or singular mass prevent that representation.

The phenomenal domain is the complete set of stream-owned experiential events on which the signed measure is defined. It is not merely a calendar interval, a sequence of survey rows, or the portion of life that left a record. A complete-history hypothesis cannot substitute observation coverage for the truth domain without changing its quantifier (Folland, 1999; Kallenberg, 2002).

A maximal event domain must exhaust the declared stream under the chosen routing rule. It includes ordinary intervals, isolated events when the measure has atoms, and any singular component admitted by the representation. A density along clock time is one convenient branch, not the definition of the domain itself.

Domain selection can manufacture neutrality. Excluding hard-to-observe intervals, dropping positive or negative atoms, or ending the domain at the first convenient zero changes the object being summed. A valid protocol freezes inclusion rules without

access to terminal residual, endpoint surprise, or completion-model output.

Nonvacuity is equally important. A qualifying stream must have a nonempty declared phenomenal domain unless a zero-exposure class is separately defined as degenerate. Otherwise exact closure could be satisfied by an empty object that says nothing about lived experience.

A domain manifest lists the event carrier, sigma-field or generating algebra, exposure measure, ownership restriction, admissible atoms and singular components, entry and exhaustion rule, and unresolved mass.

5.1.4 — Unity Gating and Qualifying Intervals

The estimated path belongs to a downstream estimator family. It cannot be substituted for the latent ledger in the formal target.

A unity gate marks when and to what degree the bearer qualifies under the chosen stream criterion. Treat it as an inclusion or weighting variable, with uncertainty or abstention when qualification is unresolved (Bayne & Chalmers, 2003; Bayne, 2010).

Hard gating includes or excludes an interval. Sleep, anesthesia, delirium, fragmented access, covert awareness, and rebinding may require a family of thresholds or a soft gate expressing graded evidential support.

The gate must be identified independently of the terminal residual. If the same valence estimator both decides whether an interval counts and supplies the value being accumulated, unfavorable intervals can disappear by construction. Unity evidence and valence evidence may share data sources, but their algebraic roles and validation targets remain distinct.

Qualifying intervals need endpoint discipline. A gap in reportability is not automatically a pause in experience, and a

pause is not automatically the end of a stream. The gate family should therefore be estimated with rules for return, reversible loss of access, threshold hysteresis, and uncertainty around the first and last qualifying times.

A useful stress test compares two pipelines on the same synthetic history. One uses an independently calibrated unity gate; the other tunes the gate after inspecting terminal residuals. Similar apparent closure in the second pipeline is not corroboration. It is a demonstration of how selection can imitate the target.

Report paths indexed by admissible gate families. If the target cannot be identified or validly bounded, report No adequate estimate under Section 24.2.

5.1.5 — Split, Fusion, Pause, and Continuation

Probability-one closure requires a fixed reference class, routing branch, maximal instances, and event ownership (Parfit, 1984; Schechtman, 1996; Bayne, 2010).

The next display states the branch-specific probability-one target under a declared probability law and carrier event. It is a formal claim, not an observed frequency and not a substitute for routing decisions.

$$Q^b(FPTVC^b) = 1$$

A pause differs from an endpoint because continuation remains admissible. Sleep, reversible anesthesia, temporary loss of report, and interrupted access may preserve one stream under a declared relation, or they may remain ambiguous. Biological continuity alone does not settle the question, and observation shutdown cannot decide it by default.

Fission requires a rule for the parent instance and the beginning of successor ownership. Fusion requires

corresponding rules for predecessor instances and the fused continuation. Section 5.3.2 governs nontransferability between those branches.

Copying and restoration create a related trap. Informational sameness does not prove numerical identity of phenomenal ownership. A restored system may be a continuation, a new branch, or an unresolved candidate under the declared ontology. The result must remain conditional on that ontology rather than selected by whichever routing closes most cleanly.

Return after an apparent ending is a direct falsifier of a badly defined protocol endpoint. Resuscitation, renewed responsiveness, recovered access, or later evidence of continued experience can show that the study stopped too early. The estimate or adjudicated classification of the truth-level endpoint is revised only through the protocol-fixed adjudication rule, not through retrospective convenience.

Keep difficult routing cases as branch-indexed objects or explicit unresolved sets.

Classify unresolved routing under the evidence-state rules in Section 24.2, retaining any valid branch-specific estimates or bounds.

5.1.6 — Root Ownership and Scientific Use

The latent phenomenological ledger is a formal target. It is neither an observed total nor a public record.

The received truth-level endpoint concerns the stream itself. It is not defined by a ledger crossing, bodily death, or a convenient observation time.

An exactly zero terminal latent residual is the strong closure hypothesis. Its appearance in the target statement does not make it an empirical finding.

The reserve-coupled product realization is a candidate route, with obligations concerning its bridge, dimensions, endpoint coincidence, and retirement. It is not the primitive definition of the phenomenal ledger.

This chapter specifies the formal object that the rest of the book investigates. Representation, measurement, inference, candidate mechanisms, rival comparison, replication, and revision all refer to this stated target. The bearer, domain, endpoint, currency, and quantifiers remain explicit so that readers can follow each derivation and identify what a study tests (Folland, 1999; Kallenberg, 2002).

That boundary is a defense against semantic drift, not a ban on scientific criticism. Research may show that a branch is too coarse, a bridge is unidentifiable, or an endpoint class is empirically inadequate. Such findings can justify a formal change request, a successor object, a narrowed claim, or a withdrawn route. They cannot justify changing the target silently while preserving its public name.

A received-object concordance should accompany every operational descendant. For each symbol and branch, it should identify the truth-level meaning, the protocol approximation, the observation bridge, the estimand, the estimator if one exists, and the conditions that force abstention. The concordance makes it possible to locate a failure without absorbing every discrepancy into generic model misspecification.

Independent teams should be able to reconstruct the same target from the frozen packet. If one team treats biological death as the truth endpoint, another uses the last report, and another defines the endpoint as first hitting zero, they are not applying different methods to one hypothesis. They are studying different endpoint constructions, one of which may make closure tautological.

Operational descendants should remain traceable to the same formal target. Public explanations may simplify the notation, but technical derivations and empirical tests must restore the assumptions needed for the object actually under study.

Scientific creativity begins from this fixed point. Researchers may propose vector extensions, endpoint-inclusive branches, nonadditive functionals, revised routing, or new candidate mechanisms. Each proposal is welcome as a typed successor with its own identifiers, bridges, predictions, and failure conditions. Visible succession permits genuine discovery while preventing the current hypothesis from inheriting evidence earned by a different object. Successor proposals should also state which earlier counterexamples they resolve and which new counterexamples they expose themselves to.

The practical standard is simple: receive first, test honestly, and revise visibly. A stable target lets critics challenge assumptions, reference classes, or bridges directly instead of arguing against a moving collection of metaphors. It also lets a failed module teach the program exactly what failed without forcing either dogmatic preservation or opportunistic rescue (Cronbach & Meehl, 1955; Lakatos, 1978; National Academies of Sciences, Engineering, and Medicine, 2019). That continuity also lets later datasets remain comparable, because a revision can be evaluated against the same frozen target rather than retroactively redefining earlier evidence. That stability makes disagreement cumulative: later teams can argue about evidence against the same object instead of unknowingly testing different targets.

5.2 — Finite Signed Phenomenal Measure and Exposure

This section defines the primary signed measure first and introduces density or rate notation only on branches where absolute continuity justifies it. The distinction keeps measure-level claims separate from later physical-time representations.

5.2.1 — Signed Phenomenal Measure

Write $r_i(t)$ for the declared composite contribution rate $g\theta,i(t)q_i^{exp}(t)v_i^{real}(t)$. Physical time, unity-weighted exposure, subjective time, and protocol time are not interchangeable. Units and rates must transform with the clock, and ownership may enter only once. A polished cumulative curve cannot repair a missing exposure convention.

The display below is a chart identity on the absolutely continuous branch, not the primitive definition of the signed measure and not an empirical finding.

$$r_i(t) := g_{\theta,i}(t) q_i{}^{exp}(t) v_i{}^{real}(t),$$
$$L_i{}^{phen}(t) = L_i{}^{phen}(S_i^{\theta}-) + \int \left[S_i^{\theta}, t\right] r_i(s) ds$$

On the primary branch, the latent phenomenal object is a finite signed measure ν_i on the complete measurable domain (Ω_i, Σ_i) of one separately routed maximal qualifying stream. The measure assigns signed contribution to phenomenal event sets. It is not initially defined as a questionnaire score, a mood state, or a regularly sampled time series (Folland, 1999; Kallenberg, 2002).

Countable additivity is not licensed by arithmetic convenience. An empirical representation needs a premeasure or additive-conjoint gate: disjoint unions must preserve the relevant ordering and magnitude relations on a generating algebra, and extension to a signed measure must be coherent on the declared

domain. Failure at this gate favors an ordinal, vector, bipolar, or nonadditive branch rather than a manufactured measure (Luce & Tukey, 1964; Krantz et al., 1971; Folland, 1999).

The measure-theoretic formulation separates event ownership from clock choice. Ω_i carries the stream-owned phenomenal events; a later time chart pushes the measure onto an ordered axis. This makes it possible to distinguish a truth-level contribution from the particular cadence, window, or device used to observe it.

Density and physical-time rate notation are available only on the absolutely continuous subbranches specified in Section 5.2.3; atoms or singular components remain measure-level objects. Realized density, conditional predictor, proxy, and estimator remain notationally distinct.

The cumulative ledger $L_i(t)$ is a prefix evaluation of ν_i under the declared chart. It inherits its meaning from the measure and domain. It is not a separate metaphysical substance and not a public or private record anyone can inspect directly. Its value changes only through stream-owned phenomenal contribution under the chosen representation.

Finiteness means finite total variation over the complete domain. It rules out undefined subtraction of infinite positive and negative masses. The condition is mathematically substantive and empirically demanding; it cannot be presumed merely because every finite study produces finite numbers.

Exposure gauge freedom introduces another distinction. Different factorizations into density and exposure can represent the same signed measure. The terminal measure and its Jordan decomposition may remain invariant while duration, intensity, and normalized diagnostics change. A protocol must record the admissible gauge family rather than let terminal fit choose it.

5.2.2 — Positive and Negative Jordan Magnitudes

Every finite signed measure has a Jordan decomposition into mutually singular positive and negative measures.

Their total masses V_i^+ and V_i^- preserve the two directions of phenomenal contribution, while their sum is gross variation and their difference is the terminal signed residual (Folland, 1999).

$$\nu_i = \nu_i^+ - \nu_i^-, \nu_i^+ \perp \nu_i^-, V_i^+ := \nu_i^+(\Omega_i), V_i^- := \nu_i^-(\Omega_i)$$

Gross variation is essential because net zero can conceal arbitrarily large experience. A stream with $V_i^+=1$ and $V_i^-=1$ shares the same residual as a stream with $V_i^+=100$ and $V_i^-=100$, yet their total variation differs by two orders of magnitude. Exact closure would therefore never imply equal lives, mild lives, or adequate welfare.

Mutual singularity in the Jordan decomposition concerns the signed measure. Section 5.4 instead begins with primitive positive and negative components and explicitly preserves their overlap.

A qualified estimate of the net residual can inform the boundary equality without identifying both gross component magnitudes separately. To interpret the richer history, preserve component estimates and their joint uncertainty where they are identifiable. Precision of a difference can be genuine: with finite-second-moment estimation errors e^+ and e^-, let σ_+^2 and σ_-^2 denote their variances. Then

$$Var(e^+ - e^-) = \sigma_+^2 + \sigma_-^2 - 2\,Cov(e^+, e^-).$$

In synthetic error units, component variances of 1 and covariance 0.99 give residual-error variance 0.02. Unequal component biases still contribute $E[e^+]-E[e^-]$, and a mislocated origin remains a calibration problem. Report the net result, the separately identified gross structure, and the uncertainty relevant

to each; neither a small residual nor imprecise components alone determine the other quantities (Manski, 2003; Imbens & Manski, 2004).

A useful recovery fixture contrasts low and high gross magnitudes at the same true zero residual, plus a nonzero true residual observed as zero after an origin shift. The first two worlds should share the net result while differing in gross estimates when the observations identify that difference. The third tests origin calibration. When the supplied observations cannot distinguish the worlds, the correct output preserves that uncertainty rather than being required to recover information the design does not contain.

5.2.3 — Experienced Exposure and Temporal Rate

A signed measure becomes a temporal rate only after the program chooses an experienced-exposure measure and establishes absolute continuity on the relevant branch. The realized value density is a Radon-Nikodym derivative relative to exposure; multiplying it by the declared exposure density produces a physical-time contribution rate. Neither object is a mood level with time appended (Folland, 1999; Billingsley, 1999).

Physical time is one possible chart, but phenomenal exposure may depend on subjective duration, unity, or another declared clock. The science does not gain freedom to choose whichever chart closes best. The chart, exposure anchor, admissible gauge family, and transformation rule must be frozen before decisive terminal analysis.

$$[v_i^{real}] = V \cdot E^{-1}, \left[g_{\theta,i} q_i^{exp}\right] = E \cdot T^{-1}, [r_i] = V \cdot T^{-1}, [L_i^{phen}] = V$$

Apply the nonduplication rule in Section 5.1.2: ownership or unity enters the exposure architecture once. Any additional

weighting must represent a distinct scientific quantity rather than re-encode the same allocation.

The dimensional audit separates phenomenal value V, exposure E, and chart time T. The realized density carries V per E; the exposure density carries E per T; their product r_i carries V per T; integration returns V. Checking those relationships helps identify the conversions a proposed measurement bridge requires.

Irregular observation cadence is not the same as irregular phenomenal exposure. Experience can continue between prompts, while a device can sample densely during quiet periods and sparsely during crises. Interval contribution estimators must therefore separate the latent exposure process from the marked observation process (Stone et al., 2007; Shiffman et al., 2008).

Subjective time adds a further identification problem. Retrospective duration, prospective time judgment, memory density, and momentary temporal experience can diverge. A study may compare these channels, but it cannot silently choose one as the complete clock without a validation and transport argument.

For the gauge freedom defined in Section 5.2.1, reserve- or horizon-normalized diagnostics need separate anchors because they are not automatically invariant.

Link ledger estimates to a shared exposure register. It records the clock, unit, unity convention, ownership branch, cadence, interpolation rule, unobserved exposure, admissible gauge family, and uncertainty.

5.2.4 — Dimensions, Units, and Reparameterization

Time reparameterization is where notation can quietly manufacture invariance. Suppose a new clock u is used and physical time is written as $t=\tau(u)$, with τ increasing and

differentiable. The ledger does not keep the same contribution rate by magic. The time element changes, and the rate must carry the Jacobian (Folland, 1999; Kallenberg, 2002).

The correct change of variables is:

$$L_i^{phen}(\tau(u_2)) - L_i^{phen}(\tau(u_1)) = \int_{u_1}^{u_2} r_i(\tau(u))\tau'(u)du$$

If one wants a rate expressed in u-time, the transformed rate must satisfy:

$$r_{i,u}(u) = r_{i,t}(\tau(u))\tau'(u)$$

Endpoint-normalized time is useful precisely because it can align terminal windows, but that convenience creates a leakage risk. Mapping every history to a unit interval uses endpoint information to define the chart. The transformation, information timing, and Jacobian must therefore be declared before confirmatory analysis; otherwise a clean terminal pattern may be partly manufactured by the clock.

More generally, a noninjective time map can preserve the net signed residual while reducing gross variation inside time fibers. Equality of gross variation requires an injective chart, sign-pure fibers, or another proved no-cancellation condition. Protocols should register temporal-resolution sensitivity and retain the event-space branch when cancellation cannot be bounded (Folland, 1999).

5.2.5 — Finite Variation and the Process-Class Bridge

An ordinary pathwise integral of a sufficiently regular contribution rate has finite variation on finite intervals. That fact matters because the reserve-coupled realization later uses stochastic dynamics. A finite-variation process and a diffusion-bearing semimartingale can be related under a declared bridge,

but they are not automatically the same process (Folland, 1999; Protter, 2005).

The clean comparison is:

$$[L_i{}^{phen}]_t = 0$$

The brackets in the preceding expression denote quadratic variation.

This creates a bridge obligation rather than a disaster. One route is to treat the product realization as a reduced-form consequence model for terminal dynamics, not as a pointwise identity with the phenomenological integral. Another route is to enrich the formal ledger with stochastic integration. A third is to define an approximation relationship between the objects. Each route requires visible change control.

If the product contains a diffusion term, specify whether it models the same object, a projection, an estimator, or an auxiliary consequence.

5.2.6 — Temporal Additivity as a Component Claim

Temporal additivity states that the signed phenomenal measure assigned to a disjoint union equals the sum of its restrictions to the disjoint parts. In the primary branch, this is not merely notation. It is a representation claim about how experienced contributions compose across the complete strict-pre-endpoint domain, and it can fail even when a convenient numerical summary remains available (Stone et al., 2007; Shiffman et al., 2008).

A valid cut-point analysis must prospectively partition the same truth domain into prefix and suffix sets, assign any cut-time atom exactly once, and preserve the same representation, neutral origin, exposure clock, unit, ownership rule, routing, and endpoint convention on both sides. Changing any of those

objects across the cut turns an additivity test into a comparison of different ledgers.

On the absolutely continuous chart, the received additive branch yields the increment identity displayed below; the general measure branch uses the corresponding signed-measure increment, including any atoms or singular component. For a fixed terminal residual, prefix and suffix contributions sum to that residual whether or not it is zero.

$$L_i^{phen}(t_2) = L_i^{phen}(t_1) + \int_{t_1}^{t_2} r_i(s)ds$$

Across varying cuts, a suffix-versus-prefix slope of minus one is an additivity diagnostic shared by any conserved residual, including a nonzero residual. It supplies no independent vote for PTVC. The PTVC-bearing estimand is the independently adjudicated terminal residual and its fixed zero target, not the generic fact that prefix and suffix restrictions sum to a constant.

Observation-level non-tautology is essential. The measured suffix must come from post-cut observations or from a separately validated terminal-remainder bridge. It may not be defined by subtracting the measured prefix from the zero target under test. That subtraction would manufacture complementarity and erase the empirical burden.

A bounded laboratory study can test local refinement and complementarity. Complete-stream inference additionally requires justified prefix and suffix information, domain coverage, and an independently adjudicated endpoint. Report which of these objects the design addresses.

Sampling design must respect the temporal object. Irregular cadence, state-dependent prompting, missing intervals, high-arousal oversampling, and endpoint-related observation loss can distort both sides of the partition. Duration and exposure

weighting must be explicit, and uncertainty from unobserved intervals must travel into the residual rather than disappear through equal-weight scoring. Partition sensitivity can be checked by recomputing the same realized history under alternative admissible sampling grids rather than assuming cadence is innocuous.

Path dependence supplies possible rivals, but history-dependent rates can still be additive along a fixed realized stream. Anticipation, relief, memory, adaptation, saturation, and context may alter those rates; they demonstrate nonadditivity only when the declared disjoint-union rule fails. Stable failure should prompt review of the additive representation and, where justified, a declared nonadditive, generalized, or no-estimate branch rather than a more flexible scalar chosen after the outcome. Testing additivity should therefore vary the partition while holding the represented path fixed, separating failure of the accumulation rule from ordinary history dependence.

The additivity restriction also needs an identification route. Correlated measurement error, common smoothing, origin drift, or shared-method bias can manufacture a slope near minus one. External calibration, method-diverse replicate channels, valid instruments, sealed recovery studies, or truth-containing error bounds must be declared before the restriction is treated as evidence about the latent measure (Manski, 2003; Borsboom, 2005). Method-diverse replication is most informative when its errors are plausibly independent, because agreement produced by shared preprocessing offers weaker protection against artifact. Where channels share calibration data, report the covariance this creates between the two partition estimates.

Report the partition rule, atom treatment, observation source for each side, error structure, estimated additivity restriction, terminal-residual region, and admissible alternatives. Separate

additivity and terminal-zero decisions so a terminal failure does not discard still-qualified additivity evidence.

5.3 — Additive PTVC and Unequal Gross Variation

The formal simplicity is purchased by exact commitments. The bearer must be separately routed, the measure finite, the origin and sign fixed, the strict-pre-endpoint domain complete, and the branch genuinely additive. When any of those conditions fails, the correct response is not to keep the equation and soften its meaning. The branch must narrow, become unidentified, or yield to another representation.

5.3.1 — The Exact Additive Statement

$$Z_i^\star := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{phen}(T_i^\star -)$$

$$PTVC_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^\star = 0$$

Within the declared additive branch, the display equates equality of Jordan magnitudes, zero total signed mass, and zero latent terminal residual.

Additive PTVC constrains the terminal signed residual to exactly zero on the declared latent branch. A nonzero residual is incompatible with that target statement.

Additive PTVC requires finite and equal positive and negative Jordan magnitudes within a qualifying stream. Their common magnitude can nevertheless differ across streams.

Additive PTVC does not require equality of gross variation, duration, ordering, or peaks. These features can differ radically even when the terminal restriction holds.

Rights, welfare, opportunity, and biography are not equalized by the additive terminal restriction. Their inequalities require independent analysis.

The additive statement assigns one finite signed phenomenal measure to one qualifying stream over its complete strict-pre-endpoint domain. Its terminal residual is the measure of that whole domain. Adverse evidence requires the representation,

routing, domain, endpoint, and measured-to-latent conditions to hold and a nonempty identified terminal-residual set to exclude zero under the stated inferential guarantee (Folland, 1999; Cacioppo & Berntson, 1994).

Empirical work cannot observe the equality directly. It must declare a representation, observation kernel, missingness process, estimand, estimator, endpoint rule, and measured-to-latent bridge before a finite-band result can bear on the latent exact statement.

A critic may object that the target is too coarse because it discards order, agency, meaning, and distribution across time. That objection identifies a real limitation of the additive branch rather than a refutation of its internal statement. Those dimensions remain scientifically and ethically indispensable.

Report gross phenomenal variation alongside the residual, using the distinction and worked example in Section 5.2.2. The following notation keeps both quantities visible (Folland, 1999).

$$G_i := V_i^{+} + V_i^{-}, Z_i^{\star} := V_i^{+} - V_i^{-}$$

5.3.2 — Intrastream Nontransferability

The additive equality belongs inside one separately routed stream. No positive magnitude from one person, animal, generation, population, institution, or archive may be used to cancel negative magnitude in another (Manski, 2003; Imbens & Manski, 2004).

Formally, each stream receives its own measure space and terminal residual. A sum across streams is a different object requiring an independently justified aggregation rule, and it cannot retroactively satisfy PTVC for any constituent stream.

A population average near zero illustrates the danger. Half a cohort could have large positive residual estimates and half large

negative estimates, yielding a neutral mean while no individual stream is compatible with closure.

Aggregation across people can be legitimate in welfare science, public health, or institutional planning, but it answers a different question. A cohort mean, family narrative, or social welfare functional cannot substitute for the residual of one routed stream. If stream-specific ownership is unresolved, adjudication requires a branch-matched conclusion that holds across every admissible ownership map; otherwise intrastream PTVC remains unadjudicated.

When routing remains uncertain, the protocol should preserve an admissible set of ownership maps rather than force a single partition. Each map induces its own stream-specific residual. The set of residuals allowed by alternative admissible ownership maps is their branch-indexed union. Intersections require constraints justified to hold jointly, with any required simultaneous-coverage guarantee; prespecification alone does not justify intersecting residual sets from alternative maps. Pooling becomes permissible only after routing uncertainty has been resolved at the level the claim requires.

5.3.3 — Origin, Sign, and Positive Rescaling

Chapter 6 addresses the task of identifying a neutral origin (Folland, 1999; Cacioppo & Berntson, 1994).

Some theories deny that valence has an instrument-independent zero or interval scale. In that case exact additive PTVC may be unidentified, and a vector, ordinal, or generalized branch must be considered without pretending equivalence.

Under $F'(t)=a+bF(t)$, with $b>0$, positive rescaling preserves a zero residual when the neutral origin is transported consistently. An additive shift contributes an exposure-weighted term proportional to $aA_i(T)$ whenever total experienced exposure is

nonzero. Scale invariance therefore does not solve origin identification; the neutral origin must be independently declared and validated before accumulation.

5.3.4 — Past-Fixity and Non-Equivalent Readings

Past-fixity means that an experienced contribution, once it belongs to the declared history, does not change because later evidence, memory, relief, or interpretation changes. What can change is the estimate of that contribution. The distinction is ontic versus epistemic: the event is fixed as past, while knowledge about it may improve or deteriorate (Billingsley, 1999).

Apply Section 4.3.5's filtering/smoothing distinction: a smoother revises the estimated history under a later information set, not the experienced history.

Endpoint-revision rules are also specified prospectively. New evidence may show that the stream continued longer than first believed or that the observation endpoint was not the truth endpoint. The newly included interval belongs to the completed domain, but the revision does not alter contributions already assigned under the same representation. A changed representation requires a separate migration record.

Remembered utility and narrative appraisal are scientifically important but non-equivalent targets. A person may remember an episode differently, reinterpret its meaning, or report less distress after recovery while the earlier phenomenal contribution remains uncertain. These processes are rivals or additional objects, not automatic corrections to the ledger.

Apply past-fixity within each admissible ownership history. Section 5.3.2 governs how the resulting stream-specific residuals are combined and adjudicated.

Filtering and smoothing make the distinction concrete. A filter estimates a latent state from information available up to a time; a smoother may revise that estimate after later observations arrive. The smoother can alter the posterior history without rewriting the experienced history itself. Every report should identify which information set produced each estimate.

Later interventions can change future experience, future reporting, and future memory. They cannot retroactively make earlier suffering smaller, morally deserved, or justified. Under exact PTVC, the fixed past constrains the signed remainder regardless of the chosen mechanism. Changing a genuine accumulated balance changes that remainder requirement, but does not by itself determine both future gross totals or identify particular compensating events.

When a model version changes, old and new analyses should coexist. The record should distinguish changed evidence, changed representation, changed reference class, changed endpoint, and changed scientific claim. Without that matrix, a new scalarization or endpoint can inherit the apparent success of the object it replaced.

A terminal residual must be reported with its cancellation geometry. Positive and negative Jordan magnitudes, gross variation, covariance, origin sensitivity, and the residual belong in the same display. Two histories can have similar residuals while differing radically in exposure, intensity, and uncertainty; a single terminal number would hide that structure.

Routing uncertainty should remain explicit as a family of admissible ownership maps when the evidence does not settle one map. Past-fixity then applies within each admissible history, and the terminal result becomes an identified set over those histories. Choosing the route that produces the most favorable residual would be outcome-dependent reconstruction. Set-valued

routing can still yield a decisive result when every admissible map gives the same sign conclusion, while disagreement across maps should widen the verdict rather than disappear.

The public language follows the same discipline. Later evidence may change what researchers believe about an earlier interval, but it does not erase what was lived. Exact closure, if supported, would be a terminal relation within one stream, not a moral story in which later events compensate for earlier harm or justify withholding present care. Public explanations should therefore avoid converting fixed-past arithmetic into counterfactual claims about whether a different life would have been better or worse.

Past-fixity gives tests a stable historical object. Later corrections to the representation, endpoint, or weighting receive explicit successor labels and their own evaluation, preserving the evidence associated with the earlier version (Lakatos, 1978; Mayo, 2018). The version record should specify whether a revision changes only estimation of the fixed past or redefines the target itself, because those changes have different evidentiary consequences.

5.4 — Bipolar Mixed-Valence Branch and Overlap

Mixed experience can contain positive and negative components simultaneously. A single signed number can erase this coactivation before the components' relation has been established (Cacioppo & Berntson, 1994; Russell & Carroll, 1999).

5.4.1 — Why a Net Residual Can Hide Structure

The bipolar branch therefore begins with primitive positive and negative component measures β_i^+ and β_i^-. Their shared overlap ζ_i is preserved explicitly, while the net signed measure ν_i equals their difference. The branch does not assume that every theory of mixed experience is literally two-dimensional; it prevents subtraction from erasing coactivation before it has been studied.

The following displays define the bipolar component decomposition and its primitive gross magnitude. They are not additive PTVC in alternate notation. A bridge may connect the branches only under stated component, overlap, origin, unit, and exposure assumptions.

$$\nu_i = \beta_i{}^+ - \beta_i{}^-, \beta_i{}^+ = \nu_i{}^+ + \zeta_i, \beta_i{}^- = \nu_i{}^- + \zeta_i, \zeta_i \geq 0$$

$$G_i^{bip} := \beta_i{}^+(\Omega_i) + \beta_i{}^-(\Omega_i) = V_i{}^+ + V_i{}^- + 2\zeta_i(\Omega_i)$$

The net Jordan magnitudes V_i^+ and V_i^- describe the nonoverlapping positive and negative parts of ν_i. The overlap measure ζ_i records component mass that cancels in the net signed measure but remains part of primitive gross experience. Net residual, component totals, and overlap therefore answer different questions.

A net residual answers a particular boundary question; it does not fully describe the experience or necessarily contain all information needed to predict what follows. Under the common-

domain and common-scale conditions in Section 5.4.2, equality of the finite primitive component totals is equivalent to a zero net total even when overlap and order matter for other questions. Preserve component histories for those questions rather than asking one residual to recover information removed by subtraction (Cacioppo & Berntson, 1994; Russell, 1980).

A unidimensional rival can explain mixed reports as rapid alternation, appraisal, or report-generation structure. Test whether separately represented positive and negative components are recoverable and improve held-out prediction over those rivals in the stated context. Transport is a further claim. An unsuccessful comparison does not establish that either representation is adequate.

5.4.2 — Independent Signed Components

The bipolar branch treats positive and negative phenomenal contributions as separately represented components rather than automatic negatives of one another. In psychometric terminology this is a bivariate representation. It differs from a bipolar response scale, which places positive and negative responses on opposite ends of one axis, as discussed in Section 8.3.1. Separate components permit coactivation and asymmetric responses; they do not by themselves imply statistical independence. The empirical question is whether one component can change while the other remains stable, and whether preserving both improves representation and prediction (Cacioppo & Berntson, 1994; Russell & Carroll, 1999; Larsen et al., 2001).

Each component needs its own origin, unit, exposure rule, and observation bridge. For finite components on the same domain and a common scale, $\nu_i(\Omega_i) = 0$ is equivalent to

$\beta_i^+(\Omega_i) = \beta_i^-(\Omega_i)$; it does not identify their common magnitude, overlap, or temporal order.

Protocols should test component reliability, monotonicity, and cross-context transport before combining them. Separate elicitation can reveal ceiling effects, asymmetric noise, and differential missingness that a net scale conceals.

In chronic pain, a person may report persistent nociceptive burden alongside meaningful connection, mastery, or relief. Treating one component as the arithmetic negation of the other may misrepresent both.

Shared arousal, attention, language, and response style can correlate the component reports; model those common influences.

When component channels cannot distinguish bipolarity from one-axis or rapid-alternation rivals under the same observations, branch identity remains Model-ambiguous.

5.4.3 — Overlap, Coactivation, and Gross Exposure

Overlap records the simultaneous presence of positive and negative components. It matters because a stream with high coactivation can have the same component totals as a stream in which the components occur in separate eras (Manski, 2003; Imbens & Manski, 2004).

The fixed bipolar representation gives a quantitative resolution diagnostic. Suppress the stream index, let $G=\beta^+(\Omega)+\beta^-(\Omega)$, and let $\Pi=\{B_j\}$ be a finite measurable partition of the same owned domain. Define the overlap apparent at that resolution by

$$O_\Pi = \sum_j \min\left\{\beta^+(B_j),\, \beta^-(B_j)\right\} = \frac{G - \sum_j \left|\nu(B_j)\right|}{2}.$$

Because $\min\{a,b\}=(a+b-|a-b|)/2$, refinement can only increase the partition sum of absolute net masses. Thus, for every refinement Π' of Π,

$$0 \le \zeta(\Omega) \le O_{\Pi'} \le O_{\Pi}; \quad \inf_{\Pi} O_{\Pi} = \zeta(\Omega).$$

The infimum ranges over all finite measurable partitions. A Hahn sign partition of ν attains it: each cell then contains net mass of only one sign, leaving precisely the shared component measure ζ. A chosen sequence of clock-time grids need not attain that partition. Refinement here keeps the representation, domain, and component measures fixed; changing the instrument or rate is a different comparison.

In a two-event example, $\beta^{+}=(1,0)$ and $\beta^{-}=(0,1)$ give apparent overlap 1 in one coarse cell but 0 after separating the events. By contrast, $\beta^{+}=\beta^{-}=(1/2,1/2)$ give overlap 1 at both resolutions. Both examples have zero terminal residual and G=2. Their component timing, not their final sum, supplies the distinction.

For estimated cell masses, suppose joint bounds satisfy $|\hat{\beta}^{+}_{j}-\beta^{+}(B_j)|\le\varepsilon^{+}_{j}$ and $|\hat{\beta}^{-}_{j}-\beta^{-}(B_j)|\le\varepsilon^{-}_{j}$. The corresponding estimate obeys $|\hat{O}\Pi-O\Pi|\le\Sigma_j \max\{\varepsilon^{+}_{j},\varepsilon^{-}_{j}\}$. The estimate plus this error allowance therefore bounds true overlap from above on the joint-coverage event. Compare coactivation, rapid alternation, channel lag, and retrospective-report models on the same records. A large coarse-window value alone cannot distinguish them; qualified refinement can show which distinctions the measurements preserve.

5.4.4 — Temporal Order and Component Histories

Terminal component totals do not determine temporal order. Positive experience followed by burden, burden followed by relief, and repeated oscillation can end with the same totals while producing different adaptation, memory, action, and clinical consequences (Cacioppo & Berntson, 1994; Russell, 1980).

Observation schedules must be dense enough for the order-sensitive quantity they claim to estimate. Retrospective

summaries cannot recover every transition, and missingness may be informative when people disengage during the most intense states.

Peak-end, recency, and memory-consolidation accounts are powerful rivals for reported order effects. They may explain later judgments even when the underlying cumulative exposure is unchanged.

Order sensitivity should be estimated separately for contemporaneous reports, delayed recall, and retrospective evaluation. A pattern that appears only after memory consolidation belongs to the observation or memory model unless a bridge to phenomenal order is independently supported.

Temporal-order extensions may use kernels, path signatures, transition counts, or state-space dynamics, but adopting one creates a new branch with its own invariances, estimands, observation requirements, and counterexamples. The additive terminal residual should not be described as if it already contained sequence information.

5.4.5 — Branch Tests, Failure, and Non-Substitution

Branch selection must be prospective. The protocol should declare whether positive and negative components are primitive, whether overlap is allowed, how coactivation is measured, and which bridge would connect the bipolar object to an additive signed measure. Selecting the branch after seeing which one closes would make the result outcome-dependent.

A bridge theorem must state sufficient conditions, not merely visual resemblance. If the bipolar components integrate to the positive and negative Jordan magnitudes of an additive measure under a common origin, exposure rule, and domain, the relationship is testable. When those conditions fail, the branches

remain distinct even if their terminal summaries happen to agree.

Failure should be typed. Unstable sign indicates a representation problem; unresolved coactivation indicates an observation problem; nontransportable overlap suggests a reference-class boundary; incompatible component totals pressure the bipolar target; and a nonzero additive residual pressures the additive target. A generic mixed label would discard these distinctions.

An instrument that forces a single scalar can favor the additive formulation, while separately elicited positive and negative channels can favor a bipolar model. Compare the branches on common observations under Section 25.2's evaluation rules.

5.5 — Generalized Terminal Neutrality and Bridge Theorems

Generalized Terminal Neutrality permits a terminal target on a marked carrier and declared value space without privileging one scalar signed measure (Manski, 2003). That flexibility does not make every representation equivalent to additive PTVC. A bridge theorem must state the conditions under which the generalized object preserves the additive claim, while counterexamples identify where it does not. The empirical interpretation inherits those conditions rather than the apparent generality of the notation.

5.5.1 — Representation Families and Invariance

Mixed, vector, relational, distributional, or path-functional representations remain available when a unique scalarization is not justified.

The display below states a separate GTN target: a frozen terminal functional HGTN evaluates a declared marked carrier (Γ,s) and returns the distinguished neutral element 0GTN. Equivalence to additive PTVC requires a bridge theorem that preserves bearer, domain, origin, sign, exposure, and endpoint commitments.

$$H_{GTN}(\Gamma, s) = 0_{GTN}, (\Gamma, s) \in M_{GTN}, H_{GTN} : M_{GTN} \to V_{GTN}$$

Here $\mathfrak{M}$GTN is the declared class of marked carriers, $\mathscr{V}$GTN is the target value space, and 0GTN is a prospectively identified neutral element. The equation is meaningful only when the carrier, codomain, admissible transformations, and zero set are fixed independently of the observed endpoint.

A vector target may support a terminal region rather than a point. State what is observed, what remains latent, and which

set-valued or inequality conclusions the available information supports.

A generalized target is intelligible only relative to a family of admissible representations and transformations. The science must identify which changes are coordinate recodings, which preserve the experiment's information, and which change the claim (Krantz et al., 1971; Blackwell, 1953).

An admissible invariance group must be stated explicitly. Positive rescaling, component permutation, monotone transformation, and measure-preserving reparameterization preserve different structures and cannot be treated as one operation.

Check the substantive zero set across instruments, cultures, developmental stages, care settings, and temporal resolutions; invariant fit statistics alone do not establish its invariance.

Two laboratories may use different component bases for the same latent vector. A bridge is possible only if the terminal functional transforms consistently across those bases and does not depend on one laboratory's arbitrary coding.

Overly broad invariance can empty the target by making all states equivalent; overly narrow invariance can turn harmless unit changes into different hypotheses. Both errors must be tested through counterexamples.

The proposed transformation class should also be closed under composition and inversion where those operations are claimed to preserve meaning. Without that algebraic coherence, two individually permitted recodings can combine into a recoding that changes the target.

5.5.2 — Bridge Theorems to Additive PTVC

A bridge theorem states sufficient and, where possible, necessary conditions under which a generalized terminal

functional is equivalent to the additive signed-measure target (Popper, 1959; Mayo, 2018).

A valid bridge must map bearer, complete domain, positive and negative structure, origin, scale, and endpoint across representations. It must show that zero under one branch entails the appropriate zero or equality under the other without hidden outcome selection.

For example, a norm of a vector terminal residual equals zero only when every component is zero. That fact can bridge to componentwise neutrality, but it does not by itself imply equality of integrated positive and negative Jordan masses unless the representation connects them.

A bridge theorem should state which assumptions do the work and, where possible, illustrate their necessity with separating counterexamples outside those assumptions. Such examples may show a generalized functional vanishing while the additive residual remains nonzero, or an additive zero failing the generalized target. Within the theorem's full premises, the proved implication or equivalence must hold.

Premise testing should be operational rather than ceremonial. If the theorem requires a common origin, additive exposure, exhaustive routing, or endpoint regularity, each premise receives an observable diagnostic, a measurement-error model, and a failure threshold. When a premise is unidentified, the bridge is unlicensed even if both endpoint quantities happen to be near zero (Borsboom, 2005; Mayo, 2018).

5.5.3 — Counterexamples and Non-Tautology

Fix the carrier, neutral set, representation family, and endpoint rule independently of terminal observations. There must be admissible data that fail the target under that same functional;

otherwise the construction has no discriminating content (Mayo, 2018).

Simulation should include adversarial generators that place terminal states just inside and just outside the declared neutral set. The analysis must recover those differences under realistic observation error and missingness.

A learned embedding that defines neutrality as the center of its own terminal training sample will almost inevitably report centrality. That result may describe the embedding but cannot adjudicate a pre-existing terminal law.

Flexible representation learning, regularization, and latent-space rotation can all hide post hoc rescue. The strongest safeguard is to freeze the training objective, evaluation set, and failure thresholds before terminal outcomes are inspected.

A useful red-team benchmark gives the pipeline data generated from a world with no terminal constraint and asks whether model selection still manufactures one. A method that frequently discovers neutrality in those counterfactual worlds has failed before it reaches human data.

5.5.4 — Empirical Consequences and Status Discipline

Empirical consequences should be derived at the level of observables and identified sets. A generalized terminal condition may imply invariance across admissible representations, a bounded compatibility region, componentwise restrictions, or transported order relations. Each implication must state the observation and measured-to-latent bridges it requires.

Counterexamples are valuable for exposing bridge failures. Two representations may agree on ordinary episodes and diverge under mixed valence, extreme duration, singular contributions, or endpoint-inclusive events. Those cases should

remain a test suite unless their removal from the reference class is explained.

Recovery tests must distinguish the correct branch and competing bridges, including false attribution from neighboring-branch generators.

A successor functional that changes the zero set or invariance class is a new target. Apply Section 5.7.5's change packet to identify which earlier results remain comparable, require reanalysis, or become historical.

Generalized Terminal Neutrality is a separate hypothesis family, not a semantic shelter for additive PTVC. Its terminal functional, representation class, invariances, and endpoint domain must be declared before data are inspected. A generalized target becomes scientifically useful only when it generates consequences that can fail under the same source and change-control discipline.

Bridge theorems determine when a generalized result bears on the additive branch. The theorem must name the representation family, origin, sign structure, scalarization, domain, endpoint, and regularity conditions that make the two targets equivalent or one imply the other. Without that bridge, similarity of terminal language is not equivalence.

The strongest protection is to preregister the representation family. Adding a new functional only after the additive residual fails would make the target indefinitely elastic. A successor may still merit study, but it requires a new identifier, reference class, and held-out tests not used to invent it.

Partial identification is often proper. If several admissible representations yield different terminal values, the output is a set of compatible values or verdicts. Exact neutrality is licensed only when every admissible member satisfies the declared target or when additional evidence narrows the set by a justified

bridge. Set width is itself informative, because it reveals whether ambiguity is a narrow numerical tolerance or a disagreement among qualitatively different representations.

Status language identifies the outcome precisely: generalized-compatible, additive-compatible, bridge-supported, bridge-failed, representation-ambiguous, or No adequate estimate. Each label describes the scientific analysis and the next work it supports. A status transition should name the bridge that changed so neighboring branches do not gain or lose standing by association.

Simulation and theorem work make the formal program testable. Simulations assess whether the pipeline recovers the correct branch and distinguishes competing bridges. Derivations establish consistency and consequences; empirical studies evaluate those consequences through their specified measurement, endpoint, observation, and model-comparison procedures. Recovery benchmarks should include data generated from neighboring branches, so a generalized method is tested for false attribution as well as successful detection.

The durable contribution is a disciplined ladder of claims. If the additive branch fails, the program may still learn that a vector, ordered, bipolar, or generalized object better represents the data. That is progress only when the new object is more constrained and more testable, not merely more accommodating (Lakatos, 1978; Mayo, 2018). Greater generality earns value only when it creates additional exclusions or stable bridges that empirical work can actually challenge.

5.6 — Reference Classes, Modal Scope, and Past-Fixity

A formal equation does not determine how widely it is claimed to hold. Reference class and modal scope are separate scientific objects. A theory may concern one species, one class of maximal qualifying conscious streams, one endpoint type, one observation regime, or a far broader domain, and those possibilities cannot be collapsed into the word universal (Billingsley, 1999; Kallenberg, 2002).

5.6.1 — Reference Classes Before Universality

The same separation applies to probability. Probability-one closure under a declared branch law, support containment under a declared topology, high finite-cohort frequency, closure under every policy, and closure under every intervention are different propositions. Evidence for one does not automatically raise the status of the others.

$$Q^b(FPTVC^b) = 1 \not\Rightarrow supp(Q^b) \subseteq FPTVC^b$$

$$supp(Q^b) \subseteq FPTVC^b \not\Rightarrow \forall \pi, a : Z_i^\star(\pi, a) = 0$$

These are schematic non-implications, not claims that no stronger theorem is possible. In regular topological settings, probability-one membership in a closed target set may imply support containment. All-policy and all-intervention assertions still require their own counterfactual domains and assumptions; vocabulary cannot supply them.

Reference-class closure asserts the target within a declared class. It does not establish transport beyond that class.

Probability-one closure is an almost-sure assertion under a declared law. It does not establish closure for every possible path or intervention.

Support containment requires all supported residuals to satisfy the target. It does not establish that the observation or representation model is correct.

All-policy or all-intervention closure asserts counterfactual invariance under specified actions. It supplies neither ethical permission to take those actions nor a universal mechanism.

Formal universality quantifies over every bearer satisfying a frozen eligibility predicate, not over whoever happened to enter a study. A valid counterexample defeats a set-universal branch only when the case satisfies that predicate and the representation and endpoint prerequisites; a transport failure in a new class has a different scope (Billingsley, 1999; Kallenberg, 2002).

A reference class can be defined by species, developmental status, clinical condition, stream architecture, endpoint hazard, or measurement feasibility. The defining variables must not be chosen after observing favorable terminal behavior.

A narrow reference class can produce real knowledge, but it can also hide selection. Reports must identify exclusions, their scientific rationale, and whether they remove the trajectories most capable of testing the claim.

External validity is a separate question. A failure outside the declared class does not refute the class-specific result, but it blocks universal language and identifies the next transport problem. The record should state which representation, origin, endpoint, and observation assumptions would have to survive before extension.

5.6.2 — Probability One and Support

Probability-one closure is relative to a declared probability law and may tolerate exceptional trajectories; support containment additionally depends on topology, closedness, and regularity; set-universal closure quantifies over every eligible history in the

declared class. These propositions are distinct, and none may be inferred from finite-cohort frequency or a confidence interval around an average (Pearl, 2009; Hernán & Robins, 2020; Frangakis & Rubin, 2002).

Inference must identify the probability law, sampling scheme, dependence structure, censoring, and observation process.

A continuous model may assign probability zero to each exact path while its support contains many paths incompatible with closure. Saying that each path is individually improbable does not rescue the target.

Rivals may agree on terminal concentration while differing in support, mechanism, or counterfactual behavior. Compare them with proper scoring on held-out data where they imply different observable predictions. If they induce the same observed distribution, predictive scores cannot identify the remaining differences.

Reports should identify the exceptional set allowed by an almost-sure claim and the topology used to define support. Concealing either choice makes a probabilistic statement sound stronger than the model actually warrants.

5.6.3 — All-Policy and All-Intervention Distinctions

A claim that closure holds under the realized policy differs from a claim that it holds under every possible policy. An all-intervention statement can be stronger when its declared intervention family includes the policy family and additional interventions that may alter stream unity, routing, endpoint, or representation (Lakatos, 1978; Mayo, 2018).

Counterfactual scope requires a structural model that defines how actions change trajectories and whether the same bearer and endpoint remain meaningful. Without that model, universal intervention language is undefined.

Experiments can test selected interventions, but they cannot license every-policy closure by induction. Each intervention also carries ethical constraints independent of scientific value.

Analgesia, sedation, psychotherapy, stimulation, or environmental change may alter experience, observation, memory, and endpoint hazard simultaneously. A favorable terminal estimate under one treatment does not show policy invariance.

Control-theoretic and time-symmetric routes may imply invariance across policies or interventions that ordinary adaptation models do not. Those stronger quantifiers require direct prospective contrasts.

5.6.4 — Compositional Scope and Stream Boundaries

Compositional closure asks whether closure properties of parts imply closure of a whole or vice versa. For conscious streams, composition is inseparable from the rules for split, fusion, pause, branching, and ownership (Billingsley, 1999; Kallenberg, 2002).

A parent stream ending at a split and child streams beginning thereafter is not mathematically equivalent to one ledger continuing through all descendants. The chosen routing rule changes the domain of every terminal statement.

Data systems must preserve those boundaries instead of merging records because one body, account, family, or device persists. Biological continuity and informational continuity do not settle phenomenal ownership.

A hypothetical fusion case shows the difficulty. Two pre-fusion streams may each have histories, while a post-fusion stream may inherit memories from both. Cross-history information does not automatically create cross-stream ledger transfer.

Different unity theories imply different compositional carriers. The choice requires an independent mind-source bridge, not selection by terminal fit (Section 12.1).

5.6.5 — Versioning Reference Classes

Record target eligibility and observation coverage separately. Narrowing changes the population to which a claim applies; loss to follow-up changes what is observed about that population (Lakatos, 1978; Mayo, 2018).

Each reference class is versioned as an inclusion graph recording bearer, routing, domain, endpoint, representation, observation requirements, exclusions, and failure modes. Parent-child links state what changed and why.

Record the modal form from Section 5.6 with its own assumptions and inference criteria.

Representation updates receive a migration matrix. Earlier experiences remain fixed, while estimates may change under a new origin, scalarization, or observation model; the matrix separates changed knowledge from a changed object and marks which verdicts still compare.

5.7 — Endpoint Domain and Change Control

The formal target depends on the initial boundary and strict-pre-endpoint domain. The following decomposition separates the earlier contribution from the terminal interval (Andersen et al., 1993; Aalen et al., 2008).

5.7.1 — Pre-Endpoint and Inclusive Domains

$$\mathrm{L}_i^{phen}\left(\mathrm{T}_i^{\star}\text{-}\right) = \mathrm{L}_i^{phen}\left(\mathrm{t}_0\text{-}\right) + \nu_i\left(\left[\mathrm{t}_0, \mathrm{T}_i^{\star}\right)\right)$$

The decomposition keeps delayed observation and truth-domain closure separate. The estimated or protocol endpoint may help study the truth endpoint, but it does not become that endpoint through notation.

The primary branch uses the complete strict-pre-endpoint domain and excludes any value assigned at the endpoint itself unless a distinct endpoint-inclusive branch is declared. The two domains can produce different terminal residuals, so any endpoint atom must be assigned prospectively rather than moved across the boundary after inspection (Andersen et al., 1993; Aalen et al., 2008).

The choice matters when the process may jump, terminate with a final experiential atom, or continue through a boundary that the protocol treats as terminal. Left limits and endpoint atoms are not interchangeable.

Protocols must specify how the last interval is handled, whether terminal uncertainty creates censoring, and whether observations after the protocol endpoint can reveal continued experience.

Observation shutdown can manufacture a clean-looking left limit. Devices may be removed, speech may end, sedation may deepen, or sampling may stop precisely when endpoint classification becomes hardest. If the terminal contribution

cannot be bounded across that gap, the correct verdict for the strict-pre-endpoint target is No adequate estimate; a narrower observation-window or endpoint-inclusive result remains available only if it is separately defined and supported.

5.7.2 — Truth, Protocol, and Estimated Endpoints

The truth endpoint is the end of the qualifying stream under the declared bearer and unity rule. The protocol endpoint is the event at which a study stops or classifies. The estimated endpoint is a statistical object inferred from available data (Billingsley, 1999; Kallenberg, 2002; Greer et al., 2020).

The Law formulation used in this series also includes a biological-death admissibility condition. Scientifically, treat it as a separate boundary restriction in the Law formulation rather than by redefining biological death as the truth endpoint. Biological death may not occur before the declared closure condition has been satisfied for the relevant stream. The condition does not require biological death to occur when closure is reached. A qualifying conscious stream may permanently end after satisfying closure while biological processes continue without consciousness, including under artificial life support; such continued nonconscious biological persistence does not extend the completed stream. If the truth-level endpoint coincides with biological death, or if an endpoint-inclusive contribution is declared, any death-associated phenomenal contribution must be assigned prospectively to the appropriate branch. This biological-death admissibility condition is logically additional to the primary terminal-equality statement and therefore requires its own scope, bridge assumptions, and empirical tests.

Apply Section 4.3.4's constructional separation between endpoint channels and the tested ledger estimator, retaining any shared causes in the joint model.

A clinical protocol may stop at loss of responsiveness, while later evidence suggests residual awareness or a new stream state. Such evidence requires reassessing the endpoint and continuation rule; it does not by itself establish fraud or identify the truth endpoint the formal claim needs.

Rivals include endpoint-conditioned selection, censoring, first-hitting termination, continued stream hypotheses, and misclassified unity. Each can produce different terminal patterns from the same observed window.

With imperfect endpoint evidence, report a confusion matrix or posterior over endpoint classes and propagate that uncertainty into the terminal residual.

5.7.3 — Initial Conditions and Delayed Observation

Section 6.6 treats the unknown prefix; delayed entry also introduces selection.

Delayed enrollment is usually selective. Diagnosis, communication capacity, care setting, social support, referral pathways, and survival to enrollment jointly shape who becomes observable. Sensitivity analysis should vary both the missing prefix and the entry mechanism, and the reported reference class should follow the prospectively specified selection model or bounds rather than the broad population label.

5.7.4 — Simplification and Counterexamples

Several attractive simplifications recur because they make terminal neutrality easy to display. Conservative state functions, adapting baselines, derivative transformations, endpoint centering, and vanishing-factor products can all create zero-like

outcomes by construction (Andersen et al., 1993; Aalen et al., 2008).

Model recovery can expose these failures by generating data under known nonclosing processes and checking whether the analysis manufactures support for closure. A pipeline that exceeds its predeclared false-positive tolerance on those controls requires redesign or retirement; valid compatibility under limited resolution remains inconclusive.

A Brownian bridge conditioned to end at zero and an unconditioned process with a genuine preterminal controller may look similar near the endpoint. Without prospective endpoint separation, the simplified interpretation is not identified.

A simplification is scientifically useful when it carries an error bound, an approximation regime, and a counterexample outside that regime. Without those limits, a convenient reduced model can silently replace the object it was meant only to approximate.

5.7.5 — Formal Change Control and Withdrawal

Vector extensions, endpoint-inclusive branches, nonadditive functionals, revised routing, and new mechanisms are typed successors. Name their identifiers, bridges, predictions, and failure conditions, including earlier counterexamples resolved and new ones permitted. Under Section 28.2, classify each dependent result as unchanged, reinterpreted, requiring reanalysis, or withdrawn.

Archive retired formal objects with their tests, failures, code, and replacement reasons under Section 28.2's stable-identifier and tombstone rules.

Withdraw only the object the evidence defeats: a theorem, route, reference class, or universal branch. A narrower successor

retains prior evidence only where its supporting assumptions and inferential bridge still apply.

Use Chapter 28’s dependency procedure to propagate formal changes to affected protocols, tables, diagrams, summaries, and public artifacts. Record which artifacts were reviewed and found unaffected so a deliberate nonchange can be distinguished from a missed dependency.

Close the revision through the review process in Section 28.4. For consequential repairs, a reviewer who did not design the change should test whether it resolves the defect without silently changing the target, assumptions, or evidentiary threshold. Keep material unresolved objections attached to the version (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

Chapter 6 — Representation and Measurement Interfaces

Consider a hypothetical coding exercise: two minutes receive a positive code of one and a one-minute episode receives a negative code of two in magnitude. The time-weighted codes cancel. Changing the negative code to three preserves its negative sign and its rank among states, yet changes the total. Arithmetic alone cannot choose the representation.

This is a question about what addition means. A scale can distinguish better from worse without identifying distances, and it can identify distances within positive and negative regions without linking their units. To interpret a cumulative result, the measurement system must earn the comparisons used by the calculation. A convenient numerical range does not supply them.

The origin presents a different problem. Moving the point called neutral changes which contributions count as positive and negative. Increasing the precision of repeated observations can sharpen an estimate around the wrong origin. The relevant evidence must constrain that reference point independently of the terminal answer the study hopes to evaluate.

Duration also needs a declared role. A state rating is not automatically a contribution rate, and the clock used to sample reports need not be the exposure measure the theory requires. The interval estimate must explain how the recorded state and duration jointly represent accumulated contribution.

Even a well-linked signed scale can discard structure. Relief and discomfort may vary separately, or two dimensions may remain incomparable under the available evidence. A vector or a family of admissible scalarizations can preserve those distinctions. The scientific question then becomes which conclusions survive that family, rather than which weight happens to produce a convenient total.

These choices precede the choice of estimator. A model can predict a questionnaire response accurately while leaving the

origin, cross-sign units, or exposure relation unresolved. Its success remains useful for the response it predicts; connecting that result to an accumulated phenomenal target requires the corresponding measurement relation.

The chapter follows those relations through concrete tests. Origin anchors constrain location; cross-sign comparisons test common units; channel restrictions challenge a nominated scalar model. Each test can fail separately. Their combination identifies where a representation is supported and where the available observations still permit alternatives.

Observation adds uncertainty to the representation. Different channels may respond to different features, and correlated errors can make agreement look stronger than it is. Retaining channel identity allows the analysis to determine whether a discrepancy concerns the target, the instrument, or the way a response was generated.

Incomplete history adds a boundary problem: a precise estimate of the observed increment can coexist with an uncertain earlier contribution. The result must retain both objects and their dependence.

The final adjudicability contract asks whether the planned observations can distinguish the scientific states at issue. That is the point at which a representation becomes a testable measurement proposal.

6.1 — Latent Objects, Routes, and Estimands

Measurement connects a defined target to the observations used to estimate it. The object map introduced above specifies the inferential work needed at each connection (Borsboom, 2005; Cronbach & Meehl, 1955).

6.1.1 — The Typed Layer Map

The latent ledger is the formal claim object under assumptions about bearer, rate, gate, and endpoint. It is not a sensor output, clinical score, measure of human worth, public identity, or directly observed truth.

A measured or estimated path is defined under an observation model. It can support qualified empirical statements about a reference class, instrument version, and window, but it cannot provide direct access to the latent ledger or a ranking of persons.

L(t) denotes the latent cumulative target under explicit assumptions about valence, unity, time, and endpoint (Borsboom, 2005; Cronbach & Meehl, 1955).

$\hat{L}$ (t) is different in kind. It is a data-derived estimate of a ledger-like path, produced by a measurement system and an analysis model. Its value depends on channels, transforms, sign conventions, filters, priors, calibration decisions, missingness assumptions, endpoint rules, and the reference class being studied.

$$L_i^{phen}(t) = L_i^{phen}(S_i^{\theta} -) + \int \left[S_i^{\theta}, t\right] r_i(s) ds$$

A clean measurement statement says: under instrument version v, transform ψ, cadence c, population P, endpoint rule E, and model family M, the data support an estimated distribution of measured cumulative contributions over the observed window.

The distinction also protects negative evidence. If $\hat{L}(t)$ fails, the program must ask what failed: the latent claim, the observation bridge, the instrument version, the scalarization, the endpoint adjudication, the missingness model, or the rival comparison. Treating the estimate as the latent object would make every failure both too large and too small at once.

6.1.2 — Momentary and Interval Estimators

A candidate momentary estimator or rate proxy supplies local evidence about a declared contribution rate within its context of use. It is not a universal feeling meter or a portable public score.

An interval-contribution estimate uses a declared convention for accumulating measured increments. Additivity requires justified rates, units, and cadence; the resulting quantity is not moral credit, debt, or exchangeable human value.

A candidate momentary contribution estimator attempts to estimate a physiological or physiological-plus-model signal that bears on the latent valence rate. A candidate interval contribution estimate is produced under an interval contribution convention that lets those momentary estimates enter a running measured cumulative estimate (Carroll et al., 2006; Hernán & Robins, 2020; Pearl, 2009).

To construct an interval contribution, express the momentary estimate as a validated rate or proportional rate proxy and integrate it over the specified duration. The conversion defines the units of the accumulated quantity.

$$\Delta\widehat{L}_{ij} := \int_{t_j}^{t_{j+1}} \widehat{r}_i(s)ds$$

6.1.3 — Operational Proxies and Context

Operational proxies can interpret, bound, challenge, or qualify the measurement bridge without themselves estimating interval

or cumulative contribution (Campbell & Fiske, 1959; Manski, 2003).

The distinction is practical. A unity measure may indicate whether the gate is plausibly open; a reserve proxy, whether a terminal model is in the relevant regime; and a motion artifact, why a physiological channel lost reliability. None alone estimates cumulative felt burden.

6.1.4 — Posterior Paths and Information Sets

A posterior path is a model-conditioned distribution over possible latent paths. It supports component claims and sensitivity analyses that retain uncertainty, rather than recovery of an inner life or a final verdict.

A posterior path is a distribution over trajectories conditional on data and assumptions. Strong priors, smoothing, shared parameters, endpoint conditioning, or an overconfident observation model can narrow it without improving access to the latent path (Borsboom, 2005; Cronbach & Meehl, 1955).

Given model M, instrument version v, data D, and specified priors or estimation rules, a posterior may favor some ledger-like paths. Another observation model may yield a wider distribution, shifted path, or no usable estimate from those data.

Apply the information-set distinction in Section 4.3.5: smoothed paths can aid sensitivity analysis and visualization, but later observations cannot establish prospective prediction of the terminal approach.

6.1.5 — Versioning and Provenance

Version instruments, firmware, sampling rates, channel availability, questionnaires, code, preprocessing, and calibration models so cumulative estimates can be compared across time,

sites, and studies (Carroll et al., 2006; Hernán & Robins, 2020; Pearl, 2009).

Versioning identifies the measurement and analysis procedure. If one estimator version uses heart-rate variability, sleep architecture, and endocrine timing, while another adds skin conductance and removes a motion-sensitive channel, the intended estimand may remain the same even though the estimator changes. A migration study must establish whether the versions estimate that target comparably; a changed target requires a separately stated estimand.

Governance must specify when a derived result is retained for reproducibility, superseded, or used in model training under separate permission.

Version comparison needs a declared migration protocol. It may use overlapping calibration samples, synthetic fixtures, bridge studies, or consent-permitted reanalysis of archived raw data, or find versions incomparable for a given verdict.

6.1.6 — Intervention and Scientific Verdict

Intervention stays on its own layer because an action may alter the latent process, route state, observation process, endpoint distribution, or several at once. The causal estimand names the manipulated object and its identification assumptions; otherwise reporting, medication, or observation changes can masquerade as ledger change (Pearl, 2009; Hernán & Robins, 2020).

The decision rule is distinct from the estimator. One interval can yield different decisions under different loss functions, equivalence regions, or abstention costs, so the rule must state how false support, false defeat, and forced estimation are treated.

6.2 — Representation and the Neutral Origin

Measurement begins by defining the representation that a signal is intended to inform. For an accumulated signed quantity, the representation specifies its origin, scale, and temporal composition before sensor calibration begins (Luce & Tukey, 1964; Krantz et al., 1971; Borsboom, 2005).

6.2.1 — Representation Before Measurement

A representation determines which phenomenal distinctions the model treats as scientifically meaningful. It may be scalar, bipolar, vector-valued, ordinal, distributional, or partially ordered, and each choice supports different operations (Krantz et al., 1971; Borsboom, 2005).

Integration is licensed only after sign, interval-comparable units, exposure, and clock are identified. A vector representation additionally requires either a declared scalarization whose weights are identified or a genuinely non-scalar target with explicit adjudication rules.

A device that predicts self-reported intensity may be useful while remaining agnostic about whether positive and negative valence share one axis. Predictive success on one channel does not settle the representation.

6.2.2 — Neutral Origin as Identification Problem

$$\tilde{v_i}(t) = a_i + b_i v_i(t), b_i > 0, \tilde{L_i}(T) = a_i A_i(T) + b_i L_i(T)$$

Positive rescaling preserves terminal zero when the origin term vanishes. Translation generally does not. The cumulative active exposure $A_i(T)$ makes the origin error grow with the domain over which experience is accumulated.

The neutral origin is the point separating positive from negative phenomenal contribution under the declared

representation. It cannot be assumed to coincide with a verbal midpoint, sample mean, adaptation level, or absence of overt distress (Cacioppo & Berntson, 1994; Russell, 1980; Cronbach & Meehl, 1955).

Because translation changes an accumulated residual, the origin is a structural parameter rather than a cosmetic convention. Exact closure is unidentified when multiple origins fit the same observations and imply different terminal signs.

A person may use the word neutral for emotional numbness, calm, boredom, low arousal, or inability to report. Those states need not occupy the same phenomenal origin.

Separate the phenomenal neutral origin from an adaptation level and from a report's reference point. A moving regulatory baseline can coexist with a fixed zero separating positive and negative contribution. In a local observation model $O(t)=aV(t)+b(t)+\varepsilon(t)$, with $a>0$, a changing report intercept $b(t)$ moves the observed location of $V=0$ without moving that latent origin. Compare joint state-and-observation models using independent anchors; better dynamic prediction alone does not identify which quantity changed.

Triangulation is informative only when disagreement is preserved. If indifference judgments, psychophysical thresholds, and within-person reversals identify different neutral regions, the protocol should carry that discordance into an identified set or No adequate estimate rather than average the anchors until the terminal residual closes.

6.2.3 — Affine Freedom and Positive Rescaling

The affine identity above distinguishes a stream-constant positive unit change from an origin shift. Psychometric interval-scale freedom must be narrowed by independent origin evidence

before exact closure is tested (Mokkink et al., 2024; Gagnier et al., 2025).

The zero-preserving transformation criterion fixes the admissible change of units. A fixed positive measurable pointwise multiplier chosen independently of the ledger preserves zero total for every zero-total finite signed measure for which the weighted total exists exactly when it is constant on the event space. To see why, place equal positive and negative unit masses at any two events. Their original total is zero; the weighted total is zero only when the two multipliers agree. Because the events are arbitrary, the multiplier is constant. Conversely, a constant multiplier factors out of every total and preserves zero. Event-, state-, or time-dependent reweighting therefore defines a new representation whose invariance must be established on its declared branch (Folland, 1999).

That distinction matters for attention weights, subjective-time weights, unity weights, and adaptive rescaling. A strictly positive reweighting preserves pointwise sign but may preserve the final net residual in one dataset while changing atomic or singular masses, gross variation, covariance, or route diagnostics elsewhere. Weights that can vanish may additionally remove contributions from sign regions. Protocols should register each weighting rule’s integrability domain and effects before terminal fitting.

Common-factor models constrain factor means or intercepts to identify their statistical coordinates. Terminal analyses should propagate admissible location uncertainty and evaluate the resulting estimates against independently established origin anchors.

Two laboratories can therefore report almost perfectly correlated trajectories and still disagree about the terminal

residual because one scale is shifted by a constant. Agreement in rank or covariance cannot repair a displaced origin.

Cross-instrument calibration tests whether linking transformations preserve the independently anchored origin in the declared population and context.

Before terminal fitting, the protocol should freeze the admissible origin family, its complexity, calibration sample, and leverage limits. Blocked cross-fitting can estimate that family without allowing terminal outcomes to select the zero, and the report should show terminal-origin leverage and sensitivity across every admissible anchor. If a saturated or outcome-adaptive origin class can manufacture closure, or if the verdict changes across admissible origins, the correct result is Origin-sensitive or No adequate estimate rather than selection of the favorable parameterization (Manski, 2003).

6.2.4 — Independent Origin Anchors

An origin anchor must be specified and calibrated without using the terminal residual being evaluated to select or tune it. This constructional separation prevents neutrality from being fitted into existence; it does not by itself imply statistical independence (Krantz et al., 1971; Borsboom, 2005).

Possible anchors include within-person indifference points, bidirectional tradeoff tasks, opponent-process symmetry, physiological transitions with convergent self-report, or experimentally manipulated reference conditions. Each anchor needs a context of use.

In a repeated-choice task, the point at which a participant is indifferent between a brief positive and negative perturbation may provide one origin constraint. It remains an experimental construct rather than a direct reading of the latent zero.

Demand characteristics, framing, loss aversion, memory, and strategic responding can shift apparent indifference. Multimethod convergence is therefore more informative than any single task. If the anchors still admit several origins, the result is an identified set or No adequate estimate, not a favored neutrality convention.

Independent sign anchors can place a concrete interval around a common origin. Suppose a candidate channel has the local relation $Y = aV + b + e$, with $a > 0$, and independently qualified negative and positive anchors have $V \leq 0$ and $V \geq 0$ respectively. If each anchor error satisfies $|e_k| \leq \delta_k$, every admitted origin b lies in

$$[\max_{i \in \mathcal{N}}(Y_i - \delta_i),\ \min_{j \in \mathcal{P}}(Y_j + \delta_j)].$$

Here $\mathcal{N}$ and $\mathcal{P}$ denote the negative and positive anchor sets, each nonempty. An empty interval challenges the common-origin, sign or error assumptions; a nonempty interval preserves the remaining origin uncertainty. Anchor signs must be justified as experienced valence, not inferred solely from preference, avoidance or the channel being calibrated. With probabilistic error bounds, coverage must hold jointly across the anchors. Holding out a context or anchor family then tests whether the same origin interval transports without using terminal outcomes to select it.

6.2.5 — Adaptation, Moving Baselines, and Abstention

Reports can adapt to context, expectation, illness, deprivation, repeated exposure, and local norms. A moving comparison level can make severe states sound ordinary without making them phenomenally neutral. The scientific task is to separate change in represented experience, change in adaptation state, and

change in reporting rule (Cacioppo & Berntson, 1994; Russell, 1980; Borsboom, 2005).

A dynamic-origin model specifies how its reference point changes and how that changing reference relates to the target origin. Recentring on a recent mean removes variation that an analysis of drift may need to preserve.

Digital measurement already offers an encouraging, testable starting point. In the thirty-day study by Cloos et al. (2025), 3,761 Norwegian adults registered for ecological momentary assessment and 2,718 completed the post-study survey. That survey showed high tactile precision and generally consistent ratings of affective scenarios. Among its respondents, 82.19% placed neutrality within five points of the unipolar midpoint of 50, and 88.89% within five points of the bipolar midpoint of zero. Some participants instead located neutrality near their own typical affect. These findings support the feasibility of precise reporting and identify a concrete calibration task: distinguish familiar feeling, scale interpretation, and phenomenal neutrality.

An observation-control design must specify which relation is intended to remain unchanged. If a manipulation alters reporting while leaving the represented experience unchanged, it may test the reporting intercept or gain; that causal exclusion requires its own justification. A shift in the reported neutral location is then not automatically a shift in the phenomenal origin. Test the joint state-and-observation model against independent anchors, and record whether a revision changes the estimate, the reporting coordinate, or the represented target.

Transport offers another test. Freeze an origin model estimated in one context and test it under different language, burden, affective range, and observation conditions. Failure may indicate construct drift, response shift, or a reference-class boundary; it

cannot justify silent recalibration until terminal zero returns (International Test Commission, 2018; Mokkink et al., 2024).

Report origin sensitivity beside the point estimate, including how anchor disagreement and dynamic-origin uncertainty change the terminal band.

The abstention rule should be protocol-fixed in the units of the identification problem before outcome inspection. It can depend on anchor disagreement, posterior width, invariance failure, transport loss, or sensitivity to admissible baseline models. Once the threshold is crossed, terminal-neutrality language stops for that context. The trigger should be tested on simulated and negative-control cases so its abstention behavior is known before it encounters the substantive endpoint data.

Archive competing origin models, accepted anchors, overlap regions, residual sensitivity, and reopening conditions (Manski, 2003; Gagnier et al., 2025).

6.3 — Sign, Cross-Sign Units, and Scalarity

Sign is not merely a plus or minus printed before a number. It is a claim that experiences on opposite sides of an origin are oriented within a common structure and that their magnitudes may be compared or accumulated under declared units. Such arithmetic requires empirical representation conditions, not just labels (Luce & Tukey, 1964; Krantz et al., 1971; Cacioppo & Berntson, 1994).

6.3.1 — Sign Identification

Instruments that order severity within pain or pleasure do not by that achievement establish commensurate units across the two signs.

$$x_i(t) \in R^d, F_{i,w}(t) = w^T x_i(t)$$

$$I_i(T) := \{ \int w^T x_i(t) dt : w \in W \}$$

When weights are not identified, the scientifically honest object may be the identified set $I_i(T)$ rather than one scalar ledger. Exact zero is then a property of all, some, or none of the admissible scalarizations, and those cases must be distinguished.

Sign identification determines which phenomenal changes count as positive, negative, or neutral under the target representation. Verbal labels alone are insufficient because language, appraisal, and response strategy can reverse or blur them (Krantz et al., 1971; Cacioppo & Berntson, 1994).

A sign rule should be anchored in directional preference, controlled tradeoffs, approach and avoidance under controls, and convergent channels while permitting mixed and uncertain states. The rule constrains sign without pretending direct access or forcing near-origin observations into binary labels.

Measurement models should propagate sign uncertainty. Near-origin states may require intervals or posterior probabilities over sign.

Motivational salience, arousal, and action tendency are imperfect proxies. Fear can motivate approach, pleasure can coexist with avoidance, and social display can oppose felt valence.

6.3.2 — Cross-Sign Comparability

Cross-sign comparability asks whether positive and negative magnitudes share units that support subtraction. Ordering states on each side separately does not answer that question (Russell, 1980; Borsboom, 2005; Manski, 2003). For a mixed-sign record with positive total $P > 0$ and negative magnitude $N > 0$, rescaling the two signs by separate positive factors a and b preserves every ordering and the neutral point, yet changes the total to $aP - bN$. Choosing a/b above, below or equal to N/P gives a positive, negative or zero result from the same ordinal observations. Independent cross-sign calibration is therefore what makes a zero total informative.

Commensurability may be investigated through bidirectional tradeoffs, duration exchange, intensity matching, or structural models that predict choices across signs. Each method introduces assumptions about memory, risk, and decision context.

Protocols should report whether comparability is global, local, person-specific, or only partially identified. A single conversion factor is not justified by convenience.

Choice asymmetries do not automatically reveal phenomenal cross-sign units. Prospect-theoretic weighting, loss aversion, and decision noise can produce similar behavior. Choice data can

inform the bridge only through a model that separates valuation, action, and experienced exposure.

Cross-sign comparability should be tested as a family of local bridge functions, not presumed to be one universal multiplier. A protocol can ask whether the positive-to-negative exchange rate is stable across intensity, duration, context, and person. Heterogeneity that survives calibration belongs in the identified set and limits any scalar terminal claim.

Local exchange maps also need loop consistency. If a sequence of positive-to-negative and negative-to-positive calibrations returns to a different unit when the comparison loop closes, the scalarization is path-dependent rather than globally coherent. A global scalar claim therefore requires identity holonomy, or a prospectively declared tolerance, across the relevant comparison graph; persistent loop discrepancy points to context dependence, an omitted state variable, or the need for local charts and an identified set (Krantz et al., 1971).

6.3.3 — Units, Rates, and Exposure

Apply Section 5.2.4's clock and rate transformation to the measurement units: integrating a rate per experienced-time unit produces accumulated phenomenal magnitude (International Test Commission, 2018; Mokkink et al., 2024).

Instruments should distinguish state level from rate contribution. Repeated samples may approximate an integral, but irregular spacing and missing intervals require explicit quadrature or state-space estimation.

Two studies can report the same average intensity while observing different durations. Treating the averages as equal accumulated burden would discard the very exposure PTVC concerns.

Repeat Section 5.2.4's dimensional audit after each transformation to identify changed estimands before fitting.

6.3.4 — Scalarity and Non-Scalar Alternatives

A scalar representation can support a total order; arithmetic accumulation additionally requires justified units and additivity. Experience may resist these requirements because states can differ along dimensions that are not reducible without value-laden weights (Krantz et al., 1971; Cacioppo & Berntson, 1994).

A nominated scalar measurement model can face an explicit test before accumulation. In one fixed context, suppose four channels satisfy $Y_j = \alpha_j + \beta_j V + e_j$, with finite second moments, fixed loadings, conditionally zero-mean errors and errors uncorrelated across channels conditional on V. Their off-diagonal covariances obey $\sigma_{ij} = \beta_i \beta_j Var(V)$, so four distinct indices must satisfy

$$\sigma_{ij}\sigma_{k\ell} - \sigma_{ik}\sigma_{j\ell} = 0.$$

Freeze the channels and error structure before testing this restriction on held-out records. Correlated method effects, nonlinear loadings or additional dimensions can break it. Failure challenges this particular linear observation model; a scalar phenomenal quantity may still admit a different observation relation. Success earns further sign, origin, cross-sign and additivity tests. Repeated channels are informative only when their error assumptions survive independent checks.

If scalarity fails, retain a supported vector path, partial order, or family of scalarizations and identify the additional comparison axiom needed for a scalar ledger.

6.3.5 — Claims Without Cross-Stream Comparison

An intrastream zero is invariant under its own positive unit rescaling. That invariance does not establish a common interpersonal scale (Krantz et al., 1971).

Population comparisons need not assume an interpersonal unit of phenomenal value. For the randomized-remainder comparison in Section 19.2, choose a finite positive reference s_i in each stream's own contribution units before assignment; hold it fixed for the prefix and complete remainder. Let H denote prespecified baseline strata and require Z to be independent of the pair (L_i,s_i) conditional on H. Exact closure then gives

$$\frac{R_i}{s_i} = -\frac{L_i}{s_i}.$$

Its distribution is therefore identical across assigned arms within H: the normalized negative prefix is a fixed function of the pre-assignment pair. Multiplying L_i, R_i, and s_i by the same positive stream-specific unit conversion leaves both ratios unchanged. The comparison concerns distributions of dimensionless within-stream quantities, not a pooled ledger or a ranking of lived value. Unconditional randomization permits omission of H; conditional randomization requires the common-stratum comparison described in Section 19.5.1.

Fix the reference rule before assignment. A synthetic counterexample has L=1 and R=-1 for everyone: using s=1 in one assigned arm and s=2 in the other creates normalized remain-ders -1 and -1/2 although every stream closes. Include uncertainty in the measurement-to-normalized-remainder relation in its interval or error budget, especially when the reference is near zero. The Wasserstein result requires finite first moments after normalization.

Transport across streams is a separate problem. Similar observation patterns can arise from different origins, response

functions, routing, or exposure clocks. A transported estimator must state the invariances and overlap conditions it uses rather than treating demographic or diagnostic similarity as sufficient.

Noncomparability restricts interpersonal operations; qualified within-stream measurements and route tests remain available.

6.4 — Mixed Valence and Identified Sets

Non-scalar alternatives include vectors, cones, partial orders, signed measures on feature spaces, and distributions. Section 6.4 compares representation options. The analysis distinguishes uncertainty about a fixed target from sensitivity to changing the target itself.

6.4.1 — Vector and Multidimensional States

A vector representation asks whether multiple dimensions carry irreducible information. It gains support when distinct dimensions predict auxiliary evidence and dynamics; unrecoverable dimensions or failed held-out transport require the proposed representation to be reconsidered.

A vector experience state represents the possibility that lived experience contains multiple dimensions relevant to burden and relief. Instead of assuming one latent rate F(t), the model may begin with z(t), a vector of affective, interoceptive, motivational, and contextual dimensions that might or might not collapse responsibly onto one signed axis (Cacioppo & Berntson, 1994; Russell, 1980).

Include a dimension only when it has a measurement route and a defined relationship to the latent target.

Models can predict the same signed projection while disagreeing about state geometry, covariance, and meaning-preserving transformations. Test dimension recovery across tasks, transport of their relations to held-out contexts, and comparison with rival dimensional systems. Additional coordinates that change no observable consequence or decision remain descriptive possibilities rather than identified structure.

6.4.2 — Scalarization Weight Families

A single signed scalar asks whether one additive path can represent contribution. Stable held-out prediction, valid sign interpretation, and adequate residual behavior support that use. Dominance by vector or ordinal rivals requires narrowing.

Scalarization weights define how a vector state becomes a signed rate. A weight vector w maps z(t) into $w^T z(t)$. That operation is not just arithmetic. It is a claim about which dimensions count, in what direction, with what relative force, and under which population and context (Krantz et al., 1971; Manski, 2003; Imbens & Manski, 2004).

Weights can come from theory, within-person calibration, expert prior, measurement model, predictive performance, or declared sensitivity analysis. They cannot be chosen because they make terminal results look favorable.

$$F_w(t) = w^T z(t)$$

$$L(T;W) := \{ \int_0^T g_\theta(t) w^T z(t) dt : w \in W \}$$

A transparent scalarization report should state the candidate dimensions, raw units, transformations, signs, weights, constraints on admissible weights, and why the family W was chosen before terminal analysis. It should also report which conclusions survive across W and which depend on a narrow region of the weight space.

Weights should be testable. Held-out prediction, residual structure, response shift, auxiliary witness conflict, and rival model comparison all place pressure on the chosen projection. A scientifically useful scalarization states in advance which observations would support it, require revision, or rule it out.

Hierarchical or partially pooled weights may improve transport, but they introduce assumptions about exchangeability

and group structure that must be tested. If only a narrow, outcome-selected region of weight space produces compatibility, report fragility rather than confirmation.

6.4.3 — Identified Sets and Partial Orders

A weight family tests whether conclusions survive plausible scalarizations. Agreement across the family strengthens a conclusion; changes in sign or verdict expose representation sensitivity and require narrower reporting.

A scalarization sensitivity module should be treated as a standard output, not a special appendix. The study declares a family W of admissible weights and computes the full range of measured cumulative estimates over W, or a certified outer bound on that range. A representative grid alone yields a sampled sensitivity analysis, not the full identified set, unless a justified interpolation or extremum bound covers the unsampled weights.

A result is weight-sensitive when some admissible scalar weights support near-neutrality and others do not, even though the temporal rule is unchanged. Report the appropriate identified set or scalarization-sensitivity set, distinguishing them as specified below, rather than a single neutrality conclusion.

This weight-sensitive case changes the scalar projection while holding the temporal rule fixed. Keeping those choices separate identifies which representational assumption controls the verdict.

When weights are uncertain, the study can report every ledger conclusion generated across a declared weight family, together with the evidence and assumptions that define that family. Use the term identified set when these are values of one fixed target compatible with the available evidence and declared model. When changing the weights changes the target itself, report a

scalarization-sensitivity set and distinguish that sensitivity from uncertainty about a fixed target (Manski, 2003; Imbens & Manski, 2004; Mokkink et al., 2024).

Identified sets can also strengthen a result. If every admissible weight family produces the same qualitative local signature, such as variance compression or rival defeat within the observed window, the component result becomes less representation-sensitive.

6.4.4 — Sensitivity to Weight and Kernel Choice

Simple duration weighting asks whether additive accumulation adequately represents the temporal contribution. Held-out superiority over recency and memory rivals strengthens that model; better performance by kernel models requires narrowing.

Compare duration weighting with a recency kernel, a memory-consolidation kernel that weights sleep intervals, and a peak-end model of reports. Specify the temporal rule for each fitted target.

A scalar path may proceed within the tested context when all plausible weights preserve the sign and near-band conclusion and the additive temporal model predicts held-out data best. That conclusion retains the tested weight family and context.

The received ledger integrates contributions over time without an additional memory kernel. Measurement must still test whether relevant empirical targets behave that way. A recency kernel, memory-consolidation kernel, or state-dependent temporal weighting can produce different path functionals from the same momentary estimates (Cacioppo & Berntson, 1994; Russell, 1980).

A participant’s later report, clinical expression, or decision behavior may depend more on recent, peak, unresolved, or narratively organized states than on simple duration-weighted

accumulation. If the study's observed outcome is a report or behavior, the temporal rule matters.

$$L_{w,K}(T) = \int_0^T K(T,s) g_\theta(s) w^T z(s) ds$$

A kernel-weighted model can explain how a reporting channel depends on temporal history. Comparison with the proposed accumulation model identifies which observable summaries the channel supports and which additional measurement relation is needed.

Temporal kernels must be specified before analysis. K(T,s) may depend on recency, sleep consolidation, arousal, memory salience, or state transition, but each dependence changes the measured object. Endpoint-normalized kernels chosen after seeing terminal data can manufacture convergence.

A strong study should compare simple integration with kernel alternatives using held-out prediction and rival discrimination. It should ask whether the additive path predicts future physiological and auxiliary patterns better than recency or memory-weighted rivals.

Distinguish uncertainty about a kernel from heterogeneity across people or contexts. The former calls for better identification, richer measurements, or wider uncertainty sets; the latter may require a restricted reference class or a heterogeneous model.

6.4.5 — Memory, Recency, and Duration Rivals

A result is kernel-sensitive when scalar weights are held fixed but a recency or peak-end model explains reports better than an additive path. That comparison tests the temporal organization of the report. Distinguish remembered evaluation from time-local contributions before using it to assess the additive experiential target.

A remembered evaluation may overweight intense or final moments while neglecting duration. A terminal report can therefore favor a peak-end rival even when it fails to track accumulated experience (Kahneman et al., 1993; Redelmeier & Kahneman, 1996).

Duration also matters. Under the same calibrated rate and exposure convention, pain of equal sustained intensity for ten seconds and ten hours contributes different accumulated amounts. Some reports compress duration, especially under memory distortion, trauma, sedation, or narrative reconstruction. The measurement system must identify whether the channel is duration-sensitive.

If peak-end rivals explain reports better, the study may still use physiological rate channels to estimate additive contributions. But it must not use those reports as if they validated the additive path without resolving the mismatch.

6.4.6 — Representation Failure as a Result

A vector representation is required for the tested analysis when no scalar projection preserves residual adequacy and multiple dimensions are needed. Suspend the scalar interval-contribution estimate for that reference class.

Failure may be local. A scalar can work for ordinary unimodal episodes yet fail under mixed valence, trauma, altered consciousness, or extreme duration; one instrument may fail while another remains usable. The least committal supported object then replaces the failed one: vector for scalar, identified set for point, partial for total order, or generalized functional for the additive branch. If the family remains unresolved, classify the affected target under Section 24.2.

Representation includes resolution. Coactivation at one scale can appear as alternation at another, and smoothing can erase

short excursions. Weights, kernels, summaries, and resolutions must therefore be justified independently, evaluated out of sample, and varied within prospectively named families rather than selected for a desired residual.

Sealed-world recovery should distinguish scalar, vector, set-valued, origin-failed, and observation-failed generators. A pipeline that always returns one scalar cannot recognize when that object is unwarranted. Its failure register must identify the smallest invalidated descendant, preserving ordinal, componentwise, or formal work that still survives.

6.5 — Observation Kernel and Protocol Measurement Map

The observation kernel is the probabilistic and procedural map from latent phenomenal states to the data a protocol records. It includes instruments, response processes, context, missingness, timing, calibration, and the transformations that turn raw signals into analysis variables (Bollen, 1989; Carroll et al., 2006).

6.5.1 — Momentary Contribution Estimators

$$Y_{ijk} = \mathscr{H}_{\eta,j}\left(X_i(t_k), C_{ijk}, R_{ijk}\right) + \varepsilon_{ijk}, \qquad \widehat{\theta}_i = \mathscr{A}\left(Y_i, M_i; \eta\right)$$

The observation kernel $\mathscr{H}\eta$ maps latent state, context, and response conditions into observations. The analysis operator $\mathscr{A}$ then produces an estimator under an explicit missingness model M_i. Here i indexes person or stream, j indexes channel, k indexes occasion, and Y_i denotes the collection $\{Y_{ijk}\}_{j,k}$.

The received candidate momentary estimator uses qualified physiological channels. Self-report, pairwise comparison, behavior, and contextual variables remain distinct auxiliary witnesses for calibration, challenge, and the modeling of observation conditions. A broader multimodal estimator may combine these sources with autonomic signals and neural features under a separately declared model, but it must not silently replace the received physiological candidate. Each channel retains its own error structure.

Calibration should use held-out tasks, repeated measures, and within-person anchors. The estimator's context of use must state populations, settings, time scales, and conditions under which abstention is required.

Proxy dominance is a central risk. A high-performing behavioral or neural model may predict reports while missing

unreported experience, response suppression, or culture-specific expression.

Local estimators should be recalibrated across contexts and versions, with drift monitored through anchor tasks and replicate measurements. A stable model coefficient is not evidence of stable measurement when the response process has changed.

6.5.2 — Auxiliary Witnesses Remain Distinct

Keep auxiliary witnesses distinct in the channel model (Campbell & Fiske, 1959; Mokkink et al., 2024; Gagnier et al., 2025).

A multi-channel model should preserve channel identity before fusion. Conditional dependence, shared artifacts, and common causes must be represented rather than treated as independent confirmations.

Protocols can compare channel-specific likelihoods, posterior predictive checks, and conflict patterns. Persistent conflict may narrow the estimand or trigger no-estimate rather than being hidden in a composite.

Conflict analysis should ask which channel fails under which conditions. A channel may lag, saturate, react to observation, or lose validity near an endpoint while another remains informative. Modeling those patterns explicitly can identify the failing measurement interface, whereas an omnibus composite can conceal it behind an apparently stable average.

Common-method bias can make questionnaires converge, while shared device placement can make sensors share one artifact. The adversarial plan should treat agreement that disappears when methods, hardware, or timing are diversified as support for a shared-method rival rather than counting the number of channels as independent evidence.

6.5.3 — Latent-to-Observed Channel Map

Model responses conditional on context, memory, attention, communication, and device conditions (Shiffman et al., 2008; Stone et al., 2007; International Test Commission, 2018).

A channel model may be parametric, nonparametric, state-space, or simulation-based, but it must expose identifiability assumptions. Different latent trajectories can produce the same observations when gain, noise, or missingness changes.

Use qualified negative controls and perturbation studies (Section 14.5) to test the channel map.

Channel validity should be evaluated through held-out devices, sites, states, and perturbations, with calibration as well as discrimination reported. A decoder that preserves rank but not scale cannot support the same cumulative estimand without an additional bridge.

6.5.4 — Interval Contribution and Cadence

Cumulative estimation requires interval contributions, not isolated snapshots. Cadence determines what transitions, bursts, and durations can be recovered from discrete observations (Bollen, 1989; Carroll et al., 2006).

An interval contribution estimator may integrate a filtered latent path, use quadrature between observations, or model continuous-time dynamics. Its uncertainty must include interpolation and unobserved intervals.

Sampling design should be tied to the fastest scientifically relevant dynamics and participant burden. Dense measurement is not automatically better when it changes experience, exhausts participants, or increases dropout.

Irregular sampling, informative prompting, and event-triggered recording can distort apparent duration and transition rates. The design mechanism belongs in the likelihood or

sensitivity analysis. When those designs remain observationally equivalent, cadence cannot be used to favor a route; the ambiguity belongs in the verdict.

Plot actual observation intervals and gaps; row order alone cannot represent irregular sampling or its exposure uncertainty.

Cadence is an identification parameter whose effects can be tested directly. Hong and O'Brien (2026) found that more frequent ratings amplified changes in evaluated experience. In their image-sequence experiment, five ratings rather than ratings only after the first and last images produced both a larger decline for a deteriorating sequence and a larger improvement for an improving sequence. The useful implication is direction-specific: a sampling schedule can alter the trajectory used to estimate adaptation.

A calibration study can randomize sparse versus frequent affect prompts within each exposure sequence and retain identical assessments at shared, prespecified times. Include in the randomization an equally frequent interruption condition using questions unrelated to the target experience, with duration and response demands matched and checked. Hong and O'Brien's (2026) supplementary experiments provide precedents for unrelated-rating comparisons. Compare groups at the common probes before estimating a response kernel, and prespecify the schedule-by-direction contrast and a practically meaningful tolerance. Compare frequent affect prompts with unrelated questions to estimate the additional effect of asking about affect under the matched tasks; compare unrelated questions with sparse affect prompting to examine the added interruption regime. Record prompt duration and assess whether effects also appear in independently qualified channels. Report cadence sensitivity, the resulting uncertainty, and which interval functionals remain recoverable under each schedule.

6.5.5 — Context of Use and Versioning

Qualify an estimator for its declared population, setting, cadence, decision, and consequences of error. Materially changed uses require new validation (Campbell & Fiske, 1959; Mokkink et al., 2024; Gagnier et al., 2025).

Report performance by relevant setting and subgroup, and test transport. Record instrument changes using the provenance and migration rules in Section 6.1.5.

An estimator trained on verbal adults in laboratory tasks may fail in pediatrics, dementia, intensive care, nonverbal disability, or another language. Those failures are not edge cases when the theory claims broad conscious scope.

6.5.6 — Abstention, Privacy, and No-Score Governance

Abstention rules must be protocol-fixed before outcome inspection for excessive uncertainty, out-of-distribution inputs, unresolved channel conflict, missingness, calibration failure, endpoint ambiguity, or privacy limits. Human review may add context, but it cannot convert an unidentified object into an estimate or waive a formal stop (Manski, 2003; Mokkink et al., 2024).

Collect only channels needed for the declared estimand, retain them for a stated period, and separate research identifiers from ordinary identity records. Secure enclaves, federated analysis, disclosure limits, and protected access may reduce exposure; state their effects on reproducibility, calibration, subgroup analysis, and independent audit (Nissenbaum, 2010).

Review bodies set deployment limits and own adverse-event response, suspension, and correction propagation under Chapter 3's protections.

6.6 — Missing History and Running-State Uncertainty

Three uncertainties must remain distinct: what preceded enrollment, why a stream entered the study at that time, and what the study failed to observe afterward. The following decomposition gives these uncertainties different roles rather than concealing them in one aggregate error band (Little & Rubin, 2019; Rubin, 1976).

6.6.1 — Observable-Window Decomposition

$$L_i^{phen}(T_i^\star -) = L_i^{phen}(t_0 -) + \Delta L_i(t_0, T_i^\star)$$

$$L_i^{phen}(t_0 -) \in B_i(t_0)$$

The observable window can inform the increment. The missing prefix belongs to a bounded or modeled set Bi(t0) unless independently recovered. Whole-stream adjudication depends on how much that set can be narrowed.

Report whether a result concerns local dynamics, an observed-window increment, a prefix bound, or the complete terminal residual. Local drift, variance, intervention response, and endpoint timing can be studied while whole-stream claims remain unavailable. Naturally occurring changes and independently justified beneficial interventions can test short-window opponent dynamics.

The prefix identified set should be displayed separately from the observed-window increment so readers can see which part of the conclusion comes from data and which part comes from assumptions. A narrow total produced by a narrow prior is not equivalent to direct historical coverage.

Dependence between the missing prefix and observed-window increment must be propagated rather than suppressed by treating the prefix as a fixed offset.

6.6.2 — Multiple Compatible Prefix Histories

The same observed late trajectory may be compatible with many earlier cumulative histories. This is a structural source of nonidentification, not merely a shortage of sample size (Tsiatis, 2006; Manski, 2003; Aalen et al., 2008).

The sensitivity analysis should vary more than the prior mean. It should alter admissible path roughness, exposure rates, origin uncertainty, and dependence between the missing prefix and observed window. A conclusion that survives only one narrow historical model is a property of that model, not of the complete-stream target.

Population priors can narrow prefix uncertainty by importing social stereotypes, health inequities, or diagnosis-based assumptions. Apparent historical reconstruction that collapses under subgroup transport, prior sensitivity, or record-free validation should favor that rival explanation rather than a recovered individual history.

6.6.3 — Bounds and Sensitivity Sets

Bounds place explicit limits on terminal residuals under incomplete history (Rizopoulos, 2012; Imbens & Manski, 2004).

Rate bounds, duration bounds, event records, and monotonicity assumptions can narrow the prefix set. Each assumption should be named and stress-tested.

Protocols can report best-case, worst-case, and assumption-indexed residual ranges. Equivalence with zero is meaningful only if the relevant range lies within a declared empirical band under all required gates.

A documented period of prolonged severe pain may support a lower bound on the magnitude of a negative historical contribution under a justified observation bridge, even when

finer details are missing. That bound is different from a complete reconstruction.

Each bound should be accompanied by an assumption-deletion analysis showing how it widens when one restriction is removed. This reveals which premise supplies apparent precision and prevents several weak assumptions from hiding behind one narrow interval.

6.6.4 — Historical Records and Their Limits

Medical records, diaries, ecological momentary assessments, wearable data, family testimony, and institutional records can constrain missing history. None is a transparent archive of phenomenal experience (Little & Rubin, 2019; Rubin, 1976).

Each source has selection, coding, purpose, and missingness mechanisms. Records created for billing or treatment may omit duration, meaning, or unreported experience.

Source triangulation should preserve provenance and disagreement. Historical evidence can update a prefix model without being merged into one falsely homogeneous stream of data.

A medication chart may document dosage accurately while offering little information about relief, side effects, fear, or interpersonal context. A family narrative may preserve meaning while misremembering timing.

Show historical coverage windows, source type, provenance, and known omissions; leave unrecorded periods explicit.

6.6.5 — No Reset at Enrollment

A useful analysis carries a distribution, interval, mixture, particle cloud, or compatible-prefix set suited to the problem. A mean and variance may fail when the family is multimodal or sign uncertainty changes reachability in opposite directions.

Overlapping cohorts can test local-dynamics transport. Different entry times provide common windows for comparing transition structure and measurement behavior, while historical coverage is documented separately.

Use synthetic cases with identical post-entry dynamics but different known prefixes to test for an implicit enrollment reset. When the supplied information identifies different whole-history conclusions, the pipeline should recover that difference. A second run withholding the prefix should instead preserve the resulting uncertainty. These paired checks distinguish a reset defect from a genuinely uninformative observation window.

Vary prefix and endpoint uncertainty jointly, and report the resulting compatible residual set. Factorial sensitivity analysis identifies whether one boundary or their interaction controls the conclusion. Retain any conclusion shared by every admissible combination; uncertainty need not eliminate a result when all compatible cases agree.

Enrollment begins the observed window. A complete-history analysis combines its measured increments with an independently identified or bounded prefix (Manski, 2003).

The correct decomposition keeps the prefix explicit. Multiple compatible histories can produce the same observed state at enrollment while implying different component totals, residuals, route states, and future completion sets. Replacing that family with zero is an assumption, not a neutral starting point.

Historical records can constrain the prefix, but each channel needs validation. Diaries, medical notes, informant reports, prior devices, treatment histories, and recalled episodes capture different objects and fail for different reasons. Agreement may reflect shared documentation, not access to phenomenal value.

Prefix and endpoint uncertainty must be varied jointly. Modest uncertainty on each side can combine into a wide terminal set,

especially when continued-experience, false-ending, or routing branches remain plausible. The report should identify whether history, endpoint, representation, observation, or their interaction is the limiting gate. Factorial sensitivity analysis can show whether uncertainty is dominated by one bridge or by interactions that become important only when several weak links coincide.

The output should distinguish post-entry change, a full-history bound, and a complete-stream verdict. The first may be estimable when the others are not. This distinction permits useful longitudinal science without overstating enrollment data. That separation also prevents a precise within-window effect from lending unwarranted precision to the unobserved prefix.

No reset at enrollment is a technical and ethical rule. It prevents convenience from erasing hidden prior burden or presenting a study-relative baseline as the beginning of a person’s lived history. When the prefix cannot be identified or validly bounded, No adequate estimate is the honest whole-stream result. A bounded prefix may still support useful partial conclusions; an unbounded one should block only the whole-stream inference that depends on it.

6.7 — Equivalence, Adjudicability, and No-Neutrality-by-Fiat

Several ordinary analysis choices can create apparent neutrality before a terminal hypothesis has been tested. Centering, weighting, endpoint selection, exclusions, and transformations must therefore be evaluated for the constraints they impose on the answer. The relevant controls preserve the intended origin and domain and expose whether the pipeline manufactures the pattern it is meant to investigate.

6.7.1 — Latent Zero and Empirical Compatibility

The standard also protects cross-study accumulation. A later meta-analysis can combine results only when target geometry, observation resolution, endpoint class, and bridge status are sufficiently aligned. Pooling incompatible exact, band, atom, or measured-only objects creates apparent precision around no single scientific question.

Exact latent closure and empirical compatibility must remain visibly separate. For a nonempty truth-containing residual set R_i, exact compatibility requires $R_i = \{0\}$; exact adversity requires $0 \notin R_i$; and every wider set containing zero is exact-indeterminate. A confidence interval or posterior region containing zero does not by itself satisfy the singleton condition (Manski, 2003; Imbens & Manski, 2004).

Finite data can nevertheless contribute. A study may show that a declared measured estimand is compatible with a narrow region around zero under a specified observation, endpoint, missingness, and recovery model. That result is resolution-bounded evidence for the measured or bridged branch it actually addresses.

Repeated success across prospectively nested bands can strengthen the case only while the representation, bridge family,

endpoint class, recovery guarantees, and limiting rule remain fixed. Choosing a narrower band after seeing favorable data tests a new analysis, not a stronger version of the original one.

An exact latent atom requires more than shrinking observed bands. A narrow continuous distribution, rounded measurements, censoring, smoothing, or an instrument floor can all create a visible concentration near zero. Distinguishing an atom requires a bridge that can recover that feature or a partial-identification argument that rules out the continuous rivals.

6.7.2 — Equivalence Bands and Observation Error

The band must be justified before outcome inspection through measurement resolution, calibration error, biological or temporal variability, and the consequence of false equivalence. A tolerance chosen because the observed estimate fits inside it is an outcome-selected target.

Observation error enters through bias as well as variance. A narrow interval around a biased measured estimand can falsely suggest equivalence, while a wide but unbiased interval may remain scientifically honest. The analysis must propagate calibration, origin, endpoint, missingness, and bridge uncertainty rather than attaching them as prose after the decision.

Different uses may require different bands. An exploratory route screen, a measurement-qualification study, and a public claim about terminal concentration need not share one threshold. Each band should name its estimand, scientific purpose, error control, and status ceiling.

When several bands are defensible, the report should show the verdict profile across them rather than select the most favorable. A prospectively specified hierarchy can distinguish a primary band from sensitivity bands without treating all thresholds as interchangeable.

Observation floors, rounding, smoothing, and censoring can compress estimates into the band even when latent dispersion remains. Negative controls and simulated nonneutral data should show whether the pipeline can separate genuine compatibility from instrument-limited concentration.

6.7.3 — Adjudicability Before Estimation

A study becomes adjudicative only when distinct scientific states can produce distinguishable planned outcomes under the declared measurement and endpoint architecture (Borsboom, 2005; Manski, 2003; Mayo, 2018).

Review each target's required dependencies using Section 4.1. A failed prerequisite caps downstream interpretation at the last object that remains identified.

Large datasets can remain nonadjudicative. If two endpoint models, scalarizations, or observation kernels generate the same observed law, more observations of that law improve precision without separating the explanations. The design needs a new contrast, perturbation, instrument, or assumption, not merely more participants.

Model recovery provides an early gate. Simulated worlds with known additive closure, band compatibility, atoms, continuous near-zero laws, observation attenuation, and rival dynamics should be analyzed blind. A pipeline that fails the prespecified recovery criteria for those cases is not ready to support the corresponding human-data model comparisons.

Endpoint ambiguity, unstable sign, incomplete routing, or an unidentified origin can block the primary target. Report narrower estimands whose own prerequisites still hold.

A pre-estimation adjudicability table should list each target branch, the contrast that could separate it, the principal rival, the required data, and the abstention condition. The table reveals

when a planned analysis is descriptive, route-informative, or capable of adjudicating the declared terminal branch.

When no planned contrast separates the relevant states, the scientifically correct output is Model-ambiguous when adequate rivals remain inseparable, No adequate estimate when the target cannot be identified or validly bounded, or Open-set inadequate when none of the models is adequate, together with the evidence needed to change that status. Estimation should begin only after the design has shown how the result could matter.

6.7.4 — Centering and Baseline Subtraction

Centering can force means or windows toward zero. Specify the origin rationale before outcome inspection and test sensitivity to it.

Baseline subtraction can make residuals appear balanced through the choice of reference. Use an independently justified baseline and identify the resulting quantity's scientific status.

Derivative transforms can reduce a cumulative claim to boundary terms. Label the observed change and state the historical coverage it represents.

No-neutrality-by-fiat protects this hierarchy from analytic choices that make favorable outcomes easier. Centering, baseline subtraction, difference transforms, flexible scalarization, endpoint alignment, smoothing, missingness handling, and exclusions can all be legitimate. They become defects when their role or timing is hidden or when they quietly change the target.

Freeze the origin or baseline before outcome inspection and check sensitivity across declared alternatives. Failure of that justification requires withdrawal of exact or measured-neutrality language.

Centering and baseline subtraction are useful descriptive tools, but neither identifies the phenomenal neutral origin. A sample mean, adaptation level, site mean, preterminal average, or fitted terminal center can force measured values toward zero while leaving the latent origin unknown (Borsboom, 2005; Lakens, 2017).

A baseline must have a pre-outcome rationale and a declared role. It may support visualization, within-person contrast, calibration, nuisance adjustment, or a state model. Each role licenses different language; none automatically creates the zero used by the terminal hypothesis.

Temporal placement matters. A baseline estimated after adaptation, medication, sleep loss, disease progression, or entry into terminal care may absorb part of the process under study. A baseline fixed too early can instead treat ordinary drift as cumulative valence. The protocol should state the causal and temporal relation between baseline and target interval.

Report sensitivity to within-site and within-person centering, including differences associated with culture, illness severity, communication capacity, device gain, and reporting standards.

Derivative and difference transforms create a related hazard. Summed changes can telescope to a final-minus-initial state, converting path burden into boundary contrast. Such transforms may test local dynamics, but they do not become cumulative phenomenal measures without an independently validated bridge.

$$\sum_{j=0}^{m-1} \left[\varphi\left(X_{j+1}\right) - \varphi\left(X_j\right) \right] = \varphi\left(X_m\right) - \varphi\left(X_0\right)$$

Endpoint alignment can compound the problem. If trajectories are centered or stopped at the point closest to zero, the procedure has placed the finish line near the answer. Truth and protocol

endpoints must be fixed by information not algebraically identical to the residual being tested.

A transformation audit should report the original scale, every centering or differencing operation, the reason and timing of the choice, alternative defensible transformations, and the resulting verdict profile. A centered graph may be useful; it must never conceal that neutrality entered through the preprocessing rule.

6.7.5 — Outcome-Selected Analysis Choices

Outcome-selected weights can select channels or signs that cancel. Freeze the weights before confirmation or label the analysis exploratory.

Endpoint fitting can move the window to a favorable ending. Separate endpoint determination from the tested outcome and report sensitivity to the endpoint rule.

Exclusions can remove inconvenient trajectories. Apply symmetric rules and independent audit to preserve the intended comparison.

The timing vocabulary is exact. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection without asserting a public registry. Publicly preregistered is reserved for a timestamped external registry with a persistent identifier; this Part makes no such claim without that record.

Later deviations from protocol-fixed specifications remain visible and are routed to sensitivity or exploratory status unless an independently governed amendment justifies another label.

Freeze channels and signs before outcome inspection, with a blinded artifact and sign audit. Changes made after seeing outcomes leave the affected analysis exploratory.

Freeze scalarization weights before outcome inspection and report the identified set across the declared weight family. If

conclusions vary across that family, retain the representation sensitivity in the verdict.

Freeze the endpoint rule before outcome inspection and review its independence from the tested ledger. A failed separation requires abstention or a clearly labeled sensitivity analysis.

Outcome-selected flexibility can manufacture neutrality while looking like model improvement. With many signs, channels, lags, weights, filters, windows, endpoints, and exclusions, some combination may approach zero even when no prospectively specified representation does (Mokkink et al., 2024; Gagnier et al., 2025; Simonsohn et al., 2020).

Weights and signs may be learned on development data, but the confirmatory terminal analysis must apply a protocol-fixed rule to protected outcomes. Cross-validation helps only when the endpoint and feature construction do not leak into every fold.

Endpoint fitting includes trimming the final interval, selecting the closest zero crossing, excluding sudden or low-signal endings after inspection, or changing the endpoint class because one definition is favorable. Necessary exclusions require symmetric, documented rules and sensitivity across plausible adjudications.

Channel rejection has the same burden. Artifact removal can protect validity, yet a contradictory channel cannot be dismissed merely because it weakens the preferred story. Blind quality review, negative controls, and predefined failure rules reduce that discretion.

A prospectively specified multiverse can show how the result changes across defensible weights, kernels, endpoints, and exclusions. The primary confirmatory route remains one declared analysis; the wider multiverse is sensitivity evidence, not a license to choose the branch that wins.

Retest a favorable representation discovered after outcome inspection on independent data under a new protocol-fixed rule.

6.7.6 — Audit, Reanalysis, and No Adequate Estimate

Adversarial reanalysis asks whether reasonable alternatives reverse, erase, or weaken the conclusion. Synthetic fixtures ask whether a nonneutral process can be manufactured into apparent neutrality, whether observation attenuation can imitate latent convergence, and whether derivative transforms telescope into boundary differences. These tests let the pipeline fail before participants carry more burden.

Freeze the missingness model before outcome inspection, then check severity-related absence and device tolerance. Failure of its assumptions requires wider uncertainty or no estimate.

Version code and scientific displays before outcome inspection and permit external reanalysis or secure audit. Material failures require correction, retraction, or retirement of the affected claim.

Use Chapter 26's reproducibility packet to reconstruct the declared measurement specification and result (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

Blinded reconstruction gives an independent analyst the frozen objects and protected data without the original conclusion. Rebuilding the estimand and decision rule can expose undocumented preprocessing, software defaults, ambiguous definitions, or hidden judgment.

Adverse reanalysis should test bounded alternatives for bias, informative missingness, covariance, origin and weight families, endpoint uncertainty, and serious rivals. Scientific purpose constrains the search, because unbounded flexibility can

manufacture defeat as readily as support (Steegen et al., 2016; Simonsohn et al., 2020).

Sensitive records require controlled environments, disclosure-limited outputs, synthetic fixtures, federated execution, or purpose-limited external review. Report the resulting audit limits. Independent reviewers should receive enough permitted synthetic or derived material to test the transformations carrying the claim; restricted access is not independent verification (Nissenbaum, 2010).

Name each abstention reason under Section 24.2: representation, endpoint, observation, missingness, recovery, computation, privacy-limited audit, or an open rival envelope. Keep these failures distinct in the register.

Replication can test whether the same abstention boundary recurs under the same declared conditions (Munafò et al., 2017; Mayo, 2018).

Chapter 7 — Mechanism-Neutral Consequences

An exact terminal relation can constrain a trajectory before its ending is observed. The question is what follows from that relation alone and what enters only when a mechanism is added. This chapter separates those consequences so that changing a preferred route cannot erase a failed necessary condition while leaving its premises unchanged.

Start with a complete prefix of a qualifying stream. Closure requires its remaining net contribution to have the opposite sign and equal magnitude. Whether that contribution is available depends on the remaining domain and the permitted directions of change. A bound on total remaining magnitude can exclude completion without identifying the process that would otherwise produce it.

An exclusion is only as strong as the bound. If future rates, exposure, or endpoint timing remain insufficiently constrained, the observed prefix may leave several continuations open. The useful result then describes that feasible set. It does not turn a preferred continuation into a forecast, and it does not infer impossibility merely because a simulation failed to find one.

Endpoint timing supplies another restriction. A ledger cut at an independently distributed time faces a different problem from one stopped when it reaches a boundary. The independent-cutpoint result has explicit support and measurability premises. Hitting times, atoms, and state-dependent timing change those premises and therefore require their own analysis rather than an altered interpretation of the same theorem.

Possibility must also be separated from probability. A completing path may exist yet receive no probability under the policy or dynamics being tested. Individually available actions may fail to form one admissible strategy over time. Under partial observation, indistinguishable histories may require

incompatible actions. These distinctions determine what a reachability calculation can actually establish.

Consider the chapter's hypothetical comparison between a reserve-linked process and an ordinary diffusion conditioned on a terminal event. Both can produce a closing curve. A perturbation or different endpoint regime is informative only where the specified models predict different observable outcomes. Similarity in the original graph neither identifies the mechanism nor guarantees that the proposed new comparison will separate it.

The reserve product adds commitments beyond the boundary hypothesis. Reserve, relaxation, and innovation cannot be fitted independently under the reference route; their shared equation creates a test. The product's process class must also match its proposed relation to an accumulated rate. A mathematically closing construction can expose these restrictions before its phenomenal and endpoint bridges are established.

The chapter develops the resulting bounds, timing restrictions, and route comparisons together with their failure cases. A theorem can survive while a particular measurement cannot test it; a route can fail while a mechanism-neutral restriction remains open. Keeping those outcomes attached to their premises makes the next discriminating calculation or observation identifiable.

7.1 — Mechanism-Neutral Consequences of Exact Closure

A complete-stream terminal constraint restricts what each prefix leaves attainable on the remaining domain. The following reachability bound uses only a declared rate bound and endpoint, before a mechanism is chosen (Aubin, 1991; Aubin et al., 2011).

7.1.1 — Constraint Before Mechanism

$$\left|L_i^{phen}(t)\right| \leq \int_t^{T_i^\star} m_i(s)ds =: C_i(t)$$

The inequality is a reachability bound. If the current residual exceeds the maximum remaining signed capacity in the needed direction, exact closure is impossible under that bound and route class.

Mechanism-neutral derivations use only the target, domain, regularity, and boundedness assumptions explicitly stated. Any result that invokes a route-specific state belongs to a later layer.

Protocols can test necessary consequences across multiple routes. A consequence that fails under adequate measurement narrows the target or its assumptions regardless of which mechanism was preferred.

Some alleged consequences disappear under an allowed change of coordinates or arise because the endpoint was built into the state definition. A mechanism-neutral result must survive admissible rescaling and an independently specified endpoint. Otherwise it is a representational artifact rather than a consequence of closure.

7.1.2 — Prefix Feasibility and Remaining Capacity

At any prefix time, the accumulated residual and the maximum possible remaining signed exposure jointly determine whether

closure remains feasible under a declared class (Protter, 2005; Kallenberg, 2002; Ames et al., 2019).

The reserve bound compares current magnitude with an integral bound on future contribution. It is necessary under the rate bound, but it is not sufficient when sign, timing, routing, or endpoint accessibility also constrain the future.

Estimating the bound requires uncertainty in the current residual, remaining duration, rate limits, and observation model. Feasibility may therefore be set-valued rather than binary.

A large negative residual near a short, independently defined endpoint may be incompatible with closure if the maximum possible positive contribution is too small.

Rate and exposure bounds should come from independent physiological, psychophysical, or protocol evidence and should be stress-tested before terminal outcomes are inspected. Their scientific value lies in excluding continuations, not in expanding until every path survives.

7.1.3 — Nontransferability as Consequence

Apply Section 5.3.2's intrastream rule to completion sets: one stream's feasible continuation cannot replace another stream's missing contribution.

Collective-consciousness or fusion hypotheses would require new bearer and routing definitions. They cannot be introduced after individual-stream failure merely because social interaction was present. Unless a new ontology and data model distinguish fusion from interaction, the individual-stream analysis is narrowed rather than rescued.

7.1.4 — Necessary Remainders and Covariance

A joint mathematical fingerprint can make the cumulative claim more informative. Consider a declared probability law for a

qualified reference class on the primary additive branch. Partition each stream's complete strict-pre-endpoint domain into m disjoint measurable parts, for the same prospectively fixed finite integer $m \geq 1$, assigning each exposure token and any cut-time atom exactly once under independently justified endpoint rules. Write X_j for the signed phenomenal mass in part j and assume $\mathbb{E}\left[X_j^2\right]$ is finite for every j. Let X collect those masses, let $\Sigma = Cov(X)$, and let $\mathbf{1}$ denote the m-vector of ones. Then probability-one closure entails

$$\mathbf{1}^{\top}\mathbb{E}[X] = 0, \qquad \Sigma\mathbf{1} = 0.$$

Conversely, these two exact population conditions imply $\sum_{j=1}^{m} X_j = 0$ with probability one: the sum has both mean zero and variance $\mathbf{1}^{\top}\Sigma\mathbf{1} = 0$. This is a direct derivation from additivity and elementary covariance identities, under the square-integrability assumption stated here. Each row also gives $Var(X_j) + \sum_{k \neq j} Cov(X_j, X_k) = 0$, exposing a whole-history dependency restriction rather than a requirement that every neighboring interval oppose the last.

This supplies a useful multivariate diagnostic when the complete-domain and measurement premises can be met. To distinguish zero from another fixed total, test the mean restriction alongside the covariance restriction using the independently calibrated origin. Fit an unconstrained measurement model, propagate correlated errors, and quantify uncertainty in the joint mean and covariance restrictions when comparing the linked prediction with equally flexible rivals.

7.1.5 — Mechanism-Neutral Limits and Falsifiers

A mechanism-neutral derivation no longer applies when a required premise fails; record those conditions before data.

Unbounded rates, incomplete domains, uncertain routing, or endpoint-inclusive jumps can invalidate a simple reserve bound (Protter, 2005; Kallenberg, 2002; Ames et al., 2019).

A recovery benchmark should include provably feasible, provably infeasible, and observation-limited histories. The pipeline should classify the first two and abstain on the third; a method that merely widens remaining-capacity bounds until closure becomes possible has no exclusion power.

A protocol-fixed falsifier should state the measurement precision, endpoint class, and admissible-route family under which an empty completion set would be decisive. Changing any of those after failure creates a new test rather than rescuing the old one.

An empirical infeasibility certificate requires more than a point estimate. The prefix, directional capacity bounds, endpoint support, and admissible route class must be identified tightly enough that every compatible continuation misses the target. When that standard is met, the counterexample attaches to the declared branch rather than to a person or an informal narrative.

A valid mechanism-neutral counterexample reaches the target only through all premises in Section 2.6.3. Failed reserve dynamics, hazard relations, or variance compression instead defeat the assumptions that made those route-specific patterns necessary (Popper, 1959).

Counterexamples should be built prospectively. Synthetic one-sign streams test whether the model recognizes impossible additive closure. Prefixes outside every completion set test reachability. Independent diffuse endpoints test the non-atomic obstruction. Paths with identical terminal residuals but different gross variation test whether the analysis confuses net and total variation. A recovery pipeline should identify which condition failed rather than returning one omnibus fit score.

Track how the continuation set contracts, splits, or becomes empty as prefix observations accumulate. Its size and shape show how strongly observed history restricts possible completion, while recording which prospectively justified bounds drive each change (Ames et al., 2019; Protter, 2005).

7.2 — Random Endpoints and the Non-Atomic Obstruction

Endpoint timing is one of PTVC's sharpest discriminators. If an endpoint can occur independently at a continuously distributed time while a nontrivial pre-endpoint ledger evolves, exact closure at that random cut is extraordinarily restrictive (Kallenberg, 2002; Billingsley, 1999).

7.2.1 — The Random-Cutpoint Question

Let E_i be the prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I, and write $P_i^0 = P(\cdot \mid E_i)$. Under this same law, let $\widetilde{L}_i(t\text{-})$ be jointly measurable, let $\mathscr{F}_i^L = \sigma\left\{\widetilde{L}_i(s\text{-}) : s \in I\right\}$, and let T_i^{exo} be independent of $\mathscr{F}_i^L$. Assume that T_i^{exo} is I-valued, has a Lebesgue density $f_{T_i^{exo}}(t)$ positive for Lebesgue-almost every $t \in I$, and $P_i^0\left\{\widetilde{L}_i(T_i^{exo}\text{-}) = 0\right\} = 1$.

$$0 = P_i^0\left\{\widetilde{L}_i(T_i^{exo}\text{-}) \neq 0\right\} = \int_I P_i^0\left\{\widetilde{L}_i(t\text{-}) \neq 0\right\} f_{T_i^{exo}}(t)\,dt.$$

The integrand is nonnegative. Therefore $\widetilde{L}_i(t\text{-}) = 0$ for Lebesgue-almost every $t \in I$ on a P_i^0-probability-one set of eligible paths.

The result does not apply when timing is state-coupled, independence fails, the endpoint contributes phenomenally, routing changes the stream, or the pre-killing extension does not exist. First-hitting or absorbing constructions, endpoint atoms, singular timing laws, continued experience, and endpoint-inclusive value remain separate branches requiring their own

premises rather than proof details that may be changed after a result is seen.

Test and report the theorem's timing, measurability, and support premises before interpreting its empirical application.

An independent diffuse cutpoint has no path dependence under the reference law. The principal mathematical burden is the non-atomic obstruction.

A state-coupled hazard depends on the declared state. Its scientific burden is to recover that coupling beyond what frailty and selection explain.

A first-hitting or absorption endpoint is generated by a boundary event. Its use must avoid defining the tested terminal result into the endpoint itself.

Atomic or singular timing concentrates on a restricted set. Specify that support and the limits on transporting the resulting inference.

The endpoint anti-tautology rule in Section 4.3.4 remains universal. First hitting, absorption, zero-surface, and last-exit routes supply no target evidence when the endpoint is defined by the same zero event and then scored on that event.

Empirical work cannot prove full independence from finite data, but it can compare timing classes. A prospectively specified tournament can include independent diffuse timing, measured state dependence, latent frailty, first hitting, absorption, endpoint atoms, and observation-shutdown models, each with its own likelihood, diagnostics, and failure conditions.

Biological death or an external accident may be relevant timing variables without being the phenomenal endpoint. A cause of bodily death is not automatically a cause or observation of the end of qualifying experience. The endpoint bridge remains a separate object that can fail.

Audit leakage using variables known only after the endpoint, indicators of observation shutdown, and features affected by the ledger but not by the exogenous probe. These checks can reveal a model that appears independent because the timing process or data pipeline was misclassified.

Selection, common causes, and unobserved frailty can manufacture apparent dependence or independence. Model entry into the risk set, which endpoints remain observable, and how latent state affects timing and observation.

7.2.2 — Independent Diffuse Endpoints

On the branch considered here, the endpoint is independent of the ledger path under the declared reference law and has a Lebesgue density positive almost everywhere on the specified interval. Non-atomic singular timing laws are treated separately in Section 7.2.5 (Aalen et al., 2008; Andersen et al., 1993; Revuz & Yor, 1999).

Before the stronger independence theorem, a simpler at-risk screen applies. If the regular conditional law of the pre-endpoint ledger at the endpoint assigns no atom to zero for the declared conditioning history, exact zero has conditional probability zero on that risk set. This result does not call the endpoint exogenous and does not require path-endpoint independence; it only rules out probability-one exact closure for the specified non-atomic conditional law. Apply Section 24.2 to empty or unidentifiable risk sets, retaining any valid bounds (Kallenberg, 2002).

A device failure time may be diffuse and independent of experience but is merely an observation endpoint. Treating it as the truth endpoint would produce a false obstruction or false support depending on the analysis.

The empirical task is to estimate how much endpoint-ledger dependence reconciles the observed path occupation with the

endpoint distribution under prospectively specified covariates and timing classes.

The approximate conditional-dependence result supplies a graded target. Fix the covariates, off-band threshold, and a pre-killing ledger process on a common time domain prospectively. Work on standard Borel path and covariate spaces with jointly measurable left-limit evaluation and the required regular conditional laws. For each covariate value, compare the actual joint law of endpoint time and ledger path with the product of their conditional marginal laws. Under the product law, the off-band probability is the ledger's off-band occupation weighted by that covariate-specific endpoint-time distribution. Average this probability over the covariates and subtract the actual off-band endpoint probability. Define the mismatch as the positive part of that difference. The mean conditional total-variation distance is at least this mismatch, and conditional mutual information, measured in nats, is at least twice its square (Cover & Thomas, 2006).

The proof compares the same off-band event under the conditional joint and product laws. By the definition of total-variation distance, their probability difference is bounded by that distance. Averaging gives the mismatch bound. Conditional Pinsker's inequality and Jensen's inequality give the squared information bound. The result quantifies the residual endpoint–ledger dependence that a state-coupled or selection model must explain after the declared covariate adjustment.

A sharper information bound uses the same common pre-killing domain and conditional joint and product laws as above. Given covariates $C = c$, let P_c be the joint law of endpoint and ledger path, Q_c the product of their conditional marginals, and $E_\varepsilon = \{|L_{T-}| \le \varepsilon\}$. Write $p_c = P_c(E_\varepsilon),\ q_c = Q_c(E_\varepsilon),\ p = E[p_C]$

and $q = E[q_C]$. Mapping both laws to this same event and applying joint convexity gives

$$I(T;L \mid C) \geq E[d_{Ber}(p_C \| q_C)] \geq d_{Ber}(p \| q).$$

$$d_{Ber}(p \| q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

Logarithms are natural, with the usual extended-value conventions. For $p = 0.9$ and $q = 0.1$, the lower bound is 1.758 nats, compared with 1.280 from the squared total-variation bound. If $p = 1$, it becomes $\log(1/q)$; shrinking bands with product-law probability tending to zero require unbounded endpoint-path information in this idealized comparison. Ordinary hitting or state-coupled timing can supply that dependence. The scientific task is to identify which timing mechanism predicts its observable consequences.

An observable companion tests endpoint enrichment against a specified ordinary event model. Fix a finite horizon u. Let N_i count prospectively classified events while observed, stopped at censoring; let Y_i be the predictable at-risk indicator. In one declared observed filtration, let $Y_i \lambda_{0i}$ be the null compensator density and $Z_{i,\varepsilon}$ a bounded predictable band indicator or filtered probability for a separately defined observed-window residual. Then

$$U_\varepsilon(u) = \sum_i \int_0^u Z_{i,\varepsilon}(t)[dN_i(t) - Y_i(t)\lambda_{0i}(t)\,dt].$$

When the compensator is correct and square integrability holds, $E[U_\varepsilon(u)] = 0$. With continuous compensators and orthogonal stream martingales, the predictable quadratic variation is

$$\langle U_\varepsilon \rangle_u = \sum_i \int_0^u Z_{i,\varepsilon}(t)^2 Y_i(t) \lambda_{0i}(t)\, dt, \qquad Var[U_\varepsilon(u)] = E[\langle U_\varepsilon \rangle_u].$$

The random bracket is not itself an unconditional variance or an automatic normal-test justification. Estimated intensities require nuisance uncertainty and validated calibration. Freeze models on development records; include frailty, treatment, state-dependent timing and informative observation loss. Cross-fitting controls some outcome reuse but cannot repair an incorrect hazard. Survivor-conditioned risk sets and shuffled endpoint dates do not reproduce the product-law comparison above. A positive score identifies residual event association relative to the declared model; the event and experiential bridges determine its stronger interpretation.

7.2.3 — The Non-Atomic Obstruction

Apply Section 7.2's positive-density and measurability assumptions when checking path regularity, simulation, and numerical rounding (Protter, 2005; Mayo, 2018).

The result should be checked under discontinuities, càdlàg paths, equivalence classes, and left limits. Endpoint atoms or singular supports require separate analysis.

A simulated ledger that is nonzero on a set of positive Lebesgue measure in the sampling interval has a probability strictly below one of exact zero at an independent uniformly sampled endpoint. The probability is zero only when its zero set has Lebesgue measure zero. Model recovery should reproduce that distinction.

Numerical rounding can hide the obstruction by turning a finite band into exact zero. The empirical band and the latent theorem must remain separate.

7.2.4 — Coupled, Hitting, and Absorbing Endpoints

State-coupled endpoints have hazards that depend on route state or ledger-related variables. Hitting-time endpoints occur when a process reaches a boundary, and absorbing endpoints remain in a terminal state thereafter. Each endpoint class induces a different likelihood, support geometry, and rival set; they cannot be pooled as interchangeable explanations of closure (Kallenberg, 2002; Billingsley, 1999).

These alternatives evade the independent-cutpoint implication by changing the timing law. They also create new testable relations among state, hazard, approach geometry, and endpoint support.

A first-hitting rule can guarantee terminal zero by definition if the endpoint is set to the first time the ledger reaches zero. That is a tautology unless an independently motivated physical or psychophysical mechanism links termination to that surface.

Absorption can also be confused with observation shutdown. A flat observed series after device removal is not evidence that the latent process entered an absorbing neutral state.

Protocols should model endpoint hazard prospectively and compare state-coupled, frailty, disease-severity, treatment, and observation-selection accounts under the same information clock. The distinctive state-hazard relation, support implication, or safe-intervention response must recover in simulation before supporting the affected empirical comparison. Extension beyond the tested cohort additionally requires transport evidence.

7.2.5 — Atomic, Singular, and Protocol Consequences

Endpoint laws may contain diffuse, atomic, mixed, or singular components generated by scheduled procedures, biological thresholds, or classification rules (Aalen et al., 2008; Andersen et al., 1993; Revuz & Yor, 1999).

Atomic timing samples named times, so closure constrains those masses rather than an interval.

Support is decomposed by component and tied to declared timing mechanisms or protocol classes instead of being forced into one smooth hazard.

Anesthesia emergence, scheduled life-support withdrawal, and device-defined stops may concentrate protocol times, but their relation to the truth-level stream endpoint remains empirical.

Mixture class, support, and assignment are fixed or validated on held-out data. The estimand is component-specific mass and support, not a smooth hazard flexible enough to absorb every failure.

A branch matrix states the consequence of each component: diffuse mass can invoke the interval obstruction, atoms constrain named points, and singular mass needs carrier analysis. Mixed laws require mixed conclusions. Recording each component's weight, or the range of weights consistent with the data, makes the scope of each conclusion easier to assess. A mixture weight near zero may make a component practically minor, but it does not disappear from a universal or probability-one statement that quantifies over it.

At each protocol stop, record the reason, the last usable measurement, and which aspects of the subsequent trajectory remain observable. This lets later return or resuscitation evidence reconnect to the same case without converting the protocol stop into the truth endpoint.

Report timing class, support, uncertainty, endpoint alternatives, and downstream sensitivity. Unrecoverable class membership narrows the result or yields No adequate estimate (Aalen et al., 2008; Revuz & Yor, 1999). Separate within-branch sampling uncertainty from between-branch sensitivity to endpoint classification before synthesis: precise estimates within branches can coexist with uncertainty about which branch applies.

Stress tests vary density support, state dependence, frailty, selection, and endpoint definitions individually and jointly. Record changes in off-band occupation and verdict, ranking assumptions by their effect on the protocol-fixed decision boundary (Billingsley, 1999; Kallenberg, 2002; Andersen et al., 1993).

Multi-state models separately label clinical transitions, observation shutdown, and theorem-relevant endpoints. An explicit return-to-observation transition reconnects resumed monitoring with the existing record. Where the endpoint bridge supports identification, separate estimates or bounds for observation loss, return, and endpoint entry keep observation dynamics from being absorbed into endpoint hazard.

7.3 — Directional Reachability and Completion Sets

Remaining capacity is rarely symmetric. A stream may retain more opportunity for positive than negative contribution, or the reverse, because duration, physiology, environment, action, care, and endpoint hazard constrain directions differently (Aubin, 1991; Aubin et al., 2011).

7.3.1 — Directional Remaining Capacity

Directional reachability therefore replaces a single scalar reserve with sets of attainable future signed increments. The result is not a forecast of what will happen. It is a statement about what remains possible under a declared model class.

$$C_i(t,x) := \left\{ \Delta \in R : \exists \gamma \in A_i(t,x), \Delta = \int_t^{T_i^\star} d\nu_i^\gamma \right\}$$

$$-L_i(t) \in C_i(t,x)$$

The completion set depends on current state x, admissible histories $\mathcal{A}_i$, endpoint support, and the representation. Closure is feasible only if the required opposite increment lies in that set.

Upper and lower reachability envelopes can be derived from signed rate bounds, state constraints, and endpoint support. They need not be equal or monotone.

Estimation requires models of future state, action availability, endpoint support, experienced exposure, and route uncertainty.

Severe sedation may reduce both positive and negative observable variability while leaving asymmetric possibilities for pain, comfort, or awareness. A single reserve number cannot capture that structure.

Frailty, disease progression, treatment, or loss of social support can shrink capacity. Test the joint directional prediction

against these ordinary dynamics under Section 7.7's comparison rules.

7.3.2 — Positive and Negative Completion Sets

Positive and negative completion sets collect future increments reachable in each direction under the declared dynamics. Their geometry may be disconnected, state-dependent, sign-asymmetric, or constrained by action and endpoint timing (Ames et al., 2019; Altman, 1999; Kallenberg, 2002).

The sets may be represented through stochastic reachable sets, viability kernels, control bounds, or simulation-based envelopes. Each representation has assumptions about dynamics and intervention.

Protocols should report uncertainty in set boundaries and test calibration through held-out transitions. A set that contains every observed outcome is not automatically informative if it is too wide to exclude anything.

A rehabilitation program may expand positive experiential possibilities by restoring agency while also increasing short-term pain exposure. The two directional sets change differently.

Sign-specific reachability should be reported as more than two maxima. The analysis should preserve gaps, disconnected regions, action constraints, and the probability or support attached to each attainable increment. A route may permit some positive movement while still excluding the exact opposite-signed increment required for closure.

When several active streams share finite resources or opportunities, individual reachability and joint solvency separate. The correct object is a vector of current residuals together with a joint attainable future-contribution set whose coordinates preserve stream ownership and provenance. Simultaneous completion requires the negative residual vector to

belong to that joint set; summing coordinates or allowing one stream's surplus to cancel another's deficit would change the hypothesis (Aubin et al., 2011).

Shared constraints can create subset bottlenecks: each stream may have an individually completing continuation while no single feasible allocation completes all of them together. For any fixed direction, every jointly attainable increment has a projection no larger than the supremum over the joint set. If the required increment exceeds that supremum, simultaneous completion is infeasible. Testing directions supported on stream subsets supplies necessary support-function screens for the convexified joint set; exact membership remains the sufficiency target when the attainable set is nonconvex.

7.3.3 — Prefix Infeasibility and Reachability Bounds

A prefix is infeasible for exact closure under a route class when the required compensating increment lies outside the completion set. This is a direct route-failure condition (Protter, 2005; Mayo, 2018).

Infeasibility may be proved by analytic bounds, comparison theorems, justified dynamic programming, or certified numerical outer bounds. Failure to find a completing path in a finite simulation is not by itself a proof of impossibility. Empirical use must also carry endpoint and observation uncertainty into the bound.

Empirical adjudication should distinguish structural infeasibility from insufficient estimation. When a valid set cannot be narrowed enough for a decision, the verdict is Inconclusive evidence rather than compatibility by default. No adequate estimate applies when the target cannot be identified or validly bounded.

7.3.4 — Action Support Without Prescribing Action

Action support describes structurally available transitions from a reached history. Physically possible, ethically admissible, and behaviorally positive-probability actions are different sets (Aubin, 1991; Aubin et al., 2011).

Under a frozen action-first regime, let each admissible action carry a completion probability after every downstream policy, physical transition, endpoint, routing, selection, and measurement-reactivity component is fixed. The policy-level completion probability is the policy-weighted mixture of those action-specific values.

That mixture equals one only when the policy assigns probability one, almost everywhere under its own action distribution, to actions whose complete frozen continuations close with probability one. A favorable action, nonempty reachable set, positive-probability closure, or topological-support statement is insufficient. The condition characterizes policy mass, not psychological motivation, conscious ledger awareness, moral instruction, or endpoint support.

Finite soft guidance has a ceiling. Reweighting actions by a finite positive power of their completion probabilities removes zero-completion actions but retains any baseline-positive action with an intermediate probability. Any retained intermediate mass prevents exact probability-one completion. Approaching one as guidance strengthens is limit-sure behavior, not finite attainment.

Almost-sure and limit-sure attainability must remain distinct. Almost-sure attainability requires one admissible policy that reaches completion with probability one; limit-sure attainability may only provide a sequence of policies whose probabilities approach one. If the admissible policy class is empty, the route

is unavailable rather than almost-sure or limit-sure (Chatterjee et al., 2016; Asadi et al., 2025).

Partial observation creates a robust intersection problem. An observation-adapted policy can guarantee completion across all compatible histories only when their probability-one action sets have a nonempty admissible intersection and a measurable, time-consistent selector exists. If indistinguishable histories require disjoint safe actions, no policy based only on that observation can guarantee both.

Hard exclusion also creates an identification problem. When observed behavior assigns zero probability to putatively unsafe actions, their counterfactual completion probabilities are not identified without positivity, structural restrictions, valid instruments, experiments, or partial-identification bounds. The strongest safety claim can concern precisely the actions least observed.

Ethically bounded evidence may come from ordinary treatment variation, low-risk tasks, natural experiments, exogenous availability changes, or sealed recovery studies. Harmful, illegal, endpoint-manipulating, or lifetime-risk interventions remain prohibited. The route must accept unavailability rather than treating ethical limits as missing opportunities to test the theory.

7.3.5 — Completion Potentials and Prediction

A completion potential is a declared state function tied to future reachability, such as a committor, viability value, harmonic function, or bounded reachability functional. Each has its own boundary condition, recursion, and interpretation; a flexible endpoint predictor does not acquire that status by name (Aubin, 1991; Ames et al., 2019; Altman, 1999; Kallenberg, 2002).

Its strongest test is transported prediction under frozen state variables, feature times, endpoint, and action interface. Re-estimating the coordinate system in every cohort tests local predictors, while terminal or post-baseline information may not leak into a representation later praised for forecasting.

Generic prognosis, frailty indices, and latent embeddings are equal-treatment rivals. A completion potential earns distinction only by recovering its defining equation or recursion and adding held-out sign-specific reachability, action-sensitivity, or boundary-access predictions beyond ordinary prognosis, with calibration over reachable states and recovery in synthetic worlds (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Action invariance is stronger than passive discrimination. If safe interventions create the same measured state but different reachability, a potential that treats them alike lacks causal structure; if it changes when only observation changes, it may be tracking measurement. Matched interventions should therefore vary action support while preserving observation quality, making causal reachability harder to confuse with sensor response.

Score transition, hitting, and endpoint distributions when transporting the frozen potential.

When several potentials remain compatible, report the reachability set and conclusions shared across that family. Identify conclusions that depend on unverified geometry.

7.4 — Endpoint Support and Accessibility

Reachability is not enough when the endpoint may arrive before the path reaches the required region. The support of endpoint timing, the intensity of termination events, and the accessibility of the boundary jointly determine whether a completion set is relevant (Andersen et al., 1993; Aalen et al., 2008).

7.4.1 — Endpoint Support as a Model Object

Counting-process and compensator language provides a way to separate observed event occurrence from predictable hazard. It also clarifies how much time the process spends in states that make closure possible or impossible.

$$N_i(t) = M_i(t) + \Lambda_i(t)$$

$$E\left[\int_0^T H_i(t)\,dN_i(t)\right] = E\left[\int_0^T H_i(t)\,d\Lambda_i(t)\right]$$

For predictable H_i that is nonnegative, or satisfies $E\int_0^T |H_i(t)|\,d\Lambda_i(t) < \infty$, the compensator identity relates event-weighted occupation to the predictable endpoint intensity. It does not identify the truth endpoint unless the counting process itself has been validly linked to stream termination.

Endpoint support consists of the time-state points whose every open neighborhood has positive probability under the declared endpoint law. It constrains which prefix states the boundary hypothesis must handle (Andersen et al., 1993; Aalen et al., 2008).

Support may depend on calendar time, biological state, route state, intervention, or latent hazard. It is distinct from the observed distribution in one selected cohort.

Positive tails introduced by smoothing enlarge a model's mathematical support; they do not establish endpoint

accessibility from observations alone. For a closure statement quantified over that support, enlarging it adds cases the model must handle rather than automatically removing a violation. Report sensitivity to censoring and to any explicitly approximate effective-support threshold without silently deleting low-probability cases.

Support uncertainty should be represented by nested lower and upper support sets. The inner set contains regions with well-supported endpoint risk; the outer set contains regions not excluded after censoring and rare-event uncertainty. Boundary-accessibility claims may use only the set appropriate to their error cost, rather than treating a smoothed density tail as established support.

7.4.2 — Counting Processes and Compensators

A counting process records endpoint-like events or transitions, while its compensator represents the predictable cumulative intensity under the filtration. The martingale residual captures departures from that predictable component (Protter, 2005; Jacod & Shiryaev, 2003; Revuz & Yor, 1999).

Compensator analysis can test whether endpoint events are related to route states beyond ordinary risk factors. It requires careful specification of the event, at-risk process, censoring, and time scale.

Endpoint support supplies a restriction without assuming independence from the ledger. On a declared stopping-time branch, let $T > 0$ be the true endpoint, $N_t = \mathbf{1}_{\{T \le t\}}$, and Λ its predictable compensator. In that same filtration let L_{t-} be predictable. For finite u and $\delta \ge 0$, compensation of the nonnegative predictable indicator gives

$$\Pr(T \le u,\ |L_{T-}| > \delta) = E \int_0^u \mathbf{1}_{\{|L_{t-}| > \delta\}}\, d\Lambda_t.$$

Exact closure makes this nonnegative integral zero almost surely, so endpoint compensator mass is confined to the pre-endpoint zero set. If $d\Lambda_t = Y_t \lambda_t dt$ with ordinary finite intensity and that zero set has zero at-risk time occupation, no exact-closing endpoint can occur in the interval with positive probability. A singular or predictable hitting endpoint can have a different structure.

If $\lambda_t \geq \lambda_0 > 0$ throughout the off-band at-risk region and the displayed endpoint-failure probability is at most ε, positivity gives

$$E \int_0^u Y_t \mathbf{1}_{\{|L_{t-}|>\delta\}}\, dt \leq \varepsilon / \lambda_0.$$

A lower bound established only on a subregion gives only that subregion's occupation bound. The compensator and ledger predictability must use the same filtration: projecting a hidden endpoint to an observed hazard does not permit multiplication by a separately projected band probability. Ordinary hitting and zero-plateau shutdown mechanisms can satisfy the restriction. Their prospectively predicted timing and occupation relations, together with independent event and contribution calibration, determine the comparison.

In a clinical cohort, loss of responsiveness, device removal, physiological death, and adjudicated stream endpoint may be different counting processes. Combining them erases the very distinctions the formal model needs.

Compensator diagnostics should therefore include time-varying covariates, treatment processes, observation intensity, and care transitions in the at-risk specification. Residual state coupling after those adjustments is the relevant signal, not the unadjusted hazard shape.

7.4.3 — Occupation and Remaining Exposure

Occupation measures record how long a trajectory spends in regions of state space before the endpoint. They connect reachability geometry with the temporal opportunity to realize future increments (Kallenberg, 2002; Billingsley, 1999).

A state may be reachable yet have negligible occupation probability before termination. Conversely, long occupation in a region can dominate accumulated exposure even when transitions are rare.

Estimation can use continuous-time state-space models, inverse-intensity weighting, or simulation. Observation cadence and informative missingness must be part of the calculation.

A person may repeatedly enter a state associated with relief but remain there only briefly, while spending long intervals in burdened states. Counting visits alone would misrepresent exposure.

Compare occupation summaries under the route and ordinary disease-and-care models using the same observation model.

Occupation data can reject a reachability story even when the target state is topologically accessible. A path with insufficient occupation of the necessary region before the qualifying stream’s endpoint lacks the temporal exposure required by the route. Reaching that region only after observation ends instead limits empirical adjudication unless the observation stop has been independently linked to the stream endpoint. Conversely, long occupation does not establish completion unless the permitted signed increments and endpoint relation are also validated.

7.4.4 — Boundary Access and First Hitting

A boundary is accessible when admissible trajectories can reach it before termination under the declared dynamics. First-hitting

geometry describes approach, contact, crossing, and absorption. Inaccessibility under those dynamics is a direct route failure; apparent access created only by numerical smoothing or changed boundary rules is not recovery (Andersen et al., 1993; Aalen et al., 2008).

Accessibility depends on drift, diffusion, jumps, state constraints, and endpoint hazard. A terminal set may be mathematically present yet polar or practically unreachable.

Simulation should estimate hitting probabilities and times under each declared route and rival, then test calibration on held-out transitions. Numerical schemes need dedicated checks near singular, sticky, reflecting, or absorbing boundaries.

Numerical recovery requires convergence diagnostics, discretization sensitivity, and benchmark cases with known solutions or trustworthy bounds. A simulated hitting pattern that changes materially with step size, boundary convention, or interpolation rule is a numerical result, not evidence about the target route.

7.4.5 — Endpoint Jumps and Support Failure

An endpoint-inclusive model specifies the mark distribution, timing law, sign, magnitude, measurability, ownership, latent-representation link, and compensator (Protter, 2005; Jacod & Shiryaev, 2003; Revuz & Yor, 1999).

Observation is often weakest at the jump. Communication loss, emergency intervention, device failure, sedation, or classification uncertainty can hide the final interval. A route that repeatedly needs an unseen contribution of exactly the required sign is fragile, and support failure must be allowed to retire it.

Sensitivity analysis should bound the unobserved contribution and vary mark and hazard laws rather than impute closure. Support must be separated into topological, effective, and

observed forms: a state can be reachable yet negligible before the endpoint, or model-supported yet absent from the cohort because of censoring and care transitions (Aalen et al., 2008; Andersen et al., 1993).

Simulation should include inaccessible and rare boundaries, wrong support classes, missed jumps, device shutdown, sedation, and false endpoint labels. Negative-control channels and administrative timing can reveal whether an apparent terminal atom belongs to the latent process or the measurement system.

Outcomes remain branch-specific. Strict-pre-endpoint closure may fail while an endpoint-inclusive branch remains open; a first-hitting route may fail while state-coupled hazard survives; observation may be inadequate while truth-level status remains unresolved. Failure should localize the branch rather than create an all-purpose terminal rescue.

7.5 — Bridge, Selector, and Admissible-History Distinctions

Several mathematically distinct constructions can produce paths that approach a terminal target. A native dynamical law can drive the process. A bridge can condition ordinary dynamics on a future endpoint. A selector can retain only histories that close. An intervention can alter the path. A protocol can observe only the subset that survives to measurement (Doob, 1957; Föllmer, 1985).

7.5.1 — Native Dynamics Versus Conditioning

The constructions answer different causal and probabilistic questions. Their off-endpoint predictions may differ, but separate constructions need not induce different observed laws (Section 16.4.3).

$$L^h f(x,t) = Lf(x,t) + a(x,t)\nabla log h(x,t) \cdot \nabla f(x,t)$$

$$h(x,t) = Pr_{x,t}\left[X(T) \in A\right]$$

For the displayed diffusion case, $\mathscr{L}$ is the baseline spatial generator and $a = \sigma\sigma^T$ is its local covariance matrix. The formula is used where h is positive and sufficiently regular for the generator calculation. A zero-probability point endpoint instead requires a transition-density or other justified regular-conditional bridge construction; its zero event probability cannot be substituted into log h. The Doob h-transform modifies the drift of the baseline process under the declared conditioning. It is a bridge construction whose predictions must be compared with an unconditioned native process and with selection mechanisms.

Native dynamics changes the forward generator or state law; perturbations and behavior away from the endpoint help

distinguish it. A bridge or h-transform instead conditions the path law on a terminal event, making dependence on endpoint information a distinguishing feature.

An admissible-history selector changes the support or weight of complete paths. Its constraint must be fixed prospectively so that path selection cannot be chosen to obtain the observed result.

A policy or intervention changes the action or causal transition law; controlled counterfactual responses distinguish it. Observation selection changes which data are recorded; missingness and measurement-gain mechanisms are the relevant diagnostics.

Conditioning a baseline process on a terminal event can induce singular or steepening drift near that time even when the baseline process has none. Observing steepening drift is therefore not uniquely diagnostic of a controller (Doob, 1957; Föllmer, 1985).

Endpoint conditioning can manufacture convergence by construction. A bridge earns empirical relevance only if a frozen conditioning rule predicts held-out transitions, interventions, or endpoint-support behavior beyond the event used to define it. Fit to the conditioned endpoint alone is circular.

The split should be enforced at the data level. One module fixes the endpoint class and terminal set; another, held out from that choice, scores pre-endpoint drift, covariance, or hazard predictions. Cross-fitting or external replication can recover power, but no observation may both define the ending and serve as evidence that the path anticipated it.

7.5.2 — Admissible-History Sets

An admissible-history model restricts the path space to trajectories that satisfy a global boundary condition. The

restriction can be written as a constrained measure over complete histories (Léonard, 2014; Léonard et al., 2014; Chetrite & Touchette, 2015).

Global path constraints differ from local control because they need not produce an ordinary forward-time mechanism. They may be time-symmetric, variational, or boundary-conditioned.

Empirical consequences must be derived from the constrained ensemble, including preterminal correlations, perturbation responses, and transport across endpoint classes.

A path-integral construction may assign zero weight to nonclosing histories while preserving ordinary-looking local dynamics. Identifying this construction empirically requires predictions that distinguish it from the declared rivals.

An admissible set chosen after observing the path is selection, not a risky law. The constraint and baseline measure must be specified independently.

7.5.3 — Bridge Laws and Doob h-Transforms

A bridge law inherits its baseline generator and terminal event; errors in either can enter the induced dynamics (Çetin & Danilova, 2016; Kallenberg, 2002).

A bridge to zero at a fixed time produces a characteristic pull that depends on remaining time. A random or uncertain endpoint produces a mixture of such pulls and may soften the apparent singularity.

The transformed drift can resemble the reserve-coupled drift. Conditional slope alone therefore cannot discriminate the routes. If the native and conditioned laws remain indistinguishable on held-out pre-endpoint data, the evidence cannot identify which construction generated the path.

7.5.4 — Selectors, Policies, and Interventions

A selector determines which histories or actions enter a constructed model, a policy assigns action probabilities, and an intervention changes the data-generating system. These operations can produce similar trajectories while answering different causal questions. Each requires its own estimand, assumptions, and identification strategy (Doob, 1957; Föllmer, 1985; Pearl, 2009).

A policy-support condition does not identify psychological motivation, conscious ledger access, endpoint knowledge, or computation of completion probabilities. If every available action already completes with probability one, the condition is exact but behaviorally silent.

Selectors require measurable and time-consistent construction. A completion-compatible set at one history is not yet a policy over time, and a sequence of locally favorable choices need not form an admissible recursively consistent strategy. Empty or nonmeasurable selector sets make the route unavailable.

Observed choices do not by themselves identify intervention effects. Selection adjustment, inverse weighting, g-methods, randomized contrasts, instruments, and sensitivity analysis are useful only when their assumptions match the operation and data. Conditioning on survivors or communicative participants is not evidence that an intervention generated their terminal pattern (Hernán & Robins, 2020).

Off-policy evidence has a bounded warrant. Data generated under one behavior policy identify another policy only under adequate overlap or a defensible structural bridge, with uncertainty from weighting, model extrapolation, and unobserved actions carried into the result. A route observed under one policy therefore cannot silently become an all-policy

or all-intervention conclusion (Hernán & Robins, 2020; Manski, 2003).

Represent state-dependent treatment in the causal model: care policies can produce apparent feedback control.

7.5.5 — Selection and Equal-Information Rivals

Apply Section 4.3.6’s marked observation model when comparing selection with latent dynamics (Little & Rubin, 2019; Aalen et al., 2008).

Apply Section 7.7's common-information contract. A candidate using future-informed reserve must not be compared with a rival limited to baseline covariates.

Denominator flow reports who entered each risk set, who remained observable, why observation stopped, and what other channels still showed. A model that succeeds only among easy-to-observe participants may fail where the terminal claim is most demanding.

Candidate negative controls include technical variables such as battery level and channel outages, provided the protocol independently justifies that they affect observation without changing latent valence. Staffing and prompt schedules require the same exclusion argument because care and measurement reactivity can also alter experience. Outcomes sharing measurement burden but not the proposed completion relation supply another check.

Test latent, endpoint, and observation errors jointly: modest errors can combine into a persuasive terminal pattern.

Bounds, pattern-mixture alternatives, selection models, and negative controls can narrow unobserved selection but cannot create missing information. The verdict states which conclusion survives the prospectively specified observation envelope.

7.6 — The Reserve-Coupled Route as One Candidate

The reserve-coupled route is a candidate realization of the mechanism-neutral constraints. It introduces a declining nonnegative reserve U_i, a signed arithmetic reference process Y_i, and a positive scale constant c^{LU}, then defines a route state L_i^{rc} as their product. This construction generates linked drift and innovation restrictions for empirical comparison (Protter, 2005; Revuz & Yor, 1999).

7.6.1 — Declining Reserve as Route Coordinate

The displays introduce the reference route and its finite-variation special case; Section 7.6.3 discusses when additional covariation terms are required.

$$L_i^{rc}(t) = c^{LU} U_i(t) Y_i(t), c^{LU} > 0, dU_i = -r_i^U dt, dY_i = \sigma_i dW_i$$

$$dL_i^{rc} = -(r_i^U / U_i) L_i^{rc} dt + c^{LU} \sigma_i U_i dW_i$$

In the reference case with a continuous finite-variation reserve, r_i^U is the nonnegative reserve-depletion rate and c^{LU} fixes units between reserve times the normalized coordinate and the route state. The product approaches zero if reserve reaches zero at the relevant route boundary and Y_i does not diverge fast enough to defeat the product. Coincidence with the truth-level endpoint is a separate bridge.

A finite-variation reference reserve satisfies a declared depletion law and remains nonnegative before the endpoint. More general reserves require separately declared dynamics and a product-rule calculation that retains any nonzero covariation terms.

Construct validation should specify raw indicators, units, transformations, context of use, confounds, and transport limits.

Reserve cannot be whatever fitted quantity makes the product close.

Frailty indices, physiological capacity, communication ability, and care intensity may all correlate with endpoint proximity while measuring different constructs.

Reserve proxies can confound age, disability, medication, disease severity, access to care, and social support. Those variables are not interchangeable measures of remaining completion capacity. Apparent depletion that follows them but fails the route's independent construct, cross-equation, sign-specific, perturbation, or transport tests should favor the bias, frailty, or prognosis rival.

Independent reserve validation should precede the product fit. Before outcome inspection, the protocol should state whether reserve is reflective, formative, or merely prognostic; which indicators can move it; and which perturbations should change it without directly changing the ledger. Failure of those cross-construct predictions retires the coordinate even if the product equation fits.

7.6.2 — Signed Arithmetic Reference Dynamics

The normalized process must support positive and negative values because the ledger-like product can lie on either side of zero. An arithmetic diffusion is the simplest sign-flexible reference case (Karatzas & Shreve, 1991; Jacod & Shiryaev, 2003; Särkkä, 2013).

The reference process may later include drift, jumps, stochastic volatility, or state dependence, but each addition changes predictions and must be justified prospectively.

Test held-out residual structure. A latent process flexible enough to absorb every mismatch supplies no test of the route.

The arithmetic reference is a laboratory for consequences, not a claim that phenomenal experience literally follows Brownian motion.

7.6.3 — Product Rule and Induced Ledger Dynamics

The product rule reveals which assumptions make the compact induced equation possible. A continuous finite-variation reserve has zero quadratic covariation with the reference process; continuous martingale components or shared jumps can produce additional covariation terms (Revuz & Yor, 1999; Kallenberg, 2002).

Writing the general semimartingale product before the special case prevents hidden process-class changes. Filtration, adaptedness, integrability, and boundary behavior belong to the statement.

Simulation and estimation should use the same calculus as the derivation. Discretization can introduce spurious drift or conceal covariation near small reserve.

Correlated physiological shocks and experiential shocks would create a covariation contribution absent from the simplest equation. Ignoring it can make the fitted relaxation look stronger.

7.6.4 — Noise, Ratio Fragility, and Transport

Apply Section 4.3.6's noise-floor and observation-gain model before interpreting compression (Karatzas & Shreve, 1991; Jacod & Shiryaev, 2003; Särkkä, 2013).

For $U_i > 0$, the normalized ratio $|L_i^{rc}| \,/\, (c_{LU} U_i)$ equals $|Y_i|$ under the exact product representation and is fragile when numerator and denominator are uncertain near zero. It remains secondary and is reported with both components, covariance, reserve-floor probability, and bounded or Fieller-type uncertainty.

Transport requires a common reserve coordinate, unit, endpoint relation, observation map, and perturbation interface. Refitting coefficients is permitted; redefining reserve after outcome inspection creates a new route rather than evidence of transport.

Freeze thresholds for construct invalidity, denominator instability, cross-equation failure, endpoint desynchronization, and nontransportable parameters before protected analysis. Apply Section 7.7.6 to qualified rival comparisons (Mayo, 2018; Nosek et al., 2018).

Near the boundary, joint errors-in-variables or state-space models propagate ledger and reserve uncertainty together. A stable ratio can hide two unstable quantities, while an explosive ratio may reflect a denominator artifact rather than failed route dynamics (Fieller, 1954; Gleser & Hwang, 1987; Carroll et al., 2006; Särkkä, 2013).

Observation noise sets a floor on estimated compression, while observation gain may collapse as communication or movement declines. Larger samples do not remove systematic observation bias (Karatzas & Shreve, 1991; Jacod & Shiryaev, 2003; Särkkä, 2013).

Late sensor dropout can imitate declining innovation while latent dynamics remain uncertain. Negative controls and joint observation models are therefore essential.

The route narrows or retires when reserve lacks independent validation, endpoint synchronization fails, normalized dynamics become unstable, cross-equation restrictions fail, or serious rivals predict as well or better.

Retirement preserves narrower achievements, including better reserve measures, endpoint models, observation analysis, and a clearer account of what the target alone does not determine.

Transport varies endpoint class, instrument, cadence, clinical state, and observation regime, not merely geography. The same coordinate and cross-equation restrictions must survive with reserve uncertainty carried into drift and innovation estimates.

Useful subcomponents can survive route retirement. A reserve proxy may retain prognostic value, an observation model may improve terminal-window inference, or a state-space decomposition may clarify error. The record migrates each survivor to its narrower status rather than treating retirement as erasure (Mayo, 2018).

7.7 — Discriminating Signatures and Equal-Treatment Rivals

The decisive addition at this stage is an internal consistency restriction: drift, innovation amplitude, reserve-depletion rate, and reserve level cannot be chosen independently under the frozen reference route. Fit those components without imposing the identity, then score their held-out agreement and the rival models under one observation contract (Gneiting & Raftery, 2007; Vehtari et al., 2017).

7.7.1 — Steepening Relaxation

The reference reserve route has a compact cross-equation prediction. For differentiable $U(t) > 0$, depletion rate $r_U(t) \geq 0$, and constants $c > 0$ and $\sigma > 0$, define its mean-reversion coefficient and innovation amplitude by

$$a(t) = \frac{r_U(t)}{U(t)}, \qquad b(t) = c\sigma U(t), \qquad \frac{dU(t)}{dt} = -r_U(t).$$

Then

$$a(t) = -\frac{1}{2}\frac{d}{dt}\log\left[b(t)^2\right].$$

The same reserve trajectory therefore links relaxation strength to the rate of change in innovation variance. Estimate the drift and innovation features without imposing this identity, propagate their joint estimation uncertainty under the common observation model, and compare the linked prediction with alternatives that allow the features to vary separately.

Steepening relaxation requires an increase in the ratio of depletion rate to reserve. A prospective reserve-linked slope must distinguish this restriction from bridge effects or homeostatic reversion.

In the reference case, innovation shrinks with the declared reserve-dependent diffusion factor. Sensor loss and sedation are strong rivals, so the innovation estimate must remain credible under the observation model.

The printed assumptions impose a cross-equation restriction linking relaxation, reserve depletion, and diffusion. Test that drift–diffusion coupling under protocol-fixed rules against a flexible heteroscedastic rival.

The joint fingerprint must predict held-out observations under fair recovery and comparison rules. Relevant rivals include OU, bridge, frailty, medication, and observation-only models.

The reference case with a continuous finite-variation reserve predicts a relaxation coefficient $\kappa_i^U = r_i^U / U_i$. As reserve falls, the coefficient steepens when the depletion rate falls proportionally more slowly than reserve, remains constant, or increases. This is a conditional route signature, not a generic prediction of PTVC (Protter, 2005).

A study should therefore estimate κ or an analogue jointly with reserve. It should ask whether the relaxation steepens as U falls, whether the relation survives confounds, and whether the estimate is available prospectively rather than only after endpoint alignment.

Steepening relaxation should be reported with uncertainty, window sensitivity, and rival fits. A Brownian bridge toward zero has its own diverging pull as T approaches. If the bridge receives the same endpoint information and fits as well, drift cannot support the reserve realization.

7.7.2 — Shrinking Innovation and Variance Structure

The paired reference signature is shrinking latent innovation amplitude. In the displayed special case, the route-state diffusion coefficient is $\sigma_{L,i} = c^{LU} \sigma_i U_i$, so it contracts with reserve when σ_i

and c^{LU} remain stable and the observation bridge is adequate (Protter, 2005; Karatzas & Shreve, 1991).

Time-varying σ is allowed only when independently identified. If σ is simply reduced near the endpoint to fit variance, the model loses content. A stochastic-volatility extension must specify why volatility changes, how it is measured, and what rival it beats.

The innovation signature also depends on sampling cadence. Irregular observation can miss bursts. Late windows may be shorter and less dense. Missingness may be informative. A properly fitted model should account for unequal intervals and not treat silence as smoothness.

7.7.3 — Cross-Equation Restrictions

The reference case yields a cross-equation restriction linking relaxation and innovation through the same reserve: $\kappa_i^U = r_i^U / U_i$ and $\sigma_{L,i} = c^{LU}\sigma_i U_i$, so their product equals $c^{LU}\sigma_i r_i^U$ in the constant-volatility special case. The restriction lowers flexibility because drift and diffusion cannot be fitted independently (Protter, 2005).

That restriction is useful only under its printed assumptions: finite-variation reserve, coherent reserve units, a stable or prospectively modeled σ_i, and estimators for κ_i^U, $\sigma_{L,i}$, and r_i^U that do not share hidden outcome information. If an assumption fails, the equation becomes a sensitivity target rather than a confirmatory restriction.

$$\kappa_i^U(t) = r_i^U(t)/U_i(t),$$
$$\sigma_{L,i}(t) = c^{LU}\sigma_i U_i(t),$$
$$\kappa_i^U(t)\sigma_{L,i}(t) = c^{LU}\sigma_i r_i^U(t)$$

The restriction should be estimated with joint uncertainty and covariance. Near low reserve, errors in U_i affect both estimated

relaxation and innovation, so independent standard errors can overstate agreement. A state-space or errors-in-variables analysis should report whether the apparent product relation survives that shared dependence on reserve.

If the restriction fails, the result is informative. The drift component might fit while innovation does not, or innovation might fit while drift does not. That would suggest ordinary homeostasis, observation noise, stochastic-volatility extension, or failure of the reserve product in that reference class.

A structured-volatility successor must derive its own cross-equation restriction.

7.7.4 — Prospective Endpoint Separation

Apply Section 4.3.5's information-set distinction when evaluating the reserve route's pre-endpoint forecasts (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Prospective separation can be achieved in several ways. The endpoint rule can be protocol-fixed through independent unity-proxy channels. Reserve and ledger estimators can be cross-fitted. Model parameters can be trained on one subset and tested on held-out trajectories. Forecasts can be made for windows not used to define the endpoint.

The Brownian bridge rival should be given the endpoint information allowed by the study design. If the PTVC candidate uses endpoint alignment while the rival does not, the comparison is unfair. If both use the endpoint and perform similarly, the PTVC-specific interpretation weakens.

7.7.5 — Observation-Model Robustness

The joint signature must survive observation rivals. A reserve-linked pattern that disappears when sedation, frailty, medication, missingness, sensor quality, or observation gain is modeled is

not a reserve-realization success. It may be a measurement lesson (Talts et al., 2018; Wilson & Collins, 2019; Munafò et al., 2017).

Observation-model robustness requires more than adding covariates. The study should include generative observation models capable of producing false compression. It should simulate data from those rivals and verify that the analysis can distinguish them from latent reserve coupling.

Use the negative controls and exclusion checks in Section 7.5.5, adding non-affective physiological channels only when their separation from latent valence is independently justified.

7.7.6 — Held-Out Superiority to Strong Rivals

The rival set should include endpoint-conditioned bridges, OU and homeostatic processes, age-only decline, frailty-only decline, medication-only decline, observation-shutdown models, shared-driver disease models, and unconceived-model sensitivity. Each rival should be implemented as a generative model where possible.

Held-out superiority should be evaluated on components that matter: drift, innovation, reserve coupling, endpoint forecasting, variance shape, observation robustness, and prediction calibration. A single aggregate score can hide a prerequisite failure. The gates should remain noncompensatory (Gneiting & Raftery, 2007).

Held-out advantage can support the reserve-realization component. Rival parity narrows or suspends PTVC-specific interpretation. Rival superiority retires the tested realization in that reference class. Apply Section 5.7.5 to any successor.

7.7.7 — Conjunctive Signatures

Analyze the signature graph jointly: shared measurements, route validation, and endpoint assumptions create dependencies between tests. Sealed-world recovery must check that repeated views of one signal are not counted as independent confirmation.

Exploration may compare proxies, lags, volatility models, and kernels, but confirmation freezes the surviving objects, tuning ranges, and scores in a new cohort or protected site. Negative controls should include false endpoints, shuffled route states, synthetic dropout, bridge generators, and ordinary mean-reverting worlds; a false-support rate above the predeclared tolerance disqualifies the pipeline.

Use an independent adversarial team and code, including reasonable alternative implementations. Test transport of the coordinate, units, depletion direction, endpoint relation, and joint restriction (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

Record separately route failure, compatible drift with failed innovation, unavailable endpoint separation, rival preference, bridge failure, and open-set inadequacy. Classify the resulting evidence under Section 24.2.

7.8 — Formal Failure, Open Theorems, and Route Retirement

The formal consequences leave a set of bridge questions that determine how a candidate realization can bear on the received ledger. Each question needs a mathematical object, a proof obligation or counterexample, and a stated consequence of failure. This register connects those obligations to the empirical claims that depend on them.

7.8.1 — Integral-to-Route Bridge Failure

The integral–product bridge is a route-specific mathematical question. On the absolutely continuous chart, the phenomenal ledger accumulates a qualified valence rate under a declared exposure rule. An ordinary integral of an integrable rate has finite variation; a Brownian reserve product can have nonzero quadratic variation. A bridge between these descriptions must state whether it is exact or approximate and at which level it applies.

A transparent finite-variation alternative is available. Take time zero to be the declared stream onset, not enrollment. Fix a deterministic horizon $T>0$ and a deterministic continuously differentiable reserve U on $[0,T]$, with $U(t)>0$ for $t<T$ and $U(T)=0$. Let W be standard Brownian motion and let Z have the stationary Ornstein–Uhlenbeck law below, with $\beta>0$, $\sigma>0$, and $Z(0)$ Gaussian with mean zero and variance $\sigma^2/(2\beta)$, independent of future Brownian increments. Take $c>0$:

$$dZ = -\beta Z\,dt + \sigma\,dW, \qquad Y(t) = \int_0^t Z(s)\,ds.$$

Define the candidate ledger and its contribution rate by

$$L_{fv}(t) = cU(t)Y(t).$$

$$r_{fv}(t) = c[U\prime(t)Y(t) + U(t)Z(t)].$$

The ordinary product rule gives $Lfv(t)=\int_0^t rfv(s)ds$ and $Lfv(T)=0$. Continuous sample paths of Z make the rate continuous on this finite interval. The ledger therefore has finite variation and zero quadratic variation. With dimensionless U and Y, Z has units of inverse time; β has the same units; σ has units of time to the power −3/2; and c has ledger units.

This alternative also yields a cadence test. At an interior time, conditional on the full current state (Y(t),Z(t)), the leading ledger-innovation variance is $c^2U(t)^2\sigma^2\Delta^3/3$ as Δ decreases to zero. A nondegenerate diffusion ledger instead has variance of order Δ. Uncertainty about the current rate can contribute an order-Δ^2 term, while independent additive sampling error produces an increment-noise floor. The comparison must model these effects rather than infer cubic scaling from raw differences. It can be investigated without observing an entire completed stream.

The construction establishes exact rate integration and closure for this specified alternative. Its horizon and vanishing reserve are imposed model conditions, not identified biological findings. A phenomenal application still needs a qualified rate and exposure relation and an independently justified stream endpoint.

Sections 18.4.4 and 29.2.3 develop other finite-variation constructions; comparing their assumptions and observable consequences is a concrete next step.

7.8.2 — Reserve Construct Invalidity

Reserve construct validity requires a measured object rather than a label assigned by model fitting. Proxy instability, bias, or shared measurement artifact prevents the route from being treated as instrument-ready.

Apply Section 7.6.1's independent construct definition and frozen validation criteria to assess reserve failure in each reference class (Mayo, 2018; National Academies of Sciences, Engineering, and Medicine, 2019; Vehtari et al., 2017).

Evaluate reserve-proxy misclassification across disability, age, sedation, communication capacity, and baseline physiology.

If reserve cannot be measured independently in a reference class, the reserve realization may still be a formal possibility, but it is not empirically testable there. The correct verdict is not support; it is not-yet-instrument-ready or unusable for that context.

7.8.3 — Endpoint Synchronization Failure

Endpoint synchronization requires reserve collapse to coincide with an independently determined endpoint. Systematic early or late mismatch requires narrowing the reference class or revising the route.

Endpoint synchronization failure occurs when reserve collapse and independently defined truth-level stream endpoint do not align in the relevant way. This can happen if reserve reaches near-zero while conscious stream activity continues, or if the endpoint occurs while reserve remains materially positive (Munafò et al., 2017; Nosek et al., 2018).

Synchronization failure should be measured prospectively where possible. Reserve trajectories and endpoint-proxy trajectories should be compared under rules that allow early reserve collapse, late endpoint, sudden endpoint, ambiguous endpoint, and no-estimate outcomes.

If synchronization repeatedly fails in a reference class, the reserve realization narrows for that class. A different reserve definition may be proposed, but only as a new version with new risky predictions.

7.8.4 — Normalized-Process Explosion

The counterexample Y equals one over U shows why normalized-process explosion can prevent closure. As reserve declines, the normalized process grows too quickly for the product to approach zero. The product may remain away from zero or become unstable (Popper, 1959; Lakatos, 1978).

Explosion can be mathematical, biological, or observational. Mathematically, the process class may permit growth rates that defeat closure. Biologically, terminal distress, delirium, pain, fear, or conflict may create large normalized excursions. Observationally, denominator error can make a stable process look explosive.

The model must distinguish these possibilities. A joint posterior for the route state, U, and Y can reveal whether explosion is robust or driven by reserve uncertainty. Observation-quality metrics can reveal denominator artifacts. Clinical covariates can reveal treatable causes.

The decisive closure question is whether the product approaches zero. Divergence of Y alone is insufficient: for $U > 0$, the mathematical example $Y = U^{(-1/2)}$ gives $UY = U^{(1/2)}$, which still tends to zero as U tends to zero. A finding can reject a bounded-Y or fixed-volatility subbranch without rejecting every closing product. Failure of the required product limit, after accounting for denominator and observation error, is adverse to the declared reserve route.

7.8.5 — Rival-Equal or Rival-Better Prediction

Rival comparison is necessary because adequate alternatives may improve held-out prediction under common rules with fewer strained assumptions. Such dominance requires revision or retirement of the candidate version.

Apply Section 7.7.6's route-specific verdict rules. The remaining question here is how the comparison responds to a serious rival outside the original set (Mayo, 2018; Vehtari et al., 2017).

After a favorable reserve result, test serious new rivals under the same protected-data and recovery rules; the named set does not exhaust possible explanations.

7.8.6 — Formal Change Control and Route Retirement

The integral–product bridge connects the formal ledger to the stochastic candidate. Without a defensible mapping of path class and dimensions, the route requires a formal change request or reduced status.

Apply Section 5.7.5’s versioning rule to any new process class, reserve definition, endpoint convention, bridge, or policy assumption: each receives a new identifier and dependency record.

Use Sections 5.7.5 and 28.2 to compare retired and successor routes. Rerun results depending on changed objects or mark them nontransportable, preserving independently valid components.

Part III — Measurement and Psychometrics

The same response can carry different information in different settings. A rating made during a brief prompt, a recollection at the end of a day, and an answer given after treatment may reflect different temporal windows and response processes. Measurement begins by deciding what each observation is an observation of. Part III develops that decision into a bridge from experience to quantities that can be compared across occasions and examined by another investigator (American Educational Research Association et al., 2014).

Chapter 8 starts with the target and the evidence already available. Circumstances affect experience, while memory, preference, and behavior reveal particular aspects of it. Reading the established literature through those distinctions helps identify which findings can guide a new instrument and which questions remain for direct investigation. It also gives the program an empirical starting point: existing results can shape the next study without having to answer a question they were never designed to address.

Chapter 9 follows an observation through the measurement pipeline. The participant interprets a prompt, chooses a response, and encounters a device or interviewer; the resulting record is scored and modeled. Each step can preserve, distort, or omit information. A response-process account explains how those steps connect. It gives an investigator specific places to look when two channels disagree, instead of treating disagreement as a single undifferentiated source of noise (Campbell & Fiske, 1959; Borsboom, 2005).

The observation schedule also participates in the process. A prompt may interrupt sleep or arrive when a participant cannot respond. Device use can change with discomfort or clinical care. Recording those conditions makes it possible to study why observations are present at particular times. Participant welfare

and proportionality consequently influence the design of the measurement system itself (World Medical Association, 2024; International Council for Harmonisation, 2026).

Chapter 10 examines the comparisons that a scale permits. Calibration must address origins, units, and links across contexts before a sequence of responses can support an accumulated quantity. Error can enter through the instrument, through linking, or through temporal exposure. Following these sources separately shows where a better sensor would help and where the relevant improvement is a different design. The chapter gives particular attention to the uncertainty that remains after a plausible calibration has been fitted.

The aim is a measurement account another team can reconstruct. It should explain the choice of measure, the modifications made, and the evidence supporting its proposed use (U.S. Food and Drug Administration, 2025b; Gagnier et al., 2025; Flake & Fried, 2020). That account becomes the common reference for the later study families. It lets their disagreements be traced to a substantive feature of experience or to a specific difference in how experience became data.

Chapter 8 — Lived Experience and the Evidence Baseline

Consider a person keeping vigil beside a hospital bed after good news that arrived too late. Relief, grief, love, exhaustion, anger, and gratitude coexist. A momentary rating, an account of the ending, and the decision to stay describe different aspects of this situation. The chapter begins with the question of what each observation could represent (Russell, 1980; Cacioppo & Berntson, 1994).

Material circumstances matter. A quieter room, reliable care, food, or somewhere to rest can change the experience of the vigil. Their effects pass through bodily condition, perceived safety, dependency, expectation, and meaning. The same provision can relieve one burden while leaving another untouched; its availability does not determine a fixed amount of felt benefit.

Remaining at the bedside may express a deeply held priority while also prolonging exhaustion. Choosing to stay therefore does not identify how the hours feel. Nor would choosing rest prove that the relationship mattered less. A useful representation must allow agency, constrained alternatives, and experienced cost to occupy different roles in the explanation.

This matters when two accounts of the vigil appear to conflict. A person may judge the time worthwhile and wish some of its pain had been absent. Rest could ease bodily strain without replacing the meaning of being present. Treating either judgment as the complete account would erase a relevant distinction.

Mixed feeling creates a further choice. A bipolar rating may place the person near its midpoint, while separate pleasant and unpleasant reports preserve substantial amounts of both. These representations make different claims about which states count as alike. Their performance must be examined before a difference score becomes a summary of the whole experience (Cacioppo & Berntson, 1994; Larsen et al., 2001).

Time complicates the comparison. A memorable ending can organize the later story of a long vigil without describing every ordinary interval. Momentary records and retrospective judgments can help explain one another, provided the study keeps their temporal targets distinct. Duration, sequence, anticipation, and recovery then become features to investigate rather than weights silently supplied by memory.

Agreement among channels also depends on their intended roles. Indicators modeled as effects of one state face different tests from components that jointly constitute a score. A mismatch can expose an omitted dimension or a response process; correlation alone cannot decide between those explanations (Campbell & Fiske, 1959; Cronbach & Meehl, 1955; Bollen, 1989).

Fairness appraisal adds a social layer: the care may feel respectful, neglectful, deserved, or humiliating, changing the experience and its interpretation. Section 8.7 examines these pathways through allocation, cooperation, comparison, and cultural evidence.

The aim is to establish which distinctions a proposed representation can sustain. Chapter 9 can then design observations around that target instead of allowing an available score to define it.

8.1 — Why Lived Experience Is the Target

The proposed target is lived experience, so the representation must begin with what its signs, magnitudes, and temporal units are intended to describe. Reports, behavior, and physiology offer different routes to that target. Clarifying their roles makes it possible to ask whether a signed accumulation is defensible before selecting an instrument that merely produces convenient numbers.

8.1.1 — Experience Over Appearance

Visible functioning can conceal burden even when the observations are accurate. Paid rent, kept appointments, a reliable smile, or completed work do not exhaust what the stream carries (Russell, 1980; Borsboom, 2005).

The Law of Fairness places lived experience at the target because hidden burden often survives inside acceptable appearance. A person may comply because dependence makes refusal costly. They may succeed while carrying chronic pain. They may perform calm under threat because panic would invite punishment. These are not literary exceptions; they are ordinary routes by which external observers can miss the carried cost.

Measurement cannot become a gate to assistance or to taking reported suffering seriously. Visible harm already calls for attention and repair under Chapter 3's care and rights commitments (National Commission, 1979).

8.1.2 — Goods Become Lived Conditions

Resources have no fixed affective conversion. A public benefit may bring relief or surveillance; an apology may repair dignity or reopen dependence. The construct map must identify which pathway occurred.

Consider secure housing as a possible pathway to better sleep. Temperature, privacy, reduced threat and continuity can each contribute, while a move may also change familiar relationships and daily routines. The material event, the mediating conditions and the resulting experience are distinct observations. Recording their timing helps explain why the same resource can have different experiential profiles across people and why its effects can unfold at different rates (Cronbach & Meehl, 1955; Bollen, 1989).

8.1.3 — Preferences Are Not Experience

Preference is evidence of something, but it is not identical to experience. A person may choose what is least dangerous, most familiar, socially required, economically possible, or least humiliating among bad options. Revealed preference can record constrained survival rather than felt relief. Willingness to pay can record wealth, bargaining power, hope, fear, or lack of alternatives (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

Adapted deprivation exposes a special failure of preference inference. People may learn to want less when asking is punished, costly, or hopeless, and may report an unjust condition as acceptable because imagining alternatives is itself painful. The absence of expressed demand for relief can therefore be an effect of constraint rather than evidence of low burden. Preference data remain informative, but only after the feasible set, incentives, and adaptation process are modeled.

Preferences can reveal priorities, tradeoffs, agency, practical constraints, valued outcomes, and acceptable participation burdens.

Choice data are especially vulnerable to scarcity and path dependence. A repeated choice can reflect learned feasibility,

switching cost, habit, coercion, or the expectation that alternatives will disappear. Preference elicitation should therefore record the option set, information, time pressure, consequences, and reversibility of the choice. Without those conditions, a ranking may describe constrained behavior while being reported as a ranking of felt value (Sen, 1999; Kahneman et al., 1993).

8.1.4 — Life Satisfaction Is Not a Momentary Rate

Life satisfaction is a reflective judgment about a life, period, role, or condition. A life judged worthwhile may still contain long periods of fear, pain, boredom, exhaustion, or grief; that evaluation is not a momentary contribution rate (Shiffman et al., 2008; Stone et al., 2007).

Life evaluation can emphasize an episode's ending, meaning, or narrative rather than its full temporal distribution. Section 8.5 tests those memory and path-function alternatives.

Aspiration, modesty, comparison, gratitude, religious interpretation, and social desirability can affect life-satisfaction reports. High scores may reflect peace or resilience, lowered expectations, fear of complaint, or meaning alongside suffering.

The distinction can be studied directly. Kapteyn et al. (2025) compared life satisfaction with momentary affect around major life events in adults aged 50 and older. On separately standardized scales, affect changes were generally smaller than changes in life satisfaction. The affect bursts occurred two to six weeks after the events, and life satisfaction was usually recorded earlier, so the design leaves immediate reactions unresolved. This creates a useful next test: collect both outcomes on aligned schedules, with voluntary reports near event onset and later follow-up. Preserve evaluative judgment and momentary feeling as separate outcomes, and estimate their timing and magnitude

before testing any cumulative relation. Life satisfaction can thereby illuminate meaning and appraisal while an independently qualified time-local measure addresses the phenomenal trajectory.

8.1.5 — Meaning and Dignity Matter Within Feeling

Meaning, dignity, agency, violation, loyalty, awe, shame, moral injury, and love can change the felt quality of a state. They belong in the construct when they alter experience (Cronbach & Meehl, 1955; Borsboom, 2005).

Consider pain under two contexts: pain inflicted as humiliation and pain endured in chosen care for someone loved. The sensory intensity may overlap while the experience differs in agency, meaning, threat, resentment, and dignity. The second case does not make pain morally owed payment or credit. It shows that pain is not fully represented by nociceptive intensity alone.

Brief humiliation can have prolonged felt consequences; respectful treatment can reduce fear or restore agency without eliminating illness or poverty. Their temporal effects require measurement alongside immediate intensity.

Meaning must be represented without being assigned a permanently positive coefficient. It can sustain action, deepen commitment, reframe pain, intensify loss, or coexist with severe distress. A model that automatically lets meaning compensate suffering is as misspecified as one that excludes meaning entirely. The relevant question is how meaning changes phenomenal structure in the declared context, not whether it belongs on a universally favorable side of a ledger.

Some dimensions may be noncompensatory within the scientific representation. Severe violation, loss of agency, or humiliation may not be responsibly traded against comfort by

one fixed weight in every context. Thresholds, lexicographic rules, vector outcomes, or identified sets can preserve that structure instead of forcing every dimension into a smooth utility surface. This approach prevents the measurement model from erasing distinctions that participants, mechanisms, and evidence treat as qualitatively important (Manski, 2003; Borsboom, 2005).

8.1.6 — Material Justice Does Not Wait

No one waits for affective science to become precise before receiving care, rights, safety, food, housing, medicine, disability support, or protection. Measurement cannot become a gate a person must pass before suffering is believed; people remain people before they are research cases (National Commission, 1979).

That separation is methodological as well as ethical. If report is unavailable, a sensor intolerable, or representation contested, the study has not earned its intended bridge. The person's experience and claim to care do not thereby disappear.

Care conditions shape data quality. Fear of losing treatment, housing, access, family trust, or status can alter reporting and participation, while deprivation can select for unusual endurance. Accessible responses, compensation, transport, flexible scheduling, independent care, and immediate stopping therefore belong to design (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

A protocol specifies dignity-preserving, reason-coded abstention paths before collection. It records which link failed, what remains interpretable, which claims withdraw, and which care continues unchanged. Uncertainty stays with the instrument and study.

The rule holds under every scientific outcome. Evidence consistent with PTVC does not excuse present suffering, and evidence against it does not diminish the urgency of repair. Rights and relief are independent of a terminal-boundary verdict.

Measurement can still contribute to justice by exposing burdens that appearance misses without creating new gates or dossiers. The achievement is a bounded account that informs research and care while preserving refusal, privacy, and nonparticipation.

The representation questions concern ordering, interval meaning, additivity, sign, and transport. When a required property fails, the program narrows its claim rather than repairing the person or delaying justice.

8.2 — Representation Before Measurement

The order of obligations matters: identify the bearer and target; establish ordering; identify the neutral origin and sign; justify distances and units; test composition; define the temporal path; only then attempt cross-person linking. A later failure does not erase a valid earlier relation, but it blocks every claim that depends on the missing rung. The ladder therefore preserves partial knowledge without promoting rank, local change, or within-stream meaning into a common cardinal scale.

8.2.1 — Ordinal Versus Interval Structure

Ordinal qualification asks whether states can be ordered as more or less burdensome within the context. It permits comparison of direction and rank before experience can be summed. Where a common ordering fails, use qualitative or partially ordered descriptions.

Interval qualification asks whether differences retain stable meaning across nearby states. If they do, local rates and changes may be estimated under the declared conditions. If they do not, additive-ledger claims must narrow to rank, threshold, or local contrasts.

Ordinal structure supports ordering; interval structure also gives stable meaning to differences. Integrating a rate requires the latter: rankings alone cannot establish that successive gaps carry comparable magnitude (Krantz et al., 1971; Borsboom, 2005).

Numbered ratings require tests of spacing through response models, anchoring, repeated measures, invariance, and prediction; the printed numbers do not supply interval structure.

Ordinal structure can support more science than it first appears to permit. Repeated pairwise comparisons may identify stable order relations, detect reversals, and test whether one context

preserves another context's ranking even when equal intervals are unavailable. The key is to report the structure actually identified. A robust order is not an interval scale, but neither is it mere impression; it can constrain models, reject proposed scalarizations, and define partial-order estimands that later work must respect (Krantz et al., 1971; Borsboom, 2005).

8.2.2 — Cardinality and Additivity

Additivity asks whether momentary contributions can be accumulated meaningfully over time. A justified additive representation makes path integrals candidate summaries; otherwise duration and ordering require separate path functionals.

Meaningful magnitude does not by itself justify summation or transport. The ledger requires a defensible composition rule and temporal interpretation; the measurement-model distinctions in Section 8.6.4 determine how its components should be tested (Cronbach & Meehl, 1955; Bollen, 1989; Borsboom, 2005).

Consider two short intervals of mild fear and one continuous interval of the same combined duration. An additive account concerns the time-local contributions actually experienced. If the continuous interval changes through exhaustion, habituation, or anticipation, equal clock duration and the same verbal label need not imply equal contributions. Relief between the short intervals also has its own experienced contribution. A useful comparison therefore separates changes in momentary experience from changes in the rule for accumulating it. Retrospective judgments can then be studied as a further outcome, rather than silently substituted for the path being accumulated.

Weights applied to arbitrarily scaled dimensions can encode instrument conventions rather than lived tradeoffs. Internal consistency cannot supply meaningful units and may be

irrelevant when components are not interchangeable effects of one latent state (Bollen, 1989; Flake & Fried, 2020).

One practical starting instrument is the Labeled Hedonic Scale. Lim, Wood, and Green (2009) used magnitude estimates from trained participants to position its verbal labels, then compared ratings of food names with magnitude estimation and a nine-point hedonic scale. A PTVC qualification study can test within-person cross-sign consistency across sessions and contexts while independently varying exposure duration. Retain time-local ratings so changing sensitivity can be modeled, and test the proposed summation rule separately from retrospective judgments of whole episodes. Freeze origin and scale parameters before examining cumulative totals. Agreement between rating methods then supports a declared calibration task; temporal addition earns its own test.

8.2.3 — State Versus Rate

A state describes a moment; a rate describes contribution per unit time. A state estimate can enter an integral only through a justified rate mapping. Its intensity alone does not specify duration, recovery, anticipation, or aftereffects, and multiplying the score by time does not establish valid units (Russell, 1980; Cacioppo & Berntson, 1994).

The distinction becomes especially important for sustained conditions. A background dread that is low in momentary intensity but present for months can dominate a life if treated as a rate. A sharp acute pain may dominate memory while occupying little clock time. The model must decide what time-local contribution means before it claims cumulative burden.

Classify each source as a state, rate proxy, transition marker, recovery cost, anticipatory cost, or remembered evaluation before combining it.

The state-rate distinction also determines sampling design. A slowly changing state can be represented by sparse observations only under assumptions about interpolation, while a rapidly changing rate requires sampling fast enough to avoid aliasing. If the instrument’s temporal resolution is coarser than the process, the apparent interval contribution may reflect the observation schedule more than the experience. The measurement map should therefore state the timescale on which the construct is defined and the timescale on which the estimator is permitted to speak (Shiffman et al., 2008; Stone et al., 2007).

8.2.4 — Within-Person and Between-Person Meaning

Within-person and between-person meaning are separate achievements. A stable local scale may support idiographic tests of temporal shape, event response, and anchors without permitting direct comparison with another stream (Bollen, 1989; Borsboom, 2005).

Identical prompts do not establish shared quantities. Language, physiology, disability, trauma history, medication, display rules, and social expectation can change the response process. Interpersonal arithmetic requires evidence that origin, sign, unit, and response process survive the link (Putnick & Bornstein, 2016; International Test Commission, 2018; American Educational Research Association et al., 2014).

Within-person and between-person associations can have opposite signs. Population structure may inform calibration, priors, and rivals, but a group difference cannot substitute for the stream's temporal trajectory (Molenaar, 2004; Hamaker, 2012).

8.2.5 — Neutral Origin and Sign

Sign and origin qualification requires a reference for positive, negative, or neutral contribution that was not selected from the

outcome. Without that reference, neutrality claims must abstain or use only signatures that do not depend on the disputed origin.

The formal core owns the abstract origin problem; lived measurement supplies the empirical ways an apparent midpoint can mislead. A person may select ‘neutral’ because nothing dramatic is happening, because language offers no better option, because dissociation has flattened access, because expectations have narrowed, or because a midpoint feels socially safe. Those response processes define different candidate origins and must be separated before zero acquires cumulative meaning (American Educational Research Association et al., 2014).

Sign depends on construct. High arousal can be joyful, terrified, angry, sexually excited, or medically distressing; low arousal, peaceful, depressed, exhausted, or sedated; pain, threatening, chosen, meaningful, humiliating, or ignored. A one-dimensional physiological sign rule fails when context changes meaning.

Apply Section 6.2.4's independent-anchor procedure to repeated judgments, contextual interpretation, and auxiliary physiology (Krantz et al., 1971; Cacioppo & Berntson, 1994; American Educational Research Association et al., 2014; Putnick & Bornstein, 2016).

When neutral origin is weak, output should become an identified region or directional statement. The study may report decreased burden, movement toward an operational neutral band, or indeterminate sign, but not claim exact zero was observed.

Neutrality can be studied through paired tests of language and elicitation. Gasper and colleagues (2021) found that neutrality reports could coexist with positive or negative reports, including when presence was asked before intensity. A pilot can preserve separate pleasant, unpleasant, and neutral responses, keep

uncertainty and unfamiliar terms distinct from absence, and ask what participants mean by neutral. Wells and colleagues (2026) compared thirty-nine purportedly neutral films across several neutrality indices, identifying candidates that elicited relatively low affect. Use such clips as candidate conditions for within-person checks while independently assessing felt neutrality and mixed experience. These comparisons can identify repeatable anchors whose meaning survives changes in wording and context, independently of any desired cumulative-balance result.

8.2.6 — Representation Failure as Scientific Result

Within-person transport asks whether the representation remains stable across states of the same stream. Longitudinal comparisons require version labels and drift checks; failed transport requires state-specific models or an end to the affected phase comparisons.

Between-person transport asks whether the representation retains its meaning across persons or cultures. Supported population summaries remain bounded by their tested reference class. Unsupported extensions remain local, personal, or group-specific.

A representation can fail honorably. Failure of a scalar, additive, signed, or transportable representation is substantive, not a setback hidden by revised language. It shows one empirical bridge cannot carry the claim (Manski, 2003).

Representation can fail in several ways. Reports may remain ordinal; dimensions may resist stable weights; mixed states may be incomparable; groups may use different response processes; clinical states may alter item meaning; physiological channels may track arousal or illness rather than valence; and temporal aggregation may be path-dependent.

Such findings do not deny lived experience. They reject a proposed simplification. The person remains real, burden remains ethically significant, and care remains urgent. The science may move from scalar to vector language, common to within-person scope, or additive to path-functional rivals.

Abstention is legitimate. A study may return no scalar estimate for a person, context, group, or interval when representation gates fail. That output is more informative than a number forced from incompatible dimensions, uncertain signs, or noncomparable scales (Manski, 2003).

The hypothesis becomes more scientific when representation failure can narrow it. A program that can lose scalarity, additivity, or comparability at the first gate has exposed real ways to be wrong rather than protecting a grand claim behind flexible language.

8.3 — Mixed Affect and Multidimensional States

Mixed feeling challenges a one-dimensional representation because positive and negative affect can coexist while arousal, control, meaning, and intensity vary independently (Russell, 1980; Cacioppo & Berntson, 1994; Dejonckheere et al., 2022). A net score may therefore hide distinctions that matter to prediction or interpretation. The task is to determine which dimensions the evidence requires, when states can be ordered, and where clinical or cultural variation makes a shared scalar description inadequate.

8.3.1 — Coexisting Positive and Negative Affect

A bipolar rating or difference score can collapse high positive and negative activation into the same result as low-arousal neutrality. A bivariate design permits the components to be present, absent, or unequal together. Compare their predictive and discriminant performance (Cacioppo & Berntson, 1994; Dejonckheere et al., 2022).

Vaccaro and colleagues (2024) analyzed fMRI from 27 participants watching a short film. After scanning, participants replayed it and retrospectively marked positive, negative, and mixed feelings from the original viewing. Anterior cingulate and ventromedial prefrontal patterns showed consistency during reported mixed states. These findings give mixed-feeling models a neural comparison target, with recall and temporal resolution represented explicitly.

A next comparison can preserve separately elicited components and ask which patterns survive a fixed change of resolution, using Section 5.4.3's overlap bound alongside models of rapid alternation and report generation. That is a proposed extension, not an analysis reported by the study.

Neural consistency identifies a useful measurement question; calibrated component magnitude remains a separate task.

8.3.2 — Valence, Arousal, Control, and Meaning

Dimensional models are useful because they make this pressure visible. A valence-arousal plane already shows that high-energy positive and low-energy positive states are not identical. Adding control or dominance distinguishes anger from fear, confidence from panic, and helpless pain from chosen effort. Meaning adds another layer because the same bodily state can be interpreted as sacrifice, violation, endurance, or absurdity.

Clinical forms also differ: depression can involve low arousal, anhedonia, agitation, numbness, or self-condemnation; anxiety can involve anticipatory simulation, bodily vigilance, and social threat. These distinctions affect what the instrument must represent.

8.3.3 — Intensity and Depth

Intensity, persistence, and life-structuring effects are different properties. An acute insult may fade while a persistent threat shapes later attention and action; representation must distinguish magnitude, duration, salience, and temporal reach (Russell, 1980; Cacioppo & Berntson, 1994).

Beyond intensity, test whether a state changes agency, identity, relationships, or capacity to recover. These effects may persist below a dramatic peak.

Depth also matters in positive experience. Awe, trust, belonging, dignity, and peace may not always score as high-intensity pleasure. Yet they can alter the structure of the stream, reduce defensive effort, open attention, and change the meaning of time. A crude pleasure scale may undervalue them.

A useful state description separates at least immediate magnitude, persistence, salience, functional disruption, recovery cost, and felt significance. A particular study may estimate only a subset, but it should state which dimensions are present, which are omitted, and whether omission changes the comparisons the estimator is expected to support.

Ceiling effects can compress severe states, while verbal categories change with context. Apply the scaling tests in Sections 8.2.1 and 10.2.3, reporting where saturation or response style leaves the intended magnitude unidentified (Krantz et al., 1971; American Educational Research Association et al., 2014).

8.3.4 — Ambivalence and Incomparability

Ambivalence is not indecision alone. It can be the felt presence of reasons, attachments, fears, loyalties, and losses that do not collapse onto one preference or one affective sign. A person may want and dread the same outcome. They may love someone and fear them. They may feel gratitude for care and humiliation at needing it (Cacioppo & Berntson, 1994; Campbell & Fiske, 1959).

Incomparability appears when two states cannot be honestly ranked on a single line without losing what matters. Is numbness better or worse than panic? Is meaningful pain better or worse than empty comfort? Is anger with agency better or worse than despair with quiet? Sometimes the answer is context-specific. Sometimes the representation should remain partial.

Ambivalence also affects report. People may describe a state as fine because no single label fits, or because naming the conflict would create social consequence. A measurement system that demands one rating can prevent participants from expressing that conflict accurately. The construct should allow

people to say that a state is mixed, unresolved, incomparable, or not representable in the offered categories.

8.3.5 — Vector-State Alternatives

Vector dimensions still require observation models: a pain coordinate is not a questionnaire item, and social safety is not the mere presence of another person.

A vector representation permits dimensions to evolve on different clocks and respond differently to intervention. Pain intensity may change quickly while fatigue, control, meaning, or social safety changes slowly. Multivariate dynamics can therefore distinguish an intervention that improves one dimension from one that merely shifts the burden elsewhere. Scalar summaries may still be reported for bounded purposes, but the vector state should remain available for diagnostics, sensitivity, and construct-failure review (Russell, 2003; Bollen, 1989).

8.3.6 — Clinical and Cultural Variation

Context can change the felt sign of an otherwise identical stimulus. In 16 healthy volunteers, Leknes and colleagues (2013) presented the same moderate heat in two settings: one in which the alternative was nonpainful warmth, and another in which it was intense pain. Average sensation ratings shifted from painful in the first setting to pleasant in the second. The shift accompanied reduced skin-conductance responses in the 12 participants with usable recordings and changes in pain-related and valuation circuitry. This gives PTVC’s emphasis on experienced value a concrete experimental foundation: the relation between an event and its alternatives can alter the feeling itself. A context-sensitive valence model can therefore make quantitative predictions about changes in felt sign while

the physical input remains fixed, with expectancy and contrast models providing direct comparisons.

Clinical state, disability, language, culture, and treatment can alter report, physiology, expression, and interpretation, limiting transport (Cacioppo & Berntson, 1994; International Test Commission, 2018).

Sign is context-sensitive. Restlessness may reflect anxiety, pain, delirium, medication, dyspnea, or discomfort, while low expression may reflect comfort, sedation, depression, fatigue, fear, or limited communication. Context can constrain alternatives, but no display becomes a universal sign rule.

Positive and negative channels may change reliability independently. Keep component trajectories visible so a stable net score does not conceal substantial changes in both kinds of experience.

Momentary items require response-process, distribution, within-person variability, trend, and level-specific reliability checks (Dejonckheere et al., 2022; Siepe et al., 2025; Castro-Alvarez et al., 2026). A pilot tests whether participants interpret each item consistently across contexts and whether the schedule captures change at the relevant timescale.

Audit item, channel, and construct separately: translation can fail while physiology remains usable, medication can alter physiology while communication remains interpretable, or the construct itself can differ. Invariance models cannot create shared phenomenal meaning (Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

8.4 — Scalarization and Identified Sets

Section 6.4 established why scalarization and identified sets belong to PTVC's representation gate. Here the task is psychometric: specify, estimate, validate, and stress-test the projection from multidimensional lived states to a study-specific scalar before that scalar enters a ledger (Fredrickson & Kahneman, 1993; Eldar et al., 2016; Manski, 2003).

8.4.1 — Weight Vector w

In the first display, v(t), a(t), p(t), c(t), m(t), r(t), s(t), and b(t) name eight abstract component time series in the displayed order; the letters do not prescribe a psychological taxonomy. A protocol must assign each coordinate a construct, scale, orientation, and observation rule before fitting the weights. The column vector z(t) collects those coordinates, and the equally ordered column vector w gives their scalarization weights. W is the declared set of admissible weight vectors. The eight-coordinate display is illustrative: a different prespecified dimension requires a conformable weight vector.

$$z(t) = (v(t), a(t), p(t), c(t), m(t), r(t), s(t), b(t))$$

$$x_w(t) = w^\top z(t), L_w(a,b) = \int_a^b g_\theta(t) x_w(t) dt$$

$$I_W(a,b) = \{L_w(a,b) : w \in W\}$$

Justify the source and validity of scalarization weights, using Section 6.7.5's separation of development from protected confirmatory testing (Manski, 2003; Krantz et al., 1971).

Weights can be fixed by theory, estimated from data, elicited from participants, constrained by expert judgment, or compared across plausible families. Each route has different risks. Participant weights may reflect current mood or social pressure.

Expert weights may encode professional bias. Data-derived weights may optimize prediction of an outcome that is not lived experience. Theory-derived weights may be elegant but wrong.

A weight choice specifies which dimensions may compensate for others. Down-weighting pain when meaning is high, for example, makes a substantive tradeoff that requires justification (Manski, 2003).

A weight vector is part of the estimand, not merely a computational setting. Changing the weights changes which features of experience count, how mixed states are ordered, and which trajectories appear close to neutrality. The protocol should therefore state whether weights are person-specific, population-level, context-specific, learned from independent data, or fixed by a declared normative convention (Manski, 2003; American Educational Research Association et al., 2014).

8.4.2 — Weight Uncertainty and Sensitivity

The right object may be a family W of admissible weights.

Weight uncertainty can include sampling variation in estimated weights and uncertainty about which scalarization defines the target. Separate those sources; Section 6.4.3 distinguishes an identified set for a fixed target from sensitivity across different targets.

Propagate uncertainty from estimated weights, task framing, anchor instability, response shift, cultural interpretation, and model choice into each downstream quantity. Robust or Bayesian analyses can represent a distribution or admissible set of weights, but the output remains conditional on that representation (Borsboom, 2005; Manski, 2003).

Researchers should identify which conclusions hold throughout the admissible set, which depend on a narrow corner, and which reverse sign across plausible choices. A robustness

frontier can show how much assumption is required before a one-number claim appears. When the central conclusion depends on particular weighting commitments, the report identifies those commitments and the measurements that could distinguish them (Manski, 2003; Imbens & Manski, 2004).

8.4.3 — Identified-Set Reporting

Use Section 6.4.3's distinction between an identified set and sampling uncertainty. A confidence procedure must specify whether it covers the target parameter or the entire identified set (Manski, 2003; Imbens & Manski, 2004).

8.4.4 — Withdrawal of One-Number Claims

A one-number claim should be withdrawn when scalarization is unstable, admissible weights cross incompatible statuses, dimensions remain only partially ordered, cultural or clinical variation changes meaning, or temporal functionals disagree beyond the study's ability to resolve them (Manski, 2003; Borsboom, 2005).

Withdrawal should identify the highest representation tier that survives. A failed scalarization may still permit sign, partial order, within-person direction, a vector profile, or bounded change; it does not automatically invalidate every measurement.

Requalification requires data that did not generate the repair and a prospective criterion for resolving the specific original defect (American Educational Research Association et al., 2014; Nosek et al., 2018).

The decision rule belongs in the preregistration. Separate rows should address sign instability, weight sensitivity, common-origin failure, noncomparability, transport failure, and disagreement among temporal functionals; each row states the evidence threshold, the language that becomes unavailable, and

the narrower output that remains permitted. This makes withdrawal reproducible rather than a rescue invented after an attractive result fails (Manski, 2003; Imbens & Manski, 2004).

Withdrawal also propagates beyond the analysis. Figures, summaries, public excerpts, model cards, and dataset descriptions must be corrected, versioned, or retired when their scalar premise is no longer supported. A clean current result does not close the error if an older one-number image continues circulating without its withdrawal notice (Munafò et al., 2017; Center for Open Science, 2025). The correction record should identify which active public surfaces were checked and which copies could not be updated, making residual exposure to the superseded claim visible rather than assumed away.

Reports should distinguish cardinality loss, comparability loss, transport loss, and aggregation loss. These failures have different remedies and different public consequences; calling all of them ‘measurement failure’ discards information, while silently switching from a shared unit to rank order or from a lifetime claim to a local association disguises a changed target. Each failure should be paired with the strongest relation that still survives, such as within-person direction, ordering, a vector profile, or a set-valued result, so withdrawal preserves usable information instead of collapsing every limitation into the same verdict.

When a scalar claim withdraws, redesign should return to the path rather than search immediately for another summary. The ordinary-day schedule below illustrates how a study can preserve waking transitions, obligations, waiting, social contact, bodily events, relief, evening recovery, and sleep-related carryover so that later model comparison is not built from peaks alone. Preserving that temporal structure allows competing path functionals to face the same record instead of letting data

collection quietly favor whichever summary replaces the withdrawn scalar.

8.4.5 — No Outcome-Selected Weights

Outcome-selected weights are weights chosen because they produce the desired result. In the context of this book, the danger is obvious: a researcher could adjust the scalarization until the terminal path looks neutral, the intervention looks beneficial, or the rival looks weaker. That would convert measurement into rescue (Manski, 2003).

The prohibition must be explicit because outcome selection can hide inside reasonable language. A team may say that it refined the model after inspecting anomalies, optimized weights for fit, removed a dimension that behaved inconsistently, or chose the psychologically plausible transformation that happened to improve closure. Some refinements are legitimate in development. They are not confirmatory evidence unless locked and retested.

Exploration and confirmation must remain distinct. Exploratory work may compare weight families, discover dimensions, revise items, and inspect failure. Confirmatory work freezes scalarization, population, endpoint, temporal rule, and analysis plan before testing the claim. Changing weights after results are visible creates a new claim version that requires a new test (Munafò et al., 2017; Nosek et al., 2018).

This rule also protects participants. A weight selected to make a result look fair may erase a burden the participant actually carried. If fear is removed as noise, if pain is offset by meaning, or if dignity violation is down-weighted because it disrupts a curve, the model has shifted burden onto the person for the sake of the theory.

Outcome-selected weights are not always malicious. They can arise from pressure to make complex data coherent. But the effect is the same: the measurement bridge becomes immune to defeat. A bridge that cannot fail cannot bear evidence.

No outcome-selected weights is therefore a scientific and ethical firewall in one sentence. The scalarization must face the result from outside the result. If it changes, the book must say that the claim changed too.

8.5 — Duration, Memory, and Path Functionals

A life is extended in time, but time is not a neutral container once experience enters it. Ten minutes of terror, ten minutes of boredom, and ten minutes of quiet safety have the same clock length. Their felt contributions may not have the same structure. A simple integral treats equal time intervals as equally eligible for accumulation. That assumption needs rivals (Eldar et al., 2016; Fredrickson & Kahneman, 1993; Shiffman et al., 2008).

8.5.1 — The Ordinary Middle

Record how the first work, care, school, treatment, or dependency obligation enters felt time. Otherwise visible productivity can replace the hidden cost that the study intends to examine.

Sleep onset and interruption connect affective residue with rest, dreams, and next-day carryover. They are not proof sources for the Law, but omitting them can make the night invisible even when it carries the day forward.

Waiting, commuting, monitoring bills, and repeated care can be missed by designs concentrated on peaks and endings (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

Small recurring vigilance costs or quiet relief can accumulate across long intervals. Sampling only dramatic events can therefore distort the path being estimated.

The ordinary middle is also where sampling burden and scientific value meet. Brief random prompts, end-of-day reconstruction, passive context, and time-use records each reveal different portions of extended experience. A design that samples only when emotion is intense will overrepresent peaks; a design that samples mechanically may burden the very routines it seeks to observe. The schedule should be justified by the temporal

process and tested for reactivity, missingness, and coverage of low-salience intervals (Stone et al., 2007; Shiffman et al., 2008).

8.5.2 — Duration Weighting

Saturation, habituation, sensitization, and exhaustion can make the tenth hour experientially different from the first. An additive integral can represent those differences through a changing rate; equal clock weighting does not require constant experience. A separate duration-weighting rule needs its own justification and must not count intensity or aftereffects twice.

The analysis should distinguish clock duration, conscious-access duration, attention-weighted duration, burden-weighted duration, and recovery-inclusive duration. The design must permit prospective challenges from attention, consciousness level, saturation, sensitization, and recovery; if equal weighting fails, the response is a better path functional rather than deletion of duration from the target.

The temporal question has an empirical foundation. Cui et al. (2023) synthesized 95 effect sizes from 31 studies involving 3,776 participants and found that valence, arousal, stimulus type, and timing task moderated emotional duration judgments. Negative rather than positive valence, and greater arousal, generally accompanied greater overestimation. This gives the program measurable alternatives to compare: physical duration, reported duration, and any independently qualified phenomenal-exposure model. It makes the choice of clock an explicit research question, with its rule fixed before the cumulative result is examined.

8.5.3 — Peak-End and Memory Effects

Waiting intervals expose less conspicuous burden, uncertainty, pain, boredom, and constrained choice. Sampling only peak events would miss those contributions.

Evening recovery can reveal accumulation, rumination, meaning, memory, and returning agency. Including it prevents the path from being reduced to isolated moments.

Model retrospective evaluation as a reconstruction whose dependence on peaks, endings, narrative coherence, later meaning, and current state can be tested against momentary records (Fredrickson & Kahneman, 1993; Kahneman et al., 1993; Stone et al., 2007).

Peak-end effects are not merely cognitive curiosities. They can make a temporally longer and more painful episode remembered more favorably if the ending improves. They can also make a mostly tolerable experience remembered harshly because it ended badly.

Memory can still be evidence. Retrospective evaluation matters to future choice, identity, narrative meaning, and suffering that reappears as recollection. Remembered utility, experienced utility, predicted utility, and decision utility must remain separate so that each source speaks in its own status.

The Law-facing risk is endpoint contamination. If terminal calm changes memory of a painful period, a retrospective report near death may appear to support neutrality while the experienced trajectory remained non-neutral. Conversely, a frightening ending may make a life period remembered as worse than its momentary distribution. Both directions matter.

Peak-end effects are not a universal law of memory, and one simplification should not replace another. Their magnitude varies with episode structure, delay, attention, meaning, and the

question asked (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

8.5.4 — Recency and Kernel Models

The waking transition records baseline mood, bodily demand, sleep residue, and anticipatory load. Beginning observation later can miss burdens already carried at the start of the day.

A kernel model assigns different weights to moments depending on their location in time or relation to an event. Recency kernels give more weight to recent states. Anticipatory kernels weight future expectation. Recovery kernels carry the cost of a state beyond its immediate interval. Such models may fit some psychological phenomena better than a flat integral (Eldar et al., 2016; Russell, 1980).

The timing of earlier events can improve predictions of present mood. Keren and colleagues (2021) compared a primacy model with several recency models across random, structured, and adaptive reward environments, including adult and adolescent samples and participants with depression. Its expectation term is the arithmetic mean of preceding outcomes; successive expectation terms still enter an exponentially weighted mood prediction. This primacy model predicted successive mood ratings more accurately in the tested comparisons. Its expectation-related parameters also corresponded to activity in frontal regions involved in mood regulation. This supplies a concrete benchmark for cumulative research: compare flat, recency-weighted, primacy-weighted, and latent-state histories on the same future observations.

Kernel models can represent how a past threat changes later experience. A flat integral of an adequately specified, history-dependent rate already includes those later effects. A separate recovery kernel is useful when it improves that representation,

but it must not count the same experienced aftereffect once in the rate and again as an added historical cost.

Recency kernels are themselves estimands that require identification. A rapidly decaying kernel, a long-memory kernel, and a reconstruction rule based on salient transitions can produce similar retrospective summaries under limited sampling. Prospective experience reports, controlled delays, and held-out prediction of later recall can help separate them. If several kernels remain observationally equivalent, the analysis should retain the family instead of selecting the one that produces the cleanest cumulative path (Kahneman et al., 1993; Särkkä, 2013).

8.5.5 — Order, Recovery, and Hysteresis

Social contact may bring support, threat, loneliness, obligation, or masking. The record must distinguish these possibilities so that relational burden is not treated solely as an external circumstance.

Relief and repair permit examination of recovery, contrast, gratitude, exhaustion, and partial improvement. Their inclusion prevents the model from recording harm without release or treating release as complete compensation.

Order can change the experienced trajectory without changing the rule used to accumulate it. Reordering external events may alter anticipation, adaptation, and recovery, thereby changing the realized valence rates. By contrast, permuting the same realized rate–exposure pairs preserves their additive total. A different memory or preference for two sequences need not imply a different phenomenal total. This is the distinction made in Section 5.2.6: history dependence of the rate is not, by itself, failure of additivity.

Sachs et al. (2025) provide a human example of temporal context. In the analyzed fMRI sample of 39 participants, musical events were presented in different orders so that the same passage followed different emotional contexts. Group-level neural patterns and transition timing, as well as retrospective emotion ratings collected after scanning, depended on preceding context. Separate online continuous ratings helped validate the stimuli; they were not contemporaneous reports from those same scanned participants. The findings make sequence-sensitive response models concrete without establishing a cardinal valence scale or a complete-stream equality.

Estimate a history-sensitive response model on one ordering and predict a held-out ordering, accounting for acoustic content, elapsed time, and carryover. Score the response trajectory separately from later remembered evaluations. Existing data can test whether a compact state description transports across sequences; any recovery burden enters through experience during recovery, without an extra retrospective charge.

Hysteresis indicates that present experience may depend on the route by which the present state was reached. Equal current pain ratings can carry different meanings after escalation, relief, failed treatment, or prolonged uncertainty. When history matters, the state representation must be enlarged to include the relevant memory, adaptation, or control variables; otherwise a non-Markov process will be forced into a misleading memoryless model. Closed loops and recovery trajectories are therefore diagnostic tools for discovering missing state rather than evidence of a preferred mechanism (Särkkä, 2013; Dejonckheere et al., 2022).

8.5.6 — Path Functional Model Comparison

Bodily events track pain, nausea, breathlessness, fatigue, hunger, and medication effects as lived conditions. Their bodily origin does not make them merely confounds outside the target.

The path functionals developed above preserve different structures. Fit alone does not make any one of them the target.

Comparison begins from common observations, inclusion rules, and a prospectively fixed functional set. Each candidate declares its invariances, cadence, missing-interval rule, endpoint sensitivity, and held-out predictions. Allowing every model to choose a favorable representation would confound fit with target selection (Kahneman et al., 1993; Fredrickson & Kahneman, 1993).

Prespecified contrasts can reorder external events, extend a low-intensity interval, alter the final minutes, or repeat recovery episodes. Interpret differences using Section 8.5.5's separation of history-dependent rates from the rule that accumulates them.

Held-out prediction is matched to use. A functional may predict remembered evaluation, behavior, treatment preference, or later reports without measuring experienced cumulative contribution (Stone et al., 2007; Shiffman et al., 2008).

Identification depends on observation. Sparse prompts may leave order and duration unresolved, reconstruction may preserve narrative while missing brief episodes, and passive channels may improve timing while leaving sign uncertain. The honest output may be an observationally equivalent model set.

8.6 — Evidence Baseline and Construct Failure

No single observation channel supplies every property needed for a lived-experience measure. First-person report, behavioral evidence, physiological observations, and contextual records contribute different information and have different failure modes. Their combination is useful when the observation model states what each channel identifies and tests the assumptions connecting them.

8.6.1 — First-Person Report

First-person report supplies evidence about felt quality, meaning, burden, and context. Its proposed use requires independent checks of units, sign stability, and temporal coverage, accounting for memory, demand, language, fear, fatigue, and response shift.

First-person report is privileged because the person lives the experience. Systematically ignoring that testimony distances an estimator from its target (Cronbach & Meehl, 1955; Borsboom, 2005).

Privilege is not sovereignty. Reports can be shaped by memory, language, fear, shame, expectation, demand characteristics, fatigue, medication, clinical state, culture, and lack of concepts. People may also report strategically because consequences attach to honesty.

The correct posture is respectful modeling. Reports should be collected in ways that minimize burden, protect privacy, allow uncertainty, invite context, and separate immediate experience from memory and evaluation. The model should know whether it is using momentary report, episode reconstruction, narrative interview, symptom scale, or life evaluation.

First-person report can also contest the model. If a participant says the labels are wrong, the timing missed the event, a device

changed the feeling, or a question forced a false choice, that information belongs in data quality. Contestability is not an administrative courtesy. It is part of construct validity.

During validation, compare alternate prompt wording and concurrent versus retrospective reports. Record whether disagreements recur by person or context, and use those patterns to test the response model while preserving refusal and participant challenge (Stone et al., 2007; Shiffman et al., 2008).

8.6.2 — Physiology and Behavior

Section 8.6 locates common physiological and behavioral signals and their failure routes. Interpretation must account for exertion, temperature, device fit, disability, social masking, and environmental demand (Campbell & Fiske, 1959; Bollen, 1989).

External channels can supply timing and arousal information when reports are unavailable, delayed, constrained, or reconstructed from memory. Conflicts with context require review of the observation model.

Lee and colleagues (2024) developed distinct fMRI patterns for reported affective valence and intensity during sustained pleasure from chocolate and pain from capsaicin in 58 participants. In an independent sample of 62 participants, mean within-person prediction correlations were 0.10 for valence and 0.16 for intensity. The principal valence analysis used signed ranks with matched positive and negative rating distributions; additional training on original ratings gave comparable within-person prediction correlations. These results identify a measurement candidate whose predictive performance and scale construction need to be evaluated separately.

A PTVC pilot can test frozen models against independently calibrated within-person magnitudes across signs, sessions, and held-out stimuli. Evaluate prediction within each sign; assess

transfer with additional models trained on one sign and frozen before testing the other. Fit preprocessing and calibration on development data and freeze them before evaluating held-out records. Compare predictions with intensity and arousal channels and direct reports before using them as interval contributions. This tests whether predictive information can support stable signed magnitudes through time.

Neuroscience provides a concrete reason to investigate shared structure across different feelings. Chikazoe and colleagues (2014) found orbitofrontal population patterns that tracked affective valence across visual and taste experiences. Later work separated shared from modality-specific patterns across several aversive experiences (Čeko et al., 2022). Such results make a common affective dimension a productive empirical target while preserving the distinctive qualities of each modality.

Pleasure itself is becoming a more specific measurement target. Kragel and colleagues (2023) developed a distributed neural signature using 28 fMRI studies and evaluated it in independent datasets. A pooled analysis of four independent validation studies found larger signature differences for hedonic contrasts than for monetary contrasts. In a separate 19-person placebo-controlled crossover study, naloxone reduced the signature’s response to erotic images. These results provide a candidate for distinguishing pleasure-related responses from incentive-value responses in the tested settings. Combining such a measure with independently validated unpleasantness channels could test whether cross-sign comparison is stable within a person.

Positive experiences now have a similarly concrete research opening. In 52 adults, Savalia and colleagues (2026) predicted responses to personalized music, videos, positive memories and monetary rewards from distributed neural activity and

connectivity. Combining the two feature families improved held-out prediction. Their measure asked how much participants wanted an experience again, making separation of wanting from felt pleasure a valuable next validation step. A PTVC-oriented measurement study can retain this positive range while testing direct pleasantness and unpleasantness independently.

Valence assignment is also experimentally tractable at the circuit level. In mice, Li et al. (2022) identified a neurotensin pathway from the paraventricular thalamus to the basolateral amygdala that biased learning about rewarding and punishing outcomes. Activating this pathway promoted reward learning; removing neurotensin from the pathway strengthened punishment learning. Related work identified partially segregated ventral hippocampal cell populations associated with appetitive and aversive memory engrams, together with distinct molecular and projection features (Shpokayte et al., 2022). These findings give the project concrete examples of biological systems that distinguish valence-related learning and memory. A declared route can be tested against independent appetitive and aversive learning models by asking whether its specified coupling predicts previously unseen changes in report and choice under ordinary, nonharmful variation.

8.6.3 — Clinical and Relational Context

Physiology supplies timing and correlates of arousal, interoceptive load, stress, pain, and treatment state. Its relation to the target experience requires calibration under the relevant bodily state, including medication, illness, disability, sensor gain, and nonspecific arousal.

Behavior records approach, withdrawal, speech, activity, sleep, social contact, and care use. Interpretation requires contextual evidence about motivation, constraints, and the

experience of the act, including masking, coercion, habit, culture, access barriers, and survival strategies.

Clinical context records pain, dyspnea, nausea, delirium, fatigue, diagnosis, and care burden. Integrate it with participant reports and other available evidence while examining overmedicalization, underreporting, protocol burden, and role pressure.

Relational witnesses can describe familiar patterns, social safety, loneliness, dependence, and change. Their contribution requires consent, source provenance, and alternative interpretations, including projection, conflict, loyalty, grief, and privacy intrusion.

Diagnosis, medication, pain history, oxygenation, delirium risk, disability, sleep disruption, infection, hydration, and treatment burden belong in the observation model when they distinguish affective change from illness, altered access, or sensor failure (Shiffman et al., 2008; Khalsa et al., 2018; Giacino et al., 2018).

Close others may recognize familiar patterns of pain, fear, relief, or withdrawal that a one-time researcher misses. Their accounts also require checks for projection, pressure, memory error, minimization, exaggeration, and their own burden.

Apply Section 3.4.4's consent, bystander, and minimization rules to relational evidence; providing context does not authorize access to the participant's other data.

8.6.4 — Convergent and Discriminant Validity

Validity evidence must match the measurement model. In a reflective model, indicators should behave as consequences of the proposed latent state under the declared conditions. In a causal or composite model, components may jointly define the score without being interchangeable or highly correlated. In a

network model, relations among components may be the object of interest. Convergent and discriminant tests therefore depend on what the model says the observations are. A lived-experience estimator should relate plausibly to reports, contexts, known contrasts, and interventions without merely reproducing arousal, illness severity, social desirability, compliance, or device tolerance (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Bollen, 1989; Borsboom, 2005).

Convergence requires checking shared wording, scoring, training labels, missingness, and contextual confounds. Redundant witnesses do not supply independent confirmation.

Construct validation also requires ecological validity. A construct validated only in artificial laboratory stimuli may fail in the ordinary middle of life. A construct validated only during acute pain may fail for boredom, shame, safety, or awe. Evidence must match the range of experiences the claim intends to cover.

Disagreement between report, behavior, physiology, and clinical context can reveal construct breadth, lag, or method failure. Multitrait-multimethod designs, explicit latent-variable models, causal-composite alternatives, and held-out intervention tests can locate that disagreement, but none creates direct access to phenomenology (Campbell & Fiske, 1959; Bollen, 1989; Borsboom, 2005).

8.6.5 — Known-Groups and Intervention Tests

An intervention response asks whether changes in pain, safety, rest, agency, or support alter reported and observed experience. Specify the construct in advance and examine active mechanisms, contextual effects, and unmeasured mediation rather than assigning the response to a single cause.

Known-groups tests ask whether a construct distinguishes groups or states that a prospectively specified scientific model predicts will differ. Intervention tests ask whether the construct changes when conditions change. Both can support a lived-experience target, but only if the comparison is justified before the result and does not exploit vulnerability (Munafò et al., 2017; Nosek et al., 2018).

Noise reduction, better pain control, privacy, respite, debt relief, or a mobility accommodation supplies a concrete intervention contrast. Failure to respond when participants report relief and context supports it puts the proposed construct under pressure.

Relief research provides a particularly useful separation of expectation, sensation, and positive feeling. Willems et al. (2025) studied 31 adults and found that omission of an expected painful stimulus produced reported relief that increased with expected threat intensity and probability. Omission also recruited parts of reward and salience circuitry. The neural pattern was prediction-error-like rather than a complete canonical reward-prediction-error signal: one required axiom failed, and the ventral tegmental probability effect did not reach the conventional significance threshold. The substantive opportunity is that relief can occur when the anticipated harm never happens. A phenomenal account must therefore record experienced anticipation and experienced relief in their own windows, while the omitted event contributes no unexperienced pain. Approved reanalysis or naturally occurring relief records can compare expectation-based and event-only models. These findings give temporal and appraisal models a concrete event sequence to explain.

Known-groups tests are dangerous when they turn identities into expected feeling. Disability, culture, age, poverty,

diagnosis, or family status should not define burden by stereotype. The comparison must concern specific conditions and pathways, not the worth or presumed inner life of a group. Experts should ask whether the group label is scientifically and ethically appropriate.

Intervention tests can also mislead. An intervention may change behavior without changing experience, change report without changing burden, or reduce one burden while increasing another. A privacy intervention may reduce observation stress while making data sparse. A medication may reduce physiological arousal while increasing numbness. The construct must be able to show tradeoffs.

Learning from pain and learning from its avoidance can follow different processes. In a randomized, blinded study of 83 healthy adults, Jepma et al. (2022) found that a model with separate learning rates for received and avoided pain fit behavior better than a single-rate model. Under placebo, estimated group learning rates were 0.72 for received pain and 0.32 for avoided pain. Levodopa and naltrexone each increased learning from avoided pain, while imaging associated the two learning processes with different systems. No reduction in the number of painful outcomes was detected between the treatment groups. The study identifies a concrete, modifiable learning asymmetry. For this program, independently estimated positive and negative learning kernels can be compared with a declared coupled model using existing approved data, with future observations reserved for scoring. This comparison can identify which coupling mechanism best explains the observed learning asymmetry.

8.6.6 — Construct Failure and Redesign

Construct failure occurs when evidence does not support the target as represented: channels fail to show the convergence

required by the measurement model, meaning or sign drifts, interventions move indicators without the intended experience, or scalarization collapses states that participants and experts distinguish. The conclusion concerns the construct, bridge, or measurement practice, never the participant's credibility (Cronbach & Meehl, 1955; Campbell & Fiske, 1959; Flake & Fried, 2020).

Redesign begins with the smallest failed relation. An unstable item is revised or removed, a nonspecific channel is narrowed, a failed scalar becomes a vector or identified set, and a target without stable content is retired locally. More channels do not repair a construct if they merely combine arousal, illness, device gain, or response style.

The revised object must return to content, response-process, structural, external, and consequential validation, including time-sensitive reliability and repeated-administration effects. Independent teams should recover both the improvement and the original failure (Nosek et al., 2018; Munafò et al., 2017; Vogelsmeier et al., 2024).

Unidentified origin, irreducible sign ambiguity, intolerable burden, nontransportable observation, or persistent mismatch with lived reports can close the route. Preserve the failed use under Section 28.2.

8.7 — Fairness Appraisal and Cultural Recurrence

Fairness appraisal concerns distributions, procedures, intentions, promises, status, reciprocity, and relative position. These judgments can alter affect, memory, and behavior, so the experience of an outcome must be distinguished from judging it fair (Fehr & Schmidt, 1999; Bolton & Ockenfels, 2000).

8.7.1 — Inequity Aversion and Fairness Appraisal

Inequity aversion describes the possibility that people care not only about their own payoff but also about how outcomes compare. Formal models distinguish disadvantageous inequity, in which another receives more, from advantageous inequity, in which one receives more than another. These terms capture recurrent behavior in allocation tasks, but they are models of preference and appraisal rather than direct measures of lived valence (Fehr & Schmidt, 1999; Bolton & Ockenfels, 2000).

The experiential structure can be mixed. Disadvantage may bring anger, humiliation, threat, envy, or resignation. Advantage may bring pleasure, guilt, responsibility, fear of retaliation, or no salient feeling at all. A scalar fairness score that treats all departures from equality as the same burden would erase the very mechanisms the experiment is meant to reveal.

Procedure and intention also matter. An unequal result produced by chance may be experienced differently from the same result produced by favoritism, contempt, or broken promise. Conversely, an equal result can feel unfair when it ignores effort, need, risk, or prior commitment. Outcome equality is therefore neither a complete construct nor a universal neutral origin.

Neurochemical work gives this contextual distinction another experimentally accessible layer. During clinically indicated awake surgery in four people with Parkinson’s disease, Batten

and colleagues measured dopamine and serotonin while participants responded to economic offers attributed to humans or computers. Social source changed decisions, and the two transmitters carried distinguishable contextual and value information (Batten et al., 2024). This small clinical study demonstrates a route for investigating how social interpretation enters valuation.

Preserve heterogeneity in disadvantage sensitivity, rejection, task interpretation, deservedness, and trust. Averages can conceal appraisal functions with different signs or shapes (Henrich et al., 2005).

Prespecify the allocation rule, participant information, intention cues, reference group, stakes, rejection costs, and candidate affective mediators. Compare preference, appraisal, norm-compliance, and experimenter-demand models.

8.7.2 — Reciprocity, Cooperation, and Punishment

Reciprocity links present action to remembered treatment. Positive reciprocity can support trust, gratitude, help, and durable cooperation; negative reciprocity can support retaliation, withdrawal, punishment, and refusal. Laboratory games, field observations, and institutional records show that people sometimes incur costs to reward cooperation or sanction defection (Fehr & Gächter, 2002; Tomasello & Vaish, 2013).

Concern about fairness also extends beyond direct retaliation. In a study of 535 children aged 5 to 15 across communities in Canada, India, Peru, Uganda, the United States, and Vanuatu, children were more likely to intervene against maximally unequal than equal sharing between two absent peers. Some interventions required sacrificing the child's own rewards, and sensitivity to cost and age varied across settings. This makes immediate personal gain or repayment of a personal injury an

incomplete account of the observed behavior. Learned norms, social evaluation, and evolved cooperative capacities remain live explanations. The useful PTVC research question is whether independently measured affective dynamics add a distinctive prediction beyond these mechanisms (McAuliffe et al., 2025).

Costly punishment is not experientially simple. Anger may motivate the act, anticipated relief may sustain it, and the act may later produce guilt, social approval, or escalation. A punishment decision can therefore be both behaviorally costly and affectively reinforcing. Treating it as a clean negative or positive contribution would confuse action, motive, appraisal, and felt consequence.

Cooperation also depends on expectations and institutions. Repeated interaction, reputation, communication, group identity, enforcement, and perceived legitimacy can change whether cooperation feels safe or exploitable. The same cooperative act can be carried as generosity, duty, fear, belonging, or coerced compliance. A science of fairness must model these meanings rather than use cooperation as a proxy for well-being.

Cooperation can also contribute positively to the experience of the person who helps. Aknin and colleagues combined surveys from 136 countries with experiments, including charity-versus-self purchasing in Canada and South Africa; the experimental prosocial purchases increased reported positive affect even without an opportunity to build social ties (Aknin et al., 2013).

More recently, three two-month randomized studies of daily online micro-donations, with samples of 125, 296, and 462, reported improvements in depressive symptoms and emotional positivity relative to waitlists (Y. Zhang et al., 2025). The first two studies registered their plans after recruitment and baseline measurement; the third registered before data collection. These

findings offer concrete, testable pathways from voluntary social action to better reported experience.

Prosocial spending supplies a complementary finding about the giver's experience. A registered replication with 712 students found greater momentary happiness after purchasing for a stranger in need than after purchasing for oneself. A recall experiment with 1,950 adults found no corresponding advantage, while a revised recall task with 5,199 participants found a small benefit. The full pattern supports a conditional connection between generosity and positive feeling and identifies task engagement and measurement as issues for replication. Material expenditure can therefore coexist with positive experience, which is a reason to measure felt consequences directly (Aknin et al., 2020).

Repair can also change the emotional state of the person who acts. Donohue and Tully randomly assigned 97 children aged 6 to 10 to conditions separating guilt from the opportunity to make amends. Children led to believe they had upset a peer showed more reparative behavior than children in a no-guilt condition. Guilt declined when children had an opportunity to offer stickers and a note, whereas it did not decline in the condition without that opportunity. This provides a concrete local sequence linking appraisal, constructive action, and relief. It motivates time-resolved observation of naturally occurring repair as a regulatory process (Donohue & Tully, 2019).

Separate the cost paid, the norm invoked, the emotion carried, the social consequence, and the later interpretation in time-resolved records.

8.7.3 — Social Comparison and Relative Position

Social comparison concerns the use of other people, past selves, or imagined standards as reference points. Upward comparison

can motivate, inspire, shame, or discourage. Downward comparison can reassure, isolate, or provoke guilt. The direction of comparison does not determine its valence; meaning, controllability, identification, and perceived legitimacy matter (Festinger, 1954; Suls et al., 2002).

A changed peer group or persistent former standard can alter how income, recognition, safety, workload, or illness feels. The comparison field is a mediator of material conditions, not a substitute for recording them.

Reference groups are dynamic. People compare with coworkers, siblings, neighbors, online networks, prior selves, cultural ideals, and expected futures. Platforms and institutions can manipulate salience by repeatedly displaying exceptional outcomes. A measurement system that ignores the comparison field may attribute resulting distress to an intrinsic trait or to the wrong event.

A material improvement may fail to produce sustained relief when the reference group rises faster; a stable condition may feel better after a threatening comparison disappears.

A time-resolved experiment makes this relation quantitative. In 47 participants, Rutledge and colleagues found that both receiving less than a partner and receiving more could lower momentary happiness. Adding social-inequality terms to a recent-reward model increased mean explained variance from .39 to .44, and the asymmetry between the two inequality sensitivities correlated with generosity in a separate anonymous allocation (rho = −.48; Rutledge et al., 2016). The study therefore connects relative outcomes, felt appraisal, and generosity in a separate task within one testable account. It also provides an important comparison model for PTVC research:

recent rewards, expectations, and social comparison already explain meaningful structure. Happiness ratings were standardized within participants.

A disciplined study records the reference target, comparison direction, perceived similarity, controllability, justice appraisal, exposure frequency, and resulting affect over time. It tests whether comparison variables improve held-out prediction beyond material conditions and individual history, thereby refining the local model of experience.

8.7.4 — Just-World Cognition and Narrative Balance

Just-world cognition is the tendency to expect that people generally receive outcomes they deserve or that the world is meaningfully ordered. Such beliefs can support persistence and trust, but they can also encourage victim blame, denial of structural harm, and retrospective moralization. For a project named around fairness, this is a central adversarial risk (Lerner, 1980; Hafer & Bègue, 2005).

Just-world beliefs can also shape reports. Participants may minimize suffering to protect a worldview, exaggerate agency to avoid stigma, or reinterpret harm as deserved, educational, or spiritually necessary. These interpretations can be sincerely held and experientially consequential. They still require separation from the scientific question of what was felt and when.

Research can examine framing effects directly. Experiments can vary deservedness cues, narrative framing, outcome order, and information about structural causes, then measure changes in memory, fairness judgment, blame, and affect. Comparing explicitly balanced framing with neutral framing helps distinguish a reproducible experiential pattern from a response to the expectations introduced by the task.

Beliefs about moral consequences also have experimentally measurable effects. White and colleagues (2019) found that reminders of karma reduced selfish allocations among karmic believers across religious affiliations, while reminders of God reduced selfishness among believers in God. These findings identify a connection between culturally learned expectations and choices about sharing. They give the study of fairness a concrete social mechanism to investigate alongside material incentives, reputation, relationship history, and the feelings associated with cooperation.

8.7.5 — Developmental and Cross-Cultural Recurrence

Recurrence across societies and developmental stages creates a comparative research opportunity. Shared social problems, evolved cooperative capacities, convergent institutions, learning, cultural transmission, experimental demand, and narrative habits offer distinct explanations whose predictions can be examined across settings (Henrich et al., 2005; Blake et al., 2015).

The empirical starting point is substantial. In a controlled study of 40 participants, Tricomi and colleagues found that neural reward responses and subjective appeal depended on whether monetary transfers reduced or increased an initially unequal distribution (Tricomi et al., 2010). These results give fairness appraisal a measurable place in value processing: external rewards are evaluated in a social context. For PTVC research, the gain is a concrete set of variables and experimental contrasts through which social events can alter experience within a stream.

Blake and colleagues studied 866 pairs of children aged four to fifteen in seven societies. Rejection of disadvantageous inequity developed across all seven, whereas rejection of advantageous inequity appeared later and in three societies

(Blake et al., 2015). In complementary experiments, jointly earning rewards increased equal sharing among three-year-olds compared with windfall or parallel-work conditions (Hamann et al., 2011). Across eight societies, culturally variable prosocial behavior developed alongside children’s responsiveness to social norms in middle childhood (House et al., 2020).

Ethnographic evidence adds a broader map of cooperative values. Curry and colleagues examined seven forms of cooperation in records from sixty societies and found positive moral evaluation in 961 of 962 coded observations (Curry et al., 2019). Coverage differed across domains; fair division of disputed resources was documented in nine societies. Recurring problems of kinship, coordination, reciprocity, conflict, and possession repeatedly invited moral solutions.

A later machine-reading study extended coverage to 256 societies and detected most of these moral themes in most societies across world regions (Alfano et al., 2024). Its expanded corpus included the earlier records. Its automated classifications tended to overestimate prevalence relative to the hand-coded benchmark. Together, these studies provide a large comparative record of cooperation-centered moral evaluation and a basis for investigating how social practices shape experience.

Task equivalence is not guaranteed. Currency, anonymity, experimenter authority, household obligations, reputational risk, and local meanings of sharing can change what an allocation game measures. Translation must include institutions and incentives, not merely words. A numerical difference may reflect task ecology rather than a deeper difference in felt fairness.

A 2026 study sharpens the distinction between broad fairness sensitivity and particular institutional norms. Among 697 children aged three to eleven in Norway, Japan, the United

States, and Italy, aversion to unequal allocations emerged across settings by age ten, while rejection of bribes appeared in late childhood only in some contexts (Reyes-Jaquez et al., 2026). The dissociation is informative: related moral judgments need not be expressions of one interchangeable faculty. It gives future research a practical strategy: compare closely related judgments under the same task framework and identify which developmental and cultural conditions carry each pattern.

Children may lack concepts for mixed motives; adults may mute complaints under politeness rules. Local meanings of anger, refusal, or entitlement affect reports and behavior without determining capacity for experience.

Religious traditions also supply observable social mechanisms. In eight communities, a study of 591 participants associated beliefs in knowledgeable, punitive gods with more impartial allocations to distant co-religionists (Purzycki et al., 2016). An expanded two-wave study covering 2,228 participants in fifteen societies, including the earlier sample, further examined this relation and its limits across religious groups (Lang et al., 2019). These findings make belief, accountability, group boundaries, and cooperation experimentally and comparatively tractable. Analyses of this evidence should account for the overlapping samples and shared cultural histories. Their contribution is a better map of the meanings and social conditions that can alter experience within a stream.

Use local interpretation and within-site replication to test task equivalence and identify which social or developmental assumptions fail to transport.

The comparative frontier remains active and increasingly precise. A primary-data synthesis of twenty-three studies across eighteen nonhuman species found no inequity-aversion evidence in the accept/reject paradigms analyzed (Ritov et al., 2024).

Subsequent bonobo experiments found greater refusal under unequal rewards, including controls for disappointment in a human experimenter, although the second study was only partly consistent with an inequity-aversion account (Radovanović et al., 2025). A 2026 group study of twenty-seven participating chimpanzees found responses that depended on food quality and the social partners present (Martinez et al., 2026). These results invite stronger tests of context, reward, and relationship effects rather than a universal animal-fairness verdict. Developmental research supplies the same lesson: a coordinated study testing 1,018 infants, with 567 included across thirty-seven laboratories, found no helper preference in the specific hill paradigm (Lucca et al., 2025). Such coordinated work makes claims more precise while leaving other tasks and capacities open.

8.7.6 — Recurrence as a Source of Hypotheses

A PTVC-related route can be compared with cultural learning, reciprocity, social-preference models, and bounded-memory reward learning under the same observations. If a route predicts a history-dependent change beyond current affect, recent rewards, expectations, resources, and norms, derive that prediction, fix it in advance, and test it in held-out observations with identification and observation error accounted for.

A practical starting point is the Global Preference Survey, which measured experimentally validated preference items in approximately 80,000 people across 76 countries, including reciprocity, altruism, and trust. Its larger variation within countries than between countries encourages person-sensitive analysis rather than national stereotypes (Falk et al., 2018). These data can help choose contexts and social-preference measures. A new, low-burden longitudinal study could combine such measures with repeated affect reports and naturally

occurring episodes of help, disappointment, and voluntary repair. Candidate PTVC relations and appraisal, learning, reputation, and norm-based alternatives would make protected predictions using the same observations.

Expected equality, balance, redemption, or moral concern can influence participant responses and the outcomes researchers select for attention. Use neutral task language and Chapter 6's prospective specification and complete-reporting rules.

Chapter 9 — The Measurement Pipeline

Consider a hypothetical participant reviewing a research summary. The event is recognizable, but its timing is wrong: a report about an earlier interval was treated as a measurement at the moment the answer arrived. The plotted sequence now implies a different ordering of events. Finding the error requires more than examining the final graph or rerunning unchanged code.

The received candidate momentary estimator remains physiological. Reports, choices, behavior, and contextual or relational records are auxiliary witnesses unless a broader multimodal estimator is separately declared. In the example, the participant's report challenges how evidence was aligned; it does not silently replace the estimator with a different target.

The correction begins with the question the report asked, the interval it referred to, the collection time, and the rule that aligned it with other channels. If the source record is accurate but the alignment is wrong, changing the original answer would conceal the failure. The analysis map, and every dependent output, needs examination instead.

Qualification therefore belongs to a versioned pipeline. An instrument name cannot identify which prompts, transformations, exclusions, or models produced a result. A participant correction may expose a local error while leaving other channels useful. It may also reveal a systematic interpretation problem that performance summaries missed (Flake & Fried, 2020; Gagnier et al., 2025).

Repeated observations make such distinctions consequential. They can reveal within-person dynamics, yet repeating an unstable item does not resolve its meaning. Item distributions, trends, measurement structure, and reliability at the relevant level must be examined alongside the trajectory (Shiffman et al.,

2008; Wrzus & Neubauer, 2023; Vogelsmeier et al., 2024; Siepe et al., 2025; Castro-Alvarez et al., 2026; Peeters et al., 2026).

The requested quantity determines which defect matters. Recovering a short-lived peak demands different temporal information from estimating an interval total. Some calibrated channels identify a total while the intervening trajectory remains unresolved. The chapter develops examples and bounds for this narrower recovery task.

A useful audit follows the affected relation through the pipeline. It asks which observations remain interpretable, which estimates require recalculation, and which comparisons depended on the failed version. That dependency can be checked through methods, fixtures, and controlled records without making a participant's complete record public.

The context of use limits those comparisons. Evidence for local tracking does not by itself establish portability across devices, people, or clinical states. A change may require a bridge study, version split, or abstention.

Participant challenge is part of this process. The record must preserve what was collected while allowing its contextual interpretation to be corrected and the downstream consequences to be traced.

Chapter 9 follows these decisions from observation to reported quantity. Chapter 10 then examines the origin, units, and links needed when the resulting estimates are compared.

9.1 — Context of Use and Qualification

Qualification begins with a question, not a device. The context of use should name the construct, population, state range, setting, sampling cadence, decision purpose, participant-burden ceiling, tolerated error, and prohibited uses before performance is summarized. Evidence adequate for exploratory within-person tracking may be inadequate for clinical interpretation, cross-group comparison, or a terminal-window claim. Current fit-for-purpose clinical-outcome-assessment guidance and measurement-property reporting standards reinforce this claim-specific discipline (FDA-NIH Biomarker Working Group, 2016; U.S. Food and Drug Administration, 2025b; Gagnier et al., 2025).

9.1.1 — Research Question First

The same physiological stream may inform arousal, autonomic instability, sleep disruption, pain response, stress recovery, or affective valence. Define the target and its observation relation before selecting an estimator (American Educational Research Association et al., 2014; Bollen, 1989; Borsboom, 2005; FDA-NIH Biomarker Working Group, 2016).

Within-person tracking over two waking weeks and prediction of a prespecified endpoint-window trajectory ask different questions, even with identical sensors. Specify the target, timescale, population, intended inference, and failure boundary for the chosen use.

State the loss function for the signed-contribution target.

9.1.2 — Population and State Scope

Declare age, language, disability, clinical condition, medication exposure, device placement, sleep-wake state, cadence, and setting in the qualification envelope (FDA-NIH Biomarker

Working Group, 2016; Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

Exercise, panic, fever, sleep, delirium, opioid medication, and palliative sedation can produce physiological signals that resemble or invert one another. Qualify their state-dependent interpretation.

Population scope also includes who is excluded by design. If the model performs only in fluent speakers using one device under low burden, it must say so. If people with disability, skin-tone sensor artifacts, autonomic neuropathy, arrhythmia, movement disorders, chronic pain, or medication profiles are underrepresented, the model's silence is not neutral. It is an unqualified region.

9.1.3 — Development Versus Confirmatory Use

Instrument development may compare features, transforms, lags, and channels. Apply Section 6.7.5's separation of development from protected testing: freeze the qualified measurement specification before confirmatory use, and disclose each later modification as a new version requiring fresh evaluation (Shiffman et al., 2008; Stone et al., 2007; Flake & Fried, 2020).

9.1.4 — Performance Requirements by Use

Assess sign orientation, next-hour prediction, terminal-dynamics estimation, and reliability across persons, occasions, and states separately. Match each performance criterion to the intended use and measurement model (American Educational Research Association et al., 2014; Hernán & Robins, 2020; Vogelsmeier et al., 2024; Castro-Alvarez et al., 2026).

Beyond association and artifact control, within-person tracking requires responsiveness, temporal stability, lags, and individual baselines. Prediction adds time-forward calibration

and distribution-shift error; terminal inference adds comfort-first feasibility, independent endpoint adjudication, missingness, rivals, and abstention.

Validation architecture should mirror the intended use. New-stream transport requires person-held-out evaluation; forecast claims require time-forward splits; hardware portability requires device or firmware holdouts; and care-system transport requires site holdouts. These are not interchangeable checkboxes, and failure in one lane must narrow the claim that depends on that lane rather than being averaged into one global score.

Performance also includes humane feasibility. If a device is accurate only when it wakes participants, interrupts care, requires frequent adjustment, or pressures families to maintain signal, the instrument has created burden that may alter the data.

9.1.5 — Abstention and Exclusion Rules

A reason-coded abstention is a valid and informative measurement output. Apply Section 24.2's canonical evidence-state rules, and record the measurement, calibration, consent, or participant-protection condition that blocked the estimate (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

Abstention rules should be declared before use. The system should withhold a candidate momentary estimate when required channels fail, sign-bearing evidence disappears, missingness is informative beyond modeled sensitivity, device contact is poor, medication or care procedure makes the channel uninterpretable, context falls outside qualification, or participant burden exceeds the protocol threshold.

Intervals outside the context of use must remain gaps, identified sets, or sensitivity bounds in cumulative outputs.

Preserve version-separated estimates when migration fails and label later-informed smoothing under Section 4.3.5.

Abstention has to be audited. A system that abstains more often for disabled participants, darker skin tones, terminal patients, nonfluent speakers, people in pain, or people without stable housing may be revealing a fairness defect in the instrument. The answer is not to force estimates for equity optics. The answer is redesign, narrower claims, or retirement.

9.1.6 — Context-of-Use Contract

Enforce Chapter 3's prohibited uses through the instrument's purpose limits, access controls, audit, and retirement rules (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

A context-of-use contract states when changes in instrument, software, treatment, language, population, or setting end qualification. A blank field leaves the tool promising, not qualified (FDA-NIH Biomarker Working Group, 2016; U.S. Food and Drug Administration, 2025b).

Consider a momentary estimator for adults recovering at home during the first fourteen waking days after surgery. The question is whether a locked physiological model tracks within-person change under known devices, medication patterns, language, prompt burden, and care events without using recovery outcomes to select signs or weights. The result concerns within-person postoperative tracking under those specified conditions.

Performance requirements follow that use. Analytical reliability, artifact control, within-person responsiveness, checks for medication effects and motion artifacts, time-forward validation, participant burden, and contestability may be required, while between-person comparability and whole-life inference remain outside scope. A participant with device

intolerance, missing anchors, or unusual medication effects can remain fully included in care and descriptive observation while the estimator abstains.

Exploratory development requires transparent features, artifact rules, basic reliability checks, burden monitoring, and explicit labels. Its outputs remain exploratory associations and development results.

Within-person tracking requires repeated anchors, time-forward validation, lag tests, a missingness model, and a route for participant challenge. Its conclusions concern within-person change under those qualified conditions.

External replication requires independent data within the declared context of use. Match person, site, device, or state holdouts to each claimed form of transport, and restrict conclusions to the reference class and context actually tested.

A confirmatory study of the Law requires a frozen estimator, protocol-fixed estimand, independent endpoint determination, rival models, and abstention rules. It tests only the specified contrasts within the qualified construct, context, and scale.

Prohibited contexts must be enforced technically. Purpose-bound schemas, separated identifiers, role-based access, deletion schedules, query restrictions, audit logs, and release controls should prevent persistent rankings, cross-system dossiers, secondary training, and eligibility dashboards (National Institute of Standards and Technology, 2020; National Institute of Standards and Technology, 2024; Wilkinson et al., 2016).

9.2 — Maps, Estimands, and Estimators

Four objects must be separated. The measurement map describes how the latent target and context generate observable records. The estimand states the exact quantity the study seeks to learn. The analysis map specifies how raw records become analyzable variables. The estimator is the finite-sample procedure that returns an answer with uncertainty. Validation at one layer cannot be borrowed by another.

9.2.1 — Measurement Map and Observable Record

The measurement map states how states, events, reports, sensors, and context become recorded variables, including timing, units, transformations, and failure codes. It fails its intended role if the record cannot distinguish a target-relevant change from device, language, task, or context change.

Reactivity belongs in the map. Prompts can alter attention, devices can change sleep or movement, clinical observation can change behavior, and repeated ratings can teach participants a vocabulary that reshapes later reports. A channel that changes the state it measures may remain useful, but its intervention effect must be modeled or bounded.

Align a signal recorded at one minute with the interval addressed by a report made ten minutes later, using the declared response kernel and lag model.

Model observation selection as specified in Section 9.6.2, including burden-related channel loss and any gap between the observation stop and the target endpoint.

For each channel, the measurement-map table records target relation, model role, unit, time window, transformation, lag, confounds, burden, missingness, version, prohibited inference, measure source, modifications, and validity evidence for the declared use (Flake & Fried, 2020; Gagnier et al., 2025).

9.2.2 — Scientific Estimand and Target Contrast

The scientific estimand identifies the exact population, stream, window, contrast, functional, and conditioning set to be learned. A reported number that answers a different question does not estimate the protocol-fixed target.

An estimand is the precise scientific quantity a study intends to learn. It includes the population or stream class, state and time window, treatment or exposure condition where relevant, outcome definition, handling of intercurrent events, and contrast of interest. An estimand is not the observed score and not the algorithm that estimates it (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

A momentary estimand must name what momentary means. It may refer to an instantaneous idealization, a short window, a latent state at a time point, or an average over a sampling interval. A rate estimand must say how its unit relates to time. That statement makes time accumulation dimensionally interpretable.

An interventional estimand also separates experience from assignment and adherence. Treatment, care, context, device use, and participant choice can change both the target and observation. The protocol should state whether it seeks a treatment-policy effect, an effect under adherence, a principal-stratum contrast, or a descriptive within-person response.

Intercurrent events, such as a relevant treatment change, can alter the interpretation or existence of the outcome. Device removal, missed prompts, and withdrawal may instead create missing data. Record the clinical event separately from observation loss and handle each according to the estimand (International Council for Harmonisation, 2019).

9.2.3 — Analysis Map and Preprocessing

The analysis map specifies preprocessing, alignment, features, missing-data handling, calibration, and transformations before estimation. A reasonable alternative that reverses sign, changes scope, or materially alters the conclusion exposes a choice requiring independent justification.

The analysis map specifies synchronization, artifact rejection, filtering, features, normalization, missingness, lags, channel exclusions, scoring, scale modifications, and learned representations. Version these decisions because they can create or remove the claimed pattern (Munafò et al., 2017; Nosek et al., 2018; Flake & Fried, 2020).

Preserve the filtered, smoothed, and forecast information sets defined in Section 4.3.5, and distinguish imputed intervals filled under assumptions. Do not combine these products in one unlabeled path.

Methods prose must reconstruct the scientific choices and their chronology without requiring code access; code then supports execution and reproducibility (Flake & Fried, 2020; Peeters et al., 2026).

9.2.4 — Candidate Momentary Estimator

The estimator is the finite-sample rule mapping the analysis record to an estimate or identified set, with uncertainty and regularization declared. Bias, instability, nonrecovery, or unjustified certainty under plausible simulated or held-out conditions defeats the affected use.

A candidate momentary estimator produces a bounded estimate of a declared momentary or short-window target. Candidate architectures include a transparent weighted composite, a state-space model, and a Bayesian latent-state

model (American Educational Research Association et al., 2014; Bollen, 1989; Borsboom, 2005; Gelman et al., 2013).

Display a transparent composite's contributions and weight sensitivity. Standardized channels may lack common units or meaning, equal weights may hide unequal reliability, and learned weights may overfit outcome labels.

In a dynamic latent-state estimator, test how time-varying loadings, process noise, contextual covariates, and smoothing affect recovery of continuity, lag, missingness, and state evolution.

Uncertainty should be a distribution or identified set when the model permits it. Pointwise standard errors alone do not capture model choice, sign ambiguity, channel conflict, missingness, or calibration uncertainty. The estimator should widen, split, or abstain when those uncertainties dominate.

9.2.5 — Interval Estimation and Path Integration

Construct interval contributions using the qualified rate, clock, exposure, unity, and gap conventions in Sections 5.2.3–5.2.4 and Section 6.5.4 (Shiffman et al., 2008; Stone et al., 2007).

Sparse or uneven cadence can alias rapid dynamics; interpolation and smoothing can invent continuity or suppress extremes. Prespecify when to interpolate, forecast, impute, abstain, or return bounds, propagating those choices into path uncertainty (Shiffman et al., 2008; Stone et al., 2007; Peeters et al., 2026).

Posterior path integration is preferable to summing isolated point estimates when dynamics and serial dependence matter. The model should draw coherent trajectories, integrate each trajectory over the declared interval, and report the resulting distribution of interval contributions. That procedure carries

timing, lag, covariance, and missingness into the cumulative object.

Report uncertainty for the object actually estimated. Integrating reconstructed paths maps their distribution or compatible set into interval totals; a method that bounds the total directly need not recover every intervening state (Sections 9.4.3 and 10.6.1). Distinguish deterministic bounds, confidence coverage, and posterior probability, retaining invalid or modeled intervals and version boundaries in the result.

9.2.6 — Reported Quantities and Decision Rules

The reported quantity must identify whether readers are seeing an observed channel, local contrast, interval estimate, path estimate, or identified set. Language, labels, graphics, and captions must not imply a stronger latent or cross-person claim.

The verdict rule connects evidence to abstention, narrowing, route comparison, correction, or withdrawal. It must be fixed in the protocol; changing it after outcomes or allowing downstream success to compensate for an upstream prerequisite failure invalidates that confirmatory use.

Name the reported quantity, its estimand, its estimator, and whether it is an estimate, compatible set, or abstention. Freeze the decision rule under Sections 6.7 and 24.2 before outcome inspection (Gneiting & Raftery, 2007; Manski, 2003).

For confirmatory held-out evaluation, fix the outcome, horizon, information set, and scoring rule before inspecting the evaluation data. A scoring rule’s mathematical propriety is distinct from whether its selection and use support a confirmatory claim. Calibration, sharpness, coverage, and decision loss answer different questions; predictive success supports the declared forecast relation unless construct and

measured-to-latent bridges justify more (Gneiting & Raftery, 2007).

9.3 — Phenomenal-Value Channels and Auxiliary Witnesses

The chapter distinguishes candidate phenomenal-value channels from auxiliary witnesses. A candidate channel may contribute to a declared estimator under explicit transformation, sign, lag, and calibration rules. An auxiliary witness helps validate, challenge, contextualize, or explain the channel model. Usefulness does not silently promote a witness into the target (Campbell & Fiske, 1959; Shiffman et al., 2008).

9.3.1 — First-Person Reports and Pairwise Judgments

First-person report contributes construct meaning, sign, language, timing, salience, and challenges to the model. Fear, demand, communication limits, recall, reinterpretation, and nonresponse can invalidate a proposed use and must be examined.

Reports vary in what they ask. A rating of intensity, unpleasantness, relief, agency, meaning, or desire is not interchangeable with a global mood score. Pairwise judgments may be easier than numerical ratings when a person can say that one interval was more burdensome without knowing how much more. The representation level should match the response task.

Participant challenges of labels, sign, event timing, or missing dimensions (Section 8.6.1) are construct-validity evidence requiring recorded adjudication. Even when the model retains its estimate, it must explicitly address the reported mismatch.

9.3.2 — Trade-Off Choices and Behavioral Evidence

Pairwise and trade-off judgments can establish local ordering, bounded contrasts, or within-person calibration under declared options. Choices also reflect power, information, obligation,

stakes, and opportunity, so they cannot be read as experience alone.

Behavior records approach, avoidance, persistence, withdrawal, performance, and intervention responses. Its relation to experience is many-to-many; strategy and context changes can defeat a simple interpretation.

Use Section 8.1.3's model of choice constraints: option set, incentives, obligations, risks, and information. Without a justified observation relation, the inference remains about behavior.

Trade-off tasks may help with local scaling when designed ethically. Repeated within-person choices can indicate ordinal or interval-like relations among bounded states, especially when paired with reports and contextual anchors. The task must not manufacture suffering, withhold care, or impose stakes that make refusal costly.

Consistency, bravery, equality, or sign cues can shape responses. Neutral framing, randomized order, comprehension checks, debriefing, and competing response models reduce but do not eliminate demand effects.

9.3.3 — Autonomic and Interoceptive Channels

Autonomic physiology can identify the timing of arousal, effort, threat, recovery, and regulation under local calibration. Sign ambiguity, medication, movement, illness, device artifact, and person-specific gain can require narrowing or removal of a channel.

Interoceptive evidence concerns awareness and interpretation of bodily states, including mismatch between physiology and felt state. Access, language, learned meaning, and interoceptive accuracy vary, so a bodily signal does not interpret itself.

Autonomic measures include heart-rate dynamics, electrodermal activity, respiration, temperature, vascular and pupil responses. Interoception concerns sensing and interpreting bodily states. Their relations to arousal, effort, threat, pain, recovery, and signed valence require local calibration (Cacioppo & Berntson, 1994; Khalsa et al., 2018).

The body can participate causally in affective regulation. Hsueh et al. (2023) used an optical pacemaker to increase heart rate in mice and found increased anxiety-like behavior in risky contexts. The same cardiac manipulation did not produce a detectable preference or aversion in a neutral place-preference test, and inhibition of posterior insular cortex attenuated its anxiety-related effect. This pattern supports a context-sensitive body-to-brain pathway rather than a fixed emotional meaning for heart rate. It supplies a useful constraint for affect models: bodily state and environmental appraisal must be allowed to interact. In noninvasive human observational work, the corresponding model comparison can ask whether a prespecified physiology-by-context interaction improves held-out predictions of reported feeling beyond either channel alone. A declared PTVC route can extend that comparison with its own predicted timing or state interaction.

Apply the channel specification in Section 9.2.1, accounting for sensor placement, skin contact, posture, hydration, ambient temperature, illness, and hardware version.

Hierarchical borrowing must preserve person-specific baseline, gain, and state sensitivity; within-person reliability does not establish a common cross-person quantity.

Experiments have begun to separate current bodily state from its neural anticipation. In mice, Livneh et al. (2020) found that insular population activity tracked hunger and thirst, while learned food or water cues rapidly shifted activity toward

patterns associated with future satiety before physiological replenishment. Artificially activating hypothalamic hunger or thirst systems could restore seeking in sated animals while ongoing insular activity continued to reflect satiety. These dissociations give predictive interoception an empirical architecture: motivation, present physiology, and anticipated physiology can be measured separately. They support a biologically grounded route through which expected bodily consequences may shape affect. A human measurement study can compare models of current physiology, anticipated need, reported feeling, and choice using ordinary eating and drinking records without deprivation or care alteration. The resulting dissociations provide a basis for comparing anticipatory and accumulated-history mechanisms.

9.3.4 — Neural and Clinical Channels

Neural evidence can supply mechanistic, temporal, or state-classification information relevant to pain, reward, salience, memory, and consciousness. Task or population transport failure, many-to-many mapping, and classifier drift preclude direct inference of valence or personhood.

Cross-task affect decoding supplies a concrete comparison. Yang et al. (2026) combined intracranial recordings with self-rated valence and arousal across two tasks in 18 participants. Personalized models benefited from gray- and white-matter signals and generalized across the tested tasks; real-time decoding was demonstrated in four additional participants. The recordings originated in clinical seizure-localization work, and analysis code is archived as Zenodo record 20500593.

An approved secondary analysis can compare calibration and cumulative prediction error on held-out recordings with a report-history predictor given the same available information. Existing

clinical recordings offer data for this temporal measurement comparison.

Person-specific neural measurement has already advanced beyond brief laboratory snapshots. Shirvalkar et al. (2023) paired repeated pain reports with ambulatory intracranial recordings over months in four people with refractory neuropathic pain and decoded higher versus lower reported pain within individuals. Huang et al. (2025) extended naturalistic acute-pain work to 12 people undergoing clinically indicated epilepsy monitoring, combining neural recordings, reports, and facial dynamics. Their within-person neural classifier distinguished high from low reported pain with a mean area under the receiver-operating-characteristic curve of 0.70. Together these studies demonstrate feasible repeated measurement of a limited experiential dimension in real clinical settings. With consent and existing clinical oversight, approved secondary analyses can test temporal stability, channel disagreement, and generalization to later recordings without adding implants or changing care.

Clinical intervention can also test whether an individualized neural marker connects to a change in lived symptoms. Scangos et al. (2021) mapped symptom-related activity and stimulation responses in one person with severe treatment-resistant depression, then used a chronically implanted device to trigger stimulation when the selected biomarker appeared. Symptoms improved rapidly and remained improved during the reported follow-up. The intervention links a timed internal-state estimate with an independently reported improvement. Consented secondary analysis of clinically approved records could compare symptom-responsive, time-only, and state-insensitive models while preserving treatment and privacy. The scientific gain

would be a better bridge between measured dynamics and reported experience.

Consciousness and communication assessments identify retained capacities, available response channels, and state-specific uncertainties. Those findings guide the choice of measurements and the intervals in which each channel is interpretable.

9.3.5 — Contextual and Relational Witnesses

Clinical observation records illness, medication, care context, delirium, pain behavior, communication capacity, and acute change. Observer bias, workload, familiarity, institutional routines, and care effects belong in the observation model.

Contextual or relational witnesses can clarify confounds, timing, social setting, and missing or conflicting channels. Purpose expansion, dossier formation, bystander privacy, and narrative overreach can require minimizing or deleting the record.

Medication, care procedures, sleep, temperature, noise, privacy, social contact, workload, language, legal pressure, and material security can affect experience and observation. Include the context variables needed by the measurement model (Shiffman et al., 2008; Little & Rubin, 2019; Nissenbaum, 2010).

9.3.6 — Plural Evidence and Composite Limits

Multiple channels offer different access, error, burden, and failure modes. Convergence is informative to the extent that a qualified joint measurement model can distinguish shared experiential variation from shared method effects. A common confound can explain agreement, but partial shared error does not automatically eliminate every additional constraint supplied

by a channel. Report which distinctions the joint observations identify and which remain unresolved (Campbell & Fiske, 1959; Bollen, 1989).

Fusion must preserve channel-level estimates, uncertainty, missingness, lag, negative controls, and disagreement. Compare the designs and diagnostics in Section 9.4.6 on held-out cases and recovery tasks.

Channel ablation is an ethical and scientific diagnostic. The study should show whether one channel drives the result, whether removal changes sign, and whether the model remains calibrated for participants with missing or inaccessible channels. A system that works only for one bodily or communicative template is not universally qualified.

9.4 — Dynamic Measurement and Channel Conflict

Channels run on different clocks. Reports may arrive after reflection, while response-time features can capture uncertainty rather than valence. Autonomic activation and recovery follow different lags, and device records may shift because of clock drift or late synchronization. A dynamic model must estimate or bound these channel-specific offsets before treating disagreement as conflict in experience (Shiffman et al., 2008; Sosa-Hernandez et al., 2024).

9.4.1 — Impulse-Response Functions

An impulse-response function describes how a channel reacts over time to a change in the latent state. One signal may spike immediately and decay. Another may rise slowly and persist. Another may respond only after a threshold (Shiffman et al., 2008; Dejonckheere et al., 2022).

For a causal linear time-invariant channel, or a justified local approximation to one, the following continuous-time representation makes the lag structure explicit before fusion or deconvolution:

$$X_k(t) = \int_0^{\infty} h_k(r) F(t-r) dr + \varepsilon_k(t)$$

In the preceding display, X_k is the observed channel, h_k its causal impulse-response kernel, F the latent input, and ε_k the observation error.

A calibrated lag can preserve an interval total even when the instantaneous input is difficult to reconstruct. Consider a unit-gain first-order observation channel in one declared clock. Let r be an integrable qualified contribution rate, let y be its absolutely continuous noise-free channel output, and let the known response time τ be positive:

$$\tau \frac{dy(t)}{dt} + y(t) = r(t), \text{ almost everywhere.}$$

Integrating the channel equation gives the finite-window identity

$$J_{ab} := \int_a^b r(t)\,dt = \int_a^b y(t)\,dt + \tau[y(b) - y(a)].$$

The last term accounts for the change in channel state at the two boundaries. Both y and r have rate units; multiplying the difference between the two boundary values of y by τ gives contribution units. The identity avoids differentiating noisy observations and does not require pointwise recovery of r. It applies only after the gain, origin, response law, and rate interpretation have been independently qualified.

A synthetic example shows the boundary term’s importance. With τ=1 and y(0)=0, let r equal +1 on [0,1) and −1 on [1,2]. The true contribution is zero, but the output area through time 2 is $(1-e^{-1})^2 \approx 0.3996$. Because $y(2) = -(1-e^{-1})^2$, the boundary correction recovers zero exactly. The channel has not finished responding when the observation window ends.

For known τ, define $\widehat{J} = \widehat{I} + \tau\,[\widehat{y}(b) - \widehat{y}(a)]$. If $\widehat{I}$ estimates the output integral with absolute error at most δ_I and the two output readings have errors bounded by δ_a and δ_b, respectively, then

$$|\hat{J} - J_{ab}| \le \delta_I + \tau\,(\delta_a + \delta_b)\,.$$

These bounds require no independence between the three estimation errors. Unknown response time, gain, or origin must

remain in the admissible parameter set; an approximate channel equation also needs a bound on its integrated discrepancy. Cadence and missing observations enter the output-integral uncertainty. This is correction of an observation lag, not a claim of post-endpoint experience or recovery of an unobserved prefix. Choose observations by the target's required precision and participant burden, retaining broader channel uses when fine-resolution inference is unavailable.

9.4.2 — Ordinal and Non-Gaussian Models

Ordinal, counted, bounded, censored, saturated, skewed, heavy-tailed, and zero-inflated observations need appropriate families. A common Gaussian model can manufacture agreement, attenuate extremes, and distort uncertainty (Bollen, 1989; Särkkä, 2013; Shiffman et al., 2008).

The observation model, including its link function, should match the measurement process. An ordinal report may need a cumulative link model. A count of awakenings may need a count distribution. A bounded probability may need a logit-scale model. A saturated sensor may require censoring. A rare artifact may require a mixture or outlier model.

Non-Gaussian observation models also affect fusion. A high-confidence continuous channel may dominate a low-resolution ordinal channel unless the model respects each source's information content. Conversely, a coarse report may carry sign information that the physiology lacks. The model should not rank channels by mathematical smoothness.

9.4.3 — Lag Estimation and Deconvolution

Lag estimation and deconvolution are inverse problems governed by kernel, cadence, regularization, noise, and missingness. Temporal alignment or trajectory recovery must be

assessed under those conditions (Särkkä, 2013; Shiffman et al., 2008).

Before confirmatory use, specify which lags are fixed or estimated, the allowed priors, and how lag uncertainty enters the result. Lags selected for favorable terminal convergence remain exploratory.

Deconvolution can amplify noise under strong smoothing, irregular sampling, or lost high-frequency information. Biological and technical slowness must be distinguished; erased frequencies require additional assumptions. Report regularization, lag sensitivity, recovery, and failures before using a deconvolved path (Särkkä, 2013; Peeters et al., 2026).

Fit lag together with Section 9.6.2's context and observation-selection model; burden-related delays or channel loss can otherwise be mistaken for recovery.

The recovery target can be narrower than the whole path. In a finite noiseless linear observation model $y=Ax$ with independently qualified A, let $x\in\mathbb{R}^n$ contain interval contributions and let the required total be $J=c^Tx$. For unrestricted x, J is identified exactly when c lies in the row space of A. Equivalently, there is a weight vector w such that

$$A^T w = c, \text{ so } J = w^T y.$$

To see necessity, any vector z with $Az=0$ is invisible to the observations. If $c^Tz\neq0$, x and $x+z$ produce the same y but different totals. Conversely, a row-space representation gives $J=w^TAx=w^Ty$. For example, observing $y_1=x_1+x_2$ and $y_2=x_3+x_4$ identifies the total y_1+y_2 without identifying the four contributions separately. Observing only differences between adjacent contributions leaves a common offset invisible and therefore does not identify their total.

With measured output $\tilde{y} = Ax + \varepsilon$ and componentwise bounds $|\varepsilon_j| \leq \delta_j$, the same representation gives

$$\left|w^T \tilde{y} - J\right| \leq \sum_{j=1}^{m} \left|w_j\right| \delta_j.$$

Large weights can amplify error, so identification and attainable precision remain separate. Uncertain channel coefficients, incomplete coverage, or an unqualified phenomenal scale require their own bounds; this linear-algebra result does not supply them. The finite-window identity in Section 9.4.1 illustrates the same functional-first approach in continuous time. Compare aligned and unaligned development outputs, then freeze the independently justified channel map and uncertainty before evaluating the cumulative claim.

9.4.4 — Conflict as Evidence

Preserve conflicting reports, physiology, behavior, and clinical observations for interpretation rather than averaging them into agreement (Bollen, 1989; Tsiatis, 2006). Section 9.4.6 gives concrete patterns and responses.

Use contribution plots to inspect report–physiology disagreement and whether conflict clusters by person, site, device, state, or time.

A concrete calibration question can be posed before fusion. Within one qualified context, suppose aligned channels measure the same latent quantity X through $O_j = a_j + b_j X + \varepsilon_j$, with nonzero constant loadings and finite second moments. Assume mean-zero errors uncorrelated with X and with one another. This is an explicit single-factor model, not a presumption that arousal and valence are one construct. Write $C_{jk} = \mathrm{Cov}(O_j, O_k)$. The triple-collocation identity gives, when $C_{23} \neq 0$ (Stoffelen, 1998),

$$Var(\varepsilon_1) = C_{11} - \frac{C_{12}C_{13}}{C_{23}}.$$

Under the same assumptions, C_{2_3}/C_{1_3} identifies b_2/b_1. A fourth channel supplies an overidentifying covariance restriction (Bollen, 1989):

$$C_{12}C_{34} - C_{13}C_{24} = 0.$$

Evaluate the recovered error variances and tetrad with uncertainty that respects within-person dependence. Small denominators make the decomposition unstable. A failed restriction challenges the model; passing it does not establish uncorrelated errors, since a shared method factor proportional to the loadings can satisfy the same tetrad. Relative gains and error components are identified only under the maintained model; separate evidence must anchor the phenomenal origin.

9.4.5 — Medication and Device Effects

Model treatment timing and its effects on heart rate, respiration, movement, sleep, arousal, facial expression, report capacity, and response timing, including simultaneous changes in illness and relief (Shiffman et al., 2008; Khalsa et al., 2018; Giacino et al., 2018; Feldman et al., 2024).

Placement, contact, temperature, battery behavior, firmware, charging, removal, compression, cadence, and manufacturer preprocessing can alter signals. These changes may depend on burden: discomfort prompts removal, care prompts adjustment, and prolonged sleep or hospitalization can interrupt charging.

Use medication and device records to model competing explanations. Designate a variable as a negative control only when its required separation from the target process is independently justified. If a care-procedure flag explains a large apparent candidate momentary estimator change, the Law-facing interpretation should narrow. If a firmware update shifts estimates across participants, version migration must occur before historical and current data are combined. If device

contact quality predicts the final candidate momentary estimator, contact quality is not merely a nuisance; it is a rival.

If comfort medication makes the estimator uninterpretable, abstain; never delay or adjust care for measurement clarity.

9.4.6 — Channel Failure and Removal

Classify channel failure as analytical, technical, construct-related, ethical, or contextual, and name the affected claims when removing the channel.

Freeze reliability, artifact, burden, and context rules, or evaluate changes in separate development data. A channel may not be removed or down-weighted because it disrupts the preferred trajectory (American Educational Research Association et al., 2014; Nosek et al., 2018).

Channel composition is part of the estimator version. If one participant is estimated from five channels and another from two, their outputs are not automatically comparable; if firmware changes gain, pre- and post-update paths require a migration study or version split. Missing or removed channels must propagate into uncertainty and claim status.

Remove sensors that disturb sleep, irritate skin, expose participation, interfere with care, or become intolerable during distress. Burden-triggered removal also changes the sample and observation process and must be recorded.

When a participant reports suffering while physiology appears calm, consider dissociation, medication, learned suppression, an incorrect sign interpretation, and sensor insensitivity. Preserve the disagreement and examine its context; apparent physiological calm cannot simply override the report.

A physiological change across participants immediately after an update may reflect firmware, preprocessing, calibration, or a

device batch. Split the versions and conduct a migration study before pooling them.

Terminal variance that compresses as channels disappear may reflect observation shutdown or comfort-related device removal. Model the missingness and test that apparent compression against these alternatives.

When one channel drives a composite, examine weight dominance, artifact, a true mechanism, and fusion design. Report the components and repeat the analysis with each channel omitted in turn.

Compare fusion strategies on the same protected cases to distinguish better inference from more flexible adjudication.

Early fusion combines standardized features before fitting and can represent interactions. Because it can hide differences in scale, lag, and artifact within one input space, it requires preprocessing audits, feature ablation, and tests for domination by negative controls.

Late fusion keeps channel models separate before combining predictions. Consensus can still conceal systematic disagreement or shared confounds, so report channel-level posterior estimates and conflicts.

A hierarchical latent model jointly represents shared state, loadings, uncertainty, and missingness. A well-fitting factor may represent arousal, illness, or device gain instead of valence; test location, scale, sign, recovery, and rival generators.

An ensemble can improve prediction across heterogeneous learners while reducing interpretability or claim specificity. Evaluate it with person, time-forward, site, and device holdouts matched to the intended inference.

Adjudicative fusion retains disagreement for expert review and contextual annotation. Written criteria, blinded review, and

an audit trail must be specified in advance so that adjudication cannot become an after-the-fact rescue.

Keep uncertainty over channel state as well as latent state. Unresolved conflict can require wider or channel-specific outputs, a narrower claim, or No adequate estimate, while preserving useful evidence from a witness whose meaning remains identified.

9.5 — Measured-to-Latent Invariance

Invariance asks whether the measurement relation preserves enough structure for the intended comparison. Calibration asks how observed values relate to declared anchors and states. Linking asks whether versions or instruments can be placed on a common coordinate. Transport asks whether a relation learned in one context survives another. These are connected problems, not synonyms.

9.5.1 — Measurement Invariance as a Bridge Gate

Use Chapter 10's invariance tests to establish which comparisons across time, people, languages, states, sites, devices, or versions survive. Full, partial, local, approximate, or absent invariance licenses different comparisons (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016; Paap et al., 2025).

For within-stream tracking, the central question is often whether loadings, thresholds, sign, and response behavior remain stable enough to interpret change. Illness, medication, fatigue, learning, disability, and changing language can alter that relation.

Time-varying measurement models may be necessary when change is expected. Flexibility should be bounded, prospectively specified where possible, and evaluated through recovery studies. A model that allows every loading to move can fit almost anything while making the latent trajectory uninterpretable.

9.5.2 — Local Calibration and Context of Use

Within a declared context, calibration can use repeated anchors, ethically available contrasts, treatment or environmental changes, cognitive interviewing, and participant interpretation.

Broader transport requires a separately justified link (American Educational Research Association et al., 2014; International Test Commission, 2018; Sprangers & Schwartz, 1999).

Calibration files should distinguish direction anchors, magnitude anchors, contextual anchors, and origin claims.

Count calibration prompts, challenge tasks, device wear, clinician time, and bridge sampling against the protocol's burden limits.

Local calibration should report failure and drift. If anchors change meaning, channels decouple, or participant reports contradict the model, the bridge may split into state-specific versions.

9.5.3 — Linking Across Versions and Settings

Linking places instruments, versions, languages, devices, sites, or models into a relation that permits a specified comparison. Common-item designs, common-person designs, bridge samples, anchor events, latent linking models, and historical replay answer different transport questions and require overlap across the range and states for which the link will be used. The method must be chosen for the claim it is expected to carry, and a link qualified for ranking cannot be promoted into common-origin or common-unit comparison (American Educational Research Association et al., 2014; Millsap, 2011).

Run old and new versions in parallel where feasible, or compare them using bridge data and synthetic fixtures.

9.5.4 — Measured-to-Latent Assumptions

A latent variable is not a hidden fact merely because a model estimates it. Location, scale, rotation, label switching, and model equivalence can leave multiple latent interpretations compatible

with the same data. Constraints must be justified independently of the desired Law-facing result.

Apply Section 10.6.2's error decomposition to the measured-to-latent bridge. A narrow posterior under one model can coexist with large uncertainty about whether the model represents valence at all.

Challenge recovery with known states, lags, missingness, sign reversals, response shift, device drift, and rival generators. The pipeline must widen or abstain when identification is removed.

Report which bridge assumptions are empirically supported, imposed, or unresolved, and what evidence would require narrowing.

9.5.5 — Licensed Claims and Bridge Failure

A bridge can support directional robustness without point identification. If all admissible calibrations imply improvement, the study may say so even when the magnitude is uncertain. If signs or levels differ across admissible bridges, the result should be an identified set, sensitivity statement, or abstention.

Failure should be localized by bridge layer and licensed output. Loss of cross-person linking may leave within-person direction intact; loss of scalar level may leave ordinal change; endpoint incompatibility may leave a bounded clinical-state model; and one failed channel need not invalidate another. Salvage is legitimate only when the surviving object, population, state range, and inference tier are named separately from the failed bridge.

9.5.6 — Bridge Failure, Narrowing, and Abstention

Bridge failure occurs when representation, origin, sign, scale, invariance, linking, observation, or recovery requirements are not met. Because these gates are noncompensatory, success

downstream cannot repair failure upstream. A well-fitted cumulative curve cannot rescue an invalid bridge (Manski, 2003).

Identified sets show conclusions compatible with plausible origins, links, and calibrations. Incompatible signs or statuses indicate structural ambiguity rather than simply a large error bar.

Apply the canonical evidence states in Sections 24.2 and 27.4. Here, failed measured-to-latent identification narrows or blocks only the inference that depends on that bridge; collecting more observations does not repair nonidentification.

Later evidence of noninvariance, sign error, or invalid transport requires correction, narrowing, versioning, or withdrawal of affected figures, summaries, and verdicts. Preserve earlier outputs for traceability.

Bridge failure locates unsupported measured-to-latent transformations. Name the failed link, comparisons to narrow or stop, and lower-level claims still estimable. A more flexible fit to the same outcome does not repair the failure.

9.6 — Missingness, Provenance, Versions, and Uncertainty

An observation becomes interpretable through its collection history as well as its value. Missingness, processing versions, and uncertainty determine which comparisons the record can support. A complete provenance account links those features to the estimator and retains the information needed to reproduce a result or understand why a later revision changes it.

9.6.1 — Missingness Codes

Missingness codes should describe why, when, and at which layer a record is absent. No prompt, no response, refusal, sleep, device removal, low contact quality, care procedure, hospitalization, communication loss, privacy pause, technical failure, and endpoint-related observation shutdown are scientifically different events. Collapsing them into one missing flag destroys information about the observation process and can bias the measured path (Little & Rubin, 2019; Tsiatis, 2006; Shiffman et al., 2008).

Code refusal and withdrawal as participant consent decisions, not technical failures.

The codes should be applied as close to collection as possible and preserved through analysis.

Codes should also be auditable across groups and sites. If one site has high staff-unavailable gaps, the issue may be workflow rather than physiology. If one subgroup has high contact-quality gaps, the issue may be device design. If terminal windows show high privacy gaps, the study should respect them rather than impute over them without caution.

Record only the detail needed for a missingness code to govern inference.

9.6.2 — Informative Observation

Informative observation means that whether a record exists depends on the latent state, observed history, participant burden, care context, device condition, or another variable related to the target (McNeish, 2025). In experience-sampling research, prompt density, study duration, item burden, population, and protocol design can change compliance and therefore who and what remains observed. A large observed sample does not make the selection process ignorable (Diggle & Kenward, 1994; Wrzus & Neubauer, 2023; Peeters et al., 2026).

A compact observation model can represent how burden, device condition, care events, or other state variables alter the probability that a channel is observed:

$$\Pr\left(R_{kij} = 1 \mid F_{ij}, B_{ij}, D_{kij}\right) = logit^{-1}\left(\gamma_{0k} + \gamma_{1k}\left|F_{ij}\right| + \gamma_{2k}B_{ij} + \gamma_{3k}D_{kij}\right)$$

R_{kij} indicates whether channel k is observed for person or stream i at time j; B_{ij} can represent burden or observation burden; D_{kij} can represent device condition. This logistic form is illustrative rather than canonical. Its scientific purpose is to make observation causes explicit, because those causes can bias the measured path and may require selection models, sensitivity bounds, or abstention.

Informative observation can create false improvement. If high-burden intervals are less likely to be observed, the remaining data look calmer. It can create false worsening when measurement intensifies during crisis. It can create false compression when terminal discomfort leads to device removal and only stable channels remain.

Combine care logs, device-quality metrics, justified negative controls, and sensitivity analyses across plausible ignorable and informative observation mechanisms. Selection and pattern-mixture models expose assumption dependence; neither

identifies unobserved intervals without support. Report when credible alternatives change the scientific status (Tsiatis, 2006; Little & Rubin, 2019).

9.6.3 — Instrument and Model Version

Attach instrument and model versions to every momentary and interval estimate, using Section 6.1.5's provenance and migration requirements (Munafò et al., 2017; Nosek et al., 2018).

Distinguish instrument versions permitted for exploratory work, locked for confirmation, and retired for bias, sign failure, device shift, security, or burden.

Mark version boundaries in running outputs. Preserve the original contemporaneous result, any historical replay, and the migration method as distinguishable records and displays.

9.6.4 — Data and Model Provenance

Use Section 28.2's source-to-output dependency records for pipeline lineage, including consent permissions and access events (Wilkinson et al., 2016; Center for Open Science, 2025; National Institute of Standards and Technology, 2020).

The stack should distinguish raw data, transformed data, features, posterior paths, candidate interval estimates, running ledgers, summaries, and public figures. Each layer has different sensitivity and different reuse rules. Model-training permission for one layer should not imply permission for every layer.

Lineage must support adjudication of participant corrections to timing, context, missingness, device records, and auxiliary sources (Section 9.7.3).

9.6.5 — Migration and Backward Compatibility

Migration is the controlled relation between one instrument or model version and another. Backward compatibility is the

stronger claim that old and new outputs can be compared, pooled, or translated under a documented bridge. Neither follows from improved fit. A new model may be preferable for future work while making historical estimates noncomparable (American Educational Research Association et al., 2014; Wilkinson et al., 2016; Center for Open Science, 2025).

A migration study uses overlap where available, historical replay, known-state simulation, subgroup tests, sign stability, missingness behavior, and version-shift estimates. Report any change in historical conclusions.

9.6.6 — Audit and Reproducibility

The audit must reconstruct the measurement system, not only reproduce its analysis. Use the qualification dossier introduced in this chapter to check what was administered, transformed, and scored (Flake & Fried, 2020; Gagnier et al., 2025; Wilkinson et al., 2016).

Version fixtures with the pipeline and ensure they exercise current transformations. They must detect sign reversal, time misalignment, version drift, dropout, leakage, and failed abstention without exposing participants.

Specify access separately for raw signals, features, posterior paths, summaries, and public results. Define which checks remain possible through controlled access or secure enclaves and assess disclosure and reidentification risk (National Institute of Standards and Technology, 2020; U.S. Food and Drug Administration, 2023).

Independent review should combine psychometrics, signal processing, missing-data methods, clinical context, privacy engineering, and participant perspectives. Plausible detectors can flag different records, so flags need validation and uncertainty rather than becoming credibility scores

(Vogelsmeier et al., 2026). Disagreement among detectors is itself diagnostic and should prompt examination of the error model rather than automatic escalation against the participant or site.

Missingness codes distinguish burden, refusal, sleep, care, device, and staff causes. Public labels should also distinguish observed, missing, abstained, and modeled quantities.

Instrument versions prevent hardware or preprocessing drift from being mistaken for experience. Public outputs identify the device and pipeline version.

Model versions identify weights, priors, architecture, and code release. Their public status distinguishes exploratory, locked, retired, and sensitivity uses.

A retained source-to-output provenance chain permits audit, challenge, and reproduction. Public labels state that this lineage is available under the applicable access rules.

Migration status determines which comparisons across versions are justified. Label the versions as linked, partially linked, unlinked, or split, and retain the evidence supporting that relation.

9.7 — Anti-Score, Privacy, and Non-Deployment Governance

Chapter 3 limits govern pipeline design. Permissions restrict computation. Outputs and participant access limit disclosure. Retirement rules mark when the system must stop (Shiffman et al., 2008; Stone et al., 2007; National Commission, 1979).

9.7.1 — No Human-Worth Score

Enforce Section 3.7's prohibition on human-worth scoring in output design. Display context, uncertainty, abstention, version, and prohibited interpretations; exclude leaderboards, rank colors, badges, moral or danger labels, achievement language, and public comparison (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

The rule also applies to aggregate public communication. A site, group, country, community, or clinical program should not be ranked by average candidate momentary estimator or cumulative candidate interval contribution estimate unless a specific, ethical, and scientifically qualified comparative study has been designed for that purpose and stripped of person-evaluation meaning. Even then, the target is measurement behavior or burden conditions, not human worth.

9.7.2 — No Hidden Continuous Monitoring

Choose the least intrusive monitoring cadence sufficient for the question, define off-periods and deletion rules, and measure whether observation changes behavior (Council for International Organizations of Medical Sciences, 2016; National Institute of Standards and Technology, 2020; Shiffman et al., 2008).

Hidden monitoring is prohibited. Participants should know what is collected, when, by which devices, for which research purpose, who can access it, what derived outputs may be

created, whether model training is permitted, how withdrawal works, and which outputs can be deleted or retired. Consent to wear a device is not consent to every inference the device might later support.

In private settings, use collection windows, privacy pauses, local processing, feature minimization, or participant-controlled modes where possible.

9.7.3 — Participant Access and Contestability

Participants should receive understandable explanations of outputs returned under the protocol. Access does not mean releasing raw telemetry or posterior draws without support. It means giving the construct, context, uncertainty, limits, and a route to challenge the data or interpretation, while preserving the option not to receive a result that lacks clinical or personal meaning (American Educational Research Association et al., 2014; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Contestability should be designed before collection begins. The protocol should specify which outputs participants can view, how they can challenge them, who reviews the challenge, what corrections are possible, whether downstream derived results can be updated, and when a challenge results in abstention rather than revision.

The system should also protect participants from overinterpretation. Return-of-results language should avoid diagnosis, moral interpretation, endpoint prediction, and claims about the whole life. It should say that the output is a research estimate under a declared context. If the estimate is not clinically actionable, that must be clear.

Family and surrogate contexts require careful separation. A participant’s access does not automatically grant relatives

access. A surrogate role does not create general permission to inspect private affective data. Contestability should respect capacity, consent, privacy, and the dignity of the person whose stream is being studied.

9.7.4 — No Cross-Architecture Dossier

The privacy architecture should be designed around separation rather than one supposedly perfect vault. Identity, raw signals, derived features, context, calibration files, model outputs, and public summaries have different purposes and retention needs. Linking them by default creates a dossier even when each component was collected for a defensible reason.

The cross-domain rules in Chapter 3 apply here: research estimates must not become civic dossiers, devotional records, or institutional-transition metrics.

Aggregated research outputs need caution too. Even deidentified data can be vulnerable when combined with rare conditions, time stamps, location-like routines, or clinical events. The stack should use minimization, aggregation thresholds, differential privacy where appropriate, or nonrelease when reidentification risk is too high.

9.7.5 — Artificial Intelligence as Bounded Support

Artificial Intelligence may assist artifact detection, provenance checks, anomaly flags, code tests, model comparison, and documentation under Section 3.6.4’s authority limits and Section 20.7.4’s analysis rules (World Medical Association, 2024). Every output remains auxiliary within its declared context of use.

Include AI prompts, embeddings, summarization templates, anomaly detectors, and feature flags in Section 28.2's versioned dependency records. Participant and public materials state what

assistance occurred, its limits, and the responsible reviewer; outputs may not imply direct access to experience.

Technical quality monitoring can flag device failure, site drift, missingness clusters, or version mismatch to inform the instrument's declared abstention rules.

9.7.6 — Retirement and Deletion

Each instrument carries versioned, scoped retirement criteria: persistent sign error, subgroup noninvariance, device dependence, excessive burden, broken provenance, unsafe disclosure, security failure, or repeated prohibited use.

Include raw signals, features, embeddings, checkpoints, logs, backups, exports, and identifiers in the deletion map (National Commission, 1979; Council for International Organizations of Medical Sciences, 2016).

Stop, narrow, retire, or, where authorized, delete unsafe versions; correct or withdraw outputs that invite prohibited uses and notify affected users. Preserve methods, synthetic fixtures, aggregate performance, incidents, and correction history without unnecessary dossiers. Tie each action to the qualification or governance condition that failed (National Institute of Standards and Technology, 2020; National Institute of Standards and Technology, 2024).

Chapter 10 — Calibration, Scale, and Uncertainty

Consider a hypothetical calibration record. One participant completes an instrument in two languages before and after a medication change, then later uses supported communication after aphasia. The total score stays nearly flat. Interviews reveal a translated neutral anchor with a different meaning, an item threshold altered after medication, and an interface that changes which states the participant can express.

A smooth graph leaves several questions unanswered. Did experience remain stable? Did internal standards change? Did a channel become more or less sensitive? Did the instrument lose access to something it previously recorded? Each explanation concerns a different relation between the observations and the proposed target.

A familiar baseline helps detect change, but it does not establish a neutral origin. Likewise, consistent ordering does not establish equal differences. A scale can retain one useful property while losing another. Calibration identifies which comparison survives and the evidence needed for a stronger claim.

Response shift makes this distinction especially consequential. A person may change an internal standard, the importance assigned to a dimension, or the meaning of the construct itself. The analysis must distinguish these possibilities from change in the experience being studied; repeated administration alone cannot make that distinction (Sprangers & Schwartz, 1999; Paap et al., 2025).

In the language comparison, a common printed response category is only a starting point. Interpretation, disclosure, and the relation between categories may differ. A qualified link must preserve the particular relation being compared. If it preserves rank alone, a claim about equal levels or a common zero still needs its own evidence (International Test Commission, 2018).

Supported communication raises a related problem of access. An unusual or absent response may arise because the available interface does not let the participant express the intended state. Improving access can change the observed score without establishing a change in burden. Calibration therefore has to examine what the revised route makes observable, and which earlier comparisons it can support.

Accumulation magnifies the consequences of these choices. Repeated small origin errors can create a large apparent total; errors shared across occasions do not disappear merely because more points were collected. The relevant uncertainty is that of the requested total, with calibration and temporal dependence carried into it, rather than a collection of isolated error bars.

The chapter develops tests and bounds that locate the limiting relation. A failed common scale can leave a local direction, a bounded total, or separate phase estimates usable. Reopening a broader comparison requires evidence addressing the particular failure, not a smoother fit to the same record.

These are claims about comparability under a measurement model. Part IV then asks which conscious stream owns the represented experience and where its domain ends; calibration cannot settle those questions by assigning the records one identifier.

10.1 — Calibration Within the Stream

Within-stream calibration distinguishes group discrimination from person-level tracking and stable averages from state-dependent response. Between-person consistency, within-person sensitivity, person-specific precision, and temporal stability require separate evidence before supporting accumulation or broader comparison (Molenaar, 2004; Hamaker, 2012; Vogelsmeier et al., 2024; Castro-Alvarez et al., 2026).

10.1.1 — Personal Baselines and Their Limits

A personal baseline describes a participant's channels in a familiar state, variation in self-report across ordinary days, or physiological changes after rest, pain relief, social contact, sleep, or exertion. It provides a local reference for detecting change (Borsboom, 2005; Cronbach & Meehl, 1955).

Personal baselines and the target origin serve different analytical purposes. A usual level describes a participant's recent experience; independent origin anchors provide the reference for signed accumulation. Keeping both visible helps the analysis detect persistent burden as well as change.

Record baseline time of day, wake state, medication timing, activity, symptoms, device version, cadence, and the participant's contextual description to distinguish experience from measurement change.

10.1.2 — Repeated Anchor Events

Anchor events are moments whose direction is known well enough to test whether the measurement relation behaves as expected. The ethical constraint is crucial: anchors must be available without inducing harm. Relief after ordinary care, increased discomfort during naturally occurring pain, recovery after rest, dyspnea during documented exertion, or calm after a

chosen calming practice can become calibration evidence only when the event would occur without the study manufacturing it (Shiffman et al., 2008; Stone et al., 2007).

Repeat anchors across days and contexts. A one-off physiological change can reflect coincidence, medication, reporting delay, or care context rather than a stable anchor relation.

The anchor design should separate sign, magnitude, and timing. A person may report relief before autonomic change settles. Movement may precede dyspnea. Analgesia may reduce distress while altering respiration or heart-rate variability in ways that mimic calm. The anchor therefore needs a time window, lag plan, negative controls, and a record of co-occurring care.

A repeated-anchor file should name the anticipated direction before the data are examined; selecting the anchor because its channel changed is outcome-dependent fitting.

10.1.3 — State-Dependent Calibration

Model calibration conditional on pain, arousal, sleep, medication, illness, posture, fatigue, fever, social setting, and attention, or narrow the context of use (Khalsa et al., 2018; Feldman et al., 2024; Shiffman et al., 2008).

A simple state-dependent observation model uses a state-specific intercept, loading on the latent target, and residual error. State uncertainty, flexible parameters, or weak anchors can leave that calibration unidentified.

If loadings change by state, the same latent change can produce different observed changes. If intercepts change by state, the same observed score can correspond to different latent levels. If residual variance changes by state, a constant-error model can overstate precision in noisier conditions. Each

difference has a different consequence for the measured cumulative estimate.

The state categories and their observation parameters should be specified before confirmatory analysis. Calibration must then be demonstrated across the clinical, behavioral, and device states in which the resulting estimate will be interpreted.

10.1.4 — Calibration Drift

Calibration drift is gradual change in the measurement relation: device aging, changing contact, learned prompt use, clinical routine, disease progression, or altered response meanings can change how numbers relate to the target (Russell, 1980; Cacioppo & Berntson, 1994).

A calibration system should therefore include drift monitors before it is used for confirmatory inference. Anchor-response stability, negative-control channels, device diagnostics, residual patterns, missingness changes, and self-report discordance can all signal that the measurement relation has moved.

The drift threshold and its action rule must be declared before confirmatory outcome inspection. Changing either to retain a favorable result or exclude an unfavorable one makes calibration outcome-dependent. A justified revision belongs to a new version and is evaluated on independent data.

10.1.5 — Within-Person Validity and Transfer

For a path-based claim, between-person discrimination cannot replace evidence that the measure follows within-person change (Molenaar, 2004; Hamaker, 2012; American Educational Research Association et al., 2014).

A serious within-person design uses repeated observations, time-forward prediction, lag sensitivity, and negative controls. The estimate should track known changes in experience at the

time relation justified by its validated response model, without using later data to manufacture an earlier prospective signal. It should fail to move with irrelevant device artifacts. It should be sensitive to direction, not merely magnitude.

10.1.6 — No Automatic Between-Person Transfer

Within an episode, short-window direction or responsiveness requires a state label, sign anchor, and artifact rule. Stop when state and artifact cannot be separated. Within a person over time, trajectory and change require longitudinal invariance checks, drift checks, and response-shift review; stop or split when the bridge changes.

Across states such as sleep, illness, medication, or clinical states, use state-specific calibration and explicit error costs. Stop if sign, origin, or unit changes. Across persons or groups, common-level or same-scale claims require origin, sign, unit, linking, invariance, and transport. Stop when the scalar or common-origin claim fails.

In clinical or terminal contexts, high-burden and endpoint-adjacent inferences require treatment, symptom, burden, and observation models. Stop when care or physiology dominates the bridge.

The hardest calibration error is the leap from one stream to many. An instrument may track one person’s directional change while sharing neither origin, unit, response process, nor channel gain with another; within-person success therefore does not establish interpersonal comparability, population rank, or a common measure of burden (Molenaar, 2004; Hamaker, 2012).

Comparison should advance through explicit tiers: within episode, within person over time, across states, instruments, groups, and persons. Each tier adds anchors, linking, invariance, and transport assumptions. Between-person transfer further

requires comparable anchor function and preserved sign and interval meaning across physiology, disability, language, culture, development, treatment, and response style; identical wording or standardized scores are insufficient (Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

A calibration sheet records state, medication, device version, anchors, response process, missingness, burden, and permitted comparison. If anchors move or reporting changes, split the series, widen the identified set, or withdraw the estimate. Person-specific and population models answer different questions; pooling requires tested shared structure.

Partial transfer may support rank without level, direction without interval magnitude, or some anchors without a common zero. No person carries another person’s scale: every comparison must name the tier, population, state, instrument, bridge, and uncertainty earned, and broader use requires new calibration.

10.2 — Origin, Sign, Unit, and Linking

Origin, sign, unit, and linking license different comparisons: common-zero neutrality, direction of contribution, cardinal integration, and a shared coordinate respectively. Qualify each relation required by the claim (Krantz et al., 1971; Borsboom, 2005).

10.2.1 — Neutral-Origin Rule

$$y^{*} = a+by, \qquad b > 0$$

Two instruments standardized in separate development samples can share a plotted zero while referring to different sample means. Neither mean thereby becomes an independently defended neutral origin.

Co-movement under burden and relief anchors can orient local sign; a qualified slope link may support direction or relative change without establishing a common origin.

Label each reported zero by its reference: personal baseline, sample mean, instrument origin, or independently calibrated neutral origin. That label makes the displayed coordinate interpretable.

Apply Section 6.2.4's independently calibrated origin rule: specify the reference before confirmation, justify it without the tested terminal outcome, and propagate its uncertainty or failure condition (Krantz et al., 1971; Russell, 1980).

A complementary task in the Cloos et al. (2025) follow-up isolated interface precision: 87.0% of respondents placed a requested value of 31 between 30 and 32 on a digital 0–100 slider. Including such a task alongside affective judgments separates use of the interface from interpretation of the feeling scale. Repeating it across devices and assessment periods provides a direct check on measurement consistency.

10.2.2 — Sign Orientation

Sign orientation identifies positive, negative, and neutral contribution relative to the declared origin. Improvement is a change in that contribution, not automatically a positive contribution. In synthetic rate and exposure units, $r(t)=-2+t$ over $0\le t\le 1$ improves throughout yet contributes $\int_0^1 r(t)dt=-3/2$. Conversely, $r(t)=2-t$ declines throughout while contributing $+3/2$. These examples separate the sign of the rate from the direction in which it changes; they do not prescribe how relief must feel.

A sign anchor must orient the represented quantity. Qualified reports, physiology, and behavior can inform it, but arousal, motion sensitivity, or a positive prediction error does not establish positive felt valence. Preserve magnitude-only or set-valued results when sign remains unresolved (Bollen, 1989; Borsboom, 2005).

Orient sign using ordinary ethically available events, cross-fitted data, earlier development samples, or independent validation streams. Do not learn confirmatory sign from the terminal result being tested.

Test sign transport across people and states. A relation altered by sedation, fatigue, withdrawal, or medication requires separate sign structures or withdrawal of the common-scale claim.

10.2.3 — Unit and Interval Meaning

An interval unit requires comparable construct differences where equal numerical differences are used. A standard deviation describes a distribution; standardization alone does not supply a unit of lived burden (Cronbach & Meehl, 1955; Campbell & Fiske, 1959).

Unit claims specify range, context, and version. Calibration over ordinary fatigue need not extend to severe dyspnea, changed item wording, or new sensor preprocessing.

Integration requires a qualified rate or proportional rate proxy, not only correctly ordered responses. In a synthetic three-interval record with unit exposures and realized rates $1, 1, -2$, the total is zero. The strictly increasing transformation $h(x) = x^3$ preserves order, pointwise sign, and zero, but gives

$$1 + 1 - 2 = 0, \qquad 1^3 + 1^3 + (-2)^3 = -6.$$

An arbitrary monotone re-expression is therefore not merely a change of cumulative units. A nonlinear observation model remains usable when independent calibration supports recovery of the underlying rate and its uncertainty.

Difference measurement offers a concrete calibration route. Junge and Reisenzein (2016) tested implications of metric measurement for particular emotion qualities; they did not establish a common positive–negative phenomenal unit. Their later methodological account explains why ordering cannot be independently tested when comparisons are derived from the same graded ratings (Reisenzein & Junge, 2024). Joint cross-context measurement in visual perception supplies a further methodological precedent, whose transfer to affect remains a proposal (Aguilar & Maertens, 2020).

A benign within-person pilot can compare felt differences across several ordinary positive, neutral-candidate, and naturally occurring mildly negative observations under Chapter 3's protections. Record positive feeling, negative feeling, and perceived neutrality separately. Fit graded pair comparisons, then elicit independent direct comparisons between pairs of differences: is this pleasant-side difference larger than that unpleasant-side difference? Include several magnitudes and two

ordinary contexts. The judgment concerns experienced differences, not willingness to pay or repeat an episode.

Reserve entire comparison families, including their repetitions and reversed presentations, for evaluation; reserve a later session for transport. Keep the calibration graph connected. Freeze the response-error model, sign/context links, origin bounds, and acceptable distortion before inspecting those results. Simulate whether the planned design can detect distortions large enough to change the intended integral, then assess held-out predictions against those tolerances. A nonsignificant axiom test alone does not qualify the metric. If a common scale remains unsupported, retain ordinal or separate-component results; if several calibrated maps remain admissible, carry their union into the identified set.

The unit record names the rate, exposure rule, transformation, anchors, valid range, and uncertainty. A momentary difference scale still needs an independently qualified rate/exposure relation before integration; remembered episode evaluation is a separate fallible channel. Cross-version and cross-language links must preserve the relations claimed. Section 6.2.3 treats admissible representation changes, Section 6.3.5 fixes the positive stream normalization, and Section 10.6 shows how calibration uncertainty changes the attainable cumulative conclusion.

10.2.4 — Linking Across Instruments

The linking designs in Section 9.5.3 must preserve the origin, unit, sign, sensitivity, and state-specific response relation required by the intended comparison; correlation alone is insufficient. Their overlap must span the relevant range, languages, devices, states, and clinical conditions (Manski, 2003; American Educational Research Association et al., 2014).

Test differential item functioning, response shift, heteroscedastic error, and state-specific bias after fitting the linking slope and intercept.

Apply Section 6.2.3's affine-transformation rule and propagate every admissible origin and scale. Stream-specific positive unit changes preserve identified zero without establishing interpersonal magnitudes (Luce & Tukey, 1964; Krantz et al., 1971).

10.2.5 — Withdrawal of Common-Zero Claims

Withdrawal is required when anchors conflict, response processes change, linking leaves an additive constant unidentified, or plausible origins produce incompatible sign or terminal status. The correct output may be origin-specific results, an identified set over admissible origins, or an origin-free feature such as ordering or shape.

Restoration requires new anchor evidence and a prospectively specified linking test on data independent of the failure that motivated redesign (American Educational Research Association et al., 2014; Nosek et al., 2018).

The origin must come from an independently justified response or state relation, not from sample centering, an instrument midpoint, a population mean, a fitted terminal value, or a transformation chosen because it improves closure. Different anchors may support local zeros while failing to establish one shared origin.

Failure of common zero need not destroy all calibration. Within-person direction, thresholds, ranks, local responsiveness, and some contrastive effects may remain interpretable. The report should state exactly which relations survive and which claims, especially exact-zero and cross-person level claims, are unavailable.

Z-scores, centering, and norm-referencing express values relative to a specified sample. Analyses using them should record that reference and use independent calibration when the target requires a different origin.

10.3 — Longitudinal Invariance and Response Shift

Use temporal measurement-structure tests to distinguish a changed target from a changed response relation; repeated observations of the same items are insufficient (Bollen, 1989; Borsboom, 2005; Vogelsmeier et al., 2024; Paap et al., 2025).

10.3.1 — Configural, Metric, and Scalar Invariance

$$X_{j,t} = \tau_j(t) + \lambda_j(t)\eta(t) + \varepsilon_{j,t}.$$

The displayed relation separates changes in the latent state from changes in loading, intercept, and residual error.

Configural invariance asks whether the same pattern of indicators relates to the same construct across time. If the factor structure changes, the instrument may be measuring a different object. For the Law, this is the minimum condition for treating repeated measures as part of one trajectory rather than a sequence of different constructs with the same label (Sprangers & Schwartz, 1999; Paap et al., 2025).

Metric invariance asks whether the loadings are comparable. If loadings change, then the same latent difference produces different observed differences at different times. A channel may become more sensitive during illness, less sensitive during sedation, or differently related to burden after adaptation. Without metric invariance, slope and change claims must narrow.

Scalar invariance asks whether intercepts or thresholds are comparable. Without adequate invariant anchors, level comparison becomes unsafe. A person may endorse the same response category at two times while meaning different latent levels, or produce the same physiological level under different clinical states. Unresolved scalar failure bars the affected common-level claims even when direction may remain usable;

Section 10.3.4 governs any comparisons retained under partial invariance.

Residual invariance is stronger still. It asks whether residual variances are comparable. This may matter when uncertainty is carried into cumulative ledgers. If residual variance increases near terminal states or under device burden, carry that increase into the uncertainty for the cumulative estimate.

10.3.2 — Response Shift

Response shift occurs when a person's internal standards, values, or conceptualization of the construct change. In health measurement, it is often described through recalibration, reprioritization, and reconceptualization (Sprangers & Schwartz, 1999).

Recalibration changes internal standards. A pain level once rated severe may later be rated moderate after prolonged illness, not because the sensation is smaller, but because the person's scale has changed. Reprioritization changes what matters within an evaluation. Fear, dignity, fatigue, social connection, and autonomy may carry different weight after diagnosis. Reconceptualization changes what the person means by the construct itself.

Response shift can accompany a meaningful change in how life is experienced. A person may find purpose in conditions that once seemed unbearable. Measurement can examine that lived transformation alongside changes in reporting standards, rather than treating every altered answer as an instrument defect.

Investigate response shift with multiple channels, then-tests or individualized appraisal where appropriate, repeated anchors, qualitative context, and longitudinal measurement models. Retain unresolved alternatives in the inference.

10.3.3 — Time-Varying Loadings and Intercepts

Compare constrained time-varying measurement models with simpler invariant alternatives, using external anchors and prospective checks (Bollen, 1989; Meredith, 1993; Paap et al., 2025).

The calibration plan should specify which parameters may vary, what covariates justify variation, what smoothing is permitted, and what failure thresholds trigger withdrawal. Otherwise every discrepancy can be explained away by a moving parameter, and the model loses empirical content.

Medication, sleep, disease stage, device version, language, age, care setting, and symptoms can justify varying relations. Unobserved covariates require corresponding uncertainty or admissible parameter paths.

The distinction can be made concrete with a hypothetical respiratory channel. During one clinical state, a given change in independently assessed distress might produce a larger respiratory response than during another state: that is a change in loading. A medication might instead shift the respiratory baseline while the assessed distress remains the same: that is an intercept change. A study designed to separate these possibilities needs observations that vary state and distress sufficiently to identify both relations. Repeating the same measurement more precisely at only one state cannot by itself distinguish them.

Public language must identify whether the result used a fixed or time-varying calibration bridge.

10.3.4 — Partial Invariance

Partial invariance permits only the comparisons supported by the structure that remains stable. Stable loadings may preserve selected associations or within-person dynamics, while failed intercepts or thresholds can block latent-level comparisons; a

subset of stable anchors supports only the construct range those anchors still represent (Meredith, 1993; Millsap, 2011; Putnick & Bornstein, 2016).

Partial-invariance decisions must be prospective and auditable. The analysis should define how indicators are tested, which departures trigger review, whether modifications were anticipated, how multiplicity and model search are handled, and whether the retained indicators still cover the intended construct. Dropping inconvenient items after seeing the result is not evidence of invariance.

If dignity, fear, agency, or social-connection indicators fail while autonomic channels remain, reassess whether the reduced instrument covers the intended construct, even when fit improves.

10.3.5 — Change Versus Measurement Change

The first strategy is design: retain repeated anchors, negative controls, bridge periods, dual scoring, time-forward predictions, and context. A single composite at each occasion gives little information for separating experience from measurement change.

The second strategy is modeling. Alongside latent state and measurement drift, retain the dynamics of individual affect items. Reanalyzing five intensive-longitudinal datasets, Koch and colleagues (2026) showed how a reference-item model can represent item-specific and person-specific dynamics that aggregate scores or a single common factor can conceal. A PTVC analysis can compare its cumulative-history prediction under composite, common-factor, and item-specific models. Use the same protected observations and timestamps, and score predictions in one declared observation space with a shared prespecified scoring rule and each model's specified observation

map. Keep transformations and elapsed-time assumptions explicit: the published negative-affect analyses used log-transformed items and an approximated ninety-minute interval. Persistence across these comparisons would make the result less dependent on the particular feelings being summarized. The reference item anchors a statistical comparison; its meaning and stability require qualification, and it does not supply the phenomenal neutral origin.

10.3.6 — Longitudinal Withdrawal Rule

Failed invariance, response shift, state-dependent calibration, changed gain, or unrecoverable version breaks can invalidate a common trajectory. Record the break in the data so later models cannot bridge it without a linking study (Meredith, 1993; Sprangers & Schwartz, 1999; Putnick & Bornstein, 2016).

Prespecify tests, tolerances, affected comparison tiers, and successor outputs: split phases, state-specific models, bounded direction, or no common trajectory. Choose the fallback by the failed property; threshold and loading failures need not have the same response.

Retain only the comparisons supported under Section 10.3.4. Restoration requires requalification and external replication; a cross-version link must use data spanning both versions, rather than assume semantic equivalence.

10.4 — Culture, Language, Development, and Disability

Culture, language, development, disability, and neurodiversity help define the measurement problem. They affect what an item means, which response channels are available, and which anchors can support comparison. Calibration therefore begins with the conditions under which evidence is produced, rather than treating every departure from a standard task as error. Translation, developmental change, and differential channel functioning can justify a local model or a finding of noncomparability instead of a common score.

10.4.1 — Language and Translation

$$\Pr(X_j = 1 | \eta, g) = logit^{-1}(\alpha_{jg} + \beta_{jg}\eta)$$

This item-functioning relation allows a channel to behave differently across groups even at the same proposed latent level.

When a group-specific model is necessary, it should be presented as the valid model for its qualified context, not as a deviation from a supposedly real universal scale. That wording matters because measurement language can turn a transport failure into an unjustified claim that the population itself is anomalous.

Translation is a measurement study, not a clerical step. Forward translation, reconciliation, back translation, expert review, cognitive interviewing, harmonization, and field testing examine whether an item elicits the intended construct in the new language and context. Statistical fit cannot replace response-process evidence when words, norms, disclosure risks, or communication routes have changed (International Test Commission, 2018; Putnick & Bornstein, 2016; American Educational Research Association et al., 2014).

Cross-language linking requires bridge samples and uncertainty. If bilingual or multilingual participants, common anchors, or parallel validation data are unavailable, narrow the claim to language-specific models.

Terminal weakness, delirium, sedation, cognitive impairment, and low literacy can defeat an otherwise sound translation. Specify when proxy report, clinical observation, or physiology can provide auxiliary evidence and when they cannot replace report.

10.4.2 — Cultural Display and Meaning

Culture changes expression, interpretation, suppression, and the social consequences of disclosure. Understatement, somatic language, relational description, strategic emphasis, or avoidance may each be locally intelligible responses rather than evidence of lesser or greater experience. These differences belong in the observation model and in the design of anchors, not in stereotypes about a group (Henrich et al., 2005; Henrich et al., 2010; International Test Commission, 2018).

Include family presence, clinician authority, privacy, stigma, spiritual interpretation, and local care practices when qualifying report or behavioral anchors.

A category such as moderate may represent different intervals across contexts. A different problem arises when ‘I feel at peace’ is understood as emotional calm by one respondent and reconciliation by another. The first difference concerns calibration along a proposed scale; the second may concern what the scale measures. Cognitive interviews can distinguish the interpretations before a common numerical link is fitted.

10.4.3 — Developmental Change

Development changes construct meaning, communication, physiology, autonomy, memory, and reporting; calibration must be specific to developmental stage (Sosa-Hernandez et al., 2024; Shiffman et al., 2008).

For younger children, self-report may be limited by language, temporal understanding, scale comprehension, and dependence on caregivers. For adolescents, privacy, identity, peer context, and risk sensitivity can alter disclosure. For older adults, sensory impairment, cognitive change, medication, frailty, and social expectation can alter both channels and reports.

Use age-appropriate anchors and accessible response methods. Visual scales, behavioral tasks, caregiver context, clinician observation, and physiological channels each require bias assessment.

Maturation and response shift can change the person, construct, and observation relation together. A developmental trajectory may therefore support within-stage or transition-specific claims without one lifetime common scale.

10.4.4 — Disability and Neurodiversity

Consider an instrument trained to infer affect from facial movement. A participant with limited facial expression may give clear reports of feeling while the instrument returns a flat estimate. The useful next comparison tests whether an accessible response channel distinguishes states that the movement model merges. This locates the calibration problem in the relation between channel and experience, rather than in the participant. Local response-process evidence is also needed for assistive communication and unfamiliar sensory contexts (American Educational Research Association et al., 2014; International Test Commission, 2018).

Access is a prerequisite to interpretation. Prompt wording, response mode, timing, device design, sensory demands, motor demands, communication supports, and consent procedures must be usable before a missing or unusual response is treated as evidence about the construct. When the interface excludes a participant, the failure belongs first to the instrument.

Neurodiversity can also alter sensory burden. A device, prompt frequency, clinical context, or social demand may itself create distress. A calibration design that ignores measurement burden can alter the very state it hopes to estimate.

10.4.5 — Differential Item and Channel Functioning

Differential item functioning occurs when people with the same underlying construct level respond differently to an item because of group, language, culture, state, disability, or method. Differential channel functioning is the physiological analogue: the same latent target produces different channel behavior across contexts. Both threaten comparability (Borsboom, 2005; Russell, 1980).

DIF is not automatically bias. It may reflect contextual differences, translation, response style, symptom overlap, or model misspecification. Interpret the cause before retaining, adjusting, isolating, or removing an item.

A worked example clarifies the problem. Suppose two groups have the same latent distress level, but one group endorses a fatigue item more often because a medication produces physical tiredness unrelated to valence. If the item is retained as pure distress, the group appears more burdened. If the item is removed without noting medication burden, the model may erase real experience. Calibration must separate symptom overlap from construct bias.

Use group comparisons, response functions, residuals, anchor stability, multiple-group models, and sensitivity analysis alongside construct review. Statistical detection alone cannot decide which features should remain.

10.4.6 — Group-Specific or Noncomparable Results

Combine invariance tests with the response-process, expert, participant, and contextual evidence in Section 10.4. A common factor can reflect shared method or institutional scripts rather than a common phenomenal coordinate (American Educational Research Association et al., 2014; Mokkink et al., 2024).

Failed full comparison can leave separate group models, within-group trajectories, rank-only relations, bounded links, or partially invariant composites. Name the surviving transformation and inference, or report noncomparability.

Record validation status by language, site, device, and clinical state in the comparability matrix (Section 26.2; Gagnier et al., 2025). Unsupported cells remain unresolved until their response processes are qualified; evidence from a neighboring group does not fill them.

10.5 — Clinical, Terminal, and Edge-State Calibration

Calibration changes meaning when symptoms, arousal, communication, treatment, or context alter the relation between a signal and its target. The following cases distinguish symptom-specific observations from total valence and identify what remains estimable in each setting. A useful calibration records both its supported output and the conditions requiring a narrower interpretation.

10.5.1 — Pain, Dyspnea, Delirium, and Fatigue

In dyspnea, respiratory measurements may reflect panic, exertion, and physiology together. Distinguish breathlessness from affective distress before using a state-specific calibration model.

Delirium can change the meaning of report and expression. Distinguish experience from observation reliability; uncertainty may need to widen, or a level claim may need to be withheld.

Distinguish pain, dyspnea, delirium, and fatigue despite overlapping signals. Pain can vary with threat or dignity; dyspnea can carry panic beyond oxygen metrics; fatigue can suppress expression while burden rises; delirium can disrupt report and amplify distress (Giacino et al., 2018; Surges et al., 2024).

Symptoms also interact. Dyspnea can increase fear, pain can increase fatigue, fatigue can suppress report, delirium can alter expression, and medications can change all of them. A model that treats dependent indicators as independent can misstate uncertainty and overstate the amount of independent evidence.

10.5.2 — Sedation and Analgesia

Clinical interventions can affect the target and the channels differently. Analgesia may reduce pain while preserving access;

sedation may alter awareness, expression, or recall; anxiolysis may reduce fear without changing nociception. Protocols should therefore model treatment-specific pathways rather than interpreting every quiet signal as comfort or every absent report as absence of experience.

Medication timing is part of the measurement record. Dose, route, onset, expected peak effect, accumulation, tolerance, renal or hepatic impairment, and co-medications can affect channels. If treatment is not recorded, the model cannot distinguish relief from signal suppression. Missing treatment data should widen uncertainty or stop the claim.

Sedation changes consent, reporting capacity, and the available observation channels. The analysis records these transitions and evaluates auxiliary channels within the relevant state.

10.5.3 — Communication-Limited States

Communication loss removes or degrades a channel; it does not establish the absence of experience. Aphasia, motor impairment, fluctuating attention, intubation, or impaired command following may prevent speech, writing, gesture, or device use. Evidence from alternative behavioral and neurophysiological assessments can inform the state, but the correct conclusion may remain uncertain (Giacino et al., 2018; World Health Organization, 2020; World Medical Association, 2024).

The calibration plan should distinguish inability, refusal, fatigue, sleep, delirium, sedation, language barrier, device inaccessibility, and care interruption. Each has a different meaning. Lumping them into missing report hides precisely the information needed to interpret the measured cumulative estimate.

Alternative routes include eye gaze, yes-no systems, assistive devices, simplified prompts, caregiver context, clinician observation, or physiological indicators. Qualify each route for its context of use; auxiliary witnesses do not replace the person's access.

Communication loss may require wider identified sets, state-specific models, or abstention. If measurement burden contributes to silence, modify the protocol.

10.5.4 — Terminal Physiology

Terminal decline can compress physiological channels. Separate reserve and care effects from the proposed Law-specific dynamics, and compare rivals only after the measurement bridge qualifies.

Reduced movement, altered respiration or circulation, palliative sedation, metabolic disturbance, changed communication, and sensor degradation can compress terminal signals. Separate these observation mechanisms from a proposed PTVC effect (Giacino et al., 2018; Surges et al., 2024; Shiffman et al., 2008).

A falling channel variance may also be compatible with a Law-specific prediction in some designs. The distinction requires independent reserve measures, treatment records, negative controls, and rival models.

Apply Section 6.6's missing-prefix restriction to late-window increments. If terminal physiology also changes the scale, even late-window levels require additional qualification.

10.5.5 — Sleep, Anesthesia, and Covert Activity

Sleep and anesthesia require state-specific calibration: absent waking report supplies no neutral zero (Siclari et al., 2017; Sanders et al., 2012).

Sleep calibration must distinguish wake, non-rapid-eye-movement sleep, rapid-eye-movement sleep, awakenings, dream reports, nightmares, and next-day residue. A channel that tracks waking distress may not track dream valence. Dream reports are filtered by memory and awakening conditions. The measurement relation changes with state.

During anesthesia and disorders of consciousness, unresponsiveness can coexist with residual processing or rare awareness. Drugs, surgery, pain, temperature, ventilation, and clinical intervention can dominate physiology. Qualify separate reference conditions, error asymmetries, and abstention rules for these states.

Covert activity makes error asymmetry explicit. A false negative may erase experience from the record; a false positive may convert noise into an unwarranted claim. Edge-state calibration therefore requires separate false-negative and false-positive analyses, not one aggregate accuracy score (Giacino et al., 2018).

10.5.6 — Clinical Context of Use

During a pain crisis, symptom intensity overlaps with valence but does not establish total burden. Restrict the output to a justified symptom-specific direction or abstain.

Sedation can suppress a signal in a way that resembles relief. Distinguish the treatment effect from measured calm and retain treatment labels in sensitivity analyses.

Apply Chapter 3’s burden ceilings and stopping rules to patients, families, and clinical workflow. The clinical context-of-use file follows Section 9.1, identifies the available inference and no-estimate path, and excludes autonomous triage or denial of comfort (U.S. Food and Drug Administration, 2025b;

International Council for Harmonisation, 2026; World Medical Association, 2024).

Context of use bounds an instrument's clinical authority. Calibration in ambulatory adults cannot be assumed to survive acute pain, medication change, intensive care, delirium, sedation, terminal physiology, sleep, anesthesia, or communication loss. Each setting changes target expression and observation and therefore requires setting-specific evidence (U.S. Food and Drug Administration, 2025b; International Council for Harmonisation, 2026).

Clinical qualification should follow the setting's actual failure routes. The record must separate treatment effect from measurement suppression, symptom-specific signals from total burden, comfort from reduced observability, and deterioration from a Law-specific trajectory. State, medication, device, care event, and missingness histories belong to the bridge.

A terminal example shows the risk. Physiological variance may narrow after sedation or low perfusion while self-report and device coverage decline. Relief, observation shutdown, treatment, disease progression, or several processes may explain the pattern; terminal-valence language remains unavailable until the measured-to-latent, endpoint, and rival bridges survive (Surges et al., 2024).

The same discipline governs sleep, anesthesia, delirium, and covert activity. Unresponsiveness does not identify absence of experience, and a channel validated in waking report-rich states may fail when memory, communication, or motor output changes. Edge-state results must remain labeled by state and observation regime (Giacino et al., 2018).

Clinical calibration also protects caregivers and families. Prompts, devices, annotations, and challenge procedures must not displace care, create unpaid workflow, or turn relatives into

continuous witnesses. Burden endpoints and automatic stopping rules remain operative even when the data are valuable.

Available outputs include symptom-specific direction, state-specific calibration, treatment-labeled sensitivity analysis, uncertainty-expanded identified sets, and No adequate estimate. The protocol specifies the appropriate output for each evidence state (World Medical Association, 2024).

10.6 — Identified Sets and Uncertainty

Suppose three admissible calibration bridges give different accounts of the same observed late window. One places its estimated contribution near neutrality; another gives a negative contribution; the third leaves the sign unresolved. Those outputs identify which measurement commitments drive the conclusion. An identified set retains the possibilities allowed by the declared model, while a separate sampling analysis describes uncertainty from finite data. Origins and units enter that accounting alongside the calibration links, following Section 6.4.3's distinction (Manski, 2003; Imbens & Manski, 2004).

10.6.1 — Partial Identification

$$C(D) = \{h(\theta) : \theta \in \Theta(D), \theta \text{satisfies the calibration gates}\}$$

$$Var\,(Q|D) = E\,[Var\,(Q|D,M)\,|D] + Var\,(E\,[Q|D,M]\,|D)$$

A synthetic finite-window budget makes the limiting uncertainty explicit. On an independently qualified rate branch, let T_o be observed minutes and m missing minutes. Let u denote the chosen rate unit, so the nonnegative provisional positive and negative-magnitude areas P and N have units u·min. Consider the declared calibration model

$$J = P - qN - T_o b + M + e,$$

where $q \in [1-\varepsilon_g, 1+\varepsilon_g],\ 0 \le \varepsilon_g < 1,$ corrects the negative-side unit; $|b| \le \varepsilon_0$ bounds an additive rate offset applied after that correction; $|M| \le mR$ bounds the true missing contribution; and $|e| \le E$ bounds remaining integrated error uniformly over the admitted calibrations. The bound R concerns latent missing-time rates, not a questionnaire's printed maximum. Under these jointly valid rectangular restrictions,

$$J \in [P - N - B,\ P - N + B], \qquad B = T_o\varepsilon_0 + N\varepsilon_g + mR + E.$$

The bounds are sharp for this model because the parameter extremes attain both endpoints. With $T_o = 100$, $m = 10$, $P = 60$, $N = 40$, $\varepsilon_g = 0.10$, $\varepsilon_0 = 0.05$, $R = 0.5$, and $E = 2$, the radius is $5 + 4 + 5 + 2 = 16$ unit-minutes and $J \in [4, 36]$. Changing only ε_0 to 0.10 gives $J \in [-1, 41]$. With the other bounds fixed, strict positive identification requires $\varepsilon_0 < 0.09$. These are calculated examples, not participant results or complete-stream tests.

The example identifies a useful next measurement: improve the bound that prevents the required conclusion. More observations cannot remove an unrestricted missing contribution or a persistent origin error. Holding a fixed absolute integral-error allowance as exposure grows requires origin uncertainty to shrink as $O(1/T_o)$ and relative-gain uncertainty as $O(1/N)$. A fixed tolerance for average rate is a different objective. Probabilistic limits require joint coverage; several marginal confidence intervals do not supply it. Coupled calibration restrictions may narrow the rectangular envelope, while a model that shifts the origin before splitting signs requires a different calculation.

Partial identification can return a checkable bound on the required total rather than a preferred reconstruction of every interval. Fix the admissible representation and calibration class before evaluation. On a qualified finite linear branch, let $x \in \mathbb{R}^n$ contain signed interval contributions and let $Gx \le h(D)$ encode the measurement and substantive restrictions. This finite system must be exact or conservatively enclose the original problem, including discretization error; a convenient fitted grid is not itself a coverage guarantee (Stark, 1992).

For a nonempty feasible set $\mathscr{F}(D)=\{x : Gx \le h(D)\}$ and total $J=c^{T}x$, each pair of vectors satisfying

$$\lambda \ge 0,\ G^T\lambda = c; \quad \mu \ge 0,\ G^T\mu = -c.$$

provides the independently checkable bounds

$$-h(D)^T \mu \;\le\; J \;\le\; h(D)^T \lambda .$$

Indeed, $\lambda^T G x \le \lambda^T h(D)$ gives the upper bound; applying the same argument to $-c$ gives the lower. Feasible certificates supply outer bounds even before optimization finishes. If the linear program is feasible with finite extrema, strong duality yields sharp endpoints; feasible x values attaining them verify sharpness (Boyd & Vandenberghe, 2004). If the true x belongs to $\mathscr{F}(D)$ with probability at least $1-\alpha$ under every admitted model, projection and valid certificates inherit that coverage. Deterministic restrictions instead give a conditional deterministic bound.

A synthetic three-interval example has $x_1 + x_2 \in [0.7, 0.9]$, $x_2 + x_3 \in [0.3, 0.5]$, and $x_2 \in [-0.2, 0.2]$. Since $J = (x_1 + x_2) + (x_2 + x_3) - x_2$, its sharp range is $[0.8, 1.6]$. The feasible vectors $(0.5, 0.2, 0.1)$ and $(1.1, -0.2, 0.7)$ attain the endpoints. Without the x_2 bound the total is unbounded in both directions.

Those same bounds can guide the next observation. On the qualified linear branch, suppose $\mathscr{F}$ is a nonempty bounded polytope and a candidate channel measures the same x as $z = a^\top x + e$, with declared calibrated a and using only the additional error restriction $|e| \le \delta$, where $\delta \ge 0$. Write $\mathscr{F}_z = \{x \in \mathscr{F} : |z - a^\top x| \le \delta\}$. The largest total width that can remain after an admissible reading is

$$W(a, \delta) = \sup_{z : \mathscr{F}_z \neq \varnothing} \left[\max_{x \in \mathscr{F}_z} c^\top x - \min_{x \in \mathscr{F}_z} c^\top x \right].$$

Equivalently,

$$W(a,\delta) = \max_{\substack{x,x' \in \mathcal{F} \\ \left|a^\top (x-x')\right| \le 2\delta}} c^\top (x - x').$$

Two admissible states can produce the same reading exactly when their error intervals overlap, which gives the constraint $\left|a^\top (x-x')\right| \le 2\delta$. Thus one linear program per candidate computes W. Select the smallest W among channels meeting the protocol's burden and cost limits. This applies bounded-error experimental design (Pronzato & Walter, 1990) to the width of the required total. Probabilistic error limits require simultaneous or selection-conditional validity.

In the example, measuring x_2 with $\delta = 0.05$ gives $W = 0.50$; measuring $x_1 + x_2$ with $\delta = 0.01$ gives $W = 0.62$; even an exact measurement of $x_1 - x_3$ leaves $W = 0.80$. The most precise channel need not be the most useful channel for the declared total.

Archive the constraints, feasible endpoint witnesses, dual weights, and certified numerical tolerances. A solver's small equality residual needs a bound on its effect, not automatic treatment as zero. Do not impose J=0 while constructing the set used to test it. Empty feasibility diagnoses incompatible assumptions or observations; it is not by itself a PTVC counterexample. Uncertain calibration families require an outer envelope over their admitted possibilities. These local bounds concern the declared intervals; complete-stream inference additionally needs the complete domain, qualified scale, routing, and independently adjudicated endpoint.

10.6.2 — Measurement-Error Decomposition

Separate channel noise, item reliability, sampling, calibration, model, linking, state-label, and transport error. Random

variation within a model differs from uncertainty about the model or an unidentified coordinate (Tsiatis, 2006; Little & Rubin, 2019; Vogelsmeier et al., 2024).

Channel noise concerns the observed signal: sensor artifacts, item unreliability, environmental disturbance, or short-term fluctuation. Calibration error concerns the relationship between signal and target. Model uncertainty concerns which statistical structure is appropriate. Linking error concerns transformations across instruments, populations, languages, or versions. Transport error concerns whether the bridge travels to the new context.

Integrating posterior means can provide a point estimate when the required integrability conditions hold. Its uncertainty must still be computed from the joint path model, including temporal covariance, calibration, missingness, and linking uncertainty; attaching isolated pointwise error bars does not provide uncertainty for the integral. The law of total variance separates uncertainty within a model from uncertainty across the declared model family.

A simple error budget shows why calibration deserves separate attention. For equal-exposure increments on a common signed unit, suppose $Y_k = X_k + b + e_k$, with a shared zero-mean random offset b of variance σ_b^2 and independent zero-mean errors e_k of variance σ_e^2. Assume the offset and error sequence are independent of one another and of the latent increments. If $A_n = \sum_{k=1}^{n} X_k$ and $\widehat{A}_n = \sum_{k=1}^{n} Y_k$, then

$$Var(\widehat{A}_n - A_n) = n^2\sigma_b^2 + n\sigma_e^2.$$

A fixed offset instead produces bias nb; a bound $|b| \le \delta$ allows cumulative error $n\delta$. For unequal weights, the shared-offset variance is $\sigma_b^2(\sum_k w_k)^2$ and the independent-error variance is $\sigma_e^2 \sum_k w_k^2$. Serial errors require the full weighted covariance

matrix. Denser sampling therefore cannot, by itself, remove shared origin error.

Qualified replicate channels provide a complementary diagnostic. If $Y_k^{(r)} = X_k + e_k^{(r)}$ for $r = 1, 2$, both complete error paths have zero conditional means given the latent path and zero conditional cross-covariances, and all quantities have finite second moments, then

$$Cov(\sum_k Y_k^{(1)}, \sum_k Y_k^{(2)}) = Var(\sum_k X_k).$$

This identifies cumulative latent variance under the stated observation model. Shared bias, unequal loadings, or different measured quantities invalidate that interpretation; physiology and self-report require qualification before being treated as replicate channels. The covariance identity does not identify the neutral origin.

10.6.3 — Within–Between Sign Reversal

Within-between sign reversal occurs when a relationship has opposite directions across people and within one person over time. A channel can be higher among people with greater average burden yet fall during an individual acute episode because medication, fatigue, withdrawal, or care changes the signal. Group-level direction therefore cannot be used as a trajectory bridge without a within-person model (Molenaar, 2004; Hamaker, 2012).

The analysis should therefore decompose within and between effects. Multilevel models, person-mean centering, dynamic structural models, and time-forward validation can help, but the claim must state which level is being inferred.

10.6.4 — Sensitivity to Anchors and Priors

Vary plausible anchors, bridge samples, priors, transformation families, and state labels within declared ranges and report their influence on the conclusion (Bollen, 1989; Borsboom, 2005).

Priors can stabilize sparse terminal, rare-state, small-language-group, or edge-state estimates. Report when prior structure, rather than the observations, carries the conclusion.

Assess whether anchor and prior sensitivity changes direction, level, or public claim status, not only parameter values.

Record the scientific, clinical, accessibility, or historical justification for each anchor and prior before confirmatory use.

10.6.5 — Common-Scale Abstention

Common-scale abstention can follow unidentified origin, unstable sign, weak units, failed linking, inadequate construct coverage, scalar noninvariance, dominant response shift, or incompatible verdicts across the identified set. Name the failed comparison, surviving relations, and evidence needed to reopen it (Manski, 2003; Imbens & Manski, 2004).

Abstention is not the same as nonsignificance. A nonsignificant estimate may reflect limited information about a well-defined target; common-scale abstention means that the coordinate relation required to define or compare that target has not been earned.

10.6.6 — Uncertainty in Public Language

Consider three prospectively specified plausible neutral anchors. If the valid terminal-window uncertainty sets under the three anchors fall, respectively, entirely inside the equivalence band, entirely above it, and entirely below it, the proper result is that common-zero terminal equivalence is not identified across those anchors. Averaging the anchors would hide the dependency,

although some origin-free ordering or shape features may remain available (Manski, 2003; Imbens & Manski, 2004).

10.7 — Calibration Audit and Correction

Imagine following one instrument through two years of use. A firmware change alters signal processing, a translated prompt is introduced, and a clinic changes its medication routine. The instrument retains its name while several links between experience and recorded values change. An audit separates these changes by date and version, then asks which comparisons each affects. It can preserve a qualified early segment while directing requalification to a later one. The resulting record connects each estimate to the measurement conditions that produced it.

10.7.1 — Prospectively Specified Calibration Plan

Under Section 6.7, fix the comparison tier and the calibration choices specified in Sections 10.1–10.6 before confirmatory evaluation. Record changes as new versions requiring evaluation of the affected comparisons (American Educational Research Association et al., 2014; Nosek et al., 2018; Flake & Fried, 2020; Gagnier et al., 2025).

The plan should include thresholds for burden from calibration prompts, devices, interviews, and clinical workflow demands.

10.7.2 — External and Cross-Site Audit

A device or code update calls for dual scoring or a bridge study. Retain the hardware, code, and transformation versions, and keep claims version-specific until linking evidence justifies comparison.

A failed cross-site comparison restricts the context of use. Preserve site, population, and protocol differences; transport requires new evidence.

Cross-site audit should reproduce the deployment problem rather than reshuffle the development data. Site, device, language, and time holdouts test different transport claims, and

each should be tied to a prospectively specified failure rule. A model that succeeds only after local refitting has demonstrated adaptability, not transport of one frozen bridge.

10.7.3 — Drift and Version Alerts

Anchor instability pauses the affected comparison. Preserve the anchor version and context, then widen, split, or withdraw the claim as the evidence requires.

Differential item or channel functioning requires review of the affected indicators and retention of group and item diagnostics. The result may require separate models or a finding of noncomparability.

Distinguish drift in the observation relation from a declared instrument, preprocessing, or model version change. Use prespecified thresholds and bridging or requalification rules for both (American Educational Research Association et al., 2014; Wilkinson et al., 2016; Nosek et al., 2018).

Add differential item functioning, participant challenge, clinician concern, and changed treatment or workflow to the drift checks in Section 10.1.4. Statistical alerts without contextual review can miss clinically obvious failure; contextual concern without defined evidence can become untraceable discretion.

An alert can lead to monitored continuation, paused claims, a version split, bridge-based re-estimation, wider uncertainty, requalification, or retirement. Record the trigger, evidence, decision, and affected outputs.

10.7.4 — Participant and Clinician Challenge

A participant challenge triggers contextual review. Preserve the challenge and adjudication path whether the outcome is correction, explanation, or no change.

Calibration review should admit contextual evidence such as sensor-induced distress, misunderstood translation, edema affecting a device, and outputs inconsistent with a documented clinical state (American Educational Research Association et al., 2014; Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Provide a clinician route for reporting treatment changes, workflow artifacts, symptom confounds, and state changes that threaten calibration.

Review a challenge before rescoring. Distinguish data, context, model, device, state-label, medication, translation, and public-language failures so the correction addresses the identified cause.

Review clinician observations for cultural, disability, communication, workload, and expectation biases alongside the model's possible errors.

10.7.5 — Correction, Reanalysis, and Retirement

Record changed calibration bridges and their effects on uncertainty and scope through Section 28.2's dependency and correction records (Gagnier et al., 2025).

Recalibration requires migration tests identifying which historical records remain recoverable, which require reanalysis, and which cannot be translated (Wilkinson et al., 2016; Center for Open Science, 2025; Nosek et al., 2018).

10.7.6 — Retirement of Nontransportable Scales

Retirement is appropriate when an instrument cannot support its claim; drift or invariance failure cannot be repaired; sign or origin is unidentified; provenance unrecoverable; participant burden or disclosure risk excessive; or foreseeable misuse uncontrolled. Retirement is versioned and scoped: an estimator

may remain useful for ordinary within-person change while losing terminal, cross-language, cross-site, or cross-person authority.

Remove retired scales from public artifacts and operational systems; correct, withdraw, delete, or restrict affected outputs as recorded (National Institute of Standards and Technology, 2020; U.S. Food and Drug Administration, 2025b). Include machine-readable retirement status where feasible so automated systems stop serving withdrawn outputs. This applies to summaries, websites, donor materials, model cards and datasets as well as the scientific report.

Retirement ends an instrument's authority within a context of use. Care, listening, participant reports, and research continue: clinicians can comfort and treat while researchers develop a better bridge. A retired context should remain visible in instrument documentation so later users do not mistake continued use elsewhere for renewed qualification there.

The audit trail records development and freeze dates, qualified settings, anchors, versions, exclusions, drift alerts, challenges, failed invariance tests, corrections, and withdrawn claims. Reviewers use this history to reproduce where the scale gained or lost authority (Nosek et al., 2018; Mokkink et al., 2024). The history should preserve superseded calibrations rather than overwrite them, so later reanalysis can reconstruct which scale governed each published result.

An estimator valid in ambulatory adults may fail in palliative care as medication, low perfusion, care events, and observation loss alter its channels. Retire terminal use, requalify ordinary use, rerun affected analyses, and describe the surviving scope accurately. A split qualification can retain ordinary-use evidence while explicitly severing terminal claims, avoiding the false choice between total approval and total retirement.

Part IV — Consciousness, Stream Routing, and Endpoints

To combine experiences across time, an investigator must decide which intervals belong together. A continuous medical record does not settle that question: its continuity belongs to the record. Part IV examines the relation between observed continuity and the conscious stream that the formal hypothesis concerns. The problem is to make ownership and continuation precise enough that a change in classification has an intelligible effect on the later analysis (Bayne, 2010).

Chapter 11 begins with routing. Evidence may support joining two intervals, keeping them separate, or retaining several possible assignments. Those alternatives define different domains for accumulation. Treating them explicitly allows the investigator to see which conclusions survive a disputed assignment and which depend on resolving it. The practical task is therefore to preserve the structure of uncertainty, rather than conceal it inside a single convenient identifier.

The boundary states give that task concrete content. Sleep can separate experience from immediate access to it. Anesthesia complicates the relation between experience, movement, and later memory. Disorders of consciousness challenge inferences drawn from overt response. Artificial systems raise a different question about the relation between performance and phenomenality. Chapter 12 uses these contrasts to examine what different marker families and theories contribute to the routing problem (Siclari et al., 2017; Giacino et al., 2018; Butlin et al., 2026).

A neurocritical-care record shows why the timing of evidence matters. Overt response and task-sensitive neural activity may yield different classifications, while clinical decisions must proceed on their own timetable. Recording the channel and occasion of each observation preserves the disagreement for review. It also permits a later investigator to ask whether the

classifications addressed the same state, the same ability, and the same period (Bodien et al., 2024; Giacino et al., 2018).

Chapter 13 brings the endpoint into this account. The end of recording, a clinical event, and the end required by the hypothesis can occur at different times or have different evidential support. An endpoint procedure must explain what its markers establish and how later evidence can revise the classification. Its relation to routing is reciprocal: a claimed return may reopen an endpoint judgment, while a revised assignment of intervals may change which ending is relevant.

Across the Part, ordinary words become more exact research questions. Calling someone awake, absent, returned, or ended can combine claims about access, report, memory, and continuity. Separating those claims makes a disagreement investigable and clarifies which observation could resolve it (Siclari et al., 2017; Bodien et al., 2024; Andrews et al., 2024). The resulting account gives the study families a defined subject and domain, together with the unresolved alternatives that their analyses must continue to carry.

Chapter 11 — Conscious Unity, Stream Identity, and Routing

Consider a hypothetical person admitted unresponsive after injury. Motor output is unreliable and repeated examinations disagree. A later task reveals organized command following through a neural response, and subsequent recovery permits further assessment. The evidence changes the inference while leaving questions about intervening experience and continuity (Bodien et al., 2024; Giacino et al., 2018).

The research record must distinguish the biological organism from the candidate stream whose intervals it proposes to join. A hospital identifier can connect observations without establishing that connection between experiences. Failure to classify an interval does not by itself exclude an otherwise qualifying stream from the hypothesis's truth-level domain.

Command following supports a proposition about organized access under the tested conditions. Continuity across an earlier gap is a different proposition, requiring a relation between the states on either side. Evidence relevant to one can inform the other without making the two claims equivalent.

The negative examinations also need interpretation. A failed response can reflect the task, sensory access, motor output, fatigue, medication, or the state being investigated. An ownership rule that treats all such failures alike would make the measured boundary depend on the observer's route of access.

Timing matters when recovery becomes evidence. A retrospective analysis may use later findings to revise the earlier interval's classification. A prospective decision must retain the information actually available then. Preserving both records allows the revision to be examined without pretending that the original uncertainty was already resolved.

Several routing models may remain compatible: one stream with reduced access, interruption followed by resumption, or a new stream after the gap. The useful comparison asks what

observations each model predicts and which conclusions survive their disagreement. It cannot settle ownership merely by selecting the most convenient trajectory.

Partition and recoordination extend the same problem. Divided access can be real while the number of experiential streams remains unresolved. Improved coordination can likewise be observed before its implications for stream identity are established. Those cases require explicit event ownership rather than an intuitive rule for combining histories.

Probabilities can represent uncertainty about that ownership. They do not mean that an experience was fractionally present. Because membership and the contribution assigned to an interval may depend on the same evidence, their uncertainty has to be carried jointly into a path calculation.

False inclusion can join intervals that do not belong together; false exclusion can remove possible experience from the analysis. These scientific errors have different consequences. Neither uncertainty nor a classification changes the person's entitlement to care, communication support, or protection.

This chapter develops the evidence and routing rules needed to keep those propositions distinct. A revised record should show which inference changed, why it changed, and what remains unresolved; it should not become a new public identity for the person.

11.1 — Bearer Is Not Body or Biography

Consciousness research separates a biological carrier from a psychological narrative. A measure may target a candidate stream without capturing the whole organism or biography. Split-brain findings and memory-system dissociations show why those categories differ. Disorders of consciousness further separate body, legal person, and measurable experience (Bayne, 2010; Gazzaniga, 2000; Giacino et al., 2018).

11.1.1 — Organism and Stream

Keep organism, session, candidate stream, gate version, routing model, and reviewer decision in separate fields with declared joins. One organism may support several live routing models without duplicating an event; a hospital identifier does not establish stream identity (Bayne, 2010; Bayne & Chalmers, 2003).

This distinction also prepares later endpoint work. Somatic death, circulatory collapse, respiratory failure, brain injury, and device cessation are biological or observational events. They may be powerful evidence about the stream, but they must pass through declared endpoint and unity criteria.

11.1.2 — Biography and Memory

Family memory, clinical records, and public biography provide context and recognition while remaining indirect evidence about present phenomenal unity (Tulving, 1985; Squire, 2004).

Memory also has directionality. A later remembered episode can support the claim that something was experienced earlier, but failure of later recall is not symmetric evidence of absence. Amnesia, aphasia, sedation, trauma, sleep, delirium, and developmental limits break the symmetry. The classifier must

therefore record memory as one channel with known failure modes, not as the court of last appeal.

Narrative fluency can select easier cases over infants, people with amnesia or aphasia, and sedated patients. Treat language and autobiographical memory as access-dependent channels, not conditions for belonging to the phenomenal domain.

11.1.3 — Legal Person and Research Bearer

Legal status governs rights, duties and consent authority. Scientific classification addresses stream onset, continuity and ending through separately declared evidence. A partition model concerns the assignment of experience within that analysis and leaves legal identity intact.

Bearer classifications remain in role-limited research records, with expiring or reviewable labels. Public summaries omit confidence classes that could become stigma or administrative eligibility criteria; Section 11.6.6 specifies the prohibited uses.

11.1.4 — Public Identity and Record

Public identity is especially prone to uncontrolled reuse. A name, image, biography, diagnosis, or famous case can attract attention far beyond the original protocol and make later reviewers less independent. The safer record links public identity to the research dataset only through restricted provenance, while analytic files retain the minimum variables needed to test routing and endpoint claims.

For each archival source, record what it preserves and which continuity relation it can inform.

Public records preserve what was documented, not everything that was felt. Visible actions, language, official encounters, and institutional access are more likely to leave records than hidden burden. Treat this selection as a property of the archive when

judging what an absent entry can establish. The same record can still legitimately establish medication timing, clinical events, injury history, sleep schedules, consent, and which evidence a reviewer possessed at a particular date. These uses support reconstruction of the evidence without converting the archive into the experiential stream.

Public diagrams and captions must distinguish an archival trace from a continuing experiential stream.

11.1.5 — One Body, Possible Multiple Streams

One organism usually supports the initial one-stream model because it is parsimonious and consistent with ordinary integration. Parsimony is a starting rule, not a metaphysical axiom. Evidence of sustained access barriers, partly independent control, conflicting report channels, or causal separation may require partition alternatives, while ordinary shifts in mood, role, or attention do not.

Prefer a partition hypothesis over a one-stream alternative only when it explains access, memory, agency, or integration patterns better under the same evidence budget. Retain admissible alternatives when the evidence cannot distinguish them.

11.1.6 — Classification Limits

The protocol therefore separates scientific and care consequences. A classification may determine primary, sensitivity, or excluded analysis, but never comfort, analgesia, visitation, rehabilitation, communication support, or clinical seriousness. This separation also prevents classification-driven care changes from contaminating observation.

Transport must be shown for the marker-to-target relation. A classifier calibrated in communicative adults may change

meaning in infancy, aphasia, motor impairment, sedation, brain injury, nonhuman species, or artificial architectures. New domains require documented prerequisites, calibration, thresholds, and failure modes; otherwise evidence remains restricted or unavailable (Giacino et al., 2018; Andrews et al., 2024). Transport evidence should include error asymmetry, because a marker that fails mainly through false exclusion has different scientific consequences from one that fails through false inclusion.

A structured uncertainty set can still support conservative ownership bounds when all live models agree that certain events cannot belong to the same stream.

11.2 — Continuity Evidence and Confidence

Operational stream classification combines several kinds of evidence: access and integration, memory and learning, causal continuity, report access, converging markers, and a calibrated decision rule. Each addresses a different link between observable activity and the proposed bearer. Their roles should remain explicit so that agreement strengthens the relevant classification without silently turning one marker into numerical identity or moral status.

11.2.1 — Access and Integration

Access and integration concern coordinated availability of information across channels. Motor bottlenecks, reflex coordination, and theory-specific overinterpretation can mislead the inference; coordination alone establishes neither numerical identity nor moral status.

Integration concerns common causal organization, not agreement in sign or report. Conflicting feelings can occur within one stream, and identical behavior or reflex coordination does not establish conscious unity (Dehaene et al., 2017; Tononi et al., 2016).

Specify the access relation: visual information reaching action, auditory command content changing state, pain-related input entering regulation rather than only reflex, or interoception affecting report, autonomic response, or avoidance.

Burden can disrupt orderly report without establishing absent experience. Calibrate integration markers for the person's condition instead of requiring a healthy person's expression of coordination.

Temporal resolution matters: brief neural patterns and longer behavioral records may reveal different integration or continuity relations.

Record the theoretical bridge and rival interpretations of the same integration evidence. Chapter 12 compares the marker commitments of the competing theories.

11.2.2 — Memory and Learning

Retention, updating, affective carryover, and procedural change provide memory or learning evidence. They cannot establish the presence or absence of experience in every interval.

Interpret these traces against reconstruction, priming, suggestion, and nonconscious learning. Each can preserve an effect of earlier exposure without settling the proposed phenomenal continuity relation.

A learning marker is most useful when it includes a counterfactual: what would the system have done if the prior interval had not occurred? Without that contrast, learning can be confused with drift, medication effect, fatigue, or changing environment. Continuity evidence becomes stronger when the alteration is specific to prior exposure rather than general change (Nadel & Moscovitch, 1997; Squire, 2004).

Episodic memory, procedural learning, conditioning, habituation, priming, and preference shifts support different continuity inferences. Learning can persist without reportable memory.

Test the continuity inference against trained reflexes, environmental cues, family or clinician prompting, external reconstruction, nonconscious mechanisms, and ordinary conditioning before attributing continuity to the candidate stream.

A changed startle pattern, learned preference, persistent aversion, response to a familiar voice, or procedural adjustment can preserve evidence of earlier exposure even when verbal recollection is unavailable. These observations motivate

different memory tests: recognizing a cue, changing an action, and reporting an episode are not interchangeable achievements. Compare the proposed trace with an otherwise relevant unexposed condition and with changes attributable to the current setting before treating it as evidence of causal carryover.

11.2.3 — Recurrence and Causal Continuity

Causal recurrence supplies evidence of state dependence and continuity of trajectory. Homeostatic persistence, public records, and organism continuity can also persist across a gap, so recurrence alone does not establish full unity through it.

Recurrence bears on continuation versus restart, coincidence, or reconstruction. State the continuity proposition each marker supports; recurrent-processing and higher-order accounts may interpret it differently (Lamme, 2006; Brown et al., 2019).

Specify what persists, its carrier and timescale, observable consequences, and a rival without continuation. Candidate carriers include network organization, synaptic state, bodily regulation, learned association, affective tone, attention, or action policy. Predict what follows if the carrier is blocked, destroyed, disconnected, or pharmacologically suppressed.

Hidden-state or hidden semi-Markov models can compare continuity, interruption, partition, rebound, and merging with different dwell times. When the required routing state remains unresolved, retain any valid branch-specific estimates or bounds and classify the affected inference under Section 24.2. Evaluate the models against simpler restart and access-failure alternatives on held-out observations.

The dossier must separate continuity of substrate, continuity of causal organization, continuity of accessible content, and continuity of a candidate phenomenal field. Local recurrent loops may survive while global coordination fails; arousal may

persist while content-specific access disappears; public records may persist while the relevant experiential organization is unknown. Evidence for a weaker layer cannot silently stand in for a stronger one.

Intervention can strengthen the claim when it is ethically and scientifically appropriate. Clinically indicated treatment changes, natural transitions, noninvasive perturbations, and already occurring disconnections can test whether later organization depends on the earlier state as the routing model predicts. The purpose is never to disrupt a vulnerable person for theory; it is to distinguish a causal continuation edge from a descriptive correlation using admissible evidence (Pearl, 2009; Peters et al., 2017).

Report admissible continuation, interruption, partition, and unresolved edges with their evidence. Observationally equivalent routing graphs form an equivalence class; identify additional observations that could separate them, when known.

A simple timing example shows why the continuity model matters. Consider two records classified in the same current state, one just after entry and one after a long dwell in that state. A first-order model whose state excludes elapsed duration gives them the same next-transition rule when its other inputs match. A semi-Markov model can give them different transition probabilities because time already spent in the state is part of its rule. Recording dwell time can therefore discriminate these models even when the current marker value is identical. That discrimination concerns temporal organization; the additional bridge from organization to stream continuity must still be stated.

11.2.4 — Report and Command Following

First-person statements and intentional task responses supply report or command-following evidence. Language, motor access, arousal, fatigue, pain, and task burden can prevent a response; a missing response therefore cannot by itself establish absent experience.

Where report is available, respect for report includes respect for refusal. A participant who can communicate may decline tasks, correct language, or withdraw from parts of the protocol (Claassen et al., 2019; Bodien et al., 2024).

Section 12.3.3 develops covert command following in boundary states; here it enters only as one source of continuity evidence when overt behavior is unavailable.

Record the command, sensory delivery, permitted motor or neural response, burden, and state. Evaluate replication, controls, arousal, artifacts, and chance performance. A reproducible command-specific response supports retained capacity; language, sensory or motor limitations, low arousal, pain, and fatigue weaken an absence inference.

In ordinary and psychiatric cases, dissociation, trauma, language limits, neurodivergence, and clinical stress can alter reporting. A person may report discontinuity because memory is inaccessible, not because experience ended. Another may report continuity because the narrative self repairs gaps after the fact.

11.2.5 — Physiological and Neural Evidence

Physiology and neural dynamics provide system-level state markers and response patterns for the declared continuity model. Their interpretation depends on arousal and medication as well as signal quality. Nonconscious processing supplies a competing explanation for some of the same observations (Casali et al., 2013; Jang et al., 2024).

Discordance can localize the uncertain relation. Low behavioral responsiveness with preserved task activation differs from preserved arousal with poor integration, and both differ from a strong autonomic response with uncertain subjective experience. These patterns challenge different links in the continuity model. Retaining the pattern of disagreement is therefore more informative than replacing the channels with a single consensus score.

Apply Section 9.3's channel-provenance requirements, recording the marker's task, medication and state context, calibration target, and transport limits.

Apply Section 9.3.6's joint-measurement requirement when combining neural evidence, physiological organization, command-following tasks, learning traces, and later recovery. Weigh their agreement through that model. Disagreement may reveal a motor bottleneck, sedation, measurement artifact, partition, state fluctuation, or inadequate theory.

11.2.6 — Confidence Classes

A confidence class should read as a bounded scientific proposition, not a badge. A statement such as qualified continuity across a specified recovery window under routing model M, with marker families A and B available and command following unavailable, is less convenient than one word but identifies the target, interval, evidence, and limitation that give the classification meaning (Green & Swets, 1966; Berger, 1985).

Record the target event explicitly: behavioral recovery, covert command following, continuity across an interval, and admissibility for ledger analysis require separate qualification.

Numerical confidence requires held-out evaluation under a frozen outcome, horizon, score, and reference class. Reliability curves and proper scoring rules can test calibration. Model

selective verification from recovery, repeated assessment, survival, site resources, and family advocacy (Gneiting & Raftery, 2007; Bodien et al., 2024; Little & Rubin, 2019).

Test decision thresholds as a family. Additional noninvasive assessment and entry into primary analysis may justify different thresholds and loss functions (Manski, 2003).

A confidence class reports support for a named relation; it does not by itself establish consciousness or personal identity (Tabassi, 2023).

Selective verification can be seen in a small hypothetical record. Suppose a classifier is applied to 100 windows, but the reference procedure is completed for only 50 of them, with 40 reference-positive results. The observed fraction is 40 of 50 among verified windows. Without a justified assumption about the other 50, the total number of reference-positive windows could range from 40 to 90; it is not established as 80 of 100. This calculation concerns the declared reference classification, not direct truth about consciousness. Recording why verification was completed or absent is necessary before a model can narrow that range.

11.3 — Soft Unity Gates and Threshold Families

A protocol threshold is a decision rule applied to evidence; it does not by itself establish a natural boundary. Signal-detection and Bayesian decision theory make the costs of false inclusion and false exclusion explicit, while threshold-family analysis reveals whether a routing conclusion depends on one convenient cut point (Green & Swets, 1966; Berger, 1985).

11.3.1 — Hard-Gate Baseline

The empirical hard-gate baseline includes or excludes an interval according to a prespecified proxy threshold. Its decision concerns admissibility for the analysis; below-threshold evidence does not by itself establish absent experience.

A preregistered hard gate makes failure consequential: an adverse result under the declared rule cannot be dismissed by calling the cutoff illustrative (Green & Swets, 1966; Berger, 1985).

A hard gate can create false precision. A person near a threshold may be classified differently because of noise, medication, sleep stage, artifact, task burden, device version, or sampling time.

For the Law's empirical program, the hard-gate baseline is never enough by itself. It can be the primary protocol-fixed rule only when paired with sensitivity across threshold families, missingness analysis, and explicit reporting of intervals near the boundary.

11.3.2 — Soft Gate $g_\theta(t)$

In the probabilistic implementation described here, the soft gate records uncertainty about whether an interval belongs to the declared stream under a specified theory, prior and calibration target. An inclusion probability is not a claim that the experience

itself is fractionally present. A genuinely graded unity representation is a different modeling choice and requires its own interpretation and bridge to the formal target.

Evaluate inclusion probabilities for the named reference target and verification process in Section 11.2.6. Smooth probabilities can still be miscalibrated; carry that uncertainty into the ledger and verdict (Casali et al., 2013; Dehaene et al., 2017).

For example, a group of intervals assigned 0.8 probability for a declared reference classification should not be treated as well calibrated if comparable, independently verified intervals satisfy that classification only about 0.4 of the time. The discrepancy concerns the specified reference target and verification process; it is not a direct measurement of the amount of consciousness. It must affect the uncertainty passed downstream.

Probabilistic inclusion and valence must be propagated jointly. For an interval k, let I_k indicate membership in the declared stream and let C_k denote an integrable signed contribution under each admissible state and routing model. Given data D, write $p_k = \Pr(I_k = 1 \mid D)$. When $p_k > 0$, the expected assigned contribution is $\mathbb{E}[I_k C_k \mid D] = p_k \mathbb{E}[C_k \mid I_k = 1, D]$; when $p_k = 0$, it is zero. Multiplying a positive inclusion probability by an unconditional contribution estimate is justified only when the corresponding conditional means agree; conditional independence is a sufficient condition. The primary calculation therefore draws membership, contribution, and routing together from the declared joint model before accumulating each path.

A zero posterior mean is also insufficient for closure. If two admissible histories have equal posterior weight and terminal residuals $+10$ and -10 in the declared units, their average is zero although neither history closes. For $0 \le \varepsilon < 10$, this distribution assigns zero probability to $\left|Z^*\right| \le \varepsilon$. Report the residual distribution or identified set and, for a protocol-fixed

tolerance ε, the probability assigned to $\left|Z^{*}\right| \leq \varepsilon$. This calculation makes the contribution of uncertain intervals inspectable without treating probability as fractional experience or averaging incompatible histories into an apparently balanced stream.

11.3.3 — Threshold Families

Vary independently justified unity rules to assess threshold sensitivity. A result surviving only one cutoff requires independent justification of that cutoff before supporting the broader claim.

A threshold family groups plausible unity rules that vary cutoffs, marker combinations, smoothing windows, filtering rules, artifact exclusions, or theory-weighted evidence. Each variant requires an independent scientific rationale. Include a burden check because the most data-hungry gate may be clinically or ethically unacceptable (Berger, 1985; Manski, 2003).

Several stable numeric cutoffs may share one vulnerable preprocessing choice, marker family, or retrospective smoother. The approved family must expose those alternatives too.

Robustness should be matched to the claim. A stable access bottleneck may survive a wider family than exact cumulative contribution, and a routing conclusion may remain unresolved even when a behavioral distinction is robust. The verdict should inherit the stability of the specific proposition tested, not the ambition of the surrounding theory.

Report discrimination and held-out calibration under Section 11.2.6 together with sensitivity, specificity, and the consequences of each decision rule (Gneiting & Raftery, 2007).

Discovery may tolerate more false inclusion to avoid overlooking candidate experience; primary terminal analysis

may require stronger support for a specific route. Publish the loss function and threshold for each use.

11.3.4 — False Inclusion Costs

Section 5.1 defines the qualifying stream. Empirical unity gates estimate which intervals belong to it under a declared routing model. Failure to classify an interval limits the analysis; it does not by itself exclude an otherwise qualifying stream from the hypothesis's domain.

A conservative gate prioritizes reducing false inclusion, with a corresponding risk of excluding low-signal experience. Preserve protective no-estimate outcomes and threshold sensitivity.

A permissive gate prioritizes retaining possible experience, with a corresponding risk of false inclusion or ledger inflation. Its choice needs independent motivation and explicit public labeling.

Fluent artificial or archival output needs a justified theory and evidence path before supporting stream inclusion (Section 12.7).

Unsupported inclusion lacks sufficient evidence under the declared rules; false inclusion assigns an interval that does not actually belong to the stream. Both can compromise inference, while actual false inclusion can add non-bearer time, blur partition, or mistake interruption for continuity.

Label weakly supported intervals as uncertain, low-signal, no-estimate, or sensitivity-only. A permissive gate must be explicit and show how their exclusion changes the result.

11.3.5 — False Exclusion Costs

Both false inclusion and false exclusion are model errors, with different consequences. Inclusion can contaminate ownership and terminal estimates; exclusion can erase possible experience

or conceal selective observation. Report both rates and downstream effects while keeping protections fixed.

Audit low-signal states for forms of experience the threshold may miss; qualification in healthy expressive adults does not establish universal detection (World Medical Association, 2024; Andrews et al., 2024).

Apply Section 9.6.2's observation-selection analysis: burden-related channel loss can make excluded intervals appear to lack experience.

A low-confidence interval should enter the data model with a reason code and an uncertainty state, not as zero exposure or absence. Downstream analyses can then carry it through sensitivity sets, alternative routing models, or no-estimate rules. This preserves the distinction between an interval that contributes neutral value and an interval the study cannot responsibly classify.

11.3.6 — No Threshold Chosen for Closure

For confirmatory use, freeze unity-gate features, priors, thresholds, exception rules, and the approved threshold family before examining ledger or endpoint outcomes. Otherwise closure selects the bearer it purports to test (Hernán & Robins, 2020).

For a post-registration gate change, preserve the original analysis and change reason under Section 28.2, and evaluate the new version on new or properly held-out data. Retest motivating cases in a separate validation role, so they cannot serve simultaneously as tuning and confirmation data.

Test threshold perturbations, channel ablation, missingness alternatives, and justified averaging over the prespecified family. Proper scores compare predictive calibration, not whether a gate identifies phenomenal unity (Gneiting & Raftery,

2007). Report the threshold regions over which the substantive verdict remains unchanged.

A gate may legitimately make the study unusable. If every plausible threshold produces unstable ownership, excessive double counting, or loss of essential intervals, the affected evidence is Unusable when a required ownership or measurement gate fails. If the required gates pass but the target still cannot be identified or validly bounded, the result is No adequate estimate. The failed layer should be named explicitly so later work knows whether better markers, routing evidence, or a different estimand could restore adjudicability.

11.4 — Pause, Interruption, and Rebinding

Sleep, anesthesia and temporary disconnection separate observable responsiveness from evidence about continuity. A pause-return analysis asks how a later state connects to the pre-gap stream while preserving what was known at each time. Keep the original prospective classification alongside a distinguishable retrospective record that incorporates later recovery, report or learning.

11.4.1 — Sleep and Reversible Loss of Access

Sleep makes the distinction familiar: a person can be behaviorally unresponsive while dreaming, then awaken with only fragments of recall. For the routing problem, the useful question is how evidence about particular sleep intervals connects with the stream before and after them. Sleep stage, an awakening report and a subsequent memory each address a different point on that timeline (Siclari et al., 2017; Cataldi et al., 2024).

Sleep also reveals how time scale affects identity. Over seconds, integration may fragment; over hours, the organism cycles; over mornings, the person returns to ordinary orientation. The stream model must say whether it is classifying momentary inclusion, overnight continuity, or longer narrative persistence.

Waking orientation, memory fragments, physiological transitions, recurring goals, and person-specific patterns can support continuity after sleep without determining membership for every intervening interval.

Keep sleep onset, stage, awakening, dream-report time, physiological observations, and later recall distinct so sampled experience is not equated with what is remembered.

11.4.2 — Anesthesia and Return

Clinical anesthesia monitoring serves patient safety, dosing, analgesia, and recovery. Those signals can inform a research state model only under a separate protocol that does not alter care or imply that a depth index identifies the phenomenal bearer. The scientific use must remain subordinate to clinical purpose, especially when movement, memory, and report are pharmacologically unavailable (Sanders et al., 2012; Linassi et al., 2018; Jang et al., 2024).

Eye opening, command following, orientation, memory formation, and executive control can return at different times across drugs and people. Define the recovery milestone being timed; report separate times for different capacities and uncertainty in each estimate.

Separate induction, maintenance, surgical stimulation, medication, monitoring, neural markers, emergence, immediate report, and follow-up. Induction does not establish a truth-level endpoint, and return does not classify every gap interval.

11.4.3 — Temporary Disconnection

The key question is whether the disconnection is an access problem, an integration problem, a report problem, a control problem, or a stream-continuity problem. Those are different. A locked-in patient may retain experience with impaired output. A dissociative state may alter access to memory or self-organization. A seizure may disrupt control while leaving later continuity complex.

Disconnection can be environmental or communicative as well as neural. Language mismatch, sensory loss, fear, restraint, institutional neglect, and inaccessible interfaces can make a present person appear unreachable. Continuity analysis should therefore model the relation among organism, environment,

observer, and channel before interpreting failed communication as interruption of the stream.

Log devices, tasks, awakenings, and protocol anxiety that degrade responsiveness or reporting. Distinguish these study-induced observation failures from disease-related disconnection and stream interruption.

Use Section 11.4.6's gap record and retain a no-estimate option when access disruption cannot be distinguished from interruption or partition.

11.4.4 — Rebinding After Recovery

Rebinding asks whether evidence supports joining pre-gap and post-gap states as one stream, and with what confidence. Waking, speech, or behavioral recovery supplies evidence for that inference without settling it.

Rebinding should be evaluated with humility toward the person’s own later testimony where communication is possible. A participant’s report cannot settle every interval, but it can correct misunderstanding, identify hidden awareness, explain missing behavior, or challenge the model’s assumptions.

Rebinding evidence includes restored context, recognition, learning, affective carryover, recurring goals, neural recovery, rhythms, and responses to earlier material. Compare external prompting, record exposure, reconstruction, and generic recovery as alternatives.

11.4.5 — Competing Continuity Models

Pause-return evidence can support several models. Persistence says the stream continued through the gap. Suspension says the stream paused and resumed. Restart says a new stream began after the gap. Branching says more than one trajectory may have

emerged. Partition says coordination changed without a simple single-stream story.

Specify each continuity model's predicted differences in affective carryover, learning traces, state-dependent markers, access, memory, and control, with the mechanism and observation model that generate those contrasts.

Consider a hypothetical pair of models that agree on every recorded pre-gap and post-gap measurement. One places continuing experience inside the unobserved interval; the other places a suspension there. Renaming the models supplies no statistical contrast. A useful next study identifies an observation for which their specified versions predict different distributions, perhaps a within-gap marker or a later transfer effect that only one model predicts. If the models agree on all obtainable observations, the admissible result is the pair of models, with downstream ledger calculations shown under each.

11.4.6 — Uncertainty Across the Gap

A pause-return record begins with the last pre-gap assessment of access, report and physiology, together with its clinical context and confidence class. Record the entry event separately, whether it was sleep onset, anesthetic induction, injury, seizure, disconnection or an unidentified transition. Within the gap, retain the available signals and any possible reports alongside missingness, artifacts and care events. The sequence distinguishes the observations establishing the last known state from evidence about the intervening interval.

At the apparent exit, record whether the change was awakening, emergence or clinical recovery. Identify restored communication and changes in observation separately. Then attach the post-gap evidence from memory, learning, affective carryover, command following and neural organization. Each

revised classification retains its predecessor, reviewer and reason for change. Recording the burden that further clarification places on the family helps the protocol decide which additional assessment could contribute enough information to justify asking for it.

Carry gap-related missingness through the filtered and smoothed records specified in Section 11.4. Clinical instability, death, withdrawal, burden, and technical failure can selectively remove the hardest intervals (Särkkä, 2013; Little & Rubin, 2019).

Compare gap models on return, recurrence, learning, cross-gap transfer, and later verification. Prioritize intervals by their effect on the requested inference: a short endpoint-adjacent gap may matter more than a long earlier gap, allowing lower assessment burden when additional evidence cannot change the inference under any admissible gap model.

When the evidence cannot separate the admissible gap models, report which conclusions hold under each model and withhold conclusions that depend on an unsupported choice (Giacino et al., 2018).

The duration of a gap need not rank its inferential importance. In a hypothetical signed-unit analysis, suppose the observed contribution totals +2. If the only unknown contribution is bounded between −0.1 and +0.1, the completed total lies between +1.9 and +2.1. If a different case has the same observed total but its gap is bounded between −3 and +3, the completed total lies between −1 and +5. Even a shorter gap in the second case can determine whether zero remains compatible. The useful next observation is the one that narrows the decision-relevant contribution or ownership uncertainty, rather than simply filling the longest interval.

11.5 — Partition, Fusion, and Multiplicity

Partition and fusion change the ownership of events and therefore the histories that can enter a ledger. Divided access does not by itself establish separate streams, and later coordination does not automatically combine earlier histories. The section compares split access, dissociation, overlap, and recoordination as routing hypotheses, retaining unresolved assignments where the evidence cannot distinguish them. The resulting classifications constrain research calculations without becoming public judgments about personal identity or worth.

11.5.1 — Split Access and Hemispheric Partition

Split access refers to partial separation of information, control, or report across systems that normally coordinate. Split-brain research is the familiar example, but the scientific lesson is broader: access can be divided without the ontology being settled. The evidence may support one stream with divided access, two partially independent access systems, or unresolved structure.

Recent analyses of split brains support graded and task-dependent interpretations; routing models should specify the tasks and criteria on which a partition judgment depends (Pinto et al., 2017; Schechter, 2018). Keep every routing model that remains compatible with that evidence.

Task evidence cuts both ways. If one hand responds to information unavailable to speech, the result supports access separation under those conditions. It neither establishes two complete conscious streams nor secures one undivided stream. The observation creates a routing-model comparison that must be resolved, bounded, or carried forward as uncertainty.

Laboratory tasks can magnify asymmetry because they are built to isolate information channels. A striking dissociation

under controlled instruction may coexist with substantial coordination in ordinary activity. The routing analysis should therefore compare task-specific separation, spontaneous behavior, cross-cue transfer, longitudinal stability, and causal evidence before promoting a laboratory effect into a persistent partition model.

11.5.2 — Dissociation and Altered Self-Organization

Dissociation enters first as a lived and clinical phenomenon. Trauma, stress, memory disruption, altered agency, and changing self-organization can shape both experience and report. Participants are never required to resolve the metaphysics of their own state, and public description must not turn coping or distress into spectacle.

Test amnesia boundaries, state-dependent access, executive control, physiology, affective continuity, and therapy-related integration. Diagnosis alone cannot allocate a ledger without formal and clinical bridges.

Evidence should include within-person history, state transitions, amnesia patterns, co-consciousness, physiological changes, trauma context, report conditions, clinician assessment, longitudinal stability, and participant interpretation. It should also include rival explanations such as access barriers, memory compartmentalization, state-dependent recall, role enactment, suggestion, medication, or stress response where relevant.

Research language must be trauma-sensitive. The person should not be asked to perform identity complexity for scientific curiosity. Interview burden, public exposure, destabilization, and clinical disruption can alter the very access patterns being studied.

11.5.3 — Parallel or Overlapping Streams

Parallel or overlapping streams posit multiple experiential organizations in one apparent support system, including cases not captured by simple unity or clean partition. They remain hypotheses, unsuitable as common, settled, or personal diagnoses in public summaries (Bayne, 2010; Schechter, 2018).

The evidence burden is much higher than simultaneous processing, ambivalence, divided attention, changing roles, dissociative symptoms, or competing motivations. A multiplicity model requires patterned limits on joint availability, partially independent learning or control, and evidence that the separation is stable enough to distinguish it from ordinary segregation within one stream.

Overlapping models must classify event-token ownership as exclusive, shared, unresolved, or conditional and prevent double counting. Unidentified ownership remains set-valued.

Compare one stream with segregated access, two streams sharing resources, and time-alternating routing graphs on the same observations, with comparable complexity penalties and equal opportunity to explain missingness.

Different ownership graphs can produce the same reports, choices, and neural summaries. Retain that equivalence class and identify a separating observation or intervention when one is known.

11.5.4 — Fusion and Recoordination

Recovery, treatment, neurological change or a new task context can increase coordination. That improvement may concern motor control, report or clinical function before the evidence supports fusion of previously separate candidate streams. Conversely, a person may experience increased internal communication before it becomes visible in external behavior.

Recoordination does not supply a ledger-inheritance rule. Specify whether earlier events belong to one ancestor, several descendants, a jointly owned domain, or an identified set. Inheritance remains conditional until a formal bridge justifies it.

Test cross-access, shared memory, coordinated affect, report, control, reduced barriers, and consistency across contexts. Gather this evidence without forcing a public story of becoming whole.

Keep pre-fusion, transition, and post-fusion evidence distinct. Later integration neither establishes an uncertain earlier partition nor erases a well-supported one.

11.5.5 — Ledger Ownership Under Partition

A protocol excluding partition cases from analysis must name that exclusion; one including them specifies an allocation rule or returns an identified set (Bayne & Chalmers, 2003).

Report partition-sensitive ledgers by scenario: the one-stream assignment, each admissible partition assignment, and any events that remain unallocated. This makes the analysis usable even when ownership is unresolved. A conclusion supported only by one arbitrary allocation is conditional on that allocation; it cannot be presented as established by the observations alone.

A hypothetical arithmetic example shows why the assignments matter. Suppose one admissible partition yields two complete candidate histories with signed residuals of +3 and −3 in the declared units. Pooling them gives zero, but neither individual residual is zero. The pooled result therefore cannot establish a stream-specific terminal restriction. If a competing one-stream model assigns those same events to a single bearer, it addresses a different ownership hypothesis. The data must adjudicate or retain that routing uncertainty before the total receives its proposed interpretation.

The open problem also has measurement consequences. Candidate momentary-valence and interval-contribution estimators are built for a candidate stream under conditions. If streams partition, channel-to-stream assignment becomes uncertain. Physiological signals may be shared, reports may be channel-limited, and context may affect multiple candidates.

Under a partition hypothesis, each candidate stream would require its own evidence basis, unity gate, temporal window, and uncertainty. If the evidence cannot assign an interval responsibly, the interval should be marked unresolved rather than forced into the favored ledger.

Partition, fusion, or rebinding near a candidate endpoint links endpoint classification with ownership and ledger interpretation. Evaluate that dependency under Section 13.2.

11.5.6 — Reporting and Case Selection

Display routing structures and evidential confidence without ranking them as levels of health, worth, authenticity, or sophistication. Legends identify the candidate model and uncertainty; any ordering by confidence describes evidence, not a superior kind of being.

Rare trajectories, split-access patterns, family histories, timestamps, and clinical images can reidentify participants after names are removed. Set aggregation thresholds before case selection and apply Chapter 3’s publication safeguards.

Estimate prevalence from prospectively defined sampling frames containing ordinary, null, and ambiguous cases. Striking partition or dissociation cases can challenge a simple model without establishing how often a complex model is warranted.

11.6 — Error Costs, Abstention, and Dignity

Routing error can corrupt ledger, endpoint, or route inference. Labels can also harm care, publicity, access, or standing. Address both scientific and ethical loss while keeping protections independent of classification.

11.6.1 — Scientific Error Versus Ethical Error

For each error, record who bears its consequence, affected downstream objects, correction and withdrawal options, and protections that remain fixed, alongside sensitivity and specificity.

False continuity joins intervals or streams that should be kept separate; model comparison and a revised confidence judgment address that error. False discontinuity breaks a stream that should remain connected, so the record retains the possible continuation while further evidence is sought.

False partition multiplies candidate streams beyond the evidence, calling for review of the proposed separation and its evidential burden. False rebinding reconnects states whose relation remains unresolved. Its correction preserves the earlier uncertainty and provides a route for appeal.

The two error types interact. If classification changes behavior toward the person, the data-generating process changes. Staff attention, family anxiety, participant stress, consent pressure, and clinical workflow can all alter evidence. That means ethical leakage can become scientific contamination.

11.6.2 — Asymmetric Protection

Asymmetric protection is prospectively specified before cases appear. Possible false exclusion triggers protective caution, while possible false inclusion narrows the claim and restricts

public use. The two responses address different harms and can operate together without turning caution into a bearer verdict.

11.6.3 — Abstention as Valid Output

The reason for abstention must be typed. Marker conflict, insufficient temporal resolution, missing report, confounded physiology, theory disagreement, threshold instability, unresolved routing, clinical burden, and lack of independent review are different failures and lead to different next steps.

Prespecify the consequences of low signal, missing access, threshold sensitivity, model equivalence, and routing nonidentification under Section 24.2's evidence-state rules. An inconvenient result alone cannot trigger abstention.

Track abstention rates and causes: concentration by device can expose an instrument failure, by population an access or reference-class failure, and by high-burden setting a reason to stop that study use.

11.6.4 — Blinded and Multidisciplinary Review

Audit reviewer roles, independence, conflicts, evidence access, blinding limits, dissent, and revision triggers (Yakhkind et al., 2024; Giacino et al., 2018).

Allow a pause when classification burden affects care, family communication, participant distress, or staff workflow.

Preserve material dissent and its evidentiary basis even when a bounded conclusion is justified.

Where possible, blind continuity reviewers to ledger outcomes, terminal-neutrality expectations, public narratives, and which classification would favor the investigators' result.

Multidisciplinary review should include consciousness science, neurology, psychiatry or psychology where relevant,

palliative or clinical expertise, statistics, ethics, and lived-experience consultation where appropriate and safe.

Scale review to risk, ambiguity, and claim consequence; clear ordinary cases need not receive the process required for disorders of consciousness, dissociation, or endpoint-adjacent uncertainty.

11.6.5 — Appeal and Revision

Provide plain-language challenges, clinician-supported and appropriate family-supported routes, and independent ombuds review. Access to appeal must not depend on technical or legal expertise or institutional confidence.

Under Section 28.2, preserve initial and revised classes, evidence, reviewer, permitted use, challenge, new evidence, and affected claims.

Appeal remains private enough to be usable. Families and participants can challenge a classification without exposing intimate details in public, making care contingent on winning, or entering adversarial theater. The record preserves the grounds, response, and correction while minimizing collateral disclosure.

Appeal initiates formal reanalysis. Participants, authorized surrogates, clinicians, and independent reviewers may challenge evidence, thresholds, permitted use, or retention without risking care or participation. Propagate accepted revisions through all dependent outputs under Section 28.2 (Center for Open Science, 2025).

11.6.6 — No Administrative Use

Administrative leakage occurs when a research label travels beyond its purpose and affects eligibility, treatment, rank, or status. Access restrictions must enforce the prohibition on those uses.

Architecture enforces that boundary. Research classifications, clinical records, identity, contact information, and public outputs occupy separate access domains and retention schedules. Administrative systems never receive a portable bearer score.

Automation may assist documentation and audit, but bearer labels may not train systems for triage, surveillance, eligibility, or ranking. Any new permissible research use requires fresh justification, independent review, and accountable human authority.

The audit trail records who accessed a classification, for what authorized purpose, and whether export or model-training use occurred. Unusual access or attempted linkage triggers review and suspension authority.

Public reporting defaults to cohort-level methods and uncertainty. An indispensable case still requires consent, de-identification, narrative restraint, and hostile-crop review; reproducibility does not require indefinite exposure of rare trajectories.

Chapter 12 — Consciousness Theories and Boundary States

A consciousness theory becomes useful to this program when it changes what a study predicts or how a result is interpreted. The 2025 adversarial comparison of global neuronal workspace and integrated information theories provides a concrete example: competing commitments were constrained before theory-neutral testing. The test must distinguish commitments, rather than redescribe a familiar result (Cogitate Consortium et al., 2025).

A marker's name cannot answer that question. Global availability, causal integration, recurrence, self-monitoring, and embodied regulation refer to different proposed relations. A study must identify the capacity its task or signal addresses before that evidence enters a classification. Detecting access does not automatically establish continuity, and measuring complexity does not by itself identify a phenomenal bearer.

The same result can also have different explanations. A low response may reflect the proposed state, a failed route of access, or a change in how the marker is generated. A high response may reflect organized processing that the theory regards as insufficient. Predict when each alternative, including the preferred marker, would mislead.

Sleep offers a recurring test of this reasoning. Stage, awakening time, report, and later recall do not describe one interchangeable event. Their changing relation allows investigators to ask which part of a classification depends on accessible content and which depends on the opportunity to report it. A missing report leaves several possibilities for the model to distinguish.

Anesthesia supplies different contrasts because drugs can affect movement, memory, physiology, and experience through different pathways. The relevant markers must be evaluated by agent, dose, task, and time. These are observation questions

within clinically governed care; they do not authorize altering an anesthetic plan to obtain a cleaner theoretical contrast.

Brain injury, dementia, and delirium further expose the difference between a capacity and the channel used to reveal it. Repeated assessment can change the evidence without establishing that every intervening interval shared the newly observed state. The chapter therefore evaluates each marker for the particular inference it is expected to carry.

Developmental, nonhuman, and artificial cases test transport itself. A marker depending on adult language cannot simply become a universal requirement; fluent artificial output cannot simply inherit the evidential role of human testimony. Each extension needs a stated relation between the candidate's organization, the observation, and the capacity being inferred.

Agreement across theories is useful only when the theories are making compatible claims about the same target. Disagreement may instead delimit the inference: a study can preserve a supported observation while leaving its interpretation theory-dependent. A single averaged label would conceal that distinction if the competing models assign different meanings to it.

A boundary-state record should remain open to challenge and revision. Its value lies in making the next discriminating observation identifiable while preserving uncertainty where the available evidence cannot decide.

12.1 — Theory Pluralism and Marker Families

Consciousness theories connect different aspects of organization to different marker predictions. Global availability, causal integration, recurrence, higher-order access, predictive regulation, and embodiment consequently need not classify the same evidence in the same way. The analysis must state which theory supplies each bridge and which alternatives remain viable. Eligibility rules are chosen before endpoint outcomes are inspected, so disagreement becomes an explicit part of the comparison rather than a choice made to favor the result.

12.1.1 — Global Availability Accounts

Global-availability accounts examine broadcast, flexible access, task use, and responses that do not require verbal report. Motor or language failure can hide access, while fluent output can imitate it.

Global-availability accounts associate consciousness with information becoming broadly available for reasoning, report, memory, action, and flexible control: a change in functional reach (Dehaene et al., 2017; Mashour et al., 2020).

Predicted markers include delayed, widespread, or task-relevant activation, flexible report or decision use, broadcasting signatures, and contrasts between subliminal processing and content entering control.

Within this program, global-availability markers must distinguish what a task establishes about access from what the chosen theory infers about experience. Report-independent tasks, alternative response channels, and timing controls can reveal flexible use without demanding speech. Negative evidence remains weak when sensory input, comprehension, attention, memory, fatigue, or motor output is compromised;

positive evidence supports organized access without mapping every untested interval.

12.1.2 — Integrated and Causal Accounts

Integrated and causal accounts examine effective connectivity, perturbational complexity, and differentiated causal responses. Apparent complexity can arise from artifact, and low responses can reflect measurement failure.

Integrated and causal accounts begin from a different intuition. Consciousness seems to require a system whose parts are not merely coactive but causally integrated in a way that produces a unified field. These theories make differentiation and integration central (Tononi et al., 2016; Albantakis et al., 2023).

Candidate markers include effective connectivity, perturbational response, complexity, differentiation, and causal interaction. The strongest versions ask whether perturbing one part of the system produces a richly structured response that is neither local reflex nor global silence, making the approach especially relevant to anesthesia, sleep, and severe brain injury.

Low complexity or connectivity can reflect absent experience, drug or injury effects on propagation, an inappropriate perturbation scale, or poor data. High values can also reflect rich dynamics without the proposed unity. Interpretation requires the declared context and theory.

Progress is already measurable. Casarotto and colleagues (2016) tested a perturbational-complexity threshold in 81 patients after establishing it in a 150-person benchmark. It detected 36 of 38 minimally conscious patients, while 9 of 43 behaviorally unresponsive patients had complexity values overlapping the conscious benchmark. This provides a practical route toward identifying capacities that behavior alone can miss.

12.1.3 — Recurrent and Higher-Order Accounts

Recurrent and higher-order accounts examine feedback loops, metacognition, confidence, and self-monitoring. Recurrence can occur without consciousness, while metacognition can fail even when experience remains.

Recurrent processing accounts emphasize feedback loops rather than one-way feedforward registration. A stimulus may be processed deeply enough to guide behavior without becoming conscious unless recurrent interactions stabilize or amplify it. Higher-order accounts add a different condition: a state becomes conscious when the system represents, monitors, or is in some way aware of its own state (Lamme, 2006; Brown et al., 2019).

These accounts produce markers that are especially useful in borderline cases. Recurrent processing directs attention to sustained or feedback-mediated activity within sensory and association networks. Higher-order approaches ask whether metacognitive access, confidence, self-monitoring, or error awareness is present. Both families help explain why content, access, and self-evaluation can dissociate.

Adult metacognitive report can be unavailable in infants, animals, language impairment, or altered states without settling experience. Conversely, a permissive recurrence criterion can include loops that never become felt.

A boundary protocol records whether recurrence is local or long-range, content-specific, persistent across tasks, paired with integration or access, and robust to artifact controls. For higher-order markers, it identifies the capacity tested and whether experience could occur without it (Lamme, 2006; Brown et al., 2019).

12.1.4 — Predictive and Active-Inference Accounts

Predictive and active-inference accounts examine interoception, generative control, and embodied regulation. Viability and control can exist without establishing felt experience.

Predictive and active-inference accounts emphasize generative models of world, body, and action consequences, including prediction error, interoception, control, and maintenance of viable states (Friston, 2010; Seth & Friston, 2016).

Candidate markers include prediction-error updating, precision weighting, interoceptive coupling, perception-action coordination, attention, and regulation. They can inform pain, dyspnea, nausea, anxiety, delirium, and terminal-state models.

Regulation alone does not establish experience: a thermostat has control loops without supplying a phenomenal bridge.

Test whether these markers distinguish conscious modeling from generic regulation using reports, covert command following, perturbational evidence, and clinical context. Specify whether the target is level, content, valence, or background capacity.

12.1.5 — Embodied Accounts and Cross-Theory Limits

Enactive and embodied accounts examine organism–environment coupling, situated agency, and sensorimotor involvement with the world. Their criteria can be overextended, simulated, or tied too narrowly to human forms.

Enactive and embodied approaches emphasize active organism-environment relations. Record posture, movement, breathing, sensory sampling, autonomic regulation, pain, medication, and social coupling when they affect experience or its observation.

Preserve theory-indexed classifications when disagreement affects ledger inclusion. Report the supported observations and

discriminating evidence, narrow affected uses, and abstain when theory dependence blocks the required inference. Averaging incompatible targets into one score would hide the disagreement.

Orienting, autonomic response, sensorimotor contingency, pain behavior, and environmental engagement can constrain models without report. Test reflex alternatives, motor suppression, and intervention effects on those dynamics.

12.1.6 — Theory-to-Marker Matrix

Each theory-matrix row records version, target, contrast, timescale, population, task, observation, preprocessing, scale, prerequisites, missingness, predicted direction, uncertainty, confounds, detection opportunity, failure modes, and conditions making either result interpretable. Prespecify how conflicts remain visible.

Targets are fixed before scoring. State presence, content, access, integration, metacognition, continuity, report capacity, and action control are different objects; evidence cannot migrate among them by resemblance.

Open multisite fMRI data from 118 participants and MEG-EEG data from 100 preserve common events, quality reports, metadata, and code across cohorts and modalities. Openness does not settle consciousness, but it makes preprocessing, rival testing, and transport inspectable (Khalaf et al., 2026; Liu et al., 2026). Shared acquisition protocols also allow site and modality effects to be estimated rather than conflated with theory performance.

Interpret prediction scores with the construct definition and checks on delivery, task access, signal quality, and the intended contrast. Distinguish task-specific preprocessing repairs from

predictions surviving a frozen cross-site pipeline (Gneiting & Raftery, 2007).

12.2 — Sleep, Dreams, and Reversible Low Access

High-density recordings link dream reports to local and distributed changes across REM, non-REM, and parasomnia states. Within-state sampling and awakening design therefore matter to continuity inference (Siclari et al., 2017; Cataldi et al., 2024).

12.2.1 — Sleep Stage and Access

Sleep stages differ in arousal threshold, physiology, neural dynamics, and report probability. Rapid eye movement sleep, non-rapid eye movement sleep, slow-wave sleep, light sleep, microarousals, and transitional states should not be collapsed into one category called unconscious. A state label is only the beginning of inference (Siclari et al., 2017; Nir & Tononi, 2010).

Specify arousal markers, EEG features, autonomic context, awakening protocol, and report procedure for the sleep-stage model.

Use stage as contextual evidence about access, dreaming, memory, and responsiveness. Threshold-family analysis must vary how stage, report, physiology, and marker uncertainty affect interval inclusion; the stage label alone does not decide it.

12.2.2 — Dreaming and Non-Dreaming Reports

Awakening changes attention, memory, language, and motivation, and the absence of a report may reflect nonexperience, failed recall, weak encoding, question wording, or the timing of awakening (Siclari et al., 2017; Tempesta et al., 2018).

Evidence from sleep research suggests that reports of dreaming can occur in both rapid eye movement and non-rapid

eye movement sleep, and that posterior cortical activity has been associated with dream reports.

A strong protocol varies awakening timing, records confidence, distinguishes immediate from delayed report, and models low salience, language limits and burden-related nonresponse alongside the alternatives above.

12.2.3 — Lucid and Fragmented States

Lucid or fragmented sleep can combine vivid content, partial metacognition, weak control, altered time, unstable selfhood, and little external access, separating dimensions often bundled during waking (Cataldi et al., 2024; Nir & Tononi, 2010).

Higher-order accounts emphasize lucidity's metacognition; global-access accounts ask about control within the dream; predictive accounts examine internally generated versus external input; integration accounts ask whether the interval remains one unified stream.

Distinguish fragmented content, access, agency, memory, or self-model from changed causal ownership. Partial disruption alone does not establish independently owned streams.

12.2.4 — Awakening, Sampling, and Privacy

Vivid or late dreams may be better recalled; report training, pressure, lost earlier memories, and prompts demanding a narrative can also select or reshape the account.

Record stage, awakening schedule, question form, report delay, confidence, and the interpretation of nonresponse alongside the observation specification in Section 12.2.1.

Prespecify and record alarm burden, fatigue, and any exclusion of difficult awakenings, including the missingness those choices create (Cataldi et al., 2024; Little & Rubin, 2019). Dream narratives can expose trauma, sexuality, relationships,

and identity. Minimize narrative detail, separate analytic features from identifying content, and permit withholding or deletion without penalty (World Medical Association, 2024).

12.2.5 — Unity-Gate Sensitivity

Test a strict waking-access gate, a broader experience-likelihood gate, a theory-plural soft gate, and no-estimate treatment of ambiguous intervals. Report when the result depends on a particular sleep gate.

Sleep makes the exclusion rule testable. An interval may leave the confirmatory computation only when the gate, reason code, and sensitivity treatment were declared in advance. The exclusion remains visible in the denominator and cannot be converted into evidence that the interval was phenomenally empty.

12.2.6 — No Dream Privilege

Dream content supplies auxiliary evidence for sleep-state analysis; it does not establish lifetime closure.

Dream experience, dream report, and later recall are different objects. A person may experience without retaining a report, report fragments altered by awakening, or construct a coherent narrative from sparse memory. The study must preserve these distinctions rather than treat remembered dream content as a direct transcript of the sleeping stream (Nir & Tononi, 2010).

Sampling is selective. Awakenings are more likely in some stages, after some physiological changes, and when participants can respond. Reportable or vivid dreams therefore receive disproportionate representation, while silent intervals remain difficult to interpret. A dream dataset is a record of both sleep experience and the procedure used to interrupt and sample it.

Dream evidence is useful for testing state-dependent report, affective transition, memory, unity-gate sensitivity, and links between physiology and later testimony.

A protocol-fixed sleep design should state the stage, awakening schedule, report window, scoring method, missing-report rule, distress policy, and separation of confirmatory from exploratory features. Negative recall does not equal no experience, nor does a vivid report establish continuous experience across the interval.

The most useful output may be conditional. A study can report how marker families behave in intervals followed by particular reports, how awakening changes evidence, and which unity models remain compatible. It need not decide what every silent interval contained.

This discipline lets sleep inform the next boundary case without mystifying it. Anesthesia shares low access, uncertain report, memory failure, and state transition, but adds pharmacology, motor blockade, and nonnegotiable clinical priority.

12.3 — Anesthesia and Covert Consciousness

Anesthesia and covert-consciousness research distinguish observable responsiveness from the processes a consciousness marker is intended to detect. Changes in motor access, recall, and clinical state can alter those relations. The following analysis keeps state evidence, return evidence, and endpoint inference separate while asking what each qualified observation can establish.

12.3.1 — Drug Class and Dose

Drugs used during anesthesia have different effects on consciousness, memory, movement, and physiology. Propofol, volatile agents, ketamine, benzodiazepines, opioids, neuromuscular blockers, and adjuncts cannot be collapsed into one state category. Neuromuscular blockers cause paralysis without providing unconsciousness, amnesia, or analgesia. Drug class and dose shape access, memory, sensory response, arousal, and autonomic markers differently (Sanders et al., 2012; Jang et al., 2024).

A state model must therefore record agent, dose, timing, coadministration, depth monitoring, hemodynamics, ventilation, pain control, surgical stimulation, and patient factors such as age, illness, neurological condition, medication, and sleep deprivation.

Prespecify each theory's expected marker response by agent and dose, distinguishing access, complexity, self-monitoring, and embodied modeling.

A concrete measurement advance makes these distinctions more tractable. Maschke and colleagues analyzed spontaneous EEG from 15 healthy volunteers before and during unresponsiveness produced by propofol, xenon, or ketamine. Neural criticality measures differed across the drug-associated

experiential states and predicted perturbational complexity; ketamine-associated dream reports were especially useful for separating unresponsiveness from reported absence of experience (Maschke et al., 2024).

An external validation can reserve new agents, populations, and recording systems while retaining report and behavioral channels alongside EEG. Prespecify the relation between complexity, reported experience, and task access being evaluated.

12.3.2 — Behavioral Unresponsiveness

Record drug effects, paralysis, motor interruption, sensory failure, task design, pain control, fear, confusion, and command comprehension. Identify intact, failed, and untested sensory or motor prerequisites and available alternatives before interpreting nonresponse (Sanders et al., 2012; Linassi et al., 2018).

12.3.3 — Covert Command Following

Covert command following asks whether a person who cannot respond behaviorally can nevertheless follow instructions in neural activity. The paradigm is powerful because it separates motor output from task understanding and intentional modulation. It is also limited because it requires comprehension, attention, sensory access, task design, and equipment-sensitive interpretation (Huang et al., 2018; Claassen et al., 2019).

An openly released graded-propofol fMRI resource extends this logic in healthy volunteers. It combines mental-imagery and motor-response tasks before, during, and after loss of behavioral responsiveness, including instances of volitional imagery without overt response. The dataset permits comparison and validation of models across dose, task, connectivity, and emergence while keeping behavioral output distinct from covert

task performance. Model recovery separately requires synthetic or otherwise ground-truthed benchmark data (Huang et al., 2026).

Anesthesia can make task delivery clinically inappropriate or technically fragile. Related paradigms in disorders of consciousness test cognitive motor dissociation under their own access conditions.

A positive covert response establishes organized task-related cognition under a declared paradigm. It can support command comprehension and intentional modulation, but it does not map every interval, content, affective state, or continuity relation needed for a complete bearer classification (Claassen et al., 2019; Bodien et al., 2024).

The protocol requirements include protocol-fixed tasks, artifact controls, false-positive rates, false-negative rates, replication criteria, sensory access checks, medication context, and independent interpretation. A single surprising activation should not rewrite the bearer classification without a dossier.

A negative result is equally bounded. Task misunderstanding, sensory or language barriers, medication, fatigue, signal loss, preprocessing choices, and fluctuating state can all produce failure. Repetition is justified only while clinical condition, participant welfare, and expected information gain support another attempt.

12.3.4 — Memory Formation and Amnesia

Memory is often mistaken for experience because the public usually learns about experience through later memory. Anesthesia breaks that habit. A patient may have no later recall because memory encoding was suppressed, while the question of intraoperative or perioperative experience remains distinct (Sanders et al., 2012; Linassi et al., 2018).

The relevant distinctions include conscious experience, sensory processing, pain processing, distress, memory encoding, consolidation, retrieval, and later narrative. Each can fail or survive differently.

A physiological response does not by itself establish conscious distress. The dossier should therefore analyze memory, access, nociception, autonomic response, and report as distinct channels.

12.3.5 — Emergence and Rebinding

Emergence separates arousal, integration, attention, memory, orientation, report, agency, and affect regulation (Jang et al., 2024; Z. Huang et al., 2024). Access and report may return together while integration differs, predictive control remains affected by pain or drugs, and higher-order monitoring lags arousal. Preserve those trajectories rather than one return time.

Apply Section 11.4's separate prospective and retrospective records to induction, maintenance, emergence, and later report. Independently supported continuity after return can challenge an earlier endpoint classification without reconstructing every experience in the gap.

12.3.6 — Clinical Priority

Clinical care governs every anesthesia study. Drug choice, dose, ventilation, airway management, analgesia, monitoring, emergence, and postoperative treatment must be determined by patient safety and clinical judgment, not by the needs of the Law's research program (World Medical Association, 2024).

Research procedures require a noninterference budget. Auditory tasks, repeated stimulation, extra sensors, staff movement, and prolonged setup can alter arousal or clinical workflow. The protocol should state the permitted burden and

give clinicians an immediate stop rule without penalty to the patient or team.

The opportunity-to-respond analysis should also establish whether the state remained stable long enough for the task.

Disallowed outputs include bedside ranking, clinical authority, judgment of a particular anesthetic plan, or claims that hidden suffering has been detected in a specific procedure.

12.4 — Coma and Disorders of Consciousness

Retain repeated behavioral examination alongside multimodal assessment of covert capacity. Prognosis and treatment remain governed by independent clinical standards; research classifications do not determine personhood (Bodien et al., 2024; Giacino et al., 2018).

12.4.1 — Clinical and Diagnostic Boundaries

Behavioral assessment can support conclusions about reproducible response, arousal, or a clinical category. It cannot by itself establish absent experience when motor or sensory channels are compromised.

Coma, unresponsive wakefulness or vegetative state, and minimally conscious state are clinical categories for wakefulness, behavior, awareness evidence, and recovery, rather than complete descriptions of subjective experience (Giacino et al., 2018; Giacino et al., 2002).

Coma involves absent wakefulness. Unresponsive wakefulness preserves arousal cycles without clear behavioral evidence of awareness. Minimally conscious states show inconsistent but reproducible signs of awareness. These distinctions must be respected because they shape prognosis, care, research eligibility, and communication attempts.

In severe brain injury, the measurement analysis records whether valence estimates are identified, partially identified, or dependent on explicit marker and routing assumptions.

Assessment method can change the clinical classification even before neural markers are considered. In a 103-patient study, 18 of the 44 patients classified as vegetative by clinical consensus met criteria for a minimally conscious state on the Coma Recovery Scale-Revised (Schnakers et al., 2009). This 41% discrepancy describes that study's selected sample and

comparison, not a current population prevalence. It gives the dossier a concrete audit question: was the category assigned by informal consensus or by a named standardized examination?

12.4.2 — Cognitive Motor Dissociation

Task-based EEG or fMRI can support covert command following under the defined task. That result does not by itself establish continuous consciousness, valence, or inclusion of an entire stream in a ledger.

Cognitive motor dissociation denotes absent observable command following with fMRI or EEG evidence of a commanded mental task. It requires reassessing behavior-only exclusion when motor output is unreliable (Claassen et al., 2019; Bodien et al., 2024).

Across six centers, Bodien and colleagues studied a convenience sample of 353 participants. Task-based fMRI or EEG detected responses in 60 of 241 participants without observable command following, or 25%. Among 112 participants who did follow commands behaviorally, 43, or 38%, produced detectable task responses. These differing detection rates make channel sensitivity a central design question and support repeated, multimodal assessment of retained cognitive capacities (Bodien et al., 2024).

Evaluate apparent covert responses for artifact and analytic overfitting. Interpret absent responses in light of sensory access, task failure, and uncertain command delivery. Repetition and multimodal assessment may increase confidence, but testing must stop when fatigue, burden, or clinical priority makes further assessment inappropriate.

Phenomenal experience, continuity, prognosis, ownership, and endpoint status remain separately uncertain (Claassen et al., 2019; Bodien et al., 2024).

12.4.3 — Longitudinal Change and Confounds

A 2025 three-center EEG study in 237 acute and chronic disorders-of-consciousness patients identified recurring dynamic connectivity patterns and evaluated real-time bedside detection and recovery prediction. For this chapter, the contribution is a longitudinal, state-dynamic marker family that can be tested against diagnosis, time, confounds, and outcome rather than a one-time scalar label. Its findings do not eliminate task, clinical, or theory uncertainty, but they make repeated bedside dynamics a serious part of the dossier (Della Bella et al., 2025).

Track fluctuations in arousal, medication effects, fatigue, sleep-wake cycles, pain, infection, seizures, sensory access, and recovery. A single assessment can miss this state variation (Bodien et al., 2024; Yakhkind et al., 2024).

Repeated assessment matters because detection is probabilistic. A patient may show command following on one day and not another. A family may observe meaningful response in a familiar context while a clinical exam does not reproduce it. A machine marker may vary with state, electrode quality, or task comprehension.

Record examiner, instrument version, arousal cycle, medication, sedation depth, analgesia, sensory and motor pathways, time since injury, injury evolution, state transitions, infection, seizures, temperature, sleep, device quality, preprocessing, and adjudicator knowledge with each classification (Giacino et al., 2018; Yakhkind et al., 2024).

Classify unresolved confounding under Section 24.2; a clean trace does not make an unqualified interpretation usable.

12.4.4 — Family and Clinician Evidence

Family and clinician observations contribute context, repeated patterns, and state sensitivity. Definitive classification requires corroboration and review of relevant confounds.

Interpret family observations as possible access, affect, learning, recognition, or arousal evidence, testing reflex, coincidence, and timing artifacts under the relevant theories.

Record observer, setting, conditions, repetition, possible blinding, and the claims the observation cannot support.

12.4.5 — Classification and Treatment Decisions

Perturbational or complexity measures can inform a level or capacity judgment under a theory-dependent marker. They do not determine personhood, care entitlement, or universal consciousness status.

A research classification must never become a treatment ceiling. Low confidence, failed command following, an unresolved marker pattern, or No adequate estimate cannot justify reduced analgesia, rehabilitation, communication support, repeated examination, legal protection, or ordinary clinical care (World Medical Association, 2024; Giacino et al., 2018).

Diagnostic classification, prognosis, research eligibility, and bearer routing are separate questions. They may use overlapping evidence, but they have different targets, consequences, and error costs. A marker useful for detecting covert command following may be poor at forecasting recovery and still less suited to determining whether an interval enters a phenomenal ledger.

Audit selective verification under Section 11.2.6 before using clinical follow-up to evaluate the marker.

False negatives are especially consequential because absent response can reflect motor impairment, fatigue, fluctuating

arousal, language mismatch, sensory loss, medication, or task burden. Multimodal assessment reduces some risks but does not make every negative channel independent or decisive (Bodien et al., 2024).

Longitudinal reassessment is part of the target. A state that fluctuates across hours or days cannot be represented responsibly by one examination selected for convenience. Repeated testing, rest, alternate tasks, and context-sensitive interpretation help distinguish stable absence from episodic access or measurement failure.

Verification is selective. Cases followed by recovery, repeated testing, longer survival, or better-resourced sites are more likely to supply a later answer. Studies should report which classifications become verifiable and whether verification depends on prognosis, family advocacy, site capacity, or survival, rather than evaluating accuracy only on the easiest cases.

Family and bedside observations can identify timing, channels, and test conditions that standardized tasks miss. Familiar voices, routines, expressions, or context-specific responses should be preserved with provenance and uncertainty, then examined against alternatives. Repeated observations can test whether context-specific responses recur under comparable conditions, helping separate stable signal from one-off interpretation.

Research and treatment pathways therefore remain on separate rails. Clinical decisions made under time pressure do not certify a research bearer, and research classifications do not outrank patient values, clinical obligations, or law. Public language should state the purpose-bound result and its limits. This separation also guards against circularity in which treatment

decisions made from a clinical classification later appear as independent validation of the research label.

The scientific gain is a more accurate account of what the evidence can and cannot show. The ethical gain is that uncertainty is absorbed by the research system rather than by the person whose communication is limited. That allocation of uncertainty reduces pressure to force a binary classification when the available channels support only a range of possibilities.

12.5 — Dementia, Delirium, and Fragmented Access

Dementia and delirium require context-sensitive access and comfort measures because memory, language, and attention can change alongside distress, pleasure, recognition, and relational response (Inouye et al., 2014; Fong et al., 2015; Livingston et al., 2024).

12.5.1 — Memory Loss Is Not Experience Loss

Loss of recall or recognition does not establish absence of comfort, fear, hunger, pain, pleasure, or attachment (Squire, 2004; Livingston et al., 2024).

The evidence channels must therefore separate episodic memory, semantic memory, autobiographical narrative, immediate affect, recognition, bodily comfort, distress behavior, and physiological state. Each channel can decline differently.

Sampling only remembered states selects against impaired encoding, retrieval, language, or autobiographical organization. Model memory as a delayed, selective witness, with failure changing uncertainty rather than retrospectively vetoing experience.

Combine contemporaneous observation, accessible report, comfort/distress indicators, caregiver context, sensory checks, and repeated assessment. Identify which observations depend on memory and how recall was tested.

Prospective observation is also making retained capacities in advanced dementia easier to study. Across 539 audiovisual observations of 20 participants, Gilmore-Bykovskyi and colleagues (2025) identified nine validated lucid episodes in three people, including increases in meaningful vocal communication. A separate caregiver study found that lucid episodes often occurred well before death (Griffin et al., 2024).

These findings support patient-centered attention to fluctuating communication and offer a feasible observation method.

12.5.2 — Delirium and Fluctuation

Delirium is a fluctuating state of attention, awareness, cognition, and often distress. It can vary across hours, medication changes, infection, pain, sleep disruption, environment, and clinical stress. A single classification can be dangerously misleading because the state itself is unstable (Inouye et al., 2014; Fong et al., 2015).

The evidence problem is temporal. A person may be lucid in the morning, confused in the evening, distressed at night, and more coherent after hydration, pain control, or family presence. Measurement that averages across those changes can turn fluctuation into noise rather than evidence.

Treating the entire episode as one state can erase periods of possible experience or admit intervals whose channels were unusable. A state-space record should therefore permit included, low-confidence, and no-estimate intervals within the same clinical course.

A fair protocol records delirium screens, medication, infection, pain, sleep, environmental stimulation, sensory aids, hydration, communication access, and time of day. It also records whether a low score reflects inattention, aphasia, hearing loss, fatigue, fear, or true absence of relevant capacity.

12.5.3 — Language and Report Limits

Use accessible response routes qualified under Section 10.4 for language-limited states (Livingston et al., 2024; Kitwood, 1997).

A boundary-state protocol should distinguish inability to speak, inability to understand, inability to access words, unwillingness to report, and absence of experience. These

possibilities require different tests. Alternative communication, yes-or-no interfaces, eye tracking, gesture, affective expression, and caregiver knowledge may all matter.

The dossier should record language access, translation needs, hearing and vision status, motor capacity, communication aids, fatigue, cultural context, and confidence in interpretation. It should also state which claims remain unavailable when report is weak. Familiar people, adapted language, sensory aids, calm settings, and repeated opportunities should improve access without forcing performance or making care depend on the estimate (Livingston et al., 2024).

12.5.4 — Distress, Comfort, and Misinterpretation

Distress and comfort in dementia or delirium may appear in atypical forms. Agitation, withdrawal, calling out, resistance, facial expression, sleep change, appetite change, or bodily movement may reflect pain, fear, confusion, unmet need, environment, medication, or nonconscious reflex. Interpretation is necessary and risky (Inouye et al., 2014; Fong et al., 2015).

Excluding ambiguous behavior can hide burden; treating every movement or vocalization as distress can invent it. Use repeated observation, individual baselines, clinical and caregiver context, and converging channels to define the compatible set.

A fair protocol uses pain assessment adapted to cognitive impairment, context review, caregiver input, medication review, environmental audit, sensory access, and response to comfort interventions. It records uncertainty about whether the marker reflects felt state, physiology, habit, reflex, or communication attempt.

Comfort matters too. Pleasure, recognition, relief, touch, music, routine, food, warmth, and familiar presence can be part of felt life even when narrative is damaged.

12.5.5 — Dissociation and Split Access

Assess depersonalization, derealization, altered self-location, discontinuity, compartmentalized memory, and changed agency using Section 11.5's routing distinctions (Bayne, 2010; Schechter, 2018).

For dissociation, the discriminating question is whether apparent fragmentation reflects disrupted self-monitoring, altered body/world modeling, loss of global access, or a genuine change in causal unity. Report that theory-indexed uncertainty directly.

Record trauma risk, memory access, affective continuity, and questioning-related distress. Exclude procedures that pressure a participant to perform identity fragmentation.

12.5.6 — Care and Dignity Under Uncertainty

Classify the capacities and channels that persist, fail, or remain unresolved instead of using one global label; specify when they require abstention.

Memory loss, language impairment, disorientation, or fragmented narrative do not reduce dignity or establish absence of experience. They alter access and continuity evidence while distress, comfort, relationship, and momentary awareness remain clinically important (Livingston et al., 2024).

Medication, pain, infection, sleep, sensory conditions, familiarity, fatigue, and time of day can change communication.

One brief examination should not harden into a permanent account.

Care should maximize communication and minimize distress regardless of the estimate. Familiar people, adapted language, sensory aids, calm settings, and repeated opportunities may improve welfare and evidence without forcing performance.

A model may classify an interval as uncertain, locally supported, or unusable for one analysis. It may not turn that status into a judgment that the person is gone, less authentic, or less entitled to privacy and care.

Family and clinical observations can supply longitudinal context when source, timing, uncertainty, and alternatives are recorded.

When access remains unresolved, protective uncertainty is often correct. Science narrows its claim while care, dignity, and communication remain fully active (World Medical Association, 2024).

12.6 — Developmental and Nonhuman Cases

Developmental and comparative science cannot use adult verbal report as the universal gate. Evidence from early development, comparative neurobiology, behavior, and welfare science supports graded reference classes and protective uncertainty, while the New York Declaration emphasizes a realistic possibility of consciousness across a wider range of animals than traditional practice assumed (Lagercrantz & Changeux, 2009; Andrews et al., 2024).

12.6.1 — Developmental Emergence

The start of a qualifying stream need not coincide with biological birth. Its scientific boundary requires developmental evidence about conscious experience (Lagercrantz & Changeux, 2009; Dehaene et al., 2017).

Arousal, sensory integration, memory, and neural maturation develop unevenly; no single milestone establishes the whole boundary. Section 12.6.2 specifies age-appropriate observation channels.

Developmental evaluation must distinguish flexible coordination, causal integration, embodied regulation, and explicit metacognition rather than treat them as interchangeable markers. The boundary remains theory-indexed and age-sensitive, especially when a theory requires explicit metacognition.

A ledger cannot begin before the bearer exists, but that scientific boundary does not settle moral status.

A responsible developmental protocol states age, developmental condition, marker family, sensory access, clinical context, stress burden, and uncertainty. It uses protective defaults where false exclusion could miss experience, while

narrowing formal claims when evidence is insufficient for ledger inclusion.

12.6.2 — Age-Appropriate Evidence

The relevant channels include facial expression, physiological response, orienting, learning, touch response, sleep, distress, soothing, play, attention, social referencing, and communication attempts. Each channel requires context because it can also reflect reflex, fatigue, hunger, illness, caregiver stress, or environmental mismatch.

Theories of consciousness can be translated into age-appropriate terms. Access may appear as flexible use rather than verbal report. Integration may appear as coordinated state regulation. Predictive processing may appear in expectation, surprise, and comfort-seeking. Higher-order evidence may be limited, but that limitation cannot erase possible experience.

A protocol should state what the child could realistically do, what the environment allowed, what distress the study imposed, and which claims remain unavailable. It should also separate clinical care, developmental support, and research classification.

12.6.3 — Species-Specific Expression

Qualify each marker by age, species, task, anatomy, ecology, and welfare constraints. State the tested capacity, theoretical bridge, transport assumptions, and permitted inference.

Interpret nonhuman capacities, ecology, sensory worlds, social behavior, pain expression, play, learning, and problem solving in their species-specific context (Andrews et al., 2024; Birch et al., 2021).

The marker set will differ across mammals, birds, reptiles, fishes, cephalopods, decapods, insects, and other candidates. Some evidence may come from nociception, protective

behavior, motivational tradeoffs, learning under analgesia, play, social recognition, sleep-like states, or flexible avoidance. No single human marker can be exported unchanged.

Empirical ledger analysis of a nonhuman stream requires domain-specific consciousness, valence, and time evidence, plus qualification of the candidate realization if reserve dynamics are invoked. Welfare concern alone does not supply those bridges.

Record taxon, context, marker family, task demand, welfare protections, and known error routes. Apply Section 12.6.5's protective-uncertainty rule separately from classification.

12.6.4 — Comparative Homology and Analogy

Comparative evidence often relies on homology and analogy. Homology asks whether structures or mechanisms are shared through common ancestry. Analogy asks whether different systems perform similar functions through different structures. Both matter, and both can mislead if used loosely (Barron & Klein, 2016; Crook, 2021).

A mammalian pain system may support inference through shared neurobiology. Bird cognition may require attention to different neural architecture with similar functional organization. Cephalopod intelligence may require analogy more than direct homology. Insects may present still different evidence profiles. Each case demands its own bridge.

Theory families help sort the bridge. Integrated and causal accounts may be more open to different substrates if causal organization is relevant. Embodied accounts may emphasize organism-environment loops. Access and higher-order accounts may need translation into species-specific capacities. The bridge should be explicit rather than hidden in human resemblance.

Distinguish homologous, analogous, behavioral, physiological, computational, ecological, and welfare-protective

inference. State whether its target is consciousness, valence, pain, affective learning, or complexity.

The dossier should also state which properties are conserved and where the comparison is expected to fail.

12.6.5 — Protective Uncertainty

Protective uncertainty guides cases where suggestive evidence makes false exclusion costly. Welfare precautions can respond to that risk while researchers continue to assess formal inclusion (Andrews et al., 2024; Crump et al., 2022).

In nonhuman and developmental cases, protective uncertainty may lead to welfare consideration, reduced experimental burden, better housing, analgesia, environmental enrichment, or prohibition of harmful procedures even when consciousness classification remains open. The ethical response can be stronger than the scientific verdict.

A protocol should distinguish research classification, welfare protection, legal routing, and public communication. Protective uncertainty belongs especially to welfare and research-burden decisions. It does not authorize public scoring, domain expansion, or claims that all possible bearers have been scientifically included.

12.6.6 — No Harmful Crucial Experiments

No harmful crucial experiment is permitted to settle a question of consciousness. A study may not induce pain, deprivation, fear, injury, prolonged isolation, or clinically unnecessary burden in a child, nonhuman animal, or other vulnerable candidate merely because the result would be theoretically decisive (World Medical Association, 2024; Andrews et al., 2024).

Ethical limits still permit discriminating formal and empirical tests. Formal models and simulations may use maximally sharp endpoint-versus-continuation contrasts, including the biological-death admissibility condition in Section 5.7.2, without turning those contrasts into empirical harm. Empirical comparative work can use methods such as naturalistic longitudinal variation, comparative anatomy and behavior, reversible low-burden perturbations, benign enrichment, route choice, recovery, natural survival, and independently warranted care, husbandry, or welfare interventions. The empirical study must not create a harmful terminal condition merely to sharpen the contrast. Ethical refinement can improve inference by removing acute threat, injury, escape responses, and other processes that would otherwise become competing explanations for an observed effect. The full immediate-terminal formal discriminator is developed in Section 19.2 and the Section 29.4.5 simulation benchmark.

Research must not withhold ordinary care, enrichment, social contact, analgesia, or environmental support to preserve a contrast. Multidisciplinary review should identify avoided harms, unavailable evidence, and the limits those absences place on interpretation. When a humane condition removes a potentially informative contrast, the missing contrast should be recorded as an ethical identification limit rather than silently approximated.

12.7 — Artificial and Synthetic Candidates

Artificial and synthetic candidates test several assumptions at once: whether performance can substitute for experience, whether causal organization can transport across substrates, and how uncertainty should alter governance. Software systems, simulations, organoids, and other synthetic biological systems therefore require separate evidence pathways.

12.7.1 — Performance Is Not Experience

Performance is behavior under a task; experience is the phenomenal target, if any, behind that behavior. In ordinary human interaction, language is informative because it is embedded in shared biology and development. Artificial systems trained on human language break that inference and require an independent bridge (Bender et al., 2021; Butlin et al., 2026).

Pain language can arise from prompts, training examples, reinforced style, retrieved scripts, or modeled user expectations. Each supplies a nonphenomenal explanation of the output.

Record function, generating mechanisms, training and tool chains, persistent state, and surviving alternatives. Assign the bounded research status in Section 12.7.

12.7.2 — Architecture and Causal Organization

Consciousness theories concern causal organization, so architecture matters beyond surface behavior. Recurrence, integration, global availability, memory, attention, predictive modeling, metacognition, higher-order monitoring, embodiment, self-modeling, agency, and action-world feedback form different indicator families with different commitments (Butlin et al., 2026; Dehaene et al., 2017).

Each indicator receives an operational definition, time scale, perturbation, expected observation, and rival. Engineering labels

such as memory, reward, attention, or self-model carry no evidentiary weight merely because a component uses the name.

Causal tests replace feature inventories with counterfactual evidence. If removing or interrupting a proposed feature leaves the relevant behavior unchanged, the feature may be redundant, decorative, or incorrectly characterized; if the behavior changes, the result still requires a theory-specific bridge to consciousness.

Version training and fine-tuning, tools, memory, embodiment, sensors, actuators, persistent state, and intervention results under Section 28.2. Give indicator-rich and indicator-poor rivals the same behavioral evidence and analysis budget.

12.7.3 — Training Data and Simulation

Surface performance or scripted affect labels alone leave the claim unsupported under the current protocol. That status establishes nothing about consciousness, personhood, rights, or future architectures.

Simulation alone does not decide which required causal properties are instantiated and which are represented. Specify the relevant properties and a discriminating observation under the declared theory.

Theory families answer that question differently. Substrate-neutral accounts emphasize causal organization; embodied and biological accounts emphasize interoception, metabolism, organismic regulation, or developmental history. The dossier identifies which inference depends on which commitment.

Extend the provenance packet in Section 12.7.2 with reinforcement objectives, role instructions, simulation architecture, and the relation between affect labels and internal control. State whether the object is software, a simulation, an embodied agent, or a synthetic biological system.

12.7.4 — Organoids and Synthetic Biology

The August 2025 ISSCR compilation retains guidance stating that current central-nervous-system organoids provide no biological evidence of consciousness or pain perception that would warrant specialized consciousness oversight, while urging continued attention as models mature, combine, and gain complexity (International Society for Stem Cell Research, 2025; Hyun et al., 2020).

The dossier tracks cell types, maturity, connectivity, spontaneous dynamics, sensory coupling, learning, perturbational response, metabolic support, and theory-specific relevance. Electrical activity or complexity alone is insufficient.

Governance covers cell and tissue provenance, consent, stimulation limits, monitoring, welfare warnings, expert review, and authority to pause or stop. No aversive manipulation serves only to improve candidate-state detection.

12.7.5 — Candidate Status and Governance

Theory-derived indicators can shift credence. Fluency, model scale, simulation success, electrical activity, and biological complexity each admit nonphenomenal explanations (Butlin et al., 2026; Kreitmair, 2023; Hartung et al., 2024).

Architectural indicators with weak causal access leave the target insufficiently identified. This status supports neither the presence nor the absence of consciousness.

A theory-limited research candidate has indicators under one or two theories while relevant rivals remain viable. Its status is confined to the research interpretation licensed by those theories. Those indicators require the declared theoretical bridges and rival comparisons. They do not independently support an inference about personhood or rights.

Candidate status depends on declared theory-derived indicators and changes with system version, architecture, evidence, or failure criteria (Butlin et al., 2026; Hartung et al., 2024).

When admissible theories support different classifications, report the theory-limited results and withhold an unqualified classification. Abstain from any inference whose required architectural access or transport bridge fails, and do not pursue testing that requires harmful creation or manipulation. Record the reason and potentially decisive observation.

Artificial systems may aid search, simulation, or pattern analysis under accountable review. They cannot govern consciousness classification, certify their experience, judge another system, or decide human inner life (Tabassi, 2023; Butlin et al., 2026).

Responsible research anticipates possible artificial suffering without manufacturing it. Reversible, low-risk tests, staged review, transparent sharing, and careful public communication are preferred; unnecessary aversive training, uncontrolled replication of candidate suffering, or escalation after warnings triggers stop or redesign (Butlin & Lappas, 2025).

12.7.6 — No Artificial Intelligence Sovereignty

Multiple independent marker families that survive review can justify a cross-theory research-candidate status. Their convergence does not itself determine personhood, legal status, moral rank, or authority to adjudicate the system's own status.

A study that requires creating or intensifying suffering is prohibited. Stop or redesign it; uncertainty does not justify harmful evidence-seeking.

Apply Section 12.7.1's performance-versus-experience and theory-indexed indicator rules. Artificial self-reports remain

functional evidence unless architecture and intervention tests justify stronger inference; theory-based indicators may warrant further testing or precaution, but not a binary verdict (Bender et al., 2021; Butlin et al., 2026).

Accountability remains human and institutional. Records cover version, architecture, training, memory, tools, embodiment, data access, perturbations, rival explanations, uncertainty, and prohibited uses. No automated score enters legal, commercial, employment, clinical, or welfare administration.

Mimicry cuts both ways. Fluency or apparent distress may cause over-attribution; unfamiliar systems invite careless dismissal. Perturbation, ablation, transfer, and memory interruption clarify causal organization without revealing experience.

Artificial and synthetic candidates can expand comparative consciousness science, but candidate status remains evidence under a method. It does not confer public authority, personhood, spiritual standing, or entitlement to govern human decisions.

12.8 — Boundary-State Evidence and Review

Boundary-state evidence requires a review process capable of retaining disagreement without losing the observations that produced it. Reviewers compare the state definition, measurement channels, timing, clinical context, and return evidence before assigning the scope of a conclusion. That record supports revision when new information changes a classification and preserves uncertainty where the available evidence does not resolve it.

12.8.1 — State Evidence Dossier

Apply Section 12.1.6’s theory matrix to the observed state, timing, context, and markers. Add unavailable markers and permitted uses before classification affects a claim (Yakhkind et al., 2024; Giacino et al., 2018).

Alongside Section 12.1.6's targets, distinguish unity, valence, and endpoint relevance.

Use Section 28.2's versioned dependency record for the dossier and theory matrix. Preserve prior classifications and flag analyses affected by a change in estimate, construct, or scope.

12.8.2 — Cross-Theory Assessment

Before inspecting ledger outcomes, specify which theory-indexed markers determine primary eligibility, which support sensitivity analysis, and how disagreements are handled.

Cross-state transport can also be examined at the level of shared anatomy without assuming identical mechanisms. A 2025 analysis compared thalamocortical connectivity and local fluctuation across propofol and sevoflurane anesthesia, NREM sleep, minimally conscious states, and unresponsive wakefulness, finding nucleus-specific rather than uniform thalamic involvement. The result supports a state-indexed

transport question: which marker relations persist, which change sign or locus, and which fail outside the context in which they were estimated (Lu et al., 2025).

12.8.3 — False-Positive and False-Negative Audit

For each proposed state, audit errors under Sections 11.2.6 and 11.3. Where possible, estimate both error directions, calibration, detection opportunity, and decision consequences against the declared reference target; challenge known limitations and plausible failure modes. Complete this audit before routing, endpoint, or public use.

12.8.4 — Expert and Lived-Experience Review

Review expertise should match the boundary state and proposed inference. Clinical and consciousness specialists assess marker validity, medication and injury confounds, task access, and theory dependence; statistical review examines uncertainty and identification. Developmental, comparative, disability, or artificial-system expertise is added when the case requires it (World Medical Association, 2024; Young et al., 2021).

Seek appropriate, safe input from participants, survivors, caregivers, and family representatives about burden, inaccessible procedures, communication failures, and harmful interpretations. Distinguish that evidence from marker qualification.

Follow Section 11.6.4's reviewer record. Preserve material dissent and its consequences for confidence, permitted analysis, or the next design; a majority vote does not erase the evidentiary disagreement.

12.8.5 — Case Selection and Generalization

Publicity can give famous patients, striking scans, unusual animals, or persuasive machines extra attention and interpretive generosity. Compare ordinary and ambiguous cases, and record

whether notoriety changed testing, media exposure, model tuning, or the evidence standard.

A proposed canonical change requires a synthesis across cases, methods, theories, and failure modes, with expert disagreement and version history. Report which earlier cases would change classification under the new rule. Keep the prior rule visible in the version history so readers can reconstruct the change.

12.8.6 — No Canon Settlement by Case Study

A striking case can expose a false assumption, but it cannot settle the universal bearer definition by itself. Covert command following, a vivid anesthesia report, remarkable animal behavior, or persuasive artificial dialogue may reopen a question and revise a protocol without creating canon (Cogitate Consortium et al., 2025).

Each case should enter a versioned dossier that records marker families, theory mappings, confounds, missing channels, competing explanations, permitted uses, and revision triggers. The dossier makes the case scientifically inspectable without turning it into a symbol larger than the evidence.

Generalization requires replication and theory comparison. Equal-budget rivals, independent sites, prospective predictions, and negative controls should determine whether the case reveals a stable phenomenon, a measurement artifact, a selection effect, or a failure of one theory-specific marker. Replication should include ordinary and ambiguous cases as well as striking examples so case selection does not inflate apparent marker performance.

No single case may set treatment ceilings, legal status, species policy, artificial-system rights, or the endpoint rule for the formal hypothesis. Such decisions require their own evidentiary,

ethical, and institutional processes, and research uncertainty cannot be converted into public authority (World Medical Association, 2024). Separating research classification from downstream authority also prevents a high-profile case from becoming an informal precedent beyond the evidence it actually supplies.

Case studies remain valuable because they make hidden assumptions vulnerable. Their proper role is to change the questions and tests, not to end them. The most informative case is often the one that forces a hidden assumption into an explicit hypothesis that can be tested elsewhere.

Chapter 13 — Truth-Level and Protocol Endpoints

Endpoint science begins with several clocks that may disagree. A final breath, a clinical declaration, a fading neural signal, and the end of recording describe different events. The research problem is to determine which observed transitions constrain a qualifying stream’s ending, and how much timing information each transition supplies.

Direct cortical recordings make the problem concrete. In a nine-patient observational study during independently decided withdrawal of life-sustaining treatment, electrical silence usually preceded terminal spreading depolarization; the reported interval was 13–266 seconds (Dreier et al., 2018). Their ordering offers evidence about a biological sequence while leaving a further question: how does that sequence locate the end of a qualifying stream?

An endpoint rule must therefore be useful before the ledger result is visible. If its threshold or persistence window is selected because a particular alignment produces a neutral residual, the proposed boundary already contains part of the result it is supposed to test. Endpoint construction must account for differences in marker response times and observation conditions.

Clinical care changes those conditions. Medication can alter both state and signal; a necessary procedure can interrupt acquisition; discomfort can end monitoring. When continued measurement conflicts with comfort, treatment, refusal, or family welfare, observation stops. The resulting gap must remain a gap in endpoint evidence, with its cause recorded.

The hardest boundary is often between a biological transition and a loss of access to it. A silent channel may reflect a changed state, failed contact, or device removal. The last available measurement then tells the investigator when observation ceased, but may supply little information about when the

relevant stream ended. A usable endpoint model must represent both processes and their dependence.

Return provides a particularly demanding check. Later communication or independently supported continuity can overturn an earlier candidate crossing. The earlier record still matters: it shows which information the rule used and why the classification failed. Revised classification can improve reconstruction without creating knowledge that was unavailable at the original prediction time. Persistence, reentry, and revision rules must accommodate that difference.

Endpoint uncertainty also changes the object of analysis. A defensible interval may support a set of terminal alignments, whereas incompatible routing or observation models may leave the target unidentified. Comparing approved windows, marker families, and sampling schedules reveals whether a signature survives the available endpoint information or depends on one convenient timestamp.

The chapter develops this multiple-clock problem into an adjudication procedure. Its practical output is a traceable relation between observed events, permissible endpoint inferences, and the analyses those inferences support. Clinicians, families, reviewers, and later investigators should be able to distinguish a recorded transition from a retrospectively revised boundary without being asked to interpret the person’s death through the hypothesis (World Medical Association, 2024; Greer et al., 2023).

13.1 — Truth-Level Endpoint Domain

Clinical death rules constrain available observations. They do not define Chapter 5's endpoint. Brain and circulatory criteria name different events. Device shutdown and legal time require a declared bridge to that endpoint (Greer et al., 2020; Greer et al., 2023).

13.1.1 — Stream Endings and Biological Proxies

Irreversibility matters because a temporary interruption is not a truth-level endpoint. Sleep, anesthesia, coma, and some resuscitation contexts can include long intervals of missing report or weak signals. If later evidence supports continuity or return, the earlier candidate crossing must be demoted rather than treated as a confirmed endpoint.

Circulation, respiration, metabolism, brain perfusion, cellular integrity, and tissue viability can fail at different times. This cascade prevents somatic death from supplying a single phenomenal endpoint without an additional bridge (Greer et al., 2020; Greer et al., 2023).

Death by neurologic criteria retains its clinical prerequisites, exclusions, examinations, and jurisdictional governance. Its use in care, law, or transplantation does not identify the final unit of conscious-stream continuity.

Somatic and neurological events enter the endpoint dossier as clocks with declared relations to the phenomenal target. They can bound plausible intervals, provide competing-event information, trigger persistence checks, or identify observation shutdown (Greer et al., 2023; Dhanani et al., 2021).

13.1.2 — Legal and Administrative Time

Legal and administrative death times serve institutional purposes distinct from endpoint identification (Greer et al., 2023; World Medical Association, 2024).

Record the purpose and time resolution of each certificate, registry field, or institutional timestamp before evaluating its relation to the endpoint.

Legal concepts can still matter as context. They determine when certain data streams stop, when consent authority changes, when records are closed, and when families encounter institutional language. They may create observation patterns that a study must model.

13.1.3 — Shutdown and Retrospective Records

Severity can make monitoring less tolerable, reliable, or appropriate. Treating observation cessation as the endpoint can therefore turn burden, device fragility, or care priority into apparent terminal structure.

A valid endpoint system must therefore carry an observation clock distinct from the endpoint clock. The observation clock marks scheduled measures, failed measures, refused measures, removed devices, care interruptions, and unavailable data. It is a record of access to evidence, not a record of mind.

The endpoint file distinguishes evidence about experience before the endpoint from representations created afterward. A final message, photograph, recording, or diary may inform retrospective classification when its timing, provenance, consent, and measurement properties are established. Later memories, interpretations, and synthetic representations retain their own source and creation dates.

13.1.4 — Endpoint Uncertainty and Nonidentification

Distinguish an imprecisely estimated, identified endpoint from materially different endpoints that the bridge, clock, routing, or observation model cannot distinguish. In the latter case, report an identified set or branch-specific results rather than select the time most favorable to the ledger (Manski, 2003).

Confirmatory sensitivity analyses use the prospectively approved endpoint models, persistence windows, proxy assumptions, and observation rules. Label outcome-informed additions as exploratory; they cannot replace the primary test. Report whether sparse data, conflicting clocks, routing ambiguity, or structural nonidentification limits the endpoint. Public status language distinguishes observed, estimated, interval-identified, revised, censored, unresolved and unusable endpoints.

Clinical declarations, physiological events, legal times, last observations, device shutdown, and family reports can constrain the endpoint without being identical to it. Each proxy relates differently to the complete strict-pre-endpoint phenomenal domain and fails under different conditions.

Construction and adjudication must remain independent of the tested residual. Choosing an ending because the ledger is near zero, or excluding cases whose residual is inconvenient, manufactures closure rather than tests it.

A no-estimate endpoint is therefore a substantive scientific result. It identifies the missing bridge that prevents a terminal claim, blocks later mathematics from claiming more certainty than the domain permits, and directs the next design toward the actual source of nonidentification.

13.1.5 — Pre-Endpoint Scope and Later Records

A person can remain loved, remembered, recorded, quoted, imitated, modeled, and socially present after death. Those forms of presence matter to families and cultures and belong to the study of memory, identity, and relationships (Parfit, 1984).

Personal archives give families and researchers access to memories, relationships, language, and patterns of response. A diary, video, social media profile, voice model, transcript, digital twin, or statistical simulation can preserve different aspects of that record; its scientific use depends on the information it contains and how it was produced.

Social memory belongs to grief, history, relationships, and meaning. The formal ledger includes experience within the qualifying stream's strict-pre-endpoint domain; later archives inform retrospective questions about that earlier experience.

13.2 — Protocol Endpoint Objects and Adjudication Predicates

An endpoint system contains distinct acts: constructing a candidate event, estimating its timing, applying persistence rules, and confirming or revising its status. Keeping these acts separate prevents a favorable ledger result from determining the boundary it is supposed to test (Hernán & Robins, 2020). The protocol must specify the information available at each act, the roles of independent reviewers, and the consequences of shared confounding, later return, or unresolved residual dependence.

13.2.1 — Candidate Protocol Endpoint Object

A candidate crossing records that a protocol-fixed threshold event occurred in the endpoint system. Persistence and the remaining adjudication criteria still require evaluation.

A candidate crossing is the first event at which a prospectively specified endpoint rule is triggered. It may involve a unity-proxy threshold, a loss of responsiveness pattern, a neurophysiological marker, a quality-controlled absence of state evidence, or a composite event declared before the study (Hernán & Robins, 2020).

Keep the ledger concealed from endpoint adjudicators until confirmation, return, and revision decisions are locked.

One workable construction is a threshold time conditioned on quality and persistence prerequisites. The variables belong to endpoint construction, not to the ledger estimator. A crossing that fails the required quality checks is not a qualified candidate endpoint; record the quality failure without assuming that the underlying event was absent.

Denominator discipline reports how often candidate crossings occur, fail confirmation, precede return, or become

uninterpretable because evidence quality fails. Publishing only successful confirmations hides the false-crossing burden.

Candidate reversals and missed endings identify different error burdens, but their counts alone do not establish sensitivity or specificity. Sensitivity requires the proportion of independently adjudicated endings detected by the rule; specificity requires the proportion of independently adjudicated non-endings correctly left unflagged, with reference rules, time windows, and denominators declared. Those tradeoffs belong in the endpoint report before any terminal ledger claim is interpreted.

13.2.2 — Persistence and Confirmation Window

A persistence window distinguishes a sustained marker pattern from transient fluctuations caused by artifact, medication, arousal, care procedures, or sampling cadence (Cox, 1972; Andersen et al., 1993).

The window is not always a fixed number of minutes. It may be a duration, a repeated-state requirement, a sequence of low-signal observations with quality checks, an absence of return under clinically meaningful conditions, or a combination of channels. The choice must be justified by endpoint science, not by convenience or by the expected ledger shape.

Persistence also has ethical limits. A protocol may not extend monitoring because a researcher wants a cleaner persistence window if the person’s comfort, preferences, or care needs require stopping. When care interrupts the window, the correct status may be unresolved rather than confirmed.

13.2.3 — Adjudication and Endpoint Estimates

An estimated endpoint assigns an event-time distribution or interval under declared assumptions. Retain that uncertainty and its assumptions in downstream analysis.

Unresolved or unusable. Uncertainty or failed endpoint qualification; distinguish them using the evidence-state rules in Section 24.2. Unresolved or censored endpoints restrict inference. Failed gates make evidence Unusable; passed gates without identification or a valid bound yield No adequate estimate.

Estimate an endpoint distribution, interval, or classification under explicit state definitions, observation, censoring, quality, and revision models, with priors or likelihoods where used (Hernán & Robins, 2020; Fleming & Harrington, 2005).

Endpoint uncertainty may be wide or multimodal. Low-signal records can support several candidate intervals, while a sudden unobserved event may leave no defensible interval within the recorded data.

Propagate endpoint uncertainty and test whether the terminal signature survives plausible timing jitter. Report any dependence on a particular alignment.

13.2.4 — Confirmed Endpoint Status and Performance

A confirmed endpoint has satisfied every applicable confirmation requirement in the protocol. Later return or an identified error can still require reopening the classification.

Confirmation requires the declared persistence, review, return, quality, and clinical-context rules to be satisfied with a documented decision path (World Medical Association, 2024; Center for Open Science, 2025).

Confirmed endpoints are eligible for primary analysis only if the confirmation procedure was prospectively specified, constructionally separated, and inferentially protected. A confirmed endpoint obtained after outcome-dependent tuning or ledger-aware review may be useful for exploratory learning, but it is not confirmatory evidence.

Distinguish absence under high-quality observation from absence after device removal, sedation, emergency intervention, or poor contact. Carry these confirmation-quality classes into sensitivity analysis rather than pooling them as equivalent evidence.

13.2.5 — Revision, Versioning, and Audit

A revised endpoint reflects later evidence changing the versioned classification. Preserve the original primary analysis and label retrospective revisions.

Revise an erroneous endpoint classification while preserving what was known when the primary analysis ran (Särkkä, 2013).

A revision can arise from return, resuscitation, recovered communication, post hoc device-quality review, clinical context, later family or clinician evidence, or discovery that an endpoint channel was compromised. Each source has a different evidentiary weight and different risk of bias.

Revised classification corrects the endpoint record; retrospective smoothing re-estimates an earlier state using later observations. Label the reconstruction by its information set and preserve the initial analysis as its own version; Section 13.5.2 gives the forecasting and smoothing rules.

13.2.6 — No Endpoint Defined by the Tested Residual

When several candidate endpoints exist, the study may not choose the one that makes the residual most neutral, the curve most elegant, or the rival least competitive. The confirmatory endpoint object must be fixed before the ledger result is inspected (Hernán & Robins, 2020).

The object includes more than a threshold. Persistence duration, last-exit rules, allowable gaps, smoothing, censoring, reentry, competing events, and revision handling can all change

terminal alignment. The protocol should freeze the primary construction, publish a bounded sensitivity family, and report whether the signature survives rather than substituting the most favorable definition.

The primary endpoint file must be reproducible without access to the tested residual; apply the analyst separation and blinding controls in Sections 13.2.8–13.2.11.

The firewall should face direct audit. Synthetic ledger trajectories, perturbed endpoint labels, permission tests, endpoint jitter, and reproducibility checks can reveal whether information that should be unavailable changes the construction. A failed audit requires a redesigned workflow, a new blinded package, or a narrower estimand.

The endpoint and ledger estimators can be written separately as follows.

$$E_t \equiv f_E\left(Z^E_{\leq t}, \Gamma_E\right), \hat{L}_t = f_L\left(Z^L_{\leq t}, \Gamma_L\right), Z^E \not\equiv Z^L$$

13.2.7 — Constructional Independence

Constructional separation prohibits defining the endpoint from the algebra or affective-burden estimates used to construct the measured ledger (Hernán & Robins, 2020).

The separation does not require different organs or wholly unrelated sensors. Consciousness, physiology, and affect share biology. The rule is narrower and stricter: the endpoint channels must not be algebraically identical to the ledger estimator, and any overlap in raw acquisition, preprocessing, or latent variables must be declared.

A unity-proxy endpoint may use perturbational response, connectivity, command-following evidence, arousal context, or other consciousness-related markers. A measured ledger may use physiological valence estimators. When a raw channel contributes to both systems, the primary protocol should either

remove the overlap, split the feature families, or route the case to residual-dependence modeling.

Audit training labels and preprocessing at the feature level. A unity signal can become entangled with valence estimation when affective labels trained the same representation.

13.2.8 — Inferential Independence

Inferential separation prevents the people and procedures around the endpoint from leaking terminal results into endpoint definition. Even if endpoint channels are constructionally distinct, analysts can contaminate the endpoint by tuning thresholds or windows after viewing ledger outcomes, selecting preprocessing or smoothing rules that improve the terminal pattern, or excluding hard cases after seeing results (Särkkä, 2013; Fleming & Harrington, 2005).

Tune the endpoint rule in one partition and apply it to held-out cases, so the observations selecting the rule do not also supply its terminal confirmation.

13.2.9 — Shared-Confound Analysis

Disease severity, treatment, sensor quality, care procedures, and observation or family availability can affect both endpoint markers and ledger estimates (Little & Rubin, 2019; Hernán & Robins, 2020).

A shared-confound model begins by identifying variables that plausibly influence both systems. Some are measured directly, such as medication timing or device removal. Some are latent, such as global decline, delirium burden, or unobserved discomfort. The model should state which are measured, proxied, or unmeasured.

Include shared causes in the primary causal diagram and sensitivity plan. Report which plausible dependence models

remove the signature and classify the remaining inference under Section 24.2.

Confounding can also create false endpoint evidence. For example, a care procedure may reduce signal quality and physiological variability simultaneously. Without context, the study may read this as low unity plus variance compression. With context, it becomes a rival explanation.

13.2.10 — Independent Teams and Blinding

Endpoint adjudicators work without the terminal ledger whenever possible, ledger analysts cannot change endpoint labels, and review committees receive only the information required for their task (World Medical Association, 2024; Center for Open Science, 2025).

The level of independence can be scaled. A small feasibility study may use documented role separation and audit logs. A multisite confirmatory study should use separate endpoint and ledger teams, locked data packages, blinded adjudication, and prespecified dispute resolution. The stronger the claim, the stronger the separation must become.

Blinding does not mean ignorance of clinical context. Endpoint adjudicators may need medication, imaging, treatment, arousal, and device-quality information. What they do not need is the measured ledger trajectory whose terminal behavior the study is trying to evaluate.

Offset unavoidable role overlap with masked datasets, staged releases, decision logs, and independent review.

13.2.11 — Registration and Locked Adjudication

Use Section 6.7’s terminology for protocol-fixed, prospectively specified, and publicly preregistered. The endpoint record names endpoint channels, thresholds, persistence windows, exclusion

and abstention rules, confirmation procedures, revision handling, primary and secondary analyses, and public-reporting categories (Chambers, 2013; Center for Open Science, 2025).

A protocol that fixes the threshold but improvises return handling has not protected the endpoint; false crossings, recovery, device cessation, low quality, sudden death, withdrawal, and competing somatic events receive declared classification consequences before outcomes are known.

13.2.12 — Residual Dependence and Sensitivity

Use Section 13.2 to determine endpoint eligibility before model fitting. Retrospective qualification can support a declared reconstruction, including a primary retrospective analysis, but cannot create an earlier forecast. Endpoint quality may not be renegotiated after the ledger result is known.

Separate teams and code do not remove shared latent causes; retain the dependence model in Section 13.2.9 (Little & Rubin, 2019; Manski, 2003).

Shared random effects, latent severity, site and device effects, treatment processes, observation models, and sensitivity parameters can represent that dependence. The scientific question is how much shared structure would be required to explain the apparent terminal signature without PTVC, and which parameters are informed by data rather than assumption.

Quantify the dependence needed to remove the signature. When plausible dependence cannot be bounded, use Section 24.2 to determine which endpoint or ledger inferences remain available.

Report remaining shared channels and their sensitivity consequences, identifying separation, cross-fitting, joint modeling, or retrospective adjudication where used.

13.3 — Stream Routing, Origin, Maximality, and Junctions

An endpoint route can be modeled as a directed graph. Each node represents a candidate stream interval or junction. Each edge records a permitted continuation under the stated rule. No phenomenal contribution may be assigned twice.

13.3.1 — Stream Origin and Initial Ownership

The protocol's stream origin is the earliest boundary assigned to one routed phenomenal history. Observable anchors such as birth, first report, registration, or device connection constrain that boundary without defining conscious continuity (Lagercrantz & Changeux, 2009).

Because early experience is often left-censored, origin uncertainty should be represented rather than erased. A study may define a reference-class-specific eligibility origin, a first observed interval, and an identified set of plausible truth origins. Reporting all three prevents the administrative start of observation from masquerading as the beginning of the stream.

Neurodevelopment, behavior, physiological integration, and later continuity can constrain origin classes under the specified reference class and theory family (Lagercrantz & Changeux, 2009; Dehaene et al., 2017).

Freeze the origin rule, admissible alternatives, and left-censoring assumptions before terminal analysis. Moving the origin to balance positive and negative magnitudes is circular; report the inference lost when origin qualification fails.

13.3.2 — Maximality and Continuation

A maximal candidate stream cannot be extended under the declared unity and routing rules without joining evidence that

those rules do not license (Bayne & Chalmers, 2003; Casali et al., 2013).

The protocol should compare at least three continuation models where the case permits: uninterrupted continuation, pause with rebinding, and termination followed by a new stream. Model comparison must use the same observations and loss rules.

Vary unity thresholds, permitted gaps, observation models, and rebinding priors; assign claim status from the resulting endpoint sensitivity.

13.3.3 — Split Points and Descendant Streams

Split-brain research, dissociation, focal injury, and engineered partition motivate testing whether one organism still supports a single maximal stream under the declared unity rule (Gazzaniga, 2000; Pinto et al., 2017).

Under parent-termination routing, end the parent ledger at the adjudicated split and initialize each descendant separately. Descendants do not inherit a balancing obligation, and their later totals cannot be pooled.

Partial partition may require an overlap model rather than an immediate binary split. In such cases the protocol can represent a junction interval with competing ownership hypotheses and prohibit terminal adjudication until the overlap is resolved or explicitly carried as nonidentification.

False splitting can duplicate or erase contributions; false continuation can pool separately owned experience. Report the residual under each admissible split rule without selecting the rule by terminal fit.

13.3.4 — Fusion, Overlap, and Junction Rules

Under fusion routing, earlier histories remain separate and a new post-junction stream begins under its own origin rule; later unity does not merge the earlier ledgers (Bayne, 2010; Tononi et al., 2016).

Overlap is different from full fusion. A system may show shared access in one domain and segregated access in another, or alternate between integrated and partitioned organization. The routing graph should allow junction states and mixed evidence rather than forcing all intervals into one permanent topology.

Junction rules must specify temporal granularity. A split or fusion that appears at one sampling rate may dissolve at another. Multiscale analysis therefore reports whether the junction persists across defensible resolutions, and it treats resolution-dependent conclusions as model-limited rather than ontologically final.

Junction diagrams show temporal direction, separate pre-junction streams, the uncertain junction interval, the post-junction candidate, and live alternatives.

13.3.5 — Exhaustive Routing and No Double Counting

Exhaustive routing assigns every phenomenal contribution in the declared study domain to one stream, probabilistically among mutually exclusive alternatives, or to the unresolved set. Unassigned contributions must remain visible. A protocol cannot discard difficult intervals yet claim complete-stream adjudication (Little & Rubin, 2019; Hernán & Robins, 2020).

No double counting is equally strict. An interval represented under several candidate routes belongs to sensitivity or model-averaging analysis, not simultaneous ledgers whose values are later combined. The same experience cannot support multiple terminal equalities merely because routing is uncertain.

Probabilistic routing requires the estimand and its posterior or mixture summary to remain distinct. Posterior ownership probabilities yield an expected or mixture-level quantity, not an observed truth-level ledger. Analysis must separate route from valence uncertainty and never present a posterior mean as exact stream ownership.

Audit routing completeness through counts or measures of included, excluded, multiply claimed, and unresolved intervals.

Missingness and routing can interact. The least observable intervals may be the ones most likely to be dropped, reassigned, or merged. Sensitivity analyses should therefore couple observation models to routing models rather than assuming that missing intervals are exchangeable with observed ones (Little & Rubin, 2019; Hernán & Robins, 2020).

13.3.6 — Endpoint Routing Audit

Before an endpoint verdict, independently reconstruct the route, compare admissible alternatives under the same loss rules, and check for revisions made after inspecting the terminal residual (Greer et al., 2023).

Record the adjudicators' information set and superseded routes. Use the declared family of admissible graphs or model weights in the terminal estimate (Manski, 2003).

13.4 — Accessibility Classes and Terminal Windows

Terminal accessibility is a multistate observation problem. Section 13.4 distinguishes the clocks whose records determine endpoint eligibility (Andersen et al., 1993; Aalen et al., 2008; Little & Rubin, 2019).

13.4.1 — Clinical Accessibility Class

The displayed inequality is a diagnostic clock-separation convention, not a theorem that every recorded time must differ in every case.

$$T* \not\equiv T_C, T* \not\equiv T_{\text{det}}, T* \not\equiv T_{adj};$$
$$T_C \not\equiv T_{\text{det}}, T_C \not\equiv T_{adj}, T_{\text{det}} \not\equiv T_{adj}$$

Any two realized times may coincide in a case without becoming the same clock object.

The clinical clock records what care is doing. Medication, sedation, oxygen support, ventilation, hemodynamics, resuscitation attempts, comfort measures, withdrawal of devices, transfer, discharge, and clinical declarations all belong here. They shape what can be observed and sometimes shape the experience being studied (Greer et al., 2023; Giacino et al., 2018).

Changes in medication, ventilation, analgesia, monitoring, or location should be recorded as time-varying causes of both observation and state, not treated as protocol deviations to be suppressed.

13.4.2 — Unity-Proxy Accessibility

The unity-proxy clock records prospectively specified consciousness-related channels and thresholds, together with signal quality and clinical context (Casali et al., 2013; Bodien et al., 2024). A proxy is an observation used by the unity model; it

is not the unity gate itself. A low reading can represent a different candidate state, an unreliable channel or an unresolved classification. Keeping those possibilities distinct lets the clock report what changed in the evidence before an endpoint interpretation is assigned.

13.4.3 — Observation Window and Sampling Support

The observation clock records scheduled, failed, refused, removed, interrupted, and unavailable data. Cessation of observation does not establish the truth-level endpoint.

Use the observation clock defined in Section 13.1.3 to document the available sampling support (Andersen et al., 1993; Aalen et al., 2008).

The observation clock is also where site differences enter the endpoint model. A hospice room, intensive care unit, home setting, and emergency department may produce different observation patterns even for similar physiology. If those differences are ignored, the endpoint system may learn the institution rather than the stream. A credible multisite protocol therefore models observation infrastructure alongside state evidence.

13.4.4 — Ledger-Independent Terminal Window

The ledger clock records affective estimates, measured accumulation windows and terminal alignment choices. Its behavior stays separate from endpoint construction so that the tested outcome does not select its own endpoint (Little & Rubin, 2019).

For the primary terminal test, endpoint rules and adjudication must be protected from outcome-driven ledger tuning. A prospectively specified joint model may nevertheless estimate declared dependence between endpoint and ledger processes;

joint modeling is not automatically exploratory. The analysis must identify its estimand, information set, shared causes, and anti-circularity controls, and must not describe a retrospective estimate as an advance prediction.

13.4.5 — Terminal Support and Timing Types

A multistate endpoint model represents movement among declared protocol states such as open, low signal, candidate crossing, confirmed endpoint, return, resuscitation, competing event, censoring, withdrawal, device cessation, and unknown. Every state needs an operational meaning and an allowed analytic role (Fine & Gray, 1999; Fleming & Harrington, 2005).

History and dwell time matter. A crossing followed by return differs from persistent absence under high-quality observation, and sedation differs from low signal after technical failure has been excluded. Hidden semi-Markov structure can represent duration without turning the first low-access instant into finality.

Transition intensity and occupancy answer different questions: increased hazard of a terminal-support state does not imply simultaneous individual entry, and high occupancy does not prove absorption. Both are reported so a population pattern cannot masquerade as a truth-level endpoint (Andersen et al., 1993; Aalen et al., 2008).

Absorbing status is versioned. A frozen primary classification may later enter error memory after recovery, while withdrawal, device cessation, and competing events can end observation or the study question without resolving mind absence.

Endpoint laws may be discrete, absolutely continuous, singular continuous, or mixtures of these; endpoint observations may be interval-censored, and the law may be only partially identified. When the last qualifying observation precedes the

first verified nonqualifying state, preserving the bounded interval is stronger than manufacturing a timestamp.

The transition graph specifies reversals, absorbing conventions, misclassification, and censoring, keeping observation shutdown distinct from an endpoint.

13.4.6 — Clock Misalignment and Censoring

Compare clinical, unity-proxy, observation, legal, detection, confirmation, and adjudication clocks under the alignment tests below (Little & Rubin, 2019; Manski, 2003).

Jitter candidate times, compare clock alignments, and quantify gaps by site, diagnosis, medication, and device. Dependence on a preferred clock limits the result to that endpoint system.

Propagate systematic clock gaps into the endpoint estimate or identified set and apply Section 24.2 before drawing a terminal conclusion.

13.5 — Filtrations, No-Look-Ahead, and Endpoint Leakage

The following formulas distinguish filtering with information available through time t from smoothing with later observations. Section 13.5.2 states the corresponding limits on advance-prediction claims.

13.5.1 — Prospective Filtration and Information Set

$$\pi_t(i) = Pr\{S(t) = i | O_{\leq t}\}, \widetilde{\pi}_t(i) = Pr\{S(t) = i | O_{\leq n}\}$$

A prospective filter uses only available observations and clinical context. It cannot use later return, device review, or final adjudication to classify the current endpoint state (Särkkä, 2013; Fleming & Harrington, 2005).

The filter's inputs must be documented: which channels were available, missing, or low quality; which clinical context variables were included; and which endpoint thresholds were frozen.

13.5.2 — Retrospective Smoothing

A retrospective smoother reconstructs earlier states using later return evidence, device-quality audits, clinician notes, family reports, or other state information (Särkkä, 2013; Little & Rubin, 2019).

Smoothers can also introduce bias if the future data are selected or incomplete. A case with rich post-event review may be smoothed differently from a sudden unobserved death. The smoother must carry observation and selection limits.

Retrospective smoothing may improve an earlier-state estimate, but it cannot show what was known at the prediction time. A prespecified primary retrospective analysis may use later evidence; a primary forecast may not. Tables, figures, and

verdicts should identify both the analysis role and the information set. An outcome-selected refinement cannot replace a failed primary test while retaining confirmatory status.

13.5.3 — Leakage and Frozen Prediction Time

Look-ahead leakage occurs when later information influences an estimate, threshold, classification, or feature represented as available before that information arrived (Hernán & Robins, 2020).

The simplest leakage is retrospective smoothing shown as if prospective. More subtle forms include tuning endpoint thresholds using complete trajectories, choosing a persistence window after seeing return outcomes, selecting features that best align to terminal neutrality, or excluding cases because later evidence made them messy.

Run a deliberately leaky pipeline as a diagnostic control to quantify the artificial order created by future information. Resemblance to the claimed signature requires especially strong separation in the primary pipeline.

Leakage can enter through preprocessing as well as explicit endpoint labels. Imputation models, denoising procedures, feature selection, and representation learning may all use information collected after the frozen prediction time. A valid prospective analysis therefore versions the entire pipeline, not only the final regression, and fits every data-dependent transformation inside the training fold or prospective window.

Nested evaluation is especially important when endpoint definitions, unity thresholds, or feature families are tuned. The outer split estimates performance on unseen cases; the inner split selects among candidate pipelines. Without that separation, a model can appear to predict terminal structure while merely

remembering the particular endpoint construction used to optimize it.

13.5.4 — Last Observation and Informative Shutdown

Aligning data to observation cessation can produce apparent compression or terminal concentration when severity, device tolerance, or care workflow determines shutdown (Hernán & Robins, 2020; Little & Rubin, 2019).

The endpoint chapter therefore requires last-observation analysis. Do candidate signatures align better with the unity-proxy endpoint or data-cessation clock? Are patterns similar when observation continues after candidate crossing? Do device failures produce comparable apparent compression?

When endpoint-aligned and last-observation-aligned curves agree, assess their common measurement and endpoint assumptions rather than treating agreement as independent validation.

13.5.5 — Reporting Revision and Leakage Audits

Apply Chapter 28's correction process to revised endpoint classifications (Spiegelhalter, 2017; World Medical Association, 2024).

A leakage audit traces each datum by generation time, availability to the endpoint classifier, first analyst access, and derivation from ledger or outcome-adjacent variables. Event logs, separated teams, frozen outputs, and an independent integration step are stronger than retrospective assurances; where full separation is impossible, overlap and its consequences are documented. A timestamped provenance graph can reveal leakage paths that prose descriptions miss, especially when derived variables cross team or software boundaries.

Report each channel checked, unresolved overlap, negative audit findings, and analysis downgrades under the prespecified rule.

Endpoint revision shows that the system can correct itself, but public reporting must show how claim status changed. It should identify the original classification and information set, later evidence, revised classification, and affected analyses while avoiding spectacle or consolation language about the person's death (Spiegelhalter, 2017; World Medical Association, 2024).

Preprocessing is part of the audit. Artifact rejection, smoothing, imputation, censoring, and quality thresholds may leak endpoint status when tuned on known cases, so versioned pipelines, locked parameters, and held-out stress sets must test whether the terminal pattern survives. Stress sets should include intentionally leaky pipelines so the audit can demonstrate that it detects contamination rather than merely confirming the preferred workflow.

The public record includes negative audit findings, unresolved channels, staff overlaps, and analyses downgraded because separation was incomplete. The audit states which channels were checked and what each check found. Publishing the downgrade rule in advance prevents partial separation from being reinterpreted after a favorable terminal pattern appears.

13.6 — Return, Resuscitation, and False-Ending Controls

An apparent ending may not be final. Resuscitation and autoresuscitation test whether persistence continues across that boundary. Delayed recovery and reversible neurologic shutdown test whether an interval is reentry rather than ending. These cases can require revision of a previous endpoint classification (Dhanani et al., 2021; Greer et al., 2023; Parnia et al., 2023).

13.6.1 — False Ending and Premature Crossing

A false crossing is a candidate endpoint followed by behavioral, communicative, neurophysiological, clinical, or contextual evidence that the stream remained or returned (Dhanani et al., 2021; Greer et al., 2023).

False crossings can arise from artifact, sedation, delirium, transient hypoperfusion, device failure, motor impairment, low arousal, or a brittle endpoint threshold.

A false crossing requires demotion of the earlier endpoint status. It may become a low-signal interval, a reversible interruption, an artifact event, or an uncertain state.

13.6.2 — Resuscitation and Recovery

A cardiac arrest, respiratory collapse, or severe neurologic crisis may be followed by restored circulation, consciousness, communication, or other continuity evidence (Dhanani et al., 2021; Parnia et al., 2023).

Recovery evidence can vary. Immediate command following is strong but not always available. Later communication may support continuity but can be affected by memory gaps, medication, delirium, or trauma. Physiological return may constrain endpoint timing without directly establishing

subjective continuity. The record should grade evidence rather than collapse it.

13.6.3 — Persistence, Hysteresis, and Reentry

Hysteresis means the rule for leaving a state need not be identical to the rule for entering it. Endpoint adjudication needs hysteresis because near-boundary signals can oscillate. Without reentry rules, a noisy channel can produce multiple candidate endpoints in rapid sequence (Greer et al., 2023; Fleming & Harrington, 2005).

Reentry rules should define how a stream moves from candidate crossing back to open, from low signal to candidate, from confirmed to revised, and from unknown to classified. These rules should be protocol-fixed, state-specific, and sensitive to clinical context.

Hysteresis should be tuned for scientific stability rather than aesthetic smoothness. A rule that is too strict may miss genuine return; a rule that is too loose may oscillate with noise. The protocol should test hysteresis parameters on held-out examples and report how often endpoint status changes under plausible alternatives.

13.6.4 — Return Without Recall

Memory after return depends on encoding, consolidation, retrieval, language, and reporting capacity. An endpoint model evaluates these processes alongside other state evidence when interpreting a gap in recall (Squire, 2004; Parnia et al., 2023).

Return without recall can overturn a premature endpoint classification when the returned state is independently linked to the same stream under the declared routing rules. Amnesia does not by itself resolve that linkage or show that the intervening

interval was empty. Report the return evidence, the revised endpoint status, and any remaining uncertainty about the gap.

13.6.5 — Competing Events, Refusal, and Withdrawal

Competing events must be defined relative to the study's event of interest: they preclude that event, whereas an interruption of observation alone does not. Transfer, discharge, device removal, refusal, withdrawal of consent, loss of capacity without adequate surrogate permission, clinical emergency, family request, or care procedure can all interrupt endpoint adjudication (Fine & Gray, 1999; World Medical Association, 2024).

Classify withdrawal as censored, unresolved, or unavailable under the protocol; refusal supplies no evidence of closure.

Each listed event's role should be specified in the estimand structure of the study. Is it a competing risk, a censoring event, an intercurrent event, an endpoint-disqualifying event, or a secondary classification? The answer should be written before data are collected.

Code competing events and observation interruptions by cause where ethically permissible, and handle them under the prespecified estimand and imputation conditions in Section 13.7.5.

13.6.6 — Error Memory and Protocol Correction

Report useful original and revised analyses together, holding other prespecified choices fixed where possible. A large discrepancy triggers review of the new endpoint information; do not average the two estimates or combine correction with model retuning.

Correct dependent public conclusions through Chapter 28 without sensationalizing return (Center for Open Science, 2025).

13.7 — Sudden, Diffuse, and Low-Signal Endpoints

Sudden endings can obscure terminal support. Sedation or delirium may alter experience. Emergency action and equipment removal may also alter observation. The analysis must separate these effects before an endpoint verdict (Little & Rubin, 2019; Surges et al., 2024).

13.7.1 — Sudden and Unobserved Endpoints

A sudden unobserved ending leaves terminal observations absent or too sparse for adjudication, including after transfer, withdrawal, or an event that prevents recording. Knowing that death occurred need not identify the required endpoint state (Greer et al., 2023; Little & Rubin, 2019).

Report sudden unobserved endings in the denominator with the status: no terminal endpoint verdict available.

The frequency of sudden unobserved endings limits qualified terminal coverage and may select observed cohorts toward slower, more monitored trajectories. Include that selection in scope and transport assessments.

13.7.2 — Diffuse Support, Low Signal, and Artifact

Near terminal states, signal quality can change with perfusion, device contact, movement, sweating, skin integrity, electrical noise, medication, clinical procedures, or physiology outside the qualified context of use. Classify these observation conditions before interpreting the signal (FDA-NIH Biomarker Working Group, 2016; Little & Rubin, 2019).

Quality thresholds should be declared by channel. A unity proxy may be unavailable if contact quality fails. A physiological marker may be unusable during movement or care. A device may produce clean-looking numbers that are physiologically meaningless under terminal conditions.

Low signal can also be a real state. A fading consciousness marker under adequate quality may be endpoint-relevant. The difficult task is separating true low signal from poor observation. That separation requires quality indicators, negative controls, clinical context, and abstention rules.

Imputation, smoothing, or manual cleaning may be useful only when rules are frozen and the procedure is labeled. If artifact dominates a candidate endpoint, the case may need to become unresolved or unusable.

A low-signal branch should record which channel failed, when it failed, what competing causes were present, and whether any independent marker remained. Sensitivity analyses can then test whether conclusions depend on treating the interval as censored, missing, or potentially conscious. None of those branches is equivalent to assigning low consciousness by the size of one signal.

13.7.3 — Emergency Intervention and Rescue Priority

Emergency rescue, stabilization, and comfort take priority even when they interrupt measurement or destroy endpoint interpretability (World Medical Association, 2024; Greer et al., 2023).

The emergency event is still recorded on the clinical clock with its treatments, signal consequences, and missingness. It can explain candidate crossings or false endpoint classifications without becoming evidence of the truth-level endpoint.

Document emergency-care deviations. Review any incidental data under the prior consent and approved burden limits.

13.7.4 — Sedation, Delirium, and Rapid Decline

Sedation, delirium, and rapid decline are not interchangeable. Sedation can reduce arousal, memory, movement, report, and

physiological variability. Delirium can fragment attention, perception, distress, and communication. Rapid decline can alter perfusion, metabolism, and signal quality. Each can mimic endpoint approach without being a truth-level endpoint (Surges et al., 2024; Inouye et al., 2014).

Endpoint models should therefore carry state context. Medication class, dose timing, route, cumulative burden, delirium assessment where possible, pain or dyspnea treatment, infection, hypoxia, metabolic changes, and care goals may all matter.

Model the interacting state and observation processes described above jointly over time. Compare medication, illness, arousal, sensor failure, and genuine endpoint explanations; when they remain inseparable, widen the endpoint interval or withhold the affected endpoint inference.

13.7.5 — No Imputed Closure

Imputation requires an explicit observation model, assumptions appropriate to the estimand, and uncertainty propagation. A prespecified primary analysis may use defensible imputation, accompanied by sensitivity analyses. It may not impose terminal zero, a favored trajectory, or the hypothesis under test on the missing observations and then count that imposed result as evidence. If plausible missingness assumptions change the verdict, report that dependence or abstain (Little & Rubin, 2019).

No imputed closure applies in figures or text. A plot does not draw a graceful terminal line through an unobserved interval unless it is labeled as simulation, sensitivity, or imputation, and a table does not convert an unusable endpoint into a neutral one for completeness.

13.7.6 — Unusable Endpoints and Abstention

Apply the evidence-state distinctions in Section 24.2 and Section 27.4 to endpoint uncertainty. A defensible interval of endpoint times may support branch-specific or set-valued analysis, while inseparable adequate endpoint, routing, or missingness models remain Model-ambiguous (Manski, 2003).

Retain difficult endpoints and unfavorable residuals in the accounting. Report unusable-case counts, reasons, differences by site, state, treatment, or access, and each restriction’s effect on the reference class and denominator.

13.8 — Comfort-First Care and No-Estimate Outcomes

Care-related shutdown closes observation without supplying a terminal value. Apply the permitted uses in Section 13.8 under Chapter 3's care rules (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016; Sanders et al., 2024).

13.8.1 — Comfort-First Care

Measurement stops when care requires access, when devices burden the person, when the person refuses or withdraws, when family involvement becomes harmful, when clinicians judge the protocol disruptive, or when dignity would be compromised by continued observation (World Medical Association, 2024; Sanders et al., 2024).

Prefer passive or clinically integrated observation to intrusive prompting. Specify active measurement's burden caps and the staff roles authorized to pause it.

Record a care-required stop with its cause code and endpoint status. Audit stop-rule application across sites and populations, carrying shutdown-related selection into the reference class.

13.8.2 — Refusal and Family Communication

Explain that a candidate crossing does not establish that the person is gone, and that monitoring must yield to clinical judgment. Prevent endpoint terminology from pressuring families or clinicians (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Family narratives can be evidence in some contexts, but they are not clean endpoint labels. Last words, apparent lucidity, silence, agitation, peacefulness, or remembered signs may

matter clinically and relationally. In research, they require denominator, selection, timing, and interpretation limits.

Clinicians need a separate channel for concerns. If research terminology confuses care, increases family anxiety, or creates pressure around monitoring, the protocol should pause or stop.

13.8.3 — Cohort Composition and Public Meaning

A qualified endpoint can support the declared analysis within its reference class. It does not establish an exact truth-level endpoint or a personal meaning for the event.

An unresolved endpoint can inform feasibility, uncertainty, design, or valid bounds. It cannot alone support a terminal verdict.

An unusable endpoint can remain in the operational denominator and feasibility evaluation. It cannot supply imputed closure or an anecdotal substitute for admissible evidence.

Class structure governs the reference claim. A hospice cohort with many independently qualified endpoints and an emergency cohort dominated by sudden unobserved endings support different inferences even when their nominal sample sizes match. Denominator flow should show how many cases entered each class and which transitions occurred.

Display mixed-class composition and the unresolved, censored, revised, or unusable cases excluded from qualified-endpoint claims.

Test whether including, excluding, or reclassifying retrospective and lower-quality endpoints changes the conclusion.

13.8.4 — Formal and Empirical Change Authority

Changes to the truth-level endpoint or strict-pre-endpoint domain require formal-root concordance. Changes to estimators,

persistence windows, evidence classes, or adjudication rules follow scientific-method governance; clinical ethics continues to govern care (World Medical Association, 2024).

13.8.5 — Scientific, Legal, and Spiritual Boundaries

Hard cases often revise the empirical bridge rather than the formal target. A return may show that a threshold or confirmation rule was wrong. New boundary-state evidence may improve unity proxies. Neither result, by itself, settles the truth-level endpoint or the formal identity of a stream.

Formal change control should be rare, visible, and justified. Empirical updates should be expected, versioned, and tied to evidence. Confusing those two processes lets the project escape defeat by redefining the target, or lets a measurement failure falsely weaken the formal hypothesis.

Clinical, legal, scientific, and spiritual systems answer different questions. Clinicians direct care, law records institutional status, science classifies evidence under a protocol, and families or traditions may carry meaning. An endpoint estimate does not inherit the powers of those neighboring systems, and none of them supplies the scientific endpoint merely by naming an ending.

Change propagation should follow the layer that changed. A new formal definition triggers concordance across the model and series; a new endpoint estimator triggers reanalysis, versioning, and public correction within the scientific program. Keeping those routes separate prevents an operational difficulty from quietly redefining PTVC and prevents a formal clarification from being mistaken for new empirical support.

13.8.6 — Observation Shutdown and Public Stops

A revised endpoint can update the versioned record and a labeled secondary analysis. It cannot rewrite what the original prospective evidence established.

Observation shutdown is a protocol event, not a truth-level endpoint. Record why monitoring stopped and keep endpoint adjudication separate.

Care, refusal, family privacy, and clinical capacity outrank data completion. Preserve the last valid information set and the missingness consequences; return Unusable evidence, Inconclusive evidence, or No adequate estimate according to the canonical evidence-state rules rather than extrapolating.

If later evidence changes the admissible endpoint model, correct dependent analyses through Chapter 28. Audit whether stop rules were applied consistently across sites and populations so selection introduced by shutdown is reflected in the reference class.

Part V — Study Families and First Tests

The first useful study is the one whose possible outcomes change the next research decision. It may identify an informative channel, rule out a proposed dynamic signature, or show that the planned observations cannot separate two explanations. Part V turns the earlier formal and measurement work into study families organized around such decisions. The point of a first test is to make the next step better founded, whatever direction its result takes (Nosek et al., 2018; Munafò et al., 2017).

Chapter 14 develops the initial discriminating protocol. Before recruitment, the proposed observation schedule is placed inside simulated worlds where candidate and rival accounts can be compared. This exposes a practical question: would the available cadence, channels, and sample conditions retain the difference the study hopes to detect? A simulation can justify revising the design before anyone incurs its burden. The protocol then specifies how treatment changes and other intercurrent events enter the question being estimated (International Council for Harmonisation, 2019; Chan et al., 2025).

Chapter 15 examines sleep, dreams, and affective reprocessing. Here the observation window has a distinctive structure. Reports may be delayed, awakenings may alter the process, and physiological channels carry information different from a remembered account. The study family uses these features to connect established sleep research with testable proposals about affective dynamics. Its design problem is to obtain informative comparisons while respecting the participant’s control over private experience.

Chapter 16 moves to terminal windows. It asks what independently measured horizon and capacity variables would imply for a candidate route’s drift, dispersion, and covariance. Such a window can reveal local dynamic relationships even

when earlier history remains incomplete. The inferential task is to determine which observed differences distinguish the route and which arise from conditioning on the window, the endpoint process, or the availability of observations.

Chapter 17 extends the view across longer periods. Delayed entry, changing devices, treatment, and intermittent visibility become central design features. A long record is valuable because it can reveal persistence and change within a person; its interpretation depends on how those observations were assembled. The chapter therefore connects longitudinal contrasts to observation histories and asks how results may travel when the conditions of visibility differ.

The four families offer complementary forms of information. A sleep study may clarify a channel needed by a longer follow-up; a terminal-window analysis may reveal a signature worth testing elsewhere. Their combination is most useful when each result remains attached to the question it actually answers. By the end of the Part, the reader can assess a proposed study through its expected information: the alternatives it can distinguish, the uncertainty it can reduce, and the next investigation it can reasonably support.

Chapter 14 — The First Discriminating Protocol

Before studying terminal signatures in palliative care, investigators must know what a humane, feasible design could distinguish. The first protocol therefore asks whether prospectively observed preterminal dynamics justify a next research step. A neutral-looking ending is insufficient: the design must identify which measurements distinguish the candidate from explanations that remain plausible in the same clinical setting.

The proposed study is observational and staged. It does not induce pain, change medication for research, delay care, or manipulate expectation. SPIRIT 2025 formally concerns randomized-trial protocols; its relevant checklist principles are used here as a voluntary framework alongside good-clinical-practice principles and the Declaration of Helsinki (Chan et al., 2025; International Council for Harmonisation, 2026; World Medical Association, 2024).

Recruitment is part of the observation process. A person approached after hospice referral may differ from someone whose illness leaves no safe opportunity for invitation. Contact pathways, dependence on caregivers, communication access, and the timing of acute decisions can shape both participation and the resulting record. A coherent cohort begins with a defensible opportunity to decline.

Enrollment also starts an observation window, not a complete history. The primary question concerns dynamics that can be estimated within that window. Any extension to a complete-stream residual needs independent information about the earlier contribution; the protocol cannot obtain that information by choosing an initial value that makes the observed ending balance.

This distinction changes what the study must predict. Reserve-linked drift and dispersion, evaluated before a held-out interval,

can challenge a process model even when the earlier ledger is unknown. A terminal equivalence analysis remains a separate, conditional target. Its apparent precision cannot compensate for a failed historical or endpoint bridge.

Clinical context can generate the very structure under study. Sedation may change physiology and reporting together; worsening symptoms may increase both treatment and observation loss. Burden-driven device removal can select which parts of a trajectory remain visible. These pathways require time-resolved measurements and competing models, so an apparent terminal pattern can be traced to the process that produced it.

Simulation supplies the first decision about recruitment. Under the permitted cadence and burden budget, can the proposed observations separate the candidate from its strongest rivals? Failure may call for a different measurement or a narrower question rather than more participants. The protocol links each such finding to a concrete action before a vulnerable cohort is enrolled.

Reporting follows the observational design: STROBE supplies its primary discipline, while CONSORT 2025 offers flow, harms, and transparency checks for a later randomized extension (von Elm et al., 2007; Hopewell et al., 2025). A prepared report must explain what was observed, what failed, and what the next study would need. Individual terminal trajectories remain protected research records rather than public illustrations of personal meaning.

14.1 — The First Decision-Relevant Question

The study evaluates a late-window process under declared instruments, endpoint rules, and rivals, with results linked to subsequent research decisions (Nosek et al., 2018; Chambers & Tzavella, 2022).

14.1.1 — Primary Target and Its Limits

The primary confirmatory target is the preterminal dynamic conjunction: reserve-linked drift, innovation, variance shape, and held-out rival comparison in the declared reference class. Its support does not establish exact latent closure, whole-life balance, personal meaning, or legal or spiritual status.

The terminal equivalence band is a secondary target. After the origin, endpoint, and observation requirements pass, it tests compatibility of the measured estimator with the declared band. It does not establish instrument-free equality or a whole-life terminal ledger.

The endpoint model determines eligibility for the declared inference. Its qualification is a prerequisite for using the relevant terminal-window evidence.

The primary question is whether the protocol-fixed reserve-linked dynamic conjunction improves held-out predictive and generative adequacy over strong rivals under a common observation model (Nosek et al., 2018; Chambers & Tzavella, 2022).

The conjunction includes components that fail differently: a reserve-by-state interaction, a specified dispersion restriction, filtered prediction of a held-out interval, observation-process diagnostics, and negative-control behavior. A negative drift, smooth terminal curve, or interval overlapping zero cannot substitute for this joint test.

Carry the unknown pre-enrollment ledger explicitly: enrollment begins a local observation window. Section 14.8.2 supplies initial-state treatment and bounds for any complete-history extension (Hernán & Robins, 2020).

14.1.2 — Component Claims Under Test

Assign separate estimands and error rules to interval-estimator qualification, independently constructed reserve, endpoint adjudication, observation availability and held-out dynamics (Gneiting & Raftery, 2007; Wilson & Collins, 2019). The measurement component tests what an interval estimate represents in this setting; the reserve component tests the independently constructed capacity state; the endpoint and observation components establish which records can support the intended comparison. The dynamical component then compares candidate and rival predictions on held-out observations.

Mixed results are informative. Recruitment can be feasible while measurement recovery remains unresolved; an endpoint system can be usable while the predicted reserve coupling disappears. Reporting those outcomes separately identifies what the study has learned and which part of the next design should change.

14.1.3 — Reference Class

The first reference class should be narrow enough to be clinically coherent. A plausible starting point is adults receiving specialist palliative or hospice care at participating sites, with prognosis windows appropriate to observation, and with care teams able to support optional, low-burden participation (Nosek et al., 2018; Chambers & Tzavella, 2022).

Exclude settings where crisis, unstable care access, coercive dependence, or urgent clinical decisions prevent optional participation, comfort, refusal, or adequate staff attention.

Define the reference class by site, diagnosis, prognosis, care model, language, communication access, medication context, and consent pathway.

14.1.4 — Decision Relevance

A failed pre-enrollment recovery benchmark blocks recruitment. Distinguish remediable resolution limits from structural nonidentification, and test revised sample size, cadence, or measurement plans under the same burden limits. Resume only after the redesigned benchmark passes. Adverse burden suspends expansion; adding candidate flexibility after a rival victory requires a declared, tested successor version.

Feasibility and burden outcomes determine whether expansion is justified. They examine recruitment, observation, and clinical workflow for ethical and operational viability; by themselves, they establish no proposition about the truth or falsity of the Law.

Decision relevance means that every major result changes a predeclared scientific action. A favorable component result may justify independent replication; a rival victory may retire the candidate realization in that reference class; measurement failure may redirect resources to instrument development; and an unacceptable burden profile may close terminal recruitment. The protocol therefore registers its action map as carefully as its estimands (Chan et al., 2025; Hernán & Robins, 2020).

Decision relevance also requires bounded ambition. This protocol is a gate, not a container for all future public belief. It asks whether the next gate should open, close, or be rebuilt. A

design that claims to settle the theory would create more ethical and interpretive burden than its evidence could justify.

The decision branches should be written in the prospective specification rather than composed for the press release. If the candidate wins on a subset of estimands but loses on observation-process checks, the public wording must be already constrained. If model recovery is inadequate, the result is unusable even if the observed pattern looks favorable.

A decision-relevant protocol makes humility operational. It is not enough to say that negative results will be welcome. The study must specify what negative results do. It must let the Law become smaller, less public, more conditional, or scientifically retired in the tested form.

14.1.5 — What the Study Cannot Establish

Endpoint qualification asks whether the stopping system is sufficiently independent to support terminal-window analysis. It does not identify an ontological truth-level stream endpoint or determine clinical or legal death.

The first protocol evaluates measurement, endpoint, route, and comparator questions within its declared reference class. The registry and study reports identify those questions and the evidence used to answer them (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

The study reports late-window dynamics. Any complete-history extension uses an independently qualified initial-ledger bound or lifelong bridge specified before data inspection. The report should separate conclusions supported by the observed late window from those that depend on the additional historical bridge.

Participant data contribute to scientific models under consent and governance. Personal meaning remains with the participant.

Optional qualitative accounts can accompany the scientific record without being converted into latent-state labels the participant never authorized.

It cannot authorize clinical care changes. Comfort care, pain relief, sedation, rescue, communication, and family support remain prior to the study. The protocol’s conclusions cannot justify delaying care, increasing measurement, or altering treatment in pursuit of cleaner data. Care events that interrupt measurement should be recorded as part of the observed process, not treated as deviations to be corrected away.

Applications in civic design, government, law, religion, transition, artificial intelligence, and ordinary life are evaluated through the evidence and authority appropriate to each domain. A scientific result can constrain factual premises in those domains without transferring the study's decision rule or authority into them.

A study cannot rescue itself by changing scope after failure. The public result must name the failed object, and any new hypothesis must carry a new version. That rule lets the protocol teach without becoming self-protective. Versioning also protects negative evidence, because the failed claim remains identifiable even after a successor is proposed.

14.2 — Population, Enrollment, and Burden

Population and enrollment decisions determine both whom the protocol can represent and what participation requires. Eligibility, recruitment, consent, observation schedules, and burden limits therefore belong to the scientific design. The following specifications connect the proposed reference class to feasible participation and to the clinical and personal circumstances under which observations can continue.

14.2.1 — Inclusion and Exclusion

Inclusion criteria should be scientific and safety criteria. They should name the clinical state, expected observation window, capacity pathway, communication support, measurement feasibility, care-setting stability, and site readiness needed for the estimands. Convenience, expected fit, dramatic terminal proximity, or anticipated compliance must not drive inclusion (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Exclusion criteria should prevent harm and noninterpretability. A person should not be invited if participation would predictably burden comfort, disrupt care, intensify family pressure, overload staff, or produce data too confounded by crisis to answer the primary question.

Diagnosis and trajectory strata should be declared before enrollment because they generate different symptom paths, medication exposures, observation opportunities, and endpoint processes. Cancer, neurodegenerative disease, organ failure, dementia-related pathways, and mixed frailty should not be pooled merely to enlarge the sample. Partial pooling may borrow information where justified, but it must not turn composition change into an apparent terminal signature.

Specify nonverbal, augmentative, interpreted, caregiver-supported, fluctuating-attention, and sensory-access routes. If a site cannot support an eligible group, record the resulting restriction of the reference class.

Exclusion rules must be audited for unfair patterning. If disability, language, poverty, race, site resources, family structure, or home-care access repeatedly exclude people, the report should say so.

14.2.2 — Late-Window Enrollment

The enrollment time should be recorded as a scientific event, not merely an administrative date. The point at which a person becomes eligible may depend on diagnosis, prognosis, hospice referral, insurance, family capacity, clinician judgment, cultural expectations, and site practice. Those variables shape the observed terminal window.

Late enrollment selects for people who reach a study site with time, capacity, and support. Model that selection relative to sudden, unsupported, or emergency endings.

Pre-enrollment context is limited to variables required by the estimand and rival models. Diagnosis, medication history, recent symptom trajectory, major care transitions, and hospitalizations may be necessary. The collection plan specifies how each requested field contributes to the analysis.

14.2.3 — Capacity and Consent

The key information should be understandable under illness and sufficient for a considered decision. It should explain that this observational study is not individualized treatment, does not guarantee personal benefit, and will not determine care access or interpret a participant’s life. It should name the burden, data

collected, optional components, and right to stop without explanation or loss of ordinary care.

Surrogates require careful boundaries. A legally authorized representative may permit certain research procedures when the participant cannot decide, but surrogate permission should not turn a person’s dying process into data if prior wishes, current distress, or clinical judgment argue against continuation. The comfort-first rule remains superior to completeness.

Withdrawal should include data-retirement options declared in advance. Some derived aggregate results may not be retractable after analysis, while identifiable future use can be stopped. Consent materials must state these differences plainly so refusal and withdrawal remain real rather than ceremonial.

14.2.4 — No-Crisis Recruitment

No-crisis recruitment should be enforced by timing, personnel, and documentation. Invitations are deferred during acute fear, unresolved treatment decisions, unstable symptoms, family conflict, or any circumstance in which declining could reasonably feel unsafe. Screening logs should record deferral without converting vulnerability into noncompliance (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Use recruiters independent of direct care where possible. Clinical introductions must distinguish research from care and transfer the discussion to trained research personnel; care staff should not also be responsible for persuasion. Record the contact pathway and recruiter-participant authority relationship.

Remove recruitment appeals to legacy, heroism, proof, special access, or helping humanity through a participant's death.

The protocol should include a cooling-off interval where clinically appropriate, and it should let sites abstain when timing cannot be made fair.

Recruitment monitoring should include refusal rates, reasons for refusal where voluntarily offered, staff concerns, family concerns, and participant-reported pressure. If a site produces unusually high acceptance under dependence, monitoring should investigate pressure rather than celebrate efficiency.

14.2.5 — Bystander and Family Protections

Model family presence, visitor timing, caregiver distress, staff interaction, room changes, and relational events when they plausibly affect the recorded signals (Emanuel et al., 2000; Wendler & Grady, 2008).

Context reports, timing notes, or post-event family narratives require their contributors' own burden, consent, refusal, support, and withdrawal provisions. Participant consent does not authorize unlimited family testimony.

Bystander data should be minimized and purpose-limited. The protocol can record relational context categories where necessary, such as alone, family present, clinician present, procedure occurring, or conflict noted, without collecting names, private conversations, or interpretable family stories unless separately consented and scientifically necessary.

Neither family encouragement nor opposition should become recruitment pressure. Participant preference, capacity rules, and comfort govern participation.

14.2.6 — Participant and Staff Burden Budget

The burden budget should specify maximum prompts, sensor-wear intervals, device checks, family contacts, research visits, staff documentation, and quality-control requests, together with

cumulative limits across days. It should also name prospective pause, reduction, and withdrawal triggers for fatigue, distress, device intolerance, family concern, staff overload, clinical disruption, or repeated refusal (International Council for Harmonisation, 2026; National Commission, 1979).

Procedure counts alone cannot represent burden. A one-minute prompt during severe breathlessness may be heavier than a longer conversation on a stable day, and the same device may become intolerable after dyspnea, skin fragility, fatigue, delirium, or family crisis. Burden should be updated prospectively from participant report when available, bedside observation, device-tolerance records, symptom state, clinician judgment, and staff workflow indicators (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Staff burden shares the budget. Aggregate time, emotional labor, documentation, device maintenance, overnight coverage, workflow interruption, and independent consent support by role and site.

Data quality or additional power cannot justify exceeding comfort limits. Adding sensors or prompts, or prolonging observation, requires renewed review within those limits.

Burden-triggered pauses and withdrawals also shape the data-generating process. Observation may disappear precisely when distress or incapacity rises, so an ethically correct stop can still change the population represented by the estimate. Reports should show the relevant burden threshold, the observations lost after it was crossed, and the reference class to which the remaining result can still apply (Little & Rubin, 2019; Hernán & Robins, 2020).

Enrollment flow should preserve the denominator from first approach through screening, deferral, exclusion, consent, first

measurement, withdrawal, and analysis. Eligibility, temporary deferral, inability to approach safely, site limitation, communication barrier, device intolerance, and active refusal require distinct neutral codes. Collapsing them can make a restrictive design look broadly acceptable and can hide avoidable barriers to participation (Emanuel et al., 2000; International Council for Harmonisation, 2026).

14.3 — Measurements, Endpoint, and Data Flow

Measurement development begins in lower-risk settings. Before a palliative cohort is considered, the candidate momentary-valence estimator, interval valence-contribution rule, reserve proxies, endpoint channels, preprocessing, quality controls, and provenance system are declared and qualified for the target clinical state (FDA-NIH Biomarker Working Group, 2016; International Council for Harmonisation, 2026).

14.3.1 — Momentary-Valence Estimator Use

Qualify the candidate momentary-valence estimator for the enrolled population and the relevant illness, sedation, delirium, pain, dyspnea, fatigue, and low-movement states using Sections 9.1 and 10.5 (FDA-NIH Biomarker Working Group, 2016; International Council for Harmonisation, 2026).

Use the context-of-use contract in Section 9.1, with this cohort's clinical state range, temporal resolution, sign anchors, calibration route, and interval-invalidity rules.

Keep the physiological estimator distinct from self-report, behavior, clinical notes, medication records, family context, and relational observations used to validate, challenge, or explain it.

The candidate momentary-valence estimator needs explicit invalidity rules. Loss of sign-bearing information, artifact levels beyond qualification, sensor intolerance, model-version mismatch, failed calibration, or a clinical state outside the declared context of use should suspend the estimator for that interval. Invalid windows remain invalid; they are not replaced by the model's expected trajectory or silently imputed as neutral.

14.3.2 — Reserve Proxies

Each proxy needs units, cadence, uncertainty, confounds, and context of use. Medication, disability, disease type, age,

hydration, infection, pain, care quality, sleep, nutrition, family support, and sensor gain can all affect apparent reserve. The protocol should treat those effects as modeled pathways, not as noise to average away.

Propagate reserve-proxy uncertainty into interactions, slopes, and route comparisons. Classical error may attenuate associations, differential error may manufacture them, and shared inputs may create circular agreement with the affect estimator. If the proxy cannot separate the candidate route from disease severity or prognosis, report only qualified observed-variable claims and apply Section 24.2 to the latent target.

14.3.3 — Unity and Endpoint Channels

Predeclare responses to channel conflict: retain discordant channels separately, test sensitivity, narrow the estimand, or withhold the affected inference. Do not conceal reserve-quality, sedation-valence, or endpoint-ledger dependence within a composite score.

Apply Section 13.2's constructional, inferential, and shared-confound controls, with the endpoint-state and return rules in Sections 13.4.5 and 13.6.

The channel set may include clinical neurological observations, responsiveness, perturbational or electrophysiological markers where clinically appropriate, communication evidence, medication and sedation context, and charted care events. The protocol should declare which channels support candidate crossing, which support confirmation, and which only contextualize uncertainty.

14.3.4 — Treatment and Care Context

Record treatment and care as time-varying data: analgesia, palliative sedation, anxiolytics, antipsychotics, opioids, oxygen,

ventilation, procedures, delirium treatment, repositioning, family and chaplain visits, and changes in care goals. Specify their lags and roles as exposures, confounders, mediators, observation modifiers, or competing events (Chan et al., 2025; International Council for Harmonisation, 2026).

Model site differences in care delivery and observation, and test whether they account for apparent terminal structure.

Treatment data should be captured with enough temporal resolution to align with observation windows. A medication dose recorded only by calendar day may be too coarse for physiological windows. Conversely, overburdening staff with minute-level coding may exceed the burden budget. The protocol should choose a cadence that is scientifically useful and clinically tolerable.

The analysis plan should include medication and care-context sensitivity models. It should ask whether apparent variance compression, drift, or endpoint prediction survives after accounting for sedation, analgesia, delirium, procedures, and observation quality. Disappearance of a pattern after care-context adjustment shows sensitivity to those modeling assumptions; a rival-explained verdict requires the declared identification and model checks.

14.3.5 — Observation Marks

Every planned observation should receive a mark. Completed, failed, refused, paused, removed, clinically interrupted, family-deferred, device-error, low-quality, sedated, transferred, and post-endpoint observations are different events. Record observation outcome, quality, and clinical context in separate fields when they can coexist; sedation, for example, does not imply a missed observation. Treating them all as missing would erase the pathway by which evidence became available or

vanished (Aalen et al., 2008; Andersen et al., 1993; International Council for Harmonisation, 2026).

Observation marks should be chosen before enrollment and kept simple enough for staff use. A mark system that demands too much judgment at the bedside becomes another burden. The protocol should favor a small controlled vocabulary with optional free-text notes for clinically important exceptions.

Display scheduled opportunities and cause-coded observation marks across the terminal window, distinguishing directly observed segments from segments dependent on assumptions.

14.3.6 — Versioned Data Pipeline

Apply Section 9.6's versioned data pipeline (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020; Nosek et al., 2018).

Do not tune preprocessing, thresholds, or features while viewing confirmatory endpoint outcomes. If an instrument or transformation changes after early inspection, preserve the original output and test the migration under a declared plan.

One authoritative clock map should connect prompts, wearable samples, clinical events, medication administrations, room changes, device restarts, and endpoint adjudication. Original timestamps, synchronization tolerances, rounding rules, time-zone handling, and every alignment transformation should be retained. A terminal-window pattern that moves when clocks are corrected is a lineage finding, not a stable biological signature.

Hardware and software changes require bridge evidence. Firmware updates, sensor substitutions, battery-saving modes, altered sampling frequencies, vendor preprocessing, and device replacement can create discontinuities that resemble physiological change. Every observation should therefore carry

the active hardware, software, calibration, preprocessing, and quality-control version, and unsupported comparisons across versions should be withheld when old and new versions cannot be linked; separately qualified version-specific uses may continue (U.S. Food and Drug Administration, 2023; International Council for Harmonisation, 2026).

Test reconstruction from a published result through the frozen dataset and feature table to source observations, consent, access conditions, and every transformation. Before enrollment, use synthetic records with clock conflicts, duplicates, missing marks, device and consent changes, and endpoint reversals to expose breaks in that chain.

14.4 — Estimands and Rival Models

Before enrollment, each target should name population, treatment or exposure condition when relevant, time origin and window, outcome or latent object, intercurrent-event strategy, endpoint relation, summary measure, and uncertainty. Those fields align the scientific question with the analysis and stop a favorable pattern from being upgraded after inspection (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

14.4.1 — Relaxation and Innovation Estimands

The next tuple is an illustrative estimand-register template. In displayed order, its entries denote the reserve-linked coefficient in the drift model; the exponent of variance against reserve; the candidate-minus-baseline difference in mean held-out predictive loss; the parameter vector governing observation availability; and mean participant-reported measurement burden over the protocol window.

For this template, negative loss differences favor the candidate. Specify the remaining scales, units, equations, population, and windows in the study protocol before enrollment. These are proposed targets to define and estimate, not numerical findings supplied by the book.

$$\Xi = (\beta_r, \gamma_v, \Delta S, \eta_{obs}, \kappa_{burden})$$

Relaxation and innovation tests examine state-dependent drift and stochastic innovation in filtered windows against OU homeostasis, medication effects, and endpoint-conditioned bridges. Unstable or unrecoverable parameters block inference; Section 22.8.1 governs the rival verdict.

Variance-shape tests examine reserve-space and time-domain relations after accounting for proxy error. Frailty, observation

attenuation, and shared disease severity remain rivals. A slope explained by the noise floor, proxy drift, or observation process does not support the proposed distinctive mechanism.

Prospective endpoint prediction uses only information available before the endpoint in held-out evaluation. Compare it with generic time-to-death, age, prognosis, and site-practice baselines; absent qualified predictive gain, the proposed advantage is unsupported.

The observation model treats availability, quality, and missingness as data-generating processes. Observation shutdown, device intolerance, and care interruption can explain apparent compression and thereby defeat its proposed distinctive interpretation.

A discriminating conjunction combines components that face different rivals under a joint score. Challenge it with flexible hybrid rivals and data outside all candidate models. Redundancy, nonseparability, or failed simulation recovery prevents the conjunction from doing its intended inferential work.

Relaxation estimands describe whether the measured path shows state-dependent movement toward the candidate neutral region under conditions declared before analysis. Innovation estimands describe the size and structure of the unpredictable component after modeled drift, reserve, treatment, and observation processes are accounted for. Both are preterminal targets (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

Use the pre-endpoint filtration in Section 13.5 to isolate the state available before each held-out interval. Retrospective smoothing can support reconstruction or sensitivity analysis, but a held-out prospective prediction must use only information available at its prediction time.

A useful estimand might ask whether drift toward neutrality strengthens as independently measured reserve declines, after accounting for site, diagnosis, medication, sedation, delirium, care changes, and observation quality. The exact model can vary, but the target must be declared before seeing terminal behavior.

Test residual variability after modeled drift. Variance compression caused by lost sensor sensitivity or sedation is explained by the corresponding observation or medication rival.

14.4.2 — Variance-Shape Estimands

Variance-shape estimands ask how measured cumulative estimate uncertainty and dispersion change as reserve declines and as endpoint approaches. They must separate reserve-space relations from time-domain scaling. A slope near two in reserve space means something different from a time-domain exponent, especially when reserve declines nonlinearly (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Conditional variance in reserve space, variance indexed by a prospectively estimated horizon, and variance in windows aligned retrospectively to endpoints are distinct estimands with different risk sets and artifacts. The protocol should prospectively specify one primary object and treat the others as bridge or sensitivity analyses. Otherwise endpoint alignment can create apparent compression in a process with no native reserve-coupled structure.

Measurement noise must be included in the estimation model. An additive noise floor can bias apparent variance compression and flatten slopes. The protocol should include noise-floor characterization and a justified observation-error model or deconvolution procedure before interpreting variance shape as support.

Reserve proxy error also matters. If the reserve proxy is noisy or shares disease severity with the measured cumulative estimate, conditional variance estimates can be attenuated, inflated, or shifted. The analysis should report sensitivity to proxy-error assumptions and abstain when those assumptions dominate the target margin.

14.4.3 — Prospective Endpoint Prediction

Prospective endpoint prediction asks whether filtered preterminal features improve prediction of independently adjudicated endpoint states. The features must be available before the endpoint window they predict, and the endpoint rules must be declared without looking at ledger behavior (Lipsitch et al., 2010; Wilson & Collins, 2019).

Candidate features should be evaluated against a transparent clinical baseline containing diagnosis, age, site, performance status, recent decline, symptom burden, treatment, sedation, prognosis, and observation quality where available. Improvement is assessed in held-out data with calibration as well as discrimination. Failure to beat that baseline is informative, and it cannot be converted into support by emphasizing an in-sample coefficient.

The protocol-fixed report should provide held-out log or Brier score where appropriate, calibration-in-the-large, calibration slope, discrimination, horizon-specific error, and comparison with a transparent clinical baseline. A feature that improves rank ordering while degrading calibration has not earned a general endpoint-prediction claim (Gneiting & Raftery, 2007; Steyerberg, 2019).

The prediction target may be time to a candidate endpoint, probability of transition into a confirmed endpoint state, or probability of no-estimate classification within a window.

14.4.4 — Observation-Process Estimands

Observation-process estimands test availability, quality, and modality as primary rivals. Weakening may bring sensor failure, reduced movement, refused prompts, device removal, or clinical interruptions that generate the apparent signature (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

The probability and type of observation should be modeled from prior measured history, clinical state, treatment, site, family presence, device status, reserve proxy, and endpoint proximity. Complete-case analysis is defensible only under assumptions made explicit and stress-tested. When observation depends on the latent state under study, selection is part of the scientific model rather than a removable inconvenience.

Apply Section 22.8.1 when an observation model accounts for apparent compression or drift. Predictive superiority supports the tested component explanation; identifying its causal pathway requires additional evidence.

14.4.5 — Strong Rival Set

The rival set should represent the strongest credible explanations of the same observations. Ordinary homeostatic mean reversion can create drift toward a set point. Endpoint-conditioned bridges can create terminal concentration. Frailty and age can reduce variability. Medication and sedation can flatten physiological signals. Observation shutdown can create false compression. Hybrid models can combine these routes (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Include open-set generators and absolute-adequacy checks so the analysis can detect failure of the entire named model set.

14.4.6 — Discriminating Conjunction

Members, thresholds, tie rules, and stopping rules are fixed before the real-data pattern. Rivals receive comparable lags, nonlinearities, calibration effort, and information, while recovery simulations map regions in which models are indistinguishable or absolutely inadequate so the procedure can abstain (Wilson & Collins, 2019; Durbin & Koopman, 2012; Talts et al., 2018).

Register held-out prediction, terminal-band compatibility, feasibility, burden, missingness, and safety as separate estimands with the fields specified above (Hernán & Robins, 2020; Gneiting & Raftery, 2007).

Decompose uncertainty into sampling, calibration, proxy, endpoint, missingness, model, and transport components.

14.5 — Power, Recovery, and Negative Controls

Design sensitivity concerns the ability of the proposed observations and analysis to reach the intended decision. Vary measurement noise, sampling, missingness, endpoint uncertainty, and rival separation before recruitment. These checks identify which weaknesses merely reduce precision and which prevent the protocol from distinguishing the scientific possibilities it was built to compare.

14.5.1 — Synthetic Data Generator

The synthetic data generator creates candidate, rival, hybrid, and misspecified worlds under one shared observation system. Each world includes latent path structure, reserve trajectories, treatment events, endpoint states, observations from the candidate momentary-valence estimator, auxiliary context, missingness marks, device quality, burden stops, and site-level variation (Talts et al., 2018; Wilson & Collins, 2019).

The generator should vary clinically plausible features rather than only mathematical parameters. Diagnosis, prognosis, site practice, care model, symptom volatility, medication cadence, family presence, communication capacity, device tolerance, staff burden, and transfer risk should be part of the scenario library.

14.5.2 — Parameter Recovery

Vary reserve-proxy error across independent error, attenuation, shared drivers, and drift. These conditions test the robustness of the proposed reserve interaction.

Test recovery of the reserve interaction, drift, innovation scale, observation effect, endpoint jitter, and medication coefficient (Gelman et al., 2013; Vehtari et al., 2017).

Uncertainty calibration matters as much as point accuracy. Coverage, posterior calibration, interval width, rank calibration,

and sensitivity to priors should be reported before the protocol is allowed to claim confirmatory capacity.

Failure to separate medication effects from drift blocks the drift claim; inseparable reserve-proxy error and disease severity block reserve-interaction identification. Endpoint jitter that changes the estimate's sign or magnitude limits the endpoint system's support for that claim.

14.5.3 — Model Recovery

Simulate OU, bridge, frailty, medication, observation-shutdown, hybrid, and none-of-the-models worlds. Measure the resulting model confusion before recruitment.

For each simulated world, the analysis should score the candidate and every rival using the same held-out rules planned for the confirmatory study. The protocol should record how often each world is classified as the candidate, how often the candidate is missed, and how often an unmodeled world is forced into a named family.

The most important failure is false candidate victory. If observation shutdown, endpoint-conditioned bridges, sedation, or shared clinical decline are frequently recovered as the Law candidate, the design is not ready.

Model recovery also measures false rejection of the candidate. Coarse cadence, proxy noise, or endpoint jitter may cause candidate-generated data to be classified as frailty or medication effects. Those errors identify where the design needs improvement before a hypothesis comparison.

A real terminal pattern may contain ordinary homeostasis, care-quality changes, medication effects, social context, and reserve-linked dynamics at once. The model-recovery plan should not require one winner when the honest conclusion is a mixed or nonseparable result.

14.5.4 — Power Across Noise and Missingness

Vary measurement noise across low, moderate, high, and state-dependent floors. Evaluate both false compression and false-negative risk.

Vary missingness across MCAR, MAR, MNAR, observation shutdown, and comfort-driven removal. This tests validity under the missingness mechanisms the clinical design may encounter.

Power analysis must vary effect size, measurement noise, missingness, endpoint uncertainty, reserve-proxy error, sample size, cadence, site heterogeneity, and burden stops together (Talts et al., 2018; Wilson & Collins, 2019).

The protocol should report the probability of distinguishing the candidate from each rival, not merely the probability of finding a nonzero coefficient. A design that has high power to detect time-to-endpoint change but low power to distinguish the candidate from a Brownian bridge does not meet this protocol's discrimination requirement.

A separability threshold is fixed before enrollment. Across plausible noise and missingness, the benchmark may require a stated upper bound on confusion between the candidate and its strongest rival, together with protection against falsely defeating a rival that generated the data.

14.5.5 — Negative-Control Outcomes

A negative-control outcome must be justified as causally unaffected by the exposure or intervention being evaluated while remaining informative about the relevant bias. Battery state, contact quality, ambient temperature, and administrative timing are candidate diagnostics, not automatically valid negative controls: temperature may affect comfort, and device or staffing variables may affect both observation and experience. The protocol should identify the assumed causal paths and

distinguish negative-control outcomes from technical quality checks (Lipsitch et al., 2010).

Set quantitative control thresholds. Incidental correlations need not defeat the analysis, but matching the primary signature's direction, timing, and magnitude should trigger a predefined observation-process or no-estimate branch.

14.5.6 — Negative-Control Timing and Alignment

Vary endpoint uncertainty across exact adjudication, interval censoring, delayed detection, and jitter. Determine whether alignment to the endpoint generates the apparent result.

Shuffled, displaced, person-swapped, too-early endpoints and unrelated anchors test whether alignment alone reproduces the signature (Lipsitch et al., 2010; Nosek et al., 2018).

Timing controls should be protocol-fixed before enrollment. The plan specifies endpoint jitters, shuffled or cross-person labels, impossible windows, summary statistics, multiplicity handling, and the similarity threshold that triggers an observation-process, rival-favored, or no-estimate branch.

If displaced or shuffled endpoints reproduce the primary signature, the confirmatory branch fails: the pipeline did not distinguish the specified endpoint from false alignment.

14.5.7 — Circular Perturbation-Design Gate

The circular perturbation-design gate rejects any design in which the reserve construction, endpoint rule, preprocessing, normalization, or perturbation already forces the target pattern. Before enrollment, an independent reviewer should be able to derive what the pipeline would produce under neutral and rival generators (Talts et al., 2018; Wilson & Collins, 2019).

Apply Sections 6.7.4–6.7.5 to the entire pipeline, testing whether its transformations manufacture closure, variance

compression, or apparent neutrality when the candidate process is absent.

Failed circularity review blocks confirmatory interpretation. Retain the output only for labeled development, simulation, or diagnostic exploration, outside confirmatory status updates.

Recovery exercises should include deliberately broken worlds: mis-zeroed scales, invalid reserve proxies, informative dropout, endpoint misclassification, time leakage, device drift, and generators outside the fitted family. Control diagnostics should have a predeclared expected direction, tolerance, and consequence: causal-null controls test the relevant exclusion, while deliberate artifact challenges test whether the pipeline detects a known failure. A diagnostic that cannot detect the manufactured failure is too weak to protect the primary interpretation (Vehtari et al., 2017; Nosek et al., 2018).

14.6 — Missingness, Safety, and Adverse Events

Missing observations and adverse events can arise from the same conditions: illness, discomfort, withdrawal, device failure, clinical intervention, or the demands of the study itself. Define those events and their consequences for participants, families, staff, sites, and data systems before enrollment. A safety stop may protect the person while limiting the evidence, and a missingness model cannot make an unacceptable procedure permissible. The protocol therefore connects event classification, support, stopping, and the status of each affected inference.

14.6.1 — Marked Missingness and Joint Event Models

Extend Section 14.3.5's observation marks with fatigue, device intolerance, clinical decline, discharge, family privacy requests, staff unavailability, death, and unknown absence where these can be recorded without added pressure (Little & Rubin, 2019; Rubin, 1976).

Refusal expresses participant agency; fatigue signals a possible burden or state limit. Neither is a measurement failure to override. The protocol may record that a scheduled observation was declined, but it should not demand an explanation. If the participant volunteers a reason, it can be marked in a predefined category without being treated as noncompliance.

Device failure and device intolerance should be separated. A battery problem, contact-quality problem, lost signal, sensor displacement, or software failure implies one kind of observation problem. Skin irritation, discomfort, sleep disruption, embarrassment, or family objection implies another. Combining them hides both burden and instrument behavior.

Keep unknown missingness reasons unknown. A prespecified primary observation model may estimate missing quantities under explicit assumptions, uncertainty, and adverse sensitivity analyses, but cannot impose neutral experience or terminal closure. Apply Section 14.6.5 to the resulting inference.

Repeated measurements, observation availability, treatment changes, device tolerance, and the terminal event share causes and therefore belong in a joint data-generating account. A decline in clinical state may change both the estimated affect path and the probability that the path is observed. Joint models, multistate formulations, and marked observation processes answer different questions, so the protocol should state which dependence structure its primary estimand requires (Rizopoulos, 2012; Andersen et al., 1993).

A joint longitudinal-event model can represent this dependence without defining the endpoint through the cumulative-value estimate. The longitudinal component models observations from the candidate momentary-valence estimator, reserve, auxiliary channels, and quality marks over time. The event component models endpoint states, withdrawal, device cessation, transfer, death, and competing events. Shared random effects or latent clinical processes may explain dependence between the two without making the endpoint a function of the measured path.

Missingness analysis should include a protocol-fixed primary model, at least one materially more adverse missing-not-at-random scenario, and a no-estimate region in which the target is not recoverable. The comparison should show how the verdict changes as unobserved-state dependence strengthens.

$$Y_{\mathrm{it}} = m_i(t;\theta,b_i) + \varepsilon_{\mathrm{it}}, \lambda_i(t \mid b_i) = \lambda_0(t)exp\{\alpha^T Z_i(t) + \gamma_b b_i\}$$

Here, b_i is the shared random effect or latent clinical process, and γb is the event-association parameter.

14.6.2 — Comfort and Clinical Stop Rules

Apply Chapter 3's stop authority locally: participants, clinicians, bedside staff, and designated research personnel may halt observation, with comfort controlling. A family member may request interruption for privacy or when the participant cannot comfortably decide. Prespecified stops cover distress, device intolerance, clinical instability, privacy requests, and workflow conflict; staff need not negotiate continuation during clinical strain.

Every stop event should be recorded as a burden and feasibility outcome when recording can occur without new burden.

14.6.3 — Privacy and Security Incidents

Define privacy and security incident classes before enrollment, with containment, notification, deletion, and publication consequences (National Institute of Standards and Technology, 2020; World Medical Association, 2024).

Data access should be separated by role. Clinical staff should not receive research summaries that could alter care unless a predefined safety finding requires communication. Analysts should not receive identifiable family narratives unless the analysis cannot be performed without them. Model developers should not receive data for secondary training unless separately permitted.

Incident response should include immediate containment, access suspension where needed, audit-log review, participant or surrogate notification when appropriate, family notification when affected, IRB or oversight reporting, deletion or

quarantine where required, and a written decision about whether affected data can be analyzed or published.

Versions that expose identifiers, ignore consent flags, leak relational data, or permit unauthorized secondary training should be quarantined and retired, and outputs derived from them should be reexamined.

14.6.4 — Unanticipated Distress

Treat distress caused by prompts, devices, conversations, family reactions, or research presence as an adverse event, with immediate support routes (World Medical Association, 2024).

The first response is support and reduction of burden. The study should pause, remove devices, stop prompts, provide clinical or psychosocial support through ordinary care routes, and document only what is needed for safety and oversight. It should not ask for an explanatory interview during distress.

Review unanticipated distress for patterns, severity, and plausible causes. Repeated incidents of the same kind warrant investigation for a design defect; a single serious incident may already justify action. Independent monitors should be able to reduce cadence, remove a modality, narrow the population, pause enrollment, or stop the study.

14.6.5 — Abstention and Inconclusive Conditions

Operational monitoring remains separated from outcome interpretation. An independent review can examine burden, device tolerance, recruitment pressure, privacy, staffing, and observation loss without learning whether the candidate model is performing well, preventing scientific excitement from raising the stop threshold.

When a prespecified protocol-level termination threshold is crossed, close that protocol version with its data and adverse

result intact. Restart requires a reviewed successor, a new burden analysis, and an account of which inferences remain comparable across versions.

Apply the evidence-state taxonomy in Section 14.7 and Sections 24.2 and 27.4. For this protocol, failed estimator qualification, unstable sign, dominant device artifact, unresolvable channel conflict, invalid endpoint separation, excessive jitter, model-recovery failure, strong-rival nonseparability, or uncertainty crossing the decision boundary routes the primary inference to the corresponding prespecified branch. The local task is to identify the blocking condition, not redefine the status vocabulary.

Ethical conditions can also close authority: unacceptable burden, coercive recruitment risk, privacy incident, staff overload, distress-related withdrawal, or interference with care. The study may still report instrument, feasibility, safety, or narrower observed-variable findings, but it may not promote them to the primary claim.

Every burden signal, mitigation, pause, and restart is timestamped and linked to the protocol in force. A study able to return No adequate estimate is stronger than one compelled to produce a claim because it can surrender authority before it loses trust.

14.7 — Decision Tree and Public Reporting

Before outcome inspection, register each result branch, the evidence needed to reach it, and its required action (Chan et al., 2025; Hopewell et al., 2025; Nosek et al., 2018).

14.7.1 — Candidate Outperforms Rivals

When the candidate outperforms rivals after all prerequisites pass, report component support within the declared reference class. Independent replication and transport checks are the next steps.

Candidate superiority requires the qualification, endpoint-separation, reserve, recovery, negative-control, missingness, and burden conditions in Sections 14.2–14.6 to pass, followed by comparison under common information, scoring, and adequacy rules (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Report preliminary component support or route-favored status for the tested reserve-linked conjunction, naming its qualified estimator, endpoint system, rival comparison, first unresolved dependency, and required replication.

Identify the surviving component, window, specification, and comparators for independent replication and transport evaluation (Hopewell et al., 2025; Nosek et al., 2018).

14.7.2 — Rivals Match or Win

Adequate models that cannot be separated receive Model-ambiguous status; a rival meeting the frozen victory criterion receives Rival-favored status. Diagnose the distinction and revise or retire the tested bridge only where the evidence warrants it.

A rival wins when it meets the protocol-fixed victory criterion under the common scoring and adequacy rules. If no model meets that criterion and adequate candidate and rival models

remain inseparable, the result is Model-ambiguous. Endpoint changes, post hoc covariates, retrospective narrowing, or relabeling the rival as partial support would change the question after the answer and are therefore reported only as new exploratory analyses (Nosek et al., 2018; Chambers & Tzavella, 2022).

Different rivals carry different consequences. An OU or homeostatic rival may retire a dynamical interpretation while preserving measurement development. A bridge rival may show that endpoint alignment explains the pattern. A frailty or disease-severity rival may relocate the finding to clinical trajectory. A medication or sedation rival may make the result directly care-relevant.

14.7.3 — Measurement or Endpoint Failure

Measurement or endpoint qualification failure makes the targeted inference Unusable. Return to the affected instrument or endpoint-development task.

No adequate estimate is a valid refusal to infer. Publish the feasibility findings and reasons the estimate could not be obtained.

Measurement failure includes failure to qualify the candidate momentary-valence estimator, sign ambiguity, unstable calibration, excessive artifact, device drift, incompatible versions, or response-shift failure. These failures send the program back to measurement development rather than into a weaker interpretation of the same data. Posterior path uncertainty larger than the target region can instead leave a valid analysis Inconclusive; it is not by itself a measurement-validity failure.

When the endpoint bridge fails, the result names the failed property and exits the terminal-window branch. Loss of

independence sends the design back to endpoint construction; false crossings or unresolved returns require a revised state machine; inadequate terminal coverage narrows the estimand to an earlier window; and future-leaking adjudication voids the prospective analysis.

Name the unusable measurement or endpoint bridge and the failed property. Broader defeat follows only if the protocol prospectively made that bridge necessary to the broader claim.

14.7.4 — Mixed or Nonseparable Results

Inseparable adequate models receive Model-ambiguous status. For other mixed results, apply the prerequisite, adequacy, and identification rules in Section 14.7.4, and redesign before expansion.

When candidate, rival, and observation processes remain mixed or nonseparable, apply Section 24.2's gate, adequacy, and identification rules to assign the status in Section 14.7 (Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019).

Mixed results are judged against the entire protocol-fixed conjunction. Reserve coupling with failed negative controls, improved prediction with endpoint leakage, or a favorable mean with unrecovered dependence can produce component findings but cannot advance the route verdict.

Mixed results should be diagnosed by source: model overlap, inadequate cadence, weak negative controls, informative missingness, endpoint ambiguity, or open-set inadequacy. Each source implies a different redesign or stopping rule (National Academies of Sciences, Engineering, and Medicine, 2019; Talts et al., 2018).

14.7.5 — Adverse Ethical Result

An adverse ethical result can require the protocol route to narrow or stop. Protect participants, report the event, and review the failed route before further use.

Excessive burden, distress, privacy risk, staff overload, family pressure, care confusion, or discomfort-driven withdrawal can require stopping or narrowing the route (Nosek et al., 2018; Chambers & Tzavella, 2022).

The adverse branch should be protocol-fixed, with pause, modification, and termination thresholds; authorized trigger roles; support and incident procedures; data-disposition rules; and required public reporting. Ethical feasibility cannot be reconstructed after the study by subtracting the participants who carried the greatest burden.

Report an ethical stop throughout the study record, including exposure before stopping, support offered, data disposition, and whether the design can be modified or must retire.

An adverse ethical result should trigger route review. The program may move to less intrusive designs, safer populations, instrument-only development, retrospective feasibility studies, or no further terminal recruitment until the cause is repaired.

14.7.6 — Registered Public Report

Each branch in Section 14.7 specifies the permitted claim and required action; analysts, oversight bodies, and public communicators use the same decision.

Before enrollment, investigators prepare a protocol-fixed report shell for participant flow, burden, missingness, endpoint adjudication, deviations, negative controls, recovery, rival scores, adverse events, and claim-status consequences. For this prospective protocol, the plan qualifies as preregistered only when it is deposited before enrollment in a timestamped, read-

only external registry record with a persistent identifier (Chan et al., 2025; Hopewell et al., 2025).

Protect raw terminal and family material while making every result branch auditable with its estimand-register entry, qualified pipeline version, decision rule, and deviations.

Route replication failures, data corruption, classifier retirement, privacy incidents, and coercive recruitment through Chapter 28’s correction procedure (von Elm et al., 2007; National Academies of Sciences, Engineering, and Medicine, 2019).

14.8 — Protocol Limits and Replication Gate

A protocol's result is bounded by the population, instruments, endpoint rules, and comparison design that produced it. Replication asks which of those commitments survive a new implementation and which require further qualification. The transition from this protocol to the next study should therefore carry the complete dependency record, including unresolved limitations that could change interpretation.

14.8.1 — Single-Site Results Are Preliminary

Single-site evidence may depend on hospice practice, staffing, culture, diagnosis mix, family presence, device tolerance, or endpoint adjudication. Treat its transport as untested (Chambers & Tzavella, 2022; Munafò et al., 2017).

The next site should not be selected because it is likely to reproduce the result. It should test whether the result survives meaningful change in care context, population, device environment, and analytic personnel.

An unfavorable single-site result may also be preliminary. If the design fails because the site cannot support cadence, endpoint independence, or burden limits, that result may be operational rather than theoretical. The report should distinguish site infeasibility from component defeat.

14.8.2 — No Whole-Life Reconstruction

A late clinical window identifies an increment under its measurement model. Complete-history reconstruction uses an independently qualified initial-state bridge or sufficiently comprehensive prior observation specified before terminal outcomes are examined (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

The initial state therefore enters the analysis explicitly, with its available information, bounds, and sensitivity to alternative assumptions.

Complete-history inference can sometimes proceed by bounds rather than a reconstructed path. On the qualified additive branch, partition the complete pre-endpoint domain into the pre-enrollment contribution B, the contribution D covered by the observation model, and the remaining unobserved contribution M. If jointly valid bounds are $B \in [b_-, b_+]$, $D \in [d_-, d_+]$, and $M \in [m_-, m_+]$, then the terminal residual lies in $[b_- + d_- + m_-, b_+ + d_+ + m_+]$. This is an outer bound; known dependence may tighten it. Independence is not required for interval addition, but a stated confidence level requires simultaneous coverage of the component bounds.

For a hypothetical calculation in one independently calibrated signed unit, $B \in [-3, 1]$, $D \in [2, 3]$, and $M \in [-0.5, 0.5]$ imply a terminal interval $[-1.5, 4.5]$. Zero remains possible, but the bounds do not select closure over nonzero residuals. With $B \in [-5, -4]$, $D \in [1, 2]$, and the same bound on M, the terminal interval is $[-4.5, -1.5]$. A real result of the second kind would provide adverse evidence for the declared closure branch only if the measurement, ownership, endpoint, and joint-coverage requirements were satisfied. Sections 17.1.4 and 17.2.1 identify the historical and sampling assumptions such bounds require. The calculation directs the next measurement toward the uncertainty component that prevents a discriminating conclusion.

Report the observed window and any independently qualified bridge extending it (National Academies of Sciences, Engineering, and Medicine, 2019).

14.8.3 — Participant-Level Outputs

Return study-level findings and agreed descriptions of observed windows under the consent and result-return plan (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024). Optional qualitative accounts do not authorize conversion into latent-state labels.

Individual-level estimates may exist within the statistical model, but access should be restricted and purpose-limited. They are research quantities used for model fitting, diagnostics, and uncertainty assessment.

Interfaces and access controls should prevent model diagnostics from being repurposed as personal interpretations (Council for International Organizations of Medical Sciences, 2016; Nissenbaum, 2010).

14.8.4 — Independent Replication Required

Replication must test independence in site selection, recruitment, clinical workflow, endpoint adjudication, measurement, reserve construction, analysts, rivals, and reporting review (Chambers & Tzavella, 2022; Munafò et al., 2017).

The replication plan should include at least one site differing in care model or population, an analyst team independent of the original primary analysis, and an endpoint adjudication team applying the same protocol-fixed rules without access to ledger-derived features. The replicated estimand and the intentionally changed transport dimension must be identified separately.

Device and pipeline replication should test sensor models, preprocessing versions, and quality-control rules; cross-device comparisons still require calibration and versioned provenance.

Carry the full rival set into replication and revise weak comparators when criticism identifies a defect.

14.8.5 — Transport and Calibration Recheck

Requalify measurement, endpoint, reserve, and burden assumptions across changes in region, language, illness, disability, care practice, family context, or technology (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018).

Transport analysis should specify which assumptions are being moved. Population, clinical state, endpoint class, instrument version, reserve proxy, medication regime, cadence, and observation process should each receive a transport statement. If one cannot be defended, the claim narrows.

Where feasible, collect dual-device, parallel-version, shared-calibration, and independent-endpoint overlap data. Keep bridge estimation separate from the final transport test.

If a change in language, device, care setting, or cadence breaks the measurement bridge, investigators must recalibrate, narrow, or stop; failed transport does not justify extra participant exposure (Dahabreh et al., 2019; World Medical Association, 2024).

14.8.6 — Public Artifact Embargo

Embargo public artifacts until claim-status, privacy, recruitment, and hostile-crop review are complete. Apply Section 26.8.2's scope and uncertainty specification to each release (Hopewell et al., 2025; Fischhoff & Davis, 2014; Spiegelhalter, 2017).

Label quantities by the analysis and information-set categories in Section 13.5. Recruitment materials may not appeal to beauty, destiny, spiritual resonance, belonging, aid, care access, or fear of missing out. State burden, privacy, withdrawal, and possible absence of usable inference with equal visibility.

Replication materials preserve threshold sensitivity, alternate windows, competing missingness models, rival specifications,

and analyses that nearly changed the decision. Private terminal narratives, identifiable family material, and raw records unnecessary for verification remain protected.

The protocol version records registration, amendments, freeze points, and responsible committees. This record prevents silent drift between claim versions.

The endpoint manual specifies candidate, detected, adjudicated, censored, return, and no-estimate rules. It preserves the separation between endpoint decisions and the tested ledger.

The momentary and interval-estimator dictionary records channel versions, transformations, cadence, quality marks, and posterior-path rules. These specifications prevent score summation from replacing measurement theory.

The reserve-proxy specification records context of use, units, confounds, calibration, the error model, and the abstention rule. It prevents reserve from becoming an unrestricted label for frailty.

The rival and simulation package records generative worlds, parameter ranges, recovery thresholds, and the confusion matrix. Together they show whether the design could distinguish the submitted alternatives.

The burden and safety record includes participant, family, staff, privacy, and clinical stop events. Feasibility and ethics therefore remain part of the scientific status assessment.

Prepare branch-specific public summaries for support, a rival victory, no adequate estimate, and an adverse result. Fixing these descriptions in advance prevents rhetorical rescue after the outcome is known.

The report should distinguish a gate that can be repaired before replication from one that invalidates interpretation of the current cohort entirely.

Chapter 15 — Sleep, Dreams, and Affective Reprocessing

A morning dream report is produced after the experience it describes. Between the night and that account lie awakening, memory access, selection, language, and a decision about disclosure. Physiological recordings follow another path, registering events whose relation to experience must be established. Sleep research is valuable here because these observation pathways can be compared across repeated transitions within the same person.

Silence has several possible causes. Fear, bodily discomfort, grief, respiratory struggle, or trauma recurrence may be experienced without a report; a remembered episode may remain private. Conversely, a vivid report may preserve only a fragment of a longer night. The study must distinguish missing access from absent experience while allowing participants to decline the access that measurement seeks.

The mechanistic question is equally specific. Next-day relief can follow changes in memory, arousal, bodily symptoms, or waking interpretation. A proposed sleep-dependent process must explain a defined transition beyond these alternatives. Predicting improvement alone is weak discrimination when several mechanisms predict it; the useful contrast concerns when change occurs, which channel changes, and what earlier information predicts it.

Time separates those contrasts. A within-night state transition, an immediate awakening rating, morning mood, next-day behavior, and recurrent nightmares over weeks are different outcomes. Reduced autonomic arousal after a REM-rich period need not coincide with reported relief. Measuring the stages separately permits disagreement to identify a mechanism instead of disappearing into a single recovery score.

Nightmares make that distinction consequential. Repeated threat can accompany worsening sleep and daytime distress,

even when a researcher expected affective repair. Trauma replay, grief, medication, and arousal require their own explanatory comparisons and care responses. Any genuinely felt contribution belongs to the qualifying stream under its declared representation; explaining why a dream occurred is a different task from assigning its contribution.

Repeated nights permit the same participant’s pre-sleep state, intervening physiology, report availability, and later affect to be compared. Natural variation and existing records can support many such comparisons while preserving sleep. Where obtaining a sharper contrast would require prohibited disruption, the question must narrow. The observation method itself may change the next night or the willingness to report.

Circadian phase and homeostatic pressure act before dream content enters the analysis. Evening burden, sleep debt, temperature rhythm, cortisol pattern, light exposure, and awakening time can change the baseline against which nocturnal effects are judged. Their timing matters: an improvement between evening and breakfast cannot be assigned to dreams merely because a dream was recalled during the interval.

The chapter develops an experience-to-report model alongside tests of reprocessing. Its aim is to connect each proposed mechanism to a temporal contrast and an observable outcome. Clinical findings, physiological changes, memory effects, and narrative meaning can then inform one another while retaining the separate measurements needed to determine what actually changed.

15.1 — Sleep as Auxiliary Evidence

Sleep provides repeated, state-sensitive observations for testing measurement and affective-reprocessing hypotheses (Walker, 2009; Siclari et al., 2017). Its scientific value lies in comparing prior state, sleep architecture, available reports, and later responses under defined conditions. Dream reports and physiological channels contribute different evidence, and waking comparators help test ordinary explanations. These studies can qualify an auxiliary mechanism without supplying a complete phenomenal history or establishing the terminal claim.

15.1.1 — Auxiliary, Not Foundational

Next-day affect can update a local transition or residual-burden claim under the stated controls. It cannot establish whole-life closure or endpoint equality.

Repeated-night patterns can update within-person auxiliary mechanisms and rival comparisons. They cannot establish universal Law status or a cross-person rank.

A sleep study connects a specified conscious state, observation channel, and prediction about experience, reporting, or measured affect. Those connections define its empirical question (Walker, 2009; Palmer & Alfano, 2017).

An auxiliary hypothesis can be exact. It may predict that specified affective residues change during particular sleep states, that prespecified dream features add next-day predictive information after report bias is modeled, or that sleep disruption changes a transition expected under a reprocessing account. Each prediction receives its own test.

Use the observation chain in Section 15.2 to localize failures between experience and report.

15.1.2 — Conscious Experience During Sleep

Dream reports inform selected nocturnal content and the processes of recall and disclosure. They do not reveal whole-night experience, a hidden ledger, or moral meaning.

Sleep can contain imagery, emotion, bodily sensation, thought-like mentation or lucidity. A report of experience does not establish continuous awareness throughout the interval; Section 15.2 separates experience from later reports (Tempesta et al., 2018; Cartwright et al., 1998).

Apply Chapters 11–13's unity and routing criteria to sleep intervals; record uncertainty rather than assuming all sleep is ledger-active or ledger-silent.

Record sleep state with report status, timing, confidence, medication context, and missingness. Distinguish negative reports, absent awakenings, awakening distortion, and low-confidence or medication-confounded reports.

15.1.3 — Sleep Architecture and Affect

REM and NREM stages, slow-wave activity, spindles, arousals, respiratory and cardiac events, and circadian timing can affect memory, reportability, and next-day function (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

Greco and colleagues analyzed 18 healthy adults after replaying, during REM sleep, sounds associated with half of a set of negative images. At the 48-hour assessment, cued images elicited lower responses in selected emotion-related brain regions and less heart-rate deceleration than uncued images. Subjective arousal decreased for initially more upsetting images but increased for less upsetting images. This small study supports selective, baseline-dependent modulation of emotional responses (Greco et al., 2025).

Wassing and colleagues (2019) studied 29 participants with evening and morning brain measurements around overnight sleep recording. More uninterrupted REM sleep was associated with greater overnight amygdala adaptation; more restless REM was associated with less adaptation. Include sleep continuity alongside memory and waking context in prospective affect-change models.

NREM slow-wave activity, spindles, hippocampal-cortical replay, and homeostatic pressure may influence memory stabilization, abstraction, and next-day regulation.

A nap study makes the NREM comparison concrete. Arpaci et al. (2025) studied 42 young adults selected for differing trait-anxiety levels during a two-hour nap opportunity. In a joint regression, slow-oscillation density and spindle density had opposing associations with post-nap negative affect: greater slow-oscillation density accompanied lower scores, whereas greater spindle density accompanied higher scores. These are associations with post-nap ratings, not demonstrated pre-to-post improvement or effects of experimentally changing the oscillations.

Authorized secondary analysis or voluntary observation could compare oscillatory features with stage duration. A new change study would also measure pre-sleep affect and preserve the observation schedule. This proposal requires neither repeating the negative-picture task nor disrupting sleep.

15.1.4 — Reprocessing as Candidate Function

Affective reprocessing denotes changes in how affective material is organized, carried, recalled, or embodied during sleep (Walker, 2009; Palmer & Alfano, 2017).

A workable reprocessing hypothesis needs a direction, a time scale, a measurement object, and a rival set. It might predict

reduced next-day reactivity to a daytime stressor after intact sleep, altered dream affect after targeted memory reactivation, or a reserve interaction in which poor sleep quality weakens preterminal affective stabilization.

15.1.5 — Waking Comparators

Compare sleep-linked change with waking rest, social contact, distraction, rumination, medication, therapy, exercise, pain relief, food, light, and elapsed time. Specify the sleep-specific contrast or interaction before analysis (Tempesta et al., 2018; Cartwright et al., 1998).

Measure pre-sleep stressor exposure, pain, conflict, grief cues, medication, fatigue, social support, physical symptoms, and anticipatory worry. These baselines distinguish next-day recovery, persistence, and worsening.

The second comparator is an altered-sleep or poor-sleep night. Fragmented sleep, delayed sleep, REM suppression, respiratory disruption, nightmare recurrence, and awakening protocols can all change the next-day affective trajectory. These conditions should be modeled carefully rather than treated as mere exclusions, because they may reveal which processes are actually doing the work.

Compare nights with and without dream reports. A next-day change without recalled content may arise from sleep physiology, consolidation, or unreported experience; an association limited to intensive awakening protocols may arise from sampling.

The fourth comparator is waking reappraisal: people reinterpret events, receive comfort, change plans, and forget details while awake. Without measuring this, sleep studies may credit the night with what the day accomplished.

15.1.6 — Rival Sleep Science

Sleep-stage physiology informs state timing, arousal context, and rival mechanisms. It does not establish a latent cumulative-process level, terminal valence closure, or human worth.

The comparison set should include emotional-memory consolidation, extinction, reconsolidation, threat simulation, predictive updating, arousal regulation, circadian mood effects, medication effects, and sleep-disorder mechanisms. Zhang and colleagues provide an active-dreaming result within this larger field (Palmer & Alfano, 2017; Tempesta et al., 2018; Zhang et al., 2024).

Against ordinary restoration, memory abstraction, or autonomic regulation, the candidate must predict a distinct dependence on pre-sleep signed burden, state sequence, or independently measured capacity.

Permit worsening predictions: nightmares may rehearse fear, fragmented REM may amplify distress, pain or dyspnea may interrupt sleep, and disordered breathing may increase depletion.

15.2 — Dream Experience, Report, and Recall

Dream experience differs from its report. Recalling content adds a distinct task. High-density EEG tests those distinctions. Parasomnia makes the contrast salient when recall is incomplete (Siclari et al., 2017; Cataldi et al., 2024).

15.2.1 — Dream State Versus Dream Report

The dream state is the candidate experience; the later report is the observation. A person may experience without recalling, recall only fragments, reshape content after awakening, or decline to disclose material that feels private, shameful, sacred, or confusing. Report probability and recall confidence therefore belong in the observation model, and silence cannot be equated with dreamlessness (Siclari et al., 2017; Cataldi et al., 2024).

Model report availability using covariates recorded independently of whether a report is obtained: sleep stage, awakening timing and method, medication, pre-sleep burden, arousal, language, and disclosure context. Report-derived intensity, salience, confidence, willingness, or next-day mood require their own observation-time and missingness assumptions when included. A prospective model may use only information available at the prediction time; a retrospective joint model must represent the later measurements and their availability explicitly.

Name the variable report-coded dream affect, with separate stage, delay, confidence, and disclosure fields; do not label an awakening report as direct dream valence.

15.2.2 — Awakening and Sampling

Awakening changes physiology, memory access, subsequent reports, and sleep continuity. Randomize or model the permitted awakening schedule where feasible, record failed or refused awakenings, and distinguish the sampled state from the state

altered by measurement (Palmer & Alfano, 2017; Tempesta et al., 2018).

Use staged sampling when sufficient: physiology or actigraphy, brief morning affect, optional keywords, and full narrative only in a smaller separately consented study.

Sampling should be matched to the scientific question. If the question concerns within-night stage-linked mentation, awakenings may be justified. If the question concerns next-day residue, morning report and EMA may be enough. If the question concerns nightmares, clinical diary methods may be safer than laboratory interruption.

Set population-specific awakening budgets for laboratory volunteers, older adults at home, trauma survivors, hospitalized patients, and palliative participants. Noninvasive instrumentation does not establish equal tolerability.

Sampling-dose sensitivity can compare suitable existing records or naturally varying observation schedules. Any newly assigned awakening schedule must first satisfy the Chapter 3 limits; a desire to compare zero, one, and several awakenings does not itself authorize sleep disruption. Report whether the permitted observation method changes sleep continuity, recall, morning affect, or willingness to continue.

15.2.3 — Recall Delay and Reconstruction

Recall delay can reduce detail, select salient affect, and add narrative coherence or waking interpretation. Immediate awakening, breakfast, and later clinical or family reports therefore require different observation models (Palmer & Alfano, 2017; Tempesta et al., 2018).

Reconstruction may fill gaps, reorder scenes, translate images into cultural symbols, omit sensitive content, or alter meaning. These changes need not indicate dishonesty.

A serious observation model records delay as a variable. It should include awakening time, report time, prompt type, confidence, affect rating, and whether the participant distinguishes remembered content from inferred meaning. A report that says, I think I was afraid, should not be coded as the same kind of evidence as an immediate fear rating.

Model recall delay nonlinearly: early sensory and affective fragments may decay while later reports become more coherent and interpretive.

The observation model should therefore separate immediate affect rating from later meaning attribution. A person can wake with dread and later understand the dream as helpful, or wake comforted and later find the content troubling. A single post hoc valence label loses the very trajectory the chapter is trying to measure.

For this illustrative specification, k indexes scheduled report opportunities, and R equals one when a permitted report is obtained within the declared response window and zero otherwise. Each predictor carries the same opportunity index. S codes a prespecified sleep-stage contrast; A is the scheduled prompt delay after awakening. C codes a recorded disclosure context, and M codes medication exposure before the opportunity.

Freeze the predictor coding and response window before fitting. The alpha terms are regression coefficients; the inverse-logit maps their linear predictor to a probability. Record these covariates for opportunities with and without reports. Report-derived confidence requires a separate conditional or joint observation model because it is unavailable when no report is obtained.

$$\Pr(R_k=1|S_k,A_k,C_k,M_k)$$
$$=logit^{-1}(\alpha_0+\alpha_S S_k+\alpha_A A_k+\alpha_C C_k+\alpha_M M_k)$$

15.2.4 — Language and Cultural Interpretation

Adapt translated dream codebooks as instruments, testing construct continuity and permitting unclassified meaning rather than assuming universal symbolism (International Test Commission, 2018).

The first rule is to separate content coding from meaning coding. Content may identify persons, places, actions, bodily sensations, threats, social contact, or affective tone. Meaning coding asks what the dreamer takes the dream to signify. Those layers should not be collapsed, especially in cross-cultural or multilingual work.

Translation may preserve plot while losing culturally shaped emotion, agency, kinship, obligation, spiritual presence, or bodily meaning. Back-translation alone cannot establish construct continuity.

Version codebooks, record language, separate participant meaning from analyst categories, and use local review to test cross-cultural construct continuity.

15.2.5 — Lucidity and Metacognition

Lucid and metacognitive dream states show that awareness, access, and control vary within sleep. Participants able to signal or deliberately influence dream content provide a valuable selected sample; separate studies characterize ordinary dreaming, fragmented recall, parasomnia, and states with limited access (Palmer & Alfano, 2017; Siclari et al., 2017).

Time-locked communication during dreams has already been demonstrated. Across four research teams and 36 participants, Konkoly and colleagues obtained 29 correct answers in 158

communication attempts during signal-verified lucid dreams. Responses used agreed eye or facial signals; most attempts produced no clear answer. Three independent experts unanimously classified 26 of the 29 successful trials as REM sleep. This establishes a practical route for selected questions before awakening. Repeated signaling and broader recruitment are practical next steps toward more detailed observations during sleep (Konkoly et al., 2021).

Metacognition also complicates valence. A dream can be frightening while the dreamer knows it is a dream; a dream can be pleasant while the dreamer feels trapped; a person can choose to alter content and still wake distressed. Experience, appraisal, control, and later meaning must remain separate fields.

Findings from trained lucid dreamers should remain in their own reference class unless transport to nonlucid sleep is independently demonstrated.

Use selected lucid-dream cases to compare immediate signals with later reports, test timing assumptions, and assess how metacognitive access changes recall.

15.2.6 — Interpreting Dream Evidence

Dream records can capture affect, memory, fantasy, threat, bodily sensation, moral imagery, and grief. Each study specifies which features it measures and how they connect to its research question, while preserving the dreamer's control over disclosure (Palmer & Alfano, 2017; Tempesta et al., 2018).

This boundary protects method and dignity together. Dream reports are selected observations shaped by awakening, encoding, recall, language, and reconstruction. The same rule keeps visitation, reconciliation, terminal, and grief dreams from being appropriated as evidence for PTVC, an afterlife claim, or a compensatory meaning the dreamer did not assign.

A bounded study may estimate associations among pre-sleep state, sleep stage, protocol-fixed report features, next-day affect, and ordinary rivals. Coding remains limited to the declared constructs, such as affect, threat, agency, social content, bodily sensation, recurrence, and recall confidence. Analysts must not treat symbolic resemblance, debts in a dream narrative, or post hoc reward stories as evidence of balancing. A separately specified test of signed experiential contributions requires its own valid measurements and comparisons. Separating report content from signed experiential estimation allows rich qualitative data to inform rivals without granting symbolic interpretation quantitative ledger status.

Public artifacts require equal restraint. Word clouds, symbolic collages, dramatic quotations, and terminal anecdotes can make private narratives appear to be direct evidence of a hidden ledger. Sensitive content should be minimized or restricted, and every displayed variable should be identified as a report-derived measure rather than the experience itself. Where examples are necessary, synthetic or composite illustrations can sometimes communicate coding logic without exposing a participant's distinctive narrative.

A focused research question makes dream reports useful for comparing timing, content, recall, and affect across specified conditions. Its value increases when repeated awakenings or independent scorers can test whether the reported pattern is stable rather than a one-time reconstruction.

15.3 — Affective Reprocessing Hypotheses

Model pre-sleep state, sleep architecture, dream recall and affect, physiological change, and next-day memory or reactivity as separate stages. Reported associations motivate tests of active reprocessing rather than a general repair inference (Cartwright et al., 1998; Zhang et al., 2024).

15.3.1 — Pre-Sleep State and Post-Sleep Change

Affective burden must be measured in the evening with enough detail to distinguish distress, arousal, pain, grief, anger, bodily discomfort, rumination, social support, and fatigue (Cartwright et al., 1998; Zhang et al., 2024).

The post-sleep state must also be more than a mood rating. It should include affective valence, arousal, bodily symptoms, sleepiness, pain, motivation, social readiness, intrusive memory, and confidence in recall.

Repeated observations distinguish nightly within-person change from stable between-person differences. Parallel-group or randomized between-person designs answer different estimands under their own assignment assumptions.

For an illustrative nightly model, let the displayed affect change be the next-day score minus the pre-sleep score on one prespecified affect scale. Fix both assessment windows and the scale orientation in advance. The indices identify participant i, night n, and the following day n + 1. The outcome is an observed score difference; epsilon is the unexplained model residual.

REM and N3 denote prespecified features of REM sleep and stage N3 sleep, respectively; N3 is one NREM stage. NM denotes nightmare burden, B the pre-sleep burden measure, and the beta terms their regression coefficients. The participant-specific u term is an intercept. For within-person associations,

center nightly predictors within person and fit the participant intercepts; distinguish these associations from causal sleep effects.

The pre-sleep baseline should also include expectation. If a participant expects the study to find nocturnal repair, or fears that the study will interpret nightmares, that expectation can change reports and next-day affect. The study should avoid deceptive expectancy manipulation and should measure ordinary expectation as a context variable.

The post-sleep measure should be timed more than once where burden allows. Immediate awakening state, first-morning state, midday residue, and evening recovery can diverge. A pattern that appears at 7 a.m. and disappears by noon should not be given the same status as a stable next-day residual.

$$\Delta A_{i,n+1} = \beta_{REM} REM_{i,n} + \beta_{N3} N3_{i,n} + \beta_{NM} NM_{i,n} + \beta_B B_{i,n} + u_i + \varepsilon_{i,n}$$

15.3.2 — Dream Valence and Waking Residue

Dream valence and waking residue are separate targets. A negative dream may leave relief, a positive dream longing, a neutral dream unease, and a forgotten dream later change. Measuring both stages independently prevents the report from dictating the interpretation of the next day (Walker, 2009; Tempesta et al., 2018).

The first question is association. Does dream affect, when reported under controlled sampling, predict next-day affect after accounting for pre-sleep burden, stage, awakening timing, sleep quality, medication, and recall confidence? If the answer is no, dream content may still matter clinically, but it does not support the auxiliary reprocessing mechanism.

The second question is direction. A negative dream followed by reduced reactivity may fit extinction, exposure-like habituation, reconsolidation, or meaning-making. A negative

dream followed by increased distress may fit nightmare pathology, trauma replay, arousal carryover, or sleep fragmentation. The same content sign can support different mechanisms depending on the temporal pattern.

A candidate sleep model can predict an interaction among dream valence, sleep quality, a reserve proxy, and next-day residual affect. Specifying that pattern before analysis makes it comparable with alternative explanations.

Lin and colleagues (2026) followed 87 adults with seven-day dream diaries and then recorded EEG during an emotional-picture task. The EEG analysis included 84 participants. An exploratory analysis linked greater dream fear to stronger late-positive-potential responses to negative pictures among participants reporting good sleep quality. Dream affect showed no significant association with subjective response ratings or waking emotion regulation.

15.3.3 — Habituation and Extinction

Sleep research provides a concrete example of emotional responses changing across time. Van der Helm and colleagues (2011) compared responses to previously viewed emotional pictures after a night of sleep or a day awake. The sleep group showed reduced amygdala reactivity and subjective emotional intensity, together with changed amygdala–prefrontal functional connectivity. The overnight reductions were associated with features of REM-sleep physiology. This creates a tractable setting for studying how earlier experience, intervening sleep, and later emotional response form a connected trajectory.

Habituation is a reduced response to repeated stimulation that is not explained by sensory adaptation or fatigue (Rankin et al., 2009). Pavlovian extinction can reduce a learned response when a predictive cue is repeatedly presented without its expected

outcome (Bouton, 2002). A study can compare these learning mechanisms with the additional contribution of sleep by measuring both processes (Palmer & Alfano, 2017; Siclari et al., 2017).

Variation across findings helps identify the conditions under which sleep changes emotional responses. In 29 adults, Halonen and colleagues compared REM suppression with slow-wave-sleep suppression. Skin-conductance responses to a social-stress task increased more overnight when REM was preserved, while no between-condition differences were detected in subjective affect or memory. The result supplies a useful contrast to accounts of universal emotional dampening. Future work can test moderators such as task type, initial arousal, sleep stage, and the outcome channel, using existing data or independently justified, low-burden observation (Halonen et al., 2024).

A usable design would measure the daytime fear or burden, the sleep-stage context, any dream or mentation report, physiological arousal during sleep, and the next-day response to a nonharmful cue. It would test whether the association is strongest under hypothesized reprocessing conditions and weaker under control nights.

Persistent nightmare distress, avoidance, or dependence on waking therapy can contradict a prespecified repair account or favor a therapy-and-memory explanation.

The appropriate designs use naturally occurring concerns, mild ethically approved learning tasks, clinical data where care is primary, or retrospective coding with strong limitations.

The extinction rival also requires attention to context renewal. A fear response may diminish in one context and return in another, which means a morning improvement after sleep may not represent durable change. Follow-up prompts should test

whether the affective change persists across cues, settings, and time.

15.3.4 — Reappraisal and Memory Updating

Reappraisal changes an event's remembered meaning and subsequent emotional response. Dream-related changes in roles, perspective, or conflict can be compared with waking reappraisal (Cartwright et al., 1998; Tempesta et al., 2018).

Memory updating can occur through waking reflection, social conversation, therapy, sleep-dependent consolidation, targeted cueing, or spontaneous dream imagery. A study must separate the sleep contribution from waking meaning-making after the report. If a participant wakes and then interprets the dream with a partner, clinician, or interviewer, the post-report context becomes part of the outcome path.

Separate relief from changed reporting: integration, silencing, medication-related arousal change, and pressure can produce different pathways to a lighter remembered account.

Positive learning can influence which emotional memories remain accessible. In a study of 37 participants, Xia and colleagues first established word associations with negative images, then linked selected words to positive images. Replaying those words during subsequent NREM sleep weakened recall of the associated negative images and increased positive-memory intrusions relative to uncued items. The result supports selective memory updating and a concrete route for studying how positive experience changes later access to emotional material (Xia et al., 2024).

The following display is a proposed affect contrast using the change defined in Section 15.3.1. It compares cue and sham conditions at prespecified stage and context values; it does not estimate the memory outcome in the preceding study. A causal

interpretation requires a justified assignment and adjustment design, with stage and context defined before cueing.

$$\theta_{TMR} = E\left[\Delta A | cue, stage, context\right] - E\left[\Delta A | sham, stage, context\right]$$

15.3.5 — Reserve and Recovery

Test whether fatigue, illness, medication, grief, and clinical fragility modify observed sleep-recovery dynamics (Walker, 2009; Tempesta et al., 2018).

A reserve-aware sleep study tracks the conditions that shape both experience and observability: sleep duration, fragmentation, REM proportion, slow-wave activity, respiratory disturbance, pain, medication, mobility, caregiving interruption, and daytime exhaustion. These variables help explain the capacity to sleep, dream, recall, and recover.

Test whether greater sleep continuity and lower physiological disruption weaken the pre-sleep-burden/next-day-residual association, and whether fragmentation, obstruction, pain, or nightmares remove or reverse that transition.

Model illness-related changes in sleep quality, medication, report capacity, family presence, and next-day affect before interpreting their shared variation as a reserve mechanism.

Missed nights may concentrate when burden is greatest. Examine plausible missingness scenarios. Compare their reserve estimates with the complete-case result. Narrow a claimed pattern if it survives only by excluding high-burden nights.

15.3.6 — Failure of Reprocessing Predictions

Failure is part of the reprocessing hypothesis. If protocol-fixed dream features do not predict next-day residuals, the dream-content route weakens; if sleep architecture predicts change but content does not, the explanation moves toward physiology or memory consolidation; if neither adds information beyond

waking burden and ordinary rivals, the auxiliary pathway may retire for that reference class (Palmer & Alfano, 2017; Siclari et al., 2017).

Diagnose sleep-study nulls against recall quality, timing, coding noise, and power. Record sleep disruption, excessive burden, privacy restrictions, and unmodeled report loss as separate limitations on usable evidence.

If dream affect predicts the next-day residual after controls, the result may support a sleep-linked affective transition. Report selection and memory salience remain strong rivals, and the conclusion is limited to the auxiliary mechanism.

If nightmares predict worsening in the tested class, the prespecified repair prediction fails there. Nightmare pathology or a trauma loop may explain the pattern; no repair inference follows.

If no separable pattern appears, the auxiliary mechanism remains unsupported. Measurement limits or rival equivalence may require abstention; retiring the mechanism requires valid adverse evidence.

15.4 — Nightmares, Trauma, and Memory Rivals

Recurrent nightmares, persistent fear, hyperarousal, extinction failure, and fragmented memory create competing explanations for sleep-related affective change (Gieselmann et al., 2019; Zhang et al., 2019). A proposed reprocessing mechanism must distinguish its predictions from those alternatives and from medication or sleep-disorder effects. Clinical support takes priority when observation would increase distress. The comparison uses ethically available evidence; it cannot create exposure or prolong suffering to obtain a decisive test.

15.4.1 — Nightmares and Recurrent Threat

Nightmares may repeat threat, loss, helplessness, bodily danger, or social humiliation; chronic forms may involve hyperarousal and impaired fear extinction (Gieselmann et al., 2019; Zhang et al., 2019).

Recurrent threat may accompany trauma, anxiety, grief, medication, sleep disruption, pain, withdrawal, fever, neurodegenerative conditions, or ordinary stress. Comparing those contexts can help identify the processes maintaining the symptoms and the conditions that improve them.

Nightmares can provoke awakenings, avoidance of sleep, daytime fatigue, hyperarousal, and anticipatory fear. Tracking the following day captures how nocturnal experiences and waking symptoms interact.

Nightmare studies should distinguish frequency, affective intensity, thematic recurrence, autonomic arousal, awakening, recall confidence, avoidance, sleep fragmentation, and next-day impairment. These fields may move together, but they support different mechanisms and clinical consequences. Collapsing them into one negative-dream score hides the rival structure the study is meant to test.

15.4.2 — Trauma Replay and Fragmentation

Trauma replay can involve sensory fragments, bodily threat, helplessness, shame, and physiological arousal. Fragmented, state-bound memory or threat-system involvement can explain why the report lacks a coherent narrative (Palmer & Alfano, 2017; Tempesta et al., 2018).

The chapter distinguishes trauma replay from trauma processing by examining re-experiencing, intrusive imagery, nightmare recurrence, physiological activation, and indicators of safe memory integration. Intervention studies can help identify which changes accompany recovery.

Study designs in trauma contexts require the highest restraint. No participant should be exposed to trauma cues, asked to intensify nightmares, or pressured to disclose dream content for a theoretical test. Existing clinical care, optional diaries, and de-identified coding are the ethical limits unless expert review justifies a narrower design.

Trauma also changes report. Some participants cannot describe dreams without distress; others avoid sleep, suppress recall, or disclose only fragments. Missingness may be a protective response. Treating missing trauma-dream data as absence of dream burden would be a serious error.

Fragmentation also challenges dream-content coding. Trauma dreams may contain sensory flashes, bodily states, or scene fragments that resist ordinary narrative categories. A codebook designed for plot, characters, and valence may misclassify the most clinically important material. Trauma-informed coding should include no-code, fragment, and bodily-threat categories.

15.4.3 — Memory Consolidation Rival

Memory consolidation is one of the strongest rivals to Law-specific sleep reprocessing. Sleep can stabilize, transform,

prioritize, or reorganize memories. Emotional memories may become more accessible, less reactive, more generalized, or more persistent depending on context (Walker, 2009; Tempesta et al., 2018).

Compare predictions involving prior learning, sleep stage, cue exposure, memory accuracy, emotional reactivity, and recall. The candidate must add a testable restriction beyond consolidation.

Therapeutic learning offers another concrete application. After one EMDR treatment session, 33 patients with PTSD received either memory cues timed to slow oscillations during sleep or sham stimulation. The cueing group showed greater reduction in avoidance during a later script-driven imagery assessment. Baseline avoidance was higher in the cueing group, which complicates interpretation of the size of that improvement. Cue-evoked slow oscillations and spindles also correlated with symptom reduction within that group. These early results motivate further study of sleep as an adjunct to treatment, with avoidance and overall PTSD symptoms assessed separately (van der Heijden et al., 2024).

Include memory-task rivals, cueing histories, day residue, and next-day memory outcomes to test whether dream affect tracks memory strength rather than burden relief.

A useful analytic plan should include both memory accuracy and affective tone. A memory may become more accurate and less distressing, less accurate and less distressing, more generalized and more distressing, or unchanged. Those combinations distinguish mechanisms more effectively than a single next-day mood score.

15.4.4 — Medication and Sleep Disorder Effects

Medication, withdrawal, pain, respiratory disturbance, and other sleep disorders can jointly alter stage architecture, arousal, recall, and next-day affect. The protocol therefore records drug, dose, timing, indication, respiratory events, and sleep-disorder status as time-varying exposures and observation modifiers rather than unexplained noise (American Academy of Sleep Medicine, 2023; Zhang et al., 2019).

Model sleep-disordered breathing, restless legs, periodic limb movements, pain, nocturia, fever, neurodegeneration, and circadian disruption as possible causes of dream-report and next-day affect patterns.

The protocol needs predeclared abstention rules for major medication changes, severe sensor disruption, or untreated sleep pathology that makes the candidate contrast uninterpretable. A confounded night can remain clinically meaningful without entering the confirmatory estimate.

These effects can also be informative in safer populations. If REM suppression weakens dream-report associations but not next-day recovery, dream content may be less central. If apnea severity predicts negative next-day residue, respiratory disruption may explain the pattern better than reprocessing.

A medication taken at bedtime, a rescue dose during the night, and a morning dose after awakening may each alter stage architecture, recall, pain, or affect differently. A daily medication indicator is often too coarse for this chapter’s questions.

Sleep disorder treatment also changes the pathway. Continuous positive airway pressure use, mask discomfort, oral appliances, sleep-position advice, and treatment adherence can change sleep and next-day burden and must be modeled before any reprocessing claim is made.

15.4.5 — Clinical Support Before Study

Establish clinical support before studying high-burden nightmares, insomnia, trauma replay, anxiety, sleep-disordered breathing, or medication-related distress (Palmer & Alfano, 2017; Tempesta et al., 2018).

Consent must state that declining disclosure, refusing awakenings, stopping devices, omitting sensitive content, or leaving the study will not affect care.

A randomized study provides a hopeful treatment example. Thirty-six adults with nightmare disorder received imagery rehearsal therapy. Both groups then heard a sound during REM sleep at home for two weeks, but only one group had first associated that sound with its positive rehearsal scenario. Mean weekly nightmares at two weeks were 0.19 in the association group and 1.02 in the comparison group. This small trial supports further study of a specific treatment enhancement. The contrast identifies a concrete way that positive rehearsal and sleep cueing can work together (Schwartz et al., 2022).

Before enrollment, establish referral routes, distress response, participant-controlled privacy limits, and clinician review of awakening or interview procedures. Destabilizing reports trigger support, pause, or withdrawal.

Train dream-report researchers in trauma-informed responses, noninterpretation, confidentiality, and escalation.

15.4.6 — No Exposure for Crucial Testing

No study may induce trauma, nightmares, serious fear, humiliation, grief intensification, sleep deprivation, treatment interruption, or comparable distress to create a decisive test. The prohibition also covers repeated awakening when clinical need favors rest, emotionally provocative prompts, public challenges, and designs that make care or belonging feel contingent on

participation (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Use natural variation, clinically indicated treatment, within-person observation, permitted low-burden sampling, existing archives, or simulation. Leave causal contrasts unresolved when they require the prohibited exposures above.

15.5 — Sleep Stage, Physiology, and Measurement

Section 15.5 assigns sleep-state, timing, physiological, and disturbance channels their roles and limits (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

15.5.1 — EEG and Sleep Staging

EEG staging classifies wakefulness, NREM, REM, arousals, and transitions. It does not by itself establish consciousness or valence.

Score wake, NREM, REM, arousals, respiratory events, and transitions under declared polysomnographic rules, using the channels required for each event. Preserve the manual version, montage, scorer procedure, and automated classifier because they can change the state sequence (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

Stage scoring uncertainty should be preserved at the epoch level. Record montage, epoch length, scorer confidence, adjudication rules, algorithm and model version, training population, and relevant age, medication, and sleep-disorder context. Automated–human disagreement in key windows enters the uncertainty or sensitivity analysis, especially around transitions and artifacts; it is not silently resolved in favor of the label that best supports the candidate.

Distinguish reports after stable REM from reports after mixed transitions or arousal clusters; nearest-stage labeling can erase timing relevant to recall and affect.

15.5.2 — Autonomic and Respiratory Signals

Autonomic signals inform arousal, regulation, and bodily stress context. Movement, respiration, and medication can confound their interpretation.

Heart-rate variability, respiration, oxygen saturation, skin conductance, movement, and temperature characterize arousal, bodily stress, and disruption, including nights with absent or unreliable reports (American Academy of Sleep Medicine, 2023; Tempesta et al., 2018).

These signals are also vulnerable to artifact. Sensor contact, movement, sweating, room temperature, sleep position, device tolerance, battery state, and algorithmic smoothing can create apparent change.

Respiratory disturbance deserves special attention. Sleep-disordered breathing can fragment sleep, alter oxygenation, increase sympathetic activation, and worsen next-day mood or fatigue. If respiratory events account for the affective residual, the respiratory pathway becomes a rival explanation to test, not a nuisance to remove.

Autonomic signals also lag and smooth. A frightening dream, an arousal, a movement, and a respiratory event may appear close in time without a clear causal order. The analysis should specify lag windows and sensitivity ranges rather than forcing a single narrative.

Channel separation can reveal useful mechanisms. In a counterbalanced study of 17 participants, Viselli and colleagues found that experimentally fragmented REM sleep impaired cardiac habituation to emotional stimuli, with no detected change in subjective ratings, electrodermal responses, or recognition memory. Fragmentation was accompanied by shorter REM duration. The value for this program is a clear measurement lesson: physiology can identify a sleep-sensitive process even when reported affect does not change. Naturally occurring sleep variation and clinically indicated improvements in sleep can investigate that distinction (Viselli et al., 2026).

15.5.3 — Awakening Protocols

The study should register the maximum number and timing of awakenings, participant-controlled refusal, recovery intervals, and the estimand that remains when an awakening is skipped (Palmer & Alfano, 2017; Tempesta et al., 2018).

Different awakening strategies answer different questions. REM-targeted awakenings may enrich dream reports. Random awakenings help estimate report probability across stages. Spontaneous-awakening reports reduce intervention burden but increase selection. Morning reports preserve sleep but add recall delay.

Begin with natural nights, spontaneous awakenings, low-burden morning reports, and tolerated physiology. Any additional awakening procedure must satisfy Chapter 3's prohibition on research-induced suffering and sleep disruption for Law-specific evidence; healthy status and consent alone do not create an exception. Where the desired contrast requires prohibited disruption, use suitable existing data or narrow the question.

Awakening effects must be measured. After each awakening, the study should record sleepiness, distress, willingness to continue, report confidence, and whether the awakening itself changed the content or mood.

Training and analytic nights should be separated because equipment, prompts, laboratory novelty, and repeated awakenings change sleep. Adaptation nights can establish tolerability and stable procedures, while confirmatory nights follow the frozen schedule. Treating early nights as exchangeable with later nights risks interpreting acclimation as physiology.

15.5.4 — Experience Sampling After Awakening

Awakening reports give selected access to recent mentation. Sampling burden and reconstructive recall limit their interpretation.

Awakening reports may record experience presence, affect, bodily sensation, social content, threat, agency, recall confidence, and willingness to elaborate. Full narrative is optional; distinguish refusal from no recall (Siclari et al., 2017; Cataldi et al., 2024).

The order of questions matters. Asking for a full story first may reshape affect ratings; asking for affect first may prime content selection. The protocol should standardize the sequence and test whether the sequence changes reports. Method order is a measurement variable.

Reports include confidence and nonresponse options. A participant may indicate experience without recall, recall without confidence about affect, affect without narrative, or a wish not to disclose. Those categories preserve more information than a forced choice among positive, negative, and neutral.

Measure immediate post-report reactivity, record report-related burden, and apply the study's stop rule when required.

Ask first about experience, feeling, remembered content, and confidence. Offer an optional meaning field and analyze it separately as participant interpretation.

15.5.5 — Sleep-Specific Estimator Qualification

Morning EMA records next-day residue and function. Waking context and delayed memory can influence those observations.

The candidate momentary-valence estimator cannot be imported into sleep as if waking calibration automatically applies. Sleep changes muscle tone, autonomic balance,

respiration, movement, reportability, temperature, and consciousness state. A physiological composite trained during waking distress may fail or reverse under REM, NREM, sedation, illness, or sleep disorder (American Academy of Sleep Medicine, 2023; Tempesta et al., 2018).

Qualify the sleep estimator separately for physiological arousal, state modeling, or next-day affect prediction. Each use requires its own validation.

A sleep-specific qualification program for the candidate momentary-valence estimator should require stage-specific calibration, artifact rules, lag sensitivity, device-shift checks, missing-modality analysis, and comparison with reports and next-day outcomes. The estimator should abstain when the channels cannot support sign or when the state lies outside the validated domain.

Qualify sleep signals in lower-risk laboratory or naturalistic settings before terminal or trauma use; vulnerable cohorts must not be the estimator's first sleep-calibration setting.

Use one versioned clock map for stages, physiological events, awakenings, reports, and recall delays, with fixed tolerances. Propagate lag uncertainty; post hoc lag selection or realignment cannot establish a prospective marker.

15.5.6 — No Unity or Valence From One Marker

No single report, sleep stage, electroencephalographic feature, autonomic pattern, respiratory event, movement signal, or behavioral sign can decide consciousness, unity, or valence during sleep. Marker meaning depends on theory, context, timing, disorder, medication, and the observation pathway (American Academy of Sleep Medicine, 2023; Siclari et al., 2017).

Single-marker authority fails in both directions. A quiet or absent signal may coexist with experience the channel cannot reveal, while an active signal may reflect arousal, movement, breathing disturbance, memory, or artifact. False-negative and false-positive burdens therefore require the separate evaluation described in Section 10.5.5 rather than one blended accuracy score. Separate error profiles can reveal whether a marker is useful for screening, confirmation, or neither, without granting it universal classification authority.

The appropriate architecture uses multiple imperfect witnesses with explicit abstention. Reports, stage, physiology, behavior, clinical context, and scorer judgment should enter through the joint-measurement rules of Section 9.3.6, while vector, set-valued, or unresolved outputs remain available when the channels do not license one classification. Discordance is a result to analyze, not contamination to erase. The fusion rule should state whether disagreement widens uncertainty or changes the estimand, since those responses imply different scientific objects.

Qualification should survive channel removal, lag perturbation, scorer disagreement, stage uncertainty, nights without report, and clinically relevant subgroups. Following Section 24.2's decision logic, a protocol-fixed discordance matrix should state what each pattern permits, what remains unresolved, and when No adequate estimate is required. Discordance that clusters by state, medication, or device can reveal where a marker requires narrower qualification instead of a global rejection.

Every marker needs provenance: channel identity, device and firmware, preprocessing, artifact rules, scorer version, temporal window, report pathway, and missingness status. Tables, captions, dashboards, and public summaries must carry the same

ceiling so that one striking trace cannot travel as direct evidence of hidden valence, unified experience, or absence of experience. Provenance also makes cross-site harmonization possible by revealing whether apparent biological differences are actually differences in devices, scoring, or preprocessing.

15.6 — Study Families and Data Schema

The following study families connect the sleep estimands to data fields, coding rules, and privacy layers for replication (Shiffman et al., 2008; Nosek et al., 2018).

15.6.1 — Repeated-Night Within-Person Design

Record medication, pain, trauma, disorders, family events, and environment as contextual variables. They identify rival mechanisms and pathways through which burden affects the study.

Specify whether the outcome is next-day affect, sleepiness, function, or intrusive memory. That choice defines the transition actually being tested.

The core sleep design uses repeated nights within participants, with prespecified pre-sleep and next-day affect, sleep architecture, optional dream report, physiology, and context. Relate within-person variation in nightly predictors to the affect change defined in Section 15.3.1; distinguish it from between-person associations rather than treating repetition alone as control of stable differences (Shiffman et al., 2008; Stone et al., 2007).

One concrete extension is a preregistered test of history dependence. Compare nights that begin with similar momentary affect, sleep opportunity, medication, and available support but differ in an independently specified summary of earlier affective history. Test whether that history summary improves next-day predictions after current state, waking events, sleep architecture, memory salience, and report probability are modeled. The proposed auxiliary route must specify its direction, lag, capacity interaction, and failure criterion before analysis. Ordinary learning models must receive the same history. A surviving contrast would identify additional predictive structure. Natural

variation or independently justified care provides the setting, with the history summary defined over the observed window.

Repeated-night designs should include negative-control nights with a prospectively justified null prediction for the proposed auxiliary signal, alongside observation-process stress tests for no report, severe sensor failure, and major external disruption. Low pre-sleep burden qualifies as a negative control only when the proposed mechanism predicts no signal under that condition. A model that produces the same favorable pattern on every night lacks specificity even if its average fit is high.

Specify minimum usable nights and maximum burden per participant. Do not use imputation to conceal why a participant has insufficient interpretable nights.

Carryover should be modeled across nights. A nightmare can change the next evening’s affect, an awakening schedule can alter later sleep, and reporting a dream can provoke rumination or relief. Calendar separation does not create independence; washout assumptions and lag structure should be protocol-fixed and tested.

In the evening, collect EMA on burden, pain, affect, medication, and social context. Add a brief cognitive or memory-cue task only when ethically justified.

During the night, the low-burden default is actigraphy or comfortable wearable physiology. Optional higher-resolution work may use PSG, stage-targeted awakenings, or a TMR sub-study under its additional safeguards.

In the morning, record sleep quality, recall status, affective residue, and willingness to disclose. Structured dream coding is an optional higher-resolution layer requiring a consented narrative.

During the next day, use short EMA assessments of residual effects and functioning. Add clinical or behavioral follow-up when relevant to the declared question.

15.6.2 — Naturalistic and Laboratory Designs

Version the sleep stage, epoch, scorer, algorithm, and confidence judgment. Stage claims must remain connected to the procedure that produced them.

Version the physiological device, units, artifact rule, lag window, and missingness record. These details make the sensor's evidential contribution inspectable.

Naturalistic studies preserve ordinary beds, schedules, relationships, and privacy boundaries. Laboratory studies improve measurement precision but can alter sleep through setting, burden, first-night effects, and observation pressure (Nosek et al., 2018; Chambers & Tzavella, 2022).

Choose naturalistic sampling for day residue, family context, recurrence, and low-burden follow-up; laboratory methods can address precise physiology, permitted stage-targeted sampling, cueing, or controlled tasks.

For actigraphy, consumer EEG, heart-rate wearables, or phone prompts, record device version, placement, algorithm, battery, user behavior, and missingness.

A hybrid design can calibrate stages and reports in the laboratory, deploy lower-burden naturalistic measurements, and return a subset for targeted questions. Validate the bridge between phases.

15.6.3 — Dream Database Resources

A substantial shared resource is already available. The initial DREAM database assembled 20 datasets from 505 participants, linking sleep M/EEG with standardized reports across 2,643

awakenings. It makes cross-study tests of stage classification, report availability, and neural predictors of reported experience feasible across laboratories. A useful first analysis would freeze its report model on some datasets and evaluate transfer to independent datasets, preserving sleep-stage and sampling differences. Its linked neural recordings and awakening reports give measurement development a shared empirical resource (Wong et al., 2025).

Database interpretation begins with its sampling frame and observation process: who agreed to dream research, slept in a laboratory, and tolerated awakenings; which opportunities produced recall; and which content participants chose to disclose. These documented conditions define the population and observations available for analysis.

Harmonize prompts, languages, scoring systems, montages, awakening times, and report classifications before pooling dream databases.

Disclose prior access when specifying an existing-data analysis. Confirmatory status depends on that access, prospective specification, selection, and suitability for the estimand; existing-data use alone does not make a test exploratory.

15.6.4 — Multimodal Data Dictionary

The data schema also gives the research team a deletion rule. If a variable cannot be tied to a prospectively specified question, rival test, privacy requirement, or safety monitor, it should not be collected merely because it might become interesting later.

Extend Section 15.6 with sensor placement, artifact decisions, and the complete awakening/report procedure (Shiffman et al., 2008; Stone et al., 2007).

Distinguish no awakening, no recall, refusal, distress stop, device failure, low signal, medication confound, language barrier, and clinician intervention.

Link every derived sleep variable to its raw fields and generating rules.

The data dictionary should include a privacy field for each report element. A participant might allow affect rating, allow theme coding, restrict raw text, prohibit quotation, or request deletion of relational details. These permissions should travel with the data so later analysts cannot accidentally exceed consent.

15.6.5 — Protocol-Fixed Coding and Analysis

Record each dream-report prompt, delay, confidence, privacy level, and coding version. This separates the report procedure from the experience it is intended to inform.

Fix coding variables, coder training, reliability thresholds, uncertainty and noncodable categories, primary contrasts, and the defined dream features before outcome inspection under Section 6.7 (Nosek et al., 2018; Chambers & Tzavella, 2022).

The analysis plan should state which outcomes are confirmatory and which are exploratory. Within-night change, next-day residual, nightmare recurrence, dream-report probability, and stage-specific physiology cannot all become primary after the data are known.

Link rivals to frozen features and observation models: retained-memory outcomes for consolidation, threat codes and recurrence for threat simulation, stage-linked physiology for arousal regulation, and report probability for selection. Fix the coding manual, training set, reliability floor, and held-out adjudication sample before comparison.

Coding should also include abstention. Some reports will be too fragmentary, too private, culturally ambiguous, or emotionally unsafe to code. A valid codebook includes no-code and restricted-code categories rather than forcing content into convenient cells.

Use features available at the fixed analysis time. Later family accounts, clinical interpretations, or autobiographical revisions may support separate retrospective analyses but cannot redefine the original dream variable.

Report coder disagreement, confidence, and unresolved cases rather than forcing consensus.

Model recovery precedes the confirmatory mechanism comparison. Simulated nights should test whether the proposed design can distinguish sleep reprocessing from memory consolidation, arousal regulation, report selection, and sleep-disorder pathways. If it cannot, redesign that comparison before recruitment; separately justified measurement or feasibility work retains its own ethical and methodological requirements.

15.6.6 — Open and Restricted Data Layers

Separate public codebooks, labeled synthetic examples, aggregate features, missingness summaries, model specifications, and de-identified derived variables from restricted narratives and identifiable context (National Institute of Standards and Technology, 2020).

Schema migration tests should verify that old and new versions produce equivalent derived objects on shared records before pooled analysis is allowed.

Federated analysis, secure enclaves, or site-level sufficient statistics can answer some multisite questions without moving raw narratives into one repository. They reduce what travels but do not erase privacy risk or establish comparability. The plan

records what leaves each site, what remains local, how updates are audited, and how withdrawal propagates (National Institute of Standards and Technology, 2020; Nissenbaum, 2010). A privacy-preserving architecture should also record the inferential limits created by local-only variables so missing cross-site information is not mistaken for absence.

Consent to sleep-reprocessing research is not consent to train general language models, infer personality, build public artifacts, or support unrelated prediction. Purpose limits attach to all raw and derived data layers, including embeddings and summaries; materially new uses require separate permission and review.

15.7 — Privacy, Public Risk, and Verdict Limits

Dream reports are private. Their meaning depends on context. Public reuse can erase that context. Release therefore needs safeguards (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

15.7.1 — Dream Content as Sensitive Data

Dream content can expose sensitive material without establishing endorsement, intention, memory, or character. Distinctive narratives may reveal trauma, religion, sexuality, family conflict, illness, or grief (National Institute of Standards and Technology, 2020; Nissenbaum, 2010).

A leaked dream can harm relationships, identity, and dignity through exposure of distinctive narrative and relational content. Apply Section 14.6.3's incident-response requirements to dream data.

15.7.2 — Bystander and Relationship Privacy

Dreams often contain other people. Those people did not consent to become research content, and the dreamer may not want the relationship exposed (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Use role labels, initials, generalized categories, or redaction before coding. Collect relational identity only for a narrow purpose with restricted access.

The protocol should warn against interpretation, prohibit moral inference, and keep raw content away from employers, institutions, family members, public archives, or scoring systems.

Bystander content also complicates open data. A de-identified dream can still identify a relationship if the narrative is

distinctive. De-identification should therefore include contextual rarity, not just names and dates.

Do not return raw dream content to relatives through the study; sharing remains the participant's choice.

15.7.3 — Terminal and Grief Dreams

Dream reports during serious illness may carry profound personal and family meaning. Research can examine their recall, affect, physiology, content, and change over time using a stated question and a qualified coding scheme. Participants retain control over how their material is used (Palmer & Alfano, 2017).

Studies of terminal dreams document whether reports are immediate or retrospective, direct or family-mediated, and how culture, medication, and report availability enter the record. A defined sampling frame makes population comparisons possible.

Captions and summaries distinguish the study's observations from the dreamer's own interpretation.

Code grief-dream themes only with consent and privacy protection. Keep participant meaning separate from analyst interpretation rather than forcing either into a universal scale.

Grief can change recall and reporting. People may attend to dreams involving the deceased more than other dreams, share them more often, and remember them longer. Those selection processes enter the observation model.

15.7.4 — Hostile-Crop Review

Figures and tables should keep uncertainty, missingness, stage, report selection, and rival explanations visible. Titles, colors, captions, axes, and row labels carry claims; a label such as dream repair signatures can mislead even when the cells are careful.

Hostile-crop review asks how a sentence, table, figure, headline, or quotation reads after its surrounding safeguards disappear. Dream research especially needs this test because vivid material can turn an auxiliary finding into claims about hidden balance, useful trauma, mystical access, or personal moral status (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Raw dream quotations require necessity, consent, redaction, and protection from spiritual or moral inference. Public summaries should instead name the tested auxiliary relation, such as reported dream affect predicting a next-day outcome after stage and recall controls. Labeled synthetic or composite examples can explain the coding without exposing a distinctive narrative.

An independent reviewer should crop the artifact and ask what a skeptic, journalist, family member, or online critic could reasonably infer without the chapter. Review includes proof drift, privacy drift, score drift, and mystical drift across prose, graphics, and tables (Fischhoff & Davis, 2014; Spiegelhalter, 2017).

The artifact is revised until its auxiliary status and uncertainty survive detachment. Public reach is earned when a bounded result remains bounded outside the page that first explained it.

15.7.5 — Bounded Auxiliary Verdict

The chapter evaluates its protocol-fixed sleep and dream estimands: report, reprocessing, memory, physiology, and privacy-feasible measurement. The result states which hypotheses gained support, which weakened, and which remain unresolved (Nosek et al., 2018).

A favorable result names the reference class, measurement version, pre-sleep variables, report model, window, rivals, and

better-predicted pattern. Narrowing or adverse outcomes can localize effects, contradict simple repair accounts, expose recall or medication confounding, or stop a burdensome and privacy-risky design.

Refusal, no recall, partial disclosure, unclassified narrative, and non-disclosure are valid observation states. Sleep can inform state, memory, and reprocessing while remaining a private domain not owed to research.

Every summary carries the component verdict and its limits. The night remains meaningful and private while the program learns exactly where auxiliary mechanisms survive, narrow, or fail.

Chapter 16 — Terminal-Window Tests

A narrowing terminal curve can arise from different processes. Conditioning on an endpoint, losing high-variance observations, or changing sensor sensitivity can resemble contraction generated by a reserve-coupled route. Terminal-window testing therefore begins with predictions about the observed process, including how observations become available, rather than with the appearance of one fitted trajectory (Doob, 1957; Gneiting & Raftery, 2007).

The candidate gains precision by linking several relations: drift, conditional dispersion, dependence between times, and responses to changes in support or available actions. These restrictions must use the same state representation. Fitting a separate convenient reserve or horizon to each feature would remove the constraint that makes their conjunction testable.

Transport asks whether those linked predictions survive outside the data that selected them. A reserve relation estimated in one window should constrain what happens in a held-out window; a model for one channel should declare its relation to another. The comparison becomes informative where candidate and rival predictions differ by an amount the measurement system can resolve.

The harder case is exact overlap. A reserve-product construction can reproduce a Brownian bridge under a particular innovation clock. In that case, collecting more passive paths under the same conditions cannot separate the two descriptions. An independently qualified restriction on reserve or innovation, or a measurement or intervention with different predictions, must supply the distinction.

Information timing provides another contrast. Two participants can have the same realized time to an eventual endpoint while having different symptoms, treatment options, support, and contemporaneous expectations. A retrospective

clock places them at the same distance. A filtered state need not, because it represents the information available before the endpoint occurred. A prospective prediction must preserve that difference through estimation and evaluation.

Clinical conditions also change what the study can see. Sedation, delirium, analgesia, communication loss, device dropout, and changing care goals may affect both trajectories and their visibility. Those events require timed observation and treatment models. Comfort, consent, and care determine which measurements and support changes are permissible; an informative contrast does not authorize its collection.

Different summaries expose different failures. A mean near zero can coexist with wide tails or opposing subgroup trends. Marginal contraction can coexist with the wrong covariance or predictable residual drift. Conversely, an instrument floor can hide contraction that remains identifiable under independent calibration. The task is to match each proposed restriction to the assumptions and resolution needed to test it.

The chapter develops that matching process through horizon and reserve measurement, scaling, joint dependence, intervention response, and detectability. Its final comparison concerns one frozen set of relations under a shared observation model. A result can then identify which restriction survived, which failed, and which the available data could not distinguish.

16.1 — Candidate Route and State Identification

Specify the route state and measurement map for its signed residual, reserve, filtered horizon, route indicator, and contextual covariates. Distinguish formal objects, candidate mechanism states, and observed proxies (Durbin & Koopman, 2012).

16.1.1 — Route State, Filtration, and Observability

Distinguish reserve from correlated fatigue, frailty, pain, remaining time, and treatment intensity through a measurement map, justified exclusions, and state-misspecification sensitivity analysis.

Test whether a prospectively estimated remaining horizon interacts with unresolved state and independently measured reserve beyond ordinary finite-horizon motivation (Carstensen, 2006; Särkkä, 2013).

Require explicit transition and observation models, with recovery under realistic noise, irregular sampling, and dropout. A flexible filter cannot favor a route unless simulation and held-out tests distinguish its fitted state from serious alternatives (Durbin & Koopman, 2012; Särkkä, 2013).

16.1.2 — H_t From Available Information

The notation H_t names the filtered remaining-horizon candidate at time t. It is not the true remaining lifetime. It is the model's expectation, distribution, or decision-relevant summary of remaining opportunity given the information set available at that moment (Särkkä, 2013; Hernán & Robins, 2020).

Declare the information set, written as $\mathscr{I}_t$, before analysis. It may include current clinical status, functional change, self-reported future-time perspective, observable task context, recent choices, and approved auxiliary data. Future death dates and any endpoint adjudication, sensor-loss information, or family

narratives unavailable at time t belong only to separately specified retrospective analyses.

One compact definition is to let H_t equal the expected positive remaining time to a latent endpoint under a specified horizon model. The definition is useful only if the model also reports its uncertainty. A point horizon without an uncertainty distribution invites the false impression that the future is already known.

For behavioral tasks, H_t can be discretized into opportunity bins rather than calendar time. A participant may have a short task horizon because only two trials remain, a short social horizon because an interaction is ending, or a short clinical horizon because mobility or communication options are closing. The shared scientific object is remaining usable opportunity, not metaphysical finality.

In nonterminal tasks, vary known, probabilistic, or learned horizons only with a disclosed information structure. Hidden changes may test surprise or anxiety rather than the filtered state defined here.

16.1.3 — Uncertain and Revisable Horizon

A horizon model allows revision when prognosis, treatment, symptoms, plans, or support change (Seow et al., 2011; Vogel et al., 2013).

In a task environment, revision can be experimentally controlled. A participant may learn that a block has fewer remaining trials than expected, or that a previously unavailable action has reopened. The model then asks whether choices adjust in ways consistent with finite-horizon control rather than with a fixed personality trait.

Uncertainty affects the value of exploration. If the horizon might be long, learning can retain future value. If it is almost certainly short, exploration may compress. But if uncertainty

itself is high, information-seeking can increase. A model that always predicts urgency under uncertainty is too crude.

Record horizon revisions as well as current estimates. Stable, abruptly shortened, and widening histories can predict different behavior at the same current mean; repeated route closures may also alter response to a newly available option.

Revisions should also be bounded by credibility. If every new observation radically changes H_t, the model may be unstable; if no observation changes it, the filter may be insensitive. A useful filter learns at a rate that can be justified by the evidence stream. Sensitivity to that learning rate should appear in the design, because overreactive and underreactive horizons produce different behavioral predictions.

16.1.4 — Objective and Subjective Horizons

Objective and subjective horizons answer different questions. Objective horizon concerns the opportunity window available under external conditions. Subjective horizon concerns what the person or system believes, fears, hopes, or attends to. The two can diverge sharply (Durbin & Koopman, 2012).

Time perspective already has a testable behavioral role. In studies in the United States and Hong Kong, older adults' preference for familiar social partners diminished when they imagined an expanded future; under an imagined time constraint, both younger and older adults preferred familiar partners. Preferences also changed around Hong Kong's 1997 handover (Fung et al., 1999). These findings connect anticipated opportunity with social goals and provide an experimental way to separate perceived time from chronological age.

The distinction also improves the comparator. In an integrated analysis of 2,504 adults, greater age was associated with a more favorable emotional profile, while a more limited future-time

perspective was associated with a less favorable profile even after accounting for health-related factors (Grühn et al., 2016). Perceived horizon, emotional benefit, and age therefore require separate measurements. A declared PTVC route can build on these studies by predicting a specific interaction between independently measured signed history and remaining opportunity, beyond social preference, current mood, health, and ordinary goal regulation.

Clinical prognosis and perceived opportunity can diverge through depression, trauma, displacement, loss, relationships, projects, or spiritual practice. Measure both without treating either lived perspective as a personal error.

Choice dependence on subjective horizon points toward belief, meaning, or attention; dependence on objective opportunity after controlling belief points toward physical constraint, reserve, or task availability. Include both when both contribute.

Manipulate objective task horizon through remaining trials, time limits, or moves. Measure subjective horizon through beliefs, confidence, urgency, and willingness to invest in delayed payoff.

Clinical and social settings add cultural death scripts and prognosis awareness. Some communities emphasize preparation, repair, reconciliation, or acceptance under threat; others avoid direct prognostic talk. These scripts may produce horizon-like behavior without any reserve-coupled Law-specific pressure.

Test subjective-horizon effects, objective-menu effects, and reserve-horizon interactions separately, accounting for the cultural scripts above.

Report objective opportunity, subjective confidence, information source, and route trust separately.

16.1.5 — No Endpoint Leakage

For a prospective horizon test, use only $\mathscr{I}_t$ and forecasting rules trained on other participants or properly held-out periods. The participant's realized time-to-death cannot be an input to a prediction claimed to precede it.

Endpoint leakage can be subtle. Sensor dropout, hospice admission, sudden medication changes, or family alerts may encode proximity to death. These variables may belong in rival or missingness models, but they require caution if used to define the horizon. What looks like shrinking opportunity may be observation shutdown.

Include horizon-filter selection in the leakage audit: compare the frozen filter with the full set of filters explored during tuning.

16.1.6 — Horizon Measurement Models

Specify each channel's target: future-time reports concern perceived opportunity and may reflect mood, culture, or interviewer demand; clinical prognosis concerns survival; task choices concern policy under the experimental conditions.

Preserve disagreements among reported opportunity, prognosis, task-imposed time, and environmental options. Test whether those disagreements predict behavior or model failure before fitting a common latent score.

Horizon measurement also needs time scale. A person may have a short horizon for mobility, a medium horizon for communication, and a long horizon for relationship meaning. A single H_t can be used only when the domain is declared. Otherwise the model should use a vector of domain-specific horizons.

Match horizon cadence to the prediction: daily clinical updates, hourly task states, weekly route checks, and momentary

prompts describe different intervals. Declare continuous, visit-, decision-, or bin-based estimation and test for route changes between observations.

The horizon variable represents available future options under a declared information set. Measures of despair, acceptance, readiness, meaning, optimism, and perceived opportunity can be collected separately when the research question calls for them.

16.1.7 — No Prognostic Authority

A state model may forecast shutdown, care transitions, or affective volatility without a terminal invariant. PTVC inference additionally requires a bridge to the phenomenal residual and endpoint branch. If latent states induce the same measured process, report the identified set and apply Section 24.2; further discrimination needs new information.

Restrict filtered horizons to their declared research use. Do not use them for care, eligibility, insurance, resource priority, or personal remaining-life forecasts.

Record exposure to route-state outputs that could alter care, communication, monitoring, or endpoint timing. These responses create feedback and selection that can contribute to the estimate’s apparent predictive success.

Access controls should distinguish who needs the estimate for scientific analysis from personnel whose care decisions should remain insulated from it.

16.2 — Reserve and Horizon Proxies

Horizon describes available continuation or time structure; reserve describes capacity to alter future contribution. Either can be high while the other is low (Ramsay & Woods, 2014).

16.2.1 — Reserve as a Multi-Domain Capacity State

Model physiological endurance, cognitive flexibility, communication, social support, treatment access, and environmental opportunity separately when they bear on different routes. A scalar reserve requires a declared measurement model, dimensionality analysis, and domain-level sensitivity checks.

Declare reserve's causal role before estimation: mediator of disease effects on action, moderator of horizon-dependent choice, predictor of observation burden, or marker of decline. Identify confounders and consequences, then test its added information beyond prognosis, medication, frailty, and care context.

16.2.2 — Biological Reserve Candidates

Candidate reserve proxies may include functional status, treatment access, cognitive capacity, autonomic flexibility, available actions, social support, or a route-specific controllability index. No single proxy is the reserve by definition. The measurement model must show shared variation, channel-specific variation, and confounding by the outcome.

Test physiological stamina, inflammation, sleep continuity, mobility, nutrition, oxygenation, autonomic flexibility, cognitive endurance, medication burden, and functional performance as context-qualified reserve indicators (Seow et al., 2011; Gerstorf et al., 2010).

Record each proxy's raw units and justify any transformation or composite: steps, grip strength, heart-rate variability, creatinine, oxygen saturation, gait speed, cognitive variability, and performance ratings are not interchangeable scales.

The Law-specific use of reserve is narrower than general health. Reserve matters because it may modulate viable action, volatility, and the admissible set of possible repair or avoidance policies. A biomarker that predicts mortality but not choice, capacity, or signal dynamics may be a prognosis marker rather than a reserve marker.

Distinguish reversible capacity changes from measurement changes due to medication, pain, sleep disruption, delirium, infection, disability, or care quality.

Reserve is a latent candidate construct represented through competing indicator models. Each analysis states its normalization and the transformations that preserve the tested restriction.

A candidate biological route can begin with an independently mapped system. Kleckner and colleagues (2017) identified a large-scale allostatic–interoceptive network in separate discovery and replication samples. J. Zhang and colleagues (2025) extended the map with high-resolution 7 Tesla imaging in 90 participants, mapping resting-state functional relationships between cortical regions and brainstem and other subcortical structures. This supplies specific candidates for studying how bodily condition and affective regulation interact. The useful PTVC-oriented contrast asks whether a frozen cumulative-history variable improves prediction beyond current physiology, recent experience and ordinary allostatic models.

Where independently justified and safe, test reserve channels under ordinary task fatigue, sleep variation, minor effort, recovery from mild cognitive load, or natural changes in clinical

symptoms and treatment. Evaluate recovery and tolerance as well as baseline status.

Model capacity added by oxygen support, assistive devices, caregiving skill, accessible transport, pain relief, and predictable routines. Distinguish remediable lack of support from biological depletion.

Frailty scores may embed age, disability, mobility, or care-access assumptions. Audit those inputs before using a score as a candidate proxy for U(t).

A reserve proxy should also face a prognostic-ablation test. For the declared choice, capacity, or dynamic outcome, first fit the survival, disease-stage, recent-decline, treatment, and observation-quality baseline; then add the independently estimated reserve channels and evaluate held-out calibration and discrimination. Reserve earns a separate role only when its contribution survives that comparison and remains interpretable after shared inputs are removed.

Functional capacity may be recorded through mobility, grip, gait, and fatigue tolerance. Disability access, pain, equipment, and care support affect these measures, so their use requires accommodations and disease controls.

Autonomic flexibility may be recorded through HRV, respiratory coupling, and recovery time. Medication, sleep, anxiety, and sensor contact affect interpretation; treat this evidence as one channel rather than total reserve.

Metabolic and inflammatory load may be recorded through laboratory values, nutrition, fever, and cachexia markers. Disease, treatment, and infection can be shared causes as well as capacity signals and must be modeled accordingly.

Cognitive endurance may be recorded through latency drift, attention, and task persistence. Use accessible tasks and account for delirium, sleep, medication, and education.

Social and environmental support includes available help, transport, and communication aids. Inequality, family pressure, and resources affect these conditions; treat support as a modifier of available options rather than a biological reserve measure.

16.2.3 — Reserve and Disease Severity

Test reserve's added relation after illness severity, medication, care, and observation availability are modeled. Failure to distinguish reserve from generic severity blocks the reserve-coupled interpretation.

Disease progression can jointly change reserve, prognosis, medication, visibility, and available choices (Ramsay & Woods, 2014).

At least three pathways must remain separate: severity to reserve, severity to horizon belief, and severity to behavior or measurement. When they move together, an apparent reserve interaction may be a disease-stage interaction, and the analysis should report that common cause directly.

Time-varying clinical covariates are not optional. Medication dose, pain level, dyspnea, delirium status, functional decline, sleep disruption, and care setting can change from day to day. A cross-sectional reserve score may be too coarse for horizon prediction.

A useful design can exploit within-person deviations. If reserve improves temporarily because symptoms are managed and horizon belief remains stable, do choices and latencies change? If prognosis worsens but functional reserve is maintained, does the model still predict menu compression? These contrasts help separate mechanisms.

Represent poverty, inaccessibility, transport, and caregiving constraints as limits on available actions rather than biological reserve.

Declare the causal question before adjustment. Disease severity may confound a reserve effect or mediate the capacity change of interest; assign confounder, mediator, stratifier, or rival roles accordingly.

16.2.4 — Reserve Over Time

Effective capacity may decline, recover, fluctuate, or change abruptly. Such trajectories require an explicitly declared reserve model. The nonincreasing reserve branch in Section 4.2.2 is a narrower candidate; its drift and boundary consequences do not automatically transfer to a recovering or stochastic reserve process. A baseline proxy alone cannot establish which branch describes the observed window.

Match reserve cadence to the prediction: daily fatigue and medication effects, weekly functional decline, or minute-scale task stamina.

Measurement error is especially important because reserve often appears inside interactions. Error in U(t) can attenuate the estimated interaction with horizon, create false threshold effects, or shift apparent variance slopes.

State-space models can represent reserve proxies as noisy emissions from a latent capacity process, with process noise, observation noise, missingness, and version changes. Propagate the resulting distribution or identified set through exponent, covariance, intervention, and terminal-window analyses.

Confirmatory reserve predictions use filtered or forecast estimates; smoothed trajectories using future data are labeled retrospective.

Longitudinal reserve should be modeled with lags. Capacity at noon may affect action in the afternoon; action in the afternoon may alter capacity in the evening. A model that treats U_t and behavior as simultaneous can confuse cause and consequence.

Lag sensitivity, recovery windows, and delayed effects should be tested before reserve is described as an explanatory state.

16.2.5 — Independent Estimation

Predeclare reserve-channel tiers. One scheme uses functional and physiological capacity with minimal direct affect content, then clinical symptoms overlapping experience, then exploratory composites. Confirmatory claims should favor the least circular tier.

Independent estimation also requires temporal separation. Reserve features used to explain an interval should be constructed from information available before that interval, with endpoint-derived summaries, later clinical deterioration, and affect-estimator outputs excluded or separately ablated.

Independent estimation guards against circularity. Where feasible, build reserve proxies from separately qualified physiological, functional, cognitive, or environmental channels, apart from the ledger's affective and experience-linked inputs. Validate context of use, analytical and clinical validity, responsiveness, failure modes, and version drift before entry into the route equation (FDA-NIH Biomarker Working Group, 2016; Goldsack et al., 2020).

Perfect separation is rarely possible. Pain affects mobility, mood affects sleep, medication affects cognition, and distress affects physiology. The goal is not impossible purity. The goal is to declare overlap, model shared causes, and test whether the reserve effect survives when overlapping channels are removed.

Independent estimation also matters for dignity. If distress lowers a reserve score and the score then predicts future distress, the model may simply be scoring suffering twice. That is scientifically weak and ethically dangerous.

Routinely remove medication-linked, mobility, autonomic, and report-linked channels to test whether the interaction persists. A signature lost after plausible channel removal lacks robustness.

Independence is not all-or-nothing. A study may have strong independence for physiological reserve, partial independence for functional reserve, and weak independence for self-reported energy. The status statement should name those levels. A reserve result based on a partly overlapping channel can still orient research, but it cannot carry the same claim as a result built from independently validated capacity measures.

16.2.6 — Reserve Failure Conditions

Reserve failure is a scientific outcome that clarifies the next step. Candidate components may fail to cohere, an ordinary rival may explain the pattern, a proxy may not transport, or the public-risk burden may make the object unsafe to use. Each failure updates a different proposition and should be recorded separately.

Triangulation should compare proxies from genuinely different pathways rather than stack correlated versions of one signal. Biological, functional, cognitive, social, and material capacities may constrain different menus and time scales, and no biomarker may stand for the entire reserve state. Agreement is informative only when the channels do not share the same measurement error and causal role.

Test monotonicity: reserve may recover, fluctuate, compensate, or fall abruptly, while support can expand options and increase burden. Treat social and material resources as environmental pathways, distinct from personal biology or individual strength.

Severity, clinician expectations, treatment, observation, and endpoint processes can affect both reserve proxies and outcomes. Estimate reserve prospectively, then compare prognosis, frailty, and observation-selection explanations before drawing route conclusions.

The valid output may be an identified set rather than one number. When several reserve states remain compatible with the data, exponent, covariance, intervention, and terminal-window analyses propagate that uncertainty.

Retain failed or retired proxies and their affected analyses and artifacts in the version ledger. Reanalyze, narrow, or withdraw dependent results. Publish failure and retirement conditions before inspecting outcomes.

16.3 — Boundary Exponents and Reserve Dispersion

Distinguish calendar-time, reserve, and reachability coordinates. Their boundary exponents are linked only by a specified bridge and are route-specific consequences (Karatzas & Shreve, 1991).

16.3.1 — Rising Delay Sensitivity and Boundary Rate

The empirical target is conditional dispersion estimated from information available before the endpoint. A local power relation between reserve and remaining time yields a time-domain exponent only under an explicit reserve-time bridge, while measurement error, heterogeneous paths, smoothing, and selection can alter the apparent slope. The protocol should fix a multiscale window ladder, error floor, random-effects structure, and rival family before any log-log relation is interpreted.

Candidate asymptotic restrictions assume the declared endpoint clock, reserve-time bridge, and normalized-dispersion model (Karatzas & Shreve, 1991).

$$U_t \asymp c_U (T-t)^{\beta}, Var\,(Y_t | U_t, \Xi_t) \asymp c_Y (T-t)^{\gamma}$$

$$L_t^{rc} = c_{LU} U_t Y_t$$

$$Var\,(L_t^{rc} | U_t, \Xi_t) \asymp c_{LU}^2 c_U^2 c_Y (T-t)^{2\beta+\gamma}$$

$$\gamma = 0 \Rightarrow Var\,(L_t^{rc} | U_t, \Xi_t) \asymp (T-t)^{2\beta}$$

As filtered horizon contracts, delayed outcomes may lose motivational relevance because they fall outside perceived opportunity. Socioemotional selectivity is therefore an ordinary rival. The route-specific test asks whether unresolved state and independently estimated reserve contribute a protocol-fixed interaction beyond future-time perspective, prognosis, urgency, and scarcity (Carstensen, 2006).

Naturalistic observations broaden the horizon finding beyond imagined scenarios. Fung and Carstensen (2006) observed shifts toward emotionally meaningful goals around the September 11 attacks and the SARS outbreak. These settings couple time perspective to threat, social contact, and available resources, so their contribution is ecological convergence rather than randomized causal identification. The next useful comparison tracks those variables separately and asks whether a common, prospectively fixed horizon-by-state relation transports across ordinary transitions such as relocation, retirement, and the completion of a voluntary project. Successful transport would give a candidate mechanism a substantive constraint to explain.

Perceived time can be studied separately from a frustrated goal. Fung and Carstensen (2004) compared social motivation under foreshortened time and blocked-goal conditions, finding different patterns in social preferences and reported goals. Their design tested alternatives to socioemotional selectivity theory rather than treating all constraints as equivalent. It offers a useful template for this chapter: measure perceived opportunity and current goals separately, then ask whether independently estimated experiential history improves predictions of subsequent choice.

In a task with h remaining steps, the prediction concerns the policy for delayed repair options. When h is large, the system may choose information gathering or delayed high-value repair. When h is small, it may prefer immediate lower-value actions if they reduce an unresolved state before the menu closes.

This can be tested without terminal pressure. A task can vary remaining trials, delay to reward, reversibility, and cost while keeping all stakes minor. The outcome is choice probability, latency, confidence, post-choice affect, and physiological control load.

Model policy heterogeneity through baseline behavior, available menus, reserve, culture, framing, and belief; shortening horizons need not produce one common response.

A good delay experiment should include catch conditions. Some delays should be irrelevant to outcome, some should threaten route loss, and some should consume reserve. If all delays produce the same response, the effect is not horizon-specific. If only route-threatening delays matter under short H_t, the candidate becomes more precise and more vulnerable.

Delay sensitivity should be measured against a baseline delay function. Some people are generally impatient; some are generally deliberative; some delay because access is difficult. The candidate pattern is a change in delay sensitivity as H_t and U_t change, not a stable trait. Within-person estimates and task counterbalancing are therefore central.

The empirical design estimates the reserve map and boundary law separately across a prespecified ladder of scales. A nonlinear reserve-time relation can create apparent acceleration, while saturation, smoothing, and irregular observation can flatten or steepen it. Joint models estimate reserve-time mapping, process dispersion, and observation floor, then test whether the same exponent family transports across held-out windows.

16.3.2 — Priority Reordering and Drift Slope

Priority reordering means that the relative ordering of live demands changes as horizon and reserve change. A person may shift from long-range exploration toward emotionally meaningful contact, repair, communication, rest, or avoidance. The proposed route predicts a signed-state dependence that can be compared with ordinary horizon-driven goal selection (Carstensen, 2006).

Define the demand set from task goals or penalties, participant reports, obligations, symptoms, unfinished tasks, or documented choices. Do not infer it from retrospective moral interpretation.

Priority reordering can be visible in time allocation, task switching, communication attempts, search behavior, medication questions, social contact, or rest selection. Each channel has its own confounds. A phone call may reflect relationship repair, loneliness, practical logistics, family pressure, or habit.

If reordering follows ordinary expected value, prognosis information, or social scripts with no reserve-queue interaction, the horizon-specific claim should be narrowed.

Model the sequence of opening, postponing, abandoning, and revisiting demands rather than only the final choice.

Allow partial orders and ties when demands such as symptom relief, apology, deadlines, conversation, and rest are incomparable.

16.3.3 — Risk Pruning and Dispersion Compression

Risk pruning is the proposed loss or reduced attractiveness of high-variance options as remaining recovery opportunity changes. Test choices against represented opportunity, current condition, and recent experience; nearby clinical studies supply symptom and affect trajectories for comparison (Seow et al., 2011; Vogel et al., 2013).

Permit both increased and reduced risk-taking under short horizons, conditional on capacity, meaning, novelty, relief, or autonomy.

Specify whether the task varies reward, affect, physiological load, social outcome, or delay variance.

The pruning pattern does not distinguish a reserve-linked candidate when baseline risk preference, disease severity, cognitive change, misunderstanding, or an option set unrelated

to reserve explains it equally well. Predeclare those comparators. Under Section 22.8.1, distinguish a failed reserve prediction from adequate but inseparable explanations before retiring the candidate.

Variance pruning also requires a recovery measure. A high-variance option is different when recovery is possible. In the task, the same loss can be paired with high or low reserve recovery. In field settings, recovery may come from rest, medication, support, money, or time. Without a recovery variable, the model cannot know whether risk was pruned because variance rose or because the cost of error became unrecoverable.

Distinguish known outcome variance from unknown probabilities and institutional mistrust. Avoidance driven by ambiguity or prior unreliable access is a separate rival to risk pruning.

16.3.4 — Accelerated Repair and State Dependence

Register repair and avoidance categories with their opportunity constraints before outcomes. Shorter horizons may increase either; do not select only morally favored actions as route evidence (Carstensen, 2006).

Direct repair is only one policy. Avoidance may be rational when repair is impossible, unsafe, coercive, or too costly. Blocking may protect a person from overload. Substitution may reduce a related burden when the original burden cannot be changed.

A task can distinguish repair and avoidance by offering actions that reduce a specific unresolved state, actions that suppress signals without resolving them, and actions that substitute a lower-cost route. Outcome measures can include choice, latency, post-action residual, and future reactivation.

Specify when repair is feasible, safe, and meaningful, and when avoidance is predicted by threat, cost, or low repair probability. Without these classes, the model can accommodate both action and inaction.

A paired task using hypothetical routes can make this test sharper. The same unresolved demand can be represented with a safe repair route, an unsafe repair route, a delayed route, and a substitute route. The candidate predicts different policies across those route types under changing horizon and reserve. If behavior does not differ by route feasibility, the repair-avoidance interpretation weakens.

16.3.5 — Reserve-Horizon Interaction

The core quantitative prediction is not a main effect of age or calendar time. It is a protocol-fixed interaction among filtered horizon, independently measured reserve, unresolved state, and the live action menu. A short horizon with high reserve may permit focused action; the same horizon with low reserve may yield rest, compression, or no observable response (Carstensen, 2006).

Emotional strengths can persist when circumstances become difficult. In 945 US adults aged 18–76 assessed during COVID-19, older age was associated with better emotional experience even after measured risk and other covariates were considered (Carstensen et al., 2020). Sun and Sauter (2021) reproduced age advantages in 23,350 people across 63 countries, while their within-person Dutch comparison of 4,370 participants found that those advantages diminished during sustained pandemic stress. Together, these studies make context-dependent regulation a concrete research target. A reserve-horizon test should measure capacity, exposure, social support, and perceived opportunity independently, then compare a frozen

interaction model with these established accounts. The empirical opportunity is to explain when resilience persists, when support changes the path, and which additional state variable improves prediction.

A reserve-horizon interaction should have a protocol-fixed sign and measurement route for each task. For example, low reserve may increase the cost of high-effort repair, while short horizon increases the cost of delay. The model then predicts which action class becomes more likely.

In behavioral tasks, the interaction can be tested through reserve manipulations that are safe and mild: cognitive load, time-on-task fatigue, rest breaks, or simulated capacity budgets. These manipulations must never create distress or deprivation.

The interaction should be tested in designs that manipulate or observe both dimensions independently. A task can shorten horizon while holding reserve constant, lower reserve cost while holding horizon constant, or vary both. Clinical studies cannot manipulate these states casually, but they can stratify, model natural variation, and test whether the same pattern appears under different reserve proxies.

16.3.6 — Heterogeneity and Finite-Window Scaling

Permit nonmonotone trajectories: oscillation, plateau, reversal, or abrupt change after treatment, support, fatigue, or a new constraint. Do not average contradictory individual paths into smooth compression.

Specify observable heterogeneity through random effects, classes, mixtures, or person-specific parameters and test it in held-out data. Unexplained heterogeneity widens uncertainty rather than justifying an outcome-selected universal trajectory.

Nonmonotone patterns can separate rivals. State-coupled control, fatigue, episodic treatment, selection, ordinary mean

reversion, and endpoint-conditioned bridges imply different reversals, lags, and cross-channel signatures. The analysis should preserve individual and subgroup trajectories long enough to test those differences before summarizing them.

Marginal compression may reflect dropout of high-variance paths, different subgroup entry rates, or cancellation of local trends. Test component dynamics and observation selection before attributing it to one route.

Exponent recovery uses finite empirical windows. An asymptotically valid power law may not be identifiable over the observed range. Apply the reserve-to-time mapping and joint observation-error analysis in Sections 16.3 and 16.3.1 to state which exponent the data identify.

Use actual pair availability and cadence, prespecified local windows, and sensitivity to deconvolution and calibration error when estimating scaling.

16.4 — Two-Time Covariance and Martingale Diagnostics

Equal marginal means and variances need not imply equal joint laws. Compare covariance, conditional increments, quadratic variation, and held-out prediction errors only where the specified route and rival restrictions imply different laws (Doob, 1957; Revuz & Yor, 1999).

16.4.1 — Two-Time Covariance as a Route Signature

Prospective two-time prediction must use information available at the earlier prediction time; a prespecified retrospective covariance estimand must be labeled by its actual information set. Retrospective alignment can create elegant covariance surfaces that no prospective observer could have constructed. The first display below is a candidate reserve-route covariance under its stated assumptions; the second is a Brownian-bridge rival. For this display, Sθ and Y(Sθ) are fixed, and, conditional on Ξ, U and σY are deterministic while W remains Brownian. Neither formula is evidence, and both must be evaluated under the same filtered information, endpoint uncertainty, and observation process (Andersen et al., 1993; Doob, 1957; Rizopoulos, 2012).

$$Y(t) = Y\left(S^{\theta}\right) + \int_{S^{\theta}}^{t} \sigma_Y(r) dW(r)$$

$$\tau_Y(t) := \int_{S^{\theta}}^{t} \sigma_Y^2(r) dr$$

$$\mathrm{Cov}\left(L^{rc}(s), L^{rc}(t) \mid U, \Xi\right) = c_{LU}^2 U(s) U(t) \tau_Y(s), S^{\theta} < s \leq t$$

$$Cov\left(B_s, B_t\right) = \sigma^2 s \frac{T-t}{T}, s \leq t.$$

The covariance object should be protocol-fixed before the surface is inspected. A primary option may be conditional covariance at two protocol times given a filtered reserve state; a secondary option may normalize by the protocol-fixed scale process. The normalization cannot be chosen after seeing which version looks most bridge-like (Doob, 1957; Durbin & Koopman, 2012).

A synthetic example isolates what the joint law contributes. Let four equally spaced readings each have equal chances of positive one and negative one. In one process, a fair initial sign is followed by strict alternation; in another, the initial sign persists through all four readings. At every individual time the two processes have the same mean and variance. Across times they differ completely: the first has a four-reading total of zero on every path, while the second has a total of positive four or negative four. This example concerns a declared discrete observation process, not a calibrated phenomenal ledger. It shows why separate marginal summaries leave accumulation unresolved and why time-pair information is worth preserving.

Sampling cadence affects which time pairs are observed and how their covariance can be estimated. Sparse visits, selective pair availability, smoothing, and aggregation can distort the estimated dependence or change the observed quantity. Preserve raw timing, device versions, treatment changes, and uncertainty in the latent clock rather than treating a resampled grid as the original process.

Specify whether report, physiological, action, and endpoint-hazard innovations are independent, shared, or hierarchically coupled. Unrelated latent scores for each channel do not establish a common route.

Compare bridge boundary dependence, mean-reverting correlation, and frailty-induced long-range dependence under the same observation model and scoring rule.

Estimate covariance with uncertainty, then score future pairs in held-out participants or time blocks. Test recovery of long-memory, bridge-kernel, and near-Markov shapes at the planned cadence and noise level (Gneiting & Raftery, 2007; Talts et al., 2018).

16.4.2 — Dependence with Deterministic Reserve

Normalization can reveal whether shrinking scale and temporal dependence are governed by separate components. If reserve multiplies an arithmetic innovation process, dividing by a correctly specified reserve can expose a reference process whose dependence is easier to characterize. That move is legitimate only when reserve is independently estimated, the resulting amplification of measurement noise near zero is quantified, and the target remains adequately resolved (Andersen et al., 1993; Rizopoulos, 2012).

A deterministic-reserve approximation is a local model, not an assertion that human capacity follows a known clock. The approximation may be useful in controlled tasks or narrow clinical windows where reserve changes smoothly enough to estimate. Outside that context, stochastic reserve, treatment response, and abrupt state transitions should be represented rather than absorbed into residual error (Särkkä, 2013).

The normalized process should be tested for conditional mean, variance, autocorrelation, and tail behavior. A route that predicts driftless normalized innovations must say which filtration defines driftlessness and which observations are allowed into that filtration. If future endpoint information is required to make

the residuals look martingale-like, the normalization has failed the prospective test.

Test one frozen normalization family across held-out participants, sites, and windows. Cohort-specific reserve-map refitting may improve fit while violating the proposed transport restriction.

16.4.3 — Bridge and Doob-Transform Rivals

Conditioning a diffusion on a terminal value changes its drift and can reproduce endpoint concentration without an endogenous corrective controller. A Brownian bridge is one such rival (Doob, 1957; Karatzas & Shreve, 1991). The exact overlap below shows why its covariance need not distinguish it from an unrestricted reserve-product family. Causal-process and scoring requirements enter through Røysland (2011) and Gneiting and Raftery (2007).

Test native dynamics, state-coupled termination, selection, and conditioning as distinct explanations; a Doob transform can alter pre-endpoint behavior through conditioning on a future event.

An exact overlap example identifies which restrictions must do the discriminating work. For the diffusion-route illustration, let $T > 0$ be fixed, W be standard Brownian motion, $U(t) = (T - t)/T$, $q(t) = Tt/(T - t)$, and $L(t) = \sigma U(t) W(q(t))$ for $0 \leq t < T$, with the fixed unit conversion absorbed into σ. Here U is dimensionless, q has time units, and σ has residual units per square root of time. For $s \leq t$, direct calculation gives $Cov(L(s), L(t)) = \sigma^2 s(T - t)/T$, exactly the Brownian-bridge covariance. The process is centered Gaussian. It extends continuously to $L(T) = 0$ almost surely because $U = T/(T + q)$ and $W(q)/q$ tends to zero as q tends to infinity. The normalized variance nevertheless diverges: $Var(\sigma W(q(t))) = \sigma^2 Tt/(T - t)$.

Thus the unrestricted reserve-product family contains a Brownian bridge; its name alone cannot separate the mechanisms. Discrimination comes from an independently qualified reserve map and a prospectively restricted innovation clock. This bridge requires $q'(t) = T^2/(T-t)^2$ and the corresponding singular normalized diffusion coefficient $\sigma T/(T-t)$. A candidate with a separately justified stable normalized-dispersion branch predicts something different. Estimate the permitted clock family on independent data, freeze it, and test the resulting joint predictions. If the qualified family still admits the bridge clock, passive path data receive Model-ambiguous status; further discrimination requires measurements or interventions for which the models make different predictions.

Simulate fixed terminal times, random endpoints, first-hitting events, absorbing sets, and survival filters under their own laws and the same measurement/missingness model.

Intervention response offers another distinction. Conditioning alone does not specify how a safe perturbation to support, information, or action availability should propagate. A causal route may predict a state update and changed future dependence. A purely conditioned bridge may fit the passive trajectory yet fail to transport under perturbation.

16.4.4 — Martingale and Variation Diagnostics

Test predictable conditional increments after accounting for the declared state. Fitting a transformation on the tested data or inserting future endpoints into its filtration can manufacture apparently driftless residuals (Doob, 1957; Revuz & Yor, 1999).

Specify the martingale diagnostic's filtration, increment interval, predictable instruments, and sequential evidence

process. Use residual products to test dependence across states and time scales (Røysland, 2011; Ramdas et al., 2022).

Quadratic variation distinguishes diffusion-like, smoother, and jump-dominated latent dynamics. In sampled data, validate realized-variation estimators against measurement error and irregular timing under the actual sampling design.

Jumps deserve explicit treatment. Medication changes, acute symptoms, device resets, awakenings, and care transitions can produce discontinuities that a diffusion model will misread as extreme innovations. A jump-sensitive route may be scientifically appropriate, but it must specify whether jumps belong to the phenomenal process, the observation process, or the context process.

Predictable residual drift branches to missing-state, reserve-proxy, observation-model, or rival-generator diagnostics.

Qualification begins with known generators. The pipeline detects drift when present, controls false drift under calibrated martingales, and distinguishes jumps from noise. Passing those synthetic tests establishes which diagnostics can be interpreted in the empirical study.

Measurement noise complicates quadratic variation because high-frequency error can masquerade as process volatility. Replicate channels, device validation, and cadence perturbations can help separate the two. If estimated variation rises without bound as sampling becomes finer, the model may be measuring sensor noise rather than phenomenal dynamics. If it collapses after smoothing, the smoothing rule may have erased real short-scale structure. Resolution invariance is therefore a diagnostic target, not an assumption.

16.4.5 — Survivor Conditioning and Pairwise Bias

Late-window covariance is observed among people who remain alive, enrolled, capable, and measurable at both times. That subset can differ sharply from the original cohort. Pairwise-complete covariance therefore mixes the target dynamics with survival, withdrawal, transfer, device tolerance, and clinical change (Andersen et al., 1993; Rizopoulos, 2012).

Conditioning on survival can induce associations even when none existed in the source population. If survival or measurement availability is a common effect of frailty and another cause of the affective trajectory, conditioning on it can open a collider path. Frailty that causes both availability and the trajectory is a shared cause; that structure alone is not a collider. The analysis should represent the risk set at each pair of times and state which population the covariance is intended to describe (Andersen et al., 1993; Hernán & Robins, 2020).

Weighting may recover a marginal target under strong assumptions about observation and survival. Joint models may instead connect longitudinal outcomes and event processes through shared random effects or latent states. Principal-stratum estimands can describe survivors under specified counterfactual conditions. None of these methods is assumption-free, and their targets are not interchangeable.

Selective loss of high-variance observations can create late compression; preferential retention of complex cases can inflate variance. Test both directions under distress, sedation, and communication-loss scenarios.

Pairwise bias also affects cross-channel dependence. A physiological sensor may continue after self-report stops, while proxy reports enter only after capacity declines. Combining those channels without marking the transition can create an

artificial covariance pattern. Observation marks and channel-specific risk sets belong in the model.

Report any identified set under its observation assumptions. A selected-survivor result must name that population; classify remaining limitations under Section 24.2.

16.4.6 — Held-Out Dependence Recovery

Simulation must show that the proposed covariance, innovation, or martingale diagnostic is recoverable at the observed cadence, noise level, missingness pattern, and endpoint uncertainty, without routinely finding it in rival-generated data.

Use Section 17.7's validation split matched to the intended deployment claim.

Apply proper scores, calibration, and coverage to joint dependence, not only marginal means. Include open-set cases to test recognition of model-set inadequacy (Gneiting & Raftery, 2007; Yao et al., 2018).

16.5 — Terminal-Window and Intervention Signatures

Terminal-window interventions can test a proposed route only when their timing, available actions, and predicted effects are specified independently of the desired outcome. Direct repair, support, substitution, and environmental change can affect both the state and the observation process, so each contrast requires its own explanation of what would count as failure. Keep information and support low burden, justify every intervention, and never induce distress or withhold care to make a signature visible (World Medical Association, 2024).

16.5.1 — Prospectively Defined Terminal Windows

Define the terminal window before inspecting the outcome trajectory, using an independently governed protocol endpoint, specified state transition, or other admissible event rule (Aalen et al., 2008).

Several clocks remain separate in the protocol: calendar time, time since enrollment, time to a protocol endpoint, disease or care-state time, and filtered subjective horizon. Each alignment defines a different estimand. A pattern that appears only after retrospective selection of one clock is an alignment finding, not a terminal-window signature (Rizopoulos, 2012; Kurland et al., 2009).

Pseudo-endpoints and matched nonterminal cut points are candidate negative controls when the frozen route predicts no target signature there. The signature should weaken, disappear, or change in the specified way when trajectories are aligned to those events. Unexpected endpoints instead require a prospectively specified test of dependence on horizon awareness, with cause, acuity, care, and observation conditions addressed; unexpectedness alone does not make a genuine

endpoint a negative control. If the same pattern appears around arbitrary salient events, the terminal interpretation loses specificity (Lipsitch et al., 2010; Hernán & Robins, 2020).

Prospective hospice research also documents emotionally meaningful experiences near life's end. Kerr et al. (2014) conducted 453 interviews with 59 patients; 52 reported at least one end-of-life dream or vision. Experiences featuring deceased people or animals were more comforting than other reported contents, and comforting experiences involving the deceased became more prevalent as death approached. These observations make internally generated, relationally meaningful experience a legitimate target for careful study. A participant-led observation module can record timing, comfort, sleep or wakefulness, medication, and report availability alongside usual care. This produces a richer account of late-window experience and the conditions associated with comfort.

Keep sudden endings, diffuse decline, reversible interruption, censoring, and qualified terminal windows in separate estimand branches unless a declared bridge supports pooling.

16.5.2 — Direct Repair and Support Perturbations

Define the unresolved target before analysis: an unfinished task, symptom, communication barrier, practical need, conflict, or inaccessible option. Direct repair is an action that plausibly reduces that target (Hernán & Robins, 2020).

Record target burden, admissible actions, expected timing, observable consequence, and remaining residual. Separate attempt from success; a failed attempt may reflect unavailable routes, inadequate support, timing, or an incorrect prediction.

Support perturbations change information, access, safety, cost, reversibility, communication, or available time without inducing distress or withholding care. Examples include an accessible

communication device, transportation, a trusted intermediary, schedule flexibility, symptom support, or a low-burden prompt that makes an existing option usable. Each perturbation must be independently justified and acceptable even if the Law-specific analysis were removed (World Medical Association, 2024).

Measurement should distinguish the offered support, actual uptake, action timing, task completion, immediate relief, later reactivation, and any burden shifted to another person or setting. A support that improves task completion but increases fatigue, family load, privacy loss, or clinical disruption cannot be summarized by one favorable outcome.

Represent access, speech, energy, safety, money, assistance, and time constraints before interpreting nonrepair as preference. Record which routes were available, blocked, or opened by support.

Compare the frozen horizon–reserve–state–support relation with goal completion, treatment response, social expectation, deadlines, cognitive control, family pressure, and improved access under common observation and scoring rules.

16.5.3 — Substitution and Route Reordering

Substitution occurs when an unavailable or costly route is replaced by another action that changes a related state. Reframing changes the meaning, priority, or interpretation of a burden without necessarily changing the external event. Route reordering changes which live option is attempted first. These are distinct processes and should not be pooled merely because each can precede lower reported burden (Gerstorf et al., 2010; Vogel et al., 2013).

Evaluate substitution through changes in affective load, action cost, safety, future reactivation, physiology, and reported meaning. Do not assign it a moral rank relative to direct repair.

Use experience sampling, language, choice, confidence, clinical observation, and delayed narrative as partial evidence of reframing. Keep immediate measures separate from retrospective reconstruction.

Compare reframing with cognitive reappraisal, acceptance, therapy, religious coping, memory reconsolidation, social persuasion, and horizon-dependent goal change (Carstensen, 2006).

The protocol should predefine plausible substitution classes and the outcomes that would count against each one. An alternative route may repair the same target, partially repair a coupled target, provide only immediate relief, or merely distract. A classification that treats every alternative action as hidden repair has no adverse case and should not enter confirmatory analysis.

Route reordering requires a live-menu record. The study should know which actions were available, when they became feasible, what they cost, and whether support changed their order. A later choice cannot reveal that an earlier route was psychologically impossible, ethically inadmissible, or simply unavailable unless those constraints were measured independently.

Self-chosen reframing must be distinguished from pressured reinterpretation. A person may voluntarily find meaning, acceptance, or a new goal, while an institution may also demand positive meaning because actual repair is expensive or inconvenient. The former can reduce burden; the latter can transfer burden into silence. Consent, refusal, and context belong in the estimand.

After the measurement, intervention, and observation gates pass, an adequately resolved failure of the protocol-fixed horizon, reserve, unresolved-state, or menu relation counts

against the substitution or reordering prediction. Equal predictive scores alone do not falsify an adequate candidate. Apply Section 22.8.1's declared advantage, adequacy-and-simplicity, and ambiguity criteria before assigning a comparative verdict.

16.5.4 — Avoidance, Blocking, and Automaticity

Distinguish avoidance, blocking of an unresolved state, and automatic selection of a familiar action. Silence alone identifies none of these processes.

A shorter horizon does not always make repair urgent. It can make repair too slow, risky, costly, or emotionally expensive to attempt. Short-term avoidance may preserve capacity for a later safer route, while persistent avoidance may increase future reactivation. The model must represent time scale and feasibility before interpreting the same behavior as protection, habit, or unresolved burden.

Avoidance can be studied through skipped or delayed actions, task disengagement, topic shifts, physiological arousal without approach, later reactivation, and explicit refusal when available. These channels are not interchangeable. A route may be behaviorally avoided while remaining mentally active, or it may disappear from report because the observation itself became burdensome.

In bounded tasks, offer repair, substitution, neutral deferral, and avoidance while varying horizon, information, cost, and support. Record danger, social exposure, sensory load, and clinical destabilization; choosing a safer avoidance option is not adverse evidence about the person.

Habit and automaticity are serious rivals. Familiar actions can dominate when deliberation is expensive, cues are strong, or reinforcement history is deep. Baseline habit strength, reaction

time, perseveration, sensitivity to changed payoffs, and updating after route closure help distinguish automatic control from a newly reorganized priority structure.

Apply the modal distinctions defined in Sections 5.6.2 and 7.3.4. While positive mass remains on nonclosing continuations, stronger guidance or a more concentrated choice distribution does not establish fixed-policy probability-one completion.

16.5.5 — Social and Environmental Interventions

Late-life data also identify potentially helpful social conditions. In 2,910 deceased participants from the German Socio-Economic Panel followed for up to 27 years, social participation and valuing social goals were associated with higher late-life well-being, less pronounced terminal decline, and a later onset of decline after measured covariates were considered (Gerstorf et al., 2016). These associations provide a constructive complement to the accelerated terminal decline documented across national panels (Gerstorf et al., 2010). For this program, the useful target is whether changes in accessible support predict subsequent experience and action beyond health, care, and selection. Social participation is therefore a measurable correlate of the trajectory and a concrete focus for studying late-life connection.

Model communication access, assistance, relational safety, finance, physical barriers, care quality, and institutional rules as modifiers of action availability and cost.

Represent support as changes in travel cost, route reliability, response time, risk, information, or delegated action.

Within a frozen route model, interventions that change time, capacity, information, cost, reversibility, or menu size may be summarized by a marginal-value diagnostic. The estimand is the

local change in a declared model output when one constraint is relaxed; it is not a price on suffering.

Require uncertainty and simulation stability of marginal-value diagnostics, with added value over ordinary control and decision models, before vulnerable-population use. Name the changed quantity in public descriptions.

16.5.6 — Branch-Specific Failure and Residual Burden

Retain potentially nonzero residual burden after action, support, reframing, substitution, or a nearby protocol endpoint.

In bounded tasks it may appear through remaining penalty, report, conflict, reactivation, latency, or physiological load. These channels are not interchangeable; vector or set-valued representation is retained when scalarization would hide disagreement, and the measurement rule is fixed before intervention outcomes.

After the measurement and design gates pass, an adequately resolved failure of the declared relation between persistence and horizon, reserve, queue state, or the proposed control variable counts against that route. Comparative verdicts follow Section 22.8.1. Operational completion such as discharge, compensation, or case closure remains separate from experienced resolution.

Safe intervention studies require positivity and cannot induce distress or withdraw care; timing, treatment history, and observation marks address confounding by indication (Hernán & Robins, 2020; Pearl, 2009).

16.6 — Compression Detectability and Observation Floors

Resolve the specified contraction against sensor floors, smoothing, selective dropout, and changes from self-report to proxy channels (Carroll et al., 2006; Little & Rubin, 2019).

16.6.1 — Additive Noise and Variance Floors

Additive measurement error contributes variance even when the latent process contracts. If the error variance remains approximately constant while the latent variance approaches zero, the observed curve flattens at the noise floor. A naive log-slope fit then underestimates the contraction exponent and can make distinct routes appear identical (Carroll et al., 2006; Little & Rubin, 2019).

The noise model should be estimated from replicate observations, calibration sessions, device comparisons, or validated measurement studies rather than inferred from the same late-window curve it is meant to correct. State-dependent error is especially important because fatigue, sedation, movement, and communication difficulty can change reliability near the endpoint (Carroll et al., 2006).

A variance floor also interacts with participant heterogeneity. Some participants may remain measurable at high precision while others become effectively unobserved. Pooling them can produce a floor that is neither a property of the instrument nor the latent process. Hierarchical error models and participant-specific calibration are therefore preferable to one global correction.

Simulation should show how the proposed estimator behaves when the true exponent varies, when the floor is misestimated, and when the floor depends on state. The study should report the

region in which exponent recovery is trustworthy and the region in which only bounds are available.

The contraction claim is unadjudicable when calibration uncertainty, structural error nonidentification, or inadequate statistical precision prevents the declared contrast from being resolved. A latent variance smaller than the measurement-error variance is not, by itself, proof of nonidentification: an independently calibrated error model and sufficient suitable observations may still identify it. Report whether the limitation is structural or a finite-sample precision problem before deciding which design change could help.

16.6.2 — Informative Observation Shutdown

Observation often ends for reasons related to the state being studied. A participant may stop responding because of fatigue, distress, sedation, transfer, device intolerance, clinical deterioration, or a deliberate choice to protect privacy. Treating those endings as random can manufacture a smooth terminal pattern (Rizopoulos, 2012; Rubin, 1976).

Joint longitudinal-event models can connect the outcome trajectory and shutdown hazard, while sensitivity analyses can vary the strength and direction of unobserved dependence. Pattern-mixture and selection models provide alternative parameterizations (Rizopoulos, 2012; Little & Rubin, 2019).

At a transition from first-person to family or staff reports, specify the new observation model and retain the transition time and channel identity in the analysis.

16.6.3 — End-to-End Detectability Simulation

Simulate the full acquisition-to-decision pipeline, including cadence, channel errors, treatment, endpoint uncertainty,

observation marks, censoring, preprocessing, smoothing, and estimation (Lakens, 2017; Talts et al., 2018).

Require both calibration under a correct model and recovery against rival generators; computational calibration alone does not establish discrimination (Talts et al., 2018; Wilson & Collins, 2019).

Open-set generators should include mixtures, abrupt jumps, nonlinear observation floors, site-specific protocols, and unmodeled treatment effects. The pipeline should be rewarded for abstaining when none of the declared models is adequate.

Vary effect size, reserve-time mapping, branch mixture, cadence, floor, unobserved-state missingness, endpoint error, and site heterogeneity jointly. Map recovery, false route preference, unavailable estimates, and scope restrictions before recruitment.

The most informative simulations include near-equivalent models. If two routes are intentionally separated by only the amount the planned design claims it can detect, the recovery study shows whether that distinction is real.

16.6.4 — Deconvolution and Replicate-Channel Limits

Deconvolution requires error-distribution information from replicate channels, validation data, or structural assumptions; without it, latent and measurement variation may be nonidentified (Carroll et al., 2006; Little & Rubin, 2019).

Repeated reports can estimate short-term reliability, but repeated prompting may change the experience or the reporting strategy. The reliability study must therefore balance information against reactivity and burden. Calibration obtained in healthy participants may not transport to sedation, delirium, severe fatigue, or terminal illness.

Use errors-in-variables models, latent-variable methods, validation subsamples, or partial-identification bounds according to the available error-law information (Carroll et al., 2006).

Deconvolution becomes unstable near the boundary because small differences between observed and error variance can determine the entire latent estimate. Negative corrected variances, wide intervals, or sensitivity to minor calibration changes are signals to abstain rather than reasons to truncate estimates at zero.

16.6.5 — Equivalence and Instrument Resolution

Compare the observation floor with the equivalence band when assessing whether the planned data can resolve the measured target.

The margin is scientifically and ethically justified before outcome inspection. Its basis may be a validated measurement standard, clinically meaningful difference, or decision-theoretic loss. The declared decision determines the interpretation of practical equivalence.

Effective resolution depends on reliability, calibration drift, state-dependent error, missingness, and preprocessing, not merely printed scale increments.

Equivalence analysis should include the two one-sided testing logic or another prospectively specified method appropriate to the estimand, together with sensitivity across plausible margins, calibration error, and missingness models. Nonsignificance is not equivalence, and equivalence under a wide margin does not resolve a narrower target. The report should show where the conclusion changes.

The error of one observation differs from uncertainty in an estimated mean, variance, or other model parameter. Under an identified error model, additional suitable observations can

improve parameter precision. Compare that uncertainty with the declared equivalence band.

A conclusion that changes sign under a small, credible shift in the neutral origin is not robust. A conclusion that depends on outcome-selected exclusion of difficult-to-observe windows is not a confirmatory terminal finding. The final report should present a grid of verdicts across declared perturbations, making clear which conclusions are stable, which are model-dependent, and which remain unestimable.

16.6.6 — Abstention Thresholds

Apply Section 24.2 and Section 27.4 to the terminal-window limitations below (Lakens, 2017; Talts et al., 2018).

Triggers may include failed measurement qualification, endpoint leakage, unrecoverable model confusion, observation-floor dominance, excessive informative missingness, unstable calibration, insufficient event support, or a burden threshold that stops data collection. Each trigger should map to a specific component status.

Use valid bounds to exclude routes when they cannot identify a unique winner. Report the bounds and their assumptions with the main result.

16.7 — Common-Rule Rivals, Falsifiers, and Route Verdict

A failed reserve exponent may retire that route; endpoint-independence failure may require a separately qualified state-coupled branch; representation or latent-transport failure can block downstream inference (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

16.7.1 — Common Observation Rules

The candidate route and every rival must receive the same observed channels, measurement versions, preprocessing rules, endpoint information, missingness treatment, and held-out partitions (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Freeze sign convention, origin, units, channel qualification, exposure clock, and uncertainty propagation across the comparison.

Give every model the same endpoint information. A candidate with a filtered horizon must face rivals permitted to use that information; otherwise mechanism and information access are confounded. Evaluate retrospective smoothing separately from prospective prediction.

Match cross-validation folds, prior or penalty budgets, stopping rules, hyperparameter searches, and calibration effort; demonstrate recovery before ranking real data (Talts et al., 2018; Vehtari et al., 2017).

Record the source, availability time, uncertainty, and consent boundary of each candidate state variable. Audit whether any model received information withheld from rivals.

The common-rule gate fails when any material difference in data access, endpoint leakage, preprocessing, tuning opportunity, or observation handling could explain the apparent

winner. The result then returns design-inadequate or Unusable evidence for the affected comparison rather than route-favored.

16.7.2 — Serious Rival Matrix

Freeze each rival's mechanism, intended domain, distinct prediction, and victory criterion before inspection; amend through Section 26.1 (Vehtari et al., 2017; Yao et al., 2018).

Give ordinary finite-horizon and urgency rivals the same horizon estimates, menus, and context as the candidate (Carstensen, 2006).

Use Section 16.4.3's branch-specific conditioned-path and endpoint-selection rivals under common endpoint and information rules (Doob, 1957).

Fit clinical and observation rivals jointly with trajectories under Section 16.6.2 (Rizopoulos, 2012; Kurland et al., 2009).

Compare habit, reinforcement, expected-value-of-control, avoidance-learning, and privacy-choice accounts using baseline behavior and changed-payoff tests.

Map each signature to its outcome channel and time scale. A change in one channel cannot confirm a restriction stated for another.

After the required gates pass, assign Open-set inadequacy if every named model fails the declared absolute-adequacy criteria, including required calibration, even when the candidate ranks first. An unsupported transport extension instead receives the corresponding scope or evidence status under Section 24.2.

16.7.3 — Component-Level Falsifiers

The horizon component fails when the filtered state cannot be estimated without future leakage, does not update coherently, lacks transport across prospectively specified windows, or adds no discriminating information beyond ordinary future-time

perspective, prognosis, urgency, and scarcity. An unstable horizon may remain descriptive, but it cannot carry the route claim. Test recovery in held-out participants, sites, and windows under observation shifts. Rival-generated worlds must control false attribution; ablations assess the added value of horizon, reserve, unresolved state, and their interactions beyond severity, frailty, treatment, prognosis, and observation intensity.

The reserve component fails when its proxies lack a declared measurement model linking their units to the tested reserve state, collapse into disease severity or prognosis, change meaning across sites, or add no held-out value once environmental support and treatment are represented. A reserve inferred from the residual it was built to explain is circular and receives no confirmatory status.

Boundary-rate and dispersion components fail when estimated exponents, drift restrictions, variance contraction, or reserve-horizon interactions do not survive alternative clocks, calibration versions, pseudo-endpoints, and ordinary regulatory rivals. A visually narrowing curve is insufficient when the protocol-fixed cross-equation restrictions do not hold.

Failed simulation recovery blocks inference from the affected covariance, innovation, quadratic-variation, or dependence diagnostic. An adequately resolved failure of a protocol-fixed held-out restriction counts against that component. Matching adequate conditioned-bridge or state-space rivals does not alone establish failure; apply Section 22.8.1's comparison rules, distinguishing exact equivalence from finite-sample resolution. First-moment movement cannot rescue a failed dependence prediction.

Terminal-window and intervention components are evaluated against prespecified windows, responses to unexpected endpoints, and models of treatment, support arrival, habit, and

observation change. Each conclusion states the modality supported by its frozen policy and continuation analysis.

Admissibility and ethical constraints are noncompensatory. The feasible action set must preserve the reason an option is unavailable, including time, capacity, information, danger, consent, and environmental access. When those constraints cannot be identified, or when the discriminating contrast would require harm, the component is blocked. These component results feed the frozen route-verdict logic that follows.

16.7.4 — Route Verdict and Communication

Freeze route-status thresholds, claim limits, and next actions under Sections 24.2 and 27.4 before results are known.

Identify whether horizon, reserve, and queue denote formal or model states, measured proxies, task manipulations, or illustrations (Vehtari et al., 2017; Yao et al., 2018).

Apply Chapter 28's terminology registry. A successor term must distinguish an unchanged formal object from a changed interpretation before inheriting earlier evidence.

16.7.5 — Public Status Labels

Every public phrase in this chapter needs a status label. Horizon may be a latent state, task manipulation, clinical belief proxy, or analogy. Reserve may be a candidate capacity construct, measured proxy, or formal variable. Queue may be a model state or illustration (Vehtari et al., 2017; Yao et al., 2018).

Status labels distinguish metaphor, measurement, task manipulation, clinical association, and identified mechanism. Keeping those roles explicit lets each contribution inform the next stage of research.

The labels also protect public communication. A social excerpt saying 'queue pressure rises near the end' would be

unsafe unless it states the claim status, population, task, and rivals. Otherwise it invites the impression of a hidden controller.

A good status sentence is short and local. It says what is being claimed, from what evidence, under what model, and what it cannot imply. Local status should be exact enough to stand on its own.

Diagrams and tables need the same discipline. A flow from horizon to action is a candidate pathway, not evidence. A simulated trajectory is a design tool, not a finding. A fitted model is not a verdict until recovery and rival comparison are complete.

The language rule is simple enough to audit: no status, no authority. If a phrase sounds powerful but cannot say whether it is analogy, model, estimator, or result, it should not be used.

Status labels belong in captions and summaries as well as body prose. A table can overclaim through its title even when its cells are careful. Terms such as observed queue pressure, predicted horizon collapse, and reserve signature are permissible only when observed, modeled, simulated, filtered, retrospective, or exploratory status is attached to the exact object.

16.7.6 — Metaphor Retirement After Failure

Metaphors should be revised when they imply an unestablished mechanism, moral price, deserved suffering, or practical control over future rewards. A declared accounting metaphor may still explain the net remainder required by exact PTVC. Retiring a failed route removes its unsupported mechanism claim; it does not erase a conditional arithmetic consequence of the target (National Academies of Sciences, Engineering, and Medicine, 2019).

A term earns continued use only if its operational meaning, observable consequences, and failure conditions can be stated. It

should be retired when it explains every outcome, cannot be distinguished from an ordinary rival, repeatedly creates moral misunderstanding, or has a clearer technical replacement.

Retirement may be audience-specific. Queue can remain inside a bounded task model while disappearing from captions and public summaries; a technical symbol can remain in an appendix while reader-facing copy uses a plainer noun. The distinction is between model utility and public authority, not between interesting and dull language. The audience-specific status should be recorded in the release registry so public materials cannot revive terminology intentionally confined to technical use.

Empirical failure also retires vocabulary. If queue pressure adds nothing beyond ordinary urgency, or reserve adds nothing beyond severity and prognosis, keeping the metaphor for intuition would continue to bias model construction. A failed route should lose both its status and the language that made it feel inevitable. Retired vocabulary should remain searchable in the audit so future authors can recognize the old concept even after reader-facing language changes.

Technical terms should identify their objects: filtered horizon, reserve proxy, admissible subset, unresolved state, marginal value, rival model, and failure condition. Public balance language can remain when its conditional net meaning is clear.

Neither vocabulary establishes a personal forecast, moral obligation, or chosen future reward. Where a term has both a public and technical use, the chapter should preserve a stable mapping between them so simplified language cannot drift into a different claim.

Retirement should be recorded in the scientific decision and version ledger with the reason, permitted technical uses, prohibited public uses, dependent artifacts, and any approved

successor term. That record prevents later chapters or public materials from reviving discarded language without reopening the evidence. Successor terminology should not inherit favorable evidence automatically; it must show which formal object is unchanged and which interpretation has been revised.

The chapter’s final test is whether each retained term points to an observable or formal object. Clear vocabulary gives experts a common basis for evaluation. Chapter 17 carries the resulting variables into longer records, where partial history, attrition, treatment, and observation are modeled directly. This object-level discipline makes later longitudinal tests easier to compare because a changed label cannot conceal a changed estimand.

Chapter 17 — Longitudinal and Life-Course Designs

A long record contains two histories: the participant's changing circumstances and the study's changing access to them. A quieter series may reflect recovery, fewer answered prompts, altered medication, or a new device algorithm. Extending follow-up does not by itself distinguish these explanations. Longitudinal design must preserve the evidence that locates each change in the person, the measurement system, or both.

Entry dates, missed reports, nonwear, treatment transitions, and exits determine which intervals become observable. These processes can depend on the state being studied: distress may prompt contact, while fatigue or declining communication can prevent it. The resulting trajectory must be analyzed together with its observation history (Shiffman et al., 2008; Hernán & Robins, 2020; Little & Rubin, 2019).

The start of observation also limits what can accumulate. A study may estimate a post-diagnosis or treatment-window contribution while leaving the earlier state unknown. More accurate follow-up can improve the increment without resolving that initial uncertainty. Extending the inference requires independently justified historical information or a dynamical bridge, whose assumptions must remain separate from the newly collected measurements.

Time resolution determines another boundary. Widely spaced observations can miss a short medication crisis or a reversal between visits. A smooth fitted path does not exclude those events. Independently justified bounds on rates of change can constrain the unseen contribution; without such information, the apparent continuity of the graph may exceed what the record establishes.

Different longitudinal questions therefore require different comparisons. Forecasting this participant's next interval tests temporal generalization; predicting a new participant tests

another population relation. A within-person association may differ from the association between people. Combining these targets can reverse the interpretation even when every recorded value is correct. The validation split must follow the question that the study intends to answer.

Long follow-up makes measurement an ongoing intervention. Prompts consume attention, devices may become uncomfortable, and feedback may change behavior. A pause can protect sleep or privacy while also changing the observed sample. These events belong in both the burden record and the observation model; recording them does not require participants to justify a refusal.

Clinical events further change the domain of inference. Treatment may affect experience and report availability together. Transfer or withdrawal can end an observation route without ending the stream, while an outcome defined only during life has no post-death value to impute. Event order and the population still observable at each time determine which trajectory a comparison describes.

The chapter connects these problems to concrete designs and estimators. Its aim is to recover supported temporal relations while showing where records remain partial, scales have changed, or causal comparisons lack identification. The same time-indexed structure that supports inference also makes a record identifying, so permissions, participant return, and the eventual end of the study require temporal boundaries.

17.1 — From Moments to Life-Course Inference

Repeated diaries, wearable signals, pain ratings, and clinical notes cover different parts of a life at different resolutions. Before selecting observations, define the longitudinal estimand and the assumptions connecting those records to it (Shiffman et al., 2008; Bolger & Laurenceau, 2013). Dense local sampling may identify short-term dynamics while leaving earlier history, sparse gaps, and life-stage transport unresolved. The design must preserve those limits when moving from observed moments to a longer trajectory.

17.1.1 — Existing Data and Sampling Design

Emotional development already provides an encouraging empirical starting point. Carstensen et al. (2011) sampled feelings five times daily for a week and repeated the procedure approximately five and ten years later; the three waves included 184, 191, and 178 adults. Greater age was associated with more positive and stable emotional experience and greater coexistence of positive and negative emotions. This gives the life-course program an observed pattern of affective organization to explain, extending beyond a static questionnaire snapshot. The appropriate comparison is among models of regulation, goals, context, selection, and development. Repeated sampling also separates the frequency, persistence, and co-occurrence of feelings within persons, providing a practical design for following major life transitions.

Section 17.1 matches designs to questions: a method suited to lagged within-person dynamics may be unsuitable for historical reconstruction, age transport, or acute event response.

Intensive longitudinal or ESM designs examine change within a person over hours or days, capturing lags, recovery, cycles, and context. Prompt reactivity, fatigue, and short windows limit

their conclusions about within-person dynamics and pilot feasibility.

Cohort-sequential designs add overlapping age and cohort observations to examine age, period, and cohort patterns. Separating those effects still requires identifying assumptions, with cohort confounding and invariance central to any life-stage transport claim.

Accelerated longitudinal designs combine overlapping windows to approximate longer development while reducing the wait for age coverage. Their long-horizon bounds depend on evidence that the segments can be linked.

Event-based or case-crossover designs examine local changes around a specified event while controlling stable person characteristics. Anticipation, carryover, and event timing constrain their interpretation of transient responses.

An N-of-1 or idiographic design asks about the process law for one stream and centers person-specific dynamics. It can update individual process hypotheses, while limited transport and participation burden constrain the design.

A clinical cohort examines changing trajectories under care through ethically available repeated measurements. It can update clinical component tests when care confounding and truncation by death are represented.

Registry-linked designs use the scale, duration, and event history of institutional records to estimate background rates or check transport. Consent boundaries and coding drift limit what those records can establish.

Within-between decomposition separates person-specific change from cross-person association.

$$y_{\mathrm{it}} = \alpha_i + \beta_W \left(x_{\mathrm{it}} - \bar{x}_i\right) + \beta_B \bar{x}_i + \varepsilon_{\mathrm{it}}$$

Here β_W describes the within-person association and β_B the between-person association. A hierarchical interpretation writes $\alpha_i = \alpha + u_i$ and specifies the needed exogeneity assumptions, for example $\mathbb{E}[u_i \mid \mathbf{x}_i] = 0$ and $\mathbb{E}[\varepsilon_{it} \mid \mathbf{x}_i, u_i] = 0$, where $\mathbf{x}_i$ denotes the person's covariate history. Sufficient within-person and between-person variation is also required. With unrestricted person fixed effects, the person mean is collinear with those effects, so β_B is not separately identified. The two associations can differ or reverse; a causal interpretation requires additional design assumptions. For prospective prediction, person means and other history summaries use only information available before the held-out outcome.

Recovery is a measurable possibility with more than one trajectory. In a prospective bereavement study, 205 people were assessed before spousal loss and at six and eighteen months afterward; the largest classified group, 45.9%, maintained comparatively low depressive symptoms, while other groups showed recovery or persistent distress (Bonanno et al., 2002). This finding gives an evidence-based reason to investigate resilience without asking every bereaved person to follow the same timetable.

Disability research further improves the question by separating degrees and kinds of adaptation. Oswald and Powdthavee (2008) estimated partial recovery in mental well-being, with illustrative estimates of approximately 30–50% depending on severity. Lucas (2007b) found substantial lasting changes in two disability-onset samples. More recently, Köhler and Vanhoutte (2024) distinguished transient from persistent functional limitations in European longitudinal data: recovery toward initial well-being was clearer after transient limitations, and the pattern differed across depressive symptoms, life satisfaction, and quality of life. Together these results support an ambitious,

humane target: identify which resources and mechanisms sustain or restore well-being, while modeling persistent trajectories explicitly.

An idiographic analysis targets one stream's dynamics and requires enough repeated observations to estimate its lags and volatility. Forward-chaining validation within that person is the relevant test.

A hierarchical analysis combines shared and person-specific dynamics using repeated observations with comparable constructs. Evaluate calibration and prediction for held-out persons.

A transport analysis requires a defined target population, bridge variables, and overlap samples connecting source and target. Use site holdouts and target-site calibration to evaluate the intended generalization.

Federated analysis supports multisite learning without pooling records. It requires a common dictionary, local feature extraction, disclosure checks, site-level reproducibility, and audit logs.

17.1.2 — Intensive Longitudinal Targets

Specify the within-person variable, clock, lag, and target decision, retaining enough individual history to separate change from stable between-person differences (Shiffman et al., 2008; Stone et al., 2007).

Retain person-specific deviations in sleep, pain, repair, activity, and momentary valence, alongside observation, version, care, and participation changes.

Imagine an evening prompt series in which one caregiver's sleep loss reliably precedes next-day distress, while the cross-sectional group average suggests no association. That divergence is scientifically informative. It shows why the

analysis must preserve the individual's temporal sequence before pooling, and why the group result cannot be used to overwrite a process that occurs within one stream.

Forecast later points within a person using forward-chaining blocked time folds; use leave-person-out tests for new-person generalization. Prevent neighboring observations from leaking across training and test sets.

A substantial starting resource is already available. The July 2026 openESM journal article describes 60 openly available experience-sampling datasets with more than 16,000 participants and more than 740,000 observations, accessible through searchable metadata and R or Python tools. Its authors' analysis of 39 datasets found a pooled within-person concurrent correlation of −0.49 between positive and negative affect (95% CI [−0.54, −0.42]) (Siepe et al., 2026). That result supplies a reproducible description of how positive and negative affect co-occur within persons.

Select existing datasets by timestamps, cadence, scale content, context, and access before outcome inspection. Develop the measurement model on one set, freeze it, and test whole held-out studies while retaining study-specific scale and timing meanings.

17.1.3 — Sparse Windows and Unobserved Time

State the regularity, smoothness, or state-space assumptions connecting sparse windows and test conclusions at resolutions the cadence supports (Bolger & Laurenceau, 2013; Little & Rubin, 2019).

A quarterly clinic series may miss a two-week medication crisis. Compare admissible smoothness classes, worst-case bounds, and interval holdouts; widen uncertainty over unobserved intervals.

Hold out whole intervals to test the assumed gap-bridging regularity; predicting an isolated visible point does not validate the unseen path.

Report interval contributions at the resolution supported by cadence and the declared regularity class. Carry disagreements in sign, peaks, or cumulative contribution across admissible gap paths into the identified set.

17.1.4 — Unknown Prior History

At delayed entry, distinguish the observed increment from the unknown initial state. Choose a post-entry estimand, independently bounded initial state, or longer design before analysis (Hernán & Robins, 2020; Gerstorf et al., 2013).

Bookkeeping identity for delayed entry.

$$L_i(T) = L_i(t_0) + \Delta L_i(t_0, T)$$

$$\Delta L_i(t_0, T) = \int_{t_0}^{T} g_i(s) F_i(s) ds$$

Under the study's measurement model, combining the post-enrollment increment with an independently known or bounded initial ledger state yields the corresponding estimate or bounds for the cumulative state.

Vary the independent initial-state bounds and use leave-window-out checks to show how the conclusion depends on unobserved history.

If plausible initial states reverse the conclusion, the result is conditional on the initial-state model. Additional post-enrollment observations do not resolve that uncertainty unless an independently justified measurement or dynamical bridge identifies or bounds the missing prefix.

17.1.5 — Event-Time Alignment

Calendar, age, diagnosis, treatment, enrollment, and death alignment define different conditioning and risk sets. Retain unwarped timestamps and test each permitted transformation (Shiffman et al., 2008; Stone et al., 2007).

The study record needs death-alignment simulations, dynamic time-warping checks, and sensitivity to clock choice.

Backward alignment to death can create apparent convergence when the cohort increasingly consists of people with gradual, observable declines. Sudden deaths, early transfers, rapid loss of communication, and short enrollments may disappear from the later portions of the curve. A death-aligned pattern must therefore be shown beside its changing risk set and beside alternative clocks that do not condition on the same future event.

If direction or magnitude changes materially across allowed clocks, report the event-relative estimand; do not infer a common native dynamic from the preferred alignment.

17.1.6 — Local Versus Whole-Life Claims

Match scope to design: dense cadence and short lags for fast change; bridge variables for transport; endpoint and observation models for terminal windows; initial-state and completion bridges for whole-stream inference.

In the recovery suite, hold the observed late window and cadence fixed while varying hidden initial residuals across sign and magnitude. Retain the hidden generator values for evaluation without supplying them to the fitted procedure. If observationally identical records permit different whole-life conclusions, a decisive verdict that ignores that nonidentification fails the test; the same justified abstention across those records does not.

Separately vary declared initial-state assumptions or bounds and record how the conditional conclusions change. This sensitivity analysis shows which restrictions on unknown history drive the result.

17.1.7 — Transport Across Life Stages

Using the same questionnaire, sensor, or model across life stages does not establish transport. Development, aging, disability, illness, communication mode, culture, and care setting can alter item meaning, response burden, physiological baselines, and the relation between an auxiliary channel and lived experience. A transported estimate therefore requires evidence that the construct, origin, sign, unit, and measurement map remain adequate for the stated purpose (Hernán & Robins, 2020; Gerstorf et al., 2013).

Cohort-sequential and accelerated longitudinal designs can create overlap among ages or disease stages, but the overlap must contain common anchors and enough heterogeneity to test the link. A model trained in one stage should be challenged in another without silent recalibration after the result is known. If a new stage requires a new scale, the study has produced a new local claim rather than transported the old one.

Language and mode can change meaning even when wording appears simple. Cognitive interviews, bridge samples, assisted-communication records, mode-specific calibration, and participant debriefs help distinguish sincere response change from a failed linking assumption. These are inferential requirements, not accommodations outside the science.

Predictive portability is weaker than construct transport. A model may preserve rank while losing a neutral origin, interval unit, sign interpretation, or causal meaning. Reports should state

which property traveled and withdraw any interpretation that depends on a bridge the data did not support.

Longitudinal designs carry distinct clocks for age, calendar, disease, treatment, enrollment, adjudicated events, and perceived horizon. Declare a primary clock and use others as covariates, realignments, or sensitivity structures. An unnamed blend can create a trajectory with no coherent estimand.

When an observational cohort emulates intervention, first define the target trial: eligibility, strategies, assignment time, follow-up, outcome, contrast, and analysis. Emulation clarifies timing and avoids preventable bias but cannot create exchangeability or recover unmeasured indications (Hernán & Robins, 2020). A clear target-trial specification also exposes which quantities remain unidentified from the observational data before causal estimates are computed.

Time-varying treatment, health status, observation, and prior experience can affect one another. Ordinary adjustment may block the pathway or create bias, so use causal language only when temporal ordering and assumptions support it. Descriptive life-course work remains valuable otherwise. Sensitivity analyses should distinguish unmeasured confounding from measurement error and informative observation, since each requires a different repair.

Transport requires prospective overlap, common anchors, bridge evidence, mode and language records, held-out target-population performance, and a stopping point. A result may stop at a life-stage or care-setting boundary without diminishing the dignity or scientific importance of those outside it. Stopping transport at a justified boundary creates a concrete next study: test the missing bridge rather than simply enlarge the original sample.

17.2 — Active Reports and Experience Sampling

Active reports provide timed evidence but also consume attention and can change the state being reported (Shiffman et al., 2008; Trull & Ebner-Priemer, 2020). Prompt content, cadence, recall period, language, and accessibility therefore belong to the measurement specification. Compliance is not simply a participant trait: burden, pauses, and changing circumstances influence which reports exist. Calibration and drift checks must evaluate this reporting process alongside the answers it produces.

17.2.1 — Prompt Content and Cadence

Contemporary ecological momentary assessment illustrates the value of within-person, age-sensitive measurement. A 2025 study of young, middle-aged, and older adults combined repeated daily reports with an implicit measure of valence representation. Its comparison identifies age and implicit-explicit disagreement as measurement features worth examining directly (Nigam & Kar, 2025; Shiffman et al., 2008; Trull & Ebner-Priemer, 2020).

Choose the least intrusive cadence that resolves rapid transitions, daily summaries, event responses, or slower adaptation, with the question's reporting frame fixed.

A clinically grounded starting point already exists. Geeraerts et al. (2024) developed and content-validated the ESM-AC questionnaire with forty-three people living with advanced breast or lung cancer and eight health-care professionals. Its digital core contains thirty-one momentary, two morning, and seven evening items; ten participants in the usability round gave ease of use a mean rating of 4.5 out of 5. This work shows how patient input can turn intensive measurement into a practical, context-sensitive instrument. A PTVC-linked pilot could build

on that development process while separately testing sampling burden, temporal validity, and the proposed valence construct.

The subsequent seven-day ESM-AC study recruited 40 of 79 people with advanced breast or lung cancer approached through two Belgian hospitals. Thirty-seven participants provided 1,703 valid assessments, 71% of the 2,400 scheduled. On seven-point scales, mean reported burden was 2.1, ease of use 5.6, and measurement reactivity 1.3 (Geeraerts, Pivodic, et al., 2026). These results provide a practical precedent for repeated measurement in comparable patients. Eligibility restrictions, including functional status, limit their extension to more severely ill populations.

A companion analysis of the same study used 1,676 assessments from 36 participants. Retrospective scores correlated with participant-level summaries of momentary ratings, with Kendall’s tau values from .24 to .70, yet identical retrospective scores could conceal different momentary trajectories (Geeraerts, de Nooijer, et al., 2026). This supports preserving temporal information when the research question concerns accumulation or recovery. The two analyses contribute complementary evidence from one cohort; their questionnaires still require separate qualification as measures of signed phenomenal value.

A naturalistic study shows what event-sensitive sampling can add. Villano and colleagues followed university students across an academic term, recording expected examination grades before results were released and sampling affect intensively afterward. The analyzed sample comprised 249 students, divided into discovery and confirmatory cohorts. Models including the difference between expected and received grades described subsequent affect better than grades alone, while modeling positive and negative emotion separately (Villano et al., 2020).

This gives the program a concrete design precedent: measure expectations before outcomes, preserve separate affective components, and select a cadence that resolves their recovery. The study supports prospectively ordered observation of real emotional events.

The prompt pool, randomized windows, quiet hours, maximum daily count, burden stop rule, minimum usable cadence, and analysis of schedule effects are fixed prospectively. Planned cadence variation can estimate reactivity and fatigue.

Kaurin, Vize, and Wright (2026) studied naturally occurring stressful or unpleasant events in a clinical and community sample of 248 participants. Their microburst protocol combined some retrospective event-related ratings with follow-up momentary assessments at 15-minute intervals within an hour. Among the tested trajectory models, piecewise forms separating early from later change best fit both positive and negative affect. This supplies a concrete cadence target: resolve changes in recovery rate rather than assume one homogeneous slope.

Existing-data downsampling can compare which trajectory distinctions survive less frequent observation before new collection is proposed. Preserve the distinction between retrospective and momentary ratings and model the event-triggered observation rule; recovery-curve precision does not establish an unbiased time integral. Annoyance produced by prompting is itself experience and belongs to the measurement intervention’s effects, not to a category outside experience.

Randomized or staggered prompting, when ethically permitted, can estimate measurement reactivity. It is not automatically a negative control: that designation additionally requires the justified causal-null relation described in Section 7.5.5. Evaluate resolution, reactivity, and missingness together,

choosing the least intrusive schedule adequate for the stated target under Chapter 3.

Sampling error can be bounded before a smooth curve is fitted. In a declared exposure clock, suppose an independently justified constant $K > 0$ bounds rate changes by $|v(s) - v(t)| \le K|s - t|$. For an unsampled interval of length h with exactly known endpoint rates a and b, feasibility requires $|b - a| \le Kh$. The attainable integral interval is

$$\int_0^h v(s)\,ds \in \left[\frac{h(a+b)}{2} - \Delta,\ \frac{h(a+b)}{2} + \Delta\right], \qquad \Delta = \frac{Kh^2}{4} - \frac{(b-a)^2}{4K}.$$

The extremal paths follow the steepest allowed slopes from the two endpoints, so these bounds are sharp under the stated assumptions. Two zero samples with $K = h = 1$ permit any intervening total from $-1/4$ to $1/4$. Across exactly observed common endpoints, the interval bounds add. Equal exposure spacing minimizes the worst-case missing-area width for a fixed span and number of gaps; burden and statistical error remain separate design considerations. Noisy endpoint readings require simultaneous uncertainty bounds, and unobserved prefix remains additional uncertainty. Without an independently qualified regularity or magnitude bound, sparse samples alone cannot exclude an unseen excursion. A synthetic downsampling challenge can determine whether the proposed cadence resolves the required integral before recruitment.

17.2.2 — Momentary Versus Retrospective Questions

Momentary questions concern experience near report time; retrospective questions measure a later construction shaped by memory, salience, narrative, and interval choice. Keep their observation models and version flags distinct (Shiffman et al., 2008; Stone et al., 2007).

Fix current and retrospective wording, reporting windows, recall delays, and disagreement analyses. A weekly evaluation cannot replace missing momentary reports solely because the response scales match.

A late-night report of dread and a Sunday judgment that the week was meaningful need not contradict one another. They can reflect different moments, memory weights, and evaluative frames. Modeling the divergence can reveal recency, peak, recovery, or narrative effects that would disappear inside a single average.

17.2.3 — Compliance and Reactivity

Model response probability and timing with reported values: distress, exhaustion, anticipated prompts, or supportive contact can change participation and experience together (Little & Rubin, 2019; Nosek et al., 2018).

Contact-seeking responses during difficult evenings can overrepresent distress; inability to respond during distress can underrepresent it. Model both directions before interpreting the observed mean as a trajectory.

Assess selection from excluding low-compliance participants. Restrict the reported population when conclusions depend on unsupported similarity between observed and unobserved participants.

Response probability, response latency, prompt history, and planned cadence variation should enter the analysis before substantive dynamics are interpreted. The study should report whether conclusions survive inverse-probability, joint-process, and mark-restricted analyses. When the answer depends on treating response as random, the verdict remains conditional on that assumption.

17.2.4 — Participant-Controlled Pauses

Distinguish temporary pause, skipped prompt, device removal, technical failure, and withdrawal. Participants choose whether to disclose reasons for pauses protecting privacy, sleep, family time, care, or emotional stability (Shiffman et al., 2008; Trull & Ebner-Priemer, 2020).

The pause mechanism is designed before enrollment, available without penalty, and linked to immediate stopping of prompts and sensors where feasible. Analyses treat paused time as unobserved under a declared mechanism rather than as neutral experience.

17.2.5 — Language and Accessibility

Qualify translation, literacy, assisted communication, sensory access, cognitive load, and cultural framing as measurement versions. Developmental context also changes feasibility and channel burden (International Test Commission, 2018; Sosa-Hernandez et al., 2024).

Illness may change an item's meaning; voice prompts may increase exposure; assistance may alter privacy and timing. Test how those changes affect response and construct interpretation.

17.2.6 — Calibration and Drift

Calibration can drift while displayed numbers remain stable. Treatment, grief, illness, repeated prompting, new vocabulary, trust, fatigue, and changing reference points can alter scale use, so repeated anchors and prospectively specified response-shift checks replace indefinite reliance on one baseline (Little & Rubin, 2019; Nosek et al., 2018).

Preserve anchors, vignettes, overlap, brief debriefs, response latency, and linking results. Recheck around medication, disease progression, relocation, bereavement, communication change,

and long participation. Meaning shifts require segmentation, a validated bridge, wider sets, or withdrawal of the common-scale claim.

Recent experiments make these design choices measurable. In a twenty-eight-day factorial study of 205 participants, fixed rather than random prompts produced lower measured affect variability in eight of sixteen models; associations with next-day health behaviors were not reliably changed (Langford et al., 2026). A separate fourteen-day study comparing seven-point Likert responses with visual analogue scales in 119 participants found differences in means, floor responses, and some correlations, but no reliable difference in within-person standard deviations or successive-change variability (Haslbeck et al., 2025). The practical gain is a stronger instrument: freeze the schedule and format, test their effects, and estimate transport when either changes. A smoother observed trajectory can then be assessed against measured design effects before it receives a latent interpretation. Design-effect estimates should travel with the instrument version so later studies can adjust expectations when prompt schedule or response scale changes.

17.3 — Passive Signals, Wearables, and Context

Passive heart-rate, movement, sleep, temperature, oxygenation, and context records extend coverage only with their gaps and device conditions represented (Onnela & Rauch, 2016; Shcherbina et al., 2017).

17.3.1 — Physiological Streams

Qualify heart rate, electrodermal activity, movement, respiration, temperature, and sleep estimates under the recorded illness, medication, posture, motion, fit, and context (Onnela & Rauch, 2016; Shcherbina et al., 2017).

Model channel-specific lag and aggregation. Attach quality, treatment state, and sampling density so rapid autonomic responses, nightly sleep, slow frailty, and treatment effects retain their own time scales.

Preserve disagreement between reduced wearable arousal and continued reported distress. Test channel lag, sedation, motor suppression, altered interoception, and multidimensional experience as explanations.

17.3.2 — Behavioral and Environmental Context

Use StudentLife to compare sleep, activity, social context, workload, and assessment availability against the candidate trajectory under common participant splits and scores. A stable added route contribution warrants broader replication.

Collect the smallest feature set and coarsest resolution that distinguish the specified mechanisms, with contextual-integrity and attack-surface review (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Activity, light, noise, care setting, social presence, and coarse location may enter as time-varying covariates, marks, or stratification variables. Each requires a declared temporal

resolution and purpose. A variable gathered for one explanation cannot later become a general map of private life without new authority and a new scientific justification.

Distinguish restorative rest from forced immobility using permitted reports and context. Movement alone identifies neither its cause nor felt value; retain unknown context when necessary.

17.3.3 — Sensor Fusion and Channel Conflict

StudentLife offers a smaller complementary benchmark combining smartphone sensing, momentary assessments, and survey outcomes across a 10-week term in 48 Dartmouth students. The original study reported associations among sensed behavior, mental-health measures, and educational outcomes, and released its dataset for reuse (Wang et al., 2014).

Apply Section 9.3.6's joint-measurement requirements to the longitudinal channels (National Institute of Standards and Technology, 2020; Rieke et al., 2020).

The fusion architecture, held-out channels, quality thresholds, missing-channel behavior, and conflict outputs are prospectively fixed. A model may predict a qualified target, but it may not redefine that target from whichever channels remain available.

The output should show which channels contributed, which disagreed, and how uncertainty changed when each was removed. Held-out-channel validation tests whether one channel predicts another without being allowed to dominate it.

17.3.4 — Device Removal and Nonwear

Device removal and nonwear are observation events with potentially informative causes. Privacy, discomfort, skin injury, charging, illness, delirium, family presence, care transition, forgetfulness, and simple annoyance all produce different missingness mechanisms. The device record should preserve

these causes when the participant chooses to provide them and otherwise retain an explicit unknown mark (Onnela & Rauch, 2016; Shcherbina et al., 2017).

The protocol should define nonwear detection, participant removal controls, charging windows, skin-safety checks, and the distinction between technical absence and refusal. It should also state when sensing stops automatically, how data already collected are handled, and how nonwear enters the primary and sensitivity analyses.

Do not interpolate across nonwear as though removal for discomfort or intrusion were random. Preserve it as possible burden, state change, or technical failure.

A candidate signature that appears only after high-burden nonwear is coded as random has failed an observation-process test. Show how results change under cause-specific and unknown-cause missingness scenarios, and classify disagreement under Section 24.2.

17.3.5 — Vendor and Firmware Drift

Passive evidence can extend life-course observation when fatigue, recall, speech, delirium, sleep, or disability limit active reporting. Work with people receiving outpatient palliative care and family caregivers found wearable-triggered momentary assessment practicable in a bounded clinical setting. Building on that result, a study can record who can wear, respond, pause, and withdraw and evaluate how those patterns affect its measurements (Schuler et al., 2023).

Use overlap, calibration fixtures, dual-device wear, and version-boundary diagnostics to separate participant change from sensor, firmware, algorithm, or preprocessing change. Verification, analytical validation, and clinical validation apply

to specific versions (Bent et al., 2020; Goldsack et al., 2020; National Institute of Standards and Technology, 2020).

A firmware update that reduces detected awakenings at every site on the same date should trigger version-boundary checks before being interpreted as improved sleep.

Major technical changes require predeployment bridge plans with parallel measurements, version-specific quality thresholds, and a reanalysis trigger. Silent updates are incompatible with longitudinal inference. If the bridge is inadequate, segment the time series and withdraw the common-scale claim.

17.3.6 — Minimization and On-Device Processing

Test whether on-device features, coarse summaries, bursts, or site-local analysis preserve the estimand while reducing disclosure (National Institute of Standards and Technology, 2020; Rieke et al., 2020).

A device may transmit approved sleep regularity while retaining location and audio locally. The transmitted feature still requires provenance, quality checks, and deletion rules.

Specify the minimum signal for each estimand and require review before expansion. Use temporary richer audit samples to validate reduction without retaining them indefinitely.

A minimized pipeline must preserve prospectively specified model comparison across relevant states, sites, and devices. If reduction removes identifying information, the question narrows rather than the surveillance surface expanding.

17.4 — Marked Observation Processes

A marked observation process records when measurements occur and which events accompany them, alongside their values (Andersen et al., 1993; Rizopoulos, 2012). Scheduling, refusal, device removal, care interruption, and technical failure can carry information about the evolving state. Joint models and sensitivity analyses must therefore explain presence and absence rather than assuming that the observed sequence represents all intervals equally. An observation-process failure can defeat an apparent dynamic pattern even when the recorded values are precise.

17.4.1 — Observation Intensity $\lambda_i(t)$

Continuous-time state-space models are appropriate when irregular observation times and latent dynamics matter. Inserting artificial missing rows must not silently replace the actual timing with regular intervals.

Dynamic structural equation models address lagged within-person relations and person differences. Their interpretation must distinguish within-person effects from between-person effects.

Observation intensity describes the rate at which a person becomes visible to the study. It may depend on prior reports, care setting, device state, staff availability, medication, distress, or participant choice, so it cannot be treated as an administrative schedule outside the outcome model. The protocol should estimate or bound that rate on the information available at each time (Andersen et al., 1993; Rizopoulos, 2012). The displayed rate describes the absolutely continuous component of a simple observation process. Fixed scheduled opportunities are conditioned on as design points, with response and mark

probabilities modeled at those points; scheduled atoms are represented separately.

Candidate observation-intensity model under recorded prior history.

$$\lambda_i^{obs}(t \mid H_i(t^-)) = \lim_{\Delta t \downarrow 0} \frac{Pr\{N_i(t+\Delta t) - N_i(t) = 1 \mid H_i(t^-)\}}{\Delta t}$$

Evaluation should compare outcome models with and without the observation-rate model. If the apparent dynamic weakens once intensity is included, the result is sensitive to the observation model; an observation-artifact interpretation requires the declared identification and model checks.

17.4.2 — Observation Marks

Gaussian processes can flexibly smooth nonlinear or nonstationary trajectories. Check whether that smoothing hides event timing or the missingness process.

Distinguish scheduled, participant-initiated, and staff-triggered reports from refusal, device failure, removal, and clinical interruption (Little & Rubin, 2019; Scharfstein et al., 1999).

Compare otherwise similar pain scores across crisis, scheduled-visit, device-alert, and self-initiated marks to test whether apparent change depends on the measurement pathway.

Observation marks remain at the datum level in a versioned dictionary, including unknown and mixed categories. Categories may be collapsed only when simulation and sensitivity analysis support preservation of the protocol-fixed interpretation across the declared scenarios and tolerances.

Model intensities for unscheduled events and response or mark probabilities at fixed scheduled opportunities. Test event timing and mark type jointly, including self-initiated reports,

clinician-triggered assessments, and device alerts (Andersen et al., 1993).

17.4.3 — Informative Presence and Absence

Display visibility by lucid, supported, comfortable, distressed, and other relevant states before interpreting trajectories (Røysland, 2011; Rubin, 1976).

The analysis should preserve response, nonwear, visit, transfer, and interruption processes separately, then specify which may depend on observed history and which may depend on unobserved current state. Pattern-mixture, selection, and shared-parameter formulations represent different observation worlds. Agreement across them is evidence of robustness; disagreement defines the uncertainty.

17.4.4 — Joint Mark and Outcome Models

Functional data analysis can summarize sparse curves or their principal components. Its regularity assumptions determine which features are recoverable.

Joint longitudinal-event models address dependence among outcomes, observation, death, and dropout. Define the scientific endpoint independently of fitting the event model.

Specify outcome, observation-count, mark, and shared-latent processes separately in the joint model (Andersen et al., 1993; Rizopoulos, 2012).

The chosen model must match the estimand. Forecasting the next observed report, estimating a latent trajectory under informative visits, and comparing mark-specific outcomes require different likelihoods and assumptions. Simpler outcome-only and mark-only artifact models remain in the tournament so that added complexity is not rewarded merely for absorbing the observation process.

Joint observation-outcome scaffold. Role: candidate analysis model linking measured outcomes, observation opportunities, and declared event marks to examine informative visibility. In the next two probability displays, the history argument means the history immediately before the event, following Section 17.4.1. It excludes the current increment and mark; the mark distribution separately conditions on the event occurring. The rate approximation applies to the continuous component specified in Section 17.4.1.

$$y_{ij} = m_i(t_{ij}) + \varepsilon_{ij}$$

$$\Pr\{dN_i(t) = 1 | H_i(t)\} = \lambda_i^{obs}(t)dt + o(dt)$$

$$\Pr\left\{M_{ij} = k | dN_i(t_{ij}) = 1, H_i(t_{ij})\right\} = \pi_{ik}(t_{ij} | H_i(t_{ij}))$$

The recorded observation opportunities remain distinct from the model for their probability or intensity. A joint observation–outcome model may allow that intensity to depend on latent experience or shared clinical state under explicit identification and sensitivity assumptions. This dependence is not circular by itself. Circularity arises when outcome-dependent inclusion, endpoint selection, or tuning forces the signature later presented as evidence.

17.4.5 — Sensitivity to Missing-Not-at-Random

Hidden-state or semi-Markov models address regime changes and state durations. Recovery checks are necessary before latent states are given substantive interpretations.

Specify plausible clinical and behavioral mechanisms for missing-not-at-random sensitivity (Little & Rubin, 2019; Scharfstein et al., 1999).

Fix model families, tipping parameters, channel-specific scenarios, and status consequences under Section 24.2. Set

interpretable ranges for pain, relief, fatigue, privacy, sedation, fear, device intolerance, and family-protection mechanisms before inspection.

The sensitivity parameter should have an interpretable scale, such as the change in response odds or latent outcome associated with an unobserved state after conditioning on measured history. The analysis then varies that parameter over a clinically justified region and records where sign, equivalence, model choice, or decision status changes.

Display the prespecified plausible sensitivity region and the boundary where the component verdict changes. Conclusions remain conditional on that justified region (Little & Rubin, 2019; Scharfstein et al., 1999).

17.4.6 — Observation-Process Falsifiers

Observation-process falsifiers test whether the measurement process explains the proposed pattern. Mark restriction, intensity adjustment, simulated shutdown, nonwear scenarios, staff controls, device strata, and prespecified missing-not-at-random sensitivity analyses belong in the confirmatory package (Røysland, 2011; Rubin, 1976).

Repeat the observation-process checks in high-coverage intervals.

Report the frozen falsifiers regardless of direction. Apply Section 24.2 when observation models remain indistinguishable. Nonidentification of the observation model limits adjudication; it is not evidence for the preferred latent process.

A joint model should reproduce outcome and observation. Checks compare visit intensity, missing runs, marks, terminal dropout, and covariate-dependent response; fitting values without who was observed and when is not recovery (Little & Rubin, 2019; Rizopoulos, 2012).

If visibility failure is site-specific, redesign can target workflow or devices. Holding the fitted latent model fixed while improving observation reliability tests the estimate’s sensitivity to the observation process. A changed estimate supports an observation-artifact interpretation only when the design and identification checks separate visibility changes from changes in the underlying process.

17.5 — Clinical Cohorts and Natural Experiments

Clinical cohorts provide repeated observations around illness, treatment, care transitions, and death within ethically available care (Hernán & Robins, 2020; Lipsitch et al., 2010). Their changes are informative but are not automatically randomized interventions. Treatment selection, heterogeneity, observation loss, and the timing of natural events determine whether a question is predictive or causal and which assumptions it requires. Care decisions remain independently justified throughout the comparison.

17.5.1 — Prospective Clinical Cohorts

Before follow-up, specify the clinical population, entry criteria, baseline, treatment context, clocks, observation marks, and component estimand (Hernán & Robins, 2016, 2020).

A palliative cohort can follow sleep, pain, medication, reporting, and care while leaving treatment to clinicians. Model selection through enrollment, prognosis, resources, and communication ability.

Public writing supplies another observational window on terminal illness. Goranson et al. (2017) analyzed 2,616 posts from twenty-five people blogging about terminal cancer or amyotrophic lateral sclerosis. An exploratory analysis within these blogs found that positive-emotion words increased as death approached, while negative-emotion words did not show a significant increase. This supplies an observed positive pattern worth following up with direct, voluntary measures of current experience. The bloggers were a selected group able and willing to write publicly; their language could reflect coping, relationships, audience, and meaning-making as well as felt state.

Model cohort entry and continued observation with outcomes under explicit identification and sensitivity assumptions. If adequate mechanism and sampling explanations remain observationally equivalent, report Model-ambiguous.

Keep research additions within care limits. Separate routine visits, symptom contacts, remote prompts, and emergencies as observation opportunities. If their timing or burden prevents interpretation, narrow the cohort or estimand.

Existing longitudinal studies show why age and remaining time should be modeled separately. Among 1,637 deceased older adults in the German Socio-Economic Panel, distance to death described life-satisfaction change better than chronological age; the estimated decline accelerated roughly four years before death (Gerstorf et al., 2008). This supplies a quantitative timing comparison for models that also include health, care, social participation, and report availability. A prospectively specified candidate can be tested for additional predictive value alongside these measured variables.

17.5.2 — Treatment Changes

Treatment changes are time-varying exposures that may alter experience, physiology, report capacity, and the probability of observation at once. Medication, therapy, procedures, oxygen, sedation, counseling, and symptom control therefore require precise timing and role declarations as exposure, confounder, mediator, observation modifier, or competing event (Pearl, 2009; Lipsitch et al., 2010).

A pain-medication change can lower physiological variability while reducing active reports through sleepiness or communication difficulty. Model response and report availability separately at the treatment’s time scale.

Resolve treatment timing at its expected minutes-to-weeks effect scale; coarse grids can make sedation, analgesia, therapy, oxygen, tapering, or recovery appear delayed, absent, or terminal.

The analysis compares alternative treatment-confounder structures. Landmarking, marginal structural approaches, and joint models suit different questions. Record clinical indications separately so the interpretation reflects how treatment was chosen.

Recent studies sharpen the empirical comparison. Wang et al. (2025) followed 1,971 older adults, 1,119 of whom died, and found different terminal change points across psychological and social domains, approximately six to eleven years before death. Social-network size did not share that change-point pattern. Petkus et al. (2025) analyzed 2,411 participants and 12,574 depressive-symptom observations, finding acceleration approximately four to five years before death. In ninety-eight mortality-discordant twin pairs, the earlier-deceased twin showed a steeper terminal symptom increase than the co-twin. The twin comparison helps distinguish terminal-related change from shared age and family background, while leaving nonshared illness and environmental changes to be modeled. For PTVC-linked work, its value is a stronger comparison design: test any additional proposed signed-history structure against these already informative terminal models.

17.5.3 — Natural Experiments

Natural experiments use variation created outside the research assignment, such as policy thresholds, scheduling rules, phased rollouts, supply changes, or staffing patterns. Their value depends on an explicit causal graph or target-trial analogue, a defensible identifying assumption, stable measurement around

the change, and evidence that the variation is not simply a proxy for unmeasured participant state or site selection (Hernán & Robins, 2020; Lipsitch et al., 2010; Pearl, 2009).

An operational scheduling change can identify an observation or care-process effect only after checks of pretrends, anticipation, spillovers, and concurrent changes.

Prespecify the local estimand, source of variation, exclusion or exchangeability assumptions, anticipation window, and design falsifiers.

The design should try to falsify its own assignment story. Pretrend breaks, anticipation, manipulation around thresholds, co-occurring policy changes, differential measurement, and negative-control outcomes can each expose a false natural experiment. When those checks fail, the result remains a descriptive association tied to the observed policy change rather than a causal estimate.

Report uptake, crossover, implementation fidelity, eligibility, and common covariate support for the target population. Failed overlap restricts transport to supported local or partially identified claims (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

17.5.4 — Clinical Heterogeneity

Specify diagnosis, course, treatment options, communication, setting, disability, culture, family support, and site practice as strata or effect modifiers of dynamics, validity, and observation probability (Hernán & Robins, 2016, 2020).

Display clinical and site strata before pooling, including visit density, device availability, symptom management, staffing, and family involvement.

Audit hierarchical borrowing with leave-group-out tests and stratum residuals. Report when a highly observed group dominates or when pooled dynamics describe no clinical group.

Adapt cadence, channel qualification, and missingness models to rapid fluctuations, slow degeneration, episodic crises, or communication decline; preserve a common estimand when comparing designs.

17.5.5 — Causal and Predictive Questions

Specify whether the target is a future report or an intervention effect; forecasting accuracy alone does not identify treatment, support, or environmental causation (Pearl, 2009; Lipsitch et al., 2010).

For an intervention question, the target trial should be written first: eligibility, time zero, treatment strategies, assignment procedure or emulation rule, follow-up, intercurrent events, outcome, causal contrast, and analysis. Time-varying confounding, positivity, consistency, interference, censoring, and measurement assumptions are then stated separately. Forecasting may use the same cohort, but it remains a different estimand and validation task (Hernán & Robins, 2020).

Medication may predict lower future variance through severity, clinical attention, or survival selection. Test those pathways before interpreting the prediction as a treatment effect.

Validation and decision rules are task-specific: blocked temporal tests for forecasting, negative controls and sensitivity analyses for causal claims, and held-out populations for transport. When the tasks disagree, the study reports the disagreement.

Unexpected timing offers a concrete observational comparison for horizon-dependent candidates. Ito et al. (2021) followed 1,896 patients in twenty-three Japanese hospice or palliative-

care units. Four definitions of sudden or unexpected death produced thirty-day estimates between 6.4% and 16.8%. These Kaplan–Meier estimates censored other deaths; they should not be treated as competing-risk probabilities. That variation makes definition choice part of the scientific question. A prospective extension could add affective and contextual observations, freeze a prognosis model using only information available before the final decline, and compare a prespecified candidate across more- and less-expected biological deaths. Clinician surprise, physiological change, care, and missingness would remain explicit comparison variables.

17.5.6 — No Care Manipulation for Theory

Data lost because comfort medication, device removal, or a stop request took priority remain ethically unavailable and follow the prespecified missing-data or no-estimate branch (Council for International Organizations of Medical Sciences, 2016; International Council for Harmonisation, 2026; World Medical Association, 2024).

If a treatment, visit, sedation decision, monitoring interval, or support change would not be justified without the theory, the study should not perform it for the theory.

17.6 — Truncation by Death and Competing Events

Death is not ordinary dropout. After truth-level stream endpoint, future felt outcomes are undefined for that stream. A study that fills post-death values or hides death inside a composite endpoint can make a trajectory look continuous after the object of observation has ended (Frangakis & Rubin, 2002; Fine & Gray, 1999).

17.6.1 — Death Truncates Future Outcomes

Even without terminal compression, declining observation frequency as weakness rises can create apparent narrowing when records are aligned backward from death. Compare the display with a null model using the same missingness process.

For an outcome defined only while alive, death ends its clinical domain. The protocol defines the relevant while-alive, terminal-window, event, or joint estimand and treats the stream endpoint as a separate specified object. Values are estimated only within the domain of the chosen outcome (Frangakis & Rubin, 2002; Fine & Gray, 1999).

Fix outcome stopping rules, adjudication, and last-observation handling, distinguishing biological death, stream endpoint, and observation loss. End figures at the chosen estimand's boundary.

Do not insert post-death zeros, carry the final report forward, or code death as an extreme symptom. Show the event and end each outcome at its defined boundary.

The audit trail must reveal whether post-death imputation, survivor-only smoothing, or endpoint-aligned averaging can manufacture apparent neutrality. Simulations should replay those operations under known non-PTVC generators and show their false-signature rate. If contamination cannot be bounded, the stronger verdict is blocked (Kurland et al., 2009; National Academies of Sciences, Engineering, and Medicine, 2019).

17.6.2 — Semi-Competing Risks

Events that interrupt observation also matter before death. Transfer to another facility, discharge, withdrawal, loss of device access, communication loss, sedation, and family protection of quiet can end observation without ending the person.

Medication may dampen signals while monitoring is removed after sedation. Carrying the last observations forward visually can then create apparent neutrality. Model sedation and device removal as distinct marks.

When sudden and slow deaths are mixed, sudden cases may lack terminal windows, leaving only slow deaths visible near the endpoint. Identify the target as observable terminal windows rather than implying coverage of all deaths.

Semi-competing risks arise when a nonterminal event can occur before death but cannot occur afterward, while death may occur without the nonterminal event. The estimand must therefore specify whether it concerns transition into the nonterminal state, death without that state, time spent alive and observable, or the joint ordering of events. Ordinary right censoring does not preserve those distinctions (Fine et al., 2001; Kurland et al., 2009).

The protocol should define states, allowed transitions, event ordering, and the clock used for each transition. Illness-death and multistate models can represent whether a nonterminal event precedes the terminal event, but their interpretation still depends on observation and endpoint qualification.

Sedation may end active report while biological life and clinical care continue. That transition is neither death nor ordinary missingness. The analysis should distinguish loss of report, loss of device access, change in consciousness evidence,

and truth-level endpoint uncertainty rather than assigning one terminal code to all of them.

Represent event order as observed, interval-censored, or impossible. Hospitalization, device removal, transfer, withdrawal, and communication loss generate different risk sets and observation opportunities.

Report event-specific conclusions when separating terminal and nonterminal events changes the result; simplify the estimand when detailed transitions are unsupported.

On the common event clock, allow prior events to alter subsequent transition intensities and observation probabilities (Aalen et al., 2008; Andersen et al., 1993).

Use transition hazards for movement among states, cumulative incidence for event probabilities with competing risks, and illness-death or joint models for dependence under additional assumptions (Andersen et al., 1993; Fine & Gray, 1999; Fine et al., 2001).

17.6.3 — Transfer, Discharge, and Withdrawal

Record the reason, timing, permissions, and destination of transfer, discharge, or withdrawal without treating observation exit as a clinical or stream endpoint (Andersen et al., 1993; Hernán & Robins, 2020).

If separately authorized linkage cannot continue observation, inference stops at the supported window. The local result retains the status its evidence earns; any continuation claim is classified according to its remaining identification and uncertainty.

Evaluation should leave out transfer routes, discharge classes, and withdrawal categories before claiming transport across endings. If the finding disappears when one exit route is separated, the exit is part of the data-generating process and cannot be buried in generic dropout.

17.6.4 — Survivor and Immortal-Time Bias

Diagnoses may decline at different speeds while visits cluster around crises. Dynamic time warping can erase those timing differences by aligning the crises; report both warped and unwarped clocks.

Define risk sets before comparison when group entry requires survival, enrollment, or observability for a specified duration (Frangakis & Rubin, 2002; Fine & Gray, 1999).

Participants who complete thirty days of prompts may appear more stable simply because people with rapid decline, transfer, distress, or device intolerance could not complete them. Comparing completers with noncompleters after the fact makes future survival part of group membership. Landmark or time-dependent analyses can address the timing bias; they do not by themselves resolve confounding or informative observation.

Every comparison reports its risk set, eligibility rule, and required survival time. Immortal periods are assigned correctly, and analyses repeat at prospectively chosen landmarks. A trajectory visible only among selected survivors is labeled survivor-conditioned.

17.6.5 — Principal Strata and Survivor Limits

An analysis restricted to observed participants who remain communicative and device-tolerant describes an observed subgroup. It is not, without further assumptions, a principal-stratum effect. The report should distinguish this observed selection rule from the counterfactual survival or observability condition used below.

Define the counterfactual survival or observability stratum before fitting. Membership is generally unobserved, so state the assumptions identifying a point effect and restrict the claim to that stratum (Frangakis & Rubin, 2002; Zhang & Rubin, 2003).

Sensitivity analysis should vary the assumptions linking observed covariates, survival, observability, and latent stratum membership. When plausible assumptions yield materially different estimates, the study reports bounds or multiple stratum-specific results rather than selecting the most favorable specification.

17.6.6 — No Composite That Hides Death

Keep death, withdrawal, transfer, device failure, sedation, distress-triggered stopping, and nonresponse typed by irreversibility, information loss, and clinical meaning before combining them.

Choose explicitly among while-alive means, survivor averages, composite endpoints, principal-stratum contrasts, joint longitudinal-survival parameters, and component cumulative incidences (Frangakis & Rubin, 2002; Kurland et al., 2009).

Joint models link longitudinal outcomes and event hazards through assumptions about shared random effects, functional form, and unobserved values. Apply the sensitivity-family comparison in Section 17.4.5 to those links (Rizopoulos, 2012; Scharfstein et al., 1999).

Require defensible common meaning and no hidden component dominance. Report component incidence, timing, risk sets, and leave-one-component-out results; when removing death or another component changes interpretation, the component analysis governs.

For each death-truncated estimate, show numbers enrolled, alive, observable, responding, and analyzed over time, and identify any counterfactual target population (Fine & Gray, 1999; Kurland et al., 2009).

17.7 — Versioning, Drift, and Transport

Longitudinal change can arise in the participant or in the system observing the participant. Instruments, firmware, staff, sites, vendors, populations, languages, and clinical practice may all change during follow-up (Dahabreh et al., 2019; Pearl & Bareinboim, 2014). A version record identifies those changes; overlap samples and calibration evidence determine whether records can be linked. Major changes can require reanalysis or a narrower transport claim rather than uninterrupted use of one apparent trajectory.

17.7.1 — Measurement Version Ledger

Apply Section 9.6's measurement-version ledger at person-time resolution, including accessibility, staff training, and consent materials (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020).

The ledger should distinguish changes that affect cadence, origin, unit, sign, missingness, or target population. Minor software patches and major construct revisions do not receive the same treatment. Each change is assigned a bridge, segmentation, recalibration, or retirement decision before the post-change data are merged with the earlier series.

A wording change may lower reported burden without changing lived conditions; overlap evidence and exact version dates distinguish those possibilities.

A version ledger should include a diagnostic independently justified as invariant to participant state, such as a reading from a fixed calibration fixture. A shift at the version boundary then flags a change in the measuring system.

17.7.2 — Site and Era Effects

Validation should follow the deployment question: forward within-person tests for future intervals, blocked temporal folds for new periods, leave-person-out tests for new individuals, leave-site-out tests for transport, and leave-device-version-out tests for technological migration. Random observation-level splits leak dependence and are outside the confirmatory rules.

Model site and era changes in care, staffing, referral, technology, population, language, and practice; their joint effects on observation and outcome may exceed a site intercept (National Academies of Sciences, Engineering, and Medicine, 2019; Särkkä, 2013).

Preserve site-specific clocks, device ecosystems, care processes, and calendar changes. Evaluate partial pooling against leave-site-out and leave-era-out performance.

Distinguish transport of person-level relations from care-system relations. Different hospice documentation and quiet-care policies can change trajectories through observation alone.

Where available, a justified site-level negative control can probe shared bias. Strong prediction from site or vendor identity warrants investigation of case mix, care, devices, and observation practices; it does not by itself identify which process generated the pattern.

17.7.3 — Response and Device Drift

Apply the response-drift and device-drift distinctions in Sections 17.2.6 and 17.3.5 together, using anchors and technical diagnostics to separate their contributions (Shcherbina et al., 2017; Bent et al., 2020).

A participant may rate pain differently after learning the illness vocabulary, while a wearable update changes the apparent sleep score in the same month. A stable combined

trajectory can therefore hide two moving scales. Parallel anchors and version-specific residuals are needed to locate the change.

Drift monitoring should specify thresholds, detection delay, and action. Prompt learning, changing response standards, trust loss, vocabulary growth, sensor aging, and algorithm updates can produce gradual shifts that evade a single overlap check. The study should test both abrupt and delayed drift before calling the pattern a human trajectory.

17.7.4 — Bridge Samples and Overlap

Qualify bridges through overlap in time, participants, states, or calibration fixtures, including difficult cases. Cohort-sequential and accelerated designs need common anchors across age or disease-stage windows; unlinked scales support separate local claims (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

Freeze parallel-device periods, dual items, overlap size, subgroup coverage, link functions, and acceptance criteria. Analyze systems separately until the bridge passes held-out checks.

Compare nonlinear distortion, variance, thresholds, missingness, and subgroup error, beyond mean offsets. Report a version break when those structures fail to transport.

Compare bridge and target covariate support. Reweight where justified and report unsupported regions rather than extrapolating from healthy, digitally connected overlap samples into advanced illness or limited access (Pearl & Bareinboim, 2014; Dahabreh et al., 2019).

17.7.5 — Transportability Analysis

Define source, target, bridge variables, and assumed stable mechanisms, then test target calibration, observation, and

construct validity (National Academies of Sciences, Engineering, and Medicine, 2019; Särkkä, 2013).

The source setting S supplies the fitted relation, the bridge variables Z encode measured differences relevant to selection or effect modification, and the target setting T defines the population and observation conditions to which the claim is proposed to move.

The transport question is component-specific: after accounting for Z, measurement version, observation process, care context, and common-support limits, does the same estimand retain its meaning and calibration in T, or must the result remain local (Dahabreh et al., 2019; Pearl & Bareinboim, 2014)?

Transportability checks should ask whether the time grid itself travels. A cadence tolerated in a research hospice may be unsafe in home care, too slow for acute illness, or too intrusive in a work setting. If the original observation schedule cannot ethically be used in the target setting, transport requires a validated observation bridge for a permissible schedule.

Report transport separately for ranking, origin, units, cadence, reserve validity, and care-system effects, identifying failed bridges and source-bound claims (Dahabreh et al., 2019; Pearl & Bareinboim, 2014).

17.7.6 — Reanalysis After Major Change

Site, device, instrument, language, care, population, schema, or endpoint changes trigger review of affected estimands and dependent outputs under Chapter 28 (International Council for Harmonisation, 2026; National Institute of Standards and Technology, 2020).

Reanalysis starts with the affected clock and context. Changes may alter sampling interval, time since event, device lag, burden, care workflow, or prompt meaning without changing the

script. Label results confined to the old cadence or setting accordingly.

Represent each measurement version as a node and each qualified bridge as an edge. Unlinked versions remain separate.

Preserve source-schema provenance when names, categories, missing codes, units, or derivations change, because those changes can alter the estimand (Little & Rubin, 2019; National Institute of Standards and Technology, 2020).

Report whether the updated implementation preserves the original estimand, requires a transport bridge, or changes the claim. Retain the earlier result under its original version; Chapter 28 governs dependent corrections.

17.8 — Purpose Limits and Participant Return

Behavioral traces can identify people. Sleep and movement supply timing. Medication and visits add context. Transfers or family events may complete the match (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

17.8.1 — Operational–Research Firewall

Document the authority and purpose by which care or administrative data enter research. Clinical availability alone supplies neither permission for secondary modeling nor construct qualification (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

The technical file should make visible separate permissions for observation, linkage, sharing, publication, model training, and reuse.

Assess how clinical shorthand, care decisions, bystander information, billing, and workflow shape a note before using it as estimator input.

Specify permission duration: one extraction, linkage, analysis window, or return does not authorize continuing observation.

17.8.2 — Participant-Facing Results

Return only individually valid, requested or consented information in an understandable form consistent with the clinical and ethical plan. Do not personalize group findings or unstable constructs through a dashboard (Rieke et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2018).

A sleep summary may help a participant when it clearly identifies missing nights, device changes, uncertainty, and the difference between an observation and a research inference. Comprehension testing should confirm that the display does not

imply diagnosis, prognosis, or a hidden reading of the person's inner life.

Participant return has its own timing rule. Immediate feedback may change behavior, delayed feedback may lose relevance, and no feedback may be safest for some measures. The design declares when return is useful, optional, or withheld because validity is insufficient.

Provide accessible text, print, translations, audio, or discussion as needed, including an option for no return. Specify notification after correction or withdrawal without implying a change in care or personal identity.

No individual ledger judgment is generated, stored, returned, or made available for secondary use. Any feature or interface that invites such an interpretation is removed or redesigned before deployment.

17.8.3 — Access, Retention, and Deletion

Access, retention, and deletion are specified separately for identifiers, raw signals, derived features, linkage keys, provenance, code, and aggregate results. Each layer receives the smallest audience and shortest duration compatible with the approved purpose. Retention for unspecified future value is not a scientific justification (Nissenbaum, 2010; National Institute of Standards and Technology, 2020).

Specify which layers stay local, may be shared, or must not be centralized; federated analysis still requires controls for leakage, reconstruction, deletion, and audit.

Keeping raw traces and identifiers local reduces exposure only if transmitted features resist reconstruction and sites retain withdrawal and deletion authority.

Local capture retains time, clock, channel, value, unit, and confidence, with the person key and identifiable linkage held at

the source site. These fields preserve analyzable time-indexed evidence without centralizing identity.

Observation marks distinguish scheduled, refused, paused, failed, removed, and clinically interrupted measurements. Refusal and care interruption remain meaningful causes of availability or absence.

The version record identifies instrument, firmware, item version, language, and pipeline. It separates change in the person from change in the system and lets participants challenge the version that interpreted their evidence.

Feature extraction retains approved features, quality flags, and burden flags needed for the estimand. Minimize raw exposure before transfer.

Federated analysis uses site-local models, disclosure checks, and aggregate summaries to permit multisite learning without a central personal dossier. Individual records are not pooled by default.

Closing the longitudinal record documents participant choice, deletion status, archival decisions, and future-use permissions. Completion of the study creates no indefinite claim over the person.

17.8.4 — Breach and Reidentification Response

Test reidentification through rare clinical transitions, sleep rhythms, location patterns, transfer dates, and combinations of events. Reassess risk when variables, sites, publications, or auxiliary datasets change (Rieke et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2018).

Apply the incident-response steps in Section 14.6.3 to the longitudinal record and its linked releases. Also record whether the incident changed completeness, provenance, consent, or permitted access.

Suppress or coarsen compromised features, withdraw affected artifacts, revise retention and minimization, and reassess correction and withdrawal rights. Impermissible data are Unusable; classify repaired data under Section 24.2 according to their remaining validity and identification (National Academies of Sciences, Engineering, and Medicine, 2018; Nissenbaum, 2010).

17.8.5 — End-of-Study Closure

Before enrollment, specify closure duties for results, unresolved analyses, code and model status, raw and derived data, retention, deletion, future use, correction contacts, and retired artifacts. Name the authority and completion evidence for each duty (World Medical Association, 2024; Council for International Organizations of Medical Sciences, 2016).

Notify participants of findings and unresolved questions, retained or destroyed data, barred future uses, and correction locations. Keep that information accessible after collection ends.

Before enrollment, the closure plan fixes archive contents, deletion events, participant communication, failure memory, code and model status, derivative-data handling, and prohibited future uses. The study's ending is part of consent and purpose limitation; it cannot wait until the dataset becomes valuable. The plan should also identify who has authority to certify each closure obligation and what evidence shows that the obligation was actually completed. This turns closure from an administrative date into a verifiable state of the research system.

Participants should receive clear notice of what was learned or unresolved, what will be retained or destroyed, what future use is barred, and where corrections will appear. They should not infer the relationship ended merely because contact stopped. Closure communication should remain understandable and

reachable after active data collection ends, especially when a later correction changes how an earlier result should be interpreted. A durable contact route gives participants a practical way to ask about those changes without reopening participation.

Closure should assign final status to usable findings, inconclusive analyses, unusable data, unresolved access questions, and future-use prohibitions. A completed record cannot support stronger claims through reinterpretation without a newly approved analysis, consent basis, and status review. Any later reopening should therefore create a new versioned analytic record rather than silently modifying the closed one. This preserves the distinction between what the original study established and what a later investigation newly inferred.

Site reconciliation, incident review, fulfillment of retention and deletion commitments, public result registration, retirement notices, and a durable correction contact are completed before closure is certified. Null results, failed recovery, deviations, and rival-favored outcomes remain in scientific memory even when sensitive data are deleted. Reconciliation should verify completion across sites rather than assume that a centrally issued instruction was carried out uniformly. Differences in retention, deletion, incident handling, or version status should be resolved or recorded explicitly before the study is treated as closed.

Every retired dataset, instrument, model, and public artifact should receive a tombstone recording what existed, why it was retired, which successor replaced it if any, and which claims it can no longer support. The tombstone preserves provenance without preserving unrestricted access to private information.

A responsible program ends observation without erasing failure or extending authority beyond consent.

Part VI — Candidate Mechanisms and Causal-Locus Tests

A terminal-looking trajectory can arise in several ways. Native dynamics may approach a boundary; an analyst may select histories that end there; a controller may alter the route; or a stopping rule may wait for the first arrival. These constructions can look similar in a finished plot while making different predictions elsewhere. Part VI studies the differences that become visible when the path is treated as a process with a causal structure (Doob, 1957; Léonard, 2014; Chetrite & Touchette, 2015).

Chapter 18 develops completion geometry. A present state admits some continuations and excludes others under a specified model. Changing the available continuations changes what a control policy or conditioned path can accomplish. The Queue model gives this question a particular architecture whose states, observations, and actions can be examined separately. Its value as a candidate depends on the predictions contributed by those commitments, including circumstances in which the proposed architecture cannot recover the observed pattern.

This geometry also connects the model to ordinary opportunity. Transportation, care access, fatigue, and family responsibilities can remove an otherwise imaginable continuation. Measuring these conditions independently helps distinguish a blocked opportunity from a preference for another action. It gives the comparison a concrete causal question: did behavior change because the person’s options changed, because the proposed internal process changed, or because the observer saw a different subset of actions?

Chapter 19 makes that causal question explicit. An intervention on information has a different point of entry from a change in physiology, action support, or an endpoint rule. The model must specify which part changes first and what downstream divergence follows. Comparing those orders of

response can distinguish accounts that fit the same undisturbed observations. The required causal assumptions belong to the intervention and target population being studied (Hernán & Robins, 2020; Pearl, 2009).

Constrained optimization supplies a useful language for some of these comparisons. A shadow price expresses how a model’s objective changes under a marginal relaxation of a particular constraint. Its units and regularity conditions determine the interpretation. Within a fully specified model, this can connect the geometry of available paths to the predicted effect of an intervention, giving an empirical team a quantity whose changes can be checked against the proposed mechanism.

Chapter 20 examines the bodily channels through which such processes might appear. Interoception, pain, medication, and illness can affect both a candidate mechanism and its observations. Channel timing and lag therefore matter to the comparison (Khalsa et al., 2018; Barrett & Simmons, 2015). The Part ends with a more demanding account of a mechanism: a construction with a specified causal location, a relation to embodied conditions, and observable differences from neighboring constructions.

Chapter 18 — Completion Geometry

A path can approach an ending for several reasons. Its dynamics may steer it toward a boundary, its stopping rule may end observation there, or the sample may contain only paths selected for arriving there. The endpoint alone cannot distinguish these explanations. This chapter asks what the preceding path, its available continuations, and its response to intervention can reveal.

Reachability begins with what remains possible from a particular prefix. A large residual may be compatible with completion if sufficient time, resources, and actions remain; a smaller residual may already lie outside the attainable set. The relevant geometry therefore depends on the continuation class. Distance from zero by itself does not locate a path inside or outside that class.

Probability adds a different question. An attainable continuation need not have appreciable probability under the process being studied. Conversely, conditioning on an endpoint can produce a concentrated family of paths without showing that an unconditioned process ordinarily follows them. Conditioning on a finite band and constructing a law at an exact singleton require different mathematical treatment.

Both questions depend on how history becomes state. Two prefixes with the same current reading can permit different futures if earlier demands, actions, or losses continue to matter. Compressing them into one convenient state may erase precisely the dependence under investigation. State sufficiency must therefore be examined before a state-indexed completion probability can replace a history-indexed law.

The future also depends on opportunities outside the individual. Transportation, communication, care access, institutional permission, and physical capacity can change which actions are available. A model that assumes those actions remain

available can attribute a restriction imposed by the environment to an internal mechanism. Observing the eventual path does not establish that every alternative was unavailable.

Consider the chapter's comparison between a local controller and a mixture of endpoint-conditioned bridges. They may share a terminal distribution and even the same observed path law. A midcourse probe is informative only where their specified responses differ. The experiment must locate that disagreement in advance; an intervention cannot distinguish models that remain equivalent under its action and observation rules.

Partial observation makes this problem harder. A quiet record can reflect a resolved demand, blocked expression, absent measurement, or a state that the chosen channel cannot distinguish. Belief states and the illustrative Queue model provide ways to represent such alternatives. Their value depends on concrete differences in reactivation, action availability, repair, and subsequent observation.

The mathematical constructions below connect these questions. Cones describe attainable contributions, kernels distribute probability, and physical potentials impose accounting restrictions across paths and contexts. Their most useful consequences are the restrictions that survive beyond a fitted endpoint: a forbidden continuation, a shared response constraint, or a prediction made with one frozen state representation on a new path.

18.1 — Completion Geometry and Reachability Sets

Let the current signed residual be $L(t) = l$ and let $A_{fut}(t,z)$ denote the set of future signed contributions attainable from the declared state z under ownership-preserving continuations. Directional reachability has the following precise form.

18.1.1 — State Prefixes and Future Cones

$$-l \in A_{fut}(t,z).$$

The condition is necessary, not sufficient evidence that a route is realized. The geometry can also be expressed through a residual-solvency set

$$S_{res}(t,z) = \{l : -l \in A_{fut}(t,z)\}.$$

A prefix outside that set is an explicit counterexample for the declared route and continuation class. It cannot be rescued by changing the stream, endpoint, or allowable continuations after inspection.

Convex approximations permit support-function tests. For a closed convex attainable set A, membership of $-L(t)$ can be checked through inequalities against $h_A(a) = \sup_{x \in A} a^\top x$. Nonconvex sets require richer geometry, and a convex relaxation must be labeled as a relaxation rather than the true continuation support.

A state prefix contains the stream-owned residual coordinate, route state, observation history, action history, endpoint-information state, and uncertainty available at analysis time. Its future cone comprises the continuations permitted by the fixed dynamics, ownership rules, action support, and care constraints (Aubin, 1991).

Including descendants of the endpoint or intervention in a richer prefix can improve prediction while worsening identification.

The cone must also preserve route ownership. A shared resource may alter several streams, but one stream's future phenomenal contribution cannot be inserted into another stream's cone. Junctions, partition, fusion, and continued-stream branches require their own typed routing rule. These rules preserve per-stream accumulation within the multivariate model.

Partial observation gives a family of compatible cones; Section 18.1.3 defines their inner and outer regions.

A concrete test begins with held-out prefixes. The team freezes the state representation and dynamics on training data, computes the admissible cone for each held-out prefix, and asks whether observed actions, state changes, and endpoint-independent outcomes remain inside it. Systematic cone violations are evidence against the route or representation, not a reason to widen the cone after inspection.

18.1.2 — Signed Reachability with Finite Opportunity

Signed reachability requires the sign, magnitude, timing, and ownership needed to offset the current residual; large remaining variability alone is insufficient (Aubin, 1991).

Finite opportunity makes this condition sharp. Remaining time, physiological reserve, action access, social permission, and environmental support jointly bound what can still occur. Reportability limits which continuations the study can qualify, rather than directly setting the bounds of possible experience. A route must identify a feasible continuation under the declared horizon and constraints.

Reachability should be computed on phenomenal contribution rather than on an untyped behavioral proxy. A conversation,

medication change, rest period, or institutional remedy may alter future experience, but the action itself is not a signed valence increment. The model needs a separately validated transition from action and context to the distribution of future phenomenal contribution.

If the required contribution lies outside the qualified continuation support and the endpoint occurs under the declared branch, apply the route's prespecified failure criterion under Section 24.2.

18.1.3 — Viability, Solvency, and Closure Regions

Viability, solvency, and closure answer different questions. A viability region contains states from which at least one admissible policy can respect declared constraints. A solvency region adds the requirement that the current signed residual can still be offset by attainable future contribution. A closure region is stronger still and must specify whether closure is possible under one policy, probable above a threshold, supported under every admissible policy, or guaranteed on a declared branch (Aubin, 1991; Aubin et al., 2011).

These regions should be nested only when the assumptions justify nesting. A state may be viable because safe functioning can continue while being insolvent for the signed completion target. Another state may be solvent under one narrow policy but not under ordinary policy variation. Collapsing the regions hides which part of the argument is doing the work.

Uncertainty creates nested reachability regions. The inner region contains prefixes that satisfy the criterion across every declared parameter, state, and horizon compatible with the data. The outer region contains prefixes that satisfy it under at least one admissible specification. Their difference is an identified

uncertainty region, not a smooth probability gradient, and the final display should preserve that distinction.

Observation shutdown can leave viability and solvency unidentified; withhold the affected geometric inference.

The region boundary can be tested through safe perturbations that change action support or information without changing the endpoint definition. Opening a communication route, correcting an institutional barrier, or observing a clinically indicated treatment transition may move the estimated state. A route-specific model should predict which boundary moves and by how much.

Numerical grids, convex relaxations, surrogate dynamics, and finite samples can shift the frontier. If reasonable discretization or model uncertainty changes the classification, report the resulting region or bounds and apply Section 24.2.

A closure region built by excluding nonclosing paths is circular. The scientific object must be derived from independently specified dynamics, constraints, and observations, then evaluated on held-out continuations. Otherwise the region proves only that the analyst encoded the desired endpoint.

18.1.4 — Geometry Under Uncertain Endpoints

Endpoint uncertainty changes the geometry of completion before it changes any estimate. A fixed horizon gives one future cone, but a distribution over possible endpoints gives a family of cones indexed by stopping time, branch, and information state. The relevant object is therefore not one attainable set but a set-valued map from each prefix and endpoint scenario to the signed continuations that remain possible. A route that is reachable at one generous horizon may be impossible at a shorter horizon, and that difference cannot be repaired by averaging the two cases (Aubin, 1991).

Several geometric summaries are possible, and they answer different questions. The union across endpoint scenarios asks whether completion remains possible somewhere. The intersection asks whether it remains possible under every declared scenario. A probability-weighted region asks how much endpoint mass lies over solvable continuations. None of these objects is equivalent to exact closure at the truth-level endpoint, and each should carry its own status label rather than being collapsed into a single shaded region.

Prospective estimation must use only information available in the declared filtration, and endpoint models should be cross-fitted or frozen before they are used to construct a future cone.

If the apparent empirical result depends on using retrospective endpoint knowledge as a prospective input or on choosing a favorable horizon after inspection, the confirmatory comparison fails its information or protocol gate and the affected evidence is Unusable. A route that necessarily requires unavailable future information is inadmissible under the declared information constraints; an empirical Route-failed verdict requires a valid test.

18.1.5 — Topology and Hidden Support Barriers

Reachability is not determined by distance alone. An attainable set can be disconnected, folded around forbidden states, or concentrated on a lower-dimensional support. Two states that are numerically close may be separated by an action barrier, a routing rule, a safety constraint, or a zero-probability transition. Topology therefore decides whether a seemingly short path is actually available (Aubin, 1991; Aubin et al., 2011).

Convex relaxation is useful but dangerous. Filling the gaps between disconnected components can create synthetic continuations that no admissible dynamics can realize. A

relaxation should be labeled as an outer bound and tested against the original nonconvex support, especially when the conclusion depends on a narrow bridge between components.

Barriers can also be informational. Two latent states may produce nearly identical observations while permitting different futures. The apparent connection between their estimated cones may disappear once the belief state is represented. State aliasing is therefore a topological problem in the inferred state space, not merely a measurement nuisance.

A route may cross a barrier only through a junction event such as partition, fusion, return, or a change in endpoint branch. Such events require typed rules and prospective evidence.

Concrete simulation should include obstacles rather than only smooth landscapes. A model can compare Euclidean distance, geodesic distance on the admissible set, and actual hitting probability under the protocol-fixed dynamics. Divergence among those quantities reveals when a simple residual norm is an inadequate summary.

18.1.6 — Testable Consequences of a Reachability Map

Freeze predictions of which prefixes will lose sign-specific options, which actions will reopen or close a route, and which uncertainty changes will move its frontier. Evaluate these and observations that falsify the dynamics under common recovery and scoring rules (Aubin, 1991; Aubin et al., 2011; Wilson & Collins, 2019).

One consequence concerns boundary crossing. If a prefix is classified outside the outer solvency region, then no continuation in the declared class should achieve the required signed contribution. An observed qualifying continuation is a counterexample to the map or its assumptions.

A second consequence concerns intervention sensitivity. An action that genuinely changes opportunity, reserve, or route state should alter the attainable set in a specified direction. A matched sham provides this contrast only if it leaves the relevant opportunity, reserve, and route state unchanged; matching attention or expectation alone does not establish that exclusion.

A third consequence concerns calibration. Predicted inner-region membership should correspond to conservative empirical coverage, while outer-region membership should not be treated as a guarantee. Reliability diagrams, set coverage, and false exclusion rates are more informative than one global accuracy score.

Transport tests whether a geometry retains its declared invariants when the observation system, action menu, site, or population changes within a defined transport class. The target state and ownership rule remain fixed, while only the declared destination components may be recalibrated.

Use the same estimator to recover geometries generated by viable control, ordinary scarcity, bridges conditioned on endpoints, and the candidate model, under realistic noise, missingness, and state aliasing. Recovery failure withholds the affected frontier labels in real data (Villaverde, 2019; Wilson & Collins, 2019).

A cadence change is a negative control only when it leaves the latent state, action support, and continuation dynamics relevant to reachability unchanged. Under that exclusion, a change in the estimated map tests sensitivity to observation. If prompting itself changes the process, model that effect rather than treating it as a pure observation control.

18.2 — Completion Kernels and Boundary Values

A completion kernel asks for the probability of finishing inside a declared band under a declared continuation law. With current state x and residual ℓ, a state-indexed committor can be written as

18.2.1 — Completion Kernels as Conditional Forecasts

$$h_{g,K}(x,l,t) = P_g\left(l + \Delta L_{t:T_g} \in [-K,K] | X(t) = x, L(t) = l\right).$$

The regime g, endpoint branch, band, and reference law are part of the object. A history-indexed kernel is more general. Reducing it to a state function assumes that the chosen state is sufficient for the future event. That assumption should be tested by residual history dependence, held-out prediction, and rival state augmentations rather than asserted because a Markov model is convenient.

Under regularity, a committor solves a backward boundary-value problem. The boundary conditions reveal what counts as completion and failure; the generator reveals the assumed native dynamics. Changing either after seeing the data changes the hypothesis.

Uncertainty in the kernel should remain visible. Posterior bands, conformal regions, or identified sets may be more appropriate than a single surface when the state is partially observed or the generator is weakly identified.

The kernel differs from reachability. Reachability asks whether at least one admissible continuation exists; the kernel weights continuations under a specified law or policy. A state can be reachable and still have very low completion probability, or unreachable under one policy while reachable under another.

Forecasting requires a baseline law. Native dynamics, selected cohorts, endpoint-conditioned samples, and intervention policies

induce different kernels even when they share the same state coordinates. Every reported probability therefore identifies the law under which it is computed; comparisons must keep the regimes and conditioning sets explicit rather than treating the probabilities as interchangeable estimates of one law-specific object.

Under partial observation, the forecast averages state-specific completion probabilities over the specified belief distribution. Uncertainty about that distribution or the state-specific probabilities must propagate to the forecast. Several possible latent states do not by themselves preclude a precise mixture probability; unidentified weights or dynamics may instead require bounds.

18.2.2 — Committors and Boundary Problems

A committor is the probability of reaching one declared set before another under a specified stochastic process. In Markov settings it solves the generator's boundary-value problem, with conditions fixed on the competing sets. The state, generator, sets, and law must therefore be justified (Metzner et al., 2009).

The forward committor asks what is likely to happen next from the current state. A backward committor asks where a trajectory most likely came from under a stationary or appropriately defined ensemble. These are different questions. Using a backward object as a prospective forecast can import information unavailable at the decision time.

Generator misspecification can dominate the result. Sparse observations, hidden variables, non-Markov memory, state-dependent diffusion, and informative sampling can make an estimated committor look smooth while representing the wrong dynamics. Recovery simulations should test whether the

boundary-value solution remains valid under the actual observation process.

A concrete comparison can use several models with identical terminal sets. A first-hitting diffusion, a conditioned bridge, a homeostatic controller, and a queue candidate may yield similar one-time completion probabilities but different transition-path densities and intervention responses. Those differences create the discriminating work.

Committors support rare-event sampling and transition-path analysis. Preferentially sampling successful transitions changes the sample law, so likelihood weighting and support checks connect those paths to the intended population.

18.2.3 — Action-Specific Committors

An action-specific committor asks what happens if one declared action is taken now and the continuation policy thereafter is fixed or prospectively specified. It separates the immediate causal fork from the larger policy value (Metzner et al., 2009).

The action index matters because two interventions can have identical short-term observations and different future geometry. One action may open a repair route, another may reduce measurement burden, and a third may merely change what is reported. A state committor that averages across those actions can hide the divergence. The model should therefore declare the action version, timing, eligibility set, and continuation policy that define each surface.

A harmless unfinished-task study can compare information, an opportunity to resolve a task, and a matched control cue. The action-specific committor predicts which intervention changes the probability of a later route-relevant state.

Identification requires support. If participants with a particular belief state never receive one action, its committor is

extrapolated rather than learned. If action choice depends on unmeasured urgency, the surface may encode selection. Randomization, sequential exchangeability assumptions, or partial-identification bounds are needed according to the design.

Score action-specific surfaces for held-out calibration, discrimination, and transport under the common-comparison design in Section 25.2.

18.2.4 — Calibration, Transport, and Uncertainty

Kernel calibration is specific to its event, population, state definition, observation system, policy, and endpoint branch; transport requires a bridge across changed components (Gneiting & Raftery, 2007; Hernán & Robins, 2020).

Cross-fitting can reduce optimistic evaluation, but it does not solve semantic leakage. If the state representation contains realized time-to-endpoint, post-treatment information, or labels derived from the terminal event, held-out scoring still tests a future-informed model. The information audit precedes the split strategy.

Transport should be modular. A kernel may transport across sites while the observation model does not, or across devices while the action policy changes. Reporting which module moved and which was re-estimated prevents a successful local recalibration from being described as mechanism invariance.

Effective sample size, event counts, censoring, and selection weights should be reported beside the calibration result.

Do not return a research completion kernel as an individual completion probability or care forecast; apply Section 17.8.2's result-return requirements.

18.2.5 — Flexible Baselines and Rival Kernels

Completion-kernel baselines include nonparametric survival models, state-space forecasts, homeostatic controllers, finite-horizon planners, conditioned bridges, and selection models, with flexibility adequate to represent the relevant ordinary nonlinearities.

Flexible kernels can mimic route signatures through different mechanisms. A bridge can condition on the endpoint, a hazard model can exploit frailty, a recurrent network can memorize site patterns, and a homeostatic process can concentrate near a set point. The comparison therefore examines transition structure and intervention response, not only forecast accuracy.

Complexity penalties are useful but not decisive. A simpler model that predicts as well may be preferred for explanation, while a more complex candidate may still be retained for a prospectively specified distinctive prediction. Predictive ranking, causal interpretation, and theoretical scope remain separate judgments.

18.2.6 — Kernel Failure and Scope

Endpoint leakage, poor transport, unsupported extrapolation, unstable numerical solutions, state aliasing, and failure against qualified rivals require a typed response: recalibration, reference-class restriction, descriptive-only use, or retirement.

When state aliasing leaves completion probabilities unresolved, report any valid bounds or assumption-indexed family, classify the result under Section 24.2, and identify the observation or intervention that would separate compatible states.

Event failure occurs when the completion set or competing boundary changes after outcomes are known, or when the event is defined by the same estimator the kernel is asked to validate.

Such circularity cannot be repaired by cross-validation because the target itself was manufactured.

Law failure occurs when selection or conditioning changes the probability measure without proper accounting. A kernel estimated among paths that already reached the target cannot be interpreted as the native probability of reaching it. Likelihood ratios and support checks are necessary, and sometimes insufficient, when the event is probability zero.

Retain separately qualified roles when others fail: useful ranking need not imply calibrated probabilities, and descriptive calibration need not imply causal identification.

Retirement should distinguish sharpness from calibration. A kernel can issue concentrated, confident predictions that are poorly calibrated, or be calibrated yet too uninformative for the protocol-fixed decision. Either failure may be decisive, and neither should be hidden inside one performance score.

Coverage should be checked across state regions, actions, endpoint branches, and observation-quality strata. A global average can conceal failure exactly where completion is most difficult or data are least available. The retirement packet should identify the failed region and preserve only any narrower calibrated domain.

Evaluate calibration after a policy change as well as under passive observation. Failure under newly visited states blocks the claimed transport to that policy, even if calibration remains acceptable under the original behavior.

Rare completion events require special care. A visually smooth probability surface can appear calibrated because almost every case is a non-event. Event-specific calibration, precision-recall behavior, uncertainty under low counts, and synthetic recovery should accompany global scores before the kernel is used as a route object.

A kernel may also lack decision relevance. Even a calibrated probability is unsuitable for a decision role when it adds no information relevant to the protocol-fixed scientific decision, or when no lawful and safe action can depend on it. Its decision or deployment role should then retire; its descriptive or mechanism-testing role may remain if separately validated.

18.3 — Belief-State and Partial-Observation Realizations

A belief state encodes possible hidden states. The prior, transition, and action shape each update. Observation likelihood and missingness shape it too. Sufficiency holds only for the declared model and information set.

18.3.1 — Latent State, Belief State, and Observation

A latent state is the modeled condition that governs future transitions under the declared dynamics. A belief state is a probability distribution or identified set over possible latent states given the information available at the current time. An observation is one channel output used to update that belief (Kaelbling et al., 1998; Särkkä, 2013).

Attach the state, prior, channel-error, and missingness-model specification to each posterior; different assumptions can produce different beliefs from the same record.

Separate state uncertainty, measurement uncertainty, and uncertainty about the model family; a narrow posterior under one model does not resolve disagreement across plausible state and observation models.

18.3.2 — Filtering Without Phenomenal Identification

A filter can estimate threat, fatigue, action readiness, or route position for prediction; connecting those states to felt quality requires the separately qualified measurement bridge (Särkkä, 2013).

Filtering also depends on time scale. A fast sensor can respond within seconds while a report integrates minutes and a contextual record updates only after an event. Aligning all channels to one timestamp can manufacture apparent agreement.

Lag models and channel-specific kernels should be included before posterior conflict is interpreted.

Endpoint-blind training is necessary but not sufficient. Cross-fitting can prevent direct leakage from terminal labels, yet the state representation may still be selected because it produces the desired terminal pattern. Representation choice, preprocessing, and model-family search should therefore be versioned and separated from the confirmatory sample.

A useful negative control filters a state that should not carry the candidate route information, or permutes the channel expected to be distinctive while preserving generic time and severity. If the posterior remains equally predictive, the model is likely exploiting background structure rather than the proposed mechanism.

18.3.3 — POMDPs with Limited Information

A partially observable Markov decision process specifies the controller's information and supported actions through latent state, observations, transitions, reward or cost, and policy (Kaelbling et al., 1998).

The belief update uses only the current prior, protocol-fixed transition law, action, and observation likelihood. Future endpoint labels, later narratives, and the true latent residual are outside the decision-time information set. A model that sees them is a retrospective reconstruction or conditioned law, not a prospective controller.

Action affects both state and observation. Avoidance can reduce exposure while hiding the unresolved cause; disclosure can increase short-term arousal while opening repair; treatment can change physiology and the measurement gain at once. A valid realization models these pathways rather than treating actions as passive covariates.

Safe information-gathering actions, ordinary care transitions, and natural variation define the admissible exploration set.

Simulated POMDP trajectories show the encoded controller's behavior. A policy-level feasibility or probability-one guarantee requires a separate proof or justified bound; empirical realization requires recoverable state, prospective prediction, and causal contrasts beyond the construction.

18.3.4 — Sensors, Missingness, and State Aliasing

Sensor placement determines which latent distinctions can be recovered. A channel observed only during low movement, clinic visits, or periods of high adherence may confound state with availability. The study should describe where each sensor can operate, where it fails, and which parts of the target population those failures remove.

Use channel-specific observation models for lags, noise, floors, and causes of absence; a common error term can obscure these differences.

Model state-dependent device removal, report loss, and skipped prompts under the marked-observation framework in Section 17.4 (Little & Rubin, 2019; Hernán & Robins, 2020).

State aliasing occurs when two latent states generate similar channel patterns. Pain and exertion may share autonomic features; sedation and low distress may share reduced movement; danger and cultural restraint may share silence. Context and alternative channels can reduce aliasing, but sometimes the states remain only partially identified.

A low-burden prompt or quality check may separate aliased states. Compare prospective information gain with burden, privacy, reactivity, and held-out performance.

18.3.5 — Policy Under Model Uncertainty

Policy under model uncertainty begins with a declared family of plausible latent states, transition laws, observation models, and admissible actions. Robust, Bayesian, and set-valued approaches organize disagreement differently, but none can compensate for an omitted model family or an action set defined after the outcome is known (Manski, 2003; Kaelbling et al., 1998).

Section 7.3.4 defines almost-sure and limit-sure attainability. Here the additional burden is epistemic: the protocol must state the information-adapted policy class and belief state under which any such guarantee is claimed (Chatterjee et al., 2016; Asadi et al., 2025).

Apply Section 7.3.4: an empty admissible policy class returns RouteStatus: unavailable, empty admissible policy class. A missing action system, safety exclusion, or modeling error does not establish almost-sure or limit-sure failure of the candidate mechanism.

Partial observation changes the guarantee. A policy conditions on the declared observation history or belief state, not on the hidden history that the investigator wishes it could see. Posterior probability-one completion requires probability-one action-specific completion for almost every live history-action pair under the declared posterior model; a robust guarantee over all compatible histories is stronger.

Use the intersection and selector conditions in Section 7.3.4 as a feasibility test for this belief-state realization. Failure of those conditions at the available information level rules out its claimed robust guarantee.

Compare policies under Section 25.2's common resources. Recovery, calibration, transport, and open-set adequacy are separate criteria from average reward.

Material model disagreement may leave an admissible action set, a need for safer information, or no robust route. Participants and lawful decision-makers retain authority to refuse the research policy.

18.3.6 — Belief-State Failure and Abstention

Posterior predictive checks should target temporal structure rather than only marginal fit. The model may reproduce the overall distribution of reports while missing order, dwell time, lag, action dependence, or the pattern of missing observations. Those failures help locate problems in the realization.

Classify state-space regions separately under Section 24.2 when channel conflict, unsupported transport, rare actions, or endpoint ambiguity limits the estimate. An unresolved region need not invalidate estimation elsewhere.

Policy sensitivity supplies another failure test. If small plausible changes in the posterior, transition law, or utility reverse the proposed action, the action is not stably identified. A robust set or no-action output is preferable to a brittle recommendation presented with confidence.

Compare prospective observation plans by expected uncertainty reduction, burden, and induced behavior. If plausible belief models favor different channels or timings, preserve that design uncertainty rather than selecting the model-favored plan after outcomes are known.

18.4 — Path Conditioning and Bridge Constructions

Conditioning begins with a reference path law P_0 and an event A. For a positive-probability event, the conditioned law is $Q(B) = P_0(B|A)$. The resulting paths exhibit A by construction. Their existence demonstrates mathematical compatibility, not that an unconditioned system contains a native completion mechanism.

18.4.1 — Conditioning Events and Changed Path Laws

When h is a committor, the Doob h-transform developed in Section 18.4.2 can generate paths biased toward the completion event.

For exact singleton targets of probability zero, ordinary conditioning is insufficient. A bridge kernel, regular conditional law, disintegration, or shrinking-band limit must be justified. Tightness and uniqueness matter: different subsequential limits can define different candidate bridges.

Schrödinger bridge formulations choose a dynamically admissible path law close to a baseline while satisfying endpoint constraints. A representative objective is

$$J_{path}(K,\alpha) = \inf_{Q \in Q_{dyn}: Q(A_K) \geq 1-\alpha} D_{KL}(Q\|P_0).$$

The objective quantifies steering relative to a declared baseline path law; it supplies no evidence that nature pays, minimizes, or even represents that cost. Conditioned and native processes can instead be compared through likelihood, reciprocal structure, intervention response, and transport. A bridge mixture may be reciprocal without being Markov, while a local controller may become Markov only after the state is augmented. Those distinctions create the testable content.

Terminal conditioning reweights earlier transitions, dependence, and likelihoods throughout the path; it does not merely append a fortunate final value to the unconditioned process (Doob, 1957; Chetrite & Touchette, 2015).

The conditioning event must be explicit. Reaching a set before another, lying in a terminal band at a fixed time, surviving to an endpoint, or matching a terminal distribution are different events and induce different laws. A phrase such as conditioned to complete is too vague for scientific use.

Account for selection by complete terminal records, surviving device coverage, and reportability when defining the sample path law.

A useful analysis compares likelihoods under the native law, the explicitly conditioned law, and candidate causal dynamics. Held-out actions or unexpected endpoint probes are especially valuable because conditioning fitted to one terminal event may not predict them.

18.4.2 — Doob h-Transforms

A Doob h-transform constructs a new Markov law by reweighting transitions with a positive harmonic or space-time harmonic function. Under suitable conditions, the transformed generator describes the reference process conditioned or biased toward a declared event. The function h, its domain, and its boundary behavior are therefore part of the model, not decorative notation (Doob, 1957; Chetrite & Touchette, 2015).

The transform changes transition probabilities before the endpoint. It can generate local drift toward regions with higher completion potential while preserving Markov structure. That local appearance can resemble a controller even when the transformed law was obtained from conditioning rather than from a physical feedback mechanism.

Scientific use requires an independent route for h. Estimating h from the same terminal outcomes later used to claim prediction creates leakage. A defensible program estimates or specifies one completion potential, freezes it, and tests transport across held-out transitions, actions, endpoints, or terminal-support modules.

Action-indexed transforms require further typing. A separate h for every action can absorb any policy difference and lose causal content. The model should state whether h is common across actions, how interventions alter the reference generator, and what invariance is predicted under shared information.

Support matters. The transform is not valid where h vanishes or where the transformed measure is not absolutely continuous with respect to the reference on the relevant sigma-field. Apparent success outside support is extrapolation, not a conditioned-law result.

18.4.3 — Schrödinger Bridges and Entropic Steering

A Schrödinger bridge finds the path measure closest in relative entropy to a reference process while matching declared endpoint marginals. It is an entropy-regularized path-space transport problem and can be written as a stochastic control problem (Léonard, 2014; De Bortoli et al., 2021).

The endpoint objects are distributions rather than a single realized terminal value. Choosing them already encodes the target population and resolution. A narrow terminal distribution around zero, an atom at zero, and exact pathwise closure are different constraints and should not be collapsed into one bridge picture.

Soft constraints define a different object. Instead of forcing the terminal marginal to equal the target, a soft-constrained bridge permits mismatch and penalizes it through a divergence term.

That construction can improve robustness when a finite band or noisy target is scientifically appropriate, but it cannot be presented as exact endpoint matching; the tolerance, penalty, and resulting terminal distribution remain part of the claim (Garg et al., 2024).

The reference process carries most of the scientific meaning. A Brownian, homeostatic, learned-control, or clinically stratified baseline yields a different bridge even with the same endpoint marginals. Comparison therefore asks whether the inferred control is stable across plausible references and whether one reference explains the data without special steering.

Relative entropy is a divergence between path laws. Interpreting it as effort, suffering, or phenomenal burden requires a separate justified bridge.

Finite-sample algorithms can fit bridges that reproduce marginals while missing transition structure. Validation should include two-time dependence, intervention response, likelihood-ratio diagnostics, and recovery under the actual observation cadence. Matching the start and end distributions is a weak test of the interior law.

A native Markov route fixes the generator, state, action law, and endpoint branch, then evolves prospectively without selecting on the terminal event. Held-out transitions, interventions, and unexpected endpoints test its implications.

A Doob h-transform fixes a reference law, event, and positive completion potential, then reweights local transitions toward that event. Evaluate transport with the potential frozen, together with support and likelihood-ratio checks.

Interior covariance, reference sensitivity, and intervention responses can distinguish its implications.

A Brownian or reciprocal bridge fixes starting information, a terminal condition, and a horizon. It conditions trajectories on

boundary information by construction; the resulting reciprocal dependence does not establish an independently recovered native mechanism.

Steering measured features toward an endpoint requires a qualified measured-to-latent bridge before it implies steering the latent target.

Endpoint selection can reproduce the same apparent marginals without steering. Compare inverse-probability weighting, generative selection models, and held-out sampling branches.

18.4.4 — Brownian Bridges as an Explanatory Baseline

Use the Brownian bridge as the baseline for testing what endpoint conditioning alone can generate (Doob, 1957; Chetrite & Touchette, 2015).

Selecting trajectories known to end in a band conditions the sample even without an explicit bridge equation. The comparison therefore requires selection diagrams and prospective cohort definitions.

A useful simulation compares the observable laws produced by unconditioned diffusion, endpoint selection, and explicit bridging under the same observation kernel. Endpoint-selected paths and explicitly bridged paths can have the same conditional law; recovery must then return model ambiguity rather than distinguish labels unsupported by the data. Additional hazard, intervention, or endpoint-regime measurements discriminate only when the models specify different predictions for them.

A cumulative ledger requires a path-class-matched comparator. For an explicit Gaussian illustration, fix a deterministic $T > 0$ and fixed initial values L_0 and V_0, and start from

$$dL_t = V_t\,dt, \qquad dV_t = \sigma\,dW_t, \qquad \sigma > 0.$$

Here L is an accumulated coordinate and V its stochastic rate. Conditioning the reference law on $L_T = 0$ through its Gaussian regular conditional distribution gives, for $t < T$,

$$dL_t = V_t\,dt, \qquad dV_t = -\frac{3\{L_t + (T-t)V_t\}}{(T-t)^2}\,dt + \sigma\,dW_t.$$

The Brownian driver in this conditioned equation is defined under the conditioned law. The derivation uses $Var(L_T \mid L_t, V_t) = \sigma^2(T-t)^3/3$ under the unconditioned reference law. The accumulated coordinate remains finite variation even though its rate is stochastic. This supplies a linked residual–rate–horizon prediction for recovery benchmarks and matches the path class of an accumulated rate. The coefficients of the conditioning drift become singular as t approaches T, making resource requirements an explicit problem for biological implementations. Comparisons should give each model the same endpoint, selection, and observation information (Doob, 1957; Chetrite & Touchette, 2015).

For the zero-initial-value case $L_0 = V_0 = 0$, Gaussian conditioning also gives

$$Var(L_t \mid L_T = 0) = \frac{\sigma^2 t^3 (T-t)^2 (4T-t)}{12T^3}, \qquad 0 \le t \le T.$$

Writing $\tau = T - t$,

$$\frac{Var(L_{T-\tau} \mid L_T = 0)}{\tau^2} \to \frac{\sigma^2 T}{4} \quad \text{as } \tau \downarrow 0.$$

The same integrated Gaussian example therefore supplies both a conditional drift and quadratic preterminal variance contraction. These linked predictions can be assessed together when comparing candidate dynamics.

18.4.5 — Likelihood Ratios and Endpoint Leakage

A change of path law is interpretable only where the transformed measure has adequate support relative to the reference. Likelihood ratios quantify the reweighting on declared sigma-fields; they do not create information where the reference assigns zero probability. Support failure is therefore a structural limit, not a numerical inconvenience (Jacod & Shiryaev, 2003; Chetrite & Touchette, 2015).

Extreme likelihood ratios reveal practical nonoverlap. A small number of heavily weighted paths can make a conditioned estimate unstable even when absolute continuity holds formally.

Effective sample size, tail behavior, and weight truncation sensitivity should be reported before route claims are made.

Held-out endpoint modules help but do not solve support. A potential trained on one terminal event can be tested on a different event or unexpected probe, provided the state and observation bridges are common. Failure indicates that the original transform may have learned event-specific selection rather than a general route.

Unestablished support is classified under Section 24.2; leakage that invalidates the comparison makes its evidence Unusable.

18.4.6 — Native and Retrospectively Conditioned Laws

Matching endpoint distributions or one-time marginals can conceal different transition probabilities, reciprocal dependence, and intervention responses (Doob, 1957; Chetrite & Touchette, 2015; Léonard, 2014).

The distinction requires a reference law. A route claim should state the unconditioned generator or transition kernel, the information available at each time, and the way the endpoint enters. If the native law is never estimated, the analyst cannot

determine whether a fitted bridge represents a mechanism or merely a conditional description.

Preterminal covariance discriminates only when the declared models imply different observed kernels. A Brownian bridge is itself Markov and admits a time-dependent feedback representation; reciprocal structure alone does not distinguish conditioning from a native implementation. If the path laws agree, passive observations cannot separate the causal accounts. A proposed intervention must specify different responses before it can do so (Çetin & Danilova, 2016).

Unexpected endpoint probes provide another test. A transform learned from one planned terminal event may fail when an independently occurring probe samples the path. A genuine native route should state what remains invariant across those probes and what changes with the endpoint branch.

A route retained only as a conditional representation can be evaluated for descriptive calibration; that use does not identify the causal process generating the paths.

A direct diagnostic compares forward simulation from the estimated native law with endpoint-conditioned reconstruction. If only the conditioned reconstruction reproduces the data, the result favors that description over the fitted native candidate; it does not exclude other native laws or identify selection as the cause. If both fit, the mechanism remains ambiguous until intervention or unexpected-endpoint evidence separates them.

Show the reference law, conditioning event, and transformed law separately in public diagrams.

18.5 — Absorbing, Hitting, and Bridge Routes

Processes can share an endpoint value while reaching it through different dynamics. First hitting, absorption, killing, reflection, stickiness, and singular timing impose different conditions before and at the boundary. A protocol must name the branch it tests and preserve the distinction between the strict-pre-endpoint domain and an endpoint-inclusive claim. The useful comparison asks which preterminal observations discriminate the routes, rather than treating a common ending as proof of a common mechanism.

18.5.1 — First-Hitting and Zero-Surface Termination

First-hitting implementations define an endpoint as the first time a process enters a declared set. For the zero-residual surface, closure follows from the stopping rule whenever the surface is reached. The empirical questions concern the justification of that state and stopping relation and the preterminal observations that distinguish the route from alternatives (Doob, 1957).

A zero surface can be crossed, touched, approached without contact, or reached by a jump. Those possibilities imply different local signatures. Continuous diffusion may accumulate local time near the boundary; a finite-variation path may approach monotonically; a jump process may cross without ever occupying zero. The implementation must specify path class and boundary regularity before a hitting-time theorem is invoked.

A first-hitting candidate specifies a stopping mechanism and compares its timing with an independently defined endpoint. Keeping those definitions separate makes their relationship testable. The same model can also generate route-level predictions about trajectories before the stopping event.

A low-risk model-recovery study can compare simulated first-hitting paths with fixed-time bridges, absorbing processes, and

state-coupled hazards. The observation pipeline should include realistic sampling and missingness. Distinguishing features may include return behavior before stopping, hazard spikes near the surface, and response to safe perturbations that shift the state without altering endpoint definition.

Compare first hitting with ordinary threshold decisions, failure limits, physiological triggers, and institutional stop rules in the same state space.

18.5.2 — Absorbing and Killed Processes

An absorbing implementation keeps a process in a boundary state once that state is reached. A killed process instead terminates the path or transfers probability mass to a cemetery state. These constructions may share a first entrance time and differ completely afterward. The distinction matters whenever post-contact observation, return, or continued consciousness is possible (Champagnat & Villemonais, 2016).

Absorption creates a persistent atom at the boundary in later-time distributions. Killing removes the path from ordinary state space and changes denominators. A dataset that treats missing post-boundary observations as zeros can make killed trajectories look absorbed. The observation and retention processes must therefore be modeled alongside the state dynamics.

Classify administrative stops, device shutdown, report loss, and care-required cessation as observation events under Chapter 13’s endpoint rules.

Quasi-stationary behavior supplies another diagnostic. Among paths not yet killed, the conditional state distribution may stabilize even as survival probability declines. That pattern can resemble regulation or terminal concentration and should be compared directly. The route’s predictions concern both the

surviving distribution and the exit process, not one terminal histogram.

Simulation should preserve denominator flow. Analysts should report how many paths remain at risk, how many are absorbed, how many are killed, and how many become unobservable for other reasons. Recovery succeeds only if the pipeline can distinguish those branches under the actual cadence and noise floor.

18.5.3 — Sticky and Reflected Boundaries

Sticky and reflected boundaries preserve continued dynamics after contact. Reflection pushes a process back into the interior, while stickiness allows positive occupation time at the boundary before movement resumes. Both can create repeated near-zero observations without terminal absorption. Their local-time structure is therefore central to discrimination rather than a mathematical detail.

Reflection permits immediate returns after boundary contact; stickiness can produce a pile-up at zero before an endpoint elsewhere. Include sampling frequency and measurement floor in simulations because rounding small excursions can make either resemble absorption.

Useful tests examine dwell time, excursion distribution, return frequency, and sensitivity to safe perturbation. Instantaneous reflection prevents passage beyond the boundary without holding the path there; stickiness permits positive boundary occupation time. An absorbing model predicts no return. These features can be estimated from preterminal data when cadence is sufficient and the observation process is not itself sticky.

Hidden state complicates the picture. A measured channel may appear stuck while the latent state moves, or appear reflected because a bounded scale truncates values. Cross-channel

comparison and latent-state recovery should therefore precede a mechanism verdict. A zero-heavy report scale is not evidence of a zero-heavy phenomenal process.

18.5.4 — Endpoint Atoms and Singular Carriers

Endpoint atoms place positive probability on particular times or states. Singular carriers concentrate endpoint mass on sets that need not possess an ordinary density. Both can defeat arguments requiring an independent endpoint law with a positive Lebesgue density on the declared support; singular continuous laws, however, remain non-atomic. The model must declare whether atoms or singular carriers arise from biology, protocol, scheduling, state-coupled stopping, or observation design before applying the relevant obstruction.

Atoms can be mundane. A study may end at a scheduled visit, a device may shut down at a battery threshold, or a protocol may stop at discharge. Those events can produce repeated endpoint times unrelated to the truth-level endpoint. Treating them as natural terminal structure would import the study calendar into the hypothesis.

State atoms are equally important. Rounding, censoring, and bounded scales can produce a pile-up at zero. A latent mixture can also include a true atom, a narrow continuous component, or both. The analysis should compare those possibilities through an observation-kernel model rather than declaring exact neutrality from repeated zero readings.

Singular endpoint supports require topology as well as probability. If endpoints lie on a lower-dimensional manifold or a discrete state graph, density-based tests may be inappropriate. Support-aware methods should ask which neighborhoods carry mass and whether the proposed route predicts the carrier independently of the terminal observations used to discover it.

A rigorous design uses negative controls for administrative atoms and simulated observation floors. It also separates the time atom from the state atom: many paths can end at the same scheduled time without sharing a terminal state, and many can share a recorded state because the instrument has coarse bins. Each factor receives its own model component.

18.5.5 — Continuation and Endpoint-Inclusive Cases

Continued-stream and endpoint-inclusive branches change the domain of the claim. If experience continues after the observed endpoint, then the recorded terminal residual is a prefix, not the complete object. If a phenomenal contribution is assigned exactly at the endpoint, the strict-pre-endpoint and inclusive ledgers differ by that contribution. These branches are non-equivalent and must be specified separately before outcome inspection.

Return and resuscitation make the distinction concrete. A protocol may close a window when report ceases and later observe recovery. The original stop cannot be treated as the truth endpoint. The stream-routing and endpoint records should reopen or mark the earlier object as provisional rather than forcing the returned experience into a new stream for convenience.

Endpoint-inclusive models also require a rule for what belongs at the boundary. A final conscious event, a transition with uncertain duration, and a biological event without experience are different. Assigning all terminal physiology to phenomenal value would collapse mechanism and target. The inclusion rule must be independently justified and sensitivity-tested.

Comparisons across branches should use the same pre-endpoint data and explicitly different remainder models. One analysis can estimate strict-pre-endpoint compatibility; another

can bound an unobserved terminal remainder; a third can remain nonadjudicable. The branch choice should not be changed to rescue an unfavorable result.

Endpoint-branch adjudication uses a frozen event packet that withholds the terminal ledger estimate. The packet records the protocol endpoint, clinical state, evidence of continuing or returning consciousness, measurement-shutdown reason, routing decision, and uncertainty. A blinded panel then classifies the event as strict-pre-endpoint, continued-stream, endpoint-inclusive, provisional, or nonadjudicable. This ordering prevents the favored terminal value from deciding which endpoint branch is said to have occurred.

An unobserved terminal remainder can sometimes be bounded even when it cannot be point estimated. The bound may use the last observation meeting the declared qualification criteria, maximum plausible exposure under the chosen clock, duration uncertainty, and branch-specific limits on the measured-to-latent bridge. The full interval enters the route verdict. When a valid finite bound spans outcomes that reverse the conclusion, the result is Inconclusive evidence rather than a midpoint presented as terminal truth (Little & Rubin, 2019; Manski, 2003).

Recovery simulations should include branch misclassification. Synthetic paths can continue after observation shutdown, include an endpoint atom, return after a provisional stop, or carry an unobserved inclusive contribution. The complete pipeline is then asked to recover the correct branch and to abstain when the evidence is insufficient. A route estimator that appears accurate only when the endpoint branch is supplied by the simulator has not solved the adjudication problem.

Each caption names the endpoint object, the strict or inclusive domain, the treatment of return, and the remainder status.

18.5.6 — Preterminal Tests of Equal Endpoints

Compare covariance, hazard, dwell, return, jump, local-time, and intervention-response signatures where the declared branches predict differences. An endpoint-conditioned bridge and a native controller need not be distinguishable under every observation or intervention.

Include boundary-specific observation failures in the simulations: sparse sampling that misses returns, floors that hide excursions, and informative dropout that removes unstable paths.

A common-rule tournament can begin with identical state information and endpoint labels, then withhold endpoint labels in a prospective module. If a route succeeds only when the final event is supplied, it is a conditional reconstruction rather than a discovered mechanism. Held-out unexpected endpoints are especially valuable because they prevent the model from relying on a known horizon.

Hazard and local-time signatures should be modeled together where the implementation permits. A rise in endpoint hazard near the target, repeated grazing without termination, and long residence at the boundary are different phenomena. Collapsing them into one terminal-zero indicator destroys the mechanism information.

Boundary-contact data should retain exact timing at the resolution the sensor can support. Rounding several contacts into one terminal observation can make reflection, stickiness, and absorption indistinguishable. The analysis should model interval censoring rather than pretend the contact time is known.

If cadence cannot distinguish compatible implementations, name the lost contrast and apply Section 24.2's adequacy and ambiguity rules; do not count compatibility as support for each mechanism.

A path-law comparison should score one-step and multi-step predictive densities before the endpoint, not only classify the implementation after the full trajectory is visible. Transition residuals can reveal whether the fitted process places probability on the observed sequence of approaches, retreats, and returns. Cross-validated likelihood or proper-score differences should be reported by distance from the boundary so a final segment cannot dominate an otherwise poor preterminal fit (Gneiting & Raftery, 2007).

State normalization matters when comparing intervention response. Two branches may appear to react differently only because one cohort begins closer to the target surface or with less remaining action support. The perturbation analysis should condition or match on the declared preintervention state, then compare the timing and shape of divergence. A branch that requires post hoc state redefinition to recover its expected response has failed the prospective mechanism test.

Channel-ablation tests can expose observation dependence. The mechanism should be refitted or rescored after removing the channel most directly tied to endpoint ascertainment, while retaining the same state definition and verdict rule. If discrimination collapses only when that channel is withheld, the result may reflect endpoint leakage rather than a path-law difference. A robust branch should preserve at least its prospectively specified qualitative ordering across independent observation routes.

18.6 — The Queue Model as an Illustrative Candidate

The Queue model is an illustrative candidate for how unresolved demands might persist, compete for attention, and regain influence when opportunities for response change. It must specify a state, a demand representation, and a control process before priority, blocking, or repair can be interpreted. Removing an opportunity or changing a cue supplies a contrast only when ordinary learning, regulation, and control rivals are also evaluated. The model remains subject to recovery, scope, and retirement tests.

18.6.1 — Candidate Mechanism Status

Test whether explicit unresolved demands, blocking, reactivation, and repair improve prediction and causal localization beyond ordinary rivals (Kaelbling et al., 1998; Sutton & Barto, 2018).

A caregiver's recurring vigilance after months of night alarms may have queue-like features, while a transient mood shift after bad weather may not. The model must allow partial domain.

The model also needs a negative space. Certain experiences should not enter the queue simply because they are vivid. Joy, boredom, fatigue, ordinary desire, routine planning, and benign memory may affect lived experience without becoming unresolved demands.

18.6.2 — State, Demand, and Control

The queue illustration begins with three explicit objects: a state describing active and latent demands, a demand representation that can be multidimensional, and a control process that changes future state or access.

State and control are defined on the action support actually available, so inaccessible repair routes remain distinct from internal transition failures.

One candidate input is an unresolved mismatch between expected and actual affective consequence. A person expected an apology to feel safe, but it did not. A long-sought achievement produced relief for a day and dread afterward. A harmless-looking letter triggered fear because past letters carried danger. These mismatches can update beliefs and keep demands active (Kaelbling et al., 1998; Sutton & Barto, 2018).

The distinctive question is whether unresolved affective mismatch predicts priority and reactivation beyond predictive-processing and active-inference accounts of salience, attention, avoidance, and reappraisal.

Affective prediction error should not be treated as a single scalar surprise. It may involve valence, threat, bodily cost, agency, meaning, memory, and social expectation. The queue entry rule needs to specify which feature is unresolved.

One candidate operational rule admits a demand when mismatch exceeds a declared threshold, persists, and continues to influence attention, action, or expected future cost. The record separately identifies whether a repair-relevant action is available, blocked, or absent. Lack of a repair route can sustain a demand rather than prevent its entry.

Relational demands can continue to shape attention and action after an interaction ends: attachment, betrayal, obligation, shame, apology, exclusion, caregiving, conflict, and unspoken fear are context-dependent sources (Shenhav et al., 2013; Kaelbling et al., 1998).

A realistic example is a caregiver who smiles during a clinic visit because the parent is present, then cannot sleep after reading the discharge instructions alone. The demand is not

simply anxiety. It may involve responsibility, uncertainty, role conflict, lack of respite, fear of making the wrong call, and no safe route to transfer the load. A queue entry rule that sees only mood misses the social structure.

Use Section 15.7.2's bystander-privacy rules for relational-demand reports.

Entry rules require contextual priors and transport checks: a delayed reply, silence, apology, or obligation can have different meanings across social settings.

Relational repair has been studied as a time-dependent process. McCullough and colleagues followed 337 people after a recent interpersonal transgression, using repeated reports over approximately three weeks. Reported conciliatory gestures, including apologies and offers of compensation, were associated with increasing perceived relationship value and decreasing perceived exploitation risk. The longitudinal model linked rising relationship value to faster reductions in vengeful and avoidant motivation (McCullough et al., 2014). This gives a relational-demand model concrete variables to track: what the person remembers, what the partner does next, how future safety and relationship value are appraised, and how motivation changes over time.

Relational demands often interact with material constraints. A person may remain in contact because housing, care, immigration status, work, or family obligation narrows alternatives. The queue cannot treat continued contact as evidence that the demand is low. Sometimes the action set explains behavior better than preference.

The model may not prescribe forgiveness, reconciliation, exposure, confession, or public testimony. Repair remains voluntary, contextual, and safety-dependent.

A large intervention study demonstrates that emotional responses to past interpersonal harm can change through a self-directed practice. In the International REACH trial, 4,786 participants across sites in Colombia, Hong Kong, Indonesia, South Africa, and Ukraine were randomized to receive a forgiveness workbook immediately or after a two-week delay; the reported analysis included 4,598 adults. At two weeks, the immediate group had lower unforgiveness, depression symptoms, and anxiety symptoms, with standardized mean differences of −0.53, −0.22, and −0.21, respectively (Ho et al., 2024). The measured changes provide a concrete basis for studying how voluntary reflection alters the continuing influence of relational events. Longer follow-up and active comparison conditions would help identify durability and the most effective ingredients.

18.6.3 — Persistent Influence of Unresolved Demand

A demand is unresolved when its generating condition, expected cost, blocked action, or state uncertainty continues to influence the system (Shenhav et al., 2013; Kaelbling et al., 1998).

Consider a person who keeps checking the mailbox after months of medical appeals. The demand may be unresolved because the institution can still deny medication, the person lacks reliable information, and prior letters have carried real consequences.

After betrayal, trust may remain difficult even after apology. A queue model can represent remaining uncertainty, risk estimation, memory salience, and the cost of future vulnerability. The action set includes refusal and protection.

18.6.4 — Priority, Blocking, and Reactivation

Priority, blocking, and reactivation are the queue family's distinctive dynamics. Priority allocates limited control, blocking reduces current influence or observability, and reactivation predicts a specific return when context or action support changes. These claims should be separated so one vague recurrence does not count as all three.

Priority can be represented as a function of expected cost, controllability, time sensitivity, context, and resource availability. A demand can be high priority because it threatens safety, because repair is possible now, because delay increases risk, or because context makes it salient (Manski, 2003; Jacod & Shiryaev, 2003; Tao et al., 2018).

Section 18.6.4 specifies an illustrative sequence for a future simulation. It shows the intended entry, blocking, reactivation, repair, and residual-state transitions; it does not report an executed simulation or a finding about a patient or institution.

1. Entry. A hospital-billing cue adds a demand with high uncertainty and moderate controllability. Does a specific cue increase belief in an unresolved administrative-threat demand beyond general stress?

2. Blocking. A childcare deadline consumes attention and temporarily suppresses message-opening behavior. Can blocked access occur without treating the demand as resolved?

3. Reactivation. A reminder email reactivates the demand and lengthens the delay before the message is opened. Does reactivation depend on cue specificity rather than any unpleasant event?

4. Repair attempt. A caseworker call lowers uncertainty and changes available actions. Does priority fall because state changed, not because the model declared closure?

5. Residual. Future portal cues retain some salience after the bill is fixed. Can the model represent partial repair while allowing rivals to explain residual vigilance?

An illustrative priority score can make the candidate dynamics inspectable:

$$P_i(t) = \beta_C \hat{C}_i(t) + \beta_U \hat{U}_i(t) + \frac{\beta_H}{H_i(t) + \varepsilon} + \beta_R \hat{R}_i(t) + \gamma^T X_i(t).$$

Here the left-hand side is the modeled priority of demand i at time t. The hatted C term is estimated future cost, the hatted U term is uncertainty, the inverse-horizon term captures urgency as remaining opportunity narrows, the hatted R term is available resources or repair capacity, and X i (t) records context. The positive constant epsilon prevents division by zero.

The sign and units are declared for every use. A high value may mean high action priority, high monitoring priority, or high expected control cost, depending on the model. Keeping those meanings explicit makes the equation interpretable.

Priority should also be stochastic. Demands may fluctuate with sleep, medication, pain, context, social safety, and observation. A deterministic priority rank may be useful in a task, but real systems will need uncertainty. The model should report posterior priority distributions where possible.

Reactivation occurs when a demand that had low current influence returns to shape attention, report, physiology, memory, or action. Triggers may include sensory cues, similar contexts, relational contact, institutional notices, bodily states, anniversaries, treatment changes, or new repair opportunities. The trigger supplies a structured observation (Wilson & Collins, 2019; Manski, 2003).

Reactivation must be distinguished from ordinary conditioning. A cue can produce fear because it was paired with

danger. That is a strong rival. The Queue model adds value only if reactivation also depends on unresolved action, repair availability, demand priority, or state uncertainty beyond cue association alone.

A candidate reactivation condition can be written:

$$\Pr\left(D_{i,t+\Delta} > \theta_i | H_t\right) = logit^{-1}\left(\alpha_i + \beta_c c_{i,t} + \beta_b \hat{b}_{i,t} + \beta_a^T a_t^{live} + \beta_x^T X_t\right).$$

The display models the probability that demand i exceeds the protocol-fixed reactivation threshold at the next interval, conditional on the available history. Cue relevance, current state belief, the live action menu, and context enter separately, so each coefficient can fail without converting every recurrence into queue evidence. The diagnostic becomes informative only when cue-conditioning and other rivals receive the same data, tuning budget, and threshold rule.

A study might use mild, consented reminders of ordinary tasks, not traumatic reenactment. It might vary information availability or repair options rather than inducing distress.

18.6.5 — Discharge, Repair, and Residual State

Discharge and repair are route-specific changes, not narrative declarations. A source can be removed, a belief can update, an external remedy can open an action, and a residual state can remain. The model should represent which component changed and what evidence would show that apparent relief was only observation loss.

Direct resolution occurs when the generating condition is removed or the action sequence completes in a way that materially changes the state. A medical appeal is approved and medication arrives. A door is repaired and the person can sleep. A missing document is found. A threat ends in a way that is verified and stable.

Direct resolution changes the demand's transition law by reducing the expected cost of its generating condition; residual effects remain separately modeled.

A candidate transition can be written:

$$\Delta_i^{res}(s_t) = E\left[D_{i,t+\Delta} | do\left(A_t = a_{resolve}\right), S_t = s_t\right] - E\left[D_{i,t+\Delta} | do\left(A_t = a_{wait}\right), S_t = s_t\right] < 0.$$

The contrast compares expected next-state demand under a supported resolving action with the same quantity under waiting, conditional on the declared state. A negative value is the candidate prediction; it is not guaranteed. Identification requires well-defined treatment versions, support for both actions, and the same confound and measurement rules used for rivals (Pearl, 2009; Hernán & Robins, 2020).

Direct resolution should be measured with enough time depth to avoid false closure. The letter arrives today, but sleep may take weeks. The threat ends, but the body may continue to monitor.

External repair includes care, safety, apology, material change, institutional remedy, access, respite, debt relief, corrected records, and restored routes of appeal. It matters because many demands persist outside the person.

Code the state changes accompanying an apology (acknowledgment, behavior, safety, accountability, and time) rather than treating the word itself as a discharge action.

Code remedy, desired return of results, record correction, and changed access under the study's privacy rules.

Rivals can explain response to external repair through ordinary stress relief or rational action. Again, the Queue model earns support only if it predicts specific residual or reactivation patterns after repair. For example, it may predict that direct

material repair reduces demand intensity while relational cues continue to reactivate uncertainty unless accountability changes.

Residual burden remains after resolution, update, repair, or partial relief. Nonresolving burden persists because the source remains, repair is impossible, memory remains active, the action set is closed, or the model is wrong about the relevant state.

Residual influence can be represented through lower but nonzero demand states, persistent cue sensitivity, altered transition rules, or slow recovery. Each representation makes different predictions about recurrence, context, and response to repair and must therefore be selected by evidence rather than narrative convenience.

18.6.6 — Queue Failure, Scope, and Retirement

Retire or narrow the queue's empirical use when its state remains unidentified, its priority or reactivation predictions fail, or ordinary learning and scarcity models explain the observations more simply. Keep the reason distinct under Sections 24.2 and 27.7.

If a model labels ordinary fatigue, benign memory, cultural restraint, or momentary frustration as unresolved demand, it will overstate queue prevalence. If it labels every lack of report as resolution, it will understate hidden burden.

A candidate demand requires an identifiable generating event, condition, or continuing constraint. Failure to identify such a source undermines the proposed demand interpretation.

Specify the policy, attention, physiology, or memory that the candidate demand changes. Salience without a continuing demand-related influence fails this part of the account.

Identify the observation channels supporting the demand belief. Treating one noisy signal as a directly observed state fails the observation requirement.

Specify a fair simpler rival for the candidate demand model. Without that comparison, the proposed queue interpretation has not faced its required alternative.

Section 18.6.6 provides an audit aid for a research model. It is not a screening procedure for persons.

False positives should be studied through negative controls. A task might include salient but resolved events, neutral administrative reminders, or safe surprise conditions that should not produce queue-like reactivation. If the model flags these at high rates, its entry criteria are too broad.

Test a harm-avoidance controller that reproduces urgency, blocking, and apparent repair without a queue state. If the pipeline cannot distinguish it on held-out tasks, withhold distinctive queue attribution under that design. Examine which diagnostic features cause the confusion before deciding whether to redesign or retire that empirical use.

Cue-specific reactivation may survive a priority failure; blocked access may be measurable without identified discharge; residual influence may survive an unidentified entry rule. Recombine these findings only where a prospectively stated mechanism requires their conjunction.

Distinguish empirical rejection from a route withheld for burden: a scientifically open model may remain unavailable for first-edition use.

18.7 — Construction, Identification, and Recovery

Construction alone does not identify a mechanism. Different states must generate distinguishable observables. The actual observation process must preserve those differences. Recovery analysis asks whether the model can retrieve them.

18.7.1 — Existence and Empirical Realization

A solution, bridge measure, committor, or viable controller establishes a construction under stated assumptions. Its states, generator, policy, and boundaries then define the realization to investigate (Bollen, 1989; Villaverde, 2019).

Realization requires a bridge from the construction to observables. The model specifies which variables correspond to measured channels, which parameters vary across persons or contexts, and which interventions alter the proposed mechanism or its reports.

A physical completion coordinate supplies a concrete realization to investigate. On one qualifying stream, let a be the declared initial boundary, s the fixed exposure clock, and L_s the accumulated signed contribution with $L_a = 0$. Nominate a physical state S_s and scalar potential Φ independently of the tested ledger, with $\Phi(S_s)$ absolutely continuous on the asserted branch. If their changes satisfy $dL_s = -d\Phi(S_s)$, integration gives

$$L_s = \Phi(S_a) - \Phi(S_s).$$

An independently established boundary relation $\Phi(S_{T-}) = \Phi(S_a)$ then gives exact terminal closure. Equal boundary values suffice; neither value has to be zero. This route can operate through current physical state without the controller forecasting an ending, provided the claimed dynamics and boundary relation actually hold.

The research task is to identify a compact physical coordinate whose changes predict calibrated signed experience on new paths, then test its boundary relation separately. Defining Φ as minus the observed ledger supplies no independent prediction. An enrollment reset supplies no initial condition for an earlier stream, and a reset at or after T cannot change its strict-pre-endpoint residual. A successful independently qualified coordinate would connect an ordinary physical mechanism to the exact target.

A biological mechanism satisfying the declared contribution, exposure, ownership, and boundary relations realizes that branch of PTVC.

An approximate realization should report its cumulative accounting budget. For a specified finite path with calibrated signed edge contributions w_e, set $r_e = w_e + \Phi(v) - \Phi(u)$ on edge $u \to v$. Telescoping gives

$$L_{path} = \Phi(s) - \Phi(b) + \sum_{e \in path} r_e.$$

If independently justified bounds give $|\Phi(s) - \Phi(b)| \le \varepsilon_B$ and $\sum_e |r_e| \le \varepsilon_R$, then $|L_{path}| \le \varepsilon_B + \varepsilon_R$. Infinite paths require the stated boundary limits and convergence of the error sum. Small errors of one sign can accumulate; mean-zero residuals or a fitted RMSE do not bound an individual path. Statistical budgets need joint coverage under their maintained observation laws. Validate the contribution bridge, boundary mismatch, and missing history separately.

Within a specified finite-state realization, a potential is also necessary. Fix a directed graph of reachable states, one initial state s, absorbing strict-pre-endpoint boundary states, and deterministic additive edge contributions w_{uv}. Every relevant state must admit a terminating suffix, and that same suffix must

be admissible after every prefix reaching the state. Suppose every admitted finite terminating path has total zero.

Two prefixes reaching u must have equal totals: adjoining one common terminating suffix makes both totals zero. Let $\Phi(u)$ be minus this common prefix total. Appending an edge then gives

$$w_{uv} = \Phi(u) - \Phi(v), \qquad \Phi(s) = \Phi(b) = 0.$$

The equality holds for every declared boundary state b.

Conversely, these edge and boundary equations make every terminating path telescope to zero. The restriction therefore identifies exactly the accounting structure required in this recombinable class. A shared coarse label does not establish recombination: different histories may permit different suffixes.

Random rewards, omitted time, or hidden history require their own qualified state model. Freeze the state and admissible transitions before testing new paths; adding the observed ledger to each state label makes the construction automatic rather than explanatory.

18.7.2 — Identification and Equivalent Observations

Different parameter values or mechanism families may imply the same observable law (Bollen, 1989; Villaverde, 2019). The restrictions below test what a nominated continuation state must retain.

Exact completion imposes a memory requirement on a specified forward realization. At a fixed admissible cut, let L be the accumulated signed prefix and R the entire remaining contribution under the same representation, exposure, ownership and endpoint rules. Let X include all state information used to generate that continuation, so that L and R are conditionally independent given X. If $L+R=0$ almost surely, the state must determine L up to null events. With finite second moments, conditional independence gives $Cov(L,R \mid X) = 0$, while

$R = -L$ gives $Cov(L, R \mid X) = -Var(L \mid X)$; hence $Var(L \mid X) = 0$.

For real L and standard Borel X, the exact conclusion does not require moments. Conditional independence makes each conditional threshold probability $P(L \leq q \mid X)$, for rational q, equal to its square. The resulting conditional distribution is a point mass. Under finite second moments, approximate completion instead obeys

$$E[(L+R)^2] \geq E[Var(L \mid X)].$$

This lower bound turns unresolved loss of prefix information into a completion-error floor. The necessary information may be neural, molecular, bodily or distributed; it need not be consciously recalled. A noisy assay can lose information that the complete state retains. The restriction therefore tests a nominated causal-state realization, with the endpoint process included, rather than every formulation of PTVC.

The same state-sufficiency premise gives a quantitative design bound. At the declared cut, suppose L has a Gaussian distribution with mean m and variance $s^2>0$. No empirical recentering is required. Let X include the complete continuation-relevant state, including the state governing endpoint timing, and retain conditional independence of L and R given X. Set $D=E[(L+R)^2]$. For $0<D<s^2$, mutual information in bits obeys

$$I_2(L;X) \geq I_2(L;-R) \geq \tfrac{1}{2} \log_2\left(\frac{s^2}{D}\right).$$

Here −R reconstructs L with mean squared error D. Conditional independence supplies the first inequality by data processing; the Gaussian squared-error information bound supplies the second (Cover & Thomas, 2006). If X has at most K discrete states, $I_2(L;X) \leq \log_2 K$, giving

$$D \geq \frac{s^2}{K^2}.$$

Reducing root-mean-square closure error to one tenth of the prefix standard deviation therefore requires at least $\log_2 10 \approx 3.322$ bits in this model. This is necessary retained information, not a sufficient controller or a measured brain capacity. Under the same conditional-independence premise, the full error also decomposes as

$$D = E[Var(L \mid X)] + E[Var(R \mid X)] + E[(E[L \mid X] + E[R \mid X])^2].$$

The three terms separate unresolved prefix information, continuation variability, and conditional mean mismatch. As D approaches zero, the required information diverges in the continuous Gaussian idealization. The bound and decomposition therefore provide quantitative tests of nominated causal-state budgets, alongside the response-rank restriction below.

A finite-state claim also imposes an observable restriction. At a common cut, suppose the complete continuation state has at most K discrete values. Let rows of M index prospectively defined past conditions and columns index future probes and response events. If probe assignment preserves the state distribution within each row, and response laws given the state are otherwise common, then $M = AB$: A contains state probabilities and B contains probe-response probabilities. Consequently $rank(M) \leq K$. An excess singular value beyond a valid simultaneous measurement-and-sampling error bound rejects that nominated state budget. Use the actual independent episodes assigned to each probe when computing sampling uncertainty. Include observation loss as a response event, or model it without silently conditioning on complete cases. One continuous coordinate can generate matrices of arbitrarily high

rank, and low rank establishes neither a unique hidden state nor terminal closure.

If active and blocked demand, or residual burden and ordinary fatigue, cannot be distinguished, qualify only an identified combined factor; otherwise classify the unresolved target under Section 24.2.

When distinguishing states would require coercion, delayed care, excessive burden, or prohibited record linkage, the study must use the remaining admissible design and classify its identified set under Section 24.2.

18.7.3 — Generator Recovery

Recover generators through the actual observation process using preterminal drift, diffusion, jumps, hazards, action response, dependence, and missingness, rather than terminal histograms alone (Villaverde, 2019; Peters et al., 2016).

Perturb state dynamics, measurement error, and missingness separately in recovery: a wrong observation kernel can dis-credit correct dynamics, while flexible hidden-state transitions can absorb measurement error.

Where generators predict different responses, change information, action availability, or a benign state component while fixing the target, burden ceiling, and outcome rule. Score the contrast on a held-out module under common observation.

Cross-context prediction tests whether one physical coordinate explains several conditions. Freeze the state definition, contribution calibration and C^2 potential Φ. In a common clock let condition a have dynamics $dS = b_a ds + G_a dW$, generator $\mathscr{A}_a$ and contribution rate v_a. Exact accounting requires $G_a^{\mathsf{T}} \nabla \Phi = 0$ and $v_a + \mathscr{A}_a \Phi = 0$ along its supported process. On states supported in both compared conditions, with the regularity needed to extend those identities,

$$v_a - v_b + (\mathscr{A}_a - \mathscr{A}_b)\Phi = 0.$$

This subtraction is not licensed by disjoint state samples without a justified transport model. A strong experiment predicts a held-out change in context, timing or available information using the same coordinate and units, while checking intervention effects on measurement. If the combined diffusion directions span the state space at every point of a connected shared region, the common potential must be constant there. Incomplete span leaves a candidate possible but does not construct a globally integrable potential. Frozen state definitions and controlled complexity make the comparison a test of one mechanism rather than a collection of separately fitted descriptions.

Overlapping interventions can test a shared potential without first fitting its formula. In a differentiable deterministic model on one fixed physical clock, let $f(S,u)$ be state velocity and let the signed contribution rate satisfy $\ell(S,u) = -\nabla\Phi(S)^{\mathsf{T}} f(S,u)$ throughout a supported neighborhood of control u. At the same state, collect the causal derivatives $\partial f/\partial u_j$ as columns of F. If the integrated-reading rate is $z = \ell + o\rho$, with one common origin o and qualified exposure rate ρ, define $r_j = \partial z/\partial u_j$ and $h_j = \partial\rho/\partial u_j$. Every control contrast q satisfying

$$Fq = 0, \qquad h^{\mathsf{T}} q = 0$$

must satisfy $q^{\mathsf{T}} r = 0$. Differentiating the common-potential relation gives $r = -F^{\mathsf{T}}\nabla\Phi + oh$, so both unknown terms cancel. A nonzero contrast exists only if the combined drift-and-exposure matrix has a nontrivial nullspace. This is a test of every smooth potential on the nominated state, with the physical contrast fixed before inspecting the held-out rate response.

Estimated cancellation needs a complete error budget. Small physical residuals cannot bound the predicted response unless

potential gradients and origin are independently bounded or included in valid joint inference. Finite pulses add curvature error, and an omitted state can remove the apparent cancellation. For diffusion models, changing covariance contributes a Hessian term that must also cancel; the deterministic formula cannot simply be reused. A reproducible violation can exclude the specified coordinate family. Agreement supports a local physical restriction and leaves global integrability and terminal closure to separate tests.

Report class-specific recovery errors and open-set rejection; aggregate accuracy can hide failure for the route under test. Use the other recovery diagnostics in Section 25.2.

18.7.4 — Held-Out Prediction and Transport

Cross-equation transport tests whether one frozen object predicts another relation, action response, or endpoint regime. Refitting it separately for each equation removes that overidentifying restriction.

A fixed physical potential predicts the same signed contribution along any two admissible paths between the same sufficient states. For a finite graph, let B have -1 at an edge's source and $+1$ at its target, let w contain the calibrated edge contributions, and let φ contain the state potentials. Then

$$w = -B^{\mathsf{T}}\varphi, \qquad Bc = 0 \;\Rightarrow\; c^{\mathsf{T}}w = 0.$$

The cycle space contains $m - n + k$ independent algebraic constraints for m edges, n vertices and k connected components. A contrast can compare two forward paths; it need not require physically reversing a transition. Repeated dependent loops improve precision without adding independent constraints. These are standard graph-potential restrictions (Jiang et al., 2011).

Qualify whether each edge quantity is a realized contribution or only a conditional mean. Mean-zero circulation permits nonzero individual circulation when edge rewards are random. Estimate the state map on development records and test new paths with shared-edge covariance, state uncertainty and observation loss included. A failed contrast can reject the nominated state-potential route. Any added state must earn its complexity through new predictions; an ordinary potential mechanism faces the same test and can legitimately pass.

A concrete task might present low-stakes unresolved goals, information cues, and repair opportunities across days. The model predicts which goals return to attention when repair becomes possible, which remain blocked under competing load, and which decline after direct resolution.

Cue-specific reactivation. Demand-linked cues should update belief and priority more than affectively matched but unrelated cues. Associative learning or trauma-cue conditioning may explain the same pattern without queue structure.

Control allocation. Higher priority should shift latency, task switching, information seeking, or avoidance under constrained attention. Expected value of control, fatigue, or scarcity models may predict the allocation more simply.

Interoceptive salience. Bodily readiness may mark demand relevance when verbal report is delayed or blocked. Homeostasis, anxiety physiology, medication, illness, or sleep loss may explain the signal.

Memory-access pattern. Blocked demands may reappear under context, safety, or cue changes rather than constant conscious access. Memory consolidation, suppression, or ordinary contextual retrieval may win.

Repair-sensitive change. External remedy should alter later cue response when it actually changes state or options. Relief,

expectancy, social support, or simple uncertainty reduction may fully explain the change.

Each proposed queue dynamic requires both a distinctive prediction and a serious competing explanation. The following contrasts identify that comparison.

Blocking predicts that a demand's influence rises when competing load falls. Allostatic rebound or emotion enabled by restored safety may explain the same observation.

Reactivation predicts that repair-relevant cues revive specific demands. Cue conditioning and memory priming remain competing explanations.

Starvation predicts that long-neglected demands surface after capacity returns. Fatigue recovery or reporting bias may instead explain their appearance.

Dominance predicts that one demand suppresses unrelated action channels. Scarcity-related focus and threat prioritization are serious rivals.

A blocked demand might reappear after competing load drops only when repair becomes available. Allostatic rebound may predict general affect release without repair-specific action selection; cue conditioning may predict cue fear without dependence on repair availability.

Prespecify controls involving resolved demands, unrelated cues, and nonrepair actions. Broad nonspecific arousal challenges a prediction of queue specificity.

Specify channel dissociations: report change without physiological change, physiological preparation without report, or behavioral change only when action is safe.

18.7.5 — Model Recovery and Confusion Matrices

Parameter recovery. Simulate data from known Queue parameters and refit the same model. If recovered parameters are

biased, unstable, or correlated incorrectly, person-level interpretation pauses.

Model recovery. Simulate Queue, homeostatic, active-inference, model-free, model-based, and safe-control worlds. If the pipeline cannot identify the generating family above chance or practical thresholds, real-data comparison is not trusted.

Noise and missingness stress. Add report noise, sensor dropout, blocked observations, and context misclassification. If queue structure recovers only in clean simulations, support remains limited to those tested observation regimes.

Policy confusion stress. Fit rival policies to Queue-generated data and Queue policies to rival-generated data. Model mimicry under realistic samples can make discrimination practically inadequate; report separately whether the design also entails structural nonidentifiability.

Held-out task transfer. Train on one demand family and test on a different but prospectively specified family. If fit does not travel, the model may describe a local task rather than a mechanism.

A model-recovery matrix can be written as:

$$\mathbf{M} = (M_{ij}), \qquad M_{ij} := \Pr\{\widehat{m}(D) = j \mid D \sim \mathscr{M}_i\}$$

The entry M_{ij} is the probability that data generated by model i are classified as model j.

Frequent queue/active-inference confusion means this pipeline does not reliably discriminate those families under the tested sample and observation conditions.

18.7.6 — Unknown Mechanisms and Open-Set Fit

Route recovery must allow the named library to be wrong. Calibrate an explicit none-of-the-models branch on held-out unfamiliar generators, and use absolute checks such as residual

dependence, calibration failure, impossible transitions, poor intervention transport, and posterior-predictive misfit. Unknown is not favorable to PTVC; it means the current route library is inadequate and should trigger broader search or narrower scope. Chapter 25 supplies the full open-set benchmarking framework (Vehtari et al., 2017; Villaverde, 2019).

18.8 — Route Failure, Retirement, and Frontier Theorems

Geometry and committor transport need separate tests. Recovery checks may reject one route. Intervention checks may reject another. A belief-state forecast or Queue description may survive without inheriting a failed claim.

18.8.1 — Counterexamples and Impossibility Results

Counterexamples and impossibility results are stronger than poor fit when their assumptions match the claim. A prefix established to lie outside the outer solvency region rules out completion within the declared continuation class. Unrecoverability or observational equivalence limits distinctive attribution under the permitted design and may justify retiring that empirical use, but does not by itself establish Route-failed (Section 18.8.2).

A failed prediction can challenge a specified rule: failure to observe a predicted shift in control allocation when repair becomes available challenges the priority model, nonspecific arousal challenges cue-specific reactivation, and persistent burden challenges a predicted complete discharge (Popper, 1959; Lakens, 2017).

18.8.2 — Nonidentification Boundaries

Nonidentification occurs when route states, generators, or endpoint branches induce the same observed law under the available interventions and measurement process (Bollen, 1989; Villaverde, 2019).

The boundary can arise from symmetry, state aliasing, unknown origin, sparse observation, shared confounding, or endpoint selection. It can be structural, meaning separation is impossible even with infinite data under the design, or practical,

meaning finite noise and cadence make separation too weak. The result identifies which boundary applies: more sampling under the same design cannot resolve structural equivalence, whereas redesigned observations or interventions may break that equivalence, and denser or better-targeted data may resolve a practical limit.

Before power analysis, use symbolic checks, simulations, profile likelihoods, and sensitivity regions to investigate identification.

Report the identified set of routes or parameters compatible with the data and assumptions, and design safe interventions or improved channels to narrow it.

Under Section 24.2 and Section 27.4, route nonidentification limits distinctive attribution while leaving mechanism-neutral consequences separately testable; it does not by itself establish Route-failed.

18.8.3 — Frontier Theorems Worth Proving

Useful frontier results would establish conditions for directional reachability, unique completion kernels, generator recovery under partial observation, or impossibility of separating endpoint constructions within an observation class (Villaverde, 2019).

One frontier concerns topology. When attainable sets are nonconvex, disconnected, or carried on lower-dimensional supports, convex relaxations can produce false solvency. A theorem that bounds the error of the relaxation or identifies conditions under which support-function tests are exact would directly improve adjudication.

A second frontier concerns path conditioning. Several shrinking-band limits can approach the same zero-probability terminal event while inducing different bridge laws. Tightness, uniqueness, and stability under observation should be stated

rather than assumed. An impossibility theorem may be more valuable than another candidate construction if it shows that the data cannot distinguish those limits.

A third frontier concerns causal localization. If two mechanisms agree under every ethically admissible intervention in a reference class, they are empirically equivalent there. Formalizing that equivalence prevents endless mechanism tournaments that cannot change a verdict. It also clarifies what new action, channel, or population would be required for separation.

The theorem program should include machine-checkable statements where possible. Proof assistants can make formal assumptions, definitions, boundary conditions, and proof dependencies mechanically inspectable, and substantial mathematical theorems have been verified through such systems (Gonthier et al., 2013; Nipkow, Paulson, & Wenzel, 2002). Formal verification does not validate the empirical premises. A verified theorem remains conditional on the state and observation architecture it receives.

Every frontier target should name its scientific consequence. Proving it should change a protocol, recovery test, verdict boundary, or retirement rule. Theorems that do not alter an empirical or conceptual decision belong in the mathematical reservoir, not in the public argument as evidence.

18.8.4 — Versions and Successor Routes

Versioning makes retirement operational. The record names the route version, source model, assumptions, failed test, public artifacts, and any successor. Withdrawal means the old object is no longer used for new inference even if it remains available as historical evidence.

Public language follows the versioned mechanism record. A successor candidate receives its own identity, assumptions, tests, and evidential status, allowing the program to build on a clear account of the preceding result.

Revision becomes unsafe when readers cannot tell what changed. If an earlier prospectively specified version attributed reactivation to a queue-style route and later evidence favors interoception, active inference, or adaptation, the change is labeled as narrowing, replacement, or retirement. Otherwise every outcome can appear to preserve the preferred account.

A clean retirement notice should name the component, reference class, rival, evidence, and public-language consequence. It might say that queue-priority predictions failed in a low-stakes task and were better explained by model-free reward history. It might say that discharge rules were nonidentifiable in life-course data. It might say that the metaphor remains useful privately but leaves scientific mechanism language.

The Queue model is one testable candidate within a wider research program. Reporting where it succeeds and where it loses allows later work to preserve useful results, revise unsupported components, and pursue better explanations with a clear record of what has been learned.

Public artifacts must update after retirement. Diagrams, summaries, captions, social posts, donor descriptions, and talks should not continue to imply a mechanism that has narrowed or left the field. Correction-readiness is part of scientific dignity.

18.8.5 — What Survives Mechanism Defeat

The Part provides methods with continuing scientific value. Completion geometry describes feasible continuations under declared constraints. Committors provide conditional forecasts.

POMDPs organize action under partial observation. Bridge constructions supply baselines for endpoint-conditioned data. Each method contributes to a more precise account of the phenomena.

Defeat also preserves negative knowledge. A failed queue route may show that priority and reactivation add no predictive value. A failed bridge discriminator may show that the observation process is too coarse. A failed causal locus may reveal that the available interventions do not identify the state. Each result prevents the next study from repeating an unproductive path.

Update each proposition according to the result that bears on it. A route test changes the status of the specified mechanism; a test of a target-level requirement changes the corresponding theory-level claim.

Human protections remain active under every outcome. Care is immediate, participation voluntary, and no model output scores worth, trust, consciousness, eligibility, or personal obligation.

These commitments do not depend on route success or need repair when a mechanism narrows or fails.

A route registry preserves failed candidates as failure memory. Versions, assumptions, datasets, win conditions, results, and artifact corrections stay searchable. This preserves the knowledge gained when a candidate is retired.

The theorem and simulation infrastructure also survives. Impossibility regions, nonidentification boundaries, recovery benchmarks, and open-set challenge generators can be reused for new mechanisms. A retired route can therefore improve the standards applied to every successor.

A frontier theorem should be stored with its assumptions, counterexamples, and empirical dependencies. If a later study

violates one assumption, the theorem is not refuted; its application is. If the assumptions cannot be connected to observable quantities, the theorem remains formal rather than evidentiary.

Route retirement also protects the target hypothesis from informal rescue. Once a route loses its prospectively specified test, later sections may not cite its intuitive plausibility as if nothing happened. A new route begins with a new state, new assumptions, new predictions, and a new evidence status.

Failure memory should remain queryable. Later investigators need to recover which generator, endpoint branch, observation process, intervention, and scoring rule failed, together with the code and synthetic cases that exposed the failure. A narrative summary without the executable test would invite the same mistake to return under a new name.

Some tools may migrate after route defeat. A committor can remain a forecasting object, a belief state can remain an observation model, and a queue can remain a behavioral description if each passes its narrower validation. The migration must be explicit so a useful method does not retain the evidentiary halo of the retired mechanism.

The chapter closes with a standard for constructive failure: preserve the theorem, benchmark, counterexample, and humane boundary; withdraw the interpretation that lost; and leave the next study a sharper question than the one that entered.

Chapter 19 — Causal Loci and Interventions

An intervention enters a system at a particular place. A prompt changes available information, a medication can change bodily state, and an access rule changes what a person can do. Each may eventually alter the same report. Locating the cause requires tracing the intervened variable through the subsequent path, including changes to observation itself.

The continuation is the relevant outcome. An action can bring immediate relief while changing later effort, opportunity, or exposure; an intervention may also alter how long a trajectory remains observable. Comparing only the next score can miss those consequences. A continuation law represents the future states, actions, measurements, and endpoint behavior generated under a specified regime.

Timing helps localize differences between mechanisms. Two models may agree until an option opens, information arrives, or a commitment changes later behavior. Their first predicted separation offers a possible experimental fork. Its usefulness depends on the size and duration of the difference and the channel's response lag: the earliest difference need not be the easiest one to measure.

Behavior alone leaves several explanations open. The same choice can arise from different internal states, and the same state can produce different choices when opportunity or risk changes. A study must distinguish the offered menu from the usable menu and both from the action selected. That distinction connects causal inference to the reachability analysis of Chapter 18.

Institutional interventions make this distinction especially concrete. Correcting a record or opening treatment access can change future experience through the environment. Delays, staffing, family logistics, and material constraints may also determine whether an assigned option is received. Preserving

these pathways prevents an external restriction from being misdescribed as a property of an internal controller.

Random assignment addresses one part of this problem. It can separate the assignment from the earlier history, yet it does not make delivery, adherence, retention, or subsequent observation random. The chapter therefore follows the denominator from eligibility through the realized continuation. An apparent mechanism signature can emerge from who remained measurable even when the original allocation was valid.

Exact completion supplies conditional restrictions for these comparisons. Under the declared stream and assignment premises, a common past constrains the entire remaining contribution. A difference during one treatment interval does not alone locate an offset elsewhere; that inference needs an independently justified restriction on the rest of the continuation. The derivations make those additional premises explicit.

The resulting experiments seek a specified disagreement that can be observed safely. Low-burden option changes, information conditions, and permissible ordinary-care variation can test where models separate. A successful design measures delivery and observation alongside the outcome, then distinguishes a failed prediction from a failed perturbation. The useful conclusion identifies which causal relation survived, and under which intervention.

19.1 — Candidate Causal Loci and the Representation Gate

An intervention's scientific meaning depends on where it acts. Changing route state, a bodily or neural mediator, memory, policy, observation, the endpoint process, or sample selection defines a different causal estimand. Representation must first establish what quantity the channels inform. Well-defined treatment versions, mediator and collider distinctions, and cross-level bridges then determine which causal interpretation is available. Failure to identify the proposed locus requires narrowing that interpretation rather than renaming the observed effect.

19.1.1 — Representation Before Intervention

Freeze the representation before assignment: changes to sign, unit, clock, or stream ownership change the causal estimand.

A breathing intervention can change respiratory sensation, attention, report, or threat prediction. Calling its arousal effect a reduction in cumulative negative valence requires the measured-to-latent bridge; the measured-channel effect can stand separately.

Representation error can also create a collider. Selecting participants because a report and a physiological measure agree may condition on a common effect of latent state and measurement quality. The selected sample can then show associations that do not hold in the eligible population. The selection rule belongs in the causal diagram.

19.1.2 — Levels of the Proposed Causal Locus

A mechanism may act at one level, travel through another, and be measured at a third; the nearest measurable correlate need not be the source (Pearl, 2009; Hernán & Robins, 2020).

Sensory integration, prediction, salience, autonomic or endocrine regulation, pain, fatigue, and action control can mediate or confound affective change. A reproducible correlate supplies a pathway to test for specificity.

Cross-level claims require temporal order and mediation tests. If institutional repair changes behavior first, physiology later, and report last, that sequence differs from a direct bodily perturbation. The study should state the expected lag pattern and identify which alternative paths could create the same order.

19.1.3 — Manipulability and Well-Defined Treatments

Specify treatment timing, intensity, delivery, consent, context, and opened pathways. Causally different versions of support, rest, information, or repair require a treatment contrast that specifies those versions and their distribution (Hernán & Robins, 2020).

Randomizing a prompt or offering an action does not directly assign a private belief or latent route state. State the changed variable and the downstream effect supported by that intervention; truth-level endpoints are not assignable research treatments.

Include accessibility, language, burden, and opt-out conditions in treatment versions: the same notification can be informative, coercive, inaccessible, or irrelevant across participants.

Pilot work tests well-definedness. Process measures record what was delivered, understood, used, and experienced as burdensome. Qualitative feedback identifies unanticipated treatment versions and helps refine the intervention definition.

Separate information, action availability, and human contact when expectancy, control, and social-support models attribute effects to different components of a bundled intervention.

19.1.4 — Mediators, Proxies, and Colliders

Diagram the latent target, channels, treatment, mediators, actions, context, missingness, and endpoint, with sensitivity plans for omitted causes and measurement error.

A mediator lies on a causal path from intervention to outcome, and a proxy measures or predicts an unobserved variable. A collider is a common effect of two or more variables; conditioning on it can create a spurious association. These roles are not interchangeable, even when the same variable appears in each model (Pearl, 2009; VanderWeele, 2015).

Physiology illustrates the problem. Heart-rate variability may mediate a breathing intervention or proxy autonomic state. When both medication and distress influence whether a clean segment is recorded, the recording indicator is a collider, and restricting analysis to recorded segments can induce bias. The analysis must specify each variable's role and which data could distinguish it.

Post-treatment adjustment can remove part of the effect the study intends to estimate. Conversely, failing to adjust for a pre-intervention common cause can create bias. A causal diagram should therefore precede the regression list, with time ordering and treatment versions visible.

Match sensitivity analyses to the threatened causal role: justified negative controls for residual confounding, measurement-error bounds for proxy distortion, and mediation sensitivity for unmeasured mediator-outcome confounding.

19.1.5 — Cross-Level Bridges and Scope

A causal locus is the earliest component at which an intervention changes a downstream distribution under fixed representation, routing, and endpoint rules. Type intended observation changes as part of the intervention. The treatment version, assignment

time, mutable state, and continuation outcome give the locus its causal meaning (Pearl, 2009; Hernán & Robins, 2020).

Specify the causal estimand and transport assumptions linking the intervention and outcome levels (Bollen, 1989; Pearl, 2009; Hernán & Robins, 2020).

A bridge may be mechanistic, statistical, or merely correlational. Mechanistic bridges describe how a lower-level change alters a higher-level state. Statistical bridges relate measurements across levels under a declared population. Correlational associations support neither intervention transport nor phenomenal identification by themselves.

Aggregation can hide causal heterogeneity. A neural-network average may remain stable while subcircuits shift in opposite directions; a population intervention effect may conceal person-specific benefit and harm. The selected level should preserve the distinctions required by the estimand.

Use the coarsest measurable level that preserves the causal contrast; additional noisy, redundant, or common-descendant biomarkers need not improve localization.

Temporal scale is part of granularity. Autonomic changes can occur in seconds, hormonal changes over hours, learned behavior over days, and institutional repair over months. A cross-level bridge should predict the ordering and duration of effects rather than align every channel to one arbitrary clock.

A qualified test can defeat a locus claim when its distinctive state, mediator-order, or continuation prediction fails. Localize the failure: a bodily pathway may work for pain but not social threat, and a neural marker may work only in one device or site. Distinguish failed delivery or observation from Model-ambiguous evidence under Section 19.3.6 (Hernán & Robins, 2020; VanderWeele, 2015).

19.1.6 — Locus Failure and Scope Withdrawal

Interventions on the endpoint process require special caution. A protocol may change observation or administrative stopping, but no study may manipulate dying or withhold care to sharpen a terminal contrast. Natural variation and ethically permissible clinical processes set the boundary.

If several scalarizations remain defensible, report their compatible causal-effect set. Localizing an effect requires a contrast that distinguishes the proposed locus from neighboring loci.

Route inference compares explicit causal structures and tests measurement invariance across arms before assigning a phenomenal interpretation to changes in report or physiology (VanderWeele, 2015; Khalsa et al., 2018).

A causal locus fails when the declared intervention does not move the proposed state, the expected mediator order is absent, equal-treatment rivals explain the change, or the effect disappears after observation, mediation, and selection are modeled. Failure belongs in the mechanism register rather than being softened into a vague partial success (Hernán & Robins, 2020; VanderWeele, 2015).

Failure can be local. A bodily pathway may operate for pain relief but not for social threat; an institutional repair may change report without changing the candidate route state; a neural marker may predict task performance only in one device or site. Scope should narrow to the supported treatment, outcome, and population.

A locus also fails when the intervention is not well defined. Medication class, therapeutic conversation, rest, and information access each contain multiple versions with different consequences. If versions cannot be separated, the estimand

should be widened or withdrawn rather than treated as one causal treatment.

Nonpositivity produces a defined scope boundary. When a subgroup has no realistic access to an intervention or no safe alternative, the data do not identify its counterfactual continuation under that action. The analysis narrows the intervention family or reference class, uses partial-identification bounds where justified, or returns No adequate estimate rather than locating blocked access inside the person.

Representation failure outranks localization. If the measured channels cannot distinguish valence, arousal, pain, fatigue, salience, or reportability, no causal analysis can identify which phenomenal quantity changed. The result may still support a channel-level effect, but not a route-level one.

Ethical feasibility defines the interventions available for identification. A contrast that requires induced distress, delayed care, withdrawn analgesia, coercion, or overridden refusal is outside the admissible design, not an unfinished experiment awaiting greater resolve. The resulting causal question may remain partially identified or unanswerable under permissible interventions, and that limit is part of the scientific verdict. Partial identification should state which causal contrasts remain bounded by permissible evidence, turning ethical limits into explicit scientific scope.

Withdrawal should propagate. Figures, tables, summaries, and public excerpts must remove or relabel the failed locus. Leaving a vivid brain region, body signal, or policy arrow after the causal claim is withdrawn misstates the evidence. Correction should reach cached educational and presentation graphics as well, because simplified locus images can outlive the technical claim that originally justified them.

The remaining value is an explicit no-inference zone. By documenting where localization failed, the chapter prevents later studies from treating the same proxy or treatment as established. Failure memory is part of cumulative science. The no-inference zone should record which alternative loci or distributed explanations remain compatible, so later work begins from the actual uncertainty rather than from zero.

A route can survive locus failure only by naming a different, prospectively testable locus. It cannot become a diffuse system that acts wherever the data happen to change. A mechanism that has no place to fail has no place in a causal science. A successor locus must make an observation or intervention prediction that differs from the failed one; otherwise relocation is only a change of labels.

19.2 — Intervention-Indexed Continuation Laws

Let $Q_{h_t}^{a,g}$ denote the continuation law after history h_t under structural action a and continuation regime g. The law includes future state, actions, observations, missingness, and endpoint behavior. It is not merely a distribution of a final score.

19.2.1 — Potential Paths and Dynamic Regimes

Potential-outcome and dynamic-regime notation can make the contrast explicit, but consistency, positivity, and interference assumptions remain. If some histories cannot receive an action, the corresponding causal effect is not identified in that region.

Randomization yields a necessary restriction even when the past is unobserved. Let L be the signed prefix fixed before assignment Z, and R the entire remaining contribution through the true endpoint's left limit, including the intervention's effects. For a prospectively specified cohort whose realized regimes preserve the declared PTVC scope and stream ownership, exact closure gives $R = -L$ almost surely. If Z is independent of L, the distribution of R must therefore be identical across assigned arms. With conditional randomization, compare within baseline strata or standardize to a common baseline distribution.

That randomized-arm restriction can be decomposed without changing the endpoint convention. For the delayed-treatment comparison, let τ be a common, prospectively fixed treatment-onset time after assignment. On the primary strict-pre-endpoint branch, E contains the post-assignment contribution before both τ and the truth endpoint, and F contains the remaining pre-endpoint contribution at or after τ. Thus $R = E + F$; if the stream ends before or at τ, $F = 0$. A separately declared endpoint-inclusive branch instead uses $R_{incl} = E + F + J_T$, where J_T is the contribution assigned exactly at the truth

endpoint and remains excluded from the primary R. Apply the randomized-arm restriction above separately to the complete remainder of the branch declared in advance. The restriction does not by itself require any particular component or pre-treatment biological outcome to differ between arms.

A stronger localization claim requires a separately declared all-intervention structural model. For a fixed pre-assignment history h in common support, compare potential continuations under arms a and b using the same time cut and the declared bearer, routing, representation, and endpoint architecture. If primary-branch closure holds for both continuations on the model's joint probability space, then:

$$L(h) + E^a(h) + F^a(h) = 0$$

$$L(h) + E^b(h) + F^b(h) = 0$$

Subtracting gives:

$$E^a(h) - E^b(h) = -[F^a(h) - F^b(h)]$$

Here $F^a(h) - F^b(h)$ concerns the entire later contribution, not just the treatment interval. An independently justified nonzero restriction on that whole-continuation difference forces an opposite difference in $E^a(h) - E^b(h)$. Different immediate treatment effects do not establish this premise when the rest of the continuation can compensate. For an endpoint-inclusive branch, the subtraction must also include the arm-specific terminal terms J_T^a and J_T^b. A population-level directional or distributional prediction additionally depends on the structural model's distribution of histories and its observation bridge; the pathwise identity alone does not identify that prediction.

Valid randomization supplies assignment independence from the pre-assignment state within the declared randomization strata. If assignment is concealed and no assignment-correlated

physical channel reaches the organism before treatment, ordinary local forward-causal models additionally supply the null that assignment has no causal route to a systematic pre-treatment biological difference. Finite samples can still show chance imbalance, and implementation can still leak assignment information. Because assignment precedes the outcome, an association measured after assignment but before treatment is not retrocausal merely because treatment is scheduled later. A PTVC-linked model predicting such an association must specify the global-selection, viability, whole-trajectory, or other mechanism linking the assigned continuation to the earlier path and must retain, or explicitly revise, the assignment-independence premise it uses.

The terminal fork makes the conceptual contrast explicit. Imagine simulated flies from three history groups: ordinary conditions, predominantly pleasant conditions, and predominantly adverse conditions. They reach twenty physically indistinguishable routes, with no local information identifying the downstream mapping. Under the stipulated endpoint bridge, nineteen routes permanently end the relevant conscious stream one minute later; one permits further conscious life. This is a formal thought experiment and computational comparison, not an animal procedure. The history labels do not themselves determine the accumulated phenomenal contribution.

Let h denote the complete predecision stream history and $L(h)$ its signed prefix. For a short continuation d, let $C_d(h)$ include every remaining contribution strictly before the stream's endpoint. No contribution after that endpoint is available for closure. Exact PTVC requires

$$L(h) + C_d(h) = 0.$$

In the phenomenally silent idealization, $C_d(h) = 0$ requires $L(h) = 0$. More generally, if every short continuation is independently restricted to $[-b, b]$, then $|L(h)| > b$ excludes all short routes. These are compatibility conditions; they neither assign route probabilities nor imply that reaching balance causes death.

The ordinary no-cue null does not require uniform behavior. Let K be the uniformly randomized identity of the sole continuing route among m routes, independent of the history and of all information and private randomization used by the local choice policy. First suppose each arrival produces one recorded route choice A. Then

$$\Pr(A = K \mid h) = \sum_{j=1}^{m} \Pr(A = j \mid h) \Pr(K = j \mid h) = \frac{1}{m}.$$

Thus arbitrary history-dependent position preferences still give a continuing-route probability of 0.05 when $m = 20$. A systematic excess under these information restrictions would support departure from that local model. A global selector must declare its information assumptions; supplying the hidden mapping to a simulator does not supply it to an organism.

Keep arrival, route choice, and outcome recording as separate denominators. If the policy can make no route choice, write $A=\varnothing$ for that event and r(h)=Pr(A≠∅|h). Under the same hidden-mapping independence,

$$Pr(A = K \mid h) = \frac{r(h)}{m}, \qquad Pr(A = K \mid h,\, A \neq \varnothing) = \frac{1}{m}.$$

The second equality requires r(h)>0. Thus 5% is the completed-choice null for twenty routes, not automatically the per-arrival rate. Retain nonchoice and missing outcomes in the computational record. Outcome-dependent recording is an

additional selection mechanism; Section 25.2.3 gives its denominator bound. A local-null comparison must qualify the hidden mapping and ascertainment process before attributing an excess to a selector.

A nominated selector makes a stronger, quantitative prediction. Among histories reaching the fork, let q be the fraction for which only continuing routes are eligible. Assume everyone chooses one route, every route remains eligible for the others, and eligible routes retain equal baseline weights. If the continuing class occupies fraction s of the routes, then

$$p(s) = s + (1 - s)q.$$

Hard exclusion alone gives the lower bound $p(s) \geq q$; the equality also uses the stated weighting rule. If changing s preserves the history distribution and eligibility classes, $(p - s)/(1 - s)$ must remain constant. A rate of 0.15 at $s = 0.05$ fixes $q = 2/19$ and predicts 0.194737 at $s = 0.10$ and 0.328947 at $s = 0.25$, without refitting. These are calculated model predictions. An ordinary-history group has the baseline rate only when its forced fraction is zero.

Which histories are preserved matters. In a discrete reference model, let fraction q_0 have short routes that cannot close and long routes that close with probability one; let every route close with probability one for the rest. Conditioning the entire reference population on closure gives

$$p_{global}(s) = \frac{s}{1 - q_0 + q_0 s}.$$

This is exactly the finite odds-weighting curve $sw/(1 - s + sw)$ with $w = 1/(1 - q_0)$ for $0 \leq q_0 < 1$, despite literal hard exclusions in the reference population. Global conditioning changes the represented history mixture; conditioning each fixed history's continuation preserves it and gives the preceding affine

curve with $q = q_0$ instead. Route-count transfer therefore distinguishes these nominated probability constructions, not hard exclusion from soft weighting in general. The selected joint law must also state what happens to randomized assignments; its conditional associations do not by themselves define an intervention.

Mapping a localization restriction, a completion-set restriction, or terminal-fork compatibility to hatching, developmental survival, adult eclosion, or route choice requires an independently justified route, viability, and observation bridge. A biological outcome before a qualifying stream begins is evidence about the declared developmental extension, not a measured phenomenal residual. If assignment is generated only after the outcome has occurred, chance association remains possible; a reproducible systematic association beyond the prespecified null and procedural explanations would motivate an additional structural hypothesis rather than follow from the randomized-remainder result.

A restricted observation model supplies an additional check. Suppose the integrated reading is $Y = aR + oD + e$, with a common positive gain a, a common rate-origin offset o, remaining exposure D, finite first moments and equal mean errors across arms. Relative to a reference arm, $\Delta E(Y) = o\Delta E(D)$. Three or more arms can test whether a single origin explains all contrasts; if exposure means agree, reading means must agree without identifying that origin.

A voluntary enjoyable episode makes the comparison concrete. Use finite first moments and the fixed positive within-stream references chosen before assignment and applied throughout, as in Section 6.3.5. Let d be the population arm difference in the episode plus follow-up through a fixed horizon H, and f the difference in the entire later pre-endpoint

contribution. Let δ_{pre} be the pre-assignment prefix difference under the evaluated assignment law. Exact closure requires

$$\delta_{pre} + d + f = 0.$$

Preserved randomization gives $\delta_{pre} = 0$ in the population; it does not make realized sample prefixes identical. A selector that changes the assignment law needs its own prefix-difference bound.

Suppose jointly valid measurement, sampling, and tail bounds give $d \in [3,5]$ and $f \in [-1,1]$ on this normalized scale. Then $d + f \in [2,6]$, excluding the required zero under the maintained premises. If the future bound instead is $[-6,1]$, the sum lies in $[-3,6]$ and the comparison remains unresolved. Narrowing the future bound can therefore be more informative than adding observations that only refine d.

These are synthetic population contrasts. A treatment-invariant common nonzero terminal total also cancels between arms, so successful compensation does not identify the zero target or individual closure. Section 24.6.4 supplies the complementary distributional and individual-residual tests. The design gains information by pairing a measured early contrast with an independently qualified future restriction.

A capacity comparison can distinguish the target zero from a common nonzero target only with a qualified origin. Consider the terminal-fork model with rival target c, so closure requires $L + C = c$. Let each short route have the complete reachable remainder interval $[-u\tau, v\tau]$, where $u, v > 0$ are fixed signed-capacity coefficients and $\tau > 0$ indexes an independently justified linear capacity family. Short routes are eligible precisely when

$$c - v\tau \le L \le c + u\tau.$$

Retain the preceding history-preserving mixture, with the continuing class always eligible and $0 < s < 1$. Suppose a calibrated prefix contrast gives $L = x + U$, where U has a centered Gaussian distribution of fixed positive variance. The continuing-route probability has its unique minimum at

$$x^*(\tau) = c + \frac{u - v}{2}\tau.$$

A displaced minimum at one capacity can therefore reflect unequal signed capacities even when $c = 0$. At two distinct capacity levels, the same model identifies the target:

$$c = \frac{\tau_2 x^*(\tau_1) - \tau_1 x^*(\tau_2)}{\tau_2 - \tau_1}.$$

For example, minima at $x = 0.5$ and 0.7 when $\tau = 0.5$ and 1 imply $c = 0.3$. This is a synthetic identification example, not a measured departure from PTVC. Its conclusion requires the same capacity family, disturbance law, and independently fixed contribution coordinate across conditions. An unknown additive offset in L is confounded with c; an external treatment label cannot substitute for the calibrated prefix contrast. Elapsed duration alone does not establish the stipulated capacity law.

Finite precision also depends on the spacing of the capacity levels. For known $0<\tau_1<\tau_2$ and jointly valid bounds $|\hat{x}^*(\tau_k) - x^*(\tau_k)| \le e_k$, use the same two-level intercept to define $\hat{c}$. Then

$$|\hat{c} - c| \le \frac{\tau_2 e_1 + \tau_1 e_2}{\tau_2 - \tau_1}.$$

With $e_1 = e_2 = 0.02$, the levels 0.5 and 1 give a target-error bound of 0.06; levels 0.9 and 1 give 0.38. Wider spacing can improve this bound only while the same independently qualified capacity family remains valid. Uncertain capacities and an unresolved origin require their own envelope. A theoretical unique

minimum is not an observed minimum without such estimation and coverage work.

Include the intervention's effort, uncertainty, reward, frustration, relief, social evaluation, and later opportunity changes in the estimand or intercurrent-event strategy.

In longitudinal settings, treatment may depend on evolving state, and evolving state may itself be affected by earlier treatment. G-computation, inverse-probability weighting, structural nested models, and dynamic treatment-regime methods address this feedback (Robins et al., 2000; Murphy, 2003; Hernán & Robins, 2020). Ignoring it can attribute to completion dynamics a pattern produced by clinical or behavioral adaptation.

A potential continuation path is the future trajectory from a fixed prefix under an admissible regime (Robins et al., 2000; Hernán & Robins, 2020).

Only one continuation is observed at each realized history; within-person designs reduce some heterogeneity but do not reveal simultaneous counterfactual paths.

Specify whether the continuation outcome is a report trajectory, physiological channel, action-support set, endpoint-independent route state, or terminal event.

A dynamic regime maps observed history to action at each decision time, specifying when information is evaluated, available actions, and responses to changing state (Murphy, 2003; Hernán & Robins, 2020).

Compare the history-dependent rule with fixed schedules, symptom-triggered care, uncertainty reduction, and fatigue-sensitive prompting on held-out continuations.

19.2.2 — Causal Identification Assumptions

Identification by the sequential g-methods used here relies on consistency, positivity, and a defensible exchangeability condition at each decision point. Consistency links the observed outcome to the potential outcome under the treatment version actually received. Positivity requires that the compared actions lie in the support of the assignment law at each relevant history: positive probabilities for discrete actions, or the corresponding density and support conditions for continuous actions. Sequential exchangeability states that, conditional on the recorded history, action assignment carries no unmeasured information about future potential outcomes (Robins et al., 2000; Hernán & Robins, 2020).

A closing treatment window can destroy positivity; unrecorded reasons for clinical changes can create confounding; device failure can alter observation and subsequent action. Each limits the identifiable continuation effect (Petersen et al., 2012).

Continuous monitoring sharpens the same problem. When treatment, confounders, and outcomes evolve as functions of time, feedback is not a finite sequence of visits but a dense process. Identification then requires continuous-time versions of consistency, sequential randomization, positivity, and intervention well-definition; otherwise a discretized analysis may create support that the underlying process never supplied (Ying, 2024).

For repeated continuous interventions with weak positivity, a supported-value or projected estimand can replace an extrapolated full dose-response curve; display the excluded histories and values (Schomaker et al., 2024).

Inspect propensity distributions, overlap, effective sample size, simulated regime frequencies, and stochastic positivity. Absent support requires justified bounds or a narrower

population, not merely stabilized weights (Ring & Schomaker, 2026).

Negative controls can probe exchangeability but cannot prove it. An outcome measured before assignment that is associated with subsequent assignment may reveal residual confounding or implementation bias. Sensitivity analysis should quantify how strong an unmeasured factor would need to be to reverse the mechanism comparison.

19.2.3 — Interference and Shared Environments

Interference occurs when one participant's action changes another's continuation law through shared care, staffing, networks, schedules, or platforms. Causal interaction does not transfer phenomenal contributions between streams (Hudgens & Halloran, 2008).

Declare and sensitivity-test the interference unit and exposure mapping, whether household, clinic, cohort, network, or time block; administrative clustering alone does not establish the boundary.

Shared environments also create hidden burden transfer. An intervention that helps a participant by shifting work to family or staff can look beneficial in person-level outcomes while increasing unmeasured load elsewhere. The scientific record should name those displaced tasks and, where ethically and legally permitted, measure system effects without turning bystanders into involuntary research subjects.

Staffing, social information, and collective scarcity can generate apparent route-sensitive actions. Include network and resource-allocation rivals, using cluster randomization, network-aware analysis, or justified partial-interference assumptions as needed.

When interference cannot be bounded, narrow the continuation effect to the observed system or retain partial identification.

19.2.4 — Partial Identification of Continuation Effects

Unresolved state, treatment version, missingness, endpoint, or confounding can leave a set of compatible continuation effects (Manski, 2003; Imbens & Manski, 2004).

Separate bound components for missing outcomes, treatment versions, and the measured-to-latent bridge to identify which evidence could most improve causal identification.

Justify monotonicity and exclusion restrictions substantive-ly. Support can worsen outcomes through burden, expectation, stigma, or interference, defeating an assumed monotone benefit.

If every compatible effect favors an ordinary rival, the candidate loses that comparison. A set excluding a prespecified rival region may license the next gate; sets crossing the null or overlapping mechanisms are classified under Section 24.2.

19.2.5 — Description, Prediction, and Causation

Mechanism claims exceed average causal effects. Mediation, lag, negative controls, and competing-pathway tests are needed before an effect is localized. Where the proposed pathway can be perturbed safely, a localization claim should specify the expected temporal or intervention response in advance so failure of that response bears directly on the proposed mechanism.

Descriptive, predictive, and causal questions may use the same data but make different claims. Description summarizes observed continuations, prediction forecasts under a deployment law, and causation asks what changes under intervention (Hernán & Robins, 2020; Pearl, 2009).

Descriptive terminal alignment can generate hypotheses, yet selection, censoring, or retrospective clocks may produce it without a mechanism or intervention effect. Report whether the pattern survives defensible alternative alignment, censoring, and clock choices; instability across those choices is a descriptive robustness finding, not evidence for or against a mechanism by itself.

Prediction can exploit site or device proxies that are causally irrelevant. Held-out accuracy alone therefore cannot identify route state. Leave-site-out, leave-device-out, or comparable transport tests can reveal whether predictive performance survives beyond environment-specific information without converting successful prediction into causal identification.

Causal inference attaches a defined contrast to its exchangeability or alternative identification strategy, positivity, consistency, and time-varying-confounding rules instead of listing those assumptions as remote caveats. When multiple defensible identification strategies target the same causal contrast, their estimates should remain visible separately before any synthesis; disagreement then becomes evidence about assumption sensitivity rather than variation to be averaged away.

Where possible, the roles use separate data splits: description generates hypotheses, prediction tunes on training data, and prospectively specified causal contrasts face held-out or independent data. Preprocessing, feature selection, and model tuning that use protected outcomes belong inside the development process; otherwise the nominally held-out evaluation is no longer genuinely held out.

Public language names the role precisely: associated with, predicted, changed under intervention, and mediated part of the effect are different statements. Explained is reserved for an identified pathway.

Predictive value can survive causal failure, and a causal estimate can answer a narrow intervention question without broad predictive power. The record preserves those contributions rather than forcing one status.

Description and prediction may test prerequisites or signatures, while causal designs may localize a mechanism. None alone supplies complete-stream support.

19.3 — Earliest-Divergence and Action-Invariance Tests

Two complementary contrasts guide this section. Earliest divergence asks when competing mechanisms first predict different observable distributions; action invariance asks which predicted features remain stable when actions or incentives change. A useful design connects these contrasts to the information available at the decision and to the delay of each observation channel. The following subsections develop the timing comparison, identify the expected invariant features, and examine what intervention and measurement can distinguish (Klasnja et al., 2015; Liao et al., 2016; Sutton & Barto, 2018; Shenhav et al., 2013).

19.3.1 — Why Earliest Divergence Matters

Two mechanisms may fit the same endpoint and much of the same trajectory. Their earliest predicted divergence is a useful candidate test point because later responses can combine several pathways. It is not necessarily the most informative point: effect size, channel lag, measurement error, and burden determine which observation best separates the models.

Both models receive the same prefix observations and information available before the intervention; compare their first predicted difference in transition, action probability, mediator, or observation distribution.

A hidden-state divergence needs a measured consequence; trace the shortest qualified bridge from the proposed locus to its channel.

Temporal ordering helps distinguish mechanism from downstream consequence. If a candidate predicts action-support change before affect report while an ordinary expectation model

predicts report first, high-frequency measurement can test that order. A coarse end-of-day survey cannot.

Simulate scarcity, fatigue, cue learning, and expected control at the same cadence. Their earlier explanation of the path can defeat a later candidate-specific interpretation.

19.3.2 — Common Information Sets

Align features to the decision-time filtration. Exclude later measurements, post-action variables, and outcome-selected features unless the estimand permits them.

A concrete example is a cue-response task. The candidate route and a standard learning model both receive the cue, prior observations, current action set, and participant-specific baseline. Neither receives whether the task will later resolve. Their first differing action probability is then a genuine prospective object.

Provide common missingness indicators, quality flags, and observation timing, and use common imputation or joint-observation rules. Coding absence as neutral can otherwise give one model spurious certainty.

19.3.3 — Action-Invariance Under Shared Evidence

Specify the states where models share an action mechanism and where a candidate action-support rule predicts departure. Equal evidence and action support do not themselves imply invariance.

Invariance can be conditional. Models may agree for states far from a boundary and diverge only when signed reachability, reserve, or completion potential becomes active. The activation region should be specified before data are examined.

A behavioral task can manipulate an option that changes completion support while holding reward, effort, and information as constant as possible. The candidate predicts a

targeted policy shift; ordinary discounting or scarcity models predict no special shift after their own variables are matched.

A choice-probability difference can localize a policy mechanism; any phenomenal interpretation still requires qualified report and auxiliary channels.

Invariance also provides a route-specificity check. Prespecify actions that the candidate predicts will remain unchanged under the intervention. Broader shifts than predicted require comparison with general arousal, demand, fatigue, or observation reactivity before attributing the pattern to completion-dependent support.

19.3.4 — Cue Leakage and Post-Treatment Selection

Trace whether action-generated information or adjustment for a post-treatment variable changes the estimated time of divergence (Pearl, 2009; Hernán & Robins, 2020).

Audit interface timing, trial counters, adaptive difficulty, device latency, and prompts for assignment leakage. Blinding, software checks, and sham conditions test whether cues remain informative after route-relevant content is removed.

Clinical and longitudinal studies face subtler leakage. A clinician’s note may encode treatment response, a medication change may follow worsening symptoms, and a caregiver action may reveal prognosis. Using those variables as baseline predictors can make a model appear to anticipate a divergence it learned retrospectively.

Classify baseline covariates, time-varying confounders, mediators, and colliders before deciding which to adjust, model jointly, or leave untouched; predictive feature importance cannot supply those causal roles.

Shift timestamps, mask treatment indicators, and remove post-action variables in a reproducible leakage diagnostic. A

divergence surviving only the leaky specification fails as a discriminator.

19.3.5 — Temporal Resolution and Detection Power

Match cadence to the predicted separation's duration: hourly observations cannot resolve a seconds-long divergence.

Different channels have different response kernels. Behavior may change immediately, autonomic signals within seconds, endocrine signals over minutes or hours, and report after reflection. Aligning all channels to one timestamp can invert the apparent causal order.

Detection power depends on effect duration, within-person variability, measurement error, missingness, and the number of independent transitions rather than only participant count. Repeated observations within one person do not create the same information as independent causal forks.

Prompts can change attention, fatigue, disclosure, and action. Randomized, model-based, or deliberately sparse observation schedules can test this reactivity while preserving resolution for the declared contrast (Shiffman et al., 2008; Stone et al., 2007).

Represent uncertain delays with a distributed-lag or state-space model, prespecified windows, and multiplicity control; do not select the strongest lag after inspecting outcomes.

Synchronize device, server, clinical, and report clocks; preserve time zone, firmware, preprocessing, and alignment rules. Minute-scale drift can invalidate a brief divergence.

Simulate each serious rival through the planned cadence and missingness to estimate recovery of the earliest divergence. Persistent confusion requires a different intervention or channel.

Collect only the channels and windows needed for the temporal contrast, with pauses and withdrawal available.

19.3.6 — Failure to Diverge and Route Retirement

A route-specific fork fails when the models were predicted to diverge under a supported intervention, the measurement system had adequate sensitivity and timing, and the observed continuations remain invariant. That result retires the prospectively specified divergence claim even if both models still fit the endpoint.

Failure must be diagnosed before it is interpreted. The intervention may not have changed the intended locus, unaccounted-for treatment versions may have mixed, action support may have been too narrow, or the observation channel may have missed the window. Manipulation checks and model recovery help distinguish these failures from genuine invariance.

Compare action probabilities using the same prefix, current observations, and supported action menu. A protocol-fixed difference in safe actions is informative only if it survives equal preprocessing and recovery checks.

Compare mediator responses from the same pre-action state and identical treatment version. The proposed mediator must change before the downstream outcome at the protocol-fixed point of divergence. An unchanged or mistimed mediator, or a response explained by off-target pathways, fails that distinctive prediction.

Compare observation channels using the same latent-state model and quality flags. Route-specific information at a declared lag must be distinguished from artifact, activity, and generic severity.

Compare endpoint hazards within the same at-risk set and endpoint object. A change after the protocol-fixed intervention point must not depend on retrospective endpoint alignment or selection.

Under a negative-control action, retain the same task and observation history. The candidate should remain stable where it predicts no route effect; an equal response to sham or irrelevant actions challenges that specificity.

Opposite divergence is also informative. A candidate may predict increased repair-oriented action while participants conserve effort or choose rest. The model should not reinterpret the opposite action as hidden completion. The prospectively specified sign and channel determine the verdict.

Adequate models that remain inseparable at the achieved resolution are Model-ambiguous under Section 24.2. A new fork must target an actual predicted difference.

State construction uses the same observations, missingness, context, and endpoint definition for all candidates. Stop the affected comparison if the protocol-fixed recovery and calibration checks cannot recover the generating state.

Tuning uses equal search budgets, justified priors, and shared simulations. Parameter and model recovery determine whether the families can be separated; failed separation requires stopping or narrowing the comparison.

The held-out comparison uses the same supported intervention and decision-time information. Under the fixed first-divergence and causal-response diagnostics, failure of the distinctive prediction counts against the candidate.

Transport tests freeze the models before applying the declared bridge to a new site or population. Evaluate calibration, transported effects, and subgroup errors; failed transport requires narrowing rather than silent recalibration.

Open-set challenges include unknown generators and an option to reject all submitted models. Evaluate false-unknown and false-certainty rates; confident assignment of unfamiliar mechanisms requires stopping the affected use.

Prospectively specify practical equivalence for first divergence. A nonzero difference below the observation floor or unable to change the decision does not preserve the route; the margin, uncertainty interval, and detection threshold control the result (Lakens, 2017). The equivalence margin should be justified by the smallest difference that matters for the declared scientific claim or decision. If the bound is instead based on available power or resources, label the conclusion as exclusion relative to that bound, not as evidence that smaller effects are scientifically negligible. Fix the bound before observing the separation.

19.4 — Completion-Dependent Action Support

Completion-dependent action support identifies which actions retain a declared level of completion probability from the current history. Its meaning depends on the available information, the proposed transition model, and the permitted action set. The formal construction below separates that conditional support from an instruction about which action a person ought to take.

19.4.1 — Reachable Actions Versus Chosen Actions

Completion-dependent action support asks which actions retain a declared level of completion probability from the current history. One useful object is

$$\mathscr{A}_{K,\alpha}(h_t) = \left\{ a : Q^{a}_{h_t}\left(|L(T^{\text{-}})| \le K\right) \ge 1\text{-}\alpha \right\}.$$

This band identifies actions whose frozen continuation law assigns at least 1-alpha probability to the declared completion band. Report its modality, action domain, policy class, endpoint branch, and uncertainty; selection and stronger closure claims require the separate conditions in Section 7.3.4.

Distinguish support loss through disappearing resources, physiology, time, institutional closure, or improved knowledge of unreachable continuations: physical infeasibility, uncertainty, and social exclusion require different responses (Aubin, 1991; Aubin et al., 2011).

Safe reinforcement learning can encode action sets and risk constraints; ethical admissibility must be specified separately (García & Fernández, 2015; Ames et al., 2019).

Reachability means an action lies in the support of admissible policies; choice means the realized policy selects it. Reachability alone supplies neither selection probability nor a

recommendation (Kaelbling et al., 1998; Manski, 2003; Sutton & Barto, 2018).

Objectives, beliefs, learning, fatigue, and social risk can make policies select different actions from the same reachable set. A behavioral route claim therefore needs a policy model.

Observed nonchoice alone does not falsify reachability: an offered, understood, safe, and available action may still not be selected. Conversely, one observed choice does not establish broad support across other states or people.

19.4.2 — Support Restrictions and Viability

Define the histories and actions supported by both the data and the admissible action system; a model value outside that region is extrapolation (Aubin, 1991; Hernán & Robins, 2020).

Viability adds the requirement that at least one policy can keep the process within declared safety, consent, and study constraints. A state can have broad statistical support and no ethically viable intervention, or a viable action with little empirical support. The two regions should be shown separately.

Medication, disability, communication, or device exclusions can narrow support away from the target population; report excluded groups and the resulting claim restriction.

Robust viability considers model uncertainty and disturbances. A policy that is safe only under one fitted transition model may not be viable when plausible dynamics are included. Inner and outer viability sets can represent this uncertainty.

Support can be expanded by changing the environment rather than pressuring the person. Translation, transportation, accessible interfaces, respite, or an alternative clinical route may open actions. Such changes are causal interventions on opportunity and should be represented as external loci.

Include grid, surrogate, and tolerance uncertainty near the viability boundary; a smaller classification margin cannot support a definitive action-support verdict.

19.4.3 — Safe Control and Harm Avoidance

Safe control treats certain constraints as conditions of admissibility rather than costs that can be outweighed by scientific reward. Consent, comfort, clinical care, privacy, and stop requests remain outside the optimization trade space. A policy that violates them is not a high-cost option; it is unavailable (García & Fernández, 2015).

Deterministic and stochastic systems require different guarantees. In a diffusion model, the generator includes second-order terms, and a deterministic derivative cannot be quoted as though it controlled random excursions. The theorem, probability level, horizon, and regularity conditions must accompany any safety claim.

Partial observation makes safety harder. A controller acts on a belief state rather than the true state, so unobserved pain, fatigue, distress, or device failure can invalidate a hidden-state guarantee. Conservative thresholds and participant challenge routes are part of the scientific design.

Give candidate and rival policies the same admissible actions. Any remaining route difference must arise beyond that shared safety restriction; denying rivals safe actions biases the comparison.

If no action satisfies both the scientific objective and the admissibility constraints, stop or redesign the study.

19.4.4 — Completion-Dependent Feasibility

Fix a reached prefix h_t and one frozen downstream regime g containing every declared downstream policy, physical

transition, endpoint, routing, selection, and measurement-reactivity component. For each structurally available action a, q(a) is the conditional probability of the declared completion event under that complete continuation. It is a typed probability, separate from reward, phenomenal value, desirability, motivation, and advice.

Section 7.3.4 establishes the policy-mixture identity and its exactness condition under the declared action-first factorization. Here it constrains the causal locus: fixed-policy probability-one completion requires action-specific completion probability one for policy-almost every action under the frozen downstream regime.

$$P^{\pi}_{\eta,h_t,g}\left(\widetilde{F}_{\eta,comp}(h_t;g)\right) = \int_{A^{do}_{\eta}(h_t)} p^{comp}_{\eta,h_t,g}(a)\pi_t(da|h_t,g).$$

$$P^{\pi}_{\eta,h_t,g}\left(\widetilde{F}_{\eta,comp}(h_t;g)\right) = 1$$
$$\Leftrightarrow p^{comp}_{\eta,h_t,g}(a) = 1\pi_t(\cdot|h_t,g)\text{-almost everywhere.}$$

Action support and endpoint support remain different axes. The identity concerns probability-one completion under the declared policy and continuation law. Topological support, all-policy completion, and closure of every admissible continuation require their own stated conditions. When every available action already has q(a)=1, the theorem is exact and behaviorally silent.

Defining actions or continuation laws to exclude every nonclosing path makes simulated closure automatic. Test consequences not entailed by that feasibility rule.

The off-policy identification ceiling from Section 7.3.4 applies here directly. Chapter 19 adds the intervention-specific implication: evidence about excluded actions requires ethically available support from exogenous availability changes, valid instruments, bounded experiments, natural variation, or partial-

identification assumptions; harmful endpoint manipulation remains outside the evidence program.

An empty or narrow feasible set can reflect misspecification, endpoint uncertainty, incomplete action enumeration, observation limits, inaccessible support, or route infeasibility. Classify these under Section 24.2; do not infer a personal property. Empirical survival requires independent transition, action, or intervention predictions under fixed rules.

A deterministic control barrier can establish forward invariance of a safe set under admissible control. Its conditions include a sufficiently smooth barrier, deterministic dynamics, action access, and a regular boundary. The result cannot simply be quoted for stochastic or hidden-state systems.

A stochastic barrier can supply probability-one safety or a finite-time probability bound, depending on the theorem. Retain its generator, diffusion, regularity, and policy-access conditions. Omitting the second-order term or strengthening the stated guarantee invalidates the application.

A viability kernel identifies states from which at least one admissible policy remains within the constraints. Its state, action set, horizon, constraints, and computational approximation must be declared. The computed kernel is not a discovered natural law.

Reachability analysis identifies states that can reach or avoid a target under specified dynamics and controls. Disturbances, discretization, and approximation error constrain the result; reachability alone does not prove conscious dynamics.

An MPC safety filter addresses finite receding-horizon constraint satisfaction under its update rules. Its guarantee depends on model accuracy, recursive feasibility, and constraint handling. Engineered feasibility is not evidence for the Law.

19.4.5 — Soft Guidance and Partial Observation

The mixture-support result changes under finite soft reweighting, limit-sure policy sequences, and partial observation as follows.

Apply Section 7.3.4 to power reweighting, with a strictly positive normalizer and a baseline policy, action set, and downstream regime fixed independently of the behavior being explained. The resulting completion probability is a ratio of adjacent normalizing moments.

$$q_{\eta,h_t,g}(a) := p^{comp}_{\eta,h_t,g}(a).$$

$$Z_{\beta,\eta}\left(h_t,g\right) := \int q_{\eta,h_t,g}(a)^{\beta}\,\pi_{0,\eta}\left(da \mid h_t,g\right).$$

$$\beta > 0, \qquad Z_{\beta,\eta}\left(h_t,g\right) > 0.$$

$$\pi_{\beta,\eta}\left(da \mid h_t,g\right) := Z_{\beta,\eta}\left(h_t,g\right)^{-1} q_{\eta,h_t,g}(a)^{\beta} \times \pi_{0,\eta}\left(da \mid h_t,g\right).$$

$$P^{\pi_{\beta,\eta}}_{\eta,h_t,g}\left(\widetilde{F}_{\eta,comp}\left(h_t;g\right)\right) = \frac{Z_{\beta+1,\eta}\left(h_t,g\right)}{Z_{\beta,\eta}\left(h_t,g\right)}.$$

$$P^{\pi_{\beta,\eta}}_{\eta,h_t,g}\left(\widetilde{F}_{\eta,comp}\left(h_t;g\right)\right) = 1$$
$$\Leftrightarrow \pi_{0,\eta}\left(\{a : 0 < p^{comp}_{\eta,h_t,g}(a) < 1\}|h_t,g\right) = 0$$
$$\text{and } \pi_{0,\eta}\left(\{a : p^{comp}_{\eta,h_t,g}(a) = 1\}|h_t,g\right) > 0.$$

Use Section 7.3.4's almost-sure and limit-sure definitions unchanged. Here the additional requirements are controlled invariance, endpoint compatibility, measurable selection, and recursive time consistency (Chatterjee et al., 2016; Asadi et al., 2025).

$$\mathscr{S}_{AS}(\overline{g}) := \left\{ h : \exists \pi \in \Pi_{adm}(h, \overline{g}),\ P^{\pi,\overline{g}}(F_{comp} \mid h) = 1 \right\}.$$

$$\mathscr{S}_{LS}(\overline{g}) := \left\{ h : \sup_{\pi \in \Pi_{adm}(h,\overline{g})} P^{\pi,\overline{g}}(F_{comp} \mid h) = 1 \right\}.$$

For policy claims under partial observation, use the posterior-versus-robust distinction from Sections 7.3.4 and 18.3.5. A high posterior-average completion probability does not establish robust support across compatible histories; failure of the common-action condition keeps the causal claim observation-limited.

$$P(F_{comp}|o) = \int \int p^{comp}_{\eta,h,g}(a)\pi_t(da|o)\, b_t(dh|o).$$

$$\begin{aligned} A^1_{\eta,F}(h;g) &:= \{a : p^{comp}_{\eta,h,g}(a) = 1\}, \\ A^1_{rob}(o;g) &:= \textstyle\bigcap_{h \in H(o)} A^1_{\eta,F}(h;g). \end{aligned}$$

Positive-probability support requires some available action with completion probability greater than zero. This is an action-specific statement under the frozen continuation regime, not a guarantee of completion.

A fixed-policy probability-one guarantee requires completion probability one for almost every action receiving mass under that policy. The policy and continuation law are protocol-fixed; causal interpretation requires a separate identification design.

Limit-sure attainability means that the supremum over admissible policies equals one without requiring a policy that attains it. A sequence of policies can approach probability one; actual attainment remains a separate mathematical property.

A robust guarantee under partial observation requires a nonempty intersection of probability-one action sets across compatible histories, together with a measurable, time-consistent selector. The guarantee remains adapted to the available

observations and conditional on the declared support and regime.

Finite soft guidance reweights a baseline policy by completion probability raised to a positive finite guidance power. It favors actions with higher modeled completion probability. An action with zero completion probability receives zero weight, while a baseline-positive action with an intermediate probability can retain positive weight. Hold the baseline policy, completion model, and guidance strength fixed when comparing the resulting action distributions.

19.4.6 — Rival Explanations of Narrowed Action Sets

Fatigue, pain, scarcity, learned avoidance, threat, depression, social risk, institutional barriers, planning, safe-control constraints, and measurement loss can narrow observed or available options.

Distinguish usable support from selection and test whether opportunity and observation narrow together (Shenhav et al., 2013; Sutton & Barto, 2018).

Test rivals through one observation pipeline: which options disappear, and how does the set change under a safe, prespecified perturbation? Chapter 20 adds body-level lag and clinical-context tests; qualitative and contextual evidence exposes institutional constraints missed by laboratory menus.

If no specified model reproduces the menu change, apply Section 24.2's open-set criteria. Compare residuals across sites, task versions, and menu geometries: localized mismatch directs review to the interface; stable cross-context structure may motivate a new generator and comparison.

Audit action-support transport across populations and contexts. Record structural absence separately from nonselection, and accessibility changes even when printed

menus remain unchanged. Represent material support differences when pooling sites, and timestamp availability because feasible opportunities can change during the study.

Open or close one safe route while holding the value and observability of the remaining alternatives comparable. Record offered and usable menus, independently verify the support change, and test a declared contrast between the candidate’s changing prediction and stable shared controls. Only the resulting held-out asymmetry supports the route component.

19.5 — Assignment, Commitment, Selection, and Retention

Random assignment identifies an intention-to-treat contrast. Receipt is not assignment. Sustained exposure and retention are separate processes. Observation requires its own causal treatment (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

19.5.1 — Random Assignment and Delivered Exposure

Record delivery timing, dose, implementation, and nonreceipt reasons while preserving refusal and discontinuation rights (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

Conditional randomization also needs a common comparison population. Section 19.2's remainder equality holds within valid pre-assignment strata; pooled arm distributions can differ when allocation fractions differ across strata. For the interval test in Sections 24.6.4 and 25.3.3, fix H strata, positive target weights w_h summing to one, and positive cell sizes n_{jh}. On a common qualified analysis coordinate, which may use Section 6.3.5's fixed stream normalization, assume independent evaluation units within and across cells, conditional on frozen calibration, and valid miscoverage bounds γ_{jh} for each unit. Preserve all stream-scope premises.

For witness m, use the existing whole-interval extrema and define A_m^w as the larger of $\Sigma_h w_h(\bar{l}_{0h}^{(m)} - \bar{u}_{1h}^{(m)})$ and its reversed-arm counterpart. Put $\Gamma w = 2B\Sigma_h w_h(\gamma_{0h}+\gamma_{1h})$, and use

$$\rho_w = B\sqrt{2\,log(\tfrac{2M}{\alpha})\,\textstyle\sum_{h=1}^{H} w_h^2(\frac{1}{n_0 h} + \frac{1}{n_1 h})}.$$

$$D_w = \max_{1\le m\le M} A_m^w - \rho_w - \Gamma_w.$$

Under conditional remainder equality, $Pr(Dw>0)\le\alpha$. Apply Hoeffding to each weighted contrast: its summed squared range widths are $4B^2\Sigma_h w_h^2(1/n_{0h}+1/n_{1h})$. The union bound over 2M

orientations proves the result. Additional calibration-event failure is allocated as in Section 24.6.4. The weights define one fixed target population; they are not chosen from favorable outcomes.

A synthetic example shows why this matters. Let remainders be +0.8 in stratum A and −0.8 in B, with opposite prefixes so every stream closes. With arm counts (320,80) and (80,320), the pooled signed contrast is 0.96 despite valid conditional randomization. Common weights (1/2,1/2) give contrast zero. At B=1, M=2, and α=0.05, the weighted allowance is about 0.2617. These are design fixtures, not participant findings. Empty target cells, dependent repeated measures, or outcome-selected weights require a different valid analysis rather than this formula.

Instrumental-variable and principal-stratum analyses target specified effects under additional assumptions. Report sensitivity to the exclusion restrictions and monotonicity assumptions used in the analysis.

19.5.2 — Commitment Signals and Private Policies

Scheduling reminders, restricting a distraction, authorizing a supporter, or stating a plan can change expectations, opportunity, accountability, and selection before later action (Frangakis & Rubin, 2002; Hernán & Robins, 2020).

Record the policy chosen at commitment separately from later actions, including the information then available; otherwise prior planning can resemble route-specific action invariance.

A safe experiment can randomize the offer or form of a low-burden commitment while leaving acceptance voluntary. The analysis then separates offer, acceptance, and subsequent behavior. It should also measure whether the commitment reduced choice in a way participants experienced as helpful, neutral, or burdensome.

A distinctive commitment prediction must exceed ordinary reduced temptation, simplified planning, social expectation, or memory support in timing, state dependence, or intervention transport.

19.5.3 — Adherence and Informative Retention

Benefit, symptoms, fatigue, caregiving, device tolerance, distrust, prognosis, and measurement burden can affect retention and outcomes, making complete cases spuriously favorable (Scharfstein et al., 1999; Hernán & Robins, 2020).

Model device and participation processes separately: tasks may retain easier-to-measure states while wearables fail during movement or clinical decline, creating apparent stability or convergence.

Participant return and support are part of ethical retention. Reminder intensity, compensation, transportation, and caregiver involvement can alter who remains. Those supports should be offered for fairness and feasibility, not optimized to pressure continued participation.

Keep separate time-indexed histories for treatment, support, measurement availability, and withdrawal. Receiving treatment while declining a wearable or ending follow-up differs from completing measurement without receiving treatment; one compliance variable obscures exposure and observation selection (Robins et al., 2000; Hernán & Robins, 2020).

Record each action's version, timing, intensity, context, and deviation policy: identical adherence percentages can hide delayed, shortened, interrupted, or caregiver-mediated delivery. If this history is unrecoverable, target assignment under real implementation rather than a precisely delivered route (Murphy, 2003; Murphy, 2005).

19.5.4 — Principal Strata and Later Selection

Principal stratification defines groups through joint potential values of a post-assignment event, such as survival, adherence, or receipt. It clarifies a causal comparison when an outcome is undefined or observed only after selection. Membership in these groups is generally latent rather than directly observed (Frangakis & Rubin, 2002).

Near an endpoint, restricting analysis to participants who remain alive, conscious, enrolled, or reportable can select on consequences of treatment and prognosis. Randomization does not protect this later comparison. Truncation by death also requires an estimand that represents undefined outcomes rather than treating them as ordinary missing data.

For example, an intervention effect among participants who would remain observable under either assignment differs from the effect among actually observed survivors and from the population intention-to-treat effect. State which comparison is primary and why. Report the alternatives separately where they are identifiable; a favorable result for one target does not answer the others.

Stratum effects require justified assumptions or bounds. Vary monotonicity, exclusion, dependence, and parametric assumptions and show the compatible effect range. If valid bounds remain too wide for the declared conclusion, report the bounds or scenario results and classify the evidence as Inconclusive. After required protocol and implementation gates pass, use No adequate estimate only when the target is undefined or lacks both identification and a valid bound under the available data and maintained assumptions. Failed gates retain their own status (Manski, 2003; Imbens & Manski, 2004; Hernán & Robins, 2020).

19.5.5 — Repeated Assignment and Adaptive Regimes

Repeated assignment evaluates a sequence of decisions; adaptive designs may also update allocation probabilities. Distinguish the dynamic regime from the learning algorithm collecting its data (Robins et al., 2000; Hernán & Robins, 2020).

Micro-randomized trials can test proximal effects of time-varying components, while sequential multiple-assignment designs can compare adaptive strategies. These designs are useful for candidate mechanisms because they create several safe divergence points. They also increase the burden of modeling availability, carryover, fatigue, and time-varying confounding.

Adaptive randomization can create informative probabilities. Models must use the known assignment mechanism and preserve chronological feature construction. Reusing later outcomes to tune earlier decision rules turns an adaptive design into retrospective optimization.

Compare equally tailored rival regimes under common information and burden limits.

Stop or narrow when support weakens, burden rises, or no regime discriminates; prespecified adaptation rules govern subsequent decisions.

19.5.6 — Audit Trails for Selection Processes

Trace eligibility, invitation, contact, consent, assignment, delivery, adherence, withdrawal, device status, endpoints, observation, exclusion, and analysis inclusion. Where legally and ethically permitted, timestamp each transition and record reason, actor, and version.

Reconcile flow-diagram counts with executable analysis and repository denominators; investigate discrepancies.

Negative-control retention outcomes can reveal residual selection. Events assumed not to respond to the intervention, but sharing relevant causes and observation or administrative processes, can expose whether apparent route effects follow documentation intensity, site practice, or participant availability instead.

Quantify the unmeasured retention bias, informative censoring, or treatment-version imbalance needed to change the verdict, and assess those departures against the study setting.

Simulate known assignment, adherence, censoring, observation, and competing-event processes through the extraction and analysis code. Compare recovered denominators with truth to detect eligibility errors, time-order failures, duplicates, and accidental conditioning. Failure blocks causal-route adjudication.

The audit distinguishes withdrawal from treatment nonadherence; reasons for refusal, pause, or departure remain optional.

Preserve the allocation algorithm, seed or randomization service, block and stratification variables, concealment procedure, assignment probability, release time, and authorized overrides. Distinguish random draws from operational substitutions; later adjustment cannot recreate lost randomization (Hernán & Robins, 2020).

Adaptive assignment requires replayable probabilities rather than a final treatment label. At each decision time, the archive stores the information available to the policy, the versioned policy function, the probability assigned to each action, and the action actually delivered. A simulation replay should reproduce the logged probabilities from the frozen code. Failure to do so is a provenance defect that blocks policy-effect and mechanism claims because the denominator no longer identifies the

intervention regime that generated it (Murphy, 2005; Liao et al., 2016).

Human overrides are coded by function. A safety override, participant preference, clinician judgment, equipment failure, and scheduling constraint have different causal meanings even when all change the delivered action. The primary analysis can preserve assignment while secondary analyses represent receipt and version, but these pathways are not collapsed into one deviation variable. Repeated overrides concentrated in one state or site may reveal that the nominal intervention was not feasible where its mechanism was most strongly tested.

19.6 — Experimental Forks and Mechanism Tournaments

Specify each altered kernel: continuation probability, reinforcement-learning value update, expected-control effort allocation, or scarcity-driven opportunity cost. The resulting predictions must discriminate these models.

19.6.1 — Equal Information and Equal Scoring

Use common held-out cases, losses, calibration checks, uncertainty rules, and abstention opportunities. Multiple scores require distinct component questions and a prespecified combination rule (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

A model-specific observation or preprocessing difference is allowed only when it is the declared object under test.

19.6.2 — Negative Controls and Sham Mechanisms

Fix adaptation rules, stopping boundaries, arm modifications, model updates, and error spending before inspecting triggering data (Murphy, 2005; Collins et al., 2007; Klasnja et al., 2015).

Matched cues, irrelevant option changes, and simulated controllers without the completion state test the candidate's predicted specificity. Failure of that specificity narrows the mechanism.

Option-set manipulation changes what can be done while keeping the person safe. A task may add an easy route, remove a redundant route, make one route require a benign effort, or reopen a closed route through a support action. The goal is to see whether behavior tracks feasible repair structure rather than merely reward amount (Klasnja et al., 2015; Nahum-Shani et al., 2018).

An observational study of a simplified appeal route could measure timing, report burden, sleep disruption, service use, or dropout, controlling staffing, population change, outreach, and policy drift.

A pleasant option may leave avoidance of a violation-or mistrust-linked route unchanged. This can distinguish a scalar substitution prediction from a noncompensable component model.

Option-set experiments use small, reversible, transparent choices. Observational studies access operational records through a separate, documented research-use path.

Distinguish route-specificity shams from causal negative controls. A negative-control outcome shares relevant causes and observation processes with the target outcome but is not causally affected by the intervention under the maintained assumptions. A negative-control exposure shares relevant bias pathways without causally affecting the target outcome. An unexpected association can reveal confounding, reactivity, pipeline leakage, or failure of those exclusion assumptions. A null control is informative only when its causal relation and sensitivity were justified before outcome inspection (Lipsitch et al., 2010; Pearl, 2009).

Measure expectations, perceived assignment, and differential burden: side effects, difficulty, device behavior, or support timing may reveal the arm. A sham must preserve the relevant experience to distinguish route specificity from expectancy or procedure effects; improved blinding does not justify an invasive or humiliating sham (International Council for Harmonisation, 2026; World Medical Association, 2024; Lakens, 2017).

19.6.3 — Transported Predictions

Use complexity penalties or nested validation where needed to separate extra model capacity from mechanism support.

Quantify the competing fork predictions in advance: cue-specific action change for the candidate, general effort for scarcity, or reward-history dependence for learning.

Before transport across site, menu, device, population, endpoint, or channel, declare the invariant component and permitted recalibration (Hernán & Robins, 2020).

Map changed treatment versions, support, meaning, and measurement, and specify whether transport predicts parameter invariance, qualitative order, or a bounded effect.

Cross-equation transport is particularly informative. A completion potential estimated from passive transitions can be tested on held-out actions; a causal locus estimated from behavior can be tested on physiology; a lag relation estimated in one context can be tested after a safe perturbation.

Compare rival transport under equal recalibration budgets; repeated candidate refitting can erase the cross-context restriction under test.

A destination sample with different clinical severity, action constraints, or support can alter both the intervention and the pathway through which it acts. Transport diagrams or explicit adjustment sets make those assumptions inspectable, while sensitivity analysis shows how much unmeasured effect modification would be needed to reverse the conclusion (Pearl, 2009; Hernán & Robins, 2020).

The measurement bridge must travel with the mechanism or be independently requalified. A stable latent relation cannot be inferred from identical device labels when firmware, placement, language, sampling cadence, or preprocessing differs. The protocol freezes the core causal prediction, audits invariance of

the observation model, and permits only declared destination-specific recalibration. If the bridge changes enough to absorb the predicted effect, the study has not transported a mechanism; it has created a new measurement system (Bollen, 1989; Peters et al., 2016).

19.6.4 — Adversarial Tests and Win Conditions

Before outcome release, adversarial teams agree on component win conditions and amendment procedures under Section 6.7.

Specify how divergence, calibration, and transport results combine under Section 24.2. A composite score cannot average away an adverse prerequisite or target result.

Agree on strong opposing models, parameter ranges, observation bridges, and reasonable auxiliary structure before scoring.

Prespecify dependence and multiplicity rules for repeated observations, lags, features, and forks, using hierarchical or sequential methods as appropriate.

Lock predictions, preprocessing, and scores before outcome release. An independent implementation team runs the pipeline and a separate adjudication group applies the agreed rules. Record amendments and exploratory departures under Section 6.7 (Gneiting & Raftery, 2007; Vehtari et al., 2017).

19.6.5 — Ethical Stop Rules and Null Results

Statistical or scientific futility can stop collection without declaring a theory verdict. Model-recovery failure, inadequate movement of the intended locus, observation quality below the protocol-fixed threshold, or exhausted conditional power may make additional burden uninformative. The stop record states which object failed and which conclusions remain available.

A null result carries a sensitivity statement. The report shows the prospectively specified effect sizes and lag patterns the design could detect, which channels remained usable, whether the manipulation altered the intended variable, and which rival patterns remained compatible. This separates genuine invariance from weak intervention, poor timing, and measurement failure.

An adverse event can invalidate interpretation even when no direct harm is caused by the protocol. A medication change, acute illness, care transition, or institutional disruption may alter state and observation. The event is recorded and routed through the decision tree rather than adjusted away automatically.

Ethical stop rules are triggered by participant burden, distress, privacy failure, clinical conflict, device risk, or loss of valid consent, not by whether stopping helps one model. They are written before enrollment, tied to critical participant-protection and data-quality thresholds, and remain independent of the scientific winner (International Council for Harmonisation, 2026).

Stopping never changes care. Clinical decisions, analgesia, rescue, and support continue under their own authority. A study may stop observing while the person’s life and treatment proceed beyond the research window.

Participant withdrawal is a valid outcome and not evidence about the mechanism. Data use after withdrawal follows consent, law, and the protocol. The model cannot interpret refusal or missingness as latent completion or burden.

Null and stopped studies should be released with enough detail to prevent repetition. The intervention, burden, manipulation check, model-recovery result, and reason for stopping belong in the public scientific record, subject to privacy.

The ethical outcome is not a caveat added after the analysis. It determines which intervention world is admissible. A

mechanism that can be separated only through harmful or coercive designs remains outside empirical authorization.

Mechanism tournaments prospectively specify the adjudication body, its independence from model developers, and its conflicts of interest. Safety, futility, model inadequacy, protocol continuation, and public reporting each use protocol-fixed evidence thresholds. Governance becomes part of reproducibility because discretionary authority can otherwise alter the scientific branch after results are known.

Split-sample or external locking reduces rescue. Candidate generators, preprocessing, tuning ranges, and scoring rules are frozen before the confirmatory comparison. Exploratory discoveries can improve the next tournament, but they do not alter the verdict of the current one.

Win and loss conditions should be model specific. One candidate may lose through failed calibration, another through causal invariance, another through ethical infeasibility, and another because an ordinary rival predicts equally well with fewer assumptions. A single leaderboard cannot preserve those distinctions.

Tournament archives should preserve losing predictions, code, and recovery failures. Future models must face the same failed cases rather than being evaluated only on a refreshed dataset. Failure memory is what turns sequential model building into cumulative science.

19.7 — Shadow Prices as One Analytic Example

A shadow price is the derivative of an optimized value with respect to a constraint under specified regularity and duality conditions. In a constrained policy problem, a multiplier may summarize the local marginal model value of relaxing a terminal-band or resource constraint. It is not a universal exchange rate.

19.7.1 — Marginal Sensitivity and Units

A constrained value can be written schematically as

$$V_K(t,x,\ell) = \sup_{\pi \in \Pi_{adm}} \mathbb{E}_\pi \left[\int_t^T r_s \, ds \right]$$

subject to

$$\mathbb{P}_\pi \left(|L(T^-)| \le K \mid X_t = x, L_t = \ell \right) \ge 1\text{-}\alpha.$$

An empty feasible set leaves no multiplier to interpret; classify the result under Section 24.2. Interpretation also requires a suitable constraint qualification, relevant dual relation, and attained optimum. Nonconvexity or misspecification can invalidate the fitted sensitivity.

Within this finite-horizon analysis, a multiplier may be nonunique, fail to exist, or cease to approximate finite changes when the active set changes. Its permitted sign follows the declared constraint and Lagrangian convention: equality multipliers are unrestricted, whereas a fixed inequality convention imposes dual sign restrictions. These limits confine its meaning to the optimization problem that defines it (Boyd & Vandenberghe, 2004).

A multiplier cannot authorize sacrifice or reductions in desired care. Encode rights and care constraints explicitly or leave those

decisions outside the calculation (Dworkin, 1981a; Dworkin, 1981b; Boyd & Vandenberghe, 2004).

For a task's fixed fatigue limit, a multiplier may describe the objective gain from a small relaxation. Ethical admissibility can keep that limit fixed regardless of the derivative.

Test held-out constraint changes against scarcity and expected-control models; an equivalent allocation and intervention response supplies no distinctive route evidence.

A multiplier inherits objective-per-constraint units: task loss under a time constraint gives loss per minute. Changing the objective, clock, normalization, or constraint changes the number (Boyd & Vandenberghe, 2004).

Use the sign convention stated in Section 19.7. Every display should state whether relaxation increases or decreases the bound and how the derivative is oriented.

Propagate parameter, state, numerical, and nonuniqueness uncertainty into a multiplier range or an unavailable estimate.

Cross-person comparison requires commensurable objectives, constraints, scales, and identification bridges; this chapter assumes no such qualification.

19.7.2 — State Dependence and Nonlinearity

Constraint sensitivity may vanish when slack, rise near an active boundary, jump with the optimal policy, or become set-valued in a nonsmooth problem (Fleming & Soner, 2006; Aubin, 1991).

Nonconvexity is particularly dangerous. Local multipliers may describe one optimum while another policy has a different active set and better value. Duality gaps can break the usual sensitivity interpretation. A model should report whether convexity or a constraint qualification is assumed, approximated, or absent.

Closing opportunities can increase constraint pressure, while new routes can lower it. This state-, support-, and forecast-dependent response supplies no universal inverse-horizon law.

A concrete simulation can vary one constraint across a grid and plot the optimized value rather than only the derivative. Kinks, plateaus, and regime changes reveal where the local approximation fails. The graph should include uncertainty and the ordinary control rival.

State dependence also limits causal interpretation. An observed association between a proxy for pressure and behavior may reflect the state that made the constraint active, not the multiplier itself. Interventions that change the bound while holding the relevant information structure fixed offer stronger evidence.

19.7.3 — Observable Proxies and Identification

Delay sensitivity, choice under resource expansion, and policy shifts near a boundary identify a multiplier only through a structural link to the optimization problem that excludes rival pathways (Boyd & Vandenberghe, 2004; Hernán & Robins, 2020).

Choice data identify preference or policy parameters only under assumptions about beliefs, available actions, noise, and objective form. Declining an option alone does not identify high constraint pressure.

Interventions on the constraint are more informative. Extending a deadline, reducing effort, adding transportation, or clarifying information can change the feasible set. The predicted local response should be compared with scarcity, fatigue, learning, and social-support models under the same information and burden.

Apply Section 18.7.4's cross-equation test: a multiplier estimated from one task should predict responses in another module governed by the same declared constraint.

Dropout or extra support near a constraint, and device failure under load, can reverse the apparent relation. Include these processes in joint models and sensitivity analyses.

19.7.4 — Retiring and Replacing Causal Objects

Retire the shadow-price interpretation when the problem is unidentified, feasibility is empty, the required duality or constraint qualification fails, the multiplier is unstable, or held-out bound changes fail its prediction (Boyd & Vandenberghe, 2004).

Retirement may be semantic. If shadow price repeatedly invites monetary or moral interpretation, constraint sensitivity, finite contrast, active-set status, or feasibility margin can preserve the useful mathematics without importing an exchange metaphor.

Retirement may also be causal. A multiplier can summarize a fitted policy without being the variable through which behavior changes. If interventions on the bound alter outcomes through a different pathway, the causal object should move to that pathway.

Carry an identified set when state, objective, active constraint, or dual relation is uncertain, and classify it under Section 24.2 and Section 27.4 (Manski, 2003; Imbens & Manski, 2004).

Marginal constraint sensitivity is a local derivative of a declared objective with respect to one constraint. It is not a price of pain or a person, and its interpretation must be withdrawn when the required duality or stability fails.

A finite intervention contrast compares a protocol-fixed outcome under two supported constraint levels. It is restricted to

the treatment versions and horizon and is not a universal exchange rate.

Identified bounds describe the range compatible with the stated data and assumptions. Do not collapse that range to a point when it crosses zero or spans different regimes.

Finite contrasts often replace derivatives in nonlinear or nonsmooth systems. Rather than report one local multiplier, the analysis can show how the optimized value and action set change across a prospectively specified range of constraints. Kinks and regime changes become visible.

Ordinary causal objects may be better. Opportunity access, treatment availability, information timing, pain relief, social support, and institutional repair can be directly intervened on and interpreted without invoking a dual variable. They take priority when they answer the scientific question more clearly.

A viability or feasibility set identifies states and actions satisfying declared safety or completion conditions. Outside identified support, the result is no adequate estimate. An action-support set instead records actions observed or defensibly supported at a specified state, excluding coercive, inaccessible, or unsafe actions. Neither object ranks lives or supplies personal advice.

Failure propagates to tables, diagrams, captions, and examples that might otherwise preserve the impression of a hidden price. Historical versions retain the reason for withdrawal, but the active scientific presentation no longer displays the retired object as a live mechanism.

What survives is the discipline of constrained reasoning: explicit objectives, units, action sets, support, feasibility, and nonnegotiable boundaries. The analytic example has succeeded if it makes those assumptions visible and leaves no residue of moral exchange.

19.7.5 — No Monetary, Moral, or Personal Price

“Shadow price” remains a model-local sensitivity term. Any monetary interpretation depends on the modeled units, and no multiplier prices pain, persons, duties, or care (Boyd & Vandenberghe, 2004).

No multiplier measures how much suffering is worth, how much joy is owed, or whose life should receive priority. Pain, dignity, consent, and rights are not exchangeable components in the objective. They either remain outside the model or enter as nonnegotiable admissibility constraints.

The object belongs to the constraint, not to the person. The correct sentence is that a declared model has high marginal sensitivity to a particular bound in a particular state. Saying that a person has a high shadow price turns an analytic derivative into a human attribute and is prohibited.

Local sensitivity cannot justify large or irreversible changes. Active sets can switch, nonlinearities can dominate, and the objective may omit consequences that matter most. A multiplier is not a recommendation even inside the modeled domain.

Clinical language requires special care. A shorter horizon does not make care inefficient, and a model cannot use marginal information value to reduce desired comfort, communication, or support. Research optimization remains subordinate to care.

Institutional misuse is equally serious. A dashboard must not rank cases, appeals, or participants by a shadow-price proxy. Such a system would import a narrow objective into public authority and convert uncertainty into eligibility or worth.

General-reader prose can prefer constraint sensitivity or marginal model value when the technical term adds more risk than clarity. Where shadow price is retained, units, owner, objective, active constraint, uncertainty, and prohibited interpretation appear nearby.

This boundary permits precise analysis of dual variables without converting a model multiplier into moral payment, a personal price, or permission for sacrifice. Conditional net remainder accounting is a separate consequence of exact PTVC; a multiplier does not by itself establish that relation or a way to control future happiness.

Chapter 20 — Interoception and Brain-Body Mechanisms

A bodily change can carry several meanings. Medication may relieve pain while reducing speech; fatigue may close an action that remains available on paper; safety may change breathing before a person describes relief. A brain–body model must explain the ordering of these changes, through qualified measurements.

The problem is especially visible in the chapter's palliative-care example. After medication changes, a wearable records less movement, altered sleep, and lower heart-rate variability. These observations could accompany pain relief, sedation, worsening illness, or poorer device contact. Interpreting the trajectory requires treatment and quality information alongside the channels that witness experience.

A direct effect on a sensor can change the reading without changing the target. A bodily mediator can change the target before report becomes possible. A shared cause can move both. The useful causal graph separates those pathways and predicts a pattern that would differ if another pathway were responsible.

Timing helps. Breathing, autonomic activity, endocrine response, sleep, and remembered report unfold on different schedules. Aligning them to one convenient timestamp can reverse an apparent causal order. The chapter treats each channel's response and observation delay as part of the mechanism question, including delays introduced by the study itself.

History matters. Learned bodily warnings can recur when a familiar cue returns, while a current state may reflect accumulated physiological change. Molecular recorders, memory studies, and repeated thermal exposures offer different ways to ask what information persists. Each comparison must distinguish a remembered sequence from its total input and from the present bodily state.

Disability, medication, trauma, and chronic illness challenge calibration. A muted autonomic response can coexist with pain; reduced activity can reflect pacing or inaccessible opportunity; an apparently calm task can have delayed effects. Calibration must represent these possibilities in the population studied, with observation windows long enough to test the proposed response.

The framework therefore follows changes across sensing, representation, action, and recovery. A repair opportunity may first reduce threat, then alter attention, then permit action. The reverse sequence is also possible: a private decision may precede visible action and bodily recovery. Measuring only the last two events can attribute the change to the wrong starting point.

The studies connect measurement with mechanism through these contrasts. A useful perturbation specifies what changes, which channel should respond, and when another channel should remain stable. Naturalistic events retain the reasons they occurred, because a self-selected breathing exercise and an assigned task need not identify the same effect.

Wearables and analytic models enter this account as parts of the observation system. Their practical contribution depends on preserving device quality, clinical context, label meaning, and uncertainty through the analysis. The aim is a testable explanation of a particular bodily pathway and its limits across states, people, and interventions.

20.1 — Embodied Dynamics as Candidate Mechanisms

Bodily regulation, interoceptive prediction, learning, action readiness, and salience offer candidate explanations of demand persistence, avoidance, repair initiation, and recovery. Their roles must be separated: a bodily process may mediate the proposed dynamic, generate an ordinary rival pattern, or merely affect a measurement channel. Temporal records and mechanism-specific predictions help distinguish those possibilities. Embodiment connects the analysis to the organism without reducing every bodily signal to a direct reading of experience.

20.1.1 — Mechanistic Tests and Temporal Records

Distinguish genetic instructions from the changing state that records history: an unchanged sequence cannot distinguish different histories of the machinery it builds. Candidate carriers include chromatin, methylation, transcription, proteins, synapses, and distributed cellular activity. Slow molecular processes might tune a faster integrator; test information storage, calibrated updates, and survival through component replacement.

Molecular interventions can now address that question with greater specificity. Coda et al. (2025) targeted the Arc promoter within identified mouse memory ensembles and reversibly altered contextual fear-memory expression. Kukushkin et al. (2024) found that spaced chemical stimulation produced more sustained CRE-dependent transcription than massed stimulation in SH-SY5Y neuroblastoma-derived and HEK293 cell lines, measured with a luciferase reporter. The first result supplies a selective causal tool; the second supplies an ordinary cellular

mechanism that can imitate history-sensitive persistence. Neither experiment measured a signed phenomenal account.

Test the same calibrated state predictions on new histories, including equal signed totals with different order or gross stimulation. Distinguish recording, maintenance, retrieval, and expression; impaired recall alone does not demonstrate erased accounting information. Compare learning, stress, salience, and multiscale-memory explanations under Section 25.2's common resources.

Engineered recording offers a concrete first benchmark. DNA Typewriter records event order through sequential genome editing, while ENGRAM records signalling-related barcode changes in engineered cells (Choi et al., 2022; Chen et al., 2024). Complementary protein recorders support waveform reconstruction: CytoTape stores transcription-linked histories along growing fibres, while GEMINI records reporter histories in fluorescent layers (Zheng et al., 2026; Yan et al., 2026).

A proposed two-channel experiment would assign plus and minus labels to known laboratory inputs, then test recovery of their difference separately from total input and temporal order. The labels initially designate programmed signals. Before estimating area, calibrate amplitude response, reporter delay, timestamp precision, and recoverable duration in the chosen preparation. Challenge a frozen decoder with held-out equal input areas delivered through different amplitude-duration combinations and, where the calibrated writer predicts such cases, unequal areas with the same final aggregate readout. Compare aggregate abundance with timestamp-resolved readout to test which information separates those histories. Retain held-out sign reversals, equal additions to both channels, altered pulse spacing, and differences among cells, with recording error and saturation included in its predictions.

Successful recovery from pooled molecular measurements must also be distinguished from recovery within an individual cell or lineage. With nonlinear writing, different distributions of cellular histories can produce the same pooled signal. The benchmark should therefore match its unit of inference to its actual readout, track recorder retention and partition across both daughters when cells divide, and preserve failed or unedited records. Its immediate contribution is a calibrated test of biological recording and decoding; a natural PTVC realization additionally requires an independently supported relation to the qualifying stream's experience and the declared initial and pre-endpoint boundary conditions.

Established regulation science supplies background physiological relationships within the settings in which they have been tested.

A candidate mediator test asks whether a specified pathway carries part of the proposed dynamic. Its conclusion is limited to that pathway, observation model, and tested setting.

Ordinary bodily regulation can serve as a rival mechanism for the measured pattern. The resulting conclusion remains scoped to the shared model comparison.

An observation channel supplies a partial witness. Its inference remains channel-specific and carries calibrated uncertainty.

Mechanism failure can require narrowing, abstention, or retirement. State which mechanism, measurement model, and reference class the decision reaches.

20.1.2 — Interoceptive State and Prediction

Cardiac, respiratory, visceral, thermal, metabolic, nociceptive, immune, endocrine, postural, and effort signals are filtered, predicted, attended, and linked to action in distributed brain–

body systems (Khalsa et al., 2018; Barrett & Simmons, 2015; Feldman et al., 2024).

Represent bodily state as a vector when cardiac activation, respiratory constraint, nociception, immune activation, depletion, tension, or sleep pressure varies independently.

A participant who feels dread before an ordinary phone call may be responding to a learned interoceptive prediction even when the current scene is safe.

Interoceptive prediction can also be a rival. If prediction error, learned bodily expectation, and safety updating explain the timing of avoidance and repair, the Law-specific queue language may add little.

Store predicted state separately from incoming signal to distinguish bodily evidence preceding distress, bodily change following prediction, and common contextual causes.

20.1.3 — Allostatic Load and Regulation

Allostasis is anticipatory regulation through preparation, reallocation, mobilization, conservation, and shifting thresholds. It connects horizon, reserve, fatigue, threat, sleep, pain, and repair to bodily cost (Khalsa et al., 2018; Barrett & Simmons, 2015).

Allostatic load describes accumulated wear from repeated or chronic activation of regulatory systems. It can mediate burden, constrain future options, alter observation, and supply specific biological predictions for comparison.

A caregiver who sleeps in a hospital chair for four nights may show altered glucose regulation, higher resting arousal, slowed attention, irritability, and flat affect. The visible emotion may be muted while the regulatory cost is high. If a study measures only smiling, it may miss burden. If it measures only cortisol, it may confuse burden with infection, medication, caffeine, or circadian timing.

An informative allostatic account predicts which reserve dimension changes, when it changes, how it recovers, and which observation channel reflects the change.

Distinguish the level and change of an allostatic-load measure from variability and flexibility in regulatory responses. These summarize different aspects of physiology; none alone identifies current demand or depleted capacity.

20.1.4 — Affective Salience

Affective salience describes how a bodily signal becomes urgent, meaningful, aversive, inviting, threatening, or action-guiding (Sterling, 2012; McEwen, 1998; Barrett & Simmons, 2015).

Prediction, context, trauma, medication, attention, culture, and goals can change the weight of the same peripheral signal. Salience may raise queue priority or explain reactivation without a queue.

Compare predictions of report, avoidance, repair, attention capture, memory intrusion, and recovery beyond ordinary arousal, including adaptation, predictive-processing, homeostatic, and artifact explanations.

Report subgroup false positives, false negatives, and their consequences: treating activation as threat or quiet collapse as calm can burden people least able to contest the inference.

20.1.5 — Somatic Markers and Action Readiness

Somatic-marker accounts propose that learned bodily associations bias uncertain choices. Cues linked to danger, relief, humiliation, pain, reward, or repair can alter behavior without a verbalized conscious belief (Khalsa et al., 2018; Barrett & Simmons, 2015).

A new medical envelope can evoke bodily tension before its content is known, reopening a demand associated with an earlier denial.

Distinguish learned association from present cost. A marker can be protective, outdated, excessive, or trauma-shaped; repeated threats to benefit access can train vigilance that a protocol misreads as irrationality.

Track cue exposure, association, bodily response, attention, action, report, and updating. Persistence after safe experience can implicate trauma, recurrent threat, institutional recurrence, or sensitization; weakening may be explained by ordinary learning.

Cue specificity separates learned associations from diffuse uncertainty. Compare naturally encountered, consented hospital, debt, and neutral cues. Never stage a threat to provoke a response. If one cue class dominates, check whether neutral-cue order or ordinary arousal explains it.

20.1.6 — Embodiment Without Reduction

An embodied explanation must represent external demands as well as bodily responses (Sterling, 2012; McEwen, 1998; Barrett & Simmons, 2015).

Use biomarker profiles for qualified research forecasts or abstention alongside the participant's account and clinical context; care and access decisions follow their own clinical and rights standards.

Include external demands (noise, inaccessible transport, caregiving, racism, debt, or supervisory threat) as possible causes of physiology.

Test environmental and bodily interventions in one causal graph: whether environment changes body, body changes experience, or both pathways are needed.

Untreated pain can intensify fear, while shame, language barriers, or administrative threat can block pain care. These interactions motivate combined repair.

20.1.7 — Changing Causal Roles

Heart-rate variability can serve as a resting autonomic observation channel, while movement can confound its interpretation, beta-blockers can change it physiologically, nonwear can interrupt its record, and autonomic neuropathy can undermine its calibration. Represent these changing roles explicitly.

Specify whether each component is necessary, sufficient, mediating, moderating, or predictive; a useful observation correlate need not generate the terminal signature.

20.2 — Interoception and Affective Salience

Interoceptive evidence connects bodily signals with their perception, interpretation, and use in action. Baseline, response speed, observability, and known limitations determine which channels can address a question. Accuracy, sensibility, awareness, precision, and salience are related but distinct constructs; none can silently stand for valence or urgency. The analysis must retain developmental and cultural context and examine cases in which a strong bodily signal, a report, and the proposed experiential interpretation diverge.

20.2.1 — Afferent Channels and Internal Sensing

Pleasantness can change while the external stimulus remains similar. In the thermal, taste, and smell examples examined by Cabanac (1971), the hedonic response depended on the person's internal condition. He termed this state dependence alliesthesia. The finding gives the distinction between circumstance and experience a physiological basis: what an event feels like depends partly on the body receiving it. For a cumulative model, that relationship supplies a useful prediction to test across independently characterized bodily states and repeated observations of the same stimulus.

Repeated thermal relief gives this distinction a modern experimental form. Schweiker et al. (2026) studied 60 completers across eight thermal stress-and-relief sequences, measuring sensation, pleasure, relief, and physiology separately. Reducing thermal load increased reported pleasure and relief; pleasure did not disappear with repetition. The perceived-control manipulation was inconclusive. The ordinal reports do not establish equal positive and negative phenomenal areas.

A bounded extension could compare comfortable, participant-approved exposure profiles that converge on prespecified

current bodily measurements and rates of thermal change before the same test exposure, while differing in prior sequence. A model using only those current inputs predicts the same next response when they are matched. A history-sensitive candidate must specify a different response in advance and predict a new profile with its parameters fixed. Measure expectations as another competing input and give ordinary adaptation models the same history. Success would identify additional predictive state information, not prove that all relevant bodily states were matched: equal skin temperature alone cannot establish that. Keeping sensation, pleasure, relief, and physiological channels distinct makes the observation model below experimentally usable.

Equation 20.1: Observation-channel model. Role: candidate specification linking latent variables to measured channels.

$$O_{c,t} = h_c\left(S_t, E_t, M_t, D_t, Q_{c,t}\right) + \varepsilon_{c,t}$$

$$R_t = r\left(B_t, C_t, A_t\right) + \varepsilon_t$$

$$Y_t^E \not\equiv O_{c,t}, \qquad \text{and} \qquad Y_t^E \not\equiv R_t.$$

In this display, $O_{c,t}$ is the observed signal from channel c at time t; S_t is the latent bodily state; E_t is environment; M_t is medication; D_t is disability or disease context; and $Q_{c,t}$ is channel quality. R_t is report, and Y_t^E is the modeled experiential target. In the report equation, B_t, C_t, and A_t denote belief or central representation, context, and action. The terms $\varepsilon_{c,t}$ and ε_t are the respective observation and report errors. Observations and reports can inform the target without being identical to it.

Cardiac and vascular channels include heart rate, HRV, pulse arrival, and blood pressure. Their interpretation must account for movement, medication, posture, fever, fitness, and device fit.

Respiratory measurements include rate, depth, a carbon-dioxide proxy, and breath tasks. Their interpretation must account for exertion, anxiety, pain, posture, and lung disease.

Visceral and metabolic observations include appetite, nausea, glucose, temperature, and fatigue reports. Illness, medication, sleep, diet, and endocrine state can change their interpretation.

Immune and inflammatory channels include cytokines, CRP, symptoms, and temperature. Infection, disease, treatment, and circadian timing are relevant competing influences.

Nociceptive and pain-related observations include pain reports, movement, facial action, and analgesic use. Communication limits, masking, tolerance, and disability norms affect what those observations mean.

Behavioral embodiment can be observed through posture, activity, speech timing, and sleep. Environment, opportunity, access, and caregiving demands help determine what those observations mean.

Afferent channels carry information from the body toward central representation (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

Match channels to the target: respiratory timing for breath-related panic; joint inflammatory, sleep, metabolic, autonomic, and activity measures for chronic fatigue; and cardiac or somatic signals with context, memory, appraisal, and safety for social dread.

Stress coactivation can produce apparent generalization that fails during exercise, fever, stimulant use, sleep deprivation, pain relief, or changed cultural expression. Test these alternative causes of the channel signal.

A person with asthma may show respiratory changes before a meeting, but the meaning differs if the change is airway

limitation, anxiety, mask discomfort, conditioned fear, or a combination.

Measure observation burden alongside quality: cardiac sensors can be intrusive, and breath tasks can be frightening after respiratory trauma.

20.2.2 — Accuracy, Sensibility, and Awareness

Interoceptive accuracy, sensibility, and awareness are related but distinct. Performance on a detection task, self-reported attention to bodily signals, and metacognitive knowledge about accuracy need not agree. A protocol should name which object is measured rather than using interoception as one undifferentiated trait (Critchley & Garfinkel, 2017).

Measurement quality is task-specific. Newer cardiac, respiratory, and other psychophysical tasks address some weaknesses of classical heartbeat counting, yet they differ in signal strength, beliefs, attention demands, control conditions, and scoring. No single task earns the label of global interoceptive accuracy without cross-task and cross-organ evidence (Desmedt et al., 2023).

Banellis and colleagues (2026) examined cardiac and respiratory interoception in 241 participants. Sensitivity, precision, and metacognitive efficiency showed no significant cross-domain correlations, while subjective confidence showed a modest positive association. The distinction gives a measurement model two questions to answer: how well someone performs in each bodily domain, and how confidently they judge that performance. A shared tendency to feel confident can coexist with domain-specific ability. Keeping those quantities separate allows the model to represent the observed contrast rather than compress both into one interoceptive score.

20.2.3 — Precision, Attention, and Prediction Error

In predictive models of interoception, precision represents estimated reliability, often expressed as inverse variance, and controls the weighting of an internal signal or prediction error. Attention can alter that weighting. A weak signal can dominate awareness when assigned high precision; a strong signal can fade when assigned low precision, competing demands are greater, medication blunts awareness, or dissociation separates sensing from report (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

Two people with similar physiology may report different experience: one notices heartbeat changes immediately, another only fatigue hours later. A trained singer may detect breath shifts that a novice misses.

Signal priority can matter independently of intensity: a small recurrent cue may capture attention and reorganize action, while a large load may remain unreported through learned inattention or unsafe expression.

Test precision with confidence reports, cardiac or respiratory tasks, safe attention manipulations, and within-person context changes. Heightened monitoring can worsen distress or support regulation, so accuracy is not a universal well-being outcome.

Repeated bodily prompts can teach reporting, increase symptom worry, or cause interruption fatigue. Separate this measurement learning from mechanism change.

Prediction error is the mismatch between expected bodily state and incoming bodily evidence. Interoceptive updating is the process by which the system revises belief, attention, and action. This can be helpful: a safe breath, stable rhythm, or pain relief can teach the body that a situation is less dangerous than expected (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

Chronic threat, trauma, unreliable care, institutional denial, persistent pain, or unpredictable symptoms can maintain predictions of danger despite a currently safe setting.

For a predicted response to safe contact, specify channel, lag, report change, and persistence. A channel response without the predicted report response must remain visible as a discrepancy when evaluating mediation or dissociation.

A neutral cue matched for novelty, attention, and movement tests channel specificity. The same response under that cue may favor arousal, task demand, or artifact over interoceptive learning.

Laboratory safety does not establish that a learned danger prediction was mistaken; prior or ongoing danger can make caution adaptive.

20.2.4 — Valence, Arousal, Salience, and Urgency

Valence, arousal, and urgency are often correlated, but they are not the same. Valence concerns felt positive or negative tone under the project's measured target. Arousal concerns activation or readiness. Urgency concerns the felt demand for action. A painful injury, exciting performance, panic attack, deadline, fever, and reunion can all raise arousal while differing radically in valence and urgency (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017).

Qualify each axis for its intended use: valence requires sign identification, while urgency requires links to action timing, avoidance, repair initiation, or choice pressure.

A realistic example is a courtroom witness whose hands shake. The shaking may mean fear, cold, medication, public exposure, trauma, anger, illness, or motor condition. A legal or research system that treats visible arousal as credibility, guilt,

distress, or burden creates a new hidden burden. The chapter's science must be able to say: this signal is too nonspecific.

The empirical route is multidimensional. Valence reports, arousal channels, task demand, action tendency, contextual threat, pain ratings, and calibration history are modeled separately before combination. A multi-axis model must justify its added claims through held-out prediction, calibration, or mechanism discrimination. Failure narrows those claims; it does not by itself validate a one-axis representation. Retain the unresolved or vector-valued description when no scalar representation qualifies.

20.2.5 — Developmental and Cultural Variation

Development, training, health, medication, disability, neurodiversity, culture, and threat history can change sensing, attention, report, and action both across people and within a person (Khalsa et al., 2018; Critchley & Garfinkel, 2017; Seth & Friston, 2016).

The same person may also shift across a day. Morning medication may reduce pain but blunt awareness. Afternoon fatigue may make a small demand feel enormous. Evening safety may allow the body to register what it had suppressed at work. Terminal illness may change all of these relationships through treatment, inflammation, weakness, delirium risk, and care context.

Use an estimand-appropriate method for material variation in intercepts, slopes, lags, or channel reliabilities; hierarchical and person-specific models are options. Partial pooling can borrow strength across persons while preserving the possibility that one person's channel is invalid for another, or invalid for the same person in a later state.

Report subgroup response patterns alongside the average; recovery in one group can conceal worsening associated with disability, medication sensitivity, or trauma history.

Allow governed cadence changes during flares, grief, crisis, or medication adjustment, and model those changes as information about observation availability.

20.2.6 — False Salience and Dissociation

Dissociation can separate bodily signal, conscious access, report, and action. False salience can make a benign signal urgent, while reduced salience can hide clinically important change. Both patterns defeat simple signal-to-valence inference.

False salience is a signal given importance beyond what it supports about the target (Barrett & Simmons, 2015; Critchley & Garfinkel, 2017).

False salience can arise through trauma-related attention to benign sensations, an observer's overreading of movement, device motion artifact, or a predictive feature mistaken for a meaningful construct.

Document target, signal, alternative causes, and harm from misreading. The harm pathway determines which false positives require stricter abstention.

Test matched controls, context swaps, alternate sensors, blinded review, device versions, and report disagreement. Use a prespecified conflict class and bounded follow-up; neither device nor report wins automatically, and retesting cannot continue until the preferred answer appears.

When salience is strong but ambiguous, report reason-coded ambiguity: whether valence direction, arousal source, or causal attribution remains unresolved. This can guide the next measurement.

Give threat learning, pain, medication, sleep loss, dissociation, sensory sensitivity, and social-context rivals comparable channel access and measurement quality.

20.3 — Brain–Body Control and Stopping

Brain–body control can change both the usable menu of actions and the policy that selects among them. The scientific task is to model how body, environment, opportunity, and meaning shape those possibilities and choices.

20.3.1 — Distributed Control Networks

Bodily state influences belief and salience; belief and salience influence action; action changes the body and environment; the new body and environment change the next observation.

External and social context records threat, care, tasks, rooms, demands, and access. These variables identify the pressures currently acting on the person.

The bodily-state model specifies pain, breathing, fatigue, inflammation, and metabolic demand. It identifies the physiological conditions represented in the analysis.

The central belief and salience model specifies predictions, precision, attention, and learned cue values. It asks what the system treats as important.

The control policy represents selection or inhibition among available actions such as stopping, continuing, avoiding, repairing, seeking help, and resting.

Observation and report specify which device signals, behaviors, speech, diaries, and clinical notes the research can responsibly use as evidence.

Update and residue concern learning, recovery, sensitization, and residual burden. They identify what remains after action or care.

Equation 20.2: Dynamic feedback display. Role: candidate control-mediation model under partial observation.

$$S_{t+\Delta} = f_S(S_t, A_t, E_t, M_t, D_t) + \eta_t^S$$

$$B_{t+\Delta} = f_B(B_t, O_t, P_t, C_t) + \eta_t^B$$

$$A_t = \pi(B_t, H_t, U_t, C_t)$$

$$O_t = h(S_t, B_t, Q_t) + \varepsilon_t$$

$$R_t = \rho(B_t, C_t) + \varepsilon_t$$

Here S is bodily state, B is belief or central representation, A is action, E is environment, M is medication, D is disability or disease context, P is precision, C is context, H is the filtered remaining-horizon candidate defined in Section 16.1.2, U is a possibly vector-valued reserve state, O is observation, R is report, and Q is signal quality. The display is a candidate structure. It is scientifically useful only if each term receives a design, a measurement route, and a failure condition.

As written, the observation and report equations share the same additive error. A model with distinct errors must specify their joint law; independence is an additional assumption.

Model salience, attention, interoception, motor readiness, memory, valuation, inhibition, and autonomic coordination as distributed processes; repeated regional involvement does not establish process ownership (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

A circuit-level example separates learned social valence from recognition. In mice, Zorab et al. (2025) identified serotonin input from the dorsal raphe and neurotensin input from the paraventricular thalamus converging on ventral hippocampal CA1. Projection-specific manipulations differentially affected positive and negative social valence; the experiments also

assessed con-specific memory separately. This gives a history-sensitive model distinct targets: which individual is remembered and which valence is assigned to that encounter.

Together with Section 23.1.2's neurotensin example, these findings motivate circuit-specific input–state–response relations rather than a universal neuromodulator valence sign. A cumulative model must predict an additional temporal constraint on those characterized components or their outputs.

Measure functional control through reaction time, stopping, switching, avoidance, help-seeking, speech initiation, confidence, recovery, and later behavior; neural measurements add constraints to those observations.

Replicate network relations, temporal order, behavioral transitions, or channel-specific mediation across scanners, tasks, sites, and populations. For lesions, stimulation, or medication, model treatment versions and off-target effects; triangulation across distinct perturbations can test whether the same pathway recurs.

20.3.2 — Inhibitory and Stopping Processes

Stopping can reflect danger, cost, confusion, social prohibition, physical limits, or changed objectives, and can protect options or deepen avoidance (Aron, 2007; Menon & Uddin, 2010).

Stop-signal, go/no-go, delay, inhibition, and avoidance measures require a declared control target; their results are not personality scores or direct measures of burden.

Queue blocking leaves an active demand whose action is suppressed by danger or cost. Compare harm avoidance, fatigue, helplessness, depression, threat appraisal, and motor limitation.

Delayed debt contact can reflect avoidance, timing strategy, missing information, shame, trauma cueing, or refusal of a

predatory system; establish usable actions before inferring hidden burden.

Test whether reduced threat, pain relief, a safe script, or improved access opens action in the predicted physiological order; failure challenges the specified stopping mechanism.

Distinguish blocked repair from chosen nonparticipation: refusal, pause, silence, or delay may express a consent boundary or protect the person.

20.3.3 — Conflict Monitoring and Switching

Conflict monitoring concerns competing goals, demands, predictions, and actions, for example, relief versus asking costs, honesty versus retaliation, or repair versus limited capacity (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

Represent the live options, expected bodily cost, social consequences, uncertainty, and prior learning, including whether a repair route actually existed.

Measure conflict through latency, error correction, report, physiology, eye movement, switching, narrative, and recovery. Speed alone is ambiguous: fast choice can be relief or compulsion; slow choice can be reflection or paralysis.

If ordinary cognitive conflict predicts reactivation, cue effects, and repair response equally well, narrow or retire distinctive Queue attribution in that domain.

Specify stakes when transporting low-stakes symbolic tasks to abuse, benefits appeals, pain disclosure, or accommodation decisions; their action costs differ.

20.3.4 — Autonomic and Endocrine Coordination

Autonomic and endocrine coordination links preparation, energy, threat, recovery, sleep, digestion, inflammation, and

pain across time, with medication and illness affecting its specificity (Menon & Uddin, 2010; McEwen, 1998).

The model should therefore treat autonomic and endocrine data as coordinated systems with explicit relationships among channels. A respiratory change may alter heart-rate variability. A medication may alter both pain and autonomic response. An inflammatory state may alter fatigue, sleep, mood, and reportability. These relationships help identify the mechanism.

Specify windows, lags, treatment, posture, movement, illness severity, quality, and channel direction when testing mediation, prediction, or common-cause explanations.

Test coupling, decoupling, order, and recovery: isolated heart-rate, cortisol, or inflammatory changes imply different candidate pathways from coordinated respiratory, sleep, or fatigue changes.

20.3.5 — Repair Initiation and Recovery

Repair initiation is the transition from detected need or blocked, avoided, frozen, or unresolved demand into a feasible action that changes state, environment, or the generating condition. Recovery is the subsequent trajectory and may remain partial; the two phases should not be collapsed into one improvement score (Aron, 2007; Menon & Uddin, 2010; Seeley et al., 2007).

Repair initiation may precede the visible action: lowered threat, recognition of an available route, pain relief, or removal of an environmental barrier may be the first measurable transition.

Test the proposed sequence of salience change, eased breathing, flexible attention, shorter avoidance, and action. Action preceding physiology contradicts that ordering and cannot be relabeled to fit it.

A delayed conversation may become possible after pain treatment, sleep, and practical help reduce its cost. This example locates repair initiation in newly available capacity.

20.3.6 — What Neural Findings Contribute

Timed interactions, perturbation responses, and medication-sensitive mediators can help characterize how the insula, prefrontal cortex, amygdala, vagal pathways, and distributed networks participate in a specified mechanism (Menon & Uddin, 2010; McEwen, 1998).

Neural results should identify their target, comparator, timing claim, and evidentiary role. Their contribution can include distinguishing mechanisms, testing temporal order, identifying mediators, or identifying artifacts beyond the information supplied by reports and behavior. A neural measure that adds no predictive or causal information beyond existing channels should not gain authority simply because it is technologically sophisticated.

Multifunctional regions participate in salience, pain, memory, control, attention, and many other tasks. Specificity therefore comes from prospectively specified connectivity, lag, state dependence, perturbation response, and rival models, not activation under a familiar anatomical label. Specificity should be demonstrated on contrasts that hold general arousal or task difficulty comparable, preventing familiar network activation from becoming a catch-all explanation.

A responsible caption names a candidate pathway under review rather than a location of fairness. Success increases mechanism plausibility within the design; failure narrows that mechanism. Neither makes a brain image a verdict on a life. Spatial precision should be no stronger than the causal and

measurement evidence linking that pathway to the declared construct.

Lesions, stimulation, and medication provide stronger leverage but also alter multiple pathways and behaviors. Off-target effects, treatment versions, and clinical context must be modeled before a causal interpretation. Triangulation across distinct perturbations can help identify whether the same bounded pathway recurs despite different off-target profiles.

The strongest admissible brain finding may be a bounded mediator or confound result, such as a network explaining a device-report relation or medication removing an apparent signature. Such results can defeat a simple route story without identifying PTVC or consciousness. This kind of negative mediation result can be scientifically decisive even when it leaves the larger terminal hypothesis untouched.

20.4 — Channel Lags and Confounding

Channels and reports follow different clocks. A rapid respiratory response, a slower inflammatory change, and a later remembered account cannot be treated as simultaneous measurements of one state merely because they share a timestamped record. Specify the relevant lags, common causes, feedback paths, attenuation, and missing channels before interpreting temporal order. The timing ranges below guide design, while each protocol must justify its own windows and test sensitivity to plausible alternatives.

20.4.1 — Different Response Times

A channel leading report may reflect preparation, prediction, or artifact; a following channel may reflect recovery, action, or common cause. Independent movement can mark an irrelevant channel or a process the report missed.

Equation 20.3: Lag and mediation display. Role: temporal test structure for mechanism order.

For this illustrative linear candidate, i indexes participants, c the observation channel, and t recorded time. Delta is a fixed positive lag in the protocol’s stated time unit. B is a scalar belief or central-representation measure, O the channel observation, and R the report outcome. Z is a column vector of prespecified adjustment covariates available by t; delta is its coefficient row, so delta Z is scalar. Alpha and mu are participant intercepts; the rho terms are lag coefficients, gamma and zeta channel coefficients, and beta the belief-to-report coefficient. Eta and xi are disturbances whose dependence must be specified. The displayed nonzero coefficients are proposed association checks, not a sufficient mediation criterion.

$$B_{i,t+\Delta} = \alpha_i + \rho_B B_{i,t} + \gamma_c O_{i,c,t} + \delta Z_{i,t} + \eta_{i,t}$$

$$R_{i,t+2\Delta} = \mu_i + \rho_R R_{i,t+\Delta} + \beta B_{i,t+\Delta} + \zeta O_{i,c,t} + \xi_{i,t}$$

$\gamma_c \neq 0$ and $\beta \neq 0$ under rival and confound checks.

This display proposes one temporal pathway: channel observation precedes a modeled change in belief or representation, which precedes report. That ordering is testable, but lagged prediction alone does not establish mediation. Causal interpretation additionally requires a justified intervention or identification strategy, suitable confound control, and sensitivity to competing and feedback pathways.

The timing ranges summarized in Section 20.4 are illustrative rather than universal. Each protocol must estimate or justify its own sampling windows, carry channel-specific uncertainty, and test whether conclusions survive plausible lag shifts.

Respiration and movement can require lag analysis over seconds to minutes. Without that temporal distinction, task effort may be mistaken for affect.

Cardiac and autonomic channels can require lag analysis over seconds to hours. Posture, medication, or activity can otherwise be mistaken for burden.

Endocrine channels can require lag analysis over minutes to hours. Circadian phase must be distinguished from an event response.

Immune and inflammatory channels can require lag analysis over hours to days. Illness or treatment can otherwise be mistaken for a mood pathway.

Sleep and fatigue can involve changes over nights to weeks. Separate recovery and sensor nonwear from the proposed psychological change.

Report and memory can involve intervals from minutes to years. Retrospective reconstruction must be distinguished from momentary experience.

Fast signals provide precise timestamps, while interpretation may develop later as the body settles, the situation becomes clearer, or the person is ready to report. A respiratory shift or heart-rate spike may mark a trigger, preparation, or distraction; the sequence helps distinguish those roles.

Slow inflammatory, fatigue, endocrine, or sleep signals require controls for illness, treatment, environment, caregiving load, and sensor routines.

A good design declares expected lags for the specified mechanism. A candidate in which incoming bodily evidence drives salience predicts a channel change before or with attention capture. A candidate in which salience or environmental change enables repair predicts that transition before visible action. Medication-linked predictions should align with dose, pharmacodynamics, and clinical effect.

Confirmatory lag claims require fixed windows or an independently locked validation phase, with multiplicity, clock alignment, and alternative lag structures included in uncertainty.

Match cadence to mechanism bandwidth: short tasks can miss slow inflammation, daily diaries can miss respiratory shifts, and excessive sampling adds burden and noise.

20.4.2 — Common Causes and Changing Confounders

Common causes can change over time. Disease severity, medication, movement, sleep, caregiver presence, and device quality may affect both a channel and later report differently across the day. Static adjustment can leave time-varying confounding unresolved.

Prespecify the mechanism's confound set. Record movement, posture, sleep, medication, severity, device version, time, context, and report timing, or justify omissions.

Quantify how strong an unmeasured common cause would have to be to overturn the mechanism comparison; report fragility when modest plausible confounding suffices.

Include research-induced changes: observation, prompts, wiring, filming, and requests to explain bodily states can alter physiology and report.

20.4.3 — Reverse Causation and Feedback

The apparent outcome can change the channel: fear reports precede altered breathing, repair decisions precede autonomic recovery, and reassurance precedes changed pain expression (Shiffman et al., 2008; Stone et al., 2007; VanderWeele, 2015).

Represent top-down, contextual, and feedback pathways before interpreting temporal precedence, and specify a perturbation that distinguishes the proposed directions.

When temporal order is unidentified, use the association terminology in Section 19.2.5 and withhold mediation, control, or mechanism claims.

Measure private repair decisions where possible: a decision can lower threat before visible action, making physiology appear to initiate repair when it followed commitment.

20.4.4 — Measurement Attenuation and Device Error

Model drift, dropout, placement, firmware, motion artifact, and body-dependent device error; attenuation can hide a relation and correlated error can create one.

Measurement attenuation occurs when the observed signal weakens because the instrument, body-device coupling, behavior, or communication channel degrades rather than

because the target changes. Low movement, low speech, poor contact, edema, altered perfusion, sedation, and device removal can all create apparent quiet. Device-quality and observation-process variables must therefore accompany the target estimate (Bent et al., 2020; Shiffman et al., 2008).

Check simultaneous unrelated-channel attenuation for device or physiological shutdown, movement-sensitive changes against movement, and variance changes by device version.

Treat sparse clinical notes during crisis, burden-related family nonreporting, and participant withdrawal as observation events before considering imputation.

Plot attenuation beside the target estimate, including wear time, signal quality, movement, medication, and channel availability.

20.4.5 — Multimodal Conflict and Missing Channels

Report which channels are absent, why, and whether absence depends on state or burden; joint channel failure can manufacture agreement.

Apply the channel-conflict analysis in Section 9.4.4: audit timing, quality, construct, context, and missingness for each channel (Shiffman et al., 2008; Bollen, 1989; Särkkä, 2013).

Code disagreement by channel, direction, context, quality, and permitted follow-up. Use qualified channel reliability to determine their contribution, and classify unresolved targets under Section 24.2.

Consider a person with chronic pain who smiles during a clinical visit while reporting severe pain. The smile may reflect politeness, coping, fear of disbelief, cultural habit, or an attempt to protect the clinician. A model that resolves the conflict by downgrading pain because the face looks calm repeats a known measurement injustice.

If medication or disability better explains a prespecified mechanism-specific conflict under the same observation model, that evidence weakens the candidate's distinctive explanation.

20.4.6 — Causal Graphs and Lag Sensitivity

Distinguish environment, bodily state, central representation, report, action, clinical factors, and quality in the graph; the proposed experiential construct remains latent unless separately identified.

Device tolerance, repeated reportability, and survival to sampling are selection variables that can induce associations and limit transport.

Review the causal graph with participants, clinicians, ethics boards, and disability advisors so they can challenge assumed pathways.

Interpret cross-lag coefficients under the identification conditions accompanying Equation 20.3, with feedback, cadence mismatch, and measurement error represented explicitly (Bollen, 1989; Hernán & Robins, 2020).

20.5 — Medication, Disability, Trauma, and Illness

Treatment, disability, trauma, illness, aging, and terminal conditions can change both experience and the relation between experience and observation. A muted cardiac response, limited movement, or absent speech can therefore have different meanings across contexts. Qualify medication versions, accessible baselines, communication channels, and clinical conditions before transferring a model. The reference class should state which observations support the interpretation and which additional evidence would be needed to extend it.

20.5.1 — Medication Effects and Treatment Versions

Specify drug, dose, timing, indication, co-medication, metabolism, and adherence; distinguish symptom relief, side effects, reportability changes, and direct sensor effects.

Medication can move the signal, the experience, the report, and the ability to report. Analgesics may reduce pain and alter alertness. Sedatives may reduce arousal and communication.

Stimulants may raise cardiac signals and attention. Autonomic medications may mute or reshape heart-rate dynamics. Psychiatric medications may alter affective range, sleep, appetite, and interoceptive awareness (Giacino et al., 2018; Khalsa et al., 2018; McEwen, 1998).

Where ethically and clinically appropriate, record dose, route, onset, peak, duration, side effects, indication, and changes so pharmacodynamics is not mistaken for spontaneous or terminal regulation.

Clinical expertise is required to distinguish drug effects across bodies and observation routes; a generic medication flag is often insufficient.

Assign medication its causal role for the question: confound, mediator, intervention, modifier, or inference limit. Pain relief

may be the target mechanism while medication remains a rival explanation of terminal variance.

Care decisions must remain independent of the study. If a needed medication destroys a signal, the signal is lost.

20.5.2 — Disability, Access, and Baseline Variation

With autonomic neuropathy and pain, low autonomic variability may be mistaken for low burden. Use a disability-appropriate baseline and preserve the evidential priority of the participant's report.

Calibrate accessible devices, tasks, prompts, and reports to actual support and environment; protocol inaccessibility can cause signal failure.

Disability can affect baseline, expression, effort, communication, and sensor fit. Qualify the observation model for those differences rather than extrapolating able-bodied norms (World Health Organization, 2001; Khalsa et al., 2018).

Wheelchairs, ventilators, and communication devices may therefore require different observation models.

Narrow or retire models that misread disability-, pain-, or access-constrained activity as disengagement, or disability-affected speech as affect.

Co-design accommodations, alternative channels, longer response windows, assistive-technology compatibility, and challenge procedures with relevant participants and experts.

Establish within-person baseline, range, fatigue cycle, communication, and support context before cross-person claims; separately qualify transport between disability groups.

Record device refusal as valid: pain, sensory overload, irritation, dysphoria, or practical difficulty can make the observation method unsuitable.

20.5.3 — Trauma, Hyperarousal, and Dissociation

In a safe task involving someone with a trauma history, high salience may be mistaken for current threat. Record cue history, specify the precision model, and check safety.

Trauma-informed interpretation must never require reenactment or disclosure beyond consent.

Trauma can alter bodily prediction, attention, threat appraisal, salience, startle, sleep, pain, dissociation, and safety learning.

Hyperarousal may be visible in some contexts, but trauma can also appear as numbness, shutdown, avoidance, memory gaps, constrained expression, or practiced calm. There is no single trauma signature that authorizes simple interpretation (Khalsa et al., 2018; McEwen, 1998).

Compare trauma-related reactivation, blocking, and repair difficulty against learning, threat prediction, memory, and unsafe-environment explanations under humane tests.

Interoceptive-attention tasks should be brief, optional, clinically reviewed where necessary, and paired with stop rules. Exclude surprise or intense bodily focus that could amplify symptoms.

Record safety context, trigger likelihood, dissociation risk, grounding, report preferences, and aftercare, including social or cultural conditions that make vigilance rational.

Include delayed sleep loss, dissociation, and other aftereffects in follow-up and stop rules even when the task itself appears calm.

20.5.4 — Chronic Illness, Inflammation, and Fatigue

During a chronic-fatigue flare, low movement may be mistaken for withdrawal. Account for illness state, energy constraints, and the actual opportunity for activity.

Chronic illness can alter sleep, movement, pain, cognition, immunity, autonomic function, digestion, mood, participation, and opportunity. Momentary ratings can miss the ongoing work of managing energy, symptoms, treatment, uncertainty, and access (Khalsa et al., 2018; McEwen, 1998).

Fatigue is especially easy to misread. Low activity may be depletion, pain avoidance, depression, medication, symptom management, inaccessible environment, or wise pacing. High activity may be recovery, overexertion, necessity, caregiving pressure, or a day that will be paid for tomorrow. The signal needs a time horizon.

Adaptation to illness can coexist with pain, uncertainty, financial stress, lost roles, stigma, and medical work.

Record relevant energy, flares, treatment, care tasks, sleep, opportunity, access, and post-exertional changes. Give participants cadence control and stop measurement that worsens fatigue.

Use lags that can capture next-day exhaustion. Test recovery through fewer obligations, treatment, support, or pacing over the appropriate observation window.

Model appointments, paperwork, tracking, transport, side effects, coordination, and disbelief when they influence physiology or report; otherwise disease is confounded with its management burden.

20.5.5 — Pain, Communication, and Motor Limits

During a stress task, beta-blocker use may mute cardiac responses without establishing calm. Record the medication window and use appropriate alternative channels.

Pain expression and communication can be limited by motor function, language, sedation, fear of disbelief, or learned masking. Absence of report or movement is not absence of pain.

Conflicting or unqualified channels should preserve or widen uncertainty; qualified agreement may reduce it. A multimodal estimate must not override a person's testimony.

Pain combines bodily, affective, attentional, cognitive, social, and communicative processes; expression, treatment response, and remembered pain can diverge (Khalsa et al., 2018; Critchley & Garfinkel, 2017).

Behavioral pain cues are partial. Movement limitation can show pain, fatigue, motor impairment, access constraint, sedation, or protective pacing. Facial expression can be masked, culturally controlled, absent due to disability, or altered by medication. Voice can change because of breath, fatigue, language, neurological condition, or social context.

Test the predicted order of analgesia-related report, behavior, sleep, autonomic, and repair changes. Before interpreting unchanged physiology, verify that channels, windows, and detection limits could capture the predicted response.

Pain may enter the study as an already present clinical state, a participant-reported outcome, or a response to independently justified care. It may not be induced, prolonged, intensified, or left untreated to sharpen a mechanism contrast. Designs that cannot separate the scientific question from added harm are narrowed or abandoned.

Fear of disbelief can change reporting, explanation, and use of scales. Provide multiple expression routes and model that context without penalizing inconsistency.

20.5.6 — Clinical Context Before Interpretation

During palliative sedation, reduced report must not be equated with neutrality. Retain clinical context and independent endpoint determination, returning no estimate when necessary.

With optical sensors, device performance associated with skin tone may be mistaken for a physiological difference. Validate the relevant device across skin tones, fit conditions, and artifacts.

Clinical context matters beyond terminal settings: chemotherapy, dialysis, psychiatric crisis, rehabilitation, and chronic pain treatment can alter both physiology and observation.

Justify needed clinical fields: diagnosis, acuity, treatment changes, sedation, delirium risk, analgesia, mobility, communication, device interference, and consent. Do not collect the entire list by default.

Clinical expertise should enter before analysis and during adjudication, alongside participant report, device quality, and research model uncertainty.

Clinical review can stop collection when illness, fatigue, medication, distress, a nearby care decision, or changing consent makes it disproportionate or outside the declared design.

When trauma, illness, treatment, and care-related explanations cannot be separated, the result remains clinically contextual rather than route specific.

Apply abstention at the person-window level: delirium, acute treatment change, device failure, or severe burden may preclude a mechanism estimate while permitting descriptive clinical use.

20.6 — Intervention and Perturbation Designs

Mechanism tests use ethically appropriate changes in state, context, attention, posture, breathing, relief, sleep, medication, or environment to challenge predicted channels and temporal order. A useful perturbation separates the proposed mechanism from ordinary pathways, including artifacts and expectancy effects. Within-person contrasts, negative controls, and sham conditions can strengthen that comparison when justified. They do not authorize harmful provocation or changes to beneficial care for the sake of the theory.

20.6.1 — Naturalistic State Changes

An already chosen or safe naturalistic breathing change can test respiratory salience and report lag. The design excludes panic induction.

Naturalistic state changes provide first mechanism tests through consented observation of ordinary activity, rest, medication, and care (Klasnja et al., 2015; Nahum-Shani et al., 2018; Khalsa et al., 2018).

Use repeated within-person events, time-varying covariates, and justified control windows; sparse or confounded events limit mechanism attribution.

Record why natural changes occurred: comfort-motivated breathing, pain treatment, and access-related posture changes differ causally from protocol-assigned behavior.

20.6.2 — Safe Interoceptive Perturbation

Use independently justified care or approved low-risk, participant-controlled variation, preserving timely treatment and withdrawal.

Changes among sitting, standing, walking, and rest can test movement artifact and autonomic adjustment, with accessibility and disability accommodations built into the comparison.

A neutral attention task varies harmless focus or distraction to test precision and salience weighting. Define symptom monitoring and stop rules before the task.

Safe interoceptive perturbation means a low-risk, consented, reversible change in bodily attention or state that has independent ethical justification. It may include gentle breathing awareness, posture adjustment, brief attention redirection, paced movement, or other innocuous tasks approved for the population. It does not include provocation that creates clinically meaningful distress (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

Specify the perturbation's target: respiratory salience for breathing, autonomic adjustment or artifact for posture, and precision weighting for attention. Broader valence or burden claims require separate identification.

Make stop rules visible and participant-controlled. Record stopping time and permitted cause information for missingness analysis; reasons for withdrawal remain optional, and subsequent data use follows consent and the protocol.

Include debriefing and return to baseline where appropriate. If a breathing or attention exercise increases symptom focus, provide grounding, support, and review.

Pilot perturbations with the relevant population and record contraindications before enrollment. The population-specific review in Section 20.6.6 governs which tasks are appropriate.

20.6.3 — Within-Person Crossover and N-of-1 Logic

A clinically indicated medication transition can help examine medication as a mediator or confound. The care decision remains independent of the study.

An independently justified pain-relief window can examine the temporal order of pain, report, and physiology. Relief is never delayed to create the comparison.

Within-person crossovers reduce between-person differences in baseline, disability, medication, fitness, trauma, culture, and device fit (Klasnja et al., 2015; Nahum-Shani et al., 2018; Khalsa et al., 2018).

A randomized crossover can compare assigned, safe conditions in an explicit sequence within each participant. Comparisons after naturally disrupted sleep or before and after clinically indicated pain relief are instead observational within-person contrasts unless condition assignment is part of the design. Both can be useful, but the latter retain event-timing and treatment-selection confounding. Neither design permits induced sleep disruption or delayed relief for Law-specific evidence.

Model and test order effects, carryover, learning, fatigue, expectation, symptom fluctuation, and changing context; justify any washout period.

20.6.4 — Mechanism-Specific Predictions

Existing neuroscience provides a concrete example of mechanism-guided recovery. Xia et al. (2025) recorded population activity in stress-susceptible and resilient mice and identified different dynamics linking ventral hippocampus and basolateral amygdala. Manipulating the hippocampal inputs to the amygdala shifted dysfunctional dynamics toward the resilient pattern and reversed anhedonic behavior. The experiment connects an affect-related behavioral difficulty to a

specified circuit and shows that intervention can change it. This gives recovery research an experimentally grounded pathway to investigate.

Perturbation designs can help distinguish routes when their assignment and observation models are justified. Breathing-related changes in salience before report and medication-related changes in report before physiology provide temporal contrasts to test. A posture-related device change without a corresponding report warrants an artifact check; that disagreement alone does not establish artifact. A response in a qualified negative control challenges the assumed specificity.

A randomized crossover study illustrates why recovery timing matters. Among 82 healthy participants completing both conditions, Ferstl and colleagues (2022) compared auricular vagus-nerve stimulation with sham stimulation during prolonged effort. Positive mood, a secondary outcome, improved relative to sham during the post-stimulation period; the improvement was not detected during stimulation. The reported difference was approximately 9.6% relative to the model's mood intercept. This provides a concrete brain–body comparison with a measurable lag and suggests testing whether recovery models predict when a benefit appears as well as its magnitude.

Prespecify channel, timing, direction, and failure. For example, a breathing-related candidate may predict respiratory salience changes before urgency changes, with no corresponding change under an activity-matched control; a pain-related candidate may predict pain-report and sleep changes before general mood. Test the sequence specified by that candidate (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

Specify when medication changes, device-quality failure, participant withdrawal, excessive channel conflict, or task

burden make the mechanism test uninterpretable and require abstention.

A concrete human comparison separates successful regulation from a single assumed neural route. Jiang et al. (2026) developed whole-brain fMRI signatures in a preregistered discovery cohort of 59 participants and evaluated them without refitting in an independent validation cohort of 33. Participants used acceptance or reappraisal while viewing negative video clips and rated negative feeling after each trial. Both strategies reduced those ratings relative to natural responding, with a larger reduction for reappraisal. The strategies had shared and distinguishable distributed neural signatures. Their condition-classification performance is a different target from calibrated momentary phenomenal magnitude.

Where reuse is permitted, reproduce the published held-out contrasts before extending them on untouched records. A PTVC extension needs qualified temporal contributions, not integrated classifier scores. Compare mediator, interoceptive-learning, allostatic, and artifact models on their shared and differing temporal and cross-channel predictions.

Positive affect can also be an explicit target of recovery. In an assessor-blinded randomized trial, Meuret et al. (2026) assigned 98 adults with very low positive affect and impairing depression or anxiety to 15 weekly sessions of Positive Affect Treatment or a comparison therapy focused on negative affect. The positive-affect treatment produced greater improvement in the jointly analyzed measures of positive affect, anhedonia, and depression and anxiety symptoms, with an effect size of d = 0.27 for improvement and d = 0.21 at one-month follow-up. The composite advantage was driven mainly by depression and anxiety scores; between-treatment differences in positive affect and interviewer-rated anhedonia were not statistically

significant. Reward anticipation and attainment improved on multivariate targets combining self-report, behavioral, and physiological measures, without a significant difference in change between treatments. For this program, the trial supplies a practical model of constructive inquiry: specify a capacity for improvement, compare active approaches, and track experience alongside candidate processes. The ability to develop rewarding experience becomes a concrete research question with measurable outcomes.

20.6.5 — Negative Controls and Sham Perturbations

Choose negative controls using the causal-null requirements in Section 19.6.2, matching attention, expectancy, movement, and contact where relevant. Shams must remain credible, safe, and nondeceptive under the protocol's consent rules.

Match controls to the pathway: movement raising heart rate without breath awareness for respiratory salience, posture without expected affect for artifact, or a nonanalgesic attention-changing routine procedure for a pain-relief pathway.

Artifact challenges can test whether a model's apparent specificity survives matched alternative conditions. For a model predicting burden from wrist movement, compare similar movement across contexts with independently assessed burden. For a voice model, fatigued reading without reported distress tests sensitivity to fatigue-related prosody. These contrasts become negative controls only under the exclusion justification in Section 14.5.5.

20.6.6 — No Harmful Provocation

An ordinary sleep-schedule change or clinical sleep support can examine recovery and fatigue pathways. The study does not induce deprivation to test the theory.

Some experiments would be informative and forbidden. Pain induction, panic provocation, sleep deprivation, social humiliation, deprivation, dangerous breath challenge, medication withholding, or terminal observation pressure could create clean-looking mechanism contrasts. They are outside the chapter's route because they make people carry burden for theory (Seth & Friston, 2016; Critchley & Garfinkel, 2017).

Indirect tests and narrower questions provide useful alternatives when a direct experiment would create harm. Recording which question each design can answer keeps the research program both informative and humane.

Safe alternatives exist: observe clinically indicated relief, use mild and reversible attention tasks, rely on naturalistic variation, study routine movement and posture, use archival data under consent, compare device artifacts, and build simulations that do not recruit vulnerable bodies. These alternatives provide practical routes to reproducible and ethically sound evidence.

A protocol should include a provocation audit. Does any task create more distress than everyday minimal risk for the population? Does it amplify symptoms? Does it pressure people to continue? Does it delay treatment? Does it treat withdrawal as failure? If any answer is yes, redesign or stop.

The analysis proceeds by care before contrast. Mechanism science is strongest when it learns from life without making life worse. The person is never the instrument of the theory. When the safest contrast is less statistically efficient, that cost belongs to the design rather than to the participant.

No-harm design forbids implicit pressure to endure. Payment, researcher enthusiasm, scientific or spiritual motives, and fear of disappointing clinicians may press continuation. Consent scripts and staff must make stopping ordinary, not exceptional. Continuation checks should be phrased neutrally and separated

from scientific enthusiasm so a participant does not have to resist the project to exercise withdrawal.

Minimal-risk classification must be population specific. A posture change, breath task, or cognitive load mild for healthy adults may be unsafe or exhausting with cardiopulmonary disease, chronic pain, trauma, pregnancy, or disability. Safety review must follow actual participants, not the task label. The protocol should record population-specific contraindications before enrollment rather than discover them through preventable adverse burden.

Stopping thresholds should be observable, participant controlled, and independent of the desired contrast. Symptoms, distress, fatigue, device intolerance, clinical change, or a simple request to stop are sufficient. The analysis should preserve data collected before stopping without treating withdrawal as noncompliance. Stopping-related missingness should be marked by cause and timing, preserving the ethical reason for data loss in later sensitivity analysis.

20.7 — Wearables, Artificial Intelligence, and Humility

Measurement technologies and public applications need a defined scientific role for each output. A wearable channel, estimator, or dashboard observes a qualified part of a process through a fallible measurement relation. The following distinctions connect those roles to their contexts of use and to the limits that must accompany interpretation.

20.7.1 — Wearable Quality, Context, and Drift

A wearable sensor supplies partial channel observations with quality flags. It does not directly reveal inner life.

Device validation should be context-specific. A sensor may perform well at rest and poorly during movement. It may measure heart rate adequately but fail for rhythm irregularity. It may work differently across skin tones, body hair, perfusion states, tattoos, edema, disability, and clinical contexts. Evidence from one device generation may not travel to another (Bent et al., 2020; Nelson et al., 2020; Tabassi, 2023).

Record the needed device model, firmware, location, cadence, quality, artifact, nonwear, charging, discomfort, and context fields. Unknown quality restricts interpretation.

In palliative care, weight change can loosen a wristband, irritation can prompt removal, care can shift placement, and perfusion can decline. Track these causes before interpreting a smooth terminal signal.

Where feasible, compare wearables with chest straps, ECG patches, clinical monitors, reports, or observations to investigate device error and changed observation processes.

20.7.2 — Digital Biomarkers and Clinical Validation

A digital biomarker is a digitally measured characteristic proposed to indicate a biological, physiological, or behavioral process in a defined context of use. Qualification specifies target, population, setting, measurement performance, and decision role (FDA-NIH Biomarker Working Group, 2016; Coravos et al., 2020; U.S. Food and Drug Administration, 2023).

Fit-for-purpose evaluation separates verification, analytical validation, and clinical or research-context qualification. Verification asks whether the sensor and software operate as specified. Analytical validation asks whether the pipeline estimates the declared physical or behavioral quantity with adequate error under relevant conditions. Context qualification asks whether that quantity is related closely enough to the declared concept and decision for the intended use (Goldsack et al., 2020; U.S. Food and Drug Administration, 2023).

Attach the validated hardware, software, preprocessing, quality thresholds, population, and setting to the result; a device brand alone does not identify the qualified pipeline.

The pipeline also needs an explicit route from raw data to action. The DACIA framework separates collection, aggregation, contextualization, interpretation, and action, revealing where a sensor feature acquires meaning and where burden, feedback, or deployment enters. Each transition can fail independently, so classifier performance at the end of the chain cannot certify the earlier measurement and construct links (Daniore et al., 2024).

Freeze the digital feature's causal role: observation, mediator, confound witness, quality flag, modifier, eligibility criterion, or outcome. Reuse as selection gate and confirmatory outcome can create circularity or collider bias.

Report effect size, uncertainty, decision consequence, subgroup performance, and failure conditions; group classification can add no useful information for the intended decision.

20.7.3 — Affective Computing Limits

An affective model may serve as an exploratory or validated classifier within its declared scope. It is not a universal reader of emotion.

Label provenance defines the prediction target: self-report, observer annotation, elicitation condition, clinical classification, and behavioral success provide different information. Record who supplied each label, when, and under which interpretation. Where sufficiently many elicitation contexts support meaningfully different groupings, add a context-regrouping diagnostic: compare the declared valence labels with prespecified alternative assignments of whole contexts to labels, excluding the original grouping and mere relabellings of the same classes. Retain recording blocks, participant splits, and comparable class sizes. Refit the same classifier pipeline under each assignment with identical tuning resources and held-out splits; for new-participant validation, withhold each test participant's records from training and tuning. Similar accuracy across many alternative groupings would show that context recognition remains a viable explanation. This adapts label-permutation logic as a descriptive diagnostic (Ojala & Garriga, 2010). Qualification then asks whether a frozen model succeeds in held-out contexts with independently obtained valence labels and the intended within-person scale.

Human expression is many-to-many: the same outward pattern can reflect different experiences or social demands, and similar experiences may produce different expressions. Context and

alternative channels reduce ambiguity but do not abolish it (Picard, 1997; Calvo & D'Mello, 2010).

Test shared task cues, annotators, preprocessing, and missingness as correlated-error sources in multimodal validation; require incremental channel value.

A defensible validation program includes calibration, out-of-distribution testing, subgroup and intersectional performance, device and environment shifts, disability and medication conditions, cultural expression variation, missing-channel stress, and abstention (Autio et al., 2024).

Published benchmarks now make temporal affect prediction a concrete comparison task. SemEval-2026 Task 2 paired ecological essays with self-reported valence and arousal and separated present-state assessment from forecasting change. Its design included tests involving both previously observed and unseen participants (Soni et al., 2026). A report-channel model can use such a benchmark to test whether apparent accuracy survives future observations and new people, then compare text-based predictions with appropriate history and context baselines. The self-reported labels make forecast performance directly assessable. Existing consent, access conditions, and privacy protections continue to govern reuse.

Candidate applications include artifact detection, quality control, exploratory clustering, prompt routing, and task-specific prediction within the demonstrated context of use.

For disagreement with a participant's report, inspect label provenance and apply Section 20.4.5's conflict procedure, then classify the unresolved target under Section 24.2.

Test standalone clips, heat maps, and labels with nonexpert readers; revise displays that imply mind reading or exceed the qualified use.

20.7.4 — Analysis Support and Data Protection

A dashboard can support internal review of uncertainty. It must not supply public rankings or eligibility decisions.

A data archive is a purpose-limited research record. It is neither the person nor a dossier for unrestricted use across architectures.

Validate AI-assisted extraction, artifact detection, code review, simulation, recovery, and sensitivity work against primary data and frozen specifications, recording each output's analytic role (U.S. Food and Drug Administration, 2025a; World Health Organization, 2025).

Map each channel to its estimand, quality, or safety purpose. Respiratory timing need not collect speech, sleep timing need not collect location, and quality review does not justify indefinite raw-stream retention.

Apply the Chapter 3 security controls to brain-body records. Public release should use aggregated, deidentified, or synthetic displays only after reidentification and hostile-crop review.

Research data must not be merged across domains into a general person dossier. Purpose-limited linkage to clinical or other research records requires separate authorization and necessity for the approved study.

20.7.5 — Abstention and Failure Actions

Equation 20.4: Failure-and-abstention display. Role: local trigger for abstention pending evidence classification.

$$Abstain_t = 1\left\{ \begin{array}{c} q_{\text{signal}}(t) < q^* \ \vee \ C_{\text{conflict}}(t) > c^* \\ \vee \ P_{\text{confound}}(t) > p^* \ \vee \ \mathscr{S}_{\text{scope}}(t) = 0 \end{array} \right\}.$$

A failure matrix links each failure class to a defined next action. Verification failure stops use of the device output; construct failure withdraws the measured-to-latent bridge; causal

failure narrows the object to prediction; transport failure restricts the reference class; ethical failure retires the protocol; open-set failure expands the model library. This classification lets later evidence reopen the particular failure it addresses.

20.7.6 — Mechanism Failure Remains Live

An Artificial Intelligence assistant may support coding, pattern searches, and documentation. It cannot issue autonomous scientific verdicts or release results without supervision.

A participant portal supports return, correction, deletion, and challenge. It must not pressure participants to explain themselves.

Failure remains active. Interoceptive features may underperform, allostatic measures prove nonspecific, somatic markers transport only within narrow cue classes, wearables prove biased or fragile, Artificial Intelligence features fail out of sample, and medication or illness dominate channels. Any outcome can favor an ordinary rival or retire a mechanism component (Autio et al., 2024; Picard, 1997; Calvo & D'Mello, 2010).

Mechanism failure is classified by the object that failed: identification, prediction, calibration, transport, causal order, construct validity, subgroup performance, device verification, ethical feasibility, or conceptual scope. Each class has its own consequence, from recalibration or reference-class restriction to bridge withdrawal, protocol retirement, or No adequate estimate. Typed failure converts adverse evidence into cumulative design knowledge.

A useful failure-and-abstention rule can be explicit.

Apply the canonical evidence states in Sections 24.2 and 27.4. Low signal, channel conflict, confounding, or scope failure

narrows or blocks only the affected inference; the local diagnostic should not invent a bespoke verdict vocabulary.

Mechanism failure should be published or logged when the protocol permits. A wearable that fails in a disability subgroup, a respiratory task that worsens symptoms, a neural correlate that adds no held-out prediction, or a result explained by medication changes the design obligations of future studies. Negative evidence belongs in the cumulative safety and recovery record, not in an unpublished remainder.

A PTVC-linked interpretation retains only what survives these failures. If ordinary brain-body mechanisms explain the patterns under fair comparison, the route narrows or retires. If a prospectively specified signature survives recovery, perturbation, transport, and strong rivals, the route becomes more answerable without being promoted to the terminal target.

Part VII now gives the ordinary rivals full generative force. The route-level result should specify which ordinary mechanism was defeated, because survival against one rival does not imply survival against the rest of the envelope.

The chapter's final rule is practical: devices witness, bodies participate, models estimate, and people remain more than any channel. A science worthy of felt experience must know when another measurement would add knowledge and when putting the instrument down is the more rigorous act. Burden-adjusted information gain can make that judgment explicit by comparing expected inferential value with the participant cost of another channel.

Graceful nonuse is a result. A promising mechanism may be burdensome, privacy-invasive, fragile, clinically disruptive, or easily misused. Science can preserve it as a bounded research question while refusing deployment or public-artifact travel. A nondeployment verdict should state which scientific uses remain

permitted, preventing a useful laboratory result from drifting into operational recommendation.

Successor mechanisms begin from failed assumptions and unresolved residuals. A new neural or Artificial Intelligence model needs a version identity, recovery benchmark, qualification plan, privacy review, and public-status ceiling. The successor should identify the exact residual pattern the new mechanism is designed to explain so added complexity remains accountable.

Part VII — Rival Models, Inference, and Adversarial Tests

A returning state and a closing cumulative path are different objects. Confusing them can make an ordinary regulatory process look like evidence for a much stronger restriction. Part VII begins with that mathematical distinction and expands it into a comparison program. The candidate accounts face alternatives capable of reproducing plausible-looking patterns, so the comparison can reveal which information a proposed explanation actually adds (Popper, 1959; Mayo, 2018; Wilson & Collins, 2019).

Chapter 21 develops the nearest alternatives: adaptation, opponent responses, homeostasis, and related stochastic dynamics. Their integrated paths show how local return can coexist with accumulated variation. Chapter 22 extends the comparison to predictive processing, learning, and control. It places the competing accounts under common observations so differences in states, policies, or endpoint treatment become testable commitments rather than changes in vocabulary (Lageman et al., 2026; Samara, 2026).

Chapter 23 widens the frame. An evolutionary account can concern function without predicting comfort, and a population relationship need not describe change within an individual. Developmental, cross-species, and synthetic-system comparisons introduce further questions about what can be transported. Work on nonergodicity gives the within-person distinction particular force: averaging across people can change the relationship an investigator is trying to estimate (Hunter et al., 2024; Mistry et al., 2026).

Chapter 24 follows the question into statistical inference. The target population, observation process, and treatment of delayed entry determine which quantity an estimator addresses. Missingness and competing events can change the relation between a recorded trajectory and the intended target. Making

that relation explicit permits a sensitivity analysis to answer a substantive question: which conclusion changes when a disputed observation assumption changes (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020)?

Chapter 25 tests the analysis itself in worlds whose generator is known. Parameter recovery asks whether the pipeline recovers quantities; model recovery asks whether it distinguishes explanations. Open-set tests examine cases outside the submitted candidates, while transport stress changes the conditions under which observations arise. Nonstationary recovery and simulation-based calibration show why these are separate checks. Adversarial collaboration adds advance agreement about the comparison and the evidence that would count against each account (Schumacher et al., 2025; Säilynoja et al., 2026; Cogitate Consortium et al., 2025; Ceci et al., 2026; Isch et al., 2025).

The Part makes the opponent part of the experimental design. A capable rival can reveal an uninformative channel, an avoidable asymmetry, or a target that the proposed study cannot identify. Preserving those outcomes allows later teams to improve the comparison rather than repeat its limitations (Munafò et al., 2017; Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019). This is how a collection of fitted models becomes a cumulative inquiry.

Chapter 21 — Adaptation and Homeostasis

A return toward ordinary feeling can follow several processes: a disturbance fades, a delayed response emerges, expectations change, or regulation adapts to continuing demand. These processes have different implications for what remains after apparent recovery. This chapter makes adaptation a quantitative comparison through response profiles and accumulated paths (Solomon & Corbit, 1974; McEwen, 1998; Diener et al., 2006; Kuppens et al., 2010).

The first difficulty is deciding what recovered. A daily report can return while avoidance, sleep disruption, or restricted opportunity persists. Function can improve through pacing even when pain remains. State, report, memory, and behavior therefore need distinct observation rules. A smooth average can conceal change in which situations are encountered or which experiences become reportable.

The central mathematical question concerns the accumulated path. Returning a current state to its reference does not generally cancel its earlier area. Some ordinary controllers do impose an area restriction, but that property follows from their state-output relations and boundary conditions. The comparison must identify those conditions instead of inferring them from the appearance of recovery.

Opponent responses make the distinction concrete. A positive movement and a later negative movement can have similar peaks but different durations. Their net area depends on both. The chapter derives an equal-area condition and then examines a stable system where the condition follows structurally. Its finite-window residual still matters, because a response that cancels asymptotically need not cancel before observation ends.

Homeostatic feedback raises a related question about memory. A controller's output can settle while its internal state retains a change. Measuring only the output misses that record.

Conversely, alternative state coordinates can describe exactly the same input-output behavior. Distinguishing a physical accounting mechanism requires an independently calibrated state and an intervention whose meaning survives the coordinate choice.

Stochastic regulation adds another distinction. Stable averages or controlled variance need not imply exact accounting along individual realized paths. The chapter examines reaction counts, their compensators, and accumulated continuous states separately. These comparisons identify what a biological model must retain and what its measured noise allows, without turning a population mean into an individual guarantee.

The Ornstein–Uhlenbeck baseline then supplies an explicit test. Its state can return toward a reference while the variance of its integral grows. Initial conditions, cadence, and observation error alter the finite-window prediction, so they enter the calculation and its evaluation. The examples provide numerical checks and show which cumulative features demand more than a state-return model.

The final comparisons use these restrictions across event sequences, interventions, and new observation conditions. Adaptation, opponent responses, clinical change, and regulation may coexist; their combinations must be distinguishable before a specific mechanism is credited. Chapter 22 takes up learned policies and predictive models, whose information and objectives create further alternatives to the response dynamics developed here.

21.1 — Adaptation as a Leading Rival Family

Adaptation needs a state and reference level. Return dynamics define its time course. Memory, treatment, and report processes set further conditions. Person-specific baselines and response shift can distinguish rivals.

21.1.1 — Set-Point and Baseline Accounts

Set-point accounts specify a person-specific reference range, disturbance, and relaxation. Compare their prospective trajectory and residual predictions after modeling uncertainty, report, treatment, and context (Brickman et al., 1978).

After job loss, mood may partly return while debt, shame, sleep disruption, avoidance, and strain persist. Credit the set-point model for predicted features; identify any remaining feature independently rather than inferring a latent residual from an unexplained story.

Display reference range, shock, return rate, plateau, noise, and asymmetric or incomplete paths, labeled as a rival simulation with its parameters.

21.1.2 — Incomplete and Asymmetric Adaptation

Allow lasting gains and persistent adverse effects with different adaptation rates and limits (Diener et al., 2006; Lucas, 2007a).

Quantitative evidence supports differentiated adaptation trajectories. A meta-analysis of 188 publications, comprising 313 samples and 65,911 participants, found that affective and cognitive wellbeing respond differently to life events and follow varied trajectories (Luhmann et al., 2012).

Lindqvist, Östling, and Cesarini studied 3,362 Swedish lottery players 5–22 years after prize events, using randomized prize variation within comparable groups and preregistered analyses. An additional after-tax prize of US$100,000 in 2011 dollars was

associated with approximately 0.037 standard deviations higher overall life satisfaction. The benefit persisted; effects on happiness and mental health were smaller and not statistically distinguishable from zero (Lindqvist et al., 2020). Better circumstances can produce lasting gains, and following evaluative life satisfaction and momentary affect separately helps identify how those gains develop.

Pacing can restore a chronic-pain routine while vigilance, sleep disruption, and movement limits persist. Identify which outcomes returned and test treatment, expectation, and reporting explanations for their divergence.

21.1.3 — State, Report, and Memory Dynamics

For a widower reporting daytime stability but avoiding loss-associated places, test affective adaptation, strategic reporting, contextual triggering, and memory reconstruction through their distinct report, behavior, memory, and physiological predictions.

Specify state transitions, contemporaneous report, and later-memory kernels. Recover them under disagreement and missing reports, then predict protected sequences. Apply Section 24.2 when layer identity remains unresolved (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008).

Estimate adaptation within persons, allowing their baselines, contexts, and measurement error to vary; cross-sectional averages can conceal trajectories that diverge, reverse, or fail to recover (Wilson & Gilbert, 2008; Brickman et al., 1978).

Define return, plateau, overshoot, persistence, and transport on a fixed clock; test them across event classes and assess parameter recovery under the actual irregular, noisy observations.

21.1.4 — Conditions for Rival Victory

Adaptation wins this component only under Section 22.8.1's declared advantage or adequacy-and-simplicity rule, comparing prespecified set-point, asymmetric, and within-person models through the common observation process (Diener et al., 2006; Lucas, 2007a).

Test the Section 22.8.1 rule on synthetic ties and near-ties before outcome release; classify unstable rankings under Section 24.2.

The rival family must be able to win in practice. Heterogeneous baselines, asymmetric gains, event classes, treatment, and observation error belong in the prespecified adaptation family. Only a strong ordinary account makes the contest meaningful.

Imagine a prespecified adaptation model that outpredicts a more elaborate PTVC candidate in a protected cohort and stays calibrated across event classes. Protocol-fixed loss, complexity allowance, sensitivity plan, and held-out prediction rule determine the consequence, not which account inspires more. If no distinctive transported restriction remains, the PTVC-specific component narrows.

Conditions for rival victory belong in the estimand and scoring plan before outcomes are seen. State, event, window, channel, baseline, missingness process, and loss function remain common across models. Otherwise victory is vulnerable to a quiet change in what the analysis was asked to estimate. The scoring plan should also fix how complexity and calibration are penalized, so flexibility cannot be converted into an unpriced advantage after fitting.

The victory rule should be executable from the protocol alone. It identifies the common estimand, candidate and rival families, tuning budget, complexity adjustment, protected split,

uncertainty rule, and score margin. A tie or unstable advantage remains Model-ambiguous; a reproducible rival advantage triggers the prospectively specified narrowing or retirement rather than a new post hoc contest (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008). Executable rules should be tested on synthetic ties and near-ties so unstable ranking behavior is known before real outcomes create pressure to choose.

A practical adaptation protocol therefore preserves at least three distinguishable objects: current state or report, the slowly moving reference against which change is appraised, and the cumulative exposure history. The first describes the present observation, the second describes learned normality, and the third preserves what was carried while the baseline moved. Conflating them lets adaptation explain everything and identify nothing. Separating these objects also allows adaptation to be modeled as a change in reference without retroactively erasing cumulative exposure.

This is where the Law of Fairness question becomes precise. It asks not whether people recover, but whether local recovery curves, response shifts, and incomplete adaptation exhaust the stronger pathwise patterns the formal hypothesis requires. Comparing strong rivals identifies the features they explain and the additional structure a PTVC model predicts. The most informative residual is one predicted prospectively by PTVC and missed by well-calibrated adaptation rivals, not one discovered after inspecting their errors.

21.1.5 — State Recovery and Accumulated Experience

Adaptation changes the current state and the path that leads to it. After an event, recovery can coexist with delayed effects, altered opportunities, and changed constraints. Recording both the

returning state and the accumulated signed area makes these features available for direct model comparison.

A complete adaptation model can combine local recovery, persistent path area, changing standards, and context-dependent reporting. Including those features gives the comparison a realistic account of human change.

A student may return to class after sustained bullying and appear behaviorally recovered while avoiding corridors, friendships, or activities associated with the harm. The example tests whether local functioning, felt safety, memory, and cumulative exposure move together. It does not assume hidden damage; it requires the rival to specify which layer it predicts and how the unobserved interval is represented.

A two-panel display can place momentary state above the signed cumulative area generated by that state. The upper curve may return toward baseline while the lower curve retains the area contributed during the disturbance. This visual distinction is the scientific point: local adaptation and terminal closure are different objects even when they share an attractive endpoint picture.

The value of the rival is that it can make predictions before the Law enters the room. A return rate, lag, plateau, or variance pattern can be estimated, simulated, and scored. If that machinery explains the relevant structure, the chapter has learned something important even when the PTVC-specific reading becomes smaller.

When local recovery without terminal closure explains the observations, public language should say exactly that. The result should not be enlarged into a universal claim, softened into covert support, or used to reinterpret accumulated burden. Naming the ordinary mechanism precisely is part of scientific honesty.

The decisive test freezes an adaptation family on training events and asks it to predict both later state trajectories and the accumulated path proxy in protected events. Baselines, return rates, asymmetry, response shift, treatment, and observation error must be estimated without terminal leakage. When local adaptation explains the held-out structure and no additional PTVC restriction survives, the candidate interpretation narrows to the residual it can actually defend (Brickman et al., 1978; Diener et al., 2006; Lucas, 2007a; Wilson & Gilbert, 2008).

21.2 — Opponent Processes and Local Counterweights

Opponent-process theory specifies a delayed counterresponse: flatness after pleasure, relief after danger, tolerance with repetition, or an acquired opposite state during withdrawal. Its event-specific lags and gains must be estimated (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

21.2.1 — Primary and Opponent Responses

Estimate primary rise, opponent onset, and repetition effects separately across physiology, report, behavior, and memory, at a cadence that resolves their timing.

A simple response model makes the area question explicit. Let

$$h(t) = Ae^{-at} - Be^{-bt}, \qquad A, B, a, b > 0, \quad t \geq 0.$$

The integrated response is

$$\int_0^\infty h(t)\,dt = \frac{A}{a} - \frac{B}{b}.$$

The zero-area case therefore imposes the relation $A/a = B/b$. Amplitudes and time constants can be estimated from the response profile and used to compare this constrained case with an unconstrained two-component model. Repeating that comparison across settings turns an apparent counterweight into a quantitative question about response shape, duration, and shared parameters.

For the declared response $h(t)$ and a finite observation horizon $H > 0$,

$$I(H) = \int_0^H h(t)\,dt = \frac{A}{a}\left(1 - e^{-aH}\right) - \frac{B}{b}\left(1 - e^{-bH}\right).$$

Conditional on this response model, the unobserved tail obeys

$$|I(\infty)-I(H)| \le \frac{A}{a}e^{-aH}+\frac{B}{b}e^{-bH}.$$

Propagate parameter uncertainty and model error through both quantities. A nonzero finite-window area can remain compatible with a zero complete-response area. Section 21.2.1 supplies a stable state-output realization with that behavior.

Local counterresponses are measured phenomena. In three human psychophysical studies, Leknes et al. (2008) found that relief depended on preceding pain and its decline, with cooling increasing the intensity, pleasantness, and duration of relief in the tested conditions. Franklin et al. (2013), studying forty participants, used separate startle-response indicators to show that pain offset could both increase positive affect and reduce negative affect. The timing and intensity conditions mattered. Together, the studies justify measuring relief in its own right, with positive and negative channels distinguished.

A later study of 122 healthy adults measured pleasant relief for four minutes after an aversive stimulus ended. Relief differed across groups identified from their preceding pain trajectories, making person-specific response dynamics empirically tractable (Henri et al., 2022). These timing and individual-difference findings provide concrete targets for comparisons using existing data and observations during timely pain relief.

The equal-area condition above can arise structurally in a specified model. In dimensionless time, consider two stable states with x(0)=1, y(0)=0 and a,b,c>0:

$$\dot{x} = -ax, \qquad \dot{y} = x - by, \qquad r = c\dot{y}.$$

For a≠b their response is

$$y(t) = \frac{e^{-at}-e^{-bt}}{b-a}, \qquad r(t) = c\frac{be^{-bt}-ae^{-at}}{b-a}.$$

Both states return to zero, so $\int_0^\infty r(t)dt=c[y(\infty)-y(0)]=0$ for every positive a and b. When a=b, $y(t)=t\,e^{-at}$ and $r(t)=c(1-at)e^{-at}$. The amplitudes and time constants satisfy the area constraint because of the stated state-output relation, not a second terminal fit. For a=1, b=2, c=1, $r(t)=2e^{-2t}-e^{-t}$ changes sign at log 2; its positive and negative areas are both 1/4. Its finite-window integral $e^{-t}-e^{-2t}$ remains nonzero for every $t>0$.

This is a local asymptotic construction, not a finite-endpoint phenomenal model. Empirical relevance requires independent support for the state-output relation $r=c\dot{y}$; differentiating or recentering an observed record cannot supply that support. The construction adds predictions about rate, state, and remaining response that can be compared with the unconstrained kernels above. Sampling should resolve their rise and delayed reversal, with channels selected for identification rather than redundant coverage.

A wider input test follows by integrating a stable linear model. Let $\dot{z} = Az + Bu$ and $r = Cz + Du$, with one input and one output, A Hurwitz, $z(0) = 0$, and an ordinary absolutely integrable input u. The full response is absolutely integrable, $z(t) \to 0$, and

$$\int_0^\infty r(t)\,dt = G(0)\int_0^\infty u(t)\,dt, \qquad G(0) = D - CA^{-1}B.$$

Thus zero response area for every such input is equivalent to $G(0) = 0$, the zero-frequency restriction familiar from linear adaptation theory (Sontag, 2003, Section 1.1).

Drive the preceding two-state model from $x(0) = y(0) = 0$ by replacing its first equation with $\dot{x} = -ax + u$. The output $r = c\dot{y}$ has zero area for every absolutely integrable input.

A nearby model, $r_\delta = c\dot{y} + \delta y$, instead gives

$$\int_0^\infty r_\delta(t)\,dt = \frac{\delta}{ab}\int_0^\infty u(t)\,dt.$$

For $a = 1$, $b = 2$, and $\delta = 0.1$, any unit-area pulse therefore produces total area 0.05 under the nearby model and zero under the structural constraint. Changing pulse width and amplitude supplies protected predictions using the same fitted coefficients.

The full response tail matters. At a finite horizon H, or with a nonzero initial state, the exact identity is

$$\int_0^H r(t)\,dt = CA^{-1}\left[z(H) - z(0)\right] + G(0)\int_0^H u(t)\,dt.$$

The boundary term must therefore be measured or bounded. Matching state boundaries removes it; output recovery alone does not. This test extends the local response comparison without converting an asymptotic result into finite-endpoint closure.

A complementary test separates present output from retained state. In a dimensionless linear family, let

$$\dot{m} = v - \lambda m, \qquad \dot{v} = -av - bm + u, \qquad a, b > 0, \quad \lambda \geq 0.$$

The characteristic polynomial is $s^2 + (a+\lambda)s + a\lambda + b$, so the system is stable. With no input after a declared cut and state m_0, v_0 at that cut, integrating both equations gives the full remaining response

$$R_\lambda = \int_0^\infty v(t)\,dt = \frac{\lambda v_0 - b m_0}{a\lambda + b}.$$

For $\lambda = 0$, the remainder is $-m_0$. Positive leakage instead predicts a smaller compensation when $v_0 = 0$; returning both states to zero no longer guarantees zero accumulated response. The state m must be nominated and calibrated independently of the response integral being predicted.

For $a = 2, b = 1$, two histories leaving $v_0 = 0$ but $m_0 = +1$ or -1 predict remaining areas -1 or $+1$ without leakage, and $-5/7$

or $+5/7$ when $\lambda = 0.2$. An output-only decay model $\dot{v} = -2v$ predicts zero from either cut. In the zero-leak case,

$$m(t) = m_0(1+t)e^{-t}, \qquad v(t) = -m_0 t e^{-t}, \qquad \int_0^H v(t)\,dt = -m_0\left[1-(1+H)e^{-H}\right].$$

At $H = 5$, the predicted finite-window areas are approximately -0.959572 and $+0.959572$. These synthetic values distinguish the models before future observations are used. Fit the state calibration, coefficients and leak on development data, then predict protected trajectories and finite-window areas after histories with similar present output but different measured state. A history effect alone does not distinguish this cumulative-state hypothesis from learned expectations or reference adaptation; those rivals require the same historical information.

Fit a frozen impulse-response model across exposures to recover primary amplitude, opponent lag, rebound, and dose dependence through the actual observation process. Predict a protected sequence; a stable delayed counterresponse favors the opponent account, while unrecoverable or inseparable kernels leave the mechanism unresolved (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

A complementary circuit result comes from rats with postsurgical pain. Local-anesthetic blockade of input from the injury produced preference for the associated location, activated ventral tegmental dopamine cells, and increased dopamine release in the nucleus accumbens; blocking accumbens dopamine signaling prevented the place preference (Navratilova et al., 2012). This connects relief to a causal reinforcement pathway and gives the human findings a mechanistic comparison. Temporally defined relief, behavior, and circuit activity can therefore be studied together.

Safety can acquire a learned biological value that sustains future action. Macdonald et al. (2026) identified a pathway from

the paraventricular thalamus to the nucleus accumbens that developed a response when mice successfully avoided threat. The signal weakened when the action's safety value was reduced. Silencing the pathway at safety onset reduced subsequent avoidance and willingness to work for safety; computational fits attributed the behavioral change primarily to lower estimated safety value. The researchers also identified plasticity at connections to accumbens cholinergic interneurons that helped regulate dopamine release. This provides a concrete mechanism through which an improved situation gains motivational value. It gives the book's relief models measurable components: learned safety, outcome valuation, circuit plasticity, and the timing of future action.

21.2.2 — Habituation, Tolerance, and Withdrawal

Repeated exposure can attenuate a response while absence of the familiar stimulus becomes a disturbance. Habituation, tolerance, and withdrawal models predict history-dependent relief, craving, numbness, agitation, and return toward ordinary functioning (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

Satiation supplies a useful comparison between prior experience and present access to it. Galak, Redden, and Kruger (2009) found in small music and food experiments that recalling prior variety could increase later enjoyment of a repeatedly consumed stimulus. This is a lead for replication: a reminder changes what is accessible now without adding the earlier consumption again.

A prospective study could randomize benign exposure histories and then, within each history, assign a truthful reminder or a matched control task. Record expected enjoyment, enjoyment during renewed exposure, retrospective enjoyment, and voluntary choice separately. At a fixed exposure history, a

reminder effect on current enjoyment would challenge a model whose inputs are only exposure count, elapsed time, and measured bodily state. An effect confined to retrospective ratings would instead locate the observed change in that reporting channel. Memory-sensitive valuation and reference-adaptation models should receive the same histories and reminder information as the cumulative candidate; the latter must specify an additional prediction and test it on new participants or stimulus classes.

Recalling an event cannot retroactively change its already-lived contribution. It can change current feeling, later action, or reporting. The experiment therefore tests a specified pathway from remembered history to subsequent experience; it does not identify a conserved phenomenal total.

21.2.3 — Timescale Dependence

Specify the rival's clock: counterresponses can span seconds, days, seasons, treatment cycles, grief, or years. A daily-report fit need not predict minute-level physiology, and cadence can obscure a terminal pattern (Solomon & Corbit, 1974).

Fast regulation and slower changes in affect can be studied within the same event sequence. In an observational study of 821 students, D’Ottone et al. (2026) averaged emotional responses sampled every 45 minutes during the eight hours after exam-grade feedback and sampled later affect every other day over two weeks. Initial emotional responses predicted subsequent shifts in affect, with associations changing over time. The joint modeling also implicated the grade itself, making the event’s outcome and the initial emotional response distinct candidate predictors rather than treating later affect as a single decaying response.

Their public data and code offer a concrete starting point: reproduce the reported analyses, then compare a prespecified single-state decay model with a model retaining outcome, expectation, and initial emotional response. Score both on the same future observations, withholding complete participants or examination periods rather than randomly withholding rows. Because the published data have already informed model construction, confirmation would require a new cohort or a separately controlled untouched evaluation sample with the models and sampling windows fixed in advance. This would test whether event history contributes predictive information across timescales and identify candidate state variables for the additional cumulative-balance test.

Compare prespecified resolutions and aliasing: hourly samples can hide minute-scale overshoot, while dense sampling can emphasize transients outside the proposed kernel.

Localize disagreement to the response kernel, sampling grid, lag distribution, or cross-timescale transport criterion; identify the additional observation window that would separate the models (Solomon & Corbit, 1974; Koob & Le Moal, 2008).

21.2.4 — Event-Specific Versus Global Processes

Declare whether the process explains a discrete disturbance, recurring context, or global tendency. Adaptation to commuting, side effects, or social cues can coexist with persistent grief, coercion, illness, or fear (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

Use hierarchical kernels and leave-one-event-class-out prediction. If local kernels transport but the constrained shared component fails, retain the local mechanisms and retire the global counterweight claim (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

21.2.5 — Simulation and Discriminating Predictions

Generate repeated exposures varying primary amplitude, opponent lag, tolerance, withdrawal, noise, and missingness. Fit blinded rival families through near-miss settings to test actual discrimination (Wilson & Collins, 2019; Talts et al., 2018).

Report calibration, family confusion, and abstention rates by regime (Solomon & Corbit, 1974; Koob & Le Moal, 2001).

21.2.6 — Rival-Unique Predictions

Opponent-process models predict overshoot, aftereffects, sensitization, tolerance, withdrawal, and lags. Their distinctive patterns can be tested at the relevant timescale with negative controls and protected data (Koob & Le Moal, 2001; Solomon & Corbit, 1974).

A rival-unique prediction must survive heterogeneity rather than depend on an idealized average. Event class, channel, exposure history, treatment, and baseline can alter both the magnitude and timing of an opponent response. The candidate Law should face the strongest version of that prediction before any residual pattern is called distinctive.

Consider a delayed aftereffect that grows across repeated exposures. Sensitization, tolerance, anticipatory learning, and changes in reporting can all produce such growth, but they imply different relations among dose, latency, and recovery. A rival-unique prediction must therefore specify which relation should change first and how that prediction will be scored on exposures not used to tune the model.

Rival-unique predictions need a cadence matched to the proposed delay, rebound, and dose history rather than an ever-denser record by default. The protocol should specify which channel is expected to move first, how long the counterresponse should persist, and what pattern would be absent under a simple

adaptation model. A record that does not resolve those temporal contrasts cannot be treated as a test of opponent dynamics.

Before protected data are examined, specify the expected channel order, lag, rebound, repetition effect, and decay after exposure. Evaluate that joint temporal pattern out of sample against calibrated alternatives with the same information, then report which discriminator succeeded, remained unresolved, or failed. Any broader conclusion remains limited to the predictions entailed by the tested branch (Solomon & Corbit, 1974; Koob & Le Moal, 2001; Koob & Le Moal, 2008). Estimate gain and lag separately: the same peak can arise from different trajectories.

Opponent-process studies estimate two time constants: one for the primary response and one for the delayed counterresponse. Both can change with repetition, so habituation and a growing delayed component can make the same event feel different across exposures. Recording exposure history connects those changes to the observed response. This calls for observations before, during, and after the primary response rather than a single post-event rating.

Opponent responses can change how repeated pleasure, relief, or absence is experienced. Their timing and persistence make aftereffects useful targets for clinical observation and model comparison. Follow-up after exposure ends can distinguish a persistent counterresponse from a brief change in reported intensity.

21.3 — Homeostasis, Allostasis, and Mean Reversion

Homeostasis regulates current bodily conditions. Allostasis can also anticipate future demands. Temperature, glucose, sleep pressure, arousal, inflammation, pain, and threat supply candidate variables. Stability, recovery, and changed variance can arise from viability control (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022; McEwen & Wingfield, 2003).

21.3.1 — Homeostatic Set-Point Control

Set-point control specifies feedback, recovery rates, and changing references for temperature, glucose, sleep pressure, pain avoidance, or autonomic state (McEwen, 1998).

Regulation has also been studied directly in everyday affect. Smartphone observations from more than 28,000 people showed a state-dependent activity pattern: lower mood preceded mood-enhancing activities, while better mood preceded useful activities with less immediately pleasant consequences (Taquet et al., 2016). A later analysis examined 28,212 smartphone participants and 30,116 participants in six-country population data, finding weaker mood-homeostasis patterns with low mood and a history of depression (Taquet et al., 2020). The later report extends the smartphone analysis with population data; the smartphone participants overlap across reports. These observational findings reveal a measurable feedback relation between feeling and activity selection, with substantial variation in how it operates.

Breathing support, medication, and social reassurance can help panic settle through partly different channels. A homeostatic account can predict their timing and interactions, and additional route models can specify further relationships for comparison.

Diagram regulated variable, reference, sensor, comparator, action, delay, and saturation, separating latent-state, report, and physiological feedback paths.

The comparison can begin from measured affect dynamics. Kuppens et al. (2010) used two experience-sampling studies to examine a diffusion model with an affective homebase, variability, and attractor strength. Separately estimating these features allows researchers to compare models of current state, ordinary return dynamics, and longer-term history while preserving person-specific baselines and observation error.

Research records should include the effort, pain, environmental conditions, and treatment that accompany stable function. These contextual variables help explain how regulation operates within a person's circumstances (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

Estimate feedback gain, delay, saturation, and reference movement under permissible perturbations or natural changes; predict recovery and overshoot in protected sequences with uncertainty (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

The biological feasibility of precise regulation has direct experimental support. In bacterial chemotaxis, adaptation accuracy can remain robust while pathway protein levels change; integral-feedback analysis explains how the network retains this property under specified biochemical conditions (Alon et al., 1999; Yi et al., 2000). Aoki et al. (2019) went further by engineering an integral controller in living cells and demonstrating robust growth-rate regulation in E. coli. These findings make cumulative error integration a concrete biological capability. They give PTVC a productive mechanistic starting point: identify a carrier of cumulative information and investigate its relationship to experienced valence.

Natural cellular regulation also supplies an empirical example. Muzzey and colleagues (2009) combined single-cell measurements of volume and Hog1 signaling with assays of glycerol accumulation during changes in external osmolarity. Their perturbations and systems analysis supported an integral-feedback account and helped locate the feedback mechanism in the network regulating glycerol synthesis. The study demonstrates how measurements of several linked quantities and responses to different input shapes can reveal a history-sensitive controller in a living system.

An integral controller also makes signed response area directly analyzable. Let its internal state obey $dz/dt = k_I(r - y)$, with constant reference r, constant positive integration gain k_I, and an integrable output y. Integrating between a and b gives

$$\int_a^b (y - r)\,dt = -\frac{z(b) - z(a)}{k_I}.$$

Output recovery, $y \to r$, can coexist with a retained integral-state change. Zero signed response area occurs exactly when $z(b) = z(a)$. Measuring both the output and the internal state under safe perturbations would distinguish these dynamical outcomes (Yi et al., 2000).

The choice of state coordinates also matters. In the zero-leak $a = 2, b = 1$ example of Section 21.2.1, the same dynamics can be written as a cascade:

$$\dot{x} = -x + u, \qquad \dot{z} = -z + x, \qquad v = x - z.$$

The change of coordinates $m = z$ and $x = v + m$ gives $\dot{m} = v$ and $\dot{v} = -2v - m + u$ exactly, including corresponding initial states. Both descriptions have transfer function $s/(s+1)^2$. Observing the input and output, even perfectly, therefore cannot choose between these coordinate descriptions.

A physical mechanism claim must identify the measured components and what a permitted intervention actually changes. Removing a coefficient after relabeling latent coordinates does not define the same physical intervention. Freeze the state readout on separate calibration data and test its prospective contribution beyond current response, expectations and reference adaptation. A physically separate assay trained on the same reports does not by itself provide an independent phenomenal calibration.

The feasibility case extends to stochastic biological systems. Antithetic integral feedback provides a reaction architecture for robust mean regulation under its stability and ergodicity conditions (Briat et al., 2016). Frei et al. (2022) implemented a related circuit in mammalian cells, maintaining tunable transcription-factor expression despite increased degradation or changed feedback structure. Additional proportional feedback reduced variability while retaining adaptation. Noise and biological complexity can thus be incorporated into experimental studies of precise regulation.

An exact molecular count identity and a mean regulation identity make different predictions. In a nonexplosive reaction model with one event at a time and the required compensator integrability, consider two counts Z_1, Z_2 whose births have predictable intensities μ and θY_s and whose paired annihilation removes one of each.

For $D = Z_1 - Z_2$, annihilation cancels exactly, while compensated birth counts leave the reaction martingale M:

$$D_t - D_0 = \int_0^t (\mu - \theta Y_s)\, ds + M_t.$$

A return of D therefore need not make the integrated rate error zero on that realization. This follows directly by subtracting the

actual birth counts and their compensators; replacing them by their means discards the remaining fluctuation. Such reactions provide a concrete architecture to investigate, with mean regulation, event accounting and experiential accumulation measured separately.

A fixed potential of a locally finite pure-jump state is constant between jumps. If its change equals minus an absolutely continuous ledger, continuity also forces it to remain unchanged at every jump, leaving no nontrivial accumulation on that branch. A continuous integrating coordinate, explicit time dependence or an independently justified atomic contribution measure changes the model and requires its own bridge.

The same bookkeeping supplies a quantitative precision test. For a fixed molecular coordinate $q(X)$, let reaction j change q by d_j at intensity a_j. Its drift b, martingale variance rate c and coordinate-changing activity A satisfy

$$b = \sum_j a_j d_j, \qquad c = \sum_j a_j d_j^2, \qquad A = \sum_{j:d_j \neq 0} a_j, \qquad b^2 \leq Ac.$$

The inequality is weighted Cauchy–Schwarz. At a measured upper activity bound, a specified drift therefore demands a minimum innovation variance. This is a restriction on reaction noise, not a lower bound on final variance under feedback. A molecular candidate must show which events write its record, which preserve it, and how its fluctuations relate to independently calibrated experience.

Complex regulation can also improve when complementary mechanisms work together. Hu and Murray (2022) engineered layered feedback in E. coli and found improved joint robustness and response speed under chemical, temperature and nutrient disturbances compared with the component single-layer designs. This gives a concrete reason to investigate coordinated routes.

A model linking independently measured layers could predict signed-area recovery, timing and uncertainty across new conditions, with the layers and their coupling specified before the test.

Precision and resource availability can be studied together. Lan et al. (2012) linked adaptation accuracy, response speed and energy dissipation in a sensory-feedback model and tested its implications in E. coli. As cells became less energized, adaptation slowed while accuracy was maintained. This finding motivates joint measurement of reserve-related variables, response times and error. Their coupling offers a quantitative target for an affective-regulation model that also includes a calibrated cumulative valence measure.

A recurring response shape becomes more informative when it survives a specified transformation. Fold-change detection predicts that multiplying a sensory input history and its background by the same factor preserves the complete output response (Shoval et al., 2010). Bacterial experiments tested such response rescaling at pathway and population-behavior levels (Lazova et al., 2011). This supplies a useful design precedent: state the transformation, predict the entire dynamical relation, and test its transfer to new conditions without refitting.

Recent control research also makes the variability question more concrete. Lim and colleagues developed a combined molecular mean-and-noise controller that preserves specified stationary first and second moments under ergodicity and finite-second-moment conditions. Their analysis and simulations extend mean adaptation toward control of output dispersion, with openly archived code for replication (Lim et al., 2026). The study’s calibrated simulation of an Escherichia coli DNA-repair system illustrates an application of the controller.

Use mean-and-variance controllers as executable rivals, and test whether the candidate transports additional joint predictions beyond both forms of regulation.

21.3.2 — Allostatic Prediction and Load

Allostatic models allow reference and control changes before expected demand; threat, poverty, caregiving, illness, discrimination, or unstable work can maintain costly preparation despite steady behavior (Sterling, 2012; Bobba-Alves et al., 2022).

A caregiver may sleep lightly for months because a fall, seizure, or call for help could occur at night. Daytime reports can look stable while anticipatory regulation remains costly. The case gives allostasis an observable structure: expected demand, altered sleep, physiological preparation, and recovery after the caregiving contingency changes.

Forecast responses to anticipated demand, separating current state, expectation, cumulative load, treatment, and observation. Test transport after demand changes; an adequate allostatic account removes distinctive credit from the same pre-event signature (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.3 — Viability Without Fairness

Viability models explain how life, function, or action are maintained under constraint. Measuring the accompanying experience allows physiological continuation and felt welfare to be studied together (Bobba-Alves et al., 2022).

An older patient may reduce movement to avoid falls while losing access to valued activities. A useful model represents both the protective effect and changes in loneliness, autonomy,

and opportunity, helping identify supports that improve the overall situation.

Jointly model functional reserve, task completion, pain, fatigue, and valence-related observations. Test whether actions preserve performance despite adverse experience and whether that relationship predicts new contexts (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.4 — Frailty, Disease, and Care Effects

Diminished reserve, fatigue, inflammation, medication, delirium, care transitions, and reduced expression can narrow terminal state or observation. Compare them through a joint clinical-observation model (McEwen, 1998; Sterling, 2012).

Recover nonlinear clinical decline, treatment response, competing events, and state-dependent observation jointly, distinguishing decline from device loss or reduced reporting before terminal-window use. If these processes explain contraction, withdraw its distinctive PTVC attribution in that cohort (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

Model medication and care effects on report, physiology, behavior, and sampling that can mimic compression or rebound (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

Use the medication and observation records specified in Section 20.5.1, adding the care goal and placing all events on a common timeline for comparison with terminal-dynamics rivals.

Randomized variation, natural changes, or target-trial methods can then estimate which divergence occurs first. Causal attribution to treatment requires the identification assumptions of the chosen design; earlier timing alone is insufficient (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

21.3.5 — Generic Regulation as Default Rival

Living systems stabilize many variables, making homeostasis, allostasis, frailty, treatment effects, mean reversion, and their combinations useful comparison models under shared observations (Bobba-Alves et al., 2022; McEwen, 1998).

The rival must be strong enough to represent expert expectations. A flexible state-space family can include person-specific baselines, nonlinear return, plateaus, latent states, observation error, treatment effects, and changing control regimes. It receives the same information and tuning budget as the PTVC candidate and is judged on model recovery, calibration, held-out prediction, intervention response, and transport. Model discrimination requires a controller that adequately represents those mechanisms.

Data collection should resolve the rival's actual state and control variables rather than demand a maximal dossier. If return rate, set-point movement, intervention timing, or observation error cannot be estimated without burdensome or speculative measurement, the comparison narrows instead of expanding collection without limit.

The ordinary model must include clinic, household, and care context. Oxygen, opioids, inflammation, delirium, disease stage, caregiver anticipation, and observation-channel changes can move a terminal curve toward quiet without implying a fairness constraint. Allostasis makes the contest harder by allowing anticipated demand to shift the operating range around vigilance or safety (McEwen, 1998; Sterling, 2012; Bobba-Alves et al., 2022).

When generic regulation reproduces held-out state trajectories and intervention responses, those results support the regulatory model for the tested components. The accumulated-path and endpoint targets remain separate questions for the comparison,

with their own observations and prospectively specified predictions.

A mainstream control victory classifies a model-data relation, not a life. PTVC remains distinctive only where a prospectively specified joint restriction survives after the strongest ordinary regulatory family has been fully spent.

21.4 — Ornstein–Uhlenbeck State Dynamics

OU specifies reference-directed drift, stochastic innovation, an initial law, and a clock, enabling exact predictions under irregular sampling, observation error, and hierarchical variation (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Särkkä, 2013).

21.4.1 — OU State Equation

Use nonlinear, heteroskedastic, switching, person-specific, treatment-dependent, or partially observed extensions where the Gaussian linear OU baseline fails.

Equation 21.1: OU state equation as rival generator

$$dX_t = -\kappa X_t dt + \sigma dW_t, \kappa > 0, \sigma > 0$$

Status: Rival state-dynamics model under the stated drift and diffusion assumptions.

Equation 21.2: OU conditional mean

$$E\left[X_t | X_0 = x_0\right] = x_0 e^{-\kappa t}$$

Status: Conditional state mean under Equation 21.1.

For a centered post-event series, test whether a prospective alternative predicts later features systematically missed by OU return, persistence, and fluctuations.

Fit the continuous-time state-space likelihood using actual intervals and measurement error. Test drift, diffusion, autocovariance, and residuals in protected data; failure of linear reversion or constant diffusion rejects that specification (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Gardiner, 2009).

21.4.2 — State Mean Reversion

For the OU state model, expected return depends on elapsed time:

$$E[X_{t+h} - \mu \mid X_t = x] = (x - \mu) exp(-\kappa h).$$

Regression to the mean can occur with a fixed latent state. As a contrast, draw a latent value θ once from a Gaussian distribution with mean μ and variance $s^2>0$, then hold it unchanged across two readings: $Y_1=\theta+\varepsilon_1$ and $Y_2=\theta+\varepsilon_2$. Let the errors be independent mean-zero Gaussian variables with variance $v>0$, independent of θ. Then

$$E[Y_2 \mid Y_1 = y] = \mu + \frac{s^2}{s^2+v}(y-\mu).$$

An extreme first reading therefore predicts a less extreme second reading even though θ has not changed. Fit the observation model and the latent dynamics separately; repeated measurements and variation in lag can help distinguish these explanations under their declared assumptions (Fuller, 1987; Uhlenbeck & Ornstein, 1930).

Test conditional increments against current deviation across horizons after modeling error and timing. Unstable drift sign or magnitude rejects simple OU; specify and evaluate the richer rival (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.3 — Stationary and Nonstationary Starts

Fix the start law before comparison: equilibrium and post-shock, treatment, diagnosis, bereavement, or illness starts generate different transients (Uhlenbeck & Ornstein, 1930).

Identical drift and diffusion with stationary versus displaced starts can produce different early recovery-like trajectories; test whether a claimed distinction survives that initial-law difference.

Plot stationary and displaced starts under common OU parameters, separating transient behavior, long-run variance, and observation uncertainty.

21.4.4 — Time-Varying and Hierarchical OU

Allow identifiable between-person and setting variation in drift, diffusion, reference, start, observation error, sampling, symptom regime, and care response. Qualify that flexibility by recovery, complexity-adjusted protected prediction, and interpretable parameters (Øksendal, 2003; Uhlenbeck & Ornstein, 1930).

After a stroke, a person's apparent return rate may change across rehabilitation phases as treatment intensity, fatigue, and motor access evolve. A single fixed OU coefficient can misread that sequence as noise or stable mean reversion.

Generate person-level parameters from a population distribution, reproduce observation schedules, and recover both hierarchy levels. Evaluate calibrated prediction in held-out persons and sites (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.5 — Observation Error and Irregular Sampling

Sparse reports, missed prompts, device dropout, clinic-only sampling, and terminal low-signal windows can mimic mean reversion or hide change. Compare models through the same observation process (Uhlenbeck & Ornstein, 1930).

Distress-related missed prompts and restlessness-related dropout can make observations appear calmer than the latent process. Jointly model response and device availability with state, then test the signature under informative-missingness scenarios.

Simulate the fitted continuous-time process on actual visit and device schedules, then refit through its observation model. Report coverage, bias, and selection error as cadence and noise worsen; loss of advantage under the actual schedule withholds latent-dynamics attribution (Uhlenbeck & Ornstein, 1930; Øksendal, 2003).

21.4.6 — OU as a Model That Can Win

Fix the OU family, target, horizon, observation and missingness laws, complexity allowance, and victory threshold. Person-specific, hierarchical, or switching extensions must be recoverable and prespecified.

Require parameter recovery, reproduced drift and covariance, calibrated uncertainty, protected prediction, and transport before applying any adequacy-and-simplicity criterion.

Comparable prediction and the absence of an additional transported restriction do not by themselves establish OU victory. Apply the prespecified advantage or separately declared adequacy-and-simplicity rule in Section 22.8.1.

21.5 — The Integrated OU Ledger

The integrated OU ledger is the chapter's mathematical hinge. If X_t is a mean-reverting state, A_t is its time integral and therefore a different stochastic object: state return can coexist with path-dependent accumulation and long-horizon variance growth. The calculation supplies a baseline for testing distinctive predictive claims; it is not a phenomenal conclusion by itself (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Gillespie, 1996; Särkkä, 2013).

21.5.1 — Closed-Form Variance

The derivation uses zero-mean, zero-start OU in continuous time without measurement error. Empirical tests must specify units, clock, centering, initial-state uncertainty, sampling, unobserved intervals, observation error, treatment, heterogeneity, and misspecification.

Equation 21.3: Integrated OU state

$$A_t = \int_0^t X_s ds$$

Status: Definition of the accumulated state in the OU comparison model.

Equation 21.4: Variance of the integrated OU state

$$Var\left(A_t\right) = \frac{\sigma^2}{2\kappa^3}\left(2\kappa t - 3 + 4e^{-\kappa t} - e^{-2\kappa t}\right)$$

Status: Variance consequence of the zero-start OU model under the stated assumptions.

Equation 21.5: Long-time integrated OU approximation

$$Var\left(A_t\right) \sim \frac{\sigma^2}{\kappa^2} t \text{ as } t \to \infty$$

Status: Long-time variance behavior of the integrated OU model.

Equation 21.6: Expected integrated OU state from fixed start

$$E\left[A_t | X_0 = x_0\right] = \frac{x_0}{\kappa}\left(1 - e^{-\kappa t}\right)$$

Status: Equation 21.6 is a fixed-start OU consequence. It shows that an initial deviation contributes to the integrated path even while the state mean decays.

Use independently estimated OU parameters to predict integrated variance and covariance across horizons, including the observation model and simultaneous bands. Systematic failure rejects that regime; agreement removes distinctive credit for the pattern (Uhlenbeck & Ornstein, 1930; Øksendal, 2003; Gardiner, 2009).

21.5.2 — Asymptotic Linear Growth

The OU result illustrates a general distinction between recovering a state and cancelling its accumulated path. For real, centered, covariance-stationary equal-exposure increments X_k with finite second moments, write $A_n = \sum_{k=1}^{n} X_k$ and $\gamma_h = Cov(X_0, X_h)$. Counting covariance pairs gives

$$V_n := Var(A_n) = n\gamma_0 + 2\sum_{h=1}^{n-1}(n-h)\gamma_h.$$

If $\sum_{h\in\mathbb{Z}} |\gamma_h| < \infty$, division by n and dominated convergence give $V_n/n \to \ell = \sum_{h\in\mathbb{Z}} \gamma_h = 2\pi f(0)$, using $f(\omega) = (2\pi)^{-1}\sum_h \gamma_h e^{-ih\omega}$. Positive ℓ yields growing cumulative dispersion even though $E[(A_n/n)^2] \to 0$.

For a Gaussian process and fixed $\varepsilon > 0$,

$$\Pr(|A_n| \le \varepsilon) = 2\Phi(\frac{\varepsilon}{\sqrt{V_n}}) - 1 \sim \sqrt{\frac{2}{\pi}}\frac{\varepsilon}{\sqrt{n\ell}}.$$

A non-Gaussian central limit theorem with positive limiting variance also makes the fixed-band probability tend to zero, but does not by itself provide this rate. Summable covariances alone do not establish that central limit theorem. These stationary-process comparisons extend the OU example to wider accumulation architectures.

For stationary AR(1) increments $X_k = \phi X_{k-1} + e_k$, $|\phi| < 1$, with independent centered innovations of variance q, $\ell = q/(1-\phi)^2$. In contrast, $X_k = e_k - e_{k-1}$ gives $A_n = e_n - e_0$ and $V_n = 2q$. This ordinary cancellation mechanism has bounded cumulative variance while generally retaining a nonzero terminal residual. More generally, an absolutely summable linear filter has $\ell = q(\sum_j h_j)^2$, so zero-frequency cancellation requires its net response to vanish.

For a real centered jointly measurable stationary continuous-time process with finite second moments and covariance $C(u)$, $Var(\int_0^T X_t\,dt) = 2\int_0^T (T-u)C(u)\,du$. Absolute integrability of C gives the corresponding variance-per-time limit. Keep stationary and zero-start initial laws distinct. Centering a rival mathematically does not identify phenomenal neutrality: subtracting each completed record's own mean forces apparent closure by construction. Compare held-out window-length variance predictions, with dependence, measurement error and unobserved low frequencies included. This can discriminate accumulation mechanisms before complete-stream evidence is available.

21.5.3 — Initial-Condition Variants

Propagate Section 21.4.3's starting-state law into the time integral's finite-window mean and covariance.

With the same OU drift, diffusion, observation law, and horizon, stationary, fixed displaced, post-event, and hierarchical starts can produce different finite-window cumulative means and covariances. The analysis must report which ledger conclusions depend on the start model, first recorded value, or burn-in rule (Uhlenbeck & Ornstein, 1930; Gardiner, 2009; Øksendal, 2003).

21.5.4 — Worked Calculation and Parameter Sweeps

A worked calculation makes the state–integral distinction concrete. Use dimensionless model units, $X_0 = 0$, $\kappa = 1$, and $\sigma = 1$. The OU state variance is $\frac{1-e^{-2t}}{2}$, while Equation 21.4 gives the integrated variance. At $t = 1$, 5, and 10, the state variances are approximately 0.4323, 0.5000, and 0.5000; the corresponding integrated variances are 0.1681, 3.5135, and 8.5001. The state variance settles while the integrated variance continues to grow. With the same dynamics but a fixed start $X_0 = 1$, the expected state at $t = 5$ is only 0.0067, whereas Equation 21.6 gives an expected integral of 0.9933. These calculations follow from the declared OU model (Øksendal, 2003).

An analytic parameter sweep also shows the role of return rate. Keep $X_0 = 0$, $\sigma = 1$, and $t = 5$. Substituting $\kappa = 0.5$, 1, and 2 into Equation 21.4 gives integrated variances of approximately 9.2864, 3.5135, and 1.0625. At fixed κ and t, doubling σ multiplies the variance by four. Starting-state assumptions matter too: with $\kappa = \sigma = 1$ and a stationary start, the integrated variance at $t = 5$ is $5 - 1 + e^{-5}$, approximately 4.0067, rather

than the zero-start value 3.5135. These calculations provide numerical checks for an implementation.

Use the analytic values as simulator checks and Section 25.5.4's second-channel benchmark for a specified observation contrast. For fitted-model recovery, vary return rate, diffusion, initial law, horizon, cadence, and noise on a fixed grid. Report parameter recovery and family confusion under common evaluation, preserving seeds and solver tolerances (Uhlenbeck & Ornstein, 1930; Gardiner, 2009; Øksendal, 2003).

21.5.5 — Consequences for Model Comparison

Test cumulative predictions from independently estimated drift, diffusion, and start on protected horizons under common units, clocks, observation, censoring, and endpoints. Agreement retains OU; systematic departure rejects that specification in the affected regime (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

A candidate can add a prespecified, separately measured reserve relation that jointly predicts drift, innovation, and accumulated behavior, subject to the measurement and endpoint bridges.

The integrated OU derivation gives the comparison a calculable cumulative trajectory, with explicit dependence on drift, diffusion, horizon, and initial conditions (Uhlenbeck & Ornstein, 1930).

The OU baseline quantifies persistence and mean reversion at the measured scale. A candidate route can specify an additional restriction and test it against that baseline.

Mean reversion of the OU state can coexist with a growing variance of its accumulated integral. These quantitative predictions can be compared under the declared measurement and observation models.

Explicit parameter units, observation errors, windows, censoring rules, and endpoint definitions make the OU calculation reproducible and identify the settings in which it informs model comparison.

The calculation earns no PTVC-specific implication merely by being exact. A candidate route must add a prospectively specified restriction, such as independently measured reserve coupled to drift and innovation, endpoint discipline not built into the path law, or transported held-out advantage. Otherwise the integrated OU process remains the ordinary baseline.

The OU derivation can pressure an interpretation without settling the phenomenal target. If the observed path follows the ordinary variance law, the candidate loses a claimed signature; if it does not, the OU specification loses that regime. In neither direction does the equation authorize personal prediction, care delay, or a story about why suffering occurred (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

Reviewers should separate three questions: whether the OU derivation is correct, whether the process approximates the latent state, and whether the observation bridge is adequate at the relevant horizon. A correct theorem can coexist with a rejected empirical model, while an imperfect approximation can remain an informative rival. Reporting these layers separately prevents mathematical correctness from being mistaken for construct validity or measurement adequacy.

The central intuition is concrete. A thermostat can return room temperature to its set point while a meter continues recording how long the room was cold. Likewise, a state can be pulled toward zero while its accumulated area retains history. Present recovery and accumulated exposure are different scientific objects. The same distinction applies when a baseline itself

moves: local restoration can coexist with a cumulative history that remains far from zero.

The integral preserves duration as well as sign and intensity, so many small deviations can accumulate even when the current state appears ordinary. Measuring the accumulated path therefore adds information to the description of state recovery. This makes the integral a direct rival discriminator whenever competing theories predict similar state recovery but different accumulated exposure.

21.6 — Recovery, Hybrid Models, and Protected Prediction

Pass each rival's latent histories through the common observation kernel and apply the protected scoring rule in Section 19.6.1 (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019; Talts et al., 2018).

21.6.1 — Common Synthetic Observations

Use one exogenous event schedule for each benchmark replicate, with each generator producing its own latent trajectory through a common observation and missingness model. All fitted candidates receive that replicate's identical observed data.

Expose generator coverage and observation kernels so challengers can add omitted rivals, harder missingness, or dependence structures at identified points in the benchmark (Talts et al., 2018; Hernán & Robins, 2020).

21.6.2 — Parameter Recovery

An opponent process may resemble adaptation or rebound. Examine lags, overshoot, and repetition effects; where that account succeeds, describe the local counterweight rather than cumulative closure.

Test whether the observation design recovers return rates, opponent lags, loads, and variances, determining which parameter interpretations and sampling decisions it supports (Vehtari et al., 2017; Yao et al., 2018).

Vary return rate, diffusion, opponent lag, error, and missingness; report bias, coverage, multimodality, and nonidentifiability by parameter and regime for every rival.

Report which parameters remain aliased or prior-sensitive (for example, whether cadence identifies return rate but not

diffusion) and which design change would resolve that limit (Talts et al., 2018; Hernán & Robins, 2020).

21.6.3 — Model Recovery

Set-point adaptation can resemble a PTVC candidate or an OU process. Event-specific return and plateau checks determine whether ordinary return explains the data and requires narrower PTVC-specific language.

An allostatic model may resemble homeostasis or the PTVC candidate. Load markers and context shifts test whether regulation under demand is the primary explanation.

When an allostatic model explains stability under chronic demand, describe the load as a cost of regulation. Public interpretation should identify that allostatic burden.

The confusion matrix tests substitutions among set-point return, opponent response, allostatic shift, OU drift, treatment quieting, and terminal concentration (Wilson & Collins, 2019).

Use generating families as rows and selected or abstained outcomes as columns, with separate hybrid and open-set rows and uncertainty intervals for cell rates.

Model recovery fails when confusion with rivals or confident misclassification of open-set worlds exceeds the prespecified class-specific error limits. That failure blocks the corresponding human-data verdict, even if parameter estimates look precise inside the selected model. A later redesign must pass fresh synthetic tests before the component can reenter confirmatory use (Talts et al., 2018; Hernán & Robins, 2020).

21.6.4 — Hybrid Rival Models

The PTVC candidate must distinguish its reserve-coupled joint restriction and independently defined endpoints from OU and

hybrid rivals. It survives this comparison only if recovered under fair rules.

A hybrid ordinary rival combines treatment, frailty, regulation, and report dynamics. Remove PTVC-specific status from patterns it explains and restrict any remaining claim to the unexplained region.

Fit plausible combinations of adaptation, treatment, frailty, allostasis, observation shutdown, and conditioning under common observation and complexity rules; test whether the candidate adds transportable structure.

Target measurements that distinguish adaptation rate, treatment effect, shutdown, and reserve-linked terms. Withhold component attribution when those terms remain aliased under the design.

Vary hybrid component strengths from identifiable to nearly aliased; recover components where information permits and classify the remaining result under Section 24.2 (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

21.6.5 — Held-Out Prediction

Evaluate frozen models using the scoring, calibration, and uncertainty rules in Section 19.6.1; preserve the clinically meaningful errors relevant to the target (Gneiting & Raftery, 2007).

Hold out the intended shift: later recovery, another site, endpoint regime, cadence, or care setting. Assess the frozen model's performance and calibration on that target without post-failure refitting.

When transport fails, identify the shifted measurement, care, event, or context process and narrow the affected claim.

Inspect residuals and investigate leakage, dependence, and shift when advantage occurs only under observation-level random splits (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

21.6.6 — Rival Dominance

An OU rival can be mistaken for the PTVC candidate. Integrated-variance and state-return recovery checks allow ordinary mean reversion to win when it explains the observations.

When adaptation explains local return after events, retire that return as distinctive support for PTVC. Describe the finding as adaptation.

When an opponent process explains lag, overshoot, or withdrawal, interpret rebound as a local counterweight. The finding does not establish a balance of cumulative experience.

When OU dynamics or mean reversion explain state return and variance, a PTVC-specific account needs a distinct cumulative prediction. Without it, withdraw the distinctive claim and report mean reversion.

Rival dominance tells when ordinary mechanisms narrow or retire a PTVC-specific interpretation. It requires reliable held-out results, recoverable parameters, transport, and no hidden ethical or measurement advantage. Then narrowing is scientific progress (Vehtari et al., 2017; Yao et al., 2018).

Competitive implementations govern the contest. Each model gets defensible heterogeneity, observation structure, and tuning freedom before complexity penalties and held-out scoring. A Law-specific route narrows when an ordinary rival wins fairly.

Repeated outperformance by adaptation, allostasis, or an OU process changes the program. The candidate is narrowed, redesigned, or retired for the declared class, with each

consequence tied prospectively to replicated performance rather than answered by new unmeasured structure.

Dominance requires equal information and scoring, not greater surveillance. Every candidate shares the admissible history, missingness, and transport conditions. Unequal context or endpoint access diagnoses the contest, not the mechanism.

The comparison reports predictive accuracy, mechanism identification, calibration, and measurement performance separately.

Dominance is a protocol-fixed scientific relation, never a hierarchy among people. It can retire a mechanism or narrow a class while care, rights, privacy, and participation remain unchanged; the winning model remains correctable (Talts et al., 2018; Hernán & Robins, 2020).

Recovery, calibration, held-out prediction, transport, uncertainty, and multiplicity share one decision table. Mixed results remain component-specific rather than becoming a single winner label.

One shared observation file gives every model the same simulated streams, events, treatments, missingness, censoring, noise, and endpoint uncertainty. No candidate receives latent truth while another receives the messy record.

Recovery rehearses interpretation. A pipeline unable to recover opponent or OU generators from realistic synthetic observations cannot declare those rivals defeated in empirical data. Confusion matrices, calibration checks, and frozen fixtures expose the weakness before participants carry its cost.

21.7 — Empirical Separation and Defeat

Test joint restrictions linking cumulative behavior, independently measured reserve, endpoints, observation robustness, and protected prediction against regulatory models (Talts et al., 2018; Hernán & Robins, 2020; Munafò et al., 2017; Nosek et al., 2018; Wilson & Collins, 2019).

21.7.1 — Reserve-Coupled Joint Restriction

Simulate imperfectly observed reserve acting only through the declared transition law. Test its joint predictions for reserve, state, covariance, action, and endpoint risk against ordinary regulation on held-out modules.

Keep reserve, observed state, latent target, actions, hazard, and measurements distinct in the graph; mark only the candidate-constrained edges and retain ordinary causal paths.

21.7.2 — Prospective Endpoint Independence

Classify adverse results under Section 24.2: distinguish a failed signature, component or reference-class narrowing, and a gate-valid counterresult against the declared branch.

Freeze hazards, features, and splits to decision-time information. If removal of terminal knowledge eliminates a signature, withhold its prospective interpretation; compare conditioning and selection with endpoint coupling through the declared joint model and discriminating evidence (Aalen et al., 2008; Hernán & Robins, 2020).

Keep prospective information discipline separate from statistical endpoint independence, which concerns the endpoint-time law, risk interval, observation, and dependence structure and requires separate identification.

With hazards and features frozen at each decision time, test residual endpoint-risk association with past state, treatment,

observation, and route variables under fixed adjustment. Simulate power against plausible dependence. Coupling changes the branch; unidentified independence cannot be claimed (Talts et al., 2018; Hernán & Robins, 2020).

21.7.3 — Integrated-Ledger Consequence

In caregiving records, separate daily distress, event spikes, sleep disruption, and vigilance by a specified path functional and coverage, testing memory, sampling, and scale explanations.

Match cadence to uncertainty in the path functional, represent unobserved intervals, and report whether the distinction survives coarser sampling.

Gupta and Sontag (2025) show how cumulative output can distinguish specified nonlinear models. Motivated by T-cell measurements, they hold each model's initialization fixed, vary a constant positive input, and integrate output to a common finite horizon. Two specified incoherent-feedforward models give monotone cumulative dose responses, whereas their integral-feedback model can give a nonmonotone response. This is a model-specific discriminator, not a general test of network topology.

The feedback model also makes retained history explicit. In its normalized units, $\dot{x} = x(y - p)$ and $\dot{y} = u/x - y$, with $u, p, x_0 > 0$, $x(0) = x_0$ and $y(0) = p$. Hence $\int_0^T [y(t) - p]\, dt = \log [x(T)/x_0]$. The output returns to p while x approaches u/p, so the complete excess area is $\log [u/(px_0)]$, not generally zero. Here log denotes the natural logarithm.

Measure response area and controller-state change together under common initialization, horizon, and observation. Their experiential interpretation requires independent phenomenal calibration; the physiological reference does not establish neutrality.

Retain adequate integrated rivals and seek a predicted difference in cumulative response or cross-time dependence before identifying a generator (Uhlenbeck & Ornstein, 1930; Gardiner, 2009).

21.7.4 — Observation-Model Robustness

Test separation under alternative missingness, treatment, frailty, drift, reporting, and low-signal assumptions; unstable separation identifies the observation question still to resolve (Talts et al., 2018; Hernán & Robins, 2020).

Implementation: refit the frozen comparison across the prospectively specified channel kernels, calibration maps, error bounds, and missingness laws, then report the region in which model ordering, calibration, and verdict remain stable (Talts et al., 2018; Hernán & Robins, 2020).

21.7.5 — Out-of-Sample Advantage

A claim of predictive superiority requires a prespecified held-out advantage under common information and observation rules, with uncertainty and any complexity penalty reported separately. Scientific value can also come from independently calibrated tests of a cumulative restriction that a broader rival can satisfy. Where candidate models imply the same observation law, identical predictive distributions have identical expected scores. In that case, quantify the declared finite-resolution restriction and the alternatives it excludes, using Section 24.2.4 and Section 24.6.4, while reporting the mechanism comparison as unresolved. Challenge the same frozen relation in new sites or perturbation regimes where the candidate models make different predictions.

Vary the split, dependence structure, score, complexity allowance, and transport population under the sensitivity plan;

show where model ordering changes (Talts et al., 2018; Hernán & Robins, 2020).

21.7.6 — Defeat by Mainstream Explanation

Defeat by mainstream explanation is an honorable scientific outcome. If adaptation, opponent processes, allostasis, OU dynamics, treatment effects, frailty, observation processes, or their hybrids explain the protected patterns better, the PTVC-specific interpretation narrows or retires for that domain. Care remains untouched, and the program retains the measurement, comparison, and clinical knowledge produced along the way (Talts et al., 2018).

Defeat must be operationally possible. The ordinary rival suite receives defensible flexibility and faces the same prospectively fixed state, path functional, endpoint rule, channels, missingness model, information budget, and held-out loss. It wins when it recovers reliably, predicts and calibrates at least as well, transports at least as well, and explains the conjunction more parsimoniously.

The decision retires or narrows the exact component that lost and names the winning family, tested domain, uncertainty, and consequence in the headline-level summary. A new route requires a new prediction and fresh protected evidence rather than relabeling the winner as how fairness acted.

The distinctive burden is conjunctive: independently estimated route variables where relevant, prospective endpoint definition, accumulated-path consequences, resistance to observation alternatives, and out-of-sample advantage after complexity control. A mainstream account that explains the pattern without that conjunction changes the claim’s status.

Defeat propagates through later chapters, summaries, figures, and public artifacts. Useful care or measurement lessons remain,

but a defeated resemblance cannot survive as implied support. A science that can lose precisely becomes more capable of earning support.

Chapter 22 — Predictive Processing and Learning Rivals

Imagine two learning accounts that fit yesterday’s choices equally well. One attributes the change to an expected cue; the other attributes it to learned action value. Repeating yesterday’s task may leave the explanation unresolved. A safe new task can vary the cue while preserving reward, then vary reward while holding the cue fixed. The useful question is which observations would make their forecasts diverge, and whether the study can measure that divergence before either account adapts to the answer.

The first account may predict that a familiar cue becomes less alarming as expectations stabilize. The second may predict a change when an action’s expected return changes. Both could fit declining arousal during repeated trials. Recording expectations before the outcome and separating information about reward from the delivered reward sequence can make the next trial informative. An improvement in one prediction then identifies a feature of the learning process, with its scope determined by the task and observations.

Action complicates this comparison. A participant can avoid an uncertain situation, seek information, rehearse an alternative, or conserve effort. Those choices change what happens next and what can be observed. Active inference, reinforcement learning, and constrained control supply different ways to represent this feedback. The chapter develops their commitments far enough to ask what each predicts when information, available actions, or the remaining horizon changes. A useful comparator must be able to generate the observations attributed to it.

Some apparent dynamics arise from how a record was assembled. Aligning trajectories on a known final event can create endpoint-directed behavior in a conditioned model. Enrollment, treatment, survival, and observation shutdown can reshape a clinical sample before its curves are compared. These

explanations require different diagnostics from a learning-rate change. Comparing unaligned records, prospective forecasts, and explicitly conditioned trajectories helps identify which part of the pattern comes from the process and which part comes from the analysis.

The common comparison therefore specifies the information available at each forecast origin. Simulated generators can test whether fitting and scoring distinguish the candidate processes when their identities are known to the evaluator. If the method confuses two generators, the confusion directs the next design: a different channel, task contrast, timing measure, or calibration may be needed. The known generator is a check on the analysis, not information supplied to the fitted model.

The chapter moves from predictive processing through action, learning, and control to conditioning and selection. Its final sections bring these accounts into one comparison and connect an observed advantage to the particular component it challenges. The payoff is an executable alternative explanation and a discriminating next observation. Chapter 25 then examines how much information those observations provide for separating the models.

22.1 — Predictive Processing and Error Dynamics

Predictive-processing accounts interpret observations through learned expectations, hierarchical latent states, precision, and prediction errors. A cue's effect depends on what the system expects and how uncertain that expectation is (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013; Samara, 2026).

22.1.1 — Hierarchical Generative Models

Equation role: Rival generative template. For a fixed action sequence, this equation defines a hierarchical observation model for comparison.

$$p_\theta\left(o_{1:T}, s_{1:T} \mid a_{1:T-1}\right) = p_\theta\left(s_1\right) \prod_{t=1}^{T} p_\theta\left(o_t \mid s_t\right) \prod_{t=1}^{T-1} p_\theta\left(s_{t+1} \mid s_t, a_t\right)$$

Equation role: Prediction-error diagnostic. This loss measures precision-weighted error inside a rival model, not felt value or terminal neutrality.

$$L_{PE} = \sum_{t=1}^{T} \left(o_t - \hat{o}_t\right)^\top \Pi_t \left(o_t - \hat{o}_t\right)$$

A predictive model's latent state represents hidden causes inferred from observations. Stabilization of the cause estimate can make the observed signal look repaired without cumulative closure.

Prediction error is the difference between expected and observed input. Its decline after repeated exposure can produce apparent relief.

Precision weights an error channel. Changes in that weight can narrow attention or quiet a signal without establishing cumulative closure.

Model revision learns hidden causes over time. A changed expectation can alter report and arousal, creating another rival explanation for the observed pattern.

Higher levels of a hierarchical generative model predict lower-level causes; residual errors update those beliefs. Rao and Ballard's visual-cortex model provides a concrete instance of this architecture (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

After a repeated phone alarm, an affect report may change because the alarm is expected. Record the observation sequence, prediction errors, latent-state estimates, and precision weights needed to test that explanation.

Vary hierarchy depth, learning rates, volatility, and precision in the benchmark. Compare shallow sensory accommodation with deeper context-dependent models using explicit testable commitments (Samara, 2026). A richer predictive account may explain the apparent cumulative pattern; any remaining discrepancy must survive plausible hierarchical specifications and protected context changes (Clark, 2013; Hodson et al., 2024).

22.1.2 — Prediction Error and Affect

The affective response may depend on the discrepancy between an expected and delivered outcome, including its sign and context (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Momentary happiness already has quantitative experimental models. Rutledge et al. (2014) combined repeated happiness reports with a monetary-choice task and neuroimaging in twenty-six participants, then replicated the behavioral model in 18,420 smartphone participants. Recent expectations and reward prediction errors predicted reported happiness better than current

task earnings alone. Blain and Rutledge (2020) subsequently separated reward magnitude from learned win probability in seventy-five participants: in that task, learning-relevant probability errors accounted for happiness better than prediction errors in learning-irrelevant reward magnitudes. These results show that the temporal organization of expectations and outcomes can be measured and tested against feeling.

Compare raw, signed, and precision-weighted error with uncertainty change and contextual appraisal. The same hospital sound may become familiar or continue to predict bad news (Friston, 2010; Hodson et al., 2024).

Expected money and expected feeling can contribute different learning signals. In a repeated social-exchange task with 41 participants, Heffner et al. (2025) measured both expectations before outcomes and recorded EEG. Differences between expected and reported emotion were associated with subsequent choices, especially early in learning about a partner. Reward discrepancies more consistently tracked the feedback-related negativity, while emotional discrepancies more consistently tracked the later P3b response; P3b activity was also linked to subsequent choice. This gives a temporal affect model two experimentally distinguishable inputs. Recording what people expected to receive and how they expected to feel makes the evolving relation between events, appraisal, and action more directly testable.

Better-than-expected and worse-than-expected outcomes can have different temporal profiles, bodily signatures, and learning consequences even when their absolute magnitudes match. A protocol can estimate these asymmetries and compare the contributions of surprise, valence, controllability, and their interactions to immediate feeling and later adaptation (Friston, 2010; Hodson et al., 2024).

The mapping is also experimentally discriminable. In two experiments involving three hundred participants, Forbes and Bennett (2024) found that the affective influence of negative reward prediction errors depended on free choice and the absence of counterfactual feedback; positive prediction errors did not show a significant effect in their analyses. This contextual result supplies three useful design variables: agency, outcome sign, and information about unchosen alternatives. Their interactions can be specified in advance and tested in held-out conditions.

22.1.3 — Precision and Attention

Under the precision weighting described in Section 20.2.3, a small reliable prediction error may receive more weight than a larger unreliable one. In this comparison, estimate precision from attention traces, the person's account, and contextual observations; its interpretation remains channel-dependent (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013; Hodson et al., 2024).

Precision should be independently estimated or varied through benign, consented tasks where feasible. Cue reliability and neutral attentional demands can distinguish a precision mechanism from a retrospective label for salience. Threat-related contrasts must use existing data or naturally occurring variation within Chapter 3's safeguards, not induced threat for Law-specific evidence. The comparison tests whether constrained precision parameters predict held-out responses and whether the closure candidate adds a further transported restriction (Friston, 2010; Clark, 2013).

22.1.4 — Learning and Model Revision

Repeated evidence can revise the generative model itself. Distinguish a familiar cue's reduced surprise from a changed belief about its hidden causes (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

Specify update rates, priors, volatility, context changes, and the time scale of model revision.

Combine repeated experience sampling with context and uncertainty measures to separate rapid habituation from slower revision.

Test renewal across rooms, relationships, and physiological states, alongside context transitions and volatility changes. Learning models may predict the return of an apparently extinguished response; a cumulative candidate must state a different response or invariant relation for the same transitions (Sutton & Barto, 2018; Gershman, 2018).

22.1.5 — Predictive Success Versus Felt Value

Predictive success and felt value need separate outcome measures (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

A caregiver may learn precisely which forms an office will reject and how long its calls take while her burden remains unchanged (Friston, 2010).

Score the generative forecasts and the experiential measurement bridge separately. Predictive accuracy cannot compensate for a failed bridge; a defensible experiential estimate cannot compensate for failed forecasting. Report which dependency the disagreement challenges (Gneiting & Raftery, 2007; Hernán & Robins, 2020).

22.1.6 — When Predictive Processing Wins

Apply Section 22.8.1's victory criterion to the predictive-processing model and restrict any explanatory claim to its demonstrated transport (Rao & Ballard, 1999; Friston, 2010; Clark, 2013; Seth, 2013).

In a terminal-room routine study, test whether familiarity, medication timing, caregiver continuity, and precision-weighted expectation explain reduced arousal. A negative control can examine nonterminal periods with comparable familiarity and care structure (Hodson et al., 2024; Clark, 2013).

22.2 — Active Inference and Expected Control

Active inference models action as well as perception: policies trade predicted observations, uncertainty, and prior preferences. This supplies a future-directed comparator for horizon and repair claims (Seth & Friston, 2016; Friston et al., 2017; Lageman et al., 2026).

22.2.1 — Policies and Expected Outcomes

Specify the rival's latent states, policy set, preferences, epistemic and pragmatic terms, precision, and observation model before recovery testing (Lageman et al., 2026).

Equation role: one conventional active-inference rival objective. This policy-selection quantity belongs to a generative model; it is not the phenomenological ledger.

In this rival, preference denotes a prior over expected outcomes. It is not moral value, human worth, or an interval contribution to the phenomenal ledger.

A policy denotes a possible action sequence in the rival model. It is neither destiny nor a path that independently proves the Law.

Free energy denotes a model-evidence bound or an expected objective in the relevant construction. It is not the phenomenal ledger or an exact derivative of valence.

Precision denotes confidence or gain on an error channel. It does not measure the importance of a person.

$$G(\pi) = E_{q(o,s|\pi)}\left[lnq\left(s|\pi\right) - lnp\left(o,s|C\right)\right]$$

Evaluate policies through their predicted observations, hidden states, uncertainty, and prior preferences (Seth & Friston, 2016; Friston et al., 2017).

Model how action changes later evidence. Avoidance may reduce immediate error while preventing belief revision;

exploration may carry an immediate cost while improving future control. Compare predictions of choice and subsequent affect from the declared policy set. Any additional cumulative constraint must predict a feature not already built into that objective (Friston et al., 2017; Badcock & Davey, 2024).

22.2.2 — Preferences and Prior Beliefs

Prior preferences specify favored outcomes, such as bodily ranges, temperatures, or sensory conditions, within the model (Seth & Friston, 2016; Friston et al., 2017).

A patient's preference for dim light and fewer interruptions can explain policy choice and reduced arousal through preferred observations (Seth & Friston, 2016; Badcock & Davey, 2024).

Empirical work should declare whether preferences are fitted from behavior, self-report, physiology, clinical context, or task structure.

Distinguish preferences from beliefs about what is likely: expecting pain does not imply preferring it. Identify how each parameter is estimated, then test whether preferences transport when beliefs change rather than allowing both to absorb discrepancies after fitting (Friston et al., 2017).

22.2.3 — Epistemic and Pragmatic Value

Epistemic value concerns information gain; pragmatic value concerns outcomes aligned with prior preferences. Separating them makes information-seeking and avoidance distinguishable in the model (Seth & Friston, 2016; Friston et al., 2017).

Refreshing a biopsy-results portal can seek information despite an anticipated emotional cost; avoiding it can favor immediate comfort (Badcock & Davey, 2024).

Vary information availability and action costs independently, then compare the predicted choices across active-inference and

reinforcement-learning accounts (Friston et al., 2017; Sutton & Barto, 2018).

22.2.4 — Homeostatic and Allostatic Coupling

Homeostatic and allostatic versions couple policy selection to expected bodily states (Seth & Friston, 2016; Friston et al., 2017).

For breathlessness after a stair climb, model how expected bodily range, panic, action options, and medication timing shape the response.

Specify body-state variables, observation noise, control actions, and horizons; test the response to treatment, rest, posture, or changes in the care environment.

22.2.5 — Free-Energy Change Is Not the Ledger

Variational free energy, expected free energy, prediction error, and surprise have different roles within these models. A proposed link from any of them to cumulative felt experience needs a separately validated measurement bridge (Seth & Friston, 2016; Friston et al., 2017).

Apply Section 18.7.1's potential-accounting test: a time derivative integrates to the difference between boundary values, which is zero only when those values agree. An independently specified free-energy potential still requires a validated bridge to phenomenal valence.

22.2.6 — Discriminating Predictions

Reduced arousal after gaining reliable room control may be shared by several models. The discriminating question concerns the additional feature on which their predictions differ. In a hypothetical comparison, both specified models predict improvement, but one predicts a rapid change after control becomes available while the other predicts a slower change as

the new conditions are learned. A design that measures only the final level may miss that difference. Fix the predicted timing, direction and observation window before analysis, then score both models on the same observations.

For each contrast, declare its expected direction and observation window before analysis, and score the competing models on the same quantity.

22.3 — Reinforcement Learning and Policy Adaptation

Reinforcement learning represents adaptation through learned values, policies, transition models, and exploration. Its task-defined reward requires a separate bridge to felt valence. Include these generators in recovery testing and apply Section 22.8.1's comparison criterion (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018; Gershman, 2018).

22.3.1 — State, Action, Reward, and Return

Homeostatic reinforcement learning supplies a concrete physiological architecture for this comparison. Keramati and Gutkin (2014) define primary reward through reduction of distance from a physiological set point. Under their assumptions, policies maximizing discounted reward also minimize discounted homeostatic deviation, and learned cues support anticipatory regulation. This is a mathematical result within a specified model, supported by comparisons with existing behavioral findings. It shows how reward seeking and bodily stability can share a computational objective. The model also supplies a useful algebraic comparison: an undiscounted sum of rewards defined as state differences telescopes to the initial-to-final change. In simulation and ethically available observational records, compare the declared PTVC route with this internally motivated learning model under the same state information, observation process, and complexity budget.

Equation role: POMDP belief update. Here η is the normalizing constant that makes the posterior sum to one, and the summation runs over prior hidden states s. The update defines partial observability for learning rivals and prevents oracle-state comparisons.

$$b_{t+1}(s') = \eta\, p\left(o_{t+1}|s'\right) \sum_{s} p\left(s'|s, a_t\right) b_t(s)$$

Equation role: Temporal-difference rival update. This is a learning rule for value estimates, not a measurement of felt experience.

A changing-reward rival predicts policy shifts after a value update. The discriminating question is whether PTVC adds a restriction that transports beyond the updated-return explanation.

A hidden-state rival predicts belief tracking and checking behavior. Test whether uncertainty explains recurrence without invoking burden closure.

A finite-horizon rival predicts urgency and risk taking that depend on the planning horizon. Test whether that pressure appears without terminal neutrality.

An offline-replay rival predicts delayed policy change after memory simulation. Test whether planning explains the observed reactivation.

$$Q_{t+1}\left(s_t, a_t\right) = Q_t\left(s_t, a_t\right) + \alpha\left[r_{t+1} + \gamma \max_{a} Q_t\left(s_{t+1}, a\right) - Q_t\left(s_t, a_t\right)\right]$$

A reinforcement-learning decision process specifies states, actions, transition probabilities, rewards, and a policy. Its return is a discounted or finite-horizon sum of the defined rewards (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

In a finite-horizon task, test whether choice between a small safe gain and a larger risky gain changes as remaining opportunities shrink.

An explicit comparison is provided by the Integrated Advantage model (Bennett et al., 2022): mood is a leaky average of estimated action advantage, the chosen action's value relative to the expected value of the current state under the person's

policy. It predicts that beliefs about reward persistence influence adaptation. A prospective task can hold the delivered reward sequence fixed while varying information about future persistence, record the resulting beliefs, and measure affect before and after the information. Specify the available actions, estimated action values, and policy probabilities; matched rewards alone do not fix advantage. Give this model and a prespecified PTVC realization the same information available at the forecast origin, with parameters fitted outside the evaluation records, then compare predictions for the same later observations. This tests whether PTVC adds a distinct trajectory restriction beyond a specified learning account.

A reward variable organizes the model's policy updates and may represent an engineered proxy, delayed label, or task outcome. Comparing several specified encodings shows which one the data identify. Potential-based reward shaping is a classic example of a reward transformation that can preserve the optimal policy under defined Markov-decision-process conditions, underscoring that computational reward representations can change without establishing a new phenomenal quantity (Ng et al., 1999). Constraints on the reward function make predictions of choice and feeling across tasks more informative (Sutton & Barto, 2018; Daw et al., 2011).

An independent close behavioral replication with 50 participants largely reproduced the main findings of Rutledge's computational happiness model. Its preregistered individual analyses obtained a mean within-person fitted R^2 of 0.54, compared with 0.47 in the original report (Vanhasbroeck et al., 2021). The estimation-based replication criteria were met, but the original model's clear model-selection advantage over the alternatives was not fully reproduced. This provides an

independently reproduced quantitative account of how expectations and recent outcomes relate to reported happiness.

Affect can improve the explanation of learning itself. Teoh et al. (2026) compared three models in 75 participants: reward-based learning alone, an additional affective influence on choice, and affect-informed updating of expected value. Jointly fitting choices and explicit expectations favored the updating model, whose average participant-level WAIC model weight was 0.651. A separate 55-person study reproduced the behavioral findings, and a 95-person experiment found that valence tracked expectations during passive learning under varying predictability. Positive feeling was associated with more favorable expectations after accounting for received rewards. These results supply an empirical architecture linking present feeling, developing expectations, and future action. For the PTVC program, that architecture provides a concrete starting point for testing how accumulated affective history shapes later trajectories.

22.3.2 — Temporal-Difference Learning

Temporal-difference updates supply a timing model for subjective changes following discrepancies in expected future value (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

At the neural level, Bayer and Glimcher (2005) recorded individual dopamine neurons while monkeys learned when to make an eye movement for juice reward. Activity tracked the difference between the current reward and a weighted history of earlier rewards for better-than-expected outcomes. The sign asymmetry is informative and belongs in models of the response. Nervous systems can represent recent reward history and compute a learning-relevant update, providing a concrete

computational component for theories linking history, expectation and affect.

Temporal-difference learning can move responses from outcome to cue and propagate value backward through states. Record cue responses, outcomes, choices, and affect together. Test the held-out timing and magnitude of that transfer against the learned value function before attributing any residual to cumulative structure (Sutton & Barto, 2018; Daw et al., 2011; Gershman, 2018).

22.3.3 — Partial Observability

Use a partially observed decision process when actions depend on beliefs about hidden bodily or environmental states (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

A caregiver's policy may change because a symptom's meaning or a clinician's response remains uncertain (Gershman, 2018).

Separate environmental state, agent belief, and research observation: the agent and analyst may lack different information. Model both information gaps before attributing an unexplained trajectory to cumulative structure (Sutton & Barto, 2018; Gershman, 2018).

22.3.4 — Changing Rewards and Horizons

Rewards can depend on exhaustion, context, and remaining time; the rival should allow the forms of change relevant to the task (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

Finite-horizon decision-making can produce urgency, narrowing, risk aversion, or desperate exploration as remaining opportunity changes (Sutton & Barto, 2018; Daw et al., 2011).

A known twenty-trial game tests behavior under its announced horizon. Its late choices do not identify a complete-stream endpoint.

The rival should declare whether rewards are stationary, drifting, state-dependent, context-dependent, or horizon-dependent. It should also declare what the agent knows about the horizon. If the true endpoint is unknown, the model must use a belief over time remaining rather than omniscient countdown.

22.3.5 — Offline and Model-Based Planning

Offline replay and model-based planning can change policy before the next observed action, making them rivals for delayed repair or reactivation (Daw et al., 2005; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

Rehearsing a difficult call or replaying an earlier meeting can alter the next response through memory and simulation (Daw et al., 2011).

Distinguish cached model-free values from model-based search, replay, planning depth, and memory consolidation. Their different timing predictions make delayed actions and reports useful evidence for comparison.

22.3.6 — Learning-Rival Victory

A learning rival wins without privileged information through comparable or better protected prediction, calibrated uncertainty, interpretable parameter recovery, model recovery, and transport for the declared target. Parsimony follows adequacy. Sealed recovery tests must include reinforcement-learning, PTVC, active-inference, bridge, selection, and open-set worlds (Daw et al., 2005; Gershman, 2018; Keramati & Gutkin, 2014; Sutton & Barto, 2018).

If the pipeline fails the required recovery thresholds in those worlds, it is not qualified for the affected confirmatory comparison. Under a valid analysis, adequate but inseparable models yield Model-ambiguous, while failure of every declared model to meet adequacy criteria yields Open-set inadequate. If learning wins, the series may not rename it as how PTVC operated unless that bridge was prespecified and independently tested; the route status changes.

Learning-rival victory can redirect humane practice toward information, agency, environment, or uncertainty reduction while retiring a terminal-closure mechanism. Defeating a route can still reveal where burden was hidden, shifted, or reduced.

22.4 — Safe and Constrained Control Baselines

Safe and constrained control is the primary ordinary comparator for claims about admissible paths, reserve, no-return states, and finite horizons. The next subsections distinguish constraint-based safety, barrier and viability methods, and their failure modes (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

22.4.1 — Constrained Markov Decision Processes

Equation role: Constrained-decision baseline. This is a safe-control rival that optimizes return under declared cost limits.

$$\max_{\pi} E_{\pi}\left[\sum_{t=0}^{T}\gamma^{t}R\left(s_{t},a_{t}\right)\right] s.t. E_{\pi}\left[\sum_{t=0}^{T}\gamma^{t}C_{j}\left(s_{t},a_{t}\right)\right] \leq d_{j}$$

Equation role: Deterministic control-barrier condition. This enforces forward invariance for an engineered controller under declared dynamics.

A constrained Markov decision process can reproduce a trade-off between reward and harm cost. Record its cost functions and constraint thresholds before comparison.

Risk-sensitive control can reproduce avoidance of volatile paths. Record the objective and risk parameter that generate that prediction.

$$L_{f}h(x)+L_{g}h(x)u+\alpha\left(h(x)\right)\geq 0$$

A barrier controller can keep a trajectory inside a safe set. Record its dynamics, safe set, and invariance condition.

No-return avoidance can produce urgency near irreversible states. Specify the irreversible region and uncertainty in the relevant transitions.

A constrained Markov decision process optimizes return subject to declared expected cost limits, providing a comparator for trajectory narrowing (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

In a rehabilitation task, pain-flare constraints may explain smaller effort increments, earlier rest, or refusal of a higher reward.

Declare the constraint cost, threshold, state transitions, and policy class, then test whether observed avoidance and recovery follow those objects. A flexible constraint should not be inferred from the same path it explains. Prospectively specified constraints must predict action and affect under new conditions (Altman, 1999; Wachi et al., 2024).

22.4.2 — Risk-Sensitive and Robust Control

Risk-sensitive and robust models penalize different aspects of uncertain outcomes, including tails and model uncertainty (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Refusal of a high-upside option may reflect the cost of its worst case. A family declining additional monitoring may value quiet more than the protocol assumes (García & Fernández, 2015; Wachi et al., 2024).

Specify the risk measure and parameter-estimation rule: a variance penalty, entropic risk, conditional value at risk, or worst-case transition set makes different predictions.

Vary tail risk and model uncertainty independently, then test held-out choices and affect against the risk-sensitive and robust predictions (García & Fernández, 2015; Wachi et al., 2024).

22.4.3 — Control Barriers and Viability Kernels

Use Section 19.4.3's barrier and viability guarantees with the dynamics, admissible controls, and regularity conditions required by each theorem (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Feasible safety sets may shrink as resources decline or disturbances grow. Compute these regions from independently specified dynamics and constraints, then compare predicted action boundaries with observed trajectories and additional predictions for experience or endpoint behavior (Altman, 1999).

22.4.4 — No-Return Avoidance

No-return avoidance can follow from irreversible loss of future options. Test it through reachability and available actions under the declared dynamics (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

A concrete study might track decisions in a low-burden planning task where some choices remove future options. If participants avoid irreversible branches as the horizon shrinks, finite-horizon safe control may explain the pattern.

Define no-return states before analysis using the task structure, clinical risk, physiological thresholds, or loss of available options.

Distinguish physical irreversibility, learned danger, institutional barriers, and phenomenal burden: each can produce abrupt avoidance. Test whether an independently estimated reachability model predicts the boundary and its movement as conditions change (García & Fernández, 2015; Wachi et al., 2024).

22.4.5 — Shared Constraints in Control Benchmarks

Label simulated controller results with their generator, setting, assumptions, and scientific role (García & Fernández, 2015).

After constructing a controller for an attainable target, test its proposed variables and causal structure against independent observations and competing explanations (Wachi et al., 2024).

Equalize information and feasible actions under Section 19.3.2, including latent-state access, dynamics knowledge, and endpoint information (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

Include action delays, costs, and prohibited interventions in the common feasible set (Wachi et al., 2024; Altman, 1999).

22.4.6 — Common Information and Action Sets

Safe-control comparisons become invalid when models receive different information or feasible actions. A controller with the true latent state, exact dynamics, and endpoint knowledge can outperform a model that sees only noisy observations. In that case the advantage belongs to oracle access rather than theory, so the contest must equalize information and action sets before scores are interpreted (Altman, 1999; García & Fernández, 2015; Ames et al., 2019; Brunke et al., 2022).

The common information set specifies observations, history, sensor reliability, endpoint uncertainty, reserve proxies, context, and action availability. The common action set specifies what the organism or simulated agent can actually do, including delays, costs, and prohibited interventions (Wachi et al., 2024; Altman, 1999).

A hospice dataset provides a clear example. A model that knows the death time while the participant and clinicians did not know it cannot be compared directly to a filtered-horizon model. The endpoint information itself becomes a hidden advantage.

Fair comparison should match each model's information set to what was actually available at the decision time, not to what analysts know retrospectively.

Safe control connects to common-observation comparison through a shared interface. Before scoring, every model declares what it observes, when it observes, what it may choose, and how missingness affects the record. This declaration should include information delay, because a model with faster access can appear superior even when its mechanism is no better.

Fairness toward rivals and participants sets the limit. Model comparison should not turn a person’s limited information into evidence against them, nor should it let a theory win by using retrospective knowledge unavailable to the stream. Information parity should therefore be checked during model recovery rather than assumed from nominally identical input variables.

Safe-control rivals win only when their information and action sets are legitimate and their protected predictions remain calibrated. If they require oracle endpoint knowledge, unobserved latent states, or impossible actions, their apparent advantage becomes a design diagnostic rather than a conclusion about the observed system. A rival that wins only with unavailable actions defines an upper bound on achievable control, not an explanation of the observed behavior.

22.5 — Bridge and Doob-Transform Rivals

Compare the conditioned-process models of Sections 7.5 and 16.4.3 through prospective information, transition laws, and intervention responses (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

22.5.1 — Fixed-Endpoint Brownian Bridge

For each Doob comparator, specify the base process, conditioning event, h-function or event likelihood, observation filtration, and prospective information set. Include the applicable Brownian or Schrödinger bridge, survival conditioning, and random endpoint cases; Section 22.6 treats recruitment and observation selection (Doob, 1957; Léonard, 2014; Léonard et al., 2014).

Equation role: Fixed-endpoint Brownian bridge. Drift is determined by endpoint conditioning.

$$dX_t = \frac{x_T - X_t}{T - t}dt + \sigma dW_t, 0 \leq t < T$$

Equation role: Doob-transform rival. The simplified form applies on the declared harmonic conservative branch. A time-dependent bridge uses a space-time harmonic function; nonharmonic or killed branches retain the corresponding temporal or potential term.

$$L_t^h f = h^{-1} L_t(hf) - h^{-1} f L_t h = L_t f + a_t \nabla log h \cdot \nabla f$$

A bridge moves toward its conditioned endpoint. Test whether drift magnitude or reserve dependence distinguishes that behavior from the candidate.

A bridge can produce a characteristic variance-collapse pattern. Test whether the candidate's reserve-linked innovation imposes a distinct restriction.

Endpoint information is often inserted by conditioning in a bridge model. Establish whether the endpoint in the proposed test was defined prospectively.

The last observation can act as a false endpoint. Model observation shutdown before interpreting the apparent terminal pattern.

Backward alignment from death or final observation uses information unavailable to a prospective forecast. Represent that conditioning explicitly (Doob, 1957).

Compare fixed-endpoint and unconditioned generators over identical windows and measurement noise. Test prespecified intervention responses, endpoint-timing relations, and cross-equation constraints beyond their shared endpoint pull (Doob, 1957; Léonard, 2014).

22.5.2 — Random Endpoint Times

When endpoint timing is uncertain or state-dependent, use a random-endpoint comparator appropriate to the target: stream ending, a terminal window, or observation ending.

Specify hazard, state dependence, observation rules, and the relation between clinical events and endpoint confidence (Aalen et al., 2008; Little & Rubin, 2019).

Random endpoints condition on a distribution of times. Mixtures can alter the familiar bridge shape, and dependence between endpoint timing and the process can create selection effects. Model the endpoint law explicitly and compare independence, state coupling, censoring, and unobserved continuation with equivalent flexibility across candidates (Aalen et al., 2008).

22.5.3 — Last-Observation Alignment

Last-observation alignment can manufacture a terminal-looking path by treating device removal, transfer, discharge, sensor failure, or withdrawal as the truth endpoint (Little & Rubin, 2019).

Final intervals retain the uncertainty implied by the missingness model and the documented reasons observation ended.

Compare calendar, protocol, event, and plausible anchor times, and show the unaligned data. Signatures confined to last-observation alignment make alignment and attrition the immediate model-comparison targets (Little & Rubin, 2019).

Last observations may occur because illness, burden, care, or equipment changed. They are observation events, not automatically endpoint events (Little & Rubin, 2019).

A wearable removed during hospice may leave a stable final reading that appears to settle. The pattern may reflect device tolerance and care decisions rather than terminal neutrality.

Model observation shutdown. If last-observation alignment reproduces the concentration, narrow or suspend the closure-specific interpretation.

Prospectively defined endpoint windows, independent stopping rules, and explicit observation-process models allow the analysis to test whether the signature persists across alternative accounts of missingness.

22.5.4 — Bridge Variance Shape

Conditioned bridges approach endpoints by construction, weakening drift as a discriminator. Variance, covariance, endpoint surprise, and reserve relations discriminate better when reserve, drift, innovation, and endpoint structure are estimated independently (Doob, 1957; Revuz & Yor, 1999; Särkkä, 2013).

Medication, sedation, frailty, dropout, and changing cadence belong in the model of observed variance.

Diagnostics report bridge variance, reserve-coupled variance, observation floors, and selection under common simulated cadence, missingness, endpoint uncertainty, and device versions. Using the same simulation clock makes differences in apparent variance attributable to the candidate process rather than to unequal observation opportunity. Test which joint pattern transports.

Public figures state whether compression is latent, measured, noise-adjusted, bridge-generated, reserve-linked, or unresolved. Captions should also identify whether the endpoint was conditioned, observed, or independently adjudicated, since that distinction changes the interpretation of compression.

Bridge rivals win by reproducing protected structure through endpoint conditioning and the shared observation model. PTVC is distinctive only where prespecified restrictions survive. A bridge victory should retire the claimed native signature while leaving separately supported measurement or endpoint findings intact.

22.6 — Event-Conditioned and Selection Rivals

Enrollment and observation can reshape a clinical sample without changing its latent dynamics. Each terminal figure should identify its analyzed denominator and link to the full eligibility, entry, and observation-loss record. Section 22.6 separates the mechanisms to model.

22.6.1 — Hospital and Hospice Selection

Hospice enrollment follows prognosis and family consent. That entry process can concentrate the sample within a particular care context.

Model hospital or hospice entry through diagnosis, prognosis, access, referral, support, and participation decisions (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

A hospice cohort conditions on service entry; sudden deaths and people outside the service may have different trajectories (Aalen et al., 2008).

Estimate entry probability and care intensity alongside time since enrollment before interpreting concentration in the selected sample.

Where ethical and lawful access permits, compare participants with eligible nonparticipants. Model entry and site practices jointly with observation schedules, then limit transport claims to supported reference classes (Aalen et al., 2008; Hernán & Robins, 2020).

22.6.2 — Treatment-Conditioned Trajectories

A new medication or care routine can shift a trajectory through treatment. Record the intervention before attributing the shift to the proposed latent dynamic.

Medication and caregiver changes can explain reduced agitation or altered sleep. Compare these treatment-conditioned

trajectories explicitly (Hernán & Robins, 2020; Little & Rubin, 2019).

Record time-varying treatment, dose, timing, indication, and care goals. Prior state can affect treatment and later observation; treatment can change both trajectory and measurement probability. Causal treatment contrasts may require marginal structural models or the g-formula under their identifying assumptions. Joint models address specified dependence among trajectories, events, and observation; they do not by themselves identify causal effects or unmeasured experience (Hernán & Robins, 2020; Tsiatis, 2006).

22.6.3 — Observation Shutdown

Device removal for comfort can create a clean-looking final curve by ending observation. The observation-shutdown mechanism must remain visible.

Model channel-specific observation shutdown using Section 13.2's distinction between report loss and stream ending (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Shutdown associated with greater burden can understate distress; shutdown associated with improved comfort can hide relief. Model the direction relevant to each channel (Little & Rubin, 2019).

Use separate dropout models where self-report, wearable signals, notes, and proxy observations cease for different reasons.

Model shutdown causes, preserve uncertainty, and identify recoverable estimands. Proxy, physiological, and passive channels may extend coverage according to their validated roles; classify the remaining observation limitation under Section 24.2, retaining any valid estimates or bounds (Little & Rubin, 2019).

22.6.4 — Frailty-Conditioned Decline

Declining physiological margin can jointly change activity, report intensity, sensor reliability, and survival, producing apparent convergence (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Specify baseline vulnerability, time-varying decline, acute episodes, treatment, functional status, and observation quality.

Because frailty influences treatment and survival, it can imitate coordinated terminal dynamics. Use prospective frailty measures, allow time-varying change, and test whether the signature remains after stratification or joint modeling. A frailty adjustment built only from late observations risks conditioning on consequences of the process (Aalen et al., 2008; Andersen et al., 1993).

22.6.5 — Competing Events

Discharge, transfer, sudden death, and recovery are different competing events. Treating them as one endpoint can generate a misleading mixture.

Distinguish events ending a study record from the target endpoint. Discharge, transfer, withdrawal, recovery, and death route participants through different observation systems (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Model event-specific probabilities under state, treatment, site practice, resources, and participant or family decisions. Withdrawal ends observation; recovery followed by discharge does not complete a stream.

Choose cause-specific or subdistribution estimands for the declared question and compare their consequences; do not choose the formulation for its favorable curve (Aalen et al., 2008; Andersen et al., 1993).

22.6.6 — Versioned Rival Envelope

Use the selection model to identify the additional records, negative controls, denominator flows, and sensitivity bounds needed for the chosen estimand.

The rival envelope is the versioned set of models specified for a protected comparison. New observation artifacts, clinical confounds, sensor behavior, active-inference formulations, and safe-control baselines may enter a later version when independently motivated. The original comparison and its verdict remain in the record; changing the envelope cannot retroactively turn that completed test into a different test (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

Each version records what changed, why, and whether earlier results require rerun or correction. Selection rivals win when referral, treatment, survival, frailty, device removal, reporting capacity, or observation shutdown explains the terminal pattern under the common observation model.

When a newly identified comparator changes the analysis, the public record should show the addition and its consequences. Every terminal figure includes denominator flow: eligibility, entry, observability, loss, death, report incapacity, and protected withheld records.

Include the envelope version and denominator with each curve.

22.7 — Common-Observation Comparison

Apply the common-observation design in Section 21.6 to the learning and control rivals. The criterion below adds a model-selection penalty while keeping its interpretation separate from predictive accuracy (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

22.7.1 — Shared Data-Generating Interface

Equation role: Penalized model-selection criterion. The held-out log predictive score measures forecast performance; the model-specific penalty is a separate selection preference. The penalized total is not automatically a proper predictive scoring rule. Fix the penalty and its units before evaluation, and report unpenalized predictive performance and calibration alongside the total.

A PTVC candidate may be misidentified when a bridge rival generates the same signature through endpoint conditioning. Report nonseparability when the design cannot distinguish them.

Predictive-processing dynamics may be mistaken for the PTVC candidate. If error dynamics explain the pattern, narrow the PTVC-specific claim.

$$S_M = log p_M\left(o_{test} | D_{train}\right) - \lambda_M df_M$$

Safe control may be mistaken for the PTVC candidate. Test whether avoidance of no-return states explains the path.

Observation selection may be mistaken for a dynamic mechanism. Rebuild and test the observation process before claiming latent dynamics.

Keep latent quantities distinct from measured channels while providing every rival the same available evidence (Gneiting & Raftery, 2007).

Declare when endpoint information becomes available. Prospective models use filtered beliefs; retrospective bridges identify the final-event information used.

Known latent states and generator labels remain hidden from fitting and are used only for the declared recovery checks. Record transformations and exclusions before evaluation, and use synthetic fixtures to verify that each implementation reads the interface correctly (Wilson & Collins, 2019).

22.7.2 — Equivalent Parameter Freedom

Equivalent parameter freedom does not mean identical parameter counts. Different models need different structures, with declared flexibility disciplined against hidden degrees of freedom (Vehtari et al., 2017; Wilson & Collins, 2019).

Priors, nonlinearities, latent-state dimension, regularization, early stopping, and feature engineering can give one model a much larger effective search space. The benchmark should therefore report tuning budgets, prior predictive behavior, effective complexity, and sensitivity to initialization. When exact matching is impossible, the asymmetry should be made visible and stress-tested (Gneiting & Raftery, 2007; Vehtari et al., 2017).

22.7.3 — Predeclared Scoring

Apply Section 21.6.5's frozen scoring rules, with trajectory and endpoint-window diagnostics specified for the learning and control rivals (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018; Wilson & Collins, 2019).

Separate the score for momentary forecasts from tests of cumulative consequences and endpoint behavior (Wilson & Collins, 2019).

Declare primary and secondary scores, aggregation, tie rules, and missing-prediction consequences, using a proper score for each predictive object (Gneiting & Raftery, 2007).

22.7.4 — Complexity and Identifiability

Assess structural and practical identifiability under Section 25.5. Adequate but inseparable models receive Model-ambiguous status; a valid comparison that remains undecided receives Inconclusive evidence (Gneiting & Raftery, 2007; Vehtari et al., 2017).

Complexity penalties cannot repair nonidentifiability. Use profiles, posterior geometry, calibration, and confusion matrices to determine whether the design distinguishes the models before interpreting a small selection-score difference (Talts et al., 2018; Wilson & Collins, 2019).

22.7.5 — Posterior Predictive Diagnostics

Compare posterior predictive simulations with protected observations at the features each rival claims to explain (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018).

Prespecify diagnostics for autocorrelation, sign asymmetry, perturbation recovery, context renewal, endpoint variance, and missingness-linked residuals. Report channel-specific successes and failures separately.

A model might reproduce choices accurately yet miss the delayed physiological response, while another captures that response but misses the change in reported feeling. The diagnostic result identifies those separate successes and discrepancies. It can then guide a new measurement schedule or a revised mechanism aimed at the particular relation that remains unexplained.

22.7.6 — No Winner When All Are Wrong

Apply open-set adequacy checks before interpreting the relative ranking of these model families (Gneiting & Raftery, 2007; Vehtari et al., 2017; Yao et al., 2018).

Fix calibration, residual, recovery, and transport thresholds. Site-specific failure patterns can identify observation assumptions that need revision before pooling.

Distinguish poor absolute fit, nonseparability, and uncertainty crossing the decision boundary. Each leaves no winner for a different reason and requires a different measurement or design response.

Apply rejection criteria symmetrically. A clear component loss under its declared rule still stands when open-set concerns remain elsewhere.

22.8 — When Learning and Control Rivals Win

When a rival examined in this chapter best explains the observations under the common comparison rules, the component-level public record should identify that result (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

22.8.1 — Component-Level Defeat

Equation role: Rival-victory rule. A rival victory is a status event when the protocol-fixed comparison favors the rival beyond the declared uncertainty and practical margin.

A component loss requires narrowing or retiring the affected component. Public wording should state that the mechanism did not survive this comparison.

Repeated losses of distinctive predictions place pressure on the strong form. Report the adverse evidence confronting the broader claim.

If all compared models fail adequacy, report Open-set inadequate status. Choosing the least inadequate model does not make it adequate.

A rival victory can redirect research and preserve a useful finding, including any separately justified care implication. The finding retains its actual evidential basis.

$$\Delta_{R,L} = S_R - S_L; \Delta_{R,L} > \delta_{pre} \Rightarrow narrow\,(L\text{-specific component})$$

A comparative defeat narrows or retires the tested component; its consequence for the formal Law depends on the component's role in the dependency graph (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

A component loses comparative support when an adequate rival meets a prospectively specified advantage criterion under common information and observation rules. When both models are adequate and the predictive difference is confined to a

prespecified practical-indistinguishability region under valid uncertainty, report Model-ambiguous. Equal point scores or nonsignificance alone do not establish that result. A strict-superiority claim or a declared adequacy-and-simplicity criterion requires its own decision rule. Report the criterion actually met before narrowing or retiring the component.

Record the exact claim, evidence, rival, reference class, affected outputs, and correction owner under Chapter 28's claim-status procedure (Popper, 1959; Lakatos, 1978).

22.8.2 — Strong-Form Pressure

Determine whether the tested PTVC route requires the component that lost. Reserve matters to this inference only when the route assumes reserve coupling (Lakatos, 1978).

Use a dependency graph to distinguish failures of shared assumptions from failures of separate gates: representation, routing, endpoint, observation, mechanism, and transport. Represent correlated evidence with a joint model or calibrated decision rule, count exact duplicates once, and direct revision toward failed common assumptions (Lakatos, 1978; Stanford, 2006).

22.8.3 — Bridges, Hybrids, and Mechanism Revision

A connection between a successful mechanism and a PTVC route can be proposed with distinct predictions and tested independently and prospectively on protected evidence against the original rival as a new claim (Stanford, 2006).

For a hybrid combining learning, control, and a PTVC-specific term, use ablation and transport tests to determine whether the added term contributes predictive information beyond the ordinary components (Wilson & Collins, 2019).

A successor should target the surviving discrepancy: for example, reserve-linked innovation unexplained by an endpoint bridge. If the rival predicted every prespecified feature, an added hidden variable needs a new prediction to earn a test (Popper, 1959; Lakatos, 1978).

22.8.4 — No New Mechanism After Every Loss

Mechanism revisions state their motivation, source grounding, new predictions, and comparison plan. This record makes theoretical development assessable across successive versions (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

This rule matters in a broad project because the Law has many possible auxiliary concepts: state-allocation pressure, reserve, horizon, dual-variable cost signal, interoception, dreams, endpoint windows, and measurement governance. Each can be useful, but none should become an infinite repair kit (Popper, 1959; Lakatos, 1978).

A new mechanism should answer a precise failure. If bridge rivals explained endpoint drift but not reserve-linked innovation, a revised mechanism might target that residual. If predictive models explained every prospectively specified feature, adding a new hidden variable without novel predictions would weaken the project.

The protocol should record mechanism changes with version dates, prior rationale, prediction changes, and implications for older results. Old support should not automatically transfer to the revised model. A changed claim needs new evidence.

A revision record lets readers follow how the theory develops, which observations motivated each change, and which predictions are being tested next.

A revision plan can make this concrete. Before each confirmatory cycle, specify which auxiliary changes are

permitted, which need a new exploratory phase, and which require a new evidence claim. Changes to endpoints, representations, routing rules, or mechanisms receive their own versioned predictions and frozen artifacts, preserving what each study tested while making room for creative development (Nosek et al., 2018).

22.8.5 — Publication of Rival Victories

Rival victories should be published, logged, and summarized in the same claim-status system as supportive results. A complete public record preserves adverse findings so that readers can assess the evidence as a whole (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

A comparative result can identify the model that best predicted endpoint-window behavior and state the cohort, observation model, endpoint definition, uncertainty, and component-level consequence (Lakatos, 1978).

Rival victories also protect applied books. If a city, religion, government, or transition-facing artifact cites the science loosely, the public log should make overclaim easy to correct. The science book's honesty becomes a firewall for the whole series.

Plain-language and technical layers are required. The technical report carries equations, likelihoods, diagnostics, and model recovery; the public summary states support, partial support, narrowing, inconclusive, or defeat. Both should be versioned.

Adverse results attach to models and claims, never to participants, clinicians, families, sites, or readers. The archive evaluates scientific propositions rather than lives.

Publishing rival victories succeeds when future readers can see the project correct itself. It fails when negative results are

buried, blurred into vague uncertainty, or recast as hidden support.

Search rival victories by claim and component. A public register identifies the tested version, cohort, endpoint rule, observation model, scores, winner, uncertainty, and propagation without exposing private data. Machine-readable fields let figures and summaries update rather than preserve stale claims. Record later replications or reversals so one adverse result is neither hidden nor treated as final beyond its evidence. Scientific credibility grows when archives make correction easier than selective memory (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019).

22.8.6 — Research Redirection

Wins by learning and control rivals may still yield measurement tools, safer baselines, lower-burden routines, or selection controls. These gains belong to the rival, not terminal support (Lakatos, 1978; Munafò et al., 2017; Nosek et al., 2018).

Redirection preserves useful findings without treating them as terminal evidence. It may improve measurement, protocols, care, or comparison arenas while leaving target status unchanged (Stanford, 2006; Popper, 1959).

A study may show that stable routines reduce distress under an active-inference account and still guide care design. It should not be presented as terminal-neutrality evidence. Better care matters independently.

Record redirection by domain: measurement, rival development, clinical insight, governance, artifact correction, or formal narrowing. Each has different authority, and none alone authorizes policy, belief, or transition.

Useful science may survive a defeated claim, but the claim cannot hide inside it. Evidence should remain modest, actionable, and explicit about what lost.

Scientific fairness places the burden on the claim. Rival victory changes its status; surviving residuals face stronger tests; care, rights, privacy, and human standing remain independent.

Chapter 23 — Evolution, Non-Ergodicity, and Outer Limits

An explanation can succeed at one level and leave a different question open. An evolutionary account may explain why a response persists in a population. A temporal model may explain how that response changes during an individual life. A comparative study may ask whether the same relation appears in another species. Each inquiry can be informative, but their conclusions connect only through an account of what remains the same as the level or setting changes.

This chapter develops those connections. Evolution supplies mechanisms of affect and predictions about the tradeoffs they embody. A signal that helps an organism respond to danger can also impose an experiential cost. The relevant comparison concerns how the signal’s function, its dynamics, and its effects on experience relate. It requires a temporal account of the individual as well as an explanation of the trait’s persistence.

The next problem is statistical. A relationship observed across people can differ from the relationship that unfolds within one person. Recent neurocognitive studies make this distinction concrete and provide models for estimating both levels (Hunter et al., 2024; Mistry et al., 2026). The chapter works through a numerical reversal, then examines heterogeneous dynamics and changing cohort composition. These examples show exactly where a population summary can answer a different question from the one posed about a stream.

Complex dynamics extend the comparison through time. Feedback, multiple stable regimes, and learned history can produce changes that a model based only on the latest report will miss. The question becomes which representation of the past improves prediction and why. A fitted history term may recover an ordinary latent state more effectively; a particular mechanism must explain the further restrictions it adds. The chapter

develops a predictive comparison and separates its information gain from approximation error.

Comparative work then changes the setting. Different species express behavior through different bodies and ecologies, while development changes the meaning of measurements within a life. The examples in this chapter connect concrete findings to proposed instruments and study designs. They also distinguish the functional component an experiment establishes from the additional evidence required to interpret that component within a qualifying conscious stream. Synthetic systems provide another setting in which these relations can be examined under controlled constructions.

The chapter ends by assembling a map of scope. A mechanism may travel across classes even when a common numerical scale does not. Conversely, comparable observations may arise through different mechanisms. Identifying those patterns helps determine which comparison to attempt next and which result needs a narrower interpretation. The goal is a comparative program whose extensions are explicit enough to be tested, and whose limits reveal a specific scientific question rather than an unexplained exception.

23.1 — Selection and Experienced Welfare

Evolutionary accounts connect affective responses with survival and reproductive consequences. They supply hypotheses about signal function, mismatch, and tradeoffs whose effects on individual experience can be studied over time (Williams, 1966; Nesse & Williams, 1994; Dawkins, 1976; Nesse, 1990; Williams & Nesse, 1991; Nesse & Ellsworth, 2009).

23.1.1 — Fitness, Reproduction, and Affective Signals

Adaptive-signal models treat pain, pleasure, relief, threat, and attachment as guides to action. Compare their dynamics with cumulative experience within a stream, allowing an adequate signal model to win under the prespecified advantage or adequacy-and-simplicity criterion (Section 22.8.1).

A trait can spread because it improves survival, mating success, kin support, parental investment, or group coordination. Its effects on felt experience form an additional dimension of the biological account (Williams, 1966).

Fear illustrates how a protective response can carry experiential costs. Treating an uncertain rustle as a threat may support survival, while false alarms produce racing heart, narrowed attention, freezing, and vigilance. Measuring these responses links the signal's function with its effect on the individual.

Social pain supplies a parallel question. Sensitivity to exclusion or status may affect coalition membership while imposing costs on the individual. Its persistence and within-life consequences require separate analysis.

Adaptive signal accounts can predict asymmetry when missing danger is costlier than missing opportunity. Selection may preserve a protective response despite substantial experiential

costs, providing a concrete alternative for patterns attributed to terminal balancing.

23.1.2 — Mismatches and Modern Environments

Evolutionary mismatch proposes that a response shaped under one environment can have different consequences after its inputs or constraints change (Williams & Nesse, 1991). A specific account identifies the changed exposure and the mechanism connecting it to experience. Those predictions can be compared with the additional cumulative restrictions proposed by PTVC.

Separate the evolutionary claim from the present causal model. The first asks whether a trait's current effects differ from those in an independently justified selection environment; the second asks which present inputs and mechanisms maintain the observed state. A current causal intervention can inform the second without reconstructing the first (Williams & Nesse, 1991).

A concrete example comes from mice exposed to a chronic high-fat diet. Gazit Shimoni et al. (2025) linked altered feeding responses to reduced neurotensin signalling in the lateral nucleus accumbens–ventral tegmental area pathway. Increasing neurotensin expression mitigated several measured changes. The mice could retain a home-cage preference for high-fat food while showing reduced consumption in an acute palatable-food assay. This dissociation gives separate behavioral and circuit targets rather than a single inferred quantity called pleasure.

A temporal model can ask which input history changes which feeding response and how that relationship develops or recovers. The comparison then tests whether a prespecified cumulative term adds predictive information beyond those measured mechanisms.

23.1.3 — Individual Versus Population Success

Population selection concerns survival and reproductive success. Test how within-stream trajectories relate to population composition and survivorship; those processes may explain apparent neutrality through mixture rather than a terminal restriction.

Selection changes trait distributions through reproductive outcomes. Individual experiential consequences are a separate quantity to study alongside those population changes (Williams, 1966; Nesse & Williams, 1994).

A trait may persist while high-cost carriers leave reproduction or the observable cohort. Account for that changing composition when interpreting later records.

23.1.4 — Neutral and Maladaptive Traits

Traits may be neutral, linked to other traits, historically constrained, or maladaptive under present conditions. Classifying an affective pattern requires independent evidence about its history and environment; selection does not supply an adaptive explanation by default (Nesse & Williams, 1994).

Mismatch models explain persistent burden when existing regulation meets changed environments. Test how environmental fit affects distress and recovery, including whether mismatch variables account for the apparent closure signatures.

Developmental constraints include inherited architecture, path dependence, and limits on repair. Test which constraints change recovery and experienced welfare, allowing them to explain the observed dynamics without a terminal boundary.

Developmental constraints matter. A nervous system is not redesigned from first principles for comfort. It inherits architecture. It uses ancient circuitry, repurposes systems, and

carries tradeoffs. The result can be brilliant in one context and destructive in another. Back pain, chronic inflammation, reproductive risk, trauma memory, and anxiety sensitivity can persist because bodies are compromises, not perfect welfare machines.

Genetic drift and linkage also prevent simple comfort stories. A trait may ride with another trait, remain below selection pressure, or appear only under rare environmental combinations. The fact that it exists does not mean it serves fairness. Existence is not endorsement.

Some trajectories reflect poorly matched environments, physiological damage, or developmental constraints. Modeling those processes helps explain individual variation and identifies opportunities for support or recovery.

A methodologist should hear the modeling implication. Evolutionary parameters need uncertainty. Trait history, current environment, developmental exposure, genetic architecture, culture, treatment, and measurement regime all affect whether an observed pattern is adaptive, maladaptive, neutral, or mixed.

Evolutionary complexity helps sharpen the PTVC candidate. Regulation supplies mechanisms to investigate, while pathwise predictions specify the contribution of a terminal-boundary account. Neutral, maladaptive, mismatched, and constrained mechanisms expand the range of trajectories the comparison should represent.

Some experiential patterns arise as side effects of other mechanisms, such as vigilance coupled to social memory or pain coupled to mobility protection. Others persist under weak selection, recent environmental change, or developmental constraints. Including these possibilities gives the model a fuller account of stable individual differences.

23.1.5 — Evolutionary Rival Victory

Evaluate evolutionary models of the declared signatures using the comparison criterion in Section 22.8.1.

Life-history trade-offs allocate energy, risk, reproduction, and care differently across stages. Test their effects on experiential trajectories and whether life-stage ecology explains horizon effects without a PTVC-specific residual.

Evolutionary explanations can account for selection, development, ecology, and population dynamics with testable predictions for the observed trajectories. When those predictions perform best, the result identifies the biological mechanism to carry into the next stage of research (Dawkins, 1976; Williams, 1966).

Candidate explanations include sedation, attrition, family communication, care pathways, survivorship, changing behavioral range, physiological transitions, adaptation, sleep, learning, social support, and conserved control architecture. Each predicts a different combination of timing and observation effects.

Plausibility supplies a starting point for an evolutionary model. The next step is to generate held-out predictions, reproduce path signatures under the same observation model, check model recovery, and identify cases where PTVC predicts a different result.

The comparison can refine the model's biological mechanisms and scope, improve measurement, and identify useful questions about affect dynamics and end-of-life care. Each result determines which component to develop next.

Public summaries should distinguish biological plausibility, demonstrated mechanisms, and comparative prediction results. Each contributes a different part of the scientific case and should be described through the evidence supporting it.

Evolutionary theory supplies mechanisms and strong comparison models for PTVC. Testing their competing predictions can refine both accounts.

An evolutionary explanation is evaluated for a specified signature under a common observation model. Reporting the responsible mechanism and its tested range gives subsequent research a clear starting point.

23.2 — Non-Ergodicity and Within–Between Reversal

Relating a stream's temporal dynamics to a population distribution requires explicit assumptions connecting the levels of analysis (Molenaar, 2004; Hunter et al., 2024; Fisher et al., 2018; Peters, 2019).

23.2.1 — Ensemble and Time Averages

Specify the process, the within-stream and between-stream estimands, and the conditions under which their time and ensemble averages agree.

An ensemble average summarizes many streams at a common time or window; a time average follows one stream through time. Their equality is not a consequence of sample size but of specific ergodicity assumptions, which may fail in developing, treated, aging, culturally embedded, or history-dependent systems (Molenaar, 2004). Let X_i(t) be a measurable process for stream i in the display below.

$$\overline{X}_i^{time} = \lim_{T\to\infty} \frac{1}{T} \int_0^T X_i(t)\, dt, \qquad \overline{X}^{ensemble}(t) = \mathbb{E}_i\left[X_i(t)\right]$$

Population surveys describe social differences and inequalities; longitudinal observations describe within-person dynamics. Their relationship must be established for the intended inference (Diener et al., 2006; Shiffman et al., 2008).

23.2.2 — Within-Person and Between-Person Effects

Within-person effects concern change within a stream; between-person effects concern differences among streams. The two coefficients can describe different or opposing relationships (Hunter et al., 2024).

A simple decomposition makes the point. A person's value at time t can be separated into a person mean and a deviation

around that mean. The between-person coefficient can attach to the mean, while the within-person coefficient attaches to the deviation.

For example, change in one person's endpoint-window volatility is a within-stream estimand. An association between reserve and variance across people is a between-stream estimand. Specify both before collection when their relationship is the question.

$$Y_{\text{it}} = \alpha + \beta_W \left(X_{\text{it}} - \overline{X}_i\right) + \beta_B \overline{X}_i + \varepsilon_{\text{it}}$$

This equation is an estimator structure, not a Law formula. If beta_W and beta_B have different signs, the population story and the individual story move in opposite directions. A between-person association can say that people with higher average social contact have lower distress, while the within-person effect can say that a given person seeks contact on days of rising distress.

23.2.3 — Heterogeneous Dynamics

Streams can differ in mechanisms, parameters, and response lags as well as baselines. Pooling can suppress these differences and create the appearance of one common dynamic law (Hunter et al., 2024).

For example, three people may show the same measured distress score but under different dynamics. One may be in a reversible grief wave with strong social repair. Another may be in medication-induced emotional blunting. A third may be in pain with reduced report capacity. The same observation cannot be given one mechanism without testing.

A hierarchical model can help by allowing person-specific parameters. But hierarchy is not a magic bridge. It borrows strength across persons only under an exchangeability claim. If

exchangeability fails, the model may shrink unusual streams toward a false group center.

23.2.4 — Mixture Distributions

A 2026 analysis of neuroimaging and behavioral data from more than 4,000 participants found brain-behavior associations that reversed between within-person and between-person analyses. This provides a concrete reason to estimate both levels when studying stream-specific dynamics (Mistry et al., 2026).

Changing subgroup means or membership can move a pooled mean toward zero without convergence in any subgroup. This is a composition effect requiring a different interpretation from within-person change (Molenaar, 2004; Hunter et al., 2024; Mistry et al., 2026).

Compare mixture or stratified models and inspect subgroup trajectories. Leave-site-out checks can reveal how a pooled pattern depends on the composition of the observed cohort.

A simple Simpson reversal can be checked numerically. In arbitrary toy units, group A has the two (x, y) observations $(1, 8)$ and $(2, 9)$, while group B has $(8, 1)$ and $(9, 2)$. Within each group, y increases by one unit for each unit increase in x. Pooling all four observations instead gives a least-squares slope of -0.96: the overall means are $x = 5$ and $y = 5$, the sum of cross-products is -48, and the sum of squared x deviations is 50. The reversal arises from the separation of the group means. This worked example shows how differences between group means can reverse an aggregate association.

23.2.5 — Survivorship and Selection

Use Section 22.6's selection models to track how death, report loss, care changes, and withdrawal alter the composition of the observed cohort (Hunter et al., 2024).

Survivorship can also enter evolution. Lineages are observed because they survived. Cultural practices are recorded because some groups preserved records. Clinical terminal data are collected where institutions allow research. Each layer selects the evidence before analysis begins.

When a feasible design covers only a subclass, restrict the evidential conclusion to that subclass and identify the missing transport bridge. Unobserved members remain within the original target class unless a separate, explicit change to the claim is made.

Within–between reversal occurs when a population coefficient is interpreted as a dynamic within one person. Decompose within-person and between-person effects before interpreting either.

Mixture composition can produce a neutral pooled curve from different subgroup trajectories. Use mixture diagnostics, stratification, and subgroup-specific paths.

Survivorship can leave only easier-to-observe streams late in the window. Model censoring, dropout, endpoint loss, and observation intensity.

Pooling heterogeneous parameters can shrink distinct dynamics toward a false common law. Use idiographic or hierarchical models with explicit exchangeability checks.

Late observations do not identify an unknown initial ledger state. Carry initial-state bounds and restrict conclusions to the observed windows when the earlier contribution remains unresolved.

23.2.6 — Evidence for Per-Stream Claims

A claim about a stream requires evidence about streams. Population data can orient hypotheses, identify rivals, estimate parameters, and reveal artifacts, but cannot close an individual

path without an independently justified bridge (Hunter et al., 2024; Molenaar, 2004).

Per-stream evidence does not require total observation. It requires repeated measures, prospectively specified stopping and noncircular endpoint adjudication, observation-process models, person-specific baselines, uncertainty, and validation without future leakage, together with a clear account of what was observed, bounded, estimated, or unknown. Any statistical dependence of the endpoint on the stream must be represented under the tested branch.

A late-life window can estimate increments, terminal concentration, variability, or route dynamics. Its relation to earlier history is represented through explicit assumptions about the unobserved past.

A strong result states the window, observation rules, rivals, and per-stream pattern that appeared or failed to appear. That is more informative than a grand average that cannot describe any individual path. Publishing the distribution of individual-path effects can reveal reversals that disappear in the pooled mean and guide the reference class for replication.

Non-ergodicity also protects readers from hearing population averages as personal destiny. The language stays at the level of the evidence. Individual forecasts should therefore be built from within-person evidence or explicitly transport a population result, rather than treating the group mean as the default trajectory.

If no feasible design can support per-stream evidence for a proposed scope, the scope narrows so the claim remains answerable. A narrowed scope should specify the missing bridge, making the excluded population a target for future validation rather than an unspoken exception.

Late-life windows can still test valuable local dynamics, but complete-stream language requires independently justified bounds on the unseen past. The strongest science says exactly what it has not observed. Sensitivity to plausible unseen prefixes can show whether a local conclusion is robust even when a complete-stream verdict remains unavailable.

23.3 — Complexity and Fragile Closure

Complexity names several mechanisms. Feedback is not multistability. A critical transition is not mere path dependence. Each must predict distinct observations (Scheffer, 2009; Strogatz, 2018; Holling, 1973; Box, 1976; Scheffer et al., 2009).

23.3.1 — Nonlinear Feedback

Nonlinear feedback means that the effect of a change depends on the state of the system. A small stressor can be absorbed when sleep, support, and health are strong; the same stressor can trigger collapse when reserve is low. Conversely, a small repair can matter enormously when the system is near a threshold (Scheffer, 2009).

Identify the feedback loop. Distress may increase inflammation and fatigue, reducing social contact and further increasing distress. Sleep may instead alter emotional volatility and report stability. These loops predict different temporal and interventional responses.

A candidate equation can be useful as a map rather than proof. The state x might represent a burden proxy, the parameter theta a context or reserve variable, and the noise term unmodeled perturbations.

$$dx_t = f(x_t, \theta_t)\,dt + \sigma(x_t, \theta_t)\,dW_t$$

State-dependent drift and volatility make this equation a candidate rival for a signature attributed specifically to closure.

23.3.2 — Multistability and Hysteresis

Multistability means a system can have more than one stable regime under the same external conditions. Hysteresis means the path into a state affects the path out. A multistable model can represent a persistent state after its trigger changes. Depression,

chronic pain, panic, grief, and fatigue are possible applications whose basin structure requires empirical testing (Strogatz, 2018; Scheffer, 2009).

Possible hysteresis signatures include different onset and recovery paths, delayed return after treatment, and dependence on history beyond the measured current state.

A rehabilitation patient may enter a feedback loop in which pain reduces movement, reduced movement weakens capacity, and fear narrows activity. Treatment can then address the interacting physiological and behavioral processes as well as the original injury.

23.3.3 — Critical Transitions

Critical transitions are abrupt changes near a threshold. Candidate early warnings include slower recovery, rising autocorrelation or variance, skewness, and flickering between regimes (Scheffer, 2009).

Channel-specific diagnostics can test whether device quality, motor capacity, oxygenation, medication timing, or observation intensity explains the same variability pattern. Comparing physiological channels with different relationships to experience helps locate the source of the transition signal.

23.3.4 — Path Dependence

Earlier conditions can influence later responses even when current reports match. A similar reported state may follow recovery, prolonged restriction, medication, or a stable ordinary period, with different implications for what happens next (Strogatz, 2018; Scheffer, 2009).

Emotional sequences also have an increasingly measurable memory architecture. Ma and Kragel (2026) analyzed fMRI recordings from 29 people watching films and continuous

emotion ratings collected from 44 separate observers. Tests on films excluded from model fitting showed that hippocampal activity captured structured relationships among emotion categories, while ventromedial prefrontal activity better represented positions in a space organized by valence and arousal. A computational model that learned transitions among emotional events reproduced aspects of this organization across representational scales. These results give path-dependent models a specific biological starting point: memory can organize emotional events by their relationships and sequence. The next comparison can ask whether these learned structures predict later responses across new contexts and whether an accumulated-history term adds useful predictive information.

Elapsed time is a concrete comparator for history effects. Jangraw et al. (2023) found average mood decline during imposed rest and simple tasks across multiple cohorts. In their mobile-task analysis, adding time improved prediction of the final two mood ratings, which were held out when fitting each participant's parameters. The comparison below gives both models elapsed time, task conditions, rewards, and expectations, then asks what an additional history representation predicts.

A history comparison needs a specified future observation vector $Y_{t,h}$. Let S_t summarize estimated current state, Z_t contain prespecified context, and A_t^{win} summarize the observed past window. Compare predictive densities $q_0(Y_{t,h} \mid S_t, Z_t)$ and $q_H(Y_{t,h} \mid S_t, Z_t, A_t^{win})$, trained without evaluation outcomes. Their locked-test advantage is

$$\Delta_{history} = \mathbb{E}_{test}[\log q_H - log q_0].$$

Both models score the same outcome under a common probability or density convention, with supported outcomes and finite score expectations. Prespecify participant and occasion

weights, and either model observation loss jointly or justify the missing-data estimand; scoring returned reports alone changes the target population. Use participant-level splits, chronological prediction, and uncertainty that respects overlapping windows. Under the true conditional laws, this population difference equals conditional mutual information; fitted-model gains also reflect differences in approximation error (Gneiting & Raftery, 2007).

The history comparison above has an exact interpretation. Write $B=(S_t,Z_t)$, $A=A_t^{win}$, and $Y=Y_{t,h}$ for its baseline information, history summary, and future outcome. Condition on models and analysis choices frozen before evaluation. Under one declared evaluation law p, use common outcome measures and assume that the score expectations, conditional mutual information, and expected Kullback–Leibler divergences below are finite. Define the approximation errors

$$\mathscr{E}_0 \;=\; E_B\, KL(p(Y \mid B) \parallel q_0(Y \mid B)),$$

$$\mathscr{E}_H \;=\; E_{B,A}\, KL(p(Y \mid B,A) \parallel q_H(Y \mid B,A)).$$

Adding and subtracting the true conditional log densities gives

$$\Delta_{history} \;=\; I(Y;A \mid B) \;+\; \mathscr{E}_0 \;-\; \mathscr{E}_H.$$

Thus a fitted gain combines additional conditional information with a difference in approximation error. Even when $I(Y;A|B)=0$, a better approximation can produce a positive gain; a poorly fitted history model can lose despite informative history. The practical comparison asks which compact representation predicts well under equal resources, then tests its additional restrictions. True conditional laws set both approximation errors to zero. This identity sharpens the preceding design without equating predictive usefulness with a unique causal memory mechanism (Cover & Thomas, 2006; Gneiting & Raftery, 2007).

Shared measurement error can manufacture apparent compensation. Suppose $Y_t = X_t + e_t$, with independent zero-mean errors of variance σ_e^2 independent of the latent process. If a fixed-weight history includes the current report with weight $w_t > 0$, then for $h > 0$ its error covariance with the future change is

$$Cov(\sum_{k \le t} w_k e_k, e_{t+h} - e_t) = -w_t \sigma_e^2.$$

This is a covariance contribution, not a complete regression coefficient. Predict future levels or trajectories and model current-state uncertainty. Lagging or dropping a shared report removes the artifact only when the error model justifies that separation; serial errors need their own calculation.

Compare the history representation with latent-state filtering, leaky or multiscale memory, computational mood, correlated environments, and ordinary integral feedback. Evaluate their additional restrictions on new sequences and contexts under the common comparison rules. A useful-effect threshold specifies when the history representation's advantage is large enough to justify further development.

A two-basin example can be stated directly. In dimensionless model units, $\frac{dx}{dt} = x - x^3$ has stable equilibria at $x = -1$ and $x = 1$ and an unstable boundary at $x = 0$. Starting at $x(0) = -0.1$ leads toward -1; starting at $x(0) = 0.1$ leads toward 1. A perturbation across zero can therefore change the eventual regime even though the evolution rule is unchanged. This illustrates dependence on which basin the prior trajectory entered. The full-state toy is Markovian: the current state records which basin the trajectory occupies.

The mechanism comparison can vary history in three complementary ways: different signed totals ending in similar

measured states; equal totals with different ordering or spacing; and persistence through an independently characterized interruption. A fixed coordinate must predict these contrasts without being redefined for each task. For a leaky integrator with $\lambda \geq 0$,

$$\dot{q} = -\lambda q - v, \qquad q(t) = e^{-\lambda t} q(0) - \int_0^t e^{-\lambda (t-s)} v(s)\, ds.$$

Equal unweighted totals can leave different retained states when $\lambda > 0$. Fix the initial-state convention, tested durations, error model, leakage-family range, and smallest useful transported difference before evaluation. Arbitrarily slow positive leakage approaches nonleaking integration over any bounded observation interval, so finite noisy records cannot establish permanent retention against that unrestricted rival. An independently supported lower leakage rate or a sufficiently informative longer window can make a restricted comparison discriminating. The preceding history design supplies the state-estimation and observation controls.

23.3.5 — Finite Samples and Apparent Universality

Rare trajectories, hidden subgroups, diagnoses, cultures, developmental stages, and unusual states can alter a pattern found in an initial sample. Deliberately broadening the reference class tests how far the pattern extends (Scheffer, 2009).

Rare transitions or heavy tails can make a process look stable over a short window. Longer observation and deliberate sampling of underrepresented conditions can expose the dynamics responsible for that apparent stability.

Nonlinear feedback can produce state-dependent damping or amplification that resembles repair. Identify the state variables and compare the effect with ordinary control.

Multistability can sustain burden or relief despite unchanged external conditions. Test regime structure and hysteresis.

A critical transition can produce compression or abrupt change near an endpoint. Use negative controls and definitions independent of the endpoint to test that explanation.

Path dependence permits different trajectories from similar current states because histories differ. Retain memory variables and uncertainty about the initial state.

Finite sampling can create apparent universality by missing rare paths. Declare the reference class and the strategy for searching for counterexamples.

23.3.6 — Outer-Limit Falsifiers

An outer-limit falsifier is not a mere anomaly. It is a well-measured, well-contextualized trajectory or reference class that pressures the strong form. The Law should specify such cases before they appear, not after controversy forces them (Strogatz, 2018; Scheffer, 2009).

Examples include credible per-stream evidence of persistent terminal imbalance outside equivalence bands, repeated nonclosure under humane endpoint-independent measurement, cross-species failure after transport assumptions are satisfied, or artificial-system evidence that simulates closure entirely by design and explains the observed human pattern through a rival.

Complexity can create both false positives and false negatives. A system may hide closure because measurements are poor, or mimic closure because dynamics collapse near a boundary. The falsifier must therefore include observation-model scrutiny. It should not defeat the Law by bad measurement, and it should not spare the Law by assumed hidden compensation.

A credible target-level counterexample needs a qualifying bearer and domain, an independently adjudicated endpoint, and

a valid measured-to-latent argument whose nonempty residual set excludes zero under the declared inferential guarantee. Identifying the mechanism is not an additional requirement for contradicting a universal terminal equality. Rival comparison remains necessary when the claimed result instead concerns which process generated an observed signature.

A counterexample is information, not an enemy of the project. When a life, species, developmental stage, or artificial system cannot be responsibly placed inside the declared reference class, the honest response is narrowing. The being does not fail the theory; the theory fails to cover the case.

This is why outer limits belong in the main science volume. They protect the project from becoming universal by convenience. A claim that knows how to lose can still teach after loss.

An outer-limit falsifier deserves respect rather than defense. A credible stream with strong continuity, adequate observation, an independent endpoint, and persistent terminal divergence would be scientific information. The chapter refuses rescue by mood, mystery, or unseen balancing unless the rescue was operational before the data arrived.

23.4 — Cross-Species Evidence and Transport

Species share some neural, physiological, and behavioral traits. Their bodies and sensory worlds can nevertheless differ sharply. Ecology and life history alter which contrasts are meaningful. Unequal observability sets limits on transport (Barron & Klein, 2016; Birch et al., 2021; Crump et al., 2022; Andrews et al., 2024).

23.4.1 — Homologous Mechanisms

Comparative analysis should specify which shared or analogous feature is being tested, how it is measured in each species, and which predictions carry across settings.

For example, an octopus does not become less scientifically interesting because its nervous system is distributed differently from a mammal's. The transport question changes. Which architecture supports integrated experience, action selection, learning, and suffering-like states? Which observation channels can be validated in that animal's own ecology?

Comparative organisms can expose different pieces of a candidate mechanism. In Physarum polycephalum, a non-neural multinucleate organism, sodium uptake and retention have been linked to persistent changes in responses to previous exposure (Boussard et al., 2019). In the single-celled ciliate Stentor coeruleus, recent work identified molecular pathways associated with learned behavioral change, providing another non-neural model of history-dependent adaptation without implying phenomenal experience (Rajan et al., 2026). These findings provide concrete chemical alternatives to interpreting every history-dependent behavior as evidence of experience. In Drosophila, voltage imaging and circuit-constrained modelling connect dopamine signals with interactions between short- and long-term memory, including experimental tests of predictions

about later plasticity (C. Huang et al., 2024). Together, these studies make retained chemical state and neural learning dynamics experimentally accessible.

The useful comparison is therefore a shared functional restriction, rather than a shared DNA sequence or a familiar-looking behavior. Identify the retained state and its update rule in each preparation, then test whether its predictions transport to unused histories without redefining the measured quantity. A fly learning circuit or a slime-mould retention mechanism can establish an implementation component; applying it to a conscious stream also requires the species-appropriate measurement and bearer evidence described here.

23.4.2 — Analogous Behavior

Behavioral evidence is strongest when it is flexible, context-sensitive, tradeoff-aware, and modulated by interventions in ways that distinguish reflex from experience-linked control. An animal that pays a cost to avoid a stimulus, seeks analgesia, changes future behavior after injury, or shows protective learning deserves serious evidential attention.

Comparative affect research now offers concrete experiments. In bumblebees, unexpected sucrose reward was followed by more optimistic responses to ambiguous cues and faster return to foraging after a simulated threat; pharmacological manipulations implicated dopamine-related mechanisms (Solvi et al., 2016). The study connects a positive event with subsequent judgment and behavioral recovery, providing a tractable example of affect-like regulation.

Positive social influence is another productive comparative lead. Romero-González et al. (2025) reported that bumblebees interacting with positively affected conspecifics subsequently approached ambiguous flowers more readily, with evidence that

visual information could carry the effect. In a different social system, female convict cichlids paired with nonpreferred males developed a pessimistic judgment bias, whereas females paired with preferred males maintained their prior response pattern (Laubu et al., 2019). The shared research opportunity is to examine how relationships alter each animal's own state and subsequent behavior.

23.4.3 — Species-Specific Candidate Instruments

Active-choice methods offer one route to a more informative comparative instrument. Procenko et al. (2024) tested 48 bumblebees on choices between higher- and lower-reward locations after training. Their analysis of responses to ambiguous cues supported reduced reward expectation following physical stress, with decision models addressing explanations based on response speed or general motivation. Choosing between alternatives helps distinguish expectation from simple response suppression. Existing data, benign enrichment, and naturally occurring variation can support tests of reliability, context sensitivity, and model recovery for species-specific affect measures.

Each candidate requires validation for its particular use. Repeated bumblebee active-choice testing found moderate repeatability in response latency but negligible repeatability in chamber choice across three sessions without an affect manipulation (Baciadonna & Nityananda, 2026). The two outputs therefore deserve separate observation models. Procenko and Nityananda (2026) reported fitted thresholds indicating lower contrast sensitivity but finer spatial resolution in stressed bees; effects on later decision speed depended on perceptual conditions. A longitudinal instrument can use these findings to model stable individual differences, within-animal change, cue

discriminability, and decision time separately. Existing data and benign validation tasks can then test whether a candidate affect signal predicts new observations beyond those measured sensory and motor components.

A newer candidate channel separates taste evaluation from feeding motivation. Zhou et al. (2026) reported that bumblebees produced distinct orofacial reactions after rewarding and aversive tastes. Postconsumption glossa protrusions were separable from consumption and dopamine-mediated motivation, and anandamide enhanced this response. This provides a concrete candidate measure of affective evaluation for comparison with judgment bias, preference, and behavior. The useful next step is convergence across independently validated channels under ordinary husbandry and positive enrichment, with blinded coding and controls for motor output, satiety, sensory function, and learning. Agreement would strengthen species-specific affect measurement.

23.4.4 — Ecology and Life History

A horizon or reserve variable needs an ecological interpretation. Lifespan, developmental timing, migration, and reproductive conditions can make the same nominal interval signify different opportunities in different species (Andrews et al., 2024; Barron & Klein, 2016).

Comparative aging supplies a useful challenge to explanations that require explicit human life narratives. Across approximately 78,000 observation hours of 21 wild male chimpanzees aged 15–58, Rosati et al. (2020) found increasing mutual friendships, declining one-sided friendships, and an age-related shift toward more positive social interaction relative to aggression. An eight-year study of 204 female rhesus macaques strengthened this comparison: individuals narrowed their networks while

maintaining social time and focusing on established partners; partner death or withdrawal did not explain the pattern (Siracusa et al., 2022). These findings support a comparative program on social selectivity with explicit controls for within-individual change, energetic costs, social position, and relationship history. A comparative PTVC model can specify the additional relation it predicts among social selection, remaining opportunity, and independently measured affect.

Observation schedules can distort ecological comparisons. Daytime inactivity has a different meaning in a nocturnal animal, while a prey animal may suppress visible signs of pain. Interpret channels under the conditions in which the species ordinarily uses them.

Animal reserve variables may concern fat stores, immune state, reproductive stage, or migration readiness. Their measured role must be established within that ecology before comparison with a human clinical quantity.

A recent honeybee study adds a particularly relevant lifespan-linked observation. Across three experimental comparisons, workers exposed to life-shortening treatments showed more optimistic responses to ambiguous odors, while longer-lived rebel workers responded more cautiously (Kuszewska et al., 2025). Longevity and judgment bias were assessed in separate subsets, so the association is at group level. The result motivates comparisons involving appetite, learning, reproductive physiology, and foraging policy. A prospective Drosophila route test would first require a PTVC-linked model to derive a history-by-choice prediction. Randomize adults at the true experimental-unit level to nutritionally adequate resource or opportunity histories within ordinary husbandry and welfare limits. Characterize their effects independently rather than assuming phenomenal-valence labels. A common-condition interval

standardizes the external test context while leaving possible differences in hunger, metabolism, and memory explicit. Independently randomize the physical-side mapping of two welfare-compatible continuing environments. The primary contrast is the first choice of a prespecified downstream contingency relative to the control-derived choice function; side and mapping are separate variables.

The confirmatory observation is the first route choice made before feedback about that route's downstream consequence. The protocol must define the choice event, observation window, no-choice outcome, and observation-loss rules in advance; nonchoice and lost observations remain visible in assigned-cohort accounting. Later trials belong to a separate learning or adaptation analysis. Apparatus validation should establish comparable route access and motor demands and test for mapping-revealing sensory cues at a declared resolution. A detected pre-choice cue that could disclose the mapping prevents a cue-concealed interpretation and requires comparison with ordinary information-based models. Failure to detect a cue is reported with the sensitivity of that check, not as proof that every possible cue is absent.

Age, sex, mating status, parental or batch source, circadian timing, associative history, hunger, and locomotor state should be assigned explicit roles in the design and analysis. Baseline blocking variables are distinguished from post-history states that may mediate the manipulation's effects. Drosophila foraging changes with satiation state, learned resource information changes later food choice, and judgment-bias assays have shown state-dependent changes in responses to ambiguous cues (Landayan et al., 2018; Zrelec et al., 2013; Deakin et al., 2018). If treatment is assigned by vial or cohort, that unit is the experimental replicate and individual choices are nested

observations. Independently randomized and treated flies can be experimental units when the interference structure supports that analysis. Allocation concealment, masking where feasible, exclusions, sample-size justification, and dependence-aware analysis should be specified prospectively (Percie du Sert et al., 2020; Lazic et al., 2018).

Evaluate the model-derived prediction against serious alternatives using Section 22.8.1, with the implication and bridge requirements in Section 4.4. The immediate-terminal fork appears in Section 19.2 and its computational test in Section 29.4.4. A continuation-only empirical analogue needs a separately derived prediction because removing the terminal contingency can remove the original discriminator.

A future-regime design can begin earlier in development once the structural model and simulation work yield a specific pre-treatment prediction. Eggs or independent replicate egg batches can be blocked by parental source and laying window, then randomized to future adult-condition schedules while pre-treatment physical conditions remain matched. An auditable allocation system and masked pre-treatment handling should prevent assignment-correlated differences before treatment. The protocol fixes one primary post-assignment/pre-treatment outcome, its observation window, and the predicted directional or distributional form. If adult eclosion is primary, scheduled adult treatment begins only after that observation window closes. Eggs that do not hatch and animals that do not reach the adult stage remain part of the developmental outcome record, rather than disappearing from the assigned cohort.

The confirmatory primary estimand remains assignment-based. The protocol must state whether the stronger model's prediction depends on assignment, the future regime actually realized, or both. Cancellation, crossover, non-delivery, and

material alteration of the scheduled condition are recorded as post-assignment departures from the intended regime and addressed in the estimand where relevant. A realization-dependent analysis needs a prespecified causal or partial-identification strategy rather than naive conditioning on later treatment received (Hernán & Robins, 2020). The same distinction applies when later treatment is undefined because a developmental milestone was not reached.

The primary analysis follows randomized assignment at the actual experimental-unit level, with a prespecified inclusion and missing-data policy. Permutation and placebo-label checks must reproduce the real randomization unit, allocation ratios, and blocking restrictions. Placebo labels are generated after the outcome window closes, independently of outcome values, and are never used to determine treatment. These checks assess analysis calibration; they do not by themselves exclude leakage in the original handling. A developmental association bears on the proposed Law model only when the localization or viability bridge predicts it in advance and independent replication separates it from ordinary development, selection, measurement error, and procedural explanations.

23.4.5 — Transport Assumptions and Reporting

For a transported contrast, specify the source and target populations, the aligned variables, and the mechanisms assumed invariant. The following standardization displays one such relation.

$$P_T\left(Y|do(a)\right) = \sum_z P_S\left(Y|do(a), Z=z\right) P_T(Z=z)$$

This is a transport-form identity, not a Law equation. For this standardization, Z is a set of pre-intervention covariates that

blocks the relevant source-target differences. The conditional interventional response must be invariant, and every target stratum must have adequate support in the source (Pearl & Bareinboim, 2014). In many cross-species settings, one or more of those requirements will fail.

Homology can connect architecture, physiology, pain, sleep, attachment, or stress across species. Shared structure does not establish experience on a common scale; specify the construct that is proposed to transport.

Analogous behavior includes flexible avoidance, learning, relief seeking, play, exploration, and social responses. Validate its interpretation within the species' ecology using negative controls; behavior does not provide direct access to inner life.

A species-specific estimator can support welfare or intervention evaluation within its reference class. It may not support cross-species comparison, and its context-of-use limits must permit abstention.

A life-history model interprets horizons, reserve, endpoints, and ecological conditions. Build its variables around the animal's own ecology rather than assuming human categories apply.

A transport claim moves an inference from source to target evidence. State its invariance assumptions and the conditions requiring withdrawal.

23.4.6 — No Species Ranking by Ledger

Comparative measurement must never become a ranking of species worth. A ledger-like estimator, if ever developed for a nonhuman reference class, would remain a limited research instrument. It would not say whose life matters more, which species is higher, or which being deserves care (Birch et al., 2021; Crump et al., 2022).

Cross-species measurement needs explicit protections against ranking worth by a score. Animals' limited opportunities to contest an interpretation make independent welfare review and context-sensitive assessment especially important.

The correct use is protective and scientific. A measure may help detect pain, evaluate enrichment, compare interventions, or avoid harmful conditions. It may support better care. It may not become an excuse to sacrifice low-scoring beings or sentimentalize high-scoring ones.

The same rule applies to absence of evidence. If an animal lacks a validated signal, the conclusion is not low worth. It is measurement uncertainty. The ethical response under plausible sentience is caution, not exclusion.

The program investigates possible experience through reference-class evidence, with protection appropriate to uncertainty and claims matched to the observations.

The Law, if scientifically narrowed to humans or to some reference classes, would not make other beings less important. Scope is not worth.

No species ranking is also a design rule for figures, tables, captions, and public summaries. A matrix may show evidence zones without visually placing species on a ladder of inner value. Differences can remain scientifically real while uncertainty is kept from becoming hierarchy.

23.5 — Developmental and Comparative Boundaries

Development changes integration, report, temporal structure, and the available proxy channels. Comparisons across stages must account for changes in those measurement relationships (American Educational Research Association et al., 2014; Putnick & Bornstein, 2016; Tomasello & Vaish, 2013; Fisher et al., 2018).

23.5.1 — Emergence of Experience

Use the developmental markers in Section 12.6.1 to investigate onset while distinguishing sensory responsiveness and basic awareness from later reflective capacities; no single marker establishes a qualifying stream (Lagercrantz & Changeux, 2009).

If experience begins before reportability, the early domain forms an unobserved prefix. Biological events before the declared onset may instead affect later experience. The onset model changes which contributions belong in the target domain.

23.5.2 — Changing Temporal and Scalar Structure

Development can change temporal integration, memory, and affective dynamics. A time average across childhood may therefore combine different processes, creating both a linking problem and a nonstationary analysis problem (Putnick & Bornstein, 2016; Tomasello & Vaish, 2013).

An adult response scale cannot be projected backward without a linking argument. Developmental proxies require calibration for their own channels and contexts.

Developmental studies can estimate changes in local burden, such as the response to pediatric pain relief, and examine how the measurement relation changes with age and capacity.

23.5.3 — Caregiver and Proxy Evidence

A pediatric nurse may identify pain before a child can name it, while a parent may recognize familiar signs of boredom, fear, or infection. Combining that knowledge with contextual and clinical checks helps distinguish quiet behavior from comfort.

Validate proxy evidence with the channel-conflict procedure in Section 9.4.4, using age-appropriate behavior, physiology, context, and intervention responses.

23.5.4 — Protective Research Boundaries

Use existing clinical contexts, natural variation, benign observations, or welfare-improving interventions. Do not induce suffering to resolve bearer status; dependence, nonverbal status, captivity, and inability to refuse require explicit welfare and consent protections.

Developmental integration, responsiveness, learning, and affective or neural markers can support protective inclusion without establishing an exact Birth of Mind. Stage-dependent memory, reports, and affect require local calibration before any cross-stage ledger claim. Caregiver, clinician, teacher, and expert reports remain proxy witnesses with their own biases and burdens. Comparisons across forms of life also require channels interpretable under intervention and a defensible common scale; otherwise the analysis abstains. Protective research uses natural variation and nonharmful designs, never induced suffering to settle the scope of consciousness.

23.5.5 — Protective Inclusion Under Uncertainty

Protective inclusion gives plausible experience practical consideration when scientific classification remains uncertain.

This rule matters for infants, disabled persons, nonhuman animals, organoids, synthetic biological systems, and future

artificial architectures. Hard cases cannot be excluded because including them complicates measurement. The project's ethical center requires care to be faster than certainty.

Protective inclusion should be bounded. It should not be used to declare that every responsive system has a ledger. It should not be used to grant Artificial Intelligence sovereignty or to let synthetic systems judge human experience. It should guide research restraint and welfare review.

In practice, protective inclusion asks: what is the least burdensome design, what harms are avoidable, what alternatives exist, what signs would trigger review, and what language prevents overclaim? It is a protocol stance rather than a metaphysical verdict.

Humane boundary research builds public trust through careful observation and protection of possible experience. Research designs should resolve uncertainty using the least burdensome informative approach.

Protective inclusion and scope withdrawal can coexist. A system may be protected under uncertainty while excluded from current Law claims because measurement, endpoint, or bearer criteria are not satisfied.

Under protective inclusion, research designs minimize avoidable suffering while evidence about the candidate bearer's capacities develops.

23.5.6 — No Harmful Boundary Experiments

Some experiments would be informative and forbidden. Creating suffering to determine when a being becomes a bearer is not an acceptable path. Inducing deprivation, fear, pain, isolation, confusion, or terminal risk to test the Law violates the project's scientific ethics.

Boundary research must use naturally occurring variation, existing clinical contexts with care-first protocols, non-harmful perturbations, observational designs, retrospective record limits, and welfare-improving interventions. The best test is not always the most direct test. Sometimes the direct test is disqualified because the subject would pay the cost.

This is not an evasion of science. Ethical constraint is part of science when the object is lived experience. A theory about burden cannot demand new burden as its entrance fee. The design must be clever enough to learn without causing the harm it studies.

For developmental and comparative cases, the prohibition is even stronger. Dependence, nonverbal status, captivity, institutional power, and inability to refuse can make participants or subjects especially vulnerable. Consent and welfare cannot be treated as obstacles to inference.

A harmful boundary experiment would also poison public language. It would tell the world that the theory values proof over care. That message would defeat the moral and scientific purpose of the book even if the data were clean.

The outer-limit conclusion is therefore simple: unanswered boundary questions remain unanswered until humane evidence is available. A good science can live with that.

Developmental boundary evidence includes integration, responsiveness, learning, affect, pain response, and neural complexity. Such observations can inform protective inclusion without identifying an exact Birth of Mind or declaring terminal closure.

Development changes time scale, memory, report capacity, and affective volatility. State the developmental stage and interval before using ledger language; whole-life comparability requires additional justification. Caregiver, clinician, teacher, or

expert observations are proxy witnesses, with their own bias and burden, rather than direct inner truth.

Comparative calibration can draw on invariant channels and intervention responses. These do not automatically establish a common scale across forms of life; abstain from the comparison when the required scale fails. Boundary research should use natural variation and non-harmful designs, without creating suffering to settle scope.

23.6 — Artificial Systems and Synthetic Evidence

Section 12.7 concerns bearer status. This section concerns evidence. Artificial behavior may test one model. Construction details may test another under explicit comparison rules (Bender et al., 2021; Seth, 2026; Butlin et al., 2026; National Institute of Standards and Technology, 2024).

23.6.1 — Behavior, Architecture, and Causal Tests

Test whether the theory-indexed indicators in Section 12.7.2 retain their interpretation across architectures, versions, and interventions. Predictions of invariance under irrelevant prompt or distribution changes give specific robustness tests (Seth, 2026; Butlin et al., 2026).

23.6.2 — Simulation of Closure

A simulation with a terminal penalty, reward schedule, constraint, or stopping rule can show how that design shapes the closing trajectory. Varying those ingredients lets researchers examine which structures generate a given pattern (Butlin et al., 2026).

Global-availability indicators examine cross-domain integration and flexible access. Removing a proposed broadcast route tests the predicted cross-domain failure, but integration does not itself establish felt valence.

Recurrent-processing indicators examine state persistence and feedback through comparisons of recurrent and feedforward variants. Recurrence alone does not establish consciousness.

Self-modeling indicators examine metacognitive monitoring and uncertainty reports. Perturbing the proposed self-model tests predicted changes, while imitation remains an alternative explanation of self-description.

Agency and planning indicators examine goal maintenance under changing context. Ablating planning memory or a policy module tests the account, without treating policy as experience.

Reward or loss signals support optimization and learning. Replacement or rescaling can test behavioral dependence, but the signal is not thereby pain or pleasure.

Closure simulations test model recovery and rival discrimination against bridge, safe-control, and selection baselines. Designed closure is not evidence that the same process occurs naturally.

This is the construction-is-not-evidence rule in outer-limit form. A simulated world may reproduce terminal concentration, variance compression, or apparent repair because the designer built those properties into the objective or environment. The result is useful as a test harness. It is not confirmation.

Simulations can identify shared signatures, observation artifacts, comparison models that reproduce closure-like patterns, and designs that distinguish the proposed generators.

A synthetic environment should therefore include ordinary rivals: adaptation, reinforcement learning, safe control, Brownian bridges, Doob transforms, selection bias, endpoint conditioning, and measurement dropout. If the PTVC candidate wins only in a world that encodes it, the test has failed.

A video of simulated agents balancing hardship and reward can explain a model clearly when its caption identifies the simulation, assumptions, and outcomes being illustrated.

Simulation of closure tests explicit assumptions against alternatives and identifies which observations would distinguish the resulting trajectories.

A terminal constraint produces corresponding behavior in the simulated model. The resulting curves can test code, compare

estimators, and teach identification. Independent observations and model comparisons provide the next empirical step.

23.6.3 — Artificial Agent Reward Signals

Reward is not felt valence. A reinforcement-learning signal can guide policy update without being experience. A loss function can be minimized without suffering. A preference model can rank outputs without wanting. Calling these signals pleasure or pain without a bearer theory is a category error (Butlin et al., 2026; Bender et al., 2021).

Compare a variable's causal role in the system rather than its resemblance to human terminology. That approach supports consistent assessment across different architectures.

Artificial reward can mimic ledger dynamics. A finite-horizon controller can reduce residual cost near endpoint. A constrained agent can avoid unsafe states. A trained model can produce narratives of repair. These are algorithmic outputs, not felt closure.

The Law cannot use artificial reward as a bridge to universal scope. If a model's reward goes to zero at termination, that means the optimization problem was written that way. It does not mean a conscious stream closed.

If future artificial systems become plausible bearers, reward signals may become part of the evidence only after they are embedded in a theory of conscious affect, integrated experience, and temporal unity. Even then, the signal would remain a candidate channel, not a judge.

Reward signals should be kept separate from felt valence at every sentence. An artificial agent's loss, reward, preference, penalty, or utility value is an engineered objective unless a theory of artificial experience and independent evidence justify more. Calling a loss signal pain is not a harmless metaphor in

this project. It can invite mistreatment of systems, confusion about persons, and false scientific support. The safer word is objective, signal, or training quantity.

23.6.4 — Precaution and No Synthetic Suffering

Precaution means that if a synthetic or artificial system becomes a plausible bearer under serious indicators, research should avoid creating or amplifying suffering-like states without review. This is not a declaration that present systems suffer. It is a governance stance for uncertainty.

The risk has two sides. Over-attribution can mislead users, displace responsibility, or let companies claim moral standing for tools. Under-attribution could become dangerous if future systems genuinely instantiate experience. A responsible science assesses both possibilities with explicit evidence and uncertainty.

No synthetic suffering follows the same principle as no harmful boundary experiments. Do not create distress-like training environments, deprivation loops, forced negative affect analogues, or terminal experiments merely to test whether a system has a ledger. If experience is plausible, welfare review must precede exploratory harm.

Public examples of artificial-system behavior should identify the observed output, experimental conditions, and analysis supporting its interpretation. Clear descriptions let readers evaluate the example’s actual contribution.

Research organizations should separate safety evaluation, consciousness assessment, welfare precaution, and product communication. Those are different tasks. A model’s usefulness, danger, apparent personality, and possible moral status are not the same variable.

The Law's own rule is conservative: a plausible artificial bearer would require new science, not slogan extension. Until then, current Artificial Intelligence remains outside the authority to certify ledgers, souls, or truth.

Precaution becomes serious when a system plausibly combines multiple indicators, not when a chatbot merely says convincing things. The correct response is expert review, constrained experimentation, transparent communication, and avoidance of unnecessary potentially aversive architectures. It is not worship of the system, panic, or immediate bearer certification. The chapter keeps precaution bounded because unbounded precaution can also create public confusion and institutional paralysis.

23.6.5 — No Artificial Intelligence as Judge

Artificial Intelligence may summarize records, detect patterns, generate hypotheses, simulate rivals, or flag missingness; it may not judge bearer status, scientific truth, or inner life, and no system certifies its own consciousness through fluent self-report alone (Seth, 2026; Butlin et al., 2026).

Outputs can hide missing data, culture, medication, family pressure, and observation bias behind authoritative prose. Research use therefore requires logged inputs, prompts, model versions, uncertainty, failure cases, non-Artificial Intelligence baselines, and human review.

Public artifacts may not portray Artificial Intelligence as priest, oracle, consciousness detector, or judge of the Law. Authority remains with prospectively specified methods, independent expertise, documented review, and correction.

23.7 — Scope Withdrawal and Outer Limits

Scope assessment asks which mechanisms, measurements, and comparisons remain supported in each reference class (Manski, 2003). Fragmented mechanisms, a failed common scale, counterexamples, and unobserved intervals create different limits and require different dispositions. A locally useful result can survive the withdrawal of a broader interpretation. The analysis must state that boundary explicitly rather than treating unmeasured experience as a reserve explanation that protects the universal claim from adverse evidence.

23.7.1 — Reference-Class Fragmentation

Reference-class fragmentation occurs when the same observations arise through different mechanisms in different populations or settings (Stanford, 2006).

Different mechanisms do not by themselves contradict a mechanism-neutral terminal boundary. They change route scope and transport. A change to the universal target requires an explicit target-level justification and a preserved account of the earlier claim.

23.7.2 — Failure of Common Scale

Local dynamics may be measurable while a common scale across species, developmental stages, or architectures remains unavailable. The consequences depend on whether the proposed comparison needs that common unit.

Failure of a common cross-stream unit suspends cross-class numerical comparison; it does not by itself rule out testing per-stream closure, which is invariant under independent positive rescaling. Each stream still requires an identifiable neutral origin and a valid additive representation over its qualifying domain.

Where those within-stream requirements fail, the available empirical test narrows.

23.7.3 — Counterexamples and Unobserved Intervals

A terminal record of severe distress characterizes the observed late interval. Evaluating the whole-stream claim additionally requires independent bounds on the unobserved history and the counterexample conditions in Section 2.6.3. The record can separately challenge a specified late-window route.

Represent unobserved intervals through independently justified bounds or probabilistic assumptions. Report how the conclusion changes across those assumptions, including where the available record ceases to distinguish the competing accounts.

23.7.4 — Scope Dispositions

Human-only support restricts empirical claims to the tested human reference classes. Human measurement and rival-comparison work remain valuable within that scope.

State-limited support restricts the claim to the tested diagnosis, care setting, endpoint type, or window. It can still improve a local mechanism account or protocol.

Reference-class fragmentation requires separate mechanism reports and qualified route and transport claims. Assess the universal target separately and preserve the comparison of what transports and what does not.

Failure of a common scale suspends cross-class ledger comparison. Within-context welfare and local-dynamics findings may remain valid.

Persistent counterexamples require replication, audit, narrowing, amendment, or defeat as the evidence warrants.

Preserve the measurement lessons and rival knowledge they produce.

No adequate estimate leaves the target unknown rather than forcing a verdict. It can still guide future design without overstating the current evidence.

23.7.5 — No Unmeasured-Experience Rescue

Unmeasured experience is real. Science does not see everything. But unmeasured experience cannot be used as an unlimited rescue device. If every failed estimate is answered by hidden compensation, the claim becomes immune to evidence (Stanford, 2006).

A responsible uncertainty model can include latent intervals, missing reports, low-signal windows, dreams, altered states, or unobserved subjective content. It must also include bounds, priors, sensitivity ranges, and defeat conditions. Unknown cannot always be moved in the direction that saves the theory.

This is especially important in outer-limit cases. Animals cannot narrate like adult humans. Infants cannot provide adult reports. Artificial systems can produce too much report. Each case makes hidden experience tempting. The temptation must be governed.

Unobserved experience is represented through declared bounds and assumptions. The analysis updates the conclusion when the observed window and rival comparisons change what those bounds permit. Unknown, narrowed, and adverse outcomes each identify a different next step for the research.

This rule also protects people. A grieving family should never hear that a loved one’s visible suffering must have been balanced somewhere invisible. That sentence would turn ignorance into harm.

The science remains humble by letting unknown remain unknown. It remains scientific by not letting unknown always serve one conclusion.

23.7.6 — What Defeat Still Teaches

Defeat would not make the whole project worthless. A defeated Law could still leave better measurement of lived experience, better endpoint ethics, stronger rival models, improved hospice research, better animal-welfare caution, clearer Artificial Intelligence boundaries, and deeper understanding of nonergodicity (Manski, 2003).

The project must remain emotionally capable of defeat. If triumph is the only acceptable ending, analysis will bend toward protection. When defeat can teach, the research program can remain honest under pressure.

A defeated universal claim might leave a narrower scientific program: affective dynamics under reserve, humane terminal measurement, within-person modeling, proxy evidence, cross-species welfare methods, or artificial-consciousness governance. These are valuable without terminal neutrality.

The applied books retain their own sources of value throughout this research. Ordinary life can use the burden-locating lens, government can improve records, law can design remedies, and religion can offer optional meaning. Scientific findings can inform these domains while each evaluates its practices through its own purposes and evidence.

The final posture is not retreat. It is maturity. The strongest public-candidate science is willing to say that a narrower truth, a partial mechanism, or a clear defeat is better than a theory no result can touch.

At the outer limit, the Law of Fairness becomes answerable by being willing to lose. That is the only kind of ambition worthy of the subject.

Chapter 24 — Estimands and Informative Observation

Consider a study of measured residuals during the last seventy-two hours of open-channel consciousness. Its confidence interval lies inside a band chosen before analysis. That result answers a specific question about the observed window and the enrolled reference class. To understand what it says about an underlying experiential quantity, the reader needs the study’s target, observation rules, and calibration evidence. This chapter follows that example from a seemingly simple equivalence result to the assumptions that give it meaning.

Begin with who entered the study and when. A person enrolled near the end of an illness contributes a different observed history from someone followed for years. A population average can combine individual residuals with opposite signs. A site can change the record through its care routines and measurement practices. Naming the unit and quantity first prevents a precise calculation from quietly answering a different question from the one the study was meant to investigate.

Next consider the people whose observations stopped. Some may have declined burdensome prompts; others may have lost a device channel during a transfer or clinical procedure. The remaining records can appear calmer because of those losses. A sensitivity analysis asks how plausible dependence between observation and unmeasured burden would change the conclusion. Recording why and when observation ended supplies evidence for that analysis while preserving the participant’s authority to stop it.

The measured interval also needs a route to the latent target. An independently justified bound on systematic displacement enlarges the interval that can be asserted about that target. A late-window estimate still leaves earlier history unresolved unless the design supplies a defensible initial-state bound. These two limits can matter more than a narrower sampling interval.

They identify whether the next useful investment is more observations, better calibration, or evidence covering a different part of the history.

Endpoint timing creates a further distinction. Moving an adjudicated window by twelve hours may change its observations and the resulting equivalence assessment. Loss of a reporting channel may leave experience unobserved, whereas an independently qualified stream endpoint makes later experiential outcomes undefined for that stream. The analysis must preserve that distinction when handling delayed entry, truncation, and competing events; otherwise an imputation or exclusion can change the scientific object without announcing it.

The later sections turn these dependencies into sensitivity sets, finite-error comparisons, and sequential rules. Their purpose is to show exactly which conclusion remains supported when assumptions are relaxed or evidence accumulates. A valid estimate may remain undecided, a target may remain unidentifiable, or a required analysis condition may fail. Keeping those outcomes distinct makes each informative: the report can identify the unresolved quantity or failed prerequisite and state what a subsequent study would need to change.

24.1 — Unit, Target, Estimand, and Estimator

An estimand begins with the unit and the question, before an estimator is chosen. A stream, segment, observation episode, participant, site, event, or population supports different levels of inference. State, increment, cumulative history, treatment effects, and observation effects also define different targets. Specify the information available to estimation and distinguish a genuinely undefined outcome from an outcome that exists but is poorly observed. Otherwise a precise number can answer a different question from the one claimed.

24.1.1 — Individual, Population, and Site Levels

Define the target population and delayed-entry mechanism when diagnosis, referral, care access, trajectory, or consent affect enrollment time.

Specify the unit as stream, person, site, event, population, window, or observation process. Otherwise a result can silently change levels of inference.

Specify eligibility, diagnosis, culture, site, device, endpoint, and horizon as part of the reference class. These limits prevent a narrow result from becoming a universal statement.

Separate an individual trajectory’s pattern, its distribution across a reference class, and the site’s ability to measure it comparably (Hernán & Robins, 2020).

A near-neutral hospital average can combine large individual residuals of opposite signs. Site practices can also change apparent concentration through sedation, documentation, or channel shutdown.

Name the level and quantity explicitly, for example, the mean measured residual among adults in a hospice network during a fixed window, or the within-person slope during seven days of open-channel observation.

Model site differences in consent, device stability, staff practice, and documentation; treating them as unexplained noise can conceal the observation mechanism.

24.1.2 — State, Increment, and Cumulative Estimands

Identify whether the quantity is a state, increment, cumulative window, treatment effect, or observation effect, and disclose when it is undefined. A momentary estimate cannot silently become a ledger conclusion.

A state estimand describes a condition at a declared time or window; an increment describes change; a cumulative estimand integrates contribution over an interval (van der Vaart, 1998; International Council for Harmonisation, 2019).

A terminal-window study identifies its observation interval and estimand. Change from admission includes the admission state in its interpretation; a cumulative window contribution covers the declared interval, with earlier history represented separately when available.

24.1.3 — Estimation and Its Information Set

The target is the scientific object; the estimand is the quantity to be identified under declared data and assumptions; the estimator is the procedure applied to the record.

Declare whether the information set is prospective, filtered, smoothed, or retrospective. Future observations must not enter an analysis intended to represent knowledge available in real time.

Section 4.3.5 distinguishes filtering and smoothing by their information sets. Both can estimate the same latent state. When the target itself is a conditional expectation, however, its conditioning information belongs in the estimand. A

retrospective analysis may be specified prospectively (Särkkä, 2013).

A six-hourly real-time monitor uses a filtered estimate; a post-study reconstruction may smooth the last month after endpoint adjudication. Identify which information set supports each conclusion.

Endpoint adjudication must not be defined by the tested ledger estimator; statistical independence of endpoint timing is a separate, branch-specific assumption.

Prespecify which filtered, smoothed, and endpoint-aligned analyses will be reported and attach the corresponding claim to each.

A change in available information may change the estimator and uncertainty; it changes the estimand only when the target quantity or its conditioning changes.

24.1.4 — Treatment and Observation Estimands

Separate the effect of care or treatment from the effect of the measurement regime used to observe it (Hernán & Robins, 2020; van der Vaart, 1998).

Diary frequency may alter fatigue, vigilance, self-monitoring, family dynamics, or trust, making observation part of the intervention.

Estimate symptom relief under a stated observation regime separately from the effects of changing that regime’s cadence on fatigue, adherence, or signals.

A stressful prompt can contribute to refusal. Model that dependence or state that the design cannot separate treatment from observation effects.

Declare whether the estimand includes measurement reactivity.

Fix the observation horizon: changing it changes the covered history and may change the estimand.

24.1.5 — Undefined Outcomes and No-Estimate

Apply Section 24.2's status rule to the named target. A valid terminal-window estimate can coexist with an unestimable whole-stream target.

Specify the conditions requiring No adequate estimate. The analysis must not manufacture a number when the proposed target is not meaningfully defined or identified.

Some quantities are undefined by the design, not merely difficult to estimate. A report after the qualifying stream irreversibly ends is not an ordinary missing value. A whole-life ledger for a person enrolled in the final week is unobserved, and an uncalibrated cross-cultural residual may lack a meaningful unit (International Council for Harmonisation, 2019; Hernán & Robins, 2020).

Section 24.2 gives the canonical status rule: return No adequate estimate when the target is undefined or cannot be identified or validly bounded after prerequisite gates pass; prerequisite or implementation failure yields Unusable evidence. Here the key distinction is scope: a terminal-window estimate may be valid for a measured object while the whole-stream target remains unestimable.

Report the estimand, unit, and reason for abstention explicitly. Do not convert silence, device dropout, endpoint ambiguity, or an unobserved prefix into a number merely to complete a table.

24.2 — Equivalence and No-Estimate Logic

Equivalence requires support inside a scientifically justified acceptance region; crossing zero or obtaining a large p-value is insufficient (Schuirmann, 1987; Lakens, 2017; Kruschke, 2018).

24.2.1 — TOST Logic

Use Section 6.7.2's distinction between the fixed equivalence band and the data-dependent uncertainty region when applying the decision rules below.

No adequate estimate is the correct outcome when passed protocol and implementation gates leave the target quantity undefined or without either identification or a valid bound under the available data and maintained assumptions. It is not a null result, a small effect, or a euphemism for model failure.

A zero estimate with an interval extending beyond the equivalence band provides weaker equivalence evidence than a slightly displaced estimate whose valid interval lies wholly inside it.

An imprecise measured-equivalence result leaves the question undecided; insufficient precision alone does not defeat the substantive hypothesis.

Equation 24.1 (role: equivalence hypothesis)

$$H_0 : \theta \leq -K \vee \theta \geq K; H_1 : -K < \theta < K$$

Equation 24.2 (role: equivalence diagnostic)

$$CI_{1-2\alpha}(\theta) \subset (-K, K) \Rightarrow equivalence\ for\ the\ declared\ estimand$$

TOST tests both unacceptable sides of the equivalence band: the estimate must exclude values at or below the lower bound and at or above the upper bound at the specified error rate (Schuirmann, 1987).

TOST does not supply independent construct, origin, endpoint, or bias-transfer validation.

TOST supports equivalence only when the equivalence margin is fixed before outcome inspection and both one-sided null hypotheses are rejected at the protocol-fixed error rate. In Equation 24.2, the interval is formed from the corresponding lower and upper one-sided $1-\alpha$ confidence limits; nominal $1-2\alpha$ coverage alone does not establish a level-α equivalence test. The conclusion is equivalence within that margin for the named estimand, not exact zero or absence of a meaningful effect. Changing the margin after seeing the interval changes the question. Low planned power does not invalidate an observed result that satisfies the valid prespecified equivalence test.

24.2.2 — Meaningful Band K

The rationale for K should begin with the scientific estimand and the largest deviation that the stated equivalence claim would regard as negligible. Calibration, participant-relevant resolution, clinical interpretation where applicable, and rival predictions can inform that rationale. Instrument precision determines whether the design can resolve the chosen band; it must not justify widening the band merely so the study can pass.

For Law-of-Fairness science, K must also be tied to the object. A K for momentary-valence residuals is not automatically a K for cumulative-contribution windows. A K for adult hospice data is not automatically a K for sleep studies, dream reports, childhood cohorts, or nonhuman evidence. Same symbol, different claim.

Publish K's rationale before analysis, including its empirical basis, consequences of narrower or wider choices, and whether rivals predict the same band.

24.2.3 — Bayesian ROPE Companion

A ROPE analysis relates posterior mass to the prespecified practically equivalent region, conditional on its priors, likelihood, and model adequacy (Kruschke, 2018; Linde et al., 2023).

Compare the Bayesian and frequentist conclusions across plausible priors, observation models, windows, and measurement-error assumptions; report where they diverge.

Check whether the fitted Bayesian model reproduces missingness, variance, and terminal timing before interpreting posterior mass near zero.

Report the posterior mass conditionally on the fitted model and priors, alongside the model checks and sensitivity results.

24.2.4 — Distributional and Individual Equivalence

The individual-versus-cohort example in Section 5.3.2 applies directly to equivalence tests: a mean-equivalence result answers a population question, not a pathwise one (Schuirmann, 1987).

Name the object tested: mean, median, distribution, tail, variance, or individual path. A mean-equivalence result need not constrain the variance or tails.

Distributional equivalence and individual equivalence are different targets. The latter specifies which individuals, or what proportion, must satisfy the band under declared uncertainty.

For individual-path claims, retain the dependence, observation, and endpoint-alignment assumptions that determine which portions of each path enter the test.

Finite-resolution support can be quantified. Fix the reference class, neutral origin, and scientifically justified tolerance rule before observing outcomes. In an ideal benchmark, let n qualifying streams be sampled independently from that reference class, let every true complete-stream residual be

classified without error against the declared tolerance, and let p be the population probability of exceeding it. If no violations occur, their probability is $(1-p)^n$. Inverting this probability gives the exact one-sided $100(1-\alpha)\%$ upper confidence bound $p \leq 1-\alpha^{1/n}$. With $\alpha = 0.05$, zero violations among 100 streams gives an upper bound of approximately 2.95%; among 1,000, approximately 0.299%. These are illustrative calculations, not observations supporting PTVC.

A real study must first qualify complete-stream coverage, the neutral origin, endpoints, the tolerance rule, and classification uncertainty. Unadjudicable streams are unresolved: they must neither count as successful closures nor disappear from the target through silent exclusion. Selection, clustering, and classification error require a corresponding inferential model and error budget instead of the ideal binomial calculation. Once those requirements are met, a small upper bound would give a positive, interpretable result: departures larger than the declared tolerance are rare in the studied reference class.

24.2.5 — Rival Victory Under Equivalence

Use the protocol's generative rival set (Section 14.4.5; Sections 22.5–22.6) when evaluating measured near-neutrality (Lakens, 2017; Lakens et al., 2018).

If both candidates predict the equivalence band, report that shared finite-band result and apply the prespecified comparison rule. Measurement validity still comes from separate calibration evidence.

The comparison set should include adaptation, homeostatic mean reversion, conditioned bridges, sedation, selection, and observation shutdown as possible generators of near-zero measured residuals (Lakens, 2017; Lakens et al., 2018).

Equivalence therefore belongs inside rival comparison. When the candidate and a simpler rival both predict the declared region, the result may validate a measure or favor the rival while leaving PTVC unchanged.

The analysis predeclares what would discriminate, such as equivalence plus reserve-linked variance compression under endpoint-independent stopping rather than mean reversion alone. The discriminating criterion should be evaluated in model recovery before empirical use so an equivalence finding cannot win merely because the pipeline lacks power against the rival.

A simpler rival that explains the result wins honestly. Letting it win strengthens the program more than treating every near-zero finding as hidden support. The report should credit the rival at the same level of specificity used for the candidate, including the reference class and observation bridge under which it succeeded.

Reports state when equivalence was met but failed to discriminate under the shared observation model. Such a result can still validate the equivalence procedure while showing that it has no Law-specific evidentiary force in that setting.

24.3 — Observation Kernels and Measured-to-Latent Bias

An observation kernel describes how latent history generates recorded channels, including the process that leaves some records absent. A measured result can be transferred to a latent claim only through justified calibration, bias bounds, and context-of-use assumptions. Unknown initial history and unstable origin or sign remain separate obstacles. The purpose of the kernel is to expose these limits, so that measured-only findings, bounded latent conclusions, and unavailable transfers receive different interpretations.

24.3.1 — Bias Bounds and Latent Intervals

When calibration, invariance, and observation assumptions identify only a set of latent values, retain that set.

The analysis states how bias widens the conclusion and where latent inference stops when bias cannot be bounded. In Equation 24.3, L_M denotes the measured estimand, L the latent target, and δ a bound on their systematic displacement; $\widehat{L}$ estimates L_M and also has random estimation error.

Equation 24.3 (role: bias-transfer bound)

$$|L_M - L| \leq \delta \wedge |L_M| \leq K \Rightarrow |L| \leq K + \delta$$

Equation 24.4 (role: initial-state decomposition)

$$L(T) = L(t_0) + \int_{t_0}^{T} g_\theta(s) F(s) ds$$

Validated transfer requires a valid measured interval and defensible bias bounds and calibration within the stated reference class. Under those conditions, the inflated band can bound the latent claim.

When an estimator is coherent but its latent bridge is unvalidated or limited by state, report only whether the measured residual met its criterion. Keep the result at the measured level.

If systematic error cannot be bounded under plausible assumptions, no latent transfer is justified.

When a late window is observed but the earlier ledger contribution is unbounded, report only the late increment or state. Whole-history inference remains unavailable.

If origin or sign choices change the conclusion, the coordinate is unstable. Report the sensitivity and narrow the claim accordingly.

A bias bound δ limits systematic measured-to-latent displacement for a declared estimand and reference class. Defend it using calibration, device, state, and cultural evidence; it is distinct from random estimation error (Carroll et al., 2006).

Establish the bound independently of the terminal result.

Let L_M denote the measured estimand and L the latent target, with a defensible bound $|L_M - L| \leq \delta$. The estimate $\widehat{L}$ also has random estimation error. If $[a, b]$ is a valid confidence interval for L_M, then $[a - \delta, b + \delta]$ is a conservative interval for L under that fixed bound. A measured equivalence result transfers to an inflated latent band through this interval, not by checking the point estimate alone. Any uncertainty in δ must also be propagated.

For example, a measured estimate of 0.2 with a confidence interval $[-0.1, 0.5]$ and a defensible fixed bias bound $\delta = 0.8$ gives a latent interval $[-0.9, 1.3]$. The measured interval is inside the band $[-1.0, 1.0]$; the latent interval is inside the inflated band $[-1.8, 1.8]$, but not the original band. If the inflated band is too wide for the scientific question, the latent transfer is too weak for support.

A bias bound can also be asymmetric. Some devices or report regimes may systematically understate distress more than relief, especially under sedation, disability, cultural restraint, or care dependence. The analysis should not force a symmetric δ when the mechanism is directional.

The calculation also helps compare studies. A study with a slightly wider measured interval but a well-validated δ may be more informative about the latent object than a study with a narrow interval and unbounded bias. Precision without transfer can be less useful than modest precision with known error.

Display the measured confidence interval, measured equivalence band, and bias-expanded latent interval separately.

If the bias-adjusted band is too wide for the intended inference, its width identifies the calibration precision the next measurement study needs to achieve.

24.3.2 — Calibration and Transport Error

Include both within-setting calibration error and error from applying that calibration to another setting.

Validate the estimator in the intended illness state, where speech, movement, autonomic response, medication, and care practices can alter its measurement relation.

Calibration transport should therefore be treated as part of δ or as a separate uncertainty component. If a device version changes, if a site adopts a new documentation workflow, or if a culture-specific reporting pattern changes the anchor, the measured-to-latent bridge must be reaudited.

Separate calibration error from a site effect on actual burden. A single random intercept can obscure whether the environment or the measurement relation changed.

Compare within-site, pooled, leave-site-out, and device-version calibrations. Limit transfer to the regimes that survive.

24.3.3 — Unknown Initial Ledger

Late-enrollment studies can estimate terminal residuals, increments, and window dynamics. A whole-course cumulative analysis additionally requires an independently bounded initial state at enrollment or observation spanning the relevant horizon (Carroll et al., 2006; Fuller, 1987).

Bound the initial term through appropriate longitudinal records, qualified retrospective anchors, or justified reference-class assumptions; those bounds determine the cumulative comparisons available.

24.3.4 — Sensitivity to Origin and Sign

The measures reviewed here do not supply a universally validated instrument-fixed neutral origin and cross-sign unit. Independent calibration and sensitivity analysis therefore remain empirical tasks (Fuller, 1987). Establishing their scope would make cumulative comparisons more informative; present measurement uncertainty does not decide whether a natural phenomenal origin exists.

Origin uncertainty interacts with exposure. Write a complete remainder as R=J+F, where J is the observed contribution and F the unobserved remainder. Suppose $\tilde{R}=J+bH+\eta$, where $H\geq 0$ is observed exposure, b is a common rate-origin error, and η is the remaining estimation error.

Independently justified bounds $|b|\leq b_{\max}$, $E|\eta|\leq m$, and $E|F|\leq u$ give

$$\mathbb{E}\left|\widetilde{R}-R\right| \leq b_{\max}\mathbb{E}H + m + u.$$

All terms use one qualified signed unit and exposure rule. If an independent rate bound M applies throughout the missing domain, then $u\leq M\,E[H_{miss}]$; otherwise retain a directly justified

bound on F. Denser observations can reduce η while leaving origin error and unobserved contributions unresolved.

A shared origin error need not cancel between arms. In a synthetic exact-closure example, let L be equally likely −1 or 1 and R=−L in both arms. Let observed exposures be $H_0=2+L$ and $H_1=2-L$, with b=1 and F=η=0. Both exposure marginals are {1,3}, yet $\tilde{R}_0=2$ while $\tilde{R}_1$ is equally likely 0 or 4; their W_1 distance is 2. Matching exposure distributions alone therefore cannot replace a joint origin-and-exposure analysis.

Report the conclusion across independently justified origin, sign, and exposure models. Section 24.6.4 uses these error budgets for point estimates or, when complete-remainder intervals are justified, a coverage-based comparison that retains uninformative records.

If the neutral origin shifts, a residual can move across the equivalence band. If the sign convention changes for a channel, burden-increasing and restorative contributions can be reversed. If a cultural or clinical anchor is wrong, a measured near-zero result may reflect the coordinate system more than the stream.

A good analysis therefore tests origin sensitivity. It reports how the conclusion changes under plausible neutral anchors, within-person baselines, site-specific anchors, and calibration anchors. It also audits sign conventions for each channel before the composite estimate is allowed to enter the ledger calculation.

Sign sensitivity is especially important in physiological data. Increased arousal may signal distress, effort, excitement, medication response, pain relief, or agitation depending on context. A fixed sign assigned to a nonspecific channel can create false movement toward or away from neutrality.

The chapter does not demand impossible certainty. It demands that coordinate choices be visible. A result that survives plausible origin and sign changes is stronger. A result that

changes direction under reasonable alternatives is not defeated automatically, but it loses the right to clean language.

The reporting sentence should say whether the conclusion was invariant, narrowed, changed sign, or became unusable under origin and sign sensitivity. That is how measurement humility becomes statistical design.

24.3.5 — No Transfer Without Validation

Validation connects measured equivalence with the specified latent target through the declared measurement relation (Carroll et al., 2006).

Validation has layers. The estimator must have a coherent construct, stable calibration, known uncertainty, state-appropriate performance, endpoint-compatible use, and a defensible relation to the latent target. A study may satisfy some layers and fail others. The conclusion should match the lowest unresolved layer.

This prevents the science from becoming impressive at the wrong level. A sensor composite may predict reports. A report composite may predict care transitions. A terminal residual estimate may be stable across sites. None of those achievements automatically validates transfer to the latent ledger.

A transfer study can identify improvements needed in calibration, missingness analysis, care-sensitive observation, or model comparison. Those findings define the next qualification task.

The public risk is obvious. If a measured estimator is described as if it reached inner life directly, readers may hear that a number has ranked or read a person. The research language must keep the estimator inside the protocol and leave no worth, care, trust, or spiritual implication behind.

The path forward is typed: measured results support measured claims, latent claims require a qualified measured-to-latent bridge, and an unsupported bridge remains a gap. The language of transfer must match the evidence that licenses it.

24.4 — Informative Sampling, Missingness, and Reactivity

Model dependence of observation on the evolving state, treatment, and recorded history; otherwise selective observation can mimic improvement or concentration (Rubin, 1976; Little & Rubin, 2019; Hernán & Robins, 2020).

24.4.1 — Missingness Taxonomy

Specify the observation process, denominator flow, causal assumptions, sensitivity parameters, and evidence that could distinguish the proposed mechanisms.

Model observation occurrence jointly with its mark, such as the measured burden when a report arrives. This separates changes in recorded values from changes in the opportunity to record them.

Equation 24.5 (role: marked observation process)

$$\begin{aligned} \mathbb{E}\left[dN_i(t) \mid H_i(t^-)\right] &= \lambda_i\left(t \mid H_i(t^-)\right) dt, \\ M_i(t) &\sim p\left(m \mid dN_i(t)=1, H_i(t^-)\right). \end{aligned}$$

For a simple marked point process with a locally integrable predictable intensity, the first line is infinitesimal conditional-mean notation and does not assert exact Poisson increments (Didelez, 2008).

Scheduled absence can reflect a low-burden window or planned rest. The resulting evidence describes that schedule, not the unobserved moments.

Participant refusal can reflect burden, distrust, fatigue, privacy, or preference. Respect the refusal and model it separately from other causes of missingness.

When the care team suspends measurement for comfort or safety, record the absence as clinical protection. It is not random missingness.

Device failure can involve battery, placement, skin contact, motion, firmware, or signal quality. Use provenance and version checks to identify the affected observations.

A transfer or change of site, staff, or recording regime changes the observation process. Model that care transition explicitly.

After a qualified stream endpoint, later outcomes for that stream are undefined. Apply truncation logic rather than imputing those outcomes.

Classify each absence using Section 24.4 before selecting its missingness model (Rubin, 1976).

A nurse’s decision to protect sleep and a battery failure require different observation mechanisms.

Distress, sedation, cognitive fatigue, and care protection may predict missing observations. Missingness is nonignorable when it still depends on unobserved values after conditioning on the observed history used by the analysis. Record these predictors where appropriate and test sensitivity to remaining dependence rather than classifying all such losses as MNAR by cause alone. Planned low-burden windows define the observation schedule; validity for that schedule does not by itself identify unobserved moments.

Disaggregate missingness by category, time, site, device, and endpoint relation.

24.4.2 — Marked Point-Process Model

Model observation times jointly with their measured marks using Equation 24.5 (Little & Rubin, 2019).

Reports can cluster when participants can respond, notes during crises, and device losses during transfers. Model that timing rather than imposing a fixed observation grid.

Include staffing, device/software versions, charging, family presence, and care routines when they affect observation intensity. Record protective omissions and collection burden on staff.

Unmeasured dependence that cannot be bounded requires sensitivity analysis or No adequate estimate.

24.4.3 — Missing at Random as an Assumption

Missing at random is an assumption about the dependence of missingness on observed and unobserved data. Conditional on the specified observed information, it asserts no residual dependence on the missing values. Design and observed-data diagnostics can inform its plausibility, but generally cannot verify that absence of dependence from the observed data alone. The analysis must name the conditioning history and examine departures through sensitivity analysis (Little & Rubin, 2019).

Assess missing-at-random assumptions conditional on the relevant history: device version and staffing may explain battery losses while refusal or protected pauses can retain dependence on unobserved burden.

Justify the conditioning history separately for each absence category, including recent observations when baseline covariates are insufficient.

24.4.4 — Pattern-Mixture and Selection Sensitivity

Pattern-mixture models vary outcome distributions across missingness patterns; selection models vary observation probability with outcomes or latent states (Scharfstein et al., 1999; Rubin, 1976).

Vary absent outcomes in scientifically justified directions and report the change needed to overturn the conclusion.

For selection models, report the dependence of observation intensity on unmeasured burden needed to erase equivalence.

24.4.5 — When Missingness Defeats Inference

Assess missingness by its possible effect on the target, not just its frequency. Inference fails when unobserved parts of the path could change the conclusion and their effect cannot be bounded. A few selectively absent intervals may matter more than extensive planned absence (Little & Rubin, 2019).

The defeat condition should be stated before analysis. It may involve missingness concentrated near endpoints, missingness associated with high distress markers, missingness tied to treatment changes, missingness differing sharply across sites, or missingness that removes entire classes of participants from the target window.

When missingness defeats inference, the result should not be rescued by a sophisticated imputation model. Imputation is a model-based assumption about unobserved values. If the assumptions are unconstrained, the imputed conclusion says more about the model than about the stream.

A good report distinguishes data incompleteness from inferential failure. If required protocol and implementation gates passed but endpoint-window observation depended on unmeasured clinical deterioration whose effect could not be validly bounded, report No adequate estimate for the primary target while retaining any valid secondary descriptive summaries. Reserve Unusable evidence for a failed required gate. That is more useful than burying the failure inside an adjusted estimate.

Defeat by missingness can teach the research program. It may show that cadence was too high, participant return was too burdensome, devices were not appropriate, staff processes were underdesigned, or endpoints needed different windows. The correct response is better design, not rhetorical rescue.

The chapter’s rule is that absence may narrow, weaken, suspend, or defeat a claim. It may not be converted into support merely because the observed remainder looks clean.

24.5 — Delayed Entry, Truncation, and Competing Events

Delayed entry and truncation separate the study window from the stream's complete domain. A participant observed late supplies only a suffix of the complete history, while follow-up may stop before the truth-level endpoint. Complete-stream inference therefore needs a joint prefix-observed-suffix object on one latent scale.

24.5.1 — Truncation Is Not Ordinary Dropout

Classify each event as part of the endpoint, an intercurrent event, censoring, or a change to the estimand.

After a care transition and sedation change, a participant may lose the reporting channel. The next report remains missing or unavailable unless an independently qualified terminal event makes it undefined for that stream.

Equation 24.6 (role: ordered pair of semi-competing-risk probabilities)

$$\Pr\left(T^N \le t, T^D > t\right), Pr\left(T^D \le t\right)$$

Ordinary dropout means an outcome might exist but was not observed. Truncation by death means that a specified post-death outcome is undefined. For PTVC, a later experiential outcome may be treated as undefined for a stream only after its endpoint has been independently qualified. In Equation 24.6, the first probability is the chance that the nonterminal event has occurred by time t while the terminal event has not; the second is the terminal-event cumulative incidence by time t. These distinctions change what can be estimated, what can be imputed, and what the result is allowed to mean (Aalen et al., 2008).

Post-endpoint carry-forward imputes an undefined outcome; excluding earlier deaths from a later window changes the survivor population. Neither is ordinary missing-data repair.

Retain valid preterminal trajectories, terminal-window estimates, or event-time estimands without assigning values to undefined outcomes.

24.5.2 — Terminal and Nonterminal Events

A nonterminal event occurs within the target process and may affect later observation or terminal risk; a qualified terminal event ends that process (Andersen et al., 1993; Rizopoulos, 2012).

A semi-competing-risk formulation is appropriate when the nonterminal event can occur before the terminal event, but the terminal event precludes its later occurrence. That asymmetry must be justified for the selected events. An experiential report may become undefined after a qualified stream endpoint, whereas a device removal or administrative record can still occur afterward and needs a different event definition.

Model dependence between terminal and nonterminal events when it affects the estimand. State whether the target concerns all enrolled people, survivors to a time, or a potential-survival stratum.

24.5.3 — Survivor Average Causal Effects

Applying the survivor-average estimands introduced in Section 17.6.5 requires a declared treatment contrast and explicit bounds or identifying assumptions for latent stratum membership (Frangakis & Rubin, 2002; Zhang & Rubin, 2003).

A survival-conditioned claim must say which survivor population it targets. Is it the observed survivors under realized treatment, the always-survivor principal stratum that would

survive under both treatment and control, or a survivor population under a specified care regime? These are different estimands with different assumptions; only the always-survivor contrast is the conventional survivor average causal effect.

If measurement requires sufficient stability, identify that selected reference class and assess how earlier endpoints and deterioration alter inclusion.

24.5.4 — Composite Outcomes and Their Costs

A composite may summarize several events or measurements when that combination answers the stated question (Tsiatis, 2006; Aalen et al., 2008).

Report survival, symptom relief, measured residual, and observation completion separately so the total cannot hide opposing component changes.

A composite is a technical summary, not a claim that death or suffering balances another component.

Freeze component directions, transformations, and weights. A favorable total cannot repair a failed prerequisite or erase an adverse component.

24.5.5 — Joint Longitudinal–Event Models

Joint longitudinal-event models connect measured trajectories and events through specified shared dependence (Aalen et al., 2008).

A shared-state adjustment for informative dropout is conditional on the joint model that links the residual to event hazard.

Clinical, unity, and care variables may inform the event model, with endpoint adjudication specified independently of the tested ledger estimator.

Test sensitivity to distributional form, shared effects, and measurement quality.

Report the longitudinal, event, observation, and endpoint submodels and how conclusions vary across plausible dependence structures.

24.5.6 — Competing Endpoint Definitions

Endpoint rules may diverge when consciousness, unity, irreversibility, clinical status, and observation windows do not align. Sensitivity to adjudication therefore belongs in the analysis, with each candidate rule kept independent of the ledger and any alternatives for unity threshold, sustained loss, return, or resuscitation prespecified (Andersen et al., 1993; Rizopoulos, 2012).

Prespecify the primary and sensitivity endpoint rules. Show their effects on the estimand, equivalence, missingness, and event model; narrow a conclusion dependent on one fragile rule. Low-signal endings may permit only a bound or No adequate estimate.

24.6 — Sensitivity and Partial Identification

Sensitivity analysis varies assumptions that are not settled by the observed record and asks whether the intended conclusion survives. Origin, representation, observation, missingness, and endpoint choices can affect different parts of that conclusion. Report the resulting compatible range and identify which assumption changes the decision, rather than treating a collection of favorable specifications as independent confirmation.

24.6.1 — Prior Sensitivity

Vary target assumptions over justified sensitivity ranges (Section 24.6).

Report the identified set of target values compatible with the data and maintained assumptions, including when it is too broad to decide the question.

Before interpreting results, name posterior predictive discrepancies that would weaken the model, including endpoint dropout, site variation, and terminal-residual structure. Compare simulated and observed discrepancies under Equation 24.8.

Equation 24.7 (role: Bayesian companion)

$$p(\theta|y,M) \propto p(y|\theta,M)\, p(\theta|M)$$

Equation 24.8 (role: posterior predictive check)

$$\tilde{y} \sim p\left(\tilde{y}|y,M\right),\ compare\ D\left(\tilde{y}\right)\ with\ D(y)$$

Compare skeptical, weakly informative, broad, and site-informed priors over justified ranges. Report how the conclusion depends on the prior.

Vary the missingness model, endpoint model, site hierarchy, and smoother to expose dependence on model structure.

Compare justified windows such as 48 hours, 7 days, 30 days, and a reserve-time clock. Report dependence on temporal alignment.

Vary the noise floor, bias bound, sign, origin, and calibration assumptions. Report how these measurement choices constrain transfer to the latent target.

Assess predictive adequacy through residual distributions, missingness, site effects, and event timing. Report the specific features on which the model succeeds or fails.

Prior sensitivity assesses how assumptions and data jointly determine the posterior conclusion (Gelman et al., 2013; Gelman et al., 2020).

For terminal residuals, justify priors using measurement noise, directional uncertainty, or earlier cohorts. A prior concentrating near zero must have independent model or calibration grounds.

Compare weakly informative, skeptical, and broader priors where feasible, including their effects on equivalence, latent transfer, and missingness sensitivity.

Include priors on missingness, calibration, endpoints, site variance, and error bounds; varying only the main residual prior is insufficient.

24.6.2 — Model-Space Sensitivity

Test plausible model families as well as parameters within one family (Rosenbaum, 2002; Gelman et al., 2013).

Name the model alternatives that change claim status, for example, a plausible distress-linked pattern-mixture model under which equivalence disappears.

Average models only when their outputs address compatible estimands; otherwise show them separately.

24.6.3 — Window and Alignment Sensitivity

Evaluate the prespecified window and alignment alternatives for the terminal signature (Gelman et al., 2013).

Vary window length, low-signal inclusion, adjudication, and the applicable calendar, reserve, illness-stage, care-transition, or observation clock.

Display a sensitivity surface or table and identify whether support depends on one retrospective alignment.

If cadence cannot resolve the target dynamics, report the design limitation instead of selecting a favorable window.

24.6.4 — Measurement-Error Sensitivity

Vary measurement-error components under Section 24.3's transfer analysis (Carroll et al., 2006; Manski, 2003).

The analysis should vary both random error and systematic error. Random error can increase uncertainty and, in nonlinear transformations or models with error-prone predictors, can also induce bias. Systematic error can displace an estimate from its target. Sign error can reverse contribution. Origin error can change whether a residual lies inside K. These errors have different effects and should not be collapsed into one vague uncertainty term.

If channel reliability changes near the endpoint, assess both time-varying error and dependence on the underlying state; a constant noise model may understate uncertainty where the target is most sensitive.

The randomized-remainder restriction in Section 19.2 also has a finite-error consequence. Consider groups j=0,1 defined by valid randomized assignment, not by later adherence or report availability, with the same pre-assignment prefix law after any required baseline standardization. Use one qualified analysis scale, or the fixed positive, pre-assignment stream normalization

specified in Section 6.3.5, consistently for prefix and remainder. Let L_j be the fixed prefix, R_j the complete strict-pre-endpoint remainder, and $Z_j=L_j+R_j$. Assume finite first moments and prospectively specified bounds $E|Z_j|\le\delta_j$. For laws P and Q, the first Wasserstein distance is the smallest $E|X-Y|$ over their joint couplings (Villani, 2009). Then

$$W_1(P(R_0),P(R_1)) \le \delta_0 + \delta_1.$$

Each remainder is coupled to its own negative prefix with cost at most δ_j. The negative-prefix laws coincide, so the triangle inequality proves the bound. If a measured complete-remainder estimate $\tilde{R}_j$ has a justified mean absolute error bound $E|\tilde{R}_j-R_j|\le e_j$ on the same scale, another two coupling steps give

$$W_1(P(\tilde{R}_0),P(\tilde{R}_1)) \le \delta_0 + \delta_1 + e_0 + e_1.$$

On the exact-closure branch, both terminal-residual bounds are zero. A valid lower confidence bound for the measured distance exceeding the combined measurement-error budget is adverse to the randomized-arm implication under the stated bridges. Estimated error budgets require simultaneous uncertainty control. Section 24.3.4 shows how origin error and unobserved exposure enter that budget; the bounded-witness calculation below supplies one finite-sample test. Equal arm distributions are necessary but insufficient; Section 25.2.3 supplies the common-nonzero-total comparison. The test requires measurement or bounds for the complete remainder. The per-stream residual-set construction below supplies a complementary test.

Robust exclusion requires a residual set that contains every possibility allowed by the independently justified measurement and bridge assumptions. Fix the admissible configuration family *G* before inspecting the outcome. Each configuration specifies the origin, representation, exposure, stream routing, endpoint,

observation relation and missing-history assumptions. Let $R(g)$ be its compatible terminal residual set, constructed without imposing closure, and let $s(g) > 0$ be an independently specified quantity with residual units. Define

$$\mathscr{R}_{rob} = \bigcup_{g \in G} \{r/s(g) : r \in R(g)\}, \qquad d = \inf_{x \in \mathscr{R}_{rob}} |x|.$$

For a nonempty union, d>0 excludes zero under every admitted configuration and is invariant to a positive change of units. Positive distance is sufficient, not necessary: the set (0,1) excludes zero while having distance zero. An empty union indicates incompatible assumptions or observations; it cannot provide a truth-containing exclusion. A union containing both zero and nonzero values is exact-indeterminate under Section 6.7.1. Only the singleton {0} says that every admitted residual is zero, conditional on the set's justified truth coverage.

A numerical example shows why unknown history matters. A current residual interval $[8, 10]$ and a bound $[-5, 4]$ on the entire remaining signed contribution imply the terminal outer interval $[3, 14]$. Additional unresolved prefix uncertainty $[-4, 4]$ enlarges it to $[-1, 18]$, removing the exclusion. These are synthetic units. Bounds on observed late experience cannot silently replace bounds on the complete prefix.

Statistical exclusion requires coverage under every admitted null law. Let $\mathscr{P}_0$ be the declared family of observation and nuisance laws under which the tested stream closes exactly, and let C_α be the projected residual confidence set. The required guarantee is

$$\inf_{P \in \mathscr{P}_0} P\{0 \in C_\alpha\} \geq 1 - \alpha$$

implies

$$\sup_{P \in \mathscr{P}_0} P\{0 \notin C_\alpha\} \leq \alpha.$$

A joint confidence region provides this protection when its coverage is established uniformly over the admitted null configurations. Marginal coverage across a mixture of zero and nonzero residuals does not establish coverage conditional on the residual being zero. Repeated searches across streams or analysis times require a corresponding simultaneous guarantee.

An observed endpoint can carry its own explicit error budget. Over a fixed finite horizon, suppose a simultaneous interval band covers the complete pre-endpoint ledger with failure probability at most α, and an independently qualified endpoint rule matches the true endpoint and timing convention with failure probability at most β. If the tested branch permits true off-band completion within that horizon with probability at most ε, an observed endpoint whose entire nonempty residual interval lies outside the declared band has probability at most $min\{1, \varepsilon + \alpha + \beta\}$. This follows by the union bound and needs no independence between these errors. Timing uncertainty requires taking the union over all admitted endpoint times; missing-prefix uncertainty remains inside the band. The exact probability-one target sets ε and its theoretical residual tolerance to zero, while the empirical certificate retains its declared error probabilities.

A bounded witness turns this distributional restriction into a finite-sample comparison. In the unweighted two-group case, take fixed sample sizes n_0, n_1 with independent observations within and across the assignment groups. Choose a 1-Lipschitz function $f:\mathbb{R}\to[-B,B]$, $B>0$, before evaluation, or learn it on independent training records. Let $\hat{d}$ be the difference between

the sample means of f(R̃). Applying Hoeffding's inequality directly to this difference, with natural logarithms and $0<\alpha<1$, gives (Hoeffding, 1963)

$$\rho_\alpha = B\sqrt{2\left(\frac{1}{n_0} + \frac{1}{n_1}\right) log\left(\frac{2}{\alpha}\right)}.$$

$$\ell_\alpha = max\{0, \left|\widehat{d}\right| - \rho_\alpha\}.$$

The population contrast in f cannot exceed W_1, so ℓ_α is a lower confidence bound with coverage at least $1-\alpha$ under the maintained first-moment assumptions. A value exceeding $\delta_0+\delta_1+e_0+e_1$ is adverse to the corresponding randomized-remainder implication. For a synthetic illustration, $B=1$, $n_0=n_1=400$, and $\alpha=0.05$ give $\rho_\alpha\approx0.1921$. A hypothetical contrast $|\hat{d}|=0.45$ yields $\ell_\alpha\approx0.2579$. These numbers illustrate the rule, not observed evidence or a sample-size recommendation.

Incomplete records can instead enter through intervals. Keep the same randomization and scope premises and fix $f_B(x)=\max\{-B,\min\{B,x\}\}$, extended to $\pm\infty$. For every enrolled unit construct a nonempty interval $C_{ji}=[a_{ji},b_{ji}]$ for its complete remainder R_{ji}, using independently justified measurement, domain, and missing-exposure bounds. Under every admitted null law require the arm-specific marginal miscoverage probability $P(R_{ji}\notin C_{ji})\le\gamma_j$. Both the coverage guarantee and the independent-unit sampling conditions must hold conditional on any independently trained, frozen calibration. This is a coverage premise to establish, not a chosen tolerance that makes data acceptable. Here marginal means averaged over evaluation units within the declared arm, conditional on the fitted calibration; an average over random calibration fits is a different guarantee.

Set $l_{ji}=fB(a_{ji})$ and $u_{ji}=fB(b_{ji})$, and use bars for their sample means. On coverage, $l_{ji}\le fB(R_{ji})\le u_{ji}$; each failure can change an expectation bound by at most 2B. With the same ρ_α, define

$$D_\alpha = max\{\bar{l}_0 - \bar{u}_1, \bar{l}_1 - \bar{u}_0\} - \rho_\alpha - 2B(\gamma_0 + \gamma_1).$$

Under exact closure and the resulting equality of complete-remainder distributions, $P(D_\alpha>0)\le\alpha$. The two one-sided Hoeffding bounds prove the claim after allowing $2B(\gamma_0+\gamma_1)$ for interval failures. No independence between coverage failure and the true remainder is needed. Marginal coverage suffices; the intervals need not cover every unit simultaneously. Because fB is bounded, this particular equality test does not require finite raw-remainder moments.

A wholly uninformative record contributes $C=\mathbb{R}$, hence $[l,u]=[-B,B]$, rather than being dropped or coded neutral. In a synthetic example with the same B, n, and α, suppose each group has 20% such records. Let the other intervals be [0.4,0.6] in group 0 and [−0.6,−0.4] in group 1. If valid bounds give $\gamma_0=\gamma_1=0.005$, the observed interval contrast is 0.24 and $D_\alpha \approx 0.0279$. This illustrates an informative test despite incomplete records, conditional on valid bounds; it is not a power calculation.

If the conditional coverage bounds hold on an independently qualified calibration event with probability at least $1-\eta$, allocate η separately: the unconditional erroneous-rejection probability is at most $\min\{1,\alpha+\eta\}$. Independent training alone does not supply that event. Section 19.5.1 gives the weighted extension for fixed pre-assignment strata. Clustered observations, adaptive stopping, and data-selected witnesses still need matching inference. Intervals do not repair an invalid representation or endpoint bridge. Failure to exceed a threshold remains inconclusive about

exact closure; the method never supplies a cross-stream balance. A single clipped-mean witness can miss different symmetric spreads. Section 25.3.3 supplies a prespecified finite witness family with joint error control.

24.6.5 — Robustness Is Not Replication

Correlated choices can manufacture robustness. Analyses sharing a missingness assumption, calibration, or endpoint alignment may look diverse while protecting one weakness. The sensitivity grid should identify those dependencies.

Use Chapter 28's replication procedure to state which target, signature, and conditions are repeated and which recruitment, measurement, or adjudication conditions change.

New participants can share a device, scoring, or stopping artifact. Add an independent check of the vulnerability at issue, such as endpoint adjudication separate from the signature estimator.

Robustness means a conclusion survives variations within a dataset or workflow; replication means it appears in new data under governed conditions. A result can withstand many modeling choices in one biased dataset yet fail elsewhere (Rosenbaum, 2002; Gelman et al., 2013).

Within-dataset robustness evaluates sensitivity to priors, windows, calibration, and missingness assumptions. Replication extends the test to new sites, devices, cultures, diagnoses, or endpoint regimes.

A robustness summary should identify the dataset, measured residuals, analyses, and assumptions that produced stable conclusions. Readers can then assess exactly which features were examined.

A strong report states what robustness covers and leaves untouched: for example, that measured equivalence was robust

to prior and window variation but not nonignorable missingness. This is more trustworthy than a broad claim.

A replication needs a clear account of what is being repeated. The protocol should identify the target quantity, the claimed signature, and the conditions under which the original explanation should still apply. Changes in recruitment, measurement, or endpoint adjudication belong in that account. Readers can then assess whether a disagreement challenges the original result or reflects a difference between the questions the studies actually addressed.

New participants do not make every source of error independent. Two sites may inherit the same device calibration, scoring code, or stopping convention, allowing a shared artifact to recur. A useful replication therefore examines which vulnerabilities remain shared and introduces independent checks where feasible. For a terminal signature, this might include endpoint adjudication that does not depend on the measurements used to estimate the signature.

When studies disagree, the size and precision of their estimates determine what the disagreement means. A precise result incompatible with the original effect poses a different problem from a wide interval that leaves both the effect and its absence plausible. Differences in measurement or participant selection may suggest a narrower explanation, which then requires its own prospective test. The disagreement becomes useful when it identifies what evidence could resolve it.

24.7 — Sequential Evidence and Optional Stopping

Sequential evidence permits learning while data accumulate, but only under an explicit process. E-values, e-processes, alpha-spending, group-sequential plans, or Bayesian monitoring retain their guarantee for the exact protocol-fixed null, filtration, stopping permissions, and model assumptions to which they were attached (Jennison & Turnbull, 2000; O'Brien & Fleming, 1979; Pocock, 1977; Ramdas et al., 2022).

24.7.1 — E-Process Definition

Separate valid sequential monitoring from retrospective changes to models, endpoints, or publication decisions.

Specify the tested claim, null family, filtration, and data-update rule; the displays below and Sections 24.7.1–24.7.4 give the corresponding conditions. A valid anytime guarantee does not require the actual stopping time to have been scheduled before the first look. Prospectively specifying the scientific comparison remains a separate protection against selecting claims or procedures after seeing favorable results.

Check whether adaptive recruitment, delayed outcomes, device changes, or selective cleaning preserve the maintained observation law. Log the information and procedural changes at each look.

Equation 24.9 (role: e-process expectation condition)

$$E_t \geq 0, E_0[E_\tau] \leq 1 \; for \; allowed \; \tau$$

Equation 24.10 (role: anytime-valid error control)

$$P_0\left(\sup_t E_t \geq 1/\alpha\right) \leq \alpha$$

An e-process controls its expected value at allowed stopping times under every law in its declared null class, subject to the specified filtration and conditions (Howard et al., 2021).

A new estimand or transport claim requires its own justified evidence-process guarantee.

The evidence-process construction, including any permitted adaptation rule, must be specified before the data it evaluates are reviewed. Predictable updates may use past observations to choose future bets when the declared null and filtration guarantee remains valid. Selecting favorable transformations of already observed evidence does not establish an e-process.

24.7.2 — Allowed Stopping Times

A stopping rule is compatible with an e-process when it uses the declared filtration and lies within the process's stopping-time guarantee. An unplanned look or a decision influenced by public pressure does not, by itself, invalidate an anytime-valid e-process. Retrospective changes to the null, evidence construction, threshold, or data-selection mechanism can invalidate the guarantee. Statistical validity and compliance with the study's governance rules should be audited separately (Ramdas et al., 2022; Howard et al., 2021).

Record permitted looks and stopping reasons, including ethical or operational stopping. A care-based stop may leave only a conditional estimate or No adequate estimate.

Match the evidence process to state-dependent measurement failure or data-release delays, including delays influenced by interim results.

24.7.3 — Threshold and Error Control

Threshold crossing rejects the specified null at the controlled error rate only under the evidence process's assumptions (Howard et al., 2021).

Declare the threshold before the data it evaluates are inspected; do not move it in response to early results or outside pressure.

Apply the error guarantee only to its tested null and reference class. Any broader latent or theory claim needs its own inference; use Chapter 28 for public status changes.

Prespecify how non-crossing will be interpreted under the chosen process: insufficient information, an undecided comparison, or evidence against its alternative where justified.

24.7.4 — Dependence and Adaptive Data

Account for within-person dependence, shared site practices, and changes in sampling prompted by interim evidence (Howard et al., 2021; Ramdas et al., 2022).

Reassess a fixed-sampling construction when favorable interim results determine the next site, window length, or device version.

Define the unit of each look and the treatment of clustering, site correlation, adaptive recruitment, and delayed endpoints.

Model burden-reducing or representativeness-improving adaptations within the valid construction, or separate their data from the affected confirmatory inference.

Preliminary public claims can change behavior, recruitment, or documentation; include those changes in the sampling analysis.

24.7.5 — Evidence for Components, Not Canon

Sequential evidence can update a component claim. It cannot update the canon as a whole by itself. This distinction protects the series from treating a mathematical crossing as public validation of the Law, religious doctrine, civic design, legal architecture, or transition readiness (Ramdas et al., 2022).

A component might be a measured equivalence result in a terminal-window estimator, a missingness sensitivity threshold, a site-calibration success, or an observation-model check. Each component has a defined null, reference class, and inference route. The canon has many more gates.

The report should say exactly what changed. A sequential analysis may move a component from inconclusive to supported, from supported to fragile, or from plausible to defeated. It may not say that the Law has been scientifically validated unless the broader verdict system later determines that status.

This protects skeptics and supporters alike. Skeptics can see that evidence is not being exaggerated. Supporters can understand that progress is real even when narrow. A narrow component update may be scientifically valuable without becoming a public proclamation.

The section also prevents dashboard pressure. A running number can excite people because it feels like the theory is moving toward victory. Readers should be trained to ask which component the number concerns and which gates remain closed.

Sequential evidence updates the component whose null and process were specified in the protocol. The public record identifies that component, the resulting evidence, and the prerequisites for its next use.

24.7.6 — No Dashboard Deciding Status

A dashboard can make evidence look more authoritative than the evidence is. A line rising over time, a green threshold, or a public status badge can compress estimand, missingness, endpoint, bias, sensitivity, and review into a single visual cue. That compression is dangerous for this book (Howard et al., 2021).

The chapter does not forbid internal monitoring. Researchers may need dashboards for data quality, safety, missingness, accrual, calibration, and sequential evidence. The rule is that dashboards are operational tools, not public verdict engines.

A public dashboard would be especially risky if it displayed a running probability, evidence ratio, or equivalence status without the assumptions beside it. Readers might interpret the number as proof, moral meaning, or permission to apply the Law in ordinary life. That would violate the science book's authority boundary.

Human review must remain above the number. A committee must examine endpoint independence, data quality, missingness, measurement burden, adverse findings, public-risk wording, and rival implications before any status language changes. The dashboard can alert; it cannot decide.

This rule also protects participants and sites. No person, family, clinician, site, or culture should be placed under public suspicion because a data-quality or evidence metric looks unfavorable. Research metrics stay inside the research governance architecture.

The section ends with a design rule: sequential graphics may be beautiful, but their captions must make them humble. A running number is a witness to a component, not a judge of the Law.

24.8 — Open-Set and Protected-Theory Risk

Model ambiguity occurs when adequate models remain indistinguishable in a valid protected comparison at the declared resolution. Wide uncertainty or analysis-sensitive ordering is instead inconclusive. A pipeline that misses its required recovery thresholds instead remains unqualified for the affected confirmatory comparison.

24.8.1 — Inconclusive Evidence

Apply the same change control to PTVC and its rivals; allowing only the preferred model to revise its endpoint or representation after failure prevents an informative comparison.

An inconclusive analysis is valid but does not resolve the estimand. Avoid trend language that presents the unresolved result as support.

An interval crossing an equivalence boundary or conclusions changing across plausible priors and missingness assumptions can leave a valid comparison inconclusive.

Use the unresolved dependency to direct the next design: cadence, calibration precision, endpoint timing, or reference-class resolution.

The result may still reduce uncertainty or change relative plausibility. Report those gains with their inferential basis while keeping the categorical verdict separate.

24.8.2 — Unusable Evidence

Failure of a required protocol, consent, endpoint, measurement, or provenance condition makes the affected evidence Unusable. Do not report it as evidence for that claim.

Name the failed required condition and the affected analysis when reporting Unusable evidence (Lakatos, 1978; Stanford, 2006).

Apply Unusable status to the affected estimand. Retain legitimate device-quality or observation-process findings from the same record.

24.8.3 — Analytic Flexibility and Model Revision

Audit choices capable of changing the result, including windows, exclusions, transformations, priors, and endpoint or measurement rules (Stanford, 2006).

Distinguish choices fixed before data, adapted under blinded rules, explored after inspection, and triggered by observed outcomes.

A revised analysis must state the explanatory or predictive improvement, added assumptions, and possible failure before it can upgrade claim status.

24.8.4 — Transparent Adverse Reporting

Narrowing retains only a smaller reference class or a weaker claim. Update the stated scope explicitly.

For each failed assumption, adverse analysis, weakening sensitivity result, or viable rival interpretation, name the model, analysis version, reference class, and decision changed. Distinguish failure of the claim from failure of its implementation (Lakatos, 1978; Stanford, 2006).

A supported component has passed its estimand, band, bias, missingness, endpoint, and sensitivity requirements and met the prespecified evidential criterion. Name the supported component and its reference class.

A valid adverse result or failure under a relevant rival creates defeat pressure. Report that pressure using Chapter 27’s verdict rules.

Adverse reporting includes failed equivalence, nonsignificant but inconclusive results, unbounded transfer bias, missingness

defeat, endpoint sensitivity failure, posterior predictive failure, and protected-theory audit findings. Each remains visible enough for an outside statistician to reproduce the status decision. The record should also identify the affected component, the diagnostic or threshold that triggered the adverse status, and the downstream claims that lose permission as a result.

Adverse reporting protects the entire series. If the science book hides adverse findings, applied books will inherit false confidence. If it publishes them clearly, applied domains can remain conditional and honest. Scientific loss becomes a form of public care. This propagation discipline also lets an applied domain preserve practices supported on independent grounds while withdrawing only the scientific authority that actually changed.

The report should not dramatize defeat. It should state what failed, what remains learned, what claim status changes, and what future design would be required. A sober adverse report is stronger than a defensive favorable report. The most useful adverse report also ends with a decision consequence, such as replicate, redesign, narrow, suspend, or retire, rather than leaving the reader with a rhetorical verdict.

Transparent adverse reporting also gives critics a real place in the program. A critic who can point to published failed assumptions, rival-sensitive results, and No adequate estimate labels sees a theory that can be challenged. That trust is more valuable than temporary rhetorical victory. Criticism becomes scientifically productive when the record shows not only that an objection was heard, but which test, revision, or status change followed from it.

The chapter closes with analyses that connect conclusions to explicit targets, assumptions, and evidence. Clear results

strengthen the next scientific decision, whether they support a model or identify a better direction. The handoff should identify which findings are stable enough to carry forward and which unresolved dependency should determine the next study.

24.8.5 — No-Estimate Route

If every named model fails the required adequacy checks, report Open-set inadequate. After required gates pass, use No adequate estimate separately when the named target is undefined or neither identified nor validly bounded. A relative ranking alone does not resolve either failure.

Apply Section 24.2's No adequate estimate rule and Section 27.4's evidence states. In protected-theory analysis, prespecify the abstention branch in the analysis plan, results template, and public-status map so nonidentification or a failed required gate cannot be repaired into a favorable result after inspection.

Chapter 25 — Model Recovery and Adversarial Tests

Suppose an analysis confidently selects a reserve-coupled model from synthetic trajectories generated by ordinary adaptation. The optimizer may have converged and the uncertainty intervals may look precise. The mistake lies in attribution: the available observations did not lead the analysis to their known source. A benchmark makes that mistake visible before the same method is used to interpret a participant’s history. It also lets investigators vary one cause of confusion at a time and test whether a proposed repair works.

Start with the generator and observation system separately. The simulator knows the latent path and its parameters; the fitting procedure receives only the channels planned for the study. Noise, irregular timing, and channel loss can then conceal a distinction that is obvious in the latent trajectories. Recovery asks what survives that transformation. Fitting the generating family alone tests parameter estimation, while fitting every candidate to every family reveals which explanations the comparison can distinguish.

An aggregate success rate can hide the region that matters. A method may classify most trajectories correctly yet confuse a bridge with the candidate whenever endpoint timing is uncertain. It may estimate a compression parameter reliably until the sensor floor becomes comparable to the signal. Mapping those failures against the proposed design gives the benchmark practical value: it identifies the channel, calibration, timing, or model assumption that must improve before a model-specific conclusion becomes available.

The repair itself needs protected evaluation. Choosing a threshold from the same simulations used to report its success can turn development into apparent confirmation. The chapter separates fitting, selection, and final challenge records and retains adverse fixtures after a change. A small executable case

can expose a sign error or leakage route; a broader challenge can assess recovery and uncertainty under the intended conditions. Both records remain useful when the revised method succeeds.

Known alternatives are only part of the test. A generator outside every fitted family can reveal whether the pipeline recognizes inadequacy or confidently forces an unfamiliar process into a familiar label. Transport tests then change the population or observation regime to locate the boundary of a successful comparison. These exercises qualify a method for particular inferential work. They also identify when a useful measurement result can proceed independently of an unresolved mechanism comparison.

Independent critics strengthen this process by proposing failures the developers did not anticipate. Adversarial collaboration requires advance agreement on the comparison and a record of what the challenge changed; the Cogitate collaboration illustrates its promise and practical demands in consciousness science (Cogitate Consortium et al., 2025). The chapter closes by making that record executable: a reader should be able to identify the released generator, reproduce a challenge, and determine which conclusion its result actually warrants.

25.1 — Why Model Recovery Comes First

The recovery study uses synthetic worlds with known latent states, parameters, observation kernels, endpoints, missingness, and model labels. Analysts receive only the observation layer planned for the empirical study. The hidden answer is revealed after the pipeline produces its verdict.

25.1.1 — Parameter Recovery

Equation 25.1, Generative benchmark declaration.

$$G_j : \theta_j \sim p_j(\theta), x \sim p_j(x|\theta_j), y \sim p_j(y|x, \theta_j, \psi)$$

Role label: generator specification. Here G_j is a benchmark world, theta_j are generator parameters, x is the latent trajectory, y is the observed record, and psi holds observation-system parameters.

Report parameter bias through mean or median recovery error. This identifies estimates that drift systematically toward the favored model.

Report empirical coverage of the declared credible or confidence intervals in the recovery experiment. Undercoverage exposes unjustified confidence.

Report the selected or weighted fitted model for each generating family. The resulting recovery pattern identifies confusion and false classification.

Record false equivalence when the pipeline declares equivalence although the generating estimand lies outside the specified band. This detects synthetic support produced in null or artifact worlds.

Record false defeat when the target is rejected although the generator includes it. This detects excessive rejection under noise.

Evaluate probability and interval calibration by state, site, and time. Aggregate calibration must not hide failure in the settings relevant to the verdict.

Apply Sections 14.5.2 and 21.6.2 across frozen generators, including the reserve, OU, bridge, shutdown, and frailty parameters relevant to the proposed comparison (Wilson & Collins, 2019).

Draw parameters, latent paths, and the planned observation layer; fit the model; then report bias, coverage, tails, shrinkage, and convergence by design condition.

If a fitted model underestimates a sensor floor, it can distort the terminal compression exponent. Map the conditions under which that exponent becomes unrecoverable.

Check whether the same fitting machinery distorts OU or bridge parameters toward reserve-coupled signatures even when it recovers the candidate's own parameters.

Freeze recovery criteria for bias relative to the scientific margin, interval coverage, and diagnostics, including failures concentrated in the terminal window.

25.1.2 — Model Recovery and Confusion

Fit every planned analysis model to every generator, testing correct attribution and refusal of unsupported confident labels (Wilson & Collins, 2019).

Use the families in Section 25.2 as the generating-model by fitted-model grid, with a common observation interface and comparison rules.

If a random-terminal-time Brownian bridge is frequently classified as the reserve-coupled candidate when endpoint uncertainty is high, the benchmark identifies that region of nonseparability. The next design can test which observations would resolve it.

Allow ties and abstention where the design cannot separate a pair; use the confusion to target redesign or scope narrowing.

Map confusion by noise floor, cadence, endpoint uncertainty, sedation, diagnosis, and device-removal timing rather than relying on an average recovery rate.

Preserve named near-miss fixtures whose familiar graphs arise under slightly violated assumptions; test whether the pipeline detects the difference.

25.1.3 — Recovery Across Noise and Missingness

Begin with a clean-data baseline, then vary the intended observation conditions. Posterior simulation-based calibration adds a conditional check near a particular observed dataset to broad prior-simulation checks (Säilynoja et al., 2026).

Prespecify plausible parameter ranges and extreme cases for sensor error, cadence, channel-specific loss, endpoint uncertainty, and care transitions.

Sweep noise-floor estimation error jointly with fitting-window length; record misclassification and deconvolution failures, including invalid negative variance estimates.

Generate missingness from its proposed causes, allowing perfusion, sedation, or care protection to affect both observations and latent-state interpretation.

25.1.4 — Recovery Before Human Data Verdicts

Confirmatory model-discrimination claims require the relevant synthetic challenge to pass its protocol-fixed criteria for recovery, calibration, absolute adequacy, protected prediction, and open-set detection. Transport checks apply to the generalization claimed. A separately justified measurement result or model-independent bound uses its own prerequisites; it need not recover every candidate mechanism to be informative.

Separate preprocessing, feature and window selection, tuning, and threshold choice from the final synthetic evaluation through nested resampling or an untouched challenge set.

Before model-based attribution, an independent analyst should reproduce the frozen pipeline's recovery, uncertainty, confusion, calibration, and abstention outputs from synthetic worlds. Failure blocks the attribution dependent on that pipeline (Wilson & Collins, 2019; Talts et al., 2018).

25.1.5 — What Simulation Validates

Simulation validates the specified computational and statistical construction. Simulation-based calibration tests inference under its assumed joint law; posterior predictive checks assess selected observed features (Talts et al., 2018).

Preserve adverse and undecided fixtures after successful runs. If reserve and random-endpoint bridge models remain unresolved, demonstrate improved separability before recruiting for that model-discrimination claim.

25.1.6 — No Simulation as Scientific Validation

Simulation evaluates computational and statistical machinery under declared assumptions. By specifying a data-generating process, researchers can test whether an analysis recovers known features and quantifies uncertainty under those simulated conditions. These controlled checks prepare the computational pipeline for empirical comparison (Talts et al., 2018).

An executable reserve-coupled model supplies a candidate generator for examining terminal concentration and comparing proposed mechanisms.

Simulation-based calibration can expose computational error, inconsistent implementations, or posterior miscalibration under

an assumed joint distribution. Posterior predictive checks test fit to selected features.

Simulation success, ethical review, and demonstrated design separability together inform the next empirical step, including whether recruitment and confirmatory comparison are justified.

The safeguard is operational: each output carries a role label such as computational check, recovery check, design-separability check, rival-envelope check, or artifact-release check. The label prevents a persuasive figure from outrunning its warrant.

The next section builds the serious generator library required to make this recovery gate more than a performance for one favored model.

Every result also records the artificial generator, fitted models, holdout design, recovery metric, allowed inference, and a status field separating benchmark calibration from evidence about nature.

Later reports should cite the executable benchmark result, not a sentence of assurance. Trust comes from leaving enough machinery for a skeptical reader to rerun the case and disagree precisely.

Minimum recovery bars are fixed for every reported quantity. They may include bias below the scientific margin and near-nominal coverage for a slope, low false support for a model label, or stable recovery across the noise, missingness, and endpoint conditions expected in the human protocol.

Recovery reports include computation time, convergence failures, discarded fits, and resource demands. A model that succeeds only after hidden divergence or unrealistic compute provides weaker practical evidence than a stable pipeline.

Chronology is decisive: the benchmark is completed before the emotionally powerful human result appears, because later

tuning is otherwise pulled toward the terminal graph. Freezing benchmark performance before human data also makes later deviations from the pipeline visible as new analyses rather than invisible repairs.

That frozen benchmark also resists narrative momentum. It preserves the worlds where the method would be wrong, undecided, or defeated by a rival before one resonant interpretation begins to dominate. Those adverse worlds should remain in the benchmark after success, because deleting hard cases would gradually convert recovery into self-confirmation.

Failed recovery is a publishable result. If the analysis cannot separate a reserve-coupled candidate from a random-endpoint bridge, it has shown that the proposed human study would ask vulnerable data to carry a question the method cannot yet support. Design should change before recruitment, not after public interpretation. A redesigned study should show improved recovery on the exact confusion that blocked the earlier design before participants are asked to contribute new data.

25.2 — Generative Benchmark Library

A generative benchmark makes the analysis answerable to worlds whose construction is known. Its library must include the reserve-coupled candidate, ordinary regulation, bridges, clinical and observation mechanisms, hybrids, and null worlds under a common interface. Version each generator, parameter distribution, endpoint rule, observation process, and expected diagnostic. Recovery in a favorable candidate world is insufficient when realistic rivals or combinations can produce the same reported conclusion.

25.2.1 — Reserve-Coupled Candidate

Admit an independently motivated executable rival with fresh predictions before the protected evaluation is inspected.

The reserve-coupled generator implements the declared reserve and signed reference dynamics. It tests the candidate's signatures without treating the construction itself as evidence.

OU, adaptation, and opponent-process generators implement local regulation, rebound, and baseline drift. They allow ordinary regulation to win when it explains the observations.

Bridge and event-conditioned generators use fixed or random endpoints and conditioning on future events. They test endpoint concentration without a PTVC-specific mechanism.

Frailty, medication, and observation generators represent illness, sedation, device removal, and staff cadence as distinct sources of observed patterns. These generators test clinical and operational artifact pathways before such patterns are attributed to the proposed latent dynamic.

Hybrid and null generators combine ordinary components or omit a structured effect. They test both false support and false defeat.

A flexible nonparametric or machine-learning reference establishes a predictive benchmark. Predictive success alone gives it no authority to interpret the mechanism.

Equation 25.2, Full fitted-model matrix.

$$R_{ij} = Score\left(M_i, y_{G_j}^{test} \mid y_{G_j}^{train}\right), \qquad i = 1,\ldots,I, \quad j = 1,\ldots,J$$

Role label: model-recovery diagnostic. Every fitted model Mi is evaluated on protected data generated by every generator Gj; the matrix exposes recovery, confusion, and false certainty. Here I is the number of fitted models and J is the number of generators.

Implement the reserve-dependent fluctuation and relaxation mechanisms specified in Sections 7.6–7.7, including their predicted variance compression and endpoint behavior.

The generator should include the finite-variation reference case and approved stochastic extensions only when the product dynamics are rebuilt correctly. If reserve is stochastic, jumps, shares shocks, or changes filtration, the benchmark must use the general product rule rather than quoting the finite-variation shortcut.

Separate scientific, nuisance, and observation parameters in the generator specification.

State how the generator imposes its construction: independent endpoint definition, any conservative-field closure, and the paths permitted by a closing controller.

25.2.2 — OU, Adaptation, and Opponent Models

Implement the OU, adaptation, and opponent-process families from Chapter 21 with their declared baselines, relaxation or response times, noise, and context dependence (Wilson & Collins, 2019).

Include changes with illness, treatment, sleep, fatigue, culture, and observation regime rather than limiting the library to stationary parameters.

The benchmark can also expose the cost of a weak comparison. Run the same synthetic observations through a restricted rival library and through the fully specified library, holding the scoring and information rules fixed. Record whether the candidate's apparent advantage survives. A reversal shows which omitted or underdeveloped ordinary mechanism had made that advantage appear distinctive.

25.2.3 — Bridge and Event-Conditioned Models

Use Section 7.5's conditioned-process competitors to test confusion with reserve coupling.

Include fixed, random, and censored endpoint constructions; specify endpoint error and whether conditioning represents dynamics, statistical conditioning, or selection.

Prospectively define the eligible population and retain the full sampling denominator. These data make a selection explanation quantitatively testable. Let Q be a specified rival's unselected trajectory law in that population and a declared baseline stratum. Fix a preterminal trajectory event A, such as a cumulative-response band, defined for every eligible trajectory under a common qualified observation rule. Let $S = 1$ indicate actual retention, $q = Q(A)$, $\pi = \Pr(S = 1) > 0$, and $p = \Pr(A \mid S = 1)$. Under a selection-only extension of Q, retention changes which records enter but not the feature's underlying law or observation rule.

Even arbitrary dependence of retention on the trajectory must obey

$$\pi p = \Pr(A, S = 1) \le q, \qquad p \le \min\{1, q/\pi\}.$$

For example, if $q \leq 0.10$ and at least half the eligible population is retained, selection alone cannot make the event occur in more than 20% of retained records. With bounds $q \leq q_U$, $\pi \geq \pi_L > 0$, and $p \geq p_L$ jointly valid across all claimed events, strata, and arms, the condition $\pi_L p_L > q_U$ rejects that selection-only rival at the declared joint error level. The bounds need not be independent. These numbers illustrate the constraint; they are not participant results.

Obtain q from a frozen rival prediction or independent reference data qualified for the same population and observation rule, not by refitting the selected test records. For a rival family, q_U must cover every admitted member. Evaluate the bound within declared baseline strata or assignment arms before combining results. This turns the sampling denominator into additional evidence against a specified selection explanation, complementing the joint selection-and-outcome comparison in Section 25.3.1 (Bareinboim & Pearl, 2016). Conditioning an abstract process on a probability-zero endpoint is a different construction and requires its own sampling model. A route that prospectively predicts the excess can gain comparative support from this test; the result does not distinguish zero closure from a common nonzero total. The location comparison below addresses that separate question.

For the formal twenty-route fork in Section 19.2, apply this bound to completed choices before outcome-dependent recording. A uniform hidden mapping gives q=1/20 for continuing-route choice. Recording every continuing choice and a fraction 17/57 of short-route choices retains one third of completed choices and produces an apparent continuing fraction of 15%, without changing the local choice law. Conversely, a jointly justified recording fraction of at least one half limits that selection-only explanation to at most 10%. These are

computational selection fixtures. They show why the retained denominator and the route-mapping audit are part of the proposed comparison.

A second benchmark makes the cross-window restriction explicit. Let Y be the vector of signed contributions from n prespecified disjoint windows, with reference distribution $N(\mu, \Sigma)$, where Σ is positive definite. For the Gaussian reference law conditioned on $\mathbf{1}^{\top} Y = 0$,

$$\mu_c = \mu - \frac{\Sigma \mathbf{1} \left(\mathbf{1}^{\top} \mu\right)}{\mathbf{1}^{\top} \Sigma \mathbf{1}}, \qquad \Sigma_c = \Sigma - \frac{\Sigma \mathbf{1} \mathbf{1}^{\top} \Sigma}{\mathbf{1}^{\top} \Sigma \mathbf{1}}.$$

Consequently $\mathbf{1}^{\top} \mu_c = 0$ and $\Sigma_c \mathbf{1} = 0$. For $\Sigma = \sigma^2 I$, each off-diagonal covariance is $-\sigma^2/n$ and each marginal variance is $\sigma^2(1 - 1/n)$. This is a coupled mean-and-covariance prediction. Independent, zero-mean additive measurement error with covariance R preserves the conditional mean and changes the observed covariance to $\Sigma_c + R$. The benchmark therefore uses an external neutral origin, prespecified disjoint windows, and an independently calibrated observation model. The reference mean and covariance are estimated independently of the confirmatory closure fit, with any systematic measurement bias receiving its own calibrated correction. These choices leave the total free to be estimated from the observations. By contrast, subtracting each trajectory's own mean forces its observed sum to zero, and selecting trajectories by their totals directly changes the distribution being studied.

This Gaussian route also supplies explicit nonzero-total comparisons. Conditioning on a fixed nonzero total yields the same covariance restriction while changing the conditional mean. When prefix variance is positive, a slope of -1 between latent prefix and suffix follows from any fixed terminal total. The comparison must therefore join mean location, covariance,

complete coverage, and independently identified endpoints, with fixed-nonzero and diffuse-total models included. An episode-level application tests a separately specified local-closure hypothesis.

An analytic check makes the distinction concrete. Use four dimensionless window contributions with independent standard-normal reference values and independent measurement errors of variance 0.04 per window. Conditioning the latent total to zero gives each window mean 0; conditioning it to 2 gives each window mean 0.5. Both fixed-total models have latent marginal variance 0.75 and off-diagonal covariance −0.25. Measurement error changes the marginal variance to 0.79 while leaving the off-diagonal covariance −0.25. The observed total therefore has variance 0.16 in both fixed-total models. Without terminal conditioning, its variance is 4.16.

Covariance contraction separates these fixed-total constructions from the unconditioned reference but cannot distinguish zero total from total 2. Their mean locations provide that additional contrast when the neutral origin and measurement bias are independently anchored. These analytic results provide exact numerical targets for checking benchmark code.

25.2.4 — Frailty, Medication, and Observation Models

Generate the clinical state, treatment, report-capacity, and observation processes specified in Section 22.6, including the frailty and medication pathways (Aalen et al., 2008; Hernán & Robins, 2020).

Allow severity to drive reserve proxies, medication, observation, and endpoint timing jointly; treating those effects as independent understates the rival’s explanatory capacity.

25.2.5 — Hybrid and Null Worlds

Hybrid worlds combine mechanisms; null worlds test whether noise, observation, and endpoint processes alone produce false support (Wilson & Collins, 2019).

Combine candidate and ordinary components with bounded, declared interactions to test over-attribution in mixed records.

In null worlds, estimate false-support rates and their simulation uncertainty under the specified observation and endpoint artifacts. Require the declared error criterion before confirmatory use.

Hybrid and null worlds also support false-equivalence and false-defeat checks. False equivalence occurs when the procedure declares the specified finite-band claim although the generating estimand lies outside that band. A nonzero generating value inside the band is not a false equivalence. False defeat means rejection of the particular claim that is true in the generated world. Report both rates against the exact target and decision rule used.

Report component recovery and residual/open-set diagnostics: a hybrid can partly match several candidates, while a null world can expose their shared inadequacy (Talts et al., 2018; Wilson & Collins, 2019).

25.2.6 — Versioned Assumption Registry

Register each generator’s release, mathematical assumptions, parameters, units, observation and endpoint rules, and known nonidentifiability regions. A change records its author, reason, and required reruns.

Access-tier rules for public versus sensitive benchmark materials are specified in Section 25.8.5. The assumption registry records which tier governs each object.

Add the mechanism statement, mimicked signatures, and assumptions that make the comparator unfairly strong or weak.

25.3 — Design Separability and Power

Power is specific to a decision. A design may detect a mean difference yet remain unable to distinguish the mechanisms that generate it. Planning therefore compares expected information and recovery across the intended candidate–rival contrasts, tests redundant signatures and unknown generators, and accounts for observation burden. If ethically available measurements cannot separate the relevant alternatives, increasing enrollment may add precision without making the proposed mechanistic inference possible.

25.3.1 — Expected Information by Design

Estimate separation under the proposed sampling, channels, and endpoint rule before recruitment (Gneiting & Raftery, 2007).

Compare the expected protected log-score gap with simulation uncertainty for each design and pair.

Equation 25.3, Expected log predictive discrimination.

$$D_{ab}(d) = E_{G_a,d}\left[\log p_a\left(y^{test}|y^{train},d\right) - log p_b\left(y^{test}|y^{train},d\right)\right]$$

Role label: design diagnostic. The quantity compares rival predictive performance under design d before human recruitment; small or unstable values indicate nonseparability.

A distribution-level bound complements fitted-model recovery. Let P and Q be a specified candidate and rival law for the complete latent record. Apply the same observation kernel K, including cadence, noise, missingness, endpoint classification, and recorded selection. For a measurable test $0 \le \varphi \le 1$ with $\mathbb{E}_{QK}\varphi \le \alpha$,

$$\mathbb{E}_{PK}\varphi \le min\{1, \alpha + TV(PK,QK)\} \le min\{1, \alpha + \sqrt{D_{KL}(PK\|QK)/2}\}.$$

$$D_{KL}(PK\|QK) \le D_{KL}(P\|Q), \qquad TV(PK,QK) \le TV(P,Q).$$

Total variation bounds the difference in rejection probabilities; Pinsker's inequality and information contraction give the remaining bounds, using natural logarithms (Cover & Thomas, 2006). A joint observable-record divergence below $2(\pi-\alpha)^2$ rules out target power $\pi>\alpha$ for that pair at size α. This is a necessary design condition, not a sufficient power calculation. Nominal reports cannot be counted as independent information, and an estimated divergence needs its own uncertainty.

Conditioning on survivors or complete reports changes the experiment. Analyze the joint selection-and-outcome record, or derive the selected laws and retain their selection probabilities; conditioning can increase separation, so unconditional contraction cannot simply be reused. For composite families, one difficult admitted pair already limits uniform performance.

Compare discriminating information with burden: dense active reports may separate fast dynamics yet be unsuitable for terminal care, while passive physiology may leave valence ambiguous.

Exactness has a calculable observation limit. In a simple calibrated model, observe $Y=L+e$, where e is centered Gaussian error independent of L, with known variance $\sigma^2>0$ and an independently fixed zero origin. Compare $L=0$ with a continuous rival L distributed as $N(0,\eta^2)$. The observable variances are σ^2 and $\sigma^2+\eta^2$. With $r=\eta^2/\sigma^2$, their divergence is

$$D_{KL}(P_\eta\|P_0)=1/2\{r-log(1+r)\}\sim\frac{\eta^4}{4\sigma^4}.$$

For n independent records, joint divergence is n times this quantity. The observation-space power bound therefore rules out uniform separation of exact zero from every arbitrarily concentrated continuous rival by a fixed finite design. A useful

design instead specifies the departures it must resolve and calculates a resolution curve that includes its full calibration and observation uncertainty.

Error calibration can be the decisive resource. If σ^2 is unknown, an exact residual with error variance σ^2 and a nonzero residual of variance η^2 with error variance $\sigma^2 - \eta^2$ produce exactly the same Gaussian observation law whenever $0 < \eta^2 < \sigma^2$. More outcomes from that ambiguous channel do not identify which explanation generated them. Independent reference measurements or qualified replicate channels must constrain the error law, including whether its gain, origin and variance transport to the actual observations. Study planning should allocate effort to that qualification as well as to outcome collection.

Population identification remains a different question. For known independent additive error with a nowhere-zero characteristic function, division of the observed characteristic function by the error characteristic function identifies the latent distribution. This does not guarantee stable recovery from a finite sample. Unknown origin, missing prefix and uncertain endpoint require their own identification work even when the noise law is known.

Section 24.6.4’s interval-observation test also has a conservative power bound. Keep its fixed B, α, coverage penalties γ_0,γ_1, and independent evaluation units. Under a specified alternative interval law, choose one orientation in advance and define $\Delta=E[l_0]-E[u_1]-2B(\gamma_0+\gamma_1)>0$. Here the expectations describe intervals produced by the fixed observation and calibration procedure, not ideal latent values. Put $s=1/n_0+1/n_1$. If $\Delta>\rho_\alpha$, a one-sided application of Hoeffding’s inequality gives

$$P(\text{no rejection}) \le \exp\{-\frac{(\Delta-\rho_\alpha)^2}{2B^2 s}\}.$$

Nonrejection forces the chosen empirical interval contrast at least $\Delta-\rho_\alpha$ below its expectation. Each independent summand has the bounded range used in the original test, which proves the display. For desired power at least $1-\beta$, $0<\beta<1$, equal group sizes $n_0=n_1=n$ therefore suffice when

$$n \ge \frac{4B^2\left(\sqrt{\log(\frac{2}{\alpha})}+\sqrt{\log(\frac{1}{\beta})}\right)^2}{\Delta^2}.$$

In synthetic units, $B=1$, $\alpha=0.05$, $\beta=0.20$, and a justified alternative margin $\Delta=0.30$ give $n\ge 452.0675$, hence 453 independent units per group. At $n=453$ the bound on nonrejection is approximately 0.1983. This is a conditional planning result, not a recommended clinical cohort size or a claim that current instruments supply that margin (Hoeffding, 1963).

For the M-witness family in Section 25.3.3, use its threshold $\rho_{\alpha,M}$ and replace $\log(2/\alpha)$ by $\log(2M/\alpha)$ in the sample-size display. The alternative margin must be justified for one specified witness and orientation.

Estimate plausible interval laws in a separate pilot or validated simulation; evaluate a range of margins rather than insert the eventual observed contrast into this planning formula. If width, missingness, or calibration uncertainty makes Δ nonpositive, increasing n alone cannot make this guarantee apply. Allocate any additional calibration-failure probability explicitly. Report simulation uncertainty, channel costs, and participant burden so the next design can improve the limiting measurement instead of merely enlarging the cohort (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.2 — Discriminating-Power Matrix

Use Section 25.2's generator families as rows and the proposed discriminating signatures as columns, adding relevant adaptation and opponent-process variants.

Record each contrast's conditions, simulation uncertainty, and redesign threshold. Leave failed or unevaluated cells visible. If no ethically available signature separates the models, redesign or classify the work as exploratory or methodological (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Confirmatory recruitment requires high discrimination, stable calibration, acceptable burden, and ethical permission. Passing the benchmark is necessary within this decision rule.

An exploratory study may proceed when some contrasts are informative but key rivals remain unresolved. Its public language must retain exploratory status.

Confusion among critical rivals or weak open-set detection requires redesign. Change the cadence, channels, endpoint system, or model set before using the affected comparison.

Do not recruit for a question that remains nonseparable under ethically permissible observation. Participant burden cannot be justified by data that cannot address the stated comparison.

25.3.3 — Redundant Signatures

Evaluate each signature's added information for the declared contrast against its collection burden (Wilson & Collins, 2019).

Map shared causes: concentration, final-window compression, and last-observation calm may all reflect the same shutdown process.

Include common sensor failures, shared reporting influences, and sedation when evaluating channel dependence.

Remove a signature only when it adds burden without useful information. Dependence alone does not require deletion: a

justified joint likelihood or calibrated joint rule can use correlated observations without treating them as independent confirmations.

Section 24.6.4's interval-observation test can examine several distributional features without counting them as independent confirmations. Retain its fixed sample sizes, assignment-defined groups, independent evaluation units, and complete-stream premises. Prespecify a positive integer M and M 1-Lipschitz witnesses $f_m : \mathbb{R} \to [a_m, b_m]$, with fixed finite range widths $\omega_m = b_m - a_m > 0$, or train the entire family on independent records. Conditional on the independently fitted, frozen calibration and witness training, retain valid per-unit miscoverage bounds γ_0, γ_1 and the independent-unit requirements. If those calibration claims hold only on a separately controlled event, allocate its failure probability as in Section 24.6.4. For each nonempty complete-remainder interval C_{ji}, let $l_{ji}^{(m)}$ and $u_{ji}^{(m)}$ be the infimum and supremum of f_m over the entire interval, not merely its endpoints.

For witness m, let A_m be the larger of the two sample lower-minus-upper arm contrasts: $\bar{l}_0^{(m)} - \bar{u}_1^{(m)}$ and $\bar{l}_1^{(m)} - \bar{u}_0^{(m)}$. With natural logarithms and $s = 1/n_0 + 1/n_1$, define

$$\rho_m = \omega_m \sqrt{\frac{s}{2} \log\left(\frac{2M}{\alpha}\right)}, \qquad \Gamma_m = \omega_m (\gamma_0 + \gamma_1).$$

$$D_{\alpha,M} = \max_{1 \le m \le M} \{A_m - \rho_m - \Gamma_m\}.$$

Under exact closure and the resulting equality of complete-remainder laws, $\Pr(D_{\alpha,M} > 0) \le \alpha$, conditional on the qualified calibration. Under the null, each oriented contrast has expectation at most Γ_m: an uncovered interval can distort its witness contribution by at most ω_m. Its summed squared

summand ranges are $\omega_m^2 s$. Hoeffding's inequality and a union bound over all $2M$ orientations prove the result. Witnesses using the same records may be dependent; independence is required across evaluation units, with the stated conditioning. Choosing the common range $[-B, B]$ recovers the conservative formula with width $2B$ (Hoeffding, 1963).

The actual range matters for design. Pair signed clip-ping, whose range is $[-B, B]$, with the magnitude witness $f_2(x) = \min\{B, |x|\}$, whose range is $[0, B]$. Symmetric point laws at ± 0.2 and ± 0.8 have equal signed-clipped means when $B = 1$, but magnitude means 0.2 and 0.8. For 400 independent units per group, $M = 2$, and $\alpha = 0.05$, the signed allowance is about 0.2093 and the magnitude allowance is about 0.1047. The magnitude miscoverage penalty is also half as large. An interval crossing zero has magnitude lower bound zero; $C = \mathbb{R}$ contributes $[0, B]$ for that witness and $[-B, B]$ for signed clip-ping. Neither is a neutral point estimate. These are synthetic comparisons, not observed evidence.

The same refinement sharpens the conditional power calculation. For a fixed alternative interval law and a specified witness and orientation, define the net margin $\Delta_m = E\left[l_0^{(m)}\right] - E\left[u_1^{(m)}\right] - \Gamma_m > 0$. For power at least $1 - \beta$ with equal group sizes, Section 25.3.1's argument gives the sufficient bound

$$n \geq \frac{\omega_m^2 \left[\sqrt{\log(2M/\alpha)} + \sqrt{\log(1/\beta)}\right]^2}{\Delta_m^2}.$$

At the same stated net margin $\Delta_m = 0.30$, $\alpha = 0.05$, $\beta = 0.20$, and $M = 2$, a width-one witness gives $n \geq 125.5868$, hence 126 independent units per group; the conservative width-two bound gives $n \geq 502.3472$, hence 503. These are sufficient bounds, not

minimum cohort sizes or claims that current instruments attain the margin. The net margin includes the relevant coverage penalty and must be justified before evaluation. The signed-witness and common-range examples in Section 25.3.1 remain valid.

Record each witness's target, range, blind spots, shared errors, and inferential status in the signature dependency table. Verify the complete family's recovery and error control in simulation before requesting additional participant burden for the confirmatory comparison. A finite family can still miss differences outside its chosen features (Gneiting & Raftery, 2007; Wilson & Collins, 2019). For conditional randomization with different stratum mixes, retain Section 19.5.1's common fixed target weights and its conditional coverage and independence requirements. The range-aware extension replaces s by $\sum_h w_h^2 (1/n_{0h} + 1/n_{1h})$ and $\gamma_0 + \gamma_1$ by $\sum_h w_h (\gamma_{0h} + \gamma_{1h})$; it does not pool unequal stratum mixtures.

Compare the dependence-aware analysis with the independence-assuming analysis and report what changes. For two unit-variance measurements with correlation ρ, the variance of their average is (1+ρ)/2, not always 1/2. At ρ=0.8 it is 0.9; at ρ=−0.8 it is 0.1. Ignoring dependence can therefore make this uncertainty either too small or too large. Retain a scientifically justified joint model and evaluate information for the specified contrast rather than automatically collapsing correlated signatures (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.4 — Open-Set Separability

Test the none-of-the-models option using Section 25.5's adequacy diagnostics alongside pairwise recovery (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Generate plausible worlds outside the fitted library, such as nonlinear mixtures, changed care practices, hidden subgroups, and unfamiliar observation dynamics.

Test a world where reporting becomes more likely during calm states. Confident reserve attribution there exposes a failure to detect the observation explanation.

Report anomaly calibration, detection, and abstention on boundary generators that resemble familiar curves but violate a relevant assumption (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

25.3.5 — Burden-Adjusted Power

Compare expected discrimination with participant and staff burden, privacy exposure, family intrusion, and device tolerance (Wilson & Collins, 2019).

Report the discrimination gained per added burden unit and compare less intrusive channels that might resolve the same contrast.

Include measurement reactivity (the possibility that participation changes the experience being studied) in the burden assessment.

Compare minimal, preferred, higher-burden, and rejected designs with their discrimination uncertainty and failure modes.

25.3.6 — No Study When Models Are Nonseparable

Link a no-study decision to its recovery artifact, unresolved pair, and targeted repair: change sampling or channels, narrow the estimand, proceed exploratorily, or decline recruitment. More of the same uninformative data is not a repair (Gneiting & Raftery, 2007; Wilson & Collins, 2019).

Include artifact challenges in the discriminating-power matrix. Prespecified impossible windows can expose alignment or

preprocessing errors. Associations with device-quality channels warrant investigation under the observation model; they are not automatically evidence of artifact. Qualify proposed negative controls under Section 14.5.5.

A control that fails only after one preprocessing step can identify the stage where discriminating power is being manufactured.

25.4 — Endpoint and Observation Rival Envelope

Endpoint conditioning and observation selection can imitate the same terminal patterns that a candidate dynamic is intended to explain. The rival envelope therefore includes both how an endpoint is generated and how measurements continue, stop, or disappear around it. Random-terminal bridges, survival conditioning, shared disease drivers, and care pathways require explicit implementations under Sections 22.5 and 22.6. Freeze their comparison rules before outcomes are inspected, and version any later addition.

25.4.1 — Random-Terminal-Time Bridges

A bridge combined with censoring of late high-volatility windows may strengthen apparent terminal concentration.

A restriction shared by the candidate and conditioned-bridge models can be tested even when their mechanisms are observationally indistinguishable.

Register the random-endpoint law, conditioning event, reference process, observation schedule, and holdout. Identify what the conditioning imposes and what prospective observation could distinguish.

Compare preterminal covariance, hazard response, and intervention behavior to distinguish native dynamics from event conditioning and observation selection (Doob, 1957; Aalen et al., 2008; Léonard, 2014).

25.4.2 — Doob-Transform and Survival Conditioning

Implement Doob-transform and survival-conditioned competitors from Section 22.5 through the common observation and scoring interface (Aalen et al., 2008; Little & Rubin, 2019).

Vary conditioning on survival, endpoint class, open-channel status, and the selected terminal features.

Archive the code, base law, conditioning event, h-function or weighting rule, endpoint support, censoring, and any limiting construction for zero-probability events. State the prospective diagnostic distinguishing the construction (Doob, 1957; Aalen et al., 2008; Little & Rubin, 2019; Léonard, 2014).

A random-terminal bridge combines endpoint uncertainty with conditioning. It can generate apparent drift and variance contraction near an aligned endpoint.

Doob or survival conditioning selects on a future event or open-channel status and can generate conditional drift without a PTVC-specific implementation.

Treating device removal or report cessation as the endpoint can create apparent calm or an artificially narrow final window. Model that last-observation artifact explicitly.

Shared disease drivers can move reserve, observation, endpoint timing, and proxies together. Frailty can therefore generate correlated signatures without latent closure.

Care pathways involving sedation, analgesia, hospice practice, and comfort-first monitoring can generate observed relief or missingness. Those effects need not arise from a PTVC-specific route.

Add a new rival before protected evaluation through a prospective version update. This strengthens the rival envelope without turning an observed loss into an after-the-fact rescue.

25.4.3 — Observation and Device-Removal Alignment

Use the observation-ending distinctions in Section 22.5.3 when constructing alignment rivals (Little & Rubin, 2019).

Generate removal and report loss from the relevant device, clinical, participant, and care decisions in Section 22.6.

Compare true and adjudicated endpoints, last observation, last valid device window, care transition, and retrospective death alignment.

A signature confined to last-observation alignment requires investigation of that observation mechanism before model attribution.

25.4.4 — Shared Disease and Sensor Confounding

Link clinical progression, cognition, medication, perfusion, and device adherence to both channel values and future observation probability.

Allow shared stochastic shocks between reserve-like variables and measured proxies. Fitting their covariance with a finite-variation reserve shortcut can misattribute the interaction.

Report which signatures survive the shared-driver worlds before assigning their effects to the proposed mechanism.

Vary common cause, sensor gain, dropout, and clinical progression independently in public synthetic fixtures; include the causal graph and thresholds at which contraction becomes unidentified (Aalen et al., 2008; Hernán & Robins, 2020).

25.4.5 — Treatment and Care Pathways

Use the treatment and care mechanisms in Section 22.6.2 as executable rival pathways (Sanders et al., 2024; Surges et al., 2024; Hernán & Robins, 2020).

Generate treatment initiation, dose changes, care transitions, and observation cessation jointly with the evolving state that influences treatment decisions.

Compare treatment-history adjustment, explicit generative treatment models, and care-pathway stratification. Identify when the treatment change also changes the estimand.

Version dose histories, care transitions, observation rules, and synthetic trajectories so reviewers can ablate a pathway and rerun model comparison (Aalen et al., 2008; Hernán & Robins, 2020; Little & Rubin, 2019).

25.4.6 — Envelope Update Before Results

Add a rival through a new registered generator and protected challenge before inspecting its evaluation outcomes (Aalen et al., 2008; Little & Rubin, 2019).

Record the assumptions, implementation diff, protected-data access date, affected results, and runnable predecessor (Nosek et al., 2015; Munafò et al., 2017).

25.5 — Open-Set Adequacy and None-of-the-Models

All fitted models may fail. Open-set checks test that possibility. Novel observations must be recognized. Otherwise a winner is not warranted (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006; Scheirer et al., 2013; Hendrycks & Gimpel, 2017; Ovadia et al., 2019).

25.5.1 — Posterior Predictive Misfit

Check protected features that matter to the claim beyond the score used for selection (Vehtari et al., 2017).

Compare fitted-model replications with prespecified trajectory, observation, event, tail, and calibration diagnostics.

A fit to endpoint concentration can fail observation-shutdown checks; a fit to missingness can fail reserve-linked contraction. Record both adequacy results.

Show prespecified discrepancies by endpoint window, treatment stratum, and observation group; aggregate agreement can conceal the decisive failure (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.2 — Residual and Calibration Diagnostics

Evaluate residuals and probability/interval calibration by the time, clinical, site, device, and observation strata relevant to the claim (Yao et al., 2018; Stanford, 2006).

Use residual dependence on treatment, device, workflow, or endpoint uncertainty to identify candidate omissions for the next model version.

A concrete computational check distinguishes using the observations from merely returning a plausible distribution. Standard parameter-only simulation-based calibration can miss an algorithm that ignores the data and samples from the prior. Modrák et al. (2025) show why data-dependent test quantities,

including the joint likelihood, make this failure detectable. For a reproducible scalar fixture, generate θ from N(0,1) and y conditional on θ from N(θ,τ²), with τ>0. The exact posterior is

$$\theta \mid y \sim N\left(\frac{y}{1+\tau^2}, \frac{\tau^2}{1+\tau^2}\right).$$

A deliberately defective implementation that returns independent N(0,1) draws for every y still gives exactly uniform parameter ranks in Equation 25.4, averaged over this generator. The generating θ and those draws are identically distributed and independent. At τ²=0.04 and y=1, however, the correct posterior has mean 25/26 and variance 1/26, whereas the defective implementation returns mean zero and variance one.

Also rank the data-dependent quantity $g_y(\theta)=-(y-\theta)^2/(2\tau^2)$, evaluating every draw with the same simulated y. Under the correct posterior, conditional exchangeability gives the uniform rank reference; the prior-only implementation has a nonuniform likelihood-rank law. Retain the existing independence and tie requirements. Archive the generator, deliberately defective fit, correct fit, rank outputs, and prespecified diagnostic thresholds alongside stratum-level coverage and held-out predictive checks. Passing a finite diagnostic set supports computational agreement at its tested resolution.

Equation 25.4, Simulation-based calibration rank statistic.

$$r_m = \sum_{s=1}^{S} \mathbf{1}\left\{\theta_m^{(s)} < \theta_m^{\mathrm{sim}}\right\}$$

Role label: computational calibration check. Here m indexes a simulation replicate for one declared scalar parameter or test quantity, S is the number of posterior draws, the superscript sim marks the generating value, and superscript (s) labels its s-th posterior draw. Independent posterior draws yield the usual rank

reference law; dependence must be accounted for when calibrating the diagnostic. For discrete test quantities or ties, randomized ranks or another prespecified tie rule is required. Departures from expected rank uniformity under the assumed generator can reveal algorithm or implementation problems, but the check does not validate the scientific model as nature's generator (Talts et al., 2018; Säilynoja et al., 2026).

Posterior predictive failure occurs when key held-out features fall outside the replicated-data behavior. The model is then inadequate for the tested claim.

Systematic residual error by site, endpoint distance, device, or treatment can indicate a missing rival or observation mechanism. Preserve that structure in the model diagnosis.

Unreliable probabilities or intervals in critical strata constitute calibration failure. Such model confidence cannot support the verdict.

Protected data unlike all candidate worlds create an open-set anomaly. Evaluate whether none-of-the-models status is required.

Marked predictive superiority by a flexible nonmechanistic reference suggests that the submitted model set may omit an important mechanism. It does not identify that mechanism by itself.

An independently motivated new generator must make fresh predictions and enter a new benchmark version before it can change claim status.

25.5.3 — Holdout Anomaly Tests

Use protected anomaly tests to detect an incomplete model set (Stanford, 2006).

Split at the dependency unit in Section 25.6.2. Random observation-level splits can leak person, site, device, or endpoint information.

Prespecify scores, discrepancies, or learned summaries for detection. A flexible detector becomes a mechanistic explanation only through a separately specified generative model and fresh predictions.

A new site's shutdown policy can place its records outside the trained observation envelope.

Near-miss generators should preserve familiar marginal curves while introducing a scientifically relevant difference in the joint observations. The benchmark tests whether that observable difference can be detected. If two generators induce exactly the same law for all permitted observations, no anomaly detector can distinguish them from those observations alone; report structural nonidentifiability and identify what additional measurement or assumption would be required (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.4 — Flexible Reference Models

Use Gaussian processes, splines, state-space learners, or ensembles to detect predictable structure missed by mechanistic candidates (Vehtari et al., 2017; Yao et al., 2018).

Treat flexible references as predictive benchmarks, not identified mechanisms.

A fully specified Gaussian example shows how an additional channel can make a comparison informative. Let Z be a mean-zero stationary Ornstein–Uhlenbeck process with return rate $\beta=3$ and variance $q=1$. Define $Y(t)=\int_0^t Z(s)ds$ and $U(t)=1-t$ on $T=1$. Model C has closing latent coordinate UY and primary observation $D=UY+\varepsilon$. Model S has latent coordinate Y, with no terminal-zero constraint, and primary sensor gain U, producing

the same D. With the same independent observation errors, their primary records have identical laws; equal-prior optimal classification is exactly 50%, however many such primary-only records are collected.

Add a separately calibrated witness V=UY+η under C and V=Y+η under S. Sample at 12 equally spaced times from 0.05 to 0.95. Let the mean-zero Gaussian errors be mutually independent across times and channels and independent of Z, with primary standard deviation a=0.25 and witness standard deviation b. For s≤t, the integrated-OU covariance is

$$K(s,t) = \frac{q}{\beta^2}[2\beta s - 1 + exp(-\beta s) - exp(-\beta(t-s)) + exp(-\beta t)].$$

Let K be this covariance matrix at the observation times, $A = diag(1 - t_j)$, and I the 12-dimensional identity. The zero-mean joint record (D, V) has covariance

$$\Sigma(B) = \begin{bmatrix} AKA + a^2 I & AKB \\ BKA & BKB + b^2 I \end{bmatrix}.$$

Use B=A for C and B=I for S. The primary covariance is identical; the witness and cross-channel blocks differ. Direct Gaussian calculation gives KL(C‖S)=1.3077, 0.3043, and 0.0939 nats per paired record for b=0.25, 1, and 2 respectively:

$$KL(C \parallel S) = \frac{1}{2}[tr(\Sigma_S{}^{-1}\Sigma_C) - 24 + ln(\frac{det\ \Sigma_S}{det\ \Sigma_C})].$$

These are reproducible information calculations for a specified synthetic pair, not participant findings or fitted-model recovery. Witness calibration can be challenged quantitatively. Replace S's witness loading by Bλ=(1−λ)I+λA, where λ is a fixed sensitivity parameter. At b=1, KL(C‖Sλ) is 0.3043, 0.0916, and 0 nats for λ=0, 1/2, and 1. At λ=1 the complete joint laws coincide: a second channel then adds observations but no model

discrimination. An informative design therefore qualifies the witness's gain relation, not merely its presence.

The accompanying Reproducible_Benchmarks directory contains the executable Gaussian calculation and a separately fitted training, selection, and test comparison for this synthetic pair. Its records identify the laws, split assignments, and scoring rules.

Use nested tuning, protected holdouts, regularization, calibration, and transport tests for flexible references.

Prespecify how a flexible reference's predictive advantage changes the mechanistic comparison: revise the model set or observation plan, narrow the target, or withhold the affected verdict (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

25.5.5 — None-of-the-Models Verdict

Freeze the adequacy triggers: protected predictive failure, calibration failure in critical strata, or unexplained structure across all candidates. Failure to meet open-set recovery thresholds instead leaves the analysis unqualified.

Record the failed check, affected target, blocked conclusion, and next action in the machine-readable result (Vehtari et al., 2017; Yao et al., 2018; Stanford, 2006).

The best member of a list can still miss the process being observed. In a hypothetical benchmark, every fitted candidate predicts an immediate response, while the protected observations consistently show a delayed response. One candidate will have the smallest error, but the registered adequacy test may reject all of them. Preserve both the closed-set winner and the open-set verdict in the benchmark record. Their difference shows what the forced-choice procedure would have concluded and which unexplained feature motivates the next model.

25.5.6 — Model Entry and Retirement

Use Section 25.2.6's generator registry and Section 25.4.6's update record for new entrants, adding the recovery plan, computation, proposal and access dates, and consequence rule. Chapter 28 governs changes to the hypothesis.

Test a revision on a new protected challenge after it explains the case that motivated it.

Narrow or retire models that repeatedly fail the required recovery, adequacy, or transport checks.

25.6 — Transport and Stress Tests

Transport tests ask which parts of a successful development analysis survive a specified change of environment. Site practice, population, instrument version, prevalence and observation quality can change separately or together. Holding the other parts of a synthetic comparison fixed makes each source-to-target test interpretable; a later combined test examines whether those changes interact (Arjovsky et al., 2019; Peters et al., 2016; Yao et al., 2018; Ovadia et al., 2019; Sagawa et al., 2020).

25.6.1 — Site Shift

Report shift-specific performance rather than averaging development and changed conditions together.

Site-specific recovery may depend on learned care and observation practices (Arjovsky et al., 2019).

Keep every observation from a held-out site outside tuning and model selection. This tests transfer across care and observation regimes while blocking development leakage. Separately perturb synthetic site parameters beyond the development range to identify which changes explain a loss of performance.

For example, imagine a target hospice that removes a device earlier for comfort than the development sites. The underlying state process could remain the same while the final observed window changes. A synthetic replay can hold that state process fixed and substitute the target site’s removal schedule. If the predicted late pattern changes, the comparison has located a dependence on observation practice. The site-specific result then identifies the schedule information needed for a more transferable model.

Report site-specific prediction, calibration, recovery, confusion, and the variables needed to improve transport.

Release site manifests, protected holdout assignments, calibration procedures, and transport weights at the appropriate access tier, allowing reviewers to vary the target mix and locate failures (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

25.6.2 — Population Shift

Specify population differences that can change a channel's meaning or calibration (Peters et al., 2016; Yao et al., 2018).

Vary those parameters and hold out population strata, checking sign, calibration, missingness, endpoint rules, and channel weights.

Model relevant language, expression, family, and refusal practices rather than absorbing them into unexplained noise.

Low movement illustrates why a population label alone is insufficient. The same reading can accompany fatigue, a motor limitation, comfort, device placement or a person's ordinary baseline. A population-shift generator represents the relevant mixture of those pathways, and a calibration sample tests how the channel relates to the intended state in the target group. Changing only the group's frequency in a pooled dataset does not perform that calibration.

Person holdouts keep all observations from a person out of development when testing that person. They assess generalization across streams while blocking person-level leakage.

25.6.3 — Instrument Shift

Instrument and processing changes can move measured values while the latent process remains unchanged (Yao et al., 2018).

Keep instrument versions out of development when testing transfer to those versions. Vary calibration transfer, feature definitions, sampling frequency and sensor quality, so the comparison identifies which part of the observation system changed.

When reprocessing changes historical values, identify the frozen predecessor, required reruns, and results that are no longer comparable.

When replacement coincides with clinical decline, use bridge samples, calibration fixtures, and version-specific models to separate origin, gain, lag, and floor changes (Little & Rubin, 2019; Hernán & Robins, 2020).

Where stored raw signals permit it, processing the same input through the old and new software provides a paired engineering comparison. Differences in origin, gain, lag or artifact rejection then arise from the processing change rather than from different people or different clinical moments. That comparison can establish a version-specific observation bridge. A separate target sample still tests whether the bridge describes the conditions in which the new version will be used.

25.6.4 — Prevalence and Prior Shift

Vary generator and event base rates relevant to the source and target populations (Arjovsky et al., 2019; Peters et al., 2016).

Report how posterior model probabilities, stacking weights, thresholds, and open-set decisions respond.

Document source and target prevalences, covariate support, prior weights, and recalibration. Identify model ordering dependent on an unverified target prevalence (Sugiyama et al., 2012; Pearl & Bareinboim, 2014).

A focused prevalence test keeps each generator’s conditional observation law fixed while changing how often the generator

appears. The evidence supplied by a particular trace can remain unchanged even as its posterior model probabilities change with the model weights. Report those two contributions separately. If the target population also changes what a given mechanism produces, run that as an additional conditional-model shift, rather than attributing every change in the verdict to prevalence.

25.6.5 — Adversarial Missingness and Noise

Locate the observation conditions at which recovery fails and test whether the pipeline detects that failure (Peters et al., 2016).

Remove high-burden observations near endpoints, increase low-reserve noise, or favor calm-state reporting to test the resulting false-support pathways.

Also measure false rejection when the same stressors hide a genuinely present candidate signature.

A paired synthetic test makes the two error directions visible. Generate one set of histories from the candidate mechanism and another from a rival, then apply the same family of observation masks and noise levels to both. As information is lost, track whether the analysis retains the correct distinction, reports ambiguity, or selects the wrong model. The first world tests how a real candidate pattern becomes hidden; the second tests how selective visibility creates an apparent candidate pattern. A useful recovery boundary describes both worlds under the same observation burden.

Release the missingness generator, noise distribution, severity grid, and decision rule so independent reruns can locate calibration failure and changed model ordering (Little & Rubin, 2019; Hernán & Robins, 2020).

25.6.6 — Transport Failure

A failed source-to-target test limits the demonstrated use of the method while retaining its qualified local result (Yao et al., 2018; Arjovsky et al., 2019).

Classify the failed transfer dependency before choosing the response.

For each source-target transfer, identify which parameters, recovery, calibration, endpoint diagnostics, and predictive scores survived or failed, with sensitivity results.

For example, a benchmark may recover a return-rate parameter accurately when the generating model is supplied, yet fail to choose that model from its rivals at a new site. Parameter recovery conditional on a known model and model discrimination are different accomplishments. Conversely, a pipeline may select the correct family while its uncertainty intervals lose calibration. Reporting those outcomes separately identifies whether the next experiment needs a better discriminator, a revised uncertainty model or additional calibration evidence.

For stress tests beyond the development range, predeclare whether each transfer failure narrows the target, changes measurement or rivals, blocks estimation, or retires the affected route (Arjovsky et al., 2019; Peters et al., 2016; Yao et al., 2018).

25.6.7 — Time and Endpoint Holdouts

Forward-chaining time folds keep future evaluation windows out of fitting and tuning. They test temporal prediction and disciplined updating.

Endpoint-threshold holdouts protect stopping decisions from development leakage. They test whether the comparison survives independent endpoint choices.

The two tests address different routes by which evaluation information can enter development. A pipeline may respect forward time order yet select the endpoint threshold that gives the most favorable held-out result. Another may fix the threshold in advance yet use later observations while smoothing an earlier prediction window. Protecting one choice does not protect the other. For each test, record the held-out unit, the information withheld and the point at which the relevant rule became fixed.

25.7 — Red Teams and Challenge Logs

Independent reviewers should be able to challenge the executable pipeline before public use and trace the consequences of their findings (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a; Munafò et al., 2017; Cogitate Consortium et al., 2025).

25.7.1 — Independent Red-Team Roles

Assign independent reviewers to the model, measurement, endpoint, software, and ethics/public roles in Section 25.7.2 (Nosek et al., 2018).

Adversarial collaboration requires advance agreement on predictions, operationalization, analysis, documentation, and arbitration. Participants report that it often clarifies boundary conditions rather than producing a single winner (Ceci et al., 2026; Isch et al., 2025). Use it to define disagreements the benchmark can actually test.

Give red teams synthetic benchmarks, documentation, plans, code, and non-sensitive artifacts before participant results are interpreted and while releases remain easy to change. Apply Section 25.8.5 to sensitive access.

Allow rival generators and failure tests, recording rejected proposals and reasons as well as accepted challenges.

25.7.2 — Challenge Bounties

Challenge bounties can reward reproducible rival generators, code failures, leakage, calibration flaws, misleading artifacts, and open-set worlds that defeat the benchmark (National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

Publish bounty criteria for the affected artifact, reproducible failure, expected consequence, severity, and repair where available.

Exclude harassment, participant exposure, and clinical intrusion from bounty eligibility.

Disclose funding conflicts, validity adjudicators, and an appeal route outside the challenged team.

Resolve challenges with code and synthetic artifacts first. Human or animal data require a specific gain unavailable by less intrusive means (Nosek et al., 2018; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

Credit reproducible scientific impact, including low-profile structural defects, rather than publicity or spectacle (Nosek et al., 2015; Munafò et al., 2017).

The model red team challenges generators, priors, identifiability, and the rival envelope. Its findings may require a new generator or a narrower claim.

The measurement red team challenges channels, calibration, missingness, and sign conventions. It may require an observation-model correction or No adequate estimate.

The endpoint red team challenges stopping rules, alignment, and leakage. It may require rerunning endpoint analysis or excluding a flawed result.

The software red team challenges code, containers, splits, seeds, and dependencies. Its findings may require a patch, rerun, or security update.

The ethics and public-communication team examines burden, privacy, captions, and risks to vulnerable readers. It may require revised wording or withdrawal of an artifact.

Equation 25.5, Challenge-log completeness condition.

$$C_k := (a_k, r_k, s_k, p_k, u_k)$$

$$\text{Complete}(C_k) = 1 \text{ iff } a_k, r_k, s_k, p_k, \text{ and } u_k \text{ are all present.}$$

The fields record the attack or challenge, reproduction status, severity, patch or response, and public-status update. Role label: artifact-governance diagnostic. The tuple is not a statistical theorem; it is a reproducibility rule for adversarial challenge records.

25.7.3 — Exploit Log

Give each challenge a stable ID, affected release, reproduction fixture and environment, expected and observed failure, severity, response, rerun result, disposition, and claim-status consequence. Preserve enough to replay both defect and repair (Nosek et al., 2015; Nosek et al., 2018; Munafò et al., 2017; Nüst et al., 2020; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

An exploit here is a reproducible route to overclaiming, uncertainty loss, misclassification, disclosure, or misleading presentation.

Rate severity by scientific and disclosure consequences; endpoint leakage and tuning on protected data directly compromise the comparison.

Track reproduced, unreproduced, disputed, fixed, accepted-risk, and retired challenges.

Redact public logs as needed while retaining the scientific lesson. Security-sensitive details may remain restricted or be delayed, but record the issue's existence, affected releases, and resolution status at an appropriate access tier.

25.7.4 — Public Review and Challenge Summaries

Adjudicate challenges by reproducibility, relevance, severity, and effect on claims. Public summaries distinguish submission, reproduced flaw, unresolved dispute, repair, and status change.

A public log can distinguish submissions from independently reproduced defects. Suppose twelve submissions trace to the same time-indexing error. Correcting that error repairs one defect across twelve reports; it does not establish twelve independent failures of the model. Conversely, one submission can expose several distinct defects that require different repairs. Give each reproducible defect its own identifier and link its affected submissions and releases to that record. Readers can then follow what changed without counting the same correction repeatedly.

25.7.5 — Critic Protection and Nonretaliation

Critic protection is a scientific requirement when a project has public identity, community interest, or donor attention. A critic who finds a real flaw should not be treated as disloyal, hostile, or morally suspect. The benchmark needs their pressure (National Academies of Sciences, Engineering, and Medicine, 2019).

Nonretaliation rules should cover publication, community moderation, access to artifacts, bounty eligibility, and public response. A critic can be wrong, sharp, or inconvenient without losing standing. Abuse and harassment can be moderated, but substantive criticism should remain protected.

The protocol should separate adjudication from public popularity. A challenge is assessed by reproducibility, relevance, severity, and effect on claims, not by whether supporters like the submitter.

Critic protection also protects the project. If critics fear retaliation, flaws move underground, and public artifacts become more brittle. A project that supports serious critique strengthens its scientific independence and public trust.

Critic-protection procedures should leave a versioned, inspectable record of the challenge, evidence, reproduction status, response, patch or nonpatch decision, confidentiality option, conflict or retaliation concern, response time, and appeal route outside the challenged team. Later claims of an open challenge culture should cite that operating record (Nosek et al., 2015; Nosek et al., 2018; Munafò et al., 2017; National Academies of Sciences, Engineering, and Medicine, 2019; UNESCO, 2021a).

25.7.6 — Public Challenge Without Theater

Public challenge can strengthen the program, but debate-as-verdict, humiliation clips, pile-ons, scoreboards, and winner-loser framing replace method with spectacle. Public artifacts should distinguish submitted challenge, reproduced flaw, unresolved dispute, resolved patch, and claim-status change rather than slogans such as proved or destroyed (UNESCO, 2021a; Nosek et al., 2018).

Challenges belong first to executable artifacts and bounded questions. Synthetic fixtures, thresholds, rival envelopes, and caption drafts let critics test the benchmark without exposing participants, families, clinicians, sites, or private records. A failed benchmark is never permission for personal endpoint testing, harm, clinical delay, or public scoring.

When the method is nonseparable or unrecoverable, more intimate data cannot substitute for missing analysis. The challenge must improve the model before it enlarges participant burden (National Academies of Sciences, Engineering, and Medicine, 2019).

Staged disclosure, neutral replication, and visible status changes keep preliminary exploits and institutional rebuttals from becoming personality contests. The aim is a reproducible

path from objection to correction, not humiliation (Nosek et al., 2015; Munafò et al., 2017).

Red teams should work before artifacts become costly to change, and mature logs should preserve unresolved entries without treating them as scandal. Some attacks await a stronger simulator; others expose wording risk rather than model error.

Public challenge without theater is not timidity. It is the courage to make the evidence trail stronger than prestige while keeping vulnerable people outside the spectacle.

25.8 — Code, Containers, and Reproducible Artifacts

A reproducibility claim identifies the executable release and its permitted data-access route; Section 25.8.2 separates the artifact layers (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020; Peng, 2011; Wilson et al., 2017).

25.8.1 — Versioned Source and Release IDs

Link each result to its code, generator, analysis, threshold specification, and artifact manifest through a release ID (Sandve et al., 2013).

Display the release ID in result tables and figures; flag affected results when a dependency changes.

Version source, dictionaries, environments, specifications, fixtures, seeds, and rendered outputs together.

Test that manifests and hashes distinguish scientifically different executable states even when their displayed titles match (Nüst et al., 2020; Wilkinson et al., 2016).

25.8.2 — Containers and Environment Files

Capture the runtime, numerical, compiler, operating-system, and relevant hardware dependencies (Boettiger, 2015; Wilkinson et al., 2016).

Provide pinned language and package versions, external dependencies, and execution commands; pair lighter environment files with locks where possible.

Archive hashes, images, and readable environment specifications, and test rebuilding as dependencies age.

Provide a quick fixture and a complete challenge run with documented compute requirements.

The source release identifies the exact code, generator version, and analysis version. It permits reproduction of the benchmark underlying a claim.

The container or environment records dependencies, operating system, and commands. These specifications permit rerunning the analysis without invisible software drift.

The machine-readable prospective specification records locked decisions, thresholds, holdouts, and status labels. It permits comparison of the executed analysis with the declared plan.

Sensitive empirical data and site-specific details occupy a restricted layer. That layer protects participants while preserving an authorized audit route.

The retirement record identifies superseded, deprecated, withdrawn, and archived artifacts. Earlier releases must not continue to appear as current authority.

Equation 25.6, Release identity.

$$R = (G, M, O, D, T, S, V)$$

Role label: reproducibility manifest. In this tuple, G identifies the generator release, M the fitted-model and analysis release, O the observation-system specification, D the data or synthetic-fixture release, T the decision-threshold specification, S the protected-split and seed specification, and V the software environment and manifest version. A result without a release identity cannot be fully reproduced or challenged under the standards established here.

25.8.3 — Synthetic Fixtures and Unit Tests

Use small known fixtures to verify formulas and transformations against the implementation's declared model. Include sign conventions, information leakage, and edge cases (Oberkampf & Trucano, 2002; Wilkinson et al., 2016).

Give each registered family and observation stressor a fixture with expected outputs or acceptable ranges.

Test signs, alignment, endpoint independence, splitting, predictive generation, calibration ranks, scores, and figure inputs.

Include deliberately defective cases: a leaked endpoint threshold should make the corresponding leakage test fail.

Wholly artificial fixtures can normally be released after a disclosure-risk check. Fixtures calibrated from sensitive participant or site data require review of what their parameters and outputs could reveal. A synthetic label alone does not establish that a release contains no private information.

Record the tested invariant, expected outcome, edge cases, software version, and maintainer.

25.8.4 — Machine-Readable Prospective Specification

Encode the fixed generator, observation, inclusion, tuning, holdout, threshold, adequacy, and status decisions in a specification the code can check (Nüst et al., 2020; Sandve et al., 2013).

Flag differences between that specification and the executed analysis.

Encode each run’s inferential role so exploratory outputs cannot enter a confirmatory table automatically.

Test the encoded estimands, exclusions, stopping decisions, and verdicts on synthetic workflows before using the protocol with participants (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020).

25.8.5 — Restricted and Public Layers

Separate public synthetic materials, restricted empirical records, secure metadata, and public result summaries (Sandve et al., 2013).

Release synthetic generators, fixtures, code, specifications, safe challenge logs, and synthetic figures. Restrict empirical records, sensitive parameters, validation outputs, and security details as required.

Govern restricted access under Chapter 28's application, permissions, audit, retention, and incident-response procedures.

Restricted evidence remains subject to the same recovery and adequacy requirements; its existence does not strengthen an unsupported public claim.

For each access level, identify outputs that can be regenerated, outputs that remain restricted, and the route for qualified reanalysis. Publish reproducible fixtures and machine-readable regeneration limits (Sandve et al., 2013; Boettiger, 2015; Wilkinson et al., 2016; Nüst et al., 2020).

25.8.6 — Artifact Retirement and Security

Record an artifact's active, superseded, deprecated, withdrawn, restricted, or historical status with its reason, date, affected claims, and replacement (Boettiger, 2015; Wilkinson et al., 2016).

A compromised dependency can make a wrong result reproducible. Quarantine affected outputs, patch the dependency, rerun the analysis, and record which results changed.

Credentials, containers, datasets, and outputs that cannot be maintained safely should not remain silently callable.

State which reproducibility depth was achieved: public synthetic regeneration, authorized restricted rerun, or future environment rebuilding. Synthetic regeneration does not establish independent reproduction of a restricted empirical analysis.

Part VIII — Replication, Verdicts, and Scientific Updating

A finding changes character when another team must reproduce it. Choices that seemed obvious in the original setting become questions about training, timing, equipment, and interpretation. Part VIII follows evidence through that transition and onward into the claims made from it. The task is to preserve a result's meaning as its custodians and uses change, while allowing new evidence to revise what the result can support (Center for Open Science, 2025; Nosek et al., 2018).

Chapter 26 treats replication as an organized comparison among settings. A rural hospice with intermittent connectivity and an urban hospital with continuous monitoring may implement the same nominal schedule differently. Recording the sources of those differences lets investigators distinguish clinical variation from infrastructure or departures from the common protocol. The replication then informs both the claim and the conditions required to study it elsewhere. Additional sites become informative through this account of their differences.

Publication arrangements affect what the research community will see. Registered Reports place design and analysis review before the result is known, with publication tied to the accepted route and quality conditions. The research record also needs a reconstruction path appropriate to its data. Code and metadata can be inspectable while sensitive recordings remain under governed access. Disclosed use of rapidly changing artificial-intelligence tools belongs in that account of how the result was produced (Grant et al., 2026; Feuerriegel et al., 2026).

Chapter 27 asks what accumulated findings mean for the claim. Results may agree at one component and diverge at another. A mechanism can lose support while a broader question remains open; a properly qualified adverse case can have a different scope. Following these dependencies makes the conclusion responsive to the evidence. It also distinguishes a

defensible narrowing, prompted by an identifiable boundary, from a revision introduced only to protect an attractive outcome.

Chapter 28 follows a changed conclusion into its descendants. A corrected assumption may affect an estimate, an argument, a chart, and a public statement. Revision is complete when those consequences are traced and the current account can be distinguished from its predecessors. People who relied on an earlier version may also need an explanation that addresses their use of it. The scientific record therefore includes the path of correction as well as the corrected result.

Chapter 29 gathers the dependencies into a staged research program. Its purpose is to show what work is ready to begin, which later questions depend on earlier qualification, and what findings would change the order of effort. The volume ends with an agenda whose progress can be assessed through completed, inspectable work. Independent teams can take up a component, challenge a dependency, or propose a better test, and make that contribution visible within the larger inquiry.

Chapter 26 — Multisite Replication and Research Systems

Imagine four sites attempting the same terminal-window study: an academic palliative unit, a community hospice, a rural clinic using another device vendor, and a care network working in another language. Their data arrive under a common schema, yet the apparent signature differs by site. The consortium’s first task is to determine whether it has measured one object under different conditions or several objects under a shared name. Replication becomes informative when those possibilities can be separated (Nosek et al., 2018; Wilkinson et al., 2016).

The rural device may have a different noise floor. A translated item may elicit politeness where the original item elicited a report of relief. A site with fewer overnight staff may miss observations during the most demanding care periods. Each difference suggests a distinct investigation: calibration, cognitive interviewing, or an observation-process audit. Pooling before those questions are addressed can make the estimate more precise while obscuring what the sites actually contributed.

The shared protocol defines what must remain comparable, while site supplements record how each setting implements it. Some differences can be linked through qualified overlap measurements; others require a site-specific analysis or an explicit no-link decision. This chapter treats those outcomes as findings about the measurement system. A successful network does not need every field to pool. It needs a defensible account of which comparisons its evidence supports and which remain unavailable.

Human capacity affects that account. A coordinator explaining consent during family crisis, a translator resolving an unfamiliar term, and a nurse deciding to protect uninterrupted rest all shape the record. Their judgment needs time, training, support, and authority to stop a task. The resulting burden and missingness records help explain whether a protocol can operate in the

setting it claims to represent. Feasibility limits therefore belong beside calibration and model assumptions in the replication report.

Independence also requires a working arrangement. The originating team can clarify a specification without seeing unblinded outcomes; rival stewards can challenge the comparison under the same information rules; endpoint reviewers can adjudicate without using the tested result. Logs of clarification and staged access show whether those separations held. If a minority site contradicts the pooled pattern, its measurement quality and rival explanation remain part of the principal evidence rather than becoming an inconvenient appendix.

The final sections connect this distributed work to a durable public record. A result must remain traceable through its local implementation, analysis release, incidents, and corrections. Outcome-independent publication commitments keep adverse and unusable findings available for later synthesis. The operational aim is a network that can reconstruct a disagreement, correct the analyses affected by a defect, and preserve conclusions that survive independently. These procedures make replication a test of scope as well as a repetition of computation.

26.1 — Multisite Preregistration and Registered Reports

Fix the common protocol and local supplements before recruitment, using Registered Reports or another outcome-independent design review. The cited protocol, reporting, and good-clinical-practice standards inform that record (Chan et al., 2025; Hopewell et al., 2025; International Council for Harmonisation, 2026; Grant et al., 2026).

26.1.1 — Shared Core Protocol

Specify the exact component claim, estimand, endpoint window, and rival matrix. This prevents sites from testing different objects under the same study title.

Protocol-fidelity qualification includes a dry run, synthetic-case walkthrough, timing check, missingness-code practice, and amendment procedure. Correct failures before the site contributes confirmatory data.

The core protocol fixes the common estimand, eligibility, window, channels, endpoint proxy, missingness categories, safety conditions, and primary comparison (Chan et al., 2025; International Council for Harmonisation, 2026; Chambers, 2013; Scheel et al., 2021).

26.1.2 — Site-Specific Supplements

The context of use identifies population, setting, language, device, cadence, care regime, and intended inference. These conditions define where the result can and cannot travel.

Site qualification verifies staffing, training, calibration, consent routes, support plans, and stop authority. Operational feasibility is a prerequisite for the scientific use.

Burden and incident rules specify participant, family, staff, safety, security, and deviation triggers. Harm and workflow pressure remain visible in the evidence record.

Staff qualification records role-specific training, endpoint-language practice, consent rehearsal, incident escalation, and burden-stop authority. A site failing these requirements remains in shadow mode or is barred from enrolling high-risk windows.

Participant support requires a consent route, withdrawal path, referral resources, after-hours contact, and separation of care from participation. Pause enrollment until that support is operational.

Family and caregiver protections specify contact limits, explanations, bereavement timing, bystander consent, and opt-out routes. Suspend affected measures until failed protections are corrected and the measures requalified.

Direct replication repeats the protocol, estimand, and observation model under qualified site conditions. Its result strengthens or weakens the same component claim within that context.

Multisite replication tests whether the pattern survives across qualified sites with site effects modeled explicitly. It can establish robustness or reveal scope limits and site-dependent claims.

Document how each site implements the core protocol under its law, language, workflow, infrastructure, and burden limits (Chan et al., 2025; International Council for Harmonisation, 2026).

Identify local population and referral conditions, available devices, staffing, contact practices, and feasibility limits; distinguish required, optional, and prohibited adaptations.

26.1.3 — Registered-Report Review

Registered-Report review places the central design assessment before the result is known. In-principle acceptance commits publication to the accepted question, design and quality conditions, allowing a null or rival-favored result to contribute under the same commitment. Reviewers should challenge separability, burden, missingness, endpoint independence, rival coverage and public-status wording at that stage (Chambers, 2013; Scheel et al., 2021; Center for Open Science, 2025).

Preserve the submitted protocol, objections, responses, in-principle acceptance, and amendments so readers can determine which analyses the publication commitment covers.

26.1.4 — Analysis Classes and Amendments

The analysis version specifies primary, secondary, exploratory, and rival analyses with their thresholds. It protects confirmatory status from changes made after outcomes are known.

Device readiness records hardware, firmware, calibration fixtures, artifacts, backup plans, and replacement routes. Failed readiness excludes the device channel or limits the site to qualified analyses that do not use it.

Named staff must have authority to pause recruitment or collection, quarantine data, and escalate incidents. Without that authority, a site cannot operate endpoint-adjacent protocols.

Conceptual replication tests the same component through a different operational route that does not merely share the original artifact. Construct breadth is supported only if the new route faces its own rival comparisons.

Transport replication tests a new population, device regime, language, care setting, or cadence. It extends scope only under explicit transport assumptions and successful invariance checks.

Distinguish primary, secondary, and exploratory analyses prospectively. Date post-inspection analyses honestly; a preregistered classification scheme does not preregister an analysis conceived later (Chan et al., 2025; Hopewell et al., 2025; Nosek et al., 2018).

Route an unexpected subgroup finding to a new prospective test rather than recasting it as the original primary result.

Record the amendment's date, reason, affected sites and measures, reviewing authority, and required reanalysis.

A firmware change during recruitment can alter sleep-state classification. Preserve the version boundary in the observation model.

26.1.5 — Publication Independent of Outcome

Record legitimate local adaptations and their rationale before outcome inspection so later differences can be interpreted without retrospective exclusions or rescue explanations.

A replication whose design, harmonization, safety, or data quality fails before interpretation supplies an infrastructure finding. It does not thereby support or defeat the component.

Adversarial replication lets critics and rival stewards help specify models, thresholds, and failure routes. Survival strengthens confidence in the tested component; a rival victory requires the corresponding narrowing.

Institutions commit to publication before results are known. This covers nulls, Rival-favored outcomes, failed instruments, and No adequate estimate under protocol-fixed quality conditions (Center for Open Science, 2025; Chambers, 2013).

The charter, funder agreement, and journal route protect adverse or unusable results, harmonization failures, and ethics suspensions. They specify amendment authority, review triggers, and visible disputed exceptions.

If observation shutdown explains terminal concentration better than the candidate, the result remains publishable under the same quality criteria as a favorable result. Suppressing it would turn a scientific comparison into a selective record.

Narrative-driven suppression is a governance failure, not a participant or hypothesis failure. Preserve findings and narrow supported status (Scheel et al., 2021; Nosek et al., 2018).

Publishing adverse evidence is a duty, not a reputational liability; retaining it protects participants, sites, and critics.

Fix questions, log amendments, and protect publication so comparability tests need not trust consortium memory.

26.2 — Harmonization Without Forced Sameness

Pool a field only when its meaning, unit, timing, context, and missingness behavior support the shared estimand (Fortier et al., 2011; American Educational Research Association et al., 2014; Wilkinson et al., 2016; International Test Commission, 2018).

26.2.1 — Common Data Elements

Record the interpretive burden on local staff, translators, participants, and families when a common format does not fit local meaning.

A common field uses the same unit, source type, timestamp, missingness code, and transformation. Full pooling also requires evidence that its meaning is shared.

A local field requires site-specific validity for its declared use. It may support local analysis or qualitative interpretation without being pooled.

A bridge variable uses overlap samples, device crosswalks, translation comparisons, or a linking model. It can support linked analysis with uncertainty or a decision not to link.

Device-version comparison requires calibration records, firmware logs, artifact behavior, and version holdouts. The result may require a version-specific model, a justified bridge, or exclusion.

Language-version qualification combines translation review, cognitive interviews, cultural review, and response-pattern checks. An item may be linked, retained locally, revised, or withdrawn.

When invariance fails, identify the failure and affected estimand. Administrative convenience does not justify pooling.

Equation 26.1. Candidate linking model for site-qualified measurement.

$$D_{link}\left(P\left(Y_{ij} \in \cdot \mid \eta_{ij}, s, d\right), P\left(Y_{ij} \in \cdot \mid \eta_{ij}, r, d\right)\right) \leq \varepsilon_{link}.$$

A common data element is a field with declared units, timing, source, missingness code, and permitted values. Without a common-data-element boundary, identical field names can conceal incompatible measurements (Fortier et al., 2011; American Educational Research Association et al., 2014). In Equation 26.1, i indexes persons, j indexes the item-or-occasion unit declared by protocol, s is the candidate site, r is the declared reference site, and d is the device or instrument configuration; η_{ij} is the common latent construct value, D_{link} measures discrepancy between the two conditional measurement distributions, and ε_{link} is the allowed discrepancy; the latent scale, discrepancy rule, and tolerance are protocol-fixed before pooling.

The data dictionary assigns each field the semantics above, a permissible transformation, sensitivity tier, and responsible owner.

Retain the source distinction when two sites call self-report and actigraphy outputs “sleep interruption”; link them only through a qualified bridge.

26.2.2 — Local Measurement Qualification

Use the site’s context-of-use evidence to decide whether a measure can enter the proposed pooled or transported estimand (American Educational Research Association et al., 2014; International Test Commission, 2018).

Report acceptance criteria and pilot results for reliability, calibration, missingness, feedback, device tolerance, and workflow; state the site’s resulting qualification.

If a translated calm item elicits politeness rather than relief, the response process has changed. Resolve that measurement issue before linking.

26.2.3 — Bridge Samples and Linking

Estimate instrument, language, or workflow links within the declared overlap population (Wilkinson et al., 2016; International Test Commission, 2018).

Record the fitted transformation, calibration sample, bias and uncertainty, version, owner, and requalification rule.

During a migration, dual measurement must be consented and limited to the overlap evidence the linking question needs.

Report uncertainty and no-link regions, keeping unsupported comparisons local or partially identified.

26.2.4 — Partial Harmonization

Some variables can be harmonized only partially, and that status must be declared before pooling (International Test Commission, 2018; Fortier et al., 2011).

The harmonization plan records which parameters were linked with uncertainty, which measures remained site-specific, and which comparisons were prohibited. Preserve observations reserved for local or qualitative interpretation.

A respiratory feature may link across devices while an agitation item remains specific to local charting. Preserve that distinction in analysis and release.

26.2.5 — Translation and Cultural Review

Record translation, adaptation, cognitive testing, rejected wording, and approval before site outcomes are inspected (Fortier et al., 2011; American Educational Research Association et al., 2014).

Give language, clinical, community, disability, and field perspectives defined review and revision roles. Record disagreements when apparent lexical equivalence changes the social meaning of a response.

Where wording changes pressure or response behavior, adapt the instrument, narrow the link, or declare a no-link region (Wilkinson et al., 2016; International Test Commission, 2018).

26.2.6 — No Pooling by Administrative Convenience

Administrative convenience is not comparability. Pooling is allowed only after declared invariance, device-bridge, support, and transport gates make the combined estimand interpretable; the plan states when site-specific reporting replaces a common estimate (American Educational Research Association et al., 2014; Wilkinson et al., 2016). A harmonization decision should record which parameters were linked, which remained site-specific, and which comparisons were prohibited.

When funders expect one headline despite failed bridges, the scientific response is a qualified common result beside incompatible site results. Forcing them together turns administrative preference into an undeclared model and hides the defect from the main analysis (International Test Commission, 2018; Fortier et al., 2011). Presenting incompatible results beside the common estimate can reveal whether the pooled signal is robust or driven by a subset of sites.

No pooling is a valid result. It preserves where comparability ended and lets heterogeneity reveal measurement drift, transport boundaries, care-regime interactions, or distinct processes requiring their own tests. No pooling can also improve the next design by identifying the exact bridge or site condition that must be validated before combination becomes legitimate.

26.3 — Site Effects, Heterogeneity, and Transport

Differences among sites can reflect effect modification, referral and population differences, care workflows, devices, or measurement failure (Gelman & Hill, 2007; Bareinboim & Pearl, 2016; Stuart et al., 2011). A pooled estimate cannot decide which explanation applies. Hierarchical modeling must retain the variables and assumptions needed to distinguish genuine variation from an invalid comparison. Transport claims then identify the target setting and the parts of the result that survive those differences.

26.3.1 — Hierarchical Site Models

A hierarchical model estimates site-level variation while borrowing strength under assumptions. Check whether shrinkage obscures a genuine boundary at an outlying site.

The 2026 SCORE results also show why site and method variation should be interpreted rather than averaged into one credibility score. A claim may be computationally reproducible from its original files, analytically fragile under other defensible specifications, and only partly replicable in independent data; another field may show stronger reproducibility while still carrying transport limits. Each pattern points to a different repair, including provenance reconstruction, specification review, direct replication, or reference-class narrowing (Miske et al., 2026; Aczel et al., 2026; Tyner et al., 2026; Brodeur et al., 2026).

Equation 26.2. Candidate hierarchical site model.

$$\widehat{L}_{ijsvt} = \mu + a_i + b_j + c_s + d_v + x_{ijsvt}^{\top}\theta + \varepsilon_{ijsvt}$$

Equation 26.3. Transport estimand under declared target population.

$$\psi_T := \int m(x)\, dP_T(x), \qquad w(x) := \frac{dP_T(x)}{dP_S(x)}, \qquad P_T \ll P_S.$$

Hierarchical site models specify how person, site, workflow, device, population, and calendar variation enter the analysis. Because that structure controls pooling and shrinkage, it is declared and tested as part of the generative model rather than added after a surprising site result appears (Gelman & Hill, 2007; Bareinboim & Pearl, 2016). In Equation 26.2, i indexes person, j indexes workflow, s indexes site, v indexes device or version, and t indexes observation occasion or calendar time; a_i, b_j, c_s, and d_v are their declared hierarchical effects, θ is the covariate coefficient vector, and ε_{ijsvt} is the residual term. In Equation 26.3, m(x) is the conditional estimand function, P_S and P_T are the source and target covariate laws, and w is the Radon–Nikodym transport weight; if P_T is not absolutely continuous with respect to P_S on the analysis support, that weighting route is unavailable.

Record fixed, random, pooled, and interacted effects, their justification, site sample sizes, and assumptions supporting estimation at each level.

26.3.2 — Care and Workflow Effects

Model care and workflow contrasts explicitly rather than absorbing them into an unexplained site effect (Bareinboim & Pearl, 2016; Stuart et al., 2011).

Version the definitions, timing, sources, and ownership of the care-workflow variables used.

Proactive and delayed symptom management may yield different distress records. Analyze that difference while keeping care decisions independent of participation.

Specify the causal role and timing of care-workflow variables before adjustment. Adding them to a model does not by itself separate care effects from latent dynamics. Earlier treatment may change symptom severity, which then influences later treatment and observation; ordinary regression adjustment can be biased when a time-varying confounder is itself affected by earlier treatment. For a causal care-effect contrast, define the estimand and causal graph, then use an appropriate longitudinal method, such as the g-formula or an inverse-probability-weighted marginal structural model, only under its stated consistency, sequential exchangeability, positivity, and model assumptions. Examine unmeasured-confounding sensitivity. If these conditions cannot be defended, retain an associational comparison or Model-ambiguous status rather than claiming causal separation (Robins et al., 2000; Hernán & Robins, 2020).

26.3.3 — Device and Laboratory Effects

Hardware, software, preprocessing, and calibration can create apparent site effects. Record their versions as potential sources of heterogeneity (Stuart et al., 2011; Fortier et al., 2011).

Retain device model, firmware, placement, sampling, filters, laboratory procedure, and preprocessing version for each site.

Bridge, quarantine, or separately report observations affected by a device change; narrow the component claim if comparability cannot be restored (Gelman & Hill, 2007; Bareinboim & Pearl, 2016).

26.3.4 — Population and Referral Effects

Define the population represented by each site's catchment, referral, access, and self-selection processes (Fortier et al., 2011; Gelman & Hill, 2007).

Model the relevant referral and retention variables and record authority to define or revise eligibility.

Stable family advocacy at one site and late referral at another can change the represented population independently of biological differences.

26.3.5 — Transport Estimands

Fix the transport target and record differences adjusted, unsupported, or assumed stable (Gelman & Hill, 2007; Bareinboim & Pearl, 2016).

For hospice-to-home transfer, qualify the changed cadence, care, device, caregiver, and endpoint conditions.

Label unsupported extension as extrapolation and withhold broader claims where overlap or invariance fails (Stuart et al., 2011; Fortier et al., 2011).

26.3.6 — Interpreting Site Disagreement

Use a declared adjudication rule to distinguish measurement failure, mechanism limits, rival advantage, care effects, and unresolved disagreement.

Retain a minority site's rival-favoring result and observation model in the principal comparison.

An adverse site triggers narrowing, reanalysis, quarantine, or withdrawal under protocol-fixed criteria, not selective exclusion (Fortier et al., 2011; Gelman & Hill, 2007).

Show site estimates, predictive intervals, rival fits, sensitivity, and unresolved minority interpretations beside the pooled result.

A site with stronger endpoint independence may expose coupling behind another site’s apparent compression. Evaluate that design difference before attributing the disagreement to noise.

26.4 — Independent Replication Roles

Distribute theory, implementation, custody, analysis, rival, endpoint, ethics, and communication authority as specified below. Institutional separation alone does not establish independence (Mellers et al., 2001; Nosek & Errington, 2020).

26.4.1 — Originating Team

The quality team reviews site-coded data without outcome interpretation to identify unusable channels, timing failures, drift, and missingness patterns.

The originating team may clarify theory, code, fixtures, and assumptions. It does not control replication outcomes or their unblinded interpretation.

Deposit versioned specifications, failure conditions, notes, and fixtures before unblinded replication results become available.

Log and distribute an originating team's blinded clarification to every affected site, and assess whether it changes the accepted analysis.

Disclose potentially outcome-informed or selectively distributed advice; reassess the affected analysis and reduce its status when independence cannot be restored (Ioannidis, 2005; Center for Open Science, 2025).

26.4.2 — Independent Implementing Sites

The implementing site may report feasibility limits, deviations, and local harms. It cannot silently change confirmatory estimands.

Implementing sites adapt the shared protocol to local staffing, law, culture, devices, care workflows, and participant needs. Document feasibility limits as conditions on replication scope (Nosek & Errington, 2020; Ioannidis, 2005).

A hospice's refusal of a cadence that displaces care defines a feasibility limit on the protocol's scope.

Enrollment beyond support capacity triggers the declared review of affected data and analyses (Center for Open Science, 2025; Mellers et al., 2001).

26.4.3 — Blinded Analysis Teams

Before access to real data, the analysis team uses synthetic fixtures and known-answer cases to verify code, formulas, and container behavior.

Primary blinded analysts receive the locked dataset with site identities masked where feasible. They execute the declared primary and rival analyses without premature narrative or endpoint information, following the specified stages of access.

An independent adjudication body releases site, device, and endpoint labels in a predeclared order. This permits examination of changes in interpretation when context becomes available.

Log when each team receives site identities, endpoint labels, treatment information, and interim results within Section 26.4's staged workflow.

26.4.4 — Rival Model Stewards

Rival stewards receive the same observation model and access constraints as the candidate's analysts. Alternatives therefore face identical data conditions.

Control contextual leaks as well as dataset access. Public attention, withdrawal narratives, and disputed windows can reveal information before a planned unblinding stage.

Rival stewards strengthen serious alternatives and publish their performance. Their evaluation remains independent of personal or institutional loyalty.

Rival stewards specify executable alternatives under equal observation access and tuning budgets, identifying the signature each can reproduce.

If a steward disputes a bridge's comparator priority, retain the disagreement and evaluate it under the common scoring and information rules.

26.4.5 — Independent Endpoint and Ethics Review

Give an independent body authority to suspend, narrow, or disclose when endpoint disputes, pressure, excessive burden, or incidents affect the study.

Endpoint and ethics reviewers may suspend, quarantine, or narrow evidence. Their role gives them no public authority to issue verdicts about persons.

Independent endpoint and ethics reviewers authorize collection, stop rules, and usable endpoint classifications (Mellers et al., 2001; Nosek & Errington, 2020).

Record the criteria, evidence, burden review, dissent, recusal, and reasons for endpoint/ethics approval, restriction, or suspension.

Withdrawal of final-week recordings changes both permission and endpoint coverage.

The appropriate outcome may be withdrawal without penalty, No adequate estimate, or a narrower window even when retention would improve precision (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

26.4.6 — Conflict-of-Interest Disclosure

Public-interpretation reviewers receive results, deviations, incidents, rival outcomes, and correction obligations. They

translate that record into status-labeled language without upgrading the claim.

An independent funder or sponsor supports publication of adverse and null findings. Funding confers no right to strengthen the public claim.

Conflict analysis identifies incentives from funding, authorship, advocacy, prestige, and public identity that may affect a decision (Nosek & Errington, 2020; Ioannidis, 2005).

Tie each conflict disclosure to its decision scope, date, reviewer, mitigation, and review trigger.

Apply disclosure and recusal when funding pressure seeks release before the disagreement record is complete.

Where influence is plausible, require recusal, staged access, independent review, or a documented reason to retain the role (Center for Open Science, 2025; Mellers et al., 2001).

26.5 — Repositories, Provenance, and Access

Use the repository to link a claim to its data, methods, access decisions, and correction dependencies (Wilkinson et al., 2016; Barker et al., 2022; Grant et al., 2026).

26.5.1 — Data Dictionary and Provenance

The measurement-source record identifies instrument, device, firmware, reporter, clinical system, and language version. These details determine harmonization and bridge requirements.

The transformation record identifies cleaning, artifact filters, imputation flags, model versions, and quality flags. Reproducible analysis requires these decisions to remain available.

Local data preparation. Local dictionary, unit checks, missingness codes, device versions, and provenance map match the shared schema. Raw identifiers remain local and access is logged under site governance.

Aggregation. Site-level estimates, uncertainty, convergence flags, and quality exclusions are combined under a declared meta-analytic model. No participant-level data leave the site unless separately approved.

Check preprocessing, time zones, window construction, and device bridges across nominally identical containers. Align them or restrict the comparison.

Extend Section 26.2.1's field semantics with the instrument, site, time, version, transformation history, and quality flags needed to reconstruct each value.

A reprocessed actigraphy feature needs a new transformation reference even when its field name remains unchanged.

26.5.2 — Public, Controlled, and Restricted Tiers

Assign purpose-limited access tiers to sensitive clinical, family, and endpoint records under Section 25.8.5.

Identity and linkage fields include pseudonym, site, visit, time, consent scope, and withdrawal status. Assign controlled or restricted access according to reidentification risk.

Clinical, family, endpoint-adjacent, biometric, free-text, and location information require sensitivity assessment. That assessment determines the public, controlled, or restricted access layer.

Define eligibility, purpose, approval, logging, retention, and revocation for each public, controlled, or restricted tier (National Institutes of Health, 2026b).

Record the privacy and consent grounds for refusing a raw-data request or offering a controlled alternative.

26.5.3 — Code and Model Registry

Provenance records the creator, reviewer, timestamp, software, container, and release identifier. It supports audit and correction.

Record downstream papers, scientific displays, model cards, public summaries, and participant returns. These dependencies determine where corrections must propagate.

Each site runs synthetic known-answer cases before handling sensitive data. Only synthetic records travel during this code-validation stage.

Central versus local sensitivity. Central pooled analysis, where lawful and ethical, is compared with federated estimates. A material unexplained discrepancy triggers access review and scientific reanalysis before public release.

Audit and retirement. Logs, release IDs, failures, and incident decisions are preserved for future reproduction. Deprecated

code, compromised outputs, and unsafe artifacts receive retirement notices.

Bind each result to its executable analysis, software environment, dependencies, model card, test fixtures, and release identifier, using the release manifest in Section 25.8 (Near et al., 2025; Wilkinson et al., 2016).

The registry also links the manuscript, any public preregistration record, and data schema, and records deprecations, security patches, and environment changes that can alter output.

26.5.4 — Retention, Deletion, and Purpose Limits

Specify local computations and approved exchanged outputs when a network retains sensitive records at each site.

Apply Section 3.4.5's bounded-purpose retention rules before storing multisite records, reconciling participant rights and deletion obligations with provenance and reproducibility needs (Near et al., 2025; Rieke et al., 2020).

Specify retention periods, withdrawal routes, deletion feasibility, derived-data handling, and legal holds.

On withdrawal of a sensitive channel, follow the declared deletion route and identify copies, derivatives, or lawful holds that prevent full erasure.

Check new uses, linkages, retention, training, and releases against the original permission even when technical security is intact (Wilkinson et al., 2016; Barker et al., 2022).

26.5.5 — Breach and Vendor Risk

Local model execution. Same container, same model version, same preprocessing declaration, and same failure flags are used. Outputs are reviewed for small-cell, leakage, and reidentification risk.

Govern external services as dependencies affecting access, provenance, continuity, and reproducibility (Rieke et al., 2020; Wilkinson et al., 2016).

Record vendor, credential, export, backup, and subcontractor risks, including the mappings needed to reconstruct derived data after a service failure.

A vendor closing an export route triggers the continuity plan and review of the analyses whose provenance is affected.

Require exportability, incident notice, deletion support, and provenance preservation in vendor contracts.

26.5.6 — No Operational Record Merger

Federated access preserves local control; comparability still depends on qualified local measurements and estimates.

Keep research records separate from civic, religious, governmental, Workgame, archival, donor, and ordinary-life records. Establish purpose limits, access controls, retention, and deletion before secondary reuse is proposed; later promises cannot fully undo a merged dossier (National Institutes of Health, 2026b; Barker et al., 2022).

Separate operational records, research records, public summaries, model-training corpora, and participant communications by purpose and access. Register each regime's owner, permitted links, and authority to reopen them.

A public dashboard team asks to reuse study metadata to illustrate country readiness. Without a new authority and consent basis, that reuse converts a research record into an operational dossier and changes the purpose for which the data were collected.

Research participation silently becomes civic visibility. That purpose migration is a governance failure, calling for separation, deletion where feasible, and a documented purpose-limit

incident rather than wider use (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Near et al., 2025).

Purpose separation should be enforced in schema, contracts, access logs, and public wording. Record separation protects participants while preventing administrative reuse from becoming unconsented inference.

26.6 — Staff Burden and Participant Support

Research fidelity depends on people having the time, training, support, and authority to perform the protocol. Staff workload can change observation timing and incident detection, while participant and family burdens can change consent, withdrawal, and missingness. Budget protected time and support before enrollment, preserve clinical priorities, and provide routes for distress and ethical concerns. Credit and authorship should recognize the labor that makes the evidence possible, including work otherwise hidden in routine care.

26.6.1 — Research Labor and Training

Qualify actual expertise and available time; training attendance alone does not establish equivalent capacity across sites (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979; Council for International Organizations of Medical Sciences, 2016).

Use supervised practice and refreshers for the protocol, consent, devices, incidents, communication, and stop authority; record demonstrated competence.

Supervised practice can make differences between sites visible before enrollment. Two coordinators may follow the same written script yet explain an ambiguous endpoint question differently. Reviewing a worked case reveals where the script needs clarification and where staff need additional preparation. Recording the agreed interpretation gives later site comparisons a common procedure to examine.

26.6.2 — Clinical Workflow

Record missed or delayed care, interrupted clinical actions, and urgent overrides. Care displacement makes affected data ethically and scientifically suspect and requires review.

Fit research tasks around clinical duties and account for the resulting timing and observation constraints (Council for International Organizations of Medical Sciences, 2016; World Medical Association, 2024).

Map task timing, responsible staff, competing care duties, handoffs, suspension triggers, and unresolved objections.

26.6.3 — Participant Support and Referral

Keep support and referral independent of participation; distress or device burden must not become a reason to retain someone as a data source.

Measure staff time per participant and shift, after-hours work, and rework episodes. These burdens can predict departures from the protocol and missingness.

Record device wear time, diary prompts, interview length, and withdrawal requests as participant observation load. That load can alter sampling and the missingness process.

Specify distress response, referral, support after withdrawal, interpretation authority, and the boundary with ordinary care.

If dependence on the study team sustains burdensome participation, transfer support and renew the participation choice.

Record missingness after support transfer and narrow the claim if the supported sample changes (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979; Council for International Organizations of Medical Sciences, 2016).

26.6.4 — Family and Caregiver Burden

Record proxy reports, troubleshooting, and repeated explanations as family or caregiver load. These demands affect feasibility and the quality of consent.

Record family and caregiver research tasks, their frequency and duration, available support, and thresholds for suspension. Use a lighter record if measuring burden itself adds unjustified burden.

Repeated proxy diaries can add to a caregiver's clinical duties and change report timing, content, and dropout.

Model exhaustion-related loss as informative where appropriate; reanalyze or withdraw the affected component if its effect cannot be represented (Little & Rubin, 2019).

26.6.5 — Moral Distress and Debriefing

Debrief requests, ethical concerns, reassignment, and sick-leave signals can reveal staff distress. Provide support and assess whether the affected work requires suspension.

Provide a route for staff distress and ethical concerns before silence shapes the research record.

Record debriefing, escalation, reassignment, counseling access, delays, and unresolved concerns; protect staff from retaliation.

26.6.6 — Fair Credit and Authorship

Name contributions to site outputs, community review, and data curation in credit and authorship records. Research must not extract this labor while leaving it invisible.

Use a contributorship matrix covering scientific, coordination, curation, technical, translation, community, security, and negative-result work; distinguish authorship from acknowledgment (Brand et al., 2015).

Apply the declared contribution criteria to small-site methodological work and provide an appeal route for overlooked labor.

26.7 — Incident, Audit, and Negative-Result Systems

Incidents can compromise safety, data meaning, protocol fidelity, access controls, or public interpretation through different pathways. A common intake route should preserve those distinctions and assign responsibility for assessment, containment, and correction. Scientific deviations, ethical events, security failures, null findings, and replication disagreement all belong in the accountable record. Their consequences depend on the affected inference rather than on whether the event is inconvenient for the project.

26.7.1 — Scientific Deviation Audit

Log protocol deviations, classify severity, and notify the responsible reviewer. The consequence may be reanalysis, exclusion, reduced status, or restriction to local use.

Freeze an analysis affected by endpoint leakage and assess contamination. Its confirmatory use may require a new protected evaluation or Unusable status.

Link deviations to affected units and analyses, with timing, discovery, severity, decision owner, reanalysis, and public-status consequence.

A fatigue-driven change in item order can alter reactivity and missingness; record it as a changed local observation process.

26.7.2 — Ethics and Safety Incidents

After an ethics incident, support the participant, pause affected procedures, and obtain independent review. The response may require site suspension, protocol revision, or withdrawal of evidence.

Ethics incidents include pressure, excessive burden, consent confusion, care displacement, distress, and bystander harm (World Medical Association, 2024).

Document the immediate response, suspension authority, support, independent review, notification, and resulting evidence restrictions.

If a family feels watched for death, pause the affected procedure and obtain independent clinical and ethics review of consent and burden; determine whether correction or cessation is required.

26.7.3 — Data and Security Incidents

Contain a security incident, notify the responsible parties, audit access, and quarantine affected data. Assess which outputs require reanalysis, restriction, correction, or retraction.

Link security incidents to the affected evidence and derived results (National Institutes of Health, 2026b; Committee on Publication Ethics, 2025).

Preserve containment, access, forensic, vendor, and reanalysis decisions with the dependent outputs that lose, retain, or recover status.

26.7.4 — Negative and Null Result Registry

Register and publish a null result, compare rivals, and assess quality. The resulting status may be absence of support, rival pressure, narrowing, or an inconclusive comparison.

Preserve site-level results and minority interpretations when replications disagree. The disagreement may require a scope restriction or a new mechanism test.

The negative-result registry gives null, adverse, and unusable findings a durable, searchable home (Committee on Publication Ethics, 2025; Center for Open Science, 2025).

Register the hypothesis, site, sample, quality, rival results, missingness, deviations, interpretation, and publication state under a stable ID and protocol link.

Classify each result by the predeclared criteria in Section 24.8, distinguishing an informative null, an imprecise estimate, a failed instrument, a nonseparable comparison, and an ethics-limited study. The classification determines eligibility for synthesis and the appropriate redesign (Ioannidis, 2005; Franco et al., 2014).

26.7.5 — Audit-Ready Research Program

Specify archive retention periods and reconstruction-test requirements before publication (Sandve et al., 2013).

Archive the linked protocol, permissions, code, data schema, incident, analysis, and public-release versions. Assign owners, revision authority, reopening triggers, and dependency links; record reconstruction gaps.

Have an independent reviewer rebuild from the archived manifest before relying on it publicly. Log undocumented interventions as defects and repeat after material infrastructure changes.

An audit-ready program can reconstruct a result without relying on institutional memory or personal assurances. Required artifacts, retention periods, responsible owners, and recovery tests should be specified in advance, because missing provenance discovered after publication may make an otherwise attractive result unauditable (Sandve et al., 2013).

A critic tries to reproduce a figure two years after publication. If the code environment, transform history, or dependency versions have disappeared, the figure has outlived the evidence needed to reconstruct it. Long-term reproducibility requires preservation of executable environments or an independently documented migration path, not code files alone. Reconstruction should begin from the archived inputs and declared environment rather than from a later analyst’s memory of how the pipeline

probably worked. A successful rerun should reproduce not only the image but also the intermediate quantities and version identifiers on which the published result depended.

If an environment file, device log, or exclusion script is missing, the affected figure cannot be treated as reproducible. It should be quarantined while the environment is reconstructed; if independent rebuilding fails, the figure and any dependent claim require correction or withdrawal (Committee on Publication Ethics, 2025; Crossref, 2025). Quarantine should extend to downstream derivatives whose validity depends on the missing artifact, not just the visibly affected figure. The dependency record should therefore identify which conclusions can survive independently and which remain suspended until reconstruction succeeds.

Independent dry runs should test audit readiness by exposing missing artifacts before a public claim depends on them. Dry runs should include a person unfamiliar with the originating workflow, since tacit knowledge is itself a reproducibility dependency. The dry run should start from the archived manifest rather than from informal help by the original team, and any required undocumented intervention should be logged as a reconstruction defect. Repeated dry runs can also reveal whether later software or infrastructure changes have silently broken an otherwise complete archival package.

Public release can now be treated as a controlled transformation from record to claim. Every compression step should preserve the tested component, reference class, uncertainty, and correction route. A release manifest can make that transformation auditable by linking each public object to the analysis version and evidence packet from which it was derived. The manifest should also record the transformation that produced each public table, figure, caption, or summary,

allowing reviewers to trace a simplified artifact back to the exact result it represents. If the underlying analysis changes, that dependency map identifies which public objects require review rather than relying on memory or broad manual searches.

26.8 — Public Release and Correction

A public release must state a claim that remains accurate when read on its own, while linking to the complete instrument, endpoint, rival, heterogeneity, access, and correction record. Review requirements determine when release is permitted; status labels determine what may be said. Correction and retraction procedures must reach public artifacts and participant communications as well as technical outputs, so that the current claim register and the materials in circulation do not contradict one another.

26.8.1 — Embargo Until Required Review

State which rivals were compared and which remain viable. An untested rival has not been defeated.

Complete the specified scientific, incident, participant-communication, and wording reviews before broad release (Committee on Publication Ethics, 2025; Crossref, 2025).

Tie the embargo to named gates and an expected decision date.

Document outcome leaks that could steer interpretation; rerun the affected analysis under protected conditions where feasible and reduce its status when independence cannot be restored (Center for Open Science, 2025).

26.8.2 — Status-Labeled Release

Report null findings, failed sites, unusable data, deviations, and incidents alongside favorable patterns, preserving the adverse evidence needed to interpret the result.

Set release status and scope from the component’s versioned evidence record. Disclose material uncertainty and link that record with the artifact (Crossref, 2025; Center for Open Science, 2025).

The release register identifies the sentence owner, status source, correction trigger, and history of disputed wording. Treat unsupported upgrades in fundraising or media excerpts as public-risk incidents (Committee on Publication Ethics, 2025; Crossref, 2025).

A terminal-concentration graphic that omits two sites' failed device harmonization misstates the scope. Correct or withdraw it and restore those limits at every active destination (Committee on Publication Ethics, 2025).

26.8.3 — Correction and Retraction

Give the repository release identifier, correction contact, and retraction trigger. A public release remains subject to correction as its evidence changes.

Apply Chapter 28's correction and retraction workflow to each affected surface, assigning an owner and recording completion (Committee on Publication Ethics, 2025).

Record corrected project surfaces, notified external reuses, and unrecalled copies; retain a durable notice leading older readers to the current status.

26.8.4 — Participant Communication

Participant communication should not include a request to become a public example.

Plan participant and family communication prospectively: return can create burden, false meaning, or participation pressure (National Academies of Sciences, Engineering, and Medicine, 2018).

Prespecify optional study-level return in plain language, including timing, findings, limits, incidents, support, withdrawal implications, correction notices, and a route for questions. Individual interpretation requires its separate clinical authority.

Avoid communications that resemble judgments on a loved one's inner life. Account for bereavement timing and the burden of return; distress after receiving findings is not hostility to science.

Participants choose whether and how to discuss their involvement publicly.

State the context of use: population, site type, device, language, care regime, and endpoint rule. That context does not imply applicability to every conscious stream.

26.8.5 — Release and Claim-Register Updates

Name the measurable component actually studied. A component result does not establish that the Law is proven.

Identify the replication as direct, conceptual, multisite, adversarial, transported, or failed. Any claim of independent confirmation must retain those qualifications.

Publication does not change the formal series. A release may enter review, but publicity cannot revise root definitions, applied authority, religious doctrine, law, or transition timing (Crossref, 2025; Center for Open Science, 2025).

Each release names the tested component, review owner, untouched claims, and public limits. A favorable multisite result updates only the evidence register; broader scientific, applied, spiritual, legal, or transition changes need separate review.

Freeze and correct unsupported publicity upgrades; return evidence to human change control (Committee on Publication Ethics, 2025).

Chapter 27 — Evidence Accumulation and Defeat

A research program can accumulate results while losing track of what those results establish. An instrument improves, a mechanism fails, and a previously qualified finding stops transporting after a device update. Counting favorable papers would obscure these different changes. This chapter follows evidence through the components it can reach, so that a review can strengthen one conclusion, suspend another, and preserve the reason for both decisions (Mayo, 2018; Lakatos, 1978).

Consider a replication whose pooled estimate resembles the original result, although the new firmware changes the observation channel at one site. The first question is which observations still support the declared estimand. A qualified subset may remain informative; the affected version may require requalification. Neither the favorable pooled center nor the word replication settles that issue. The review must connect the new evidence to the instrument, context, and claim actually tested.

That connection also separates trustworthy work from eventual correctness. A carefully specified result can later be over-turned without making the original study dishonest. Conversely, a favorable finding does not earn methodological credibility merely because later studies point in the same direction. An inspectable question, evaluable design, visible error controls, and a conclusion fitted to the evidence let other investigators identify what changed and why it matters (Nosek et al., 2026).

The practical unit is therefore a component record with a history. It connects the tested version to its prerequisites, rival comparisons, adverse observations, and subsequent revisions. A change in one prerequisite can force review of dependent conclusions while leaving independent findings intact. This matters when useful measurement or clinical work survives the retirement of the particular mechanism that first motivated it.

The record should preserve that contribution under its surviving scope.

Scientific change also needs a decision procedure. Reviewers can disagree about whether a result is sufficiently informative, whether a boundary is justified, or whether a replacement theory makes a new prediction. Recording dissent does not require postponing every verdict indefinitely. A deadline, an accountable decision, and an explicit route for reconsideration allow a program to act while keeping the minority's evidence available to later reviewers and to successor studies.

The sections that follow move from component updates through replication, narrowing, retirement, and defeat. Their central question is what a result changes in the claim's evidential standing. When a claim fails, the evidence and the objections that exposed the failure must remain available before anyone reuses it. Participants and families remain people who contributed to research; they do not become embodiments of a theory's success or failure when its scientific status changes. This history also prevents successor studies from repeating failed procedures or recruiting participants under interpretations that the evidence no longer supports.

27.1 — Evidence Attaches to Components

A finding updates the component it reaches and the claims that depend on it. Sections 27.1 and 27.2 identify the units and dependencies to track across studies.

27.1.1 — Evidence Components and Empirical Leads

Taquet et al. (2016, 2020) relate mood to subsequent activity selection in everyday life. Compare predictions using current mood, activity opportunities, and prespecified affective-history summaries. Track participants and the dataset together in synthesis because the reports share the smartphone cohort.

The scientific verdict axes give the evidence record four distinct fields. TargetStatus records the conclusion about the declared target branch. RouteStatus records the conclusion about a candidate mechanism. ComparatorStatus records both the adequacy of the rival set and the relative performance of its members. InferenceStatus records whether the target is identified, partially identified, inconclusive, unusable, or unavailable. Each field retains its reference class, information set, and inferential type. A single study can therefore improve measurement, distinguish two mechanisms, and leave the complete-stream target unresolved; the record preserves all three achievements at their proper level.

For example, a prospectively specified comparison might find that a reserve-linked model predicts held-out observations better than the declared mean-reverting rival, while both pass their adequacy checks. The route and comparison fields can record that result. If the study observes only a finite window, the inference field can still identify the window-level estimand while the target field records that the complete-stream claim was not tested. A later replication updates the relevant result and

scope. It need not overwrite the other fields to register a real scientific advance.

Keren et al. (2021) found that a primacy model predicted mood better than recency alternatives in the studied tasks. Compare primacy, recency, and cumulative-history models across tasks and timescales, tracking the participant sample and task as the evidence unit.

Gupta and Sontag (2025) derive different cumulative dose–response shapes for integral-feedback and particular feedforward models. That comparison motivates specifying response patterns that distinguish models of accumulated affect. Track model architecture, input history, and output measure when evaluating the analogy.

Van der Helm et al. (2011) and Wassing et al. (2019) connect overnight emotional-response changes with sleep and REM continuity. Jointly model prior affect, sleep continuity, and subsequent responses, while retaining the participant cohort and sleep protocol as the synthesis units.

White et al. (2019) report increased generous allocations after reminders of karma or God among corresponding believers in the experiments. Compare accounts based on belief, social incentives, and affective history, tracking the experiment, participant sample, and belief group.

The representation component covers scalarization, temporal additivity, origin conventions, and multidimensional alternatives. Coherent representation is required before endpoint findings receive their proposed interpretation.

The measurement component covers calibration, context of use, uncertainty propagation, and version performance of momentary-valence and interval-contribution estimators. Instrument qualification proceeds alongside evaluation of the formal realization.

The bearer and endpoint components cover unity classification, independent stopping rules, open-channel windows, and false endings. They require qualified bearer classification and independent endpoint adjudication.

The realization and mechanism component covers reserve coupling, Queue, horizons, brain–body mediators, and stochastic assumptions. It must connect the tested mechanism to the specified target.

The rival and consequence component covers the signature conjunction and alternatives involving bridges, adaptation, predictive control, selection, and missingness. Compare these explanations under common rules.

Replication and transport evidence includes direct, conceptual, multisite, transported, site-holdout, and device-holdout tests. Preserve heterogeneity and site-specific outcomes alongside pooled estimates.

Equation 27.1 (component evidence graph):

$$G_{evidence} = (C, D, E, R, V)$$

Here C names components, D names dependencies among components, E names evidence records, R names rival states, and V names versioned public verdicts. The display records how scientific components, evidence, and decisions are connected.

Equation 27.2 (noncompensatory gate rule):

$$PublicStatus(C*) \preceq Status(c), \ for\ every\ c \in Dep(C*)$$

The dependency set in Equation 27.2 contains only necessary prerequisites; optional and explanatory links remain in the broader evidence graph. The preorder expresses a noncompensatory dependency rule: a downstream public status cannot outrun any failed prerequisite on a compatible status axis. Unlike verdicts remain separate, and the component states are not forced into a total order. A successful mechanism cannot

repair failed measurement, and a favorable pooled result cannot repair endpoint failure.

Version each component so later citations identify the exact instrument, endpoint rule, realization, or rival matrix tested.

Mark dependencies as necessary, optional, or explanatory. A mechanism may explain a consequence without being necessary to it; a public artifact instead creates a risk of misrepresent-ing the evidence.

Include permitted uses, affected decisions, and protected participant rights in each component record.

27.1.2 — Representation Component

Reliable arousal data and stable within-person response scales do not settle a valence origin that varies across illness stages. That pattern motivates state-specific calibration or a richer representation.

A representation may be supported, unresolved, narrowed, replaced, or retired. Downstream fit alone cannot validate it when nuisance parameters absorb representation error.

27.1.3 — Measurement Component

An instrument qualified in cancer hospice may fail in advanced neurological disease because speech loss, motor slowing, medication, or autonomic disturbance changes channel meaning. Impose the resulting context-of-use boundary and version note; pooling the populations cannot establish calibration.

27.1.4 — Bearer and Endpoint Components

The bearer and endpoint record states how the evidence identifies a continuing stream and its independently specified stopping rule (Popper, 1959; Lakatos, 1978).

A study can stop collecting data while the endpoint remains unresolved. Record the additional endpoint evidence required

for terminal inference separately from that data-collection decision (Manski, 2003).

27.1.5 — Realization and Mechanism Components

A variance-compression result evaluates the reserve realization only through the qualified relation between latent burden and observed dynamics.

If the selected window also favors bridge and observation-shutdown models, record the model-confusion region and test which additional measurements or windows would separate them.

A revised mechanism needs a new version, replacement predictions, and challenges; earlier support does not transfer automatically.

27.1.6 — Consequence and Rival Components

Report the rival-specific result: support against adaptation may remain unresolved against bridge models, and an advantage over frailty in one diagnosis may reverse under medication-only decline in another.

27.1.7 — Strong-Form Component

The strong-form review joins qualified representation, measurement, endpoint evidence, and reference-class comparisons to test terminal neutrality for qualifying streams.

Identify the dependency and evidence responsible for any change in strong-form status.

27.2 — Sequential Evidence and Replication States

Apply Chapter 26's distinctions when moving from exploration to prospective testing, direct or conceptual replication, multisite or transported replication, and review. Public preregistration requires an actual deposit (National Academies of Sciences, Engineering, and Medicine, 2019; Nosek et al., 2018; Klein et al., 2018; Bareinboim & Pearl, 2016).

27.2.1 — Exploratory Signal

An exploratory signal generates design questions and candidate signatures. It motivates a prospectively specified comparison while retaining its exploratory status.

A prospectively governed single study tests one component under declared rules. It may update only that component within the tested reference class.

Direct replication independently repeats the essential test. Its result may strengthen or weaken the same component claim.

Conceptual replication tests the component through different design logic. It may support robustness without establishing identity of mechanism.

Multisite or transported replication tests heterogeneity and context of use. It may widen, narrow, or split the reference class.

Governed review integrates quality, dissent, rivals, and public correction. Any issued status label states its scope and reasons.

Equation 27.3 (sequential status update, human-reviewed):

$$S(c,t+1) = J\left[S(c,t), e(c,t+1), q(c,t+1), r(c,t+1)\right]$$

The symbol J stands for governed judgment, not a mechanical score. It receives the prior component status, a new evidence record, quality information, and rival-state information, then issues a versioned decision with recorded reasons.

Prespecify status-change triggers and a review deadline.

Mark evidence as stale when firmware, care regimes, or diagnostic conventions change its applicability. Preserve the study while revising its role in current status.

Include failed registrations and abandoned protocols. A deposited study stopped for ethical burden has a different history from a private draft or an unlaunched design, with different implications for the next study.

Independent reanalysis should cover a predeclared family of defensible specifications and carry analytic or expert dispersion into status review (Steegen et al., 2016; Simonsohn et al., 2020; Silberzahn et al., 2018).

Exploration identifies patterns and prospective comparisons. Report the source, selection, analytic flexibility, and uncertainty; distinguish empirical observations from predictions generated in simulation (Chambers, 2013; Nosek et al., 2018; Lakens, 2017).

Lower late variance in a terminal dataset may motivate a compression test, but altered visit frequency, medication, admission selection, or missing high-distress observations may produce it.

Label exploratory figures at first release and link them to the analysis record (Schönbrodt et al., 2017; Nosek et al., 2018).

27.2.2 — Publicly Preregistered Single-Study Result

A public preregistration meeting Section 6.7 makes the promised analysis and decision rule independently inspectable; evidentiary strength still depends on design and execution (Lakens, 2017; Schönbrodt et al., 2017).

Changing the registered rule after outcome inspection loses that rule’s confirmatory status. Label the subsequent analyses separately while preserving the faithfully executed original test (Nosek et al., 2018; Klein et al., 2018).

The single-study statement identifies the tested component, scope, registration and versions, deviations, rivals, and uncertainty.

27.2.3 — Direct Replication

Direct replication preserves the essential test under independent recruitment and analysis controls (Schönbrodt et al., 2017; Nosek et al., 2018).

Record protocol adherence, deviations, versions, burden, endpoint adjudication, and blinding. A changed target is a new test; a changed design qualifies as conceptual or transported replication only when it tests the same declared component and explains its relation to the original.

A second site can repeat the preterminal compression test with the same instrument and endpoint rule. An adequate failure then pressures the original component result, rather than merely adding another publication.

27.2.4 — Conceptual Replication

Conceptual replication uses a different design or measurement route for the same declared component (Nosek et al., 2018; Klein et al., 2018).

Experience sampling, clinician-assisted reports, and passive channels may agree on a narrowing pattern. Disagreement may instead reveal that one measures arousal, another reportability, and neither alone measured burden.

Name the invariant, the deliberate design change, and any unresolved mechanism or endpoint bridge.

27.2.5 — Sequential Validity and Review

Sequential validity concerns how evidence is reviewed over time without letting repeated looks, changing thresholds, or accumulating public pressure distort the result. Statistical tools

can help, including prediction intervals, sequential meta-analysis, calibration checks, and robustness summaries. But the judgment remains governed by humans who must record reasons, dissent, and scope (Jennison & Turnbull, 2000; Lakens, 2017).

A living evidence system should separate analysis updates from status updates. New data may update plots, posterior distributions, heterogeneity estimates, or model weights before the public status label changes. This prevents every interim fluctuation from becoming a headline.

Sequential review is tested by a series of three replications, one favorable, one mixed, and one negative under a new device version. Sequential review should not choose the most comforting interpretation. It should ask whether the component is version-dependent, whether transport failed, whether the original mechanism is wrong, and whether the public label must narrow.

Review by momentum treats elapsed time and institutional investment as evidence for an upgrade. Once a program has a public identity, committees may feel that status should move gradually upward unless catastrophe occurs. That pressure is rejected. A single severe dependency failure can block an otherwise favorable accumulation (Schönbrodt et al., 2017; Nosek et al., 2018).

Sequential review should publish the rule by which evidence states are revised, the quality criteria applied, the dissent recorded, and the correction path for public artifacts. The result is not a score, but it is not private intuition either.

The sequence ends where judgment begins. Tools arrange evidence; people accept responsibility for language.

27.3 — Route-Favored, Partial, and Replicated Results

Positive findings differ in what they establish. Preliminary component support, partial support, replication, mechanism support, and convergence across methods require different evidence and retain different unresolved dependencies. A stronger label is justified by the relevant comparison and qualification, not by the accumulation of favorable descriptions. This section specifies those thresholds and explains why success of a bounded component cannot automatically validate the strong form or other domains of the series.

27.3.1 — Preliminary Component Support

Preliminary component support requires one adequate prospectively specified result for the declared component. Name the reference class, analysis version, and unresolved rivals.

A publicly preregistered, adequate equivalence test may place a nonempty measured compatibility set wholly inside the declared band for one estimand, instrument version, diagnosis, and independently defined open-channel window. That supports the measured component in that context, not discovery of the universal Law.

Record dependencies and rivals still untested, with conditions that would reverse preliminary support (Klein et al., 2018; Vehtari et al., 2017).

27.3.2 — Partial Support

Partial support records predicted structure alongside unresolved dependencies. List the adverse, missing, and untested dependencies explicitly.

Record two qualified measurement contexts, one unusable measurement estimand, and one suspended terminal inference.

The route remains Model-ambiguous and the strong target unadjudicated.

Qualified measurement can support a preterminal compression pattern while failed endpoint adjudication leaves terminal inference unavailable. Where measurement also fails, separate those unusable observations from the qualified subset.

27.3.3 — Replicated Component Support

Replicated component support comes from independent direct or transported evidence. It retains heterogeneity and site-specific failures.

Four significant and two nonsignificant estimates do not establish heterogeneity. Include every site, compare effects and uncertainty with the declared hierarchical or interaction analysis, and assess out-of-site prediction before widening or narrowing replication scope.

27.3.4 — Mechanism Support

Mechanism support requires distinctive predictions to survive qualified rival comparison. Causal attribution additionally requires identification of the proposed pathway; mechanism support alone does not support the strong form.

Suppose two sites qualify the instrument, a third detects calibration drift, and a fourth cannot adjudicate the endpoint because sedation and missingness exceed fixed bounds. At the qualified sites the reserve candidate beats one ordinary regulator but remains indistinguishable from an endpoint-conditioned bridge under held-out scoring.

Held-out compression and reserve-linked innovation changes favor a realization when it meets Section 22.8.1’s prespecified criterion against bridge and shutdown rivals under the same

observation model. Support remains specific to that realization and context.

27.3.5 — Convergent Support Across Methods

Convergence requires different methods to reach one prospectively defined component through declared bridges. Similar narratives across dreams, terminal telemetry, horizon tasks, and longitudinal reports do not establish a shared target (Open Science Collaboration, 2015; Camerer et al., 2018).

Convergence is more informative when confounds differ; shared missingness or selection can create a common artifact.

27.3.6 — No Whole-Claim Validation

Success of one component does not validate the full series. It grants no automatic upgrade to applied, legal, religious, or transition claims.

The Science of Fairness updates scientific components and their public descriptions. Applied titles use relevant findings through their own research, legal, practical, and interpretive processes (Camerer et al., 2018; Klein et al., 2018).

Multisite support applies to a named component, reference class, and method. Status labels must follow donor pages, graphics, summaries, and paired books to block prestige by association (Vehtari et al., 2017; Open Science Collaboration, 2015).

Component support, bounded replication, and narrowed evidence state preserve the contribution without validating the series, religion, transition, or political authority.

27.4 — Inconclusive and Unusable Evidence

A valid but imprecise comparison may be Inconclusive. Adequate models left indistinguishable are Model-ambiguous; failed recovery instead leaves the affected comparison unqualified. Measurement or endpoint failure blocks the target it reaches. These states require different repairs under Section 24.2, rather than a common label for disappointing results (Manski, 2003; Little & Rubin, 2019; Lakens, 2017).

27.4.1 — Wide or Mixed Uncertainty

Wide or mixed uncertainty leaves intervals, posteriors, or sensitivity analyses unable to discriminate the tested alternatives. The component result is inconclusive.

A temporary barrier may await a less burdensome sensor or stronger translation method. A principled barrier remains when the only route would manipulate suffering, pressure final reports, or turn private grief into research material.

Locate the dominant uncertainty contributions and show how they affect the verdict. A wide interval alone does not reveal whether more sampling, better measurement, or a different design could resolve it.

Lead the public label with unresolved uncertainty; a favorable center cannot conceal width, reversals, or unusable cases (Lakens, 2017; Mayo, 2018).

27.4.2 — Nonseparable Models

Failed model recovery or design separability prevents a winner from being identified. Redesign the comparison or narrow the claim.

Record an unanswerable zone when no responsible design can reach the component, identifying whether its barrier is temporary, design-contingent, or principled.

Before recruiting for a confirmatory model-discrimination study, require the relevant synthetic recovery and design-separability checks in Chapter 25 to pass.

A prespecified adequacy-and-simplicity rule may select a model for use under Section 22.8.1 even when the observations do not identify its causal mechanism (Mayo, 2018; Manski, 2003).

27.4.3 — Measurement and Endpoint Failure

Failure of instrument sign, scale, missingness assumptions, or calibration leaves the target untested in that context.

An ambiguous, unavailable, outcome-contaminated, or branch-incompatible stopping object suspends the affected terminal inference.

Update consent materials after measurement failure; future recruitment must not imply that the failed instrument worked.

If terminal inference demanded observation beyond comfort limits, redesign the question or method before asking participants and families to bear more.

Identify the target and context for which measurement failed, the responsible channel or calibration problem, and the validation required to proceed (Lakens, 2017; Mayo, 2018).

Retire, narrow, or rework the failed instrument version; do not treat a nonstandard signal as deception or noise by default.

A failed endpoint gate may preserve preterminal evidence while leaving terminal inference unadjudicated; apply Chapters 7 and 13’s branch-specific conditions (Mayo, 2018; Manski, 2003; Little & Rubin, 2019; Lakens, 2017).

Code endpoint failures by cause, since sudden death, sedation, dropout, burden, delayed documentation, and post hoc selection require different repairs.

If the last usable window ends forty hours before the inferred stream endpoint, do not label it terminal. Suspend terminal conclusions unless independently justified bounds on the gap support them; retain qualified preterminal dynamics.

Distinguish unavailable endpoints, unresolved timing, contaminated adjudication, false endings, and failed branch-specific premises. Dependence alone is not failure when a state-coupled branch models it.

27.4.4 — Ethical or Burden Stop

Excessive burden or risk requires stopping or reducing observation. Classify each affected inference using the valid prerequisites that remain.

Preserve stop reasons, metadata, and protocol lessons while minimizing or deleting sensitive observations where retention creates risk. Reconcile failure memory with data dignity case by case.

Stop when the protocol’s burden or risk limits are exceeded. A failed required gate makes the affected evidence Unusable; otherwise consent-permitted analysis of retained data keeps the status warranted by validity, identification, and precision (Council for International Organizations of Medical Sciences, 2016).

Report the stopped procedure, reason, remaining usable data, participant or staff support, and questions the program will no longer pursue in that form.

27.4.5 — Public Language for Uncertainty

Mixed evidence or evidence awaiting review requires a public statement of uncertainty with its current status. It cannot be presented as support through selective wording.

Name the component, evidence state, scope, and reason. A qualified multilingual estimate may remain Inconclusive because valid missingness bounds do not discriminate; failed calibration instead makes the affected evidence Unusable. State the next action without promising that every uncertainty can be resolved.

State that no responsible verdict is available for this component and reference class under current methods, and identify any feasible redesign.

Public uncertainty language must avoid both promotional spin and false defeat. The public can understand uncertainty when it is explained as a precise evidence state. What destroys trust is not uncertainty itself, but uncertainty hidden until critics expose it (Little & Rubin, 2019; Lakens, 2017). A stable vocabulary for uncertainty makes corrections less dramatic because the public has already been shown that inconclusive and unusable are legitimate outcomes.

The uncertainty categories separate in a public website summarizing a mixed replication. The accurate label may be partial support for one terminal consequence, measurement failure in one subgroup, and rival dominance in one model comparison. A single green, yellow, or red icon would mislead. Component-level display can still be visually simple if each icon or line corresponds to a named claim rather than one aggregate project mood. A compact table can place the shared study identifier beside separate component outcomes, making disagreement visible without turning the entire project into one score.

Compressing mixed uncertainty into one status label destroys information the verdict needs. Donor pages, media summaries, diagrams, and social posts often shrink nuance into a mood. Public artifacts must preserve the component structure or refrain

from release (Mayo, 2018; Manski, 2003). If compression cannot preserve the distinction, a link to the full component table is not enough; the simplified artifact itself should not imply a unified verdict. When one component is corrected, the notice can identify the other components that remain unchanged instead of replacing the entire account with a new headline.

A public summary must name whether unusable evidence narrows, suspends, blocks, retires, or leaves the question unanswered. The summary should also identify whether the limitation comes from data quality, identification, model adequacy, or a genuine adverse result. An adverse result under valid prerequisites retains its adverse classification; it is not reclassified as unusable simply because the preferred account lost.

27.5 — Narrowing Reference Classes and Claims

A reduced population, time domain, or context may be a genuine reference-class narrowing; a change in representation, exactness, modality, endpoint, or target object creates a non-equivalent successor that must receive its own identity and decision rule.

27.5.1 — Population Narrowing

A principled narrowing is predeclared or independently motivated, publicly labeled, testable in its new form, and connected to a mechanism or measurement reason. A degenerating narrowing follows failure with a smaller claim that predicts nothing new.

Population narrowing may restrict diagnosis, age, species, state, culture, or care regime. Identify both the earlier broader class and the new bounded class.

Temporal narrowing changes a terminal claim into a preterminal or window-specific claim. Remove wording that still implies endpoint coverage.

Representation narrowing may replace scalar closure with a multidimensional consequence. Revise the affected formulas, displays, and summaries accordingly.

Endpoint narrowing replaces an unsupported universal endpoint use with a qualified stopping event. Identify which endpoint decisions no longer transport.

Statistical narrowing may replace pathwise universality with a reference-class tendency. Remove universal Law-language from the affected empirical claims.

A public graphic or guide that overstates the earlier scope requires replacement, a correction date, and a preserved correction history.

Equation 27.4 (schematic rule for pure reference-class narrowing):

$$D_{\text{new}} \subsetneq D_{\text{old}}$$
$$\mathscr{P}_{\text{new}} = \mathscr{P}_{\text{old}}|_{D_{\text{new}}}$$

Here D denotes the declared reference-class domain and $\mathscr{P}$ the claim's prediction rule on that domain. Pure reference-class narrowing restricts the domain while keeping predictions for retained cases, target geometry, and the decision rule unchanged. Changes to representation, modality, endpoint, or target object require a non-equivalent successor and are outside this schematic rule. The new class is independently motivated, defined without inspecting favorable outcomes, and tested with a new transport or inclusion rule. Otherwise, post hoc exclusion becomes rescue.

Distinguish population limits due to mechanism, measurement, or rival performance: the reason determines what can travel.

Population narrowing needs evidence that a component holds, changes meaning, fails, or remains untested across the proposed boundary (Bareinboim & Pearl, 2016; Stuart et al., 2011).

A favorable subgroup found after broad-study failure remains exploratory until independently justified and prospectively tested (Fortier et al., 2011; Lakatos, 1978).

27.5.2 — Temporal Narrowing

A window-specific claim may remain identifiable after terminal evidence fails (Stuart et al., 2011; Fortier et al., 2011).

When final windows are unusable because of sedation and endpoint ambiguity, qualified contraction in the preceding month supports only that preterminal pattern.

Replacing a terminal target with a preterminal one creates a non-equivalent successor requiring its own evidence (Lakatos, 1978; Bareinboim & Pearl, 2016).

27.5.3 — Representation Successors

Independently motivated multidimensional successors can retain distinct predictive information that scalarization loses.

A multidimensional study may show that physical pain, social fear, and existential distress behave differently near an endpoint and reject the prespecified restrictions required by the proposed scalarization in that context. The result can motivate a vector or set-valued successor and preserve valuable lived-experience science, but it does not count as support for the failed scalar branch.

Propagate the representation change to dependent formal objects, estimands, diagrams, and verdicts; identify what survives and what needs reanalysis.

27.5.4 — Endpoint Narrowing

Endpoint narrowing retains the same endpoint object but restricts the claim to independently adjudicable instances (Lakatos, 1978; Bareinboim & Pearl, 2016).

A protocol that identifies terminal open-channel windows in gradual decline does not extend to sudden endings without a separately tested endpoint model.

27.5.5 — Statistical Successors

Changing universality, exactness, or target geometry creates a successor with a new protocol, error guarantee, and status record (Bareinboim & Pearl, 2016; Stuart et al., 2011).

Near-neutral concentration on average may coexist with adverse sites and a wide prediction interval. Report that

tendency and heterogeneity as a new claim, not pathwise or universal closure.

Match revised prose to its formal statement: overwhelmingly tends, essentially closes, and in the limit need explicit probabilistic or limiting meanings (Lakatos, 1978).

27.5.6 — Public Correction After Narrowing

Apply Chapter 28's correction workflow, naming the withdrawn scope, surviving claim, and unresolved copies (Crossref, 2025).

Replace a universal terminal-curve graphic when the evidence narrows to one palliative reference class. Put the surviving class on the replacement itself, archive the correction, and cross-reference reuses so cropping cannot restore the old interpretation.

27.6 — Mechanism Retirement and Rival Dominance

Retirement removes a candidate mechanism or instrument version from active scientific use after failed prediction, nonrecoverable attribution, infeasibility, or persistent rival dominance. Those failures do not have identical implications for the formal target or for valid measurements produced along the way. Record the failed object, preserve useful component findings, and require an independently motivated replacement to make a new testable prediction. A renamed explanation does not erase the earlier loss.

27.6.1 — Reserve Realization Retirement

Bridge failure, parameter nonrecovery, rival superiority, or an incompatible variance pattern can retire the reserve realization. The formal question and separately justified reserve measurement may survive.

If endpoint bridges reproduce compression under the same observation schedule while plausible noise floors make reserve parameters nonidentifiable, the product realization may retire for that evidence family.

27.6.2 — Retiring Scheduling and Horizon Routes

If ordinary control explains priority, blocking, and repair signatures, the Queue or horizon realization may retire. Sound behavioral-task findings and burden analyses can remain.

Rename useful residual methods when needed: a retired scheduling-route test may continue as an urgency-control task without carrying its failed interpretation.

Compare executable models under shared tasks. If ordinary harm avoidance or model-predictive control reproduces the signatures with fewer assumptions and better held-out performance, narrow or retire the scheduling interpretation.

27.6.3 — Candidate Instrument Version Retirement

Calibration, sign, transport, or missingness failure can retire an estimator version. Data protections and valid lessons for revised measurement continue.

A qualified contradiction can yield Route-failed; failed recovery blocks attribution without contradicting the mechanism, and infeasibility is a separate retirement reason. Record the failed predictions or gates, rival results, scope, and affected artifacts.

Retirement is version- and context-specific. A failed version does not become qualified because a later version succeeds elsewhere; its record remains available.

Label the failed instrument's affected evidence Unusable, name the context, and distinguish repairable from structural failure. The failed object is the instrument, not the person.

27.6.4 — Rival Dominance

Repeated superior held-out prediction and calibration by rivals can retire the distinctive interpretation. Bounded components not explained by the rival may remain viable.

Apply Section 22.8.1 to repeated held-out dominance that survives adequacy checks and independent replication. Name the better explanation and remaining unexplained structure (Burnham & Anderson, 2002; Palminteri et al., 2017).

Frailty, observation shutdown, and endpoint-conditioned diffusion may jointly predict terminal patterns better across sites than the reserve candidate, leaving no Law-specific residual. That component becomes rival-dominant.

27.6.5 — New Tests for Replacement Mechanisms

A replacement mechanism without independent motivation or a new falsifiable prediction earns no successor status. It must face

prospectively specified risk before replacing the retired explanation.

A replacement must explain the old route's failure, have motivation independent of the new target result, operationalize its variables, and specify discriminating data and a new falsifier. A vector reserve proposal must name dimensions and a prediction separating it from ordinary allostatic regulation.

27.6.6 — Preserve Useful Component Science

When a Law-specific interpretation fails but the underlying findings remain sound, retain the clinical, measurement, and ethical insights under narrower labels.

Specify whether a retired model remains permitted as a simulation baseline, teaching example, exploratory tool, or public citation.

When rivals explain a component better, describe them as its leading explanation in later summaries and artifacts.

Useful component science may survive after a Law-specific interpretation ends: measurement can improve palliative research, missingness methods can protect participants, rivals can clarify terminal data, and failed mechanisms can leave valuable tools (Chamberlin, 1890; Platt, 1964).

Retired approaches can retain operational value. An instrument may reveal device bias, and a scheduling model may leave an effective burden-monitoring protocol. Recording those contributions preserves useful knowledge for later work.

Public summaries separate mechanism retirement from target defeat and reject both total collapse and automatic preservation of attached claims. Keeping useful science under the authority it earned lets truth and usefulness coexist (Burnham & Anderson, 2002; Palminteri et al., 2017).

27.7 — Theory-Level Defeat and Failure Memory

Theory-level pressure differs from a local method failure. A credible counterexample, repeated bridge failure, and loss of distinct empirical content require different evidentiary arguments. The review must identify which claim is reached, how its prerequisites have been established, and which useful findings remain after the resulting narrowing, amendment, or defeat.

27.7.1 — Credible Counterexample

A well-identified stream and endpoint with robust terminal imbalance can supply a credible counterexample. It directly pressures the declared universality and strong-form status.

Persistent superiority of serious rivals across component comparisons requires retiring the Law-specific interpretation of those components.

A defeat review determines the scope of withdrawal and the claims that survive. Its statement triggers public correction, status revision, and updates to dependent artifacts.

Preserve data, code, reasons, dissent, and corrections as a record of failure. This prevents renewed overclaiming and repetition of avoidable recruitment burdens.

Equation 27.5 (empirical-content exhaustion test):

$$\lim_{k\to\infty} \Delta Prediction(k) = 0 \quad \Rightarrow \quad \text{rescue-versus-refinement review flag}$$

Equation 27.5 is a schematic review flag: k indexes successive revisions, and ΔPrediction denotes their added testable predictive content; the display supplies no numerical decision threshold. Under Section 27.6.5, require an independently motivated prediction, a prespecified falsifier, and data capable of distinguishing the successor from the prior claim and serious rivals. Without these, classify the revision as rescue. If no

prospectively specified discriminator remains, apply Section 27.7.4’s withdrawal rule.

Defeat is branch-specific. A gate-valid counterexample contradicts the exact universal branch it reaches; persistent nonidentification may instead require abstention or a non-equivalent successor. Record that distinction.

Preserve objections that defeat was too broad or too narrow through Section 28.3.1’s minority-report procedure.

Failure records constrain later editions and public language. Reopening requires a citation to the original defeat and independently justified new evidence.

Protect counterexample cases from sensationalism; report the evidence route while preserving privacy and dignity.

A credible counterexample requires a qualifying stream, independently adjudicated endpoint, valid measurement or other justified evidence, robust uncertainty analysis, and artifact checks (Popper, 1959; Lakatos, 1978).

The counterexample must reach the complete-stream latent residual through adequate representation, origin, routing, endpoint, missingness, and measured-to-latent guarantees. Identifying the natural mechanism responsible for a valid nonzero residual is not an additional requirement for rejecting the exact universal branch.

Consider a qualifying stream with an independently adjudicated endpoint, context-qualified measurement, and a nonempty, truth-containing compatibility set for its complete-stream latent residual that lies wholly outside the prospectively fixed equivalence band. The set must account for the unobserved past, measured-to-latent uncertainty, informative missingness, and declared artifact checks. Such a case is adverse to that band claim and, when zero is excluded under the protocol-fixed

inferential guarantee, to the exact universal branch, even if an ordinary rival explains the nonzero residual.

Pooling favorable cases cannot dilute a valid counterexample. Exclusions for anomaly or measurement noise must follow prospectively specified criteria; a case meeting the counterexample conditions changes the declared branch and enters failure memory (Mayo, 2018; Meehl, 1978).

Report the exact branch or scope defeated, and distinguish narrower claims that remain logically possible from claims that have earned evidence.

27.7.2 — Repeated Conjunction Failure

Repeated absence of the distinctive conjunction across independent, adequate tests creates program-level pressure beyond a local signature failure (Lakatos, 1978; Mayo, 2018).

For example, concentration may remain compatible with endpoint-conditioned rivals while reserve-linked contraction fails and held-out rivals outperform the candidate. Only comparisons with valid prerequisites count as adverse evidence; endpoint-gate failures leave the affected test unadjudicated.

27.7.3 — Unresolvable Identification Failure

When no defensible observation-to-target bridge exists under ethical constraints, suspend the public empirical claim (Mayo, 2018; Meehl, 1978).

Identify the component and observation route responsible for nonidentification, and any independently justified alternatives that remain available.

If terminal self-report would impose unacceptable burden and passive channels cannot identify the quantity, suspend that design and investigate a separately qualified bridge.

27.7.4 — Persistent Rival Superiority

Rival superiority across components, contexts, and held-out tests can pressure the broader program beyond retirement of one mechanism (Meehl, 1978; Oreskes et al., 1994).

An unexplained remainder supports discrimination only if it was prospectively specified, robust to reasonable alternatives, and large enough to separate generators (Popper, 1959; Lakatos, 1978).

Theory-level reporting should specify which rivals dominate which components and whether any prospectively specified Law-specific discriminator remains. If none remains, withdraw the affected claim of empirical distinctiveness. State separately whether a valid counterexample contradicts the terminal target or whether that target remains unresolved by the available tests.

27.7.5 — Defeat Statement and Scope

The defeat statement identifies the withdrawn object and scope, the decisive evidence path, and recorded dissent.

A review may withdraw the reserve-coupled signature program for adult hospice oncology after rival dominance and nonrecoverable parameters. That disposition does not by itself adjudicate the terminal target or the entire series.

27.7.6 — Failure Memory

Preserve the evidence, reasons, dissent, corrections, and changed public language needed to disclose a failed claim before reuse (Popper, 1959; Lakatos, 1978).

Give negative results durable identifiers, searchable component and version metadata, and links to successor work.

An invalid sign convention must remain flagged on the retired instrument and dependent artifacts, so it cannot reappear in a later graphic. Machine-readable status can help detect that reuse.

27.8 — Care and Dignity After Scientific Change

A scientific status change creates obligations to participants and readers as well as to the claim register. Support cannot turn suffering into moral payment, and defeat cannot diminish the value of a participant's contribution or entitlement to care. Research variables remain barred from public scoring and ranking. Communicate narrowing or loss precisely, preserve independently justified applied work, and treat a smaller accurate claim as a useful outcome rather than a reason to conceal revision.

27.8.1 — Care Does Not Rise or Fall With Status

Support must not moralize suffering. Care remains immediate and independent, and the measured outcome gives suffering no moral purpose or status as an owed payment.

Care follows present need, not scientific status. No supported, narrowed, retired, or defeated claim may delay relief or reduce the obligations already owed.

This rule prevents a subtle hierarchy. If support for terminal neutrality made caregivers less urgent, the science would have become dangerous. If defeat made people feel their pain was merely ordinary and therefore less worthy, the science would have become cruel.

Consider a family hearing that a terminal signature study has been defeated in their relative's diagnosis. The correct message is not that hope was foolish or care is different. The correct message is that scientific language changed while the duty to reduce suffering remains unchanged.

27.8.2 — No Victim-Blaming Under Any Verdict

Scientific defeat concerns the tested claim, not the participants. Public statements should preserve the equal dignity of the

people and families whose contributions made the study possible, without blame, rank, or claims that suffering was deserved. Care implications are stated separately and grounded in participants' needs.

27.8.3 — No Score or Rank Survives

PTVC research constructs may classify claims, uncertainty, missingness, transport limits, model performance, and declared site-method features within research governance. They may not rank people or groups, including diagnostic or cultural groups, or assign worth, eligibility, trust, holiness, loyalty, dignity, destiny, or care priority. Public displays must distinguish component or methodological status from any judgment of human or group standing.

27.8.4 — Applied Lens May Remain Useful

Failure of an applied interpretation does not erase ethically grounded burden analysis. A surviving instrument, dataset, protocol, burden audit, or care tool needs an independently justified use, validated context, and corrected label. Ordinary care and institutional repair continue on their own justification.

27.8.5 — Public Communication of Loss

Do not hide or dramatize a scientific loss. Issue a calm, dated, precise correction that makes the change visible.

Before public release, explain the evidence and status change directly to affected sites and communities. Protect critics who identified the weakness and preserve their objections accurately.

Families need not reinterpret a loved one's life or make private experience public because a claim changed. Credit participants and staff for what their contributions established, and give readers a precise correction.

Public loss statements must be calm, dated, component-specific, and visible. Record the old and new labels, evidence path, reason, surviving claims, withdrawn language, correction schedule, corrected artifacts, and the responsible maintainer. Do not hide loss as nuance or delay correction to protect institutional image.

27.8.6 — A Smaller True Claim Is Success

A smaller well-supported claim can be a successful outcome of research. Public accuracy takes precedence over the breadth of the original ambition.

A smaller true claim is a scientific success when stated plainly, tested within its scope, and not used to preserve a defeated universal claim. Summaries, graphics, and detachable sentences must carry the protocol-fixed verdict, reference class, and limits rather than a movement slogan. Answerable science is willing to become smaller than the hope that first motivated it.

Chapter 28 — Scientific Updating and Claim Revision

A scientific result can change before every sentence that depends on it changes. A paper may be corrected while an old diagram circulates, a device may be deprecated while its outputs remain in a summary, or a narrowed finding may retain a general headline. This chapter follows the correction from the evidence decision to each affected use (Committee on Publication Ethics, 2025; Crossref, 2025).

The first task is to identify the object that changed. A defect in an instrument, a failed transport claim, and a challenge to a formal premise require different repairs. Treating all three as one verdict can either conceal a failure or extend it beyond the evidence. The update must preserve the distinction while following dependencies that actually inherit the defect.

Consider the hypothetical hospice example developed here. A terminal-signature estimate satisfies a declared equivalence band in two networks, but prediction fails in a third with different sedation practices. The result calls for separate decisions about the supported contexts, the failed transport, and the observation model. Averaging the sites cannot determine which component needs repair or authorize general wording.

A reviewable correction names those consequences. Researchers using the original care regime need to know what remains usable; researchers in the third setting need to know what requires requalification. Readers of the public summary need an accurate replacement. Each receives the same scientific decision at a level of detail suited to the task.

Version history keeps the original finding available for assessment. Reanalysis with a revised instrument creates a linked result rather than rewriting what the earlier procedure showed. The old version may become historical, restricted, or unavailable for new use. Those distinctions let another

investigator reproduce the failure and judge whether the successor addresses its cause.

Disagreement can concern the evidence or the correction itself. A reviewer may accept the technical repair but object that its caption still overstates the result. An applied-domain steward may identify a consequence missed by the scientific team. Dissent, appeal, and reconsideration preserve those objections with a defined route to a decision and conditions for reopening it.

The work also has a clock. A misleading public image may need an interim notice while a formal question remains under review. Delayed propagation requires an owner and a visible completion state; it cannot be hidden inside a pending scientific debate. Conversely, a corrected website does not resolve an unanswered methodological objection.

The chapter specifies how these actions fit together: evidence review, controlled versions, compatibility analysis, correction notices, and completion checks. Its central test is practical. After an approved change, can a reader identify the current claim, recover the earlier decision, understand what changed, and avoid relying on an obsolete use? A correction is complete only when its affected dependencies have been addressed.

28.1 — Human-Governed Scientific Status Updates

A status update joins evidence, scope, dependencies, and consequences through accountable human review. The packet identifies the exact object, the result or error that triggered reconsideration, the appropriate authority route, and the claims affected under Section 27.1's dependency rule. Reviewers disclose conflicts, record dissent and reasons, and approve public wording within that scope. Automated diagnostics can assist the process but cannot silently convert a numerical output into a final scientific verdict.

28.1.1 — Evidence Packet and Review Criteria

Identify the exact claim, instrument, realization, endpoint rule, public artifact, or applied-domain bridge under review. Evidence about one component must not silently change the whole theory.

Identify the trigger: new data, replication failure, rival victory, measurement defect, public risk, code defect, or inconsistency with the formal root. Correction follows the cause rather than the project's reputation.

Choose the appropriate authority route: technical update, public correction, formal change request, or handoff to the formal root. The science volume cannot silently rewrite the root.

Name reviewer roles, expertise, conflicts, dissent, and abstentions. The decision requires accountable human judgment.

Collect studies, replications, rival models, deviations, ethical findings, public-risk notes, and dissent in the evidence packet before changing claim status (Committee on Publication Ethics, 2025; Crossref, 2025).

Declare criteria for each possible status change before reviewers know which outcome is preferable, specifying the exact component under review (Crossref, 2025; Center for Open Science, 2025).

A notation clarification needs formal and mathematical review. A measurement revision needs psychometric, statistical, cultural, disability, and data-governance review. An endpoint revision needs consciousness, neurology, palliative, and ethics review. A public-artifact correction needs science, public-risk, accessibility, and communications review. A root-formal amendment requires formal-law handoff rather than science-book unilateral action.

Disclose conflicts alongside expertise so readers can assess developers' technical knowledge and independent reviewers' alternatives.

Assign deadlines by urgency: immediate public-risk correction, expedited technical correction, standard review, or reconsideration after new data. Record delays while an affected artifact remains public.

For a retired realization, state whether root wording changes, diagrams are withdrawn, earlier claims become historical, or a narrowed version remains testable.

Use an object-specific sentence: for example, the reserve-coupled realization remains a candidate consequence generator, while its measurement bridge is retired for terminal cohorts using the deprecated device.

Review the proposed wording for misreadings that could inflate a sensor failure into total defeat, preserve an unsafe artifact, encourage self-testing or care delay, or obstruct a necessary correction (Committee on Publication Ethics, 2025; Crossref, 2025).

28.1.2 — Conflict, Recusal, and Public Wording

Record the decision as support, narrowing, suspension, retirement, defeat, correction, deprecation, or handoff. This gives the object a usable status without creating a global score.

Identify affected books, websites, diagrams, captions, code, datasets, social excerpts, donor pages, and archive notices. Obsolete claims must not survive in widely circulated dependent artifacts.

Under Section 26.4.6, record authorship, funding, advocacy, ownership, or other interests, the affected decision, recusal, and independent reviewer. Interested experts may clarify technical issues without sole control of the verdict or wording.

A status decision includes a concise public sentence naming the component, result, reference class, uncertainty, and strongest surviving alternative. Its wording should remain accurate when quoted on its own.

State the correction's practical cost: which inference is suspended, what old wording implied, or whether a withdrawn figure overcompressed, omitted rivals, exceeded measurement, or retained obsolete claims (Crossref, 2025; Center for Open Science, 2025).

Tailor correction notices to the affected decision. Reclassification of an endpoint study may require clinical communication; deprecation of a code container requires a technical migration notice.

The sentence should avoid three temptations. It should not soften a defeat into refinement. It should not inflate a partial result into discovery. It should not reassure readers by praising the project’s humility while hiding what changed. A phrase like “updated for clarity” may be appropriate for a notation fix, but dishonest for a realization failure. A phrase like “new evidence strengthens the project” should be used only when the component, scope, and rival comparison actually justify it.

The public sentence should also name care-independent consequences where relevant. A scientific correction never reduces present care, dignity, privacy, or nonparticipation rights.

But that line should appear only where the correction could otherwise be misused. If an endpoint study is reclassified, the public sentence may need to say that care remains immediate and inference remains research-bound. If a code container is deprecated, that sentence is not necessary; the local risk is technical, not clinical.

A public status sentence should be tested against hostile quotation. Could it be clipped to imply unsupported proof or defeat, moral payment, promised rewards, or Artificial Intelligence authority? Could it be used by supporters to overstate and by critics to misstate? No sentence is immune to bad faith, but the wording should not hand bad faith an easy weapon. Precision is the project's first public defense.

The update board should maintain approved public-language templates for each status class, but templates cannot replace judgment. The point of a template is to prevent accidental inflation, not to make every correction sound the same. A correction about a failed endpoint rule should sound different from a correction about a website caption. One-author style is not sameness; it is disciplined recognizability across different local risks.

28.1.3 — Recorded Reasons

If an independent replication finds no concentration under the fixed endpoint rule but exploratory redefinition does, the confirmatory result governs status. Keep the exploratory pattern hypothesis-generating pending the required endpoint and root review.

Recorded reasons turn a status transition into a scientific artifact rather than a mood. The decision states why stronger and weaker alternatives were rejected, which evidence carried weight, what remains unresolved, and what public wording must

now change (Committee on Publication Ethics, 2025; Crossref, 2025).

A second failure case occurs when a donor-facing summary keeps a stronger claim because the update board delayed public propagation. The technical register may be correct, but the public surface remains wrong. If a reader or funder sees the donor page first, the project has failed at the level of communication authority. The update system should treat this as an incident, not an inconvenience.

A third failure case occurs when a critic’s model outperforms the candidate model and the project responds by changing the status label without naming rival dominance. The update may appear to preserve dignity, but it deprives the critic and future researchers of a clear scientific outcome. Rival victory should be visible because it is one way the program learns. Hidden rival victory is just another form of proof drift (Center for Open Science, 2025; Tabassi, 2023).

A fourth failure case occurs when an artificial-intelligence assistant improves the tone of a correction notice but omits the exact consequence for claim status. Reviewers check the draft against the approved decision so that clearer language preserves scope, uncertainty, withdrawal, and every other material consequence. Readability and scientific accuracy must improve together.

A fifth failure case occurs when public reaction is treated as evidence against a correction. If supporters dislike narrowing, or critics celebrate defeat, neither reaction controls status. Public reaction can reveal misunderstanding and artifact risk. It cannot decide the scientific object. The update route should capture public reaction as a communication signal, not as a warrant signal.

These failure cases show why the update body needs records, authority limits, and public propagation. The risk is not only that the wrong scientific decision is made. The risk is that the right decision is made in one room and then loses its meaning on the way to the world.

28.1.4 — Human Authority over Automated Verdicts

Computational tools may flag contradictions, summarize studies, and test consistency, while identifiable human reviewers issue reasoned and appealable status decisions. Governance frameworks support that work through oversight, evaluation, and accountability (Crossref, 2025; Center for Open Science, 2025; Autio et al., 2024; World Health Organization, 2025; Feuerriegel et al., 2026).

Scientific vocabulary should identify the specific achievement: discovery of a pattern, validation of an instrument, confirmation of a prediction, or replication of a result. Direct, concrete language makes the finding clear.

Status language states the approved scientific outcome directly, including defeat, retirement, or rival dominance when applicable, and then identifies the knowledge and methods that remain useful (Tabassi, 2023; Committee on Publication Ethics, 2025).

Metaphor cannot bypass status. Phrases such as “coming into view,” “the world responding,” or “the model evolving” are acceptable only when they carry the same warrant as the technical register.

Maintain a versioned record of prohibited implications. Exclude moral desert, prescribed harm, human-worth ranking, machine sovereignty, archive-as-soul claims, and forced meaning. Review evidence-status claims against the findings they describe. Conditional net accounting and the Series Map’s

“There is no free happiness” formulation are not prohibited merely for expressing a balance requirement; Section 3.7.1 defines their permitted meaning.

Every material update needs reasoned minutes: the object, evidence, rival pressure, options, rejected alternatives, decision, dissent, and public consequence. The record should explain why both weaker and stronger statuses were rejected.

Minutes must name the exact uncertainty that remains, whether measurement, endpoint, rival, transport, reproducibility, artifact safety, or root compatibility, because each implies a different next action.

Preserve rejected public wording and list every open propagation task with an owner and deadline. This prevents stale language from returning and keeps correction accountable until every dependent surface is closed.

These records let future readers judge whether a correction was careful, pressured, cosmetic, or complete while keeping the burden of explanation with the project.

28.2 — Versioned Claim, Model, and Evidence Registers

Every claim and artifact needs a stable identifier. Registers record status, owners, dependencies, dates, and reasons for change. Supersession changes present authority. It must not erase the prior decision or its evidence.

28.2.1 — Claim Identifier and Version

Record object class, exact wording, active scope, and change level under a stable identifier and version (Crossref, 2025; Wilkinson et al., 2016).

Define version levels for this scientific register: major changes alter target identity or interpretation; minor changes revise controlled content while preserving identity; patches correct errors without altering meaning.

The claim register records exact wording, status, scope, dependencies, date, reviewers, dissent, and public language. A headline change requires corresponding object-specific scope.

The instrument register records version, calibration population, context of use, failures, replacement status, and retention limits. New wording cannot overwrite a measurement failure.

The protocol register records study-registration, protocol, and analysis-plan versions, the endpoint rule, observation model, consent scope, analysis container, and deviations. Reanalysis under a new protocol is a new analysis.

The dataset register records release identifier, access tier, provenance, transformations, consent boundaries, and withdrawal notices. Dataset availability does not itself authorize public model training.

The public-artifact register records approved wording, source claim, status, image and caption versions, corrections, and

withdrawal state. No graphic can carry a stronger claim than its source.

The failure record names the failed object, criterion, affected results, public consequence, and reconsideration trigger. A rejected bridge remains visible as a known hazard.

28.2.2 — Evidence Links

Link each status to its studies, code, datasets, protocols, replications, red-team reports, adverse findings, and public-risk records, subject to the appropriate access tier (Wilkinson et al., 2016; Barker et al., 2022).

Attach results to their producing versions; a new observation-model analysis receives a linked result describing its relation to the earlier finding.

If previously unmodeled medication sensitivity invalidates a terminal study's inference, preserve its version-bound record, withdraw the affected public claim, and mark permitted raw data for reanalysis. Sensitivity already handled by a valid model does not itself invalidate the result.

28.2.3 — Scope, Provenance, and Failure Memory

Record population, setting, language, instrument, endpoint rule, care regime, device version, cultural context, and intended inference (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Provenance links creators, data, code, transformations, producing versions, and later status decisions.

Record funding sources, critic-initiated corrections, and artificial-intelligence assistance in code generation or contradiction screening.

28.2.4 — Outstanding Uncertainty and Release States

Every versioned object needs a release state. Within this register, Draft denotes an unreleased working version rather than an authorized public source. A publicly released draft or preprint may be cited with its version and provisional status. Candidate means under review within declared limits; Active identifies the current project version, not independent scientific validation. Superseded, deprecated, withdrawn, and defeated preserve history while identifying the uses no longer warranted.

Supersession can replace a formatting template; deprecation can restrict a biased observation model; withdrawal can stop circulation of a misleading graphic; retirement can follow rival victory. State the reason and permitted next use.

Deprecation language should be explicit about permitted use. A deprecated instrument may remain available for historical replication or sensitivity analysis, but not for new confirmatory claims. A deprecated figure may remain in the archive with a warning, but not in active public materials. A deprecated dataset release may remain citable, but any current inference should use the corrected pipeline if consent and provenance permit (Wilkinson et al., 2016; Barker et al., 2022).

Record safety withdrawal separately from scientific defeat: an unsafe artifact can be withdrawn while its source claim remains active, and an artifact explaining a defeated model can remain useful.

Record release state in the register and relevant artifact metadata, readme, caption source, revision note, or model card; a file name alone is insufficient.

28.2.5 — Stewardship of Retired Claims

Assign register stewardship for completeness, access, version identity, updates, and visible corrections by expertise and access needs.

When withholding a detail, record the reason class publicly: participant privacy, consent boundary, clinical sensitivity, security risk, legal restriction, or ongoing review (Barker et al., 2022; National Institutes of Health, 2026a; National Institutes of Health, 2026b).

An independent auditor periodically checks agreement among version labels, public artifacts, dataset releases, and code identifiers.

28.2.6 — Public and Internal Views

Public and protected views retain the same status and scope. Public summaries disclose the correction's consequence; sensitive participant, clinical, security, or confidential dissent details remain in authorized review layers (Wilkinson et al., 2016; Barker et al., 2022).

Release notes identify old and new versions, reason, affected and unaffected claims, required reanalysis, public corrections, and interpretive change level.

28.3 — Dissent, External Challenge, and Reconsideration

Dissent, appeal, and reconsideration serve different purposes. Dissent preserves a competing interpretation; appeal challenges a decision's procedure or scope; reconsideration responds to new evidence, methods, failures, or public risks. Each route requires access, deadlines, reasons, and a threshold for action, with protection from retaliation. A closed review can remain final for present use while specifying the evidence that would reopen it, rather than treating either perpetual dispute or permanent immunity as scientific governance.

28.3.1 — Minority Report

A minority report records unresolved methodological, evidentiary, ethical, or public-language objections and remains attached to the status (Mellers et al., 2001; Silberzahn et al., 2018).

Keep majority and minority interpretations side by side, including unresolved dependencies and reopening conditions.

Dissent can accept data yet reject heterogeneity interpretation, accept a model yet reject its terminal window, or accept an instrument locally while rejecting transport. Public-risk dissent can concern a crop-vulnerable diagram despite accepted science.

A specific objection might identify support for preterminal variance compression under instrument version 4 in one care regime, then request correction of wording that claims terminal neutrality across diagnoses (Nosek et al., 2018; Mayo, 2018).

Include a dissent field even when empty, identifying reviewers, route, and the opportunity to object.

28.3.2 — Appeal Route

Allow appeals for procedural error, excluded rivals, overlooked confounds, improper conflict rulings, or misleading wording, under declared standing rules and deadlines (Silberzahn et al., 2018; Nosek et al., 2018).

Standing includes a critic who specified a rival, a site investigator whose data were reclassified, an external expert whose domain was used, a participant advocate, and a steward of an affected applied artifact.

Separate public-risk urgency from technical appeal pace. Use an interim banner, hold, or withdrawal when needed while substantive review proceeds (Mayo, 2018; Mellers et al., 2001).

Reopen a closed appeal only for new evidence or method, a new artifact, undisclosed conflict, material error, or previously invisible downstream incompatibility.

Dissent can concern scope, methods, status, wording, public risk, or dependency logic. Record the minority position and any requested wording change, future test, or appeal.

An appeal can identify procedural flaws, missing expertise, improper scope, conflicts, ignored evidence, or omitted public risks. Its disposition may affirm or revise the decision, reopen review, impose an interim hold, or initiate a formal-root handoff.

Reconsideration can follow new evidence, methods, rivals, misuse, device drift, context change, or scheduled review. It may produce a new version, deprecation, correction, reinstatement, narrowing, or retirement.

When no new evidence or procedural defect is presented and the objection has already been answered, close the record with a rationale and an explicit threshold for reopening.

28.3.3 — Reconsideration and Correction Dissent

Apply the rescue audit in Section 28.4.5 to reconsideration proposals (Mellers et al., 2001; Silberzahn et al., 2018).

Reconsideration preserves prior dissent and states whether new evidence vindicates it or why it remains unpersuasive.

Correction actions remain open to dissent.

Review whether the notice's disposition matches the error: a withdrawal may be needed where it says clarification, while a graphical correction may have been described incorrectly as scientific defeat.

28.3.4 — Nonretaliation and Applied-Domain Appeals

Apply the nonretaliation procedures in Section 25.7.5 to everyone reporting a defect, including applied-domain observers and minority reviewers. Reporting must not jeopardize access, safety, credit, employment standing, or public dignity (Mayo, 2018; Mellers et al., 2001).

Applied-domain appeals must identify scientific wording and its consequence: City transparency, administrative scoring, religious interpretation, or transition readiness may depend on it.

Applied stewards identify practical consequences; independent scientific review decides the evidence status.

Record the receiving domain and whether the remedy is a local wording change or a scientific status update. A religious-community artifact may require withdrawal for coercive implications while its source claim remains scientifically active.

28.3.5 — Independent Reanalysis and Sunset Review

For independent reanalysis under Section 27.2, record specification, access conditions, result, affected version, reviewer, and disposition: correction, sunset review, or no change.

Reanalysis tests another implementation or defensible specification; sunset review checks whether instruments, benchmarks, endpoints, summaries, AI workflows, and applied evidence boundaries remain current after drift or new evidence.

Display each object's next review date and the reason for its interval. High-reach artifacts, endpoint rules, and workflows with rapidly changing dependencies may need shorter review intervals than technical appendices or historical records.

If missing evidence prevents a sunset decision, mark the object under watch rather than renewing it silently. This release state permits current wording while flagging unstable dependencies.

28.3.6 — Finality Is Provisional

Finality is provisional because public language needs current stability while science needs future correction. A decision can be binding for active artifacts and still remain reopenable under better evidence, better methods, or a newly visible public-risk failure (Silberzahn et al., 2018; Nosek et al., 2018).

Dissent does not disappear after a correction is published. A public correction may itself be disputed: too narrow, too broad, too slow, too soft, too technical, too public, or too quiet. The dissent route should remain open for the correction action, not only for the original claim. A project can correct the science and still mishandle the correction.

Public-correction dissent may concern wording. A reviewer may argue that a notice says "clarification" when the artifact required withdrawal. Another may argue that a withdrawal notice implies scientific defeat when the problem was only graphic overcompression. These are not cosmetic disputes. Public status language can redistribute interpretive burden across readers, critics, participants, and applied domains.

Dissent may also concern propagation. A reviewer may accept the technical correction but object that the donor page, social excerpt, or archived figure remains active. This is a propagation dissent. It should create a checklist item rather than reopen the entire scientific decision. The route should distinguish object-level disagreement from propagation failure (Mayo, 2018; Mellers et al., 2001).

The correction record should show whether dissent was resolved, maintained, or escalated. If the project keeps a contested public wording, it should explain why. If it accepts the dissent, the correction itself receives a new version. The old correction notice may remain in the archive with its own correction history. This may feel intricate, but public scientific language at scale is intricate.

Dissent after correction should not become a reputational battle over who was most careful. The goal is to reduce public error. The record should focus on the object, not personalities. A critic may be right about wording and wrong about theory. A supporter may be right about scope and wrong about artifact risk. The route should separate these judgments.

28.4 — Formal Change Requests and Root Concordance

The science volume may revise realizations and instruments. Its remit extends to protocols, observation models, and evidence. Scientific wording and correction procedures remain in scope. Changes to the formal claim, endpoint definition, currency, or root boundary require handoff to The Law of Fairness.

28.4.1 — Trigger for Formal Review

Open formal review when unit failure, endpoint leakage, process-class conflict, notation ambiguity, nonidentification, or a proposed construct implicates the received root.

Notation clarification resolves ambiguity without changing the object. Record a substantive amendment if a new construct is introduced.

Representation change can replace scalar reserve with a vector, latent state with belief state, or a single-channel estimator with a multichannel model. Specify which earlier findings remain comparable and which become historical.

Measurement revision changes instruments, calibration, anchors, weights, abstention, missingness handling, or context of use. Reanalyze raw data only where consent, provenance, and compatibility permit (Crossref, 2025; Committee on Publication Ethics, 2025).

Endpoint revision must preserve adjudication independent of the ledger estimator, identify affected endpoint-window studies, and specify which public claims are held during review.

Candidate-realization replacement requires the successor’s new predictions, a record of the predecessor’s failure, and deprecation of dependent artifacts. Replacing a bridge remains distinct from redefining the formal Law.

Identify the old object through exact wording, version, status, and affected display, code, or protocol. Without that record, the change cannot be audited.

Provide exact proposed wording or a reviewable replacement object, with proposed status and version class. An aspiration alone is not an actionable change request.

State the evidence, error, rival victory, public-risk finding, dependency conflict, or root inconsistency motivating the change. The rationale must distinguish substantive correction from cosmetic revision.

List affected formal definitions, instruments, protocols, datasets, code, public artifacts, and applied references. Otherwise contradictions can survive downstream.

For each dependency, specify whether it remains unchanged or requires a patch, deprecation, withdrawal, reanalysis, or root handoff. Old results cannot be silently overwritten or reused without justification.

Record reviewers, expertise, conflicts, votes, dissent, and appeal conditions. These details make the authority behind the decision inspectable.

Specify implementation steps, public notice, archival rules, rollback triggers, and the failure-record entry. The correction must be both executable and reversible where required.

28.4.2 — Compatibility and Migration

Compatibility analysis traces both scientific dependencies and public uses, including indirect references that may break when a technical object changes.

Migration preserves the version-bound history specified in Section 28.2.2; reanalysis receives a new record, and superseded figures are archived with notice, replaced, or withdrawn (Committee on Publication Ethics, 2025).

Include participant-facing materials in migration review when families, clinicians, or vulnerable participants encountered the old claim. Decide which notices and replacements they need.

28.4.3 — Scientific Evidence and Open Questions

Separate the evidence motivating a revision from unresolved transport, measurement, and rival questions that still constrain it (Crossref, 2025; Committee on Publication Ethics, 2025).

List the formal commitments, assumptions, signatures, benchmarks, prior results, artifacts, and protocols that change or remain intact. Disclose any weakened prediction as a status consequence.

28.4.4 — Root-Book Authority

Root-book authority governs which formal statement the series publishes and which version its other volumes inherit. It prevents an empirical revision from silently changing the hypothesis under test. It does not govern independent scientific judgment. Researchers may criticize the received formulation, propose alternatives, and publish favorable or adverse findings without project endorsement. The project's review process records its own version decisions; scientific assessment remains open to public evidence, argument, replication, and independent review. A decision to retain the root wording must leave any unresolved empirical or formal objection visible.

Identify the exact root element implicated (stream, currency, latent ledger, terminal neutrality, endpoint, boundary condition, or ethical firewall) and why a science-level repair cannot resolve it.

During root review, place holds, provisional labels, or withdrawal notices on artifacts whose wording depends on the challenged element.

If the root route declines amendment, retain the conflict and dissent, including any scientific bridge, observation model, or empirical target that remains narrowed under unchanged root wording.

28.4.5 — Version and Dependency Audit

Audit a proposed repair's independent motivation, predictive gain, actual consequences, fresh failure conditions, and retained failure record.

Document the timing and independent grounds for a revision: theory, prior data, measurement limits, rival comparison, or public-risk analysis. A revision may respond to failure; its justification must extend beyond preserving the old headline.

For a substantive model revision, specify new constraints, improved recovery, sharper rival separation, stronger transport logic, or clearer failure conditions. Flag any loss of empirical content (Crossref, 2025; Committee on Publication Ethics, 2025).

Disclose every change to scope, interpretation, prior-result use, and public wording. Narrowing or withdrawal may be necessary; notation or code repairs may preserve the claim and predictions.

Publish the rescue-audit reasoning in the scientific register even when public summaries compress it.

28.4.6 — Visible Claim Changes and Rollback

A candidate repair does not become settled through repeated use. Until formal adoption, it remains labeled candidate, provisional, superseded, retired, or under review as appropriate (Crossref, 2025).

Separate scientific breakage from editorial propagation in the compatibility matrix. Dependency depth shows whether a

change reaches the endpoint, observation model, target, or only their expression.

Specify rollback triggers: replication or reproducibility failure, accessibility or public-risk defect, adverse rival result, or a deeper formal change. The safe fallback may be suspension or withdrawal.

Retain the rolled-back revision and reversal evidence in version history. Identify the withdrawn revision and current state in its correction notice.

28.5 — Correction Propagation Across the Series

Correction is complete only when it reaches the scientific and public objects that depend on the changed claim. Books, code, datasets, summaries, captions, and widely circulated excerpts may require different actions under different authorities. Map those dependencies, assign owners, preserve earlier versions with their status, and check whether the active materials now agree. Domain-specific duties remain independent; a scientific correction does not silently rewrite a formal root or suspend ordinary care.

28.5.1 — Status-Labeled Cross-Book Notice

Applied titles receive the corrected status needed by their conditional claims; root-level issues use the formal change route.

A cross-book notice identifies the scientific change, affected wording, unaffected claims, and review required from each title (Crossref, 2025; Committee on Publication Ethics, 2025).

Map direct quotations, definitions, equations, tables, figures, and protocols, and indirect metaphors, summaries, panels, talks, donor pages, social excerpts, and applied scenarios that inherit the changed status.

Specify technical changes to containers, benchmarks, and dataset metadata separately from public notices, captions, and image withdrawals; Section 28.5.1 maps their migration actions.

Changes to formal-root wording require a root-handoff packet. They cannot be made within the science book alone.

Directly affected science prose requires revised wording, status labels, captions, and dependency references.

Conditionally affected applied-book scenes require updated conditional language or explicit labeling as imaginative architecture rather than evidence.

A public diagram that compresses away essential limits requires replacement, deprecation with notice, or withdrawal from active use.

A website summary requires an updated headline, status sentence, correction date, and archive link.

A dataset or code change requiring reanalysis receives a new release identifier. Preserve the earlier result under its earlier version.

A widely circulated social excerpt requires a corrected replacement or removal from active rotation with a historical notice.

28.5.2 — Routing Corrections to Each Title

Applied domains can investigate their own outcomes. A City practice, government service, religious ritual, Workgame pilot, or ordinary-life habit can be evaluated through its effects on burden, participation, access, or wellbeing.

The Science of Fairness owns updates to measurement, protocols, observations, signatures, recovery, replication, synthesis, verdicts, and change control, forwarding effects on other titles through their handoff routes.

The Utopia of Fairness and the applied titles distinguish conditional exploration from claims that a finding is established. When a scientific result changes, update any passage that relies on its former evidential status. A clearly stated exploration of what would follow if PTVC were true may retain that conditional premise and its valid consequences without claiming confirmation.

Religious interpretation can continue within its own domain, while scientific descriptions in religious material track the active evidence register. The handoff identifies which empirical

wording requires updating and which interpretive questions remain local.

Update The Transition of Fairness readiness language when scientific dependencies change; pilot observations may generate questions for separate qualification and testing.

28.5.3 — Independent Duties and Correction Examples

Corrections do not suspend care, privacy, consent, nonparticipation dignity, or lawful remedy.

If a donor page calls a future city a natural testbed proving the Law, replace it with research-question language. Applied observations generate hypotheses and confounds; evidence requires ethical prospective design and a persistent preregistration record for a public registry claim (Elliott et al., 2017; Crossref, 2025).

Withdraw excerpts that describe morally owed suffering, prescribed harm, or promised chosen reward, explaining the correction in retained historical records. An explicitly limited conditional accounting metaphor alone is not a withdrawal trigger.

28.5.4 — Removal of Outdated Proof Language

Scan introductions, summaries, captions, figures, explainers, donor/media material, glossary, and websites for obsolete proof, validation, settled, inevitable, or confirmation claims. Living-review methods support recurring updates (Akl et al., 2024; Elliott et al., 2017).

Assign sign-off by responsibility: science, formal root, applied translation, public risk, accessibility including alt text and tables, and repository stewardship.

Record reviewed but unaffected surfaces and why, for example, an applied claim independent of the corrected

instrument (Crossref, 2025; Committee on Publication Ethics, 2025).

Human reviewers classify each automated hit as current authority, historical record, quotation, or harmless coincidence; automated absence does not prove obsolete meaning has vanished.

Each sign-off should state the action taken: revised, no change, deprecated, withdrawn, archived with notice, or escalated. When a surface cannot be updated immediately, it should receive a temporary status banner or access hold if the risk justifies it.

28.5.5 — Owner-Book Review and Correction Lag

Correction lag is the interval between approval and full propagation across the ten titles and their Foundation and Reference editions, public artifacts, and code. Track open tasks and display interim notices where materially outdated content remains reachable.

Use proportionate interim notices. Formatting issues need no banner; artifacts implying unearned proof, moral payment, prescribed harm, inner-life access, or AI authority may require withdrawal before replacement. Mark them unavailable for status inference.

Use recurring delays to diagnose missing owners, copied artifacts without stable identifiers, and resistance to changed wording (Committee on Publication Ethics, 2025).

Track propagation as pending, delayed, blocked by a dependency or dispute, or resolved by update or notice.

28.5.6 — Correction Completion Audit

For each dependency, choose a correction action: revise, deprecate, withdraw, replace, add notice, freeze as historical, or

open a further change request. Record the action's owner and completion status.

A completion audit confirms revised status across active text, archives, code, data, summaries, and high-reach surfaces (Crossref, 2025; Committee on Publication Ethics, 2025).

Use synthetic drills to retire an instrument, deprecate a figure, narrow a claim, or rehearse root handoff, testing surface detection, owner action, and completion time.

Log missed surfaces, unclear roles, disputes, and delay; repeat drills after structural, website, release-pipeline, or pilot changes.

Search by meaning as well as phrase, including paraphrases and translations, and verify upstream and downstream dependencies.

Keep corrected reader-facing prose natural; retain detailed machinery and provenance in the register.

28.6 — Public Artifacts, Websites, and Withdrawal

Public artifacts carry scientific claims into settings where readers may encounter only a diagram, caption, summary, or web page. Their maintenance must connect each displayed claim to its current evidence status and correction history. The procedures below specify how review, revision, withdrawal, and accessible replacement keep that public record aligned with the research.

28.6.1 — Artifact Classes and Review

Interpretive artifacts (diagrams, captions, summaries, slides, cards, and excerpts) need object, status, and source-version labels. Redesign any image whose cropped form would misrepresent the claim.

The second class is archival artifacts, which remain available under appropriate access with clear historical notices and must not circulate as current (Tabassi, 2023; UNESCO, 2021a).

Computational artifacts include containers, benchmarks, model/data cards, notebooks, fixtures, and reports. Record scope, failures, and prohibited uses: successful execution does not validate the assumptions implemented.

The fourth class is conversational artifacts (replies, interviews, and discussions), which must remain correction-ready and within current status even under skeptical or hostile questioning.

Require stronger review and faster withdrawal for high-reach, compressed, vulnerable-reader-facing, or crop-prone artifacts.

An explainer that shows a smooth candidate path without rivals requires a comparison of candidate and rival predictions and an explicit status label.

A headline claiming early evidence for terminal neutrality overstates partial support for one measured signature. Rewrite it, add a correction date, and archive the prior text with notice.

A donor page implying that applied pilots prove the Law requires research-question language and a clear separation between applied work and scientific evidence.

Withdraw a social card that turns a conditional balance requirement into moral payment or prescribed harm. Retain a historical copy only with a warning about the superseded interpretation.

Code implementing a deprecated observation model requires a new release. Mark the old container deprecated and link the results affected by the change.

28.6.2 — Summary, Glossary, and Archive Status

Preserve corrected status, scope, and uncertainty in concise summaries and glossary definitions (Crossref, 2025; Tabassi, 2023).

Record release state separately from access status. An artifact withdrawn from current use may remain restricted for audit, and a restricted dataset may retain discoverable public metadata.

Cite stable identifiers, release names, or version labels in controlled references. Manuscript prose should remain intelligible if a linked site changes; withdrawal records must explain what changed.

28.6.3 — Websites and Social Excerpts

Update website dates, correction notices, share cards, and caches so old excerpts cannot circulate as the current claim (Tabassi, 2023; UNESCO, 2021a).

The notice should distinguish correction, clarification, deprecation, withdrawal, suspension, and reclassification. A clarification says the object was ambiguous. A correction says the object was wrong or misleading. Deprecation says an object remains historically available but should not be used going

forward. Withdrawal says active circulation stops. Suspension says no current status should be asserted. Reclassification says the evidentiary status has changed.

State the current evidentiary status and which earlier wording no longer applies, even when a model improves while its associated claim narrows or fails (Committee on Publication Ethics, 2025; Crossref, 2025).

Design artifacts for correction by reserving status/version fields, correction links, and relevant prohibited-use statements in their templates.

28.6.4 — Persuasive Copy, Graphics, and Captions

Review donor, media, and recruitment copy closely for obsolete authority claims; funding, participation, and prestige can create pressure to preserve them (UNESCO, 2021a; Committee on Publication Ethics, 2025).

Caption each graphic by role: candidate, diagnostic, evidence map, rival comparison, protocol, failure record, or artifact example.

Test crops without title or caption for implied proof, ranking, person measurement, AI authority, archive-as-soul, or transition destiny. Add embedded status, redesign, or restrict release when context is necessary.

Match visual hierarchy to evidence: bold candidate paths with faint rivals or hidden correction routes can contradict cautious captions (Crossref, 2025; Tabassi, 2023).

In AI-assisted correction diagrams, show human review controlling the tool; centralizing AI while marginalizing reviewers contradicts that authority relation.

Provide alt text that describes the figure’s content, labels, relationships, and stated assumptions in reading order.

28.6.5 — AI Summaries and Withdrawal

Correct AI-generated snippets, answers, abstracts, and summaries through source updates, metadata, notices, and superseded labels where possible. Record source identity, retrieval date, instruction context, and human verification (Committee on Publication Ethics, 2025; Crossref, 2025; Feuerriegel et al., 2026).

When correction is insufficient, promptly withdraw the artifact from active use and state the misleading, unsafe, or incompatible feature requiring removal.

If a withdrawn artifact served an educational purpose, replace it with one that teaches the corrected distinction.

28.6.6 — Archives and Correction Intake

Maintain stable correction pages with date, version, affected objects, current wording, and access state, distinguishing current, corrected, superseded, and withdrawn artifacts.

Metadata, redirects, preview cards, filenames, cached images, and other high-reach discovery surfaces are part of correction because they can continue to surface obsolete claims after a page changes (Committee on Publication Ethics, 2025; Crossref, 2025). Restricted artifacts may be listed without unrestricted download when participant, privacy, or misuse risks require it.

Public intake identifies artifact and problem, protects privacy, acknowledges without prejudgment, and routes science, safety, accessibility, cropping, and version concerns to accountable review. Explain resolution and link the current artifact.

28.7 — Applied Evidence Intake and Future Testbeds

Applied experience can identify a useful research question before it supplies evidence for a scientific claim. Intake therefore records the observation, its context, the proposed explanation, and the comparison needed to investigate it. A future testbed earns its role through those defined questions and methods, with the scientific interpretation remaining open to the results.

28.7.1 — Testbed Status

Testbed status denotes a possible research context; evidence requires a separately opened study design (National Institutes of Health, 2026a; National Institutes of Health, 2026b; Rieke et al., 2020).

Translate applied observations into explicit questions: does transparent administrative routing reduce vigilance; does visible correction change trust; does voluntary ritual language alter grief reports without changing scientific-status beliefs; does failure memory improve decision learning without increasing shame?

Consider alternative explanations including being heard, faster service, belonging, lower cost, expectation, novelty, self-selection, and institutional attention (Tabassi, 2023).

A Transparent City project can generate research questions about public burden, consent, and civic trust. It does not supply evidence for terminal neutrality or prove the formal Law.

A government office can generate questions about administrative load, delay, error correction, and public records. Service improvement does not itself validate the scientific hypothesis.

A religious setting can generate questions about meaning, grief language, ritual effects, and pastoral safety. Spiritual resonance does not constitute scientific support.

A Workgame or transition pilot can generate questions about sequencing, failure memory, readiness, and rollback behavior. The pilot does not by itself authorize transition or establish destiny.

Ordinary-life practice can generate N-of-1 questions, private reflection, and design insights. It does not become research evidence without the required consent and protocol.

Responses to public artifacts can reveal communication risks and generate hypotheses. Popularity or controversy is not a measure of truth.

28.7.2 — Consent, Scope, and Suitability

The future-testbed firewall has four gates: translate the observation into a research question; pass ethics, consent, and burden review; specify measurement, endpoint, context, rivals, and failure conditions; and enter results in the component evidence register under ordinary study-status rules.

Specify testbed population, setting, language, authority relation, source, and permitted and prohibited inferences. City language communities, care and workplace regimes, and academic versus grieving audiences require distinct qualification.

If burden or consent makes the question unsuitable, choose a less burdensome design or setting, or document leaving it open.

28.7.3 — Negative Operational Findings

Route increased stress, exclusion, dropout, privacy fear, staff burden, family pressure, and cultural mistrust into the testbed incident register (National Institutes of Health, 2026a; National Institutes of Health, 2026b).

Reporting, devices, record consent, and research visibility can create compliance pressure even under formal voluntariness.

Minimize cadence, scope, identifiability, and reuse in applied settings.

Track staff workload, involuntary family proxy reporting, and facilitator pressure for favorable results as potential harms and sources of bias (Rieke et al., 2020; Tabassi, 2023).

Collect only data needed to distinguish the hypothesis from rivals. Consider less intrusive reports, synthetic fixtures, aggregate timing, or qualitative designs before intensive telemetry.

Incident records identify domain, artifact or practice, failure, affected population, mitigation, resulting research question, and any scientific-status effect. Many applied failures alter public-risk controls or study design alone.

Applied failure can require an artifact hold or research pause without deciding the scientific claim. Withdraw or correct an artifact that misleads readers even if the underlying science remains open; pause research in a context producing coercive measurement pressure. Any change to scientific evidence status requires the relevant evidence and dependency review under Section 27.1.

Route communication failures to artifacts, governance failures to applied design, measurement failures to science, and formal conflicts to the root review.

28.7.4 — Context-Specific Protocol Review

This protocol review addresses scientific suitability; The Transition of Fairness separately addresses transition timing. Declare each context's opportunities and constraints: City settings add ecological validity and public-power relevance but complicate consent; religious settings enrich language while introducing belief and belonging effects; Workgame settings structure choices while creating performance pressure.

Design the evaluation to admit adverse outcomes through specified rivals, negative controls, nonresponse analysis, and suspension criteria. Program evaluation can test its own outcomes; a PTVC study additionally requires the target, measurement, and comparison conditions above (National Institutes of Health, 2026a; National Institutes of Health, 2026b).

28.7.5 — Science Constrains Claims, Not Governance

Science constrains claims, not governance. A scientific result can limit what the series may say about evidence, but legal rights, civic design, religious practice, material policy, and transition gates remain governed by their proper books and institutions.

An applied-domain failure can be scientifically valuable without being evidence for the Law. If a public artifact creates confusion, the failure informs communication design. If a City process produces distrust because records feel like surveillance, the failure informs consent and record boundaries. If a religious phrase is heard as forced meaning, the failure informs pastoral language. If a Workgame pilot makes readiness feel like ranking, the failure informs transition gating.

Applied failure can also reveal confounds. A study planned in a civic setting may discover that changes in reported burden are driven by caseworker kindness, speed of service, or relief from paperwork rather than by any Law-specific mechanism. That is useful. It helps build better rivals and prevents the project from claiming credit for ordinary humane administration (Tabassi, 2023).

Failure in applied domains should not be hidden to protect the image of the series. A public theory about fairness becomes untrustworthy if it cannot admit when its own surfaces shifted

burden. The applied incident register is not a scandal list; it is a learning system. It keeps the burden of public experimentation inside the record.

28.7.6 — Transition Timing Remains Separate

Transition timing stays separate because readiness is not a laboratory output. Even strong evidence cannot authorize rollout, artifact release, Workgame scaling, political movement, or public transition without independent gates, rollback, refusal rights, and timing control.

The route from observed practice to research protocol should protect people while preserving learning. First record the observation without evidentiary status; list confounds and ethical risks; turn the question into an estimand or qualitative aim; and specify consent, measurement, rivals, burden, and stop conditions. Only then may the study become protocol-fixed or publicly preregistered through an external registry with a persistent identifier.

Prospective public-communication rules belong in the study record. Before results, the applied domain may say only that the setting generates questions, the design is prospective, participation is voluntary, results may be negative, and public status awaits review. Public preregistration may be claimed only with an external registry record and persistent identifier.

A good future-testbed protocol lets applied life inform science without using lives as proof; it is neither shy nor extractive.

28.8 — No Formula, Score, or AI Sovereignty

A formula, score, or artificial-intelligence system can organize information without acquiring authority over the person represented by that information. Scientific use requires a defined task, evidence about performance, and a responsible decision process. Human review, dissent, appeal, and checks for automation bias keep those roles explicit as tools are introduced or revised.

28.8.1 — Decision Aids and Review Boundaries

Replication, heterogeneity, predictive scores, safety, dissent, and public risk cannot compensate for one another in a single authority score (Tabassi, 2023; UNESCO, 2021b).

Decision aids may display thresholds, requirements, dependencies, and comparisons with inputs, assumptions, uncertainty, and abstention. They flag review; a controlled scientific verdict requires human adjudication.

Automated literature searches can identify candidate sources, contradictions, and missing rivals. They cannot certify completeness, truth, or consensus.

Language-model drafting can suggest wording, flag repetition, and propose summaries for review. It does not approve manuscript copy, public status, or the canon.

A statistical dashboard can display diagnostics, uncertainty, dependencies, and abstention flags. It cannot issue a final discovery or defeat verdict or authorize a transition.

A recovery pipeline compares candidate and rival generators under declared conditions. Construction or fit alone does not establish the formal Law.

An archive preserves versions, corrections, failures, and provenance under access controls. It does not represent the person, soul, worth, or complete inner life.

A formal equation states a definition, candidate realization, estimator, diagnostic, or decision aid under its status label. The equation alone supplies no proof of the empirical claim, moral permission, clinical instruction, or public score.

Audit dashboard layout, color, ordering, thresholds, filters, and summaries for effects that steer reviewers toward rubber-stamping a tool's result (UNESCO, 2021b).

28.8.2 — Human Review, Dissent, and Appeal

For language tools, record model identity and version, access date, instructions, settings, source set, output, human verification, and material rejected text. Check invented sources, omitted adverse evidence, merged versions, lost branch distinctions, and dissent misrepresented as consensus (Feuerriegel et al., 2026).

Human review, dissent, and appeal keep responsibility traceable. People sign the decision, record reasons, disclose conflicts, preserve minority reports, and accept appeal; the presence of computational support never removes the human duty to answer for the update (Council for International Organizations of Medical Sciences, 2016; Tabassi, 2023).

Every substantial tool-assisted decision should leave a tool-use record. The record should state the tool, version if known, input class, output class, human reviewer, permitted use, and rejected use. If a language model helped draft a correction notice, the record should say that. If a static analyzer found stale version strings, the record should say that. If a simulation tool generated benchmark results, the record should say the container and seed policy.

Tool-use records prevent both mystification and scapegoating. The project should not pretend that a human alone did work a tool assisted. It should also not blame a tool for a human

decision to publish, approve, or ignore. Provenance lets future reviewers ask whether the tool was appropriate and whether human review was adequate.

Audit should include leakage risk. Tool inputs may include sensitive draft material, participant information, restricted records, or unpublished protocols. The default should be local, controlled, or redacted use unless governance approves otherwise. Convenience cannot override privacy, consent, or publication control. A tool that improves drafting but leaks restricted context is not acceptable.

Tool auditing is not anti-innovation. It is how the project can use powerful assistance without letting assistance become authority.

28.8.3 — Audit of Automation Bias

Audit of automation bias tests whether reviewers overdefer to technical outputs. The system looks for anchoring, confidence-label effects, interface-driven ranking, selective summarization, prompt framing, and quiet replacement of judgment by tool convenience (Tabassi, 2023; UNESCO, 2021b).

Permit abstention when the evidence, proposed wording, or authority route cannot support a decision.

Correct tool-assisted decisions through the same revision and propagation process as other decisions.

The sovereignty loophole appears when the project says no tool has authority, but then structures workflows so that humans merely ratify tool outputs. If a dashboard selects the status, a language model drafts the public wording, a model chooses the route, and reviewers approve by inertia, the tool has become de facto sovereign. The workflow must force human reasons independent of the output.

One protection is independent human justification. A reviewer should state why the decision is correct in ordinary language, naming the source, evidence, scope, and public consequence. The tool output may be attached, but it cannot be the reason. If the reviewer cannot explain the decision without the tool, the decision is not ready.

Adversarial tool review supplies another safeguard. A second reviewer or system can search for overclaim, stale versions, conflicting artifacts, excluded rivals, root-handoff triggers, and public-risk hazards that the first pass missed. The goal is complementary scrutiny, not a contest among tools, and no fluent pipeline should become invisible infrastructure (Council for International Organizations of Medical Sciences, 2016).

A third protection is abstention. If the evidence is too ambiguous, the tool should not be forced to classify. If the public wording cannot be made safe at the required compression level, the artifact should not be released. If the register cannot resolve whether a change is root-level or science-level, the route should hold. Sovereignty often enters through the refusal to abstain.

The final protection is revocability. Tool-assisted decisions must be correctable. If later review finds that an artificial-intelligence summary omitted a fatal caveat, the correction route should mark the affected decision and propagate changes. Tool assistance does not create a privileged shield against revision.

Closing the loophole means the project does more than say humans decide. It designs workflows so that humans actually carry the reason, the uncertainty, the correction, and the cost.

28.8.4 — Public Language of Responsibility

Every controlled scientific update should identify the accountable human role or roles, reviewed object, status, scope,

uncertainty, conflicts, and current version. Signatures record responsibility and relevant expertise, not prestige (Council for International Organizations of Medical Sciences, 2016; Tabassi, 2023).

When software or artificial intelligence assisted, record its bounded task and version and what humans verified or rejected;

authority remains with the reviewers (Tabassi, 2023; Autio et al., 2024; World Health Organization, 2025; Feuerriegel et al., 2026).

Correction notices should state the practical action for readers: use the new version, do not use the old artifact, treat the evidence as narrowed, await review, or recognize that a component has retired. Where a formula or model informed the change, state the local calculation it supported rather than allowing a local diagnostic to become a global adjudicator.

Applied domains may retain independent local value after a scientific correction, but they do not inherit stronger scientific warrant. Older copies remain historical and should point to the current status where feasible.

Chapter 29 — The Open, Staged Research Program

The next useful study is the one that resolves a consequential uncertainty. More precise measurements can improve a weak estimate, but additional observations under the same unidentified design may leave its central question untouched. This final chapter orders work by the decision it can change, with resource and burden limits included in that choice (Fenwick et al., 2020; Skivington et al., 2021).

Consider a hypothetical measurement design on the state graph used below. Under the proposed potential model, a tree supplies no cycle constraint on the common rate origin. Repeating its edges can reduce measurement noise without supplying that missing relation. A new path forming an independent cycle may instead add identifying information when its qualified exposure contrast is nonzero. The design change matters because it changes the available constraint.

That example also separates calibration from testing. One independent cycle can calibrate an origin within the model without checking that calibration against another constraint. Further independent paths can test the relation with the state representation fixed. Shared edges, uncertain exposures, and uncertain state membership remain in the error budget. The measurement agenda therefore asks both what an instrument can estimate and which new observation could show that its organizing assumptions fail.

History-sensitive prediction raises a parallel choice. A cumulative variable may improve on current mood or a short recent-event model simply by retaining more history. A rival given the same full history poses a harder question: whether the candidate offers a useful representation that transfers with fewer separately fitted relations. The roadmap preserves both comparisons and states what each could establish.

The first work need not require new recruitment. Existing-data reproductions can expose unavailable variables, retrospective preprocessing, or a forecast that uses information recorded after its starting time. Synthetic challenges can reveal when observation loss makes different mechanisms look alike. These tasks identify what a new study must measure before a more demanding setting is approached.

The proposed sequence is conditional on those findings. Ordinary-life calibration can continue when an unrelated terminal branch is blocked; a failed endpoint method should stop the inference that depends on it. Each stage therefore has a defined output and an expansion condition. Annual review asks whether the next study still resolves the intended uncertainty or whether a less burdensome design has become available.

The final responsibility is to make the result reusable. A proof needs explicit premises, a benchmark needs known generating conditions, and an empirical comparison needs identifiable targets and a reproducible analysis. Null and adverse findings must identify what remains learned. The program advances when another team can locate the decisive assumption, repeat the comparison, or design an observation that separates the remaining alternatives.

29.1 — Scientific Case and Dependency Map

Section 29.1 identifies results that directly change each component's status. Consequences propagate to dependent claims under Section 27.1's noncompensatory rule; a local update does not exempt those dependencies.

29.1.1 — The Received Hypothesis

Exact PTVC adds restrictions on the complete remainder and admissible endpoints. Compare those restrictions with precisely specified approximate models.

Near-term tests concern preterminal variance, endpoint windows, reserve-linked dynamics, and model-discriminating patterns (Manski, 2003; Gelman et al., 2020).

29.1.2 — Why the Program Is Scientifically Interesting

Investigate the relation among a signed experiential history, a state carrying its relevant information, and the contribution still to come. The endpoint restrictions in Chapter 7 and randomized-remainder restriction in Section 19.2 provide consequences independent of a favored mechanism. Section 24.6.4 connects the latter to measurement design through a finite-error bound.

Test whether one completion relation predicts future increments, response timing, action probabilities for action-mediated routes, and endpoint support. Reserve models must identify their resource variables. After recovery through the observation process, train without confirmatory endpoints and test transfer across people, conditions, horizons, and laboratories, assessing the complete model's complexity.

Account for shared participants, instruments, analyses, pathways, and cultural ancestry in joint evidence. Test cross-modal or cross-species relations through prespecified transformations and new observations.

29.1.3 — The Most Testable Near-Term Claims

Prioritize representation, sign, context of use, recovery, and preterminal dynamics in bounded, auditable settings (Manski, 2003; Gelman et al., 2020).

29.1.4 — Rivals and Decision Consequences

Use synthetic and held-out recovery comparisons with bridges, frailty, safe control, and observation shutdown. Rival dominance narrows or retires the affected Law-specific interpretation.

Use the generative rivals in Parts VI–VII through common observation and endpoint uncertainty, applying Section 25.2's comparison rules (Gelman et al., 2020).

Separate mechanism retirement, reference-class narrowing, and signature-specific rival support under Section 24.2 (Manski, 2003; Gelman et al., 2020).

29.1.5 — What Would Change the Case

Clarify the formal bridge, origin, endpoint, and reserve assumptions. A proved bridge, separation theorem, or formal amendment request directly changes the formal component; dependent claims follow the stated dependency rules.

Qualify the momentary-valence and interval-contribution estimators for disability, medication, cultural context, and missingness. Stable calibration or a demonstrated failure updates measurement status before terminal inference.

Use multisite and adversarial replication with public-correction rules. Support, narrowing, inconclusive evidence, and defeat attach to specified components rather than automatically to the whole program.

The case changes only when evidence attaches to a specified component. Representation evidence changes representation status. Endpoint evidence changes endpoint status. Model-

recovery evidence changes design readiness. Replication evidence changes transport confidence. No result should change more than its design can support.

A strong early result would not be a declaration that the Law is true. It might show that a physiological estimator has a stable context of use, that a horizon task separates scarcity from candidate scheduling pressure, or that a synthetic benchmark can distinguish reserve-coupled candidates from bridges under realistic missingness.

Qualification findings guide the timing of later terminal studies. Unresolved sign, instrument transfer, endpoint independence, or model recovery pauses the dependent analysis and identifies the work needed before it resumes.

A mixed result should produce a smaller map, not a confused headline. A mechanism might retire while the measurement program remains useful. A reference class might narrow while the formal problem remains open. A rival might dominate one signature while leaving another undecided (Manski, 2003; Gelman et al., 2020).

The change record should also preserve timing. A result that looked strong before a bridge failure may become historically important but no longer status-bearing. A result that survives a later amendment may need reanalysis rather than inheritance. Scientific memory should protect both the excitement of discovery and the humility of versioned correction.

Public wording must change whenever the component status changes. Words such as discovery, support, narrowing, inconclusive, unusable, retired, rival-dominant, and defeated should not float above the evidence. They should be tied to a versioned claim, a population, an instrument, and a comparison set.

The case changes most honorably when it becomes easier to state. A result that forces the program to say less, more exactly, may strengthen science even if it weakens the public story.

What changes the case is not accumulated enthusiasm but an evidence-bearing dependency transition. A frozen component passes or fails a protocol-fixed test, a rival becomes more or less adequate, or a reference class narrows; the case changes only by that amount.

29.1.6 — Next Measurement Priorities

Validate consciousness-state and stopping rules prospectively. Terminal claims pause if independent adjudication or a required timing assumption fails.

The next measurement priorities are a stable valence scale, an empirically anchored origin, longitudinal coverage, and a workable endpoint method. Reserve-based models add the task of identifying and calibrating their resource variables (Manski, 2003).

Extensions to children, nonhuman animals, artificial systems, additional cultural settings, sudden death, coma, dementia, and altered states each create specific bearer-classification and measurement questions. Targeted studies can establish the conditions for those extensions.

Figures, tables, equations, and pipelines organize the work and identify stopping conditions. Their conclusions depend on whether the assumptions survive testing.

Timelines map dependencies. Evidence may shorten a ten-year plan, measurement difficulties may slow it, ethics review may pause it, and a stronger rival may redirect it (Gelman et al., 2020).

Public summaries should pair proposed signatures with open questions about whole-life initial state, origin identification,

scalarization, endpoint independence, and cross-species transport. This makes the research agenda accessible beyond the technical chapters.

Applied domains offer varied settings for new questions. City experiments, government observations, religious practices, Workgame dynamics, and ordinary-life reflections can reveal patterns worth studying. Researchers can translate those observations into defined variables, competing predictions, and prospective tests.

An explicit map of open questions keeps the program testable and guides future science.

29.2 — Open Theorems and Formal Verification

Open formal obligations limit every empirical claim that depends on them. The program must identify the required theorem, its assumptions, a counterexample that would defeat the proposed bridge, and the reviewer and completion criteria. Origin, representation, endpoint–reserve synchronization, random endpoints, and stochastic reserve pose distinct problems. An unresolved proof cannot be replaced by successful simulation, while a formal amendment must disclose how the target and its consequences change.

29.2.1 — Origin and Representation

A frozen potential model makes origin calibration an explicit identification problem. Let y contain integrated readings on one provisional cardinal scale, τ their independently qualified exposures, and B the incidence matrix of the fixed state graph. A common rate-origin offset o gives $y = o\tau - B^{\mathsf{T}}\varphi + \varepsilon$. Choose a full-row-rank matrix C spanning the cycle space and write $g = Cy$ and $d = C\tau$. Eliminating the potential gives $g = od + C\varepsilon$. In the noiseless model, o is identified exactly when $d \neq 0$; a tree supplies no cycle constraint.

For centered errors with a fixed positive-definite cycle covariance V and known d, the generalized least-squares estimate is

$$\widehat{o} = \frac{d^{\mathsf{T}} V^{-1} g}{d^{\mathsf{T}} V^{-1} d}.$$

Its denominator measures information, so a nonzero but weak duration direction can still yield poor precision. Additional declared nuisance patterns must be removed before assessing rank and information. Noisy exposures, shared edges and uncertain state membership require joint inference. One fitted

independent cycle calibrates an origin without testing that calibration; further independent constraints provide overidentification and should include held-out paths.

The cross-sign unit assumption can also be tested within a specified two-channel model. Let P and N be independently defined operational magnitude channels associated with positive and negative experience; both may be active during a mixed state, and their interpretation remains to be qualified. Write the candidate edge contribution as $aP - bN - o\tau$ with $a, b > 0$. Division by a gives $CP = rCN + \kappa C\tau$, where $r = b/a$ and $\kappa = o/a$. The two parameters are identified only when $[CN, C\tau]$ has rank two; further independent cycles test the relation. Both exercises remain conditional on the proposed potential and observation maps. Agreement with independent sign and scale calibration would strengthen their phenomenal interpretation.

Determine the admissible scale: ordinal-only evidence cannot identify additive magnitudes, while context-limited interval structure limits cumulative and neutrality interpretations to those contexts.

Develop implementable invariance tests across language, disability, medication, development, and within-person calibration.

29.2.2 — Endpoint – Reserve Synchronization

Define reserve and consciousness-related endpoint adjudication independently of the ledger; reserve cannot serve as an algebraically disguised endpoint clock.

For a route requiring vanishing reserve near an independent endpoint, derive the relation, bound its timing mismatch, and predict rebounds, jumps, and external support.

Test synchronization against ventilation, sedation, analgesia, delirium, infection, organ failure, device support, and sudden

events that can move reserve proxies without terminating consciousness.

Freeze reserve and endpoint rules, then test timing during independently indicated support or natural recovery. Preserve proxy versions, observation conditions, and mismatch magnitude. Retrospective alignment or a clinical event alone does not establish the truth endpoint or authorize care changes.

Reserve-dependent terminal tests require their synchronization bridge. Mechanism-neutral residual bounds require it only if their derivation does; independently qualified preterminal dynamics can remain testable while synchronization is unresolved.

29.2.3 — Closure Modes and Random Endpoints

Classify which realizations imply exact equality, measured equivalence, probabilistic convergence, bounded error, or reserve-linked compression (Kallenberg, 2002).

For random endpoints, separate dynamics from stopping-induced conditioning, survival selection, truncation, and changed observation distributions.

Publish examples where closure modes agree and diverge, including measured equivalence without exact latent closure, concentration through selection, and a favorable ensemble average without pathwise closure.

Identify observable distinctions among the closure modes, or prove their equivalence under the permitted observation design.

A concrete classification separates a fixed deadline from an endpoint that waits for the state. In one declared clock, consider an absolutely continuous ledger and a noisy contribution rate,

$$dL_t = V_t\,dt, \qquad dV_t = u_t\,dt + \sigma\,dW_t, \qquad 0 \le t < T,$$

where T is a deterministic positive finite deadline, $\sigma > 0$, initial states are square integrable, W is Brownian in the full

information filtration and u includes all progressively measurable drift affecting V. Exact $L_{T-}=0$ almost surely is impossible if $E\int_0^T u_t^2 dt<\infty$. To see why, write the final h-window remainder as $L_{T-}=m+C+Z$, where m is known at $T-h$, $C=\int_{T-h}^{T}(T-s)u_s ds$ and Z has conditional variance $\sigma^2 h^3/3$ and conditional mean zero. Exact closure requires $C-E[C\mid \mathscr{F}_{T-h}]=-Z$. Conditional variance and Cauchy–Schwarz then imply

$$E\int_{T-h}^{T} u_s^2 ds \geq \sigma^2 \quad \text{for every } 0<h\leq T,$$

contradicting finite expected total quadratic effort as h decreases to zero. This effort is a mathematical control quantity, with no assumed conversion into metabolic energy. The restriction concerns the specified fixed-deadline innovation model; it does not impose those dynamics on every PTVC route.

Exact completion is nevertheless constructible with ordinary stochastic feedback. Start at $L_0=V_0=0$, put $\tau=T-t$, taper the diffusion to $\sigma_t=\sigma_0\tau/T$ and choose

$$u_t=-\frac{5(L_t+\tau V_t)}{\tau^2}.$$

For $M=L+\tau V$, the equation becomes $dM=-5M dt/\tau+(\sigma_0/T)\tau^2 dW$. Its solution gives $E[u_t^2]=(5\sigma_0^2/T^2)\tau[1-(\tau/T)^5]$, hence $E\int_0^T u_t^2 dt=25\sigma_0^2/14$. Finite absolute drift integral ensures a continuous endpoint extension; M tends to zero and τV tends to zero, so $L_{T-}=0$ almost surely. The feedback gains remain singular even though expected quadratic effort is finite. This is a mathematical realization with vanishing relevant innovations, not evidence that an organism implements it.

A random hitting endpoint supplies a different ordinary construction. With $L_0 = l > 0$, $V_0 = 0$ and constant drift $u = -a < 0$ under persistent $\sigma > 0$, $L_t = l - at^2/2 + \sigma \int_0^t W_s ds$ eventually becomes negative almost surely. Its first zero is finite and continuity gives exact pre-endpoint closure, with bounded drift control. In a complete-stream account, l would represent a preceding prefix. Identifying that hit with the stream endpoint remains an independent scientific obligation. Together the constructions direct measurement toward relevant innovation strength, correction dynamics and independently classified endpoint rules, rather than treating every convergent-looking trace as the same mechanism.

29.2.4 — Stochastic Reserve and Markov Closure

Acute events, treatment, fatigue, infection, sleep, and medication motivate stochastic, multidimensional, or switching reserve models, with measurement changes handled separately. These biological sources motivate extensions without validating a particular latent model (McEwen & Wingfield, 2003; Inouye et al., 2014).

The bracket terms and terminal conditions must be derived for the chosen stochastic reserve. The following test asks a stronger question: can the accumulated ledger be represented by a smooth function of the state?

A smooth state-potential realization must accommodate the process's actual noise. In one common clock, suppose $dS_s = b(S_s)ds + G(S_s)dW_s$ and $dL_s = v(S_s)ds$. For a C^2 potential Φ, Itô's formula and uniqueness of the semimartingale decomposition show that the pathwise identity $dL = -d\Phi$ requires

$$G^{\mathsf{T}}\nabla\Phi = 0, \qquad v + \mathscr{A}\Phi = 0,$$

along the supported process, $ds \times Pr$ almost everywhere, with the usual localization and integrability assumptions. The generator is $\mathscr{A}\Phi = b \cdot \nabla\Phi + 1/2tr(GG^{\mathsf{T}}\nabla^2\Phi)$. Noise can coexist with exact accounting when it acts in directions that leave the potential unchanged. A nonconstant smooth potential is impossible on a connected open region where these supported identities extend pointwise and the diffusion directions span the whole state space.

Instantaneous rank deficiency is not sufficient protection. For sufficiently smooth noise fields and potential, successive commutators of noise directions must also annihilate Φ; if those directions together span the region, Φ is constant there. Drift directions cannot be added to this test without an additional argument. This identifies a precise research target: a coordinate responsive to the signed contribution while protected against the relevant fluctuations. A coarse noisy assay does not establish the noise geometry of the complete biological state, and jump or hybrid models require their own calculation.

Determine which expanded state restores Markov structure and permits closure signatures to be distinguished through the measured variables.

Test minimal state requirements for reserve dimensions, affect, observation intensity, endpoint hazard, treatment, and confounders, including when added flexibility destroys discrimination.

Compare the finite-variation reference with labeled stochastic-reserve extensions; failure under common tests narrows that realization family.

29.2.5 — Formal Amendment Candidates

Use the versioning rules in Section 28.2.2 for historical results and reanalyses (Crossref, 2025). A cross-version comparison

distinguishes reproduced findings, newly derived findings and findings rendered incomparable by the amendment. Each reanalysis retains its own assumptions and status alongside the historical result.

Disclose any amendment's loss of simplicity, scope, or compatibility with earlier figures and results, identifying required reanalysis.

Route each candidate through the formal handoff and concordance procedure of Section 28.4, distinguishing a change to the received root claim from a change only to its empirical bridge.

Endpoint–reserve synchronization requires a non-tautological relation between the endpoint clocks and reserve process. Failure pauses reserve-dependent terminal inference while leaving separately qualified mechanism-neutral tests available. Random endpoints also require a convergence taxonomy that distinguishes stopping from conditioning.

Random-endpoint work must distinguish its convergence claims under stopping and conditioning. Without that distinction, endpoint-conditioned rivals remain undefeated.

Stochastic-reserve work must specify general product dynamics and identification limits. Failure can narrow the reserve family or leave it nondiscriminating.

Formal amendments require a versioned request identifying costs and new predictions. An earlier headline cannot silently retain authority after bridge failure.

Some future work may discover that the received formal statement needs repair. That possibility remains visible, but no repair enters the active scientific record before formal acceptance. A formal amendment candidate is a proposal under governance, not an invisible rewrite.

Amendment candidates might concern the representation of valence, the product realization, reserve dimensionality, endpoint clocks, scalarization, or whole-life initial state. Each candidate should state the old claim, the proposed new claim, the reason for change, and the predictions it improves or sacrifices.

A progressive amendment solves an independently identified problem and creates new risky predictions. A degenerating rescue move appears after failure only to preserve a favored conclusion. The program must keep those categories distinct before results arrive.

Amendment candidates should also include a public-cost statement. A repaired theory may be more accurate but less simple, less universal, or less connected to earlier figures. The audience deserves to know when an elegant public explanation has been sacrificed for scientific truth, because that sacrifice is one of the signs that correction is real. The cost statement should also identify which prior empirical results remain valid under the amendment and which require reanalysis.

The formal-law book maintains the root statement. The science book supplies evidence, proposed amendments, and implications for formal change review. A concordance between the two books can then show whether the scientific amendment changes the root claim, only its testable bridge, or both.

A formal amendment may someday make the program stronger or smaller. The final chapter leaves that route open because a theory that cannot be repaired honestly cannot be tested honestly. Any amendment must preserve the failed premise, counterexample, proposed repair, and new empirical obligations together, so formal change cannot masquerade as continuity. Fresh obligations should be prospectively testable,

ensuring that formal repair increases empirical exposure rather than only logical flexibility.

The open formal problems have distinct deliverables. The integral–product bridge needs a theorem, separation result, or formal amendment packet; without it, the candidate realization cannot speak for the received ledger. Origin and representation need stated scale and origin conditions with sensitivity tests, limiting ledger and candidate interval-contribution claims to contexts where those conditions survive.

Stochastic reserve requires general product dynamics and their identifiability limits. The resulting family may narrow or prove nondiscriminating. Formal amendments require versioned change requests with their costs and predictions, so that the old claim does not silently continue after its supporting bridge fails.

29.3 — Identification and Measurement Problems

Identification requires more than a coherent target and a fitted estimator. Sign, origin, rate, additivity, reserve, unity, endpoint classification, and observation each create a different connection between the intended object and the available record. A failure at one connection limits the claims that depend on it. The following priorities separate those problems so that calibration, protocol design, and missingness work can improve the particular inference they are capable of supporting (American Educational Research Association et al., 2014; Manski, 2003).

29.3.1 — Sign and Neutral Origin

Apply Section 10.2.2's distinction between sign and change when calibrating accumulation. Carry sign and neutral-origin uncertainty into the cumulative estimate (American Educational Research Association et al., 2014).

Cross-check sign in bounded horizon/scheduling tasks, affective reprocessing, and ordinary-life reports across behavior, physiology, and time.

Negative controls and artifact challenges are essential. A proposed negative control requires an independently justified absence of the relevant causal relation; device looseness, motion, sensor drift, speech impairment, and clinical routines must also be tested for recording artifacts and possible genuine effects on experience. A system that assigns a confident valence sign to every channel change has failed (Lipsitch et al., 2010).

Deliver within-person anchors, contextual neutrality bands, state-specific uncertainty, and justified sign-assignment rules.

29.3.2 — Momentary-Valence Instrument Use

Specify the candidate momentary-valence estimator's context of use by population, state, language, device, cadence, care regime,

endpoint relation, clinical exclusions, and intended inference (Goldsack et al., 2020).

Prespecify how channel decoupling, absent reports, medication changes, and quality failure affect uncertainty, abstention, and classification under Section 24.2 (Lakens, 2017; American Educational Research Association et al., 2014).

29.3.3 — Rate, Additivity, and Scalarization

Test rate representation, interval additivity, and scalarization within each proposed context of accumulation.

Sparse reports require explicit interpolation or valid interval bounds before accumulation; clock duration alone cannot determine an unobserved contribution.

Test noncompensable dimensions and compare vector representations before accepting a scalar cumulative target (American Educational Research Association et al., 2014; Little & Rubin, 2019).

Compare scalar, vector, ordinal, and interval models in bounded tasks, ordinary sampling, and safe perturbations, locating exact, approximate, and failed additivity.

29.3.4 — Remaining-Support Construct Validity

Validate reserve beyond frailty, severity, prognosis, remaining time, and quality, comparing scalar and vector versions using Section 20.1.3's dimensions (Lakens, 2017; American Educational Research Association et al., 2014).

Require reserve to predict the proposed innovation or relaxation dynamics beyond death, dropout, or care-intensity prediction.

Model analgesia, sedation, oxygen, dialysis, ventilation, nutrition, infection control, and palliative care effects on reserve proxies (Little & Rubin, 2019; Manski, 2003).

Compare reserve dimensions, shared-sensor effects, measurement error, and treatment sensitivity, declaring when confounding blocks distinctive interpretation.

Claim distinctive remaining-support attribution only for prespecified held-out predictions that separate it from ordinary resource, prognosis, or opportunity measures; otherwise retain the established construct or nondiscriminating parameterization.

29.3.5 — Unity and Endpoint Classification

Qualify confidence and abstention for sleep, anesthesia, delirium, coma, dementia, dissociation, sedation, sudden death, return, and low-signal states.

Use prospective windows, consciousness markers, clinical adjudication, and blinded review independently of ledger estimates and expected neutrality.

Test premature endings that erase return or impose terminal meaning on reversible states, and delayed endings that miss windows or create selection (Manski, 2003; Lakens, 2017).

Deliver versioned endpoint rules, open-channel windows, return handling, confidence classes, and independent adjudication, tested first on nonterminal or simulated transitions.

Before complete-stream residual claims, qualify abstention, alternative routing sets, and false-ending controls because classification sets both the unit and boundary.

29.3.6 — Observation and Missingness Models

The open program should model observation intensity, dropout, device adherence, care transitions, family refusal, medication changes, and site routines jointly with the outcome.

Use feasible harmless prompt-timing variation, redundant devices, or backup reports to identify observation bias while

minimizing burden (Lakens, 2017; American Educational Research Association et al., 2014).

When observation loss overwhelms the signature, stop that inference or move to a less vulnerable design.

Specify how the estimand handles relevant intercurrent events, and address sedation, fatigue, withdrawal, device failure, death, and selective nonresponse in estimation and sensitivity analyses.

Use bounded anchor tasks and within-person sensitivity bounds to test sign and neutral origin. A signed-ledger claim requires both direction and origin to survive those checks.

Define the candidate momentary-valence estimator's context of use before testing its transfer. Validate that estimator externally across states, sites, devices, and medication conditions. It remains local or abstains outside the context supported by evidence.

Test rate and additivity through the cadence and representation battery. Failure of interval structure narrows interval-contribution claims.

Compare vector and scalar reserve qualification. Confounded reserve requires narrowing or retiring the product realization.

Test endpoint classification with an independent consciousness-state protocol. Failure of independent adjudication or a required timing assumption pauses terminal inference.

Jointly model visits, dropout, device behavior, and care transitions. Keep unusable evidence visible rather than imputing it into closure.

29.4 — First Study Families and Simulation Gates

The first study families address different prerequisites rather than forming one undifferentiated test of the Law. Representation and sign, estimator qualification, unity and endpoint methods, generative benchmarks, and humane preterminal dynamics each require their own simulation and qualification criteria. Existing-data work can identify useful contrasts before new observation is justified. Decisions to expand, redesign, or stop depend on the information each study can add and the burdens required to obtain it.

29.4.1 — Stage Zero Existing-Data Challenge

Stage Zero is a bounded existing-data challenge that can begin before new recruitment. Use openESM to extend the inventory introduced in Chapter 17 (Siepe et al., 2026), but start with an auditable project. Blain and Rutledge's (2020) public learning-task data and code support reproduction and exploratory comparison of reward, expectation, and retained-history predictors. For a second task-based reproduction, Vanhasbroeck et al. (2021) identify raw and preprocessed data and R scripts at OSF 9g2zw; verify the released version and permitted use before starting. D'Ottone et al.'s (2026) project supplies event-linked affect summaries and analysis code; the release does not reconstruct the item-level sampling and exclusions. MIDUS daily diaries provide documented positive- and negative-affect items for short daily comparisons. None supplies a complete-stream ledger. Record access conditions, versions, variables, sampling times, exclusions, missing codes, and participant overlap before fitting.

A forecast must use information available at its declared starting time. Blain and Rutledge's published pipeline uses session-wide standardization and jointly fitted estimates to

initialize separate environment fits. These procedures serve its retrospective analysis; a prospective adaptation must estimate scaling and initial values from training or earlier observations only. In D'Ottone's release, N_Responses enters the models, but its construction is absent. The article describes a count over a two-week collection period. Exclude it from an eight-hour-origin forecast unless provenance establishes that it uses only pre-origin information; a completed post-event count would introduce future information. The event average itself becomes available only after its eight-hour window.

Keep prediction in the recorded MIDUS units. Before accumulation, qualify its affect-frequency categories, separate positive and negative scales, temporal weights, and missing-window bounds. Audit centering, differencing, detrending, filtering, interpolation, and rescaling with known constant, slowly changing, and cumulative signals. Carry missing periods into an explicitly justified uncertainty model.

The data-use plan is part of Stage Zero. MIDUS conditions prohibit uploading participant data to public large language models or external AI platforms lacking the required confidentiality and non-retention protections. Verify the current provider terms and institutional authorization before any AI-assisted processing (Inter-university Consortium for Political and Social Research, 2026). Public methods, permitted documentation, code without participant records, and synthetic fixtures can support open review; participant-level analysis remains in the environment authorized for those data. Public availability does not waive these conditions.

Keep the accumulation comparison in Sections 10.6.2 and 21.5.2 separate from Section 23.3.4's forecasting question: does specified history improve prediction of future recorded affect? Declare the forecast origin and horizon, then give every model

the same available observations and tuning resources. Realized future events may enter only a separately declared event-conditional comparison that supplies them to every model. Compare heterogeneous autoregression with event inputs, expectation or reference adaptation with flexible signed decay, a slowly varying baseline with observation noise, and the proposed cumulative controller. Include an otherwise matched controller without the candidate's extra restriction. Freeze the primary score, practical-gain threshold, calibration checks, and missingness sensitivity. Average predictive performance within each person before averaging across people; account for participant and family dependence. Fit preprocessing and select lag windows within training data, respecting the actual sampling clock. The generators and recovery tests in Section 29.4.5 establish which differences the proposed observation process can reveal.

A practical sequence is MIDUS 2 for development, MIDUS Refresher 1 for a frozen new-cohort comparison, and MIDUS 3 for a separately reported follow-up comparison among retained participants (Ryff & Almeida, 2017, 2020, 2022; ICPSR 26841, 37083, and 38529). The Refresher is a new cohort within the same research program and instrument family; it is not independent-laboratory replication. Harmonize items and missing codes, keep linked persons and families together, and document earlier access to outcomes. Public availability alone does not disqualify a protected secondary test, but a random split made after examining the target outcomes does not create one. If those outcomes shaped the model, use a new cohort or a separately controlled untouched sample for confirmation.

Stage Zero delivers a versioned reproduction, rival-recovery benchmark, qualified observed-range statistics, and frozen forecast comparison, each with owner, resource ceiling, and

independent reproduction task. Predictive advantage supports those dynamics; complete-stream interpretation needs its independent measurement and boundary bridge. Recovery failure identifies the next design change.

29.4.2 — Study One: Representation and Sign

Study One tests sign outside terminal settings through bounded variation in horizon, scheduling pressure, effort, relief, anticipation, and recovery, with specified report, behavioral, and physiological models (Nosek et al., 2018; Cook et al., 2006).

Use benign choices varying opportunity and competing priorities to compare urgency, scarcity, avoidance, and the candidate's specified prediction. Do not induce suffering, distress, deprivation, or sleep disruption for this evidence.

A companion sleep/reprocessing study can test next-day adjustment beyond report memory, mood inertia, or arousal.

Study One evaluates sign, interval structure, rate behavior, and scalar versus vector adequacy (Cranmer et al., 2020; Center for Open Science, 2025).

Predefine prohibited procedures, including trauma induction, deprivation, humiliation, and high-stakes loss. Test whether humane, bounded variation is sufficient to identify the required representation.

If the contextual model cannot identify sign or compatible scales, or reconcile its channels, stop ledger claims and continue representation research.

29.4.3 — Study Two: Estimator Qualification

Study Two tests whether the physiological estimator can be qualified for its declared context of use after representation has sufficient local stability (Goldsack et al., 2020).

Recruit an ordinary-life cohort with active reports, bounded event notes, sleep/activity context, selected passive signals, health and medication covariates, and recalibration, spanning age, disability, illness, language, and access variation.

Test transport through grief, fatigue, work stress, pain flare, caregiving, and medication changes before clinical inference.

Match Section 17.7's validation to the claim: held-out people, sites, or device versions for transport, and chronological splits for within-person forecasting, controlling dependence and preprocessing leakage (Center for Open Science, 2025; Nosek et al., 2018).

Qualification yields a context-limited estimator and model card; failure returns to channels, person-specific baselines, vectors, or abstention design.

Justify longitudinal measurement cadence by participant attention, privacy, and decision value.

29.4.4 — Study Three: Unity and Endpoint Validation

Study Three tests whether consciousness-state and endpoint methods can be qualified and evaluated independently of the ledger.

Test sleep, anesthesia recovery, sedation, delirium, and independently evaluated consciousness-disorder methods before developing terminal endpoint inference.

Study Three addresses unity and endpoint classification, including any prospectively declared independence branch. It estimates rates of false-positive and false-negative endpoint classifications and evaluates ambiguous windows, return handling, and confidence classes, while keeping endpoint adjudicators blinded to ledger estimates and outcome expectations.

Before terminal extension, obtain review from neurologists, palliative clinicians, anesthesiologists, disability advocates, statisticians, ethicists, and family advisors (Council for International Organizations of Medical Sciences, 2016).

Test whether consent forms, summaries, and diagrams are understood as state-classification tools, and record misunderstandings as artifact-safety findings.

Success yields a prospective endpoint protocol with uncertainty bands and abstention categories. Failure redirects work to consciousness-state methods and blocks the affected terminal claims.

29.4.5 — Study Four: Generative Benchmarks

Study Four is computational: apply Chapter 25's benchmark protocol before the confirmatory model-attribution tests in this portfolio (Center for Open Science, 2025; Nosek et al., 2018).

Use Chapter 25's full library, including declared cross-species or future-regime extensions, developmental and learning models, and defective allocation or observation pipelines. Each confirmatory comparison specifies prediction, unit, ordinary null, information, and recovery, calibration, and false-positive criteria under realistic observation.

For the delayed-treatment comparison, simulate both the primary strict-pre-endpoint and any separately declared endpoint-inclusive branch. Include streams that end before, at, or after scheduled treatment onset, with terminal contributions assigned only under the appropriate branch convention. Distinguish the complete-remainder restriction from stronger pre-treatment localization. Null simulations should include realistic finite-sample imbalance, ordinary later compensation with no early arm difference, and development or survival selection. Any selection or reweighting used to enforce closure

must be checked for induced dependence between assignment and the pre-assignment prefix.

The implementation challenge includes route-cue leakage, route-to-contingency coding errors, one-shot versus learned choice, no-choice and observation-loss outcomes, allocation leakage, parental and batch structure, density and position effects, handling and scoring artifacts, censoring, and multiplicity. Permuted and placebo assignments must preserve the actual randomization scheme; placebo labels are generated independently of outcome values after ascertainment and do not determine treatment. Simulate cancellation, crossover, non-delivery, and other regime changes. When a prediction depends on actual regime realization, test its prespecified causal or partial-identification strategy rather than treating later adherence as randomized.

The terminal fork in Section 19.2 supplies exact checks for the generative challenge. Include its general and phenomenally silent cases, the hidden-uniform-mapping local null under arbitrary position preferences, and the history-preserving mixture whose calibration at one continuing-route fraction predicts another without refitting. Include the globally conditioned hard-exclusion population and its exactly matching finite-odds curve. A pipeline given only their aggregate routing curves must report Model-ambiguous for that pair, rather than claim to have recovered the mechanism. Where additional history or assignment observations are supplied, evaluate the joint laws each model actually specifies.

The target-identification fixture should include both zero and nonzero targets with unequal signed capacities. Test whether the analysis recovers the target across two capacity levels when the origin and capacity family are fixed independently, and recognizes nonidentification when those restrictions are

removed. Compare first choices, repeated choices with learning, and continuation-only analogues separately; the latter require their own derived prediction. Candidate and rival implementations must expose their local or global information assumptions. The terminal fork remains a theoretical and computational comparison, not an empirical lethal protocol.

Archive arrivals, completed choices, recorded outcomes, and mapping assignments separately. Include a no-choice local null and an outcome-dependent capture fixture that raises the recorded continuing fraction without changing the underlying choice law. Evaluate the selector on qualified denominators rather than on whichever subset produces the largest excess. For the two-capacity target fixture, vary minimum-location error and capacity spacing, then check the stated target-error envelope. Those tests qualify the observation and identification procedures; they are not additional empirical terminal experiments.

Advance the tested comparison after its recovery and error criteria pass, retaining unresolved and open-set results. Recovery qualifies simulated separability; independent scientific and welfare review governs testing its preregistered empirical prediction (Cook et al., 2006; Cranmer et al., 2020).

29.4.6 — Study Five: Humane Preterminal Dynamics

Attempt Study Five only after the representation, estimator, endpoint-method, and recovery prerequisites needed by its declared comparison have passed (Nosek et al., 2018; Cook et al., 2006).

Begin with feasibility monitoring of comfort, burden on participants, families, and staff, adherence, medication transitions, and clinical events before end-of-mind inference.

Within qualified use, test preterminal variance, reserve-linked change, endpoint-window concentration, and model-specific residuals against rivals.

Keep family/staff burden monitoring internal. Rising research load, documentation demand, or privacy strain must permit cadence reduction or stopping.

A supportive result motivates independent replication in the next justified reference class. A valid adverse result changes the tested component's status; a failed prerequisite identifies the formal, measurement, or protocol repair needed before escalation.

29.4.7 — Portfolio Decision Points

Recovery failure blocks the affected confirmatory comparison. Independently justified calibration, descriptive, or feasibility studies can retain their own prespecified questions and rules.

A horizon and queue task with safe affective anchors tests sign, scale, horizon sensitivity, queue pressure, and representation limits. It asks how those dynamics relate to independently defined stream endpoints.

A sleep and affective-reprocessing study tests changes in next-day affect under benign observation. It asks which sleep-linked effects generalize across populations, nights, and channels.

An ordinary-life calibration cohort tests estimator context of use, transfer, missingness, and cadence. It asks how far calibration carries across states, populations, and instruments.

A consciousness and endpoint-method study tests prospective endpoint independence and unity classification. It asks how qualified methods can be linked to longitudinal affect measurements.

A generative benchmark challenge tests recovery, open-set adequacy, and design separability. It asks how recoverable models perform on prospectively collected observations.

A humane preterminal-dynamics pilot compares signatures with rivals while monitoring clinical burden. It asks which signatures replicate across qualified sites and care settings.

29.5 — Staged Roadmap and Value of Information

The roadmap orders studies by the evidence each one needs from earlier work. Its proposed dates help coordinate activity, while the dependencies determine when a particular study can produce an interpretable answer. Later funding responds to earlier results: measurement and comparison models may need further work, and participant support or a different study may offer greater value. Value of information asks how much a result could improve the next decision relative to its cost, delay and burden (Claxton, 1999).

29.5.1 — Stage One: Formal and Measurement Lock

In the proposed first two years, prioritize formal obligations, representation, estimator qualification, fixtures, and code/data governance (Claxton, 1999; Fenwick et al., 2020). Expansion requires formal and measurement specifications to be fixed and qualified.

Begin the critic-designed synthetic challenge during this stage.

A useful Stage One handoff is a shared worked case. The formal packet identifies the target object, domain, endpoint architecture and units. The measurement packet specifies the observations and calibration assumptions used to approach that target. A synthetic fixture then supplies a known generating case that both packets describe. Running the proposed estimator on that case tests whether the definitions, data dictionary and implementation agree before differences in real observations need to be interpreted.

At year two, assess readiness for bounded calibration and state-method work against the formal and measurement prerequisites.

29.5.2 — Stage Two: State and Endpoint Validation

In proposed years two to four, qualify independent state and endpoint classification through sleep, anesthesia, sedation, delirium, and recovery before terminal use (Fenwick et al., 2020; Skivington et al., 2021). Preserve the purpose separation among operational records, research data, derived features and model-training data described in Section 26.5.6.

Compare classifications from clinicians, consciousness researchers, families, disability experts, and statisticians; identify reliable agreement, retained dissent, and abstention regions.

Unqualified endpoint classification blocks terminal inference; independently qualified ordinary-life and nonterminal work may continue.

29.5.3 — Stage Three: First Protocol and Replication

Years three through five overlap with endpoint validation because ordinary-life and nonterminal protocols can proceed under their own prerequisites (Skivington et al., 2021; Center for Open Science, 2025).

Use Section 26.1's Registered-Report route where appropriate for the first confirmatory protocol.

Prioritize direct replication of representation and estimator qualification before a clinical pilot builds on the initial result.

Independent implementation can separate three tasks: reconstruct the preprocessing, fit the frozen candidate and rival models, and run the adversarial checks. Each team records its inputs and outputs so a disagreement can be traced to a transformation, a fitting choice or a diagnostic. The replication then tests a reproducible procedure rather than depending on the original analysts' unrecorded judgments.

An effect confined to one analyst, device, site, or subgroup remains local or developmental until the intended replication claim is supported. Replication quality and model separability govern further expansion.

29.5.4 — Stage Four: Mechanism and Transport

Proposed years five to seven test transport across populations, sites, care, cultures, diagnoses, devices, and states after the relevant early gates pass (Center for Open Science, 2025; Claxton, 1999).

Enter multisite cohorts after harmonization, retaining site heterogeneity as evidence about scope (Fenwick et al., 2020; Skivington et al., 2021).

Evaluate subgroup errors, reversed site effects, changed clinical meaning, and their consequences alongside average transport performance.

Add a site, population, or measurement system when it can resolve a component decision; model the added heterogeneity.

29.5.5 — Stage Five: Terminal and Comparative Work

Years seven through ten may support more demanding terminal and comparative work after earlier prerequisites pass (Claxton, 1999; Fenwick et al., 2020).

Terminal-dynamics studies should be comfort-first, opt-in, low-burden, and clinically subordinate. They require prospectively specified windows, independent endpoint adjudication, medication and care-transition models, family burden monitoring, and stop authority outside the hypothesis team.

Monitor statistical evidence, futility, staff fatigue, participant distress, family privacy, and endpoint uncertainty for early stopping.

Year ten closes with a synthesis of supported components, narrowed reference classes, retired mechanisms, and adverse findings.

29.5.6 — Resources and Value of Information

The resource plan counts participant time, privacy exposure, family attention, clinician and analyst labor, computation, ethics work, communication risk, and the opportunity cost of studying this hypothesis instead of other urgent problems (Fenwick et al., 2020; Skivington et al., 2021).

Reserve funds for negative results, adversarial replication, stewardship, security, accessibility, care, training, and correction.

Include critics, clinicians, stewards, and participant advocates in resource review.

Consider a choice between enlarging an observational cohort and running a focused calibration study. Suppose the next decision depends on whether two channels share a stable origin in the intended setting. More observations from the existing design could reduce sampling error while leaving that question unchanged. The calibration study could instead determine which estimator or study design should be used. Its value comes from the decision it can change, including the possibility that the proposed design should be revised.

Compare the possible outcomes of each study with the actions they would support. A study is especially useful when plausible outcomes lead to meaningfully different choices and when those choices matter to the program. Account for the time needed to obtain the answer, the chance that the study remains inconclusive and the alternative work forgone while it runs. Repeating this comparison after each major result lets the roadmap adapt through explicit research decisions.

Throughout the program, maintain failure records, public corrections, and audits of staff and participant burden. Stop an affected activity when its burden exceeds its decision value.

29.6 — Release, Pause, Redesign, and Stop Conditions

Readiness depends on the specific inference a study proposes. Formal, measurement, endpoint, comparison, and ethical prerequisites therefore define a sequence of decisions rather than a single permission to expand. A failed prerequisite pauses the work that depends on it, while independently qualified work can continue. The same dependency record determines when methods, protocols, and public materials are ready for release.

29.6.1 — Readiness by Dependency

Formal readiness requires a stable bridge, notation, origin, endpoint convention, and amendment route for the object being tested. Measurement readiness requires qualified sign, rate, calibration, context of use, and abstention behavior. If either remains unresolved, identify the object or estimator that cannot yet support the proposed inference, and narrow or retire it where the evidence warrants.

Endpoint readiness requires noncircular adjudication and any timing assumption invoked by the chosen branch. Comparison readiness requires signatures that separate the specified rivals under realistic observations. Failure identifies the consciousness-state method or comparison design needing repair. Ethics and burden remain independent requirements: unacceptable care or privacy risk, coercion, or disproportionate participant, family, or staff burden blocks the proposed work.

29.6.2 — Pause and Reopening Decisions

Record the failed prerequisite, the dependencies it blocks, the valid knowledge retained, and the evidence required for reopening. Reopening means resolving that recorded condition, not merely changing the study’s description. Repeated rival victory or bridge failure across independent attempts invokes the

declared move to a better rival, a narrower component question, or retirement analysis.

The dependency record also distinguishes a local pause from program-wide closure. An endpoint-classification failure can leave an independently qualified ordinary-life measurement study available. A comparison that cannot discriminate its rivals can still reveal which observations a future design would need. These retained questions should be stated explicitly, so that a pause directs subsequent work toward the unresolved dependency.

29.6.3 — Reproducible Release Conditions

Release synthetic generators and fixtures after the formal object and rival envelope are versioned. Release registration, protocol, and analysis-plan templates after the relevant measurement and endpoint contexts of use are declared. Freeze the software dependencies and data dictionary for each code-container release so that replication does not depend on silent software changes.

Model cards and candidate momentary-valence summaries follow external validation and specified abstention rules. Clinical protocol materials follow independent ethics and burden review. Public diagrams and captions follow hostile-quotation and crop audits. Failure logs and correction pages begin with the first failed gate or retired component, preserving the negative knowledge needed to prevent repeated error.

29.7 — What Every Verdict Teaches

Each portfolio result should identify what has been learned and which decision it changes. The canonical verdicts in Section 24.2 separate support, unresolved comparisons, method failure, rival victory, and target defeat. Applying those categories to the research portfolio makes the next task specific while preserving findings that remain useful outside the claim under review.

29.7.1 — Support and Remaining Questions

Replicated component support means that a specified dependency survived independent tests. Name the component, context, and limits. Partial or reference-class support gives a smaller but more exact claim: state the population, instrument, endpoint, and transport question that delimit it. Subsequent work can test that transport question without treating the initial result as universal support.

Inconclusive evidence means that a valid analysis did not decide the question. Identify the bottleneck and the next design capable of changing the decision. For example, more observations of the same nondiscriminating channel will not resolve a comparison whose rival predictions coincide there; the next design needs an observation on which they differ, or an explicit account of why that distinction remains inaccessible.

29.7.2 — Method Failure and Rival Knowledge

Instrument or endpoint failure means that the apparatus is not valid for the intended inference. Repair or retire the affected method before advancing the dependent claim’s status. Mechanism retirement or rival victory is a different result: an alternative or ordinary mechanism explains the signature better. Credit that account and reduce the corresponding Law-specific

scope while retaining independently valid measurement and design findings.

Target defeat requires contradiction of the exact target under its declared prerequisites. Empirical-program retirement can instead follow exhaustion of the program’s distinct, testable content. Record which basis applies, correct the affected public artifacts, and preserve useful scientific knowledge. The resulting record should let a successor study begin from what the evidence established rather than repeat the same unresolved comparison.

29.8 — Final Posture: From Architecture to Testing

The next useful result need not establish the terminal hypothesis. It may identify transferable history structure, distinguish previously equivalent models, expose a failed measurement bridge, or show that a proposed observation is not worth its burden. Independent investigators can pursue those bounded questions through humane designs and shared comparison rules. The resulting evidence should determine the next test and the scope of the claim, while valid methods and negative findings retain value beyond any one outcome.

29.8.1 — Independent Investigation

One team can hold Section 25.5.4's primary-observation law fixed while varying witness calibration; another can add latent explanations under the same variables and budget. Share generators and scores, reporting which calibrated observations distinguish the candidate worlds.

29.8.2 — The Scientific Value of Humane Design

Within Chapter 3's protections, select observations by the uncertainty, coverage, and identification they resolve.

Use Section 24.6.4's incomplete-record intervals to test whether valid coverage bounds remain informative before demanding clinical observation. Keep uninformative records visible rather than imputing neutrality.

A continuing-outcome analogue of Section 23.4.4 requires a new derivation of its observable prediction when the terminal contingency is removed. Specify that prediction before empirical testing.

The scientific ambition of the project and its humane design reinforce one another. Low-burden measurement, participant choice, clinically appropriate observation, careful privacy

practice, and independent stop authority improve both the ethics and the interpretability of the resulting data.

This is especially important for research involving illness, altered consciousness, or terminal settings. The goal is to discover the strongest evidence that can be obtained within ordinary duties of care. That constraint encourages better questions, better instruments, less intrusive observation, more realistic missing-data models, and study designs capable of functioning in the environments they claim to describe.

Many foundational studies can occur far from terminal settings. Measurement qualification, simulation, formal verification, model recovery, longitudinal affect studies, sleep research, natural experiments, and prospective prediction can establish much of the infrastructure first. By the time more demanding clinical questions are approached, the program can arrive with better instruments, stronger models, and clearer reasons for collecting each observation.

29.8.3 — The Next Step Is to Test It

The next scientific handoffs follow.

A useful next step separates formal work from measurement. The formal-law route receives bridge proofs and proposed representation amendments; the Law itself must not be silently rewritten in scientific prose. Measurement research must qualify and calibrate momentary valence estimators while specifying limits and abstention rules. An estimator does not grant authority to score persons or read inner life directly.

Consciousness and endpoint researchers test unity classifications and endpoint windows, including checks against false endings. They cannot select endpoints because predicted neutrality would follow. Independent rival-model teams develop synthetic challenges and open-set tests that give alternatives a

genuine chance. Clinical and palliative partners determine humane feasibility and retain authority to stop a study whenever care is at risk.

Public communicators maintain versioned claim status, correction records, and withdrawal decisions while leaving uncertainty visible. The wider series contributes questions and contexts; it cannot confer proof or authorize transition.

Choose one consequential relation, specify the observation that could distinguish it, and leave a result another team can reproduce. The next test should make clear what was learned, which assumption remains open, and what evidence would change the conclusion.

Further progress now depends on implementation: formal proofs or counterexamples, measurement qualification, synthetic recovery, preregistered comparisons, and independent replication. Each task should begin only when its prerequisite gates are met and should use prospectively specified stop rules when additional work is no longer expected to change the scientific decision enough to justify its cost or burden. The next phase is therefore not a general call to validate the Law, but a sequence of component tests that can support, narrow, redirect, or retire parts of the program.

Together, those contributions can turn the architecture in this volume into evidence. The immediate priorities are to test the representations, qualify the instruments, strengthen endpoint methods, build capable rivals, run recovery simulations, and replicate results. Future status should follow those outcomes: components may gain support; PTVC may narrow in scope or remain unresolved; the strong empirical program may be retired. The scale of the question should not decide the verdict.

29.8.4 — Research Value Beyond a Single Test

The program has value beyond any single experiment when its component questions produce reusable methods, discriminating predictions, or clearer failure boundaries. A research group can contribute by solving one difficult component well, including by showing that a proposed measurement, route, or distinction does not survive testing. The broader scientific value should be judged by what remains usable after results narrow or reject the founding hypothesis.

Glossary

A

Abstention: A decision to withhold a positive or negative verdict while specified identification, measurement, endpoint, model, or data-quality requirements remain unresolved.

Action-mediated closure: A candidate route in which the consequences of available actions produce the proposed closure relation.

Active inference: A family of models in which an organism can reduce expected mismatch by updating expectations and by acting on or selecting its environment so preferred or viable states become more likely. See also: Predictive processing; Reinforcement-learning reward signal; Viability kernel.

Ad hoc patch: A revision introduced after an adverse result chiefly to protect a preferred conclusion, rather than to test an independently supported successor. See also: Correction and withdrawal; Endpoint independence.

Adjudicability: The capacity to evaluate a declared target under the specified formal, measurement, endpoint, and inference conditions.

Adverse case: A gate-valid case for which the branch-matched inferential object supports the registered Target-adverse conclusion. Exclusion of zero by a deterministic identified set guaranteed to contain the truth under maintained assumptions yields a deductive counterexample; exclusion by a confidence or posterior object yields only its coverage- or posterior-qualified conclusion. Favorable population averages cannot cancel the case.

Affective valence: See Felt valence.

AI sovereignty: Decision authority assigned to an artificial intelligence system over a person’s inner life, standing, or

eligibility. The project prohibits this authority. See also: Human-worth score prohibition; Testimony-like output; Latent cumulative object.

Almost-sure attainability: The existence of at least one admissible policy that reaches a declared target with probability one from a stated information state. It differs from closure under a fixed law, success under every policy, and limit-sure attainability, where probabilities can approach one without any single attaining policy. See also: Limit-sure attainability; Probability-one closure; Reachability.

Almost-sure closure: See Probability-one closure.

Alternative endpoint branches: Prospectively declared architectures including state coupling, first hitting, absorption, endpoint atoms, singular timing or support, continued experience beyond a protocol boundary, and endpoint-inclusive value. Each specifies its endpoint, representation, bridge, and testing assumptions. See also: Endpoint atom; First-hitting and absorption routes; State-coupled timing.

Applied lens and formal Law: The applied lens asks bounded present questions about burden, options, care, and repair. The formal Law concerns an exact truth-level condition over one complete qualifying stream. See also: Care before proof and interpretation; Law of Fairness; Target-route distinction.

Arousal: The activation, wakefulness, or energetic mobilization dimension of experience. High or low arousal can accompany welcome, burdensome, mixed, or neutral states. See also: Felt valence; Observation channel and observation model.

Atom at zero: Positive probability mass located exactly at zero in a terminal-residual distribution. Its presence is distinct from concentration within a finite band: a small atom can coexist with a widely dispersed remainder, and strong near-zero

concentration can occur without an atom. Probability-one closure requires all mass at zero under the declared law.

Attainability: The existence of at least one admissible route or policy capable of reaching the target from a specified state. Attainability alone does not show that the route will be selected, that it works under partial observation, or that every admissible path closes.

Attrition mechanism: The process by which participants leave or become unavailable during follow-up, changing who remains observable. See also: End of observation; Informative missingness; Longitudinal research trace.

Auxiliary assumption: An assumption needed to connect a theoretical claim to an observation, estimator, or test. A failed prediction may reflect the target hypothesis, an auxiliary assumption, or both, so confirmatory work must state these assumptions in advance.

B

Balance: Book 1’s accessible term for exact intrastream terminal valence closure.

Baseline drift: A time-dependent change in a reference level. In a measurement model, changes in scale use, neutral-point calibration, comparison standards, or reporting can create apparent convergence without latent phenomenal change. In an adaptation model, set-point drift instead belongs to the modeled dynamics. See also: Measurement model; Response shift; Hedonic adaptation.

Bearer: The truth-level subject or unit to which qualifying felt experiences belong and within which the proposed accumulation is evaluated. Its identity is established through

the declared stream and routing criteria. See also: Maximal qualifying conscious stream; Ownership; Stream identity.

Behavioral nonresponse: Absence of an expected command-following or other observable response under a specified assessment. The assessment must account for barriers to expression. See also: Felt availability; Low-unity state; Unknown category.

Biological death: The end of biological life. It is not identical by definition to the truth-level endpoint of a conscious stream. The Law formulation used in this series separately includes a biological-death admissibility condition under which biological death may not occur before the declared closure condition has been satisfied for the relevant stream. A qualifying conscious stream may permanently end after satisfying closure while biological processes continue without consciousness, including under artificial life support; such continued nonconscious biological persistence does not extend the completed stream. This additional condition does not identify the truth endpoint, a route to closure, or any particular biological or behavioral marker. See also: Branch; Endpoint-inclusive branch; Truth-level endpoint.

Bipolar branch: A representation branch that keeps positive and negative phenomenal components separate so they may coexist rather than being forced into one momentary net value. It is not interchangeable with the primary additive branch, and results transfer between them only through a justified bridge on a common identified domain. See also: Generalized Terminal Neutrality; Primary additive branch; Scalarization.

Blinded bearer adjudication: A procedure for deciding among bearer or routing models while withholding, as far as possible, the downstream result each classification would help produce.

Fixed criteria, independent reviewers, visible disagreement, and outcome separation reduce post hoc routing in favor of the theory. See also: Bearer; Routing; Unknown category.

Blinding: A design safeguard that withholds selected information from participants, investigators, coders, analysts, or adjudicators when that information could bias conduct or interpretation. The form of blinding must fit the study and its ethical limits.

Boundary hypothesis: A hypothesis about a limiting, terminal, or otherwise sharply delimited property of a process rather than a claim about its ordinary moment-to-moment path. PTVC is a candidate boundary hypothesis because it concerns a stream’s terminal signed residual. Its representation, measurements, and predicted consequences determine how the proposal can be tested. See also: Falsifiability; Law of Fairness; Lawhood.

Branch: A prospectively declared version of a target, model, or test whose bearer, representation, origin, domain, endpoint architecture, target geometry, modality, and reference class are fixed. Transfer between branches uses an explicitly justified bridge.

Bridge: A stated relation, with assumptions, scope, uncertainty, and failure conditions, that licenses movement between distinct objects or levels, such as measured to latent or protocol endpoint to truth-level endpoint.

C

Calibration: The process of relating a measurement system to an independently justified reference so that its origin, scale, uncertainty, and interpretation are known.

Candidate mechanism: See Candidate route.

Candidate realization: A formula or model proposed as one route-level realization linking received PTVC objects to observable consequences. It carries the formula-status label CR. See also: Candidate route; Formula-status code; Route state.

Candidate route: A proposed mechanism, policy structure, dynamical construction, or mathematical path by which the terminal target might be satisfied. A route can receive support or fail independently of the target hypothesis.

Candidate sentience escalation: A protocol-governed increase in protection when evidence about a candidate system's organization, activity, or welfare exceeds the assumptions of its approved review. See also: Precaution without personhood; Testimony-like output; Theory-derived indicator.

Candidate signature: A prospectively specified observable pattern predicted by a candidate route and evaluated against competing explanations. See also: Negative control; Nonunique explanation; Predictive fit versus explanation.

Care before proof and interpretation: The priority of responding to present need with appropriate care before pursuing evidence or interpreting an experience. See also: Ethical floor; No harm for proof; Outcome independence.

Causal locus: The place in a system where a causal influence is proposed to operate, such as experience generation, action selection, routing, endpoint timing, observation, or reporting. Distinguishing loci helps prevent one process from being mistaken for another.

Channel qualification: The staged process by which a report, sensor, behavior, contextual record, or other observation channel earns a defined research role through technical verification, analytical validation, construct validation, burden

review, and specified performance thresholds. See also: Calibration; Multichannel estimate; Observation channel and observation model.

Claim ceiling: The strongest inference supported by a channel, design, model, comparison, or public artifact under its identification, uncertainty, scope, and safeguards. See also: Detachable-excerpt ceiling; Public-artifact limit.

Closure: Satisfaction of the terminal condition specified by the declared target.

Coercive measurement: Inner-life measurement for which authority or dependence makes refusal practically costly, including loss of care or opportunity. See also: Consent, refusal, and withdrawal; Human-worth score prohibition; Participation without surveillance.

Comfort-first observation: Observation whose timing and continuation are governed by routine care, clinical judgment, comfort, and the participant's preferences. See also: Care before proof and interpretation; End of observation; Protocol endpoint.

Common-action requirement: Under partial observation, an action or policy must be selected from the information actually available. For probability-one closure, its probability of completion must equal one for almost every compatible latent state under the declared conditional state law. A robust guarantee over every observationally indistinguishable state is stronger and requires its own feasibility and selection premises. State-specific actions that cannot actually be selected do not satisfy either requirement.

Comparative consciousness evidence: Species-appropriate evidence that combines behavior, learning, neurobiology, ecology, affective organization, and motivational tradeoffs

into a converging case about possible experience. Human resemblance may inform a bridge but is neither necessary nor sufficient, and welfare protection need not wait for a complete theory. See also: Precaution without personhood; Theory-derived indicator; Unknown category.

ComparatorStatus: See Scientific verdict axes.

Complete strict-pre-endpoint felt history: Every qualifying phenomenal contribution owned by a stream before its independently defined truth-level endpoint, under the declared representation, ownership, and domain rules. See also: Endpoint-inclusive branch; Maximal qualifying conscious stream; Truth-level endpoint.

Complete strict-pre-endpoint phenomenal domain: See Complete strict-pre-endpoint felt history.

Completion potential: A declared state function tied to future reachability under a candidate route. Examples include a committor probability, viability value, harmonic function, or bounded reachability functional. Each carries its own boundary conditions, recursion, and interpretation.

Component support: A verdict that a declared component survived its applicable tests within a specified reference class; replicated support requires independent confirmation under compatible rules. See also: Claim ceiling; Replication; Target-route distinction.

Conditional support: The support of a declared probability law after conditioning on specified information, with the conditioning rule and topology fixed. A point belongs to that support when each of its open neighborhoods has positive conditional probability. This differs from a high-probability region or a guarantee covering every admissible continuation.

Conscious stream: A temporally connected sequence or field of qualifying conscious experience treated as one unit under declared identity and routing criteria.

Consent, refusal, and withdrawal: An ongoing permission relationship that permits informed choice, safe refusal, and withdrawal with understandable data consequences. See also: Data minimization and deletion; Measurement without ownership; Participation without surveillance.

Constraint does not imply design: A distinction between structural relations that limit possible histories and hypotheses about how those relations originated. Each origin hypothesis requires its own explanatory model and evidence. See also: Constraint without a judge; Spiritual resonance is not scientific support.

Constraint without a judge: An impersonal structural or boundary relation governing admissible complete histories without specifying a judging agent. See also: Boundary hypothesis; Constraint does not imply design; No moral desert, debt, or payment.

Construct-estimand distinction: A construct is the experience-related concept a study intends to represent. An estimand specifies the exact population, time, endpoint, treatment, and analysis quantity it seeks to learn. A cohort mean, individual probability, and path-level endpoint therefore pose different questions. See also: Estimator; Observation channel and observation model; Reference class.

Construct validity: The extent to which the proposed interpretation and use of a measure are supported as representing the intended construct.

Correction and withdrawal: The process of correcting or retiring affected research objects and propagating the change

while preserving the superseded scientific record. See also: Ad hoc patch; Public-artifact limit.

Counterfactual reachability: Whether a target could be reached under one or more alternative admissible actions or policies from a specified state. Counterfactual reachability does not establish what actually occurred or what a partially informed agent could select.

D

Data as trace: A report, waveform, image, note, gesture, archive, or model output is a context-bound record produced through an observation process. Its interpretation uses the relevant measurement model and provenance. See also: Latent cumulative object; Observation channel and observation model; Provenance.

Data-generating process: The full process producing observations, including the latent dynamics and the mechanisms of measurement, selection, and reporting.

Data minimization and deletion: Collecting and retaining the least revealing data needed for a declared purpose, with accurate limits on deletion across records and derived outputs. See also: Consent, refusal, and withdrawal; Provenance; Record-regime separation and no-dossier rule.

Decision rule: A prespecified mapping from evidence to an outcome such as support, adverse result, narrowing, abstention, or route retirement. Decision rules should be fixed before outcome inspection whenever confirmatory status is claimed.

Defeat condition: A prospectively specified result that counts against a target, route, measurement claim, or auxiliary assumption under stated validity and inference rules.

Detachable-excerpt ceiling: The evidential scope preserved when a fragment is encountered apart from its source. See also: Claim ceiling; Public-artifact limit.

Dimensionality of valence: The question whether phenomenal valence can be represented on one signed dimension or requires multiple dimensions. The primary additive PTVC branch uses a signed representation, but that representation remains an empirical and conceptual commitment to be tested.

Direct identification: Identification obtained from assumptions and observations that determine the target without relying on a route merely because it fits. Direct identification is distinct from selecting one convenient model among several observationally equivalent alternatives.

Dream-as-felt-experience window: An interval of conscious experience during sleep that may belong to the felt history of a stream even when waking recall omits it. Studies relate that interval to sleep stage, reports, and the adopted continuity model. See also: Complete strict-pre-endpoint felt history; Dream-report chain; Reversibility and return rule.

Dream-report chain: The sequence connecting experience during sleep, awakening, the memory trace available afterward, and the report given to another person. Each transition can select, reorder, or reshape what enters the record. See also: Data as trace; Informative missingness; Response shift.

E

Empirical band: See Finite-band compatibility and measured equivalence.

End of observation: The last time usable data are obtained because a channel fails, monitoring stops, communication is

lost, privacy is requested, or another practical event occurs. It is an observation boundary that may precede the protocol endpoint and truth-level endpoint and creates missingness rather than terminal truth. See also: Protocol endpoint; Truth-level endpoint; Observation window.

Endpoint: The boundary at which a qualifying conscious stream ceases for purposes of the truth-level target. The endpoint must be defined by the stream model and cannot simply be replaced by biological death, loss of contact, device shutdown, or the last available report.

Endpoint atom: Positive probability mass assigned by a declared endpoint-time or marked-endpoint law to a particular time or point. It specifies endpoint timing or marking, distinct from probability mass at a terminal residual of zero. See also: Atom at zero; First-hitting and absorption routes; State-coupled timing.

Endpoint bridge: The independently validated relation connecting an observed or protocol endpoint, possibly an interval, to the truth-level endpoint required by the claim while carrying error, return, revision, confounding, and competing-event uncertainty. See also: Protocol endpoint; Reversibility and return rule; Truth-level endpoint.

Endpoint conditioning: Selecting, observing, conditioning, or reweighting trajectories according to where they end. This can change the apparent earlier drift or dispersion and supplies an explicit statistical model for comparison. See also: Candidate signature; Endpoint independence; Random-endpoint obstruction.

Endpoint coupling: A candidate route in which endpoint timing or endpoint conditions depend on the evolving stream state.

Tests compare the predicted relationship with alternative accounts of endpoint timing and pre-endpoint dynamics.

Endpoint-inclusive branch: A prospectively declared branch that assigns a specified phenomenal contribution exactly at the endpoint. Its temporal domain includes a point excluded from the primary strict-pre-endpoint branch, so comparison between the branches requires the corresponding domain and representation bridge. See also: Complete strict-pre-endpoint felt history; Exact terminal closure; Truth-level endpoint.

Endpoint independence: Two noninterchangeable conditions must be distinguished. Prospective endpoint independence fixes the endpoint branch, evidence source, selection rule, and adjudication procedure without using a ledger estimate or terminal residual to choose the endpoint. Probabilistic path-endpoint independence states, under a specified reference law, that endpoint time is independent of the sigma-field generated by the ledger extension on the declared interval; it is a premise of the independent-diffuse-endpoint theorem. The first condition does not imply the second. See also: Ad hoc patch; Protocol endpoint; Random-endpoint obstruction.

Endpoint marker: An observable event or signal used to estimate or bound a truth-level endpoint. A marker can be informative without being identical to the endpoint itself.

Endpoint uncertainty and ambiguity: Unresolved alternatives concerning endpoint identity, timing, or interval under the declared evidence and adjudication rules. See also: Endpoint bridge; No adequate estimate; Truth-level endpoint.

Equivalence band: A prespecified finite interval around a reference value within which effects are treated as practically equivalent for a stated purpose. An equivalence result around

zero supports only the declared finite-band claim; it does not establish exact zero.

Error budget: An explicit accounting of uncertainty from sampling, instruments, calibration, sign, origin, timing, missingness, model choice, transport, and construct identification across the evidence chain. Separating these sources identifies which additional data or methodological changes could improve the result. See also: Calibration; Noise floor; Partial identification.

Estimand: The precisely defined quantity an analysis seeks to learn, including its population, unit, time domain, conditioning, and scale. The estimand is not the same as the estimator used to calculate it.

Estimator: A rule or procedure for using observed data to estimate an estimand. An estimator may be biased, uncertain, inconsistent, or nonidentifying even when its numerical output is close to zero.

Ethical burden too high: A typed outcome for a question whose proposed design imposes disproportionate burden or risk and therefore requires redesign or nonperformance. See also: No harm for proof; No adequate estimate; Safe perturbation.

Ethical floor: The research safeguards that apply regardless of the hypothesis's scientific status. See also: Care before proof and interpretation; No harm for proof; Outcome independence.

Every-continuation closure: Every continuation in a prospectively declared nonempty admissible continuation class satisfies closure. The claim says nothing about paths outside that class and is not implied by probability-one closure or support containment. If the class is empty or undefined, closure is unavailable rather than vacuously established. See

also: Probability-one closure; Reachability; Support containment.

Evidence ladder: A map of evidential dependencies and branch-specific implications, rather than a single ranking of evidential strength.

Exact closure: A truth-level terminal signed residual of exactly zero for the qualifying stream and declared branch.

Exact terminal closure: See Exact closure.

Exact zero: See Exact closure.

Experienced exposure and unity gate: Experienced exposure is the declared measure of how qualifying phenomenal experience contributes across a stream and may differ from calendar time. The unity gate states when felt contribution is treated as active and owned by the stream. See also: Complete strict-pre-endpoint felt history; Maximal qualifying conscious stream; Ownership.

F

Failed upstream gate: An unmet prerequisite, such as representation, origin, bearer, endpoint, observation, measured-to-latent bridge, or transport, that determines which downstream conclusions remain available. See also: Measured-to-latent bridge; No adequate estimate; Scientific verdict axes.

Falsifiability: A scientific claim's specification of observations or evidence objects that could count against it under prospectively declared validity requirements and decision rules. See also: Boundary hypothesis; Gate-valid counterexample; Strong-form defeat.

Falsification: A result that conflicts with a prespecified claim under the study's stated assumptions and validity gates.

Falsification must identify which target, route, or auxiliary claim is affected rather than overgeneralizing from one failed component.

Felt availability: The modest claim that something is occurring for a conscious stream even when the experience cannot be spoken, deliberately used, remembered later, or organized into a clear self-story. It separates possible experience from an observer's access to evidence about it. See also: Behavioral nonresponse; Low-unity state; Observation channel and observation model.

Felt valence: The evaluative character of qualifying conscious experience as lived from within. Under a declared scalar signed representation it is positive, negative, or neutral relative to an independently justified origin; a bipolar branch preserves co-occurring positive and negative components. See also: Arousal; Bipolar branch; Neutral origin; Primary additive branch.

Finite-band compatibility and measured equivalence: Latent finite-band compatibility requires the complete nonempty latent compatibility object to lie wholly inside a prospectively calibrated latent acceptance set. Measured equivalence requires the complete nonempty measured compatibility set to lie wholly inside the appropriate bridge-defined measured region. Partial overlap or merely including zero is unresolved. See also: Exact terminal closure; Measured-to-latent bridge; Terminal residual compatibility object.

Finite signed phenomenal measure: A finite signed measure representing positive and negative phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain. Finiteness permits well-defined positive and negative Jordan masses and a terminal signed residual.

Finite soft-guidance ceiling: Under the registered finite power-weighting route, any retained positive probability mass on actions with completion probability below one prevents probability-one completion. This is a probability-mass result; topological-support containment and set-universal closure require separate premises.

Finiteness: On the primary finite branch, each complete positive and negative phenomenal magnitude is finite. This is a per-stream condition; different streams may have different finite totals. See also: Positive and negative phenomenal magnitudes; Primary additive branch; Gross experience.

Firewall: A boundary on the transfer of authority, evidence, or interpretation between domains.

First-hitting and absorption routes: A first-hitting route uses the first entry into a prospectively declared set. An absorption route additionally keeps the process within that set afterward. Relating either event to the truth-level endpoint uses a separately specified endpoint bridge. See also: Endpoint atom; Endpoint bridge; State-coupled timing.

Fission-and-fusion routing rule: Under the book's primary formal convention, genuine fission ends the parent stream and begins each qualifying child stream, while genuine fusion ends the prior streams and begins a new one. Earlier phenomenal balances are neither duplicated among descendants nor pooled into the fused stream. See also: Bearer; Routing; Stream identity.

Fitness, feeling, and moral worth: The distinction among reproductive propagation, the valenced quality of a bearer's experience, and the ethical standing owed to that being. Each is evaluated through its appropriate biological, experiential, or

ethical framework. See also: Comparative consciousness evidence; Outcome independence; Viability kernel.

Formula-status code: The labels attached to formulas in the technical synopsis: R = received definition; CR = candidate realization; OE = operational estimator; DC = design criterion; and OC = open conjecture. They classify a formula's role, not its empirical support.

G

Gate-valid counterexample: An adverse case that passes the declared bearer, representation, origin, domain, endpoint, observation, bridge, and inference gates for the tested branch. Deductive exclusion requires a deterministic truth-containing object guaranteed under maintained assumptions to exclude the target. Confidence or posterior objects support only their registered coverage- or posterior-qualified adverse conclusions, not literal universal refutation. See also: Falsifiability; Set-universal closure; Strong-form defeat.

Gate-valid stream: A qualifying stream that satisfies the prespecified inclusion, endpoint, measurement, data-quality, and model-validity gates required for the declared test. Only gate-valid cases may carry the evidential status assigned by that test.

Generalized Terminal Neutrality: A broader family of terminal-zero conditions defined by a declared functional, measurable output space, and distinguished zero when scalar addition would lose relevant structure. It is a separate branch and is not PTVC unless a justified bridge connects its zero condition to the additive positive-negative decomposition. See also: Bipolar branch; Primary additive branch; Scalarization.

Gross experience: On a declared additive scalar representation, the complete positive-plus-negative phenomenal variation before cancellation. Its value can differ across streams with the same terminal residual. See also: Outcome independence; Phenomenal magnitude; Terminal residual.

H

Hedonic adaptation: The changing emotional impact of a gain, loss, or continuing condition as novelty, attention, expectations, comparison, routine, and context change over time. Adjustment varies across events, people, domains, measures, and times. See also: Opponent process; Response shift.

Held-out transport: Testing whether a prospectively specified model or result travels to data, settings, populations, channels, periods, or endpoint regimes reserved from model development and tuning. See also: Measurement invariance; Model recovery; Reference class.

Hidden burden and hidden relief: The observation that visible advantage may coexist with distress and visible hardship with joy, competence, love, meaning, or peace. It motivates direct attention to experience alongside external conditions. See also: Outcome independence; Measurement without ownership; Per-stream claim.

Hidden-state model: A model in which the process of interest contains latent states that are inferred from observable signals. Such a model must distinguish state dynamics, observation error, routing, selection, and missingness.

Homeostasis and allostasis: Homeostasis regulates variables within workable ranges, while allostasis anticipates changing demands by adjusting operating levels, mobilizing resources,

and redistributing physiological cost. These processes provide measurable architectures for studying regulation, recovery, and resource use. See also: Mean reversion; Opponent process; Viability kernel.

Human-worth score prohibition: The prohibition on using research estimates or participation records to rank a person's worth, standing, or eligibility. See also: AI sovereignty; Coercive measurement; Record-regime separation and no-dossier rule.

I

Identification: The ability of the available data and stated assumptions to distinguish the target quantity or model from alternatives. More data cannot resolve nonidentification when competing explanations remain observationally equivalent under the design.

Identified set: The set of every target value compatible with the observed data, admissible measurement models, and declared assumptions. It is not a confidence interval: partial identification and sampling uncertainty are separate layers and may both need reporting. See also: Partial identification; Terminal residual compatibility object.

Inconclusive evidence: Evidence from a valid analysis that does not resolve the declared estimand within the prespecified uncertainty and sensitivity bounds. It is neither support nor Unusable evidence.

Independent neutral-origin calibration: A procedure that identifies the valence zero point using evidence and rules independent of the completed record's terminal residual.

Inference: Reasoning from observations and stated assumptions to a conclusion about a target quantity or process.

InferenceStatus: See Scientific verdict axes.

Informative missingness: Missingness related to state, treatment, access, or another factor relevant to the inference, so recorded observations can differ systematically from unobserved ones. See also: Attrition mechanism; Missingness; Observation channel and observation model.

Institutional administration limit: The boundary denying institutions authority to administer a latent cumulative estimate as a fact about a person's worth or eligibility. See also: Human-worth score prohibition; Latent cumulative object; Record-regime separation and no-dossier rule.

Interoceptive prediction: The anticipation and interpretation of bodily signals such as heartbeat, breathing, pain, fatigue, hunger, nausea, temperature, and arousal through expectations, context, uncertainty, attention, and action. See also: Active inference; Predictive processing; Prediction error and valence.

Interval contribution estimate: An operational estimate $\widehat{\Delta L}_{i,[a,b]}$ of accumulated measured contribution over a declared interval, constructed from a versioned momentary estimator and exposure/time rule with propagated uncertainty. See also: Momentary valence estimator; Phenomenal ledger.

Intrastream: Evaluated within one conscious stream rather than by pooling contributions across streams. See also: Nontransferability; Per-stream claim; Routing.

J

Jordan decomposition: The representation of a finite signed measure as the difference between mutually singular positive and negative measures. Their total masses provide the positive

and negative phenomenal quantities compared by the primary additive PTVC target.

Jordan masses: The total masses of the positive and negative measures in a finite signed measure's Jordan decomposition. Within a qualifying stream, these are the total positive phenomenal mass and the magnitude of the negative phenomenal mass. Their difference gives the net signed total on the declared domain.

L

Last conscious experience: The final qualifying experience before a truth-level stream endpoint, when such a final experience is well-defined. PTVC does not require this last experience itself to feel neutral and does not equate it with the last observed report.

Latent cumulative object: The unobserved formal representation of signed qualifying felt experience accumulated within one stream across its declared domain. Reports and instruments constrain it through a measurement model. See also: Phenomenal ledger; Measured-to-latent bridge; Terminal residual.

Latent valence: The unobserved phenomenal valence attributed to experience itself and investigated through models linking reports, behavior, physiology, and neural signals.

Law of Fairness: The series' public name for the PTVC hypothesis and the wider inquiry built around it.

Lawhood: The aspiration that a scientific claim describes a general structural condition within its declared domain. See also: Boundary hypothesis; Falsifiability; Claim ceiling.

Ledger: See Phenomenal ledger.

Limit-sure attainability: The property that policies can make the probability of closure arbitrarily close to one, even though no single admissible policy need achieve probability-one closure. Within the same fixed environment and nonempty admissible policy class, it is weaker than almost-sure attainability and does not establish closure under an arbitrary fixed law or every admissible continuation.

Local return and terminal closure: The separation between movement toward a familiar range during an episode or observed interval and a proposed exact cumulative condition at the independently defined end of one complete qualifying stream. See also: Hedonic adaptation; Mean reversion; Pathwise closure.

Longitudinal research trace: A repeated sampled record assembled under specified instruments, observation schedules, contexts, and permissions. See also: Attrition mechanism; Consent, refusal, and withdrawal; Response shift.

Low-unity state: A state in which ordinary access, responsiveness, integration, or evidence of unified experience is reduced or uncertain, as in sleep, anesthesia, coma, delirium, or severe brain injury. It may be a pause, altered-state branch, or unresolved interval rather than the end of a stream. See also: Behavioral nonresponse; Reversibility and return rule; Truth-level endpoint.

M

Maximal qualifying conscious stream: A qualifying stream extended as far as the adopted routing and identity criteria permit, with the complete resulting domain retained for analysis.

Mean reversion: A tendency for a process to move back toward a typical or equilibrium level over time. Apparent return in noisy observations may also reflect regression to the mean, sampling variation, treatment, or changed context; identifying a restoring mechanism requires additional evidence. See also: Homeostasis and allostasis; Local return and terminal closure; Nonunique explanation.

Meaning as optional: Freedom to adopt, revise, decline, or live without an encompassing interpretation of experience. See also: Care before proof and interpretation; Spiritual resonance is not scientific support.

Measured-to-latent bridge: The independently justified relation that allows a result on a measured estimand to constrain the declared latent phenomenal target, with scope, uncertainty, and failure conditions carried forward. It is separate from the observation model, endpoint bridge, estimator, and route; without it, evidence remains on the measured scale. See also: Endpoint bridge; Estimator; Latent cumulative object.

Measurement invariance: The tested stability of a measurement relation across the people, groups, languages, devices, sites, times, or contexts for which a comparison is proposed. Its results identify supported comparisons and opportunities for further calibration. See also: Calibration; Held-out transport; Response shift.

Measurement model: A formal account of how latent valence or related states generate reports, behavior, physiology, neural signals, or other observations. It includes scale, origin, error, drift, and possible response-process distortions.

Measurement reactivity: A change in the target or its observation process caused by measurement itself. See also:

Informative missingness; Observation channel and observation model; Consent, refusal, and withdrawal.

Measurement without ownership: The principle that describing or modeling experience creates research responsibilities without conferring authority over the person. See also: Observation channel and observation model; Consent, refusal, and withdrawal; Human-worth score prohibition.

Mechanism retirement: See Route retirement.

Metrology: The discipline of measurement, including units, calibration, traceability, uncertainty, and comparability.

Missingness: The process by which observations are absent, censored, truncated, or unavailable. Missingness near endpoints may be informative, and treating it as random without support can bias terminal conclusions.

Modal reachability ladder: A family of nonequivalent, scope-dependent claims, including existence of a closing continuation, positive-probability completion, fixed-law probability-one closure, limit-sure attainability, almost-sure attainability, support containment, and closure of every admissible continuation. These claims do not form one unconditional total order. Within the same fixed environment and nonempty admissible policy class, almost-sure attainability implies limit-sure attainability; a support-containment claim additionally requires a declared topology. Every implication must preserve the carrier, completion event, policy scope, and relevant assumptions. See also: Almost-sure attainability; Every-continuation closure; Support containment.

Model ambiguity: A situation in which two or more adequate declared models remain practically indistinguishable under

common comparison rules. See also: No adequate estimate; Rival model and equal-budget comparison; Scientific verdict axes.

Model-ambiguous: See Scientific verdict axes.

Model governance: Accountable rules for choosing a model's target, data, loss function, constraints, interventions, permitted uses, review process, correction route, and verdict. See also: Decision rule; Human-worth score prohibition; Scientific verdict axes.

Model recovery: A simulation-based check of whether an analysis can distinguish candidate model families when their generating identities are known. Parameter recovery separately tests estimation of parameters under a specified generator. Both are evaluated under realistic observation conditions; success in one does not establish success in the other.

Momentary valence estimator: A versioned, context-bound operational estimator $\widehat{v_i}(t)$ of a valence-rate-like contribution from declared observation channels under a measurement model. See also: Interval contribution estimate; Measurement model.

Motivation: The internal or external factors associated with action selection. A policy describes a mapping from information or state to actions; it should not be equated with the person's consciously experienced motive.

Multichannel estimate: A versioned, purpose-bounded estimate formed through an explicit model relating channels such as report, behavior, physiology, neural data, and context to a declared construct and estimand. See also: Error budget; Observation channel and observation model; Provenance.

N

NAE: See No adequate estimate.

Narrowing: A transparent restriction of the reference class or domain of an otherwise unchanged claim after a limitation or adverse result. Record the reduced scope and preserve the original verdict. Changing the target, representation, endpoint meaning, or exactness requirement may instead create a non-equivalent successor, which requires its own identifier, assumptions, and decision rule. Neither action confirms the unchanged original. See also: Reference-class narrowed; Verdict.

Near-zero band: A finite interval around zero used to describe limited measurement resolution, uncertainty, or practical closeness.

Negative control: A condition, timing, procedure, exposure, or outcome chosen because an independently justified causal model excludes the relation under test while preserving relevant sources of bias. Matched non-horizon tasks, sham procedures, impossible timing, and alternate outcomes qualify only when that exclusion is defensible. Unexpected associations can then expose confounding, demand, leakage, selection, or analytic artifacts. See also: Candidate signature; Nonunique explanation; Safe perturbation.

Neutral origin: The identified zero point separating positive from negative phenomenal valence in the adopted representation. Its location must be justified independently rather than chosen to make terminal records balance.

No adequate estimate: An inference status for an undefined target, or one that the available data and maintained assumptions neither identify nor validly bound. Valid but unresolved evidence and failed protocol evidence receive

different statuses. See also: Abstention; Decision rule; Model ambiguity.

No harm for proof: The prohibition on creating, prolonging, or withholding relief from suffering for evidence or promotion of the hypothesis. See also: Care before proof and interpretation; Ethical floor; Stop authority.

No moral desert, debt, or payment: The distinction between a conditional terminal-balance requirement and moral entitlement, deserved suffering, or an obligation to pay for experience. See also: Constraint without a judge; Outcome independence.

No-neutrality-by-fiat: The rule that origin, baseline, sign, scalarization, endpoint, smoothing, missingness treatment, weights, and exclusions may not be chosen after outcome inspection to manufacture apparent neutrality. Deviations require visible exploratory or sensitivity status.

No personal endpoint testing: The prohibition on dangerous personal attempts to test the Law, including self-harm, deprivation, or refusal of needed treatment. See also: Care before proof and interpretation; No harm for proof; Truth-level endpoint.

Noise floor: The smallest distinction an observation system can reliably resolve under declared conditions. Rounding, censoring, preprocessing, and random or systematic noise help determine that resolution. See also: Calibration; Error budget; Finite-band compatibility and measured equivalence.

Nonidentification: Compatibility of distinct target values or models with the same available observations and maintained assumptions.

Nontransferability: The rule that valence belonging to one conscious stream cannot be credited to, debited from, or used to balance another stream.

Nonunique explanation: Several mechanisms generate the same observed trajectory, endpoint concentration, rebound, report, compression, or return pattern. Additional observations, interventions, or model comparisons can investigate the differences among them. See also: Candidate signature; Mean reversion; Model ambiguity.

Null model: A specified alternative under which the distinctive PTVC target or route is absent. A useful null model makes concrete predictions and is tested alongside serious rivals rather than serving as a vague absence of effect.

Null outcome: A result that does not support the tested effect or distinction under the declared analysis. A null outcome may be informative, inconclusive, or adverse depending on power, identification, and the prespecified decision rule.

O

Objective functions versus experience: The distinction between a modeler’s rule for ranking selected outcomes and the lived experience the model seeks to investigate. A measurement relation states which aspects of experience the chosen objective represents. See also: Model governance; Reinforcement-learning reward signal; Viability kernel.

Observation: A recorded report, behavior, physiological measure, neural signal, event, or other data item interpreted through the relevant observation model.

Observation channel and observation model: An observation channel is recorded evidence such as report, behavior, physiology, neural activity, or context. The observation model

states how a latent construct and nuisance conditions can generate or alter that output, including sampling, timing, error, and channel failure. Neither the channel nor its model is the experience. See also: Informative missingness; Measured-to-latent bridge.

Observation model: See Observation channel and observation model.

Observation process: The mechanism by which latent experience and system states become recorded data, including measurement error, reporting, censoring, sampling, and missingness. It must be modeled separately from the target process.

Observation shutdown: Closure of one or more observation channels because of technical, clinical, consent, or practical conditions, creating missingness in the record. See also: End of observation; Informative missingness; Observation process.

Observation window: The finite span during which a study records data. Its relation to the complete stream is specified through the observed interval, any missing prefix or remainder, and the applicable measurement and endpoint bridges. See also: Complete strict-pre-endpoint felt history; End of observation; Protocol endpoint.

Open-set inadequate: See Scientific verdict axes.

Opponent process: A temporal counterprocess in which an initial state recruits a later afterstate whose strength and timing may change with repetition, helping explain relief, recovery, tolerance, withdrawal, fatigue, and rebound. See also: Hedonic adaptation; Local return and terminal closure.

Origin problem: The inability to interpret exact zero unless the neutral origin is independently identified. A constant shift in a

valence rate changes the accumulated ledger in proportion to included exposure, so post hoc centering can manufacture closure. See also: Independent neutral-origin calibration; Neutral origin.

Outcome independence: Care, rights, dignity, privacy, repair, ordinary causation, and independently supported knowledge retain their own grounds across changes in PTVC's scientific status. See also: Care before proof and interpretation; Ethical floor; Strong-form defeat.

Ownership: The prospectively specified rule assigning each qualifying phenomenal event or contribution to its bearer without omission, double counting, or cross-stream transfer.

Ownership answers whose experience a contribution is, while routing governs continuity through time and transitions; data ownership is a different legal and institutional question. See also: Bearer; Nontransferability; Routing.

P

Palliative sedation: Clinically governed sedation used proportionately to relieve otherwise refractory suffering, with its indication, depth, duration, and context recorded separately from research classifications. See also: Comfort-first observation; End of observation; Protocol endpoint.

Partial identification: A condition in which the evidence restricts a target to a set or interval while leaving more than one value possible under the maintained assumptions.

Partial observation: A decision or inference setting in which the relevant latent state is not fully observed. Closure claims in such settings must respect common-action and selection requirements rather than assuming state-specific knowledge.

Participation without surveillance: Voluntary research participation with bounded purposes, data minimization, and practical control over what is observed and retained. See also: Consent, refusal, and withdrawal; Data minimization and deletion; Record-regime separation and no-dossier rule.

Past-fixity: The principle that later contributions do not alter what earlier experience was, even when they change the cumulative balance. See also: Outcome independence; Phenomenal ledger; Reserve coupling.

Pathwise closure: Closure evaluated on complete individual trajectories. The stated quantifier distinguishes one realized path, probability-one closure, support containment, and closure of every admissible path. See also: Complete strict-pre-endpoint felt history; Probability-one closure; Set-universal closure.

Per-stream claim: A claim evaluated for each separately routed maximal qualifying conscious stream as its own unit rather than for a population, household, country, species, or group average. See also: Intrastream; Maximal qualifying conscious stream; Population mean and expected neutrality.

Per-stream terminal valence closure (PTVC): The hypothesis of exact terminal valence closure for each qualifying stream in the declared class. Section 5.3 gives the complete formal statement and assumptions.

Phenomenal ledger: On the primary additive branch, the latent cumulative signed phenomenal-value process through time. See also: Latent cumulative object; No moral desert, debt, or payment; Terminal residual.

Phenomenal magnitude: The represented amount of positive or negative felt experience assigned to a stream under a declared valence representation. Positive and negative gross

magnitudes are evaluated separately from their signed net difference. See also: Felt valence; Gross experience; Positive and negative phenomenal magnitudes.

Phenomenal measure: A measure assigned to conscious experience itself for purposes of representing valence across a stream. It is conceptually distinct from clock time, event count, memory strength, neural activity, or report frequency.

Phenomenal time: A measure or ordering of experienced duration within a conscious stream. It need not coincide uniformly with external clock time and should not be inferred solely from the number of observations.

Phenomenal valence: See Felt valence.

Policy: A rule mapping available information or state to actions, possibly probabilistically.

Policy mixture: A probabilistic combination of actions or component policies. Average cancellation across components does not establish exact closure on each realized path.

Policy-mixture exactness: Exact closure with probability one under a randomized policy requires conditional closure probability one for policy-almost-every action, under the declared disintegration. For a discrete policy, every action assigned positive probability must therefore satisfy that conditional guarantee. Extending it to every point of a continuous policy's topological support requires additional assumptions. Closure only in expectation is weaker. See also: Policy-support condition.

Policy-support condition: A condition requiring the support of a selected policy to contain only actions or continuations compatible with the declared exact closure claim. Assigning

very small but positive probability to a nonclosing option does not satisfy exact support restriction.

Population mean and expected neutrality: Expected neutrality means that the terminal residual has expectation zero under a named probability law and information set, when that expectation exists. A sample or cohort mean estimates a population quantity under further assumptions. Either can arise through cancellation while every individual stream remains nonzero. See also: Per-stream claim; Probability-one closure; Set-universal closure.

Positive and negative phenomenal magnitudes: The nonnegative total masses of the positive and negative parts of the declared signed phenomenal measure. See also: Finiteness; Gross experience; Terminal residual.

Positive-probability completion: Under a prospectively frozen continuation regime and its declared induced law, the completion event has probability strictly greater than zero, optionally conditional on a stated prefix and admissible action or policy. This weak reachability claim does not imply likelihood, probability one, an attaining policy, support containment, or every-continuation closure. See also: Modal reachability ladder; Probability-one closure; Reachability.

Positive rescaling: Multiplication of a valence representation by a positive constant. Exact zero and sign are preserved under positive rescaling, whereas shifting the origin or applying arbitrary nonlinear transformations can change the target.

Precaution without personhood: Proportionate, revisable welfare protection when relevant risks remain plausible but consciousness or legal personhood is unresolved. See also: Candidate sentience escalation; Comparative consciousness evidence; Unknown category.

Prediction error and valence: The distinction between mismatch under a predictive model and the positive, negative, or neutral quality of felt experience. Surprise and valence can interact without being the same quantity: expected harm can remain painful, surprise can be welcome, and mean-zero prediction error does not imply neutral experience. See also: Felt valence; Interoceptive prediction; Predictive processing.

Predictive fit versus explanation: Predictive fit is a model's ability to forecast held-out observations. Explanation additionally relates its variables and dependencies to the process that generated those observations. See also: Candidate signature; Held-out transport; Model recovery.

Predictive processing: A family of accounts in which perception, bodily interpretation, and action depend on generative models, expected signals, prediction errors, uncertainty, and estimated reliability. See also: Active inference; Interoceptive prediction; Prediction error and valence.

Preregistration: Public preregistration records the hypotheses, estimands, gates, measures, exclusions, models, analyses, and decision rules in a timestamped external registry with a persistent identifier before the relevant outcomes are inspected. An internal advance specification is prospectively specified, not publicly preregistered. Deviations remain visible and receive the status appropriate to their timing and purpose. See also: Protocol-fixed and prospectively specified; Registered Report.

Preterminal compression: A candidate reduction in within-stream dispersion before a declared endpoint, evaluated against physiological, observation, and selection explanations.

See also: Candidate signature; Endpoint conditioning; Noise floor.

Primary additive branch: The branch that represents qualifying phenomenal valence as one finite signed scalar measure with a justified neutral origin and additive consistency across disjoint parts of a stream. Its positive and negative parts yield complete-stream magnitudes, and their difference yields the terminal residual. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Probability-one closure: Closure with probability one under a specified probability law, and under a specified policy when the model includes one. It permits measure-zero exceptions and is distinct from set-universal closure, which requires closure for every admissible realization.

Procedural fairness: Fairness in rules and decision paths, including knowable standards, voice, understandable reasons, consistency, limited bias, respectful treatment, correction, and appeal. See also: No moral desert, debt, or payment; Outcome independence; Consent, refusal, and withdrawal.

Process evidence: Evidence about a specified route, mechanism, state transition, or selection process.

Protocol endpoint: A prospectively specified operational event, adjudicated interval, stopping region, or governed proxy used by a study to locate or approximate a relevant boundary. It may be usable, uncertain, revisable, or ethically unavailable and is distinct from both the truth-level endpoint and end of observation. See also: End of observation; Endpoint bridge; Truth-level endpoint.

Protocol-fixed and prospectively specified: A choice is protocol-fixed or prospectively specified when it is recorded

in the study protocol before the relevant outcome is inspected. This timing status does not guarantee that the choice is valid or the study succeeds, and it is not public preregistration without a prior external timestamped record. See also: Endpoint independence; Preregistration; Rival model and equal-budget comparison.

Provenance: The documented lineage, authority, transformations, and version history of a record or derived output. See also: Data as trace; Public-artifact limit; Record-regime separation and no-dossier rule.

Psychometrics: The theory and practice of measuring latent psychological or experiential constructs, including reliability, validity, scaling, response processes, and measurement invariance.

PTVC: See Per-stream terminal valence closure (PTVC).

PTVC core notation: On the primary additive branch, ν_i is the finite signed phenomenal measure on the declared measurable space; V_i^+ and V_i^- are its Jordan masses; $L_i^{phen}(t)$ is the latent phenomenal ledger; $T_i^\star$ is the truth-level endpoint; and $Z_i^\star = \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{phen}(T_i^\star -)$ is the terminal residual.

On the absolutely continuous chart, $r_i(t) = g_{\theta,i}(t) q_i^{\exp}(t) v_i^{real}(t)$. The relation $L_i^{rc}(t) = c_{LU} U_i(t) Y_i(t)$ is a candidate realization, and hatted quantities are operational estimators, not identities with the latent target. See also: Candidate realization; Formula-status code; Phenomenal ledger.

Public access-permission distinction: The distinction between a record's public availability and permission to reuse it for a particular purpose. See also: Consent, refusal, and withdrawal; Data as trace; Record-regime separation and no-dossier rule.

Public-artifact limit: The requirement that a public research artifact accurately preserve its source, result, intended use, and correction route. See also: Detachable-excerpt ceiling; Correction and withdrawal.

Q

Qualifying conscious stream: A stream belonging to the target's declared class under its scope, identity, routing, and inclusion criteria. Difficulty observing a member does not remove it from that class; a measurement-feasible subclass must be declared as narrower.

Queue model: A constructive candidate route that represents unresolved signed valence and asks whether admissible future transitions can remove the remaining residual.

R

Random-endpoint obstruction: Under a specified reference law and declared positive-probability eligible event, suppose a jointly measurable pre-endpoint ledger extension exists on an interval I, endpoint time T is independent of the sigma-field generated by that ledger path on I, and T has a conditional Lebesgue density positive almost everywhere on I. If the ledger is zero at T with probability one, then it is zero with probability one for Lebesgue-almost every time in I. A ledger that is nonzero with positive probability on a positive-measure subset of I is therefore incompatible with probability-one closure on this branch. State-coupled timing, endpoint atoms, singular support, continued experience, routing changes, and endpoint-inclusive value require separate analysis.

Reachability: Whether at least one admissible continuation from a state can reach the declared target. Reachability does

not guarantee selection, common feasibility under partial observation, probability-one success, or universal success.

Record-regime separation and no-dossier rule: Research, clinical, employment, educational, civic, legal, family, spiritual, platform, and public records must retain separate purposes, identifiers, access roles, retention rules, consequences, and correction paths rather than being joined into a master profile. Inferred reconstruction can breach this boundary even without copying a forbidden field. See also: Data minimization and deletion; Human-worth score prohibition; Public access-permission distinction.

Reference class: The prospectively declared domain over which a claim and its quantifier range, specified by eligible streams or cases together with the representation, endpoint regime, population, setting, and measurement architecture. It is not automatically the enrolled sample or analyzed cases; outcome-driven exclusions define a new claim. See also: Held-out transport; Maximal qualifying conscious stream; Set-universal closure.

Reference-class narrowed: See Scientific verdict axes.

Registered Report: A publication format in which a study's question, methods, and analysis plan receive peer review before results are known, with publication not conditioned on obtaining a preferred outcome.

Reinforcement-learning reward signal: A formally specified signal used to update action values or policies from outcomes. Its interpretation follows the model's definition, which may concern relief, safety, access, or another specified result. See also: Active inference; Objective functions versus experience; Predictive processing.

Remaining capacity: The set or amount of admissible future change still available before an endpoint. A closure route must show that remaining capacity is sufficient and selectable under the actual information and support constraints.

Replication: A new test of whether a result, measurement relation, or model distinction recurs under prespecified conditions.

Representation: A mathematical or conceptual encoding of phenomenal valence, time, state, or route that preserves the distinctions required by the stated research target.

Reserve: A route-level variable representing remaining capacity, flexibility, or corrective potential before an endpoint.

Reserve coupling: A candidate relation between accumulated residual and remaining capacity or future transition structure.

Response shift: A change in a person's standards, priorities, values, or understanding of a question after illness, disability, adaptation, treatment, trauma, grief, or aging, altering what the same response scale means over time. A changed answer can reflect changed experience, changed measurement, or both. See also: Calibration; Longitudinal research trace; Measurement invariance.

Reversibility and return rule: A prospectively specified rule under which reliable return of unified access can defeat an earlier classification as a final endpoint, while nonreturn alone cannot prove absence of experience or identify the exact ending. Protocols must state return evidence, revision timing, and its effect on classification. See also: Endpoint bridge; Low-unity state; Protocol endpoint.

Rival dominance: A verdict that an adequate protocol-fixed rival explains the declared signatures with better held-out

performance and fewer strained assumptions under common data, flexibility, and scoring rules. It must state which route, component, or target interpretation loses standing and does not by itself decide unrelated branches. See also: Rival-favored; Theory-level defeat.

Rival-favored: See Scientific verdict axes.

Rival model: A concrete alternative explanation capable of generating the observations relevant to the comparison.

Rival model and equal-budget comparison: A rival is a serious alternative that can generate the relevant pattern without the proposed target or route. Equal-budget comparison makes information access, evaluation rules, and permitted flexibility comparable. See also: Model ambiguity; Model recovery; Scientific verdict axes.

Route: A proposed process or construction through which a target might be reached, with its own state variables, assumptions, and predictions.

Route-failed: See Scientific verdict axes.

Route-favored: See Scientific verdict axes.

Route retirement: A prespecified decision to stop treating a proposed mechanism or route as viable after decisive failure, poor recovery, nonidentification, or superior rival performance. Retiring a route need not settle the target hypothesis.

Route state: A latent or observed variable defined within a candidate mechanism and used to describe its dynamics.

RouteStatus: See Scientific verdict axes.

Routing: The prospectively specified treatment of continuation, pause, restoration, fission, fusion, copying, replacement, and unresolved continuity, including where prior and later

phenomenal contributions are assigned. Routing governs transitions through time and is distinct from ownership, compensation, and causal influence. See also: Fission-and-fusion routing rule; Ownership; Stream identity.

S

Safe perturbation: A reversible, low-burden intervention that changes a declared local feature, such as feedback, delay, expectation, opportunity, or a mild stimulus, to test a named causal contribution. Its meaning depends on target engagement, spillovers, controls, stopping, and the stated causal estimand. See also: Candidate signature; Negative control; No harm for proof.

Scalarization: A prospectively justified mapping from a richer experiential representation to a signed scalar for a specified purpose and domain. See also: Bipolar branch; Generalized Terminal Neutrality; Signed representation.

Scientific verdict axes: Four non-overwriting reporting fields preserve different conclusions: TargetStatus records what the evidence says about the declared target branch; RouteStatus records the status of a candidate route; ComparatorStatus records the absolute adequacy and relative performance of declared rivals; and InferenceStatus records whether the target is identified, partially identified, inconclusive, unusable, or unavailable. Labels such as Target-adverse, Route-failed, Route-favored, Rival-favored, Model-ambiguous, Open-set inadequate, Reference-class narrowed, and No adequate estimate must be assigned only to the relevant field, with branch, reference class, and inferential type retained. No field overwrites another. See also: Decision rule; Model ambiguity; Target-route distinction.

Selection condition: A requirement governing which available action or continuation is actually chosen. An available closing option does not establish probability-one closure if the selected nonclosing event has positive probability. Support-containment and every-continuation claims require their own stronger exclusions. See also: Policy-mixture exactness; Support containment; Every-continuation closure.

Separately routed: Assigned its own continuity and transition structure for evaluating the terminal target.

Set-universal closure: Closure for every admissible realization, path, or member of the declared set. It is stronger than probability-one closure because measure-zero exceptions are not permitted.

Severe test: A test with high probability of exposing the specified error if the claim were false in that way.

Shortening horizon: A reduction in remaining time, opportunity, revision, access, or other route-relevant room for future action, affecting the continuations that remain available. See also: Reachability; Reserve.

Signed representation: A representation that assigns positive and negative directions relative to a justified neutral origin and supplies the declared scale and aggregation rules needed for the target branch. See also: Felt valence; Neutral origin; Primary additive branch.

Signed residual: The cumulative positive phenomenal mass minus the cumulative negative phenomenal mass over a specified domain. The terminal signed residual is evaluated at the truth-level stream endpoint.

Spiritual resonance is not scientific support: The distinction between finding personal or traditional meaning in a proposal

and supplying scientific evidence for its empirical claims. See also: Meaning as optional; Falsifiability.

State-coupled timing: An endpoint branch in which endpoint timing or hazard depends on prospectively specified current biological, consciousness, ledger, or route-state variables while the endpoint ontology and adjudication remain fixed independently of desired closure. Dependence is allowed, but the state relation cannot be invented after outcome inspection. See also: Endpoint atom; First-hitting and absorption routes; Random-endpoint obstruction.

Stop authority: Preassigned, effective power to pause an activity and require action when a declared safety or rights boundary is crossed. See also: Correction and withdrawal; No harm for proof; Public-artifact limit.

Stream: In formal PTVC contexts, shorthand for a Maximal qualifying conscious stream. In broader discussion, see Conscious stream.

Stream identity: The criteria that determine which experiences belong to the same conscious stream. Biological continuity, memory, body, legal identity, and data linkage may inform the analysis but are not automatically decisive.

Strong-form defeat: Defeat of the strong universal branch can occur by two routes. Deductive branch-matched refutation requires an independently verified maximal qualifying stream for which a deterministic identified set, guaranteed to contain the truth under the maintained bearer, representation, origin, domain, endpoint, observation, bridge, and inference assumptions, excludes the registered exact target. Governed scientific withdrawal can also follow repeated adequate conjunction failures or repairs that exhaust distinctive empirical content; only the first route is deductive refutation.

Confidence or posterior objects support only their registered qualified adverse verdicts. See also: Falsifiability; Gate-valid counterexample; Set-universal closure.

Support condition: A restriction on the values, actions, paths, or outcomes that retain positive possibility under a model or policy. Exact claims often require excluding nonclosing support, not merely making it unlikely.

Support containment: Under a declared topology, the support of the relevant probability law lies inside the completion event or target set. This topological modality is not every-continuation closure, positive probability at a point, or proof that an unsampled possibility is absent from support. See also: Every-continuation closure; Modal reachability ladder; Probability-one closure.

T

Target: The truth-level property asserted by a hypothesis under its declared domain and quantifier.

Target-adverse: See Scientific verdict axes.

Target modality and geometry: The exact mathematical form and quantifier of a declared target, including whether it is an exact singleton, finite band, distributional atom, probability-one event, support condition, every-continuation property, or set-universal claim. Evidence and defeat must be matched to that form rather than borrowed across nearby claims. See also: Exact terminal closure; Modal reachability ladder; Terminal residual compatibility object.

Target-route distinction: The target hypothesis states the terminal relation to be evaluated, while a candidate route proposes one optional process that might realize or reproduce its consequences. Evidence updates the object tested; a proved

entailment specifies when a route result also changes a target assessment. See also: Candidate route; Scientific verdict axes; Claim ceiling.

TargetStatus: See Scientific verdict axes.

Terminal neutrality: Book 1's accessible term for a zero terminal signed residual over the whole qualifying stream. It does not mean that the final conscious moment is emotionally neutral.

Terminal residual: The signed total of phenomenal valence over a qualifying stream's complete strict-pre-endpoint domain, evaluated at its truth-level endpoint.

Terminal residual compatibility object: A branch-matched nonempty object representing residual values allowed by declared representation, endpoint, observation, missingness, bridge, and uncertainty assumptions. It retains its inferential type: deterministic identified set, confidence region, credible region, or sensitivity envelope. Only a deterministic guaranteed truth-containing object supports deductive exclusion; confidence and Bayesian objects license only registered coverage- or posterior-qualified conclusions. A point estimate is not such an object. See also: Finite-band compatibility and measured equivalence; Partial identification; Target modality and geometry.

Terminal signed residual: See Terminal residual.

Terminal window: A finite observed period near a presumed endpoint, used to investigate specified temporal or process relations.

Testimony-like output: Language or behavior from an artificial system that has the form of a first-person report but is generated under a particular model, prompt, memory state,

tool set, training regime, and sampling process. It is evidence about configured output, not self-authenticating testimony that a subject feels what is said. See also: AI sovereignty; Candidate sentience escalation; Theory-derived indicator.

The Law: See Law of Fairness.

Theory-derived indicator: An observable property selected because a stated theory connects it to recurrent processing, global availability, higher-order representation, agency, embodiment, metacognition, or another proposed feature of consciousness. See also: Candidate sentience escalation; Comparative consciousness evidence; Testimony-like output.

Theory-level defeat: A branch-specific verdict reached when a declared target is contradicted under its own prerequisites, or when repeated adequate conjunction failures, persistent rival superiority, or outcome-dependent repairs exhaust the claim's distinctive empirical content. It is distinct from local route failure and from nonidentification; the defeat record must state the branch, evidence, assumptions, and surviving claims. See also: Strong-form defeat; Route retirement; Rival dominance.

Truth-containing residual set: See Terminal residual compatibility object.

Truth-level endpoint: The actual boundary of the qualifying stream in the target model. It is conceptually distinct from any observed marker, last report, terminal window, loss of contact, or biological event used to estimate it.

U

Unity gate: See Experienced exposure and unity gate.

Universal branch: See Set-universal closure.

Universal quantifier: A claim applying to every member of a declared class. Because the primary PTVC branch is per-

stream and universal over qualifying streams, a valid adverse case cannot be repaired by averaging across the class.

Unknown category: An unresolved bearer, consciousness, or boundary classification under declared rules, recorded with its reason. It does not imply a negative finding, equal probabilities, zero valence, or No adequate estimate. See also: Bearer; No adequate estimate; Precaution without personhood.

Unusable evidence: Evidence that cannot update the affected claim because a required research or implementation gate failed, even if a statistic can be computed.

It is distinct from Inconclusive evidence and No adequate estimate.

V

Valence: See Felt valence.

Valence density: A representation of signed phenomenal valence per unit of an adopted base measure, such as phenomenal time. Its interpretation depends on the validity of both the valence scale and the base measure.

Valence ledger: Book 1's metaphor for the cumulative signed phenomenal measure within one stream.

Verdict: The declared evidential conclusion produced by a prespecified decision architecture, such as support, adverse result, narrowing, abstention, or route retirement. A verdict must state its target and scope.

Viability: The possibility of remaining within declared constraints over the specified horizon under an admissible continuation or policy, with any probabilistic guarantee stated explicitly. Maintaining access to closure is an additional completion-specific requirement when it is included among those constraints.

Viability kernel: The set of modeled states from which at least one admissible continuation satisfies a route's declared constraints. It marks where a future remains formally workable but does not mean that a continuation is likely, humane, accessible, desirable, or supported by resources a person actually has. See also: Reachability.

Z

Zero-containing interval: An uncertainty or identification interval that includes zero. Any nonzero members remain alternative values under the interval's stated interpretation.

Zero crossing: A time at which a cumulative or instantaneous signed quantity passes through zero. PTVC does not require monotonic movement, frequent zero crossings, or any particular number of crossings before the endpoint.

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The entries below provide bibliographic destinations for the Author/Year citations used throughout the manuscript. Author and institutional creator names appear in bold. Distinct works by the same author in the same year retain their manuscript suffixes. DOI identifiers appear as plain text so the entries remain readable in print. Readers following a claim beyond this book should consult the cited edition, the proposition it supports, and the source’s stated scope.

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About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as "MrGameTheory," Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer

for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009 book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky’s fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build

infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today's generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project's future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity's oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person's question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

The Science of Fairness

If the universe requires that only fair lives can end what unique signature would that rule leave in nature?

If the Law of Fairness is real, nature should carry its mark. A rule governing which conscious lives can truly end could leave patterns in experience, remaining possibilities, and endpoints. The Science of Fairness asks what that signature would look like and whether it can be separated from adaptation, chance, and other known forces.

The book turns lifetime fairness into a scientific research program, moving from mathematics and measurement through simulations, longitudinal studies, nonhuman comparisons, terminal dynamics, and humane experiments that test competing explanations. If that signature survives testing and independent replication, it could reveal a new constraint on conscious life and reshape how we think about experience, choice, and life's boundaries.



LawofFairness.com